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**NISTIR 8271 DRAFT SUPPLEMENT**

# Face Recognition Technology Evaluation (FRTE)

## Part 2: Identification

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*Information Access Division*
*Information Technology Laboratory*

This document is a draft supplement of NIST Interagency Report 8271

2026/09/04

National Institute of
Standards and Technology
U.S. Department of Commerce

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September, 2026

U.S. Department of Commerce
*Howard Lutnick, Secretary*

National Institute of Standards and Technology
*Arvind Raman, Under Secretary of Commerce for Standards and Technology and NIST Director*

# RELEASE NOTES

**2026-09-04**: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-fourth draft update to NIST Interagency Report 8271. It contains results for seven new developer: Innov8tif Solutions, Integrated Software and Technologies Ltd, Virtisha AI, Intpixel Labs Private Ltd, Thakaa Advanced Company for Information Technology, TrueSight Laboratories Tech LLC, and UniverseAI.
- ▷ The document also includes results for algorithms for eight returning developers: Iris ID Systems, Miaxis Biometrics, Nominder, Panasonic R+D Center Singapore, ROC, Sensetime Group, Shanghai Jiao Tong University, and Vsoft.

**2026-08-04**: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-third draft update to NIST Interagency Report 8271. It contains results for three new developer: Intozi Tech Pvt Ltd, Techainer, and Youverse.
- ▷ The document also includes results for algorithms for six returning developers: Compal Electronics, Dermalog, Idemia, NEC, Onfido, and Technology Control Company.

**2026-07-09**: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-second draft update to NIST Interagency Report 8271. It contains results for four new developer: AI Implementation Group, Facehawk Ltd, Fissionsoft CO Ltd, Lebentech Biometrics, NextMeta Labs, STCON LLC, Stellar Technologies, and Xailient.
- ▷ The document also includes results for algorithms for three returning developers: Hyundai Motor Company, IDENTY, Viante.AI, and WINK.

**2026-05-12**: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-first draft update to NIST Interagency Report 8271. It contains results for three new developer: Naver Cloudi, Shufti, and Sparktek.
- ▷ The document also includes results for algorithms for three returning developers: DAON, Panasonic R+D Center Singapore, and Tripleize.

**2026-03-30**: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty draft update to NIST Interagency Report 8271. It contains results for six new developer: Iris ID Systems, Lishui Shuyan Huirong Technology, Miaxis Biometrics, Onfido, Technology Control Company, and Shanghai Jiao Tong University.
- ▷ The document also includes results for algorithms for six returning developers: AllGoVision, Dermalog, Paravision, PT Ice House, Sensetime Group, and ROC.

**2026-02-13**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty ninth draft update to NIST Interagency Report 8271. It contains results for one new developer: RealSense.
- ▷ The document also includes results for algorithms for eight returning developers: Adiance Technologies Pvt Ltd, GTEL OTS JSC, IDENTY, Protenet Soluções em Segurança, Toshiba, STCON LLC, Viante.AI, Viettel Artificial Intelligence.

**2026-01-09**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty eighth draft update to NIST Interagency Report 8271. It contains results for three new developer: Biometrik Systems Private Limited, FAST-NUCES Lahore Campus, and Sparktek International Pte Ltd.

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- ▷ The document also includes results for algorithms for seven returning developers: Clearview AI Inc, Compal Electronics, Hyundai Motor Company, Innovatrics, NEC, Panasonic R+D Center Singapore, and VDF Software.

**2025-09-30**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty seventh draft update to NIST Interagency Report 8271. It contains results for three new developer: LightVision Inc, Protenet Soluções em Segurança, and Toss.
- ▷ The document also includes results for algorithms for one returning developers: Dermalog.

**2025-08-27**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty sixth draft update to NIST Interagency Report 8271. It contains results for one new developer: Private Identity LLC.
- ▷ The document also includes results for algorithms for thirteen returning developers: AllGoVision, DAON, Incode Technologies Inc, Metsakuur, NextMedia Labs, Neurotechnology, Omnigarde, Panasonic R+D Center Singapore, Pentacomp Systemy Informatyczne, ROC, TigerIT Americas LLC, Toshiba, and VinBigData.

**2025-07-17**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty fifth draft update to NIST Interagency Report 8271. It contains results for one new developers: IDENTY.
- ▷ The document also includes results for algorithms for six returning developers: Hyundai Motor Company, Idemia, Innovatrics, Paravision, STCON LLC, and Universidade de Coimbra.

**2025-05-28**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty fourth draft update to NIST Interagency Report 8271. It contains results for three new developers: Adiance Technologies, RNIT Solutions and Services, and VDF Software.
- ▷ The document also includes results for algorithms for four returning developers: Cubox, Dermalog, QazSmartVision.AI, and Samsung S1.

**2025-04-25**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty third draft update to NIST Interagency Report 8271. It contains results for five new developers: Compal Electronics, FCI Vietnam, GTEL OTS JSC, i2V Systems, and ioNetworks.
- ▷ The document also includes results for algorithms for five returning developers: Intel Research Group, Mobbeel Solutions, Pyramid Cyber Security + Forensic (P), Viante.AI, and Viettel Cyberspace Center.

**2025-03-18**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty second draft update to NIST Interagency Report 8271. It contains results for two new developers: Algorithms Inc, and Maverick Research.
- ▷ The document also includes results for algorithms for four returning developers: AFR Engine, Akuvox, EI Networks, and VinBigData.

**2025-03-05**: The 1:N track of the FRTE remains open.

- ▷ This document is the forty first draft update to NIST Interagency Report 8271. It contains results for three new developers: Awidit Systems, Hyundai Motor Company, and Kloudspot.
- ▷ The document also includes results for algorithms for seven returning developers: Innovatrics, Inspur (Beijing) Electronic Information Industry, NEC, Nominder, ROC, STCON, and Viettel Group

**2025-02-05**: The 1:N track of the FRTE remains open.

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- ▷ This document is the fortieth draft update to NIST Interagency Report 8271. It contains results for three new developers: Keyless Technologies, Pentacomp Systemy Informatyczne, and TTI Group.
- ▷ The document also includes results for algorithms for four returning developers: DAON, Hangzhuo AIlu Network Information Technology, Unissey and useB.

**2024-12-18**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty nine draft update to NIST Interagency Report 8271. It contains results for two new developers: Pyramid Cyber Security + Forensic, and Shennona.
- ▷ The document also includes results for algorithms for nine returning developers: Alchera, Cubox, IMDS Software, Paravision, QazSmartVision.AI, Samsung S1, Sensetime, Vietnam Posts and Telecommunications Group, and Vision-Box.

**2024-11-08**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty eighth draft update to NIST Interagency Report 8271. It contains results for six new developers: FacePhi, Metsakuur, Nextmeta Labs, Scylla Technologies, Seventh Sense Artificial Intelligence, and Vietnam Payment Solution.
- ▷ The document also includes results for algorithms for four returning developers: EI Networks, Innovatrics, Nominder, STCON, Viettel Group, VinBigData, and WINK.

**2024-09-18**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty seventh draft update to NIST Interagency Report 8271. It contains results for three new developers: Antheus Technologia, Beijing Hisign Technology, and PT Ice House.
- ▷ The document also includes results for algorithms for four returning developers: Cognitec Systems, Neurotechnology, Qnap Security, and useB.
- ▷ In addition, many graphs for algorithms submitted between June 2023 and August 2024 have been included for the first time. They had incorrectly been omitted from several reports issued in that timeframe.

**2024-08-09**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty sixth draft update to NIST Interagency Report 8271. It contains results for one new developer: NextMedia Labs, OptiExacta and Universidade de Coimbra.
- ▷ The document also includes results for algorithms for seven returning developers: AFR Engine, Incode, Panasonic R+D Center Singapore and QazSmartVision.AI.

**2024-07-03**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty fifth draft update to NIST Interagency Report 8271. It contains results for one new developer: Sansap Technology.
- ▷ The document also includes results for algorithms for seven returning developers: Alchera, Clearview AI, Cyberlink Corp, Dermalog, Nominder, Omnigarde, and ROC.

**2024-05-30**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty forth draft update to NIST Interagency Report 8271. It contains results for seven new developers: ALTTEKGlobal, Aratek Biometrics, BioID Technologies, EI Networks Private, Kogniza Technology, STCON, and VinBigData.
- ▷ The document also includes results for algorithms for five returning developers: AadiSysTech, Cognitec Systems, Hangzhuo AIlu Network Information Technology, Innovatrics, and Mobbeel Solutions.

**2024-04-29**: The 1:N track of the FRTE remains open.

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- ▷ This document is the thirty third draft update to NIST Interagency Report 8271. It contains results for one new developer: FPT Smart Cloud
- ▷ The document also includes results for algorithms for one returning developer: Vision-Box

**2024-03-26**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty second draft update to NIST Interagency Report 8271. It contains results for one new developer: QazSmartVision.AI
- ▷ The document also includes results for algorithms for eight returning developers: AFR Engine, Corsight AI, Dactionable Technologies, Idemia, IMDS Software, Megvii/Face++, Panasonic R+D Center, and Unissey.

**2024-02-21**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty first draft update to NIST Interagency Report 8271. It contains results for one new developer: Yuan High-Tech Development.
- ▷ The document also includes results for algorithms for two returning developers: Clearview AI, and useB.

**2024-01-22**: The 1:N track of the FRTE remains open.

- ▷ This document is the thirtieth draft update to NIST Interagency Report 8271. It contains results for two new developer: Sparsh CCTV, and VNIS Joint Stock.
- ▷ The document also includes results for algorithms for ten returning developer: Dermalog, ICM Airport Technics, Innovatrics, Maxvision, Viettel Cyberspace Center, NEC, Neurotechnology, Nominder, ROC, and WINK.

**2023-12-18**: The 1:N track of the FRTE remains open.

- ▷ This document is the twenty nineth draft update to NIST Interagency Report 8271. It contains results for one new developer: Mobbeel Solutions.
- ▷ The document also includes results for algorithms for seven returning developer: AadiSysTech, Alchera, Aware, Cognitec Systems, Inspur (Beijing) Electronic Information, Qnap Security, and Vietnam Posts and Telecommunications Group.

**2023-11-20**: The 1:N track of the FRTE remains open.

- ▷ This document is the twenty eighth draft update to NIST Interagency Report 8271. It contains results for four new developer: Dactionable Technologies, IMDS Software Inc, Luxand Inc, and Unissey.
- ▷ The document also includes results for algorithms for seven returning developer: AllGoVision, Hangzhuo AIlu Network Information Technology, Kakao Brain, Megvii/Face++, Recognito, Verigram, and Viante.AI.

**2023-09-29**: The 1:N track of the FRTE remains open.

- ▷ This document is the twenty seventh draft update to NIST Interagency Report 8271. It contains results for one new developer: Nominder.
- ▷ The document also includes results for algorithms for one returning developer: useB.

**2023-09-09**: The 1:N track of the FRTE remains open.

- ▷ We have renamed and split FRVT into two sets of technology evaluations: The Face Recognition Technology Evaluation (FRTE) is the umbrella for the ongoing 1:1, 1:N and twins disambiguation tracks; the Face Analysis Technology Evaluation (FATE) holds the morph and attacked detection tracks, the quality assessment tracks, and the new age estimation tracks.

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- ▷ This document is the twenty sixth draft update to NIST Interagency Report 8271. It contains results for two new developers: ICM Airport Technics and Papilon Savunma.
- ▷ The document also includes results for algorithms for two returning developers: Canon and Tripleize.

**2023-08-16**: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty fifth draft update to NIST Interagency Report 8271. It contains results for two new developers: Panasonic and Sumsub.
- ▷ The document also includes results for algorithms for two returning developers: Maxvision and Innovatrics.

**2023-07-28**: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty fourth draft update to NIST Interagency Report 8271. It contains results for four new developers: AFR Engine, Corsight, Wink and Viettel Cyberspace Center.
- ▷ The document also includes results for algorithms for seven returning developers: Alchera, Cubox (cubox.aero), NEC, Rank One Computing, Samsung S-1, Tech5 SA, and Verihubs Inteligensia.

**2023-07-05**: The 1:N track of the FRVT remains open.

- ▷ This document corrects mugshot-mugshot results for algorithms submitted to FRVT 2023-02 to 2023-06. The prior report's FNIR values were uniformly approximately 0.003 higher due to a bug introduced into a script by changes to the FRVT API.

**2023-07-03**: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for eight first-time participants: Advance.AI, AFIS and Biometrics Consulting, Inspur (Beijing) Electronic Information Industry, Know U Tech, Omnigarde, Serendipity, useB, and Verigram.
- ▷ The document also includes results for algorithms for four returning developers: Hangzhuo AIlu Network Information Technology, Megvii/Face++, Paravision, and Qnap Security.

**2023-04-20**: The 1:N track of the FRVT is closed until 2023-05-01.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Armatura LLC.
- ▷ The document also includes results for algorithms for two returning developers: HyperVerge, Maxvision.

**2023-03-29**: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Recognito.
- ▷ The document also includes results for algorithms for one returning developer: AllGoVision Technologies.

**2023-03-14**: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty second draft update to NIST Interagency Report 8271. It contains results for no first-time participants.
- ▷ The document also includes results for algorithms from nine returning developers: Cloudwalk - Moontime Smart Technology, Line Corporation, Intema-LGL Group Neurotechnology, NEC, Paravision, Samsung S1, Veridas Digital Authentication Solutions, and Thales Group

**2023-02-06**: The 1:N track of the FRVT remains open.

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- ▷ This document is the twenty first draft update to NIST Interagency Report 8271. It contains results for no first-time participants.
- ▷ The document also includes results for algorithms from seven returning developers: Dermalog, DiluSense Technology, Hangzhuo AIlu Network Information Technology, Idemia, Innovatrics, Rank One Computing, and Sensetime Group.

**2022-12-15**: The 1:N track of the FRVT remains open.

- ▷ This document is the twentieth draft update to NIST Interagency Report 8271. It contains results for one first-time participant: First Credit Bureau Kazakhstan.
- ▷ The document also includes results for algorithms from five returning developers: Gorilla Technology, Pangiam, Qnap Scurity, SQIsoft, Vixvizion (formerly known as Imagus).

**2022-11-09**: The 1:N track of the FRVT remains open.

- ▷ This document is the nineteenth draft update to NIST Interagency Report 8271. It contains results for four first-time participant: Mukh, Turing Technology VIP, Verijelas and Verihubs Inteligensia
- ▷ The document also includes results for algorithms from two returning developers: Maxvision and Samsung S1.

**2022-09-23**: The 1:N track of the FRVT remains open.

- ▷ This document is the eighteenth draft update to NIST Interagency Report 8271. It contains results for two first-time participants: Intema-LGL Group and T4iSB.
- ▷ The document also includes results for algorithms from two returning developers: Cloudwalk - Moontime Smart Technology, Dermalog, Griaule, Hangzhuo AIlu Network Information Technology, Intellivision, Line Corporation, NEC, Sensetime Group, Realnetworks Inc and Vietnam Posts and Telecommunications Group.

**2022-07-28**: The 1:N track of the FRVT remains open.

- ▷ This document is the seventeenth draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Maxvision.
- ▷ The document also includes results for algorithms from two returning developers: Rank One Computing, and Viettel Group.
- ▷ We have replaced the probe set used in the visa-border benchmark. It was previously comprised of 80 000 images; it now has size 1 212 892 - see amended entries in Table 1. False negative identification rates have increased.
- ▷ We have added images to the probe set used in the visa-kiosk benchmark. It was previously comprised of 21 016 mates and the same number of non-mates; it now has 31 579 mates and 45 460 non-mates - see amended and entries in Table 1. False negative identification rates are improved (reduced) slightly.

**2022-06-08**: The 1:N track of the FRVT remains open.

- ▷ This document is the seventeenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by three first-time participants: Digidata, DiluSense Technology, and Vietnam Posts and Telecommunications Group.
- ▷ The document also includes results for algorithms from five returning developers: Canon Inc, Imagus Technology, Neurotechnology, Thales, and Samsung S1.

**2022-04-28**: The 1:N track of the FRVT remains open.

- ▷ This document is the sixteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by one first-time participants: Hangzhuo AIlu Network Information Technology.

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- ▷ The document also includes results for algorithms from three returning developers: HyperVerge Inc, Qnap Security, and Realnetworks Inc.
- ▷ The 1:N results page has been updated.

**2022-03-30**: The 1:N track of the FRVT remains open.

- ▷ This document is the sixteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by two first-time participants: Intellivision, and Pangiam.
- ▷ The document also includes results for algorithms from three returning developers: Fujitsu Research and Development Center, Idemia, and Gorilla Technology.
- ▷ The 1:N results page has been updated.

**2022-02-23**: The 1:N track of the FRVT remains open.

- ▷ This document is the fifteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by four first-time participants: Cloudwalk - Moontime Smart Technology, Decatur Industries Inc, NotionTag Technologies Private Limited, and Reveal Media Ltd.
- ▷ The document also includes results for algorithms from three returning developers: Cognitec Systems GmbH, Sensetime Group, and Viettel Group
- ▷ The 1:N results page has been updated.

**2022-01-20**: The 1:N track of the FRVT remains open.

- ▷ This document is the fourteenth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by two first-time participants: Daon and SQIsoft.
- ▷ The document also includes results for algorithms from five returning developers: Cyberlink Corp, NEC, Neurotechnology, Paravision, and Rank One Computing.
- ▷ The 1:N results page has been updated.

**2021-12-16**: The 1:N track of the FRVT remains open.

- ▷ This document is the thirteenth draft update to NIST Interagency Report 8271. It includes results for algorithms from six returning developers: Dahua Technology, Imagus Technology, Line Corporation, N-Tech Lab, Qnap Security, and Realnetworks Inc.
- ▷ The 1:N results page has been updated.

**2021-11-22**: The 1:N track of the FRVT remains open.

- ▷ This document is the twelfth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by three first-time participants Clearview AI, Griaule, and Mantra Softech India.
- ▷ This document and the 1:N results page also include results for algorithms from six returning developers: Acer Incorporated, Canon, Dermalog, Samsung S1, VisionLabs, and Veridas Digital Authentication.

**2021-10-28**: The 1:N track of the FRVT remains open.

- ▷ This document is the eleventh draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by three first-time participants (20Face, Fujitsu Research and Development Center, and Vision-Box), and five returning participants (Alchera, Gorilla Technology, Tevian, Thales-Cogent, and Visidon). Visidon
- ▷ Both the main 1:N results page and the small-gallery paperless travel page have been updated.

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**2021-09-21**: The 1:N track of the FRVT remains open. Three news items:

- This document is the tenth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by six first-time developers: Cubox, Fincore, HyperVerge, Qnap Security, Staqu Technologies, and Tripleize (Aize, 3-ize).
- It includes results also for four returning developers: Cognitec Systems, Incode Technologies, Innovatrics, Neurotechnology, and Rank One Computing.

**2021-08-02**: The 1:N track of the FRVT remains open. Three news items:

- This document is the nineth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by eight participants: Cyberlink Corp, NEC Corp, N-Tech Lab, Realnetworks Inc., Sensetime Group, Veridas Digital, Viettel Group, and Vigilant Solutions.
- Algorithms submitted since July 24 will be included in the next update scheduled for September 9, 2021.
- A new report, NIST Interagency Report 8381 - FRVT Part 7: Identification for Paperless Travel and Immigration, has been released [PDF, webpage]. It documents the use of FRVT 1:N algorithms in positive access control and immigration status update travel applications where the enrolled population size is as low as 420 people for aircraft boarding, and 42 000 for an airport security line. These population sizes are much smaller than those used in the main 1:N evaluation. Going forward, we will update the report and webpage with results for new algorithms.

**2021-07-07**: The 1:N track of the FRVT remains open. One update:

- This document is the eighth draft update to NIST Interagency Report 8271. It include results for an algorithm from one participant: Kakao Enterprises.

**2021-06-22**: The 1:N track of the FRVT remains open. Three updates:

- This is the seventh draft of the update to NIST Interagency Report 8271. It includes results for algorithms from three new participants: Line Corporation, Rendip, and Samsung S1 Corp.
- We have also added results for algorithms from five returning developers: Imagus Technology, Kneron, Tevian, Visidon, and Xforward AI Technology.
- The algorithm-specific report cards (examples: 1, 2, and 3) now include figures showing how low threshold values can be used to reduce candidate list lengths for human review, while (usually) elevating miss rates (FNIR) only modestly. The reports also feature some minor additions and clarifications.

**2021-03-26**: The 1:N track of the FRVT remains open. Three updates:

- This is the sixth draft of the update to NIST Interagency Report 8271. It includes results for algorithms from three returning developers: Neurotechnology, Guangzhou Pixel Solutions, and Tech5 SA.
- We have added results on the webpage and in the report for a new ageing dataset in which border crossing photos are searched against a gallery of border crossing photos collected between 10 and 15 years prior to the mated search photos. See section 2 for a description of the images. Table 1 has a new entry describing the experiment.
- We will mostly discontinue running the mugshot ageing test, reserving it for algorithms that show high accuracy on the new border-crossing set.

**2021-03-26**: Regarding the fifth draft of the update to NIST Interagency Report 8271:

- In addition have added results for first algorithms from two new participants: Viettel Group and Veridas Digital Authentication Solutions.
- We have added results for algorithms from two returning developers: Idemia and Cognitec Systems.

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- In addition to the report, the results page and its hyperlinked report cards have been updated.

**2021-02-08**: Regarding the fourth draft of the update to NIST Interagency Report 8271:

- We have added results for eight algorithms submitted by eight developers: Cyberlink, Dermalog, Imagus, Paravision, Sensetime, Trueface, Vigilant Solutions, and X-Forward AI. With the exception of Trueface, all of these developers have participated previously.
- We anticipate updating this report again in the first week of March 2021.
- The main results page has been revised with tabs for the investigative and lights-out identification tables, and a new tab dedicated to speed and resource consumption.
- The report cards (example here) hyperlinked from the results page have been revised to improve content and format.

**2020-12-14**: Regarding third draft of the update to NIST Interagency Report 8271:

- We have added results for fifteen algorithms submitted by thirteen developers. The four first-time participants are: Acer, Akurat Satu Indonesia, Canon, and Xforward AI Technology. The ten returning developers are: AllGoVision, Cyberlink Corp, Dahua Technology, Deepglint, Guangzhou Pixel Solutions, IIT Vision, Innovatrics, Rank One Computing, Scanovate, Sensetime Group, Synesis, and VisionLabs.
- We have added two new datasets to the evaluation: First a set of "visa-border" photos, representing search of an airport immigration lane photo against a database of closely ISO standard portraits; second a "visa-kiosk" set representing search of a photo collected in a registered traveller kiosk against the same ISO portrait gallery. The images are described in section 2.1.
- As in previous reports, we include results for searching mugshots against a mugshot gallery containing a single image of each of 12 million people. However we have suspending running searches against a gallery in which multiple lifetime photos per person are present, because this is computationally expensive. We retain a N = 3 million search test dedicated to ageing in which mugshots taken up to 18 years after the first photograph are searched - see Table 13.
- Tables containing computational resource information, Table 2. . ., now include duration of the finalization step, in which search algorithms can, at their option, build fast-search data structures.
- We have linked revised per-algorithm PDF report cards from the main results page.
- We have regenerated all figures and tables to drop algorithms submitted before June 2018. Results for prior algorithms appear in archived editions of this report.
- Going forward, we anticipate producing more frequent updates to this report. Developers may submit one algorithm to this evaluation every four calendar months.

**2020-03-24**: Regarding the second draft of the update to NIST Interagency Report 8271:

- Adds results for three algorithms from three developers, Dermalog, Innovatrics, and Synesis.
- Adds Table 13 on ageing showing the increase in false negative rates with time elapsed between two photos. Some of the results were contained in graphs in prior editions of this report, but the table adds results for some newly submitted algorithms.
- Adjusts frontal mugshot results (for recent and lifetime consolidated galleries) to include the effect of removing some images that should not have been included in image test sets. These images were mostly profile views, images of tattoos containing faces, images of faces on tee shirts, and images of photographs on walls behind the intended subject. This affects many tables and reduces false negative identification rates for all algorithms. The reduction is larger for "recent" enrollments than for "lifetime consolidated" ones with the consequence that accuracy on recent images is now superior.

**2020-02-26**: Regarding the first draft of the update to NIST Interagency Report 8271:

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- Adds results for 38 algorithms from 31 different developers, eleven of whom are entirely new to the 1:N track of FRVT. These are Allgovision, Cyberlink, Deepsea Tencent, Farbar F8, Imperial College London, Intsys MSU, Kedacom, Kneron, Pixelall, and Scanovate.

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Specific hardware and software products identified in this report were used in order to perform the evaluations described in this document. In no case does identification of any commercial product, trade name, or vendor, imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the products and equipment identified are necessarily the best available for the purpose.

## INSTITUTIONAL REVIEW BOARD

The National Institute of Standards and Technology's Research Protections Office reviewed the protocol for this project and determined it is not human subjects research as defined in Department of Commerce Regulations, 15 CFR 27, also known as the Common Rule for the Protection of Human Subjects (45 CFR 46, Subpart A).

## ACKNOWLEDGMENTS

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# Executive Summary

This document is a draft revision of the September 2019 report NIST Interagency Report 8271. That report gave extensive documentation of face recognition applied to mugshots. This report extends that by adding more two more challenging datasets containing images with serious departures from canonical frontal image standards. The report also adds results for algorithms submitted to NIST since in 2019 and 2020. The algorithms, which implement one-to-many identification of faces appearing in two-dimensional images, are prototypes from the research and development laboratories of mostly commercial suppliers, and are submitted to NIST as compiled black-box libraries implementing a NIST-specified C++ test interface. The report therefore does not describe how algorithms operate. The report lists accuracy results alongside developer names and will therefore be useful for comparison of face recognition algorithms and assessment of absolute capability. The report is accompanied by a webpage with sortable results.

The evaluation uses six datasets: frontal mugshots, profile view mugshots, desktop webcam photos, visa-like immigration application photos, immigration lane photos, and registered traveler kiosk photos. These datasets are sequestered at NIST, meaning that developers do not have access to them for training or testing. This aspect is important because face recognition algorithms are very often deployed without the developer having access to the customers image data. A possible exception to this would be in a cloud-based application where the operational image data is uploaded to a cloud operated by a face recognition developer.

The major result in NIST IR 8271 was that massive gains in accuracy have been achieved in the years 2013 to 2018 and these far exceed improvements made in the prior period, 2010 to 2013. While the industry gains were broad - at least 30 developers' algorithms outperformed the most accurate algorithm from late 2013, there remains a wide range of capability. While this report shows accuracy gains only over the period 2018-2020, the most accurate algorithm reported here is substantially more accurate than anything reported in NIST IR 8271. This is evidence that face recognition development continues apace, and that FRVT reports are but a snapshot of contemporary capability.

From discussion with developers, the accuracy gains stem from the adoption of deep convolutional neural networks. As such, face recognition has undergone an industrial revolution, with algorithms increasingly tolerant of poorly illuminated and other low quality images, and poorly posed subjects. One related result is that a few algorithms correctly match side-view photographs to galleries of frontal photos, with search accuracy approaching that of the best c. 2010 algorithms operating on purely frontal images. The capability to recognize under a 90-degree change in viewpoint - pose invariance - has been a long-sought milestone in face recognition research.

With good quality portrait photos, the most accurate algorithms will find matching entries, when present, in galleries containing 12 million individuals, with rank one miss rates of approaching 0.1%. The remaining errors are in large part attributable to long-run ageing, facial injury and poor image quality. Given this impressive achievement - close to perfect recognition - an advocate might claim that cooperative face recognition is a solved problem, a statement that can be refuted with the following context and caveats:

- **Mugshots vs. less constrained captures:** The low error rates reported here are attained using mostly excellent cooperative live-capture mugshot images collected with an attendant present. Recognition in other circumstances, particularly those without a dedicated photographic environment and human or automated quality control checks, will lead to declines in accuracy. This is documented here for side-view images, poorer quality webcam images, and, particularly, for newly introduced ATM-style kiosk photos that were not originally intended for automated face recognition. In this case, recognition error rates are much higher, often in excess of 20% even with the more accurate algorithms which variously remain intolerant of face cropping (at image edge) and of large downward head pitch.
- **Algorithm accuracy spectrum:** Recognition accuracy is very strongly dependent on the algorithm and, more

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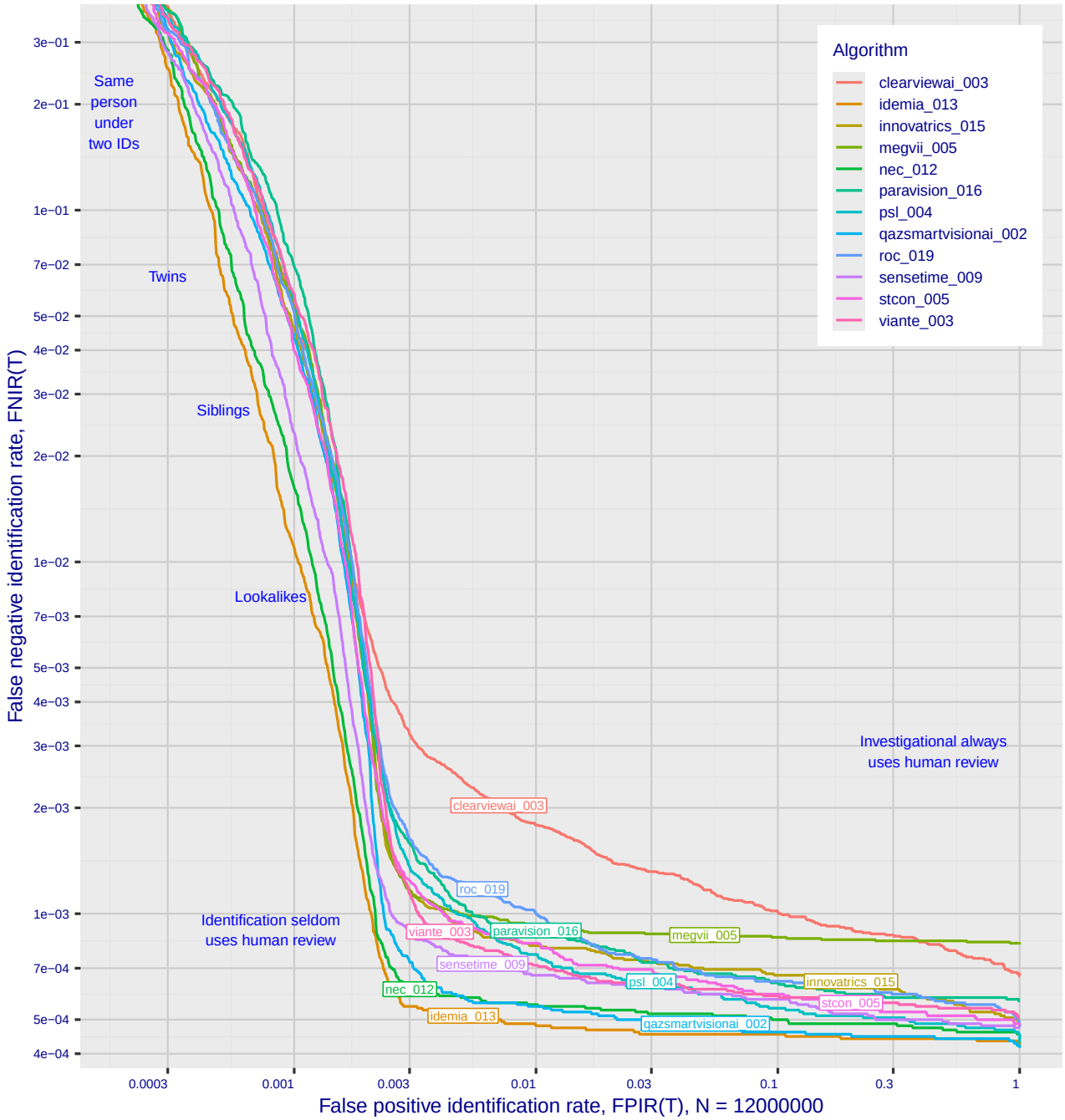


*Figure 1: Identification miss rates across the false positive range. N = 12 million individuals are enrolled with one recent image.*

generally, on the developer of the algorithm. False negative error rates in a particular scenario range from a few tenths of one percent to beyond fifty percent. This is tabulated exhaustively later: For example Table 23 shows accuracy across datasets. Figure 1 here compares algorithms on mugshot searches in a consolidated gallery of 12 million subjects and 12 million photos. Many algorithms do not achieve the low error rates noted above, and while many of those may still be useful and valuable to end-users, only the most accurate excel on poor quality images and those collected long after the initial enrollment sample.

▷ **Versioning:** While results for up to ten algorithms from each developer are reported here, the intra-provider

accuracy variations are usually smaller than the inter-provider variations. That said different versions give an order of magnitude fewer misses. Some developers demonstrate speed-accuracy tradeoffs[1]. See Figs. 18, 19.

- **Low similarity scores:** In thousands of mugshot cases the correct gallery image is returned at rank 1 but its similarity score is nevertheless low, below some operationally required score threshold. This is not so important when face recognition is used for "lead generation" in investigational applications because human reviewers are specifically required to review potentially long candidate lists and the threshold is effectively 0. In applications where search volumes are higher and labor is not available to review the results from searches, a higher threshold must be applied. This reduces the length of candidate lists and false positive identification rates at the expense of increased false negative miss rates. The tradeoff between the two error rates is reported extensively later.

- **Population size:** As the number of enrolled subjects grows, some mates are displaced from rank one, decreasing accuracy. As tabulated later for N up to 12 million, false negative rates generally rise slowly with population size. This enables use of face recognition in very large populations. However in most positive and negative identification applications[2], a score threshold is set to limit the rate at which non-mate searches produce false positives. This has the consequence that some mated searches will report the mate below threshold, i.e. a miss, even if it is at rank 1. The utility of this is that many non-mated searches will return no candidate identities at all. As the error-tradeoff characteristic shows, investigational miss rates on the right side are very low but then rise steadily (in the center region) as threshold is increased to support "lights-out" applications, and ultimately rise quickly (left side) as discussed below. Thus, if we demand that just one in one thousand non-mate searches produce any false positives, the most accurate algorithms there (Sensetime-004 and NEC-3) would fail on between 3 and 5% of mated searches. Even though the graph shows results for the most accurate algorithms, all but two would fail to find the mate in more than 8% of mated searches. While the two most accurate algorithms produce a relatively flat error tradeoff until the threshold is raised to limit false positives to about 1 in 400 non-mated searches[3]

  Thereafter, as the threshold is raised to further reduce false positives, miss rates rise rapidly. This means that low false positive identification rates are inaccesible with these algorithms, a result that does not apply for ten-finger identification algorithms. The rapid rise occurs because the lower mate scores are mixed with very high non-mate scores, the low scores from poor image quality and ageing, the high non-mates from the presence of lookalikes persons (doppelgangers), twins (discussed next) and, ultimately, the presence of a few unconsolidated subjects i.e. persons present under multiple IDs.

- **False negatives from ageing:** A large source of error in long-run applications where subjects are not re-enrolled on a set schedule is ageing. Changes in facial appearance increase with the time elapsed between photographs. These will depress similarity scores and eventually cause false negatives. All faces age and while this usually proceeds in a graceful and progressive manner, drug use can accelerate this [28]. Elective surgery may be effective in delaying it although this has not been formally quantified with face recognition. As ageing is essentially unavoidable, it can only be mitigated by scheduled re-capture, as in passport re-issuance. To quantify ageing effects, we used the more accurate algorithms to enroll the earliest image of 3.1 million adults and then search

[1] For example, NEC-0 prepares templates much faster than NEC-2 but gives twenty times more misses. Dermalog-5 executes a template search much more quickly than Dermalog-6 but is also much less accurate.

[2] In a positive identification application such as a registered traveler system, a user is making an implicit claim to be enrolled in the system - most users will be. In a negative application, such as with deportees, the implicit claim is that the subejct is not enrolled - most will not be.

[3] The gallery size here is 12 million people, one image per person. Given 331 201 non-mated searches, an exhaustive implementation of one-too-many search would execute almost 4 trillion comparisons. At a false positive identification rate of 0.0025 the number of false positives is, to first order, 828 corresponding to single-comparison false match rate of 828 / 4 trillion = 2.1 $10^{-10}$ i.e. about 1 in 5 billion. Strictly this FMR computation is meaningful only for algorithms that implement 1:N search using N 1:1 comparisons, which is not always the case.

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with 10.3 million newer photos taken up to 18 years after the the initial enrollment photo. Figure 2 puts ageing into context by contrasting it with the increase in false negatives that occurs when the number of individuals in an enrollment database becomes larger and the chance of a false positive increases such that higher thresholds may become necessary[4].

The Figure shows, from to bottom, increases in false negative identification rates (FNIR) with the algorithm being tested. This applies to increases due to N on the left side, and increases due to ageing on the right side. The relative spacing of the dots shows that for all algorithms the dependency of FNIR on $N$ (up to 12 million) is considerably less than on $\Delta T$ (up to 18 years). The figure additionally shows the most accurate nine algorithms for each year in which they were submitted, limiting to only one per developer.

In the inset table, accuracy is seen to degrade progressively with time, as mate scores decline and non-mates displace mates from rank 1 position. More accurate algorithms tend to be less sensitive to ageing. The more accurate algorithms give fewer errors after 18 years of ageing than middle tier algorithms give after four. Note also we do not quantify an ageing rate - more formal methods [2] borrowed from the longitudinal analysis literature have been published for doing so (given suitable repeated measures data). See Figures 148, 216 and 249.

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[4]Some algorithms implement stragegies to automatically adjust scores to account for increased population size. This relieves the system owner of having to increase thresholds as N increases.

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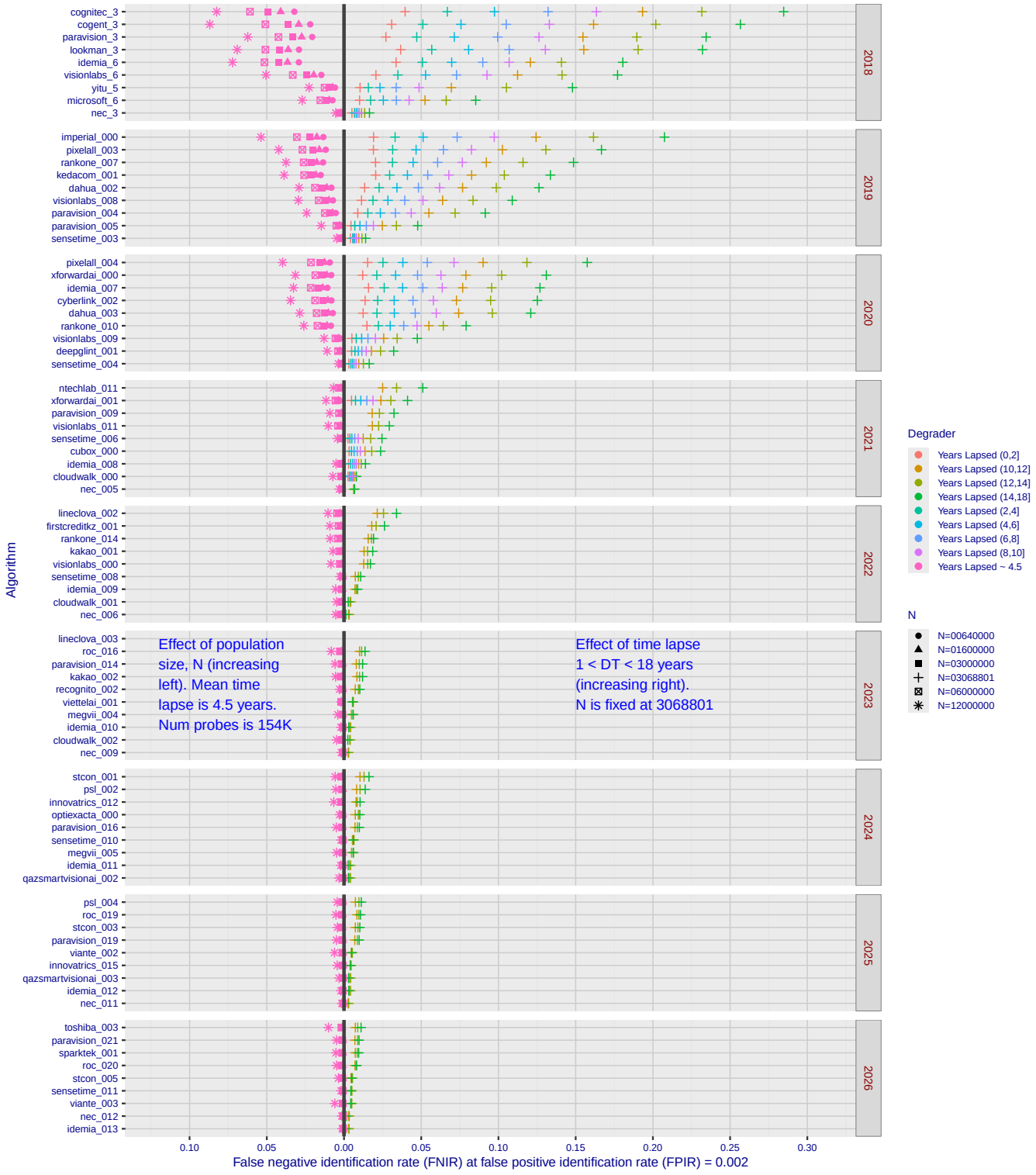


*Figure 2: Identification miss rates as a function of enrolled population size, $N$, and time-lapse, $\Delta T$.*

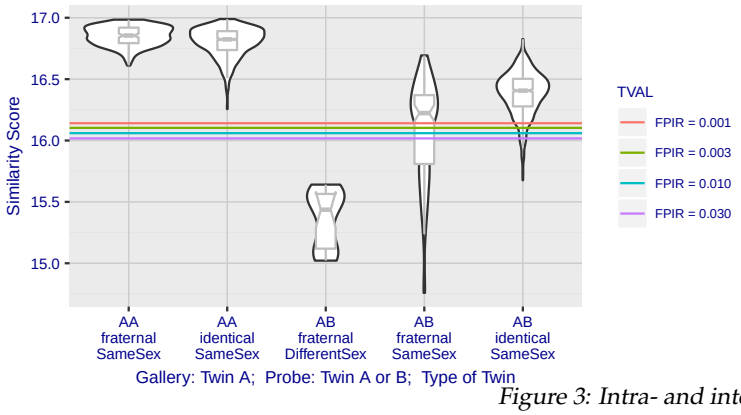


*Figure 3: Intra- and inter-twin scores*

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- **False positives from twins:** By enrolling 640 000 mugshots, adding photos of one twin, and then searching photos of those subjects and their twin the inset figure shows, for one typical algorithm, the similarity is generally greater when searching twins against themselves (A) than when searching twins against their sibling (B) but very often still above even stringent thresholds i.e. those corresponding to one in one thousand searches producing a false positive. Thus twins will very often produce a high-scoring non-match on a candidate list and a false alarm in an online identification system. The plot of Fig. 3 shows that fraternal twins are sometimes correctly rejected at those thresholds - including most different sex twins (at center). Figure **??** shows substantially similar behavior for all algorithms tested. In an investigative search, a twin would typically appear at rank 1, or rank 2 if their sibling happened to also be the gallery. Twins (and triplets etc.) constituted 3.3% of all live births [17] in recent years[5], and because that number is higher today than when the individuals in current adult databases were born, the false positives that arise from twins are now, and will increasingly be, an operational problem. Relative to the United States, twins are born with considerable regional variation. For example they are much less common in East Asia, and much more common in Sub-Saharan Africa [21].

  The presence of twins in the mugshot database is inevitable given its size, around 12.3 million people. As this is not an insignificant sample of the domestic United States population, people with other familial ties will be present also. The data was collected over an extended period and because location information is not available, we are unable to estimate the proportion of the domestic population that is present in the dataset. However, if we assume twins are neither more or less disposed to arrest than the general population, we can estimate that hundreds of thousands of individuals in the dataset are twins. This will affect false positive rates because we randomly set aside 331 201 individuals for nonmate searches, and some proportion of those will be twins with siblings in the gallery.

- **Database integrity:** An operational error rate should be added to all false negative rates in this report reflecting the proportion of images in a real database that are un-matchable. Such anomalies arise from images that: do not contain a face; include multiple persons; cannot be decoded; are rotated by 90° or 180°; depict a face on clothing; and others introduced by a long tail of various clerical errors. While the mugshot trials in this report have been constructed to minimize such effects, they are a real problem in actual operations.

This report is being updated continuously as new algorithms are submitted to FRVT, and run on new datasets. Participation in the one-to-many identification track is independent of participation in the one-to-one verification track of FRVT.

[5]See the CDC's National Vital Statistics Report for 2017: https://www.cdc.gov/nchs/data/nvsr/nvsr67/nvsr67_08-508.pdf

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## Scope and Context

*Audience:* This report is intended for developers, integrators, end users, policy makers and others who have some familiarity with biometrics applications. The methods and metrics documented here will be of interest to organizations engaged in tests of face recognition algorithms. Some of these have been incorporated in the ISO/IEC 19795 Part 1 Biometric Testing and Reporting Framework standard, now nearing publication.

*Prior benchmarks:* Automated face recognition accuracy has improved massively in the two decades since initial commercialization of the various technologies. NIST has tracked that improvement through its conduct of regular independent, free, open, and public evaluations. These have fostered improvements in the state of the art. This report serves as an update to the NIST Interagency Report 8271 on performance of face identification algorithms, published in September 2019.

*Demographics:* In December 2019, NIST published a first report on demographic dependencies in face recognition, NIST Interagency Report 8280 that documented age, sex and race differentials in one-to-one and one-to-many false positive and false negative rates.

*Scope:* NIST IR 8271 documented recognition results for four databases containing in excess of 30.2 million still photographs of 14.4 million individuals. That constituted the largest public and independent evaluation of face recognition ever conducted. It includes results for accuracy, speed, investigative vs. identification applications, scalability to large populations, use of multiple images per person, images of cooperative and non-cooperative subjects.

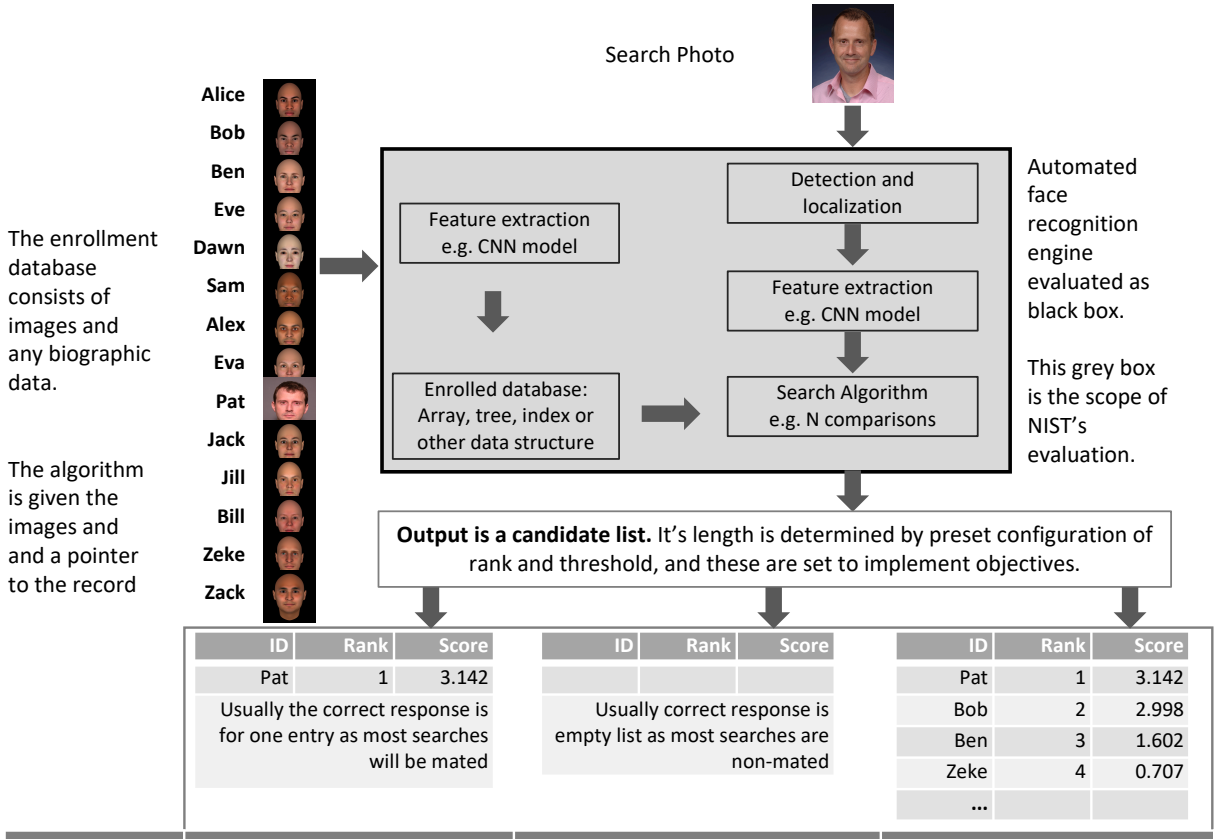
The report also includes results for ageing, recognition of twins, and recognition of profile-view images against frontal galleries. It otherwise does not address causes of recognition failure, neither image-specific problems nor subject-specific factors including demographics. Separate reports on demographic dependencies in face recognition will be published in the future. Additionally out of scope are: performance of live human-in-the-loop transactional systems like automated border control gates; human recognition accuracy as used in forensic applications; and recognition of persons in video sequences (which NIST evaluated separately [9]). Some of those applications share core matching technologies that *are* tested in this report.

*Images:* Five kinds of images are employed; these are either compared with images of the same kind, or against others from different capture environments as follows. The primary dataset is a set of law enforcement mugshot images (Fig. 5) which are enrolled and then searched with three kinds of images: other mugshots (i.e. within-domain); profile-view photographs (90 degree cross-view); and lower quality webcam images (Fig. 6) collected in similar detention operations (cross-domain). Additionally we compare high quality visa-like photos collected in immigration offices, with: medium quality border crossing images collected in primary immigration lanes; poor quality images collected in ATM-like registered traveller kiosks.

*Participation and industry coverage:* The report includes performance figures for prototype algorithms from the research laboratories of commercial developers and a few universities. This represents a substantial majority of the face recognition industry, but only a tiny minority of the academic community. Participation was open worldwide. While there is no charge for participation, developers incur some software engineering expense in implementing their algorithms behind the NIST application programming interface (API). The test is a black-box test where the function of the algorithm, and the intellectual property associated with it, is hidden inside pre-compiled libraries.

*Recent technology development:* Most face recognition research with deep convolutional neural networks (CNNs) has been aimed at achieving invariance to pose, illumination and expression variations that characterize photojournalism and social media images. The initial research [18, 22] employed large numbers of images of relatively few ($\sim 10^4$) individuals to learn invariance. Inevitably much larger populations ($\sim 10^7$) were employed for training [11, 20] but the benchmark, Labeled Faces in the Wild with (essentially) an equal error rate metric [12], represents an easy task,

one-to-one verification at very high false match rates. While a larger scale identification benchmark duly followed, Megaface [15], its primary metric, rank one hit rate, contrasts with the high threshold discrimination task required in most large-population applications of face recognition, namely credential de-duplication, and background checks. There, identification in galleries containing up to $10^8$ individuals must be performed using a) very few images per individual and b) stringent thresholds to afford very low false positive identification rates. This track of FRVT was launched to measure the capability of the new technologies, including in these two cases. FRVT has included open-set identification tests since 2002, reporting both false negative and positive identification rates [7].




| | Positive identification | Negative identification | Post-event investigation |
|---|---|---|---|
| **Example** | Access to a gym or cruise ship | Watchlist e.g. Detection of deportee or duplicate drivers license applications | Crime scene photos, or of detainee without ID documents. |
| **Claim of identity** | Implicit claim to be enrolled | Implicit claim to not be enrolled | No claim: Inquiry |
| **Threshold** | High, to implement security objective | High, to limit false positives | Zero |
| **Num. candidates** | 1 | 0 | L, set by request to algorithm |
| **Human role** | Review candidate to assist user in resolution of false negatives, or to detect impostor | Review candidate to determine false positive or correct hit | Review multiple candidates, refer possible hits to examiners see [26] |
| **Intended human involvement frequency** | Rare – approx. the false negative identification rate plus prior probability of impostor | Rare – approx. the false positive identification rate plus prior probability of an actual mate | Always |
| **Performance metric of interest** | **FNIR at low FPIR. See sec. 3.1, 3.2 and Tables 10, 19** | **FNIR at low FPIR. See sec. 3.1, 3.2 and Tables 10, 19** | **FNIR at ranks 1... 50, say. FPIR = 1. See sec. 3.2 and Table 12, 14, 16** |

*Performance metrics for applications:* This report documents the performance of one-to-many face recognition algorithms. The word "performance" here refers to recognition accuracy and computational resource usage, as measured

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by executing those algorithms on massive sequestered datasets.

This report includes extensive tabulation of recognition error rates germane to the main use-cases for face search technology. The Figure below, inspired by the Figure 1 in [23] differentiates different applications of the technolgy. The last row directs readers to the main tables relevant to those applications, respectively threshold-based and rank-based metrics that are special cases of the metrics given in section 3. The terms negative identification and positive identification are taken from the ISO/IEC 2382-37:2017 standardized biometrics vocabulary.

The algorithms are specifically configured for these applications by setting thresholds and candidate list lengths. Both rank-based metrics and threshold-based metrics include tradeoffs. In investigation, overall accuracy will be reduced if labor is only available to review a few candidates from the automated system. Note that when a fixed number of candidates are returned, the false positive identification rate of the automated face recognition engine will be 100%, because a probe image of anyone not enrolled will still return candidates. In identification applications where false positives must be limited to satisfy reviewer labor availability or a security objective, higher false negative rates are implied. This report includes extensive quantification of this threshold-based tradeoff. See Sec. 3

*Template diversity:* The FRVT is designed to evaluate black-box technologies with the consequence that the templates that hold features extracted from face images are entirely proprietary opaque binary data that embed considerable intellectual property of the developer. Despite migration to CNN-based technologies there is no consensus on the optimal feature vector dimension. This is evidenced by template sizes ranging from below 100 bytes to more than four kilobytes. This diversity of approaches, suggests there is no prospect of a standard template something that would require a common feature set to be extracted from faces. Interoperability in automated face recognition remains solidly based on images and documentary standards for those, in particular the ICAO portrait [27] specification deriving from the ISO/IEC 19794-5 Token frontal [24] standard, which are similar to certain ANSI/NIST Type 10 [26] formats.

*Training:* The algorithms submitted to NIST have been developed using image datasets that developers do not disclose. The development will often include application of machine learning techniques and will additionally involve iterative training and testing cycles. NIST itself does not perform any training and does not refine or alter the algorithm in any way. Thus the model, data files, and libraries that define an algorithm are fixed for the duration of the tests. This reflects typical operational reality where recognition software, once installed, is fixed and constant until upgraded. This situation persists because on-site training of algorithms on customer data is atypical essentially because training is not a turnkey process.

*Automated search and human review:* Virtually all applications using automated face search require human review of the outputs at some frequency: Always for investigational applications; rarely in positive identification applications, after rejection (false or otherwise); and rarely in negative identification applications, after an alarm (false or otherwise). The human role is usually to compare a reference image with the query image or the live-subject if present, to render either a definitive decision on "exclusion" (different subjects), or "identification" (same subject), or a declaration that one or both images have "no value" and that no decision can be made. Note that automated face recognition algorithms are not built to do exclusion - low scores from a face comparison arise from different faces *and* poor quality images of the same face.

Human reviewers make recognition errors [5, 19, 25] and are sensitive to image acquisition and quality. Accurate human review is supported by high resolution - as specified in the Type 50, 51 acquisition profiles of the ANSI/NIST Type 10 record [26], and by multiple non-frontal views as specified in the same standard. These often afford views of the ear. Organizations involved in image collection should consider supporting human adjudication by collecting high-resolution frontal and non-frontal views, preparing low resolution versions for automated face recognition [24], and retaining both for any subsequent resolution of candidate matches. Along these lines, the ISO/IEC Joint Technical

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Committee 1 subcommittee 37 on biometrics has just initiated projects on image quality assessment and face-aware capture.

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# Release Notes

*FRVT Activities*: Since February 2017, NIST has been evaluating one-to-one verification algorithms on an ongoing basis. NIST then restarted FRVT's one-to-many track in February 2018, inviting participants to send up to prototype algorithms. Both tracks allows developers to submit updated algorithms to NIST at any time but no more frequently than four calendar months. This more closely aligns development and evaluation schedules. Results are posted to the web within a few weeks of submission. Details and full report are linked from the Ongoing FRVT site.

*FRVT Reports*: The results of the FRVT appear in the series NIST Interagency Reports tabulated below. The reports were developed separately and released on different schedules. In prior years NIST has mostly reported FRVT results as a single report; this had the disadvantage that results from completed sub-studies were not published until all other studies were complete.

| Date | Link | Title | No. |
|---|---|---|---|
| 2014-03-20 | PDF | FRVT Performance of Automated Age Estimation Algorithms | 7995 |
| 2015-04-20 | PDF | Face Recognition Vendor Test (FRVT) Performance of Automated Gender Classification Algorithms | 8052 |
| 2014-05-21 | PDF | FRVT Performance of face identification algorithms | 8009 |
| 2017-03-07 | PDF | Face In Video Evaluation (FIVE) Face Recognition of Non-Cooperative Subjects | 8173 |
| 2017-11-23 | PDF | The 2017 IARPA Face Recognition Prize Challenge (FRPC) | 8197 |
| 2018-11-27 | PDF | Face Recognition Vendor Test - Part 2: Identification | 8271 |
| 2019-09-11 | PDF | Face Recognition Vendor Test - Part 2: Identification | 8271 |
| 2019-12-11 | PDF | Face Recognition Vendor Test - Part 3: Demographic Effects | 8280 |
| 2020-01-03 | WWW | Face Recognition Vendor Test (FRVT) - Part 1 Verification | Draft |

**Details appear on pages linked from `https://www.nist.gov/programs-projects/face-projects`.**

*Appendices*: This report is accompanied by appendices which present exhaustive results on a per-algorithm basis. These are machine-generated and are included because the authors believe that visualization of such data is broadly informative and vital to understanding the context of the report.

*Typesetting*: Virtually all of the tabulated content in this report was produced automatically. This involved the use of scripting tools to generate directly type-settable LaTeX content. This improves timeliness, flexibility, maintainability, and reduces transcription errors.

*Graphics*: Many of the Figures in this report were produced using the ggplot2 package running under R, the capabilities of which extend beyond those evident in this document.

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# 1 Introduction

One-to-many identification represents the largest market for face recognition technology. Algorithms are used across the world in a diverse range of biometric applications: detection of duplicates in databases, detection of fraudulent applications for credentials such as passports and driving licenses, token-less access control, surveillance, social media tagging, lookalike discovery, criminal investigation, and forensic clustering.

This report contains a breadth of performance measurements relevant to many applications. Performance here refers to accuracy and resource consumption. In most applications, the core accuracy of a facial recognition algorithm is the most important performance variable. Resource consumption will be important also as it drives the amount of hardware, power, and cooling necessary to accommodate high volume workflows. Algorithms consume processing time, they require computer memory, and their static template data requires storage space. This report documents these variables.

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## 1.1 Open-set searches

FRVT tested open-set identification algorithms. Real-world applications are almost always "open-set", meaning that some searches have an enrolled mate, but some do not. For example, some subjects have truly not been issued a visa or drivers license before; some law enforcement searches are from first-time arrestees[6]. In an "open-set" application, algorithms make no prior assumption about whether or not to return a high-scoring result, and for a mated search, the ideal behaviour is that the search produces the correct mate at high score and first rank. For a non-mate search, the ideal behavior is that the search produces zero high-scoring candidates.

Many academic benchmarks execute only closed-set searches. The proportion of mates found in the rank one position is the default accuracy metric. This hit rate metric ignores the score with which a mate is found; weak hits count as much as strong hits. This ignores the real-world imperative that in many applications it is necessary to elevate a threshold to reduce the number of false positives.

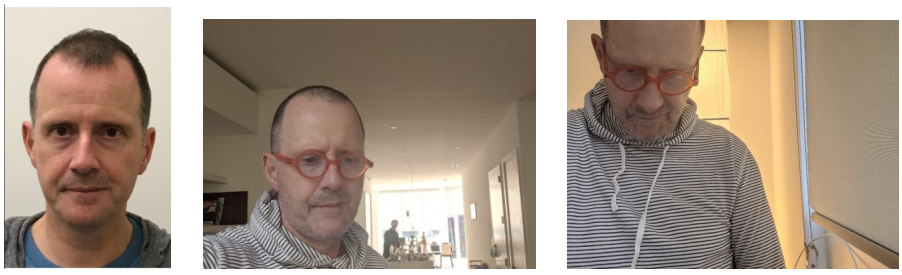
[6]Operationally closed-set applications are rare because it is usually not the case that all searches have an enrolled mate. One counter-example, however, is a cruise ship in which all passengers are enrolled and all searches should produce exactly one identity. Another example is forensic identification of dental records from an aircraft crash.

# 2 Evaluation datasets

This report documents accuracy for four kinds of images - mugshots, webcam, profiles and wild - as described in the following sections.

## 2.1 Immigration-related images

This report includes benchmark tests sharing a common enrollment of high quality frontal portrait images collected while subject make applications for various immigration benefits. We then search that with two kinds of images, webcam images collected during in-bound immigration and also images collected from registered travelers using a ATM-style kiosk. These are described below and depicted in Figure 4.



a) Application b) Immigration Lane c) Kiosk

*Figure 4: Example photos.*

- **Application reference photos**: The images are collected in an attended interview setting using dedicated capture equipment and lighting. The images, at size 300x300 pixels, are smaller than normally indicated by ISO. The images are all high-quality frontal portraits collected in immigration offices and with a white background. As such, potential quality related drivers of high false match rates (such as blur) can be expected to be absent. The images are encoded as ISO/IEC 10918-1 i.e. JPEG. Older images had a compression ration of about 16:1, while newer images, since 2010, are more lightly compressed at 4:1. When these images are provided as input into the algorithm, they are labeled with the type "iso". This report enrols 1 600 000 application images, one per person.

- **Border crossing photos**: Most images are have width 320 and height 240 pixels. They are JPEG compressed at 16:1 i.e. filesize just below 15KB. The images present challenges for face recognition in that subjects often exhibit non-zero yaw and pitch (associated with the rotational degrees of freedom of the camera mount), low contrast (due to varying and intense background lights), and poor spatial resolution (due to inexpensive cameras). There are often subjects standing in the background, usually at very low resolution (see Figure 4b). In such cases, algorithms should detect all faces and determine which is the largest and most centered. When these images are provided as input into the algorithm, they are labeled with the type "wild".

- **Kiosk photos**: These photos were collected from subjects whose attention was focused on interaction with an immigration kiosk. They images were not intended for use with automated face recognition. The camera is situated above a display which the user touches, and is triggered either without directing the subject to look at it, or without waiting for the subject to comply. The images are therefore characterized by pitch-down pose, sometimes exceeding 45 degrees, as in Figure 4c. Yaw-angle variation is mild, with most images close to frontal. The images

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have width 320 pixels and height 240 pixels and therefore tall individuals are sometimes cropped. This is often just above the eyes and can occur at the nose or mouth. Conversely, short individuals are sometimes cropped such that only the top part of the face is visible. In a quite small number of cases, there other subjects standing just behind the primary subject such that algoriths should detect all faces and determine which is the largest and most centered. Background ceiling lighting is often visible and this sometimes leads to under-exposure of the face When these images are provided as input into the algorithm, they are labeled with the type "wild".

## 2.2 Law enforcement images

The main mugshot dataset used is referred to as the FRVT 2018 set. This set was collected over the period 2002 to 2017 in routine United States law enforcement operations. This set yields three subsets

- **Mugshots**: Mugshots comprise about 86% of the database. They have reasonable compliance with the ANSI/NIST ITL1-2011 Type 10 standard's subject acquisition profiles levels 10-20 for frontal images [26]. The most common departure from the standard's requirements is the presence of mild pose variations around frontal - the images of Figure 5 are typical. The images vary in size, with many being 480x600 pixels with JPEG compression applied to produce filesizes of between 18 and 36KB with many images outside this range, implying that about 0.5 bits are being encoded per pixel. When these images are provided as input into the algorithm, they are labeled with the type "mugshot". Example images appear in Fig. 5

  NIST Interagency Report 8238 includes a comparison of this set of mugshots with the smaller and easier sets of mugshots used in tests run in 2010 and 2014.

- **Profile images**: Profile-view images have been collected in law enforcement for more than 100 years, as human capability is improved with orthogonal information. The profile images used in this report were collected during the same session as the frontal mugshot photograph, in the same standardized photographic setup. These would not therefore be used with automated face recognition. A small subset, 200 000 images, were set aside for testing. When these images are provided as input into the algorithm, they are labeled with the type "wild". Example images appear in Fig. 7

- **Webcam images**: The remaining 14% of the images were collected using an inexpensive webcam attached to a flexible operator-directed mount. These images are all of size 240x240 pixels, that are in considerable violation of most quality-related clauses of all face recognition standards. As evident in the figure, the most common defects are non-frontal pose (associated with the rotational degrees of freedom of the camera mount), low contrast (due to varying and intense background lights), and poor spatial resolution (due to inexpensive camera optics) - see examples in Fig 6. The images are overly JPEG compressed, to between 4 and 7KB, implying that only 0.5 to 1 bits are being encoded per color pixel. When these images are provided as input into the algorithm, they are labeled with the type "wild". Example images appear in Fig. 6

These are drawn from NIST Special Database 32 which may be downloaded here.

These images were partitioned in galleries and probesets for the various experiment listed in Table 1.

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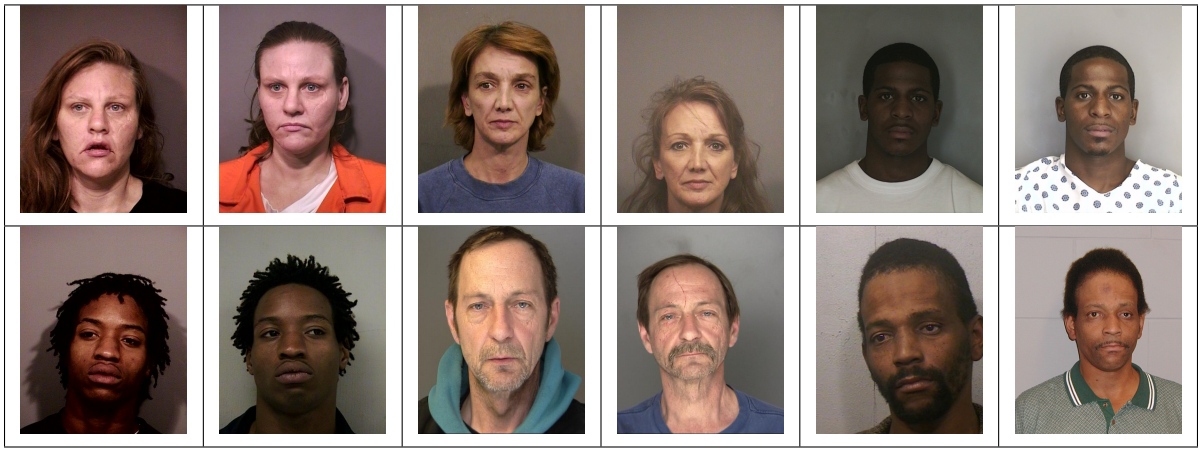

*Figure 5: Six mated mugshot pairs representative of the FRVT-2014 (LEO) and FRVT-2018 datasets. The images are collected live, i.e. not scanned from paper. Image source: NIST Special Database 32 the Multiple Encounter Deceased Subjects dataset.*

*Figure 6: Twelve webcam images representative of probes against the FRVT-2018 mugshot gallery. The first eight images are four mated pairs. Such images present challenges to recognition including pose, non-uniform illumination, low contrast, compression, cropping, and low spatial sampling rate. Image source: NIST Special Database 32 the Multiple Encounter Deceased Subjects dataset.*

*Figure 7:* **[Profile views]** *The three images are a frontal enrollment, subsequent frontal probe, and same-session ninety degree profile view. While collection of both frontal and profile views has been typical in law enforcement for more than a century, the recognition of profile to frontal views has essentially been impossible. However, reasonbly high accuracy results is now possible - see section E.*

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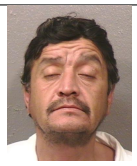
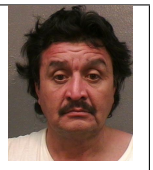
| Image | |  | |  |
|---|---|---|---|---|
| Encounter | 1 | ... | $K_i - 1$ | $K_i$ |
| Capture Time | $T_1$ | ... | $T_{K_i-1}$ | $T_{K_i}$ |
| Role RECENT | Not used | Not used | Enrolled | Search |
| Role LIFETIME | Enrolled | Enrolled | Enrolled | Search |

*Figure 8: Depiction of the "recent" and "lifetime" enrollment types. Image source: NIST Special Database 32*

## 2.3 Enrollment strategies

Many operational applications include collection and enrollment of biometric data from subjects on more than one occasion. This might be done on a regular basis, as might occur in credential (re-)issuance, or irregularly, as might happen in a criminal recidivist situation [4]. The number of images per person will depend on the application area. In civil identity credentialing (e.g. passports, driver's licenses), the images will be acquired approximately uniformly over time (e.g. ten years for a passport). While the distribution of dates for such images of a person might be assumed uniform, a number of factors might undermine this assumption[7]. In criminal applications, the number of images would depend on the number of arrests. The distribution of dates for arrest records for a person (i.e. the recidivism distribution) has been modeled using the exponential distribution but is recognized to be more complicated[8].

In any case, the 2010 NIST evaluation of face recognition showed that considerable accuracy benefits accrue with retention and use of *all* historical images [6].

To this end, the FRVT API document provides $K \geq 1$ images of an individual to the enrollment software. The software is tasked with producing a single proprietary undocumented "black-box" template[9] from the $K$ images. This affords the algorithm an ability to generate a *model* of the individual, rather than to simply extract features from each image on a sequential basis.

As depicted in Figure 8, the $i$-th individual in the FRVT 2018 dataset has $K_i$ images. These are labelled as $x_k$ for $k = 1 \ldots K_i$ in chronological order of capture date. To measure the utility of having multiple enrollment images, this report evaluates three kinds of enrollment:

- ▷ **Recent**: Only the second most recent image, $x_{K_i-1}$ is enrolled. This strategy of enrollment mimics the operational policy of retaining the imagery from the most recent encounter. This might be done operationally to ameliorate the effects of face ageing. Obviously retaining only the most recent image should only be done if the identity of the person is trusted to be correct. For example, in an access control situation retention of the most recent successful *authentication* image would be hazardous if it could be a false positive.

- ▷ **Lifetime-consolidated**: All but the most recent image are enrolled, $x_1 \ldots x_{K_i-1}$. This subject-centric strategy might be adopted if quality variations exist where an older image might be more suitable for matching, despite the ageing effect.

[7] For example, a person might skip applying for a passport for one cycle, letting it expire. In addition, a person might submit identical images (from the same photography session) to consecutive passport applications at five year intervals.
[8] A number of distributions have been considered to model recidivism, see for example [3].
[9] There are no formal face template standards. Template standards only exist for fingerprint minutiae - see ISO/IEC 19794-2:2011.

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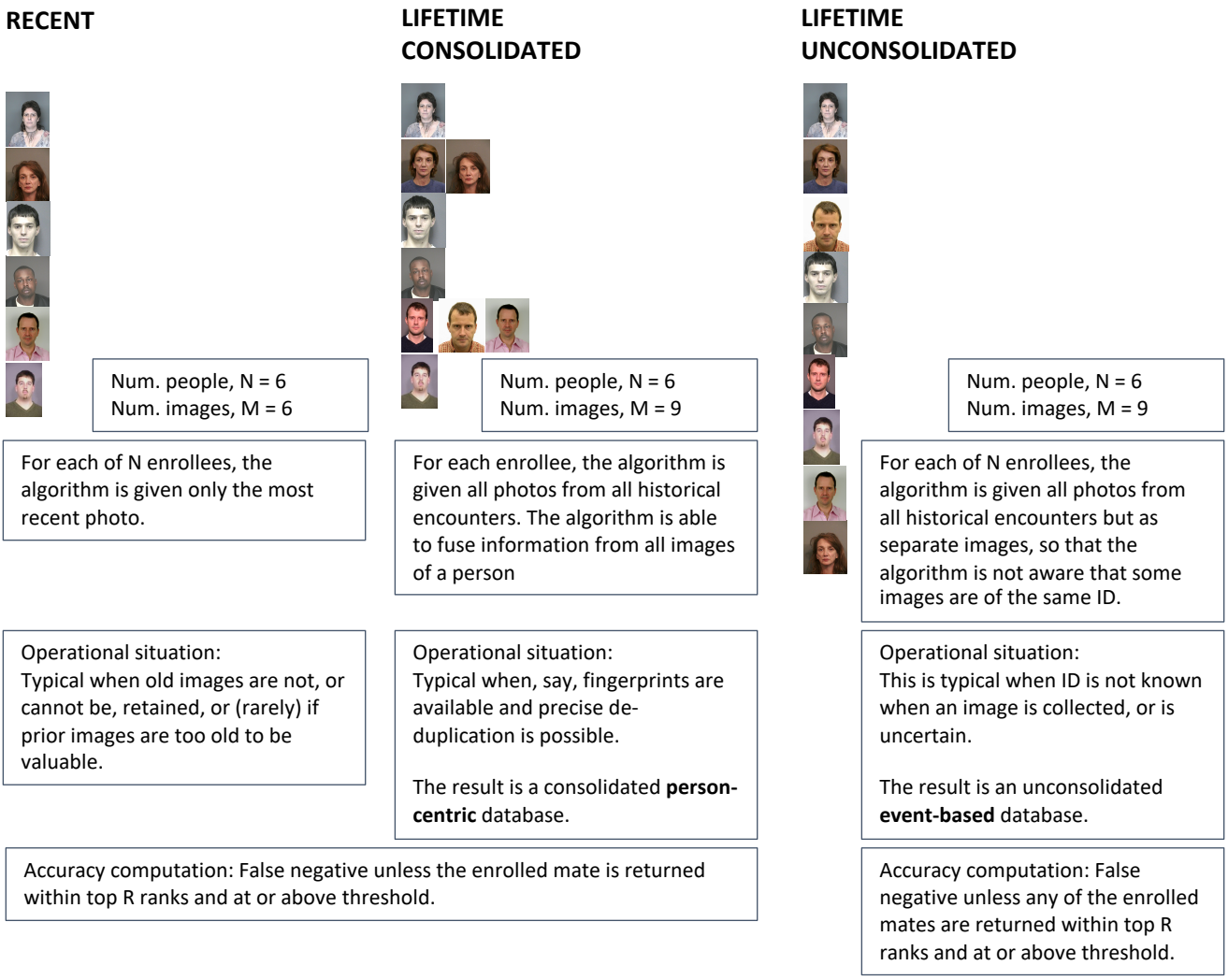


*Figure 9:* **Enrollment strategies**. *The figure shows the three kinds of enrollment databases examined in this report. Image source: NIST Special Database 32*

| | ENROLLMENT | | | | SEARCH | | | |
|---|---|---|---|---|---|---|---|---|
| | TYPE SEE | POPULATION | | | MATE | | NON-MATE | |
| | SECTION 2.3 | FILTER | N-SUBJECTS | N-IMAGES | N-SUBJECTS | N-IMAGES | N-SUBJECTS | N-IMAGES |
| **Mugshot trials from enrollment of single images** | | | | | | | | |
| 1 | RECENT | NATURAL | 640 000 | 640 000 | 154 549 | 154 549 | 331 254 | 331 254 |
| 2 | RECENT | NATURAL | 1 600 000 | 1 600 000 | | | | |
| 3 | RECENT | NATURAL | 3 000 000 | 3 000 000 | | | | |
| 4 | RECENT | NATURAL | 6 000 000 | 6 000 000 | | | | |
| 5 | RECENT | NATURAL | 12 000 000 | 12 000 000 | | | | |
| **Cross-domain** | | | | | | | | |
| 13 | MUGSHOTS AS ON ROW 2 | | | | 82 106 WEBCAM | 82 106 WEBCAM | 331 254 WEBCAM | 331 254 WEBCAM |
| **Cross-view** | | | | | | | | |
| 14 | MUGSHOTS AS ON ROW 2 | | | | 100 000 PROFILE | 100 000 PROFILE | 100 000 PROFILE | 100 000 PROFILE |
| **Mugshot ageing** | | | | | | | | |
| 17 | OLDEST | NATURAL | 3 068 801 | 3 068 801 | 2 853 221 | 10 951 064 | 0 | 0 |
| **Border crossing ageing** | | | | | | | | |
| 18 | OLDEST | NATURAL | 1 600 000 | 1 600 000 | 903 655 | 1 922 393 | 1 393 076 | 1 680 000 |
| **Visa-border** | | | | | | | | |
| 19 | PRIOR | NATURAL | 1 600 000 VISA | 1 600 000 VISA | 577 444 BORDER | 1 212 892 BORDER | 79 769 BORDER | 80 000 BORDER |
| 20 | VISA AS ON ROW 18 | | | | 14 004 BORDER | 31 579 BORDER | 42 474 BORDER | 45 460 BORDER |

*Table 1:* **Enrollment and search sets**. *Each row summarizes one identification trial. Unless stated otherwise, all entries refer to mugshot images. The term "natural" means that subjects were selected without heed to demographics, i.e. in the distribution native to this dataset. The probe images were collected in a different calendar year to the enrollment image. Missing values in rows 2-12 are the same as in row 1.*

- **Lifetime-unconsolidated**: Again all but the most recent image are enrolled $x_1 \ldots x_{K_i-1}$ but now separately, with different identifiers, such that the algorithm is not aware that the images are from the same face. This kind of event- or encounter-centric enrollment is very common when operational constraints preclude reliable consolidation of the historical encounters into a single identity. This aspect also prevents the recognition algorithm from a) building a holistic model of identity (as is common in speaker recognition systems) and b) implementing fusion, for example template-level fusion of feature vectors, or post-search score-level fusion. The result is that searches will typically yield more than one image of a person in the top ranks. This has consequences for appropriate metrics, as detailed in section 3.2.1

  NIST first evaluated this kind of enrollment in mid 2018, and the results tables include some comparison of accuracy available from all three enrollment styles.

In all cases, the most recent image, $x_{K_i}$, is reserved as the search image. For the 1.6 million subject enrollment partition of the FRVT 2018 data, $1 \leq K_i \leq 33$ with $K_i = 1$ in 80.1% of the individuals, $K_i = 2$ in 13.4%, $K_i = 3$ in 3.7%, $K_i = 4$ in 1.4%, $K_i = 5$ in 0.6%, $K_i = 6$ in 0.3%, and $K_i > 6$ is 0.2% for everyone else. This distribution is substantially dependent on United States recidivism rates.

We did not evaluate the case of retaining only the highest quality image, since automated quality assessment is out of scope for this report. We do not anticipate that such strategies will prove beneficial when the quality assessment apparatus is imperfect and unvalidated.

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# 3 Performance metrics

This section gives specific definitions for accuracy and timing metrics. Tests of open-set biometric algorithms must quantify frequency of two error conditions:

- ▷ **False positives**: Type I errors occur when search data from a person who has never been seen before is incorrectly associated with one or more enrollees' data.
- ▷ **Misses**: Type II errors arise when a search of an enrolled person's biometric does not return the correct identity.

Many practitioners prefer to talk about "hit rates" instead of "miss rates" - the first is simply one minus the other as detailed below. Sections 3.1 and 3.2 define metrics for the Type I and Type II performance variables.

Additionally, because recognition algorithms sometimes fail to produce a template from an image, or fail to execute a one-to-many search, the occurrence of such events must be recorded. Further because algorithms might elect to not produce a template from, for example, a poor quality image, these failure rates must be combined with the recognition error rates to support algorithm comparison. This is addressed in section 3.5.

Finally, section 3.7 discusses measurement of computation duration, and section 3.8 addresses the uncertainty associated with various measurements. Template size measurement is included with the results.

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## 3.1 Quantifying false positives

It is typical for a search to be conducted into an enrolled population of $N$ identities, and for the algorithm to be configured to return the closest $L$ candidate identities. These candidates are ranked by their score, in descending order, with all scores required to be greater than or equal to zero. A human analyst might examine either all $L$ candidates, or just the top $R \leq L$ identities, or only those with score greater than threshold, $T$. The workload associated with such examination is discussed later, in 3.6.

False alarm performance is quantified in two related ways. These express how many searches produces false positives, and then, how many false positives are produced in a search.

**False positive identification rate**: The first quantity, FPIR, is the proportion of non-mate searches that produce an adverse outcome:

$$\text{FPIR}(N,T) = \frac{\text{Num. non-mate searches where one or more enrolled candidates are returned with score at or above threshold}}{\text{Num. non-mate searches attempted.}} \tag{1}$$

Under this definition, FPIR can be computed from the highest non-mate candidate produced in a search - it is not necessary to consider candidates at rank 2 and above. FPIR is the primary measure of Type I errors in this report.

**Selectivity**: However, note that in any given search, several non-mate may be returned above threshold. In order to quantify such events, a second quantity, selectivity (SEL), is defined as the *number* of non-mates returned on a candidate list, averaged over all searches.

$$\text{SEL}(N,T) = \frac{\text{Num. non-mate enrolled candidates returned with score at or above threshold}}{\text{Num. non-mate searches attempted.}} \tag{2}$$

where $0 \leq$ SEL(N, T) $\leq$L. Both of these metrics are useful operationally. FPIR is useful for targeting how often an

adverse false positive outcome can occur, while SEL as a number is related to workload associated with adjudicating candidate lists. The relationship between the two quantities is complicated - it depends on whether an algorithm concentrates the false alarms in the results of a few searches or whether it disburses them across many. This was detailed in FRVT 2014, NISTIR 8009. It has not yet been detailed in FRVT 2018.

## 3.2 Quantifying hits and misses

If $L$ candidates are returned in a search, a shorter candidate list can be prepared by taking the top $R \leq L$ candidates for which the score is above some threshold, $T \geq 0$. This reduction of the candidate list is done because thresholds may be applied, and only short lists might be reviewed (according to policy or labor availability, for example). It is useful then to state accuracy in terms of $R$ and $T$, so we define a "miss rate" with the general name **false negative identification rate** (FNIR), as follows:

$$\text{FNIR}(N, R, T) = \frac{\text{Num. mate searches with enrolled mate found outside top R ranks or score below threshold}}{\text{Num. mate searches attempted.}} \quad (3)$$

This formulation is simple for evaluation in that it does not distinguish between causes of misses. Thus a mate that is not reported on a candidate list is treated the same as a miss arising from face finding failure, algorithm intolerance of poor quality, or software crashes. Thus if the algorithm fails to produce a candidate list, either because the search failed, or because a search template was not made, the result is regarded as a miss, adding to FNIR.

*Hit rates, and true positive identification rates*: While FNIR states the "miss rate" as how often the correct candidate is either not above threshold or not at good rank, many communities prefer to talk of "hit rates". This is simply the **true positive identification rate**(TPIR) which is the complement of FNIR giving a positive statement of how often mated searches are successful:

$$\text{TPIR}(N, R, T) = 1 - \text{FNIR}(N, R, T) \quad (4)$$

This report does not report true positive "hit" rates, preferring false negative miss rates for two reasons. First, costs rise linearly with error rates. For example, if we double FNIR in an access control system, then we double user inconvenience and delay. If we express that as decrease of TPIR from, say 98.5% to 97%, then we mentally have to invert the scale to see a doubling in costs. More subtly, readers don't perceive differences in numbers near 100% well, becoming inured to the "high nineties" effect where numbers close to 100 are perceived indifferently.

**Reliability** is a corresponding term, typically being identical to TPIR, and often cited in automated (fingerprint) identification system (AFIS) evaluations.

An important special case is the **cumulative match characteristic**(CMC) which summarizes accuracy of mated-searches only. It ignores similarity scores by relaxing the threshold requirement, and just reports the fraction of mated searches returning the mate at rank R or better.

$$\text{CMC}(N, R) = 1 - \text{FNIR}(N, R, 0) \quad (5)$$

We primarily cite the complement of this quantity, $\text{FNIR}(N, R, 0)$, the fraction of mates *not* in the top R ranks.

The **rank one hit rate** is the fraction of mated searches yielding the correct candidate at best rank, i.e. CMC(N, 1). While this quantity is the most common summary indicator of an algorithm's efficacy, it is not dependent on similarity scores, so it does not distinguish between strong (high scoring) and weak hits. It also ignores that an adjudicating reviewer is often willing to look at many candidates.

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### 3.2.1 False negative rates for unconsolidated galleries

As detailed in section 2.3 a common type of gallery, here referred to as the lifetime unconsolidate type, is populated with all images of an individual without any association between them. That is, the gallery construction algorithm is not provided with any ID labels that would support processing of a person's images jointly. This constrasts with the lifetime consolidate type where an algorithm may explicitly fuse features from multiple images of a person, or select a best image. In such cases, where the number of enrolled images is a random variable, we define two false negative rates as follows.

The first demands that the algorithm place any of the $K_i$ mates in the top $R \geq 1$ ranks. The proportion of searches for which this does not occur forms a false negative identification rate:

$$\text{FNIR}_{\text{any}}(N, R, T) = 1 - \frac{\text{Num. mate searches where any enrolled mate is found in the top R ranks and at-or-above threshold}}{\text{Num. mate searches attempted.}} \tag{6}$$

The second demands that the algorithm place all $K_i$ mates in the top $R \geq K_i$ ranks. The proportion of searches for which this does not occur forms a false negative identification rate:

$$\text{FNIR}_{\text{all}}(N, R, T) = 1 - \frac{\text{Num. mate searches where all enrolled mates are found in the top R ranks and at-or-above threshold}}{\text{Num. mate searches attempted.}} \tag{7}$$

Placing all mates in the top ranks is a more difficult task than correctly retrieving any image, so it holds that: $\text{FNIR}_{\text{all}} \geq \text{FNIR}_{\text{any}}$. This is evident in the results presented for November 2018 algorithms in Tables starting at **??**.

The information retrieval community might prefer to compute and plot *precision* and *recall*; this is a valid approach, but we advance the two metrics above because they relate to our normal definition of consolidated FNIR, and they cover the two extreme use-cases of wanting any hit vs. all hits.

## 3.3 DET interpretation

In biometrics, a false negative occurs when an algorithm fails to match two samples of one person – a Type II error. Correspondingly, a false positive occurs when samples from two persons are improperly associated – a Type I error.

Matches are declared by a biometric system when the native comparison score from the recognition algorithm meets some threshold. Comparison scores can be either similarity scores, in which case higher values indicate that the samples are more likely to come from the same person, or dissimilarity scores, in which case higher values indicate different people. Similarity scores are traditionally computed by fingerprint and face recognition algorithms, while dissimilarities are used in iris recognition. In some cases, the dissimilarity score is a distance possessing metric properties. In any case, scores can be either mate scores, coming from a comparison of one person's samples, or nonmate scores, coming from comparison of different persons' samples.

The words "genuine" or "authentic" are synonyms for mate, and the word "impostor" is used as a synonym for nonmate. The words "mate" and "nonmate" are traditionally used in identification applications (such as law enforcement search, or background checks) while genuine and impostor are used in verification applications (such as access control).

An error tradeoff characteristic represents the tradeoff between Type II and Type I classification errors. For identification this plots false negative vs. false positive identification rates i.e. FNIR vs. FPIR parametrically with T. Such plots

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are often called detection error tradeoff (DET) characteristics or receiver operating characteristic (ROC). These serve the same function − to show error tradeoff − but differ, for example, in plotting the complement of an error rate (e.g. TPIR = 1 − FNIR) and in transforming the axes, most commonly using logarithms, to show multiple decades of FPIR. More rarely, the function might be the inverse of the Gaussian cumulative distribution function.

The slides of Figures 10 through 15 discuss presentation and interpretation of DETs used in this document for reporting face identification accuracy. Further detail is provided in formal biometrics testing standards, see the various parts of ISO/IEC 19795 Biometrics Testing and Reporting. More terms, including and beyond those to do with accuracy, appear in ISO/IEC 2382-37 Information technology – Vocabulary – Part 37: Harmonized biometric vocabulary.

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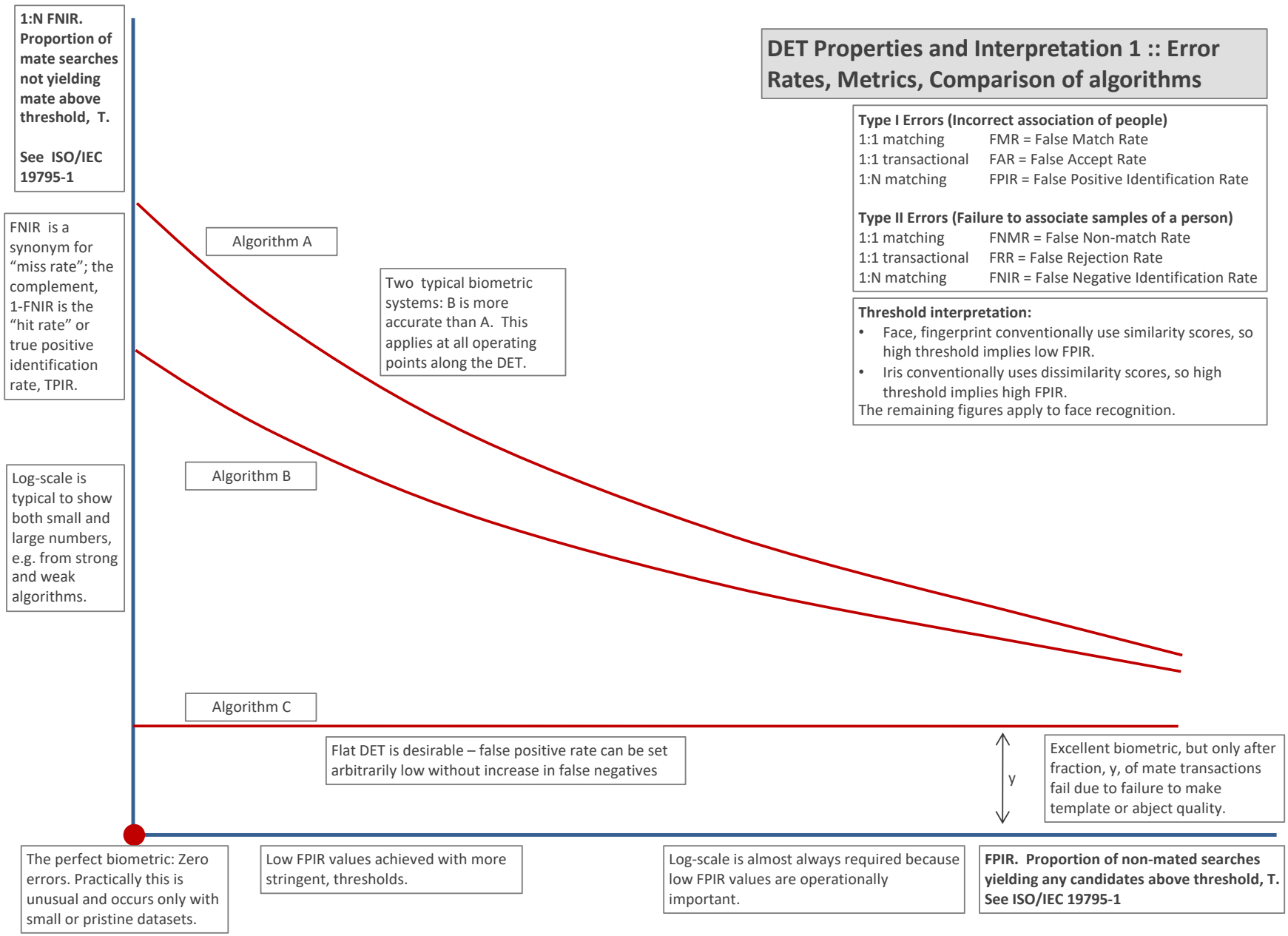


*Figure 10:* **DET as the primary performance reporting mechanism**.

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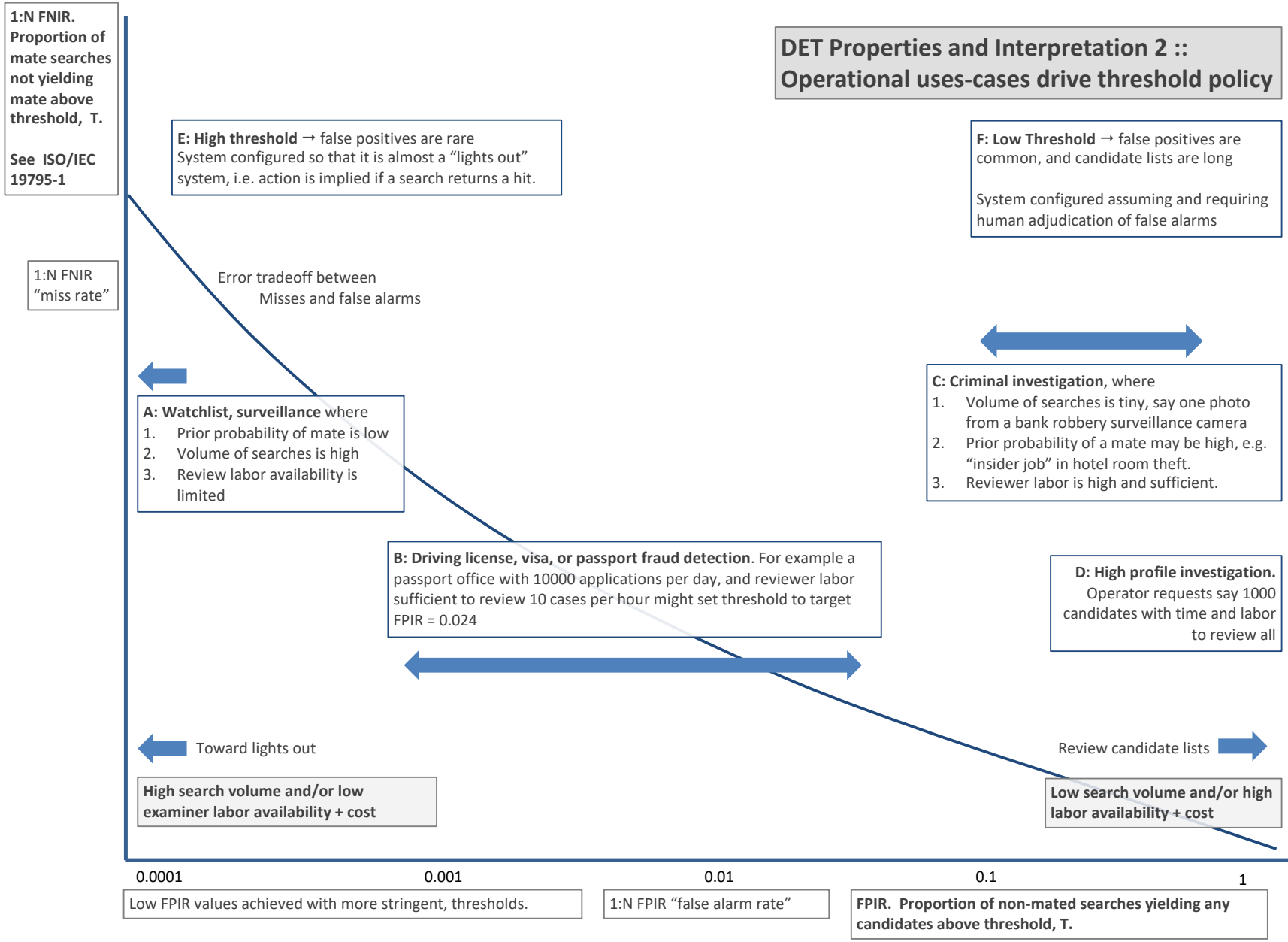


*Figure 11:* **DET as the primary performance reporting mechanism**.



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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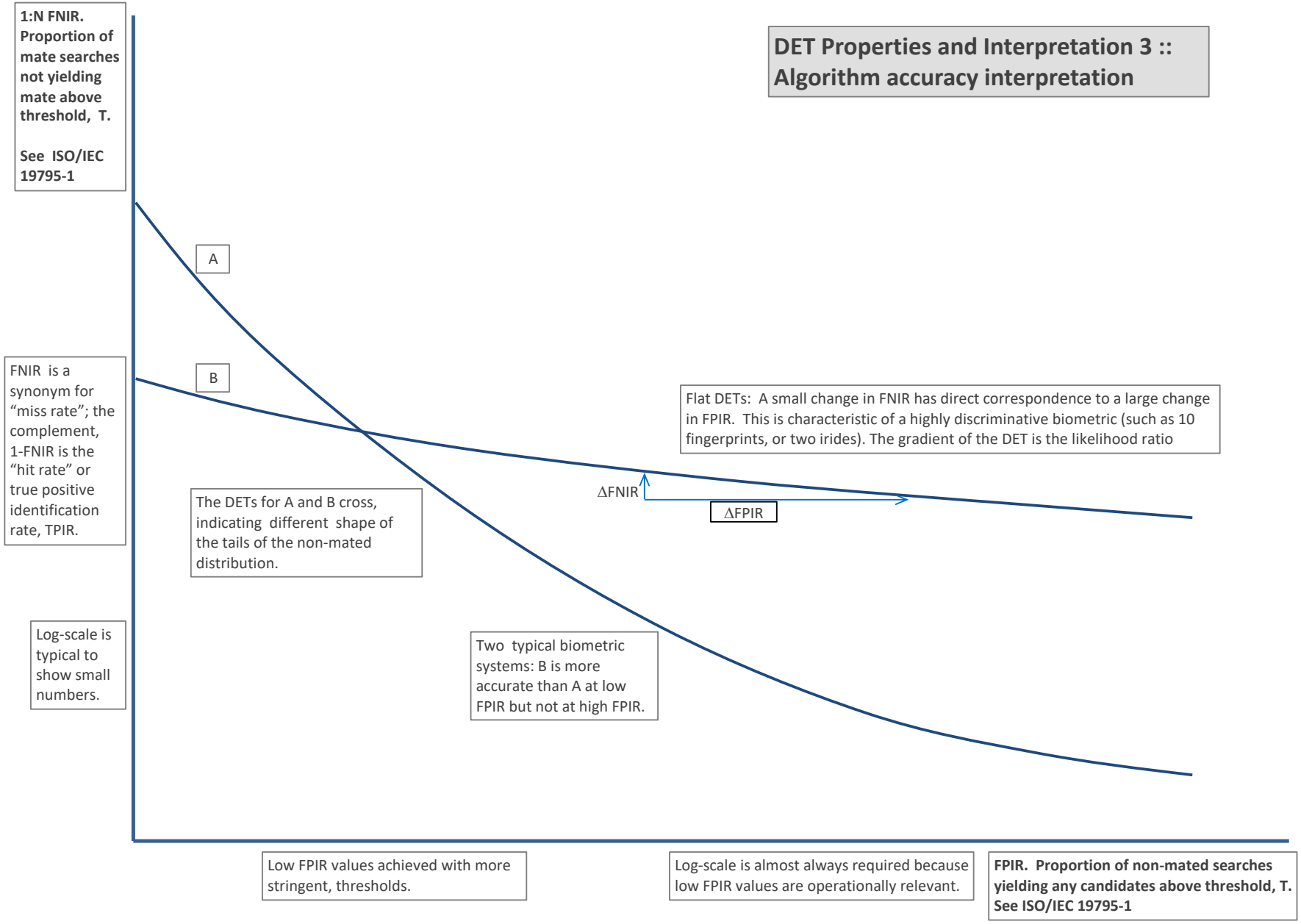


*Figure 12:* **DET as the primary performance reporting mechanism**.



2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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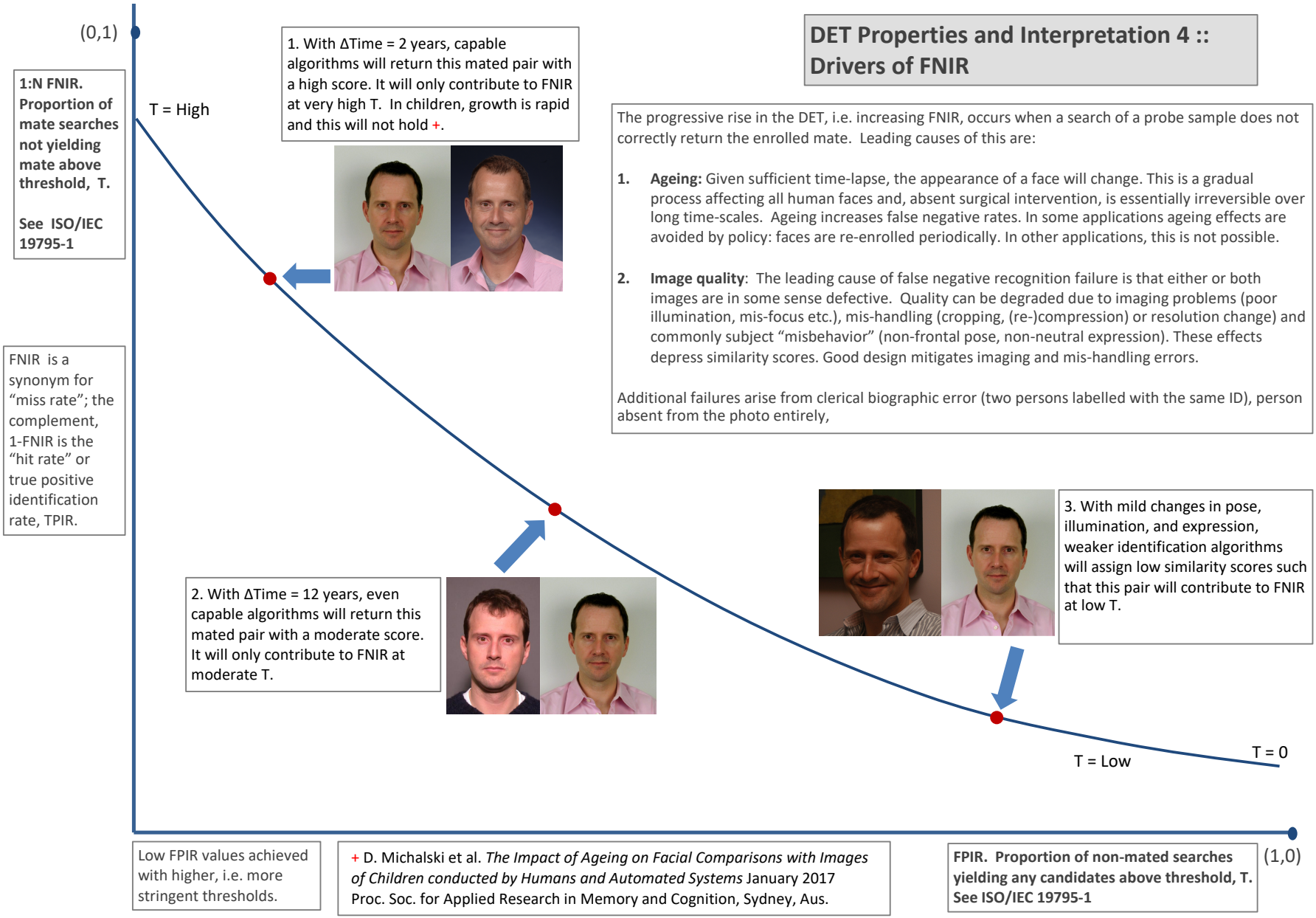


*Figure 13:* **DET as the primary performance reporting mechanism**.

2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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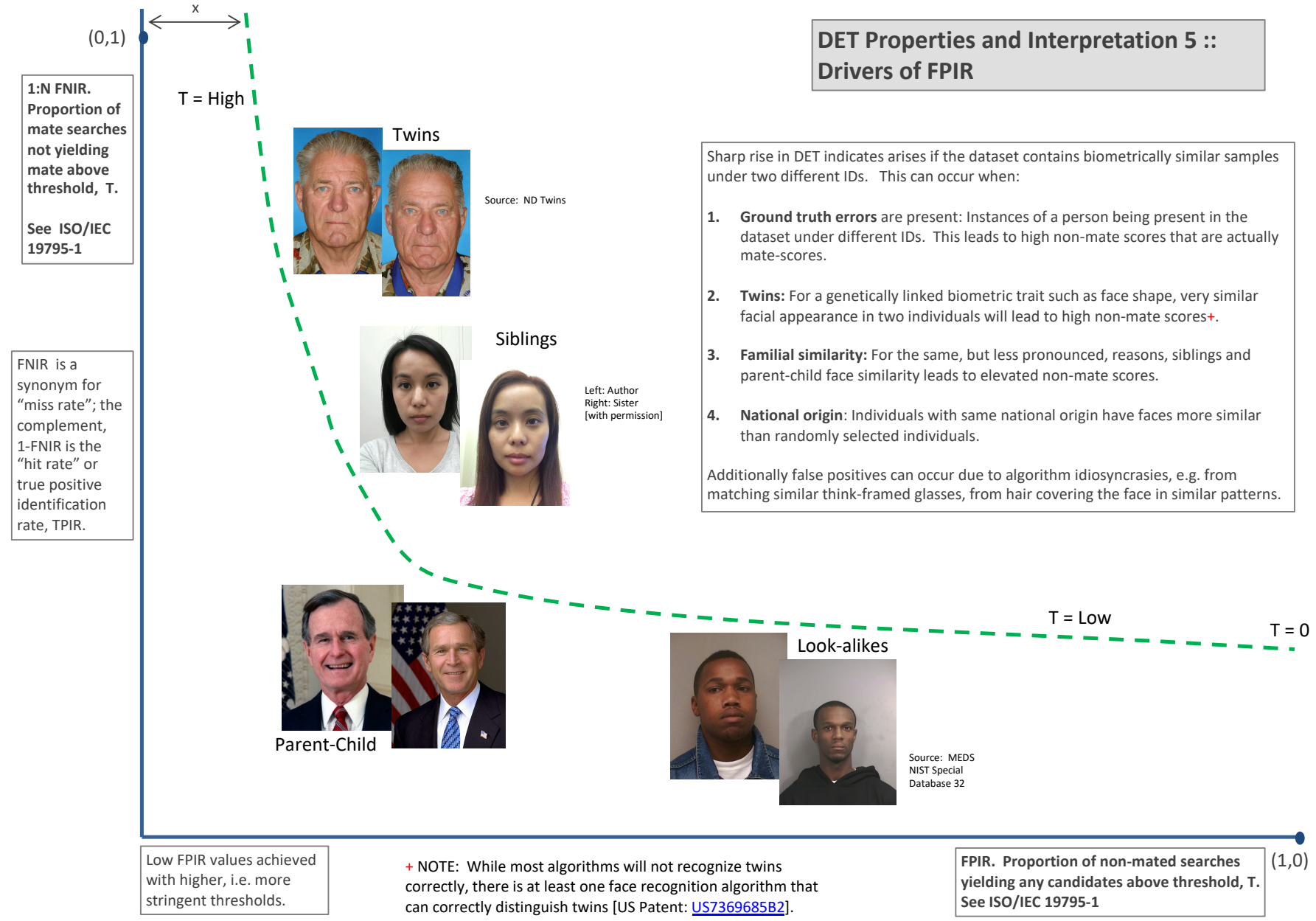


*Figure 14:* **DET as the primary performance reporting mechanism**.

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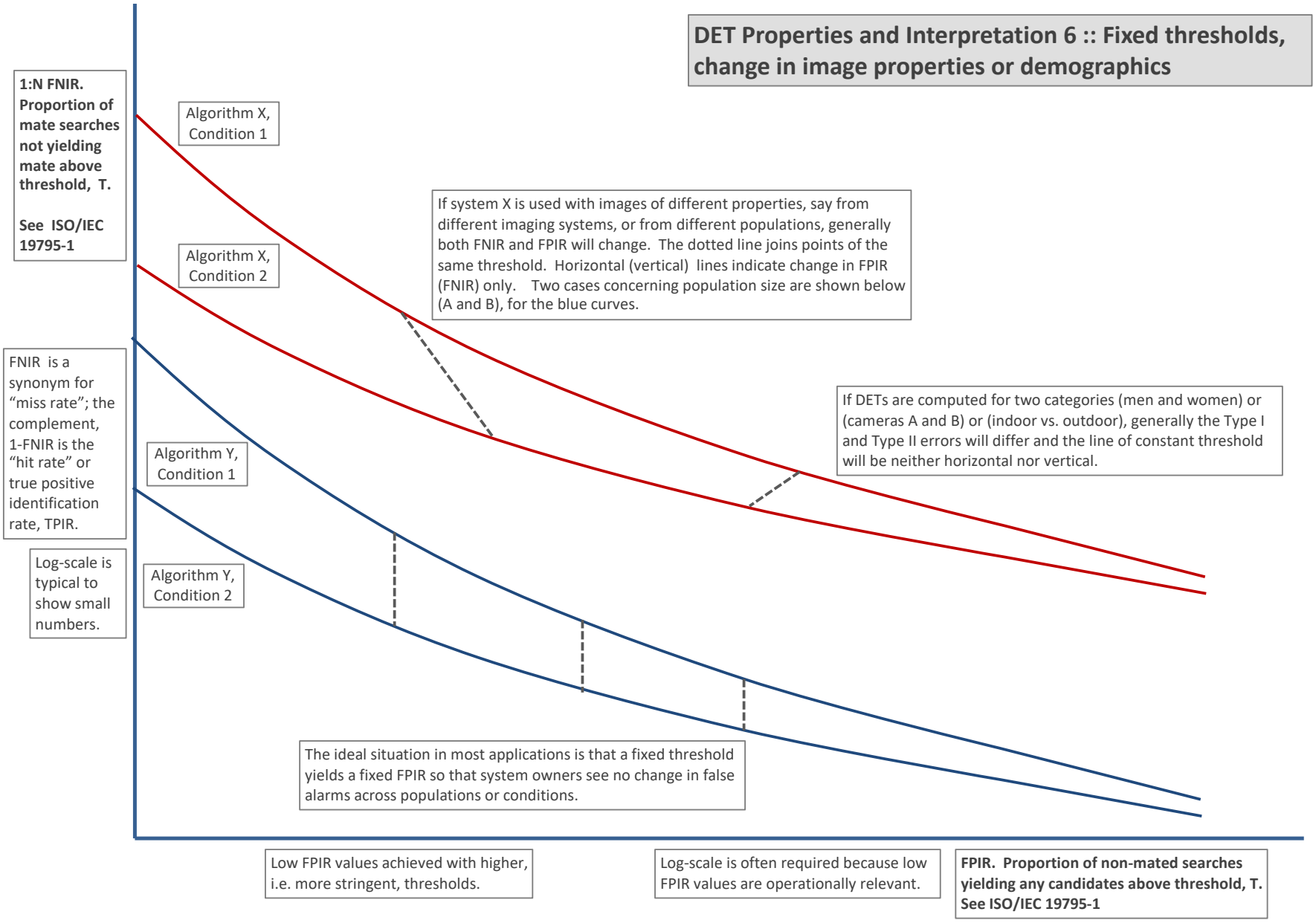


*Figure 15:* **DET as the primary performance reporting mechanism**.



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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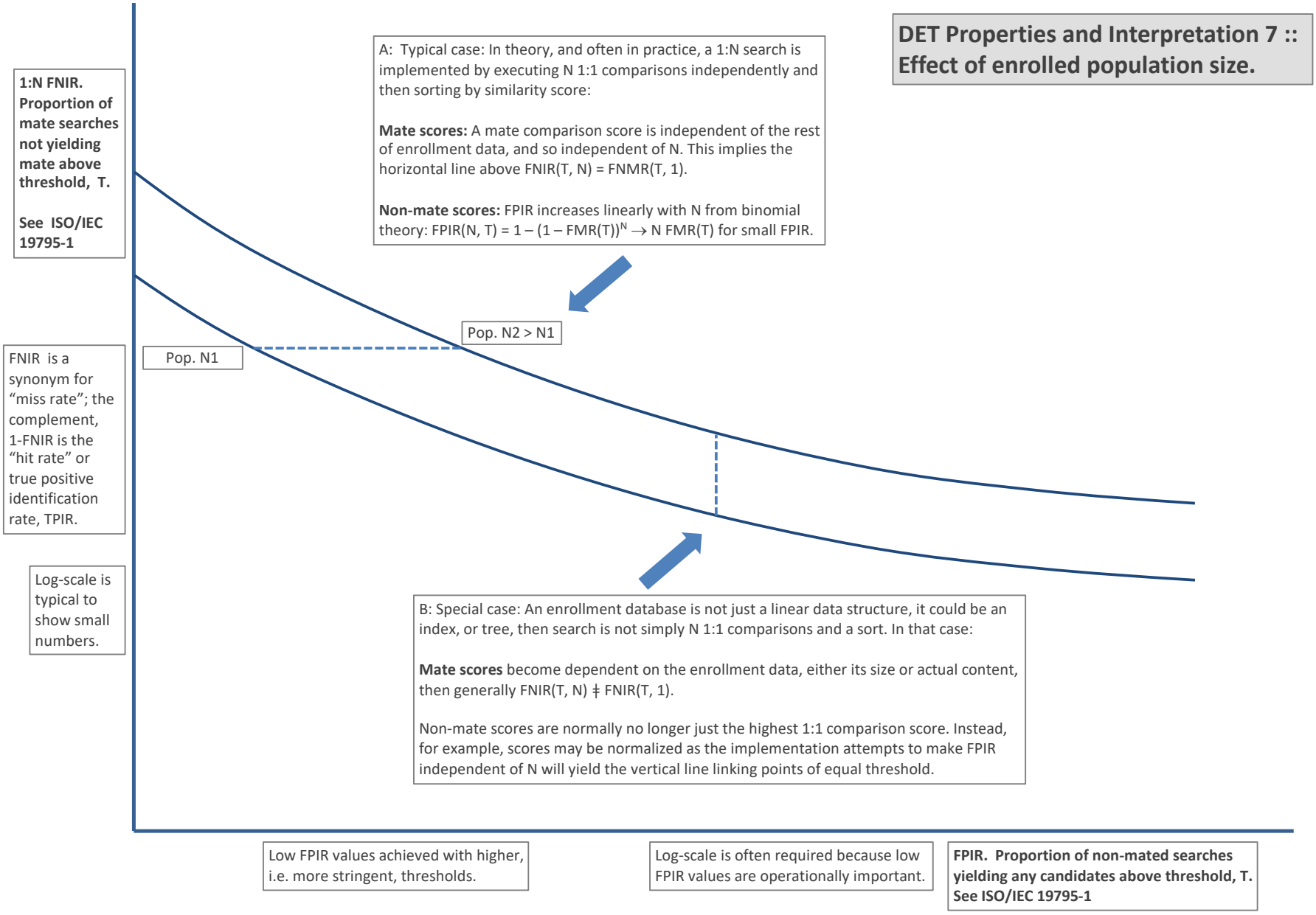


*Figure 16:* **DET as the primary performance reporting mechanism**.

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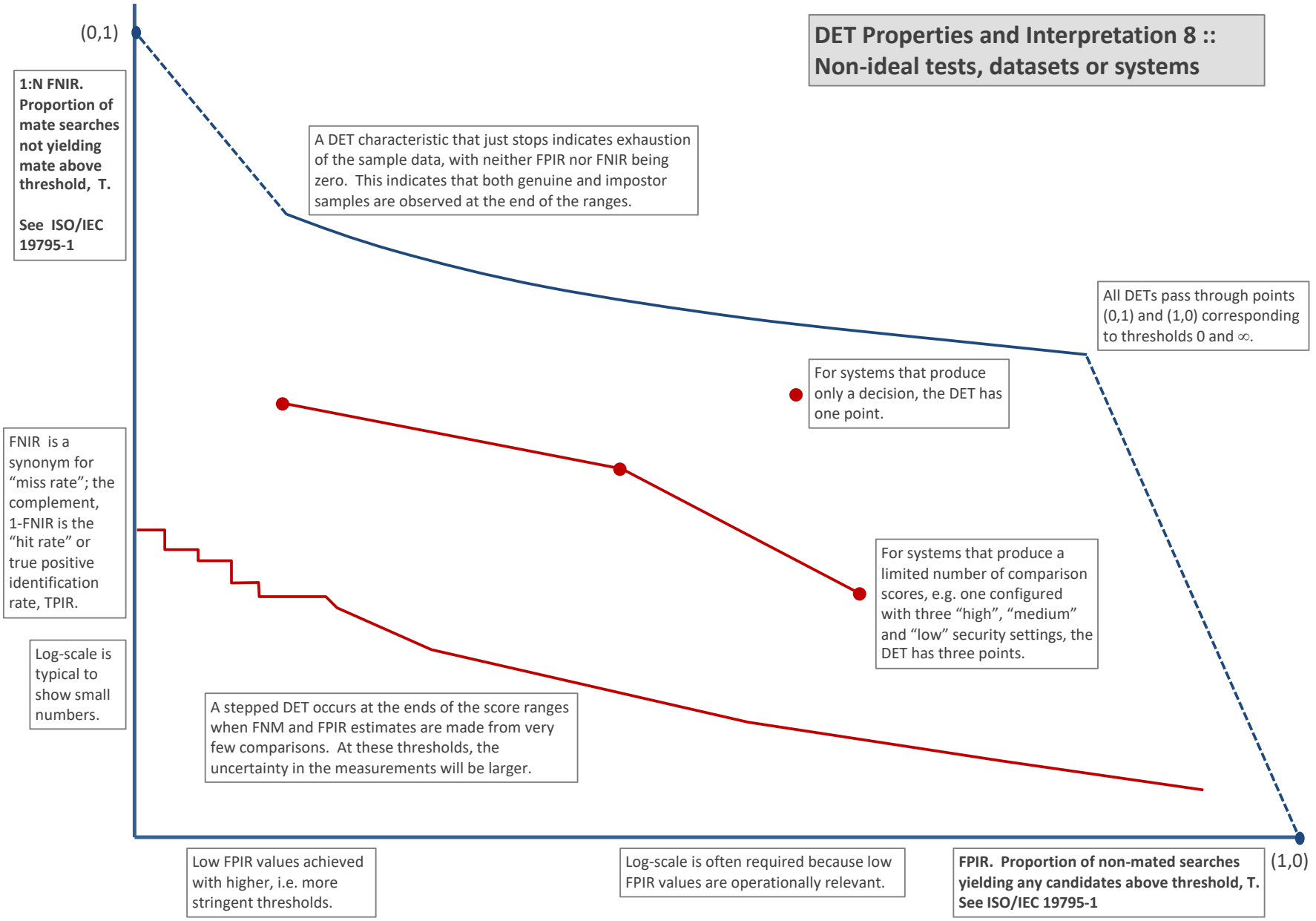


*Figure 17:* **DET as the primary performance reporting mechanism**.



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

### 3.4 Best practice testing requires execution of searches with and without mates

FRVT embeds 1:N searches of two kinds: Those for which there is an enrolled mate, and those for which there is not. The respective numbers for these types of searches appear in Table 1. However, it is common to conduct only mated searches[10]. The cumulative match characteristic is computed from candidate lists produced in mated searches. Even if the CMC is the only metric of interest, the actual trials executed in a test should nevertheless include searches for which no mate exists. As detailed in Table 1 the FRVT reserved disjoint populations of subjects for executing true non-mate searches.

### 3.5 Failure to extract features

During enrollment some algorithms fail to convert a face image to a template. The proportion of failures is the failure-to-enroll rate, denoted by FTE. Similarly, some search images are not converted to templates. The corresponding proportion is termed failure-to-extract, denoted by FTX.

We do not report FTX because we assume that the same underlying algorithm is used for template generation for enrollment and search.

Failure to extract rates are incorporated into FNIR and FPIR measurements as follows.

- **Enrollment templates**: Any failed enrollment is regarded as producing a zero length template. Algorithms are required by the API [10] to transparently process zero length templates. The effect of template generation failure on search accuracy depends on whether subsequent searches are mated, or non-mated: Mated searches will fail giving elevated FNIR; non-mated searches will not produce false positives so, to first order, FPIR will be reduced by a factor of $1-\text{FTE}$.
- **Search templates and 1:N search**: In cases where the algorithm fails to produce a search template from input imagery, the result is taken to be a candidate list whose entries have no hypothesized identities and zero score. The effect of template generation failure on search accuracy depends on whether searches are mated, or non-mated: Mated searches will fail giving elevated FNIR; Non-mated searches will not produce false positives, so FPIR will be reduced. Thus given a measurement of false negative and positive rates made over only those where failures-to-extract did not occur, those rates - call them $\text{FNIR}^\dagger$ and $\text{FPIR}^\dagger$ - could be adjusted by an explicit measurement of FTX as follows

$$\text{FNIR} = \text{FTX} + (1 - \text{FTX})\text{FNIR}^\dagger \tag{8}$$

$$\text{FPIR} = (1 - \text{FTX})\text{FPIR}^\dagger \tag{9}$$

This approach is the correct treatment for positive-identification applications such as access control where cooperative users are enrolled and make attempts at recognition. This approach is not appropriate to negative identification applications, such as visa fraud detection, in which hostile individuals may attempt to evade detection by submitting poor quality samples. In those cases, template generation failures should be investigated as though a false alarm had occurred.

[10] For example, the Megaface benchmark. This is bad practice for several reasons: First, if a developer knows, or can reasonably assume, that a mate always exists, then unrealistic gaming of the test is possible. A second reason is that it does not put FPIR on equal footing with FNIR and that matters because in most applications, not all searches have mates - not everyone has been previously enrolled in a driving license issuance or a criminal justice system - so addressing between-class separation becomes necessary.

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### 3.6 Fixed length candidate lists, threshold independent workload

Suppose an automated face identification algorithm returns L candidates, and a human reviewer is retained to examine up to R candidates, where $R \leq L$ might be set by policy, preference or labor availability. For now, assume also that the reviewer is not provided with, or ignores, similarity scores, and thresholds are not applied. Given the algorithm typically places mates at low (good) ranks, the number of candidates a reviewer can be expected to review can be derived as follows. Note that the reviewer will:

- ▷ Always inspect the first ranked image — Frac. reviewed = 1
- ▷ Then inspect those candidates where mate not confirmed at rank 1 — Frac. reviewed = 1-CMC(1)
- ▷ Then inspect those candidates where mate not confirmed at rank 1 or 2 — Frac. reviewed = 1-CMC(2)

etc. Thus if the reviewer will stop after a maximum of $R$ candidates, the expected number of candidate reviews is

$$M(R) = 1 + (1 - CMC(1)) + (1 - CMC(2)) + \ldots + (1 - CMC(R-1)) \tag{10}$$

$$= R - \sum_{r=1}^{R-1} CMC(r) \tag{11}$$

A recognition algorithm that front-loads the cumulative match characteristic will offer reduced workload for the reviewer. This workload is defined only over the searches for which a mate exists. In the cases where there truly is no mate, the reviewer would review all $R$ candidates. Thus, if the proportion of searches for which a mate does exist is $\beta$, which in the law enforcement context would be the recidivism rate [3], the full expression for workload becomes:

$$M(R) = \beta \left( R - \sum_{r=1}^{R-1} CMC(r) \right) \quad + \quad (1 - \beta)R \tag{12}$$

$$= R - \beta \sum_{r=1}^{R-1} CMC(r) \tag{13}$$

### 3.7 Timing measurement

Algorithms were submitted to NIST as implementations of the application programming interface(API) specified by NIST in the Evaluation Plan [10]. The API includes functions for initialization, template generation, finalization, search, gallery insert, and gallery delete. Two template generation functions are required, one for the preparation of an enrollment template, and one for a search template.

In NIST's test harness, all functions were wrapped by calls to the C++ std::chrono::high resolution clock which on the dedicated timing machine counts 1ns clock ticks. Precision is somewhat worse than that however.

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## 3.8 Uncertainty estimation

### 3.8.1 Random error

This study leverages operational datasets for measurement of recognition error rates. This affords several advantages. First, large numbers of searches are conducted (see Table 1) giving precision to the measurements. Moreover, for the two mugshot datasets, these do not involve reuse of individuals so binomial statistics can be expected to apply to recognition error counts. In that case, an observed count of a particular recognition outcome (i.e. a false negative or false positive) in $M$ trials will sustain 95% confidence that the actual error rate is no larger than some value.

As an example, the minimum number of mugshot searches conducted in this report is $M$ =154 549, and for an observed FNIR around 0.002, the measurement supports a conclusion that the actual FNIR is no higher than 0.00228 at 99% confidence level. On the false positive side, we tabulate FNIR at FPIR values as low as 0.001. Given estimates based on 331 254 non-mate trials, the actual FPIR values will be below 0.00115 at 99% confidence. In conclusion, large scale evaluation, without reuse of subjects, supports tight uncertainty bounds on the measured error rates.

### 3.8.2 Systematic error

The FRVT 2018 dataset includes anomalies discovered as a result of inspecting images involved in recognition failures from the most accurate algorithms. Two kinds of failure occur: False negatives (which, for the purpose here, include failures to make templates) and false positives.

**False negative errors**: We reviewed 600 false negative pairs for which either or both of the leading two algorithms did not put the correct mate in the top 50 candidates. Given 154 549 searches, this number represents 0.39% of the total, resulting in FNIR $\sim 0.0039$. Of the 600 pairs:

- **A: Poor quality**: About 20% of the pairs included images of very low quality, often greyscale, low resolution, blurred, low contrast, partially cropped, interlaced, or noisy scans of paper images. Additionally, in a few cases, the face is injured or occluded by bandages or heavy cosmetics.
- **B: Ground truth identity label bugs**: About 15% of the pairs are not actually mated. We only assigned this outcome when a pair is clearly not mated.
- **C: Profile views**: About 35% included an image of a profile (side) view of the face, or, more rarely, an image that was rotated 90 degrees in-plane (roll).
- **D: Tattoos**: About 30% included an image of a tattoo that contained a face image. These arise from mis-labelling in the parent dataset metadata.
- **E: Ageing**: There is considerable time-lapse between the two captures.

All these estimates are approximate. Of these, the tattoo and mislabled images can never be matched. These constitute an accuracy floor in the sample implying that FNIR cannot be below 0.0018[11]. The profile-views, low-quality images, and images with considerable ageing can, in principle, be successfully matched - indeed some algorithms do so - so are not part of the accuracy floor.

[11] This value is the sum of two partial false negative rates: $\text{FNIR}_B$ = 0.15 * 0.0039 plus $\text{FNIR}_D$ = 0.3 * 0.0039

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For the microsoft-4 algorithm the lowest miss rate from (recent entry in Table 51) is FNIR(640 000, 50, 0) = 0.0018. This is close to the value estimated from the inspection of misses. It is below the 0.0039 figure because the algorithm does match some profile and poor quality images, that the yitu-2 algorithm does not.

For many tables (e.g. Table 51), the FNIR values obtained for the FRVT-2018 mugshots could be corrected by reducing them by 0.0018. The best values would then be indistinct from zero. The results in this report *were not* adjusted to account for this systematic error.

**False positive errors**: As shown in Figure 1 and discussed in Figure 14 many of the DET characteristics in this report exhibit a pronounced turn upward at low false positive rates. The shape can be caused by identity labelling errors in the ground truth of a dataset, specifically persons present in the database under two IDs such that some proportion of non-mate pairs are actually mated. To look for such possibilities, we merged the highest 1000 non-mate pairs produced by three different algorithms which resulted in 1839 unique pairs. This constitutes 0.56% of all non-mate searches. We assert that it is *very* difficult for human reviewers to assign the pairs into the following three categories: twins; doppelgangers; or ground-truth errors (instances of the same person under two IDs). Given this difficulty we made no attempt to correct any possible ground truth errors except by removing 57 pairs in the following categories:

- **A: Profile views**: Thirteen pairs included one or two profile-view images. As described in Figure 466, these can cause false positives.
- **B: Same-session photographs**: For twelve pairs, the images were identical or trivially altered (e.g. cropped) versions of the same photo. These were present under a different ID likely due to some clerical or procedural mistake.
- **C: Tattoos of faces**: There were fourteen instances of tattoo photographs that contained faces causing false matches.
- **D: T-shirt faces**: There were six instances of T-shirt photographs (of Bob Marley and Che Guevara) being detected instead of the face and causing false positives.
- **E: Background faces**: There were twelve instances of one subject appearing in the background of two otherwise correct portrait photos.

Note we did not remove any images where there was a chance that the pair was actually a different person.

In any case, the results in this report have not been adjusted for this systematic error.

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# 4 Results

This section gives extensive results for algorithms submitted to FRVT 2018. Three page "report cards" for each algorithm are contained in a separate supplement. Performance metrics were described in section 3. The main results are summarized in tabular form with more exhaustive data included as DET, CMC and related graphs in appendices as follows:

- The three tables 2-4 list algorithms alongside full developer names, acceptance date, size of the provided configuration data, template size and generation time, and search duration data.
  - The **template generation duration** is most important to applications that require fast response. For example, an eGate taking more than two seconds to produce a template might be unacceptable. Note that GPUs may be of utility in expediting this operation for some algorithms, though at additional expense. Two additional factors should be considered[12][13].
  - The **search duration** is the time taken for a search of a search template into a gallery of $N$ enrollment templates. This performance variable, together with the volume of searches, is influential on the amount of hardware needed to sustain an operational deployment. This is measured here with the algorithm running on a single core of a contemporary CPU. Search is most simply implemented as N computations of a distance metric followed by a sort operation to find the closest enrollments. However, considerable optimization of this process is possible, up to and including fast-search algorithms that, by various means, avoid computation of all $N$ distances.
  - The **template size** is the size of the extracted feature vector (or vectors) and any needed header information. Large template sizes may be influential on bus or network bandwidth, storage requirements, and on search duration. While the template itself is an opaque data blob, the feature dimensionality might be estimated by assuming a four-bytes-per-float encoding. There is a wide range of encodings. For the more accurate algorithm, sizes range from 256 bytes to about 2KB bytes, indicating essentially no consensus on face modeling and template design.
  - The **template size multiplier** column shows how, given $k$ input images, the size of the template grows. Most implementations internally extract features from each image and concatenate them, and implement some score-level fusion logic during search. Other implementations, including many of the most accurate algorithms, produce templates whose size does not grow with $k$. This could be achieved via selection of the best quality image - but this is not optimal in handling ageing where the oldest image could be the best quality. Another mechanism would be feature-level fusion where information is fused from all $k$ inputs. In any case, as a black-box test, the fusion scheme is proprietary and unknown.
  - The size of the **configuration data** is the total size of all files resident in a vendor-provided directory that contains arbitrary read-only files such as parameters, recognition models (e.g caffe). Generally a large value for this quantity may prohibit the use of the algorithm on a resource-constrained device.

[12] The FRVT 2018 API prohibited threading, so some gains from parallelism may be available on multiple-cores or multiple processors, if the feature extraction code could be distributed across them.

[13] Note also that factors of two or more may be realizable by exploiting modern vector processing instructions on CPUs. It is not clear in our measurements whether all developers exploited Intel's AVX2 instructions, for example. Our machine was so equipped, but we insisted that the same compiled library should also run on older machines lacking that instruction. The more sophisticated implementations may have detected AVX2 presence and branched accordingly. The less sophisticated may be defaulted to the reduced instruction set. Readers should see the FRVT 2018 API document for the specific chip details.

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▷ Tables 51-52 report core rank-based accuracy for mugshot images. The population size is limited to N = 1.6 million identities because this is the largest gallery size on which all algorithms were executed. Notable observations from these tables are as follows:

- **Accuracy gains since 2018**: NIST Interagency Report 8238 documented massive gains over those reported in the FRVT 2014 report, NIST Interagency Report 8009. Further gains are documented in this report. Comparing the most accurate algorithm in November 2018, NEC-3, the value of FNIR(N, L, T) reduced from 0.0031 to 0.0024 for the Sensetime-004 algorithm with N = 12 million recent images. The tables show broader gains: many developers have made advances since 2018 with between two and five-fold reduction in errors.
- **Wide range in accuracy**: The rank-1 miss rates vary from FNIR(N, 1, 0) = 0.0012 for sensetime-004 up to about 0.5 for the very fast but inaccurate microfocus-x algorithms. Among the developers who are superior to NEC in 2013, the range is from 0.002 to 0.035 for camvi-3. This large accuracy range is consistent with the buyer-beware maxim, and indicates that face recognition software is far from being commoditized.

▷ Tables 60-61 report threshold-based error rates, FNIR(N, L, T), for N = 1.6 million for mugshot-mugshot accuracy on FRVT 2014, FRVT 2018, and also (in pink) mugshot-webcam accuracy using FRVT 2018 enrollments. Notable observations from these tables are as follows:

- **Order of magnitude accuracy gains since 2014**: As with rank-based results, the gains in accuracy are substantial, though somewhat reduced. At FPIR = 0.01, the best improvement over NEC in 2014 is a 27 fold reduction in FNIR using the NEC_2 algorithm. At FPIR = 0.001, the largest gain is a six-fold reduction in FNIR via the NEC_3 algorithm.
- **Broad gains across the industry**: About 19 companies realize accuracy better than the NEC benchmark from 2014. This is somewhat lower than the 28 developers who succeeded on the rank-1 metric. This may be due to the ubiquity of, and emphasis on, the rank-1 metric in many published algorithm development papers.
- **Webcam images**: Searches of webcam images give FNIR(N, T) values around 2 to 3 times higher than mugshot searches. Notably the leading developers with mugshots are approximately the same with poorer quality webcams. But some developers e.g. Camvi, Megvii, TongYi, and Neurotechnology do improve their relative rankings on webcams, perhaps indicating their algorithms were tailored to less constrained images.

▷ Tables 37, 44, 45 and show, respectively, high-threshold, rank 1, and rank 50 FNIR values for all algorithms performing searches into five different gallery sizes, N = 640 000, N = 1 600 000, N = 3 000 000, N = 6 000 000 and 12 000 000. The FPIR = 0.001 table is included to inform high-volume duplicate detection applications. The Rank-1 table is included as a primary accuracy indicator. The Rank-50 table is included to inform agencies who routinely produce 50 candidates for human-review. The notable results are:

- **Slow growth in rank-based miss rates:** FNIR(N, R) generally grows as a power law, $aN^b$. From the straight lines of many graphs of Figure 20 this is clearly a reasonable model for most, but not all, algorithms. The coefficient $a$ can be interpreted as FNIR in a gallery of size 1. The more important coefficient $b$ indicates scalability, and often, $b \ll 1$, implies very benign growth in FNIR. The coefficients of the models appear in the Tables 44 and 45.
- **Slow growth in threshold-based miss rates:** FNIR(N, T) also generally grows as a power law, $aN^b$ except at the high threshold values corresponding to low FPIR values. This is visible in the plots of Figure 84 which

show straight lines except for FPIR = 0.001, which increase more rapidly with N above 3 000 000. Each trace in those figures shows FNIR(N, T) at fixed FPIR with both N and T varying. Thus at large N, it is usually necessary to elevate T to maintain fixed FPIR. This causes increased FNIR. Why that would no-longer obey a power-law is not known. However, if we expect large galleries to contain individuals with familial relations to the non-mate search images - in the most extreme case, twins - then suppression of false positives becomes more difficult. This is discussed in the Figures starting at Fig. 10

▷ Figure **??** shows false positives from twins against their enrolled siblings, broken out by type of twin: fraternal or identical. The Figure is based on the enrollment of 104 single images on one of a pair of twins, and then the search of 2354 second images. Note that the dataset is heavily skewed towards identical twins which is not representative of the true population. There is also a skew towards same sex fraternal twin pairs compared to different sex fraternal twin pairs again not representative of the true population.

The notable results are:

- For all algorithms tested, the 1087 mated searches (Twin A vs. Twin A) produce scores almost always above typical operational thresholds, with (not shown) matches at rank 1. The images are of good quality, so this is the result expected from the rest of this report.
- For the 1066 identical twin searches (AB), almost all produce the twin at rank 1, with a few producing the mate at further down the candidate lists rank and low score.
- For the 169 fraternal searches (AB) from same sex pairs, most algorithms give a large number of very high scores, implying false positives at all thresholds. However, there there are long tails containing lower scores that are correctly below threshold. In general, scores that are higher in this distribution are all rank 1 whereas the lower scores have much higher ranks.
- (Not shown) Of the 169, there are 24 fraternal searches (AB) involving different sex twins. Here most algorithms correctly report scores well below the lowest threshold, and usually not on the candidate list at all.

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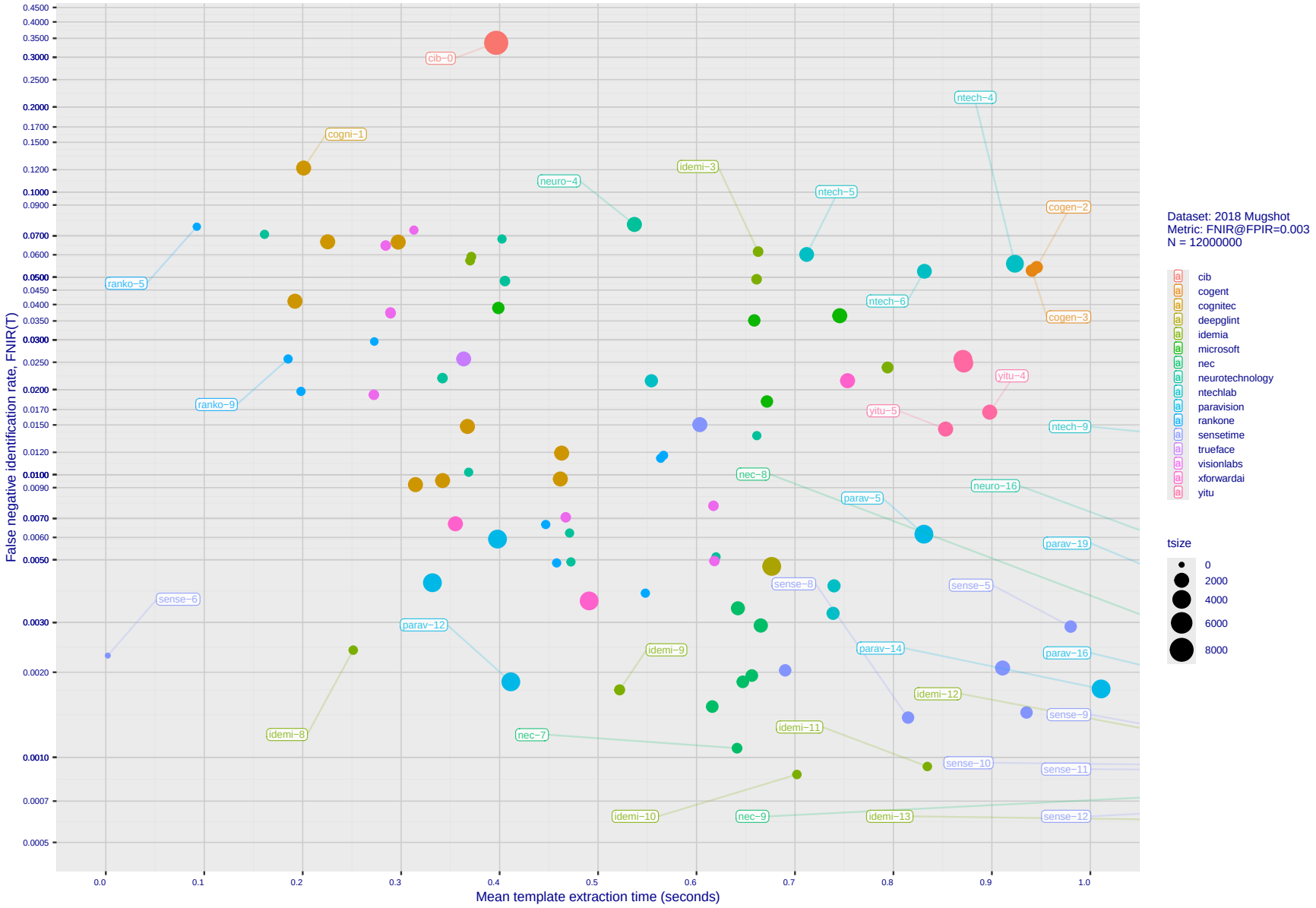


*Figure 18:* **[Mugshot Dataset] Speed-accuracy tradeoff**. *For developers of the more accurate algorithms the plot shows the tradeoff of high-threshold recognition miss-rates, FNIR(N, N, T) for FPIR(N, T) = 0.003, and template generation time. Developers are coded by color. Template size is encoded by the size of the circle. Some labels are quite distant from the respective point, to avoid superposing text. Without any other influences, the assumption would be that taking time to localize the face, and extract features, would lead to better accuracy. The most notable result, for NEC, is that their slower algorithms are much more accurate than the version that extract features in fewer than 90 milliseconds.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
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T = 0 → Investigation
T > 0 → Identification

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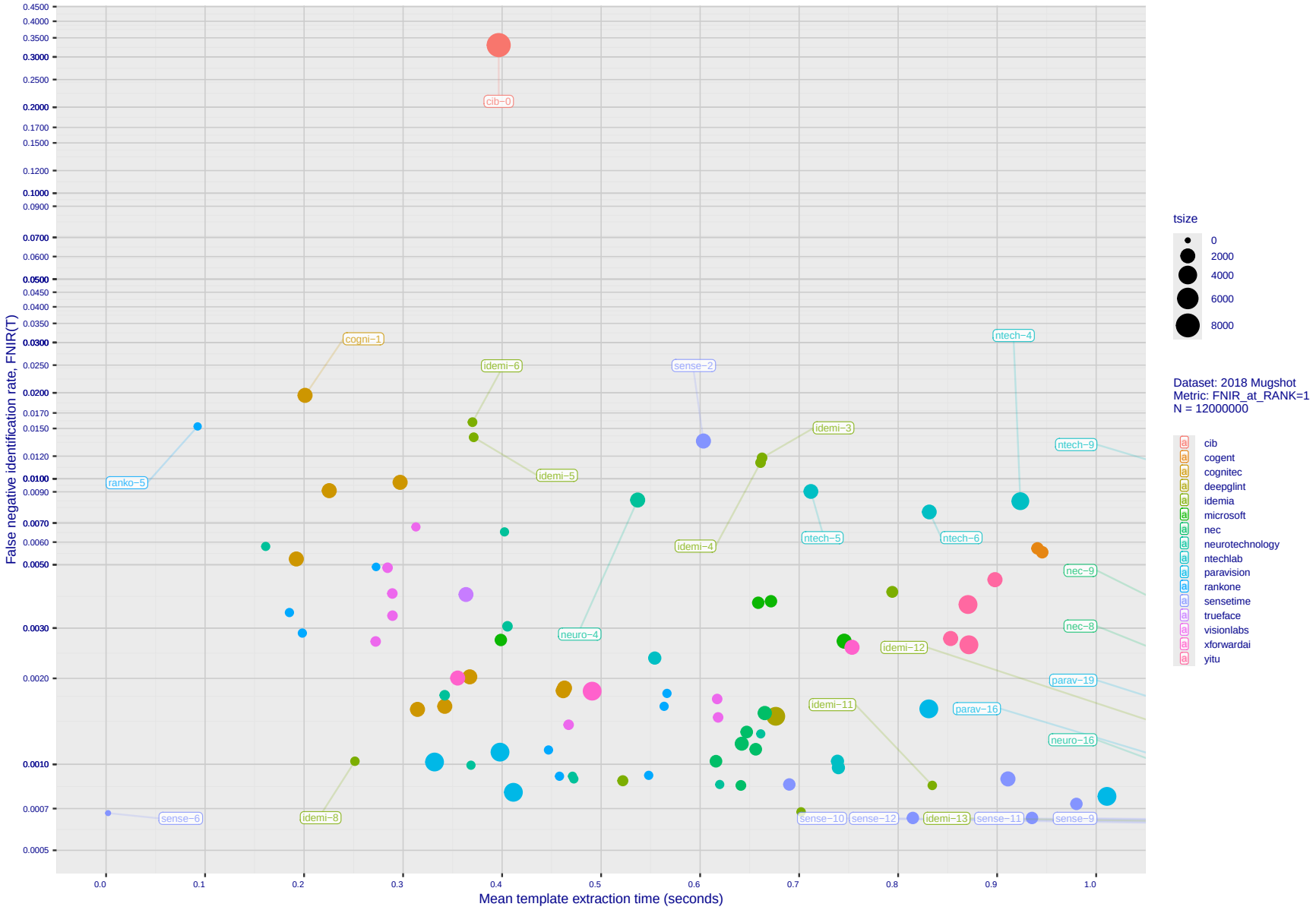


*Figure 19:* **[Mugshot Dataset] Speed-accuracy tradeoff**. *For developers of the more accurate algorithms the plot shows the tradeoff of rank-one recognition miss-rates, FNIR(N, 1, 0), and template generation time. Developers are coded by color. Template size is encoded by the size of the circle. Some labels are quite distant from the respective point, to avoid superposing text. Without any other influences, the assumption would be that taking time to localize the face, and extract features, would lead to better accuracy. This occurs for NEC with their slower algorithm being much accurate than the version that extract features in fewer than 90 milliseconds.*

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | N=1.6M | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 1 | 20Face | NL | 20face | 000 | 2021-10-01 | 112 | 319 | [298]2048 | - | [28]184 | [125]9 | [485]5679 | [504]6341 | - | - | - | |
| 2 | 3Divi | RU | 3divi | 5 | 2018-10-26 | 186 | 51 | [480]4096 | k | [279]638 | [418]28 | [198]538 | [199]537 | [191]1377 | [181]2614 | [174]5530 | $^{402}0.07N^{1.1}$ |
| 3 | 3Divi | RU | 3divi | 6 | 2018-10-26 | 187 | 51 | [85]528 | k | [280]640 | [56]5 | [28]33 | [25]33 | - | - | - | |
| 4 | AFIS and Biometrics Consulting | US | afisbiometrics | 000 | 2023-06-07 | 533 | 34 | [38]265 | - | [287]650 | [8]2 | [361]1718 | [348]1563 | [304]3063 | [309]6541 | [308]13588 | $^{316}0.46N^{1.1}$ |
| 5 | AFR Engine | US | afrengine | 000 | 2023-07-14 | 566 | 24 | [172]2048 | - | [374]818 | [194]12 | [468]4542 | [476]4973 | [409]9473 | [409]20213 | [404]45388 | $^{327}1.38N^{1.1}$ |
| 6 | AFR Engine | US | afrengine | 001 | 2024-03-15 | 565 | 24 | [349]2048 | - | [421]973 | [240]13 | [464]4505 | [466]4444 | [401]8555 | [404]18109 | [401]42844 | $^{362}0.89N^{1.1}$ |
| 7 | AFR Engine | US | afrengine | 002 | 2024-07-22 | 566 | 24 | [291]2048 | - | [443]1110 | [215]12 | [456]4324 | [461]4120 | [403]8653 | [402]17832 | [394]35825 | $^{382}0.61N^{1.1}$ |
| 8 | AFR Engine | US | afrengine | 003 | 2025-02-18 | 678 | 24 | [227]2048 | - | [425]1006 | [244]13 | [446]3803 | [455]4034 | [386]7389 | [401]17642 | [393]35554 | $^{344}0.90N^{1.1}$ |
| 9 | AI Implementation Group | UZ | aigroup | 000 | 2026-05-12 | 368 | 387 | [444]2078 | - | [185]479 | [371]19 | [307]1153 | [307]1164 | [261]2227 | [258]4525 | [246]8959 | $^{195}0.56N^{1.0}$ |
| 10 | ALTTEKGlobal | VN | alttekglobal | 000 | 2024-04-12 | 634 | 2988 | [241]2048 | - | [442]1108 | [218]13 | [105]255 | [105]257 | - | - | - | |
| 11 | AadiSysTech | IN | aadi | 000 | 2023-06-27 | 95 | 83 | [62]512 | - | [91]346 | [33]3 | [385]2024 | [368]1754 | - | - | - | |
| 12 | AadiSysTech | IN | aadi | 001 | 2023-12-08 | 307 | 787 | [224]2048 | - | [15]127 | [144]10 | [428]3144 | [425]2850 | - | - | - | |
| 13 | AadiSysTech | IN | aadi | 002 | 2024-04-17 | 95 | 567 | [248]2048 | - | [10]96 | [241]13 | [421]2992 | [424]2794 | - | - | - | |
| 14 | Acer Incorporated | TW | acer | 000 | 2020-08-12 | 35 | 67 | [71]512 | - | [36]198 | [45]4 | [121]295 | [120]295 | [103]623 | [168]2302 | [164]4915 | $^{447}0.00N^{1.3}$ |
| 15 | Acer Incorporated | TW | acer | 001 | 2021-11-08 | 42 | 610 | [171]2048 | - | [8]92 | [122]9 | [173]444 | [211]575 | - | - | - | |
| 16 | Adiance Technologies Pvt Ltd | IN | adiance | 001 | 2025-04-28 | 261 | 332 | [288]2048 | - | [398]887 | [392]22 | [391]2096 | [388]2048 | - | - | - | |
| 17 | Adiance Technologies Pvt Ltd | IN | adiance | 002 | 2026-01-13 | 265 | 367 | [328]2048 | - | [515]1470 | [482]47 | [412]2618 | [419]2610 | [361]4872 | [357]9811 | [357]19756 | $^{112}1.54N^{1.0}$ |
| 18 | Advance.AI | SG | advance | 000 | 2023-05-30 | 271 | 825 | [271]2048 | - | [245]573 | [173]12 | [396]2268 | [401]2271 | [347]4338 | [341]8690 | [342]17842 | $^{174}1.15N^{1.0}$ |
| 19 | Akurat Satu Indonesia | ID | ptakuratsatu | 000 | 2020-10-23 | 0 | 572 | [88]538 | - | [408]929 | [547]28633 | [17]18 | [14]16 | [7]17 | [6]17 | [5]17 | $^{3}6827.74N^{0.1}$ |
| 20 | Akuvox | CN | akuvox | 001 | 2025-02-14 | 0 | 213 | [531]4843 | - | [14]114 | [429]30 | [402]2434 | [420]2613 | [358]4738 | [358]10025 | [356]19681 | $^{149}1.39N^{1.0}$ |
| 21 | Alchera Inc | KR | alchera | 2 | 2018-10-30 | 7 | 14 | [309]2048 | k | [13]114 | [498]63 | [418]2923 | [428]2929 | - | - | - | |
| 22 | Alchera Inc | KR | alchera | 3 | 2018-10-30 | 251 | 14 | [339]2048 | k | [224]531 | [499]63 | [420]2955 | [429]2956 | [376]6546 | [384]15013 | [392]35262 | $^{439}0.10N^{1.2}$ |
| 23 | Alchera Inc | KR | alchera | 004 | 2021-09-17 | 476 | 24 | [215]2048 | - | [361]786 | [453]35 | [490]5801 | [512]6851 | - | - | - | |
| 24 | Alchera Inc | KR | alchera | 005 | 2023-07-18 | 977 | 482 | [247]2048 | - | [221]527 | [455]35 | [494]6123 | [494]6035 | - | - | - | |
| 25 | Alchera Inc | KR | alchera | 006 | 2023-11-22 | 977 | 482 | [180]2048 | - | [232]546 | [172]12 | [512]8046 | [513]6865 | [420]11609 | [427]27328 | [423]59209 | $^{191}3.40N^{1.0}$ |
| 26 | Alchera Inc | KR | alchera | 007 | 2024-06-17 | 978 | 482 | [289]2048 | - | [239]558 | [188]12 | [540]20981 | [550]20564 | - | - | - | |
| 27 | Alchera Inc | KR | alchera | 008 | 2024-10-25 | 2037 | 443 | [273]2048 | - | [507]1429 | [168]12 | [503]6350 | [496]6134 | [423]11940 | [420]25274 | [420]56159 | $^{371}1.03N^{1.1}$ |
| 28 | Algorithms Inc | AE | algorithmsinc | 000 | 2025-02-26 | 255 | 79 | [313]2048 | - | [165]452 | [269]14 | [261]903 | [264]924 | [257]2153 | [240]4199 | [236]8654 | $^{424}0.06N^{1.2}$ |
| 29 | Alivia / Innovation Sys | RU | isystems | 3 | 2018-10-30 | 350 | 784 | [263]2048 | 1 | [376]825 | [338]16 | [149]385 | [159]389 | [153]979 | [149]1822 | [253]9348 | $^{449}0.00N^{1.3}$ |
| 30 | AllGoVision | IN | allgovision | 000 | 2019-07-30 | 168 | 150 | [376]2048 | k | [134]404 | [199]12 | [432]3226 | [436]3193 | [370]6129 | [370]12449 | [366]25835 | $^{220}1.40N^{1.0}$ |
| 31 | AllGoVision | IN | allgovision | 001 | 2020-07-14 | 283 | 126 | [358]2048 | - | [356]777 | [231]13 | [430]3174 | [435]3183 | [368]6073 | [367]12284 | [364]25701 | $^{215}1.42N^{1.0}$ |
| 32 | AllGoVision | IN | allgovision | 002 | 2023-03-13 | 284 | 797 | [200]2048 | - | [171]459 | [251]13 | [425]3112 | [443]3454 | - | - | - | |
| 33 | AllGoVision | IN | allgovision | 003 | 2023-09-19 | 264 | 799 | [231]2048 | - | [447]1128 | [204]12 | [429]3171 | [433]3100 | [366]5906 | [373]13318 | - | $^{303}0.97N^{1.1}$ |
| 34 | AllGoVision | IN | allgovision | 004 | 2025-07-11 | 704 | 34 | [363]2048 | - | [300]669 | [198]12 | [337]1480 | [338]1492 | - | - | - | |
| 35 | AllGoVision | IN | allgovision | 005 | 2026-03-12 | 455 | 83 | [206]2048 | - | [451]1138 | [293]14 | [356]1666 | [371]1827 | [100]605 | [140]1734 | [89]2408 | $^{17}10459.37N^{0.3}$ |
| 36 | Anke Investments | CN | anke | 0 | 2018-10-30 | 779 | 27 | [442]2072 | k | [154]429 | [329]16 | [223]675 | [239]748 | [199]1483 | [196]2968 | [192]6148 | $^{318}0.21N^{1.1}$ |
| 37 | Anke Investments | CN | anke | 1 | 2018-10-30 | 779 | 27 | [443]2072 | k | [155]430 | [309]15 | [228]707 | [243]769 | - | - | - | |
| 38 | Anke Investments | CN | anke | 002 | 2019-06-27 | 341 | 401 | [421]2056 | k | [270]623 | [264]13 | [215]624 | [226]682 | [182]1306 | [172]2403 | [169]5082 | $^{200}0.30N^{1.0}$ |
| 39 | Antheus Technologia | BR | antheus | 000 | 2024-08-19 | 264 | 185 | [169]2048 | - | [486]1273 | [210]12 | [367]1781 | [369]1788 | - | - | - | |
| 40 | Aratek Biometrics Co Ltd | CN | aratek | 001 | 2024-04-18 | 248 | 15 | [344]2048 | - | [43]232 | [183]12 | [516]8371 | [523]8210 | [449]16858 | [444]33012 | [431]68839 | $^{305}2.54N^{1.1}$ |
| 41 | Armatura LLC | US | armatura | 000 | 2023-04-13 | 0 | 935 | [537]6144 | - | [291]655 | [500]66 | [498]6211 | [497]6138 | [418]11533 | [414]23189 | [408]46629 | $^{116}3.59N^{1.0}$ |
| 42 | Aware | US | aware | 5 | 2018-10-30 | 368 | 27 | [463]3100 | k | [364]792 | [449]34 | [40]95 | [48]98 | [40]203 | [36]371 | [18]252 | $^{22}4.13N^{0.7}$ |
| 43 | Aware | US | aware | 6 | 2018-10-30 | 368 | 27 | [5]124 | k | [363]789 | [13]2 | [73]158 | [72]162 | - | - | - | |
| 44 | Aware | US | aware | 007 | 2023-11-27 | 132 | 18 | [54]480 | - | [334]719 | [147]11 | [59]131 | [58]131 | [57]320 | [56]655 | [52]1324 | $^{404}0.02N^{1.1}$ |
| 45 | Awidit Systems | IN | awiros | 000 | 2025-02-10 | 478 | 584 | [186]2048 | - | [210]508 | [243]13 | [353]1628 | [377]1931 | - | - | - | |
| 46 | Ayonix | JP | ayonix | 1 | 2018-10-29 | 74 | 2 | [133]1036 | k | [5]12 | [155]11 | [116]279 | [113]279 | - | - | - | |
| 47 | Ayonix | JP | ayonix | 2 | 2018-10-30 | 74 | 2 | [134]1036 | 1 | [4]11 | [282]14 | [115]279 | [112]276 | [87]535 | [89]1087 | [86]2284 | $^{247}0.11N^{1.0}$ |
| 48 | Beijing Hisign Technology | CN | hisign | 001 | 2024-08-09 | 466 | 152 | [447]2080 | - | [269]620 | [320]15 | [266]926 | [308]1179 | [226]1855 | [228]3910 | [220]7639 | $^{42}2.25N^{0.9}$ |
| 49 | BioID Technologies SA | CH | bioidtechswiss | 001 | 2024-04-19 | 0 | 345 | [74]512 | - | [264]614 | [42]4 | [113]278 | [119]294 | [80]470 | [80]944 | [83]2141 | $^{263}0.10N^{1.0}$ |
| 50 | Biometrik Systems Private Limited | IN | biometrik | 001 | 2025-11-24 | 167 | 82 | [359]2048 | - | [516]1492 | [304]15 | [345]1543 | [336]1486 | [291]2813 | [287]5694 | [290]11840 | $^{239}0.59N^{1.0}$ |
| 51 | Camvi Technologies | US | camvitech | 4 | 2018-10-30 | 233 | 220 | [109]1024 | 1 | [312]686 | [441]31 | [29]33 | [24]32 | [15]38 | [14]40 | [9]48 | $^{4}8492.66N^{0.1}$ |
| 52 | Camvi Technologies | US | camvitech | 5 | 2018-10-30 | 257 | 220 | [110]1024 | 1 | [348]751 | [438]31 | [27]31 | [22]30 | - | - | - | |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 2: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| | DEVELOPER FULL NAME | DEVELOPER LOCATION | SHORT NAME | SEQ. NUM. | VALIDATION DATE | CONFIG[1] DATA (MB) | LIB[1] DATA (MB) | TEMPLATE GENERATION SIZE (B) | MULT[3] | TIME (MS)[4] | FINALIZE[2] TIME (S) N=1.6M | SEARCH DURATION[5] MILLISEC L=1 N=1.6M | L=50 N=1.6M | L=50 N=3M | L=50 N=6M | L=50 N=12M | POWER LAW ($\mu$s) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 53 | Canon Inc | CN | cib | 000 | 2020-10-19 | 426 | 127 | [550]8196 | - | [126]396 | [515]113 | (435)3316 | (448)3604 | (379)6738 | (377)13495 | (369)27114 | [83]$2.33N^{1.0}$ |
| 54 | Canon Inc | JP | canon | 001 | 2021-10-27 | 1139 | 91 | [473]4096 | - | [186]480 | [387]21 | (484)5489 | (509)6789 | (432)12741 | (423)25650 | (414)51922 | [134]$3.82N^{1.0}$ |
| 55 | Canon Inc | JP | canon | 002 | 2022-04-26 | 1231 | 111 | [539]6200 | - | [242]565 | [494]58 | (510)7673 | (507)6416 | (425)12178 | (418)24806 | (412)49653 | [169]$3.28N^{1.0}$ |
| 56 | Canon Inc | JP | canon | 003 | 2023-09-05 | 2462 | 78 | [523]4148 | - | [471]1211 | [425]29 | (382)2012 | (380)1966 | (331)3819 | (328)7679 | (335)16096 | [241]$0.80N^{1.0}$ |
| 57 | Clearview AI Inc | US | clearviewai | 000 | 2021-11-12 | 358 | 316 | [483]4096 | - | [124]393 | [434]30 | (217)639 | (222)657 | (169)1134 | (154)1939 | (146)3889 | [34]$1.59N^{0.9}$ |
| 58 | Clearview AI Inc | US | clearviewai | 001 | 2024-02-16 | 696 | 499 | [254]2048 | - | [344]743 | [379]20 | (156)394 | (156)384 | (124)722 | (118)1451 | (110)2919 | [120]$0.22N^{1.0}$ |
| 59 | Clearview AI Inc | US | clearviewai | 002 | 2024-06-18 | 696 | 497 | [319]2048 | - | [360]786 | [383]20 | (194)517 | (192)516 | (151)969 | (155)1946 | (149)3931 | [125]$0.29N^{1.0}$ |
| 60 | Clearview AI Inc | US | clearviewai | 003 | 2025-11-24 | 696 | 527 | [333]2048 | - | [441]1108 | [376]20 | (185)498 | (143)362 | (127)743 | (147)1780 | (142)3720 | [405]$0.04N^{1.1}$ |
| 61 | Cloudwalk - Hengrui AI Technology | CN | hr | 000 | 2021-02-10 | 501 | 392 | [304]2048 | - | [322]701 | [298]15 | (166)420 | (111)276 | (89)539 | (101)1268 | (122)3177 | [411]$0.03N^{1.1}$ |
| 62 | Cloudwalk Moontime Smart Technology | CN | cloudwalk | 000 | 2022-01-31 | 716 | 573 | [317]2048 | - | [395]869 | [143]10 | (171)440 | (150)371 | (91)547 | (87)1065 | (109)2902 | [47]$0.53N^{0.9}$ |
| 63 | Cloudwalk Moontime Smart Technology | CN | cloudwalk | 001 | 2022-07-27 | 797 | 574 | [314]2048 | - | [223]529 | [288]14 | (109)273 | (137)349 | (109)658 | (122)1492 | (119)3026 | [325]$0.10N^{1.1}$ |
| 64 | Cloudwalk Moontime Smart Technology | CN | cloudwalk | 002 | 2023-02-24 | 431 | 514 | [189]2048 | - | [225]531 | [295]14 | (238)767 | (247)826 | (204)1586 | (207)3311 | (191)6070 | [225]$0.35N^{1.0}$ |
| 65 | Cognitec Systems GmbH | DE | cognitec | 2 | 2018-10-30 | 463 | 26 | [398]2052 | k | [42]225 | [408]27 | (363)1733 | (368)1763 | (325)3660 | (320)7279 | (311)13895 | [211]$0.83N^{1.0}$ |
| 66 | Cognitec Systems GmbH | DE | cognitec | 3 | 2018-10-30 | 465 | 26 | [405]2052 | k | [66]297 | [326]16 | (362)1719 | (370)1791 | (323)3638 | (319)7277 | (320)14904 | [270]$0.66N^{1.0}$ |
| 67 | Cognitec Systems GmbH | DE | cognitec | 004 | 2021-03-08 | 384 | 60 | [401]2052 | - | [34]192 | [260]13 | (358)1673 | (363)1727 | (294)2904 | (289)5801 | (287)11707 | [73]$1.15N^{1.0}$ |
| 68 | Cognitec Systems GmbH | DE | cognitec | 005 | 2021-07-30 | 460 | 61 | [392]2052 | - | [106]367 | [128]9 | (347)1556 | (346)1551 | (296)2916 | (310)6561 | (312)13958 | [337]$0.38N^{1.1}$ |
| 69 | Cognitec Systems GmbH | DE | cognitec | 006 | 2022-02-10 | 689 | 61 | [406]2052 | - | [177]463 | [140]10 | (282)1006 | (281)1002 | (249)2097 | (248)4312 | (219)7624 | [309]$0.30N^{1.1}$ |
| 70 | Cognitec Systems GmbH | DE | cognitec | 007 | 2023-12-01 | 319 | 194 | [400]2052 | - | [73]314 | [279]14 | (292)1069 | (292)1051 | (262)2232 | (243)4244 | (226)7943 | [170]$0.55N^{1.0}$ |
| 71 | Cognitec Systems GmbH | DE | cognitec | 008 | 2024-04-29 | 359 | 39 | [260]2048 | - | [90]342 | [268]14 | (316)1258 | (288)1017 | (272)2433 | (267)4887 | (272)9888 | [389]$0.15N^{1.1}$ |
| 72 | Cognitec Systems GmbH | DE | cognitec | 009 | 2024-08-29 | 359 | 39 | [368]2048 | - | [174]461 | [249]13 | (289)1045 | (290)1025 | (250)2117 | (244)4267 | (229)8147 | [242]$0.42N^{1.0}$ |
| 73 | Compal Electronics | TW | compal | 000 | 2025-03-20 | 980 | 182 | [111]1024 | - | [483]1256 | [75]7 | (475)4960 | (486)5503 | (381)6929 | (316)6915 | (207)6911 | [18]$17496.15N^{0.4}$ |
| 74 | Compal Electronics | TW | compal | 001 | 2025-11-25 | 605 | 199 | [382]2049 | - | [407]910 | [358]17 | (542)22511 | (551)22297 | - | - | - | |
| 75 | Compal Electronics | TW | compal | 002 | 2026-07-09 | 605 | 198 | [381]2049 | - | [390]855 | [337]16 | (413)2619 | (418)2581 | - | - | - | |
| 76 | Corsight AI | IL | corsight | 000 | 2023-07-13 | 1277 | 31 | [445]2080 | - | [233]546 | [167]12 | (274)960 | (317)1299 | (254)2145 | (224)3828 | (273)9998 | [335]$0.26N^{1.1}$ |
| 77 | Corsight AI | IL | corsight | 001 | 2024-03-14 | 1172 | 30 | [446]2080 | - | [228]543 | [250]13 | (283)1011 | (286)1011 | (238)1951 | (232)4004 | (258)9391 | [342]$0.23N^{1.1}$ |
| 78 | Cubox | KR | cubox | 000 | 2021-08-24 | 529 | 298 | [331]2048 | - | [158]440 | [138]10 | (437)3410 | (459)4076 | (389)7605 | (389)15871 | - | [314]$1.16N^{1.1}$ |
| 79 | Cubox | KR | cubox | 001 | 2023-07-05 | 1654 | 342 | [195]2048 | - | [271]624 | [190]12 | (433)3268 | (444)3476 | (369)6097 | (368)12320 | (368)27074 | [155]$1.78N^{1.0}$ |
| 80 | Cubox | KR | cubox | 002 | 2024-11-29 | 1273 | 342 | [168]2048 | - | [418]962 | [182]12 | (447)3809 | (450)3859 | (384)7095 | (379)13774 | (375)28747 | [66]$3.02N^{1.0}$ |
| 81 | Cubox | KR | cubox | 003 | 2025-05-14 | 219 | 342 | [244]2048 | - | [109]369 | [223]13 | (449)3920 | (451)3915 | (385)7262 | (385)15013 | (379)29855 | [199]$1.80N^{1.0}$ |
| 82 | Cyberlink Corp | TW | cyberlink | 000 | 2019-06-12 | 217 | 93 | [397]2052 | 1 | [290]654 | [430]30 | (225)696 | (229)701 | (192)1379 | (182)2639 | (194)6214 | [254]$0.28N^{1.0}$ |
| 83 | Cyberlink Corp | TW | cyberlink | 001 | 2019-10-07 | 459 | 102 | [402]2052 | 1 | [149]423 | [419]28 | (226)698 | (228)700 | (187)1350 | (284)5524 | (294)12031 | [446]$0.00N^{1.3}$ |
| 84 | Cyberlink Corp | TW | cyberlink | 002 | 2020-07-31 | 333 | 109 | [521]4140 | - | [337]724 | [534]6875 | (327)1353 | (437)3198 | (371)6138 | (366)12205 | (301)13106 | [28]$16.71N^{0.8}$ |
| 85 | Cyberlink Corp | TW | cyberlink | 003 | 2021-01-05 | 333 | 100 | [541]6212 | - | [112]370 | [457]35 | (259)886 | (234)723 | (196)1415 | (193)2886 | (175)5643 | [368]$0.12N^{1.1}$ |
| 86 | Cyberlink Corp | TW | cyberlink | 004 | 2021-07-16 | 371 | 100 | [542]6212 | - | [120]386 | [394]23 | (265)924 | (190)504 | (149)923 | (117)1448 | (127)3350 | [46]$0.73N^{0.9}$ |
| 87 | Cyberlink Corp | TW | cyberlink | 005 | 2022-01-07 | 371 | 100 | [540]6212 | - | [340]733 | [436]30 | (187)501 | (186)498 | (176)1193 | (183)2672 | (179)5693 | [434]$0.03N^{1.2}$ |
| 88 | Cyberlink Corp | TW | cyberlink | 006 | 2024-06-12 | 1163 | 99 | [553]8260 | - | [335]722 | [491]54 | (387)2068 | (340)1498 | (330)3804 | (330)7796 | (330)15790 | [418]$0.13N^{1.1}$ |
| 89 | DAON | IE | daon | 000 | 2021-12-23 | 274 | 2 | [438]2069 | - | [220]526 | [99]8 | (286)1029 | (217)625 | (197)1454 | (202)3097 | (197)6316 | [435]$0.03N^{1.2}$ |
| 90 | DAON | IE | daon | 001 | 2025-01-03 | 289 | 64 | [302]2048 | - | [294]661 | [179]12 | (315)1242 | (312)1211 | - | - | - | |
| 91 | DAON | IE | daon | 002 | 2025-07-07 | 288 | 65 | [276]2048 | - | [347]751 | [169]12 | (323)1318 | (347)1552 | (275)2462 | (281)5452 | (280)11007 | [251]$0.53N^{1.0}$ |
| 92 | DAON | IE | daon | 003 | 2026-04-14 | 909 | 65 | [481]4096 | - | [548]2055 | [417]28 | (441)3584 | (447)3588 | (378)6731 | (375)13471 | (378)29819 | [230]$1.50N^{1.0}$ |
| 93 | Dactionable Technologies | VN | datech | 000 | 2023-10-24 | 603 | 178 | [474]4096 | - | [460]1176 | [524]1282 | (88)194 | (109)262 | (84)499 | (84)998 | (80)2029 | [350]$0.05N^{1.1}$ |
| 94 | Dactionable Technologies | VN | datech | 001 | 2024-03-14 | 709 | 178 | [488]4096 | - | [415]953 | [505]75 | (233)753 | (240)751 | (202)1537 | (194)2946 | (185)5802 | [175]$0.38N^{1.0}$ |
| 95 | Dahua Technology Co | CN | dahua | 0 | 2018-10-29 | 276 | 167 | [179]2048 | k | [115]374 | [390]22 | - | (106)258 | - | - | - | |
| 96 | Dahua Technology Co | CN | dahua | 1 | 2018-10-29 | 276 | 167 | [347]2048 | k | [108]369 | [413]28 | - | (104)257 | (98)602 | (97)1202 | (114)3007 | [421]$0.02N^{1.2}$ |
| 97 | Dahua Technology Co | CN | dahua | 002 | 2019-12-02 | 607 | 137 | [188]2048 | k | [311]685 | [374]19 | (100)243 | (110)269 | (174)1189 | (195)2950 | (201)6732 | [454]$0.00N^{1.5}$ |
| 98 | Dahua Technology Co | CN | dahua | 003 | 2020-11-18 | 889 | 154 | [256]2048 | - | [336]723 | [360]18 | (117)283 | (103)249 | (79)468 | (79)935 | (76)1871 | [86]$0.16N^{1.0}$ |
| 99 | Dahua Technology Co | CN | dahua | 004 | 2021-11-18 | 812 | 116 | [265]2048 | - | [351]758 | [146]11 | (167)423 | (165)411 | (143)871 | (128)1568 | (121)3174 | [229]$0.17N^{1.0}$ |
| 100 | Decatur Industries Inc | US | decatur | 000 | 2022-02-09 | 411 | 383 | [411]2052 | - | [394]863 | [130]9 | (365)1761 | (384)2023 | (319)3361 | (321)7283 | (317)14592 | [139]$1.06N^{1.0}$ |
| 101 | Deepglint | CN | deepglint | 001 | 2019-11-15 | 448 | 265 | [494]4096 | - | [304]676 | [452]35 | (224)677 | (339)1495 | (218)1724 | (187)2747 | (195)6246 | [25]$25.27N^{0.8}$ |
| 102 | Dermalog | DE | dermalog | 5 | 2018-10-26 | 0 | 440 | [9]128 | 1 | [222]528 | [529]3155 | (1)0 | (1)0 | (1)0 | (1)0 | (1)0 | [5]$66.21N^{0.2}$ |
| 103 | Dermalog | DE | dermalog | 6 | 2018-10-26 | 0 | 453 | [16]256 | 1 | [207]507 | [17]2 | (66)142 | (64)144 | (49)269 | (48)531 | (50)1294 | [279]$0.05N^{1.0}$ |
| 104 | Dermalog | DE | dermalog | 007 | 2020-02-12 | 0 | 424 | [8]128 | 1 | [139]410 | [3]1 | (49)98 | (44)96 | (43)218 | (39)429 | (41)1013 | [387]$0.01N^{1.1}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 3: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | | N=1.6M | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 105 | Dermalog | DE | dermalog | 008 | 2021-01-25 | 0 | 531 | [61]512 | - | [37]198 | [47]4 | [97]227 | [99]246 | [77]462 | [78]924 | [75]1849 | $^{93}0.15N^{1.0}$ |
| 106 | Dermalog | DE | dermalog | 009 | 2021-11-09 | 0 | 318 | [73]512 | - | [26]180 | [31]3 | [98]227 | [100]246 | [76]461 | [77]923 | [74]1846 | $^{88}0.16N^{1.0}$ |
| 107 | Dermalog | DE | dermalog | 010 | 2022-07-25 | 0 | 514 | [63]512 | - | [79]322 | [41]3 | [99]241 | [90]210 | [68]394 | [68]789 | [65]1580 | $^{91}0.13N^{1.0}$ |
| 108 | Dermalog | DE | dermalog | 011 | 2022-12-12 | 0 | 272 | [11]128 | - | [49]248 | [4]1 | [39]87 | [38]88 | [28]165 | [27]331 | [27]664 | $^{105}0.05N^{1.0}$ |
| 109 | Dermalog | DE | dermalog | 012 | 2023-12-19 | 0 | 287 | [7]128 | - | [25]177 | [2]1 | [44]95 | [47]97 | [30]177 | [29]353 | [29]709 | $^{74}0.07N^{1.0}$ |
| 110 | Dermalog | DE | dermalog | 013 | 2024-06-18 | 0 | 287 | [6]128 | - | [27]182 | [1]1 | [43]95 | [43]95 | [33]180 | [31]361 | [33]724 | $^{126}0.05N^{1.0}$ |
| 111 | Dermalog | DE | dermalog | 014 | 2025-05-09 | 0 | 289 | [10]128 | - | [24]177 | [530]3293 | [2]0 | [2]0 | [2]0 | [2]1 | [2]1 | $^{13}26.41N^{0.2}$ |
| 112 | Dermalog | DE | dermalog | 015 | 2025-09-12 | 0 | 683 | [120]1024 | - | [429]1033 | [118]9 | [96]217 | [95]218 | [70]410 | [71]823 | [68]1647 | $^{111}0.13N^{1.0}$ |
| 113 | Dermalog | DE | dermalog | 016 | 2026-02-04 | 0 | 888 | [108]1024 | - | [542]1908 | [70]7 | [81]181 | [83]190 | [61]341 | [69]817 | [62]1526 | $^{312}0.06N^{1.1}$ |
| 114 | Dermalog | DE | dermalog | 017 | 2026-06-25 | 0 | 889 | [114]1024 | - | [541]1906 | [92]8 | [80]180 | [75]181 | [60]340 | [73]842 | [64]1579 | $^{345}0.04N^{1.1}$ |
| 115 | Digidata | IN | digidata | 000 | 2022-06-03 | 248 | 33 | [307]2048 | - | [77]320 | [527]2444 | [3]0 | [86]198 | - | - | - | |
| 116 | DiluSense Technology | CN | dilusense | 000 | 2022-05-26 | 311 | 56 | [278]2048 | - | [40]219 | [405]26 | [373]1904 | [341]1498 | [308]3116 | [298]6126 | [296]12046 | $^{248}0.61N^{1.0}$ |
| 117 | DiluSense Technology | CN | dilusense | 001 | 2022-12-24 | 599 | 5 | [485]4096 | - | [392]860 | [460]38 | [422]3026 | [431]3042 | [359]4738 | [351]9473 | [350]19026 | $^{43}5.36N^{0.9}$ |
| 118 | EI Networks Private Ltd | IN | einetworksindia | 000 | 2024-04-17 | 444 | 128 | [225]2048 | - | [92]347 | [369]19 | [178]470 | [134]342 | [148]915 | [151]1844 | [141]3719 | $^{423}0.03N^{1.2}$ |
| 119 | EI Networks Private Ltd | IN | einetworksindia | 001 | 2024-10-24 | 444 | 197 | [370]2048 | - | [88]341 | [365]18 | [159]402 | [135]343 | [108]656 | [138]1714 | [139]3659 | $^{412}0.03N^{1.1}$ |
| 120 | EI Networks Private Ltd | IN | einetworksindia | 002 | 2025-02-21 | 1186 | 141 | [306]2048 | - | [435]1076 | [382]20 | [127]327 | [157]387 | [106]651 | [104]1310 | [106]2872 | $^{190}0.17N^{1.0}$ |
| 121 | FAST-NUCES Lahore Campus | PK | fastlahore | 000 | 2025-09-26 | 186 | 138 | [343]2048 | - | [57]278 | [301]15 | [400]2432 | [402]2272 | - | - | - | |
| 122 | FCI Vietnam | VN | fci | 000 | 2025-04-02 | 1452 | 444 | [213]2048 | - | [514]1466 | [217]12 | [535]13182 | [543]13125 | - | - | - | |
| 123 | FPT Smart Cloud | VN | fpt | 000 | 2024-03-28 | 595 | 444 | [191]2048 | - | [387]850 | [277]14 | [534]13085 | [542]13114 | - | - | - | |
| 124 | FacePhi | ES | facephi | 000 | 2024-10-03 | 134 | 182 | [66]512 | - | [56]275 | [63]6 | [94]214 | [395]2087 | [334]3854 | [329]7745 | [326]15540 | $^{444}0.01N^{1.3}$ |
| 125 | Facehawk Ltd | GB | facehawk | 001 | 2026-06-17 | 169 | 35 | [439]2070 | - | [100]360 | [486]49 | [326]1348 | [327]1353 | - | - | - | |
| 126 | FarBar Inc | TW | f8 | 001 | 2019-10-03 | 266 | 19 | [362]2048 | k | [369]810 | [275]14 | - | - | - | - | - | |
| 127 | Fincore | GB | fincore | 000 | 2021-08-18 | 250 | 224 | [252]2048 | - | [159]443 | [114]9 | [275]971 | [206]560 | - | - | - | |
| 128 | First Credit Bureau Kazakhstan | KZ | firstcreditkz | 001 | 2022-11-22 | 548 | 24 | [40]288 | - | [201]495 | [14]2 | [32]46 | [29]38 | [20]72 | [20]150 | [20]297 | $^{143}0.02N^{1.0}$ |
| 129 | Fissionsoft CO Ltd | TH | fissionsfr | 001 | 2026-04-24 | 138 | 244 | [353]2048 | - | [522]1704 | [521]883 | [4]1 | [3]1 | - | - | - | |
| 130 | Fujitsu Research and Development Center | CN | fujitsulab | 000 | 2021-10-12 | 497 | 337 | [121]1032 | - | [262]613 | [59]5 | [371]1864 | [356]1657 | [310]3140 | [304]6320 | [299]12723 | $^{198}0.78N^{1.0}$ |
| 131 | Fujitsu Research and Development Center | CN | fujitsulab | 001 | 2022-03-15 | 675 | 386 | [131]1032 | - | [227]535 | [115]9 | [370]1854 | [364]1740 | [318]3324 | [314]6730 | [322]14943 | $^{330}0.45N^{1.1}$ |
| 132 | GTEL OTS JSC | VN | gtelots | 000 | 2025-04-14 | 246 | 58 | [67]512 | - | [362]788 | [545]21132 | [8]3 | [6]3 | - | - | - | |
| 133 | GTEL OTS JSC | VN | gtelots | 001 | 2026-02-02 | 432 | 57 | [139]1088 | - | [416]955 | [465]39 | [7]2 | [5]3 | - | - | - | |
| 134 | Gorilla Technology | TW | gorilla | 2 | 2018-10-29 | 91 | 1252 | [143]1132 | k | [85]338 | [399]24 | [67]145 | [65]146 | [51]293 | [53]612 | [60]1509 | $^{381}0.02N^{1.1}$ |
| 135 | Gorilla Technology | TW | gorilla | 3 | 2018-10-26 | 94 | 1252 | [450]2156 | k | [240]559 | [539]12020 | - | [386]2047 | - | - | - | |
| 136 | Gorilla Technology | TW | gorilla | 004 | 2020-01-06 | 182 | 1244 | [451]2192 | k | [122]388 | [467]41 | [118]286 | [116]285 | [175]1191 | [174]2416 | [168]5036 | $^{445}0.00N^{1.3}$ |
| 137 | Gorilla Technology | TW | gorilla | 005 | 2021-02-22 | 306 | 1420 | [543]6288 | - | [69]303 | [506]78 | [293]1070 | [245]799 | [201]1514 | [255]4454 | [241]8820 | $^{427}0.05N^{1.2}$ |
| 138 | Gorilla Technology | TW | gorilla | 006 | 2021-09-30 | 377 | 691 | [554]8336 | - | [148]422 | [514]99 | [342]1504 | [352]1612 | [271]2422 | [252]4422 | [254]9363 | $^{197}0.59N^{1.0}$ |
| 139 | Gorilla Technology | TW | gorilla | 007 | 2022-02-16 | 392 | 322 | [544]6290 | - | [74]315 | [513]89 | [236]765 | [238]745 | [195]1408 | [191]2823 | [182]5764 | $^{136}0.42N^{1.0}$ |
| 140 | Gorilla Technology | TW | gorilla | 008 | 2022-10-31 | 321 | 290 | [527]4242 | - | [216]521 | [490]54 | [190]513 | [272]972 | [213]1699 | [178]2469 | [202]6768 | $^{260}0.30N^{1.0}$ |
| 141 | Griaule | US | griaule | 000 | 2021-11-01 | 0 | 584 | [393]2052 | - | [48]247 | [91]8 | [472]4868 | [499]6150 | [417]11473 | [413]22952 | [406]46070 | $^{84}3.89N^{1.0}$ |
| 142 | Griaule | US | griaule | 001 | 2022-07-26 | 0 | 615 | [408]2052 | - | [306]684 | [164]12 | [492]5866 | [493]6010 | [421]11627 | [416]24207 | [407]46343 | $^{182}3.02N^{1.0}$ |
| 143 | Guangzhou Pixel Solutions Co | CN | pixelall | 002 | 2019-07-01 | 0 | 165 | [455]2560 | k | [32]190 | [303]15 | [322]1296 | [324]1334 | [281]2526 | [274]5136 | [281]11045 | $^{249}0.52N^{1.0}$ |
| 144 | Guangzhou Pixel Solutions Co | CN | pixelall | 003 | 2019-11-05 | 0 | 690 | [456]2560 | k | [324]703 | [391]22 | [318]1273 | [319]1307 | [277]2474 | [275]5198 | [282]11141 | $^{283}0.46N^{1.0}$ |
| 145 | Guangzhou Pixel Solutions Co | CN | pixelall | 004 | 2020-07-02 | 0 | 538 | [454]2560 | k | [163]449 | [356]17 | [317]1259 | [318]1300 | [276]2465 | [282]5492 | [283]11443 | $^{331}0.34N^{1.1}$ |
| 146 | Guangzhou Pixel Solutions Co | CN | pixelall | 005 | 2021-03-23 | 0 | 717 | [452]2560 | - | [384]840 | [151]11 | [351]1606 | [343]1528 | [284]2609 | [269]4926 | [289]11770 | $^{167}0.73N^{1.0}$ |
| 147 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 000 | 2022-03-18 | 855 | 97 | [118]1024 | - | [169]457 | [152]11 | [411]2609 | [400]2251 | [351]4612 | [359]10075 | [339]17156 | $^{224}1.01N^{1.0}$ |
| 148 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 001 | 2022-08-18 | 273 | 162 | [364]2048 | - | [116]377 | [175]12 | [467]4537 | [470]4511 | [400]8462 | [406]19463 | [400]42527 | $^{360}0.92N^{1.1}$ |
| 149 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 002 | 2023-01-12 | 302 | 293 | [329]2048 | - | [238]555 | [201]12 | [462]4481 | [469]4509 | [398]8413 | [394]16568 | [398]39517 | $^{274}1.59N^{1.0}$ |
| 150 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 003 | 2023-05-22 | 1434 | 293 | [486]4096 | - | [303]675 | [397]24 | [526]10145 | [530]9351 | [452]17922 | [450]40619 | [438]83432 | $^{301}3.03N^{1.0}$ |
| 151 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 004 | 2023-10-13 | 1902 | 293 | [535]5120 | - | [417]956 | [422]29 | [509]7508 | [518]7639 | [437]13673 | [422]25457 | [430]66435 | $^{332}1.83N^{1.1}$ |
| 152 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 005 | 2024-04-26 | 2201 | 293 | [549]7168 | - | [505]1389 | [469]41 | [528]11242 | [535]10996 | - | - | - | |
| 153 | Hangzhuo AIlu Network Information Technology | CN | hzailu | 006 | 2025-01-03 | 2866 | 293 | [548]7168 | - | [509]1455 | [462]38 | [532]12514 | [538]11557 | - | - | - | |
| 154 | Hikvision Research Institute | CN | hikvision | 5 | 2018-10-29 | 593 | 9 | [151]1408 | 1 | [260]607 | [321]16 | [258]883 | [257]895 | [234]1908 | [222]3792 | [257]9387 | $^{403}0.10N^{1.1}$ |
| 155 | Hikvision Research Institute | CN | hikvision | 6 | 2018-10-29 | 593 | 9 | [150]1408 | 1 | [255]598 | [327]16 | [253]871 | [255]877 | - | - | - | |
| 156 | HyperVerge Inc | IN | hyperverge | 001 | 2021-08-11 | 1791 | 212 | [107]1024 | - | [196]490 | [54]5 | [227]703 | [225]681 | [186]1346 | [184]2681 | [178]5680 | $^{214}0.32N^{1.0}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 4: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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2026/09/04 10:23:32

FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 157 | HyperVerge Inc | IN | hyperverge | 002 | 2022-04-13 | 1140 | 1118 | [116]1024 | - | [157]440 | [121]9 | (181)484 | (193)518 | (164)1099 | (136)1690 | (77)1953 | [26]$6.15N^{0.8}$ |
| 158 | HyperVerge Inc | IN | hyperverge | 003 | 2023-04-04 | 2089 | 1118 | [199]2048 | - | [166]452 | [233]13 | (237)766 | (263)921 | (233)1900 | (203)3198 | (190)6065 | [140]$0.48N^{1.0}$ |
| 159 | Hyundai Motor Company | KR | hyundai | 000 | 2025-02-03 | 352 | 104 | [222]2048 | - | [103]366 | [256]13 | (453)4258 | (458)4068 | (391)7815 | (387)15618 | (389)33716 | [264]$1.52N^{1.0}$ |
| 160 | Hyundai Motor Company | KR | hyundai | 001 | 2025-06-23 | 176 | 104 | [221]2048 | - | [95]351 | [196]12 | (457)4329 | (464)4382 | (392)7827 | (392)16459 | (387)33035 | [163]$2.22N^{1.0}$ |
| 161 | Hyundai Motor Company | KR | hyundai | 002 | 2025-11-14 | 472 | 104 | [257]2048 | - | [529]1802 | [316]15 | (517)8376 | (522)8199 | (442)15300 | (439)31477 | (444)214557 | [450]$0.04N^{1.4}$ |
| 162 | Hyundai Motor Company | KR | hyundai | 003 | 2026-06-17 | 472 | 104 | [270]2048 | - | [539]1860 | [407]27 | (471)4817 | (474)4839 | (406)9006 | (403)18000 | (395)36450 | [113]$2.83N^{1.0}$ |
| 163 | ICM Airport Technics | AU | icm | 000 | 2023-08-14 | 991 | 1 | [216]2048 | - | [31]186 | [197]12 | (511)7990 | (520)7943 | - | - | - | |
| 164 | ICM Airport Technics | AU | icm | 001 | 2023-12-12 | 858 | 1 | [338]2048 | - | [19]164 | [192]12 | (507)7124 | (515)7219 | (443)15370 | (434)29562 | (433)70229 | [380]$1.10N^{1.1}$ |
| 165 | IDENTY | GB | identy | 000 | 2025-06-18 | 376 | 38 | [272]2048 | - | [296]662 | [459]37 | (140)366 | (131)330 | - | - | - | |
| 166 | IDENTY | GB | identy | 001 | 2026-01-22 | 1109 | 43 | [460]3072 | - | [510]1459 | [432]30 | (200)546 | (221)638 | (86)511 | (82)971 | (79)2024 | [21]$53.38N^{0.6}$ |
| 167 | IDENTY | GB | identy | 002 | 2026-06-03 | 1216 | 77 | [461]3072 | - | [525]1720 | [437]30 | (210)587 | (210)571 | (165)1100 | (165)2285 | (160)4761 | [258]$0.22N^{1.0}$ |
| 168 | IMDS Software Inc | CA | imds | 000 | 2023-10-20 | 364 | 344 | [220]2048 | - | [65]296 | [189]12 | (524)10077 | (534)10387 | - | - | - | |
| 169 | IMDS Software Inc | CA | imds | 001 | 2024-03-14 | 362 | 344 | [235]2048 | - | [411]943 | [314]15 | (182)485 | (183)489 | (150)924 | (159)1970 | (153)4100 | [278]$0.17N^{1.0}$ |
| 170 | IMDS Software Inc | CA | imds | 002 | 2024-11-01 | 364 | 344 | [211]2048 | - | [71]311 | [214]12 | (214)622 | (216)622 | (168)1132 | (157)1962 | (152)4000 | [49]$0.86N^{0.9}$ |
| 171 | Idemia | FR | idemia | 5 | 2018-10-29 | 417 | 48 | [47]352 | 1 | [113]371 | [55]5 | (61)137 | (61)138 | (73)437 | (63)724 | (66)1630 | [436]$0.01N^{1.2}$ |
| 172 | Idemia | FR | idemia | 6 | 2018-10-29 | 417 | 48 | [48]352 | 1 | [110]370 | [51]4 | (62)137 | (60)138 | (74)442 | (72)827 | (67)1646 | [438]$0.01N^{1.2}$ |
| 173 | Idemia | FR | idemia | 007 | 2020-01-17 | 738 | 113 | [106]860 | 1 | [365]794 | [276]14 | (68)151 | (66)152 | (113)683 | (121)1481 | (117)3022 | [452]$0.00N^{1.4}$ |
| 174 | Idemia | FR | idemia | 008 | 2021-03-15 | 378 | 65 | [44]300 | - | [50]251 | [32]3 | (60)137 | (57)131 | (46)247 | (45)501 | (42)1013 | [141]$0.07N^{1.0}$ |
| 175 | Idemia | FR | idemia | 009 | 2022-03-01 | 735 | 68 | [101]636 | - | [217]521 | [81]7 | (92)211 | (96)221 | (81)473 | (88)1070 | (87)2352 | [413]$0.02N^{1.1}$ |
| 176 | Idemia | FR | idemia | 010 | 2023-01-11 | 942 | 71 | [45]300 | - | [321]701 | [24]2 | (90)197 | (87)200 | (66)380 | (66)766 | (63)1561 | [207]$0.09N^{1.0}$ |
| 177 | Idemia | FR | idemia | 011 | 2024-03-05 | 776 | 63 | [46]300 | - | [382]834 | [22]2 | (89)196 | (84)193 | (65)368 | (65)746 | (61)1525 | [231]$0.08N^{1.0}$ |
| 178 | Idemia | FR | idemia | 012 | 2025-07-02 | 758 | 66 | [144]1212 | - | [482]1255 | [86]7 | (130)343 | (132)335 | (104)643 | (127)1544 | (144)3849 | [422]$0.02N^{1.2}$ |
| 179 | Idemia | FR | idemia | 013 | 2026-07-01 | 678 | 68 | [102]716 | - | [547]2022 | [67]6 | (169)430 | (171)427 | (134)810 | (161)2033 | (147)3894 | [366]$0.08N^{1.1}$ |
| 180 | Imagus Technology Pty | AU | imagus | 005 | 2021-01-15 | 222 | 311 | [262]2048 | - | [200]491 | [274]14 | (124)302 | (125)313 | (107)651 | (109)1361 | (93)2461 | [400]$0.03N^{1.1}$ |
| 181 | Imagus Technology Pty | AU | imagus | 006 | 2021-05-27 | 248 | 369 | [209]2048 | - | [204]505 | [131]9 | (125)303 | (97)234 | (83)499 | (102)1273 | (100)2727 | [433]$0.01N^{1.2}$ |
| 182 | Imagus Technology Pty | AU | imagus | 007 | 2021-11-16 | 248 | 366 | [365]2048 | - | [555]32200 | [108]9 | (123)301 | (98)238 | (75)442 | (76)881 | (73)1765 | [75]$0.16N^{1.0}$ |
| 183 | Imagus Technology Pty | AU | imagus | 008 | 2022-05-26 | 204 | 335 | [190]2048 | - | [58]284 | [354]17 | (206)560 | (175)441 | - | - | - | |
| 184 | Imperial College London | GB | imperial | 000 | 2019-08-28 | 461 | 15 | [301]2048 | 1 | [246]577 | [226]13 | (136)360 | (155)379 | (208)1626 | (233)4057 | (277)10291 | [455]$0.00N^{1.5}$ |
| 185 | Incode Technologies Inc | US | incode | 2 | 2018-10-29 | 71 | 31 | [229]2048 | 1 | [62]289 | [312]15 | (164)411 | (161)404 | - | - | - | |
| 186 | Incode Technologies Inc | US | incode | 3 | 2018-10-29 | 133 | 31 | [295]2048 | 1 | [319]697 | [296]15 | (163)408 | (166)412 | (139)847 | (132)1608 | (158)4486 | [393]$0.05N^{1.1}$ |
| 187 | Incode Technologies Inc | US | incode | 004 | 2019-06-24 | 254 | 50 | [176]2048 | 1 | [184]475 | [170]12 | (139)365 | (153)378 | (198)1482 | (134)1660 | (113)2954 | [324]$0.12N^{1.1}$ |
| 188 | Incode Technologies Inc | US | incode | 005 | 2021-07-29 | 259 | 21 | [274]2048 | - | [59]285 | [136]10 | (168)427 | (177)454 | (144)890 | (150)1843 | (136)3640 | [367]$0.07N^{1.1}$ |
| 189 | Incode Technologies Inc | US | incode | 006 | 2024-07-15 | 2309 | 41 | [551]8196 | - | [433]1071 | [480]47 | (360)1706 | (360)1713 | (264)2241 | (291)5891 | (263)9528 | [39]$3.19N^{0.9}$ |
| 190 | Incode Technologies Inc | US | incode | 007 | 2025-07-30 | 1766 | 69 | [538]6148 | - | [432]1047 | [454]35 | (308)1172 | (315)1256 | (230)1873 | (256)4462 | (227)7970 | [183]$0.53N^{1.0}$ |
| 191 | Innov8tif Solutions | MY | innov8tif | 000 | 2026-07-28 | 115 | 1007 | [469]4096 | - | [175]461 | [424]29 | (376)1925 | (372)1853 | - | - | - | |
| 192 | Innovatrics | SK | innovatrics | 4 | 2018-10-30 | 0 | 400 | [138]1076 | k | [129]399 | [536]10902 | (14)8 | (9)8 | (5)11 | (3)9 | (4)13 | [11]$668.38N^{0.2}$ |
| 193 | Innovatrics | SK | innovatrics | 005 | 2019-09-30 | 0 | 455 | [86]538 | 1 | [377]827 | [538]11897 | (13)8 | (10)8 | (4)9 | (4)9 | (3)9 | [1]$4055.65N^{0.1}$ |
| 194 | Innovatrics | SK | innovatrics | 007 | 2021-08-16 | 175 | 58 | [87]538 | - | [249]577 | [281]14 | (53)105 | (50)100 | (37)188 | (38)378 | (37)788 | [62]$0.09N^{1.0}$ |
| 195 | Innovatrics | SK | innovatrics | 008 | 2023-01-12 | 609 | 102 | [518]4136 | - | [284]647 | [471]42 | (247)823 | (250)864 | (206)1610 | (206)3271 | (211)6943 | [177]$0.43N^{1.0}$ |
| 196 | Innovatrics | SK | innovatrics | 009 | 2023-08-01 | 590 | 108 | [520]4136 | - | [182]474 | [496]60 | (264)924 | (273)975 | (219)1759 | (215)3496 | (209)6926 | [63]$0.81N^{1.0}$ |
| 197 | Innovatrics | SK | innovatrics | 010 | 2024-01-09 | 797 | 42 | [519]4136 | - | [253]595 | [493]56 | (251)850 | (261)907 | (217)1724 | (213)3444 | (215)7470 | [216]$0.41N^{1.0}$ |
| 198 | Innovatrics | SK | innovatrics | 011 | 2024-05-14 | 628 | 48 | [525]4209 | - | [257]600 | [510]86 | (525)10112 | (533)10087 | (455)18636 | (446)37591 | (436)75468 | [132]$5.61N^{1.0}$ |
| 199 | Innovatrics | SK | innovatrics | 012 | 2024-09-20 | 0 | 1938 | [530]4779 | - | [511]1462 | [412]28 | (321)1295 | (321)1312 | (265)2250 | (262)4635 | (269)9711 | [58]$1.14N^{1.0}$ |
| 200 | Innovatrics | SK | innovatrics | 013 | 2025-01-28 | 0 | 2784 | [545]6845 | - | [449]1133 | [463]38 | (340)1495 | (350)1581 | (297)2929 | (290)5827 | (286)11658 | [59]$1.40N^{1.0}$ |
| 201 | Innovatrics | SK | innovatrics | 014 | 2025-05-28 | 0 | 3178 | [546]6870 | - | [500]1359 | [464]39 | (328)1405 | (329)1405 | (282)2556 | (270)4929 | (260)9479 | [52]$1.49N^{1.0}$ |
| 202 | Innovatrics | SK | innovatrics | 015 | 2025-09-29 | 0 | 3057 | [547]6870 | - | [550]2124 | [487]50 | (349)1580 | (349)1568 | (299)2944 | (292)5942 | (292)11998 | [130]$0.89N^{1.0}$ |
| 203 | Inspur (Beijing) Electronic Information Industry Co | CN | inspur | 000 | 2023-05-30 | 356 | 88 | [280]2048 | - | [404]898 | [342]16 | (440)3574 | (438)3248 | (377)6610 | (374)13468 | (370)27332 | [298]$1.04N^{1.0}$ |
| 204 | Inspur (Beijing) Electronic Information Industry Co | CN | inspur | 001 | 2023-12-07 | 437 | 88 | [323]2048 | - | [427]1025 | [336]16 | (439)3483 | (439)3254 | (375)6520 | (372)13141 | (365)25818 | [233]$1.36N^{1.0}$ |
| 205 | Inspur (Beijing) Electronic Information Industry Co | CN | inspur | 002 | 2025-02-11 | 434 | 88 | [374]2048 | - | [475]1222 | [335]16 | (434)3313 | (440)3259 | (372)6169 | (380)13863 | (371)27477 | [311]$0.98N^{1.1}$ |
| 206 | Integrated Software and Technologies Ltd | BD | istl | 001 | 2026-08-07 | 264 | 26 | [435]2060 | - | [254]598 | [389]22 | (299)1106 | (298)1107 | (258)2157 | (241)4215 | (237)8701 | [176]$0.56N^{1.0}$ |
| 207 | Intel Research Group | IS | intelresearch | 001 | 2025-03-31 | 309 | 118 | [223]2048 | - | [472]1219 | [254]13 | (260)886 | (252)866 | (209)1652 | (209)3339 | (206)6862 | [178]$0.44N^{1.0}$ |
| 208 | Intellivision | IN | intellivision | 001 | 2022-03-08 | 62 | 130 | [428]2056 | - | [80]325 | [381]20 | (150)388 | (548)18764 | - | - | - | |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 5: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 209 | Intellivision | IN | intellivision | 002 | 2022-07-28 | 114 | 128 | [425]2056 | - | [68]301 | [238]13 | [539]20542 | [549]18784 | - | - | - | |
| 210 | Intema-LGL Group | LU | intema | 000 | 2022-08-24 | 1042 | 20 | [65]512 | - | [274]629 | [543]13809 | [25]27 | [26]35 | [16]39 | [16]46 | [10]51 | [8]$3478.46N^{0.2}$ |
| 211 | Intema-LGL Group | LU | intema | 001 | 2023-02-22 | 723 | 20 | [76]512 | - | [367]802 | [532]4808 | [21]23 | [19]24 | [12]31 | [10]38 | [13]54 | [14]$567.10N^{0.3}$ |
| 212 | Intozi Tech Pvt Ltd | IN | intozi | 001 | 2026-07-08 | 168 | 92 | [193]2048 | - | [521]1615 | [447]34 | [300]1111 | [301]1121 | [251]2133 | [246]4295 | [235]8626 | [187]$0.55N^{1.0}$ |
| 213 | Intpixel Labs Private Ltd | IN | vogic | 000 | 2026-07-30 | 231 | 22 | [512]4104 | - | [203]501 | [489]52 | [397]2292 | [405]2331 | [343]4282 | [342]8743 | [341]17693 | [146]$1.25N^{1.0}$ |
| 214 | IrexAI | US | irex | 000 | 2021-02-09 | 724 | 46 | [462]3080 | - | [226]532 | [373]19 | [255]873 | [212]600 | [167]1120 | [179]2477 | [187]5863 | [349]$0.13N^{1.1}$ |
| 215 | Iris ID Systems | US | irisid | 001 | 2026-02-20 | 469 | 98 | [383]2049 | - | [419]968 | [257]13 | [394]2230 | [403]2277 | [344]4284 | [343]8811 | [343]17877 | [250]$0.87N^{1.0}$ |
| 216 | Iris ID Systems | US | irisid | 002 | 2026-08-05 | 718 | 98 | [385]2049 | - | [526]1732 | [291]14 | [527]11134 | [536]11093 | [456]21213 | [452]42047 | [439]84414 | [115]$6.56N^{1.0}$ |
| 217 | Kakao Brain | KR | kakao | 000 | 2021-06-23 | 404 | 124 | [403]2052 | - | [130]399 | [94]8 | [256]876 | [92]215 | [85]510 | [81]971 | [78]1955 | [355]$0.05N^{1.1}$ |
| 218 | Kakao Brain | KR | kakao | 001 | 2022-06-08 | 615 | 102 | [187]2048 | - | [211]509 | [362]18 | [177]469 | [167]418 | [152]977 | [135]1661 | [148]3898 | [310]$0.13N^{1.1}$ |
| 219 | Kakao Brain | KR | kakao | 002 | 2023-11-07 | 749 | 102 | [177]2048 | - | [252]594 | [181]12 | [180]478 | [180]466 | [160]1055 | [176]2443 | [163]4801 | [401]$0.06N^{1.1}$ |
| 220 | Kedacom International Pte | SG | kedacom | 001 | 2019-09-16 | 239 | 36 | [41]292 | 1 | [206]507 | [19]2 | [235]764 | [241]760 | [236]1940 | [197]2983 | [200]6623 | [257]$0.31N^{1.0}$ |
| 221 | Keyless Technologies | GB | keyless | 000 | 2025-01-16 | 538 | 35 | [341]2048 | - | [256]600 | [160]11 | [131]355 | [139]356 | [135]811 | - | - | [386]$0.05N^{1.1}$ |
| 222 | Kloudspot Inc | US | kloudspot | 001 | 2025-01-21 | 575 | 237 | [394]2052 | - | [452]1138 | [225]13 | [438]3435 | [442]3438 | [374]6477 | [378]13716 | [374]28709 | [296]$1.11N^{1.0}$ |
| 223 | Kneron | US | kneron | 000 | 2020-03-03 | 366 | 13 | [355]2048 | k | [218]523 | [237]13 | [410]2535 | [415]2506 | [360]4752 | [356]9696 | [359]20926 | [259]$0.95N^{1.0}$ |
| 224 | Kneron | US | kneron | 001 | 2021-06-10 | 270 | 69 | [297]2048 | - | [63]291 | [116]9 | [427]3141 | [421]2642 | - | - | - | |
| 225 | Know U Tech LLC | US | knowutech | 000 | 2023-06-05 | 789 | 34 | [104]781 | - | [380]831 | [57]5 | [445]3773 | [441]3411 | [373]6413 | [371]12847 | [372]27685 | [217]$1.50N^{1.0}$ |
| 226 | Kogniza Technology | US | kogniza | 000 | 2024-05-14 | 0 | 385 | [152]1536 | - | [494]1336 | [120]9 | [508]7385 | [516]7272 | [438]13740 | [430]28092 | [421]57099 | [210]$3.33N^{1.0}$ |
| 227 | Lebentech Biometrics | MX | lebentech | 000 | 2026-06-15 | 0 | 144 | [453]2560 | - | [44]236 | [475]44 | [47]96 | [59]134 | - | - | - | |
| 228 | LightVision Inc | KR | lightvision | 000 | 2025-09-11 | 260 | 61 | [324]2048 | - | [393]862 | [361]18 | [480]5166 | [481]5161 | [410]9698 | [407]19472 | [396]39233 | [127]$2.93N^{1.0}$ |
| 229 | Line Corporation | JP | line | 000 | 2021-06-02 | 138 | 397 | [219]2048 | - | [76]317 | [101]8 | [470]4637 | [484]5418 | [412]10144 | - | - | [79]$3.65N^{1.0}$ |
| 230 | Line Corporation | JP | line | 001 | 2021-11-21 | 471 | 396 | [367]2048 | - | [188]485 | [102]8 | [368]1815 | [379]1934 | [324]3647 | [327]7675 | - | [290]$0.64N^{1.0}$ |
| 231 | Line Corporation | JP | lineclova | 002 | 2022-07-29 | 560 | 72 | [261]2048 | - | [152]425 | [248]13 | [107]262 | [127]321 | [99]604 | [124]1534 | [118]3023 | [374]$0.05N^{1.1}$ |
| 232 | Line Corporation | JP | lineclova | 003 | 2023-01-19 | 574 | 397 | [299]2048 | - | [141]412 | [82]7 | [69]156 | [85]195 | [158]1018 | [156]1956 | [130]3405 | [391]$0.05N^{1.1}$ |
| 233 | Lishui Shuyan Huirong Technology | CN | lssytech | 000 | 2026-03-02 | 1583 | 193 | [501]4096 | - | [504]1382 | [445]33 | [222]667 | [256]892 | [223]1811 | [189]2780 | [188]5907 | [267]$0.29N^{1.0}$ |
| 234 | Lomonosov Moscow State University | RU | intsysmsu | 000 | 2019-08-19 | 375 | 168 | [279]2048 | 1 | [263]614 | [252]13 | [170]430 | [173]431 | [142]860 | [139]1730 | [172]5353 | [419]$0.03N^{1.1}$ |
| 235 | Lookman Electroplast Industries | IN | lookman | 3 | 2018-10-28 | 203 | 24 | [42]292 | 1 | [84]336 | [30]3 | [231]739 | [237]745 | [194]1394 | [190]2817 | [230]8286 | [373]$0.13N^{1.1}$ |
| 236 | Lookman Electroplast Industries | IN | lookman | 4 | 2018-10-28 | 184 | 24 | [92]548 | 1 | [78]320 | [50]4 | [277]981 | [279]998 | - | - | - | |
| 237 | Lookman Electroplast Industries | IN | lookman | 005 | 2019-09-16 | 239 | 36 | [93]548 | 1 | [205]506 | [46]4 | [281]1005 | [282]1008 | [283]2597 | [280]5446 | [244]8939 | [370]$0.19N^{1.1}$ |
| 238 | Luxand Inc | US | luxand | 001 | 2023-10-17 | 83 | 70 | [332]2048 | - | [150]425 | [180]12 | [409]2523 | [423]2713 | [355]4690 | [353]9543 | [352]19507 | [65]$2.11N^{1.0}$ |
| 239 | Mantra Softech India | IN | mantra | 000 | 2021-10-28 | 460 | 61 | [395]2052 | - | [143]412 | [135]10 | [263]916 | [262]910 | [215]1714 | [212]3411 | [204]6841 | [89]$0.57N^{1.0}$ |
| 240 | Maverick Research | AE | maverickresearch | 000 | 2025-02-25 | 250 | 96 | [198]2048 | - | [423]995 | [176]12 | [146]372 | [176]443 | - | - | - | |
| 241 | Maxvision | CN | maxvision | 000 | 2022-06-17 | 167 | 60 | [175]2048 | - | [16]139 | [555]- | [477]5044 | [309]1195 | [259]2163 | [250]4380 | [243]8914 | [196]$0.55N^{1.0}$ |
| 242 | Maxvision | CN | maxvision | 001 | 2022-10-28 | 228 | 63 | [357]2048 | - | [94]351 | [246]13 | [309]1173 | [296]1061 | [263]2233 | [247]4311 | [248]9084 | [256]$0.42N^{1.0}$ |
| 243 | Maxvision | CN | maxvision | 002 | 2023-04-12 | 391 | 63 | [281]2048 | - | [272]626 | [318]15 | [310]1180 | [293]1052 | [260]2213 | [242]4234 | [252]9310 | [291]$0.37N^{1.0}$ |
| 244 | Maxvision | CN | maxvision | 003 | 2023-08-11 | 228 | 62 | [228]2048 | - | [93]348 | [166]12 | [290]1049 | [294]1052 | [241]2006 | [239]4177 | [255]9366 | [284]$0.37N^{1.0}$ |
| 245 | Maxvision | CN | maxvision | 004 | 2023-12-12 | 308 | 62 | [380]2048 | - | [467]1202 | [235]13 | [296]1094 | [297]1088 | [243]2044 | [234]4067 | [242]8836 | [201]$0.51N^{1.0}$ |
| 246 | Megvii/Face++ | CN | megvii | 1 | 2018-10-28 | 1703 | 41 | [489]4096 | 1 | [275]631 | [442]32 | [203]552 | [207]561 | [179]1222 | [170]2321 | [189]5968 | [390]$0.08N^{1.1}$ |
| 247 | Megvii/Face++ | CN | megvii | 2 | 2018-10-28 | 1735 | 42 | [496]4096 | 1 | [276]635 | [440]31 | [204]553 | [204]558 | - | - | - | |
| 248 | Megvii/Face++ | CN | megvii | 003 | 2023-05-11 | 1335 | 43 | [388]2049 | - | [485]1263 | [330]16 | [378]1972 | [373]1854 | [298]2942 | [315]6803 | [314]14061 | [300]$0.52N^{1.0}$ |
| 249 | Megvii/Face++ | CN | megvii | 004 | 2023-10-18 | 1605 | 43 | [389]2049 | - | [492]1303 | [280]14 | [364]1735 | [358]1694 | [315]3217 | [307]6495 | [319]14865 | [295]$0.56N^{1.0}$ |
| 250 | Megvii/Face++ | CN | megvii | 005 | 2024-02-20 | 1605 | 43 | [384]2049 | - | [498]1348 | [292]14 | [366]1764 | [362]1721 | [317]3261 | [311]6584 | [306]13376 | [162]$0.90N^{1.0}$ |
| 251 | Metsakuur | KR | metsakuurcompany | 000 | 2024-10-24 | 0 | 1011 | [413]2056 | - | [379]830 | [162]12 | [423]3068 | [511]6828 | [453]18336 | [447]38444 | [440]89809 | [441]$0.13N^{1.2}$ |
| 252 | Metsakuur | KR | metsakuurcompany | 001 | 2025-07-30 | 0 | 299 | [286]2048 | - | [195]490 | [216]12 | [514]8191 | [521]8027 | [441]15224 | [438]31411 | [432]69959 | [334]$1.96N^{1.1}$ |
| 253 | Miaxis Biometrics | CN | miaxis | 001 | 2025-02-02 | 712 | 87 | [386]2049 | - | [546]1953 | [359]18 | [518]8566 | [526]8573 | [447]16097 | [443]32220 | [429]65943 | [133]$4.83N^{1.0}$ |
| 254 | Miaxis Biometrics | CN | miaxis | 002 | 2026-07-24 | 721 | 168 | [387]2049 | - | [527]1771 | [351]17 | [491]5855 | [489]5740 | [416]10637 | [412]21716 | [402]43897 | [147]$3.10N^{1.0}$ |
| 255 | MicroFocus | GB | microfocus | 5 | 2018-10-29 | 94 | 26 | [18]256 | k | [53]262 | [25]2 | [82]182 | [81]186 | [64]354 | [61]708 | [57]1425 | [119]$0.11N^{1.0}$ |
| 256 | MicroFocus | GB | microfocus | 6 | 2018-10-29 | 94 | 26 | [23]256 | k | [52]262 | [27]2 | [83]183 | [80]186 | - | - | - | |
| 257 | Microsoft | US | microsoft | 5 | 2018-10-29 | 381 | 155 | [117]1024 | 1 | [293]658 | [159]11 | [350]1606 | [357]1673 | [306]3076 | [303]6302 | [303]13160 | [194]$0.79N^{1.0}$ |
| 258 | Microsoft | US | microsoft | 6 | 2018-10-29 | 478 | 155 | [119]1024 | 1 | [301]671 | [302]15 | [354]1642 | [355]1618 | [326]3710 | [305]6401 | [300]12892 | [238]$0.68N^{1.0}$ |
| 259 | Mobbeel Solutions | ES | mobbl | 000 | 2023-11-30 | 168 | 157 | [318]2048 | - | [247]577 | [549]34082 | [377]1958 | [410]2444 | [408]9181 | [361]11242 | [216]7502 | [35]$4.56N^{0.9}$ |
| 260 | Mobbeel Solutions | ES | mobbl | 001 | 2024-05-14 | 168 | 158 | [245]2048 | - | [244]568 | [546]22755 | [450]3993 | [472]4689 | - | - | - | |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 6: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 261 | Mobbeel Solutions | ES | mobbl | 002 | 2025-04-09 | 344 | 555 | [253]2048 | - | [440]1088 | [178]12 | (15)11 | (11)11 | - | - | - | |
| 262 | Mukh Technologies | US | mukh | 002 | 2022-09-16 | 693 | 442 | [217]2048 | - | [358]780 | [531]4261 | (10)5 | (37)82 | (27)137 | (26)319 | (26)612 | $[103]0.05N^{1.0}$ |
| 263 | N-Tech Lab | RU | ntech | 5 | 2018-10-30 | 1685 | 113 | [164]1940 | k | [330]711 | [492]55 | (102)243 | (102)246 | (88)538 | (90)1100 | (105)2867 | $[414]0.02N^{1.1}$ |
| 264 | N-Tech Lab | RU | ntech | 6 | 2018-10-30 | 1686 | 117 | [165]1940 | k | [381]831 | [497]63 | (101)243 | (101)246 | (90)546 | (91)1104 | (107)2873 | $[416]0.02N^{1.1}$ |
| 265 | N-Tech Lab | RU | ntechlab | 007 | 2019-06-25 | 2450 | 51 | [468]3348 | k | [366]795 | [503]73 | (155)393 | (170)427 | (131)780 | (145)1768 | (134)3499 | $[255]0.16N^{1.0}$ |
| 266 | N-Tech Lab | RU | ntechlab | 008 | 2020-01-06 | 1111 | 51 | [149]1300 | k | [237]554 | [458]36 | (79)179 | (78)184 | (62)341 | (59)683 | (56)1395 | $[104]0.11N^{1.0}$ |
| 267 | N-Tech Lab | RU | ntechlab | 009 | 2021-03-01 | 1208 | 42 | [148]1300 | - | [544]1936 | [456]35 | (87)193 | (79)184 | (58)336 | (58)676 | (70)1704 | $[326]0.05N^{1.1}$ |
| 268 | N-Tech Lab | RU | ntechlab | 010 | 2021-06-24 | 351 | 213 | [146]1280 | - | [342]738 | [62]6 | (221)666 | (174)435 | (136)821 | (133)1645 | (126)3337 | $[172]0.22N^{1.0}$ |
| 269 | N-Tech Lab | RU | ntechlab | 011 | 2021-12-07 | 679 | 208 | [145]1280 | - | [343]739 | [66]6 | (248)827 | (182)483 | (147)912 | (153)1869 | (166)5003 | $[379]0.07N^{1.1}$ |
| 270 | NEC | JP | nec | 2 | 2018-10-30 | 705 | 35 | [159]1616 | k | [282]642 | [368]18 | (160)405 | (164)409 | (162)1072 | (143)1755 | (157)4255 | $[396]0.06N^{1.1}$ |
| 271 | NEC | JP | nec | 3 | 2018-10-30 | 774 | 110 | [161]1712 | k | [297]665 | [385]21 | (12)7 | (8)7 | (6)14 | (13)40 | (15)82 | $[428]0.00N^{1.2}$ |
| 272 | NEC | JP | nec | 004 | 2021-07-19 | 971 | 63 | [142]1104 | - | [283]647 | [69]7 | (176)468 | (138)351 | (110)662 | (106)1330 | (97)2685 | $[128]0.20N^{1.0}$ |
| 273 | NEC | JP | nec | 005 | 2021-12-13 | 922 | 88 | [141]1104 | - | [265]615 | [76]7 | (162)407 | (202)551 | (157)1017 | (162)2091 | (155)4242 | $[164]0.28N^{1.0}$ |
| 274 | NEC | JP | nec | 006 | 2022-08-10 | 701 | 54 | [140]1104 | - | [292]656 | [113]9 | (132)358 | (163)409 | (129)765 | (126)1543 | (145)3878 | $[339]0.10N^{1.1}$ |
| 275 | NEC | JP | nec | 007 | 2023-03-03 | 632 | 56 | [99]560 | - | [281]641 | [64]6 | (95)216 | (91)214 | (72)423 | (74)853 | (71)1720 | $[384]0.03N^{1.1}$ |
| 276 | NEC | JP | nec | 008 | 2023-07-03 | 1380 | 50 | [160]1648 | - | [503]1381 | [153]11 | (197)532 | (201)548 | (193)1384 | (85)999 | (159)4757 | $[36]1.39N^{0.9}$ |
| 277 | NEC | JP | nec | 009 | 2023-12-21 | 1688 | 51 | [97]560 | - | [512]1462 | [60]5 | (64)138 | (62)139 | (47)261 | (46)523 | (45)1056 | $[137]0.08N^{1.0}$ |
| 278 | NEC | JP | nec | 010 | 2025-01-24 | 2015 | 53 | [98]560 | - | [502]1373 | [61]5 | (63)138 | (73)177 | (50)270 | (62)712 | (58)1455 | $[261]0.06N^{1.0}$ |
| 279 | NEC | JP | nec | 011 | 2025-10-16 | 1966 | 52 | [96]560 | - | [552]2362 | [71]7 | (84)184 | (77)184 | (63)346 | (60)692 | (55)1389 | $[131]0.10N^{1.0}$ |
| 280 | NEC | JP | nec | 012 | 2026-07-01 | 1966 | 52 | [100]560 | - | [553]2560 | [68]6 | (74)158 | (71)159 | (56)299 | (64)738 | (51)1308 | $[322]0.05N^{1.1}$ |
| 281 | Naver Cloud | KR | navercloud | 000 | 2026-03-30 | 1079 | 210 | [292]2048 | - | [490]1300 | [350]17 | (165)415 | (172)428 | (132)783 | (129)1570 | (137)3647 | $[275]0.15N^{1.0}$ |
| 282 | Neurotechnology | LT | neurotech | 5 | 2018-10-30 | 266 | 53 | [20]256 | k | [133]402 | [26]2 | (249)835 | (248)839 | (210)1690 | (205)3219 | (245)8955 | $[351]0.19N^{1.1}$ |
| 283 | Neurotechnology | LT | neurotech | 6 | 2018-10-30 | 564 | 53 | [19]256 | k | [338]726 | [23]2 | (250)839 | (249)842 | - | - | - | |
| 284 | Neurotechnology | LT | neurotech | 007 | 2019-10-03 | 57 | 51 | [17]256 | k | [18]161 | [20]2 | (301)1118 | (300)1110 | (253)2143 | (251)4397 | (247)9045 | $[193]0.55N^{1.0}$ |
| 285 | Neurotechnology | LT | neurotechnology | 008 | 2021-03-22 | 355 | 49 | [80]514 | - | [135]405 | [49]4 | (298)1100 | (306)1149 | (266)2266 | (261)4573 | (265)9586 | $[205]0.55N^{1.0}$ |
| 286 | Neurotechnology | LT | neurotechnology | 009 | 2021-09-01 | 246 | 82 | [78]513 | - | [87]341 | [28]3 | (269)941 | (291)1049 | (239)1977 | (245)4270 | (238)8756 | $[304]0.32N^{1.1}$ |
| 287 | Neurotechnology | LT | neurotechnology | 010 | 2022-01-07 | 247 | 83 | [21]256 | - | [295]661 | [7]2 | (278)988 | (276)984 | (231)1897 | (230)3977 | (228)8048 | $[269]0.36N^{1.0}$ |
| 288 | Neurotechnology | LT | neurotechnology | 012 | 2022-06-07 | 247 | 84 | [22]256 | - | [107]368 | [29]3 | (288)1036 | (270)945 | (224)1828 | (221)3787 | (224)7838 | $[288]0.32N^{1.0}$ |
| 289 | Neurotechnology | LT | neurotechnology | 013 | 2023-02-03 | 364 | 87 | [24]256 | - | [179]471 | [10]2 | (474)4935 | (473)4812 | (415)10461 | (436)30539 | (435)73417 | $[448]0.03N^{1.3}$ |
| 290 | Neurotechnology | LT | neurotechnology | 014 | 2023-12-27 | 362 | 95 | [25]256 | - | [180]472 | [5]2 | (93)212 | (93)215 | (71)414 | (75)854 | (72)1763 | $[273]0.08N^{1.0}$ |
| 291 | Neurotechnology | LT | neurotechnology | 015 | 2024-08-27 | 357 | 100 | [15]256 | - | [268]619 | [18]2 | (91)203 | (89)207 | (69)398 | (70)821 | (69)1690 | $[271]0.07N^{1.0}$ |
| 292 | Neurotechnology | LT | neurotechnology | 016 | 2025-08-01 | 788 | 100 | [70]512 | - | [453]1146 | [43]4 | (120)293 | (117)287 | (96)594 | (95)1187 | (90)2408 | $[307]0.09N^{1.1}$ |
| 293 | Newland Computer Co | CN | newland | 2 | 2018-10-30 | 96 | 27 | [361]2048 | - | [389]855 | [305]15 | (520)8741 | (529)8854 | (451)17892 | (449)39356 | - | $[383]1.32N^{1.1}$ |
| 294 | NextMeta Labs | IN | nextmetalabs | 001 | 2024-09-20 | 353 | 8204 | [354]2048 | - | [414]948 | [253]13 | (488)5732 | (491)5897 | - | - | - | |
| 295 | NextMeta Labs | IN | nextmetalabs | 002 | 2025-07-07 | 1079 | 6593 | [230]2048 | - | [464]1197 | [222]13 | (504)6479 | (502)6259 | (422)11828 | (425)26060 | (415)52936 | $[315]1.84N^{1.1}$ |
| 296 | NextMeta Labs | IN | nextmetalabs | 003 | 2026-06-18 | 1079 | 4076 | [327]2048 | - | [537]1852 | [311]15 | (543)28540 | (552)28460 | - | - | - | |
| 297 | Noblis | US | noblis | 1 | 2018-10-30 | 114 | 176 | [282]2048 | 1 | [38]206 | [299]15 | (319)1273 | (316)1272 | - | - | - | |
| 298 | Noblis | US | noblis | 2 | 2018-10-30 | 153 | 176 | [536]6144 | 1 | [215]517 | [472]43 | (408)2513 | (416)2522 | (363)5649 | (369)12432 | (403)44262 | $[442]0.04N^{1.3}$ |
| 299 | Nominder | LT | nominder | 000 | 2023-09-07 | 534 | 100 | [27]258 | - | [118]384 | [6]2 | (35)48 | (33)49 | (23)92 | (23)186 | (22)376 | $[181]0.02N^{1.0}$ |
| 300 | Nominder | LT | nominder | 001 | 2024-01-08 | 243 | 114 | [26]258 | - | [123]390 | [15]2 | (34)48 | (32)48 | (22)92 | (22)186 | (21)375 | $[186]0.02N^{1.0}$ |
| 301 | Nominder | LT | nominder | 002 | 2024-06-05 | 243 | 111 | [29]258 | - | [121]388 | [21]2 | (37)61 | (30)47 | (24)107 | (25)234 | (24)483 | $[409]0.00N^{1.1}$ |
| 302 | Nominder | LT | nominder | 003 | 2024-10-07 | 244 | 111 | [30]258 | - | [320]697 | [12]2 | (33)46 | (31)47 | (21)89 | (21)180 | (23)400 | $[292]0.02N^{1.0}$ |
| 303 | Nominder | LT | nominder | 004 | 2025-02-10 | 312 | 111 | [28]258 | - | [488]1294 | [16]2 | (36)50 | (34)50 | (26)116 | (24)193 | (25)507 | $[364]0.01N^{1.1}$ |
| 304 | Nominder | LT | nominder | 005 | 2026-08-07 | 472 | 99 | [79]514 | - | [543]1912 | [52]4 | (54)110 | (52)110 | (41)208 | (43)476 | (40)990 | $[340]0.03N^{1.1}$ |
| 305 | NotionTag Technologies Private Limited | IN | notiontag | 000 | 2022-01-14 | 265 | 945 | [449]2120 | - | [167]453 | [141]10 | (519)8619 | (527)8705 | (448)16652 | (448)38794 | (441)90607 | $[394]1.15N^{1.1}$ |
| 306 | Omnigarde | US | omnigarde | 000 | 2023-05-22 | 417 | 34 | [81]517 | - | [326]705 | [39]3 | (379)1983 | (378)1933 | (320)3430 | (318)7224 | (327)15597 | $[244]0.75N^{1.0}$ |
| 307 | Omnigarde | US | omnigarde | 001 | 2024-06-25 | 971 | 34 | [82]517 | - | [468]1205 | [37]3 | (384)2017 | (393)2079 | (338)3941 | (335)7908 | (334)15974 | $[123]1.20N^{1.0}$ |
| 308 | Omnigarde | US | omnigarde | 002 | 2025-07-28 | 1285 | 34 | [84]525 | - | [459]1168 | [38]3 | (407)2512 | (417)2534 | (357)4734 | (354)9553 | (351)19203 | $[78]1.72N^{1.0}$ |
| 309 | Onfido | GB | onfido | 000 | 2026-03-03 | 832 | 247 | [479]4096 | - | [465]1199 | [431]30 | (505)7075 | (517)7357 | (436)13562 | (432)28651 | (422)58683 | $[280]2.49N^{1.0}$ |
| 310 | Onfido | GB | onfido | 001 | 2026-07-21 | 832 | 332 | [472]4096 | - | [316]690 | [433]30 | (436)3368 | (446)3566 | (380)6862 | (381)13992 | (373)28042 | $[286]1.18N^{1.0}$ |
| 311 | OptiExacta | IN | optiexacta | 000 | 2024-07-23 | 505 | 2446 | [457]2564 | - | [455]1149 | [306]15 | (271)954 | (269)944 | (220)1780 | (217)3583 | (213)7238 | $[236]0.38N^{1.0}$ |
| 312 | PT Ice House | ID | icehouse | 000 | 2024-08-30 | 62 | 58 | [373]2048 | - | [7]46 | [187]12 | (537)15996 | (546)16117 | - | - | - | |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 7: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

This publication is available free of charge from: https://doi.org/10.6028/NIST.IR.8271

















| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 N=1.6M | L=50 N=1.6M | L=50 N=3M | L=50 N=6M | L=50 N=12M | POWER LAW ($\mu$s) |
| 313 | PT Ice House | ID | icehouse | 001 | 2026-02-27 | 280 | 174 | [320]2048 | - | [138]408 | [357]17 | (144)371 | (168)423 | (133)792 | (130)1585 | (123)3202 | [282]$0.14N^{1.0}$ |
| 314 | Panasonic R+D Center Singapore | SG | psl | 000 | 2023-08-08 | 795 | 607 | [552]8248 | - | [438]1085 | [474]43 | (444)3678 | (449)3669 | (383)7080 | (376)13494 | (367)26559 | [188]$1.74N^{1.0}$ |
| 315 | Panasonic R+D Center Singapore | SG | psl | 001 | 2024-02-22 | 894 | 607 | [522]4144 | - | [444]1114 | [404]25 | (395)2250 | (404)2288 | (346)4316 | (339)8577 | (354)19561 | [262]$0.86N^{1.0}$ |
| 316 | Panasonic R+D Center Singapore | SG | psl | 002 | 2024-07-26 | 992 | 607 | [524]4148 | - | [491]1300 | [395]23 | (414)2731 | (422)2704 | (362)4953 | (352)9531 | (358)20015 | [72]$1.95N^{1.0}$ |
| 317 | Panasonic R+D Center Singapore | SG | psl | 003 | 2025-07-07 | 1585 | 607 | [464]3124 | - | [470]1207 | [367]18 | (392)2146 | (407)2399 | (350)4575 | (348)9364 | (340)17474 | [227]$0.98N^{1.0}$ |
| 318 | Panasonic R+D Center Singapore | SG | psl | 004 | 2025-11-28 | 1088 | 607 | [466]3124 | - | [533]1824 | [401]24 | (431)3213 | (434)3150 | (367)5926 | (365)12019 | (380)31019 | [341]$0.74N^{1.1}$ |
| 319 | Panasonic R+D Center Singapore | SG | psl | 005 | 2026-03-30 | 1005 | 607 | [467]3124 | - | [528]1773 | [396]24 | (478)5063 | (479)5122 | (407)9094 | (398)17200 | (388)33623 | [30]$17.94N^{0.9}$ |
| 320 | Panasonic R+D Center Singapore | SG | psl | 006 | 2026-07-30 | 1005 | 607 | [465]3124 | - | [534]1826 | [402]25 | (476)5007 | (478)5084 | (405)8952 | (400)17421 | (390)33895 | [31]$17.46N^{0.9}$ |
| 321 | Pangiam | US | pangiam | 000 | 2022-02-22 | 453 | 23 | [334]2048 | - | [277]636 | [355]17 | (111)276 | (126)319 | (97)601 | (98)1210 | (92)2443 | [121]$0.18N^{1.0}$ |
| 322 | Pangiam | US | pangiam | 001 | 2022-11-17 | 991 | 36 | [167]2048 | - | [248]577 | [271]14 | (110)275 | (154)378 | (126)724 | (116)1436 | (140)3676 | [347]$0.08N^{1.1}$ |
| 323 | Papilon Savunma | TR | papsav1923 | 000 | 2023-08-22 | 923 | 24 | [39]288 | - | [354]767 | [9]2 | (31)37 | (28)38 | (18)72 | (19)143 | (19)286 | [85]$0.02N^{1.0}$ |
| 324 | Paravision (EverAI) | US | everai | 2 | 2018-10-30 | 224 | 304 | [238]2048 | 1 | [104]366 | [435]30 | (114)278 | (115)283 | - | - | - | |
| 325 | Paravision (EverAI) | US | everai | 3 | 2018-10-30 | 438 | 304 | [312]2048 | 1 | [333]717 | [416]28 | (112)278 | (114)281 | (93)572 | (92)1146 | (85)2278 | [243]$0.12N^{1.0}$ |
| 326 | Paravision (EverAI) | US | everai-paravision | 004 | 2019-06-19 | 527 | 128 | [484]4096 | 1 | [302]672 | [478]45 | (205)559 | (205)559 | (285)2611 | (306)6445 | (316)14519 | [453]$0.00N^{1.5}$ |
| 327 | Paravision (EverAI) | US | paravision | 005 | 2019-12-11 | 543 | 154 | [495]4096 | 1 | [378]830 | [484]48 | (207)561 | (208)564 | (161)1056 | (166)2298 | (165)4966 | [320]$0.16N^{1.1}$ |
| 328 | Paravision (EverAI) | US | paravision | 007 | 2021-02-01 | 529 | 235 | [498]4096 | - | [127]397 | [485]48 | (240)786 | (203)558 | (163)1086 | (163)2111 | (156)4254 | [44]$1.11N^{0.9}$ |
| 329 | Paravision | US | paravision | 009 | 2021-12-14 | 672 | 300 | [508]4100 | - | [82]331 | [507]82 | (463)4495 | (462)4230 | (396)8037 | (393)16532 | (381)31422 | [246]$1.62N^{1.0}$ |
| 330 | Paravision | US | paravision | 012 | 2023-02-07 | 762 | 182 | [506]4100 | - | [140]411 | [509]85 | (455)4313 | (471)4534 | (402)8651 | (395)16582 | (384)32114 | [82]$2.87N^{1.0}$ |
| 331 | Paravision | US | paravision | 014 | 2023-06-08 | 1501 | 182 | [503]4100 | - | [426]1010 | [512]86 | (465)4524 | (467)4454 | (394)7993 | (391)16169 | (382)31691 | [212]$1.88N^{1.0}$ |
| 332 | Paravision | US | paravision | 016 | 2024-12-04 | 1546 | 250 | [504]4100 | - | [466]1200 | [508]84 | (461)4436 | (454)4006 | (393)7861 | (396)16711 | (386)32464 | [204]$1.94N^{1.0}$ |
| 333 | Paravision | US | paravision | 019 | 2025-07-03 | 1558 | 250 | [505]4100 | - | [506]1415 | [511]86 | (454)4304 | (463)4301 | (390)7639 | (386)15454 | (383)31978 | [218]$1.78N^{1.0}$ |
| 334 | Paravision | US | paravision | 021 | 2026-03-12 | 1607 | 342 | [507]4100 | - | [549]2070 | [516]119 | (506)7086 | (514)7065 | (435)13368 | (429)27551 | (418)54944 | [268]$2.52N^{1.0}$ |
| 335 | Pentacomp Systemy Informatyczne | PL | pentacomp | 000 | 2025-01-10 | 255 | 187 | [346]2048 | - | [181]473 | [242]13 | (189)511 | (197)534 | - | - | - | |
| 336 | Pentacomp Systemy Informatyczne | PL | pentacomp | 001 | 2025-07-28 | 255 | 187 | [173]2048 | - | [172]459 | [278]14 | (339)1485 | (332)1447 | (270)2408 | (264)4708 | (261)9479 | [40]$2.83N^{0.9}$ |
| 337 | Private Identity LLC | US | privid | 000 | 2025-08-01 | 0 | 975 | [185]2048 | - | [489]1296 | [517]226 | (18)20 | (15)19 | - | - | - | |
| 338 | Protenet Soluções em Segurança | BR | protenet | 000 | 2025-09-17 | 182 | 173 | [322]2048 | - | [346]748 | [519]389 | (78)178 | (74)178 | - | - | - | |
| 339 | Protenet Soluções em Segurança | BR | protenet | 001 | 2026-01-27 | 182 | 134 | [269]2048 | - | [520]1613 | [409]27 | (295)1082 | (299)1107 | (247)2069 | (237)4146 | (231)8327 | [117]$0.63N^{1.0}$ |
| 340 | Pyramid Cyber Security + Forensic (P) | IN | pyramid | 000 | 2024-11-19 | 1016 | 141 | [350]2048 | - | [383]837 | [375]20 | (179)473 | (184)489 | (138)844 | (131)1587 | (131)3408 | [54]$0.50N^{1.0}$ |
| 341 | Pyramid Cyber Security + Forensic (P) | IN | pyramid | 001 | 2025-04-01 | 1023 | 140 | [204]2048 | - | [391]857 | [378]20 | (128)336 | (136)345 | (105)649 | (103)1303 | (116)3008 | [287]$0.12N^{1.0}$ |
| 342 | QazSmartVision.AI | KZ | qazsmartvisionai | 000 | 2024-03-15 | 547 | 95 | [55]512 | - | [323]701 | [72]7 | (72)157 | (68)157 | (54)296 | (50)597 | (49)1220 | [192]$0.07N^{1.0}$ |
| 343 | QazSmartVision.AI | KZ | qazsmartvisionai | 001 | 2024-07-16 | 1383 | 98 | [90]544 | - | [480]1245 | [79]7 | (57)124 | (55)125 | (44)234 | (42)473 | (39)957 | [144]$0.07N^{1.0}$ |
| 344 | QazSmartVision.AI | KZ | qazsmartvisionai | 002 | 2024-12-03 | 1333 | 98 | [89]544 | - | [474]1221 | [74]7 | (70)156 | (69)157 | (53)294 | (49)591 | (54)1369 | [289]$0.05N^{1.0}$ |
| 345 | QazSmartVision.AI | KZ | qazsmartvisionai | 003 | 2025-04-24 | 1358 | 98 | [91]544 | - | [479]1238 | [73]7 | (71)156 | (67)156 | (55)297 | (51)598 | (47)1204 | [157]$0.08N^{1.0}$ |
| 346 | Qnap Security | TW | qnap | 000 | 2021-07-28 | 182 | 15 | [239]2048 | - | [131]401 | [123]9 | (331)1427 | (367)1763 | - | - | - | |
| 347 | Qnap Security | TW | qnap | 001 | 2021-12-09 | 191 | 13 | [284]2048 | - | [190]487 | [95]8 | (334)1449 | (331)1429 | (321)3472 | (322)7375 | (324)15159 | [425]$0.11N^{1.2}$ |
| 348 | Qnap Security | TW | qnap | 002 | 2022-04-15 | 338 | 32 | [321]2048 | - | [193]489 | [349]17 | (273)958 | (267)938 | (227)1866 | (249)4359 | (239)8792 | [395]$0.12N^{1.1}$ |
| 349 | Qnap Security | TW | qnap | 003 | 2022-12-09 | 239 | 60 | [232]2048 | - | [46]244 | [266]13 | (357)1671 | (385)2039 | (339)4010 | (337)8379 | (321)14911 | [222]$0.87N^{1.0}$ |
| 350 | Qnap Security | TW | qnap | 004 | 2023-05-05 | 362 | 60 | [375]2048 | - | [102]365 | [158]11 | (372)1885 | (396)2096 | (314)3191 | (312)6601 | (315)14490 | [60]$1.66N^{1.0}$ |
| 351 | Qnap Security | TW | qnap | 005 | 2023-11-16 | 954 | 60 | [342]2048 | - | [420]971 | [186]12 | (341)1499 | (398)2151 | (316)3232 | (336)7969 | (325)15499 | [165]$1.02N^{1.0}$ |
| 352 | Qnap Security | TW | qnap | 006 | 2024-08-09 | 1178 | 60 | [404]2052 | - | [424]1005 | [221]13 | (375)1908 | (374)1894 | (307)3098 | (300)6263 | (313)14003 | [228]$0.72N^{1.0}$ |
| 353 | Quantasoft | CZ | quantasoft | 1 | 2018-10-30 | 276 | 452 | [378]2048 | k | [119]385 | [65]6 | (536)15422 | (544)14858 | (440)14717 | - | (344)18323 | |
| 354 | RNIT Solutions and Servies Ltd | IN | rnit | 000 | 2025-05-08 | 490 | 33 | [43]296 | - | [473]1220 | [11]2 | (152)389 | (158)387 | (128)752 | (123)1524 | (120)3098 | [240]$0.16N^{1.0}$ |
| 355 | ROC | US | rankone | 4 | 2018-10-09 | 0 | 101 | [4]85 | k | [6]36 | [77]7 | (50)101 | (51)101 | (39)190 | - | - | [76]$0.07N^{1.0}$ |
| 356 | ROC | US | rankone | 5 | 2018-10-24 | 0 | 101 | [12]133 | k | [9]92 | [78]7 | (65)140 | (63)144 | (48)266 | (47)525 | (44)1049 | [70]$0.11N^{1.0}$ |
| 357 | ROC | US | rankone | 006 | 2019-06-03 | 0 | 133 | [13]165 | k | [47]245 | [93]8 | - | - | - | - | - | |
| 358 | ROC | US | rankone | 007 | 2019-11-12 | 0 | 137 | [14]165 | k | [55]272 | [84]7 | (55)116 | (53)115 | (42)215 | (41)439 | (38)877 | [118]$0.07N^{1.0}$ |
| 359 | ROC | US | rankone | 009 | 2020-06-26 | 0 | 105 | [31]260 | k | [30]185 | [150]11 | (41)95 | (45)96 | (34)181 | (33)362 | (34)727 | [92]$0.06N^{1.0}$ |
| 360 | ROC | US | rankone | 010 | 2020-11-05 | 0 | 135 | [36]261 | - | [35]198 | [142]10 | (42)95 | (40)95 | (31)178 | (30)357 | (31)714 | [87]$0.06N^{1.0}$ |
| 361 | ROC | US | rankone | 011 | 2021-08-27 | 0 | 175 | [34]261 | - | [243]566 | [103]8 | (46)96 | (41)95 | (36)183 | (35)370 | (30)714 | [101]$0.06N^{1.0}$ |
| 362 | ROC | US | rankone | 012 | 2021-12-27 | 0 | 257 | [32]261 | - | [241]563 | [87]8 | (45)95 | (42)95 | (32)179 | (32)361 | (32)718 | [90]$0.06N^{1.0}$ |
| 363 | ROC | US | rankone | 013 | 2022-07-21 | 0 | 223 | [37]261 | - | [161]446 | [332]16 | (51)101 | (39)89 | (29)167 | (28)335 | (28)672 | [98]$0.06N^{1.0}$ |
| 364 | ROC | US | rankone | 014 | 2022-12-21 | 0 | 237 | [33]261 | - | [170]457 | [90]8 | (52)101 | (49)100 | (38)188 | (37)376 | (36)775 | [142]$0.06N^{1.0}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 8: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | N=1.6M | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 365 | ROC | US | rankone | 015 | 2023-07-20 | 0 | 282 | [35]261 | - | [234]548 | [107]9 | [48]97 | [46]97 | [35]183 | [34]366 | [35]734 | [108]$0.06N^{1.0}$ |
| 366 | ROC | US | roc | 016 | 2023-12-20 | 0 | 659 | [49]392 | - | [235]552 | [96]8 | [58]131 | [56]130 | [45]245 | [44]491 | [46]1084 | [235]$0.05N^{1.0}$ |
| 367 | ROC | US | roc | 017 | 2024-06-24 | 0 | 552 | [50]396 | - | [288]651 | [110]9 | [76]158 | [70]159 | [52]294 | [52]601 | [48]1204 | [124]$0.09N^{1.0}$ |
| 368 | ROC | US | roc | 018 | 2025-01-31 | 0 | 910 | [105]792 | - | [478]1235 | [139]10 | [119]290 | [118]290 | [92]550 | [99]1222 | [94]2527 | [321]$0.08N^{1.1}$ |
| 369 | ROC | US | roc | 019 | 2025-08-04 | 0 | 924 | [52]444 | - | [448]1131 | [548]31194 | [85]185 | [76]183 | [59]340 | [57]669 | [53]1338 | [55]$0.19N^{1.0}$ |
| 370 | ROC | US | roc | 020 | 2026-02-06 | 0 | 982 | [51]444 | - | [540]1893 | [553]43943 | [151]388 | [160]389 | [123]719 | [115]1429 | [104]2849 | [56]$0.36N^{1.0}$ |
| 371 | ROC | US | roc | 021 | 2026-08-05 | 0 | 980 | [53]448 | - | [523]1711 | [552]37620 | [141]367 | [152]377 | [112]681 | [107]1352 | [99]2708 | [61]$0.33N^{1.0}$ |
| 372 | RealSense | US | realsense | 001 | 2026-01-26 | 354 | 113 | [326]2048 | - | [545]1946 | [315]15 | [481]5187 | [480]5144 | [411]9698 | [408]19548 | [397]39427 | [109]$3.09N^{1.0}$ |
| 373 | Realnetworks Inc | US | realnetworks | 2 | 2018-10-30 | 105 | 104 | [514]4104 | k | [45]241 | [415]28 | [381]2008 | [387]2048 | [342]4194 | [340]8642 | [323]15035 | [166]$1.08N^{1.0}$ |
| 374 | Realnetworks Inc | US | realnetworks | 003 | 2019-06-12 | 93 | 102 | [162]1848 | k | [23]173 | [220]13 | [305]1145 | [302]1132 | [252]2142 | [276]5241 | [278]10495 | [363]$0.21N^{1.1}$ |
| 375 | Realnetworks Inc | US | realnetworks | 004 | 2019-10-17 | 94 | 102 | [163]1848 | 1 | [21]171 | [149]11 | [304]1143 | [304]1137 | [255]2149 | [266]4740 | [268]9693 | [299]$0.36N^{1.0}$ |
| 376 | Realnetworks Inc | US | realnetworks | 005 | 2021-06-23 | 168 | 209 | [423]2056 | - | [83]332 | [112]9 | [355]1654 | [354]1616 | [301]3030 | [296]6068 | [297]12134 | [96]$1.01N^{1.0}$ |
| 377 | Realnetworks Inc | US | realnetworks | 006 | 2021-12-02 | 250 | 56 | [418]2056 | - | [22]172 | [97]8 | [243]791 | [195]531 | [154]996 | [160]1998 | [151]3991 | [95]$0.33N^{1.0}$ |
| 378 | Realnetworks Inc | US | realnetworks | 007 | 2022-04-11 | 455 | 99 | [427]2056 | - | [89]342 | [352]17 | [246]815 | [224]674 | [205]1597 | [219]3683 | [181]5729 | [365]$0.13N^{1.1}$ |
| 379 | Realnetworks Inc | US | realnetworks | 008 | 2022-08-29 | 557 | 99 | [419]2056 | - | [219]524 | [177]12 | [199]538 | [223]673 | [181]1264 | [198]3011 | [184]5801 | [333]$0.18N^{1.1}$ |
| 380 | Recognito | AE | recognito | 000 | 2023-02-17 | 350 | 242 | [147]1280 | - | [355]775 | [126]9 | [239]780 | [213]605 | [171]1152 | [169]2320 | [180]5707 | [369]$0.10N^{1.1}$ |
| 381 | Recognito | AE | recognito | 001 | 2023-06-23 | 536 | 242 | [459]2564 | - | [439]1086 | [308]15 | [302]1128 | [277]989 | [248]2078 | [231]3978 | [154]4107 | [33]$2.45N^{0.9}$ |
| 382 | Recognito | AE | recognito | 002 | 2023-11-09 | 505 | 61 | [458]2564 | - | [434]1075 | [323]16 | [272]957 | [271]945 | [221]1791 | [253]4437 | [249]9161 | [406]$0.10N^{1.1}$ |
| 383 | Remark Holdings | CN | remarkai | 0 | 2018-10-30 | 187 | 847 | [356]2048 | k | [251]593 | [285]14 | [487]5685 | [488]5723 | - | - | - | |
| 384 | Remark Holdings | CN | remarkai | 000 | 2019-06-12 | 234 | 1092 | [311]2048 | k | [286]650 | [207]12 | [489]5776 | [487]5703 | [419]11604 | [442]32133 | [442]91436 | [443]$0.05N^{1.3}$ |
| 385 | Remark Holdings | CN | remarkai | 1 | 2018-10-30 | 187 | 847 | [287]2048 | k | [153]427 | [294]14 | [486]5680 | [490]5761 | [429]12475 | [433]28726 | [424]59618 | [430]$0.37N^{1.2}$ |
| 386 | Rendip | SG | rendip | 000 | 2021-05-21 | 0 | 416 | [196]2048 | - | [261]611 | [127]9 | [122]299 | [146]368 | [118]697 | [119]1452 | [112]2926 | [338]$0.08N^{1.1}$ |
| 387 | Reveal Media | GB | revealmedia | 000 | 2022-02-02 | 287 | 196 | [407]2052 | - | [117]383 | [134]10 | [398]2322 | [383]2019 | [333]3838 | [331]7816 | [338]16559 | [253]$0.78N^{1.0}$ |
| 388 | SQIsoft | KR | sqisoft | 001 | 2021-12-20 | 271 | 377 | [422]2056 | - | [72]312 | [124]9 | [374]1907 | [322]1319 | [274]2456 | [268]4906 | [271]9755 | [77]$0.90N^{1.0}$ |
| 389 | SQIsoft | KR | sqisoft | 002 | 2022-10-26 | 354 | 593 | [430]2056 | - | [151]425 | [245]13 | [338]1480 | [351]1596 | [302]3044 | [297]6111 | [302]13138 | [226]$0.68N^{1.0}$ |
| 390 | STCON LLC | SA | stcon | 000 | 2024-04-10 | 1999 | 62 | [478]4096 | - | [368]804 | [384]20 | [424]3092 | [445]3527 | - | - | - | |
| 391 | STCON LLC | SA | stcon | 001 | 2024-09-23 | 2973 | 49 | [492]4096 | - | [495]1343 | [423]29 | [244]794 | [280]998 | [200]1495 | [199]3016 | [222]7739 | [50]$1.36N^{0.9}$ |
| 392 | STCON LLC | SA | stcon | 002 | 2025-01-31 | 2973 | 49 | [490]4096 | - | [501]1369 | [439]31 | [242]790 | [244]791 | [225]1838 | [218]3602 | [212]7108 | [354]$0.18N^{1.1}$ |
| 393 | STCON LLC | SA | stcon | 003 | 2025-06-24 | 2973 | 49 | [500]4096 | - | [499]1359 | [428]29 | [280]1005 | [274]977 | [228]1870 | [223]3795 | [214]7362 | [153]$0.52N^{1.0}$ |
| 394 | STCON LLC | SA | stcon | 004 | 2026-01-26 | 1995 | 55 | [470]4096 | - | [518]1571 | [470]41 | [311]1221 | [313]1218 | [267]2285 | [260]4570 | [250]9203 | [100]$0.75N^{1.0}$ |
| 395 | STCON LLC | SA | stcon | 005 | 2026-05-25 | 2055 | 55 | [511]4104 | - | [517]1535 | [479]45 | [291]1050 | [260]905 | [242]2028 | [226]3845 | [218]7619 | [319]$0.27N^{1.1}$ |
| 396 | Samsung S1 Corp | KR | s1 | 000 | 2021-06-03 | 257 | 196 | [497]4096 | - | [189]487 | [377]20 | [479]5149 | [510]6794 | [434]13032 | [426]26372 | [419]55723 | [234]$2.82N^{1.0}$ |
| 397 | Samsung S1 Corp | KR | s1 | 001 | 2021-11-01 | 240 | 198 | [268]2048 | - | [198]491 | [104]8 | [399]2366 | [414]2491 | [356]4718 | [355]9614 | [363]24472 | [353]$0.53N^{1.1}$ |
| 398 | Samsung S1 Corp | KR | s1 | 002 | 2022-05-04 | 244 | 93 | [377]2048 | - | [192]488 | [322]16 | [313]1234 | [333]1450 | [287]2729 | [293]5945 | [291]11923 | [276]$0.51N^{1.0}$ |
| 399 | Samsung S1 Corp | KR | s1 | 003 | 2022-09-27 | 471 | 93 | [285]2048 | - | [191]487 | [263]13 | [352]1620 | [334]1454 | [288]2740 | [295]6025 | [285]11644 | [252]$0.57N^{1.0}$ |
| 400 | Samsung S1 Corp | KR | s1 | 004 | 2023-02-03 | 471 | 58 | [226]2048 | - | [194]489 | [239]13 | [346]1554 | [335]1481 | [312]3184 | [313]6701 | [295]12044 | [306]$0.47N^{1.1}$ |
| 401 | Samsung S1 Corp | KR | s1 | 005 | 2023-07-03 | 770 | 58 | [369]2048 | - | [385]847 | [213]12 | [333]1445 | [337]1491 | [292]2844 | [286]5691 | [304]13208 | [313]$0.44N^{1.1}$ |
| 402 | Samsung S1 Corp | KR | s1 | 006 | 2024-11-29 | 768 | 172 | [208]2048 | - | [386]849 | [224]13 | [344]1534 | [353]1613 | [305]3063 | [301]6271 | [288]11735 | [51]$1.84N^{1.0}$ |
| 403 | Samsung S1 Corp | KR | s1 | 007 | 2025-04-29 | 1439 | 172 | [330]2048 | - | [437]1083 | [230]13 | [336]1465 | [342]1520 | [293]2848 | [285]5688 | [305]13337 | [281]$0.53N^{1.0}$ |
| 404 | Sansap Technology | IN | sansap | 001 | 2024-06-07 | 0 | 357 | [236]2048 | - | [309]684 | [247]13 | [297]1098 | [285]1010 | [245]2066 | [238]4160 | [232]8363 | [285]$0.35N^{1.0}$ |
| 405 | Scanovate | IL | scanovate | 000 | 2020-01-15 | 250 | 446 | [212]2048 | - | [325]705 | [289]14 | [329]1419 | [330]1412 | [300]3008 | [364]11616 | [293]12012 | [431]$0.10N^{1.2}$ |
| 406 | Scanovate | IL | scanovate | 001 | 2020-09-10 | 250 | 446 | [336]2048 | - | [230]545 | [236]13 | [343]1520 | [323]1320 | [279]2502 | [273]5047 | [275]10163 | [179]$0.65N^{1.0}$ |
| 407 | Scylla Technologies Inc | US | scylla | 001 | 2024-10-09 | 450 | 1789 | [194]2048 | - | [401]891 | [520]467 | [522]9267 | [531]9373 | [450]17883 | [445]37069 | [437]76927 | [302]$2.84N^{1.0}$ |
| 408 | Sensetime Group | CN | sensetime | 0 | 2018-10-30 | 525 | 6 | [515]4104 | k | [317]693 | [468]41 | [186]498 | [187]501 | [177]1212 | [164]2281 | [167]5032 | [375]$0.09N^{1.1}$ |
| 409 | Sensetime Group | CN | sensetime | 1 | 2018-10-30 | 525 | 6 | [510]4104 | k | [273]628 | [483]48 | [192]516 | [188]502 | [170]1146 | [167]2301 | [161]4765 | [372]$0.09N^{1.1}$ |
| 410 | Sensetime Group | CN | sensetime | 002 | 2019-06-03 | 523 | 6 | [416]2056 | k | [258]603 | [364]18 | [133]359 | [147]370 | [232]1897 | [257]4508 | [264]9543 | [456]$0.00N^{1.5}$ |
| 411 | Sensetime Group | CN | sensetime | 003 | 2019-12-02 | 769 | 76 | [420]2056 | 1 | [406]910 | [372]19 | [473]4885 | [477]4989 | [427]12325 | [417]24712 | [410]49445 | [399]$0.67N^{1.1}$ |
| 412 | Sensetime Group | CN | sensetime | 004 | 2020-08-10 | 456 | 29 | [123]1032 | - | [315]690 | [205]12 | [406]2490 | [411]2477 | [353]4654 | [350]9402 | [355]19651 | [185]$1.22N^{1.0}$ |
| 413 | Sensetime Group | CN | sensetime | 005 | 2020-12-17 | 631 | 39 | [130]1032 | - | [422]980 | [148]11 | - | [453]3939 | [387]7398 | [382]14768 | [349]19016 | [29]$14.03N^{0.9}$ |
| 414 | Sensetime Group | CN | sensetime | 006 | 2021-07-26 | 526 | 54 | [3]- | - | [1]- | [85]7 | - | [409]2422 | [349]4527 | [345]9128 | [345]18640 | [138]$1.35N^{1.0}$ |
| 415 | Sensetime Group | CN | sensetime | 007 | 2022-01-15 | 526 | 37 | [127]1032 | - | [409]935 | [105]8 | [401]2432 | [408]2406 | [348]4513 | [344]8998 | [347]18796 | [154]$1.28N^{1.0}$ |
| 416 | Sensetime Group | CN | sensetime | 008 | 2022-08-17 | 567 | 37 | [126]1032 | - | [371]814 | [119]9 | [403]2444 | [382]2012 | [329]3785 | [326]7551 | [331]15799 | [161]$1.05N^{1.0}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 9: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | N=1.6M | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 417 | Sensetime Group | CN | sensetime | 009 | 2023-01-04 | 883 | 59 | [122]1032 | - | [493]1320 | [106]9 | [386]2031 | [389]2062 | [328]3747 | [325]7543 | [329]15781 | $^{158}1.06N^{1.0}$ |
| 418 | Sensetime Group | CN | sensetime | 010 | 2024-12-03 | 1148 | 60 | [128]1032 | - | [456]1156 | [89]8 | [106]262 | [107]259 | [82]481 | [83]981 | [81]2031 | $^{173}0.13N^{1.0}$ |
| 419 | Sensetime Group | CN | sensetime | 011 | 2026-03-12 | 842 | 62 | [125]1032 | - | [477]1228 | [137]10 | [134]359 | [142]361 | [111]679 | [108]1360 | [102]2734 | $^{106}0.22N^{1.0}$ |
| 420 | Sensetime Group | CN | sensetime | 012 | 2026-07-31 | 1223 | 62 | [132]1036 | - | [519]1583 | [272]14 | [145]371 | [148]370 | [117]696 | [120]1454 | [128]3374 | $^{328}0.10N^{1.1}$ |
| 421 | Serendipity | UZ | serendipity | 000 | 2023-06-20 | 0 | 366 | [192]2048 | - | [402]894 | [227]13 | [158]400 | [149]371 | [137]827 | [96]1188 | [88]2386 | $^{45}0.60N^{0.9}$ |
| 422 | Seventh Sense Artificial Intelligence | SG | seventhsense | 001 | 2024-09-12 | 1410 | 633 | [487]4096 | - | [445]1127 | [522]890 | [6]2 | [16]22 | [9]23 | [8]32 | [7]40 | $^{15}381.33N^{0.3}$ |
| 423 | Shaman Software | US | shaman | 6 | 2018-10-26 | 0 | 200 | [243]2048 | k | [327]706 | [286]14 | [212]603 | [214]612 | - | - | - | |
| 424 | Shaman Software | US | shaman | 7 | 2018-10-26 | 0 | 200 | [255]2048 | k | [328]707 | [290]14 | [211]602 | [215]614 | [173]1187 | [177]2448 | [170]5083 | $^{245}0.25N^{1.0}$ |
| 425 | Shanghai Jiao Tong University | CN | sjtu | 001 | 2026-02-10 | 1417 | 239 | [528]4608 | - | [536]1841 | [446]33 | [515]8200 | [524]8315 | [445]15765 | [441]31878 | [427]64177 | $^{159}4.36N^{1.0}$ |
| 426 | Shanghai Jiao Tong University | CN | sjtu | 002 | 2026-08-03 | 873 | 395 | [529]4608 | - | [535]1833 | [451]34 | [513]8190 | [525]8349 | [446]15902 | [440]31717 | [426]64136 | $^{151}4.46N^{1.0}$ |
| 427 | Shanghai Yitu Technology | CN | yitu | 4 | 2018-10-30 | 2119 | 136 | [441]2070 | 1 | [403]897 | [477]45 | [320]1288 | [311]1203 | [273]2440 | [277]5241 | [267]9671 | $^{232}0.52N^{1.0}$ |
| 428 | Shanghai Yitu Technology | CN | yitu | 5 | 2018-10-30 | 2043 | 136 | [440]2070 | 1 | [388]853 | [476]44 | [314]1237 | [310]1199 | [280]2513 | [272]5013 | [266]9620 | $^{219}0.55N^{1.0}$ |
| 429 | Shennona Co Ltd | TW | shennona | 000 | 2024-11-20 | 977 | 72 | [112]1024 | - | [487]1290 | [80]7 | [482]5286 | [482]5244 | - | - | - | |
| 430 | Shufti | GB | shufti | 001 | 2026-03-30 | 432 | 2228 | [399]2052 | - | [299]667 | [525]1302 | [5]1 | [4]3 | - | - | - | |
| 431 | Smilart | DE | smilart | 4 | 2018-10-30 | 65 | 89 | [58]512 | k | [20]167 | [48]4 | [538]16137 | [545]15633 | - | - | - | |
| 432 | Smilart | DE | smilart | 5 | 2018-10-30 | 562 | 89 | [174]2048 | k | [164]450 | [283]14 | - | - | - | - | - | |
| 433 | Sparktek International Pte Ltd | CN | sparktek | 000 | 2025-11-24 | 785 | 116 | [170]2048 | - | [285]649 | [353]17 | [184]495 | [181]467 | [159]1031 | [152]1856 | [138]3658 | $^{156}0.26N^{1.0}$ |
| 434 | Sparktek International Pte Ltd | CN | sparktek | 001 | 2026-04-15 | 2005 | 116 | [526]4224 | - | [524]1713 | [448]34 | [380]2006 | [381]2003 | [340]4118 | [360]10935 | [360]21982 | $^{440}0.06N^{1.2}$ |
| 435 | Sparsh CCTV | IN | sparsh | 001 | 2024-01-12 | 0 | 357 | [372]2048 | - | [307]684 | [154]11 | [287]1036 | [284]1008 | [235]1908 | [229]3927 | [223]7764 | $^{171}0.51N^{1.0}$ |
| 436 | Staqu Technologies | IN | staqu | 000 | 2021-08-30 | 1018 | 690 | [499]4096 | - | [308]684 | [400]24 | [466]4527 | [475]4933 | - | - | - | |
| 437 | Stellar Technologies | US | stellartecs | 001 | 2026-05-01 | 1397 | 32 | [471]4096 | - | [496]1348 | [443]32 | [419]2925 | [427]2923 | [364]5652 | [363]11313 | [362]23169 | $^{206}1.36N^{1.0}$ |
| 438 | Sum and Substance | GB | sumsub | 000 | 2023-08-18 | 169 | 108 | [267]2048 | - | [64]292 | [206]12 | [147]380 | [151]372 | - | - | - | |
| 439 | Synesis | RU | synesis | 3 | 2018-10-30 | 237 | 150 | [502]4096 | k | [11]99 | [421]29 | [241]789 | [246]801 | [237]1941 | [227]3888 | [240]8810 | $^{415}0.07N^{1.1}$ |
| 440 | Synesis | RU | synesis | 003 | 2019-07-04 | 143 | 17 | [345]2048 | k | [39]211 | [195]12 | [188]507 | [189]502 | [268]2297 | [259]4564 | [259]9452 | $^{451}0.00N^{1.4}$ |
| 441 | Synesis | RU | synesis | 005 | 2020-09-08 | 494 | 24 | [513]4104 | - | [350]756 | [398]24 | [257]877 | [251]865 | [311]3182 | [263]4658 | [270]9750 | $^{432}0.06N^{1.2}$ |
| 442 | T4iSB | BR | t4isb | 000 | 2022-08-17 | 228 | 15 | [290]2048 | - | [339]729 | [258]13 | [103]250 | [123]311 | - | - | - | |
| 443 | TTI Group | VN | ttigroup | 000 | 2024-12-16 | 455 | 444 | [178]2048 | - | [399]887 | [184]12 | [533]13054 | [541]12894 | - | - | - | |
| 444 | Tech5 SA | CH | tech5 | 001 | 2019-08-19 | 1394 | 116 | [153]1536 | k | [400]887 | [133]10 | [148]383 | [242]766 | [289]2767 | [299]6149 | [193]6178 | $^{398}0.12N^{1.1}$ |
| 445 | Tech5 SA | CH | tech5 | 002 | 2021-04-07 | 727 | 112 | [77]513 | - | [314]687 | [44]4 | [448]3816 | [508]6689 | [430]12541 | [419]25145 | [413]50239 | $^{94}4.18N^{1.0}$ |
| 446 | Tech5 SA | CH | tech5 | 003 | 2023-07-14 | 0 | 357 | [207]2048 | - | [289]653 | [208]12 | [285]1025 | [287]1013 | [244]2050 | [235]4126 | [225]7935 | $^{223}0.45N^{1.0}$ |
| 447 | Techainer | VN | techainer | 000 | 2026-06-30 | 627 | 1071 | [360]2048 | - | [97]357 | [341]16 | [202]547 | [236]734 | [180]1245 | [180]2496 | - | $^{272}0.17N^{1.0}$ |
| 448 | Technology Control Company | SA | tcc | 000 | 2026-02-25 | 54 | 1161 | [115]1024 | - | [105]367 | [100]8 | [201]546 | [200]546 | - | - | - | |
| 449 | Technology Control Company | SA | tcc | 001 | 2026-07-15 | 963 | 1209 | [259]2048 | - | [144]415 | [297]15 | [213]608 | [191]509 | - | - | - | |
| 450 | Tencent Deepsea Lab | CN | deepsea | 001 | 2019-07-29 | 250 | 323 | [240]2048 | 1 | [341]737 | [203]12 | [284]1021 | [289]1020 | [290]2774 | [288]5767 | [298]12341 | $^{437}0.06N^{1.2}$ |
| 451 | Tevian | RU | tevian | 5 | 2018-10-30 | 773 | 15 | [250]2048 | 1 | [136]405 | [300]15 | [161]405 | [162]408 | [140]854 | [144]1757 | [129]3380 | $^{297}0.14N^{1.0}$ |
| 452 | Tevian | RU | tevian | 006 | 2021-04-16 | 769 | 19 | [129]1032 | - | [231]546 | [132]10 | [129]343 | [121]295 | [94]578 | [94]1187 | [103]2741 | $^{348}0.06N^{1.1}$ |
| 453 | Tevian | RU | tevian | 007 | 2021-10-12 | 703 | 19 | [124]1032 | - | [345]747 | [53]4 | [143]369 | [122]298 | [95]579 | [93]1179 | [91]2418 | $^{266}0.11N^{1.0}$ |
| 454 | Thakaa Advanced Company for Information Technology | SA | thakaait | 000 | 2026-08-10 | 1232 | 480 | [415]2056 | - | [431]1047 | [340]16 | [452]4040 | [456]4043 | [395]8017 | [390]15990 | [385]32356 | $^{237}1.67N^{1.0}$ |
| 455 | Thales Group | US | cogent | 2 | 2018-10-30 | 681 | 39 | [136]1043 | k | [412]945 | [410]27 | [383]2017 | [397]2144 | [345]4298 | [338]8472 | [337]16429 | $^{180}1.08N^{1.0}$ |
| 456 | Thales Group | US | cogent | 3 | 2018-10-30 | 681 | 39 | [135]1043 | k | [410]940 | [129]9 | [312]1230 | [320]1311 | [286]2687 | [278]5398 | [276]10184 | $^{209}0.62N^{1.0}$ |
| 457 | Thales Group | US | cogent | 004 | 2021-02-10 | 1376 | 59 | [412]2053 | - | [413]947 | [270]14 | [416]2903 | [375]1911 | [322]3566 | [323]7498 | [336]16370 | $^{293}0.64N^{1.0}$ |
| 458 | Thales Group | US | cogent | 005 | 2021-09-13 | 1043 | 56 | [137]1062 | - | [145]418 | [58]5 | [219]648 | [278]996 | [229]1872 | [225]3845 | [217]7555 | $^{213}0.44N^{1.0}$ |
| 459 | Thales Group | US | cogent | 006 | 2022-05-14 | 508 | 70 | [94]550 | - | [142]412 | [88]8 | [209]587 | [230]706 | [185]1328 | [188]2767 | [177]5675 | $^{168}0.37N^{1.0}$ |
| 460 | Thales Group | US | cogent | 007 | 2023-01-30 | 597 | 72 | [95]550 | - | [187]485 | [40]3 | [193]516 | [231]708 | [184]1321 | [192]2839 | [183]5789 | $^{202}0.34N^{1.0}$ |
| 461 | TigerIT Americas LLC | US | tiger | 2 | 2018-10-29 | 416 | 518 | [391]2052 | k | [176]461 | [307]15 | [369]1816 | [376]1921 | [332]3833 | [324]7526 | [318]14820 | $^{221}0.83N^{1.0}$ |
| 462 | TigerIT Americas LLC | US | tiger | 3 | 2018-10-30 | 416 | 518 | [409]2052 | k | [173]461 | [551]37431 | [86]191 | [82]189 | - | - | - | |
| 463 | TigerIT Americas LLC | US | tiger | 004 | 2025-08-04 | 248 | 194 | [325]2048 | - | [168]456 | [287]14 | [276]976 | [275]982 | [240]1997 | [220]3773 | [233]8432 | $^{388}0.13N^{1.1}$ |
| 464 | Toshiba | JP | toshiba | 0 | 2018-10-30 | 961 | 105 | [158]1548 | k | [397]876 | [163]12 | [495]6153 | [501]6236 | [426]12221 | [421]25355 | [411]49448 | $^{426}0.36N^{1.2}$ |
| 465 | Toshiba | JP | toshiba | 1 | 2018-10-30 | 961 | 105 | [437]2060 | k | [396]875 | [554]44701 | [493]6007 | [505]6355 | - | - | - | |
| 466 | Toshiba | JP | toshiba | 002 | 2025-07-08 | 1080 | 75 | [417]2056 | - | [497]1348 | [200]12 | [359]1685 | [366]1757 | [313]3185 | [317]6968 | [309]13797 | $^{277}0.60N^{1.0}$ |
| 467 | Toshiba | JP | toshiba | 003 | 2026-01-09 | 1216 | 83 | [448]2084 | - | [551]2164 | [495]59 | [451]4038 | [452]3922 | [388]7488 | [383]14853 | [376]28943 | $^{64}3.32N^{1.0}$ |
| 468 | Toss | KR | toss | 000 | 2025-09-22 | 970 | 91 | [234]2048 | - | [428]1028 | [363]18 | [208]572 | [209]571 | [166]1119 | [173]2413 | [176]5647 | $^{343}0.13N^{1.1}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 10: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| | DEVELOPER FULL NAME | DEVELOPER LOCATION | SHORT NAME | SEQ. NUM. | VALIDATION DATE | CONFIG[1] DATA (MB) | LIB[1] DATA (MB) | TEMPLATE GENERATION SIZE (B) | MULT[3] | TIME (MS)[4] | FINALIZE[2] TIME (S) | SEARCH DURATION[5] MILLISEC L=1 N=1.6M | L=50 N=1.6M | L=50 N=3M | L=50 N=6M | L=50 N=12M | POWER LAW ($\mu$s) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 469 | Tripleize | JP | aize | 001 | 2021-08-06 | 262 | 150 | [277]2048 | - | [41]222 | [117]9 | [415]2870 | [432]3080 | - | - | - | |
| 470 | Tripleize | JP | aize | 002 | 2023-08-30 | 251 | 32 | [300]2048 | - | [29]185 | [209]12 | [426]3119 | [430]3037 | [365]5662 | [362]11312 | [361]22897 | [81]$2.00N^{1.0}$ |
| 471 | Tripleize | JP | aize | 006 | 2026-04-08 | 270 | 27 | [436]2060 | - | [202]501 | [317]15 | [483]5346 | [483]5326 | [413]10144 | [410]20404 | [405]46059 | [294]$1.76N^{1.0}$ |
| 472 | TrueSight Laboratories Tech LLC | AE | tsight | 000 | 2026-08-03 | 1189 | 106 | [113]1024 | - | [538]1859 | [313]15 | [104]254 | [108]259 | [78]465 | [86]1049 | [84]2257 | [317]$0.07N^{1.1}$ |
| 473 | Trueface.ai | US | trueface | 000 | 2021-01-27 | 247 | 119 | [166]2000 | - | [101]363 | [232]13 | [108]271 | [130]327 | [101]614 | [100]1239 | [96]2678 | [203]$0.15N^{1.0}$ |
| 474 | TuringTech.vip | CN | turingtechvip | 001 | 2022-09-29 | 151 | 161 | [308]2048 | - | [310]685 | [265]13 | [541]22085 | [547]17480 | - | - | - | |
| 475 | Unissey | FR | unissey | 001 | 2023-10-24 | 0 | 1052 | [475]4096 | - | [111]370 | [535]7732 | [16]15 | [12]14 | - | - | - | |
| 476 | Unissey | FR | unissey | 002 | 2024-02-23 | 0 | 1055 | [482]4096 | - | [114]371 | [533]5962 | [56]119 | [54]117 | - | - | - | |
| 477 | Unissey | FR | unissey | 003 | 2025-01-17 | 0 | 1100 | [493]4096 | - | [98]359 | [481]47 | [126]318 | [124]312 | [121]701 | [125]1536 | [108]2892 | [376]$0.05N^{1.1}$ |
| 478 | UniverseAI | SG | universeai | 000 | 2026-08-11 | 333 | 1362 | [83]520 | - | [81]325 | [523]1125 | [26]30 | [36]73 | [25]111 | [40]432 | [43]1021 | [9]$3042.36N^{0.2}$ |
| 479 | Universidade de Coimbra | PT | visteam | 000 | 2024-07-26 | 569 | 46 | [351]2048 | - | [462]1187 | [219]13 | [442]3619 | [457]4053 | [382]7043 | [388]15723 | [377]28968 | [148]$2.11N^{1.0}$ |
| 480 | Universidade de Coimbra | PT | visteam | 001 | 2025-05-28 | 569 | 46 | [205]2048 | - | [469]1207 | [228]13 | [443]3635 | [460]4113 | - | - | - | |
| 481 | VDF Software | CA | vdfsoftware | 001 | 2025-04-25 | 5949 | 0 | [266]2048 | - | [86]339 | [333]16 | [417]2912 | [426]2917 | - | - | - | |
| 482 | VDF Software | CA | vdfsoftware | 002 | 2025-10-01 | 5989 | 5 | [396]2052 | - | [430]1044 | [393]23 | [521]8765 | [528]8737 | [397]8209 | [397]16739 | [434]71934 | [48]$10.02N^{0.9}$ |
| 483 | VNIS Joint Stock | VN | vnis | 000 | 2024-01-05 | 278 | 2988 | [305]2048 | - | [156]435 | [193]12 | [220]648 | [218]633 | - | - | - | |
| 484 | Veridas Digital Authentication Solutions S.L. | ES | veridas | 001 | 2021-03-05 | 347 | 875 | [233]2048 | - | [212]509 | [234]13 | [460]4407 | [485]5469 | [414]10350 | [411]20655 | [399]41264 | [97]$3.40N^{1.0}$ |
| 485 | Veridas Digital Authentication Solutions S.L. | ES | veridas | 002 | 2021-07-06 | 347 | 870 | [184]2048 | - | [208]507 | [145]10 | [175]459 | [128]325 | [116]685 | [110]1365 | [101]2730 | [323]$0.09N^{1.1}$ |
| 486 | Veridas Digital Authentication Solutions S.L. | ES | veridas | 003 | 2021-11-09 | 346 | 870 | [366]2048 | - | [209]508 | [109]9 | [174]457 | [129]327 | [119]699 | [113]1401 | [150]3954 | [429]$0.02N^{1.2}$ |
| 487 | Veridas Digital Authentication Solutions S.L. | ES | veridas | 004 | 2023-02-03 | 1074 | 608 | [202]2048 | - | [250]579 | [261]13 | [172]440 | [169]426 | [102]618 | [54]620 | [143]3775 | [80]$0.22N^{1.0}$ |
| 488 | Verigram | KZ | verigram | 000 | 2023-05-17 | 1553 | 20 | [477]4096 | - | [305]681 | [501]68 | [229]710 | [227]688 | [122]706 | [105]1312 | [98]2696 | [23]$16.28N^{0.7}$ |
| 489 | Verigram | KZ | verigram | 001 | 2023-10-26 | 1553 | 20 | [509]4104 | - | [318]696 | [488]51 | [267]929 | [265]929 | [183]1315 | [185]2698 | [186]5822 | [41]$1.67N^{0.9}$ |
| 490 | Verijelas | ID | verijelas | 000 | 2022-10-11 | 248 | 11 | [218]2048 | - | [70]310 | [273]14 | [19]20 | [20]26 | - | - | - | |
| 491 | Viante.AI | AE | viante | 000 | 2023-10-02 | 1234 | 19 | [431]2056 | - | [436]1079 | [171]12 | [324]1336 | [314]1249 | [327]3733 | [279]5411 | [279]10878 | [361]$0.27N^{1.1}$ |
| 492 | Viante.AI | AE | viante | 001 | 2025-04-18 | 2541 | 19 | [424]2056 | - | [458]1160 | [191]12 | [294]1072 | [283]1008 | [256]2151 | [254]4443 | [256]9382 | [377]$0.16N^{1.1}$ |
| 493 | Viante.AI | AE | viante | 002 | 2025-12-16 | 1234 | 19 | [426]2056 | - | [532]1819 | [346]17 | [330]1425 | [344]1531 | [303]3052 | [302]6290 | [310]13812 | [358]$0.31N^{1.1}$ |
| 494 | Viante.AI | AE | viante | 003 | 2026-05-25 | 1234 | 19 | [414]2056 | - | [531]1812 | [325]16 | [335]1449 | [345]1539 | [309]3117 | [308]6531 | [307]13442 | [346]$0.34N^{1.1}$ |
| 495 | Vietnam Payment Solution | VN | vnpay | 001 | 2024-10-15 | 566 | 3948 | [296]2048 | - | [454]1148 | [161]12 | [216]638 | [220]637 | [178]1214 | [175]2440 | [198]6391 | [357]$0.13N^{1.1}$ |
| 496 | Vietnam Posts and Telecommunications Group | VN | vnpt | 001 | 2022-05-05 | 361 | 235 | [379]2048 | - | [199]491 | [380]20 | [245]813 | [253]867 | [207]1626 | [210]3346 | [203]6808 | [184]$0.43N^{1.0}$ |
| 497 | Vietnam Posts and Telecommunications Group | VN | vnpt | 002 | 2022-09-08 | 547 | 235 | [340]2048 | - | [278]636 | [345]16 | [252]857 | [258]897 | [214]1700 | [208]3320 | [199]6622 | [114]$0.52N^{1.0}$ |
| 498 | Vietnam Posts and Telecommunications Group | VN | vnpt | 003 | 2023-06-26 | 547 | 235 | [214]2048 | - | [372]816 | [324]16 | [262]904 | [266]933 | [216]1719 | [216]3498 | [210]6933 | [110]$0.55N^{1.0}$ |
| 499 | Vietnam Posts and Telecommunications Group | VN | vnpt | 004 | 2023-12-05 | 1088 | 351 | [283]2048 | - | [484]1263 | [339]16 | [270]941 | [259]904 | [212]1698 | [214]3485 | [208]6918 | [145]$0.50N^{1.0}$ |
| 500 | Vietnam Posts and Telecommunications Group | VN | vnpt | 005 | 2024-11-12 | 1088 | 351 | [310]2048 | - | [463]1189 | [328]16 | [268]940 | [268]939 | [211]1697 | [211]3404 | [205]6845 | [67]$0.69N^{1.0}$ |
| 501 | Viettel Artificial Intelligence | VN | viettelai | 000 | 2023-07-11 | 277 | 312 | [534]4861 | - | [75]315 | [411]27 | [154]392 | [198]537 | [156]1012 | [142]1753 | [132]3437 | [37]$1.28N^{0.9}$ |
| 502 | Viettel Artificial Intelligence | VN | viettelai | 001 | 2023-12-18 | 515 | 313 | [532]4858 | - | [329]710 | [414]28 | [183]494 | [178]459 | [155]1010 | [158]1967 | - | [152]$0.27N^{1.0}$ |
| 503 | Viettel Artificial Intelligence | VN | viettelai | 002 | 2025-04-14 | 515 | 259 | [533]4858 | - | [331]714 | [406]27 | [157]397 | [196]532 | [146]905 | [146]1771 | [115]3007 | [32]$1.40N^{0.9}$ |
| 504 | Viettel Artificial Intelligence | VN | viettelai | 003 | 2026-02-02 | 1471 | 198 | [390]2052 | - | [554]3044 | [319]15 | [196]522 | [179]465 | [145]892 | [148]1781 | [135]3624 | [71]$0.35N^{1.0}$ |
| 505 | Viettel Group | VN | vts | 000 | 2021-03-12 | 250 | 257 | [371]2048 | - | [162]447 | [526]2295 | [9]3 | [7]4 | [3]6 | [5]11 | - | [20]$0.61N^{0.6}$ |
| 506 | Viettel Group | VN | vts | 001 | 2021-07-16 | 352 | 600 | [197]2048 | - | [160]444 | [388]21 | [390]2081 | [413]2487 | [352]4644 | [347]9313 | [346]18713 | [99]$1.53N^{1.0}$ |
| 507 | Viettel Group | VN | vts | 002 | 2022-02-08 | 244 | 600 | [294]2048 | - | [405]903 | [426]29 | [405]2485 | [412]2485 | [354]4678 | [349]9370 | [348]18833 | [107]$1.49N^{1.0}$ |
| 508 | Viettel Group | VN | vts | 003 | 2022-07-14 | 493 | 468 | [237]2048 | - | [125]393 | [450]34 | [404]2482 | [391]2075 | [335]3894 | [332]7883 | [333]15891 | [129]$1.18N^{1.0}$ |
| 509 | Viettel Group | VN | vts | 004 | 2024-09-03 | 491 | 887 | [352]2048 | - | [146]420 | [420]29 | [389]2073 | [392]2075 | [337]3925 | [333]7883 | [328]15770 | [122]$1.19N^{1.0}$ |
| 510 | Viettel Group | VN | vts | 005 | 2025-01-21 | 491 | 887 | [348]2048 | - | [132]402 | [427]29 | [388]2070 | [390]2071 | [336]3917 | [334]7893 | [332]15828 | [135]$1.16N^{1.0}$ |
| 511 | Vigilant Solutions | US | vigilant | 5 | 2018-10-30 | 335 | 122 | [155]1544 | k | [352]762 | [370]19 | - | [361]1720 | - | - | - | |
| 512 | Vigilant Solutions | US | vigilant | 6 | 2018-10-30 | 337 | 122 | [154]1544 | k | [373]816 | [386]21 | - | [359]1713 | - | - | - | |
| 513 | Vigilant Solutions | US | vigilantsolutions | 007 | 2021-01-08 | 340 | 51 | [156]1544 | - | [236]552 | [344]16 | [348]1570 | [326]1352 | [295]2911 | [294]5966 | [284]11466 | [359]$0.27N^{1.1}$ |
| 514 | Vigilant Solutions | US | vigilantsolutions | 008 | 2021-07-23 | 340 | 51 | [157]1544 | - | [99]360 | [255]13 | [332]1442 | [295]1061 | [269]2330 | [283]5520 | [262]9499 | [407]$0.11N^{1.1}$ |
| 515 | VinBigData | VN | vinbigdata | 001 | 2024-05-07 | 1534 | 268 | [491]4096 | - | [450]1136 | [403]25 | [279]990 | [328]1384 | [222]1810 | [265]4715 | [251]9281 | [68]$0.91N^{1.0}$ |
| 516 | VinBigData | VN | vinbigdata | 002 | 2024-10-16 | 1080 | 268 | [183]2048 | - | [476]1222 | [165]12 | [234]756 | [233]721 | [189]1369 | [186]2745 | [173]5514 | [160]$0.38N^{1.0}$ |
| 517 | VinBigData | VN | vinbigdata | 003 | 2025-02-19 | 1080 | 268 | [315]2048 | - | [457]1157 | [229]13 | [230]738 | [235]731 | [188]1366 | [204]3199 | [196]6303 | [329]$0.20N^{1.1}$ |
| 518 | VinBigData | VN | vinbigdata | 004 | 2025-07-24 | 1080 | 270 | [251]2048 | - | [461]1179 | [259]13 | [232]748 | [232]717 | [190]1377 | [200]3024 | [221]7726 | [397]$0.09N^{1.1}$ |
| 519 | Virtisha AI | IN | virtisha | 000 | 2026-07-22 | 469 | 37 | [429]2056 | - | [375]822 | [348]17 | [306]1145 | [305]1146 | - | - | - | |
| 520 | Visidon | FI | visidon | 1 | 2018-10-30 | 166 | 42 | [410]2052 | k | [298]667 | [310]15 | [459]4370 | [468]4472 | [399]8454 | [399]17262 | [391]34288 | [150]$2.40N^{1.0}$ |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

*Table 11: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.*

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| | DEVELOPER | DEVELOPER | SHORT | SEQ. | VALIDATION | CONFIG[1] | LIB[1] | TEMPLATE GENERATION | | | FINALIZE[2] | SEARCH DURATION[5] MILLISEC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | FULL NAME | LOCATION | NAME | NUM. | DATE | DATA (MB) | DATA (MB) | SIZE (B) | MULT[3] | TIME (MS)[4] | TIME (S) | L=1 | L=50 | L=50 | L=50 | L=50 | POWER LAW |
| | | | | | | | | | | | N=1.6M | N=1.6M | N=1.6M | N=3M | N=6M | N=12M | ($\mu$s) |
| 521 | Visidon | FI | vd | 002 | 2021-05-18 | 248 | 42 | [2]- | - | [3]- | [111]9 | - | [406]2336 | - | - | - | |
| 522 | Visidon | FI | vd | 003 | 2021-10-12 | 497 | 43 | [1]- | - | [2]- | [98]8 | - | [394]2082 | - | - | - | |
| 523 | Vision-Box | PT | visionbox | 000 | 2021-09-17 | 252 | 274 | [434]2059 | - | [313]687 | [343]16 | [191]515 | [140]359 | [141]855 | [55]631 | [82]2096 | $^{27}2.46N^{0.8}$ |
| 524 | Vision-Box | PT | visionbox | 001 | 2024-03-19 | 301 | 191 | [517]4116 | - | [67]297 | [473]43 | [218]640 | [219]634 | [172]1181 | [171]2380 | [171]5276 | $^{265}0.23N^{1.0}$ |
| 525 | Vision-Box | PT | visionbox | 002 | 2024-10-31 | 468 | 166 | [516]4116 | - | [128]397 | [466]40 | [254]872 | [254]874 | [203]1573 | [201]3057 | [162]4801 | $^{38}1.76N^{0.9}$ |
| 526 | VisionLabs | RU | visionlabs | 6 | 2018-10-30 | 360 | 17 | [57]512 | 1 | [61]289 | [544]20290 | [30]36 | [27]36 | [17]39 | [15]44 | [12]53 | $^{10}3211.93N^{0.2}$ |
| 527 | VisionLabs | RU | visionlabs | 7 | 2018-10-30 | 360 | 17 | [68]512 | 1 | [60]289 | [550]34666 | [38]63 | [35]63 | [19]72 | [18]80 | [16]115 | $^{12}2076.32N^{0.2}$ |
| 528 | VisionLabs | RU | visionlabs | 008 | 2019-06-18 | 348 | 17 | [64]512 | 1 | [54]272 | [541]12747 | [22]23 | [18]24 | [10]26 | [7]29 | [6]33 | $^{6}2539.61N^{0.2}$ |
| 529 | VisionLabs | RU | visionlabs | 009 | 2020-08-04 | 689 | 20 | [60]512 | - | [178]467 | [542]13245 | [23]23 | [21]29 | [13]34 | [17]61 | [17]145 | $^{19}8.88N^{0.6}$ |
| 530 | VisionLabs | RU | visionlabs | 010 | 2021-02-05 | 1042 | 20 | [69]512 | - | [266]617 | [537]11837 | [20]21 | [23]32 | [14]36 | [12]39 | [8]43 | $^{7}3183.79N^{0.2}$ |
| 531 | VisionLabs | RU | visionlabs | 011 | 2021-10-20 | 1042 | 20 | [56]512 | - | [267]618 | [540]12255 | [24]23 | [17]23 | [11]26 | [9]34 | [11]51 | $^{16}301.26N^{0.3}$ |
| 532 | Vixvizon | AU | vixvizion | 009 | 2022-11-28 | 580 | 460 | [249]2048 | - | [214]516 | [267]14 | [153]389 | [133]339 | [125]724 | [141]1750 | [95]2622 | $^{308}0.11N^{1.1}$ |
| 533 | Vocord | RU | vocord | 5 | 2018-10-30 | 1035 | 185 | [103]768 | k | [357]780 | [83]7 | [75]158 | [88]204 | [67]383 | [67]767 | [59]1466 | $^{102}0.12N^{1.0}$ |
| 534 | Vocord | RU | vocord | 6 | 2018-10-30 | 1035 | 185 | [355]10240 | k | [359]785 | [518]243 | [77]170 | [94]216 | - | - | - | |
| 535 | Vsoft | BR | vsoft | 001 | 2026-03-16 | 9 | 108 | [433]2057 | - | [33]191 | [502]72 | [529]11496 | [540]12598 | - | - | - | |
| 536 | Vsoft | BR | vsoft | 002 | 2026-07-31 | 476 | 108 | [432]2057 | - | [147]420 | [504]74 | [531]11666 | [539]12583 | - | - | - | |
| 537 | WINK | US | wink | 000 | 2023-07-18 | 507 | 552 | [59]512 | - | [12]101 | [36]3 | [135]360 | [141]359 | - | - | - | |
| 538 | WINK | US | wink | 001 | 2023-12-21 | 685 | 552 | [75]512 | - | [17]144 | [35]3 | [137]364 | [144]363 | [114]683 | [111]1373 | [111]2921 | $^{208}0.17N^{1.0}$ |
| 539 | WINK | US | wink | 002 | 2024-10-18 | 376 | 552 | [72]512 | - | [370]811 | [34]3 | [138]364 | [145]363 | [115]684 | [112]1375 | [133]3459 | $^{336}0.09N^{1.1}$ |
| 540 | WINK | US | wink | 003 | 2026-05-08 | 178 | 90 | [201]2048 | - | [530]1803 | [331]16 | [325]1341 | [325]1335 | [278]2477 | [271]4939 | [274]9998 | $^{189}0.63N^{1.0}$ |
| 541 | Xailient | AU | xailient | 000 | 2026-05-04 | 363 | 230 | [258]2048 | - | [332]714 | [347]17 | [142]369 | [185]495 | [130]775 | [137]1711 | [124]3244 | $^{53}0.50N^{1.0}$ |
| 542 | Xforward AI Technology | CN | xforwardai | 000 | 2020-07-24 | 236 | 171 | [264]2048 | - | [349]753 | [262]13 | [469]4603 | [519]7647 | [444]15723 | [415]23900 | [416]53729 | $^{410}0.56N^{1.1}$ |
| 543 | Xforward AI Technology | CN | xforwardai | 001 | 2021-01-21 | 332 | 50 | [316]2048 | - | [96]355 | [334]16 | [458]4338 | [465]4384 | [404]8798 | [405]18553 | [409]48993 | $^{420}0.32N^{1.1}$ |
| 544 | Xforward AI Technology | CN | xforwardai | 002 | 2021-05-24 | 691 | 50 | [476]4096 | - | [197]490 | [366]18 | [502]6340 | [506]6400 | [431]12659 | [437]31077 | [428]65158 | $^{417}0.52N^{1.1}$ |
| 545 | Youverse | PT | youverse | 001 | 2026-07-17 | 458 | 286 | [337]2048 | - | [446]1128 | [444]32 | [303]1129 | [303]1136 | [246]2068 | [236]4138 | [234]8464 | $^{69}0.84N^{1.0}$ |
| 546 | Yuan High-Tech Development | TW | yuan | 000 | 2024-01-23 | 356 | 206 | [303]2048 | - | [137]408 | [174]12 | [195]522 | [194]521 | [120]700 | [114]1411 | [125]3308 | $^{57}0.38N^{1.0}$ |
| 547 | i2V Systems Pvt Ltd | IN | i2v | 001 | 2025-04-17 | 331 | 249 | [181]2048 | - | [508]1433 | [212]12 | [393]2207 | [399]2161 | [341]4133 | [346]9212 | [353]19528 | $^{352}0.46N^{1.1}$ |
| 548 | ioNetworks Inc | TW | ionetworks | 001 | 2025-05-09 | 302 | 99 | [210]2048 | - | [353]766 | [528]2884 | [11]6 | [13]14 | [8]22 | [11]39 | [14]73 | $^{24}0.35N^{0.7}$ |
| 549 | useB | KR | useb | 000 | 2023-05-08 | 979 | 482 | [246]2048 | - | [259]606 | [461]38 | [500]6280 | [500]6158 | - | - | - | |
| 550 | useB | KR | useb | 001 | 2023-09-12 | 977 | 482 | [293]2048 | - | [229]545 | [156]11 | [496]6159 | [492]5936 | [424]12125 | [431]28466 | [425]62969 | $^{408}0.58N^{1.1}$ |
| 551 | useB | KR | useb | 002 | 2024-01-19 | 977 | 482 | [275]2048 | - | [213]513 | [157]11 | [499]6231 | [495]6094 | [439]13842 | [435]29794 | [417]54153 | $^{385}0.88N^{1.1}$ |
| 552 | useB | KR | useb | 003 | 2024-08-22 | 2023 | 482 | [242]2048 | - | [481]1248 | [211]12 | [501]6295 | [503]6316 | [433]12799 | [424]25794 | - | $^{378}0.94N^{1.1}$ |
| 553 | useB | KR | useb | 004 | 2024-12-23 | 2037 | 443 | [335]2048 | - | [513]1464 | [202]12 | [497]6168 | [498]6139 | [428]12391 | [428]27431 | - | $^{392}0.81N^{1.1}$ |
| 554 | verihubs-inteligensia | ID | verihubs-inteligensia | 000 | 2022-09-29 | 204 | 75 | [182]2048 | - | [51]260 | [284]14 | [523]9715 | [532]9677 | [454]18417 | [451]41160 | [443]95915 | $^{356}2.05N^{1.1}$ |
| 555 | verihubs-inteligensia | ID | verihubs-inteligensia | 001 | 2023-06-27 | 352 | 16 | [203]2048 | - | [183]475 | [185]12 | [530]11526 | [537]11418 | - | - | - | |

| Notes | |
|---|---|
| 1 | Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas). |
| 2 | Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates. |
| 3 | This multiplier expresses the increase in template size when $k$ images are passed to the template generation function. |
| 4 | All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however. |
| 5 | Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 467. However in certain cases the model is not correct and should not be used numerically. |

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 1 | 3DIVI-005 | [98]0.0207 | [97]0.0304 | [97]0.0415 | [97]0.0533 | [97]0.0646 | [374]0.0735 | [374]0.0884 | [375]0.1148 | [101]0.1580 | [98]0.2316 | [98]0.3033 | [98]0.3740 | [98]0.4285 | [371]0.4742 | [372]0.5329 | [369]0.5975 |
| 2 | ADIANCE-002 | | | | | | [245]0.0075 | [245]0.0085 | [247]0.0089 | | | | | | [251]0.0959 | [250]0.1174 | [251]0.1525 |
| 3 | ADVANCE-000 | | | | | | [261]0.0087 | [265]0.0099 | [264]0.0111 | | | | | | [279]0.1275 | [277]0.1527 | [273]0.1920 |
| 4 | AFISBIOMETRICS-000 | | | | | | [296]0.0118 | [299]0.0146 | [303]0.0185 | | | | | | [325]0.2012 | [324]0.2397 | [325]0.3004 |
| 5 | AFRENGINE-000 | | | | | | [303]0.0129 | [301]0.0150 | [304]0.0187 | | | | | | [250]0.0958 | [246]0.1152 | [244]0.1469 |
| 6 | AFRENGINE-001 | | | | | | [166]0.0048 | [147]0.0050 | [159]0.0039 | | | | | | [96]0.0200 | [94]0.0233 | [96]0.0313 |
| 7 | AFRENGINE-002 | | | | | | [133]0.0045 | [122]0.0048 | [167]0.0041 | | | | | | [84]0.0168 | [81]0.0192 | [85]0.0273 |
| 8 | AFRENGINE-003 | | | | | | [185]0.0052 | [177]0.0056 | [195]0.0052 | | | | | | [108]0.0239 | [108]0.0286 | [108]0.0398 |
| 9 | AIGROUP-000 | | | | | | [213]0.0058 | [210]0.0065 | [215]0.0063 | | | | | | [229]0.0694 | [228]0.0850 | [226]0.1131 |
| 10 | AIZE-002 | | | | | | [321]0.0152 | [316]0.0174 | [307]0.0203 | | | | | | [338]0.2500 | [332]0.2854 | [330]0.3438 |
| 11 | AIZE-006 | | | | | | [343]0.0213 | [341]0.0248 | [336]0.0312 | | | | | | [339]0.2510 | [334]0.2920 | [335]0.3605 |
| 12 | AKUVOX-001 | | | | | | [248]0.0079 | [257]0.0093 | [256]0.0102 | | | | | | [259]0.1009 | [255]0.1240 | [255]0.1609 |
| 13 | ALCHERA-006 | | | | | | [158]0.0046 | [152]0.0051 | [162]0.0040 | | | | | | [171]0.0419 | [170]0.0535 | [173]0.0737 |
| 14 | ALCHERA-008 | | | | | | [125]0.0044 | [130]0.0049 | [135]0.0035 | | | | | | [121]0.0280 | [120]0.0348 | [121]0.0469 |
| 15 | ALGORITHMSINC-000 | | | | | | [312]0.0137 | [310]0.0164 | [310]0.0205 | | | | | | [283]0.1349 | [283]0.1654 | [284]0.2137 |
| 16 | ALLGOVISION-005 | | | | | | [289]0.0108 | [284]0.0126 | [289]0.0153 | | | | | | [300]0.1524 | [299]0.1848 | [297]0.2309 |
| 17 | ANKE-000 | [95]0.0162 | [95]0.0245 | [95]0.0333 | [95]0.0428 | [95]0.0515 | [372]0.0615 | [372]0.0780 | [371]0.1028 | [96]0.1132 | [96]0.1761 | [96]0.2402 | [96]0.3057 | [95]0.3640 | [367]0.4200 | [367]0.4928 | [367]0.5680 |
| 18 | ANKE-002 | [49]0.0055 | [50]0.0074 | [50]0.0090 | [49]0.0103 | [48]0.0116 | [310]0.0135 | [309]0.0162 | [306]0.0202 | [54]0.0329 | [54]0.0560 | [56]0.0843 | [57]0.1169 | [57]0.1481 | [318]0.1820 | [319]0.2280 | [318]0.2831 |
| 19 | ARATEK-001 | | | | | | [178]0.0050 | [173]0.0055 | [188]0.0049 | | | | | | [225]0.0686 | [226]0.0843 | [225]0.1082 |
| 20 | ARMATURA-000 | | | | | | [98]0.0041 | [109]0.0047 | [143]0.0037 | | | | | | [141]0.0344 | [141]0.0428 | [138]0.0555 |
| 21 | AWARE-005 | [106]0.0328 | [106]0.0519 | [106]0.0712 | [104]0.0910 | [104]0.1078 | [381]0.1235 | [381]0.1457 | [382]0.1831 | [106]0.3605 | [107]0.4949 | [107]0.5948 | [107]0.6783 | [108]0.7393 | [383]0.7905 | [383]0.8408 | [384]0.8831 |
| 22 | AWARE-006 | [110]0.0702 | [110]0.1110 | [110]0.1502 | [110]0.1899 | [110]0.2253 | [390]0.2614 | [389]0.3045 | [390]0.3659 | | | | | | | | |
| 23 | AWARE-007 | | | | | | [328]0.0162 | [322]0.0189 | [325]0.0257 | | | | | | [263]0.1053 | [257]0.1246 | [261]0.1676 |
| 24 | AYONIX-002 | [113]0.3360 | [114]0.4389 | [114]0.5144 | [114]0.5814 | [114]0.6340 | [394]0.6818 | [394]0.7297 | [395]0.7774 | [110]0.8288 | [111]0.9013 | [111]0.9375 | [111]0.9603 | [111]0.9744 | [388]0.9837 | [388]0.9893 | [388]0.9927 |
| 25 | BIOIDTECHSWISS-001 | | | | | | [156]0.0046 | [154]0.0051 | [129]0.0034 | | | | | | [146]0.0355 | [142]0.0429 | [137]0.0548 |
| 26 | BIOMETRIK-001 | | | | | | [356]0.0281 | [354]0.0319 | [354]0.0413 | | | | | | [349]0.2812 | [345]0.3249 | [343]0.3882 |
| 27 | CAMVI-004 | [109]0.0623 | [109]0.0944 | [109]0.1243 | [109]0.1548 | [108]0.1812 | [387]0.2056 | [387]0.2344 | [385]0.2672 | [91]0.0810 | [91]0.1267 | [88]0.1721 | [88]0.2203 | [88]0.2619 | [357]0.3040 | [356]0.3543 | [353]0.4124 |
| 28 | CAMVI-005 | [111]0.0849 | [111]0.1255 | [111]0.1631 | [111]0.1989 | [111]0.2298 | [389]0.2585 | [388]0.2915 | [389]0.3246 | | | | | | | | |
| 29 | CANON-001 | | | | | | [186]0.0052 | [181]0.0057 | [169]0.0042 | | | | | | [189]0.0491 | [187]0.0606 | [193]0.0826 |
| 30 | CANON-002 | | | | | | [220]0.0062 | [223]0.0070 | [226]0.0070 | | | | | | [181]0.0472 | [181]0.0582 | [188]0.0792 |
| 31 | CANON-003 | | | | | | [168]0.0049 | [167]0.0053 | [171]0.0043 | | | | | | [98]0.0209 | [99]0.0257 | [101]0.0358 |
| 32 | CIB-000 | [14]0.0022 | [14]0.0030 | [15]0.0037 | [15]0.0044 | [17]0.0049 | [210]0.0057 | [219]0.0069 | [214]0.0062 | [25]0.0139 | [26]0.0240 | [27]0.0373 | [28]0.0525 | [28]0.0689 | [239]0.0859 | [243]0.1109 | [243]0.1454 |
| 33 | CLEARVIEWAI-000 | [4]0.0017 | [4]0.0023 | [4]0.0028 | [9]0.0034 | [11]0.0039 | [152]0.0046 | [178]0.0056 | [185]0.0047 | [16]0.0066 | [18]0.0121 | [18]0.0194 | [19]0.0287 | [19]0.0385 | [191]0.0493 | [205]0.0662 | [204]0.0873 |
| 34 | CLEARVIEWAI-001 | | | | | | [131]0.0044 | [117]0.0047 | [117]0.0032 | | | | | | [158]0.0389 | [159]0.0497 | [161]0.0676 |
| 35 | CLEARVIEWAI-002 | | | | | | [127]0.0044 | [120]0.0048 | [137]0.0036 | | | | | | [175]0.0438 | [176]0.0566 | [184]0.0778 |
| 36 | CLEARVIEWAI-003 | | | | | | [94]0.0041 | [70]0.0042 | [83]0.0024 | | | | | | [105]0.0224 | [107]0.0285 | [105]0.0370 |
| 37 | CLOUDWALK-HR-000 | [8]0.0019 | [7]0.0024 | [8]0.0029 | [6]0.0032 | [5]0.0032 | [44]0.0036 | [67]0.0041 | [58]0.0020 | [1]0.0029 | [1]0.0041 | [1]0.0054 | [1]0.0064 | [2]0.0073 | [34]0.0085 | [34]0.0102 | [34]0.0112 |
| 38 | CLOUDWALK-MT-000 | | | | | | [60]0.0037 | [32]0.0038 | [26]0.0013 | | | | | | [26]0.0065 | [24]0.0072 | [24]0.0075 |
| 39 | CLOUDWALK-MT-001 | | | | | | [52]0.0037 | [22]0.0037 | [16]0.0012 | | | | | | [16]0.0045 | [11]0.0051 | [10]0.0042 |
| 40 | CLOUDWALK-MT-002 | | | | | | [50]0.0036 | [28]0.0038 | [17]0.0012 | | | | | | [9]0.0042 | [9]0.0048 | [9]0.0034 |
| 41 | COGENT-000 | [91]0.0128 | [91]0.0184 | [93]0.0250 | [93]0.0327 | [92]0.0407 | [366]0.0488 | [366]0.0611 | [365]0.0794 | [78]0.0559 | [79]0.0923 | [76]0.1342 | [76]0.1812 | [75]0.2243 | [343]0.2675 | [343]0.3240 | [349]0.3992 |
| 42 | COGENT-001 | [90]0.0128 | [90]0.0184 | [92]0.0250 | [92]0.0327 | [93]0.0407 | [367]0.0488 | [365]0.0611 | [364]0.0794 | [77]0.0559 | [78]0.0923 | [77]0.1342 | [77]0.1812 | [76]0.2243 | [344]0.2675 | [342]0.3240 | [350]0.3992 |
| 43 | COGENT-002 | [69]0.0081 | [66]0.0105 | [63]0.0123 | [64]0.0137 | [62]0.0157 | [330]0.0175 | [328]0.0215 | [328]0.0280 | [69]0.0499 | [68]0.0827 | [67]0.1207 | [67]0.1639 | [67]0.2037 | [332]0.2432 | [336]0.2972 | [336]0.3638 |
| 44 | COGENT-003 | [71]0.0082 | [67]0.0108 | [65]0.0128 | [67]0.0145 | [66]0.0168 | [338]0.0191 | [339]0.0239 | [337]0.0312 | [80]0.0582 | [80]0.0971 | [80]0.1417 | [80]0.1918 | [80]0.2380 | [352]0.2836 | [354]0.3440 | [356]0.4207 |

*Table 13:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = 1, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 45 | COGENT-004 | [59] 0.0066 | [53] 0.0080 | [45] 0.0085 | [39] 0.0080 | [31] 0.0083 | [273] 0.0092 | [273] 0.0106 | [278] 0.0130 | [63] 0.0410 | [65] 0.0720 | [65] 0.1099 | [65] 0.1539 | [64] 0.1974 | [333] 0.2443 | [339] 0.3043 | [339] 0.3757 |
| 46 | COGENT-006 | | | | | | [140] 0.0045 | [142] 0.0049 | [157] 0.0038 | | | | | | [155] 0.0370 | [149] 0.0448 | [149] 0.0602 |
| 47 | COGENT-007 | | | | | | [129] 0.0044 | [133] 0.0049 | [138] 0.0036 | | | | | | [138] 0.0332 | [138] 0.0415 | [142] 0.0566 |
| 48 | COGNITEC-000 | [105] 0.0265 | [103] 0.0423 | [103] 0.0588 | [103] 0.0757 | [102] 0.0894 | [379] 0.1014 | [379] 0.1169 | [378] 0.1381 | [100] 0.1522 | [99] 0.2330 | [99] 0.3051 | [99] 0.3751 | [99] 0.4300 | [372] 0.4779 | [371] 0.5307 | [368] 0.5913 |
| 49 | COGNITEC-001 | [93] 0.0149 | [94] 0.0228 | [94] 0.0312 | [94] 0.0399 | [94] 0.0479 | [369] 0.0546 | [368] 0.0656 | [366] 0.0806 | [93] 0.0963 | [93] 0.1562 | [93] 0.2157 | [93] 0.2771 | [93] 0.3287 | [365] 0.3771 | [364] 0.4343 | [363] 0.4959 |
| 50 | COGNITEC-002 | [77] 0.0101 | [80] 0.0138 | [81] 0.0170 | [81] 0.0201 | [81] 0.0237 | [354] 0.0264 | [351] 0.0309 | [350] 0.0389 | [72] 0.0517 | [71] 0.0879 | [72] 0.1269 | [73] 0.1707 | [71] 0.2098 | [335] 0.2463 | [333] 0.2919 | [333] 0.3535 |
| 51 | COGNITEC-003 | [78] 0.0104 | [81] 0.0140 | [82] 0.0174 | [82] 0.0205 | [82] 0.0238 | [355] 0.0266 | [352] 0.0311 | [352] 0.0401 | [71] 0.0504 | [70] 0.0855 | [69] 0.1235 | [69] 0.1662 | [68] 0.2045 | [331] 0.2403 | [331] 0.2854 | [331] 0.3451 |
| 52 | COGNITEC-004 | [64] 0.0073 | [63] 0.0099 | [62] 0.0118 | [59] 0.0130 | [59] 0.0147 | [329] 0.0163 | [323] 0.0189 | [322] 0.0239 | [53] 0.0325 | [53] 0.0548 | [52] 0.0798 | [51] 0.1074 | [50] 0.1325 | [306] 0.1591 | [303] 0.1952 | [302] 0.2414 |
| 53 | COGNITEC-006 | | | | | | [253] 0.0081 | [246] 0.0086 | [248] 0.0090 | | | | | | [233] 0.0777 | [232] 0.0926 | [233] 0.1274 |
| 54 | COGNITEC-007 | | | | | | [229] 0.0066 | [227] 0.0072 | [219] 0.0066 | | | | | | [198] 0.0507 | [193] 0.0616 | [194] 0.0826 |
| 55 | COGNITEC-008 | | | | | | [238] 0.0067 | [229] 0.0074 | [229] 0.0071 | | | | | | [199] 0.0508 | [189] 0.0607 | [192] 0.0836 |
| 56 | COGNITEC-009 | | | | | | [239] 0.0069 | [231] 0.0075 | [231] 0.0073 | | | | | | [200] 0.0510 | [190] 0.0609 | [199] 0.0838 |
| 57 | COMPAL-000 | | | | | | [391] 0.3198 | [390] 0.3198 | [388] 0.3206 | | | | | | [362] 0.3559 | [358] 0.3655 | [342] 0.3844 |
| 58 | CORSIGHT-000 | | | | | | [182] 0.0051 | [182] 0.0057 | [172] 0.0043 | | | | | | [202] 0.0523 | [200] 0.0641 | [195] 0.0829 |
| 59 | CORSIGHT-001 | | | | | | [170] 0.0049 | [166] 0.0053 | [176] 0.0044 | | | | | | [209] 0.0556 | [209] 0.0688 | [207] 0.0893 |
| 60 | CUBOX-000 | [7] 0.0019 | [5] 0.0024 | [5] 0.0028 | [4] 0.0031 | [4] 0.0032 | [54] 0.0037 | [94] 0.0044 | [92] 0.0027 | [6] 0.0039 | [6] 0.0059 | [7] 0.0083 | [8] 0.0111 | [8] 0.0141 | [93] 0.0185 | [97] 0.0252 | [100] 0.0339 |
| 61 | CUBOX-001 | | | | | | [87] 0.0040 | [89] 0.0044 | [96] 0.0028 | | | | | | [104] 0.0223 | [104] 0.0276 | [103] 0.0361 |
| 62 | CUBOX-002 | | | | | | [82] 0.0039 | [91] 0.0044 | [123] 0.0033 | | | | | | [107] 0.0237 | [109] 0.0293 | [107] 0.0396 |
| 63 | CUBOX-003 | | | | | | [109] 0.0043 | [119] 0.0048 | [166] 0.0041 | | | | | | [114] 0.0254 | [111] 0.0317 | [112] 0.0422 |
| 64 | CYBERLINK-002 | [50] 0.0055 | [45] 0.0068 | [41] 0.0075 | [35] 0.0078 | [32] 0.0084 | [275] 0.0094 | [275] 0.0107 | [267] 0.0114 | [32] 0.0180 | [33] 0.0302 | [33] 0.0460 | [32] 0.0643 | [33] 0.0837 | [264] 0.1058 | [266] 0.1370 | [267] 0.1787 |
| 65 | CYBERLINK-003 | [35] 0.0041 | [34] 0.0052 | [27] 0.0057 | [25] 0.0058 | [25] 0.0061 | [236] 0.0068 | [237] 0.0078 | [240] 0.0078 | [19] 0.0109 | [19] 0.0175 | [20] 0.0259 | [21] 0.0356 | [21] 0.0468 | [212] 0.0594 | [219] 0.0787 | [223] 0.1072 |
| 66 | CYBERLINK-006 | | | | | | [90] 0.0040 | [85] 0.0043 | [77] 0.0023 | | | | | | [94] 0.0188 | [91] 0.0219 | [90] 0.0291 |
| 67 | DAHUA-002 | [30] 0.0035 | [28] 0.0047 | [28] 0.0058 | [27] 0.0067 | [28] 0.0074 | [255] 0.0082 | [266] 0.0100 | [261] 0.0108 | [30] 0.0169 | [32] 0.0294 | [31] 0.0449 | [30] 0.0635 | [30] 0.0817 | [260] 0.1013 | [261] 0.1291 | [258] 0.1638 |
| 68 | DAHUA-003 | [19] 0.0026 | [19] 0.0036 | [19] 0.0043 | [20] 0.0050 | [20] 0.0055 | [219] 0.0062 | [240] 0.0080 | [232] 0.0073 | [29] 0.0160 | [30] 0.0280 | [29] 0.0432 | [29] 0.0615 | [29] 0.0794 | [256] 0.0987 | [259] 0.1270 | [254] 0.1587 |
| 69 | DAON-002 | | | | | | [224] 0.0064 | [235] 0.0077 | [236] 0.0076 | | | | | | [228] 0.0693 | [229] 0.0889 | [229] 0.1147 |
| 70 | DAON-003 | | | | | | [137] 0.0045 | [136] 0.0049 | [124] 0.0033 | | | | | | [129] 0.0308 | [128] 0.0379 | [122] 0.0487 |
| 71 | DATECH-001 | | | | | | [188] 0.0052 | [193] 0.0059 | [192] 0.0050 | | | | | | [221] 0.0641 | [220] 0.0794 | [217] 0.1011 |
| 72 | DEEPGLINT-001 | [17] 0.0024 | [16] 0.0032 | [14] 0.0037 | [13] 0.0040 | [13] 0.0043 | [172] 0.0049 | [197] 0.0060 | [197] 0.0052 | [12] 0.0058 | [10] 0.0087 | [11] 0.0119 | [11] 0.0155 | [11] 0.0199 | [111] 0.0249 | [115] 0.0338 | [119] 0.0463 |
| 73 | DEEPSEA-001 | [70] 0.0081 | [70] 0.0116 | [73] 0.0149 | [76] 0.0182 | [76] 0.0216 | [352] 0.0260 | [356] 0.0332 | [356] 0.0432 | [66] 0.0458 | [66] 0.0752 | [64] 0.1086 | [63] 0.1460 | [63] 0.1812 | [327] 0.2186 | [327] 0.2663 | [326] 0.3213 |
| 74 | DERMALOG-006 | [82] 0.0113 | [82] 0.0142 | [78] 0.0163 | [77] 0.0183 | [74] 0.0200 | [345] 0.0218 | [342] 0.0251 | [340] 0.0329 | [75] 0.0545 | [73] 0.0889 | [73] 0.1271 | [72] 0.1697 | [70] 0.2090 | [337] 0.2498 | [338] 0.3028 | [337] 0.3670 |
| 75 | DERMALOG-007 | [88] 0.0125 | [88] 0.0170 | [88] 0.0214 | [88] 0.0264 | [87] 0.0309 | [361] 0.0356 | [362] 0.0432 | [362] 0.0579 | [92] 0.0910 | [92] 0.1453 | [92] 0.2009 | [92] 0.2602 | [92] 0.3134 | [364] 0.3649 | [363] 0.4289 | [364] 0.5007 |
| 76 | DERMALOG-008 | [52] 0.0057 | [52] 0.0077 | [54] 0.0095 | [54] 0.0110 | [53] 0.0128 | [319] 0.0148 | [319] 0.0180 | [320] 0.0223 | [70] 0.0501 | [69] 0.0850 | [70] 0.1247 | [71] 0.1692 | [72] 0.2105 | [340] 0.2541 | [340] 0.3102 | [340] 0.3762 |
| 77 | DERMALOG-010 | | | | | | [206] 0.0056 | [192] 0.0059 | [170] 0.0043 | | | | | | [201] 0.0519 | [201] 0.0643 | [201] 0.0843 |
| 78 | DERMALOG-011 | | | | | | [266] 0.0089 | [276] 0.0107 | [276] 0.0128 | | | | | | [301] 0.1543 | [296] 0.1833 | [298] 0.2314 |
| 79 | DERMALOG-012 | | | | | | [270] 0.0090 | [274] 0.0107 | [281] 0.0133 | | | | | | [297] 0.1497 | [292] 0.1783 | [290] 0.2238 |
| 80 | DERMALOG-013 | | | | | | [268] 0.0089 | [272] 0.0106 | [280] 0.0132 | | | | | | [290] 0.1516 | [293] 0.1803 | [292] 0.2264 |
| 81 | DERMALOG-014 | | | | | | [347] 0.0232 | [347] 0.0279 | [346] 0.0358 | | | | | | [302] 0.1544 | [295] 0.1833 | [295] 0.2295 |
| 82 | DERMALOG-015 | | | | | | [159] 0.0047 | [131] 0.0049 | [146] 0.0037 | | | | | | [160] 0.0392 | [155] 0.0481 | [153] 0.0634 |
| 83 | DERMALOG-016 | | | | | | [76] 0.0039 | [80] 0.0043 | [101] 0.0029 | | | | | | [102] 0.0222 | [105] 0.0282 | [109] 0.0403 |
| 84 | DERMALOG-017 | | | | | | [75] 0.0039 | [82] 0.0043 | [100] 0.0029 | | | | | | [112] 0.0250 | [110] 0.0307 | [111] 0.0408 |
| 85 | DILUSENSE-000 | | | | | | [298] 0.0123 | [296] 0.0146 | [301] 0.0180 | | | | | | [317] 0.1814 | [313] 0.2149 | [310] 0.2644 |
| 86 | DILUSENSE-001 | | | | | | [227] 0.0065 | [217] 0.0069 | [224] 0.0069 | | | | | | [219] 0.0632 | [216] 0.0778 | [216] 0.1006 |
| 87 | EINETWORKSINDIA-000 | | | | | | [272] 0.0091 | [271] 0.0105 | [268] 0.0115 | | | | | | [271] 0.1139 | [268] 0.1413 | [266] 0.1785 |
| 88 | EINETWORKSINDIA-001 | | | | | | [271] 0.0090 | [267] 0.0103 | [266] 0.0113 | | | | | | [275] 0.1196 | [273] 0.1476 | [270] 0.1889 |

*Table 14:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 89 | EINETWORKSINDIA-002 | | | | | | [141]0.0045 | [148]0.0050 | [125]0.0033 | | | | | | [151]0.0361 | [151]0.0463 | [152]0.0634 |
| 90 | FACEPHI-000 | | | | | | [199]0.0054 | [189]0.0059 | [193]0.0050 | | | | | | [180]0.0468 | [177]0.0575 | [177]0.0748 |
| 91 | FIRSTCREDITKZ-001 | | | | | | [208]0.0057 | [211]0.0066 | [203]0.0055 | | | | | | [110]0.0240 | [106]0.0284 | [104]0.0368 |
| 92 | FUJITSULAB-001 | | | | | | [269]0.0089 | [262]0.0098 | [263]0.0111 | | | | | | [289]0.1403 | [288]0.1723 | [285]0.2165 |
| 93 | GORILLA-002 | [100]0.0213 | [100]0.0359 | [101]0.0528 | [102]0.0716 | [103]0.0895 | [380]0.1088 | [380]0.1367 | [380]0.1765 | [103]0.1828 | [104]0.2787 | [104]0.3654 | [104]0.4485 | [104]0.5168 | [376]0.5823 | [376]0.6508 | [376]0.7180 |
| 94 | GORILLA-005 | [38]0.0044 | [47]0.0070 | [58]0.0102 | [62]0.0136 | [67]0.0170 | [341]0.0204 | [346]0.0272 | [349]0.0373 | [79]0.0566 | [81]0.0973 | [82]0.1432 | [81]0.1937 | [81]0.2398 | [354]0.2862 | [353]0.3437 | [354]0.4150 |
| 95 | GORILLA-007 | | | | | | [291]0.0108 | [287]0.0128 | [282]0.0145 | | | | | | [320]0.1862 | [318]0.2198 | [314]0.2716 |
| 96 | GORILLA-008 | | | | | | [243]0.0071 | [241]0.0081 | [246]0.0089 | | | | | | [304]0.1557 | [298]0.1847 | [300]0.2340 |
| 97 | GRIAULE-001 | | | | | | [150]0.0046 | [145]0.0050 | [151]0.0038 | | | | | | [163]0.0402 | [158]0.0487 | [154]0.0636 |
| 98 | HISIGN-001 | | | | | | [222]0.0063 | [222]0.0070 | [225]0.0070 | | | | | | [216]0.0613 | [214]0.0754 | [213]0.0969 |
| 99 | HYPERVERGE-003 | | | | | | [191]0.0053 | [187]0.0058 | [211]0.0059 | | | | | | [227]0.0689 | [223]0.0826 | [224]0.1081 |
| 100 | HYUNDAI-ROBOTICSLAB-000 | | | | | | [267]0.0089 | [264]0.0099 | [270]0.0119 | | | | | | [242]0.0872 | [239]0.1082 | [239]0.1405 |
| 101 | HYUNDAI-ROBOTICSLAB-001 | | | | | | [258]0.0084 | [255]0.0092 | [258]0.0104 | | | | | | [245]0.0928 | [245]0.1151 | [245]0.1482 |
| 102 | HYUNDAI-ROBOTICSLAB-002 | | | | | | [145]0.0045 | [127]0.0048 | [132]0.0034 | | | | | | [184]0.0481 | [186]0.0598 | [185]0.0784 |
| 103 | HYUNDAI-ROBOTICSLAB-003 | | | | | | [142]0.0045 | [124]0.0048 | [131]0.0034 | | | | | | [182]0.0473 | [183]0.0589 | [181]0.0771 |
| 104 | HZAILU-001 | | | | | | [211]0.0058 | [190]0.0059 | [190]0.0049 | | | | | | [203]0.0524 | [198]0.0630 | [187]0.0791 |
| 105 | HZAILU-002 | | | | | | [212]0.0058 | [209]0.0064 | [196]0.0052 | | | | | | [188]0.0488 | [184]0.0597 | [172]0.0728 |
| 106 | HZAILU-003 | | | | | | [223]0.0063 | [224]0.0071 | [222]0.0067 | | | | | | [222]0.0668 | [227]0.0850 | [227]0.1136 |
| 107 | HZAILU-004 | | | | | | [179]0.0050 | [174]0.0055 | [187]0.0049 | | | | | | [196]0.0503 | [202]0.0655 | [209]0.0902 |
| 108 | I2V-001 | | | | | | [237]0.0068 | [225]0.0071 | [221]0.0069 | | | | | | [370]0.4618 | [369]0.5191 | [370]0.6022 |
| 109 | ICEHOUSE-001 | | | | | | [203]0.0056 | [198]0.0060 | [212]0.0059 | | | | | | [237]0.0826 | [236]0.1022 | [236]0.1331 |
| 110 | ICM-001 | | | | | | [297]0.0122 | [293]0.0138 | [293]0.0164 | | | | | | [324]0.1997 | [323]0.2375 | [321]0.2940 |
| 111 | IDEMIA-003 | [81]0.0110 | [86]0.0151 | [86]0.0196 | [85]0.0238 | [84]0.0281 | [359]0.0313 | [359]0.0368 | [358]0.0504 | [87]0.0717 | [86]0.1147 | [86]0.1614 | [86]0.2113 | [85]0.2553 | [356]0.2976 | [355]0.3537 | [357]0.4334 |
| 112 | IDEMIA-004 | [80]0.0107 | [84]0.0148 | [85]0.0192 | [84]0.0233 | [83]0.0277 | [358]0.0312 | [358]0.0367 | [359]0.0512 | [58]0.0373 | [55]0.0587 | [54]0.0833 | [53]0.1100 | [52]0.1340 | [305]0.1580 | [302]0.1911 | [304]0.2482 |
| 113 | IDEMIA-005 | [84]0.0118 | [87]0.0167 | [90]0.0218 | [89]0.0270 | [88]0.0317 | [362]0.0357 | [361]0.0425 | [361]0.0579 | [65]0.0440 | [64]0.0689 | [60]0.0964 | [59]0.1254 | [58]0.1513 | [315]0.1762 | [310]0.2113 | [312]0.2698 |
| 114 | IDEMIA-006 | [87]0.0124 | [89]0.0171 | [89]0.0218 | [87]0.0263 | [86]0.0302 | [360]0.0321 | [357]0.0356 | [357]0.0471 | [62]0.0409 | [59]0.0620 | [57]0.0850 | [52]0.1097 | [49]0.1309 | [294]0.1486 | [290]0.1738 | [289]0.2200 |
| 115 | IDEMIA-007 | [47]0.0050 | [48]0.0071 | [48]0.0089 | [50]0.0106 | [51]0.0124 | [314]0.0142 | [315]0.0171 | [318]0.0220 | [36]0.0202 | [36]0.0335 | [34]0.0491 | [33]0.0663 | [31]0.0825 | [258]0.0999 | [256]0.1240 | [260]0.1645 |
| 116 | IDEMIA-008 | [5]0.0018 | [6]0.0024 | [6]0.0029 | [5]0.0032 | [7]0.0035 | [77]0.0039 | [106]0.0046 | [121]0.0033 | [3]0.0034 | [3]0.0051 | [5]0.0069 | [5]0.0087 | [5]0.0102 | [67]0.0123 | [63]0.0146 | [63]0.0186 |
| 117 | IDEMIA-009 | | | | | | [88]0.0040 | [73]0.0042 | [81]0.0024 | | | | | | [37]0.0094 | [35]0.0103 | [36]0.0123 |
| 118 | IDEMIA-010 | | | | | | [18]0.0034 | [11]0.0036 | [14]0.0011 | | | | | | [15]0.0045 | [14]0.0052 | [13]0.0043 |
| 119 | IDEMIA-011 | | | | | | [11]0.0033 | [7]0.0035 | [13]0.0011 | | | | | | [10]0.0043 | [12]0.0051 | [15]0.0044 |
| 120 | IDEMIA-012 | | | | | | [16]0.0033 | [6]0.0034 | [11]0.0011 | | | | | | [12]0.0043 | [10]0.0049 | [16]0.0048 |
| 121 | IDEMIA-013 | | | | | | [4]0.0031 | [1]0.0033 | [1]0.0007 | | | | | | [7]0.0031 | [7]0.0034 | [7]0.0013 |
| 122 | IDENTY-001 | | | | | | | [399]1.0000 | | | | | | | | | |
| 123 | IDENTY-002 | | | | | | [148]0.0046 | [146]0.0050 | [153]0.0038 | | | | | | [115]0.0261 | [112]0.0321 | [113]0.0431 |
| 124 | IMAGUS-005 | [33]0.0039 | [33]0.0052 | [31]0.0061 | [29]0.0067 | [30]0.0077 | [264]0.0088 | [268]0.0103 | [262]0.0109 | [39]0.0212 | [39]0.0357 | [40]0.0539 | [40]0.0755 | [38]0.0967 | [274]0.1183 | [274]0.1485 | [271]0.1893 |
| 125 | IMAGUS-008 | | | | | | [384]0.1625 | [382]0.1704 | [381]0.1823 | | | | | | | | |
| 126 | IMDS-SOFTWARE-001 | | | | | | [189]0.0052 | [196]0.0060 | [194]0.0050 | | | | | | [210]0.0569 | [210]0.0693 | [210]0.0903 |
| 127 | IMDS-SOFTWARE-002 | | | | | | [197]0.0054 | [191]0.0059 | [189]0.0049 | | | | | | [207]0.0541 | [204]0.0661 | [200]0.0838 |
| 128 | IMPERIAL-000 | [34]0.0040 | [35]0.0054 | [36]0.0067 | [38]0.0079 | [40]0.0093 | [292]0.0112 | [294]0.0139 | [298]0.0178 | [49]0.0286 | [51]0.0503 | [51]0.0779 | [54]0.1116 | [56]0.1455 | [319]0.1844 | [322]0.2341 | [322]0.2951 |
| 129 | INCODE-003 | [94]0.0155 | [96]0.0247 | [96]0.0348 | [96]0.0463 | [96]0.0571 | [373]0.0674 | [373]0.0856 | [374]0.1114 | [102]0.1627 | [102]0.2507 | [102]0.3322 | [100]0.4122 | [100]0.4772 | [375]0.5368 | [375]0.6059 | [375]0.6766 |
| 130 | INCODE-004 | [56]0.0061 | [59]0.0087 | [59]0.0110 | [61]0.0136 | [64]0.0161 | [333]0.0185 | [337]0.0236 | [335]0.0309 | [73]0.0532 | [74]0.0908 | [75]0.1334 | [75]0.1809 | [77]0.2245 | [345]0.2675 | [346]0.3249 | [346]0.3932 |
| 131 | INCODE-006 | | | | | | [78]0.0039 | [99]0.0045 | [97]0.0028 | | | | | | [81]0.0160 | [88]0.0212 | [94]0.0298 |
| 132 | INCODE-007 | | | | | | [58]0.0037 | [63]0.0041 | [69]0.0021 | | | | | | [57]0.0114 | [71]0.0155 | [77]0.0229 |

*Table 15:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 133 | INNOVATRICS-004 | [114]0.3594 | [113]0.3629 | [113]0.3688 | [112]0.3754 | [112]0.3813 | [392]0.3870 | [392]0.3960 | [392]0.4135 | [107]0.4234 | [106]0.4642 | [106]0.5073 | [106]0.5522 | [105]0.5902 | [379]0.6274 | [377]0.6736 | [377]0.7253 |
| 134 | INNOVATRICS-005 | [41]0.0046 | [41]0.0063 | [42]0.0078 | [45]0.0092 | [45]0.0106 | [299]0.0124 | [300]0.0149 | [299]0.0178 | [55]0.0343 | [56]0.0590 | [58]0.0886 | [58]0.1222 | [59]0.1544 | [322]0.1881 | [321]0.2321 | [319]0.2874 |
| 135 | INNOVATRICS-008 | | | | | | [162]0.0047 | [134]0.0049 | [111]0.0031 | | | | | | [130]0.0309 | [121]0.0353 | [118]0.0463 |
| 136 | INNOVATRICS-009 | | | | | | [112]0.0043 | [96]0.0044 | [91]0.0027 | | | | | | [78]0.0154 | [76]0.0170 | [72]0.0214 |
| 137 | INNOVATRICS-010 | | | | | | [113]0.0043 | [101]0.0045 | [104]0.0030 | | | | | | [83]0.0167 | [80]0.0192 | [83]0.0265 |
| 138 | INNOVATRICS-011 | | | | | | [101]0.0042 | [95]0.0044 | [78]0.0023 | | | | | | [54]0.0109 | [41]0.0122 | [42]0.0153 |
| 139 | INNOVATRICS-012 | | | | | | [86]0.0040 | [75]0.0042 | [79]0.0023 | | | | | | [41]0.0099 | [37]0.0110 | [37]0.0138 |
| 140 | INNOVATRICS-013 | | | | | | [51]0.0036 | [48]0.0040 | [39]0.0017 | | | | | | [44]0.0100 | [39]0.0116 | [39]0.0142 |
| 141 | INNOVATRICS-014 | | | | | | [23]0.0034 | [19]0.0037 | [23]0.0013 | | | | | | [18]0.0055 | [18]0.0059 | [18]0.0061 |
| 142 | INNOVATRICS-015 | | | | | | [19]0.0034 | [13]0.0036 | [25]0.0013 | | | | | | [17]0.0051 | [17]0.0056 | [17]0.0058 |
| 143 | INSPUR-000 | | | | | | [167]0.0048 | [165]0.0053 | [160]0.0039 | | | | | | [206]0.0539 | [206]0.0665 | [203]0.0865 |
| 144 | INSPUR-001 | | | | | | [151]0.0046 | [149]0.0050 | [139]0.0036 | | | | | | [183]0.0478 | [182]0.0588 | [180]0.0763 |
| 145 | INSPUR-002 | | | | | | [119]0.0043 | [121]0.0048 | [113]0.0031 | | | | | | [190]0.0493 | [195]0.0619 | [191]0.0820 |
| 146 | INTELLIVISION-002 | | | | | | [370]0.0577 | [369]0.0694 | [368]0.0881 | | | | | | | | |
| 147 | INTELRESEARCH-001 | | | | | | [95]0.0041 | [87]0.0043 | [84]0.0024 | | | | | | [73]0.0132 | [65]0.0149 | [60]0.0181 |
| 148 | INTEMA-000 | | | | | | [89]0.0040 | [79]0.0043 | [86]0.0024 | | | | | | [95]0.0193 | [95]0.0235 | [93]0.0294 |
| 149 | INTEMA-001 | | | | | | [84]0.0039 | [55]0.0040 | [57]0.0020 | | | | | | [80]0.0157 | [82]0.0194 | [81]0.0246 |
| 150 | INTOZI-001 | | | | | | [247]0.0076 | [244]0.0083 | [242]0.0081 | | | | | | [266]0.1061 | [260]0.1285 | [259]0.1640 |
| 151 | IONETWORKS-001 | | | | | | [353]0.0261 | [353]0.0313 | [353]0.0404 | | | | | | [269]0.1113 | [264]0.1349 | [265]0.1723 |
| 152 | IREX-000 | [24]0.0031 | [24]0.0042 | [25]0.0051 | [26]0.0060 | [26]0.0068 | [250]0.0080 | [260]0.0095 | [260]0.0107 | [52]0.0313 | [52]0.0539 | [53]0.0815 | [56]0.1137 | [55]0.1442 | [314]0.1755 | [317]0.2181 | [315]0.2718 |
| 153 | IRISID-001 | | | | | | [246]0.0076 | [247]0.0087 | [245]0.0085 | | | | | | [257]0.0995 | [253]0.1189 | [253]0.1531 |
| 154 | IRISID-002 | | | | | | [241]0.0070 | [232]0.0076 | [220]0.0067 | | | | | | [232]0.0770 | [233]0.0950 | [232]0.1213 |
| 155 | ISTL-001 | | | | | | [284]0.0082 | [288]0.0093 | [275]0.0127 | | | | | | [246]0.0933 | [244]0.1138 | [246]0.1498 |
| 156 | ISYSTEMS-002 | [76]0.0101 | [79]0.0135 | [80]0.0169 | [79]0.0197 | [80]0.0228 | [350]0.0256 | [350]0.0304 | [351]0.0398 | [90]0.0779 | [90]0.1258 | [91]0.1759 | [90]0.2299 | [90]0.2758 | [360]0.3204 | [361]0.3763 | [359]0.4401 |
| 157 | ISYSTEMS-003 | [75]0.0089 | [69]0.0115 | [69]0.0139 | [69]0.0158 | [70]0.0177 | [340]0.0198 | [336]0.0234 | [333]0.0303 | [84]0.0647 | [84]0.1056 | [84]0.1502 | [84]0.1986 | [83]0.2402 | [350]0.2819 | [349]0.3351 | [348]0.3976 |
| 158 | KAKAO-001 | | | | | | [81]0.0039 | [81]0.0043 | [73]0.0022 | | | | | | [90]0.0182 | [92]0.0220 | [91]0.0291 |
| 159 | KAKAO-002 | | | | | | [56]0.0037 | [43]0.0039 | [43]0.0018 | | | | | | [62]0.0119 | [56]0.0141 | [54]0.0174 |
| 160 | KEDACOM-001 | [83]0.0116 | [75]0.0130 | [67]0.0135 | [60]0.0133 | [57]0.0135 | [313]0.0141 | [302]0.0151 | [297]0.0176 | [41]0.0241 | [41]0.0360 | [39]0.0513 | [34]0.0689 | [34]0.0866 | [265]0.1060 | [262]0.1327 | [262]0.1694 |
| 161 | KLOUDSPOT-001 | | | | | | [293]0.0113 | [288]0.0130 | [284]0.0148 | | | | | | [313]0.1730 | [309]0.2071 | [306]0.2534 |
| 162 | KNOWUTECH-000 | | | | | | [217]0.0061 | [226]0.0072 | [227]0.0070 | | | | | | [235]0.0805 | [234]0.0995 | [235]0.1329 |
| 163 | KOGNIZA-000 | | | | | | [164]0.0048 | [169]0.0053 | [179]0.0045 | | | | | | [187]0.0486 | [192]0.0615 | [182]0.0776 |
| 164 | LIGHTVISION-000 | | | | | | [311]0.0136 | [308]0.0158 | [308]0.0204 | | | | | | [308]0.1646 | [304]0.1959 | [303]0.2422 |
| 165 | LINECLOVA-002 | | | | | | [146]0.0045 | [132]0.0049 | [103]0.0030 | | | | | | [128]0.0307 | [126]0.0374 | [125]0.0497 |
| 166 | LINECLOVA-003 | | | | | | [123]0.0044 | [123]0.0048 | [107]0.0030 | | | | | | [147]0.0356 | [147]0.0443 | [146]0.0586 |
| 167 | LOOKMAN-003 | [86]0.0123 | [83]0.0144 | [77]0.0158 | [70]0.0168 | [71]0.0178 | [335]0.0188 | [327]0.0212 | [326]0.0260 | [64]0.0438 | [62]0.0687 | [61]0.0978 | [61]0.1296 | [60]0.1581 | [321]0.1879 | [320]0.2294 | [317]0.2756 |
| 168 | LOOKMAN-005 | [85]0.0118 | [77]0.0134 | [70]0.0142 | [66]0.0144 | [61]0.0150 | [326]0.0160 | [317]0.0176 | [315]0.0213 | [51]0.0310 | [49]0.0480 | [46]0.0698 | [46]0.0954 | [46]0.1216 | [295]0.1491 | [301]0.1890 | [301]0.2381 |
| 169 | LSSYTECH-000 | | | | | | [73]0.0039 | [71]0.0042 | [62]0.0021 | | | | | | [59]0.0116 | [61]0.0143 | [59]0.0180 |
| 170 | LUXAND-001 | | | | | | [315]0.0144 | [313]0.0166 | [312]0.0207 | | | | | | [348]0.2802 | [344]0.3247 | [345]0.3916 |
| 171 | MAXVISION-000 | | | | | | [302]0.0128 | [299]0.0146 | [296]0.0169 | | | | | | [311]0.1706 | [306]0.2023 | [307]0.2550 |
| 172 | MAXVISION-001 | | | | | | [154]0.0046 | [138]0.0049 | [126]0.0033 | | | | | | [144]0.0351 | [144]0.0435 | [143]0.0581 |
| 173 | MAXVISION-002 | | | | | | [160]0.0047 | [137]0.0049 | [108]0.0030 | | | | | | [134]0.0318 | [130]0.0385 | [126]0.0500 |
| 174 | MAXVISION-003 | | | | | | [161]0.0047 | [163]0.0052 | [154]0.0038 | | | | | | [186]0.0485 | [188]0.0606 | [190]0.0812 |
| 175 | MAXVISION-004 | | | | | | [69]0.0038 | [74]0.0042 | [60]0.0020 | | | | | | [91]0.0182 | [90]0.0217 | [92]0.0293 |
| 176 | MEGVII-003 | | | | | | [181]0.0051 | [199]0.0060 | [198]0.0052 | | | | | | [118]0.0277 | [127]0.0376 | [132]0.0520 |

*Table 16:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 177 | MEGVII-004 | | | | | | 47 0.0036 | 57 0.0040 | 49 0.0018 | | | | | | 24 0.0062 | 28 0.0078 | 27 0.0084 |
| 178 | MEGVII-005 | | | | | | 48 0.0036 | 56 0.0040 | 46 0.0018 | | | | | | 28 0.0067 | 30 0.0084 | 29 0.0094 |
| 179 | METSAKUURCOMPANY-000 | | | | | | 202 0.0055 | 207 0.0062 | 208 0.0057 | | | | | | 218 0.0632 | 221 0.0795 | 220 0.1043 |
| 180 | METSAKUURCOMPANY-001 | | | | | | 221 0.0063 | 216 0.0069 | 221 0.0067 | | | | | | 236 0.0821 | 237 0.1024 | 238 0.1363 |
| 181 | MIAXIS-001 | | | | | | 226 0.0065 | 213 0.0067 | 213 0.0062 | | | | | | 211 0.0584 | 211 0.0717 | 211 0.0920 |
| 182 | MIAXIS-002 | | | | | | 225 0.0064 | 218 0.0069 | 216 0.0063 | | | | | | 214 0.0611 | 215 0.0758 | 214 0.0970 |
| 183 | MICROFOCUS-005 | 115 0.4269 | 113 0.5527 | 113 0.6355 | 116 0.7024 | 116 0.7503 | 396 0.7876 | 396 0.8234 | 397 0.8601 | 111 0.8338 | 112 0.9113 | 112 0.9468 | 112 0.9667 | 112 0.9771 | 387 0.9836 | 387 0.9880 | 387 0.9924 |
| 184 | MICROSOFT-003 | 28 0.0034 | 32 0.0050 | 33 0.0064 | 36 0.0078 | 38 0.0092 | 287 0.0107 | 292 0.0135 | 294 0.0166 | 50 0.0288 | 50 0.0503 | 50 0.0763 | 50 0.1067 | 54 0.1359 | 309 0.1680 | 312 0.2116 | 311 0.2644 |
| 185 | MICROSOFT-004 | 25 0.0032 | 27 0.0047 | 29 0.0060 | 32 0.0075 | 35 0.0087 | 282 0.0103 | 289 0.0131 | 291 0.0159 | 47 0.0268 | 48 0.0470 | 49 0.0716 | 48 0.1007 | 48 0.1291 | 307 0.1610 | 308 0.2052 | 308 0.2590 |
| 186 | MICROSOFT-005 | 22 0.0031 | 29 0.0047 | 35 0.0066 | 43 0.0084 | 43 0.0103 | 306 0.0131 | 311 0.0164 | 302 0.0185 | 43 0.0243 | 44 0.0432 | 44 0.0658 | 44 0.0913 | 45 0.1172 | 292 0.1476 | 300 0.1874 | 293 0.2272 |
| 187 | MICROSOFT-006 | 26 0.0032 | 31 0.0049 | 34 0.0065 | 42 0.0081 | 42 0.0096 | 295 0.0117 | 295 0.0144 | 292 0.0160 | 24 0.0134 | 24 0.0233 | 25 0.0346 | 23 0.0462 | 22 0.0578 | 230 0.0713 | 230 0.0903 | 230 0.1156 |
| 188 | MOBBL-000 | | | | | | 294 0.0116 | 290 0.0132 | 290 0.0156 | | | | | | 312 0.1709 | 305 0.2020 | 305 0.2507 |
| 189 | MUKH-002 | | | | | | 382 0.1394 | 383 0.1754 | 383 0.2335 | | | | | | 386 0.9761 | 386 0.9840 | 386 0.9899 |
| 190 | NAVERCLOUD-000 | | | | | | 110 0.0043 | 110 0.0047 | 99 0.0029 | | | | | | 131 0.0311 | 135 0.0396 | 136 0.0539 |
| 191 | NEC-000 | 97 0.0195 | 99 0.0316 | 99 0.0445 | 99 0.0581 | 98 0.0699 | 376 0.0817 | 376 0.0998 | 376 0.1237 | 89 0.0759 | 89 0.1245 | 89 0.1729 | 89 0.2240 | 89 0.2671 | 359 0.3117 | 357 0.3639 | 358 0.4348 |
| 192 | NEC-001 | 104 0.0246 | 102 0.0382 | 100 0.0524 | 100 0.0672 | 101 0.0793 | 378 0.0904 | 377 0.1076 | 377 0.1317 | 94 0.1019 | 94 0.1623 | 94 0.2214 | 94 0.2834 | 94 0.3341 | 366 0.3844 | 366 0.4440 | 365 0.5183 |
| 193 | NEC-002 | 27 0.0033 | 22 0.0041 | 18 0.0043 | 16 0.0044 | 15 0.0045 | 171 0.0049 | 176 0.0056 | 168 0.0041 | 15 0.0066 | 11 0.0090 | 10 0.0111 | 10 0.0131 | 9 0.0149 | 85 0.0171 | 85 0.0207 | 84 0.0267 |
| 194 | NEC-003 | 31 0.0036 | 26 0.0046 | 24 0.0051 | 24 0.0055 | 24 0.0059 | 230 0.0067 | 234 0.0077 | 235 0.0073 | 9 0.0056 | 9 0.0076 | 9 0.0091 | 7 0.0105 | 6 0.0119 | 75 0.0137 | 73 0.0162 | 71 0.0209 |
| 195 | NEC-004 | 32 0.0039 | 25 0.0045 | 22 0.0047 | 18 0.0046 | 14 0.0044 | 157 0.0046 | 164 0.0052 | 142 0.0036 | 7 0.0046 | 5 0.0057 | 2 0.0063 | 2 0.0066 | 1 0.0069 | 32 0.0076 | 32 0.0090 | 32 0.0105 |
| 196 | NEC-005 | | | | | | 64 0.0037 | 61 0.0041 | 62 0.0020 | | | | | | 33 0.0080 | 33 0.0091 | 33 0.0107 |
| 197 | NEC-006 | | | | | | 80 0.0039 | 76 0.0042 | 71 0.0021 | | | | | | 6 0.0030 | 6 0.0033 | 6 0.0012 |
| 198 | NEC-007 | | | | | | 22 0.0034 | 17 0.0036 | 31 0.0016 | | | | | | 4 0.0027 | 4 0.0031 | 4 0.0010 |
| 199 | NEC-008 | | | | | | 55 0.0037 | 36 0.0038 | 44 0.0018 | | | | | | 5 0.0030 | 5 0.0033 | 5 0.0012 |
| 200 | NEC-009 | | | | | | 5 0.0032 | 5 0.0034 | 6 0.0010 | | | | | | 3 0.0027 | 2 0.0030 | 2 0.0007 |
| 201 | NEC-010 | | | | | | 3 0.0031 | 2 0.0034 | 8 0.0010 | | | | | | 2 0.0027 | 1 0.0030 | 3 0.0007 |
| 202 | NEC-011 | | | | | | 1 0.0031 | 4 0.0034 | 5 0.0009 | | | | | | 1 0.0027 | 3 0.0030 | 1 0.0007 |
| 203 | NEC-012 | | | | | | 2 0.0031 | 3 0.0034 | 3 0.0009 | | | | | | 8 0.0031 | 8 0.0036 | 8 0.0018 |
| 204 | NEUROTECHNOLOGY-003 | 101 0.0234 | 101 0.0379 | 102 0.0549 | 101 0.0682 | 100 0.0720 | 375 0.0747 | 375 0.0886 | 373 0.1066 | 109 0.6802 | 109 0.8187 | 110 0.8920 | 110 0.9355 | 110 0.9594 | 385 0.9738 | 385 0.9828 | 385 0.9885 |
| 205 | NEUROTECHNOLOGY-004 | 79 0.0104 | 78 0.0134 | 76 0.0156 | 73 0.0173 | 72 0.0195 | 342 0.0212 | 340 0.0245 | 338 0.0320 | 83 0.0642 | 82 0.1015 | 81 0.1426 | 79 0.1881 | 78 0.2299 | 346 0.2722 | 347 0.3269 | 347 0.3943 |
| 206 | NEUROTECHNOLOGY-005 | 74 0.0089 | 71 0.0116 | 68 0.0136 | 68 0.0152 | 69 0.0173 | 339 0.0196 | 335 0.0233 | 334 0.0306 | 76 0.0556 | 76 0.0913 | 74 0.1315 | 74 0.1766 | 74 0.2192 | 342 0.2617 | 341 0.3174 | 341 0.3843 |
| 207 | NEUROTECHNOLOGY-007 | 66 0.0078 | 65 0.0103 | 64 0.0124 | 65 0.0140 | 63 0.0161 | 332 0.0185 | 332 0.0225 | 331 0.0290 | 82 0.0641 | 85 0.1069 | 85 0.1546 | 85 0.2075 | 86 0.2572 | 358 0.3081 | 360 0.3713 | 360 0.4421 |
| 208 | NEUROTECHNOLOGY-010 | | | | | | 192 0.0053 | 205 0.0061 | 200 0.0053 | | | | | | 240 0.0863 | 238 0.1050 | 237 0.1333 |
| 209 | NEUROTECHNOLOGY-012 | | | | | | 132 0.0044 | 151 0.0051 | 147 0.0038 | | | | | | 220 0.0638 | 217 0.0783 | 218 0.1027 |
| 210 | NEUROTECHNOLOGY-013 | | | | | | 108 0.0043 | 113 0.0047 | 114 0.0032 | | | | | | 167 0.0406 | 160 0.0498 | 156 0.0654 |
| 211 | NEUROTECHNOLOGY-014 | | | | | | 118 0.0043 | 102 0.0046 | 89 0.0027 | | | | | | 132 0.0312 | 129 0.0380 | 127 0.0501 |
| 212 | NEUROTECHNOLOGY-015 | | | | | | 100 0.0042 | 93 0.0044 | 75 0.0023 | | | | | | 135 0.0322 | 133 0.0391 | 128 0.0502 |
| 213 | NEUROTECHNOLOGY-016 | | | | | | 99 0.0041 | 90 0.0044 | 94 0.0028 | | | | | | 143 0.0349 | 146 0.0438 | 144 0.0584 |
| 214 | NEXTMETALABS-002 | | | | | | 115 0.0043 | 129 0.0049 | 128 0.0033 | | | | | | 159 0.0391 | 163 0.0500 | 166 0.0691 |
| 215 | NOBLIS-002 | 112 0.1520 | 112 0.2419 | 112 0.3296 | 113 0.4114 | 113 0.4856 | 393 0.5528 | 393 0.6061 | 393 0.6532 | 113 0.9984 | 113 0.9996 | 113 0.9998 | 113 0.9999 | 113 0.9999 | 389 1.0000 | 390 1.0000 | 396 1.0000 |
| 216 | NOMINDER-000 | | | | | | 174 0.0050 | 184 0.0057 | 156 0.0038 | | | | | | 176 0.0443 | 173 0.0545 | 171 0.0727 |
| 217 | NOMINDER-001 | | | | | | 187 0.0052 | 180 0.0056 | 178 0.0044 | | | | | | 174 0.0436 | 172 0.0544 | 168 0.0709 |
| 218 | NOMINDER-002 | | | | | | 165 0.0048 | 172 0.0055 | 164 0.0041 | | | | | | 172 0.0425 | 168 0.0521 | 165 0.0690 |
| 219 | NOMINDER-003 | | | | | | 139 0.0045 | 155 0.0051 | 149 0.0038 | | | | | | 162 0.0401 | 162 0.0499 | 157 0.0657 |
| 220 | NOMINDER-004 | | | | | | 134 0.0045 | 156 0.0051 | 165 0.0041 | | | | | | 166 0.0406 | 165 0.0505 | 158 0.0662 |

*Table 17:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 221 | NOMINDER-005 | | | | | | 120 0.0043 | 143 0.0049 | 150 0.0038 | | | | | | 123 0.0283 | 124 0.0357 | 123 0.0488 |
| 222 | NTECHLAB-003 | 63 0.0078 | 76 0.0131 | 87 0.0202 | 90 0.0295 | 91 0.0405 | 368 0.0543 | 371 0.0761 | 372 0.1035 | 68 0.0491 | 72 0.0881 | 79 0.1384 | 83 0.1985 | 87 0.2594 | 361 0.3270 | 362 0.4065 | 362 0.4891 |
| 223 | NTECHLAB-004 | 62 0.0068 | 68 0.0110 | 79 0.0167 | 86 0.0239 | 89 0.0330 | 365 0.0447 | 367 0.0641 | 369 0.0891 | 60 0.0379 | 63 0.0688 | 66 0.1108 | 66 0.1629 | 73 0.2192 | 353 0.2846 | 359 0.3657 | 361 0.4524 |
| 224 | NTECHLAB-006 | 51 0.0056 | 62 0.0095 | 72 0.0148 | 83 0.0218 | 85 0.0301 | 363 0.0413 | 364 0.0591 | 367 0.0814 | 56 0.0349 | 60 0.0636 | 63 0.1023 | 64 0.1506 | 66 0.2024 | 341 0.2617 | 350 0.3374 | 355 0.4185 |
| 225 | NTECHLAB-007 | 37 0.0044 | 43 0.0066 | 49 0.0089 | 57 0.0118 | 60 0.0150 | 336 0.0189 | 343 0.0255 | 343 0.0342 | 45 0.0256 | 46 0.0450 | 48 0.0705 | 49 0.1012 | 51 0.1334 | 310 0.1692 | 314 0.2170 | 316 0.2752 |
| 226 | NTECHLAB-008 | 18 0.0025 | 21 0.0038 | 26 0.0052 | 31 0.0074 | 44 0.0104 | 316 0.0146 | 338 0.0236 | 344 0.0348 | 26 0.0143 | 28 0.0267 | 32 0.0459 | 37 0.0733 | 40 0.1062 | 291 0.1469 | 307 0.2044 | 313 0.2698 |
| 227 | NTECHLAB-009 | 13 0.0022 | 15 0.0031 | 16 0.0038 | 17 0.0045 | 19 0.0055 | 232 0.0067 | 249 0.0088 | 255 0.0100 | 18 0.0073 | 17 0.0117 | 17 0.0170 | 17 0.0238 | 18 0.0319 | 170 0.0419 | 179 0.0577 | 196 0.0833 |
| 228 | NTECHLAB-011 | | | | | | 207 0.0056 | 212 0.0066 | 233 0.0073 | | | | | | 145 0.0351 | 152 0.0475 | 170 0.0724 |
| 229 | OMNIGARDE-000 | | | | | | 127 0.0050 | 185 0.0057 | 186 0.0048 | | | | | | 169 0.0413 | 169 0.0531 | 175 0.0741 |
| 230 | OMNIGARDE-001 | | | | | | 74 0.0039 | 84 0.0043 | 85 0.0024 | | | | | | 88 0.0178 | 89 0.0215 | 86 0.0275 |
| 231 | OMNIGARDE-002 | | | | | | 92 0.0040 | 92 0.0044 | 82 0.0024 | | | | | | 64 0.0120 | 62 0.0146 | 61 0.0182 |
| 232 | ONFIDO-000 | | | | | | 325 0.0157 | 326 0.0197 | 329 0.0285 | | | | | | 276 0.1225 | 280 0.1607 | 294 0.2273 |
| 233 | ONFIDO-001 | | | | | | 309 0.0134 | 312 0.0166 | 314 0.0211 | | | | | | 243 0.0896 | 247 0.1159 | 257 0.1636 |
| 234 | OPTIEXACTA-000 | | | | | | 61 0.0037 | 69 0.0042 | 66 0.0021 | | | | | | 39 0.0099 | 46 0.0125 | 47 0.0157 |
| 235 | PANGIAM-000 | | | | | | 180 0.0051 | 175 0.0055 | 182 0.0046 | | | | | | 195 0.0503 | 194 0.0617 | 189 0.0810 |
| 236 | PANGIAM-001 | | | | | | 307 0.0132 | 305 0.0153 | 288 0.0153 | | | | | | 208 0.0545 | 208 0.0685 | 208 0.0894 |
| 237 | PAPSAV1923-000 | | | | | | 176 0.0050 | 179 0.0056 | 161 0.0040 | | | | | | 122 0.0280 | 117 0.0344 | 116 0.0447 |
| 238 | PARAVISION-002 | 53 0.0058 | 58 0.0083 | 60 0.0111 | 63 0.0137 | 65 0.0162 | 334 0.0187 | 334 0.0229 | 332 0.0295 | | | | | | | | |
| 239 | PARAVISION-003 | 44 0.0048 | 44 0.0067 | 51 0.0090 | 52 0.0109 | 54 0.0128 | 317 0.0148 | 318 0.0178 | 317 0.0219 | 57 0.0354 | 58 0.0618 | 59 0.0931 | 60 0.1290 | 61 0.1625 | 323 0.1964 | 325 0.2408 | 320 0.2924 |
| 240 | PARAVISION-004 | 16 0.0024 | 17 0.0032 | 17 0.0040 | 19 0.0047 | 18 0.0053 | 218 0.0061 | 228 0.0073 | 230 0.0072 | 20 0.0118 | 23 0.0209 | 24 0.0327 | 24 0.0465 | 24 0.0613 | 234 0.0779 | 235 0.1008 | 234 0.1285 |
| 241 | PARAVISION-005 | 12 0.0021 | 13 0.0028 | 13 0.0035 | 14 0.0041 | 16 0.0046 | 198 0.0054 | 214 0.0067 | 228 0.0070 | 11 0.0057 | 12 0.0093 | 12 0.0144 | 14 0.0207 | 15 0.0278 | 153 0.0368 | 166 0.0508 | 169 0.0715 |
| 242 | PARAVISION-007 | 6 0.0019 | 8 0.0025 | 7 0.0029 | 8 0.0033 | 8 0.0036 | 104 0.0042 | 140 0.0049 | 109 0.0030 | 10 0.0057 | 13 0.0094 | 14 0.0144 | 13 0.0206 | 14 0.0275 | 148 0.0357 | 157 0.0485 | 155 0.0652 |
| 243 | PARAVISION-009 | | | | | | 96 0.0041 | 103 0.0046 | 88 0.0026 | | | | | | 124 0.0283 | 125 0.0371 | 134 0.0525 |
| 244 | PARAVISION-012 | | | | | | 59 0.0037 | 64 0.0041 | 59 0.0020 | | | | | | 74 0.0137 | 74 0.0167 | 75 0.0219 |
| 245 | PARAVISION-014 | | | | | | 46 0.0036 | 49 0.0040 | 48 0.0018 | | | | | | 61 0.0119 | 64 0.0148 | 67 0.0195 |
| 246 | PARAVISION-016 | | | | | | 42 0.0036 | 44 0.0039 | 36 0.0016 | | | | | | 45 0.0100 | 47 0.0125 | 45 0.0156 |
| 247 | PARAVISION-019 | | | | | | 38 0.0035 | 41 0.0039 | 29 0.0015 | | | | | | 46 0.0100 | 48 0.0125 | 46 0.0157 |
| 248 | PARAVISION-021 | | | | | | 36 0.0035 | 40 0.0039 | 27 0.0014 | | | | | | 42 0.0099 | 44 0.0124 | 44 0.0156 |
| 249 | PENTACOMP-000 | | | | | | 398 0.9396 | 398 0.9391 | 398 0.9364 | | | | | | | | |
| 250 | PENTACOMP-001 | | | | | | 274 0.0094 | 278 0.0108 | 279 0.0130 | | | | | | 282 0.1323 | 281 0.1612 | 278 0.2032 |
| 251 | PIXELALL-002 | 72 0.0085 | 73 0.0119 | 71 0.0147 | 72 0.0172 | 73 0.0198 | 346 0.0225 | 345 0.0270 | 345 0.0349 | 97 0.1193 | 97 0.1900 | 92 0.2601 | 92 0.3332 | 97 0.3955 | 369 0.4565 | 370 0.5268 | 371 0.6030 |
| 252 | PIXELALL-003 | 46 0.0050 | 42 0.0063 | 39 0.0072 | 34 0.0077 | 33 0.0085 | 276 0.0095 | 280 0.0113 | 271 0.0119 | 44 0.0248 | 43 0.0418 | 43 0.0622 | 43 0.0861 | 43 0.1104 | 285 0.1364 | 287 0.1723 | 286 0.2167 |
| 253 | PIXELALL-004 | 45 0.0049 | 40 0.0063 | 40 0.0072 | 37 0.0079 | 36 0.0089 | 284 0.0103 | 285 0.0127 | 283 0.0146 | 38 0.0211 | 40 0.0360 | 42 0.0553 | 42 0.0792 | 39 0.1045 | 281 0.1317 | 286 0.1700 | 291 0.2246 |
| 254 | PROTENET-001 | | | | | | 281 0.0099 | 282 0.0113 | 285 0.0149 | | | | | | 286 0.1375 | 282 0.1652 | 281 0.2095 |
| 255 | PSL-000 | | | | | | 65 0.0038 | 77 0.0042 | 76 0.0023 | | | | | | 113 0.0252 | 113 0.0324 | 115 0.0436 |
| 256 | PSL-001 | | | | | | 45 0.0036 | 42 0.0039 | 56 0.0019 | | | | | | 97 0.0202 | 96 0.0248 | 99 0.0337 |
| 257 | PSL-002 | | | | | | 34 0.0035 | 35 0.0038 | 40 0.0017 | | | | | | 69 0.0124 | 72 0.0160 | 74 0.0218 |
| 258 | PSL-003 | | | | | | 29 0.0035 | 29 0.0038 | 34 0.0016 | | | | | | 60 0.0118 | 69 0.0152 | 70 0.0197 |
| 259 | PSL-004 | | | | | | 28 0.0034 | 33 0.0038 | 41 0.0017 | | | | | | 51 0.0105 | 60 0.0142 | 65 0.0190 |
| 260 | PSL-005 | | | | | | 53 0.0037 | 52 0.0040 | 51 0.0018 | | | | | | 68 0.0123 | 66 0.0149 | 69 0.0197 |
| 261 | PSL-006 | | | | | | 21 0.0034 | 27 0.0037 | 32 0.0016 | | | | | | 35 0.0088 | 38 0.0114 | 40 0.0149 |
| 262 | PTAKURATSATU-000 | 54 0.0061 | 55 0.0082 | 55 0.0097 | 53 0.0109 | 49 0.0120 | 305 0.0131 | 298 0.0146 | 300 0.0180 | 59 0.0375 | 57 0.0596 | 55 0.0842 | 55 0.1116 | 53 0.1357 | 303 0.1553 | 294 0.1820 | 299 0.2326 |
| 263 | PYRAMID-000 | | | | | | 143 0.0045 | 159 0.0052 | 158 0.0039 | | | | | | 136 0.0325 | 139 0.0415 | 147 0.0588 |
| 264 | PYRAMID-001 | | | | | | 163 0.0047 | 168 0.0053 | 152 0.0038 | | | | | | 157 0.0384 | 156 0.0482 | 162 0.0676 |

*Table 18:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 265 | QAZSMARTVISIONAI-000 | | | | | | 25 0.0034 | 24 0.0037 | 28 0.0015 | | | | | | 47 0.0102 | 42 0.0123 | 38 0.0142 |
| 266 | QAZSMARTVISIONAI-001 | | | | | | 13 0.0033 | 14 0.0036 | 10 0.0010 | | | | | | 27 0.0066 | 27 0.0077 | 26 0.0081 |
| 267 | QAZSMARTVISIONAI-002 | | | | | | 6 0.0032 | 8 0.0035 | 2 0.0009 | | | | | | 11 0.0043 | 13 0.0051 | 11 0.0042 |
| 268 | QAZSMARTVISIONAI-003 | | | | | | 8 0.0033 | 9 0.0035 | 7 0.0010 | | | | | | 14 0.0045 | 15 0.0052 | 12 0.0043 |
| 269 | QNAP-004 | | | | | | 285 0.0105 | 291 0.0133 | 295 0.0167 | | | | | | 278 0.1243 | 278 0.1556 | 279 0.2050 |
| 270 | QNAP-005 | | | | | | 260 0.0084 | 254 0.0091 | 265 0.0113 | | | | | | 215 0.0612 | 212 0.0724 | 219 0.1031 |
| 271 | QNAP-006 | | | | | | 242 0.0071 | 243 0.0082 | 243 0.0084 | | | | | | 223 0.0676 | 224 0.0838 | 228 0.1137 |
| 272 | RANKONE-002 | 99 0.0212 | 98 0.0313 | 98 0.0431 | 98 0.0562 | 99 0.0712 | 377 0.0881 | 328 0.1130 | 329 0.1543 | 95 0.1111 | 95 0.1707 | 95 0.2305 | 95 0.2968 | 96 0.3646 | 368 0.4345 | 368 0.5172 | 372 0.6110 |
| 273 | RANKONE-004 | 108 0.0424 | 107 0.0643 | 107 0.0875 | 107 0.1127 | 107 0.1364 | 383 0.1579 | 384 0.1914 | 384 0.2378 | 104 0.1855 | 103 0.2681 | 103 0.3431 | 101 0.4155 | 101 0.4785 | 374 0.5350 | 374 0.5980 | 374 0.6722 |
| 274 | RANKONE-005 | 92 0.0136 | 93 0.0192 | 91 0.0246 | 91 0.0303 | 90 0.0362 | 364 0.0422 | 363 0.0521 | 363 0.0694 | 81 0.0582 | 75 0.0910 | 71 0.1260 | 68 0.1645 | 65 0.2005 | 330 0.2353 | 330 0.2816 | 332 0.3522 |
| 275 | RANKONE-007 | 67 0.0078 | 64 0.0099 | 61 0.0113 | 58 0.0123 | 58 0.0139 | 324 0.0156 | 325 0.0191 | 323 0.0242 | 42 0.0242 | 42 0.0376 | 41 0.0542 | 38 0.0737 | 37 0.0935 | 270 0.1130 | 269 0.1416 | 269 0.1811 |
| 276 | RANKONE-009 | 48 0.0054 | 49 0.0072 | 46 0.0085 | 47 0.0098 | 47 0.0113 | 304 0.0130 | 314 0.0169 | 319 0.0220 | 37 0.0208 | 38 0.0345 | 37 0.0504 | 36 0.0706 | 36 0.0930 | 273 0.1174 | 276 0.1504 | 276 0.2002 |
| 277 | RANKONE-010 | 42 0.0047 | 38 0.0061 | 38 0.0070 | 33 0.0076 | 34 0.0087 | 279 0.0098 | 281 0.0113 | 272 0.0120 | 31 0.0177 | 29 0.0269 | 26 0.0368 | 26 0.0479 | 23 0.0590 | 226 0.0688 | 222 0.0803 | 215 0.0991 |
| 278 | RANKONE-011 | 23 0.0031 | 23 0.0041 | 23 0.0047 | 23 0.0053 | 22 0.0058 | 233 0.0067 | 233 0.0077 | 234 0.0073 | 23 0.0127 | 20 0.0194 | 21 0.0265 | 20 0.0345 | 20 0.0422 | 193 0.0499 | 191 0.0611 | 179 0.0756 |
| 279 | RANKONE-012 | | | | | | 228 0.0065 | 220 0.0069 | 199 0.0053 | | | | | | 178 0.0460 | 171 0.0540 | 160 0.0672 |
| 280 | RANKONE-013 | | | | | | 183 0.0051 | 157 0.0051 | 136 0.0035 | | | | | | 127 0.0306 | 122 0.0355 | 110 0.0405 |
| 281 | RANKONE-014 | | | | | | 124 0.0044 | 108 0.0047 | 98 0.0029 | | | | | | 103 0.0222 | 98 0.0255 | 89 0.0287 |
| 282 | RANKONE-015 | | | | | | 97 0.0041 | 88 0.0044 | 80 0.0024 | | | | | | 82 0.0162 | 79 0.0187 | 73 0.0218 |
| 283 | REALNETWORKS-002 | 107 0.0381 | 108 0.0687 | 108 0.1062 | 108 0.1495 | 109 0.1963 | 388 0.2513 | 391 0.3206 | 391 0.3927 | 105 0.2153 | 105 0.3323 | 105 0.4444 | 105 0.5485 | 106 0.6355 | 381 0.7132 | 382 0.7855 | 382 0.8437 |
| 284 | REALNETWORKS-003 | 103 0.0245 | 105 0.0437 | 105 0.0686 | 106 0.0975 | 106 0.1312 | 386 0.1719 | 386 0.2294 | 387 0.2907 | 98 0.1468 | 100 0.2370 | 101 0.3313 | 103 0.4269 | 103 0.5142 | 378 0.5979 | 379 0.6815 | 379 0.7567 |
| 285 | REALNETWORKS-004 | 102 0.0244 | 104 0.0428 | 104 0.0663 | 105 0.0939 | 105 0.1251 | 385 0.1634 | 385 0.2170 | 386 0.2785 | 99 0.1484 | 101 0.2377 | 100 0.3303 | 102 0.4249 | 102 0.5106 | 377 0.5924 | 378 0.6758 | 378 0.7534 |
| 286 | REALNETWORKS-006 | | | | | | 240 0.0069 | 236 0.0077 | 241 0.0080 | | | | | | 261 0.1022 | 258 0.1253 | 256 0.1622 |
| 287 | REALNETWORKS-008 | | | | | | 173 0.0049 | 171 0.0054 | 183 0.0047 | | | | | | 179 0.0462 | 180 0.0577 | 176 0.0745 |
| 288 | REALSENSE-001 | | | | | | 122 0.0044 | 105 0.0046 | 144 0.0037 | | | | | | 106 0.0233 | 102 0.0269 | 98 0.0333 |
| 289 | RECOGNITO-000 | | | | | | 105 0.0042 | 115 0.0047 | 102 0.0030 | | | | | | 86 0.0176 | 93 0.0221 | 95 0.0306 |
| 290 | RECOGNITO-001 | | | | | | 62 0.0037 | 66 0.0041 | 63 0.0021 | | | | | | 50 0.0105 | 53 0.0135 | 53 0.0172 |
| 291 | RECOGNITO-002 | | | | | | 63 0.0037 | 68 0.0042 | 65 0.0021 | | | | | | 40 0.0099 | 45 0.0125 | 48 0.0157 |
| 292 | RNIT-000 | | | | | | 201 0.0054 | 204 0.0061 | 180 0.0046 | | | | | | 290 0.1456 | 291 0.1739 | 288 0.2192 |
| 293 | ROC-016 | | | | | | 70 0.0038 | 72 0.0042 | 45 0.0018 | | | | | | 71 0.0130 | 68 0.0152 | 64 0.0187 |
| 294 | ROC-017 | | | | | | 91 0.0040 | 83 0.0043 | 90 0.0027 | | | | | | 89 0.0181 | 86 0.0209 | 80 0.0238 |
| 295 | ROC-018 | | | | | | 66 0.0038 | 47 0.0039 | 53 0.0018 | | | | | | 72 0.0131 | 67 0.0151 | 52 0.0171 |
| 296 | ROC-019 | | | | | | 33 0.0035 | 31 0.0038 | 37 0.0016 | | | | | | 65 0.0121 | 58 0.0142 | 51 0.0164 |
| 297 | ROC-020 | | | | | | 30 0.0035 | 25 0.0037 | 20 0.0012 | | | | | | 38 0.0095 | 36 0.0108 | 35 0.0120 |
| 298 | ROC-021 | | | | | | 20 0.0034 | 18 0.0036 | 4 0.0009 | | | | | | 25 0.0064 | 25 0.0073 | 22 0.0071 |
| 299 | S1-002 | | | | | | 153 0.0046 | 153 0.0051 | 155 0.0038 | | | | | | 185 0.0482 | 185 0.0597 | 186 0.0788 |
| 300 | S1-003 | | | | | | 209 0.0057 | 208 0.0063 | 207 0.0056 | | | | | | 224 0.0681 | 225 0.0839 | 222 0.1061 |
| 301 | S1-004 | | | | | | 102 0.0042 | 100 0.0045 | 116 0.0032 | | | | | | 150 0.0360 | 148 0.0448 | 148 0.0598 |
| 302 | S1-005 | | | | | | 106 0.0042 | 98 0.0045 | 106 0.0030 | | | | | | 164 0.0403 | 167 0.0515 | 167 0.0695 |
| 303 | S1-006 | | | | | | 68 0.0038 | 59 0.0041 | 74 0.0022 | | | | | | 87 0.0176 | 87 0.0211 | 88 0.0280 |
| 304 | S1-007 | | | | | | 67 0.0038 | 54 0.0040 | 64 0.0021 | | | | | | 77 0.0152 | 77 0.0177 | 79 0.0230 |
| 305 | SANSAP-001 | | | | | | 200 0.0054 | 202 0.0061 | 191 0.0050 | | | | | | 161 0.0395 | 154 0.0479 | 151 0.0632 |
| 306 | SCANOVATE-001 | 68 0.0079 | 72 0.0117 | 75 0.0151 | 78 0.0185 | 78 0.0221 | 351 0.0259 | 355 0.0321 | 355 0.0427 | 88 0.0727 | 88 0.1169 | 87 0.1650 | 87 0.2115 | 84 0.2528 | 352 0.2925 | 352 0.3437 | 352 0.4084 |
| 307 | SCYLLA-001 | | | | | | 278 0.0097 | 279 0.0112 | 274 0.0124 | | | | | | 280 0.1292 | 279 0.1600 | 280 0.2093 |
| 308 | SENSETIME-002 | 96 0.0186 | 92 0.0191 | 84 0.0183 | 75 0.0179 | 68 0.0173 | 308 0.0133 | 251 0.0089 | 210 0.0059 | 40 0.0220 | 25 0.0236 | 19 0.0237 | 18 0.0240 | 12 0.0245 | 101 0.0219 | 83 0.0195 | 76 0.0222 |

*Table 19:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 309 | SENSETIME-003 | [11] 0.0021 | [12] 0.0028 | [11] 0.0031 | [7] 0.0033 | [6] 0.0035 | [85] 0.0040 | [116] 0.0047 | [119] 0.0033 | [8] 0.0046 | [8] 0.0064 | [6] 0.0076 | [4] 0.0086 | [4] 0.0101 | [66] 0.0122 | [70] 0.0155 | [68] 0.0196 |
| 310 | SENSETIME-004 | [3] 0.0016 | [3] 0.0022 | [3] 0.0025 | [3] 0.0028 | [3] 0.0030 | [37] 0.0035 | [86] 0.0043 | [87] 0.0025 | [4] 0.0036 | [4] 0.0052 | [3] 0.0066 | [3] 0.0081 | [3] 0.0099 | [70] 0.0126 | [75] 0.0169 | [78] 0.0230 |
| 311 | SENSETIME-005 | [2] 0.0015 | [2] 0.0020 | [2] 0.0024 | [2] 0.0026 | [2] 0.0029 | [32] 0.0035 | [78] 0.0043 | [95] 0.0028 | [5] 0.0036 | [7] 0.0059 | [8] 0.0089 | [9] 0.0128 | [10] 0.0177 | [109] 0.0240 | [118] 0.0345 | [124] 0.0493 |
| 312 | SENSETIME-006 | [1] 0.0015 | [1] 0.0019 | [1] 0.0022 | [1] 0.0025 | [1] 0.0027 | [9] 0.0033 | [58] 0.0040 | [68] 0.0021 | [2] 0.0031 | [2] 0.0049 | [4] 0.0068 | [6] 0.0097 | [7] 0.0132 | [92] 0.0184 | [101] 0.0262 | [102] 0.0359 |
| 313 | SENSETIME-007 | | | | | | [31] 0.0035 | [30] 0.0038 | [30] 0.0015 | | | | | | [56] 0.0112 | [55] 0.0140 | [55] 0.0176 |
| 314 | SENSETIME-008 | | | | | | [28] 0.0034 | [37] 0.0039 | [38] 0.0017 | | | | | | [49] 0.0103 | [50] 0.0127 | [49] 0.0163 |
| 315 | SENSETIME-010 | | | | | | [10] 0.0033 | [10] 0.0036 | [19] 0.0012 | | | | | | [31] 0.0071 | [31] 0.0086 | [31] 0.0099 |
| 316 | SENSETIME-011 | | | | | | [15] 0.0033 | [15] 0.0036 | [12] 0.0011 | | | | | | [20] 0.0056 | [19] 0.0065 | [19] 0.0065 |
| 317 | SENSETIME-012 | | | | | | [41] 0.0036 | [38] 0.0039 | [42] 0.0017 | | | | | | [36] 0.0092 | [51] 0.0128 | [56] 0.0177 |
| 318 | SERENDIPITY-000 | | | | | | [257] 0.0084 | [256] 0.0092 | [252] 0.0094 | | | | | | [254] 0.0981 | [251] 0.1180 | [250] 0.1519 |
| 319 | SEVENTHSENSE-001 | | | | | | [301] 0.0128 | [309] 0.0156 | [311] 0.0206 | | | | | | [156] 0.0383 | [153] 0.0476 | [164] 0.0688 |
| 320 | SIAT-002 | [117] 0.8309 | [117] 0.8310 | [117] 0.8311 | [117] 0.8306 | [117] 0.8296 | [397] 0.8302 | [397] 0.8300 | [396] 0.8301 | [112] 0.8340 | [110] 0.8368 | [109] 0.8404 | [109] 0.8445 | [109] 0.8480 | [384] 0.8532 | [384] 0.8595 | [383] 0.8691 |
| 321 | SJTU-001 | | | | | | [49] 0.0036 | [50] 0.0040 | [61] 0.0020 | | | | | | [63] 0.0120 | [59] 0.0142 | [57] 0.0178 |
| 322 | SJTU-002 | | | | | | [24] 0.0034 | [26] 0.0037 | [24] 0.0013 | | | | | | [29] 0.0068 | [26] 0.0076 | [28] 0.0087 |
| 323 | SPARKTEK-000 | | | | | | [216] 0.0060 | [215] 0.0068 | [202] 0.0054 | | | | | | [120] 0.0279 | [116] 0.0344 | [117] 0.0451 |
| 324 | SPARKTEK-001 | | | | | | [71] 0.0039 | [60] 0.0041 | [55] 0.0019 | | | | | | [48] 0.0102 | [40] 0.0121 | [41] 0.0149 |
| 325 | SPARSH-001 | | | | | | [147] 0.0046 | [139] 0.0049 | [118] 0.0032 | | | | | | [152] 0.0362 | [145] 0.0435 | [140] 0.0557 |
| 326 | STCON-001 | | | | | | [57] 0.0037 | [65] 0.0041 | [50] 0.0018 | | | | | | [76] 0.0146 | [78] 0.0186 | [82] 0.0254 |
| 327 | STCON-002 | | | | | | [35] 0.0035 | [46] 0.0039 | [35] 0.0016 | | | | | | [58] 0.0115 | [57] 0.0142 | [62] 0.0182 |
| 328 | STCON-003 | | | | | | [39] 0.0035 | [39] 0.0039 | [33] 0.0016 | | | | | | [43] 0.0100 | [43] 0.0124 | [43] 0.0154 |
| 329 | STCON-004 | | | | | | [12] 0.0033 | [23] 0.0037 | [15] 0.0011 | | | | | | [21] 0.0057 | [21] 0.0068 | [21] 0.0070 |
| 330 | STCON-005 | | | | | | [14] 0.0033 | [21] 0.0037 | [21] 0.0012 | | | | | | [22] 0.0059 | [22] 0.0070 | [25] 0.0076 |
| 331 | STELLARTECS-001 | | | | | | [265] 0.0088 | [263] 0.0099 | [259] 0.0107 | | | | | | [267] 0.1104 | [265] 0.1354 | [264] 0.1723 |
| 332 | SYNESIS-003 | [89] 0.0125 | [85] 0.0151 | [83] 0.0174 | [80] 0.0199 | [79] 0.0223 | [348] 0.0240 | [348] 0.0279 | [341] 0.0331 | [85] 0.0658 | [83] 0.1052 | [83] 0.1483 | [82] 0.1968 | [82] 0.2399 | [351] 0.2834 | [351] 0.3405 | [351] 0.4046 |
| 333 | SYNESIS-005 | [40] 0.0044 | [37] 0.0058 | [37] 0.0070 | [40] 0.0080 | [37] 0.0091 | [283] 0.0103 | [283] 0.0125 | [287] 0.0152 | [46] 0.0262 | [45] 0.0444 | [45] 0.0666 | [45] 0.0923 | [44] 0.1156 | [288] 0.1399 | [289] 0.1736 | [287] 0.2185 |
| 334 | T4ISB-000 | | | | | | [371] 0.0606 | [370] 0.0748 | [370] 0.0970 | | | | | | | | |
| 335 | TECH5-001 | [57] 0.0061 | [61] 0.0093 | [66] 0.0128 | [71] 0.0171 | [77] 0.0221 | [357] 0.0289 | [360] 0.0412 | [360] 0.0560 | [86] 0.0660 | [87] 0.1156 | [90] 0.1733 | [91] 0.2385 | [91] 0.2998 | [363] 0.3629 | [365] 0.4424 | [366] 0.5284 |
| 336 | TECH5-003 | | | | | | [155] 0.0046 | [144] 0.0050 | [115] 0.0032 | | | | | | [142] 0.0345 | [137] 0.0414 | [135] 0.0529 |
| 337 | TECHAINER-000 | | | | | | [249] 0.0079 | [250] 0.0089 | [251] 0.0093 | | | | | | [255] 0.0985 | [254] 0.1197 | [247] 0.1514 |
| 338 | THAKAAIT-000 | | | | | | [251] 0.0080 | [253] 0.0090 | [250] 0.0091 | | | | | | [277] 0.1231 | [275] 0.1490 | [272] 0.1911 |
| 339 | TIGER-004 | | | | | | [286] 0.0106 | [286] 0.0127 | [280] 0.0152 | | | | | | [284] 0.1350 | [284] 0.1654 | [283] 0.2136 |
| 340 | TOSHIBA-001 | [73] 0.0086 | [74] 0.0119 | [74] 0.0150 | [74] 0.0178 | [75] 0.0209 | [349] 0.0241 | [349] 0.0292 | [348] 0.0365 | | | | | | | | |
| 341 | TOSHIBA-002 | | | | | | [79] 0.0039 | [51] 0.0040 | [70] 0.0021 | | | | | | [52] 0.0105 | [52] 0.0130 | [58] 0.0179 |
| 342 | TOSHIBA-003 | | | | | | [72] 0.0039 | [53] 0.0040 | [52] 0.0018 | | | | | | [55] 0.0110 | [54] 0.0135 | [66] 0.0193 |
| 343 | TOSS-000 | | | | | | [43] 0.0036 | [45] 0.0039 | [54] 0.0019 | | | | | | [53] 0.0109 | [49] 0.0126 | [50] 0.0163 |
| 344 | TRUEFACE-000 | [36] 0.0043 | [36] 0.0057 | [30] 0.0061 | [28] 0.0067 | [27] 0.0073 | [259] 0.0084 | [261] 0.0097 | [254] 0.0099 | [35] 0.0200 | [37] 0.0338 | [38] 0.0504 | [35] 0.0705 | [35] 0.0904 | [268] 0.1112 | [267] 0.1401 | [268] 0.1792 |
| 345 | TSIGHT-000 | | | | | | [7] 0.0032 | [12] 0.0036 | [9] 0.0010 | | | | | | [13] 0.0044 | [16] 0.0052 | [14] 0.0044 |
| 346 | UNISSEY-003 | | | | | | [256] 0.0083 | [252] 0.0090 | [249] 0.0091 | | | | | | [252] 0.0963 | [252] 0.1181 | [248] 0.1516 |
| 347 | UNIVERSEAI-000 | | | | | | [135] 0.0045 | [118] 0.0048 | [112] 0.0031 | | | | | | [133] 0.0317 | [134] 0.0393 | [131] 0.0516 |
| 348 | USEB-001 | | | | | | [128] 0.0044 | [125] 0.0048 | [134] 0.0035 | | | | | | [126] 0.0305 | [132] 0.0389 | [130] 0.0516 |
| 349 | USEB-002 | | | | | | [138] 0.0045 | [135] 0.0049 | [130] 0.0034 | | | | | | [137] 0.0328 | [140] 0.0417 | [139] 0.0556 |
| 350 | USEB-003 | | | | | | [126] 0.0044 | [112] 0.0047 | [120] 0.0033 | | | | | | [125] 0.0305 | [131] 0.0387 | [133] 0.0522 |
| 351 | USEB-004 | | | | | | [107] 0.0043 | [114] 0.0047 | [133] 0.0035 | | | | | | [119] 0.0279 | [119] 0.0347 | [120] 0.0467 |
| 352 | VDFSOFTWARE-002 | | | | | | [331] 0.0182 | [329] 0.0221 | [327] 0.0276 | | | | | | [328] 0.2342 | [328] 0.2744 | [328] 0.3298 |

*Table 20:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 353 | VERIDAS-001 | [58]0.0063 | [56]0.0083 | [56]0.0099 | [56]0.0113 | [56]0.0132 | [318]0.0148 | [320]0.0184 | [316]0.0219 | [61]0.0403 | [61]0.0684 | [62]0.1012 | [62]0.1386 | [62]0.1741 | [326]0.2113 | [326]0.2611 | [327]0.3233 |
| 354 | VERIDAS-004 | | | | | | [190]0.0052 | [195]0.0059 | [174]0.0043 | | | | | | [197]0.0506 | [199]0.0634 | [198]0.0836 |
| 355 | VERIGRAM-001 | | | | | | [184]0.0051 | [170]0.0053 | [209]0.0058 | | | | | | [140]0.0336 | [136]0.0413 | [141]0.0561 |
| 356 | VIANTE-000 | | | | | | [93]0.0040 | [97]0.0045 | [93]0.0028 | | | | | | [99]0.0210 | [100]0.0259 | [97]0.0315 |
| 357 | VIANTE-001 | | | | | | [40]0.0036 | [34]0.0038 | [47]0.0018 | | | | | | [30]0.0070 | [29]0.0082 | [30]0.0095 |
| 358 | VIANTE-002 | | | | | | [17]0.0034 | [16]0.0036 | [18]0.0012 | | | | | | [23]0.0061 | [23]0.0071 | [23]0.0074 |
| 359 | VIANTE-003 | | | | | | [27]0.0034 | [20]0.0037 | [22]0.0012 | | | | | | [19]0.0056 | [20]0.0066 | [20]0.0066 |
| 360 | VIETTELAI-000 | | | | | | [244]0.0072 | [230]0.0074 | [239]0.0077 | | | | | | [373]0.5122 | [373]0.5689 | [373]0.6493 |
| 361 | VIETTELAI-001 | | | | | | [204]0.0056 | [200]0.0060 | [204]0.0055 | | | | | | | | |
| 362 | VIETTELAI-002 | | | | | | [205]0.0056 | [201]0.0060 | [205]0.0055 | | | | | | [244]0.0902 | [242]0.1106 | [241]0.1438 |
| 363 | VIETTELAI-003 | | | | | | [117]0.0043 | [128]0.0048 | [127]0.0033 | | | | | | [177]0.0447 | [178]0.0575 | [183]0.0778 |
| 364 | VINBIGDATA-001 | | | | | | [238]0.0068 | [239]0.0079 | [238]0.0077 | | | | | | [287]0.1383 | [285]0.1670 | [282]0.2134 |
| 365 | VINBIGDATA-002 | | | | | | [114]0.0043 | [107]0.0047 | [105]0.0030 | | | | | | [100]0.0213 | [103]0.0271 | [106]0.0372 |
| 366 | VINBIGDATA-003 | | | | | | [116]0.0043 | [111]0.0047 | [110]0.0031 | | | | | | [117]0.0273 | [123]0.0357 | [129]0.0503 |
| 367 | VINBIGDATA-004 | | | | | | [83]0.0039 | [62]0.0041 | [72]0.0022 | | | | | | [79]0.0156 | [84]0.0197 | [87]0.0280 |
| 368 | VISIONBOX-001 | | | | | | [280]0.0099 | [269]0.0103 | [269]0.0117 | | | | | | [380]0.6689 | [380]0.7251 | [380]0.7917 |
| 369 | VISIONBOX-002 | | | | | | [252]0.0080 | [248]0.0088 | [253]0.0097 | | | | | | [247]0.0937 | [249]0.1170 | [252]0.1531 |
| 370 | VISIONLABS-004 | [43]0.0048 | [46]0.0069 | [52]0.0091 | [55]0.0111 | [55]0.0130 | [322]0.0152 | [321]0.0187 | [324]0.0242 | [74]0.0540 | [77]0.0916 | [78]0.1358 | [78]0.1855 | [79]0.2303 | [347]0.2745 | [348]0.3312 | [344]0.3913 |
| 371 | VISIONLABS-005 | [39]0.0044 | [39]0.0063 | [43]0.0081 | [46]0.0095 | [46]0.0109 | [300]0.0125 | [303]0.0151 | [305]0.0187 | [67]0.0479 | [67]0.0812 | [68]0.1212 | [70]0.1664 | [69]0.2078 | [336]0.2473 | [337]0.2999 | [334]0.3577 |
| 372 | VISIONLABS-006 | [29]0.0035 | [30]0.0048 | [32]0.0061 | [30]0.0069 | [29]0.0077 | [262]0.0087 | [270]0.0105 | [273]0.0120 | [48]0.0273 | [47]0.0465 | [47]0.0702 | [47]0.0970 | [47]0.1228 | [293]0.1486 | [297]0.1847 | [296]0.2295 |
| 373 | VISIONLABS-008 | [21]0.0028 | [20]0.0037 | [21]0.0047 | [22]0.0053 | [23]0.0058 | [231]0.0067 | [242]0.0081 | [244]0.0085 | [27]0.0143 | [27]0.0241 | [28]0.0373 | [27]0.0519 | [27]0.0677 | [238]0.0850 | [241]0.1104 | [242]0.1444 |
| 374 | VISIONLABS-009 | [10]0.0020 | [10]0.0026 | [10]0.0030 | [10]0.0034 | [10]0.0038 | [130]0.0044 | [161]0.0052 | [181]0.0046 | [14]0.0065 | [15]0.0105 | [15]0.0156 | [15]0.0217 | [16]0.0289 | [154]0.0368 | [161]0.0499 | [163]0.0681 |
| 375 | VISIONLABS-010 | [9]0.0020 | [9]0.0025 | [9]0.0030 | [11]0.0034 | [9]0.0036 | [111]0.0043 | [150]0.0051 | [184]0.0047 | [17]0.0069 | [16]0.0113 | [16]0.0170 | [16]0.0238 | [17]0.0316 | [168]0.0411 | [175]0.0557 | [174]0.0740 |
| 376 | VISIONLABS-011 | | | | | | [103]0.0042 | [104]0.0046 | [140]0.0036 | | | | | | [116]0.0270 | [114]0.0337 | [114]0.0432 |
| 377 | VIXVIZION-009 | | | | | | [327]0.0161 | [324]0.0190 | [321]0.0238 | | | | | | [316]0.1787 | [311]0.2116 | [309]0.2595 |
| 378 | VNPAY-001 | | | | | | [193]0.0053 | [186]0.0058 | [173]0.0043 | | | | | | [194]0.0501 | [196]0.0619 | [192]0.0822 |
| 379 | VNPT-002 | | | | | | [194]0.0053 | [194]0.0059 | [177]0.0044 | | | | | | [205]0.0534 | [207]0.0670 | [205]0.0882 |
| 380 | VNPT-003 | | | | | | [175]0.0050 | [183]0.0057 | [163]0.0041 | | | | | | [204]0.0532 | [203]0.0658 | [206]0.0883 |
| 381 | VNPT-004 | | | | | | [169]0.0049 | [162]0.0052 | [148]0.0038 | | | | | | [173]0.0432 | [174]0.0549 | [178]0.0754 |
| 382 | VNPT-005 | | | | | | [136]0.0045 | [126]0.0048 | [122]0.0033 | | | | | | [149]0.0357 | [150]0.0456 | [150]0.0623 |
| 383 | VOGIC-000 | | | | | | [277]0.0096 | [277]0.0108 | [277]0.0128 | | | | | | [272]0.1162 | [272]0.1425 | [275]0.1932 |
| 384 | VTS-000 | [116]0.5878 | [116]0.6312 | [116]0.6602 | [115]0.6863 | [115]0.7073 | [395]0.7246 | [395]0.7458 | [394]0.7747 | [108]0.5929 | [108]0.6397 | [108]0.6729 | [108]0.7034 | [107]0.7279 | [382]0.7493 | [381]0.7739 | [381]0.8076 |
| 385 | VTS-003 | | | | | | [195]0.0054 | [188]0.0059 | [201]0.0054 | | | | | | [213]0.0597 | [213]0.0731 | [212]0.0950 |
| 386 | VTS-004 | | | | | | [149]0.0046 | [158]0.0051 | [143]0.0037 | | | | | | [165]0.0405 | [164]0.0505 | [159]0.0671 |
| 387 | VTS-005 | | | | | | [121]0.0044 | [141]0.0049 | [141]0.0036 | | | | | | [192]0.0497 | [197]0.0625 | [202]0.0849 |
| 388 | WINK-001 | | | | | | [337]0.0190 | [330]0.0223 | [330]0.0286 | | | | | | [329]0.2344 | [329]0.2754 | [329]0.3370 |
| 389 | WINK-002 | | | | | | [263]0.0087 | [259]0.0095 | [257]0.0103 | | | | | | [253]0.0966 | [248]0.1168 | [249]0.1518 |
| 390 | WINK-003 | | | | | | [215]0.0059 | [206]0.0062 | [218]0.0066 | | | | | | [231]0.0738 | [231]0.0911 | [231]0.1190 |
| 391 | XAILIENT-000 | | | | | | [344]0.0214 | [344]0.0265 | [347]0.0360 | | | | | | [334]0.2457 | [335]0.2925 | [338]0.3694 |
| 392 | XFORWARDAI-000 | [20]0.0027 | [18]0.0034 | [20]0.0044 | [21]0.0052 | [21]0.0058 | [234]0.0067 | [238]0.0079 | [237]0.0076 | [28]0.0157 | [31]0.0281 | [30]0.0443 | [31]0.0635 | [32]0.0834 | [262]0.1050 | [263]0.1330 | [263]0.1714 |
| 393 | XFORWARDAI-001 | [15]0.0023 | [11]0.0028 | [12]0.0034 | [12]0.0037 | [12]0.0039 | [144]0.0045 | [160]0.0052 | [175]0.0043 | [13]0.0060 | [14]0.0096 | [13]0.0144 | [12]0.0200 | [13]0.0260 | [139]0.0334 | [143]0.0435 | [145]0.0586 |
| 394 | YITU-002 | [60]0.0066 | [57]0.0083 | [53]0.0094 | [48]0.0101 | [50]0.0121 | [320]0.0150 | [331]0.0223 | [339]0.0328 | [33]0.0189 | [34]0.0317 | [35]0.0494 | [39]0.0750 | [41]0.1066 | [296]0.1494 | [315]0.2171 | [323]0.2958 |
| 395 | YITU-003 | [63]0.0072 | [60]0.0089 | [57]0.0100 | [51]0.0107 | [52]0.0125 | [323]0.0153 | [333]0.0226 | [342]0.0334 | [34]0.0194 | [35]0.0321 | [36]0.0500 | [41]0.0756 | [42]0.1071 | [298]0.1500 | [316]0.2177 | [324]0.2964 |
| 396 | YITU-004 | [55]0.0061 | [51]0.0075 | [44]0.0081 | [41]0.0081 | [39]0.0092 | [288]0.0107 | [306]0.0154 | [313]0.0207 | [22]0.0125 | [22]0.0204 | [23]0.0314 | [25]0.0469 | [26]0.0671 | [249]0.0955 | [271]0.1421 | [277]0.2006 |

*Table 21:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*



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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISS RATES | | INVESTIGATION, FNIR(N, R = 1, T = 0) | | | | | | | | IDENTIFICATION, FNIR(N, R = 1, T $\geq$ 0) FOR FPIR = 0.001 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # | ALGORITHM | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] | (0, 2] | (2, 4] | (4, 6] | (6, 8] | (8, 10] | (10, 12] | (12, 14] | (14, 18] |
| 397 | YITU-005 | [61]0.0067 | [54]0.0080 | [47]0.0087 | [44]0.0085 | [41]0.0094 | [290]0.0108 | [304]0.0151 | [309]0.0204 | [21]0.0124 | [21]0.0198 | [22]0.0308 | [22]0.0462 | [25]0.0667 | [248]0.0953 | [270]0.1418 | [274]0.1930 |
| 398 | YOUVERSE-001 | | | | | | [196]0.0054 | [203]0.0061 | [206]0.0055 | | | | | | [217]0.0620 | [218]0.0786 | [221]0.1046 |
| 399 | YUAN-000 | | | | | | [214]0.0059 | [221]0.0070 | [217]0.0063 | | | | | | [241]0.0864 | [240]0.1083 | [240]0.1406 |

*Table 22:* **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 1 | 20FACE-000 | [561] 0.055 | [539] 0.085 | [458] 0.733 | [477] 0.056 | [398] 0.239 | [471] 0.243 | [560] 0.347 | [519] 0.450 | [535] 1.000 | [466] 0.424 | [311] 0.772 | [466] 0.938 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 2 | 3DIVI-003 | [571] 0.083 | [569] 0.206 | | [499] 0.141 | | [509] 0.474 | [569] 0.400 | [542] 0.626 | | [487] 0.605 | | [431] 0.821 | 0.002 | 0.005 | | | | |
| 3 | 3DIVI-004 | [519] 0.017 | [522] 0.062 | | [461] 0.035 | | [477] 0.279 | [527] 0.168 | [504] 0.343 | | [443] 0.277 | | [387] 0.607 | 0.002 | 0.005 | | | | |
| 4 | 3DIVI-005 | [520] 0.017 | [526] 0.062 | [512] 0.929 | [531] 0.821 | | [478] 0.279 | [524] 0.166 | [502] 0.339 | [410] 0.996 | [507] 0.864 | | [385] 0.597 | 0.002 | 0.005 | 0.442 | | | |
| 5 | 3DIVI-006 | [531] 0.024 | [534] 0.074 | | [469] 0.047 | | [491] 0.312 | [526] 0.167 | [503] 0.342 | | [444] 0.283 | | [391] 0.615 | 0.002 | 0.005 | | | | |
| 6 | AADI-000 | [615] 0.999 | [612] 1.000 | [594] 1.000 | [577] 1.000 | [434] 1.000 | [553] 1.000 | [611] 1.000 | [619] 1.000 | [607] 1.000 | [546] 1.000 | [471] 1.000 | [589] 1.000 | 0.028 | 0.154 | 0.990 | | 0.274 | |
| 7 | AADI-001 | [611] 0.789 | [607] 0.944 | [570] 0.999 | [535] 0.967 | [429] 0.988 | [536] 0.985 | | | [492] 1.000 | | | | 0.160 | 0.278 | 0.991 | | 0.473 | |
| 8 | AADI-002 | [610] 0.699 | [606] 0.915 | [569] 0.999 | [533] 0.898 | [426] 0.955 | [535] 0.982 | [599] 0.939 | [594] 0.992 | [453] 0.999 | [525] 0.995 | [358] 0.999 | [497] 0.998 | 0.008 | 0.047 | 0.919 | | 0.047 | |
| 9 | ACER-000 | [489] 0.010 | [480] 0.036 | [490] 0.825 | [441] 0.025 | | [455] 0.209 | [514] 0.145 | [473] 0.246 | [334] 0.981 | [433] 0.201 | | [361] 0.490 | 0.000 | 0.000 | 0.042 | | | |
| 10 | ACER-001 | [431] 0.005 | [405] 0.020 | [386] 0.416 | [383] 0.008 | [331] 0.050 | [311] 0.098 | [445] 0.056 | [382] 0.109 | [464] 0.999 | [368] 0.068 | [277] 0.406 | [359] 0.479 | 0.001 | 0.001 | 0.041 | | 0.000 | |
| 11 | ADIANCE-001 | [618] 1.000 | [611] 1.000 | [611] 1.000 | [532] 0.823 | [432] 1.000 | [529] 0.837 | [613] 1.000 | [600] 1.000 | [582] 1.000 | [503] 0.833 | [369] 1.000 | [463] 0.931 | 0.001 | 0.000 | 0.043 | | 0.000 | |
| 12 | ADIANCE-002 | [237] 0.001 | [470] 0.031 | [247] 0.098 | [468] 0.045 | [412] 0.441 | [449] 0.197 | [281] 0.012 | [585] 0.955 | [476] 1.000 | [517] 0.974 | [415] 1.000 | [493] 0.996 | 0.000 | 0.000 | 0.031 | | 0.000 | |
| 13 | ADVANCE-000 | [301] 0.002 | [305] 0.013 | [303] 0.177 | [267] 0.003 | [343] 0.060 | [254] 0.081 | [338] 0.020 | [487] 0.270 | [316] 0.973 | [419] 0.163 | [323] 0.884 | [305] 0.326 | 0.000 | 0.000 | 0.043 | | 0.000 | |
| 14 | AFISBIOMETRICS-000 | [244] 0.001 | [176] 0.009 | [263] 0.115 | [270] 0.003 | [238] 0.015 | [265] 0.083 | [390] 0.029 | [345] 0.081 | [187] 0.511 | [314] 0.034 | [206] 0.154 | [232] 0.172 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 15 | AFRENGINE-000 | [389] 0.003 | [491] 0.038 | [262] 0.114 | [460] 0.035 | [295] 0.028 | [379] 0.121 | [334] 0.019 | [488] 0.271 | [149] 0.360 | [435] 0.206 | [305] 0.743 | [300] 0.308 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 16 | AFRENGINE-001 | [262] 0.001 | [300] 0.013 | [164] 0.060 | [390] 0.010 | [253] 0.018 | [221] 0.073 | [127] 0.003 | [198] 0.031 | [115] 0.227 | [246] 0.018 | [174] 0.113 | [167] 0.120 | 0.001 | 0.001 | 0.042 | | 0.009 | |
| 17 | AFRENGINE-002 | [261] 0.001 | [287] 0.012 | [145] 0.058 | [176] 0.001 | [167] 0.008 | [105] 0.054 | [116] 0.003 | [168] 0.026 | [144] 0.344 | [179] 0.008 | [331] 0.926 | [140] 0.102 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 18 | AFRENGINE-003 | [366] 0.002 | [306] 0.013 | [165] 0.061 | [178] 0.002 | [158] 0.007 | [152] 0.060 | [174] 0.004 | [211] 0.033 | [85] 0.183 | [204] 0.011 | [166] 0.097 | [157] 0.111 | 0.002 | 0.000 | 0.044 | | 0.001 | |
| 19 | AIGROUP-000 | [215] 0.001 | [511] 0.051 | [216] 0.081 | [486] 0.076 | [413] 0.444 | [483] 0.288 | [258] 0.010 | [593] 0.991 | [416] 0.997 | [497] 0.693 | [500] 1.000 | [461] 0.929 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 20 | AIZE-001 | [437] 0.005 | [424] 0.022 | [445] 0.679 | [423] 0.016 | [333] 0.050 | [422] 0.165 | [468] 0.076 | [419] 0.143 | [384] 0.994 | [390] 0.101 | [266] 0.364 | [325] 0.387 | 0.001 | 0.001 | 0.047 | | 0.000 | |
| 21 | AIZE-002 | [397] 0.003 | [394] 0.019 | [358] 0.311 | [366] 0.007 | [326] 0.043 | [304] 0.096 | [431] 0.051 | [590] 0.981 | [523] 1.000 | [456] 0.355 | [384] 1.000 | [519] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 22 | AIZE-006 | [407] 0.004 | [465] 0.030 | [362] 0.319 | [386] 0.008 | [380] 0.144 | [367] 0.114 | [420] 0.045 | [596] 0.996 | [434] 0.998 | [450] 0.304 | [561] 1.000 | [445] 0.885 | 0.000 | 0.000 | 0.036 | | 0.000 | |
| 23 | AKUVOX-001 | [268] 0.001 | [290] 0.012 | [266] 0.118 | [218] 0.002 | [271] 0.022 | [192] 0.069 | [288] 0.013 | [269] 0.050 | [300] 0.964 | [254] 0.019 | [302] 0.737 | [194] 0.141 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 24 | ALCHERA-000 | [516] 0.016 | [508] 0.047 | [498] 0.869 | [468] 0.046 | | [486] 0.292 | [511] 0.138 | [457] 0.216 | [439] 0.999 | [425] 0.176 | | [427] 0.803 | 0.006 | 0.014 | 0.328 | | | |
| 25 | ALCHERA-001 | [614] 0.987 | [613] 1.000 | | [538] 1.000 | | [610] 1.000 | [607] 0.999 | [603] 1.000 | | [608] 1.000 | | [539] 1.000 | 0.006 | 0.013 | 0.324 | | | |
| 26 | ALCHERA-002 | [574] 0.095 | [562] 0.166 | [530] 0.954 | [525] 0.668 | | [506] 0.446 | [578] 0.485 | [537] 0.591 | [472] 1.000 | [502] 0.827 | | [428] 0.811 | 0.001 | 0.002 | 0.106 | | | |
| 27 | ALCHERA-003 | [488] 0.010 | [478] 0.035 | [460] 0.738 | [424] 0.016 | | [453] 0.206 | [517] 0.155 | [469] 0.239 | [455] 0.999 | [424] 0.172 | | [352] 0.464 | 0.001 | 0.002 | 0.106 | | | |
| 28 | ALCHERA-004 | [493] 0.011 | [484] 0.038 | [370] 0.338 | [425] 0.017 | [360] 0.088 | [412] 0.144 | [568] 0.394 | [532] 0.529 | [371] 0.991 | [467] 0.424 | [297] 0.708 | [376] 0.546 | 0.001 | 0.001 | 0.046 | | 0.000 | |
| 29 | ALCHERA-005 | [460] 0.007 | [453] 0.027 | [336] 0.265 | [406] 0.012 | [355] 0.081 | [374] 0.118 | [556] 0.314 | [517] 0.439 | [348] 0.985 | [451] 0.305 | [327] 0.908 | [348] 0.455 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 30 | ALCHERA-006 | [160] 0.001 | [195] 0.009 | [242] 0.096 | [73] 0.001 | [163] 0.008 | [99] 0.052 | [184] 0.005 | [155] 0.025 | [337] 0.982 | [115] 0.006 | [286] 0.493 | [113] 0.094 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 31 | ALCHERA-007 | [154] 0.001 | [191] 0.009 | [239] 0.095 | [97] 0.001 | [166] 0.008 | [96] 0.052 | [143] 0.004 | [132] 0.023 | [340] 0.982 | [96] 0.005 | [306] 0.743 | [89] 0.087 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 32 | ALCHERA-008 | [182] 0.001 | [186] 0.009 | [88] 0.049 | [54] 0.001 | [364] 0.096 | [92] 0.052 | [134] 0.003 | [125] 0.022 | [80] 0.177 | [200] 0.010 | [373] 1.000 | [247] 0.186 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 33 | ALGORITHMSINC-000 | [293] 0.002 | [240] 0.011 | [305] 0.181 | [279] 0.003 | [365] 0.107 | [267] 0.083 | [317] 0.016 | [408] 0.130 | [185] 0.507 | [392] 0.108 | [389] 1.000 | [342] 0.433 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 34 | ALLGOVISION-000 | [496] 0.011 | [474] 0.033 | [501] 0.893 | [435] 0.021 | | [481] 0.282 | [482] 0.088 | [436] 0.166 | [367] 0.990 | [395] 0.117 | | [370] 0.526 | 0.002 | 0.003 | 0.122 | | | |
| 35 | ALLGOVISION-001 | [474] 0.009 | [490] 0.038 | [439] 0.657 | [432] 0.021 | | [468] 0.241 | [489] 0.101 | [461] 0.221 | [350] 0.986 | [413] 0.150 | | [362] 0.491 | 0.001 | 0.001 | 0.042 | | | |
| 36 | ALLGOVISION-002 | [410] 0.004 | [433] 0.023 | [429] 0.580 | [385] 0.008 | [384] 0.161 | [400] 0.132 | [459] 0.065 | [531] 0.520 | [288] 0.950 | [437] 0.232 | [566] 1.000 | [410] 0.716 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 37 | ALLGOVISION-003 | [321] 0.002 | [320] 0.014 | [302] 0.175 | [265] 0.003 | [239] 0.015 | [264] 0.082 | [321] 0.017 | [298] 0.059 | [206] 0.566 | [267] 0.021 | [196] 0.147 | [358] 0.478 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 38 | ALLGOVISION-004 | [485] 0.010 | [449] 0.026 | [365] 0.321 | [411] 0.013 | [342] 0.059 | [389] 0.127 | [494] 0.114 | [449] 0.190 | [239] 0.747 | [396] 0.121 | [279] 0.418 | [335] 0.412 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 39 | ALLGOVISION-005 | [316] 0.002 | [307] 0.013 | [227] 0.090 | [227] 0.002 | [346] 0.064 | [229] 0.074 | [343] 0.020 | [307] 0.064 | [157] 0.393 | [333] 0.043 | [376] 1.000 | [308] 0.324 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 40 | ALTTEKGLOBAL-000 | [596] 0.274 | [575] 0.255 | [413] 0.526 | [515] 0.380 | [410] 0.419 | [518] 0.632 | [550] 0.278 | [484] 0.267 | [213] 0.617 | [461] 0.383 | [290] 0.536 | [396] 0.641 | 0.000 | 0.001 | 0.048 | | 0.001 | |
| 41 | ANKE-000 | [507] 0.013 | [488] 0.038 | [516] 0.930 | [566] 1.000 | | [619] 1.000 | [496] 0.117 | [459] 0.220 | [383] 0.994 | [619] 1.000 | | [558] 1.000 | 0.000 | 0.001 | 0.080 | | | |
| 42 | ANKE-001 | [508] 0.013 | [486] 0.038 | [524] 0.945 | [560] 1.000 | | [590] 1.000 | [499] 0.119 | [460] 0.220 | [391] 0.994 | [586] 1.000 | | [512] 1.000 | 0.000 | 0.001 | 0.080 | | | |
| 43 | ANKE-002 | [374] 0.002 | [367] 0.016 | [411] 0.516 | [329] 0.005 | | [375] 0.119 | [395] 0.031 | [339] 0.079 | [287] 0.947 | [316] 0.034 | | [273] 0.245 | 0.001 | 0.001 | 0.049 | | | |
| 44 | ANTHEUS-000 | [459] 0.007 | [463] 0.029 | [351] 0.297 | [444] 0.027 | [382] 0.152 | [429] 0.167 | [410] 0.041 | [549] 0.664 | [246] 0.793 | [506] 0.862 | [592] 1.000 | [479] 0.972 | 0.003 | 0.006 | 0.042 | | 0.000 | |
| 45 | ARATEK-001 | [131] 0.001 | [147] 0.008 | [293] 0.152 | [219] 0.002 | [177] 0.009 | [177] 0.065 | [239] 0.008 | [266] 0.050 | [536] 1.000 | [383] 0.086 | [341] 0.971 | [580] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 46 | ARMATURA-000 | [290] 0.002 | [303] 0.013 | [159] 0.060 | [189] 0.002 | [107] 0.005 | [157] 0.062 | [160] 0.004 | [158] 0.025 | [105] 0.214 | [118] 0.006 | [96] 0.048 | [102] 0.090 | 0.001 | 0.001 | 0.056 | | 0.001 | |

*Table 23:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 47 | AWARE-003 | [543] 0.030 | [541] 0.090 | [543] 0.966 | [513] 0.316 | | [484] 0.290 | [505] 0.128 | [496] 0.298 | [345] 0.984 | [468] 0.428 | | [371] 0.530 | 0.004 | 0.003 | 0.874 | | | |
| 48 | AWARE-004 | [566] 0.068 | [565] 0.176 | [553] 0.975 | [495] 0.122 | | [502] 0.414 | [548] 0.268 | [528] 0.509 | [478] 1.000 | [462] 0.397 | | [429] 0.816 | 0.003 | 0.003 | 0.776 | | | |
| 49 | AWARE-005 | [544] 0.031 | [528] 0.067 | [554] 0.978 | [472] 0.048 | | [490] 0.308 | [563] 0.363 | [476] 0.253 | [484] 1.000 | [441] 0.255 | | [453] 0.916 | 0.001 | 0.002 | 0.189 | | | |
| 50 | AWARE-006 | [568] 0.069 | [554] 0.128 | [557] 0.983 | [494] 0.111 | | [504] 0.421 | [549] 0.275 | [510] 0.398 | [467] 0.999 | [458] 0.368 | | [415] 0.749 | 0.001 | 0.002 | 0.189 | | | |
| 51 | AWARE-007 | [352] 0.002 | [369] 0.016 | [372] 0.339 | [360] 0.007 | [321] 0.038 | [369] 0.116 | [316] 0.016 | [320] 0.070 | [375] 0.992 | [337] 0.044 | [344] 0.982 | [278] 0.250 | 0.000 | 0.000 | 0.043 | | 0.000 | |
| 52 | AWIROS-000 | [463] 0.007 | [469] 0.031 | [515] 0.930 | [429] 0.018 | [368] 0.109 | [426] 0.165 | [552] 0.291 | [524] 0.471 | [504] 1.000 | [518] 0.976 | [418] 1.000 | [502] 0.999 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 53 | AYONIX-000 | [603] 0.450 | [601] 0.685 | [568] 0.996 | [523] 0.607 | | [530] 0.867 | [592] 0.811 | [583] 0.939 | [425] 0.998 | [514] 0.954 | | [482] 0.982 | 0.010 | 0.031 | 0.939 | | | |
| 54 | AYONIX-001 | [597] 0.341 | [587] 0.527 | [562] 0.993 | [537] 0.994 | | [526] 0.778 | [594] 0.824 | [572] 0.920 | [462] 0.999 | [528] 0.999 | | [476] 0.969 | 0.010 | 0.031 | 0.939 | | | |
| 55 | AYONIX-002 | [598] 0.341 | [588] 0.527 | [563] 0.993 | [518] 0.464 | | [525] 0.778 | [593] 0.824 | [578] 0.920 | [465] 0.999 | [510] 0.915 | | [477] 0.969 | 0.010 | 0.031 | 0.939 | | | |
| 56 | BIOIDTECHSWISS-001 | [134] 0.001 | [372] 0.017 | [170] 0.062 | [342] 0.005 | [261] 0.018 | [430] 0.168 | [154] 0.004 | [194] 0.030 | [355] 0.987 | [197] 0.010 | [136] 0.067 | [241] 0.179 | 0.000 | 0.009 | 0.042 | | 0.004 | |
| 57 | BIOMETRIK-001 | [423] 0.004 | [356] 0.015 | [459] 0.734 | [371] 0.007 | [319] 0.038 | [324] 0.101 | [433] 0.052 | [374] 0.098 | [479] 1.000 | [351] 0.054 | [263] 0.341 | [392] 0.618 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 58 | CAMVI-003 | [560] 0.052 | [542] 0.090 | [505] 0.910 | [489] 0.093 | | [497] 0.360 | [463] 0.071 | [410] 0.132 | [308] 0.970 | [394] 0.114 | | [330] 0.402 | 0.006 | 0.013 | 0.675 | | | |
| 59 | CAMVI-004 | [558] 0.047 | [535] 0.077 | [462] 0.741 | [485] 0.072 | | [488] 0.296 | [464] 0.071 | [412] 0.136 | [459] 0.999 | [389] 0.100 | | [423] 0.787 | 0.000 | 0.000 | 0.000 | | | |
| 60 | CAMVI-005 | [565] 0.065 | [550] 0.103 | [464] 0.743 | [492] 0.098 | | [496] 0.341 | [487] 0.099 | [446] 0.179 | [473] 1.000 | [416] 0.156 | | [498] 0.999 | 0.000 | 0.000 | 0.000 | | | |
| 61 | CANON-001 | [106] 0.001 | [67] 0.006 | [210] 0.077 | [170] 0.001 | [133] 0.007 | [164] 0.062 | [193] 0.005 | [146] 0.023 | [147] 0.357 | [162] 0.008 | [140] 0.068 | [189] 0.139 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 62 | CANON-002 | [138] 0.001 | [71] 0.006 | [240] 0.095 | [137] 0.001 | [140] 0.007 | [150] 0.059 | [170] 0.004 | [114] 0.020 | [159] 0.400 | [224] 0.013 | [149] 0.075 | [250] 0.188 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 63 | CANON-003 | [53] 0.001 | [144] 0.008 | [186] 0.067 | [293] 0.004 | [129] 0.006 | [87] 0.051 | [91] 0.002 | [69] 0.012 | [95] 0.199 | [114] 0.006 | [51] 0.030 | [52] 0.076 | 0.000 | 0.003 | 0.041 | | 0.003 | |
| 64 | CIB-000 | [233] 0.001 | [155] 0.008 | [226] 0.090 | [221] 0.002 | [209] 0.011 | [199] 0.069 | [277] 0.012 | [257] 0.045 | [497] 1.000 | [244] 0.017 | [188] 0.141 | [448] 0.894 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 65 | CLEARVIEWAI-000 | [107] 0.001 | [105] 0.007 | [104] 0.051 | [164] 0.001 | [117] 0.006 | [131] 0.056 | [202] 0.005 | [162] 0.025 | [317] 0.974 | [163] 0.008 | [118] 0.057 | [288] 0.268 | 0.000 | 0.000 | 0.037 | | 0.000 | |
| 66 | CLEARVIEWAI-001 | [110] 0.001 | [116] 0.007 | [44] 0.044 | [72] 0.001 | [50] 0.004 | [115] 0.054 | [140] 0.003 | [113] 0.020 | [70] 0.167 | [105] 0.005 | [86] 0.042 | [68] 0.079 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 67 | CLEARVIEWAI-002 | [119] 0.001 | [118] 0.007 | [46] 0.044 | [79] 0.001 | [51] 0.004 | [107] 0.054 | [152] 0.004 | [123] 0.022 | [76] 0.170 | [106] 0.006 | [85] 0.041 | [62] 0.078 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 68 | CLEARVIEWAI-003 | [113] 0.001 | [108] 0.007 | [35] 0.043 | [74] 0.001 | [46] 0.003 | [119] 0.055 | [89] 0.002 | [88] 0.015 | [46] 0.128 | [90] 0.005 | [73] 0.038 | [65] 0.079 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 69 | CLOUDWALK-HR-000 | [223] 0.001 | [215] 0.010 | [115] 0.053 | [210] 0.002 | [128] 0.006 | [136] 0.057 | [78] 0.002 | [77] 0.013 | [43] 0.123 | [91] 0.005 | [59] 0.033 | [130] 0.099 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 70 | CLOUDWALK-MT-000 | [273] 0.001 | [255] 0.011 | [60] 0.047 | [115] 0.001 | [56] 0.004 | [83] 0.051 | [76] 0.002 | [76] 0.013 | [28] 0.099 | [49] 0.002 | [18] 0.018 | [45] 0.072 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 71 | CLOUDWALK-MT-001 | [271] 0.001 | [257] 0.011 | [30] 0.043 | [21] 0.001 | [25] 0.003 | [19] 0.042 | [71] 0.002 | [61] 0.012 | [6] 0.059 | [12] 0.001 | [14] 0.015 | [12] 0.056 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 72 | CLOUDWALK-MT-002 | [272] 0.001 | [254] 0.011 | [21] 0.041 | [18] 0.001 | [18] 0.002 | [17] 0.041 | [68] 0.002 | [57] 0.011 | [3] 0.053 | [8] 0.001 | [7] 0.013 | [10] 0.055 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 73 | COGENT-000 | [486] 0.010 | [506] 0.046 | [541] 0.965 | | | | [436] 0.052 | [415] 0.140 | [396] 0.995 | | | | 0.000 | 0.000 | 0.000 | | | |
| 74 | COGENT-001 | [487] 0.010 | [505] 0.046 | [542] 0.965 | | | | [435] 0.052 | [416] 0.140 | [397] 0.995 | | | | 0.000 | 0.000 | 0.000 | | | |
| 75 | COGENT-002 | [401] 0.003 | [408] 0.020 | [510] 0.924 | | | | [417] 0.044 | [372] 0.098 | [422] 0.998 | | | | 0.000 | 0.000 | 0.000 | | | |
| 76 | COGENT-003 | [403] 0.003 | [415] 0.021 | [521] 0.938 | | | | [421] 0.045 | [366] 0.095 | [427] 0.998 | | | | 0.000 | 0.000 | 0.000 | | | |
| 77 | COGENT-004 | [303] 0.002 | [313] 0.013 | [509] 0.921 | [304] 0.004 | [265] 0.019 | [363] 0.113 | [396] 0.033 | [270] 0.051 | [420] 0.997 | [271] 0.022 | [180] 0.126 | [349] 0.456 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 78 | COGENT-005 | [258] 0.001 | [234] 0.010 | [264] 0.116 | [222] 0.002 | [203] 0.010 | [377] 0.120 | [248] 0.009 | [230] 0.037 | [360] 0.989 | [206] 0.011 | [156] 0.082 | [451] 0.905 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 79 | COGENT-006 | [158] 0.001 | [120] 0.007 | [140] 0.056 | [148] 0.001 | [150] 0.007 | [325] 0.101 | [163] 0.004 | [147] 0.023 | [118] 0.229 | [131] 0.006 | [280] 0.422 | [180] 0.130 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 80 | COGENT-007 | [109] 0.001 | [153] 0.008 | [120] 0.053 | [140] 0.001 | [124] 0.006 | [331] 0.103 | [135] 0.003 | [121] 0.021 | [86] 0.185 | [135] 0.006 | [94] 0.047 | [200] 0.144 | 0.000 | 0.000 | 0.043 | | 0.000 | |
| 81 | COGNITEC-000 | [533] 0.025 | [522] 0.059 | [539] 0.963 | | | | [521] 0.160 | [497] 0.303 | [373] 0.992 | | | | 0.003 | 0.002 | 0.924 | | | |
| 82 | COGNITEC-001 | [498] 0.011 | [476] 0.034 | [533] 0.957 | | | | [488] 0.101 | [465] 0.230 | [557] 1.000 | | | | 0.003 | 0.002 | 0.924 | | | |
| 83 | COGNITEC-002 | [439] 0.005 | [443] 0.025 | [526] 0.948 | | | | [438] 0.053 | [445] 0.178 | [499] 1.000 | | | | 0.003 | 0.002 | 0.924 | | | |
| 84 | COGNITEC-003 | [444] 0.006 | [442] 0.025 | [514] 0.929 | | | | [434] 0.052 | [434] 0.162 | [509] 1.000 | | | | 0.004 | 0.002 | 0.878 | | | |
| 85 | COGNITEC-004 | [388] 0.003 | [365] 0.016 | [487] 0.811 | [409] 0.013 | [340] 0.057 | [411] 0.143 | [394] 0.031 | [370] 0.097 | [364] 0.990 | [366] 0.068 | [257] 0.316 | [293] 0.288 | 0.002 | 0.001 | 0.635 | | 0.006 | |
| 86 | COGNITEC-005 | [254] 0.001 | [225] 0.010 | [454] 0.710 | [436] 0.021 | [318] 0.037 | [368] 0.115 | [254] 0.009 | [245] 0.041 | [606] 1.000 | [331] 0.041 | [209] 0.157 | [240] 0.179 | 0.002 | 0.001 | 0.614 | | 0.017 | |
| 87 | COGNITEC-006 | [238] 0.001 | [209] 0.010 | [451] 0.699 | [369] 0.007 | [285] 0.024 | [357] 0.111 | [233] 0.007 | [241] 0.040 | [618] 1.000 | [301] 0.030 | [216] 0.171 | [404] 0.681 | 0.002 | 0.001 | 0.568 | | 0.003 | |
| 88 | COGNITEC-007 | [197] 0.001 | [169] 0.009 | [340] 0.275 | [344] 0.005 | [274] 0.022 | [312] 0.098 | [200] 0.005 | [216] 0.034 | [216] 0.621 | [284] 0.026 | [233] 0.210 | [220] 0.160 | 0.000 | 0.001 | 0.100 | | 0.002 | |
| 89 | COGNITEC-008 | [183] 0.001 | [187] 0.009 | [321] 0.216 | [322] 0.004 | [260] 0.018 | [300] 0.095 | [206] 0.006 | [221] 0.035 | [205] 0.564 | [285] 0.026 | [191] 0.142 | [212] 0.155 | 0.000 | 0.001 | 0.091 | | 0.001 | |
| 90 | COGNITEC-009 | [229] 0.001 | [197] 0.009 | [323] 0.219 | [317] 0.004 | [246] 0.016 | [327] 0.102 | [210] 0.006 | [222] 0.035 | [204] 0.554 | [280] 0.025 | [178] 0.124 | [214] 0.156 | 0.001 | 0.001 | 0.106 | | 0.001 | |
| 91 | COMPAL-000 | [270] 0.001 | [393] 0.019 | [169] 0.062 | [482] 0.065 | [345] 0.063 | [384] 0.125 | [236] 0.008 | [237] 0.038 | | [370] 0.072 | [169] 0.104 | [202] 0.147 | 0.000 | 0.012 | 0.000 | | 0.067 | |
| 92 | COMPAL-001 | [456] 0.007 | [334] 0.014 | [163] 0.060 | [359] 0.006 | [200] 0.010 | [178] 0.065 | [282] 0.012 | [210] 0.033 | [165] 0.422 | [221] 0.013 | [160] 0.090 | [162] 0.117 | 0.004 | 0.002 | 0.049 | | 0.005 | |

*Table 24:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 93 | COMPAL-002 | 455 0.007 | 335 0.014 | 161 0.060 | 358 0.006 | 199 0.010 | 175 0.065 | 283 0.012 | 206 0.033 | 164 0.419 | 223 0.013 | 159 0.090 | 164 0.118 | 0.004 | 0.002 | 0.049 | | 0.005 | |
| 94 | CORSIGHT-000 | 114 0.001 | 104 0.007 | 181 0.065 | 139 0.001 | 132 0.007 | 135 0.057 | 211 0.006 | 186 0.029 | 189 0.517 | 188 0.009 | 137 0.067 | 384 0.594 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 95 | CORSIGHT-001 | 118 0.001 | 103 0.007 | 162 0.060 | 130 0.001 | 118 0.006 | 124 0.055 | 219 0.006 | 212 0.034 | 176 0.460 | 173 0.008 | 124 0.062 | 188 0.138 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 96 | CUBOX-000 | 204 0.001 | 218 0.010 | 67 0.048 | 180 0.002 | 78 0.004 | 67 0.049 | 104 0.002 | 104 0.019 | 64 0.158 | 81 0.004 | 47 0.028 | 46 0.073 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 97 | CUBOX-001 | 166 0.001 | 158 0.008 | 87 0.049 | 123 0.001 | 85 0.004 | 52 0.047 | 111 0.003 | 118 0.021 | 104 0.212 | 141 0.007 | 142 0.068 | 80 0.085 | 0.001 | 0.001 | 0.041 | | 0.000 | |
| 98 | CUBOX-002 | 226 0.001 | 264 0.011 | 99 0.050 | 120 0.001 | 68 0.004 | 39 0.046 | 125 0.003 | 132 0.022 | 81 0.181 | 84 0.004 | 75 0.038 | 58 0.077 | 0.001 | 0.001 | 0.040 | | 0.000 | |
| 99 | CUBOX-003 | 242 0.001 | 281 0.012 | 134 0.056 | 134 0.001 | 88 0.005 | 64 0.048 | 131 0.003 | 138 0.023 | 108 0.219 | 119 0.006 | 105 0.051 | 75 0.082 | 0.001 | 0.001 | 0.040 | | 0.000 | |
| 100 | CYBERLINK-000 | 405 0.004 | 404 0.020 | 455 0.713 | 375 0.007 | | 402 0.134 | 446 0.056 | 391 0.116 | 400 0.995 | 363 0.063 | | 314 0.339 | 0.001 | 0.001 | 0.063 | | | |
| 101 | CYBERLINK-001 | 399 0.003 | 386 0.018 | 456 0.728 | 364 0.007 | | 401 0.133 | 439 0.053 | 388 0.109 | 395 0.995 | 360 0.062 | | 399 0.652 | 0.000 | 0.000 | 0.040 | | | |
| 102 | CYBERLINK-002 | 362 0.002 | 283 0.012 | 427 0.572 | 297 0.004 | | 346 0.107 | 304 0.014 | 284 0.053 | 359 0.988 | 278 0.024 | | 294 0.288 | 0.001 | 0.000 | 0.042 | | | |
| 103 | CYBERLINK-003 | 240 0.001 | 184 0.009 | 400 0.468 | 258 0.003 | 210 0.012 | 261 0.082 | 234 0.007 | 220 0.035 | 314 0.972 | 213 0.012 | 168 0.100 | 320 0.368 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 104 | CYBERLINK-004 | 256 0.001 | 274 0.011 | 387 0.416 | 257 0.003 | 204 0.011 | 336 0.104 | 226 0.007 | 224 0.036 | 540 1.000 | 219 0.013 | 170 0.109 | 472 0.954 | 0.000 | 0.000 | 0.011 | | 0.000 | |
| 105 | CYBERLINK-005 | 278 0.001 | 245 0.011 | 313 0.200 | 231 0.002 | 193 0.010 | 313 0.098 | 260 0.010 | 242 0.041 | 482 1.000 | 228 0.014 | 158 0.089 | 460 0.926 | 0.000 | 0.000 | 0.034 | | 0.000 | |
| 106 | CYBERLINK-006 | 130 0.001 | 96 0.007 | 221 0.085 | 85 0.001 | 71 0.004 | 205 0.071 | 102 0.002 | 100 0.018 | 351 0.986 | 85 0.004 | 147 0.073 | 246 0.186 | 0.000 | 0.000 | 0.034 | | 0.000 | |
| 107 | DAHUA-000 | 479 0.009 | 448 0.026 | | | | | 478 0.086 | 411 0.135 | | | | | 0.004 | 0.003 | | | | |
| 108 | DAHUA-001 | 450 0.006 | 436 0.024 | 450 0.699 | | | | 466 0.072 | 399 0.122 | 330 0.980 | | | | 0.002 | 0.002 | 0.346 | | | |
| 109 | DAHUA-002 | 281 0.001 | 279 0.012 | 350 0.296 | 252 0.003 | | 268 0.084 | 305 0.015 | 261 0.046 | 217 0.634 | 241 0.017 | | 218 0.159 | 0.001 | 0.000 | 0.099 | | | |
| 110 | DAHUA-003 | 144 0.001 | 109 0.007 | 311 0.197 | 212 0.002 | 184 0.009 | 220 0.073 | 294 0.014 | 244 0.041 | 208 0.574 | 216 0.013 | 154 0.081 | 183 0.134 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 111 | DAHUA-004 | 102 0.001 | 126 0.008 | 277 0.134 | 184 0.002 | 139 0.007 | 193 0.069 | 224 0.007 | 166 0.026 | 181 0.479 | 175 0.008 | 106 0.051 | 161 0.113 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 112 | DAON-000 | 409 0.004 | 378 0.017 | 414 0.525 | 330 0.005 | 269 0.020 | 385 0.125 | 355 0.022 | 302 0.061 | 480 1.000 | 281 0.025 | 218 0.173 | 432 0.846 | 0.002 | 0.002 | 0.108 | | 0.001 | |
| 113 | DAON-001 | 534 0.025 | 519 0.057 | 527 0.951 | 465 0.040 | 329 0.046 | 266 0.083 | 412 0.043 | 414 0.140 | 289 0.952 | 364 0.064 | 205 0.154 | 222 0.161 | 0.014 | 0.042 | 0.950 | | 0.034 | |
| 114 | DAON-002 | 170 0.001 | 199 0.009 | 257 0.109 | 208 0.002 | 225 0.013 | 185 0.067 | 227 0.007 | 254 0.044 | | 233 0.015 | 186 0.141 | 527 1.000 | 0.000 | 0.001 | 0.000 | | 0.000 | |
| 115 | DAON-003 | 99 0.001 | 167 0.009 | 152 0.059 | 147 0.001 | 205 0.011 | 151 0.060 | 130 0.003 | 142 0.023 | 139 0.319 | 133 0.006 | 138 0.067 | 121 0.096 | 0.000 | 0.001 | 0.040 | | 0.000 | |
| 116 | DATECH-000 | 184 0.001 | 159 0.008 | 299 0.168 | 162 0.001 | 327 0.045 | 117 0.055 | 235 0.007 | 235 0.038 | 409 0.996 | 237 0.016 | 330 0.926 | 302 0.311 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 117 | DATECH-001 | 117 0.001 | 152 0.008 | 276 0.132 | 127 0.001 | 336 0.052 | 132 0.056 | 220 0.006 | 303 0.062 | 380 0.993 | 327 0.039 | 357 0.998 | 318 0.360 | 0.000 | 0.000 | 0.038 | | 0.000 | |
| 118 | DECATUR-000 | 318 0.002 | 277 0.011 | 324 0.220 | 311 0.004 | 264 0.019 | 353 0.109 | 358 0.023 | 311 0.066 | 223 0.671 | 289 0.027 | 217 0.173 | 271 0.239 | 0.001 | 0.000 | 0.044 | | 0.001 | |
| 119 | DEEPGLINT-001 | 217 0.001 | 102 0.007 | 307 0.190 | 242 0.002 | | 222 0.073 | 124 0.003 | 79 0.014 | 469 1.000 | 136 0.006 | | 217 0.159 | 0.000 | 0.000 | 0.038 | | | |
| 120 | DEEPSEA-001 | 414 0.004 | 363 0.016 | 488 0.812 | 389 0.010 | | 409 0.140 | 423 0.046 | 376 0.101 | 347 0.984 | 327 0.077 | | 310 0.326 | 0.000 | 0.001 | 0.047 | | | |
| 121 | DERMALOG-003 | 580 0.126 | 572 0.217 | | 512 0.296 | | 516 0.560 | 577 0.482 | 546 0.655 | | 494 0.677 | | 442 0.870 | 0.002 | 0.002 | 0.103 | | | |
| 122 | DERMALOG-004 | 529 0.125 | 570 0.215 | 513 0.929 | 498 0.135 | | 508 0.467 | 576 0.479 | 542 0.657 | 401 0.995 | 485 0.603 | | 439 0.856 | 0.001 | 0.002 | 0.107 | | | |
| 123 | DERMALOG-005 | 512 0.015 | 483 0.037 | 449 0.697 | 506 0.242 | | 500 0.384 | 481 0.088 | 426 0.154 | 366 0.990 | 449 0.300 | | 390 0.614 | 0.001 | 0.002 | 0.102 | | | |
| 124 | DERMALOG-006 | 469 0.008 | 441 0.024 | 436 0.615 | 393 0.010 | | 421 0.155 | 432 0.051 | 380 0.105 | 333 0.981 | 358 0.059 | | 307 0.318 | 0.003 | 0.006 | 0.181 | | | |
| 125 | DERMALOG-007 | 476 0.009 | 451 0.027 | 443 0.671 | 419 0.014 | | 431 0.170 | 479 0.086 | 424 0.152 | 365 0.990 | 388 0.099 | | 379 0.557 | 0.001 | 0.002 | 0.102 | | | |
| 126 | DERMALOG-008 | 375 0.003 | 347 0.015 | 408 0.511 | 361 0.007 | 303 0.029 | 408 0.139 | 419 0.045 | 363 0.094 | 516 1.000 | 354 0.057 | 272 0.382 | 467 0.940 | 0.000 | 0.000 | 0.002 | | 0.000 | |
| 127 | DERMALOG-009 | 373 0.002 | 333 0.014 | 296 0.157 | 370 0.007 | 431 0.999 | 339 0.106 | 341 0.020 | 312 0.066 | 500 1.000 | 303 0.031 | 361 0.999 | 434 0.840 | 0.001 | 0.001 | 0.018 | | 0.003 | |
| 128 | DERMALOG-010 | 328 0.002 | 271 0.011 | 135 0.056 | 463 0.038 | 375 0.124 | 364 0.113 | 223 0.007 | 288 0.055 | 448 0.999 | 385 0.089 | 368 1.000 | 368 0.522 | 0.001 | 0.001 | 0.018 | | 0.003 | |
| 129 | DERMALOG-011 | 243 0.001 | 227 0.010 | 223 0.086 | 277 0.003 | 305 0.031 | 288 0.092 | 348 0.021 | 351 0.087 | 433 0.998 | 402 0.129 | 348 0.991 | 418 0.764 | 0.000 | 0.000 | 0.013 | | 0.000 | |
| 130 | DERMALOG-012 | 311 0.002 | 311 0.013 | 243 0.096 | 262 0.003 | 231 0.014 | 211 0.071 | 347 0.021 | 314 0.067 | 183 0.505 | 273 0.023 | 197 0.147 | 334 0.409 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 131 | DERMALOG-013 | 294 0.002 | 305 0.013 | 244 0.096 | 272 0.003 | 232 0.014 | 213 0.071 | 346 0.021 | 317 0.068 | 278 0.908 | 274 0.024 | 204 0.153 | 373 0.541 | 0.000 | 0.000 | 0.038 | | 0.000 | |
| 132 | DERMALOG-014 | 395 0.003 | 362 0.016 | 270 0.123 | 388 0.009 | 278 0.023 | 347 0.107 | 350 0.021 | 318 0.069 | 277 0.907 | 293 0.028 | 208 0.156 | 341 0.432 | 0.000 | 0.000 | 0.038 | | 0.000 | |
| 133 | DERMALOG-015 | 241 0.001 | 252 0.011 | 139 0.056 | 207 0.002 | 144 0.007 | 128 0.056 | 186 0.005 | 199 0.031 | 136 0.313 | 182 0.009 | 130 0.065 | 216 0.158 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 134 | DERMALOG-016 | 208 0.001 | 232 0.010 | 73 0.048 | 181 0.002 | 91 0.005 | 80 0.050 | 84 0.002 | 111 0.020 | 131 0.288 | 76 0.004 | 67 0.035 | 374 0.544 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 135 | DERMALOG-017 | 214 0.001 | 231 0.010 | 74 0.048 | 183 0.002 | 90 0.005 | 79 0.050 | 120 0.003 | 143 0.023 | 110 0.220 | 117 0.006 | 81 0.041 | 263 0.220 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 136 | DIGIDATA-000 | 607 0.590 | 589 0.548 | 502 0.894 | 524 0.642 | 422 0.707 | 528 0.813 | 583 0.610 | 536 0.577 | 388 0.994 | 490 0.646 | 314 0.789 | 432 0.824 | 0.002 | 0.001 | 0.070 | | 0.001 | |
| 137 | DILUSENSE-000 | 330 0.002 | 288 0.012 | 346 0.289 | 378 0.008 | 299 0.028 | 317 0.099 | 392 0.030 | 338 0.078 | 219 0.651 | 325 0.039 | 296 0.664 | 257 0.203 | 0.001 | 0.001 | 0.219 | | 0.006 | |
| 138 | DILUSENSE-001 | 218 0.001 | 205 0.010 | 330 0.241 | 351 0.006 | 244 0.016 | 280 0.088 | 242 0.008 | 217 0.034 | 171 0.450 | 234 0.015 | 288 0.511 | 182 0.137 | 0.001 | 0.001 | 0.219 | | 0.007 | |

*Table 25:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 139 | EINETWORKSINDIA-000 | 276 0.001 | 173 0.009 | 383 0.386 | 286 0.004 | 227 0.013 | 334 0.104 | 309 0.015 | 255 0.044 | 419 0.997 | 258 0.020 | 284 0.455 | 219 0.159 | 0.001 | 0.001 | 0.055 | | 0.002 | |
| 140 | EINETWORKSINDIA-001 | 265 0.001 | 172 0.009 | 379 0.375 | 261 0.003 | 233 0.014 | 305 0.096 | 315 0.016 | 263 0.048 | 417 0.997 | 371 0.072 | 339 0.957 | 422 0.782 | 0.001 | 0.000 | 0.043 | | 0.000 | |
| 141 | EINETWORKSINDIA-002 | 126 0.001 | 81 0.007 | 92 0.050 | 63 0.001 | 311 0.033 | 78 0.050 | 149 0.004 | 128 0.022 | 89 0.188 | 160 0.007 | 337 0.953 | 175 0.127 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 142 | EYEDEA-003 | 570 0.080 | 559 0.148 | 537 0.960 | 493 0.101 | | 499 0.379 | 566 0.387 | 534 0.543 | 392 0.994 | 482 0.570 | | 425 0.792 | 0.001 | 0.003 | 0.161 | | | |
| 143 | F8-001 | 502 0.012 | | 441 0.665 | 605 1.000 | | 574 1.000 | 523 0.166 | | 437 0.998 | | | | 0.004 | 1.000 | 0.158 | | | |
| 144 | FACEHAWK-001 | 542 0.030 | 563 0.168 | 382 0.382 | 476 0.056 | 406 0.337 | 464 0.228 | 502 0.123 | 545 0.654 | 258 0.851 | 504 0.844 | 551 1.000 | 487 0.990 | 0.013 | 0.122 | 0.045 | | 0.001 | |
| 145 | FACEPHI-000 | 284 0.002 | 339 0.014 | 231 0.093 | 271 0.003 | 187 0.009 | 335 0.104 | 218 0.006 | 218 0.034 | 150 0.367 | 194 0.009 | 133 0.066 | 184 0.135 | 0.001 | 0.007 | 0.049 | | 0.001 | |
| 146 | FASTLAHORE-000 | 537 0.026 | 561 0.154 | 520 0.938 | 507 0.277 | 435 1.000 | 511 0.499 | 538 0.214 | 614 1.000 | 508 1.000 | 542 1.000 | 477 1.000 | 562 1.000 | 0.000 | 0.064 | 0.000 | | 0.000 | |
| 147 | FCI-000 | 285 0.002 | 295 0.012 | 166 0.061 | 132 0.001 | 341 0.057 | 84 0.051 | 214 0.006 | 169 0.026 | 292 0.957 | 180 0.008 | 312 0.777 | 142 0.103 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 148 | FINCORE-000 | 492 0.010 | 477 0.034 | 471 0.764 | 453 0.032 | 373 0.117 | 446 0.191 | 510 0.133 | 458 0.217 | 488 1.000 | 427 0.187 | 292 0.598 | 350 0.458 | 0.000 | 0.001 | 0.043 | | 0.000 | |
| 149 | FIRSTCREDITKZ-001 | 149 0.001 | 157 0.008 | 220 0.084 | 214 0.002 | 195 0.010 | 173 0.065 | 108 0.003 | 105 0.019 | 129 0.282 | 156 0.007 | 123 0.061 | 124 0.097 | 0.000 | 0.001 | 0.047 | | 0.001 | |
| 150 | FISSIONSFR-001 | 575 0.098 | 556 0.131 | 440 0.659 | 502 0.157 | 402 0.267 | 501 0.408 | 500 0.121 | 444 0.177 | 260 0.859 | 430 0.193 | 335 0.950 | 360 0.490 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 151 | FPT-000 | 292 0.002 | 341 0.014 | 190 0.068 | 333 0.005 | 394 0.210 | 320 0.100 | 252 0.009 | 506 0.372 | 438 0.999 | 457 0.360 | 589 1.000 | 464 0.935 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 152 | FUJITSULAB-000 | 332 0.002 | 326 0.014 | 392 0.434 | 303 0.004 | 276 0.023 | 314 0.098 | 342 0.020 | 291 0.056 | | 275 0.024 | 220 0.177 | 272 0.240 | 0.000 | 0.001 | 0.016 | | 0.000 | |
| 153 | FUJITSULAB-001 | 283 0.002 | 318 0.013 | 395 0.449 | 306 0.004 | 289 0.026 | 342 0.106 | 329 0.018 | 293 0.058 | 327 0.992 | 277 0.024 | 303 0.739 | 277 0.247 | 0.000 | 0.003 | 0.150 | | 0.002 | |
| 154 | GLORY-000 | 584 0.178 | 579 0.320 | 567 0.993 | 505 0.228 | | 519 0.678 | 564 0.367 | 535 0.547 | 394 0.995 | 472 0.453 | | 433 0.839 | 0.011 | 0.013 | 0.985 | | | |
| 155 | GLORY-001 | 581 0.126 | 576 0.267 | 561 0.992 | 503 0.178 | | 517 0.594 | 553 0.305 | 533 0.537 | 381 0.993 | 464 0.408 | | 430 0.819 | 0.011 | 0.013 | 0.988 | | | |
| 156 | GORILLA-001 | 562 0.060 | 546 0.095 | 518 0.936 | 483 0.069 | | 494 0.329 | 570 0.406 | 521 0.453 | 534 1.000 | 473 0.468 | | 598 1.000 | 0.001 | 0.001 | 0.069 | | | |
| 157 | GORILLA-002 | 526 0.019 | 501 0.044 | 467 0.751 | 443 0.027 | | 461 0.214 | 532 0.188 | 485 0.268 | 542 1.000 | 440 0.250 | | 507 1.000 | 0.001 | 0.001 | 0.069 | | | |
| 158 | GORILLA-003 | 547 0.036 | 530 0.070 | 489 0.818 | 470 0.048 | | 474 0.265 | 557 0.317 | 516 0.434 | 558 1.000 | 463 0.407 | | 542 1.000 | 0.001 | 0.001 | 0.069 | | | |
| 159 | GORILLA-004 | 445 0.006 | 437 0.024 | 447 0.693 | 401 0.012 | | 425 0.162 | 485 0.089 | 432 0.160 | 293 0.958 | 407 0.135 | | 344 0.438 | 0.000 | 0.001 | 0.042 | | | |
| 160 | GORILLA-005 | 387 0.003 | 387 0.018 | 312 0.199 | 350 0.006 | | 382 0.124 | 451 0.058 | 418 0.142 | 228 0.697 | 384 0.088 | | 305 0.315 | 0.000 | 0.000 | 0.040 | | | |
| 161 | GORILLA-006 | 269 0.001 | 285 0.012 | 258 0.112 | 276 0.003 | 254 0.018 | 338 0.105 | 378 0.027 | 355 0.089 | 197 0.525 | 291 0.028 | 214 0.166 | 262 0.218 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 162 | GORILLA-007 | 260 0.001 | 242 0.011 | 251 0.103 | 238 0.002 | 248 0.016 | 279 0.088 | 375 0.027 | 337 0.077 | 199 0.529 | 283 0.026 | 244 0.264 | 238 0.178 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 163 | GORILLA-008 | 225 0.001 | 212 0.010 | 208 0.075 | 194 0.002 | 217 0.012 | 262 0.082 | 359 0.024 | 346 0.083 | 173 0.457 | 300 0.030 | 260 0.319 | 236 0.178 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 164 | GRIAULE-000 | 355 0.002 | 319 0.014 | 363 0.319 | 396 0.011 | 308 0.031 | 387 0.126 | 339 0.020 | 306 0.063 | 398 0.995 | 310 0.033 | 225 0.185 | 254 0.198 | 0.000 | 0.002 | 0.090 | | 0.001 | |
| 165 | GRIAULE-001 | 143 0.001 | 141 0.008 | 268 0.123 | 103 0.001 | 280 0.023 | 174 0.065 | 171 0.004 | 180 0.028 | 261 0.863 | 150 0.007 | 351 0.995 | 128 0.099 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 166 | GTELOTS-000 | 588 0.194 | 571 0.217 | 426 0.558 | 508 0.278 | 414 0.446 | 515 0.560 | 536 0.206 | 478 0.256 | 232 0.727 | 447 0.292 | 499 1.000 | 386 0.603 | 0.001 | 0.000 | 0.043 | | 0.000 | |
| 167 | GTELOTS-001 | 514 0.015 | 400 0.020 | 301 0.171 | 464 0.039 | 339 0.055 | 450 0.197 | 424 0.046 | 354 0.088 | 227 0.680 | 372 0.072 | 251 0.287 | 297 0.294 | 0.001 | 0.001 | 0.043 | | 0.000 | |
| 168 | HIK-003 | 499 0.011 | 455 0.027 | 446 0.685 | 404 0.012 | | 418 0.151 | 490 0.102 | 428 0.158 | 306 0.968 | 410 0.142 | | 347 0.445 | 0.000 | 0.000 | 0.048 | | | |
| 169 | HIK-004 | 497 0.011 | 452 0.027 | 461 0.740 | 402 0.012 | | 420 0.152 | 486 0.098 | 425 0.153 | 318 0.976 | 408 0.137 | | 343 0.434 | 0.000 | 0.000 | 0.048 | | | |
| 170 | HIK-005 | 421 0.004 | 371 0.017 | 417 0.530 | 367 0.007 | | 356 0.111 | 414 0.043 | 336 0.077 | 466 0.999 | 365 0.068 | | 372 0.541 | 0.000 | 0.000 | 0.000 | | | |
| 171 | HIK-006 | 422 0.004 | 370 0.017 | 418 0.530 | | | | 425 0.047 | 350 0.086 | 532 1.000 | | | | 0.000 | 0.000 | 0.000 | | | |
| 172 | HISIGN-001 | 152 0.001 | 160 0.008 | 292 0.150 | 204 0.002 | 249 0.016 | 237 0.076 | 231 0.007 | 229 0.037 | 361 0.990 | 225 0.013 | 319 0.824 | 199 0.143 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 173 | HYPERVERGE-001 | 200 0.001 | 262 0.011 | 136 0.056 | 177 0.002 | 137 0.007 | 156 0.061 | 165 0.004 | 196 0.031 | 102 0.210 | 152 0.007 | 111 0.053 | 135 0.101 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 174 | HYPERVERGE-002 | 195 0.001 | 253 0.011 | 106 0.052 | 153 0.001 | 122 0.006 | 145 0.058 | 138 0.003 | 174 0.027 | 96 0.201 | 113 0.006 | 98 0.048 | 112 0.093 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 175 | HYPERVERGE-003 | 212 0.001 | 278 0.011 | 183 0.065 | 202 0.002 | 162 0.008 | 161 0.062 | 230 0.007 | 247 0.042 | 125 0.274 | 208 0.012 | 129 0.065 | 146 0.106 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 176 | HYUNDAI-ROBOTICSLAB-000 | 367 0.002 | 354 0.015 | 310 0.195 | 430 0.020 | 379 0.133 | 489 0.298 | 292 0.013 | 520 0.452 | 224 0.672 | 476 0.492 | 380 1.000 | 441 0.870 | 0.001 | 0.001 | 0.045 | | 0.009 | |
| 177 | HYUNDAI-ROBOTICSLAB-001 | 267 0.001 | 416 0.021 | 308 0.192 | 447 0.029 | 399 0.252 | 441 0.178 | 284 0.013 | 559 0.774 | 256 0.839 | 498 0.699 | 404 1.000 | 465 0.937 | 0.000 | 0.001 | 0.041 | | 0.001 | |
| 178 | HYUNDAI-ROBOTICSLAB-002 | 100 0.001 | 91 0.007 | 94 0.050 | 158 0.001 | 370 0.112 | 172 0.064 | 196 0.005 | 448 0.189 | 117 0.228 | 367 0.068 | 381 1.000 | 346 0.440 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 179 | HYUNDAI-ROBOTICSLAB-003 | 85 0.001 | 66 0.006 | 97 0.050 | 125 0.001 | 335 0.051 | 158 0.062 | 192 0.005 | 215 0.034 | 112 0.221 | 253 0.019 | 354 0.995 | 270 0.236 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 180 | HZAILU-000 | 329 0.002 | 317 0.013 | 328 0.235 | 259 0.003 | 241 0.015 | 283 0.090 | 337 0.019 | 271 0.051 | 303 0.967 | 262 0.020 | 256 0.316 | 207 0.153 | 0.001 | 0.001 | 0.054 | | 0.001 | |
| 181 | HZAILU-001 | 299 0.002 | 259 0.011 | 241 0.095 | 232 0.002 | 371 0.113 | 287 0.092 | 243 0.008 | 447 0.183 | 354 0.986 | 432 0.196 | 530 1.000 | 403 0.679 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 182 | HZAILU-002 | 302 0.002 | 260 0.011 | 259 0.112 | 201 0.002 | 196 0.010 | 236 0.076 | 238 0.008 | 238 0.039 | 229 0.701 | 209 0.012 | 135 0.066 | 312 0.330 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 183 | HZAILU-003 | 251 0.001 | 248 0.011 | 234 0.094 | 145 0.001 | 145 0.007 | 217 0.072 | 240 0.008 | 234 0.038 | 201 0.545 | 193 0.009 | 126 0.062 | 166 0.119 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 184 | HZAILU-004 | 175 0.001 | 228 0.010 | 214 0.079 | 107 0.001 | 97 0.005 | 209 0.071 | 164 0.004 | 200 0.031 | 156 0.390 | 108 0.006 | 91 0.045 | 139 0.102 | 0.001 | 0.000 | 0.041 | | 0.000 | |

*Table 26:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T → FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 185 | HZAILU-005 | 173 0.001 | 221 0.010 | 184 0.065 | 90 0.001 | 59 0.004 | 186 0.067 | 132 0.003 | 154 0.025 | 141 0.329 | 72 0.004 | 58 0.033 | 115 0.094 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 186 | HZAILU-006 | 168 0.001 | 220 0.010 | 173 0.063 | 82 0.001 | 54 0.004 | 183 0.066 | 129 0.003 | 129 0.022 | 140 0.325 | 69 0.004 | 52 0.030 | 106 0.091 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 187 | I2V-001 | 378 0.003 | 355 0.015 | 230 0.092 | 338 0.005 | 251 0.017 | 182 0.066 | 428 0.049 | 586 0.966 | 477 1.000 | 509 0.905 | 459 1.000 | 483 0.983 | 0.000 | 0.001 | 0.040 | | 0.000 | |
| 188 | ICEHOUSE-000 | 424 0.004 | 460 0.028 | 427 0.793 | 391 0.010 | 376 0.125 | 423 0.158 | 484 0.089 | 571 0.852 | 481 1.000 | 393 0.111 | 582 1.000 | 355 0.470 | 0.001 | 0.002 | 0.053 | | 0.000 | |
| 189 | ICEHOUSE-001 | 192 0.001 | 193 0.009 | 193 0.069 | 249 0.002 | 185 0.009 | 165 0.063 | 255 0.009 | 246 0.041 | 146 0.348 | 220 0.013 | 167 0.098 | 242 0.179 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 190 | ICM-000 | 433 0.005 | 467 0.031 | 531 0.957 | 399 0.012 | 323 0.040 | 434 0.174 | 368 0.026 | 362 0.092 | 301 0.965 | 320 0.036 | 241 0.241 | 273 0.242 | 0.005 | 0.014 | 0.949 | | 0.006 | |
| 191 | ICM-001 | 384 0.003 | 366 0.016 | 380 0.375 | 339 0.005 | 300 0.028 | 361 0.112 | 389 0.029 | 474 0.251 | 494 1.000 | 452 0.307 | 386 1.000 | 443 0.878 | 0.001 | 0.002 | 0.096 | | 0.001 | |
| 192 | IDEMIA-003 | 454 0.007 | 475 0.034 | 532 0.957 | 428 0.018 | | 457 0.210 | 426 0.047 | 435 0.165 | | 400 0.123 | | 420 0.766 | 0.000 | 0.000 | 0.041 | | | |
| 193 | IDEMIA-004 | 449 0.006 | 471 0.032 | 525 0.947 | 427 0.018 | | 456 0.210 | 407 0.037 | 395 0.118 | 315 0.973 | 399 0.123 | | 419 0.766 | 0.000 | 0.000 | 0.041 | | | |
| 194 | IDEMIA-005 | 468 0.008 | 493 0.039 | 529 0.953 | 438 0.021 | | 462 0.217 | 416 0.044 | 423 0.150 | 323 0.977 | 403 0.130 | | 444 0.879 | 0.000 | 0.000 | 0.041 | | | |
| 195 | IDEMIA-006 | 482 0.009 | 532 0.072 | 546 0.969 | 449 0.030 | | 472 0.253 | 413 0.043 | 463 0.226 | 338 0.982 | 411 0.144 | | 412 0.733 | 0.000 | 0.000 | 0.041 | | | |
| 196 | IDEMIA-007 | 359 0.002 | 353 0.015 | 614 1.000 | 352 0.006 | 314 0.036 | 397 0.131 | 328 0.018 | 287 0.055 | 597 1.000 | 346 0.052 | 223 0.182 | 599 1.000 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 197 | IDEMIA-008 | 96 0.001 | 82 0.007 | 194 0.069 | 174 0.001 | 141 0.007 | 235 0.075 | 70 0.002 | 74 0.013 | 93 0.195 | 89 0.005 | 68 0.036 | 147 0.106 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 198 | IDEMIA-009 | 69 0.001 | 74 0.006 | 129 0.055 | 109 0.001 | 96 0.005 | 90 0.051 | 52 0.001 | 48 0.011 | 49 0.131 | 55 0.003 | 39 0.027 | 49 0.074 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 199 | IDEMIA-010 | 41 0.001 | 69 0.006 | 69 0.048 | 62 0.001 | 55 0.004 | 69 0.049 | 10 0.001 | 21 0.008 | 38 0.120 | 37 0.002 | 42 0.028 | 40 0.070 | 0.000 | 0.000 | 0.037 | | 0.000 | |
| 200 | IDEMIA-011 | 91 0.001 | 128 0.008 | 26 0.042 | 58 0.001 | 43 0.003 | 23 0.043 | 19 0.001 | 41 0.009 | 11 0.070 | 20 0.002 | 33 0.025 | 16 0.061 | 0.000 | 0.000 | 0.036 | | 0.000 | |
| 201 | IDEMIA-012 | 97 0.001 | 114 0.007 | 15 0.041 | 41 0.001 | 24 0.003 | 15 0.041 | 17 0.001 | 28 0.009 | 9 0.069 | 35 0.002 | 17 0.016 | 19 0.061 | 0.000 | 0.000 | 0.036 | | 0.000 | |
| 202 | IDEMIA-013 | 2 0.000 | 9 0.005 | 3 0.039 | 19 0.001 | 8 0.002 | 11 0.039 | 1 0.000 | 1 0.006 | 128 0.282 | 5 0.001 | 5 0.010 | 9 0.055 | 0.000 | 0.000 | 0.030 | | 0.000 | |
| 203 | IDENTY-000 | 619 1.000 | 610 0.999 | 599 1.000 | 536 0.991 | 433 1.000 | 537 0.992 | 615 1.000 | 598 0.999 | 605 1.000 | 521 0.991 | 370 1.000 | 490 0.992 | 0.001 | 0.001 | 0.052 | | 0.000 | |
| 204 | IDENTY-001 | 193 0.001 | 224 0.010 | 156 0.060 | 113 0.001 | 119 0.006 | 206 0.071 | 113 0.003 | 110 0.020 | 67 0.162 | 88 0.005 | 99 0.048 | 119 0.094 | 0.001 | 0.001 | 0.052 | | 0.000 | |
| 205 | IDENTY-002 | 90 0.001 | 84 0.007 | 98 0.050 | 99 0.001 | 94 0.005 | 81 0.050 | 109 0.003 | 94 0.016 | 63 0.152 | 86 0.005 | 93 0.047 | 70 0.080 | 0.000 | 0.000 | 0.045 | | 0.000 | |
| 206 | IMAGUS-002 | 593 0.220 | 577 0.301 | 559 0.988 | | | | 588 0.749 | 565 0.816 | 537 1.000 | | | | 0.004 | 0.008 | 0.550 | | | |
| 207 | IMAGUS-003 | 600 0.356 | 585 0.513 | 564 0.993 | | | | 591 0.807 | 575 0.909 | 506 1.000 | | | | 0.004 | 0.008 | 0.550 | | | |
| 208 | IMAGUS-005 | 298 0.002 | 284 0.012 | 359 0.311 | 349 0.006 | 272 0.022 | 399 0.132 | 332 0.018 | 310 0.066 | 255 0.836 | 294 0.029 | 213 0.161 | 267 0.231 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 209 | IMAGUS-006 | 305 0.002 | 323 0.014 | 345 0.285 | 307 0.004 | 267 0.019 | 360 0.112 | 336 0.019 | 319 0.069 | 273 0.896 | 292 0.028 | 212 0.161 | 282 0.260 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 210 | IMAGUS-007 | 307 0.002 | 314 0.013 | 360 0.313 | 298 0.004 | 273 0.022 | 372 0.117 | 357 0.022 | 328 0.073 | 271 0.892 | 304 0.031 | 215 0.169 | 287 0.265 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 211 | IMAGUS-008 | 573 0.086 | 544 0.093 | 352 0.297 | 437 0.021 | 356 0.081 | 376 0.119 | 601 0.974 | 560 0.774 | 404 0.996 | 479 0.520 | 371 1.000 | 367 0.518 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 212 | IMDS-SOFTWARE-000 | 124 0.001 | 113 0.007 | 199 0.071 | 172 0.001 | 170 0.008 | 224 0.073 | 217 0.006 | 209 0.033 | 518 1.000 | 217 0.013 | 236 0.216 | 528 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 213 | IMDS-SOFTWARE-001 | 127 0.001 | 89 0.007 | 175 0.064 | 166 0.001 | 189 0.009 | 306 0.096 | 221 0.006 | 213 0.034 | 521 1.000 | 256 0.019 | 281 0.424 | 509 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 214 | IMDS-SOFTWARE-002 | 125 0.001 | 112 0.007 | 200 0.071 | 173 0.001 | 171 0.008 | 223 0.073 | 216 0.006 | 208 0.033 | 519 1.000 | 218 0.013 | 235 0.216 | 529 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 215 | IMPERIAL-000 | 351 0.002 | 348 0.015 | 338 0.272 | 321 0.004 | | 307 0.097 | 366 0.026 | 315 0.068 | 443 0.999 | 332 0.042 | | 276 0.245 | 0.000 | 0.000 | 0.000 | | | |
| 216 | INCODE-000 | 559 0.049 | 548 0.100 | 528 0.951 | | | | 554 0.310 | 514 0.420 | 430 0.998 | | | | 0.001 | 0.004 | 0.173 | | | |
| 217 | INCODE-001 | 517 0.016 | 507 0.046 | 468 0.759 | | | | 537 0.211 | 492 0.296 | 531 1.000 | | | | 0.001 | 0.004 | 0.173 | | | |
| 218 | INCODE-002 | 521 0.017 | 509 0.048 | 492 0.841 | | | | 529 0.184 | 486 0.269 | 382 0.993 | | | | 0.000 | 0.001 | 0.066 | | | |
| 219 | INCODE-003 | 506 0.013 | 495 0.040 | 469 0.762 | | | | 525 0.167 | 481 0.264 | 463 0.999 | | | | 0.000 | 0.001 | 0.066 | | | |
| 220 | INCODE-004 | 400 0.003 | 383 0.017 | 401 0.468 | 384 0.008 | | 404 0.135 | 442 0.053 | 398 0.120 | 393 0.994 | 362 0.063 | | 303 0.313 | 0.000 | 0.001 | 0.066 | | | |
| 221 | INCODE-005 | 255 0.001 | 272 0.011 | 280 0.138 | 243 0.002 | 223 0.013 | 252 0.079 | 271 0.011 | 252 0.043 | 193 0.522 | 243 0.017 | 194 0.145 | 213 0.155 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 222 | INCODE-006 | 101 0.001 | 148 0.008 | 121 0.053 | 224 0.002 | 98 0.005 | 146 0.058 | 55 0.001 | 50 0.011 | 65 0.159 | 63 0.003 | 35 0.026 | 67 0.079 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 223 | INCODE-007 | 37 0.001 | 32 0.006 | 127 0.054 | 80 0.001 | 63 0.004 | 133 0.056 | 49 0.001 | 29 0.009 | 59 0.146 | 32 0.002 | 48 0.028 | 98 0.089 | 0.000 | 0.000 | 0.043 | | 0.000 | |
| 224 | INNOV8TIF-000 | 572 0.086 | 555 0.130 | 503 0.903 | 496 0.127 | 411 0.426 | 505 0.426 | 572 0.414 | 529 0.517 | 372 0.991 | 483 0.581 | 329 0.922 | 424 0.792 | 0.000 | 0.002 | 0.089 | | 0.001 | |
| 225 | INNOVATRICS-002 | 556 0.045 | 533 0.074 | 496 0.851 | | | | 546 0.233 | 498 0.310 | 538 1.000 | | | | 0.000 | 0.001 | 0.046 | | | |
| 226 | INNOVATRICS-003 | 536 0.026 | 517 0.055 | 494 0.843 | | | | 541 0.221 | 493 0.297 | 474 1.000 | | | | 0.000 | 0.001 | 0.046 | | | |
| 227 | INNOVATRICS-004 | 504 0.012 | 497 0.040 | 534 0.957 | | | | 507 0.131 | 462 0.222 | 328 0.980 | | | | 0.000 | 0.001 | 0.046 | | | |
| 228 | INNOVATRICS-005 | 350 0.002 | 332 0.014 | 385 0.400 | 327 0.005 | | 352 0.109 | 397 0.034 | 356 0.089 | 257 0.845 | 341 0.047 | | 279 0.251 | 0.000 | 0.001 | 0.041 | | | |
| 229 | INNOVATRICS-007 | 259 0.001 | 267 0.011 | 329 0.239 | 246 0.002 | 226 0.013 | 239 0.077 | 280 0.012 | 274 0.051 | 235 0.740 | 242 0.017 | 163 0.093 | 209 0.154 | 0.000 | 0.001 | 0.041 | | 0.000 | |
| 230 | INNOVATRICS-008 | 174 0.001 | 164 0.008 | 205 0.072 | 193 0.002 | 270 0.021 | 218 0.072 | 185 0.005 | 225 0.036 | 513 1.000 | 222 0.013 | 324 0.886 | 190 0.139 | 0.000 | 0.000 | 0.004 | | 0.000 | |

*Table 27:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T → FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 231 | INNOVATRICS-009 | [189] 0.001 | [230] 0.010 | [198] 0.070 | [187] 0.002 | [125] 0.006 | [215] 0.072 | [114] 0.003 | [156] 0.025 | [142] 0.338 | [148] 0.007 | [104] 0.050 | [148] 0.106 | 0.000 | 0.001 | 0.040 | | 0.001 | |
| 232 | INNOVATRICS-010 | [203] 0.001 | [233] 0.010 | [207] 0.074 | [179] 0.002 | [125] 0.006 | [216] 0.072 | [126] 0.003 | [157] 0.025 | [145] 0.347 | [155] 0.007 | [114] 0.054 | [156] 0.110 | 0.000 | 0.001 | 0.040 | | 0.001 | |
| 233 | INNOVATRICS-011 | [112] 0.001 | [106] 0.007 | [196] 0.070 | [118] 0.001 | [108] 0.005 | [77] 0.050 | [81] 0.002 | [83] 0.014 | [194] 0.523 | [75] 0.004 | [77] 0.039 | [88] 0.087 | 0.000 | 0.000 | 0.012 | | 0.000 | |
| 234 | INNOVATRICS-012 | [111] 0.001 | [65] 0.006 | [133] 0.056 | [61] 0.001 | [72] 0.004 | [63] 0.048 | [72] 0.002 | [52] 0.011 | [73] 0.168 | [57] 0.003 | [38] 0.027 | [77] 0.083 | 0.000 | 0.000 | 0.016 | | 0.000 | |
| 235 | INNOVATRICS-013 | [64] 0.001 | [57] 0.006 | [93] 0.050 | [111] 0.001 | [74] 0.004 | [32] 0.045 | [58] 0.001 | [91] 0.015 | [61] 0.147 | [123] 0.006 | [100] 0.048 | [95] 0.088 | 0.000 | 0.000 | 0.016 | | 0.000 | |
| 236 | INNOVATRICS-014 | [49] 0.001 | [13] 0.005 | [54] 0.045 | [26] 0.001 | [21] 0.003 | [62] 0.048 | [36] 0.001 | [17] 0.008 | [33] 0.106 | [27] 0.002 | [15] 0.015 | [28] 0.066 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 237 | INNOVATRICS-015 | [48] 0.001 | [12] 0.005 | [45] 0.044 | [16] 0.001 | [16] 0.002 | [46] 0.046 | [24] 0.001 | [14] 0.007 | [45] 0.126 | [21] 0.002 | [13] 0.015 | [63] 0.078 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 238 | INSPUR-000 | [73] 0.001 | [595] 0.606 | [105] 0.052 | [448] 0.030 | [409] 0.395 | [448] 0.195 | [213] 0.006 | [576] 0.915 | [487] 1.000 | [513] 0.936 | [467] 1.000 | [496] 0.998 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 239 | INSPUR-001 | [66] 0.001 | [61] 0.006 | [124] 0.054 | [149] 0.001 | [114] 0.006 | [126] 0.056 | [191] 0.005 | [172] 0.027 | [501] 1.000 | [172] 0.008 | [119] 0.057 | [462] 0.931 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 240 | INSPUR-002 | [57] 0.001 | [55] 0.006 | [80] 0.049 | [155] 0.001 | [103] 0.005 | [111] 0.054 | [198] 0.005 | [189] 0.029 | [451] 0.999 | [166] 0.008 | [110] 0.052 | [474] 0.957 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 241 | INTELIGENSIA-000 | [306] 0.002 | [295] 0.012 | [314] 0.201 | [318] 0.004 | [312] 0.033 | [380] 0.124 | [360] 0.024 | [335] 0.077 | [245] 0.784 | [350] 0.053 | [239] 0.235 | [280] 0.255 | 0.001 | 0.000 | 0.046 | | 0.001 | |
| 242 | INTELIGENSIA-001 | [92] 0.001 | [87] 0.007 | [142] 0.057 | [192] 0.002 | [168] 0.008 | [332] 0.103 | [197] 0.005 | [178] 0.028 | [172] 0.451 | [165] 0.008 | [139] 0.067 | [179] 0.129 | 0.001 | 0.000 | 0.046 | | 0.001 | |
| 243 | INTELLIVISION-001 | [548] 0.036 | [549] 0.102 | [548] 0.971 | [478] 0.057 | [396] 0.222 | [495] 0.333 | [551] 0.279 | [511] 0.404 | [485] 1.000 | [454] 0.328 | [307] 0.749 | [406] 0.685 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 244 | INTELLIVISION-002 | [490] 0.010 | [468] 0.031 | [522] 0.942 | [426] 0.018 | [354] 0.080 | [451] 0.200 | [516] 0.154 | [450] 0.196 | [446] 0.999 | [406] 0.134 | [283] 0.437 | [351] 0.460 | 0.001 | 0.000 | 0.043 | | 0.000 | |
| 245 | INTELRESEARCH-001 | [80] 0.001 | [93] 0.007 | [187] 0.067 | [151] 0.001 | [111] 0.005 | [102] 0.053 | [66] 0.001 | [67] 0.012 | [126] 0.276 | [93] 0.005 | [53] 0.030 | [72] 0.080 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 246 | INTEMA-000 | [123] 0.001 | [162] 0.008 | [63] 0.047 | [116] 0.001 | [113] 0.005 | [82] 0.051 | [87] 0.002 | [96] 0.017 | [520] 1.000 | [92] 0.005 | [252] 0.288 | [73] 0.081 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 247 | INTEMA-001 | [12] 0.000 | [592] 0.601 | [24] 0.042 | [65] 0.001 | [211] 0.012 | [42] 0.046 | [38] 0.001 | [539] 0.603 | [22] 0.092 | [73] 0.004 | [298] 0.715 | [56] 0.076 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 248 | INTOZI-001 | [239] 0.001 | [439] 0.024 | [295] 0.156 | [481] 0.064 | [403] 0.277 | [469] 0.242 | [310] 0.015 | [582] 0.938 | [510] 1.000 | [526] 0.998 | [509] 1.000 | [557] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 249 | INTSYSMSU-000 | [582] 0.145 | [434] 0.023 | [425] 0.557 | [484] 0.072 | | [398] 0.132 | [605] 0.998 | [599] 1.000 | [475] 1.000 | [527] 0.999 | | [500] 0.999 | 0.000 | 0.000 | 0.050 | | | |
| 250 | IONETWORKS-001 | [402] 0.003 | [296] 0.012 | [267] 0.118 | [308] 0.004 | [266] 0.019 | [257] 0.082 | [300] 0.014 | [259] 0.045 | [483] 1.000 | [235] 0.016 | [181] 0.129 | [296] 0.294 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 251 | IREX-000 | [417] 0.004 | [201] 0.010 | [444] 0.677 | [240] 0.002 | [213] 0.012 | [259] 0.082 | [384] 0.028 | [301] 0.060 | [291] 0.956 | [336] 0.044 | [255] 0.302 | [231] 0.170 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 252 | IRISID-001 | [209] 0.001 | [170] 0.009 | [249] 0.103 | [264] 0.003 | [242] 0.016 | [303] 0.096 | [293] 0.013 | [272] 0.051 | [166] 0.422 | [282] 0.025 | [198] 0.148 | [225] 0.162 | 0.000 | 0.001 | 0.046 | | 0.001 | |
| 253 | IRISID-002 | [165] 0.001 | [165] 0.008 | [246] 0.097 | [256] 0.003 | [235] 0.014 | [294] 0.093 | [266] 0.010 | [258] 0.045 | [160] 0.407 | [261] 0.020 | [183] 0.133 | [227] 0.164 | 0.000 | 0.001 | 0.044 | | 0.001 | |
| 254 | ISTL-001 | [354] 0.002 | [385] 0.017 | [248] 0.098 | [260] 0.003 | [361] 0.089 | [204] 0.070 | [286] 0.013 | [327] 0.073 | [152] 0.378 | [343] 0.048 | [367] 1.000 | [333] 0.407 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 255 | ISYSTEMS-002 | [448] 0.006 | [447] 0.026 | [493] 0.842 | | | | [470] 0.078 | [403] 0.126 | [421] 0.998 | | | | 0.002 | 0.002 | 0.142 | | | |
| 256 | ISYSTEMS-003 | [434] 0.005 | [430] 0.023 | [475] 0.789 | | | | [452] 0.058 | [386] 0.107 | [486] 1.000 | | | | 0.002 | 0.002 | 0.142 | | | |
| 257 | KAKAO-000 | [220] 0.001 | [230] 0.011 | [255] 0.109 | [244] 0.002 | [220] 0.013 | [246] 0.078 | [307] 0.015 | [290] 0.056 | [172] 0.462 | [257] 0.019 | [187] 0.141 | [215] 0.158 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 258 | KAKAO-001 | [199] 0.001 | [188] 0.009 | [57] 0.047 | [83] 0.001 | [82] 0.004 | [54] 0.047 | [97] 0.002 | [99] 0.017 | [62] 0.150 | [80] 0.004 | [87] 0.042 | [48] 0.074 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 259 | KAKAO-002 | [188] 0.001 | [175] 0.009 | [52] 0.045 | [48] 0.001 | [49] 0.004 | [31] 0.045 | [79] 0.002 | [84] 0.014 | [50] 0.134 | [58] 0.003 | [72] 0.037 | [43] 0.070 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 260 | KEDACOM-001 | [464] 0.007 | [479] 0.036 | [550] 0.972 | [456] 0.034 | | [465] 0.237 | [356] 0.022 | [325] 0.072 | [352] 0.986 | [352] 0.055 | | [299] 0.305 | 0.000 | 0.000 | 0.000 | | | |
| 261 | KEYLESS-000 | [98] 0.001 | [54] 0.006 | [185] 0.066 | [129] 0.001 | [313] 0.034 | [225] 0.073 | [205] 0.006 | [185] 0.029 | [356] 0.987 | [207] 0.011 | [350] 0.994 | [204] 0.149 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 262 | KLOUDSPOT-001 | [344] 0.002 | [411] 0.021 | [288] 0.145 | [379] 0.008 | [385] 0.170 | [318] 0.100 | [361] 0.024 | | [202] 0.550 | | | | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 263 | KNERON-000 | [442] 0.006 | [454] 0.027 | [425] 0.547 | [445] 0.028 | | [447] 0.195 | | | | | | | 0.000 | 0.000 | 0.000 | | | |
| 264 | KNERON-001 | [541] 0.029 | [597] 0.621 | [327] 0.228 | [500] 0.144 | [393] 0.207 | [480] 0.280 | | | | | | | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 265 | KNOWUTECH-000 | [120] 0.001 | [94] 0.007 | [195] 0.069 | [185] 0.002 | [173] 0.008 | [203] 0.070 | [250] 0.009 | [236] 0.038 | [134] 0.310 | [214] 0.012 | [161] 0.091 | [177] 0.129 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 266 | KOGNIZA-000 | [247] 0.001 | [237] 0.010 | [213] 0.078 | [102] 0.001 | [64] 0.004 | [130] 0.056 | [207] 0.006 | [179] 0.028 | [170] 0.444 | [101] 0.005 | [103] 0.050 | [90] 0.087 | 0.001 | 0.001 | 0.041 | | 0.000 | |
| 267 | LEBENTECH-000 | [605] 0.540 | [598] 0.644 | [565] 0.993 | [511] 0.294 | [418] 0.517 | [514] 0.560 | [610] 1.000 | [573] 0.882 | [545] 1.000 | [480] 0.527 | [379] 1.000 | [473] 0.954 | 0.007 | 0.012 | 0.076 | | 0.007 | |
| 268 | LIGHTVISION-000 | [382] 0.003 | [458] 0.028 | [431] 0.585 | [324] 0.005 | [378] 0.129 | [310] 0.098 | [377] 0.027 | [561] 0.797 | [491] 1.000 | [311] 0.033 | [397] 1.000 | [260] 0.214 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 269 | LINE-000 | [333] 0.002 | [324] 0.014 | [320] 0.214 | [340] 0.005 | [301] 0.029 | [343] 0.107 | [393] 0.030 | [367] 0.095 | | [339] 0.046 | [247] 0.278 | [619] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 270 | LINE-001 | [122] 0.001 | [117] 0.007 | [112] 0.053 | [203] 0.002 | [174] 0.008 | [275] 0.085 | [176] 0.004 | [171] 0.027 | [517] 1.000 | [187] 0.009 | [143] 0.072 | [526] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 271 | LINECLOVA-002 | [178] 0.001 | [123] 0.008 | [155] 0.059 | [188] 0.002 | [206] 0.011 | [143] 0.058 | [146] 0.004 | [405] 0.130 | [332] 0.980 | [329] 0.040 | [377] 1.000 | [408] 0.700 | 0.000 | 0.001 | 0.040 | | 0.001 | |
| 272 | LINECLOVA-003 | [71] 0.001 | [594] 0.601 | [225] 0.089 | [112] 0.001 | [178] 0.009 | [273] 0.085 | [100] 0.002 | [540] 0.606 | | [107] 0.006 | [342] 0.974 | [155] 0.110 | 0.000 | 0.000 | 0.024 | | 0.000 | |
| 273 | LOOKMAN-003 | [473] 0.008 | [489] 0.038 | | [459] 0.035 | | [467] 0.239 | [415] 0.043 | [390] 0.112 | | [382] 0.084 | | [317] 0.355 | 0.000 | 0.000 | | | | |
| 274 | LOOKMAN-004 | [475] 0.009 | [494] 0.039 | [552] 0.973 | | | | [418] 0.044 | [383] 0.105 | [320] 0.977 | | | | 0.000 | 0.000 | 0.000 | | | |
| 275 | LOOKMAN-005 | [467] 0.008 | [482] 0.036 | [551] 0.972 | [458] 0.035 | | [466] 0.237 | [391] 0.030 | [348] 0.086 | [322] 0.977 | [361] 0.062 | | [301] 0.308 | 0.000 | 0.000 | 0.000 | | | |
| 276 | LSSYTECH-000 | [59] 0.001 | [79] 0.007 | [154] 0.059 | [68] 0.001 | [65] 0.004 | [141] 0.057 | [48] 0.001 | [38] 0.009 | [71] 0.167 | [51] 0.003 | [34] 0.026 | [69] 0.080 | 0.000 | 0.000 | 0.040 | | 0.000 | |

*Table 28:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 277 | LUXAND-001 | 386 0.003 | 407 0.020 | 356 0.305 | 398 0.012 | 348 0.068 | 391 0.127 | 448 0.057 | 508 0.385 | 402 0.995 | 491 0.662 | 328 0.909 | 450 0.903 | 0.001 | 0.002 | 0.088 | | 0.001 | |
| 278 | MANTRA-000 | 264 0.001 | 219 0.010 | 453 0.705 | 362 0.007 | 286 0.024 | 358 0.112 | 261 0.010 | 243 0.041 | 561 1.000 | 296 0.029 | 201 0.152 | 506 1.000 | 0.002 | 0.001 | 0.591 | | 0.003 | |
| 279 | MAVERICKRESEARCH-000 | 491 0.010 | 444 0.025 | 389 0.419 | 415 0.014 | 328 0.045 | 413 0.144 | 474 0.084 | 442 0.176 | 374 0.992 | 397 0.122 | 273 0.394 | 375 0.546 | 0.004 | 0.001 | 0.055 | | 0.005 | |
| 280 | MAXVISION-000 | 346 0.002 | 342 0.015 | 364 0.319 | 312 0.004 | 334 0.051 | 321 0.101 | 385 0.028 | 466 0.237 | 242 0.764 | 412 0.149 | 355 0.997 | 378 0.557 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 281 | MAXVISION-001 | 148 0.001 | 129 0.008 | 123 0.054 | 126 0.001 | 258 0.018 | 140 0.057 | 158 0.004 | 160 0.025 | 101 0.210 | 144 0.007 | 336 0.951 | 134 0.100 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 282 | MAXVISION-002 | 153 0.001 | 100 0.007 | 114 0.053 | 146 0.001 | 188 0.009 | 197 0.069 | 141 0.004 | 131 0.022 | 88 0.188 | 140 0.007 | 325 0.895 | 144 0.104 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 283 | MAXVISION-003 | 164 0.001 | 127 0.008 | 125 0.054 | 160 0.001 | 221 0.013 | 147 0.059 | 204 0.006 | 239 0.039 | 119 0.237 | 205 0.011 | 264 0.348 | 158 0.111 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 284 | MAXVISION-004 | 147 0.001 | 183 0.009 | 43 0.044 | 56 0.001 | 317 0.037 | 125 0.055 | 90 0.002 | 109 0.020 | 40 0.122 | 71 0.004 | 346 0.990 | 76 0.082 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 285 | MEGVII-001 | 501 0.012 | 382 0.017 | | 587 1.000 | | | 465 0.072 | 371 0.097 | | | | | 0.002 | 0.000 | | | | |
| 286 | MEGVII-002 | 503 0.012 | 384 0.017 | 393 0.444 | 598 1.000 | | | 469 0.077 | 369 0.096 | 436 0.998 | | | | 0.002 | 0.000 | 0.033 | | | |
| 287 | MEGVII-003 | 162 0.001 | 146 0.008 | 101 0.050 | 101 0.001 | 79 0.004 | 137 0.057 | 95 0.002 | 55 0.011 | 58 0.145 | 62 0.003 | 41 0.027 | 61 0.078 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 288 | MEGVII-004 | 145 0.001 | 136 0.008 | 41 0.044 | 76 0.001 | 28 0.003 | 112 0.054 | 32 0.001 | 24 0.008 | 12 0.072 | 11 0.001 | 4 0.009 | 27 0.065 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 289 | MEGVII-005 | 146 0.001 | 138 0.008 | 40 0.044 | 47 0.001 | 131 0.007 | 22 0.042 | 33 0.001 | 40 0.009 | 82 0.181 | 111 0.006 | 388 1.000 | 163 0.117 | 0.001 | 0.000 | 0.044 | | 0.000 | |
| 290 | METSAKUURCOMPANY-000 | 129 0.001 | 95 0.007 | 171 0.063 | 182 0.002 | 358 0.083 | 148 0.059 | 229 0.007 | 223 0.036 | 449 0.999 | 379 0.081 | 353 0.995 | 405 0.683 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 291 | METSAKUURCOMPANY-001 | 194 0.001 | 133 0.008 | 245 0.097 | 196 0.002 | 172 0.008 | 219 0.072 | 251 0.009 | 197 0.031 | 470 1.000 | 198 0.010 | 145 0.073 | 436 0.841 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 292 | MIAXIS-001 | 252 0.001 | 211 0.010 | 322 0.218 | 234 0.002 | 212 0.012 | 350 0.108 | 239 0.008 | 205 0.033 | 290 0.954 | 230 0.014 | 203 0.153 | 235 0.176 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 293 | MIAXIS-002 | 253 0.001 | 276 0.011 | 291 0.148 | 236 0.002 | 219 0.013 | 330 0.102 | 241 0.008 | 226 0.036 | 200 0.536 | 240 0.017 | 226 0.192 | 226 0.163 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 294 | MICROFOCUS-003 | 609 0.594 | 605 0.781 | | 530 0.708 | | 533 0.907 | 598 0.931 | 589 0.979 | | 520 0.982 | | 489 0.991 | 0.001 | 0.005 | | | | |
| 295 | MICROFOCUS-004 | 606 0.576 | 604 0.758 | | 529 0.701 | | 532 0.904 | 606 0.999 | 587 0.975 | | 516 0.974 | | 485 0.989 | 0.001 | 0.005 | | | | |
| 296 | MICROFOCUS-005 | 601 0.424 | 593 0.601 | | 520 0.494 | | 524 0.777 | 595 0.835 | 580 0.928 | | 512 0.935 | | 484 0.985 | 0.001 | 0.005 | | | | |
| 297 | MICROFOCUS-006 | 602 0.427 | 591 0.583 | | 519 0.490 | | 527 0.782 | 602 0.978 | 579 0.923 | | 511 0.923 | | 478 0.971 | 0.001 | 0.005 | | | | |
| 298 | MICROSOFT-003 | 289 0.002 | 291 0.012 | | 296 0.004 | | 354 0.109 | 380 0.027 | 359 0.091 | | 321 0.036 | | 269 0.233 | 0.000 | 0.001 | | | | |
| 299 | MICROSOFT-004 | 275 0.001 | 289 0.012 | | 285 0.004 | | 355 0.109 | 367 0.026 | 352 0.087 | | 312 0.033 | | 264 0.222 | 0.000 | 0.001 | | | | |
| 300 | MICROSOFT-005 | 331 0.002 | 251 0.011 | 278 0.134 | 269 0.003 | | 315 0.099 | 364 0.025 | 322 0.070 | 209 0.582 | 286 0.027 | | 243 0.180 | 0.000 | 0.001 | 0.049 | | | |
| 301 | MICROSOFT-006 | 338 0.002 | 275 0.011 | 283 0.140 | 291 0.004 | | 319 0.100 | 276 0.012 | 227 0.037 | 153 0.379 | 307 0.032 | | 237 0.178 | 0.000 | 0.001 | 0.049 | | | |
| 302 | MOBBL-000 | 365 0.002 | 315 0.013 | 391 0.431 | 320 0.004 | 288 0.026 | 366 0.114 | 383 0.028 | 333 0.077 | 346 0.984 | 319 0.036 | 293 0.621 | 266 0.228 | 0.000 | 0.002 | 0.088 | | 0.001 | |
| 303 | MOBBL-001 | 500 0.012 | 397 0.019 | 394 0.446 | 446 0.029 | 330 0.047 | 439 0.178 | 399 0.035 | 342 0.080 | 342 0.983 | 357 0.058 | 294 0.622 | 291 0.275 | 0.000 | 0.002 | 0.088 | | 0.001 | |
| 304 | MOBBL-002 | 591 0.212 | 568 0.193 | 452 0.701 | 509 0.282 | 417 0.500 | 513 0.550 | 540 0.215 | 455 0.207 | 363 0.990 | 445 0.285 | 356 0.997 | 380 0.558 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 305 | MUKH-002 | 535 0.025 | 481 0.036 | 437 0.633 | 400 0.012 | 353 0.079 | 393 0.129 | 581 0.594 | 472 0.242 | 498 1.000 | 422 0.170 | 304 0.741 | 326 0.389 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 306 | NAVERCLOUD-000 | 163 0.001 | 80 0.007 | 146 0.058 | 167 0.001 | 89 0.005 | 171 0.064 | 137 0.003 | 115 0.020 | 311 0.971 | 95 0.005 | 76 0.039 | 114 0.094 | 0.000 | 0.001 | 0.041 | | 0.001 | |
| 307 | NEC-000 | 518 0.017 | 499 0.041 | 536 0.958 | 448 0.025 | | 470 0.243 | 472 0.079 | 417 0.140 | 326 0.979 | | | 357 0.474 | 0.001 | 0.002 | 0.890 | | | |
| 308 | NEC-001 | 527 0.021 | 518 0.056 | 544 0.967 | 454 0.033 | | 476 0.277 | 492 0.105 | 452 0.197 | 353 0.986 | 405 0.133 | | 354 0.468 | 0.005 | 0.003 | 0.934 | | | |
| 309 | NEC-002 | 169 0.001 | 178 0.009 | 375 0.355 | 281 0.003 | | 370 0.117 | 101 0.002 | 112 0.020 | 460 0.999 | 169 0.008 | | 402 0.676 | 0.000 | 0.001 | 0.041 | | | |
| 310 | NEC-003 | 250 0.001 | 216 0.010 | 374 0.345 | 290 0.004 | 218 0.013 | 378 0.120 | 96 0.002 | 98 0.017 | 251 0.822 | 174 0.008 | 69 0.036 | 401 0.668 | 0.000 | 0.001 | 0.041 | | 0.001 | |
| 311 | NEC-004 | 207 0.001 | 171 0.009 | 419 0.533 | 266 0.003 | 155 0.007 | 234 0.075 | 63 0.001 | 73 0.013 | 214 0.618 | 82 0.004 | 20 0.019 | 132 0.100 | 0.000 | 0.001 | 0.041 | | 0.001 | |
| 312 | NEC-005 | 141 0.001 | 130 0.008 | 202 0.071 | 197 0.002 | 93 0.005 | 227 0.073 | 57 0.001 | 62 0.012 | 222 0.669 | 65 0.003 | 19 0.019 | 127 0.099 | 0.000 | 0.001 | 0.040 | | 0.001 | |
| 313 | NEC-006 | 171 0.001 | 154 0.008 | 130 0.055 | 209 0.002 | 101 0.005 | 180 0.065 | 75 0.002 | 102 0.018 | 175 0.457 | 70 0.004 | 37 0.026 | 118 0.094 | 0.000 | 0.001 | 0.040 | | 0.001 | |
| 314 | NEC-007 | 87 0.001 | 56 0.006 | 76 0.048 | 106 0.001 | 27 0.003 | 72 0.049 | 35 0.001 | 31 0.009 | 54 0.137 | 29 0.002 | 6 0.010 | 26 0.065 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 315 | NEC-008 | 105 0.001 | 77 0.006 | 42 0.044 | 96 0.001 | 20 0.003 | 51 0.047 | 44 0.001 | 27 0.009 | 20 0.089 | 24 0.002 | 3 0.008 | 21 0.062 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 316 | NEC-009 | 68 0.001 | 53 0.006 | 12 0.040 | 77 0.001 | 10 0.002 | 28 0.044 | 14 0.001 | 10 0.007 | 10 0.070 | 9 0.001 | 2 0.005 | 7 0.052 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 317 | NEC-010 | 29 0.000 | 10 0.005 | 4 0.039 | 67 0.001 | 7 0.002 | 24 0.043 | 7 0.001 | 5 0.006 | 8 0.067 | 7 0.001 | 1 0.004 | 2 0.050 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 318 | NEC-011 | 14 0.000 | 2 0.005 | 7 0.039 | 7 0.001 | 215 0.012 | 9 0.037 | 6 0.001 | 7 0.006 | 55 0.140 | 2 0.001 | 313 0.789 | 5 0.052 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 319 | NEC-012 | 16 0.000 | 3 0.005 | 9 0.039 | 11 0.001 | 2 0.002 | 8 0.037 | 4 0.001 | 2 0.006 | 5 0.056 | 1 0.001 | 89 0.044 | 1 0.049 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 320 | NEUROTECHNOLOGY-003 | 529 0.022 | 500 0.042 | 538 0.960 | | | | 585 0.631 | 483 0.266 | 576 1.000 | | | | 0.000 | 0.001 | 0.131 | | | |
| 321 | NEUROTECHNOLOGY-004 | 436 0.005 | 403 0.020 | 547 0.970 | | | | 458 0.062 | 392 0.117 | 387 0.994 | | | | 0.000 | 0.001 | 0.131 | | | |
| 322 | NEUROTECHNOLOGY-005 | 415 0.004 | 440 0.024 | 500 0.892 | | | | 443 0.054 | 406 0.130 | 426 0.998 | | | | 0.000 | 0.000 | 0.030 | | | |

*Table 29:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T → FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 323 | NEUROTECHNOLOGY-006 | 522 0.018 | 504 0.045 | 434 0.602 | | | | 547 0.249 | 512 0.418 | | | | | 0.000 | 0.000 | | | | |
| 324 | NEUROTECHNOLOGY-007 | 404 0.004 | 413 0.021 | 429 0.794 | 387 0.009 | | 442 0.180 | 457 0.062 | 440 0.173 | 495 1.000 | 455 0.339 | | 563 1.000 | 0.001 | 0.001 | 0.041 | | | |
| 325 | NEUROTECHNOLOGY-008 | 326 0.002 | 331 0.014 | 397 0.451 | 300 0.004 | 282 0.023 | 326 0.101 | 437 0.053 | 341 0.080 | 512 1.000 | 318 0.035 | 254 0.293 | 256 0.203 | 0.000 | 0.001 | 0.052 | | 0.001 | |
| 326 | NEUROTECHNOLOGY-009 | 213 0.001 | 246 0.011 | 300 0.169 | 229 0.002 | 224 0.013 | 251 0.079 | 308 0.015 | 278 0.052 | 210 0.583 | 260 0.020 | 202 0.153 | 228 0.165 | 0.001 | 0.000 | 0.046 | | 0.000 | |
| 327 | NEUROTECHNOLOGY-010 | 157 0.001 | 200 0.009 | 157 0.060 | 157 0.001 | 152 0.007 | 190 0.068 | 259 0.010 | 231 0.037 | 123 0.269 | 199 0.010 | 151 0.075 | 171 0.126 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 328 | NEUROTECHNOLOGY-012 | 84 0.001 | 139 0.008 | 109 0.052 | 75 0.001 | 109 0.005 | 138 0.057 | 222 0.006 | 204 0.032 | 295 0.959 | 181 0.008 | 122 0.061 | 454 0.916 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 329 | NEUROTECHNOLOGY-013 | 86 0.001 | 135 0.008 | 66 0.047 | 69 0.001 | 84 0.004 | 127 0.056 | 159 0.004 | 140 0.023 | 137 0.316 | 134 0.006 | 92 0.046 | 395 0.641 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 330 | NEUROTECHNOLOGY-014 | 55 0.001 | 68 0.006 | 81 0.049 | 87 0.001 | 87 0.005 | 121 0.055 | 122 0.003 | 130 0.022 | 106 0.215 | 125 0.006 | 108 0.052 | 117 0.094 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 331 | NEUROTECHNOLOGY-015 | 43 0.001 | 45 0.006 | 65 0.047 | 42 0.001 | 73 0.004 | 57 0.048 | 128 0.003 | 150 0.024 | 75 0.170 | 126 0.006 | 102 0.049 | 87 0.086 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 332 | NEUROTECHNOLOGY-016 | 33 0.001 | 29 0.006 | 56 0.046 | 30 0.001 | 62 0.004 | 65 0.048 | 121 0.003 | 120 0.021 | 83 0.182 | 151 0.007 | 112 0.054 | 111 0.093 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 333 | NEWLAND-002 | 569 0.078 | 553 0.117 | 517 0.935 | | | | 573 0.437 | 522 0.466 | 450 0.999 | | | | 0.007 | 0.012 | 0.200 | | | |
| 334 | NEXTMETALABS-001 | 545 0.031 | 543 0.090 | 406 0.510 | 501 0.154 | 420 0.550 | 503 0.419 | 543 0.233 | 601 1.000 | 514 1.000 | 612 1.000 | 534 1.000 | 559 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 335 | NEXTMETALABS-002 | 72 0.001 | 98 0.007 | 96 0.050 | 230 0.002 | 389 0.182 | 241 0.077 | 150 0.004 | 402 0.125 | 94 0.198 | 404 0.131 | 382 1.000 | 365 0.510 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 336 | NEXTMETALABS-003 | 70 0.001 | 50 0.006 | 95 0.050 | 104 0.001 | 110 0.005 | 198 0.069 | 151 0.004 | 133 0.022 | 92 0.194 | 124 0.006 | 109 0.052 | 150 0.107 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 337 | NOBLIS-001 | 595 0.249 | 586 0.522 | 566 0.993 | | | | 608 1.000 | 610 1.000 | 528 1.000 | | | | 0.000 | 0.000 | 0.000 | | | |
| 338 | NOBLIS-002 | 585 0.179 | 582 0.392 | 556 0.981 | | | | 604 0.997 | 608 1.000 | 526 1.000 | | | | 0.000 | 0.000 | 0.000 | | | |
| 339 | NOMINDER-000 | 89 0.001 | 166 0.009 | 284 0.142 | 309 0.004 | 383 0.155 | 284 0.090 | 177 0.004 | 489 0.280 | 284 0.929 | 459 0.370 | 400 1.000 | 426 0.799 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 340 | NOMINDER-001 | 228 0.001 | 261 0.011 | 252 0.104 | 91 0.001 | 268 0.020 | 144 0.058 | 190 0.005 | 349 0.086 | 225 0.675 | 167 0.008 | 378 1.000 | 208 0.154 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 341 | NOMINDER-002 | 230 0.001 | 249 0.011 | 103 0.051 | 95 0.001 | 147 0.007 | 60 0.048 | 188 0.005 | 188 0.029 | 113 0.223 | 137 0.006 | 322 0.867 | 100 0.089 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 342 | NOMINDER-003 | 231 0.001 | 243 0.011 | 83 0.049 | 86 0.001 | 123 0.006 | 35 0.045 | 181 0.005 | 176 0.027 | 98 0.208 | 121 0.006 | 318 0.816 | 83 0.085 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 343 | NOMINDER-004 | 210 0.001 | 222 0.010 | 85 0.049 | 94 0.001 | 92 0.005 | 59 0.048 | 175 0.004 | 165 0.026 | 99 0.208 | 129 0.006 | 261 0.331 | 99 0.089 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 344 | NOMINDER-005 | 221 0.001 | 256 0.011 | 68 0.048 | 105 0.001 | 99 0.005 | 36 0.045 | 118 0.003 | 124 0.022 | 74 0.169 | 112 0.006 | 265 0.356 | 97 0.088 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 345 | NOTIONTAG-000 | 349 0.002 | 293 0.012 | 309 0.195 | 301 0.004 | 243 0.016 | 299 0.095 | 323 0.017 | 296 0.059 | 212 0.606 | 268 0.021 | 200 0.150 | 234 0.176 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 346 | NTECHLAB-003 | 447 0.006 | 429 0.023 | 403 0.498 | | | | 441 0.053 | 393 0.118 | 254 0.835 | | | | 0.000 | 0.000 | 0.040 | | | |
| 347 | NTECHLAB-004 | 429 0.005 | 395 0.019 | 404 0.500 | 377 0.008 | | 394 0.129 | 409 0.040 | 381 0.105 | 253 0.831 | 349 0.053 | | 284 0.263 | 0.000 | 0.000 | 0.040 | | | |
| 348 | NTECHLAB-005 | 427 0.005 | 388 0.018 | 376 0.360 | 382 0.008 | | 373 0.118 | 411 0.042 | 378 0.102 | 243 0.769 | 373 0.073 | | 295 0.294 | 0.000 | 0.000 | 0.040 | | | |
| 349 | NTECHLAB-006 | 416 0.004 | 375 0.017 | 373 0.340 | 373 0.007 | | 365 0.113 | 403 0.036 | 364 0.094 | 241 0.751 | 355 0.057 | | 283 0.260 | 0.000 | 0.000 | 0.040 | | | |
| 350 | NTECHLAB-007 | 363 0.002 | 298 0.012 | 361 0.318 | 319 0.004 | | 344 0.107 | 363 0.025 | 313 0.067 | 240 0.747 | 308 0.032 | | 265 0.223 | 0.000 | 0.000 | 0.042 | | | |
| 351 | NTECHLAB-008 | 263 0.001 | 202 0.010 | 289 0.147 | 280 0.003 | | 269 0.084 | 298 0.014 | 260 0.045 | 195 0.524 | 313 0.033 | | 244 0.183 | 0.000 | 0.000 | 0.044 | | | |
| 352 | NTECHLAB-009 | 172 0.001 | 140 0.008 | 272 0.128 | 225 0.002 | 228 0.013 | 230 0.074 | 183 0.005 | 134 0.022 | 167 0.423 | 231 0.015 | 171 0.109 | 195 0.142 | 0.000 | 0.000 | 0.041 | | 0.001 | |
| 353 | NTECHLAB-010 | 115 0.001 | 156 0.008 | 209 0.075 | 199 0.002 | 164 0.008 | 139 0.057 | 94 0.002 | 86 0.015 | 121 0.243 | 142 0.007 | 121 0.059 | 126 0.098 | 0.001 | 0.001 | 0.043 | | 0.000 | |
| 354 | NTECHLAB-011 | 88 0.001 | 92 0.007 | 167 0.062 | 163 0.001 | 159 0.007 | 91 0.051 | 107 0.003 | 85 0.015 | 107 0.219 | 186 0.009 | 148 0.074 | 105 0.091 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 355 | OMNIGARDE-000 | 93 0.001 | 83 0.007 | 172 0.063 | 154 0.001 | 136 0.007 | 188 0.068 | 155 0.004 | 141 0.023 | 120 0.243 | 157 0.007 | 115 0.055 | 185 0.136 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 356 | OMNIGARDE-001 | 45 0.001 | 44 0.006 | 61 0.047 | 70 0.001 | 61 0.004 | 120 0.055 | 80 0.002 | 90 0.015 | 53 0.135 | 67 0.003 | 54 0.030 | 60 0.078 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 357 | OMNIGARDE-002 | 60 0.001 | 47 0.006 | 62 0.047 | 57 0.001 | 69 0.004 | 129 0.056 | 56 0.001 | 47 0.011 | 35 0.111 | 45 0.002 | 24 0.022 | 66 0.079 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 358 | ONFIDO-000 | 322 0.002 | 204 0.010 | 465 0.748 | 283 0.003 | 281 0.023 | 298 0.094 | 290 0.013 | 219 0.035 | 312 0.971 | 298 0.029 | 242 0.244 | 252 0.194 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 359 | ONFIDO-001 | 257 0.001 | 125 0.008 | 371 0.338 | 223 0.002 | 255 0.018 | 255 0.081 | 268 0.010 | 144 0.023 | 232 0.744 | 227 0.014 | 192 0.142 | 206 0.152 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 360 | OPTIEXACTA-000 | 39 0.001 | 51 0.006 | 90 0.049 | 121 0.001 | 57 0.004 | 73 0.049 | 29 0.001 | 23 0.008 | 52 0.135 | 59 0.003 | 61 0.033 | 42 0.070 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 361 | PANGIAM-000 | 142 0.001 | 131 0.008 | 174 0.064 | 211 0.002 | 157 0.007 | 176 0.065 | 208 0.006 | 191 0.030 | 135 0.310 | 192 0.009 | 185 0.136 | 145 0.105 | 0.000 | 0.001 | 0.044 | | 0.001 | |
| 362 | PANGIAM-001 | 451 0.007 | 310 0.013 | 188 0.067 | 131 0.001 | 190 0.009 | 170 0.064 | 272 0.011 | 190 0.030 | 151 0.376 | 185 0.009 | 321 0.860 | 193 0.141 | 0.003 | 0.000 | 0.040 | | 0.000 | |
| 363 | PAPSAV1923-000 | 132 0.001 | 145 0.008 | 203 0.071 | 326 0.005 | 197 0.010 | 167 0.063 | 117 0.003 | 117 0.021 | 114 0.226 | 195 0.009 | 101 0.049 | 116 0.094 | 0.000 | 0.001 | 0.052 | | 0.003 | |
| 364 | PARAVISION-000 | 523 0.019 | 488 0.038 | 416 0.529 | 517 0.423 | | 512 0.529 | 483 0.088 | 437 0.170 | 454 0.999 | 474 0.470 | | 459 0.926 | 0.000 | 0.000 | 0.000 | | | |
| 365 | PARAVISION-001 | 406 0.004 | 409 0.020 | 366 0.322 | 516 0.414 | | 510 0.484 | 427 0.049 | 404 0.128 | 440 0.999 | 471 0.444 | | 413 0.739 | 0.000 | 0.000 | 0.000 | | | |
| 366 | PARAVISION-002 | 413 0.004 | 420 0.022 | 368 0.328 | 421 0.015 | | 436 0.175 | 429 0.050 | 396 0.119 | 341 0.983 | 378 0.080 | | 363 0.497 | 0.000 | 0.000 | 0.032 | | | |
| 367 | PARAVISION-003 | 396 0.003 | 398 0.019 | 331 0.244 | 422 0.015 | | 425 0.167 | 398 0.035 | 368 0.096 | 389 0.994 | 356 0.058 | | 298 0.296 | 0.000 | 0.000 | 0.032 | | | |
| 368 | PARAVISION-004 | 248 0.001 | 226 0.010 | 233 0.094 | 347 0.006 | | 359 0.112 | 262 0.010 | 232 0.038 | 539 1.000 | 249 0.018 | | 452 0.908 | 0.000 | 0.000 | 0.032 | | | |

*Table 30:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 369 | PARAVISION-005 | 234 0.001 | 210 0.010 | 192 0.068 | 365 0.007 | | 341 0.106 | 153 0.004 | 151 0.024 | 329 0.980 | 203 0.011 | | 181 0.132 | 0.000 | 0.000 | 0.038 | | | |
| 370 | PARAVISION-007 | 137 0.001 | 151 0.008 | 131 0.055 | 335 0.005 | 198 0.010 | 322 0.101 | 147 0.004 | 159 0.025 | 511 1.000 | 183 0.009 | 173 0.113 | 556 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 371 | PARAVISION-009 | 83 0.001 | 111 0.007 | 137 0.056 | 156 0.001 | 70 0.004 | 106 0.054 | 99 0.002 | 106 0.019 | 234 0.732 | 60 0.003 | 60 0.033 | 47 0.073 | 0.000 | 0.001 | 0.025 | | 0.000 | |
| 372 | PARAVISION-012 | 62 0.001 | 101 0.007 | 100 0.050 | 152 0.001 | 58 0.004 | 93 0.052 | 50 0.001 | 68 0.012 | 178 0.469 | 50 0.002 | 32 0.025 | 37 0.068 | 0.000 | 0.001 | 0.025 | | 0.000 | |
| 373 | PARAVISION-014 | 56 0.001 | 99 0.007 | 77 0.048 | 64 0.001 | 45 0.003 | 49 0.047 | 41 0.001 | 56 0.011 | 198 0.527 | 34 0.002 | 62 0.033 | 20 0.061 | 0.000 | 0.001 | 0.025 | | 0.000 | |
| 374 | PARAVISION-016 | 63 0.001 | 75 0.006 | 53 0.045 | 46 0.001 | 40 0.003 | 85 0.051 | 31 0.001 | 53 0.011 | 36 0.112 | 22 0.002 | 43 0.028 | 23 0.065 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 375 | PARAVISION-019 | 27 0.000 | 38 0.006 | 50 0.045 | 35 0.001 | 30 0.003 | 29 0.044 | 28 0.001 | 43 0.010 | 29 0.176 | 19 0.002 | 49 0.028 | 18 0.061 | 0.000 | 0.000 | 0.003 | | 0.000 | |
| 376 | PARAVISION-021 | 13 0.000 | 39 0.006 | 55 0.046 | 32 0.001 | 29 0.003 | 27 0.044 | 26 0.001 | 46 0.011 | 44 0.124 | 16 0.001 | 46 0.028 | 17 0.061 | 0.000 | 0.000 | 0.024 | | 0.000 | |
| 377 | PENTACOMP-000 | 613 0.975 | 609 0.978 | 549 0.972 | 534 0.948 | 428 0.980 | 534 0.951 | 603 0.996 | 592 0.998 | 369 0.990 | 515 0.955 | 473 1.000 | 475 0.960 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 378 | PENTACOMP-001 | 274 0.001 | 282 0.012 | 285 0.142 | 239 0.002 | 374 0.123 | 194 0.069 | 319 0.016 | 438 0.172 | 188 0.517 | 376 0.076 | 596 1.000 | 340 0.430 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 379 | PIXELALL-002 | 419 0.004 | 423 0.022 | 485 0.808 | 395 0.011 | | 444 0.187 | 491 0.105 | 509 0.388 | 533 1.000 | 484 0.602 | | 601 1.000 | 0.000 | 0.000 | 0.000 | | | |
| 380 | PIXELALL-003 | 324 0.002 | 330 0.014 | 407 0.510 | 357 0.006 | | 417 0.151 | 351 0.021 | 326 0.073 | 471 1.000 | 324 0.037 | | 377 0.554 | 0.000 | 0.000 | 0.000 | | | |
| 381 | PIXELALL-004 | 310 0.002 | 343 0.015 | 412 0.518 | 343 0.005 | | 419 0.152 | 331 0.018 | 340 0.079 | 502 1.000 | 344 0.051 | | 491 0.994 | 0.000 | 0.000 | 0.000 | | | |
| 382 | PIXELALL-005 | 288 0.002 | 250 0.011 | 334 0.256 | 403 0.012 | 296 0.028 | 415 0.146 | 274 0.012 | 267 0.050 | 525 1.000 | 287 0.027 | 228 0.203 | 505 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 383 | PRIVID-000 | 528 0.021 | 414 0.021 | 222 0.086 | 462 0.037 | 407 0.369 | 443 0.181 | 372 0.027 | 567 0.825 | 266 0.882 | 486 0.603 | 562 1.000 | 480 0.972 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 384 | PROTENET-000 | 564 0.063 | 562 0.190 | 432 0.597 | 490 0.096 | 415 0.448 | 507 0.462 | 589 0.794 | 568 0.834 | 327 0.980 | 501 0.804 | 519 1.000 | 446 0.890 | 0.015 | 0.111 | 0.216 | | 0.015 | |
| 385 | PROTENET-001 | 341 0.002 | 358 0.016 | 269 0.123 | 278 0.003 | 390 0.187 | 323 0.101 | 345 0.021 | 382 0.105 | 203 0.551 | 398 0.123 | 430 1.000 | 388 0.610 | 0.001 | 0.002 | 0.040 | | 0.000 | |
| 386 | PSL-000 | 26 0.000 | 43 0.006 | 113 0.053 | 81 0.001 | 77 0.004 | 101 0.053 | 98 0.002 | 119 0.021 | 97 0.208 | 102 0.005 | 113 0.054 | 125 0.097 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 387 | PSL-001 | 18 0.000 | 21 0.005 | 20 0.041 | 8 0.001 | 26 0.003 | 30 0.045 | 88 0.002 | 95 0.017 | 41 0.122 | 64 0.003 | 45 0.028 | 50 0.075 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 388 | PSL-002 | 6 0.000 | 6 0.005 | 18 0.041 | 13 0.001 | 13 0.002 | 41 0.046 | 39 0.001 | 54 0.011 | 25 0.095 | 39 0.002 | 29 0.023 | 64 0.079 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 389 | PSL-003 | 9 0.000 | 15 0.005 | 31 0.043 | 27 0.001 | 14 0.002 | 50 0.047 | 42 0.001 | 51 0.011 | 31 0.102 | 36 0.002 | 27 0.022 | 141 0.102 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 390 | PSL-004 | 7 0.000 | 11 0.005 | 29 0.043 | 28 0.001 | 17 0.002 | 55 0.047 | 30 0.001 | 36 0.009 | 30 0.101 | 23 0.002 | 28 0.022 | 55 0.076 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 391 | PSL-005 | 11 0.000 | 36 0.006 | 33 0.043 | 23 0.001 | 52 0.004 | 38 0.046 | 40 0.001 | 33 0.009 | 16 0.083 | 28 0.002 | 31 0.024 | 25 0.065 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 392 | PSL-006 | 8 0.000 | 8 0.005 | 22 0.041 | 15 0.001 | 15 0.002 | 34 0.045 | 20 0.001 | 22 0.008 | 19 0.085 | 15 0.001 | 21 0.019 | 30 0.067 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 393 | PTAKURATSATU-000 | 383 0.003 | 374 0.017 | 433 0.600 | 341 0.005 | 292 0.027 | 337 0.105 | 402 0.036 | 401 0.124 | 281 0.923 | 340 0.046 | 231 0.206 | 268 0.232 | 0.000 | 0.001 | 0.039 | | 0.000 | |
| 394 | PYRAMID-000 | 108 0.001 | 72 0.006 | 110 0.052 | 88 0.001 | 153 0.007 | 122 0.055 | 144 0.004 | 116 0.021 | 90 0.190 | 146 0.007 | 282 0.433 | 120 0.095 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 395 | PYRAMID-001 | 140 0.001 | 473 0.032 | 111 0.053 | 439 0.021 | 275 0.022 | 166 0.063 | 173 0.004 | 262 0.047 | 91 0.193 | 288 0.027 | 150 0.075 | 131 0.100 | 0.000 | 0.037 | 0.041 | | 0.018 | |
| 396 | QAZSMARTVISIONAI-000 | 21 0.000 | 25 0.005 | 25 0.042 | 24 0.001 | 36 0.003 | 21 0.042 | 43 0.001 | 44 0.010 | 24 0.094 | 48 0.002 | 82 0.041 | 31 0.067 | 0.000 | 0.000 | 0.035 | | 0.000 | |
| 397 | QAZSMARTVISIONAI-001 | 5 0.000 | 16 0.005 | 10 0.040 | 10 0.001 | 9 0.002 | 10 0.039 | 9 0.001 | 19 0.008 | 7 0.065 | 26 0.002 | 36 0.026 | 13 0.058 | 0.000 | 0.000 | 0.031 | | 0.000 | |
| 398 | QAZSMARTVISIONAI-002 | 1 0.000 | 5 0.005 | 1 0.039 | 5 0.001 | 4 0.002 | 5 0.037 | 3 0.001 | 6 0.006 | 2 0.052 | 6 0.001 | 9 0.014 | 8 0.052 | 0.000 | 0.000 | 0.031 | | 0.000 | |
| 399 | QAZSMARTVISIONAI-003 | 4 0.000 | 1 0.005 | 5 0.039 | 4 0.001 | 5 0.002 | 4 0.036 | 5 0.001 | 4 0.006 | 4 0.053 | 4 0.001 | 16 0.016 | 4 0.052 | 0.000 | 0.000 | 0.031 | | 0.000 | |
| 400 | QNAP-000 | 465 0.007 | 457 0.027 | 410 0.516 | 413 0.013 | 337 0.054 | 422 0.158 | 506 0.129 | 468 0.238 | 544 1.000 | 428 0.191 | 291 0.539 | 495 0.998 | 0.001 | 0.000 | 0.054 | | 0.000 | |
| 401 | QNAP-001 | 411 0.004 | 421 0.022 | 402 0.492 | 356 0.006 | 324 0.041 | 362 0.112 | 440 0.053 | 413 0.137 | 282 0.927 | 380 0.081 | 268 0.368 | 313 0.331 | 0.000 | 0.000 | 0.004 | | 0.000 | |
| 402 | QNAP-002 | 426 0.005 | 412 0.021 | 298 0.163 | 310 0.004 | 307 0.031 | 386 0.125 | 369 0.026 | 384 0.106 | 244 0.769 | 348 0.052 | 249 0.281 | 289 0.272 | 0.001 | 0.004 | 0.057 | | 0.001 | |
| 403 | QNAP-003 | 372 0.002 | 380 0.017 | 286 0.142 | 380 0.008 | 344 0.061 | 293 0.093 | 335 0.019 | 569 0.835 | 326 0.992 | 478 0.502 | 441 1.000 | 440 0.865 | 0.000 | 0.001 | 0.002 | | 0.001 | |
| 404 | QNAP-004 | 291 0.002 | 338 0.014 | 256 0.109 | 332 0.005 | 325 0.041 | 277 0.086 | 295 0.014 | 574 0.890 | 411 0.996 | 477 0.501 | 523 1.000 | 456 0.922 | 0.000 | 0.001 | 0.002 | | 0.001 | |
| 405 | QNAP-005 | 390 0.003 | 445 0.026 | 228 0.091 | 372 0.007 | 262 0.019 | 272 0.085 | 264 0.010 | 309 0.066 | 148 0.359 | 276 0.024 | 172 0.110 | 173 0.127 | 0.005 | 0.015 | 0.054 | | 0.005 | |
| 406 | QNAP-006 | 224 0.001 | 263 0.011 | 238 0.095 | 213 0.002 | 234 0.014 | 189 0.068 | 244 0.008 | 250 0.042 | 155 0.388 | 255 0.019 | 182 0.131 | 186 0.137 | 0.000 | 0.000 | 0.046 | | 0.000 | |
| 407 | QUANTASOFT-001 | 592 0.217 | 603 0.727 | | | | | 586 0.638 | | | | | | 0.000 | 0.000 | | | | |
| 408 | RANKONE-002 | 524 0.019 | 531 0.071 | | | | | 498 0.117 | 479 0.261 | | | | | 0.000 | 0.000 | | | | |
| 409 | RANKONE-003 | 525 0.019 | 529 0.068 | | | | | 497 0.117 | 477 0.255 | | | | | 0.000 | 0.000 | | | | |
| 410 | RANKONE-004 | 554 0.041 | 552 0.141 | | | | | 533 0.193 | 515 0.426 | | | | | 0.000 | 0.000 | | | | |
| 411 | RANKONE-005 | 478 0.009 | 498 0.041 | 558 0.986 | | | | 454 0.059 | 441 0.173 | 428 0.998 | | | | 0.000 | 0.000 | 0.489 | | | |
| 412 | RANKONE-006 | 430 0.005 | | 481 0.795 | | | | 404 0.037 | | 321 0.977 | | | | 0.002 | | 0.167 | | | |
| 413 | RANKONE-007 | 392 0.003 | 390 0.019 | 478 0.793 | | | | 354 0.022 | 365 0.095 | 302 0.967 | | | | 0.001 | 0.001 | 0.102 | | | |
| 414 | RANKONE-009 | 342 0.002 | 301 0.013 | 420 0.544 | 346 0.006 | | 403 0.134 | 326 0.017 | 331 0.076 | 307 0.969 | 359 0.062 | | 311 0.328 | 0.000 | 0.000 | 0.000 | | | |

*Table 31:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 415 | RANKONE-010 | 334 0.002 | 213 0.010 | 377 0.367 | 331 0.005 | 290 0.027 | 388 0.126 | 291 0.013 | 294 0.058 | 249 0.799 | 347 0.052 | 232 0.208 | 281 0.259 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 416 | RANKONE-011 | 227 0.001 | 273 0.011 | 319 0.214 | 289 0.004 | 263 0.019 | 263 0.082 | 245 0.008 | 264 0.048 | | 323 0.037 | 222 0.182 | 481 0.977 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 417 | RANKONE-012 | 196 0.001 | 236 0.010 | 265 0.117 | 268 0.003 | 230 0.014 | 196 0.069 | 232 0.007 | 285 0.053 | | 297 0.029 | 193 0.144 | 353 0.465 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 418 | RANKONE-013 | 94 0.001 | 86 0.007 | 182 0.065 | 136 0.001 | 175 0.008 | 104 0.054 | 166 0.004 | 214 0.034 | 407 0.996 | 250 0.018 | 189 0.141 | 196 0.142 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 419 | RANKONE-014 | 74 0.001 | 64 0.006 | 143 0.057 | 92 0.001 | 112 0.005 | 74 0.050 | 119 0.003 | 152 0.024 | | 189 0.009 | 155 0.081 | 582 1.000 | 0.000 | 0.000 | 0.006 | | 0.000 | |
| 420 | RANKONE-015 | 77 0.001 | 59 0.006 | 108 0.052 | 43 0.001 | 106 0.005 | 109 0.054 | 103 0.002 | 103 0.018 | | 145 0.007 | 88 0.042 | 154 0.109 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 421 | REALNETWORKS-000 | 553 0.040 | 537 0.078 | | | | | 544 0.233 | 501 0.319 | | | | | 0.001 | 0.000 | | | | |
| 422 | REALNETWORKS-001 | 552 0.040 | 538 0.078 | | | | | 545 0.233 | 500 0.319 | | | | | 0.001 | 0.000 | | | | |
| 423 | REALNETWORKS-002 | 549 0.039 | 536 0.078 | | | | | 542 0.231 | 499 0.315 | | | | | 0.001 | 0.000 | | | | |
| 424 | REALNETWORKS-003 | 532 0.024 | 525 0.062 | 472 0.768 | 452 0.031 | | 454 0.209 | 520 0.159 | 482 0.266 | 435 0.998 | 420 0.164 | | 364 0.500 | 0.001 | 0.000 | 0.009 | | | |
| 425 | REALNETWORKS-004 | 530 0.023 | 523 0.059 | 480 0.794 | 451 0.031 | | 460 0.213 | 519 0.158 | 480 0.263 | 456 0.999 | 423 0.170 | | 389 0.613 | 0.001 | 0.000 | 0.009 | | | |
| 426 | REALNETWORKS-005 | 340 0.002 | 316 0.013 | 390 0.427 | 315 0.004 | 277 0.023 | 329 0.102 | 379 0.027 | 329 0.074 | 309 0.970 | 322 0.037 | 237 0.223 | 261 0.215 | 0.000 | 0.000 | 0.006 | | 0.000 | |
| 427 | REALNETWORKS-006 | 202 0.001 | 214 0.010 | 342 0.279 | 248 0.002 | 201 0.010 | 247 0.078 | 299 0.014 | 282 0.053 | 331 0.980 | 236 0.016 | 177 0.120 | 210 0.154 | 0.000 | 0.000 | 0.009 | | 0.000 | |
| 428 | REALNETWORKS-007 | 185 0.001 | 198 0.009 | 335 0.258 | 190 0.002 | 179 0.009 | 214 0.072 | 256 0.009 | 253 0.043 | 325 0.979 | 211 0.012 | 285 0.463 | 192 0.140 | 0.000 | 0.000 | 0.009 | | 0.000 | |
| 429 | REALNETWORKS-008 | 135 0.001 | 132 0.008 | 211 0.078 | 198 0.002 | 149 0.007 | 285 0.091 | 209 0.006 | 187 0.029 | 304 0.967 | 177 0.008 | 141 0.068 | 176 0.129 | 0.000 | 0.000 | 0.042 | | 0.000 | |
| 430 | REALSENSE-001 | 78 0.001 | 90 0.007 | 176 0.064 | 135 0.001 | 102 0.005 | 88 0.051 | 106 0.003 | 75 0.013 | 111 0.221 | 77 0.004 | 44 0.028 | 59 0.077 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 431 | RECOGNITO-000 | 58 0.001 | 70 0.006 | 117 0.053 | 59 0.001 | 293 0.028 | 33 0.045 | 74 0.002 | 59 0.012 | 78 0.175 | 158 0.007 | 301 0.730 | 108 0.092 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 432 | RECOGNITO-001 | 52 0.001 | 46 0.006 | 25 0.048 | 143 0.001 | 60 0.004 | 76 0.050 | 46 0.001 | 30 0.009 | 275 0.899 | 66 0.003 | 63 0.033 | 51 0.075 | 0.000 | 0.000 | 0.032 | | 0.000 | |
| 433 | RECOGNITO-002 | 40 0.001 | 52 0.006 | 89 0.049 | 31 0.001 | 252 0.017 | 26 0.044 | 27 0.001 | 25 0.008 | 51 0.135 | 54 0.003 | 287 0.510 | 35 0.068 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 434 | REMARKAI-000 | 394 0.003 | 389 0.018 | 438 0.656 | 376 0.008 | | 416 0.148 | 444 0.055 | 397 0.120 | 452 0.999 | 369 0.069 | | 411 0.717 | 0.000 | 0.000 | 0.000 | | | |
| 435 | REMARKAI-000 | 472 0.008 | 466 0.030 | | | | | 504 0.127 | 453 0.203 | | | | | 0.000 | 0.001 | | | | |
| 436 | REMARKAI-002 | 470 0.008 | 464 0.029 | 483 0.800 | | | | 503 0.123 | 451 0.196 | 370 0.991 | | | | 0.000 | 0.001 | 0.017 | | | |
| 437 | RENDIP-000 | 236 0.001 | 346 0.015 | 388 0.417 | 353 0.006 | 294 0.028 | 270 0.084 | 273 0.012 | 295 0.059 | 272 0.892 | 269 0.022 | 224 0.185 | 229 0.167 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 438 | REVEALMEDIA-000 | 282 0.001 | 208 0.010 | 339 0.266 | 226 0.002 | 216 0.012 | 228 0.074 | 275 0.012 | 249 0.042 | 226 0.677 | 266 0.021 | 164 0.093 | 198 0.143 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 439 | RNIT-000 | 222 0.001 | 134 0.008 | 261 0.114 | 161 0.001 | 135 0.007 | 134 0.056 | 269 0.011 | 167 0.026 | 186 0.510 | 212 0.012 | 240 0.239 | 169 0.123 | 0.000 | 0.001 | 0.076 | | 0.000 | |
| 440 | ROC-016 | 79 0.001 | 62 0.006 | 91 0.050 | 34 0.001 | 104 0.005 | 48 0.047 | 93 0.002 | 97 0.017 | | 161 0.008 | 90 0.045 | 151 0.108 | 0.000 | 0.000 | 0.001 | | 0.000 | |
| 441 | ROC-017 | 65 0.001 | 63 0.006 | 82 0.049 | 52 0.001 | 76 0.004 | 40 0.046 | 112 0.003 | 127 0.022 | 77 0.174 | 139 0.006 | 78 0.039 | 107 0.092 | 0.000 | 0.000 | 0.032 | | 0.000 | |
| 442 | ROC-018 | 44 0.001 | 23 0.005 | 49 0.045 | 25 0.001 | 47 0.003 | 25 0.043 | 85 0.002 | 101 0.018 | 57 0.142 | 109 0.006 | 71 0.037 | 79 0.085 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 443 | ROC-019 | 28 0.000 | 14 0.005 | 27 0.042 | 22 0.001 | 37 0.003 | 18 0.042 | 47 0.001 | 34 0.009 | 27 0.098 | 43 0.002 | 40 0.027 | 41 0.070 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 444 | ROC-020 | 35 0.001 | 17 0.005 | 28 0.042 | 12 0.001 | 38 0.003 | 13 0.040 | 51 0.001 | 72 0.013 | 29 0.101 | 56 0.003 | 84 0.041 | 34 0.067 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 445 | ROC-021 | 22 0.000 | 7 0.005 | 19 0.041 | 17 0.001 | 23 0.003 | 12 0.040 | 15 0.001 | 16 0.007 | 15 0.083 | 25 0.002 | 146 0.073 | 15 0.060 | 0.000 | 0.000 | 0.003 | | 0.000 | |
| 446 | S1-000 | 345 0.002 | 373 0.017 | 333 0.250 | 345 0.005 | 287 0.025 | 282 0.090 | 382 0.028 | 347 0.085 | 546 1.000 | 342 0.047 | 449 1.000 | 581 1.000 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 447 | S1-001 | 385 0.003 | 327 0.014 | 317 0.206 | 254 0.003 | 256 0.018 | 238 0.077 | 312 0.015 | 277 0.052 | 349 0.985 | 252 0.019 | 184 0.136 | 203 0.148 | 0.001 | 0.000 | 0.035 | | 0.000 | |
| 448 | S1-002 | 205 0.001 | 194 0.009 | 219 0.083 | 124 0.001 | 202 0.010 | 123 0.055 | 199 0.005 | 193 0.031 | 87 0.186 | 153 0.007 | 315 0.792 | 435 0.841 | 0.000 | 0.000 | 0.028 | | 0.000 | |
| 449 | S1-003 | 219 0.001 | 203 0.010 | 250 0.103 | 168 0.001 | 146 0.007 | 153 0.060 | 246 0.008 | 228 0.037 | 493 1.000 | 229 0.014 | 274 0.396 | 518 1.000 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 450 | S1-004 | 186 0.001 | 190 0.009 | 118 0.053 | 89 0.001 | 100 0.005 | 71 0.049 | 148 0.004 | 153 0.025 | 415 0.997 | 128 0.006 | 128 0.064 | 544 1.000 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 451 | S1-005 | 198 0.001 | 182 0.009 | 71 0.048 | 50 0.001 | 83 0.004 | 58 0.048 | 161 0.004 | 135 0.022 | 390 0.994 | 116 0.006 | 132 0.065 | 449 0.895 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 452 | S1-006 | 180 0.001 | 179 0.009 | 59 0.047 | 117 0.001 | 86 0.005 | 70 0.049 | 86 0.002 | 93 0.016 | 458 0.999 | 97 0.005 | 80 0.040 | 339 0.423 | 0.000 | 0.000 | 0.033 | | 0.000 | |
| 453 | S1-007 | 151 0.001 | 137 0.008 | 58 0.047 | 108 0.001 | 66 0.004 | 44 0.046 | 73 0.002 | 81 0.014 | 69 0.165 | 79 0.004 | 64 0.034 | 57 0.077 | 0.000 | 0.001 | 0.039 | | 0.000 | |
| 454 | SANSAP-001 | 136 0.001 | 207 0.010 | 189 0.068 | 165 0.001 | 399 0.087 | 159 0.061 | 162 0.004 | 526 0.480 | 294 0.958 | 279 0.024 | 374 1.000 | 292 0.280 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 455 | SCANOVATE-000 | 428 0.005 | 502 0.045 | 424 0.555 | 457 0.035 | | 458 0.211 | 462 0.066 | 471 0.240 | 270 0.892 | 436 0.215 | | 328 0.400 | 0.000 | 0.001 | 0.057 | | | |
| 456 | SCANOVATE-001 | 435 0.005 | 496 0.040 | 430 0.580 | 450 0.031 | | 440 0.178 | 473 0.081 | 464 0.227 | 279 0.910 | 429 0.192 | | 332 0.404 | 0.000 | 0.001 | 0.044 | | | |
| 457 | SCYLLA-001 | 327 0.002 | 459 0.028 | 254 0.107 | 434 0.021 | 397 0.231 | 414 0.146 | 320 0.016 | 595 0.995 | 174 0.457 | 469 0.431 | | 470 0.953 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 458 | SENSETIME-000 | 357 0.002 | 361 0.016 | 413 0.523 | | | | 344 0.020 | 305 0.063 | 568 1.000 | | | | 0.004 | 0.000 | 0.042 | | | |
| 459 | SENSETIME-001 | 358 0.002 | 360 0.016 | | | | | 352 0.022 | 308 0.064 | | | | | 0.004 | 0.000 | | | | |
| 460 | SENSETIME-002 | 509 0.013 | 399 0.020 | 381 0.377 | 394 0.011 | | 333 0.104 | 303 0.014 | 181 0.028 | 386 0.994 | 306 0.032 | | 369 0.523 | 0.009 | 0.000 | 0.040 | | | |

*Table 32:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 461 | SENSETIME-003 | [81] 0.001 | [85] 0.007 | [281] 0.140 | [253] 0.003 | | [286] 0.091 | [65] 0.001 | [65] 0.012 | [179] 0.471 | [170] 0.008 | | [182] 0.133 | 0.000 | 0.000 | 0.041 | | | |
| 462 | SENSETIME-004 | [76] 0.001 | [97] 0.007 | [168] 0.062 | [237] 0.002 | | [271] 0.084 | [60] 0.001 | [70] 0.013 | [109] 0.220 | [120] 0.006 | | [160] 0.113 | 0.000 | 0.000 | 0.041 | | | |
| 463 | SENSETIME-005 | [32] 0.001 | [58] 0.006 | [79] 0.048 | [235] 0.002 | [142] 0.007 | [258] 0.082 | [83] 0.002 | [80] 0.014 | [68] 0.163 | [143] 0.007 | [107] 0.051 | [143] 0.104 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 464 | SENSETIME-006 | [30] 0.001 | [41] 0.006 | [47] 0.044 | [84] 0.001 | [48] 0.004 | [169] 0.064 | [62] 0.001 | [66] 0.012 | [431] 0.998 | [74] 0.004 | [65] 0.034 | [109] 0.093 | 0.000 | 0.000 | 0.025 | | 0.000 | |
| 465 | SENSETIME-007 | [23] 0.000 | [31] 0.006 | [23] 0.041 | [49] 0.001 | [34] 0.003 | [160] 0.062 | [37] 0.001 | [39] 0.009 | [461] 0.999 | [53] 0.003 | [30] 0.024 | [81] 0.085 | 0.000 | 0.000 | 0.025 | | 0.000 | |
| 466 | SENSETIME-008 | [24] 0.000 | [27] 0.006 | [34] 0.043 | [33] 0.001 | [33] 0.003 | [184] 0.067 | [34] 0.001 | [32] 0.009 | [158] 0.398 | [44] 0.002 | [23] 0.021 | [71] 0.080 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 467 | SENSETIME-009 | [17] 0.000 | [19] 0.005 | [17] 0.041 | [40] 0.001 | [31] 0.003 | [244] 0.078 | [11] 0.001 | [11] 0.007 | [262] 0.866 | [18] 0.001 | [11] 0.015 | [85] 0.085 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 468 | SENSETIME-010 | [10] 0.000 | [22] 0.005 | [11] 0.040 | [39] 0.001 | [35] 0.003 | [245] 0.078 | [13] 0.001 | [20] 0.008 | [26] 0.096 | [30] 0.002 | [22] 0.020 | [94] 0.087 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 469 | SENSETIME-011 | [19] 0.000 | [20] 0.005 | [16] 0.041 | [38] 0.001 | [32] 0.003 | [243] 0.078 | [12] 0.001 | [12] 0.007 | [263] 0.866 | [17] 0.001 | [12] 0.015 | [84] 0.085 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 470 | SENSETIME-012 | [15] 0.000 | [18] 0.005 | [37] 0.043 | [36] 0.001 | [41] 0.003 | [249] 0.078 | [8] 0.001 | [8] 0.006 | [21] 0.090 | [13] 0.001 | [25] 0.022 | [82] 0.085 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 471 | SERENDIPITY-000 | [232] 0.001 | [344] 0.015 | [348] 0.291 | [314] 0.004 | [366] 0.108 | [226] 0.073 | [279] 0.012 | [456] 0.213 | [238] 0.745 | [247] 0.018 | [390] 1.000 | [197] 0.142 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 472 | SEVENTHSENSE-001 | [309] 0.002 | [181] 0.009 | [180] 0.064 | [217] 0.002 | [310] 0.032 | [202] 0.070 | [172] 0.004 | [251] 0.043 | [116] 0.228 | [196] 0.010 | [295] 0.643 | [174] 0.127 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 473 | SHAMAN-003 | [577] 0.124 | [564] 0.172 | | | | | [575] 0.451 | [538] 0.597 | | | | | 0.020 | 0.011 | | | | |
| 474 | SHAMAN-004 | [594] 0.222 | [578] 0.319 | | | | | [584] 0.615 | [557] 0.754 | | | | | 0.020 | 0.011 | | | | |
| 475 | SHAMAN-006 | [551] 0.039 | [521] 0.058 | [519] 0.938 | | | | [512] 0.141 | [467] 0.237 | [313] 0.971 | | | | 0.020 | 0.011 | 0.869 | | | |
| 476 | SHAMAN-007 | [550] 0.039 | [520] 0.057 | | | | | [513] 0.141 | [470] 0.240 | | | | | 0.020 | 0.010 | | | | |
| 477 | SHENNONA-000 | [505] 0.012 | [540] 0.086 | [224] 0.088 | [414] 0.014 | [401] 0.257 | [297] 0.094 | [353] 0.022 | [433] 0.160 | [154] 0.383 | [418] 0.160 | [517] 1.000 | [381] 0.558 | 0.007 | 0.063 | 0.041 | | 0.000 | |
| 478 | SHUFTI-001 | [494] 0.011 | [472] 0.032 | [332] 0.246 | [455] 0.033 | [347] 0.065 | [432] 0.171 | [386] 0.028 | [379] 0.104 | [218] 0.650 | [353] 0.056 | [271] 0.382 | [274] 0.243 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 479 | SIAT-001 | [308] 0.002 | [580] 0.333 | | [316] 0.004 | | [316] 0.099 | [325] 0.017 | [505] 0.365 | | [302] 0.031 | | | 0.000 | 0.000 | | | | |
| 480 | SIAT-002 | [312] 0.002 | [583] 0.446 | | [514] 0.348 | | [328] 0.102 | [349] 0.021 | [525] 0.478 | | [460] 0.372 | | [457] 0.923 | 0.000 | 0.000 | | | | |
| 481 | SJTU-001 | [25] 0.000 | [35] 0.006 | [141] 0.057 | [119] 0.001 | [176] 0.009 | [154] 0.060 | [64] 0.001 | [60] 0.012 | [133] 0.306 | [110] 0.006 | [175] 0.114 | [138] 0.101 | 0.000 | 0.000 | 0.030 | | 0.000 | |
| 482 | SJTU-002 | [20] 0.000 | [24] 0.005 | [102] 0.050 | [114] 0.001 | [191] 0.009 | [149] 0.059 | [18] 0.001 | [35] 0.009 | [280] 0.911 | [99] 0.005 | [289] 0.516 | [172] 0.126 | 0.000 | 0.000 | 0.030 | | 0.000 | |
| 483 | SMILART-004 | [612] 0.965 | [608] 0.974 | | | | | [600] 0.968 | [588] 0.976 | | | | | 0.011 | 0.013 | | | | |
| 484 | SMILART-005 | | | | | | | | | | | | | 0.011 | 0.013 | | | | |
| 485 | SPARKTEK-000 | [116] 0.001 | [119] 0.007 | [294] 0.154 | [200] 0.002 | [151] 0.007 | [195] 0.069 | [115] 0.003 | [87] 0.015 | [162] 0.414 | [127] 0.006 | [97] 0.048 | [149] 0.106 | 0.000 | 0.000 | 0.062 | | 0.001 | |
| 486 | SPARKTEK-001 | [54] 0.001 | [76] 0.006 | [179] 0.064 | [60] 0.001 | [44] 0.003 | [113] 0.054 | [23] 0.001 | [26] 0.009 | [100] 0.210 | [41] 0.002 | [26] 0.022 | [54] 0.076 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 487 | SPARSH-001 | [103] 0.001 | [238] 0.010 | [147] 0.058 | [205] 0.002 | [357] 0.082 | [179] 0.065 | [157] 0.004 | [558] 0.759 | [319] 0.976 | [421] 0.166 | [604] 1.000 | [393] 0.633 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 488 | SQISOFT-001 | [412] 0.004 | [396] 0.019 | [339] 0.273 | [337] 0.005 | [291] 0.027 | [308] 0.097 | [508] 0.132 | [475] 0.252 | [247] 0.794 | [328] 0.040 | [299] 0.317 | [337] 0.420 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 489 | SQISOFT-002 | [177] 0.001 | [446] 0.026 | [215] 0.080 | [334] 0.005 | [404] 0.282 | [250] 0.078 | [388] 0.029 | | [276] 0.903 | [489] 0.621 | | [471] 0.953 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 490 | STAQU-000 | [457] 0.007 | [406] 0.020 | [435] 0.609 | [431] 0.020 | [338] 0.055 | [424] 0.159 | [455] 0.061 | [518] 0.443 | [489] 1.000 | [481] 0.535 | [340] 0.961 | [541] 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 491 | STCON-000 | [587] 0.183 | [600] 0.675 | [344] 0.284 | [527] 0.674 | [421] 0.704 | [521] 0.709 | [530] 0.184 | [551] 0.678 | [138] 0.318 | [493] 0.674 | [333] 0.941 | [409] 0.709 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 492 | STCON-001 | [46] 0.001 | [34] 0.006 | [51] 0.045 | [29] 0.001 | [183] 0.009 | [20] 0.042 | [61] 0.001 | [63] 0.012 | [48] 0.131 | [52] 0.003 | [269] 0.370 | [38] 0.069 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 493 | STCON-002 | [36] 0.001 | [30] 0.006 | [39] 0.043 | [20] 0.001 | [194] 0.010 | [16] 0.041 | [53] 0.001 | [49] 0.011 | [37] 0.115 | [46] 0.002 | [278] 0.415 | [33] 0.067 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 494 | STCON-003 | [38] 0.001 | [33] 0.006 | [38] 0.043 | [14] 0.001 | [19] 0.003 | [14] 0.041 | [45] 0.001 | [42] 0.010 | [34] 0.110 | [40] 0.002 | [55] 0.031 | [24] 0.065 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 495 | STCON-004 | [34] 0.001 | [26] 0.006 | [13] 0.041 | [6] 0.001 | [6] 0.002 | [7] 0.037 | [21] 0.001 | [18] 0.008 | [18] 0.083 | [14] 0.001 | [10] 0.014 | [11] 0.055 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 496 | STCON-005 | [31] 0.001 | [28] 0.006 | [14] 0.041 | [2] 0.000 | [3] 0.002 | [6] 0.037 | [22] 0.001 | [15] 0.007 | [17] 0.083 | [10] 0.001 | [50] 0.029 | [6] 0.052 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 497 | STELLARTECS-001 | [323] 0.002 | [280] 0.012 | [341] 0.277 | [195] 0.002 | [245] 0.016 | [231] 0.075 | [301] 0.014 | [273] 0.051 | [283] 0.927 | [264] 0.021 | [308] 0.750 | [211] 0.155 | 0.001 | 0.001 | 0.042 | | 0.000 | |
| 498 | SUMSUB-000 | [557] 0.045 | [551] 0.106 | [560] 0.989 | [491] 0.097 | [430] 0.999 | [492] 0.313 | [574] 0.445 | [555] 0.742 | [515] 1.000 | [505] 0.852 | [557] 1.000 | [494] 0.996 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 499 | SYNESIS-003 | [515] 0.016 | [431] 0.023 | [491] 0.825 | [407] 0.013 | | [406] 0.136 | [460] 0.065 | [400] 0.123 | [296] 0.960 | [375] 0.075 | | [304] 0.314 | 0.000 | 0.001 | 0.063 | | | |
| 500 | SYNESIS-003 | [583] 0.170 | [573] 0.235 | | | | | [580] 0.582 | [544] 0.646 | | | | | 0.006 | 0.015 | | | | |
| 501 | SYNESIS-005 | [471] 0.008 | [304] 0.013 | [463] 0.741 | [274] 0.003 | | [289] 0.092 | [362] 0.024 | [323] 0.072 | [343] 0.983 | [309] 0.032 | | [259] 0.214 | 0.001 | 0.000 | 0.135 | | | |
| 502 | T4ISB-000 | [484] 0.010 | [435] 0.023 | [395] 0.456 | [250] 0.003 | [372] 0.115 | [256] 0.081 | [313] 0.015 | [281] 0.053 | [184] 0.505 | [263] 0.021 | [309] 0.759 | [223] 0.161 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 503 | TCC-000 | [555] 0.045 | [584] 0.448 | [523] 0.944 | [521] 0.595 | [427] 0.978 | [531] 0.879 | [596] 0.868 | [592] 0.990 | [507] 1.000 | [523] 0.992 | [455] 1.000 | [503] 1.000 | 0.000 | 0.000 | 0.002 | | 0.000 | |
| 504 | TCC-001 | [443] 0.006 | [503] 0.045 | [409] 0.515 | [474] 0.054 | [388] 0.178 | [459] 0.212 | [522] 0.161 | [591] 0.987 | [505] 1.000 | [522] 0.991 | [429] 1.000 | [504] 1.000 | 0.001 | 0.001 | 0.033 | | 0.000 | |
| 505 | TECH5-001 | [408] 0.004 | [372] 0.017 | [428] 0.579 | [363] 0.007 | | [345] 0.107 | [447] 0.057 | [581] 0.935 | [543] 1.000 | [439] 0.244 | | [492] 0.994 | 0.000 | 0.000 | 0.006 | | | |
| 506 | TECH5-002 | [364] 0.002 | [247] 0.011 | [355] 0.304 | [273] 0.003 | [302] 0.029 | [281] 0.089 | [373] 0.027 | [321] 0.070 | [250] 0.803 | [326] 0.039 | [229] 0.205 | [345] 0.440 | 0.001 | 0.000 | 0.041 | | 0.000 | |

*Table 33:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T → FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 507 | TECH5-003 | [104]0.001 | [258]0.011 | [148]0.058 | [215]0.002 | [362]0.089 | [181]0.066 | [145]0.004 | [562]0.801 | [339]0.982 | [426]0.179 | [503]1.000 | [400]0.656 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 508 | TECHAINER-000 | [336]0.002 | [345]0.015 | [229]0.092 | [292]0.004 | [222]0.013 | [201]0.070 | [289]0.013 | [283]0.053 | [182]0.479 | [245]0.018 | [207]0.155 | [201]0.145 | 0.001 | 0.007 | 0.042 | | 0.002 | |
| 509 | TEVIAN-003 | [511]0.015 | [512]0.052 | | | | | [528]0.177 | [494]0.298 | | | | | 0.001 | 0.002 | | | | |
| 510 | TEVIAN-004 | [495]0.011 | [487]0.038 | | | | | [495]0.117 | [443]0.176 | | | | | 0.001 | 0.002 | | | | |
| 511 | TEVIAN-005 | [461]0.007 | [461]0.028 | [399]0.460 | | | | [480]0.087 | [420]0.144 | [299]0.961 | | | | 0.001 | 0.002 | 0.116 | | | |
| 512 | TEVIAN-006 | [347]0.002 | [267]0.011 | [260]0.113 | [255]0.003 | [229]0.013 | [208]0.071 | [257]0.009 | [202]0.032 | [163]0.418 | [238]0.016 | [162]0.093 | [469]0.951 | 0.001 | 0.000 | 0.062 | | 0.000 | |
| 513 | TEVIAN-007 | [277]0.001 | [196]0.009 | [218]0.082 | [216]0.002 | [192]0.009 | [187]0.067 | [189]0.005 | [126]0.022 | [132]0.293 | [190]0.009 | [131]0.065 | [168]0.122 | 0.000 | 0.000 | 0.062 | | 0.000 | |
| 514 | THAKAAIT-000 | [211]0.001 | [180]0.009 | [357]0.307 | [287]0.004 | [240]0.015 | [295]0.093 | [311]0.015 | [279]0.052 | [424]0.998 | [251]0.018 | [248]0.279 | [221]0.160 | 0.000 | 0.001 | 0.067 | | 0.001 | |
| 515 | TIGER-000 | [563]0.061 | [547]0.095 | | | | | [567]0.390 | [527]0.500 | | | | | 0.000 | 0.000 | | | | |
| 516 | TIGER-002 | [441]0.005 | [429]0.023 | [405]0.509 | | | | [477]0.086 | [430]0.158 | [445]0.999 | | | | 0.000 | 0.000 | 0.056 | | | |
| 517 | TIGER-003 | [440]0.005 | [428]0.023 | | | | | [476]0.086 | [429]0.158 | | | | | 0.000 | 0.000 | | | | |
| 518 | TIGER-004 | [360]0.002 | [376]0.017 | [253]0.106 | [374]0.007 | [352]0.076 | [232]0.075 | [322]0.017 | [324]0.072 | [161]0.410 | [387]0.095 | [385]1.000 | [329]0.402 | 0.001 | 0.009 | 0.041 | | 0.004 | |
| 519 | TONGYITRANS-000 | [453]0.007 | [426]0.022 | | | | | [467]0.074 | [389]0.112 | | | | | 0.003 | 0.001 | | | | |
| 520 | TONGYITRANS-001 | [452]0.007 | [425]0.022 | | | | | [461]0.066 | [377]0.101 | | | | | 0.003 | 0.001 | | | | |
| 521 | TOSHIBA-000 | [420]0.004 | [418]0.022 | [420]0.764 | | | | [456]0.062 | [394]0.118 | [399]0.995 | | | | 0.000 | 0.000 | 0.070 | | | |
| 522 | TOSHIBA-001 | [425]0.005 | [422]0.022 | | | | | [450]0.057 | [360]0.092 | | | | | 0.000 | 0.000 | | | | |
| 523 | TOSHIBA-002 | [176]0.001 | [241]0.011 | [36]0.043 | [45]0.001 | [22]0.003 | [47]0.047 | [67]0.001 | [89]0.015 | [23]0.094 | [31]0.002 | [56]0.031 | [29]0.066 | 0.001 | 0.000 | 0.040 | | 0.000 | |
| 524 | TOSHIBA-003 | [121]0.001 | [149]0.008 | [32]0.043 | [37]0.001 | [42]0.003 | [68]0.049 | [59]0.001 | [64]0.012 | [42]0.123 | [38]0.002 | [219]0.176 | [86]0.085 | 0.000 | 0.000 | 0.039 | | 0.000 | |
| 525 | TOSS-000 | [187]0.001 | [174]0.009 | [48]0.045 | [51]0.001 | [53]0.004 | [37]0.046 | [77]0.002 | [82]0.014 | [39]0.121 | [68]0.003 | [70]0.037 | [39]0.069 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 526 | TRUEFACE-000 | [391]0.003 | [321]0.014 | [325]0.221 | [368]0.007 | [284]0.024 | [290]0.092 | [330]0.018 | [304]0.062 | [265]0.881 | [299]0.030 | [227]0.194 | [249]0.188 | 0.001 | 0.001 | 0.047 | | 0.003 | |
| 527 | TSIGHT-000 | [3]0.000 | [4]0.005 | [2]0.039 | [3]0.000 | [1]0.002 | [3]0.036 | [2]0.001 | [3]0.006 | [1]0.051 | [3]0.001 | [8]0.013 | [3]0.050 | 0.000 | 0.000 | 0.031 | | 0.000 | |
| 528 | TTIGROUP-000 | [295]0.002 | [336]0.014 | [191]0.068 | [302]0.004 | [395]0.213 | [292]0.093 | [247]0.009 | [495]0.298 | [432]0.998 | [446]0.286 | [387]1.000 | [438]0.855 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 529 | TURINGTECHVIP-001 | [480]0.009 | [450]0.026 | [197]0.070 | [467]0.045 | [392]0.199 | [463]0.220 | [558]0.345 | [570]0.850 | [378]0.993 | [519]0.978 | [619]1.000 | [501]0.999 | 0.001 | 0.003 | 0.044 | | 0.000 | |
| 530 | UNISSEY-001 | [590]0.210 | [602]0.692 | [396]0.449 | [528]0.690 | [425]0.731 | [522]0.737 | [539]0.214 | [553]0.700 | [215]0.619 | [496]0.693 | [363]0.999 | [414]0.748 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 531 | UNISSEY-002 | [586]0.181 | [599]0.673 | [349]0.291 | [526]0.674 | [423]0.710 | [520]0.703 | [531]0.187 | [552]0.685 | [207]0.571 | [500]0.724 | [364]1.000 | [447]0.891 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 532 | UNISSEY-003 | [398]0.003 | [357]0.016 | [275]0.131 | [416]0.014 | [279]0.023 | [233]0.075 | [302]0.014 | [268]0.050 | [196]0.524 | [295]0.029 | [230]0.205 | [233]0.173 | 0.001 | 0.006 | 0.042 | | 0.011 | |
| 533 | UNIVERSEAI-000 | [95]0.001 | [78]0.007 | [86]0.049 | [275]0.003 | [169]0.008 | [159]0.062 | [136]0.003 | [108]0.020 | [72]0.168 | [176]0.008 | [165]0.094 | [137]0.101 | 0.001 | 0.001 | 0.043 | | 0.002 | |
| 534 | USEB-000 | [546]0.034 | [515]0.054 | [378]0.373 | [433]0.021 | [386]0.172 | [405]0.135 | [587]0.715 | [554]0.722 | [305]0.968 | [475]0.490 | [316]0.804 | [397]0.648 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 535 | USEB-001 | [156]0.001 | [192]0.009 | [235]0.094 | [98]0.001 | [160]0.008 | [98]0.052 | [142]0.004 | [139]0.023 | [335]0.982 | [94]0.005 | [299]0.728 | [91]0.087 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 536 | USEB-002 | [206]0.001 | [206]0.010 | [239]0.095 | [100]0.001 | [161]0.008 | [100]0.052 | [156]0.004 | [148]0.023 | [336]0.982 | [104]0.005 | [300]0.728 | [96]0.088 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 537 | USEB-003 | [155]0.001 | [189]0.009 | [236]0.094 | [78]0.001 | [134]0.007 | [89]0.051 | [139]0.003 | [122]0.021 | [310]0.971 | [103]0.005 | [276]0.398 | [92]0.087 | 0.000 | 0.001 | 0.042 | | 0.000 | |
| 538 | USEB-004 | [179]0.001 | [185]0.009 | [84]0.049 | [53]0.001 | [363]0.096 | [93]0.052 | [133]0.003 | [136]0.022 | [84]0.182 | [201]0.010 | [372]1.000 | [248]0.186 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 539 | VD-000 | [604]0.474 | [590]0.551 | | | | | [597]0.917 | [584]0.946 | | | | | 0.011 | 0.013 | | | | |
| 540 | VD-001 | [539]0.027 | [514]0.053 | | | | | [534]0.201 | [490]0.281 | | | | | 0.005 | 0.001 | | | | |
| 541 | VD-002 | [481]0.009 | [456]0.027 | [499]0.892 | [412]0.013 | [332]0.050 | [437]0.176 | [471]0.078 | [422]0.148 | [403]0.995 | [386]0.095 | [267]0.367 | [322]0.372 | 0.004 | 0.003 | 0.156 | | 0.002 | |
| 542 | VD-003 | [462]0.007 | [419]0.022 | [423]0.770 | [381]0.008 | [304]0.030 | [407]0.137 | [422]0.045 | [375]0.100 | [447]0.999 | [345]0.051 | [243]0.244 | [306]0.315 | 0.003 | 0.003 | 0.144 | | 0.002 | |
| 543 | VDFSOFTWARE-001 | [589]0.206 | [560]0.179 | [482]0.799 | [392]0.010 | [377]0.128 | [371]0.117 | [609]1.000 | [530]0.518 | [527]1.000 | [495]0.680 | [365]1.000 | [468]0.948 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 544 | VDFSOFTWARE-002 | [319]0.002 | [244]0.011 | [290]0.148 | [305]0.004 | [350]0.071 | [278]0.087 | [408]0.038 | [385]0.107 | [230]0.713 | [374]0.075 | [383]1.000 | [338]0.423 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 545 | VERIDAS-001 | [370]0.002 | [328]0.014 | [422]0.545 | [355]0.006 | [297]0.028 | [395]0.131 | [405]0.037 | [345]0.082 | [358]0.987 | [334]0.044 | [245]0.266 | [286]0.264 | 0.000 | 0.002 | 0.093 | | 0.001 | |
| 546 | VERIDAS-002 | [369]0.002 | [329]0.014 | [421]0.545 | [354]0.006 | [298]0.028 | [396]0.131 | [406]0.037 | [344]0.082 | [357]0.987 | [335]0.044 | [246]0.266 | [285]0.264 | 0.000 | 0.002 | 0.093 | | 0.001 | |
| 547 | VERIDAS-003 | [279]0.001 | [266]0.011 | [347]0.289 | [299]0.004 | [247]0.016 | [349]0.108 | [318]0.016 | [289]0.055 | [412]0.997 | [259]0.020 | [199]0.150 | [239]0.178 | 0.000 | 0.002 | 0.093 | | 0.001 | |
| 548 | VERIDAS-004 | [190]0.001 | [145]0.008 | [304]0.177 | [251]0.003 | [186]0.009 | [309]0.097 | [201]0.005 | [161]0.025 | [324]0.978 | [168]0.008 | [120]0.058 | [165]0.118 | 0.000 | 0.002 | 0.094 | | 0.001 | |
| 549 | VERIGRAM-000 | [356]0.002 | [359]0.016 | [122]0.053 | [206]0.002 | [156]0.007 | [95]0.052 | [180]0.005 | [182]0.028 | [103]0.212 | [132]0.006 | [157]0.088 | [78]0.083 | 0.001 | 0.004 | 0.046 | | 0.001 | |
| 550 | VERIGRAM-001 | [353]0.002 | [351]0.015 | [116]0.053 | [191]0.002 | [148]0.007 | [53]0.047 | [182]0.005 | [240]0.039 | [211]0.584 | [130]0.006 | [345]0.988 | [74]0.082 | 0.001 | 0.003 | 0.045 | | 0.001 | |
| 551 | VERIJELAS-000 | [599]0.354 | [581]0.369 | [545]0.968 | [488]0.086 | [391]0.191 | [487]0.292 | [590]0.799 | [564]0.813 | [442]0.999 | [453]0.324 | [332]0.933 | [383]0.589 | 0.002 | 0.001 | 0.070 | | 0.001 | |
| 552 | VIANTE-000 | [167]0.001 | [150]0.008 | [204]0.072 | [144]0.001 | [75]0.004 | [118]0.055 | [105]0.003 | [78]0.014 | [60]0.147 | [138]0.006 | [74]0.038 | [93]0.087 | 0.001 | 0.000 | 0.073 | | 0.001 | |

*Table 34:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 553 | VIANTE-001 | 159 0.001 | 142 0.008 | 201 0.071 | 55 0.001 | 39 0.003 | 66 0.049 | 54 0.001 | 37 0.009 | 32 0.105 | 47 0.002 | 253 0.293 | 32 0.067 | 0.001 | 0.000 | 0.073 | | 0.001 | |
| 554 | VIANTE-002 | 47 0.001 | 49 0.006 | 8 0.039 | 1 0.000 | 11 0.002 | 2 0.036 | 25 0.001 | 13 0.007 | 14 0.078 | 42 0.002 | 234 0.211 | 22 0.063 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 555 | VIANTE-003 | 42 0.001 | 48 0.006 | 6 0.039 | 9 0.001 | 12 0.002 | 1 0.036 | 16 0.001 | 9 0.007 | 13 0.076 | 33 0.002 | 176 0.117 | 14 0.060 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 556 | VIETTELAI-000 | 418 0.004 | 392 0.019 | 206 0.073 | 282 0.003 | 309 0.032 | 212 0.071 | 453 0.058 | 556 0.753 | 530 1.000 | 470 0.435 | 352 0.995 | 596 1.000 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 557 | VIETTELAI-001 | 314 0.002 | 270 0.011 | 149 0.058 | 417 0.014 | 316 0.036 | 276 0.086 | | | | | | | 0.000 | 0.000 | 0.040 | | 0.001 | |
| 558 | VIETTELAI-002 | 313 0.002 | 462 0.028 | 150 0.058 | 336 0.005 | 322 0.040 | 291 0.092 | 267 0.010 | 543 0.631 | 124 0.269 | 465 0.421 | 362 0.999 | 416 0.750 | 0.000 | 0.000 | 0.040 | | 0.001 | |
| 559 | VIETTELAI-003 | 75 0.001 | 60 0.006 | 70 0.048 | 44 0.001 | 95 0.005 | 75 0.050 | 169 0.004 | 149 0.023 | 252 0.823 | 122 0.006 | 211 0.160 | 104 0.090 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 560 | VIGILANTSOLUTIONS-003 | 567 0.069 | 560 0.151 | 535 0.957 | | | | 571 0.408 | 548 0.660 | 441 0.999 | | | | 0.000 | 0.001 | 0.127 | | | |
| 561 | VIGILANTSOLUTIONS-004 | 578 0.125 | 574 0.244 | 540 0.965 | | | | 579 0.549 | 566 0.817 | 406 0.996 | | | | 0.000 | 0.001 | 0.127 | | | |
| 562 | VIGILANTSOLUTIONS-005 | 477 0.009 | | 507 0.919 | | | | 565 0.386 | | 541 1.000 | | | | 0.000 | 0.001 | 0.127 | | | |
| 563 | VIGILANTSOLUTIONS-006 | 483 0.010 | | 508 0.920 | | | | 561 0.349 | | 529 1.000 | | | | 0.000 | 0.001 | 0.127 | | | |
| 564 | VIGILANTSOLUTIONS-007 | 393 0.003 | 379 0.017 | 511 0.924 | 408 0.013 | 349 0.068 | 435 0.175 | 387 0.028 | 353 0.088 | 405 0.996 | 381 0.081 | 270 0.371 | 327 0.391 | 0.000 | 0.001 | 0.127 | | 0.001 | |
| 565 | VIGILANTSOLUTIONS-008 | 379 0.003 | 381 0.017 | 506 0.912 | 418 0.014 | 351 0.072 | 438 0.178 | 340 0.020 | 334 0.077 | 444 0.999 | 391 0.104 | 275 0.398 | 366 0.511 | 0.000 | 0.001 | 0.127 | | 0.001 | |
| 566 | VINBIGDATA-001 | 150 0.001 | 110 0.007 | 232 0.093 | 175 0.001 | 207 0.011 | 142 0.057 | 314 0.016 | 299 0.059 | 285 0.931 | 248 0.018 | 334 0.944 | 290 0.274 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 567 | VINBIGDATA-002 | 61 0.001 | 37 0.006 | 126 0.054 | 93 0.001 | 116 0.006 | 86 0.051 | 82 0.002 | 58 0.012 | 66 0.160 | 87 0.005 | 258 0.317 | 53 0.076 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 568 | VINBIGDATA-003 | 51 0.001 | 42 0.006 | 78 0.048 | 71 0.001 | 80 0.004 | 45 0.046 | 92 0.002 | 71 0.013 | 56 0.141 | 78 0.004 | 79 0.039 | 44 0.070 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 569 | VINBIGDATA-004 | 50 0.001 | 40 0.006 | 64 0.047 | 66 0.001 | 67 0.004 | 43 0.046 | 69 0.002 | 45 0.010 | 47 0.130 | 61 0.003 | 57 0.032 | 36 0.068 | 0.000 | 0.000 | 0.041 | | 0.000 | |
| 570 | VIRTISHA-000 | 513 0.015 | 552 0.111 | 273 0.128 | 471 0.048 | 400 0.255 | 433 0.173 | 374 0.027 | 550 0.674 | 190 0.517 | 508 0.899 | 446 1.000 | 486 0.990 | 0.009 | 0.083 | 0.042 | | 0.000 | |
| 571 | VISIONBOX-000 | 296 0.002 | 268 0.011 | 466 0.750 | 325 0.005 | 250 0.017 | 248 0.078 | 327 0.017 | 292 0.057 | 368 0.990 | 272 0.023 | 195 0.146 | 224 0.162 | 0.000 | 0.001 | 0.043 | | 0.001 | |
| 572 | VISIONBOX-001 | 380 0.003 | 350 0.015 | 495 0.848 | 284 0.003 | 259 0.018 | 410 0.140 | 509 0.133 | 300 0.060 | 522 1.000 | 414 0.152 | 359 0.999 | 458 0.924 | 0.002 | 0.000 | 0.264 | | 0.001 | |
| 573 | VISIONBOX-002 | 280 0.001 | 229 0.010 | 287 0.143 | 220 0.002 | 208 0.011 | 163 0.062 | 270 0.011 | 248 0.042 | 233 0.730 | 232 0.015 | 179 0.125 | 191 0.140 | 0.001 | 0.001 | 0.056 | | 0.001 | |
| 574 | VISIONLABS-004 | 381 0.003 | 402 0.020 | 369 0.335 | | | | 449 0.057 | 431 0.159 | 269 0.888 | | | | 0.001 | 0.001 | 0.046 | | | |
| 575 | VISIONLABS-005 | 361 0.002 | 391 0.019 | 367 0.326 | | | | 430 0.050 | 421 0.147 | 268 0.886 | | | | 0.001 | 0.001 | 0.046 | | | |
| 576 | VISIONLABS-006 | 320 0.002 | 352 0.015 | 316 0.202 | 294 0.004 | | 302 0.096 | 371 0.026 | 358 0.090 | 220 0.668 | | | | 0.001 | 0.001 | 0.051 | | | |
| 577 | VISIONLABS-007 | 317 0.002 | 349 0.015 | 315 0.202 | 288 0.004 | | 301 0.095 | 370 0.026 | 357 0.090 | 221 0.668 | 305 0.031 | | 245 0.185 | 0.001 | 0.001 | 0.051 | | | |
| 578 | VISIONLABS-008 | 304 0.002 | 322 0.014 | 274 0.131 | 233 0.002 | | 253 0.081 | 285 0.013 | 276 0.051 | 180 0.475 | 239 0.017 | | 205 0.151 | 0.001 | 0.000 | 0.075 | | | |
| 579 | VISIONLABS-009 | 128 0.001 | 163 0.008 | 217 0.081 | 159 0.001 | | 207 0.071 | 167 0.004 | 163 0.025 | 248 0.797 | 178 0.008 | | 159 0.113 | 0.000 | 0.000 | 0.060 | | | |
| 580 | VISIONLABS-010 | 201 0.001 | 235 0.010 | 151 0.058 | 150 0.001 | 121 0.006 | 200 0.069 | 187 0.005 | 177 0.027 | | 164 0.008 | 116 0.055 | 152 0.109 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 581 | VISIONLABS-011 | 139 0.001 | 177 0.009 | 119 0.053 | 141 0.001 | 81 0.004 | 168 0.063 | 123 0.003 | 107 0.020 | | 83 0.004 | 66 0.034 | 103 0.090 | 0.000 | 0.000 | 0.032 | | 0.000 | |
| 582 | VISTEAM-000 | 432 0.005 | 401 0.020 | 353 0.301 | 397 0.011 | 367 0.108 | 381 0.124 | 475 0.085 | 454 0.205 | 267 0.886 | 401 0.128 | 375 1.000 | 316 0.345 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 583 | VISTEAM-001 | 438 0.005 | 417 0.021 | 354 0.302 | 410 0.013 | 369 0.110 | 392 0.127 | 493 0.109 | 523 0.469 | 264 0.878 | 415 0.154 | 526 1.000 | 323 0.374 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 584 | VIXVIZION-009 | 335 0.002 | 302 0.013 | 318 0.210 | 323 0.005 | 306 0.031 | 348 0.107 | 376 0.027 | 332 0.077 | 236 0.742 | 330 0.041 | 250 0.286 | 356 0.472 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 585 | VNIS-000 | 576 0.105 | 558 0.143 | 442 0.671 | 497 0.127 | 416 0.448 | 479 0.279 | 555 0.311 | 513 0.418 | 379 0.993 | 499 0.701 | 320 0.838 | 407 0.696 | 0.012 | 0.019 | 0.098 | | 0.062 | |
| 586 | VNPAY-001 | 216 0.001 | 124 0.008 | 177 0.064 | 328 0.005 | 214 0.012 | 274 0.085 | 203 0.006 | 170 0.026 | 127 0.280 | 210 0.012 | 134 0.066 | 153 0.109 | 0.001 | 0.001 | 0.048 | | 0.006 | |
| 587 | VNPT-001 | 325 0.002 | 337 0.014 | 279 0.135 | 247 0.002 | 283 0.023 | 210 0.071 | 292 0.014 | 316 0.068 | 231 0.715 | 317 0.035 | 347 0.990 | 319 0.362 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 588 | VNPT-002 | 286 0.002 | 293 0.012 | 144 0.058 | 133 0.001 | 130 0.006 | 108 0.054 | 215 0.006 | 201 0.032 | 130 0.284 | 159 0.007 | 144 0.072 | 122 0.096 | 0.001 | 0.000 | 0.042 | | 0.000 | |
| 589 | VNPT-003 | 245 0.001 | 122 0.007 | 160 0.060 | 138 0.001 | 126 0.006 | 116 0.054 | 212 0.006 | 183 0.028 | 297 0.960 | 154 0.007 | 125 0.062 | 123 0.096 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 590 | VNPT-004 | 246 0.001 | 121 0.007 | 153 0.059 | 128 0.001 | 115 0.006 | 114 0.054 | 194 0.005 | 164 0.026 | 274 0.899 | 149 0.007 | 117 0.057 | 110 0.093 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 591 | VNPT-005 | 249 0.001 | 115 0.007 | 132 0.056 | 110 0.001 | 105 0.005 | 97 0.052 | 168 0.004 | 145 0.023 | 259 0.853 | 100 0.005 | 95 0.047 | 101 0.090 | 0.001 | 0.000 | 0.041 | | 0.000 | |
| 592 | VOCORD-003 | 446 0.006 | 438 0.024 | 484 0.802 | 480 0.061 | | 445 0.188 | 501 0.122 | 429 0.155 | 429 0.998 | 417 0.157 | | 331 0.404 | 0.001 | 0.011 | 0.425 | | | |
| 593 | VOCORD-004 | 466 0.008 | 410 0.021 | 476 0.789 | 405 0.012 | | 390 0.127 | 562 0.353 | 439 0.173 | 496 1.000 | 431 0.193 | | 488 0.991 | 0.000 | 0.000 | 0.000 | | | |
| 594 | VOCORD-005 | 458 0.007 | 432 0.023 | 486 0.810 | 475 0.055 | | 452 0.206 | 518 0.156 | 407 0.130 | 418 0.997 | 409 0.138 | | 324 0.381 | 0.001 | 0.009 | 0.554 | | | |
| 595 | VOCORD-006 | 616 1.000 | 619 1.000 | 582 1.000 | 554 1.000 | | 583 1.000 | 619 1.000 | 606 1.000 | 580 1.000 | 579 1.000 | | 513 1.000 | 0.001 | 0.009 | 0.554 | | | |
| 596 | VOGIC-000 | 368 0.002 | 492 0.039 | 306 0.185 | 487 0.081 | 419 0.539 | 493 0.317 | 296 0.014 | 572 0.855 | 413 0.997 | 524 0.993 | 487 1.000 | 499 0.999 | 0.000 | 0.000 | 0.032 | | 0.000 | |
| 597 | VSOFT-001 | 510 0.014 | 513 0.052 | 457 0.729 | 442 0.026 | 381 0.149 | 475 0.265 | 515 0.153 | 491 0.288 | 362 0.990 | 434 0.202 | 360 0.999 | 382 0.573 | 0.000 | 0.002 | 0.094 | | 0.001 | |
| 598 | VSOFT-002 | 339 0.002 | 294 0.012 | 384 0.393 | 313 0.004 | 315 0.036 | 351 0.108 | 381 0.028 | 373 0.098 | 414 0.997 | 315 0.034 | 349 0.992 | 255 0.201 | 0.000 | 0.002 | 0.094 | | 0.001 | |

*Table 35:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| # | ALGORITHM | INVESTIGATION MODE | | | | | | IDENTIFICATION MODE | | | | | | FAILURE TO EXTRACT | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK ONE MISS RATE, FNIR(N, 0, 1) | | | | | | HIGH T → FPIR = 0.001, FNIR(N, T, L) | | | | | | FEATURES | | | | | |
| | | N=1.6M | | | | | | N=1.6M | | | | | | | | | | | |
| | GALLERY | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | VISA | MUGSHOT | MUGSHOT | MUGSHOT | VISA | BORDER | KIOSK |
| | PROBE | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK | MUGSHOT | WEBCAM | PROFILE | BORDER | BOR¿10YR | KIOSK |
| 599 | VTS-000 | 608 0.594 | 596 0.608 | 504 0.908 | 522 0.607 | 421 0.724 | 523 0.739 | 582 0.598 | 541 0.619 | 457 0.999 | 488 0.613 | 310 0.760 | 417 0.761 | 0.000 | 0.001 | 0.047 | | 0.000 | |
| 600 | VTS-001 | 235 0.001 | 217 0.010 | 297 0.158 | 348 0.006 | 257 0.018 | 242 0.077 | 287 0.013 | 275 0.051 | 385 0.994 | 270 0.022 | 190 0.141 | 251 0.192 | 0.000 | 0.000 | 0.040 | | 0.000 | |
| 601 | VTS-002 | 297 0.002 | 312 0.013 | 326 0.224 | 420 0.014 | 320 0.038 | 383 0.125 | 365 0.026 | 330 0.075 | 468 1.000 | 338 0.045 | 238 0.231 | 336 0.417 | 0.000 | 0.000 | 0.029 | | 0.000 | |
| 602 | VTS-003 | 133 0.001 | 107 0.007 | 178 0.064 | 186 0.002 | 181 0.009 | 103 0.053 | 228 0.007 | 207 0.033 | 503 1.000 | 226 0.014 | 338 0.954 | 394 0.635 | 0.000 | 0.001 | 0.029 | | 0.000 | |
| 603 | VTS-004 | 82 0.001 | 88 0.007 | 128 0.055 | 142 0.001 | 154 0.007 | 61 0.048 | 178 0.004 | 175 0.027 | 524 1.000 | 184 0.009 | 366 1.000 | 258 0.203 | 0.000 | 0.001 | 0.029 | | 0.000 | |
| 604 | VTS-005 | 67 0.001 | 73 0.006 | 107 0.052 | 122 0.001 | 120 0.006 | 56 0.047 | 195 0.005 | 193 0.030 | 423 0.998 | 147 0.007 | 317 0.811 | 133 0.100 | 0.000 | 0.001 | 0.029 | | 0.000 | |
| 605 | WINK-000 | 540 0.029 | 545 0.095 | 555 0.979 | 473 0.049 | 387 0.176 | 498 0.366 | 535 0.203 | 507 0.377 | 344 0.983 | 442 0.261 | 606 1.000 | 398 0.649 | 0.007 | 0.001 | 0.979 | | 0.000 | |
| 606 | WINK-001 | 376 0.003 | 510 0.049 | 474 0.785 | 504 0.206 | 405 0.296 | 423 0.253 | 401 0.036 | 409 0.131 | 408 0.996 | 438 0.237 | 326 0.903 | 321 0.372 | 0.001 | 0.036 | 0.043 | | 0.169 | |
| 607 | WINK-002 | 181 0.001 | 516 0.055 | 448 0.696 | 510 0.285 | 408 0.386 | 485 0.291 | 278 0.012 | 361 0.092 | 298 0.961 | 448 0.294 | 343 0.978 | 315 0.342 | 0.000 | 0.048 | 0.041 | | 0.247 | |
| 608 | WINK-003 | 343 0.002 | 368 0.016 | 343 0.281 | 228 0.002 | 180 0.009 | 340 0.106 | 249 0.009 | 233 0.038 | 490 1.000 | 202 0.010 | 152 0.077 | 421 0.774 | 0.001 | 0.002 | 0.058 | | 0.001 | |
| 609 | XAILIENT-000 | 371 0.002 | 265 0.011 | 282 0.140 | 241 0.002 | 236 0.015 | 191 0.068 | 400 0.035 | 297 0.059 | 192 0.521 | 290 0.027 | 221 0.181 | 253 0.196 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 610 | XFORWARDAI-000 | 332 0.002 | 325 0.014 | 212 0.078 | 295 0.004 | 237 0.015 | 296 0.094 | 306 0.015 | 286 0.053 | 168 0.434 | 265 0.021 | 210 0.159 | 230 0.169 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 611 | XFORWARDAI-001 | 315 0.002 | 299 0.013 | 138 0.056 | 263 0.003 | 182 0.009 | 260 0.082 | 179 0.005 | 184 0.028 | 169 0.441 | 171 0.008 | 127 0.062 | 170 0.123 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 612 | XFORWARDAI-002 | 300 0.002 | 286 0.012 | 72 0.048 | 245 0.002 | 138 0.007 | 240 0.077 | 110 0.003 | 92 0.016 | 191 0.519 | 98 0.005 | 83 0.041 | 129 0.099 | 0.000 | 0.000 | 0.000 | | 0.000 | |
| 613 | YISHENG-001 | 538 0.026 | 524 0.060 | | 479 0.058 | | 482 0.287 | 559 0.346 | 563 0.808 | | 492 0.666 | | 455 0.919 | 0.002 | 0.005 | | | | |
| 614 | YITU-002 | 287 0.002 | 223 0.010 | | | | | 324 0.017 | 265 0.049 | | | | | 0.000 | 0.000 | | | | |
| 615 | YITU-003 | 377 0.003 | 364 0.016 | | | | | 333 0.018 | 280 0.052 | | | | | 0.003 | 0.001 | | | | |
| 616 | YITU-004 | 191 0.001 | 161 0.008 | 497 0.864 | | | | 253 0.009 | 173 0.027 | 286 0.936 | | | | 0.000 | 0.000 | 0.000 | | | |
| 617 | YITU-005 | 348 0.002 | 340 0.014 | | | | | 263 0.010 | 203 0.032 | | | | | 0.003 | 0.001 | | | | |
| 618 | YOUVERSE-001 | 161 0.001 | 168 0.009 | 158 0.060 | 171 0.001 | 143 0.007 | 110 0.054 | 225 0.007 | 192 0.030 | 122 0.253 | 191 0.009 | 153 0.081 | 136 0.101 | 0.000 | 0.001 | 0.041 | | 0.000 | |
| 619 | YUAN-000 | 266 0.001 | 308 0.013 | 271 0.128 | 169 0.001 | 165 0.008 | 162 0.062 | 265 0.010 | 256 0.045 | 143 0.342 | 215 0.012 | 262 0.333 | 178 0.129 | 0.001 | 0.006 | 0.071 | | 0.000 | |

*Table 36:* **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L) | | ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS | | | | |
|---|---|---|---|---|---|---|
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 1 | 3DIVI-005 | 525 0.1353 | 524 0.1661 | 462 0.1907 | 446 0.2339 | 436 0.2995 |
| 2 | ACER-000 | 516 0.1182 | 514 0.1452 | 456 0.1711 | 440 0.2071 | 429 0.2534 |
| 3 | ADIANCE-002 | 278 0.0089 | 281 0.0124 | 269 0.0160 | 261 0.0291 | 260 0.1106 |
| 4 | ADVANCE-000 | 336 0.0138 | 338 0.0195 | 320 0.0243 | 305 0.0401 | 306 0.1278 |
| 5 | AFISBIOMETRICS-000 | 387 0.0201 | 390 0.0295 | 367 0.0378 | 356 0.0559 | 285 0.1173 |
| 6 | AFRENGINE-000 | 332 0.0134 | 334 0.0185 | 321 0.0245 | 326 0.0461 | 376 0.1717 |
| 7 | AFRENGINE-001 | 138 0.0025 | 122 0.0030 | 122 0.0040 | 139 0.0115 | 188 0.0919 |
| 8 | AFRENGINE-002 | 127 0.0023 | 116 0.0028 | 107 0.0034 | 147 0.0117 | 176 0.0901 |
| 9 | AFRENGINE-003 | 197 0.0036 | 174 0.0044 | 156 0.0054 | 138 0.0115 | 191 0.0921 |
| 10 | AIGROUP-000 | 255 0.0062 | 258 0.0095 | 246 0.0127 | 247 0.0269 | 296 0.1239 |
| 11 | AIZE-002 | 434 0.0374 | 431 0.0508 | 400 0.0653 | 399 0.1022 | 414 0.2232 |
| 12 | AIZE-006 | 427 0.0354 | 420 0.0453 | 395 0.0588 | 384 0.0867 | 383 0.1786 |
| 13 | AKUVOX-001 | 284 0.0093 | 288 0.0130 | 273 0.0168 | 273 0.0317 | 293 0.1234 |
| 14 | ALCHERA-003 | 514 0.1172 | 517 0.1550 | 458 0.1834 | 449 0.2389 | 445 0.3457 |
| 15 | ALCHERA-006 | 185 0.0034 | 184 0.0046 | 173 0.0060 | 196 0.0166 | 237 0.1034 |
| 16 | ALCHERA-008 | 135 0.0025 | 134 0.0032 | 128 0.0042 | 115 0.0098 | 138 0.0815 |
| 17 | ALGORITHMSINC-000 | 327 0.0132 | 317 0.0160 | 313 0.0235 | 306 0.0404 | 277 0.1156 |
| 18 | ALLGOVISION-000 | 483 0.0684 | 482 0.0878 | 437 0.1081 | 424 0.1386 | 409 0.2126 |
| 19 | ALLGOVISION-001 | 490 0.0782 | 489 0.1014 | 445 0.1215 | 432 0.1581 | 417 0.2270 |
| 20 | ALLGOVISION-005 | 349 0.0147 | 343 0.0204 | 327 0.0263 | 310 0.0410 | 271 0.1144 |
| 21 | ANKE-000 | 499 0.0939 | 496 0.1166 | 451 0.1394 | 437 0.1767 | 427 0.2486 |
| 22 | ANKE-002 | 395 0.0225 | 395 0.0315 | 373 0.0402 | 359 0.0602 | 345 0.1463 |
| 23 | ARATEK-001 | 235 0.0053 | 239 0.0078 | 233 0.0109 | 292 0.0367 | 338 0.1433 |
| 24 | ARMATURA-000 | 172 0.0031 | 160 0.0039 | 145 0.0051 | 111 0.0095 | 74 0.0613 |
| 25 | AWARE-003 | 511 0.1094 | 505 0.1279 | 452 0.1438 | 435 0.1754 | 420 0.2325 |
| 26 | AWARE-005 | 565 0.3375 | 563 0.3633 | 470 0.3963 | 459 0.4470 | 428 0.2519 |
| 27 | AWARE-007 | 316 0.0117 | 316 0.0158 | 295 0.0199 | 275 0.0319 | 273 0.1146 |
| 28 | AYONIX-002 | 593 0.7859 | 593 0.8241 | 477 0.8500 | 465 0.8697 | 458 0.8927 |
| 29 | BIOIDTECHSWISS-001 | 155 0.0028 | 154 0.0038 | 150 0.0053 | 157 0.0125 | 222 0.0991 |
| 30 | BIOMETRIK-001 | 436 0.0385 | 433 0.0520 | 410 0.0675 | 398 0.0984 | 405 0.2062 |
| 31 | CAMVI-004 | 430 0.0363 | 464 0.0713 | 431 0.0976 | 451 0.2503 | 434 0.2679 |
| 32 | CANON-001 | 196 0.0036 | 193 0.0051 | 191 0.0070 | 186 0.0154 | 190 0.0920 |
| 33 | CANON-002 | 178 0.0033 | 170 0.0043 | 167 0.0058 | 153 0.0121 | 134 0.0805 |
| 34 | CANON-003 | 84 0.0015 | 91 0.0021 | 92 0.0028 | 76 0.0068 | 88 0.0660 |
| 35 | CIB-000 | 273 0.0082 | 277 0.0121 | 266 0.0156 | 267 0.0299 | 447 0.3606 |
| 36 | CLEARVIEWAI-000 | 200 0.0037 | 202 0.0055 | 197 0.0074 | 190 0.0155 | 204 0.0968 |
| 37 | CLEARVIEWAI-001 | 132 0.0024 | 140 0.0035 | 142 0.0049 | 114 0.0096 | 57 0.0556 |
| 38 | CLEARVIEWAI-002 | 147 0.0027 | 152 0.0038 | 157 0.0054 | 125 0.0102 | 73 0.0605 |
| 39 | CLEARVIEWAI-003 | 92 0.0016 | 89 0.0021 | 91 0.0027 | 68 0.0065 | 49 0.0522 |
| 40 | CLOUDWALK-HR-000 | 86 0.0015 | 78 0.0017 | 71 0.0020 | 77 0.0068 | 102 0.0698 |
| 41 | CLOUDWALK-MT-000 | 89 0.0016 | 76 0.0017 | 66 0.0018 | 36 0.0046 | 35 0.0463 |
| 42 | CLOUDWALK-MT-001 | 80 0.0015 | 71 0.0016 | 57 0.0017 | 43 0.0048 | 56 0.0551 |
| 43 | CLOUDWALK-MT-002 | 81 0.0015 | 68 0.0015 | 53 0.0016 | 42 0.0047 | 68 0.0583 |
| 44 | COGENT-000 | 451 0.0426 | 436 0.0523 | 411 0.0688 | 409 0.1106 | 391 0.1872 |
| 45 | COGENT-001 | 452 0.0426 | 435 0.0523 | 412 0.0688 | 410 0.1106 | 390 0.1872 |
| 46 | COGENT-002 | 415 0.0318 | 417 0.0441 | 394 0.0587 | 404 0.1077 | 407 0.2080 |
| 47 | COGENT-003 | 416 0.0324 | 421 0.0453 | 405 0.0667 | 416 0.1241 | 421 0.2341 |
| 48 | COGENT-004 | 392 0.0206 | 396 0.0327 | 389 0.0523 | 411 0.1135 | 408 0.2116 |
| 49 | COGENT-005 | 248 0.0061 | 248 0.0087 | 239 0.0119 | 266 0.0299 | 292 0.1230 |
| 50 | COGENT-006 | 159 0.0029 | 163 0.0040 | 154 0.0053 | 145 0.0117 | 148 0.0827 |
| 51 | COGENT-007 | 136 0.0025 | 135 0.0033 | 138 0.0046 | 130 0.0107 | 122 0.0753 |
| 52 | COGNITEC-000 | 528 0.1374 | 521 0.1602 | 461 0.1858 | 442 0.2154 | 435 0.2784 |
| 53 | COGNITEC-001 | 493 0.0803 | 488 0.1013 | 444 0.1204 | 427 0.1495 | 410 0.2159 |
| 54 | COGNITEC-002 | 444 0.0402 | 438 0.0527 | 401 0.0654 | 388 0.0910 | 388 0.1810 |
| 55 | COGNITEC-003 | 441 0.0395 | 434 0.0520 | 398 0.0642 | 385 0.0874 | 366 0.1648 |
| 56 | COGNITEC-004 | 394 0.0219 | 394 0.0310 | 369 0.0385 | 349 0.0537 | 257 0.1100 |
| 57 | COGNITEC-005 | 246 0.0060 | 254 0.0093 | 256 0.0140 | 255 0.0284 | 202 0.0964 |
| 58 | COGNITEC-006 | 228 0.0049 | 233 0.0073 | 237 0.0113 | 237 0.0251 | 185 0.0916 |
| 59 | COGNITEC-007 | 194 0.0036 | 200 0.0053 | 204 0.0079 | 189 0.0155 | 137 0.0812 |
| 60 | COGNITEC-008 | 199 0.0036 | 206 0.0056 | 201 0.0077 | 184 0.0152 | 143 0.0822 |
| 61 | COGNITEC-009 | 202 0.0038 | 210 0.0059 | 206 0.0080 | 193 0.0158 | 151 0.0829 |
| 62 | COMPAL-000 | 236 0.0053 | 236 0.0075 | 551 1.0000 | 551 1.0000 | 463 1.0000 |
| 63 | CORSIGHT-000 | 211 0.0042 | 211 0.0059 | 207 0.0080 | 200 0.0176 | 238 0.1039 |
| 64 | CORSIGHT-001 | 218 0.0045 | 219 0.0064 | 210 0.0084 | 202 0.0180 | 196 0.0945 |
| 65 | CUBOX-001 | 112 0.0020 | 111 0.0027 | 104 0.0034 | 81 0.0071 | 64 0.0575 |
| 66 | CUBOX-002 | 123 0.0023 | 125 0.0030 | 115 0.0037 | 82 0.0072 | 43 0.0493 |
| 67 | CUBOX-003 | 133 0.0024 | 131 0.0032 | 124 0.0040 | 90 0.0074 | 44 0.0506 |
| 68 | CYBERLINK-000 | 446 0.0411 | 446 0.0561 | 417 0.0704 | 401 0.1028 | 403 0.2047 |
| 69 | CYBERLINK-001 | 437 0.0389 | 439 0.0532 | 414 0.0691 | 396 0.0969 | 384 0.1791 |
| 70 | CYBERLINK-002 | 299 0.0102 | 304 0.0145 | 294 0.0198 | 304 0.0395 | 300 0.1252 |
| 71 | CYBERLINK-003 | 234 0.0052 | 234 0.0074 | 223 0.0097 | 231 0.0232 | 294 0.1234 |
| 72 | CYBERLINK-004 | 225 0.0048 | 226 0.0068 | 226 0.0099 | 324 0.0458 | 350 0.1509 |

*Table 37:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES BELOW THRESHOLD, T | | ENROL MOST RECENT | | | | |
|---|---|---|---|---|---|---|
| FNIR(N, T> 0, R >L) | | DATASET: FRVT 2018 MUGSHOTS | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 73 | CYBERLINK-005 | [256]0.0063 | [260]0.0096 | [254]0.0134 | [301]0.0391 | [360]0.1563 |
| 74 | CYBERLINK-006 | [101]0.0018 | [102]0.0023 | [100]0.0030 | [94]0.0078 | [63]0.0571 |
| 75 | DAHUA-001 | [469]0.0566 | [466]0.0724 | [425]0.0868 | [412]0.1138 | [381]0.1773 |
| 76 | DAHUA-002 | [306]0.0105 | [305]0.0148 | [287]0.0188 | [260]0.0288 | [274]0.1150 |
| 77 | DAHUA-003 | [290]0.0097 | [294]0.0135 | [281]0.0177 | [263]0.0293 | [266]0.1127 |
| 78 | DAHUA-004 | [217]0.0044 | [224]0.0065 | [212]0.0086 | [195]0.0160 | [155]0.0850 |
| 79 | DAON-000 | [356]0.0157 | [355]0.0223 | [340]0.0289 | [355]0.0558 | [372]0.1699 |
| 80 | DAON-002 | [226]0.0048 | [227]0.0069 | [220]0.0095 | [220]0.0212 | [246]0.1072 |
| 81 | DAON-003 | [126]0.0023 | [130]0.0031 | [130]0.0042 | [119]0.0100 | [91]0.0673 |
| 82 | DATECH-000 | [232]0.0051 | [235]0.0074 | [227]0.0099 | [256]0.0284 | [299]0.1244 |
| 83 | DATECH-001 | [222]0.0046 | [220]0.0064 | [221]0.0096 | [238]0.0253 | [272]0.1145 |
| 84 | DECATUR-000 | [359]0.0170 | [358]0.0226 | [343]0.0302 | [325]0.0460 | [337]0.1430 |
| 85 | DEEPGLINT-001 | [128]0.0023 | [124]0.0030 | [123]0.0040 | [148]0.0118 | [189]0.0919 |
| 86 | DEEPSEA-001 | [425]0.0343 | [423]0.0459 | [393]0.0583 | [382]0.0799 | [375]0.1705 |
| 87 | DERMALOG-005 | [487]0.0696 | [481]0.0876 | [439]0.1135 | [431]0.1550 | [422]0.2356 |
| 88 | DERMALOG-006 | [439]0.0392 | [432]0.0514 | [399]0.0651 | [394]0.0951 | [368]0.1664 |
| 89 | DERMALOG-007 | [484]0.0687 | [479]0.0859 | [438]0.1104 | [428]0.1501 | [419]0.2296 |
| 90 | DERMALOG-008 | [421]0.0334 | [419]0.0452 | [397]0.0622 | [402]0.1057 | [418]0.2273 |
| 91 | DERMALOG-009 | [345]0.0144 | [341]0.0202 | [328]0.0265 | [313]0.0413 | [325]0.1371 |
| 92 | DERMALOG-010 | [227]0.0048 | [223]0.0065 | [211]0.0084 | [214]0.0203 | [205]0.0968 |
| 93 | DERMALOG-011 | [347]0.0145 | [348]0.0212 | [337]0.0276 | [322]0.0457 | [288]0.1188 |
| 94 | DERMALOG-012 | [339]0.0141 | [347]0.0207 | [332]0.0272 | [316]0.0425 | [251]0.1083 |
| 95 | DERMALOG-013 | [341]0.0142 | [346]0.0207 | [336]0.0275 | [317]0.0435 | [258]0.1105 |
| 96 | DERMALOG-014 | [344]0.0144 | [350]0.0214 | [341]0.0291 | [329]0.0472 | [282]0.1168 |
| 97 | DERMALOG-015 | [183]0.0033 | [186]0.0047 | [179]0.0062 | [158]0.0125 | [165]0.0871 |
| 98 | DERMALOG-016 | [95]0.0016 | [84]0.0020 | [84]0.0026 | [89]0.0074 | [80]0.0632 |
| 99 | DERMALOG-017 | [125]0.0023 | [120]0.0029 | [113]0.0037 | [96]0.0079 | [86]0.0656 |
| 100 | DILUSENSE-000 | [390]0.0205 | [392]0.0301 | [366]0.0374 | [351]0.0540 | [335]0.1426 |
| 101 | DILUSENSE-001 | [240]0.0057 | [242]0.0082 | [231]0.0106 | | [278]0.1158 |
| 102 | EINETWORKSINDIA-000 | [310]0.0107 | [309]0.0150 | [297]0.0203 | [298]0.0385 | [332]0.1402 |
| 103 | EINETWORKSINDIA-001 | [313]0.0110 | [315]0.0157 | [303]0.0218 | [321]0.0451 | [361]0.1569 |
| 104 | EINETWORKSINDIA-002 | [142]0.0026 | [149]0.0037 | [146]0.0052 | [129]0.0107 | [132]0.0797 |
| 105 | FACEPHI-000 | [219]0.0045 | [218]0.0063 | [205]0.0079 | [172]0.0141 | [87]0.0657 |
| 106 | FIRSTCREDITKZ-001 | [107]0.0019 | [108]0.0026 | [111]0.0036 | [106]0.0090 | [123]0.0756 |
| 107 | FUJITSULAB-000 | [346]0.0144 | [342]0.0203 | [335]0.0274 | [350]0.0538 | [379]0.1736 |
| 108 | FUJITSULAB-001 | [319]0.0122 | [329]0.0178 | [322]0.0247 | [362]0.0642 | [406]0.2076 |
| 109 | GORILLA-002 | [531]0.1534 | [532]0.1877 | [466]0.2172 | [453]0.2586 | [444]0.3357 |
| 110 | GORILLA-004 | [486]0.0696 | [485]0.0889 | [434]0.1045 | [423]0.1367 | [397]0.1967 |
| 111 | GORILLA-005 | [456]0.0450 | [451]0.0579 | [416]0.0701 | [397]0.0971 | [346]0.1471 |
| 112 | GORILLA-006 | [377]0.0192 | [378]0.0271 | [350]0.0328 | [340]0.0513 | [263]0.1110 |
| 113 | GORILLA-007 | [374]0.0187 | [375]0.0267 | [357]0.0345 | [343]0.0516 | [265]0.1126 |
| 114 | GORILLA-008 | [358]0.0167 | [359]0.0235 | [344]0.0305 | [327]0.0465 | [264]0.1117 |
| 115 | GRIAULE-000 | [340]0.0141 | [339]0.0198 | [323]0.0250 | [307]0.0404 | [339]0.1437 |
| 116 | GRIAULE-001 | [166]0.0030 | [171]0.0043 | [176]0.0061 | [181]0.0150 | [180]0.0907 |
| 117 | HIK-003 | [494]0.0824 | [490]0.1023 | [443]0.1194 | [430]0.1513 | [424]0.2401 |
| 118 | HIK-004 | [491]0.0792 | [486]0.0984 | [440]0.1142 | [426]0.1460 | [425]0.2403 |
| 119 | HIK-005 | [412]0.0309 | [414]0.0432 | [392]0.0555 | [387]0.0894 | [399]0.2000 |
| 120 | HISIGN-001 | [231]0.0051 | [231]0.0073 | [218]0.0093 | [211]0.0189 | [227]0.1011 |
| 121 | HYPERVERGE-001 | [164]0.0030 | [165]0.0041 | [162]0.0056 | [136]0.0113 | [162]0.0869 |
| 122 | HYPERVERGE-002 | [139]0.0025 | [138]0.0034 | [131]0.0043 | [61]0.0060 | [50]0.0535 |
| 123 | HYPERVERGE-003 | [233]0.0052 | [230]0.0071 | [222]0.0096 | [207]0.0185 | [216]0.0983 |
| 124 | HYUNDAI-ROBOTICSLAB-000 | [291]0.0097 | [292]0.0135 | [274]0.0169 | [254]0.0284 | [231]0.1016 |
| 125 | HYUNDAI-ROBOTICSLAB-001 | [281]0.0091 | [284]0.0125 | [276]0.0171 | [257]0.0284 | [244]0.1071 |
| 126 | HYUNDAI-ROBOTICSLAB-002 | [186]0.0034 | [196]0.0052 | [190]0.0070 | [164]0.0130 | [113]0.0720 |
| 127 | HYUNDAI-ROBOTICSLAB-003 | [180]0.0033 | [192]0.0051 | [187]0.0068 | [160]0.0127 | [104]0.0700 |
| 128 | HZAILU-000 | [338]0.0140 | [337]0.0194 | [324]0.0252 | [308]0.0408 | [284]0.1170 |
| 129 | HZAILU-001 | [253]0.0062 | [243]0.0082 | [230]0.0105 | [216]0.0204 | [239]0.1049 |
| 130 | HZAILU-002 | [241]0.0057 | [238]0.0076 | [225]0.0098 | [205]0.0184 | [182]0.0911 |
| 131 | HZAILU-003 | [239]0.0057 | [240]0.0078 | [232]0.0108 | [201]0.0179 | [203]0.0966 |
| 132 | HZAILU-004 | [165]0.0030 | [164]0.0041 | [169]0.0058 | [134]0.0111 | [105]0.0702 |
| 133 | I2V-001 | [355]0.0157 | [428]0.0488 | [441]0.1152 | [448]0.2377 | [446]0.3600 |
| 134 | ICEHOUSE-001 | [261]0.0066 | [255]0.0093 | [247]0.0128 | [225]0.0223 | [210]0.0976 |
| 135 | ICM-001 | [391]0.0206 | [389]0.0287 | [374]0.0405 | [381]0.0792 | [416]0.2254 |
| 136 | IDEMIA-003 | [423]0.0343 | [426]0.0468 | [426]0.0869 | [454]0.2689 | [452]0.4184 |
| 137 | IDEMIA-004 | [411]0.0296 | [407]0.0370 | [378]0.0441 | [360]0.0607 | [358]0.1545 |
| 138 | IDEMIA-005 | [429]0.0356 | [416]0.0436 | [391]0.0532 | [377]0.0745 | [386]0.1797 |
| 139 | IDEMIA-006 | [426]0.0348 | [413]0.0429 | [388]0.0520 | [372]0.0722 | [404]0.2062 |
| 140 | IDEMIA-007 | [328]0.0133 | [328]0.0177 | [306]0.0224 | [288]0.0353 | [331]0.1399 |
| 141 | IDEMIA-008 | [70]0.0013 | [70]0.0016 | [74]0.0020 | [46]0.0049 | [37]0.0466 |
| 142 | IDEMIA-009 | [51]0.0010 | [52]0.0012 | [45]0.0014 | [56]0.0058 | [53]0.0547 |
| 143 | IDEMIA-010 | [14]0.0007 | [10]0.0007 | [11]0.0008 | [12]0.0023 | [6]0.0190 |
| 144 | IDEMIA-011 | [32]0.0008 | [19]0.0009 | [16]0.0009 | [11]0.0022 | [13]0.0258 |

*Table 38:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N \geq 3\,000\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES BELOW THRESHOLD, T | | ENROL MOST RECENT | | | | |
|---|---|---|---|---|---|---|
| FNIR(N, T> 0, R >L) | | DATASET: FRVT 2018 MUGSHOTS | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 145 | IDEMIA-012 | [27]0.0008 | [17]0.0008 | [15]0.0009 | [9]0.0021 | [15]0.0266 |
| 146 | IDEMIA-013 | [1]0.0005 | [1]0.0005 | [1]0.0005 | [1]0.0012 | [1]0.0111 |
| 147 | IDENTY-001 | [114]0.0021 | [113]0.0027 | | [467]0.9606 | [461]0.9688 |
| 148 | IDENTY-002 | [104]0.0019 | [109]0.0026 | [114]0.0037 | [124]0.0102 | [178]0.0906 |
| 149 | IMAGUS-005 | [330]0.0133 | [332]0.0181 | [311]0.0234 | [291]0.0364 | [242]0.1064 |
| 150 | IMAGUS-006 | [331]0.0134 | [336]0.0187 | [317]0.0241 | [303]0.0392 | [275]0.1156 |
| 151 | IMAGUS-007 | [353]0.0157 | [357]0.0224 | [338]0.0280 | [318]0.0441 | [286]0.1176 |
| 152 | IMDS-SOFTWARE-001 | [220]0.0045 | [221]0.0064 | [215]0.0091 | [215]0.0204 | [289]0.1190 |
| 153 | IMDS-SOFTWARE-002 | [216]0.0044 | [216]0.0063 | [213]0.0087 | [212]0.0192 | [250]0.1083 |
| 154 | IMPERIAL-000 | [368]0.0183 | [366]0.0256 | [363]0.0354 | [375]0.0730 | [385]0.1791 |
| 155 | INCODE-003 | [523]0.1321 | [525]0.1668 | [463]0.1951 | [444]0.2319 | [437]0.3041 |
| 156 | INCODE-004 | [442]0.0400 | [442]0.0535 | [404]0.0658 | [390]0.0914 | [363]0.1616 |
| 157 | INCODE-005 | [271]0.0079 | [271]0.0109 | [257]0.0141 | [235]0.0244 | [181]0.0909 |
| 158 | INCODE-006 | [54]0.0011 | [55]0.0013 | [59]0.0017 | [32]0.0041 | [40]0.0485 |
| 159 | INCODE-007 | [35]0.0009 | [49]0.0011 | [44]0.0014 | [25]0.0036 | [54]0.0549 |
| 160 | INNOVATRICS-007 | [279]0.0090 | [280]0.0122 | [265]0.0156 | [239]0.0256 | [252]0.1089 |
| 161 | INNOVATRICS-008 | [184]0.0034 | [185]0.0046 | [180]0.0062 | [213]0.0203 | [253]0.1090 |
| 162 | INNOVATRICS-009 | [129]0.0023 | [114]0.0028 | [110]0.0035 | [123]0.0102 | [111]0.0714 |
| 163 | INNOVATRICS-010 | [130]0.0023 | [126]0.0030 | [116]0.0037 | [122]0.0102 | [131]0.0783 |
| 164 | INNOVATRICS-011 | [83]0.0015 | [81]0.0018 | [80]0.0023 | [80]0.0071 | [96]0.0685 |
| 165 | INNOVATRICS-012 | [73]0.0014 | [72]0.0016 | [70]0.0019 | [69]0.0065 | [82]0.0649 |
| 166 | INNOVATRICS-013 | [62]0.0012 | [58]0.0014 | [62]0.0018 | [65]0.0062 | [55]0.0550 |
| 167 | INNOVATRICS-014 | [34]0.0008 | [36]0.0010 | [27]0.0011 | [48]0.0050 | [41]0.0486 |
| 168 | INNOVATRICS-015 | [29]0.0008 | [24]0.0009 | [20]0.0010 | [41]0.0047 | [38]0.0467 |
| 169 | INSPUR-000 | [212]0.0042 | [213]0.0060 | [209]0.0083 | [204]0.0181 | [211]0.0977 |
| 170 | INSPUR-001 | [192]0.0035 | [191]0.0050 | [189]0.0070 | [167]0.0132 | [156]0.0851 |
| 171 | INSPUR-002 | [191]0.0034 | [198]0.0052 | [188]0.0069 | [163]0.0129 | [119]0.0742 |
| 172 | INTELRESEARCH-001 | [60]0.0011 | [66]0.0015 | [73]0.0020 | [91]0.0074 | [125]0.0765 |
| 173 | INTEMA-000 | [91]0.0016 | [87]0.0021 | [95]0.0028 | [109]0.0094 | [118]0.0741 |
| 174 | INTEMA-001 | [53]0.0010 | [38]0.0010 | [67]0.0019 | [75]0.0068 | [92]0.0675 |
| 175 | INTOZI-001 | [303]0.0103 | [310]0.0150 | [291]0.0193 | [276]0.0319 | [313]0.1310 |
| 176 | INTSYSMSU-000 | [608]0.9982 | [605]0.9984 | [481]0.9985 | [469]0.9987 | [462]0.9988 |
| 177 | IONETWORKS-001 | [289]0.0095 | [300]0.0142 | [299]0.0207 | [323]0.0457 | [359]0.1556 |
| 178 | IREX-000 | [375]0.0187 | [384]0.0277 | [370]0.0388 | [365]0.0674 | [347]0.1476 |
| 179 | IRISID-001 | [285]0.0094 | [293]0.0135 | [279]0.0175 | [268]0.0299 | [249]0.1081 |
| 180 | IRISID-002 | [266]0.0071 | [266]0.0099 | [248]0.0129 | [232]0.0235 | [213]0.0981 |
| 181 | ISTL-001 | [287]0.0094 | [286]0.0129 | [270]0.0161 | [248]0.0272 | [219]0.0987 |
| 182 | ISYSTEMS-002 | [471]0.0579 | [470]0.0779 | [430]0.0963 | [422]0.1360 | [413]0.2221 |
| 183 | ISYSTEMS-003 | [454]0.0434 | [452]0.0585 | [420]0.0792 | [415]0.1224 | [415]0.2243 |
| 184 | KAKAO-000 | [307]0.0105 | [307]0.0148 | [289]0.0192 | [279]0.0321 | [226]0.1007 |
| 185 | KAKAO-001 | [97]0.0017 | [97]0.0022 | [96]0.0029 | [97]0.0081 | [100]0.0689 |
| 186 | KAKAO-002 | [75]0.0014 | [79]0.0018 | [77]0.0022 | [54]0.0057 | [58]0.0559 |
| 187 | KEDACOM-001 | [363]0.0177 | [356]0.0223 | [325]0.0262 | [314]0.0419 | [318]0.1337 |
| 188 | KLOUDSPOT-001 | [361]0.0176 | [361]0.0243 | [356]0.0343 | [367]0.0685 | [315]0.1319 |
| 189 | KNOWUTECH-000 | [251]0.0061 | [250]0.0089 | [238]0.0118 | [221]0.0214 | [214]0.0982 |
| 190 | KOGNIZA-000 | [210]0.0041 | [207]0.0056 | [196]0.0073 | [175]0.0144 | [114]0.0723 |
| 191 | LIGHTVISION-000 | [378]0.0193 | [377]0.0270 | [364]0.0355 | [358]0.0591 | [380]0.1742 |
| 192 | LINECLOVA-002 | [137]0.0025 | [146]0.0036 | [137]0.0046 | [146]0.0117 | [141]0.0821 |
| 193 | LINECLOVA-003 | [121]0.0022 | [100]0.0022 | [136]0.0045 | [189]0.0154 | [217]0.0986 |
| 194 | LOOKMAN-003 | [424]0.0343 | [415]0.0433 | [386]0.0509 | [371]0.0712 | [352]0.1530 |
| 195 | LOOKMAN-005 | [398]0.0237 | [391]0.0298 | [362]0.0352 | [339]0.0509 | [317]0.1331 |
| 196 | LSSYTECH-000 | [40]0.0009 | [48]0.0011 | [46]0.0014 | [476]1.0000 | [81]0.0633 |
| 197 | LUXAND-001 | [459]0.0465 | [448]0.0573 | [409]0.0674 | [395]0.0954 | [370]0.1681 |
| 198 | MANTRA-000 | [252]0.0062 | [261]0.0097 | [259]0.0147 | [270]0.0304 | [234]0.1032 |
| 199 | MAXVISION-000 | [388]0.0202 | [385]0.0278 | [360]0.0351 | [341]0.0514 | [319]0.1337 |
| 200 | MAXVISION-001 | [154]0.0028 | [158]0.0039 | [144]0.0051 | [150]0.0119 | [171]0.0891 |
| 201 | MAXVISION-002 | [150]0.0027 | [141]0.0035 | [139]0.0047 | [152]0.0120 | [197]0.0946 |
| 202 | MAXVISION-003 | [206]0.0039 | [204]0.0055 | [192]0.0071 | [177]0.0145 | [99]0.0688 |
| 203 | MAXVISION-004 | [93]0.0016 | [90]0.0021 | [86]0.0027 | [99]0.0084 | [120]0.0743 |
| 204 | MEGVII-001 | [467]0.0558 | [465]0.0719 | [424]0.0861 | [418]0.1281 | [433]0.2600 |
| 205 | MEGVII-003 | [98]0.0017 | [95]0.0022 | [94]0.0028 | [98]0.0082 | [129]0.0781 |
| 206 | MEGVII-004 | [42]0.0009 | [32]0.0010 | [24]0.0011 | [31]0.0040 | [48]0.0521 |
| 207 | MEGVII-005 | [43]0.0009 | [33]0.0010 | [25]0.0011 | [64]0.0061 | [51]0.0540 |
| 208 | METSAKUURCOMPANY-000 | [229]0.0049 | [229]0.0070 | [217]0.0093 | [219]0.0209 | [243]0.1065 |
| 209 | METSAKUURCOMPANY-001 | [254]0.0062 | [251]0.0089 | [244]0.0126 | [334]0.0498 | [377]0.1724 |
| 210 | MIAXIS-001 | [238]0.0055 | [237]0.0076 | [228]0.0099 | [208]0.0188 | [209]0.0970 |
| 211 | MIAXIS-002 | [245]0.0060 | [241]0.0080 | [229]0.0105 | [210]0.0189 | [212]0.0978 |
| 212 | MICROFOCUS-005 | [603]0.9732 | [595]0.8352 | [478]0.8550 | [466]0.8748 | [459]0.8941 |
| 213 | MICROSOFT-003 | [381]0.0196 | [380]0.0274 | [359]0.0351 | [347]0.0526 | [341]0.1440 |
| 214 | MICROSOFT-004 | [366]0.0182 | [367]0.0256 | [351]0.0329 | [338]0.0509 | [334]0.1415 |
| 215 | MICROSOFT-005 | [364]0.0180 | [364]0.0254 | [348]0.0317 | [335]0.0499 | [324]0.1368 |
| 216 | MICROSOFT-006 | [280]0.0090 | [276]0.0119 | [267]0.0158 | [262]0.0292 | [330]0.1395 |

*Table 39:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N \geq 3\,000\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES BELOW THRESHOLD, T | | ENROL MOST RECENT | | | | |
|---|---|---|---|---|---|---|
| FNIR(N, T > 0, R >L) | | DATASET: FRVT 2018 MUGSHOTS | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 217 | MOBBL-000 | 379 0.0195 | 383 0.0276 | 365 0.0364 | 315 0.0424 | 28 0.0424 |
| 218 | MUKH-002 | 581 0.5039 | 581 0.5941 | 475 0.6673 | 463 0.7313 | 457 0.8275 |
| 219 | NAVERCLOUD-000 | 140 0.0025 | 139 0.0033 | 134 0.0044 | 121 0.0101 | 117 0.0740 |
| 220 | NEC-000 | 475 0.0634 | 472 0.0786 | 429 0.0928 | 413 0.1150 | 389 0.1862 |
| 221 | NEC-001 | 495 0.0859 | 492 0.1050 | 446 0.1237 | 429 0.1511 | 412 0.2196 |
| 222 | NEC-002 | 99 0.0018 | 101 0.0023 | 99 0.0030 | 151 0.0119 | 70 0.0594 |
| 223 | NEC-003 | 106 0.0019 | 96 0.0022 | 83 0.0026 | 51 0.0053 | 42 0.0492 |
| 224 | NEC-004 | 72 0.0013 | 63 0.0015 | 56 0.0016 | 22 0.0033 | 20 0.0326 |
| 225 | NEC-005 | 61 0.0012 | 57 0.0013 | 52 0.0015 | 62 0.0061 | 18 0.0304 |
| 226 | NEC-006 | 79 0.0015 | 75 0.0017 | 78 0.0022 | 118 0.0099 | 62 0.0570 |
| 227 | NEC-007 | 45 0.0009 | 35 0.0010 | 26 0.0011 | 13 0.0025 | 14 0.0265 |
| 228 | NEC-008 | 52 0.0010 | 44 0.0011 | 35 0.0012 | 18 0.0030 | 17 0.0292 |
| 229 | NEC-009 | 19 0.0007 | 14 0.0008 | 14 0.0008 | 10 0.0022 | 7 0.0191 |
| 230 | NEC-010 | 7 0.0006 | 7 0.0006 | 7 0.0007 | 6 0.0019 | 5 0.0167 |
| 231 | NEC-011 | 6 0.0005 | 6 0.0006 | 6 0.0006 | 4 0.0018 | 3 0.0164 |
| 232 | NEC-012 | 5 0.0005 | 4 0.0006 | 2 0.0006 | 2 0.0013 | 4 0.0164 |
| 233 | NEUROTECHNOLOGY-003 | 584 0.5681 | 585 0.6312 | 476 0.6951 | 464 0.7551 | 456 0.8014 |
| 234 | NEUROTECHNOLOGY-004 | 458 0.0462 | 458 0.0625 | 418 0.0772 | 408 0.1101 | 401 0.2008 |
| 235 | NEUROTECHNOLOGY-005 | 438 0.0390 | 443 0.0535 | 407 0.0672 | 391 0.0934 | 392 0.1874 |
| 236 | NEUROTECHNOLOGY-007 | 453 0.0433 | 457 0.0619 | 421 0.0799 | 420 0.1317 | 423 0.2391 |
| 237 | NEUROTECHNOLOGY-008 | 422 0.0335 | 437 0.0527 | 427 0.0889 | 436 0.1766 | 443 0.3286 |
| 238 | NEUROTECHNOLOGY-009 | 305 0.0104 | 308 0.0148 | 290 0.0192 | 277 0.0320 | 256 0.1099 |
| 239 | NEUROTECHNOLOGY-010 | 260 0.0066 | 259 0.0096 | 253 0.0135 | 320 0.0445 | 378 0.1724 |
| 240 | NEUROTECHNOLOGY-012 | 214 0.0044 | 222 0.0065 | 219 0.0094 | 243 0.0261 | 321 0.1340 |
| 241 | NEUROTECHNOLOGY-013 | 141 0.0026 | 159 0.0039 | 158 0.0054 | 217 0.0205 | 290 0.1199 |
| 242 | NEUROTECHNOLOGY-014 | 113 0.0020 | 122 0.0029 | 129 0.0042 | 128 0.0103 | 163 0.0870 |
| 243 | NEUROTECHNOLOGY-015 | 109 0.0020 | 128 0.0030 | 126 0.0041 | 113 0.0096 | 200 0.0951 |
| 244 | NEUROTECHNOLOGY-016 | 111 0.0020 | 121 0.0029 | 120 0.0039 | 102 0.0087 | 135 0.0807 |
| 245 | NEXTMETALABS-002 | 153 0.0027 | 150 0.0037 | 153 0.0053 | 141 0.0116 | 139 0.0818 |
| 246 | NOMINDER-000 | 174 0.0032 | 177 0.0045 | 175 0.0060 | 185 0.0152 | 179 0.0906 |
| 247 | NOMINDER-001 | 195 0.0036 | 190 0.0049 | 183 0.0064 | 176 0.0145 | 198 0.0947 |
| 248 | NOMINDER-002 | 193 0.0035 | 188 0.0047 | 181 0.0063 | 173 0.0141 | 183 0.0913 |
| 249 | NOMINDER-003 | 176 0.0033 | 181 0.0045 | 174 0.0060 | 156 0.0125 | 175 0.0900 |
| 250 | NOMINDER-004 | 173 0.0031 | 175 0.0044 | 170 0.0058 | 166 0.0130 | 195 0.0943 |
| 251 | NOMINDER-005 | 119 0.0022 | 118 0.0029 | 119 0.0039 | 131 0.0109 | 153 0.0833 |
| 252 | NOTIONTAG-000 | 321 0.0124 | 323 0.0171 | 307 0.0224 | 289 0.0354 | 304 0.1267 |
| 253 | NTECHLAB-003 | 448 0.0417 | 441 0.0534 | 406 0.0667 | 386 0.0893 | 349 0.1508 |
| 254 | NTECHLAB-004 | 413 0.0309 | 409 0.0402 | 387 0.0514 | 370 0.0705 | 340 0.1440 |
| 255 | NTECHLAB-005 | 419 0.0331 | 411 0.0420 | 390 0.0530 | 378 0.0748 | 344 0.1459 |
| 256 | NTECHLAB-006 | 408 0.0285 | 403 0.0364 | 381 0.0464 | 364 0.0660 | 343 0.1445 |
| 257 | NTECHLAB-007 | 371 0.0184 | 363 0.0252 | 346 0.0314 | 332 0.0492 | 312 0.1303 |
| 258 | NTECHLAB-008 | 301 0.0103 | 298 0.0141 | 285 0.0184 | 253 0.0282 | 223 0.0991 |
| 259 | NTECHLAB-009 | 181 0.0033 | 183 0.0045 | 171 0.0059 | 154 0.0121 | 116 0.0731 |
| 260 | NTECHLAB-010 | 93 0.0016 | 94 0.0022 | 88 0.0027 | 87 0.0074 | 108 0.0707 |
| 261 | NTECHLAB-011 | 103 0.0019 | 107 0.0026 | 108 0.0034 | 83 0.0072 | 77 0.0622 |
| 262 | OMNIGARDE-000 | 157 0.0028 | 155 0.0039 | 148 0.0052 | 161 0.0127 | 167 0.0875 |
| 263 | OMNIGARDE-001 | 74 0.0014 | 80 0.0018 | 81 0.0024 | 71 0.0066 | 110 0.0711 |
| 264 | OMNIGARDE-002 | 56 0.0011 | 56 0.0013 | 55 0.0016 | 59 0.0060 | 109 0.0708 |
| 265 | ONFIDO-000 | 296 0.0100 | 290 0.0131 | 280 0.0175 | 269 0.0300 | 291 0.1225 |
| 266 | ONFIDO-001 | 267 0.0074 | 268 0.0104 | 253 0.0134 | 246 0.0267 | 305 0.1271 |
| 267 | OPTIEXACTA-000 | 26 0.0008 | 29 0.0009 | 29 0.0011 | 14 0.0027 | 24 0.0372 |
| 268 | PANGIAM-000 | 205 0.0039 | 208 0.0057 | 200 0.0076 | 191 0.0156 | 166 0.0873 |
| 269 | PANGIAM-001 | 288 0.0094 | 272 0.0110 | 249 0.0130 | 228 0.0229 | 160 0.0862 |
| 270 | PAPSAV1923-000 | 118 0.0021 | 117 0.0029 | 118 0.0038 | 101 0.0087 | 107 0.0705 |
| 271 | PARAVISION-003 | 401 0.0257 | 398 0.0347 | 377 0.0441 | 363 0.0646 | 355 0.1539 |
| 272 | PARAVISION-004 | 263 0.0070 | 262 0.0098 | 252 0.0133 | 244 0.0263 | 301 0.1253 |
| 273 | PARAVISION-005 | 158 0.0029 | 153 0.0038 | 152 0.0053 | 199 0.0171 | 236 0.1034 |
| 274 | PARAVISION-007 | 143 0.0026 | 147 0.0036 | 147 0.0052 | 218 0.0207 | 255 0.1094 |
| 275 | PARAVISION-009 | 96 0.0016 | 99 0.0022 | 105 0.0034 | 110 0.0094 | 157 0.0854 |
| 276 | PARAVISION-012 | 49 0.0009 | 50 0.0012 | 51 0.0015 | 63 0.0061 | 126 0.0767 |
| 277 | PARAVISION-014 | 39 0.0009 | 41 0.0011 | 40 0.0013 | 53 0.0056 | 112 0.0717 |
| 278 | PARAVISION-016 | 37 0.0009 | 31 0.0010 | 38 0.0013 | 45 0.0049 | 103 0.0699 |
| 279 | PARAVISION-019 | 28 0.0008 | 28 0.0009 | 31 0.0012 | 40 0.0047 | 101 0.0696 |
| 280 | PARAVISION-021 | 22 0.0007 | 26 0.0009 | 30 0.0011 | 38 0.0046 | 93 0.0678 |
| 281 | PENTACOMP-001 | 335 0.0136 | 319 0.0162 | 316 0.0238 | 312 0.0412 | 276 0.1156 |
| 282 | PIXELALL-002 | 488 0.0713 | 491 0.1049 | 454 0.1472 | 450 0.2487 | 450 0.3902 |
| 283 | PIXELALL-003 | 351 0.0155 | 351 0.0215 | 339 0.0285 | 328 0.0470 | 269 0.1135 |
| 284 | PIXELALL-004 | 324 0.0126 | 331 0.0179 | 318 0.0242 | 357 0.0573 | 369 0.1678 |
| 285 | PIXELALL-005 | 274 0.0084 | 274 0.0117 | 272 0.0168 | 297 0.0380 | 308 0.1290 |
| 286 | PROTENET-001 | 348 0.0146 | 345 0.0206 | 333 0.0272 | 403 0.1060 | 323 0.1359 |
| 287 | PSL-000 | 77 0.0014 | 98 0.0022 | 98 0.0030 | 95 0.0078 | 193 0.0937 |
| 288 | PSL-001 | 85 0.0015 | 88 0.0021 | 93 0.0028 | 93 0.0075 | 89 0.0663 |

*Table 40:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N \geq 3\,000\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column.*

| MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L) | | ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS | | | | |
|---|---|---|---|---|---|---|
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 289 | PSL-002 | 31 0.0008 | 39 0.0011 | 41 0.0013 | 57 0.0058 | 98 0.0687 |
| 290 | PSL-003 | 38 0.0009 | 42 0.0011 | 50 0.0015 | 60 0.0060 | 95 0.0685 |
| 291 | PSL-004 | 21 0.0007 | 30 0.0009 | 33 0.0012 | 37 0.0046 | 59 0.0560 |
| 292 | PSL-005 | 47 0.0009 | 40 0.0011 | 43 0.0014 | 73 0.0067 | 154 0.0846 |
| 293 | PSL-006 | 16 0.0007 | 20 0.0009 | 21 0.0010 | 29 0.0037 | 76 0.0618 |
| 294 | PTAKURATSATU-000 | 405 0.0272 | 402 0.0363 | 380 0.0455 | 344 0.0520 | 47 0.0520 |
| 295 | PYRAMID-000 | 144 0.0026 | 144 0.0036 | 141 0.0047 | 107 0.0092 | 97 0.0686 |
| 296 | PYRAMID-001 | 169 0.0031 | 173 0.0044 | 168 0.0058 | 135 0.0111 | 128 0.0780 |
| 297 | QAZSMARTVISIONAI-000 | 36 0.0009 | 43 0.0011 | 39 0.0013 | 34 0.0043 | 67 0.0582 |
| 298 | QAZSMARTVISIONAI-001 | 8 0.0006 | 9 0.0007 | 9 0.0008 | 23 0.0035 | 34 0.0453 |
| 299 | QAZSMARTVISIONAI-002 | 4 0.0005 | 3 0.0006 | 5 0.0006 | 21 0.0033 | 29 0.0444 |
| 300 | QAZSMARTVISIONAI-003 | 3 0.0005 | 5 0.0006 | 4 0.0006 | 19 0.0032 | 33 0.0453 |
| 301 | QNAP-001 | 443 0.0400 | 440 0.0533 | 403 0.0658 | 389 0.0912 | 363 0.1592 |
| 302 | QNAP-002 | 382 0.0197 | 369 0.0261 | 349 0.0324 | 331 0.0487 | 320 0.1338 |
| 303 | QNAP-003 | 334 0.0136 | 335 0.0186 | 314 0.0235 | 294 0.0376 | 333 0.1411 |
| 304 | QNAP-004 | 283 0.0092 | 295 0.0136 | 283 0.0180 | 286 0.0342 | 314 0.1318 |
| 305 | QNAP-005 | 269 0.0075 | 264 0.0098 | 241 0.0124 | 224 0.0219 | 262 0.1108 |
| 306 | QNAP-006 | 243 0.0059 | 244 0.0083 | 234 0.0111 | 226 0.0224 | 310 0.1300 |
| 307 | QUANTASOFT-001 | 586 0.6384 | 586 0.6384 | 474 0.6384 | | 454 0.6384 |
| 308 | RANKONE-002 | 504 0.0970 | 498 0.1170 | 445 0.1349 | 433 0.1697 | 432 0.2553 |
| 309 | RANKONE-003 | 505 0.0970 | 497 0.1170 | 447 0.1349 | 434 0.1697 | 431 0.2553 |
| 310 | RANKONE-005 | 460 0.0469 | 454 0.0588 | 413 0.0690 | 392 0.0934 | 394 0.1921 |
| 311 | RANKONE-007 | 357 0.0164 | 354 0.0218 | 326 0.0262 | 296 0.0378 | 267 0.1129 |
| 312 | RANKONE-009 | 325 0.0129 | 326 0.0174 | 309 0.0226 | 285 0.0341 | 187 0.0918 |
| 313 | RANKONE-010 | 300 0.0102 | 291 0.0133 | 275 0.0170 | 242 0.0261 | 130 0.0782 |
| 314 | RANKONE-011 | 244 0.0059 | 245 0.0084 | 235 0.0112 | 245 0.0266 | 268 0.1132 |
| 315 | RANKONE-012 | 237 0.0054 | 232 0.0073 | 224 0.0097 | 222 0.0216 | 261 0.1107 |
| 316 | RANKONE-013 | 168 0.0030 | 166 0.0042 | 163 0.0056 | 155 0.0124 | 164 0.0871 |
| 317 | RANKONE-014 | 116 0.0021 | 119 0.0029 | 121 0.0039 | 126 0.0102 | 83 0.0652 |
| 318 | RANKONE-015 | 102 0.0018 | 103 0.0024 | 102 0.0031 | 78 0.0071 | 79 0.0628 |
| 319 | REALNETWORKS-002 | 542 0.1939 | 542 0.2308 | 469 0.2647 | 458 0.3110 | 441 0.3185 |
| 320 | REALNETWORKS-003 | 522 0.1296 | 520 0.1591 | 460 0.1855 | 443 0.2243 | 438 0.3074 |
| 321 | REALNETWORKS-004 | 521 0.1276 | 519 0.1578 | 459 0.1854 | 445 0.2326 | 440 0.3177 |
| 322 | REALNETWORKS-005 | 383 0.0198 | 379 0.0273 | 361 0.0352 | 354 0.0557 | 336 0.1428 |
| 323 | REALNETWORKS-006 | 286 0.0094 | 299 0.0142 | 282 0.0178 | 271 0.0305 | 221 0.0988 |
| 324 | REALNETWORKS-007 | 258 0.0064 | 256 0.0093 | 240 0.0121 | 229 0.0230 | 184 0.0913 |
| 325 | REALNETWORKS-008 | 209 0.0041 | 209 0.0058 | 203 0.0079 | 168 0.0135 | 140 0.0821 |
| 326 | REALSENSE-001 | 100 0.0018 | 106 0.0026 | 109 0.0035 | 132 0.0110 | 194 0.0941 |
| 327 | RECOGNITO-000 | 69 0.0013 | 74 0.0017 | 79 0.0023 | 49 0.0053 | 45 0.0516 |
| 328 | RECOGNITO-001 | 44 0.0009 | 46 0.0011 | 42 0.0014 | 17 0.0029 | 22 0.0368 |
| 329 | RECOGNITO-002 | 24 0.0008 | 27 0.0009 | 28 0.0011 | 15 0.0027 | 25 0.0372 |
| 330 | REMARKAI-000 | 445 0.0402 | 444 0.0549 | 408 0.0673 | 400 0.1025 | 400 0.2000 |
| 331 | RENDIP-000 | 272 0.0081 | 273 0.0117 | 261 0.0153 | 252 0.0274 | 287 0.1179 |
| 332 | REVEALMEDIA-000 | 275 0.0086 | 275 0.0118 | 262 0.0154 | 251 0.0274 | 230 0.1016 |
| 333 | RNIT-000 | 247 0.0061 | 269 0.0109 | 342 0.0292 | 414 0.1179 | 430 0.2553 |
| 334 | ROC-016 | 90 0.0016 | 93 0.0021 | 87 0.0027 | 103 0.0088 | 78 0.0626 |
| 335 | ROC-017 | 108 0.0020 | 112 0.0027 | 112 0.0036 | 105 0.0088 | 85 0.0655 |
| 336 | ROC-018 | 88 0.0015 | 85 0.0020 | 85 0.0026 | 79 0.0071 | 72 0.0599 |
| 337 | ROC-019 | 33 0.0008 | 47 0.0011 | 48 0.0014 | 66 0.0063 | 46 0.0519 |
| 338 | ROC-020 | 48 0.0009 | 51 0.0012 | 58 0.0017 | 44 0.0048 | 36 0.0465 |
| 339 | ROC-021 | 15 0.0007 | 15 0.0008 | 18 0.0010 | 35 0.0045 | 27 0.0408 |
| 340 | S1-000 | 386 0.0200 | 382 0.0276 | 368 0.0378 | 361 0.0627 | 374 0.1704 |
| 341 | S1-001 | 314 0.0112 | 312 0.0153 | 292 0.0195 | 299 0.0388 | 302 0.1253 |
| 342 | S1-002 | 198 0.0036 | 199 0.0053 | 194 0.0073 | 241 0.0261 | 307 0.1282 |
| 343 | S1-003 | 242 0.0057 | 246 0.0084 | 236 0.0112 | 249 0.0273 | 309 0.1295 |
| 344 | S1-004 | 148 0.0027 | 148 0.0037 | 151 0.0053 | 194 0.0159 | 218 0.0986 |
| 345 | S1-005 | 156 0.0028 | 161 0.0040 | 155 0.0053 | 174 0.0143 | 172 0.0896 |
| 346 | S1-006 | 87 0.0015 | 86 0.0020 | 89 0.0027 | 137 0.0114 | 146 0.0824 |
| 347 | S1-007 | 71 0.0013 | 73 0.0016 | 76 0.0021 | 74 0.0067 | 71 0.0598 |
| 348 | SANSAP-001 | 161 0.0029 | 162 0.0040 | 160 0.0055 | 192 0.0157 | 209 0.0975 |
| 349 | SCANOVATE-000 | 463 0.0494 | 462 0.0664 | 422 0.0801 | 406 0.1094 | 259 0.1105 |
| 350 | SCANOVATE-001 | 474 0.0626 | 473 0.0812 | 432 0.0989 | 419 0.1289 | 395 0.1958 |
| 351 | SCYLLA-001 | 320 0.0123 | 320 0.0164 | 300 0.0211 | 283 0.0340 | 303 0.1261 |
| 352 | SENSETIME-000 | 352 0.0155 | 344 0.0204 | 329 0.0266 | 300 0.0389 | 270 0.1135 |
| 353 | SENSETIME-001 | 354 0.0157 | 352 0.0216 | 334 0.0273 | 311 0.0410 | 295 0.1238 |
| 354 | SENSETIME-002 | 342 0.0142 | 303 0.0145 | 260 0.0149 | 230 0.0230 | 84 0.0654 |
| 355 | SENSETIME-003 | 66 0.0012 | 65 0.0015 | 61 0.0018 | 47 0.0050 | 31 0.0448 |
| 356 | SENSETIME-004 | 64 0.0012 | 60 0.0014 | 60 0.0017 | 28 0.0037 | 21 0.0350 |
| 357 | SENSETIME-005 | 68 0.0013 | 83 0.0019 | 90 0.0027 | 100 0.0085 | 32 0.0451 |
| 358 | SENSETIME-006 | 55 0.0011 | 62 0.0015 | 72 0.0020 | 33 0.0043 | 23 0.0369 |
| 359 | SENSETIME-007 | 41 0.0009 | 37 0.0010 | 37 0.0012 | 20 0.0032 | 19 0.0312 |
| 360 | SENSETIME-008 | 30 0.0008 | 34 0.0010 | 34 0.0012 | 16 0.0027 | 16 0.0285 |

*Table 41:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES BELOW THRESHOLD, T | | ENROL MOST RECENT | | | | |
|---|---|---|---|---|---|---|
| FNIR(N, T> 0, R >L) | | DATASET: FRVT 2018 MUGSHOTS | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 361 | SENSETIME-009 | [11] 0.0006 | [11] 0.0007 | [12] 0.0008 | [8] 0.0020 | [11] 0.0234 |
| 362 | SENSETIME-010 | [10] 0.0006 | [13] 0.0007 | [10] 0.0008 | [5] 0.0019 | [12] 0.0241 |
| 363 | SENSETIME-011 | [12] 0.0006 | [12] 0.0007 | [13] 0.0008 | [7] 0.0020 | [10] 0.0234 |
| 364 | SENSETIME-012 | [9] 0.0006 | [8] 0.0007 | [8] 0.0007 | [3] 0.0014 | [8] 0.0195 |
| 365 | SERENDIPITY-000 | [227] 0.0088 | [279] 0.0121 | [264] 0.0156 | [250] 0.0274 | [228] 0.1013 |
| 366 | SEVENTHSENSE-001 | [163] 0.0029 | [172] 0.0044 | [186] 0.0067 | [182] 0.0150 | [149] 0.0828 |
| 367 | SHAMAN-007 | [518] 0.1209 | [513] 0.1410 | [455] 0.1577 | [439] 0.1864 | [426] 0.2404 |
| 368 | SIAT-001 | [329] 0.0133 | [325] 0.0173 | [308] 0.0225 | [280] 0.0337 | [199] 0.0950 |
| 369 | SIAT-002 | [350] 0.0151 | [349] 0.0213 | [331] 0.0270 | [302] 0.0392 | [283] 0.1169 |
| 370 | SJTU-001 | [59] 0.0011 | [64] 0.0015 | [64] 0.0018 | [55] 0.0058 | [52] 0.0543 |
| 371 | SPARKTEK-000 | [115] 0.0021 | [115] 0.0028 | [117] 0.0037 | [108] 0.0093 | [152] 0.0831 |
| 372 | SPARKTEK-001 | [25] 0.0008 | [23] 0.0009 | [36] 0.0012 | [70] 0.0066 | [61] 0.0567 |
| 373 | SPARSH-001 | [160] 0.0029 | [159] 0.0039 | [161] 0.0055 | [165] 0.0130 | [170] 0.0890 |
| 374 | SQISOFT-001 | [497] 0.0918 | [508] 0.1319 | [457] 0.1778 | [447] 0.2345 | [439] 0.3174 |
| 375 | SQISOFT-002 | [360] 0.0173 | [388] 0.0286 | [375] 0.0411 | [376] 0.0736 | [322] 0.1347 |
| 376 | STCON-001 | [65] 0.0012 | [61] 0.0015 | [68] 0.0019 | [58] 0.0058 | [90] 0.0670 |
| 377 | STCON-002 | [50] 0.0010 | [53] 0.0012 | [54] 0.0016 | [39] 0.0046 | [65] 0.0579 |
| 378 | STCON-003 | [46] 0.0009 | [45] 0.0011 | [47] 0.0014 | [30] 0.0040 | [69] 0.0590 |
| 379 | STCON-004 | [23] 0.0007 | [21] 0.0009 | [19] 0.0010 | [27] 0.0036 | [30] 0.0445 |
| 380 | STCON-005 | [18] 0.0007 | [22] 0.0009 | [22] 0.0010 | [26] 0.0036 | [26] 0.0400 |
| 381 | STELLARTECS-001 | [304] 0.0104 | [301] 0.0144 | [293] 0.0196 | [284] 0.0341 | [280] 0.1161 |
| 382 | SYNESIS-003 | [464] 0.0496 | [460] 0.0649 | [423] 0.0801 | [405] 0.1092 | [393] 0.1913 |
| 383 | SYNESIS-003 | [582] 0.5339 | [580] 0.5817 | [473] 0.6099 | [462] 0.6473 | [455] 0.6801 |
| 384 | SYNESIS-005 | [362] 0.0177 | [362] 0.0245 | [347] 0.0316 | [342] 0.0515 | [362] 0.1577 |
| 385 | TECH5-001 | [447] 0.0417 | [447] 0.0571 | [428] 0.0907 | [441] 0.2103 | [448] 0.3723 |
| 386 | TECH5-002 | [376] 0.0190 | [373] 0.0266 | [355] 0.0343 | [348] 0.0534 | [364] 0.1604 |
| 387 | TECH5-003 | [145] 0.0026 | [145] 0.0036 | [143] 0.0050 | [159] 0.0125 | [192] 0.0931 |
| 388 | TECHAINER-000 | [293] 0.0099 | [289] 0.0131 | [278] 0.0172 | [265] 0.0297 | [240] 0.1049 |
| 389 | TEVIAN-005 | [485] 0.0687 | [480] 0.0870 | [435] 0.1053 | [417] 0.1279 | [387] 0.1798 |
| 390 | TEVIAN-006 | [268] 0.0074 | [257] 0.0095 | [245] 0.0127 | [240] 0.0258 | [311] 0.1302 |
| 391 | TEVIAN-007 | [190] 0.0034 | [189] 0.0048 | [178] 0.0062 | [183] 0.0151 | [201] 0.0954 |
| 392 | THAKAAIT-000 | [312] 0.0107 | [311] 0.0151 | [298] 0.0207 | [293] 0.0376 | [316] 0.1331 |
| 393 | TIGER-002 | [477] 0.0643 | [477] 0.0857 | [433] 0.1027 | [421] 0.1317 | [411] 0.2175 |
| 394 | TIGER-004 | [337] 0.0140 | [322] 0.0168 | [319] 0.0242 | [309] 0.0410 | [279] 0.1159 |
| 395 | TOSHIBA-000 | [457] 0.0457 | [456] 0.0616 | [419] 0.0776 | [407] 0.1097 | [402] 0.2014 |
| 396 | TOSHIBA-002 | [67] 0.0012 | [67] 0.0015 | [65] 0.0018 | [84] 0.0073 | [115] 0.0727 |
| 397 | TOSHIBA-003 | [63] 0.0012 | [59] 0.0014 | [63] 0.0018 | [127] 0.0102 | [174] 0.0898 |
| 398 | TOSS-000 | [76] 0.0014 | [77] 0.0017 | [75] 0.0020 | [52] 0.0056 | [39] 0.0476 |
| 399 | TRUEFACE-000 | [326] 0.0130 | [330] 0.0178 | [312] 0.0235 | [295] 0.0377 | [327] 0.1382 |
| 400 | TSIGHT-000 | [2] 0.0005 | [2] 0.0005 | [3] 0.0006 | [587] 1.0000 | [514] 1.0000 |
| 401 | UNISSEY-003 | [311] 0.0107 | [302] 0.0144 | [284] 0.0183 | [272] 0.0306 | [254] 0.1090 |
| 402 | UNIVERSEAI-000 | [131] 0.0024 | [136] 0.0033 | [132] 0.0044 | [116] 0.0099 | [145] 0.0823 |
| 403 | USEB-001 | [151] 0.0027 | [142] 0.0036 | [140] 0.0047 | [149] 0.0118 | [144] 0.0822 |
| 404 | USEB-002 | [162] 0.0029 | [156] 0.0039 | [149] 0.0052 | [188] 0.0154 | [177] 0.0901 |
| 405 | USEB-003 | [146] 0.0026 | [139] 0.0034 | [133] 0.0044 | [133] 0.0111 | [158] 0.0854 |
| 406 | USEB-004 | [134] 0.0025 | [133] 0.0032 | [127] 0.0042 | [112] 0.0095 | [127] 0.0774 |
| 407 | VD-001 | [533] 0.1639 | [534] 0.2010 | [468] 0.2344 | [455] 0.2719 | [442] 0.3253 |
| 408 | VDFSOFTWARE-002 | [404] 0.0267 | [408] 0.0376 | [383] 0.0482 | [368] 0.0685 | [351] 0.1523 |
| 409 | VERIDAS-001 | [407] 0.0275 | [405] 0.0369 | [384] 0.0488 | [380] 0.0749 | [353] 0.1538 |
| 410 | VERIDAS-002 | [406] 0.0275 | [406] 0.0369 | [385] 0.0488 | [379] 0.0749 | [354] 0.1538 |
| 411 | VERIDAS-003 | [315] 0.0114 | [318] 0.0162 | [302] 0.0215 | [319] 0.0443 | [356] 0.1540 |
| 412 | VERIDAS-004 | [204] 0.0038 | [201] 0.0054 | [195] 0.0073 | [86] 0.0073 | [9] 0.0229 |
| 413 | VERIGRAM-000 | [187] 0.0034 | [180] 0.0045 | [135] 0.0045 | [50] 0.0053 | [2] 0.0111 |
| 414 | VERIGRAM-001 | [188] 0.0034 | [182] 0.0045 | [159] 0.0054 | [162] 0.0129 | [136] 0.0811 |
| 415 | VIANTE-000 | [105] 0.0019 | [105] 0.0025 | [106] 0.0034 | [144] 0.0116 | [124] 0.0763 |
| 416 | VIANTE-001 | [57] 0.0011 | [54] 0.0012 | [49] 0.0015 | [72] 0.0067 | [75] 0.0614 |
| 417 | VIANTE-002 | [20] 0.0007 | [25] 0.0009 | [32] 0.0012 | [85] 0.0073 | [60] 0.0567 |
| 418 | VIANTE-003 | [13] 0.0007 | [16] 0.0008 | [17] 0.0009 | [88] 0.0074 | [66] 0.0581 |
| 419 | VIETTELAI-000 | [396] 0.0227 | [453] 0.0585 | [450] 0.1394 | [452] 0.2506 | [449] 0.3742 |
| 420 | VIETTELAI-002 | [264] 0.0070 | [267] 0.0102 | [263] 0.0156 | [374] 0.0724 | [396] 0.1964 |
| 421 | VIETTELAI-003 | [175] 0.0032 | [169] 0.0043 | [172] 0.0059 | [209] 0.0188 | [232] 0.1019 |
| 422 | VIGILANTSOLUTIONS-008 | [343] 0.0143 | [340] 0.0202 | [330] 0.0266 | [330] 0.0485 | [281] 0.1161 |
| 423 | VINBIGDATA-001 | [308] 0.0106 | [314] 0.0157 | [301] 0.0213 | [282] 0.0339 | [215] 0.0983 |
| 424 | VINBIGDATA-002 | [78] 0.0015 | [82] 0.0019 | [82] 0.0024 | [120] 0.0101 | [147] 0.0824 |
| 425 | VINBIGDATA-003 | [82] 0.0015 | [92] 0.0021 | [97] 0.0029 | [117] 0.0099 | [159] 0.0860 |
| 426 | VINBIGDATA-004 | [58] 0.0011 | [69] 0.0015 | [69] 0.0019 | [104] 0.0088 | [121] 0.0744 |
| 427 | VISIONBOX-000 | [317] 0.0119 | [327] 0.0174 | [315] 0.0236 | | [460] 0.9538 |
| 428 | VISIONBOX-001 | [489] 0.0757 | [509] 0.1326 | [464] 0.2095 | [457] 0.3064 | [451] 0.4116 |
| 429 | VISIONBOX-002 | [270] 0.0078 | [270] 0.0109 | [258] 0.0145 | [234] 0.0242 | [247] 0.1079 |
| 430 | VISIONLABS-004 | [450] 0.0424 | [449] 0.0575 | [415] 0.0697 | [393] 0.0940 | [382] 0.1785 |
| 431 | VISIONLABS-005 | [432] 0.0366 | [430] 0.0498 | [396] 0.0620 | [383] 0.0831 | [373] 0.1701 |
| 432 | VISIONLABS-006 | [373] 0.0186 | [371] 0.0262 | [353] 0.0331 | [346] 0.0525 | [328] 0.1392 |

*Table 42:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES BELOW THRESHOLD, T | | ENROL MOST RECENT | | | | |
|---|---|---|---|---|---|---|
| FNIR(N, T> 0, R >L) | | DATASET: FRVT 2018 MUGSHOTS | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M |
| 433 | VISIONLABS-007 | [372]0.0186 | [370]0.0262 | [352]0.0330 | [345]0.0523 | [329]0.1393 |
| 434 | VISIONLABS-008 | [282]0.0092 | [285]0.0127 | [271]0.0162 | [259]0.0288 | [298]0.1244 |
| 435 | VISIONLABS-009 | [170]0.0031 | [167]0.0042 | [163]0.0056 | [169]0.0136 | [168]0.0878 |
| 436 | VISIONLABS-010 | [189]0.0034 | [187]0.0047 | [185]0.0066 | [178]0.0145 | [186]0.0917 |
| 437 | VISIONLABS-011 | [117]0.0021 | [123]0.0030 | [125]0.0040 | [143]0.0116 | [142]0.0822 |
| 438 | VISTEAM-000 | [480]0.0656 | [475]0.0852 | [436]0.1072 | [425]0.1387 | [398]0.1987 |
| 439 | VIXVIZION-009 | [385]0.0199 | [376]0.0270 | [358]0.0345 | [352]0.0541 | [326]0.1374 |
| 440 | VNPAY-001 | [207]0.0039 | [203]0.0055 | [198]0.0075 | [180]0.0149 | [150]0.0829 |
| 441 | VNPT-001 | [295]0.0100 | [297]0.0140 | [286]0.0186 | [264]0.0293 | [233]0.1025 |
| 442 | VNPT-002 | [224]0.0047 | [215]0.0062 | [208]0.0080 | [198]0.0169 | [225]0.1002 |
| 443 | VNPT-003 | [221]0.0046 | [212]0.0060 | [202]0.0078 | [197]0.0168 | [220]0.0987 |
| 444 | VNPT-004 | [203]0.0038 | [194]0.0051 | [182]0.0064 | [179]0.0148 | [169]0.0885 |
| 445 | VNPT-005 | [177]0.0033 | [168]0.0042 | [164]0.0056 | [140]0.0115 | [161]0.0866 |
| 446 | VOCORD-005 | [515]0.1174 | [518]0.1559 | [465]0.2152 | [456]0.3062 | [453]0.4354 |
| 447 | VOGIC-000 | [292]0.0098 | [296]0.0136 | [296]0.0200 | [333]0.0498 | [348]0.1492 |
| 448 | VTS-001 | [294]0.0099 | [287]0.0130 | [277]0.0172 | [274]0.0319 | [297]0.1240 |
| 449 | VTS-002 | [365]0.0181 | [365]0.0255 | [354]0.0341 | [353]0.0546 | [342]0.1444 |
| 450 | VTS-003 | [230]0.0050 | [228]0.0069 | [216]0.0092 | [206]0.0184 | [229]0.1014 |
| 451 | VTS-004 | [171]0.0031 | [178]0.0045 | [177]0.0061 | [171]0.0140 | [173]0.0896 |
| 452 | VTS-005 | [182]0.0033 | [195]0.0051 | [193]0.0071 | [170]0.0138 | [94]0.0680 |
| 453 | WINK-001 | [399]0.0244 | [401]0.0359 | [382]0.0481 | [373]0.0724 | [371]0.1692 |
| 454 | WINK-002 | [276]0.0086 | [278]0.0121 | [268]0.0159 | [258]0.0285 | [206]0.0968 |
| 455 | WINK-003 | [250]0.0061 | [249]0.0088 | [243]0.0126 | [339]0.0511 | [357]0.1544 |
| 456 | XAILIENT-000 | [402]0.0260 | [400]0.0354 | [379]0.0454 | [366]0.0678 | [367]0.1650 |
| 457 | XFORWARDAI-000 | [302]0.0103 | [306]0.0148 | [288]0.0192 | [278]0.0321 | [241]0.1054 |
| 458 | XFORWARDAI-001 | [179]0.0033 | [179]0.0045 | [166]0.0057 | [142]0.0116 | [133]0.0797 |
| 459 | XFORWARDAI-002 | [124]0.0023 | [110]0.0027 | [103]0.0031 | [92]0.0074 | [106]0.0702 |
| 460 | YITU-002 | [323]0.0125 | [324]0.0173 | [305]0.0223 | [281]0.0337 | [245]0.1071 |
| 461 | YITU-003 | [333]0.0134 | [333]0.0181 | [310]0.0232 | [287]0.0345 | [248]0.1079 |
| 462 | YITU-004 | [257]0.0064 | [253]0.0092 | [242]0.0124 | [223]0.0219 | [208]0.0973 |
| 463 | YITU-005 | [265]0.0071 | [263]0.0098 | [250]0.0131 | [233]0.0242 | [224]0.0993 |
| 464 | YOUVERSE-001 | [223]0.0047 | [225]0.0067 | [214]0.0090 | [203]0.0180 | [235]0.1033 |
| 465 | YUAN-000 | [262]0.0067 | [265]0.0099 | [251]0.0132 | [227]0.0227 | [529]1.0000 |

*Table 43:* **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR *= 0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N \geq 3\,000\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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| MISSES AT GIVEN RANK | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T= 0, R) | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 1 | 3DIVI-005 | 522 0.0134 | 520 0.0173 | 457 0.0207 | 446 0.0250 | 440 0.0298 | 382 0.0003 $N^{0.275}$ 397 | 493 0.0038 | 492 0.0046 | 445 0.0054 | 436 0.0065 | 430 0.0079 | 77 0.0001 $N^{0.251}$ 419 |
| 2 | ACER-000 | 483 0.0077 | 489 0.0102 | 445 0.0124 | 438 0.0153 | 430 0.0191 | 256 0.0001 $N^{0.308}$ 436 | 413 0.0016 | 442 0.0022 | 419 0.0027 | 414 0.0034 | 410 0.0042 | 23 0.0000 $N^{0.324}$ 441 |
| 3 | ADIANCE-002 | 236 0.0011 | 237 0.0012 | 231 0.0013 | 235 0.0015 | 239 0.0018 | 198 0.0001 $N^{0.181}$ 272 | 230 0.0008 | 220 0.0008 | 220 0.0009 | 215 0.0009 | 220 0.0010 | 217 0.0004 $N^{0.056}$ 283 |
| 4 | ADVANCE-000 | 291 0.0014 | 301 0.0016 | 298 0.0018 | 291 0.0022 | 289 0.0025 | 186 0.0001 $N^{0.206}$ 321 | 313 0.0010 | 310 0.0011 | 298 0.0011 | 297 0.0012 | 294 0.0013 | 261 0.0004 $N^{0.064}$ 308 |
| 5 | AFISBIOMETRICS-000 | 197 0.0010 | 244 0.0012 | 267 0.0016 | 279 0.0020 | 288 0.0025 | 18 0.0000 $N^{0.336}$ 446 | 89 0.0005 | 100 0.0006 | 128 0.0006 | 129 0.0007 | 132 0.0007 | 82 0.0001 $N^{0.097}$ 353 |
| 6 | AFRENGINE-000 | 385 0.0023 | 389 0.0029 | 329 0.0034 | 369 0.0041 | 369 0.0052 | 97 0.0001 $N^{0.275}$ 400 | 381 0.0013 | 373 0.0014 | 358 0.0015 | 347 0.0016 | 340 0.0017 | 257 0.0004 $N^{0.084}$ 334 |
| 7 | AFRENGINE-001 | 275 0.0013 | 262 0.0013 | 251 0.0014 | 242 0.0016 | 233 0.0018 | 372 0.0003 $N^{0.106}$ 99 | 370 0.0012 | 348 0.0013 | 331 0.0013 | 315 0.0013 | 297 0.0013 | 416 0.0011 $N^{0.007}$ 57 |
| 8 | AFRENGINE-002 | 276 0.0013 | 261 0.0013 | 243 0.0014 | 230 0.0015 | 224 0.0017 | 405 0.0004 $N^{0.087}$ 57 | 371 0.0012 | 349 0.0013 | 332 0.0013 | 316 0.0013 | 295 0.0013 | 421 0.0012 $N^{0.005}$ 41 |
| 9 | AFRENGINE-003 | 382 0.0023 | 366 0.0023 | 339 0.0024 | 320 0.0025 | 300 0.0027 | 438 0.0010 $N^{0.063}$ 23 | 455 0.0022 | 439 0.0022 | 405 0.0022 | 392 0.0022 | 376 0.0022 | 444 0.0021 $N^{0.002}$ 12 |
| 10 | AIGROUP-000 | 210 0.0010 | 215 0.0011 | 209 0.0012 | 207 0.0014 | 211 0.0016 | 229 0.0001 $N^{0.161}$ 231 | 190 0.0007 | 188 0.0007 | 178 0.0007 | 168 0.0008 | 166 0.0008 | 294 0.0005 $N^{0.029}$ 204 |
| 11 | AIZE-002 | 395 0.0025 | 397 0.0031 | 383 0.0036 | 376 0.0043 | 371 0.0052 | 185 0.0001 $N^{0.251}$ 377 | 380 0.0013 | 386 0.0014 | 366 0.0015 | 356 0.0017 | 348 0.0018 | 154 0.0003 $N^{0.106}$ 357 |
| 12 | AIZE-006 | 408 0.0029 | 407 0.0036 | 398 0.0044 | 388 0.0054 | 382 0.0065 | 134 0.0001 $N^{0.278}$ 406 | 373 0.0013 | 368 0.0013 | 364 0.0015 | 360 0.0017 | 359 0.0019 | 93 0.0002 $N^{0.146}$ 380 |
| 13 | AKUVOX-001 | 272 0.0013 | 268 0.0014 | 261 0.0016 | 259 0.0018 | 253 0.0020 | 292 0.0002 $N^{0.154}$ 211 | 307 0.0010 | 305 0.0011 | 293 0.0011 | 273 0.0011 | 265 0.0011 | 348 0.0007 $N^{0.031}$ 212 |
| 14 | ALCHERA-003 | 482 0.0077 | 488 0.0102 | 443 0.0120 | 437 0.0145 | 429 0.0177 | 308 0.0002 $N^{0.282}$ 414 | 469 0.0025 | 468 0.0030 | 429 0.0033 | 421 0.0040 | 412 0.0046 | 88 0.0002 $N^{0.206}$ 411 |
| 15 | ALCHERA-006 | 169 0.0009 | 160 0.0009 | 160 0.0010 | 152 0.0011 | 163 0.0013 | 270 0.0001 $N^{0.132}$ 170 | 187 0.0007 | 183 0.0007 | 177 0.0007 | 173 0.0008 | 163 0.0008 | 280 0.0005 $N^{0.032}$ 220 |
| 16 | ALCHERA-008 | 193 0.0009 | 182 0.0010 | 162 0.0010 | 165 0.0011 | 156 0.0012 | 354 0.0003 $N^{0.095}$ 78 | 253 0.0009 | 239 0.0009 | 226 0.0009 | 212 0.0009 | 201 0.0009 | 389 0.0008 $N^{0.004}$ 32 |
| 17 | ALGORITHMSINC-000 | 259 0.0012 | 293 0.0016 | 294 0.0018 | 300 0.0022 | 312 0.0029 | 30 0.0000 $N^{0.293}$ 423 | 326 0.0011 | 363 0.0013 | 367 0.0015 | 371 0.0018 | 379 0.0023 | 30 0.0000 $N^{0.256}$ 421 |
| 18 | ALLGOVISION-000 | 501 0.0097 | 496 0.0111 | 444 0.0124 | 435 0.0142 | 428 0.0162 | 436 0.0009 $N^{0.176}$ 263 | 517 0.0059 | 511 0.0064 | 452 0.0067 | 439 0.0072 | 429 0.0078 | 439 0.0017 $N^{0.091}$ 345 |
| 19 | ALLGOVISION-001 | 467 0.0066 | 474 0.0086 | 438 0.0103 | 430 0.0124 | 425 0.0154 | 275 0.0001 $N^{0.287}$ 420 | 444 0.0020 | 449 0.0023 | 420 0.0028 | 407 0.0032 | 407 0.0040 | 56 0.0001 $N^{0.238}$ 415 |
| 20 | ALLGOVISION-005 | 322 0.0015 | 316 0.0017 | 308 0.0020 | 301 0.0023 | 294 0.0026 | 238 0.0001 $N^{0.191}$ 300 | 334 0.0011 | 326 0.0011 | 314 0.0012 | 317 0.0013 | 308 0.0014 | 204 0.0004 $N^{0.079}$ 328 |
| 21 | ANKE-000 | 504 0.0099 | 507 0.0129 | 453 0.0152 | 443 0.0185 | 435 0.0222 | 352 0.0003 $N^{0.275}$ 398 | 479 0.0029 | 481 0.0037 | 440 0.0044 | 427 0.0053 | 420 0.0063 | 61 0.0001 $N^{0.261}$ 424 |
| 22 | ANKE-002 | 371 0.0021 | 374 0.0024 | 363 0.0029 | 355 0.0034 | 351 0.0040 | 200 0.0001 $N^{0.228}$ 347 | 366 0.0012 | 356 0.0013 | 342 0.0014 | 335 0.0014 | 332 0.0016 | 226 0.0004 $N^{0.085}$ 335 |
| 23 | ARATEK-001 | 131 0.0007 | 131 0.0008 | 119 0.0008 | 126 0.0010 | 130 0.0011 | 181 0.0001 $N^{0.159}$ 222 | 139 0.0006 | 125 0.0006 | 123 0.0006 | 120 0.0006 | 115 0.0006 | 224 0.0004 $N^{0.031}$ 216 |
| 24 | ARMATURA-000 | 318 0.0015 | 290 0.0015 | 265 0.0016 | 251 0.0017 | 236 0.0018 | 422 0.0007 $N^{0.061}$ 20 | 395 0.0015 | 387 0.0015 | 359 0.0015 | 338 0.0015 | 319 0.0015 | 433 0.0014 $N^{0.004}$ 27 |
| 25 | AWARE-003 | 542 0.0235 | 543 0.0303 | 469 0.0358 | 458 0.0427 | 454 0.0503 | 429 0.0007 $N^{0.260}$ 385 | 510 0.0052 | 521 0.0072 | 463 0.0088 | 452 0.0109 | 449 0.0140 | 45 0.0001 $N^{0.335}$ 446 |
| 26 | AWARE-005 | 545 0.0241 | 544 0.0308 | 470 0.0363 | 459 0.0431 | 444 0.0309 | 459 0.0055 $N^{0.120}$ 133 | 515 0.0058 | 526 0.0079 | 465 0.0098 | 455 0.0125 | 432 0.0086 | 337 0.0006 $N^{0.176}$ 400 |
| 27 | AWARE-007 | 340 0.0017 | 352 0.0021 | 349 0.0025 | 345 0.0031 | 346 0.0038 | 50 0.0000 $N^{0.282}$ 413 | 203 0.0008 | 227 0.0009 | 262 0.0010 | 281 0.0011 | 302 0.0013 | 46 0.0001 $N^{0.187}$ 404 |
| 28 | AYONIX-002 | 597 0.2933 | 598 0.3412 | 480 0.3734 | 468 0.4099 | 461 0.4463 | 464 0.0439 $N^{0.143}$ 195 | 590 0.0947 | 592 0.1271 | 480 0.1522 | 467 0.1825 | 459 0.2147 | 448 0.0023 $N^{0.279}$ 431 |
| 29 | BIOIDTECHSWISS-001 | 143 0.0008 | 134 0.0008 | 121 0.0009 | 111 0.0009 | 109 0.0011 | 285 0.0002 $N^{0.118}$ 130 | 157 0.0006 | 143 0.0006 | 140 0.0007 | 136 0.0007 | 134 0.0007 | 271 0.0004 $N^{0.027}$ 188 |
| 30 | BIOMETRIK-001 | 416 0.0033 | 423 0.0044 | 407 0.0052 | 394 0.0062 | 390 0.0075 | 176 0.0001 $N^{0.278}$ 405 | 356 0.0012 | 375 0.0014 | 372 0.0016 | 367 0.0018 | 362 0.0021 | 64 0.0001 $N^{0.191}$ 407 |
| 31 | CAMVI-004 | 515 0.0122 | 558 0.0466 | 474 0.0717 | 467 0.2360 | 459 0.2365 | 9 0.0000 $N^{1.059}$ 460 | 553 0.0116 | 575 0.0462 | 476 0.0713 | 469 0.2358 | 461 0.2362 | 10 0.0000 $N^{1.075}$ 461 |
| 32 | CANON-001 | 111 0.0007 | 106 0.0007 | 112 0.0008 | 118 0.0010 | 114 0.0011 | 170 0.0001 $N^{0.159}$ 223 | 98 0.0005 | 98 0.0006 | 99 0.0006 | 102 0.0006 | 101 0.0006 | 143 0.0003 $N^{0.042}$ 254 |
| 33 | CANON-002 | 127 0.0007 | 138 0.0008 | 151 0.0009 | 158 0.0011 | 160 0.0012 | 82 0.0001 $N^{0.196}$ 306 | 103 0.0005 | 94 0.0006 | 93 0.0006 | 87 0.0006 | 94 0.0006 | 220 0.0004 $N^{0.025}$ 181 |
| 34 | CANON-003 | 52 0.0005 | 53 0.0006 | 65 0.0007 | 70 0.0008 | 60 0.0009 | 87 0.0001 $N^{0.173}$ 249 | 40 0.0005 | 42 0.0005 | 31 0.0005 | 31 0.0005 | 35 0.0005 | 196 0.0004 $N^{0.017}$ 137 |
| 35 | CIB-000 | 232 0.0011 | 233 0.0012 | 233 0.0013 | 239 0.0016 | 460 0.3301 | 6 0.0000 $N^{1.598}$ 462 | 219 0.0008 | 209 0.0008 | 208 0.0008 | 211 0.0009 | 463 0.3295 | 6 0.0000 $N^{1.644}$ 462 |
| 36 | CLEARVIEWAI-000 | 99 0.0007 | 107 0.0007 | 110 0.0008 | 125 0.0010 | 124 0.0011 | 94 0.0001 $N^{0.183}$ 283 | 107 0.0006 | 95 0.0006 | 96 0.0006 | 100 0.0006 | 97 0.0006 | 189 0.0004 $N^{0.031}$ 211 |
| 37 | CLEARVIEWAI-001 | 124 0.0007 | 110 0.0007 | 102 0.0008 | 97 0.0009 | 92 0.0010 | 268 0.0001 $N^{0.121}$ 141 | 165 0.0007 | 157 0.0007 | 138 0.0007 | 133 0.0007 | 128 0.0007 | 322 0.0006 $N^{0.011}$ 85 |
| 38 | CLEARVIEWAI-002 | 126 0.0007 | 119 0.0008 | 111 0.0008 | 110 0.0009 | 104 0.0010 | 245 0.0001 $N^{0.132}$ 168 | 167 0.0007 | 149 0.0007 | 146 0.0007 | 135 0.0007 | 126 0.0007 | 326 0.0006 $N^{0.008}$ 67 |
| 39 | CLEARVIEWAI-003 | 128 0.0007 | 113 0.0008 | 103 0.0008 | 93 0.0009 | 88 0.0010 | 295 0.0002 $N^{0.107}$ 104 | 160 0.0006 | 156 0.0007 | 139 0.0007 | 127 0.0007 | 118 0.0007 | 336 0.0006 $N^{0.003}$ 19 |
| 40 | CLOUDWALK-HR-000 | 243 0.0011 | 223 0.0011 | 204 0.0012 | 183 0.0012 | 178 0.0014 | 408 0.0004 $N^{0.069}$ 29 | 312 0.0010 | 299 0.0010 | 285 0.0011 | 271 0.0011 | 253 0.0011 | 385 0.0008 $N^{0.016}$ 132 |
| 41 | CLOUDWALK-MT-000 | 295 0.0014 | 273 0.0014 | 253 0.0015 | 229 0.0015 | 212 0.0016 | 431 0.0008 $N^{0.045}$ 11 | 387 0.0014 | 376 0.0014 | 346 0.0014 | 327 0.0014 | 310 0.0014 | 429 0.0014 $N^{0.002}$ 13 |
| 42 | CLOUDWALK-MT-001 | 296 0.0014 | 271 0.0014 | 247 0.0014 | 224 0.0015 | 207 0.0016 | 434 0.0009 $N^{0.036}$ 8 | 386 0.0014 | 377 0.0014 | 345 0.0014 | 326 0.0014 | 309 0.0014 | 427 0.0013 $N^{0.003}$ 23 |
| 43 | CLOUDWALK-MT-002 | 298 0.0014 | 272 0.0014 | 246 0.0014 | 223 0.0015 | 202 0.0016 | 438 0.0009 $N^{0.034}$ 6 | 388 0.0014 | 374 0.0014 | 348 0.0014 | 330 0.0014 | 312 0.0014 | 430 0.0014 $N^{0.002}$ 14 |
| 44 | COGENT-000 | 502 0.0098 | 486 0.0101 | 440 0.0106 | 426 0.0112 | 418 0.0122 | 455 0.0036 $N^{0.073}$ 31 | 430 0.0018 | 435 0.0021 | 411 0.0024 | 410 0.0033 | 435 0.0092 | 15 0.0000 $N^{0.507}$ 453 |
| 45 | COGENT-001 | 503 0.0098 | 487 0.0101 | 439 0.0106 | 425 0.0112 | 417 0.0122 | 456 0.0036 $N^{0.073}$ 32 | 431 0.0018 | 434 0.0021 | 412 0.0024 | 411 0.0033 | 436 0.0092 | 16 0.0000 $N^{0.507}$ 452 |
| 46 | COGENT-002 | 396 0.0025 | 401 0.0033 | 387 0.0038 | 379 0.0046 | 375 0.0055 | 151 0.0001 $N^{0.264}$ 391 | 344 0.0011 | 347 0.0013 | 340 0.0014 | 345 0.0016 | 346 0.0018 | 73 0.0001 $N^{0.163}$ 394 |
| 47 | COGENT-003 | 402 0.0027 | 403 0.0035 | 390 0.0040 | 382 0.0048 | 378 0.0057 | 208 0.0001 $N^{0.249}$ 375 | 353 0.0012 | 372 0.0014 | 369 0.0015 | 357 0.0017 | 357 0.0019 | 79 0.0001 $N^{0.165}$ 395 |
| 48 | COGENT-004 | 301 0.0014 | 303 0.0016 | 293 0.0018 | 290 0.0021 | 281 0.0025 | 225 0.0001 $N^{0.190}$ 299 | 291 0.0010 | 297 0.0010 | 295 0.0011 | 285 0.0011 | 278 0.0012 | 231 0.0004 $N^{0.067}$ 313 |
| 49 | COGENT-005 | 258 0.0012 | 258 0.0013 | 254 0.0015 | 246 0.0016 | 234 0.0018 | 320 0.0002 $N^{0.134}$ 175 | 293 0.0010 | 280 0.0010 | 264 0.0010 | 249 0.0010 | 244 0.0010 | 358 0.0007 $N^{0.023}$ 174 |
| 50 | COGENT-006 | 160 0.0008 | 158 0.0009 | 154 0.0009 | 145 0.0011 | 141 0.0012 | 294 0.0002 $N^{0.120}$ 134 | 177 0.0007 | 182 0.0007 | 175 0.0007 | 155 0.0007 | 154 0.0008 | 292 0.0005 $N^{0.028}$ 194 |
| 51 | COGENT-007 | 107 0.0007 | 109 0.0007 | 127 0.0009 | 124 0.0010 | 128 0.0011 | 126 0.0001 $N^{0.172}$ 244 | 130 0.0006 | 127 0.0006 | 119 0.0006 | 115 0.0006 | 112 0.0006 | 193 0.0004 $N^{0.035}$ 233 |
| 52 | COGNITEC-000 | 534 0.0192 | 533 0.0249 | 465 0.0294 | 455 0.0349 | 448 0.0414 | 420 0.0006 $N^{0.262}$ 390 | 505 0.0046 | 509 0.0062 | 459 0.0074 | 450 0.0093 | 444 0.0118 | 48 0.0001 $N^{0.319}$ 439 |
| 53 | COGNITEC-001 | 495 0.0087 | 498 0.0114 | 450 0.0136 | 441 0.0163 | 432 0.0196 | 327 0.0002 $N^{0.278}$ 404 | 472 0.0027 | 470 0.0031 | 435 0.0036 | 425 0.0043 | 415 0.0051 | 78 0.0001 $N^{0.224}$ 414 |
| 54 | COGNITEC-002 | 443 0.0045 | 439 0.0053 | 416 0.0063 | 404 0.0075 | 401 0.0091 | 297 0.0002 $N^{0.244}$ 370 | 449 0.0021 | 447 0.0023 | 415 0.0025 | 404 0.0027 | 394 0.0030 | 181 0.0004 $N^{0.131}$ 371 |
| 55 | COGNITEC-003 | 447 0.0049 | 444 0.0059 | 421 0.0069 | 411 0.0082 | 404 0.0097 | 334 0.0002 $N^{0.232}$ 352 | 468 0.0025 | 459 0.0028 | 421 0.0029 | 406 0.0032 | 400 0.0034 | 335 0.0006 $N^{0.105}$ 356 |
| 56 | COGNITEC-004 | 384 0.0023 | 388 0.0028 | 378 0.0034 | 371 0.0042 | 370 0.0052 | 90 0.0001 $N^{0.280}$ 409 | 286 0.0010 | 303 0.0010 | 307 0.0011 | 319 0.0013 | 323 0.0015 | 71 0.0001 $N^{0.160}$ 391 |
| 57 | COGNITEC-005 | 240 0.0011 | 254 0.0013 | 256 0.0015 | 252 0.0017 | 255 0.0020 | 121 0.0001 $N^{0.210}$ 327 | 197 0.0007 | 200 0.0008 | 195 0.0008 | 199 0.0008 | 190 0.0009 | 197 0.0004 $N^{0.053}$ 280 |
| 58 | COGNITEC-006 | 230 0.0011 | 238 0.0012 | 232 0.0013 | 238 0.0016 | 241 0.0018 | 162 0.0001 $N^{0.194}$ 304 | 200 0.0007 | 195 0.0008 | 193 0.0008 | 193 0.0008 | 195 0.0009 | 200 0.0004 $N^{0.052}$ 279 |
| 59 | COGNITEC-007 | 180 0.0009 | 197 0.0010 | 188 0.0011 | 196 0.0013 | 204 0.0016 | 141 0.0001 $N^{0.186}$ 290 | 150 0.0006 | 153 0.0007 | 150 0.0007 | 147 0.0007 | 147 0.0007 | 141 0.0003 $N^{0.056}$ 285 |
| 60 | COGNITEC-008 | 168 0.0008 | 183 0.0010 | 181 0.0011 | 195 0.0013 | 208 0.0016 | 65 0.0000 $N^{0.218}$ 334 | 144 0.0006 | 140 0.0006 | 137 0.0006 | 128 0.0007 | 124 0.0007 | 252 0.0004 $N^{0.028}$ 196 |
| 61 | COGNITEC-009 | 225 0.0010 | 229 0.0012 | 224 0.0013 | 228 0.0015 | 238 0.0018 | 174 0.0001 $N^{0.189}$ 295 | 226 0.0008 | 213 0.0008 | 209 0.0008 | 201 0.0008 | 194 0.0009 | 331 0.0006 $N^{0.022}$ 166 |
| 62 | COMPAL-000 | 277 0.0013 | 270 0.0014 | 493 1.0000 | 534 1.0000 | 487 1.0000 | 3 0.0000 $N^{2.726}$ 466 | 340 0.0011 | 329 0.0011 | 557 1.0000 | 559 1.0000 | 539 1.0000 | 3 0.0000 $N^{2.800}$ 465 |
| 63 | CORSIGHT-000 | 117 0.0007 | 114 0.0008 | 120 0.0008 | 132 0.0010 | 133 0.0011 | 123 0.0001 $N^{0.173}$ 255 | 142 0.0006 | 139 0.0006 | 125 0.0006 | 112 0.0006 | 104 0.0006 | 325 0.0006 $N^{0.006}$ 56 |
| 64 | CORSIGHT-001 | 122 0.0007 | 118 0.0008 | 126 0.0009 | 120 0.0010 | 122 0.0011 | 190 0.0001 $N^{0.155}$ 214 | 135 0.0006 | 130 0.0006 | 126 0.0006 | 121 0.0006 | 121 0.0007 | 183 0.0004 $N^{0.038}$ 244 |
| 65 | CUBOX-001 | 176 0.0009 | 166 0.0009 | 152 0.0009 | 150 0.0011 | 142 0.0012 | 321 0.0002 $N^{0.107}$ 103 | 231 0.0008 | 214 0.0008 | 203 0.0008 | 190 0.0008 | 180 0.0008 | 375 0.0008 $N^{0.004}$ 34 |
| 66 | CUBOX-002 | 246 0.0011 | 226 0.0011 | 215 0.0012 | 198 0.0013 | 182 0.0014 | 389 0.0004 $N^{0.082}$ 45 | 319 0.0010 | 307 0.0011 | 287 0.0011 | 269 0.0011 | 249 0.0011 | 406 0.0010 $N^{0.007}$ 59 |
| 67 | CUBOX-003 | 262 0.0012 | 242 0.0012 | 225 0.0013 | 210 0.0014 | 201 0.0016 | 397 0.0004 $N^{0.085}$ 54 | 341 0.0011 | 325 0.0011 | 311 0.0011 | 293 0.0012 | 274 0.0012 | 400 0.0009 $N^{0.014}$ 116 |
| 68 | CYBERLINK-000 | 412 0.0031 | 405 0.0036 | 395 0.0042 | 385 0.0050 | 380 0.0059 | 280 0.0001 $N^{0.227}$ 345 | 424 0.0017 | 419 0.0019 | 399 0.0020 | 391 0.0022 | 382 0.0024 | 233 0.0004 $N^{0.108}$ 359 |
| 69 | CYBERLINK-001 | 399 0.0026 | 399 0.0032 | 388 0.0038 | 381 0.0047 | 377 0.0057 | 143 0.0001 $N^{0.266}$ 392 | 369 0.0012 | 366 0.0013 | 352 0.0014 | 351 0.0016 | 351 0.0018 | 104 0.0002 $N^{0.135}$ 375 |
| 70 | CYBERLINK-002 | 370 0.0021 | 362 0.0023 | 345 0.0025 | 329 0.0027 | 321 0.0031 | 379 0.0003 $N^{0.138}$ 184 | 416 0.0017 | 407 0.0017 | 385 0.0018 | 370 0.0018 | 354 0.0019 | 402 0.0009 $N^{0.043}$ 257 |
| 71 | CYBERLINK-003 | 241 0.0011 | 240 0.0012 | 235 0.0013 | 219 0.0015 | 218 0.0017 | 299 0.0002 $N^{0.139}$ 188 | 207 0.0008 | 204 0.0008 | 201 0.0008 | 207 0.0009 | 210 0.0009 | 145 0.0003 $N^{0.067}$ 314 |
| 72 | CYBERLINK-004 | 270 0.0012 | 256 0.0013 | 249 0.0014 | 244 0.0016 | 227 0.0018 | 347 0.0002 $N^{0.122}$ 144 | 308 0.0010 | 304 0.0011 | 289 0.0011 | 275 0.0011 | 267 0.0011 | 345 0.0007 $N^{0.032}$ 218 |

*Table 44:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK FNIR(N, T= 0, R) | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 73 | CYBERLINK-005 | 284 0.0013 | 278 0.0015 | 258 0.0015 | 255 0.0017 | 248 0.0019 | 355 0.0003 $N^{0.121\ 142}$ | 318 0.0010 | 313 0.0011 | 302 0.0011 | 288 0.0011 | 276 0.0012 | 327 0.0006 $N^{0.042\ 252}$ |
| 74 | CYBERLINK-006 | 139 0.0008 | 130 0.0008 | 109 0.0008 | 104 0.0009 | 98 0.0010 | 310 0.0002 $N^{0.104\ 94}$ | 168 0.0007 | 162 0.0007 | 148 0.0007 | 140 0.0007 | 130 0.0007 | 321 0.0006 $N^{0.012\ 96}$ |
| 75 | DAHUA-001 | 454 0.0051 | 450 0.0064 | 427 0.0077 | 416 0.0090 | 409 0.0109 | 289 0.0002 $N^{0.261\ 386}$ | 466 0.0025 | 460 0.0028 | 423 0.0030 | 409 0.0033 | 402 0.0036 | 286 0.0005 $N^{0.124\ 369}$ |
| 76 | DAHUA-002 | 281 0.0013 | 281 0.0015 | 277 0.0017 | 272 0.0019 | 274 0.0023 | 217 0.0001 $N^{0.188\ 294}$ | 279 0.0009 | 279 0.0010 | 273 0.0010 | 263 0.0011 | 261 0.0011 | 253 0.0004 $N^{0.059\ 292}$ |
| 77 | DAHUA-003 | 106 0.0007 | 144 0.0008 | 176 0.0011 | 185 0.0012 | 194 0.0015 | 25 0.0000 $N^{0.270\ 396}$ | 81 0.0005 | 77 0.0005 | 72 0.0005 | 85 0.0006 | 87 0.0006 | 133 0.0003 $N^{0.044\ 260}$ |
| 78 | DAHUA-004 | 105 0.0007 | 102 0.0007 | 107 0.0008 | 100 0.0009 | 116 0.0011 | 157 0.0001 $N^{0.161\ 228}$ | 91 0.0005 | 91 0.0005 | 90 0.0005 | 92 0.0006 | 91 0.0006 | 152 0.0003 $N^{0.037\ 241}$ |
| 79 | DAON-000 | 426 0.0035 | 409 0.0037 | 389 0.0039 | 370 0.0042 | 364 0.0045 | 441 0.0011 $N^{0.084\ 52}$ | 483 0.0032 | 475 0.0033 | 427 0.0033 | 413 0.0033 | 399 0.0034 | 450 0.0027 $N^{0.014\ 115}$ |
| 80 | DAON-002 | 148 0.0008 | 170 0.0009 | 184 0.0011 | 193 0.0013 | 209 0.0016 | 36 0.0000 $N^{0.239\ 364}$ | 132 0.0006 | 137 0.0006 | 131 0.0006 | 131 0.0007 | 129 0.0007 | 150 0.0003 $N^{0.047\ 268}$ |
| 81 | DAON-003 | 96 0.0006 | 99 0.0007 | 95 0.0008 | 99 0.0009 | 95 0.0010 | 161 0.0001 $N^{0.158\ 220}$ | 88 0.0005 | 82 0.0005 | 79 0.0005 | 77 0.0005 | 77 0.0006 | 191 0.0004 $N^{0.026\ 186}$ |
| 82 | DATECH-000 | 150 0.0008 | 184 0.0010 | 183 0.0011 | 191 0.0013 | 206 0.0016 | 58 0.0000 $N^{0.223\ 341}$ | 147 0.0006 | 174 0.0007 | 160 0.0007 | 154 0.0007 | 150 0.0007 | 147 0.0003 $N^{0.055\ 281}$ |
| 83 | DATECH-001 | 113 0.0007 | 117 0.0008 | 116 0.0008 | 127 0.0010 | 123 0.0011 | 160 0.0001 $N^{0.163\ 236}$ | 119 0.0006 | 117 0.0006 | 110 0.0006 | 108 0.0006 | 105 0.0006 | 174 0.0003 $N^{0.036\ 239}$ |
| 84 | DECATUR-000 | 312 0.0015 | 318 0.0017 | 315 0.0021 | 312 0.0024 | 313 0.0029 | 115 0.0001 $N^{0.236\ 358}$ | 263 0.0009 | 268 0.0009 | 263 0.0010 | 276 0.0011 | 281 0.0012 | 120 0.0002 $N^{0.097\ 351}$ |
| 85 | DEEPGLINT-001 | 227 0.0011 | 217 0.0011 | 202 0.0012 | 192 0.0013 | 192 0.0015 | 333 0.0002 $N^{0.115\ 124}$ | 278 0.0009 | 261 0.0009 | 250 0.0010 | 235 0.0010 | 230 0.0010 | 346 0.0007 $N^{0.023\ 175}$ |
| 86 | DEEPSEA-001 | 407 0.0029 | 414 0.0040 | 402 0.0048 | 395 0.0062 | 392 0.0077 | 39 0.0000 $N^{0.335\ 445}$ | 256 0.0009 | 291 0.0010 | 319 0.0012 | 321 0.0014 | 334 0.0016 | 44 0.0001 $N^{0.204\ 410}$ |
| 87 | DERMALOG-005 | 513 0.0112 | 512 0.0147 | 455 0.0198 | 451 0.0287 | 452 0.0445 | 22 0.0000 $N^{0.475\ 453}$ | 543 0.0092 | 546 0.0120 | 471 0.0169 | 460 0.0252 | 455 0.0403 | 18 0.0000 $N^{0.510\ 454}$ |
| 88 | DERMALOG-006 | 477 0.0073 | 469 0.0079 | 433 0.0084 | 417 0.0091 | 407 0.0101 | 449 0.0017 $N^{0.110\ 109}$ | 518 0.0061 | 508 0.0062 | 449 0.0063 | 434 0.0064 | 422 0.0066 | 456 0.0042 $N^{0.027\ 192}$ |
| 89 | DERMALOG-007 | 481 0.0076 | 476 0.0089 | 437 0.0099 | 428 0.0115 | 421 0.0136 | 419 0.0005 $N^{0.197\ 307}$ | 506 0.0048 | 496 0.0050 | 444 0.0053 | 428 0.0055 | 419 0.0059 | 440 0.0018 $N^{0.072\ 320}$ |
| 90 | DERMALOG-008 | 368 0.0021 | 375 0.0025 | 367 0.0030 | 364 0.0036 | 362 0.0045 | 96 0.0001 $N^{0.268\ 394}$ | 339 0.0011 | 336 0.0012 | 329 0.0012 | 322 0.0014 | 324 0.0015 | 116 0.0002 $N^{0.113\ 363}$ |
| 91 | DERMALOG-009 | 381 0.0022 | 373 0.0024 | 353 0.0027 | 341 0.0029 | 329 0.0033 | 385 0.0004 $N^{0.137\ 180}$ | 429 0.0018 | 421 0.0019 | 391 0.0019 | 376 0.0019 | 364 0.0020 | 418 0.0012 $N^{0.034\ 227}$ |
| 92 | DERMALOG-010 | 350 0.0018 | 328 0.0018 | 306 0.0019 | 280 0.0020 | 265 0.0021 | 430 0.0008 $N^{0.064\ 24}$ | 417 0.0017 | 405 0.0017 | 380 0.0017 | 355 0.0017 | 342 0.0017 | 437 0.0015 $N^{0.010\ 83}$ |
| 93 | DERMALOG-011 | 237 0.0011 | 243 0.0012 | 252 0.0015 | 253 0.0017 | 254 0.0020 | 95 0.0001 $N^{0.219\ 335}$ | 176 0.0007 | 176 0.0007 | 184 0.0008 | 183 0.0008 | 177 0.0008 | 136 0.0003 $N^{0.065\ 311}$ |
| 94 | DERMALOG-012 | 302 0.0014 | 311 0.0017 | 300 0.0019 | 281 0.0021 | 277 0.0024 | 278 0.0001 $N^{0.172\ 247}$ | 337 0.0011 | 324 0.0011 | 308 0.0011 | 294 0.0012 | 287 0.0012 | 356 0.0007 $N^{0.032\ 219}$ |
| 95 | DERMALOG-013 | 280 0.0013 | 294 0.0016 | 283 0.0018 | 275 0.0020 | 271 0.0023 | 241 0.0001 $N^{0.182\ 278}$ | 300 0.0010 | 296 0.0010 | 282 0.0010 | 268 0.0011 | 264 0.0011 | 340 0.0006 $N^{0.034\ 229}$ |
| 96 | DERMALOG-014 | 355 0.0019 | 395 0.0030 | 405 0.0050 | 421 0.0095 | 433 0.0197 | 11 0.0000 $N^{0.807\ 457}$ | 409 0.0016 | 454 0.0026 | 442 0.0045 | 449 0.0089 | 451 0.0190 | 11 0.0000 $N^{0.855\ 457}$ |
| 97 | DERMALOG-015 | 244 0.0011 | 241 0.0012 | 220 0.0012 | 208 0.0014 | 200 0.0015 | 361 0.0003 $N^{0.105\ 95}$ | 311 0.0010 | 295 0.0010 | 281 0.0010 | 259 0.0010 | 247 0.0011 | 401 0.0009 $N^{0.008\ 64}$ |
| 98 | DERMALOG-016 | 233 0.0011 | 208 0.0011 | 187 0.0011 | 178 0.0012 | 170 0.0013 | 399 0.0004 $N^{0.075\ 35}$ | 303 0.0010 | 292 0.0010 | 267 0.0010 | 253 0.0010 | 241 0.0010 | 404 0.0010 $N^{0.004\ 38}$ |
| 99 | DERMALOG-017 | 234 0.0011 | 214 0.0011 | 195 0.0011 | 176 0.0012 | 172 0.0013 | 396 0.0004 $N^{0.076\ 36}$ | 306 0.0010 | 290 0.0010 | 270 0.0010 | 255 0.0010 | 242 0.0010 | 405 0.0010 $N^{0.004\ 37}$ |
| 100 | DILUSENSE-000 | 324 0.0016 | 330 0.0018 | 320 0.0021 | 315 0.0025 | 315 0.0030 | 179 0.0001 $N^{0.219\ 337}$ | 239 0.0009 | 241 0.0009 | 243 0.0009 | 251 0.0010 | 256 0.0011 | 127 0.0003 $N^{0.086\ 338}$ |
| 101 | DILUSENSE-001 | 207 0.0010 | 218 0.0011 | 216 0.0012 | | 214 0.0016 | 203 0.0001 $N^{0.172\ 245}$ | 213 0.0008 | 202 0.0008 | 196 0.0008 | | 188 0.0009 | 289 0.0005 $N^{0.036\ 238}$ |
| 102 | EINETWORKSINDIA-000 | 273 0.0013 | 276 0.0014 | 274 0.0016 | 268 0.0019 | 267 0.0022 | 235 0.0001 $N^{0.180\ 269}$ | 321 0.0011 | 308 0.0011 | 296 0.0011 | 277 0.0011 | 263 0.0011 | 378 0.0008 $N^{0.021\ 158}$ |
| 103 | EINETWORKSINDIA-001 | 267 0.0012 | 263 0.0014 | 260 0.0016 | 257 0.0017 | 256 0.0020 | 246 0.0001 $N^{0.172\ 248}$ | 288 0.0010 | 274 0.0010 | 266 0.0010 | 257 0.0010 | 251 0.0011 | 324 0.0006 $N^{0.038\ 243}$ |
| 104 | EINETWORKSINDIA-002 | 132 0.0007 | 126 0.0008 | 114 0.0008 | 108 0.0009 | 103 0.0010 | 276 0.0001 $N^{0.120\ 135}$ | 151 0.0006 | 146 0.0006 | 135 0.0006 | 124 0.0006 | 123 0.0007 | 308 0.0005 $N^{0.014\ 113}$ |
| 105 | FACEPHI-000 | 306 0.0015 | 284 0.0015 | 264 0.0016 | 249 0.0017 | 243 0.0019 | 417 0.0005 $N^{0.077\ 39}$ | 379 0.0013 | 369 0.0014 | 341 0.0014 | 325 0.0014 | 313 0.0014 | 414 0.0011 $N^{0.017\ 138}$ |
| 106 | FIRSTCREDITKZ-001 | 158 0.0008 | 149 0.0008 | 140 0.0009 | 130 0.0010 | 119 0.0011 | 323 0.0002 $N^{0.101\ 87}$ | 223 0.0008 | 211 0.0008 | 202 0.0008 | 189 0.0008 | 175 0.0008 | 368 0.0008 $N^{0.006\ 47}$ |
| 107 | FUJITSULAB-000 | 325 0.0016 | 332 0.0018 | 324 0.0022 | 321 0.0026 | 314 0.0030 | 172 0.0001 $N^{0.222\ 339}$ | 292 0.0010 | 283 0.0010 | 290 0.0011 | 282 0.0011 | 288 0.0012 | 165 0.0003 $N^{0.079\ 326}$ |
| 108 | FUJITSULAB-001 | 282 0.0013 | 283 0.0015 | 279 0.0017 | 273 0.0019 | 270 0.0023 | 243 0.0001 $N^{0.180\ 270}$ | 289 0.0010 | 293 0.0010 | 283 0.0010 | 278 0.0011 | 272 0.0012 | 259 0.0004 $N^{0.060\ 299}$ |
| 109 | GORILLA-002 | 524 0.0145 | 526 0.0194 | 461 0.0235 | 450 0.0285 | 445 0.0348 | 359 0.0003 $N^{0.299\ 429}$ | 480 0.0030 | 487 0.0039 | 443 0.0047 | 432 0.0060 | 428 0.0078 | 35 0.0000 $N^{0.321\ 440}$ |
| 110 | GORILLA-004 | 442 0.0045 | 445 0.0059 | 423 0.0072 | 415 0.0088 | 410 0.0111 | 145 0.0001 $N^{0.307\ 435}$ | 357 0.0012 | 385 0.0014 | 387 0.0018 | 390 0.0021 | 388 0.0026 | 29 0.0000 $N^{0.269\ 429}$ |
| 111 | GORILLA-005 | 379 0.0022 | 387 0.0028 | 382 0.0035 | 374 0.0043 | 372 0.0053 | 56 0.0000 $N^{0.301\ 432}$ | 247 0.0009 | 260 0.0009 | 276 0.0010 | 303 0.0012 | 311 0.0014 | 60 0.0001 $N^{0.168\ 397}$ |
| 112 | GORILLA-006 | 245 0.0011 | 269 0.0014 | 276 0.0017 | 277 0.0020 | 287 0.0025 | 34 0.0000 $N^{0.275\ 399}$ | 175 0.0007 | 191 0.0007 | 191 0.0008 | 189 0.0008 | 192 0.0009 | 114 0.0002 $N^{0.081\ 332}$ |
| 113 | GORILLA-007 | 226 0.0010 | 260 0.0013 | 259 0.0016 | 276 0.0020 | 283 0.0025 | 26 0.0000 $N^{0.295\ 426}$ | 173 0.0007 | 170 0.0007 | 166 0.0007 | 165 0.0008 | 178 0.0008 | 137 0.0003 $N^{0.063\ 304}$ |
| 114 | GORILLA-008 | 198 0.0010 | 225 0.0011 | 230 0.0013 | 241 0.0016 | 242 0.0019 | 64 0.0000 $N^{0.229\ 349}$ | 170 0.0007 | 171 0.0007 | 165 0.0007 | 157 0.0007 | 159 0.0008 | 172 0.0003 $N^{0.050\ 273}$ |
| 115 | GRIAULE-000 | 353 0.0018 | 355 0.0021 | 341 0.0024 | 324 0.0027 | 322 0.0031 | 305 0.0002 $N^{0.175\ 259}$ | 368 0.0012 | 355 0.0013 | 350 0.0014 | 339 0.0015 | 325 0.0015 | 267 0.0004 $N^{0.078\ 324}$ |
| 116 | GRIAULE-001 | 141 0.0008 | 143 0.0008 | 138 0.0009 | 137 0.0010 | 135 0.0012 | 222 0.0001 $N^{0.144\ 196}$ | 154 0.0006 | 164 0.0007 | 149 0.0007 | 139 0.0007 | 139 0.0007 | 269 0.0004 $N^{0.029\ 201}$ |
| 117 | HIK-003 | 496 0.0089 | 499 0.0115 | 451 0.0137 | 442 0.0167 | 434 0.0201 | 330 0.0002 $N^{0.278\ 408}$ | 456 0.0022 | 465 0.0029 | 433 0.0034 | 424 0.0042 | 417 0.0052 | 41 0.0001 $N^{0.283\ 433}$ |
| 118 | HIK-004 | 494 0.0086 | 497 0.0111 | 448 0.0132 | 439 0.0160 | 431 0.0192 | 328 0.0002 $N^{0.276\ 402}$ | 460 0.0023 | 461 0.0028 | 430 0.0033 | 422 0.0040 | 414 0.0050 | 47 0.0001 $N^{0.267\ 426}$ |
| 119 | HIK-005 | 421 0.0034 | 421 0.0044 | 406 0.0051 | 396 0.0063 | 393 0.0079 | 165 0.0001 $N^{0.283\ 415}$ | 349 0.0011 | 379 0.0014 | 363 0.0015 | 366 0.0018 | 369 0.0021 | 57 0.0001 $N^{0.198\ 409}$ |
| 120 | HISIGN-001 | 130 0.0007 | 152 0.0009 | 166 0.0010 | 171 0.0012 | 174 0.0013 | 75 0.0000 $N^{0.206\ 318}$ | 131 0.0006 | 130 0.0006 | 120 0.0006 | 119 0.0006 | 113 0.0006 | 186 0.0004 $N^{0.037\ 240}$ |
| 121 | HYPERVERGE-001 | 204 0.0010 | 200 0.0010 | 182 0.0011 | 174 0.0012 | 177 0.0014 | 329 0.0002 $N^{0.111\ 115}$ | 281 0.0009 | 266 0.0009 | 246 0.0009 | 227 0.0009 | 214 0.0009 | 390 0.0009 $N^{0.006\ 54}$ |
| 122 | HYPERVERGE-002 | 208 0.0010 | 195 0.0010 | 179 0.0011 | 162 0.0011 | 146 0.0012 | 406 0.0004 $N^{0.062\ 22}$ | 268 0.0009 | 252 0.0009 | 242 0.0009 | 226 0.0009 | 216 0.0009 | 376 0.0008 $N^{0.012\ 93}$ |
| 123 | HYPERVERGE-003 | 214 0.0010 | 212 0.0011 | 200 0.0012 | 194 0.0013 | 198 0.0015 | 291 0.0002 $N^{0.136\ 178}$ | 269 0.0009 | 263 0.0009 | 248 0.0009 | 229 0.0009 | 217 0.0010 | 374 0.0008 $N^{0.013\ 104}$ |
| 124 | HYUNDAI-ROBOTICSLAB-000 | 380 0.0022 | 367 0.0023 | 348 0.0025 | 328 0.0027 | 319 0.0030 | 418 0.0005 $N^{0.107\ 101}$ | 432 0.0018 | 423 0.0019 | 396 0.0019 | 377 0.0019 | 363 0.0020 | 423 0.0012 $N^{0.030\ 208}$ |
| 125 | HYUNDAI-ROBOTICSLAB-001 | 268 0.0012 | 267 0.0014 | 263 0.0016 | 260 0.0018 | 258 0.0021 | 230 0.0001 $N^{0.178\ 266}$ | 249 0.0009 | 242 0.0009 | 247 0.0009 | 244 0.0010 | 248 0.0011 | 175 0.0003 $N^{0.067\ 315}$ |
| 126 | HYUNDAI-ROBOTICSLAB-002 | 93 0.0006 | 100 0.0007 | 96 0.0008 | 98 0.0009 | 112 0.0011 | 98 0.0001 $N^{0.178\ 267}$ | 109 0.0006 | 97 0.0006 | 100 0.0006 | 95 0.0006 | 93 0.0006 | 245 0.0004 $N^{0.021\ 156}$ |
| 127 | HYUNDAI-ROBOTICSLAB-003 | 79 0.0006 | 85 0.0007 | 89 0.0007 | 87 0.0009 | 101 0.0010 | 67 0.0000 $N^{0.190\ 298}$ | 74 0.0005 | 70 0.0005 | 73 0.0005 | 68 0.0005 | 64 0.0005 | 201 0.0004 $N^{0.023\ 168}$ |
| 128 | HZAILU-000 | 329 0.0016 | 329 0.0018 | 319 0.0021 | 316 0.0025 | 317 0.0030 | 184 0.0001 $N^{0.216\ 331}$ | 325 0.0011 | 316 0.0011 | 300 0.0011 | 292 0.0012 | 284 0.0012 | 332 0.0006 $N^{0.042\ 251}$ |
| 129 | HZAILU-001 | 309 0.0015 | 299 0.0016 | 278 0.0017 | 265 0.0018 | 261 0.0021 | 378 0.0003 $N^{0.114\ 122}$ | 331 0.0011 | 333 0.0011 | 321 0.0012 | 309 0.0012 | 299 0.0013 | 282 0.0005 $N^{0.063\ 301}$ |
| 130 | HZAILU-002 | 310 0.0015 | 302 0.0016 | 284 0.0018 | 274 0.0020 | 269 0.0022 | 344 0.0002 $N^{0.137\ 182}$ | 332 0.0011 | 334 0.0011 | 318 0.0012 | 311 0.0013 | 300 0.0013 | 281 0.0005 $N^{0.063\ 302}$ |
| 131 | HZAILU-003 | 255 0.0012 | 251 0.0013 | 245 0.0014 | 245 0.0016 | 240 0.0018 | 281 0.0001 $N^{0.153\ 210}$ | 245 0.0009 | 251 0.0009 | 259 0.0010 | 245 0.0010 | 243 0.0010 | 205 0.0004 $N^{0.064\ 307}$ |
| 132 | HZAILU-004 | 182 0.0009 | 175 0.0009 | 165 0.0010 | 166 0.0011 | 158 0.0012 | 318 0.0002 $N^{0.111\ 114}$ | 242 0.0009 | 230 0.0009 | 214 0.0009 | 206 0.0009 | 196 0.0009 | 369 0.0008 $N^{0.009\ 76}$ |
| 133 | I2V-001 | 392 0.0025 | 378 0.0026 | 354 0.0027 | 333 0.0028 | 316 0.0030 | 440 0.0011 $N^{0.059\ 19}$ | 462 0.0024 | 451 0.0024 | 410 0.0024 | 397 0.0024 | 384 0.0024 | 446 0.0022 $N^{0.006\ 53}$ |
| 134 | ICEHOUSE-001 | 189 0.0009 | 192 0.0010 | 193 0.0011 | 190 0.0013 | 185 0.0014 | 253 0.0001 $N^{0.149\ 204}$ | 186 0.0007 | 189 0.0007 | 187 0.0008 | 174 0.0008 | 168 0.0008 | 255 0.0004 $N^{0.039\ 245}$ |
| 135 | ICM-001 | 394 0.0025 | 384 0.0028 | 368 0.0030 | 351 0.0033 | 345 0.0038 | 394 0.0004 $N^{0.141\ 191}$ | 428 0.0018 | 424 0.0019 | 398 0.0020 | 386 0.0020 | 374 0.0021 | 389 0.0009 $N^{0.056\ 284}$ |
| 136 | IDEMIA-003 | 450 0.0050 | 454 0.0066 | 430 0.0080 | 423 0.0098 | 416 0.0118 | 212 0.0001 $N^{0.293\ 424}$ | 443 0.0020 | 452 0.0024 | 418 0.0027 | 408 0.0033 | 403 0.0038 | 67 0.0001 $N^{0.224\ 413}$ |
| 137 | IDEMIA-004 | 453 0.0051 | 449 0.0063 | 426 0.0076 | 420 0.0094 | 412 0.0114 | 240 0.0001 $N^{0.281\ 410}$ | 400 0.0015 | 411 0.0018 | 407 0.0022 | 400 0.0027 | 397 0.0033 | 33 0.0000 $N^{0.272\ 430}$ |
| 138 | IDEMIA-005 | 461 0.0061 | 468 0.0077 | 435 0.0093 | 427 0.0115 | 422 0.0140 | 263 0.0001 $N^{0.287\ 419}$ | 441 0.0019 | 458 0.0027 | 426 0.0033 | 423 0.0041 | 416 0.0051 | 26 0.0000 $N^{0.329\ 444}$ |
| 139 | IDEMIA-006 | 476 0.0072 | 482 0.0092 | 442 0.0109 | 432 0.0132 | 426 0.0158 | 322 0.0002 $N^{0.267\ 393}$ | 464 0.0025 | 477 0.0034 | 439 0.0043 | 429 0.0055 | 426 0.0073 | 21 0.0000 $N^{0.369\ 449}$ |
| 140 | IDEMIA-007 | 346 0.0017 | 359 0.0022 | 352 0.0026 | 349 0.0032 | 352 0.0040 | 52 0.0000 $N^{0.284\ 416}$ | 198 0.0007 | 216 0.0008 | 224 0.0009 | 254 0.0010 | 275 0.0012 | 66 0.0001 $N^{0.153\ 385}$ |
| 141 | IDEMIA-008 | 85 0.0006 | 96 0.0007 | 100 0.0008 | 95 0.0009 | 97 0.0010 | 99 0.0001 $N^{0.176\ 261}$ | 94 0.0005 | 92 0.0005 | 88 0.0005 | 89 0.0006 | 84 0.0006 | 209 0.0004 $N^{0.026\ 183}$ |
| 142 | IDEMIA-009 | 73 0.0006 | 69 0.0006 | 69 0.0007 | 61 0.0008 | 62 0.0009 | 169 0.0001 $N^{0.146\ 199}$ | 69 0.0005 | 64 0.0005 | 65 0.0005 | 62 0.0005 | 57 0.0005 | 243 0.0004 $N^{0.013\ 101}$ |
| 143 | IDEMIA-010 | 42 0.0005 | 41 0.0005 | 42 0.0006 | 35 0.0006 | 20 0.0007 | 274 0.0001 $N^{0.095\ 76}$ | 66 0.0005 | 66 0.0005 | 59 0.0005 | 54 0.0005 | 48 0.0005 | 285 0.0005 $N^{0.004\ 29}$ |
| 144 | IDEMIA-011 | 102 0.0007 | 91 0.0007 | 81 0.0007 | 64 0.0008 | 54 0.0008 | 343 0.0002 $N^{0.077\ 37}$ | 164 0.0006 | 150 0.0007 | 147 0.0007 | 134 0.0007 | 127 0.0007 | 320 0.0006 $N^{0.011\ 89}$ |

*Table 45:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK FNIR(N, T= 0, R) | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 145 | IDEMIA-012 | [103]0.0007 | [97]0.0007 | [90]0.0007 | [69]0.0008 | [53]0.0008 | [356]0.0003 N[0.070 30] | [152]0.0006 | [144]0.0006 | [136]0.0006 | [126]0.0007 | [117]0.0007 | [313]0.0005 N[0.011 92] |
| 146 | IDEMIA-013 | [5]0.0004 | [2]0.0004 | [2]0.0005 | [1]0.0005 | [1]0.0006 | [204]0.0001 N[0.106 97] | [13]0.0004 | [6]0.0004 | [6]0.0004 | [4]0.0004 | [2]0.0004 | [250]0.0004 N[0.000 6] |
| 147 | IDENTY-001 | [202]0.0010 | [193]0.0010 | [484]1.0000 | [470]0.5187 | [463]0.5726 | [4]0.0000 N[2.612 464] | [264]0.0009 | [248]0.0009 | [484]0.9999 | [468]0.1882 | [460]0.2328 | [4]0.0000 N[2.262 464] |
| 148 | IDENTY-002 | [91]0.0006 | [90]0.0007 | [87]0.0007 | [92]0.0009 | [87]0.0010 | [158]0.0001 N[0.156 215] | [106]0.0006 | [110]0.0006 | [102]0.0006 | [97]0.0006 | [96]0.0006 | [229]0.0004 N[0.025 182] |
| 149 | IMAGUS-005 | [299]0.0014 | [298]0.0016 | [290]0.0018 | [285]0.0021 | [280]0.0024 | [223]0.0001 N[0.189 296] | [261]0.0009 | [278]0.0010 | [268]0.0010 | [267]0.0011 | [283]0.0012 | [131]0.0003 N[0.089 342] |
| 150 | IMAGUS-006 | [305]0.0014 | [305]0.0016 | [291]0.0018 | [284]0.0021 | [285]0.0025 | [242]0.0001 N[0.185 288] | [305]0.0010 | [315]0.0011 | [304]0.0011 | [287]0.0011 | [293]0.0012 | [251]0.0004 N[0.065 310] |
| 151 | IMAGUS-007 | [290]0.0014 | [307]0.0016 | [302]0.0019 | [307]0.0023 | [295]0.0026 | [130]0.0001 N[0.224 343] | [232]0.0008 | [245]0.0009 | [233]0.0009 | [233]0.0010 | [259]0.0011 | [125]0.0003 N[0.088 341] |
| 152 | IMDS-SOFTWARE-001 | [114]0.0007 | [127]0.0008 | [148]0.0009 | [154]0.0011 | [159]0.0012 | [74]0.0000 N[0.201 311] | [118]0.0006 | [115]0.0006 | [109]0.0006 | [104]0.0006 | [107]0.0006 | [168]0.0003 N[0.039 246] |
| 153 | IMDS-SOFTWARE-002 | [110]0.0007 | [125]0.0008 | [135]0.0009 | [146]0.0011 | [154]0.0012 | [73]0.0000 N[0.200 310] | [99]0.0005 | [101]0.0006 | [101]0.0006 | [96]0.0006 | [100]0.0006 | [163]0.0003 N[0.037 242] |
| 154 | IMPERIAL-000 | [352]0.0018 | [351]0.0021 | [336]0.0023 | [327]0.0027 | [324]0.0031 | [293]0.0002 N[0.181 275] | [374]0.0013 | [360]0.0013 | [344]0.0014 | [333]0.0014 | [320]0.0015 | [339]0.0006 N[0.052 278] |
| 155 | INCODE-003 | [500]0.0096 | [506]0.0127 | [452]0.0151 | [444]0.0188 | [436]0.0230 | [307]0.0002 N[0.299 430] | [459]0.0023 | [466]0.0029 | [434]0.0035 | [426]0.0044 | [418]0.0054 | [37]0.0000 N[0.295 437] |
| 156 | INCODE-004 | [398]0.0026 | [400]0.0032 | [386]0.0038 | [380]0.0046 | [376]0.0056 | [136]0.0001 N[0.268 395] | [393]0.0014 | [390]0.0015 | [374]0.0016 | [363]0.0017 | [352]0.0018 | [226]0.0004 N[0.094 348] |
| 157 | INCODE-005 | [252]0.0012 | [255]0.0013 | [248]0.0014 | [247]0.0016 | [250]0.0019 | [226]0.0001 N[0.174 256] | [244]0.0009 | [243]0.0009 | [239]0.0009 | [238]0.0010 | [236]0.0010 | [234]0.0004 N[0.056 286] |
| 158 | INCODE-006 | [118]0.0007 | [101]0.0007 | [93]0.0008 | [78]0.0008 | [71]0.0009 | [326]0.0002 N[0.088 59] | [171]0.0007 | [168]0.0007 | [158]0.0007 | [145]0.0007 | [136]0.0007 | [330]0.0006 N[0.009 79] |
| 159 | INCODE-007 | [34]0.0005 | [37]0.0005 | [39]0.0006 | [26]0.0006 | [31]0.0007 | [201]0.0001 N[0.120 136] | [30]0.0005 | [31]0.0005 | [25]0.0005 | [30]0.0005 | [34]0.0005 | [158]0.0003 N[0.025 180] |
| 160 | INNOVATRICS-007 | [260]0.0012 | [259]0.0013 | [257]0.0015 | [254]0.0017 | [257]0.0020 | [228]0.0001 N[0.176 264] | [240]0.0009 | [235]0.0009 | [228]0.0009 | [228]0.0009 | [229]0.0010 | [249]0.0004 N[0.052 277] |
| 161 | INNOVATRICS-008 | [175]0.0009 | [174]0.0009 | [171]0.0010 | [169]0.0011 | [168]0.0013 | [267]0.0001 N[0.137 181] | [180]0.0007 | [177]0.0007 | [167]0.0007 | [161]0.0007 | [155]0.0008 | [298]0.0005 N[0.027 189] |
| 162 | INNOVATRICS-009 | [194]0.0010 | [189]0.0010 | [180]0.0011 | [177]0.0012 | [173]0.0013 | [316]0.0002 N[0.117 128] | [246]0.0009 | [234]0.0009 | [217]0.0009 | [208]0.0009 | [200]0.0009 | [370]0.0008 N[0.010 82] |
| 163 | INNOVATRICS-010 | [212]0.0010 | [203]0.0010 | [192]0.0011 | [173]0.0012 | [171]0.0013 | [363]0.0003 N[0.095 73] | [255]0.0009 | [236]0.0009 | [227]0.0009 | [220]0.0009 | [215]0.0009 | [338]0.0006 N[0.024 177] |
| 164 | INNOVATRICS-011 | [116]0.0007 | [112]0.0008 | [117]0.0008 | [103]0.0009 | [105]0.0010 | [237]0.0001 N[0.133 173] | [133]0.0006 | [133]0.0006 | [127]0.0006 | [123]0.0006 | [122]0.0007 | [164]0.0003 N[0.043 258] |
| 165 | INNOVATRICS-012 | [108]0.0007 | [111]0.0007 | [108]0.0008 | [112]0.0009 | [110]0.0011 | [188]0.0001 N[0.153 209] | [134]0.0006 | [128]0.0006 | [121]0.0006 | [116]0.0006 | [110]0.0006 | [223]0.0004 N[0.031 217] |
| 166 | INNOVATRICS-013 | [55]0.0005 | [64]0.0006 | [63]0.0007 | [71]0.0008 | [75]0.0009 | [61]0.0000 N[0.186 289] | [34]0.0005 | [45]0.0005 | [44]0.0005 | [37]0.0005 | [31]0.0005 | [192]0.0004 N[0.018 142] |
| 167 | INNOVATRICS-014 | [43]0.0005 | [49]0.0006 | [48]0.0006 | [50]0.0007 | [46]0.0008 | [155]0.0001 N[0.142 192] | [31]0.0005 | [37]0.0005 | [32]0.0005 | [49]0.0005 | [54]0.0005 | [126]0.0003 N[0.039 247] |
| 168 | INNOVATRICS-015 | [44]0.0005 | [48]0.0006 | [45]0.0006 | [44]0.0007 | [38]0.0007 | [213]0.0001 N[0.120 139] | [29]0.0005 | [29]0.0005 | [30]0.0005 | [29]0.0005 | [39]0.0005 | [144]0.0003 N[0.028 197] |
| 169 | INSPUR-000 | [68]0.0006 | [73]0.0006 | [71]0.0007 | [80]0.0008 | [96]0.0010 | [40]0.0000 N[0.205 317] | [50]0.0005 | [54]0.0005 | [58]0.0005 | [56]0.0005 | [51]0.0005 | [170]0.0003 N[0.024 179] |
| 170 | INSPUR-001 | [66]0.0006 | [66]0.0006 | [82]0.0007 | [84]0.0008 | [82]0.0010 | [55]0.0000 N[0.195 305] | [49]0.0005 | [57]0.0005 | [50]0.0005 | [41]0.0005 | [40]0.0005 | [208]0.0004 N[0.017 140] |
| 171 | INSPUR-002 | [53]0.0005 | [57]0.0006 | [67]0.0007 | [62]0.0008 | [63]0.0009 | [86]0.0001 N[0.173 254] | [36]0.0005 | [32]0.0005 | [36]0.0005 | [27]0.0005 | [25]0.0005 | [215]0.0004 N[0.014 110] |
| 172 | INTELRESEARCH-001 | [76]0.0006 | [80]0.0006 | [73]0.0007 | [72]0.0008 | [76]0.0009 | [132]0.0001 N[0.158 218] | [73]0.0005 | [65]0.0005 | [62]0.0005 | [63]0.0005 | [62]0.0005 | [227]0.0004 N[0.017 141] |
| 173 | INTEMA-000 | [123]0.0007 | [123]0.0008 | [122]0.0009 | [123]0.0010 | [147]0.0012 | [129]0.0001 N[0.173 250] | [158]0.0006 | [166]0.0007 | [180]0.0008 | [171]0.0008 | [206]0.0009 | [80]0.0001 N[0.116 365] |
| 174 | INTEMA-001 | [16]0.0005 | [12]0.0005 | [31]0.0006 | [42]0.0007 | [74]0.0009 | [21]0.0000 N[0.237 361] | [21]0.0004 | [16]0.0004 | [43]0.0005 | [55]0.0005 | [140]0.0007 | [42]0.0001 N[0.151 384] |
| 175 | INTOZI-001 | [224]0.0010 | [239]0.0012 | [244]0.0014 | [250]0.0017 | [251]0.0020 | [84]0.0001 N[0.224 342] | [149]0.0006 | [165]0.0007 | [159]0.0007 | [152]0.0007 | [171]0.0008 | [107]0.0002 N[0.081 331] |
| 176 | INTSYSMSU-000 | [583]0.1392 | [582]0.1454 | [476]0.1495 | [464]0.1541 | [456]0.1588 | [466]0.0765 N[0.045 12] | [592]0.1095 | [590]0.1160 | [479]0.1203 | [466]0.1248 | [458]0.1293 | [466]0.0516 N[0.057 287] |
| 177 | IONETWORKS-001 | [334]0.0016 | [402]0.0034 | [419]0.0066 | [436]0.0142 | [438]0.0263 | [8]0.0000 N[0.974 459] | [391]0.0014 | [471]0.0031 | [448]0.0062 | [457]0.0138 | [453]0.0257 | [8]0.0000 N[1.013 459] |
| 178 | IREX-000 | [432]0.0040 | [417]0.0040 | [391]0.0041 | [372]0.0042 | [360]0.0044 | [451]0.0024 N[0.035 7] | [495]0.0039 | [486]0.0039 | [438]0.0039 | [418]0.0039 | [406]0.0039 | [453]0.0038 N[0.002 16] |
| 179 | IRISID-001 | [184]0.0009 | [209]0.0011 | [214]0.0012 | [216]0.0015 | [238]0.0018 | [59]0.0000 N[0.229 348] | [124]0.0006 | [120]0.0006 | [129]0.0006 | [141]0.0007 | [142]0.0007 | [95]0.0002 N[0.085 337] |
| 180 | IRISID-002 | [146]0.0008 | [165]0.0009 | [174]0.0010 | [187]0.0012 | [199]0.0015 | [42]0.0000 N[0.230 350] | [102]0.0005 | [103]0.0006 | [104]0.0006 | [109]0.0006 | [114]0.0006 | [118]0.0002 N[0.060 298] |
| 181 | ISTL-001 | [364]0.0020 | [354]0.0021 | [330]0.0023 | [311]0.0024 | [298]0.0027 | [416]0.0005 N[0.102 92] | [425]0.0017 | [410]0.0018 | [388]0.0018 | [368]0.0018 | [350]0.0018 | [431]0.0014 N[0.018 143] |
| 182 | ISYSTEMS-002 | [449]0.0050 | [448]0.0061 | [422]0.0070 | [409]0.0080 | [402]0.0093 | [373]0.0003 N[0.209 324] | [481]0.0031 | [474]0.0032 | [431]0.0034 | [415]0.0035 | [404]0.0038 | [419]0.0012 N[0.072 319] |
| 183 | ISYSTEMS-003 | [439]0.0043 | [434]0.0049 | [408]0.0055 | [397]0.0063 | [389]0.0073 | [398]0.0004 N[0.181 273] | [477]0.0029 | [469]0.0031 | [424]0.0031 | [412]0.0033 | [401]0.0035 | [424]0.0012 N[0.063 303] |
| 184 | KAKAO-000 | [188]0.0009 | [220]0.0011 | [223]0.0013 | [236]0.0016 | [245]0.0019 | [41]0.0000 N[0.245 371] | [122]0.0006 | [134]0.0006 | [133]0.0006 | [142]0.0007 | [141]0.0007 | [105]0.0002 N[0.079 327] |
| 185 | KAKAO-001 | [215]0.0010 | [199]0.0010 | [186]0.0011 | [172]0.0012 | [165]0.0013 | [383]0.0003 N[0.079 42] | [275]0.0009 | [265]0.0009 | [251]0.0010 | [234]0.0010 | [223]0.0010 | [364]0.0007 N[0.016 135] |
| 186 | KAKAO-002 | [206]0.0010 | [188]0.0010 | [177]0.0011 | [167]0.0011 | [151]0.0012 | [387]0.0004 N[0.074 34] | [273]0.0009 | [257]0.0009 | [240]0.0009 | [230]0.0010 | [218]0.0010 | [363]0.0007 N[0.016 131] |
| 187 | KEDACOM-001 | [475]0.0072 | [464]0.0074 | [425]0.0076 | [408]0.0079 | [394]0.0083 | [459]0.0037 N[0.049 14] | [527]0.0068 | [517]0.0068 | [455]0.0069 | [439]0.0069 | [423]0.0070 | [460]0.0060 N[0.010 80] |
| 188 | KLOUDSPOT-001 | [344]0.0017 | [344]0.0020 | [340]0.0024 | [338]0.0028 | [336]0.0035 | [124]0.0001 N[0.241 367] | [315]0.0010 | [309]0.0011 | [299]0.0011 | [291]0.0011 | [290]0.0012 | [297]0.0005 N[0.055 282] |
| 189 | KNERON-000 | [440]0.0044 | [442]0.0056 | [417]0.0064 | [405]0.0075 | [399]0.0089 | [311]0.0002 N[0.239 363] | [500]0.0044 | [501]0.0056 | [450]0.0064 | [442]0.0075 | [434]0.0089 | [98]0.0002 N[0.239 416] |
| 190 | KNOWUTECH-000 | [97]0.0007 | [120]0.0008 | [143]0.0009 | [149]0.0011 | [150]0.0012 | [46]0.0000 N[0.216 332] | [64]0.0005 | [74]0.0005 | [75]0.0005 | [66]0.0005 | [71]0.0005 | [149]0.0003 N[0.034 228] |
| 191 | KOGNIZA-000 | [261]0.0012 | [247]0.0013 | [241]0.0014 | [233]0.0015 | [219]0.0017 | [353]0.0003 N[0.115 125] | [324]0.0011 | [318]0.0011 | [303]0.0011 | [280]0.0011 | [266]0.0011 | [384]0.0008 N[0.019 147] |
| 192 | LIGHTVISION-000 | [389]0.0024 | [382]0.0027 | [369]0.0030 | [354]0.0034 | [350]0.0040 | [348]0.0002 N[0.170 242] | [440]0.0019 | [427]0.0019 | [397]0.0019 | [382]0.0020 | [368]0.0021 | [432]0.0014 N[0.024 178] |
| 193 | LINECLOVA-002 | [186]0.0009 | [178]0.0009 | [169]0.0010 | [164]0.0011 | [162]0.0013 | [331]0.0002 N[0.106 98] | [229]0.0008 | [215]0.0008 | [206]0.0008 | [195]0.0008 | [185]0.0009 | [335]0.0007 N[0.011 88] |
| 194 | LINECLOVA-003 | [83]0.0006 | [71]0.0006 | [88]0.0007 | [85]0.0009 | [81]0.0010 | [102]0.0001 N[0.171 243] | [37]0.0005 | [25]0.0005 | [53]0.0005 | [57]0.0005 | [60]0.0005 | [113]0.0002 N[0.050 271] |
| 195 | LOOKMAN-003 | [487]0.0080 | [473]0.0084 | [434]0.0087 | [419]0.0093 | [406]0.0101 | [453]0.0027 N[0.079 41] | [531]0.0069 | [520]0.0070 | [458]0.0071 | [440]0.0072 | [427]0.0074 | [458]0.0051 N[0.023 170] |
| 196 | LOOKMAN-005 | [479]0.0075 | [467]0.0077 | [429]0.0080 | [412]0.0082 | [398]0.0088 | [454]0.0035 N[0.056 15] | [528]0.0068 | [518]0.0069 | [457]0.0069 | [438]0.0070 | [424]0.0071 | [459]0.0056 N[0.014 112] |
| 197 | LSSYTECH-000 | [63]0.0006 | [59]0.0006 | [59]0.0006 | [612]1.0000 | [51]0.0008 | [7]0.0000 N[1.127 461] | [97]0.0005 | [93]0.0005 | [85]0.0005 | [481]1.0000 | [72]0.0006 | [7]0.0000 N[1.040 460] |
| 198 | LUXAND-001 | [388]0.0024 | [386]0.0028 | [375]0.0032 | [365]0.0037 | [357]0.0043 | [288]0.0002 N[0.203 313] | [405]0.0015 | [401]0.0016 | [379]0.0017 | [364]0.0017 | [353]0.0019 | [342]0.0006 N[0.064 305] |
| 199 | MANTRA-000 | [253]0.0012 | [264]0.0014 | [262]0.0016 | [263]0.0018 | [266]0.0022 | [135]0.0001 N[0.210 325] | [227]0.0008 | [223]0.0008 | [218]0.0009 | [216]0.0009 | [222]0.0010 | [203]0.0004 N[0.058 291] |
| 200 | MAXVISION-000 | [342]0.0017 | [346]0.0021 | [337]0.0024 | [335]0.0028 | [335]0.0035 | [139]0.0001 N[0.237 359] | [265]0.0009 | [285]0.0010 | [288]0.0011 | [300]0.0012 | [306]0.0013 | [85]0.0002 N[0.132 372] |
| 201 | MAXVISION-001 | [149]0.0008 | [148]0.0008 | [145]0.0009 | [133]0.0010 | [131]0.0011 | [286]0.0002 N[0.122 146] | [192]0.0007 | [179]0.0007 | [176]0.0007 | [169]0.0008 | [156]0.0008 | [311]0.0005 N[0.021 162] |
| 202 | MAXVISION-002 | [154]0.0008 | [153]0.0009 | [150]0.0009 | [138]0.0010 | [137]0.0012 | [284]0.0001 N[0.124 151] | [211]0.0008 | [201]0.0008 | [192]0.0008 | [176]0.0008 | [161]0.0008 | [371]0.0008 N[0.002 18] |
| 203 | MAXVISION-003 | [171]0.0009 | [164]0.0009 | [158]0.0010 | [157]0.0011 | [153]0.0012 | [287]0.0002 N[0.126 153] | [184]0.0007 | [180]0.0007 | [179]0.0007 | [170]0.0008 | [158]0.0008 | [291]0.0005 N[0.029 203] |
| 204 | MAXVISION-004 | [161]0.0008 | [147]0.0008 | [132]0.0009 | [113]0.0009 | [108]0.0011 | [358]0.0003 N[0.082 46] | [218]0.0008 | [205]0.0008 | [194]0.0008 | [181]0.0008 | [169]0.0008 | [381]0.0008 N[0.000 8] |
| 205 | MEGVII-001 | [508]0.0104 | [501]0.0116 | [446]0.0127 | [434]0.0140 | [422]0.0158 | [447]0.0015 N[0.142 194] | [534]0.0077 | [528]0.0080 | [462]0.0082 | [447]0.0085 | [433]0.0088 | [455]0.0041 N[0.046 264] |
| 206 | MEGVII-003 | [173]0.0009 | [162]0.0009 | [142]0.0009 | [128]0.0010 | [121]0.0011 | [360]0.0003 N[0.084 51] | [224]0.0008 | [210]0.0008 | [205]0.0008 | [191]0.0008 | [181]0.0008 | [357]0.0007 N[0.010 81] |
| 207 | MEGVII-004 | [156]0.0008 | [145]0.0008 | [124]0.0009 | [101]0.0009 | [85]0.0010 | [391]0.0004 N[0.059 17] | [222]0.0008 | [206]0.0008 | [199]0.0008 | [186]0.0008 | [174]0.0008 | [366]0.0007 N[0.006 49] |
| 208 | MEGVII-005 | [158]0.0008 | [146]0.0008 | [123]0.0009 | [102]0.0009 | [84]0.0010 | [392]0.0004 N[0.059 18] | [221]0.0008 | [208]0.0008 | [197]0.0008 | [185]0.0008 | [173]0.0008 | [365]0.0007 N[0.006 48] |
| 209 | METSAKUURCOMPANY-000 | [112]0.0007 | [129]0.0008 | [130]0.0009 | [140]0.0010 | [161]0.0013 | [69]0.0000 N[0.202 312] | [82]0.0005 | [84]0.0005 | [80]0.0005 | [78]0.0005 | [78]0.0006 | [177]0.0003 N[0.029 202] |
| 210 | METSAKUURCOMPANY-001 | [183]0.0009 | [194]0.0010 | [185]0.0011 | [188]0.0013 | [195]0.0015 | [202]0.0001 N[0.166 239] | [185]0.0007 | [192]0.0008 | [185]0.0008 | [179]0.0008 | [172]0.0008 | [238]0.0004 N[0.042 253] |
| 211 | MIAXIS-001 | [250]0.0011 | [252]0.0013 | [240]0.0014 | [234]0.0015 | [221]0.0017 | [314]0.0002 N[0.135 177] | [277]0.0009 | [277]0.0010 | [260]0.0010 | [247]0.0010 | [240]0.0010 | [333]0.0006 N[0.033 224] |
| 212 | MIAXIS-002 | [254]0.0012 | [253]0.0013 | [242]0.0014 | [237]0.0016 | [228]0.0018 | [304]0.0002 N[0.140 189] | [296]0.0010 | [289]0.0010 | [272]0.0010 | [258]0.0010 | [252]0.0011 | [347]0.0007 N[0.028 195] |
| 213 | MICROFOCUS-005 | [601]0.3698 | [601]0.4240 | [481]0.4608 | [469]0.4998 | [462]0.5390 | [465]0.0673 N[0.128 159] | [596]0.1298 | [595]0.1721 | [481]0.2043 | [470]0.2423 | [462]0.2808 | [454]0.0039 N[0.263 425] |
| 214 | MICROSOFT-003 | [257]0.0012 | [289]0.0015 | [289]0.0018 | [294]0.0022 | [299]0.0027 | [35]0.0000 N[0.276 403] | [113]0.0006 | [129]0.0006 | [163]0.0007 | [180]0.0008 | [189]0.0009 | [50]0.0001 N[0.158 389] |
| 215 | MICROSOFT-004 | [251]0.0011 | [275]0.0014 | [282]0.0017 | [282]0.0021 | [297]0.0027 | [31]0.0000 N[0.287 421] | [104]0.0006 | [116]0.0006 | [134]0.0006 | [146]0.0007 | [176]0.0008 | [59]0.0001 N[0.139 378] |
| 216 | MICROSOFT-005 | [303]0.0014 | [331]0.0018 | [334]0.0023 | [342]0.0030 | [342]0.0037 | [23]0.0000 N[0.326 442] | [123]0.0006 | [138]0.0006 | [157]0.0007 | [164]0.0008 | [184]0.0009 | [63]0.0001 N[0.137 377] |

*Table 46:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK FNIR(N, T= 0, R) | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 217 | MICROSOFT-006 | 317 0.0015 | 338 0.0019 | 346 0.0025 | 343 0.0030 | 343 0.0037 | 29 0.0000 N^0.310 437 | 127 0.0006 | 152 0.0007 | 164 0.0007 | 194 0.0008 | 237 0.0010 | 38 0.0000 N^0.188 405 |
| 218 | MOBBL-000 | 365 0.0020 | 365 0.0023 | 351 0.0026 | 332 0.0027 | 301 0.0027 | 415 0.0005 N^0.107 102 | 361 0.0012 | 350 0.0013 | 334 0.0013 | 323 0.0014 | 307 0.0014 | 343 0.0007 N^0.046 265 |
| 219 | MUKH-002 | 538 0.0200 | 535 0.0255 | 466 0.0301 | 456 0.0358 | 451 0.0427 | 421 0.0006 N^0.258 383 | 509 0.0051 | 516 0.0066 | 460 0.0079 | 451 0.0097 | 445 0.0120 | 70 0.0001 N^0.293 436 |
| 220 | NAVERCLOUD-000 | 152 0.0008 | 163 0.0009 | 153 0.0009 | 136 0.0010 | 138 0.0012 | 290 0.0002 N^0.121 143 | 199 0.0007 | 193 0.0008 | 181 0.0008 | 163 0.0008 | 153 0.0008 | 351 0.0007 N^0.006 55 |
| 221 | NEC-000 | 518 0.0128 | 518 0.0167 | 456 0.0199 | 445 0.0241 | 439 0.0291 | 370 0.0003 N^0.281 411 | 470 0.0026 | 479 0.0035 | 441 0.0044 | 430 0.0056 | 425 0.0071 | 27 0.0000 N^0.344 447 |
| 222 | NEC-001 | 531 0.0177 | 527 0.0206 | 460 0.0230 | 449 0.0262 | 441 0.0300 | 448 0.0016 N^0.181 274 | 551 0.0106 | 540 0.0109 | 466 0.0113 | 453 0.0118 | 446 0.0125 | 457 0.0049 N^0.057 288 |
| 223 | NEC-002 | 166 0.0008 | 169 0.0009 | 162 0.0010 | 156 0.0011 | 139 0.0012 | 302 0.0002 N^0.117 127 | 178 0.0007 | 186 0.0007 | 174 0.0007 | 162 0.0008 | 162 0.0008 | 272 0.0004 N^0.034 230 |
| 224 | NEC-003 | 266 0.0012 | 250 0.0013 | 236 0.0013 | 212 0.0014 | 197 0.0015 | 411 0.0005 N^0.074 33 | 347 0.0011 | 328 0.0011 | 313 0.0011 | 295 0.0012 | 279 0.0012 | 403 0.0009 N^0.013 105 |
| 225 | NEC-004 | 221 0.0010 | 207 0.0011 | 197 0.0011 | 179 0.0012 | 169 0.0013 | 386 0.0004 N^0.078 40 | 270 0.0009 | 262 0.0009 | 249 0.0009 | 241 0.0010 | 225 0.0010 | 349 0.0007 N^0.023 169 |
| 226 | NEC-005 | 147 0.0008 | 141 0.0008 | 131 0.0009 | 114 0.0009 | 100 0.0010 | 336 0.0002 N^0.092 66 | 181 0.0007 | 178 0.0007 | 169 0.0007 | 160 0.0007 | 151 0.0008 | 312 0.0005 N^0.020 151 |
| 227 | NEC-006 | 174 0.0009 | 171 0.0009 | 155 0.0009 | 148 0.0011 | 129 0.0011 | 346 0.0002 N^0.095 77 | 208 0.0008 | 199 0.0008 | 198 0.0008 | 192 0.0008 | 186 0.0009 | 288 0.0005 N^0.036 236 |
| 228 | NEC-007 | 92 0.0006 | 87 0.0007 | 78 0.0007 | 65 0.0008 | 56 0.0008 | 303 0.0002 N^0.095 75 | 126 0.0006 | 122 0.0006 | 116 0.0006 | 105 0.0006 | 103 0.0006 | 266 0.0004 N^0.021 161 |
| 229 | NEC-008 | 120 0.0007 | 105 0.0007 | 97 0.0008 | 88 0.0009 | 67 0.0009 | 337 0.0002 N^0.085 53 | 143 0.0006 | 141 0.0006 | 132 0.0006 | 118 0.0006 | 111 0.0006 | 302 0.0005 N^0.016 128 |
| 230 | NEC-009 | 78 0.0006 | 68 0.0006 | 60 0.0006 | 51 0.0007 | 42 0.0008 | 306 0.0002 N^0.088 60 | 128 0.0006 | 118 0.0006 | 106 0.0006 | 98 0.0006 | 90 0.0006 | 317 0.0006 N^0.003 24 |
| 231 | NEC-010 | 33 0.0005 | 29 0.0005 | 26 0.0005 | 17 0.0006 | 16 0.0007 | 252 0.0001 N^0.100 86 | 27 0.0004 | 34 0.0005 | 35 0.0005 | 43 0.0005 | 37 0.0005 | 140 0.0003 N^0.031 210 |
| 232 | NEC-011 | 9 0.0004 | 14 0.0005 | 12 0.0005 | 8 0.0005 | 6 0.0006 | 209 0.0001 N^0.110 112 | 7 0.0004 | 13 0.0004 | 9 0.0004 | 11 0.0004 | 12 0.0004 | 156 0.0003 N^0.019 150 |
| 233 | NEC-012 | 10 0.0004 | 16 0.0005 | 11 0.0005 | 11 0.0006 | 8 0.0006 | 196 0.0001 N^0.115 123 | 10 0.0004 | 9 0.0004 | 13 0.0004 | 14 0.0004 | 15 0.0004 | 157 0.0003 N^0.019 149 |
| 234 | NEUROTECHNOLOGY-003 | 530 0.0175 | 529 0.0222 | 462 0.0259 | 452 0.0302 | 446 0.0358 | 428 0.0007 N^0.242 368 | 496 0.0039 | 500 0.0054 | 456 0.0069 | 448 0.0087 | 442 0.0109 | 32 0.0000 N^0.348 448 |
| 235 | NEUROTECHNOLOGY-004 | 435 0.0043 | 436 0.0052 | 415 0.0061 | 402 0.0071 | 396 0.0084 | 317 0.0002 N^0.231 351 | 436 0.0019 | 440 0.0022 | 413 0.0025 | 405 0.0028 | 395 0.0031 | 101 0.0002 N^0.173 398 |
| 236 | NEUROTECHNOLOGY-005 | 415 0.0033 | 415 0.0040 | 399 0.0046 | 389 0.0054 | 383 0.0065 | 271 0.0001 N^0.235 357 | 437 0.0019 | 432 0.0021 | 404 0.0022 | 394 0.0023 | 387 0.0025 | 305 0.0005 N^0.097 354 |
| 237 | NEUROTECHNOLOGY-007 | 404 0.0028 | 404 0.0036 | 392 0.0041 | 383 0.0049 | 379 0.0058 | 227 0.0001 N^0.242 369 | 418 0.0017 | 412 0.0018 | 393 0.0019 | 383 0.0020 | 377 0.0022 | 299 0.0005 N^0.091 343 |
| 238 | NEUROTECHNOLOGY-008 | 316 0.0015 | 326 0.0018 | 316 0.0021 | 319 0.0025 | 318 0.0030 | 110 0.0001 N^0.240 366 | 284 0.0009 | 271 0.0010 | 261 0.0010 | 260 0.0010 | 263 0.0011 | 256 0.0004 N^0.058 290 |
| 239 | NEUROTECHNOLOGY-009 | 196 0.0010 | 213 0.0011 | 212 0.0012 | 222 0.0015 | 226 0.0017 | 101 0.0001 N^0.207 322 | 193 0.0007 | 194 0.0008 | 186 0.0008 | 182 0.0008 | 179 0.0008 | 241 0.0004 N^0.043 255 |
| 240 | NEUROTECHNOLOGY-010 | 144 0.0008 | 157 0.0009 | 159 0.0010 | 161 0.0011 | 166 0.0013 | 168 0.0001 N^0.170 240 | 163 0.0006 | 161 0.0007 | 145 0.0007 | 138 0.0007 | 135 0.0007 | 293 0.0005 N^0.021 163 |
| 241 | NEUROTECHNOLOGY-012 | 94 0.0006 | 84 0.0007 | 74 0.0007 | 77 0.0008 | 90 0.0010 | 177 0.0001 N^0.149 203 | 129 0.0006 | 119 0.0006 | 107 0.0006 | 101 0.0006 | 99 0.0006 | 290 0.0005 N^0.014 111 |
| 242 | NEUROTECHNOLOGY-013 | 89 0.0006 | 86 0.0007 | 77 0.0007 | 73 0.0008 | 70 0.0009 | 236 0.0001 N^0.124 152 | 138 0.0006 | 126 0.0006 | 118 0.0006 | 114 0.0006 | 108 0.0006 | 264 0.0004 N^0.023 173 |
| 243 | NEUROTECHNOLOGY-014 | 58 0.0006 | 55 0.0006 | 57 0.0006 | 57 0.0007 | 66 0.0009 | 107 0.0001 N^0.162 233 | 71 0.0005 | 71 0.0005 | 76 0.0005 | 75 0.0005 | 83 0.0006 | 129 0.0003 N^0.045 262 |
| 244 | NEUROTECHNOLOGY-015 | 38 0.0005 | 43 0.0005 | 46 0.0006 | 48 0.0007 | 58 0.0009 | 72 0.0000 N^0.175 258 | 51 0.0005 | 52 0.0005 | 70 0.0005 | 84 0.0006 | 92 0.0006 | 87 0.0002 N^0.080 330 |
| 245 | NEUROTECHNOLOGY-016 | 39 0.0005 | 33 0.0005 | 33 0.0006 | 41 0.0006 | 47 0.0008 | 109 0.0001 N^0.154 212 | 44 0.0005 | 51 0.0005 | 64 0.0005 | 72 0.0005 | 67 0.0005 | 108 0.0002 N^0.058 289 |
| 246 | NEXTMETALABS-002 | 64 0.0006 | 72 0.0006 | 66 0.0007 | 58 0.0007 | 64 0.0009 | 138 0.0001 N^0.152 208 | 43 0.0005 | 47 0.0005 | 52 0.0005 | 44 0.0005 | 50 0.0005 | 160 0.0003 N^0.026 187 |
| 247 | NOMINDER-000 | 86 0.0006 | 89 0.0007 | 92 0.0007 | 91 0.0009 | 102 0.0010 | 103 0.0001 N^0.174 257 | 70 0.0005 | 63 0.0005 | 69 0.0005 | 69 0.0005 | 68 0.0005 | 155 0.0003 N^0.031 213 |
| 248 | NOMINDER-001 | 247 0.0011 | 228 0.0011 | 208 0.0012 | 201 0.0013 | 193 0.0015 | 368 0.0003 N^0.096 81 | 317 0.0010 | 302 0.0010 | 284 0.0010 | 262 0.0010 | 246 0.0011 | 409 0.0010 N^0.005 45 |
| 249 | NOMINDER-002 | 248 0.0011 | 230 0.0012 | 213 0.0012 | 203 0.0013 | 186 0.0014 | 390 0.0004 N^0.082 47 | 314 0.0010 | 301 0.0010 | 280 0.0010 | 265 0.0011 | 250 0.0011 | 397 0.0009 N^0.009 77 |
| 250 | NOMINDER-003 | 249 0.0011 | 231 0.0012 | 210 0.0012 | 206 0.0013 | 189 0.0014 | 388 0.0004 N^0.084 50 | 327 0.0011 | 312 0.0011 | 292 0.0011 | 270 0.0011 | 255 0.0011 | 412 0.0010 N^0.003 22 |
| 251 | NOMINDER-004 | 223 0.0010 | 210 0.0011 | 189 0.0011 | 182 0.0012 | 176 0.0014 | 369 0.0003 N^0.091 65 | 290 0.0010 | 276 0.0010 | 256 0.0010 | 243 0.0010 | 228 0.0010 | 393 0.0009 N^0.007 61 |
| 252 | NOMINDER-005 | 238 0.0011 | 221 0.0011 | 205 0.0012 | 197 0.0013 | 187 0.0014 | 362 0.0003 N^0.100 85 | 302 0.0010 | 294 0.0010 | 271 0.0010 | 256 0.0010 | 239 0.0010 | 407 0.0010 N^0.004 40 |
| 253 | NOTIONTAG-000 | 360 0.0019 | 349 0.0021 | 327 0.0022 | 318 0.0025 | 308 0.0029 | 374 0.0003 N^0.135 176 | 407 0.0016 | 399 0.0016 | 375 0.0016 | 354 0.0016 | 341 0.0017 | 411 0.0010 N^0.032 222 |
| 254 | NTECHLAB-003 | 441 0.0045 | 447 0.0060 | 424 0.0074 | 418 0.0092 | 411 0.0112 | 122 0.0001 N^0.315 438 | 364 0.0012 | 392 0.0015 | 381 0.0017 | 387 0.0021 | 386 0.0025 | 39 0.0000 N^0.241 417 |
| 255 | NTECHLAB-004 | 425 0.0035 | 429 0.0046 | 411 0.0056 | 400 0.0069 | 395 0.0083 | 127 0.0001 N^0.296 427 | 320 0.0010 | 345 0.0012 | 353 0.0014 | 348 0.0016 | 360 0.0019 | 49 0.0001 N^0.207 412 |
| 256 | NTECHLAB-005 | 418 0.0034 | 427 0.0046 | 412 0.0057 | 403 0.0071 | 400 0.0090 | 49 0.0000 N^0.337 447 | 212 0.0008 | 288 0.0010 | 312 0.0011 | 336 0.0015 | 347 0.0018 | 20 0.0000 N^0.288 434 |
| 257 | NTECHLAB-006 | 406 0.0029 | 416 0.0040 | 403 0.0049 | 393 0.0061 | 391 0.0077 | 43 0.0000 N^0.330 444 | 188 0.0007 | 231 0.0009 | 278 0.0010 | 304 0.0012 | 328 0.0015 | 24 0.0000 N^0.260 423 |
| 258 | NTECHLAB-007 | 351 0.0018 | 363 0.0023 | 355 0.0027 | 353 0.0034 | 353 0.0041 | 62 0.0000 N^0.278 407 | 205 0.0008 | 224 0.0008 | 235 0.0009 | 264 0.0011 | 280 0.0012 | 69 0.0001 N^0.150 383 |
| 259 | NTECHLAB-008 | 235 0.0011 | 263 0.0014 | 272 0.0016 | 278 0.0020 | 276 0.0024 | 32 0.0000 N^0.275 401 | 156 0.0006 | 163 0.0007 | 161 0.0007 | 178 0.0008 | 182 0.0008 | 94 0.0002 N^0.096 350 |
| 260 | NTECHLAB-009 | 153 0.0008 | 172 0.0009 | 170 0.0010 | 175 0.0012 | 180 0.0014 | 117 0.0001 N^0.187 292 | 125 0.0006 | 123 0.0006 | 124 0.0006 | 130 0.0007 | 131 0.0007 | 121 0.0002 N^0.064 306 |
| 261 | NTECHLAB-010 | 119 0.0007 | 115 0.0008 | 106 0.0008 | 106 0.0009 | 99 0.0010 | 247 0.0001 N^0.130 166 | 162 0.0006 | 159 0.0007 | 144 0.0007 | 132 0.0007 | 125 0.0007 | 325 0.0006 N^0.008 71 |
| 262 | NTECHLAB-011 | 90 0.0006 | 88 0.0007 | 83 0.0007 | 83 0.0008 | 86 0.0010 | 173 0.0001 N^0.151 206 | 87 0.0005 | 89 0.0005 | 87 0.0005 | 82 0.0006 | 80 0.0006 | 184 0.0004 N^0.028 199 |
| 263 | OMNIGARDE-000 | 80 0.0006 | 93 0.0007 | 98 0.0008 | 109 0.0009 | 117 0.0011 | 54 0.0000 N^0.203 314 | 46 0.0005 | 62 0.0005 | 66 0.0005 | 67 0.0005 | 65 0.0005 | 123 0.0002 N^0.049 270 |
| 264 | OMNIGARDE-001 | 51 0.0005 | 45 0.0005 | 50 0.0006 | 56 0.0007 | 52 0.0008 | 116 0.0001 N^0.157 216 | 24 0.0004 | 23 0.0004 | 22 0.0005 | 33 0.0005 | 38 0.0005 | 132 0.0003 N^0.035 231 |
| 265 | OMNIGARDE-002 | 54 0.0005 | 60 0.0006 | 58 0.0006 | 55 0.0007 | 57 0.0009 | 133 0.0001 N^0.151 207 | 23 0.0004 | 19 0.0004 | 37 0.0005 | 48 0.0005 | 52 0.0005 | 109 0.0002 N^0.052 275 |
| 266 | ONFIDO-000 | 297 0.0014 | 322 0.0018 | 318 0.0021 | 323 0.0026 | 325 0.0032 | 38 0.0000 N^0.281 412 | 161 0.0006 | 173 0.0007 | 204 0.0008 | 219 0.0009 | 234 0.0010 | 58 0.0001 N^0.155 387 |
| 267 | ONFIDO-001 | 220 0.0010 | 257 0.0013 | 280 0.0017 | 296 0.0022 | 307 0.0028 | 16 0.0000 N^0.348 448 | 92 0.0005 | 107 0.0006 | 115 0.0006 | 125 0.0007 | 137 0.0007 | 83 0.0001 N^0.097 352 |
| 268 | OPTIEXACTA-000 | 41 0.0005 | 39 0.0005 | 37 0.0006 | 33 0.0006 | 26 0.0007 | 260 0.0001 N^0.101 90 | 60 0.0005 | 59 0.0005 | 49 0.0005 | 39 0.0005 | 33 0.0005 | 277 0.0005 N^0.004 30 |
| 269 | PANGIAM-000 | 145 0.0008 | 142 0.0008 | 146 0.0009 | 142 0.0010 | 157 0.0012 | 182 0.0001 N^0.162 232 | 153 0.0006 | 160 0.0007 | 152 0.0007 | 144 0.0007 | 143 0.0007 | 199 0.0004 N^0.041 250 |
| 270 | PANGIAM-001 | 466 0.0065 | 451 0.0065 | 420 0.0066 | 399 0.0068 | 386 0.0069 | 458 0.0049 N^0.021 4 | 522 0.0064 | 512 0.0064 | 451 0.0064 | 435 0.0064 | 421 0.0064 | 461 0.0063 N^0.002 11 |
| 271 | PAPSAV1923-000 | 140 0.0008 | 132 0.0008 | 118 0.0008 | 96 0.0009 | 93 0.0010 | 325 0.0002 N^0.095 74 | 194 0.0007 | 190 0.0007 | 171 0.0007 | 153 0.0007 | 145 0.0007 | 361 0.0007 N^0.000 3 |
| 272 | PARAVISION-003 | 397 0.0026 | 396 0.0031 | 380 0.0034 | 367 0.0041 | 366 0.0047 | 282 0.0001 N^0.212 329 | 410 0.0016 | 404 0.0017 | 386 0.0018 | 375 0.0019 | 365 0.0020 | 295 0.0005 N^0.087 339 |
| 273 | PARAVISION-004 | 256 0.0012 | 248 0.0013 | 239 0.0014 | 231 0.0015 | 229 0.0018 | 309 0.0002 N^0.139 185 | 280 0.0009 | 269 0.0009 | 252 0.0010 | 239 0.0010 | 235 0.0010 | 341 0.0006 N^0.027 190 |
| 274 | PARAVISION-005 | 242 0.0011 | 234 0.0012 | 219 0.0012 | 211 0.0014 | 205 0.0016 | 332 0.0002 N^0.120 138 | 285 0.0009 | 272 0.0010 | 257 0.0010 | 242 0.0010 | 233 0.0010 | 359 0.0007 N^0.020 155 |
| 275 | PARAVISION-007 | 137 0.0008 | 137 0.0008 | 134 0.0009 | 116 0.0010 | 120 0.0011 | 266 0.0001 N^0.126 154 | 159 0.0006 | 155 0.0007 | 142 0.0007 | 143 0.0007 | 138 0.0007 | 270 0.0004 N^0.028 193 |
| 276 | PARAVISION-009 | 87 0.0006 | 83 0.0006 | 85 0.0007 | 86 0.0009 | 94 0.0010 | 106 0.0001 N^0.172 246 | 116 0.0006 | 112 0.0006 | 113 0.0006 | 111 0.0006 | 116 0.0006 | 134 0.0003 N^0.051 274 |
| 277 | PARAVISION-012 | 74 0.0006 | 62 0.0006 | 49 0.0006 | 49 0.0007 | 48 0.0008 | 254 0.0001 N^0.111 113 | 112 0.0006 | 99 0.0006 | 92 0.0006 | 88 0.0006 | 81 0.0006 | 307 0.0005 N^0.005 43 |
| 278 | PARAVISION-014 | 69 0.0006 | 56 0.0006 | 53 0.0006 | 53 0.0007 | 44 0.0008 | 259 0.0001 N^0.109 108 | 101 0.0005 | 96 0.0006 | 91 0.0006 | 86 0.0006 | 76 0.0006 | 296 0.0005 N^0.008 69 |
| 279 | PARAVISION-016 | 71 0.0006 | 63 0.0006 | 55 0.0006 | 47 0.0007 | 43 0.0008 | 273 0.0001 N^0.101 91 | 115 0.0006 | 106 0.0006 | 97 0.0006 | 90 0.0006 | 82 0.0006 | 309 0.0005 N^0.003 25 |
| 280 | PARAVISION-019 | 27 0.0005 | 27 0.0005 | 27 0.0005 | 25 0.0006 | 25 0.0007 | 197 0.0001 N^0.121 140 | 28 0.0005 | 36 0.0005 | 38 0.0005 | 34 0.0005 | 27 0.0005 | 173 0.0003 N^0.021 160 |
| 281 | PARAVISION-021 | 15 0.0005 | 13 0.0005 | 17 0.0005 | 21 0.0006 | 21 0.0007 | 128 0.0001 N^0.139 186 | 19 0.0004 | 21 0.0004 | 18 0.0004 | 16 0.0004 | 13 0.0004 | 240 0.0004 N^0.004 39 |
| 282 | PENTACOMP-001 | 267 0.0012 | 274 0.0014 | 269 0.0016 | 267 0.0019 | 272 0.0023 | 150 0.0001 N^0.208 323 | 235 0.0008 | 240 0.0009 | 230 0.0009 | 240 0.0010 | 245 0.0011 | 148 0.0003 N^0.073 321 |
| 283 | PIXELALL-002 | 419 0.0034 | 419 0.0041 | 401 0.0048 | 392 0.0059 | 388 0.0071 | 221 0.0001 N^0.256 380 | 383 0.0014 | 398 0.0016 | 382 0.0017 | 384 0.0020 | 380 0.0023 | 72 0.0001 N^0.185 403 |
| 284 | PIXELALL-003 | 323 0.0015 | 324 0.0018 | 313 0.0021 | 310 0.0024 | 309 0.0029 | 191 0.0001 N^0.213 330 | 304 0.0010 | 306 0.0011 | 294 0.0011 | 286 0.0011 | 286 0.0012 | 273 0.0004 N^0.060 297 |
| 285 | PIXELALL-004 | 283 0.0013 | 310 0.0017 | 304 0.0019 | 309 0.0023 | 303 0.0027 | 85 0.0001 N^0.245 372 | 282 0.0009 | 281 0.0010 | 274 0.0010 | 272 0.0011 | 270 0.0011 | 206 0.0004 N^0.069 316 |
| 286 | PIXELALL-005 | 293 0.0014 | 288 0.0015 | 273 0.0016 | 262 0.0018 | 259 0.0021 | 345 0.0002 N^0.132 171 | 355 0.0012 | 343 0.0012 | 327 0.0012 | 308 0.0012 | 296 0.0013 | 383 0.0008 N^0.026 185 |
| 287 | PROTENET-001 | 349 0.0018 | 341 0.0020 | 326 0.0022 | 462 0.0685 | 311 0.0029 | 12 0.0000 N^0.620 456 | 383 0.0014 | 382 0.0014 | 356 0.0015 | 463 0.0676 | 333 0.0016 | 13 0.0000 N^0.569 456 |
| 288 | PSL-000 | 17 0.0005 | 26 0.0005 | 30 0.0005 | 40 0.0006 | 45 0.0008 | 66 0.0000 N^0.173 251 | 12 0.0004 | 7 0.0004 | 15 0.0004 | 12 0.0004 | 16 0.0004 | 185 0.0004 N^0.012 99 |

*Table 47:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK FNIR(N, T= 0, R) | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 289 | PSL-001 | 20 0.0005 | 18 0.0005 | 25 0.0005 | 32 0.0006 | 35 0.0007 | 112 0.0001 $N^{0.149\ 202}$ | 11 0.0004 | 14 0.0004 | 8 0.0004 | 15 0.0004 | 16 0.0004 | 153 0.0003 $N^{0.021\ 157}$ |
| 290 | PSL-002 | 7 0.0004 | 6 0.0004 | 15 0.0005 | 19 0.0006 | 29 0.0007 | 80 0.0000 $N^{0.161\ 230}$ | 8 0.0004 | 5 0.0004 | 4 0.0004 | 3 0.0004 | 7 0.0004 | 188 0.0004 $N^{0.010\ 84}$ |
| 291 | PSL-003 | 11 0.0004 | 9 0.0004 | 9 0.0005 | 10 0.0006 | 13 0.0006 | 156 0.0001 $N^{0.128\ 160}$ | 9 0.0004 | 10 0.0004 | 12 0.0004 | 9 0.0004 | 8 0.0004 | 207 0.0004 $N^{0.009\ 74}$ |
| 292 | PSL-004 | 6 0.0004 | 7 0.0004 | 7 0.0005 | 7 0.0005 | 7 0.0006 | 194 0.0001 $N^{0.115\ 126}$ | 14 0.0004 | 11 0.0004 | 11 0.0004 | 8 0.0004 | 6 0.0004 | 258 0.0004 $N^{-0.000\ 2}$ |
| 293 | PSL-005 | 14 0.0005 | 11 0.0005 | 21 0.0005 | 27 0.0006 | 34 0.0007 | 78 0.0000 $N^{0.164\ 238}$ | 6 0.0004 | 12 0.0004 | 10 0.0004 | 7 0.0004 | 11 0.0004 | 166 0.0003 $N^{0.017\ 139}$ |
| 294 | PSL-006 | 8 0.0004 | 8 0.0004 | 5 0.0005 | 5 0.0005 | 5 0.0006 | 199 0.0001 $N^{0.112\ 118}$ | 5 0.0004 | 8 0.0004 | 5 0.0004 | 6 0.0004 | 5 0.0004 | 190 0.0004 $N^{0.011\ 87}$ |
| 295 | PTAKURATSATU-000 | 377 0.0021 | 383 0.0027 | 377 0.0033 | 362 0.0036 | 338 0.0036 | 313 0.0002 $N^{0.188\ 293}$ | 352 0.0012 | 352 0.0013 | 360 0.0015 | 350 0.0016 | 336 0.0016 | 115 0.0002 $N^{0.121\ 366}$ |
| 296 | PYRAMID-000 | 104 0.0007 | 108 0.0007 | 113 0.0008 | 117 0.0010 | 111 0.0011 | 167 0.0001 $N^{0.160\ 225}$ | 136 0.0006 | 131 0.0006 | 122 0.0006 | 113 0.0006 | 106 0.0006 | 279 0.0005 $N^{0.019\ 146}$ |
| 297 | PYRAMID-001 | 142 0.0008 | 140 0.0008 | 129 0.0009 | 121 0.0010 | 118 0.0011 | 277 0.0001 $N^{0.123\ 149}$ | 166 0.0007 | 158 0.0007 | 141 0.0007 | 137 0.0007 | 133 0.0007 | 301 0.0005 $N^{0.020\ 152}$ |
| 298 | QAZSMARTVISIONAI-000 | 25 0.0005 | 21 0.0005 | 20 0.0005 | 22 0.0006 | 23 0.0007 | 178 0.0001 $N^{0.128\ 156}$ | 16 0.0004 | 15 0.0004 | 19 0.0004 | 22 0.0005 | 20 0.0005 | 139 0.0003 $N^{0.028\ 198}$ |
| 299 | QAZSMARTVISIONAI-001 | 3 0.0004 | 5 0.0004 | 6 0.0005 | 9 0.0006 | 10 0.0006 | 111 0.0001 $N^{0.141\ 190}$ | 3 0.0004 | 4 0.0004 | 7 0.0004 | 5 0.0004 | 4 0.0004 | 182 0.0004 $N^{0.011\ 90}$ |
| 300 | QAZSMARTVISIONAI-002 | 1 0.0004 | 1 0.0004 | 1 0.0005 | 2 0.0005 | 3 0.0006 | 159 0.0001 $N^{0.123\ 148}$ | 1 0.0004 | 2 0.0004 | 1 0.0004 | 1 0.0004 | 1 0.0004 | 219 0.0004 $N^{0.004\ 36}$ |
| 301 | QAZSMARTVISIONAI-003 | 4 0.0004 | 4 0.0004 | 4 0.0005 | 3 0.0005 | 2 0.0006 | 195 0.0001 $N^{0.112\ 119}$ | 2 0.0004 | 3 0.0004 | 2 0.0004 | 2 0.0004 | 3 0.0004 | 180 0.0004 $N^{0.011\ 86}$ |
| 302 | QNAP-001 | 414 0.0031 | 411 0.0038 | 396 0.0043 | 384 0.0050 | 381 0.0059 | 300 0.0002 $N^{0.216\ 333}$ | 435 0.0019 | 429 0.0020 | 403 0.0021 | 393 0.0022 | 383 0.0024 | 328 0.0006 $N^{0.085\ 336}$ |
| 303 | QNAP-002 | 436 0.0043 | 426 0.0045 | 400 0.0048 | 386 0.0051 | 374 0.0055 | 444 0.0014 $N^{0.085\ 55}$ | 490 0.0038 | 484 0.0038 | 437 0.0039 | 420 0.0039 | 408 0.0040 | 451 0.0028 $N^{0.021\ 159}$ |
| 304 | QNAP-003 | 376 0.0021 | 372 0.0024 | 356 0.0027 | 347 0.0031 | 339 0.0036 | 312 0.0002 $N^{0.182\ 280}$ | 335 0.0011 | 341 0.0012 | 333 0.0013 | 331 0.0014 | 337 0.0016 | 97 0.0002 $N^{0.133\ 374}$ |
| 305 | QNAP-004 | 288 0.0014 | 291 0.0016 | 288 0.0018 | 283 0.0021 | 284 0.0025 | 187 0.0001 $N^{0.204\ 316}$ | 297 0.0010 | 287 0.0010 | 275 0.0010 | 274 0.0011 | 269 0.0011 | 306 0.0005 $N^{0.047\ 267}$ |
| 306 | QNAP-005 | 405 0.0028 | 390 0.0029 | 366 0.0030 | 348 0.0032 | 334 0.0034 | 439 0.0011 $N^{0.069\ 27}$ | 473 0.0027 | 457 0.0027 | 417 0.0027 | 403 0.0027 | 391 0.0027 | 449 0.0025 $N^{0.006\ 50}$ |
| 307 | QNAP-006 | 228 0.0011 | 224 0.0011 | 218 0.0012 | 217 0.0015 | 225 0.0017 | 210 0.0001 $N^{0.173\ 253}$ | 228 0.0008 | 217 0.0008 | 213 0.0009 | 204 0.0009 | 202 0.0009 | 310 0.0005 $N^{0.030\ 207}$ |
| 308 | QUANTASOFT-001 | 596 0.2174 | 592 0.2174 | 479 0.2174 | | 457 0.2174 | 467 0.2174 $N^{0.000\ 1}$ | 593 0.1113 | 589 0.1113 | 478 0.1113 | | 457 0.1113 | 467 0.1113 $N^{0.000\ 4}$ |
| 309 | RANKONE-002 | 525 0.0151 | 524 0.0191 | 459 0.0221 | 448 0.0259 | 443 0.0300 | 425 0.0007 $N^{0.233\ 354}$ | 502 0.0045 | 504 0.0057 | 453 0.0068 | 445 0.0081 | 439 0.0098 | 74 0.0001 $N^{0.268\ 428}$ |
| 310 | RANKONE-003 | 526 0.0151 | 525 0.0191 | 458 0.0221 | 447 0.0259 | 442 0.0300 | 426 0.0007 $N^{0.233\ 355}$ | 503 0.0045 | 502 0.0057 | 454 0.0068 | 446 0.0081 | 440 0.0098 | 75 0.0001 $N^{0.268\ 427}$ |
| 311 | RANKONE-005 | 474 0.0072 | 478 0.0090 | 441 0.0106 | 431 0.0129 | 424 0.0153 | 335 0.0002 $N^{0.259\ 384}$ | 461 0.0023 | 462 0.0028 | 428 0.0033 | 419 0.0039 | 413 0.0047 | 62 0.0001 $N^{0.242\ 418}$ |
| 312 | RANKONE-007 | 393 0.0025 | 392 0.0030 | 381 0.0035 | 368 0.0041 | 368 0.0049 | 224 0.0001 $N^{0.233\ 353}$ | 346 0.0011 | 353 0.0013 | 349 0.0014 | 346 0.0016 | 345 0.0018 | 81 0.0001 $N^{0.154\ 386}$ |
| 313 | RANKONE-009 | 336 0.0016 | 342 0.0020 | 335 0.0023 | 337 0.0028 | 333 0.0034 | 92 0.0001 $N^{0.253\ 379}$ | 294 0.0010 | 300 0.0010 | 301 0.0011 | 296 0.0012 | 291 0.0012 | 171 0.0003 $N^{0.079\ 325}$ |
| 314 | RANKONE-010 | 339 0.0017 | 334 0.0019 | 321 0.0022 | 317 0.0025 | 310 0.0029 | 255 0.0001 $N^{0.192\ 302}$ | 322 0.0011 | 320 0.0011 | 316 0.0012 | 302 0.0012 | 303 0.0013 | 213 0.0004 $N^{0.076\ 322}$ |
| 315 | RANKONE-011 | 213 0.0010 | 227 0.0011 | 227 0.0013 | 225 0.0015 | 232 0.0018 | 164 0.0001 $N^{0.191\ 301}$ | 214 0.0008 | 203 0.0008 | 200 0.0008 | 197 0.0008 | 191 0.0009 | 304 0.0005 $N^{0.033\ 223}$ |
| 316 | RANKONE-012 | 185 0.0009 | 196 0.0010 | 203 0.0012 | 200 0.0013 | 210 0.0016 | 142 0.0001 $N^{0.187\ 291}$ | 196 0.0007 | 197 0.0008 | 189 0.0008 | 175 0.0008 | 167 0.0008 | 314 0.0006 $N^{0.023\ 171}$ |
| 317 | RANKONE-013 | 88 0.0006 | 94 0.0007 | 105 0.0008 | 115 0.0010 | 127 0.0011 | 51 0.0000 $N^{0.206\ 319}$ | 84 0.0005 | 78 0.0005 | 81 0.0005 | 76 0.0005 | 79 0.0006 | 176 0.0003 $N^{0.028\ 200}$ |
| 318 | RANKONE-014 | 67 0.0006 | 74 0.0006 | 68 0.0007 | 68 0.0008 | 68 0.0009 | 113 0.0001 $N^{0.164\ 237}$ | 83 0.0005 | 85 0.0005 | 77 0.0005 | 71 0.0005 | 66 0.0005 | 244 0.0004 $N^{0.016\ 127}$ |
| 319 | RANKONE-015 | 82 0.0006 | 77 0.0006 | 72 0.0007 | 74 0.0008 | 72 0.0009 | 180 0.0001 $N^{0.146\ 197}$ | 90 0.0005 | 88 0.0005 | 84 0.0005 | 79 0.0005 | 75 0.0006 | 247 0.0004 $N^{0.017\ 136}$ |
| 320 | REALNETWORKS-002 | 554 0.0297 | 549 0.0391 | 422 0.0467 | 461 0.0560 | 455 0.0577 | 443 0.0013 $N^{0.237\ 360}$ | 511 0.0053 | 522 0.0075 | 464 0.0095 | 454 0.0124 | 447 0.0130 | 53 0.0001 $N^{0.324\ 442}$ |
| 321 | REALNETWORKS-003 | 532 0.0179 | 532 0.0238 | 464 0.0287 | 454 0.0349 | 450 0.0420 | 395 0.0004 $N^{0.291\ 422}$ | 491 0.0038 | 495 0.0050 | 447 0.0061 | 443 0.0077 | 438 0.0098 | 40 0.0000 $N^{0.325\ 443}$ |
| 322 | REALNETWORKS-004 | 529 0.0172 | 530 0.0233 | 463 0.0280 | 453 0.0344 | 447 0.0413 | 380 0.0003 $N^{0.299\ 431}$ | 488 0.0036 | 493 0.0047 | 446 0.0058 | 441 0.0074 | 437 0.0095 | 34 0.0000 $N^{0.335\ 445}$ |
| 323 | REALNETWORKS-005 | 335 0.0016 | 340 0.0019 | 329 0.0022 | 322 0.0026 | 331 0.0033 | 118 0.0001 $N^{0.240\ 365}$ | 217 0.0008 | 229 0.0009 | 244 0.0009 | 252 0.0010 | 258 0.0011 | 96 0.0002 $N^{0.111\ 361}$ |
| 324 | REALNETWORKS-006 | 190 0.0009 | 202 0.0010 | 211 0.0012 | 215 0.0015 | 220 0.0017 | 91 0.0001 $N^{0.210\ 328}$ | 148 0.0006 | 148 0.0007 | 156 0.0007 | 166 0.0008 | 170 0.0008 | 99 0.0002 $N^{0.091\ 344}$ |
| 325 | REALNETWORKS-007 | 181 0.0009 | 185 0.0010 | 178 0.0011 | 181 0.0012 | 188 0.0014 | 207 0.0001 $N^{0.163\ 234}$ | 140 0.0006 | 142 0.0006 | 153 0.0007 | 150 0.0007 | 157 0.0008 | 100 0.0002 $N^{0.087\ 340}$ |
| 326 | REALNETWORKS-008 | 115 0.0007 | 135 0.0008 | 139 0.0009 | 134 0.0010 | 145 0.0012 | 108 0.0001 $N^{0.182\ 281}$ | 75 0.0005 | 76 0.0005 | 94 0.0006 | 91 0.0006 | 95 0.0006 | 111 0.0002 $N^{0.061\ 300}$ |
| 327 | REALSENSE-001 | 75 0.0006 | 78 0.0006 | 80 0.0007 | 79 0.0008 | 91 0.0010 | 77 0.0000 $N^{0.185\ 287}$ | 68 0.0005 | 68 0.0005 | 74 0.0005 | 73 0.0005 | 69 0.0005 | 146 0.0003 $N^{0.035\ 232}$ |
| 328 | RECOGNITO-000 | 49 0.0005 | 58 0.0006 | 51 0.0006 | 54 0.0007 | 50 0.0008 | 137 0.0001 $N^{0.148\ 201}$ | 57 0.0005 | 43 0.0005 | 48 0.0005 | 45 0.0005 | 43 0.0005 | 222 0.0004 $N^{0.015\ 117}$ |
| 329 | RECOGNITO-001 | 61 0.0006 | 52 0.0006 | 44 0.0006 | 43 0.0007 | 36 0.0007 | 279 0.0001 $N^{0.097\ 83}$ | 79 0.0005 | 75 0.0005 | 68 0.0005 | 65 0.0005 | 59 0.0005 | 276 0.0005 $N^{0.009\ 73}$ |
| 330 | RECOGNITO-002 | 40 0.0005 | 40 0.0005 | 38 0.0006 | 31 0.0006 | 24 0.0007 | 261 0.0001 $N^{0.101\ 89}$ | 61 0.0005 | 60 0.0005 | 51 0.0005 | 40 0.0005 | 36 0.0005 | 278 0.0005 $N^{0.004\ 31}$ |
| 331 | REMARKAI-000 | 386 0.0024 | 394 0.0030 | 385 0.0037 | 378 0.0045 | 373 0.0055 | 88 0.0001 $N^{0.286\ 418}$ | 328 0.0011 | 339 0.0012 | 330 0.0013 | 332 0.0014 | 335 0.0016 | 91 0.0002 $N^{0.137\ 376}$ |
| 332 | RENDIP-000 | 231 0.0011 | 236 0.0012 | 237 0.0014 | 240 0.0016 | 244 0.0019 | 140 0.0001 $N^{0.198\ 308}$ | 220 0.0008 | 221 0.0008 | 211 0.0008 | 203 0.0009 | 204 0.0009 | 303 0.0005 $N^{0.035\ 234}$ |
| 333 | REVEALMEDIA-000 | 279 0.0013 | 282 0.0015 | 271 0.0016 | 271 0.0019 | 262 0.0021 | 283 0.0001 $N^{0.163\ 235}$ | 233 0.0008 | 232 0.0009 | 223 0.0009 | 217 0.0009 | 219 0.0010 | 263 0.0004 $N^{0.048\ 269}$ |
| 334 | RNIT-000 | 203 0.0010 | 222 0.0011 | 222 0.0013 | 214 0.0014 | 215 0.0016 | 189 0.0001 $N^{0.180\ 271}$ | 248 0.0009 | 264 0.0009 | 265 0.0010 | 261 0.0010 | 257 0.0011 | 142 0.0003 $N^{0.079\ 329}$ |
| 335 | ROC-016 | 70 0.0006 | 79 0.0006 | 75 0.0007 | 76 0.0008 | 69 0.0009 | 120 0.0001 $N^{0.161\ 229}$ | 85 0.0005 | 79 0.0005 | 89 0.0005 | 80 0.0005 | 70 0.0005 | 210 0.0004 $N^{0.024\ 176}$ |
| 336 | ROC-017 | 57 0.0006 | 65 0.0006 | 76 0.0007 | 89 0.0009 | 89 0.0010 | 37 0.0000 $N^{0.210\ 326}$ | 72 0.0005 | 81 0.0005 | 78 0.0005 | 74 0.0005 | 73 0.0006 | 159 0.0003 $N^{0.033\ 225}$ |
| 337 | ROC-018 | 48 0.0005 | 44 0.0005 | 54 0.0006 | 63 0.0008 | 61 0.0009 | 63 0.0000 $N^{0.181\ 276}$ | 63 0.0005 | 61 0.0005 | 57 0.0005 | 52 0.0005 | 47 0.0005 | 268 0.0004 $N^{0.007\ 63}$ |
| 338 | ROC-019 | 31 0.0005 | 28 0.0005 | 35 0.0006 | 36 0.0006 | 40 0.0008 | 119 0.0001 $N^{0.147\ 200}$ | 48 0.0005 | 41 0.0005 | 33 0.0005 | 25 0.0005 | 22 0.0005 | 283 0.0005 $N^{0.000\ 7}$ |
| 339 | ROC-020 | 37 0.0005 | 35 0.0005 | 34 0.0006 | 39 0.0006 | 41 0.0008 | 166 0.0001 $N^{0.136\ 179}$ | 47 0.0005 | 40 0.0005 | 29 0.0005 | 23 0.0005 | 28 0.0005 | 248 0.0004 $N^{0.007\ 62}$ |
| 340 | ROC-021 | 18 0.0005 | 22 0.0005 | 24 0.0005 | 24 0.0006 | 28 0.0007 | 152 0.0001 $N^{0.134\ 174}$ | 32 0.0005 | 28 0.0005 | 21 0.0005 | 20 0.0005 | 18 0.0005 | 246 0.0004 $N^{0.006\ 52}$ |
| 341 | S1-000 | 343 0.0017 | 345 0.0021 | 344 0.0024 | 339 0.0029 | 332 0.0034 | 149 0.0001 $N^{0.234\ 356}$ | 333 0.0011 | 330 0.0011 | 320 0.0012 | 310 0.0012 | 305 0.0013 | 242 0.0004 $N^{0.071\ 318}$ |
| 342 | S1-001 | 401 0.0027 | 385 0.0028 | 370 0.0030 | 350 0.0033 | 341 0.0037 | 424 0.0007 $N^{0.103\ 93}$ | 442 0.0019 | 428 0.0020 | 400 0.0020 | 388 0.0021 | 373 0.0021 | 428 0.0013 $N^{0.027\ 191}$ |
| 343 | S1-002 | 216 0.0010 | 205 0.0011 | 198 0.0011 | 186 0.0012 | 184 0.0014 | 340 0.0002 $N^{0.110\ 110}$ | 283 0.0009 | 273 0.0010 | 254 0.0010 | 232 0.0010 | 227 0.0010 | 372 0.0008 $N^{0.015\ 124}$ |
| 344 | S1-003 | 217 0.0010 | 219 0.0011 | 207 0.0012 | 205 0.0013 | 196 0.0015 | 301 0.0002 $N^{0.131\ 167}$ | 241 0.0009 | 247 0.0009 | 229 0.0009 | 224 0.0009 | 212 0.0009 | 316 0.0006 $N^{0.033\ 226}$ |
| 345 | S1-004 | 201 0.0010 | 186 0.0010 | 172 0.0010 | 168 0.0011 | 164 0.0013 | 364 0.0003 $N^{0.089\ 63}$ | 257 0.0009 | 244 0.0009 | 234 0.0009 | 221 0.0009 | 209 0.0009 | 362 0.0007 $N^{0.015\ 119}$ |
| 346 | S1-005 | 205 0.0010 | 198 0.0010 | 190 0.0011 | 180 0.0012 | 175 0.0013 | 341 0.0002 $N^{0.107\ 105}$ | 274 0.0009 | 258 0.0009 | 241 0.0009 | 225 0.0009 | 213 0.0009 | 382 0.0008 $N^{0.009\ 72}$ |
| 347 | S1-006 | 187 0.0009 | 180 0.0010 | 168 0.0010 | 159 0.0011 | 144 0.0012 | 366 0.0003 $N^{0.086\ 56}$ | 250 0.0009 | 238 0.0009 | 225 0.0009 | 210 0.0009 | 203 0.0009 | 373 0.0008 $N^{0.009\ 75}$ |
| 348 | S1-007 | 162 0.0008 | 151 0.0009 | 137 0.0009 | 122 0.0010 | 115 0.0011 | 349 0.0002 $N^{0.088\ 62}$ | 209 0.0008 | 198 0.0008 | 190 0.0008 | 177 0.0008 | 165 0.0008 | 352 0.0007 $N^{0.008\ 68}$ |
| 349 | SANSAP-001 | 134 0.0007 | 136 0.0008 | 136 0.0009 | 143 0.0010 | 152 0.0012 | 125 0.0001 $N^{0.176\ 262}$ | 114 0.0006 | 114 0.0006 | 111 0.0006 | 99 0.0006 | 109 0.0006 | 162 0.0003 $N^{0.039\ 248}$ |
| 350 | SCANOVATE-000 | 424 0.0035 | 428 0.0046 | 410 0.0055 | 401 0.0069 | 387 0.0070 | 258 0.0001 $N^{0.252\ 378}$ | 309 0.0010 | 340 0.0012 | 338 0.0014 | 355 0.0017 | 339 0.0017 | 65 0.0001 $N^{0.180\ 401}$ |
| 351 | SCANOVATE-001 | 429 0.0037 | 435 0.0050 | 414 0.0060 | 406 0.0076 | 403 0.0095 | 89 0.0001 $N^{0.317\ 439}$ | 299 0.0010 | 337 0.0012 | 339 0.0014 | 362 0.0017 | 371 0.0021 | 28 0.0000 $N^{0.259\ 422}$ |
| 352 | SCYLLA-001 | 321 0.0015 | 327 0.0018 | 310 0.0020 | 304 0.0023 | 304 0.0028 | 233 0.0001 $N^{0.194\ 303}$ | 295 0.0010 | 298 0.0010 | 291 0.0011 | 279 0.0011 | 285 0.0012 | 235 0.0004 $N^{0.066\ 312}$ |
| 353 | SENSETIME-000 | 366 0.0020 | 357 0.0022 | 343 0.0024 | 325 0.0027 | 320 0.0031 | 376 0.0003 $N^{0.139\ 187}$ | 404 0.0015 | 400 0.0016 | 377 0.0016 | 361 0.0017 | 349 0.0018 | 350 0.0007 $N^{0.059\ 295}$ |
| 354 | SENSETIME-001 | 367 0.0020 | 358 0.0022 | 338 0.0024 | 330 0.0027 | 337 0.0036 | 296 0.0002 $N^{0.183\ 284}$ | 399 0.0015 | 395 0.0015 | 373 0.0016 | 359 0.0017 | 378 0.0023 | 124 0.0003 $N^{0.128\ 370}$ |
| 355 | SENSETIME-002 | 520 0.0133 | 509 0.0133 | 449 0.0133 | 433 0.0134 | 420 0.0136 | 462 0.0120 $N^{0.007\ 2}$ | 558 0.0132 | 548 0.0132 | 468 0.0132 | 456 0.0132 | 448 0.0132 | 463 0.0131 $N^{0.001\ 10}$ |
| 356 | SENSETIME-003 | 84 0.0006 | 81 0.0006 | 70 0.0007 | 67 0.0008 | 65 0.0009 | 220 0.0001 $N^{0.127\ 155}$ | 120 0.0006 | 113 0.0006 | 105 0.0006 | 103 0.0006 | 98 0.0006 | 262 0.0004 $N^{0.020\ 154}$ |
| 357 | SENSETIME-004 | 81 0.0006 | 76 0.0006 | 62 0.0006 | 66 0.0008 | 59 0.0009 | 232 0.0001 $N^{0.122\ 145}$ | 52 0.0005 | 67 0.0005 | 60 0.0005 | 58 0.0005 | 61 0.0005 | 151 0.0003 $N^{0.031\ 214}$ |
| 358 | SENSETIME-005 | 28 0.0005 | 32 0.0005 | 32 0.0006 | 34 0.0006 | 37 0.0007 | 153 0.0001 $N^{0.137\ 183}$ | 35 0.0005 | 24 0.0005 | 24 0.0005 | 19 0.0005 | 19 0.0005 | 260 0.0004 $N^{0.004\ 28}$ |
| 359 | SENSETIME-006 | 29 0.0005 | 30 0.0005 | 22 0.0005 | 23 0.0006 | 18 0.0007 | 216 0.0001 $N^{0.112\ 120}$ | 33 0.0005 | 35 0.0005 | 26 0.0005 | 32 0.0005 | 30 0.0005 | 178 0.0003 $N^{0.019\ 148}$ |
| 360 | SENSETIME-007 | 26 0.0005 | 23 0.0005 | 18 0.0005 | 14 0.0006 | 15 0.0006 | 244 0.0001 $N^{0.101\ 88}$ | 38 0.0005 | 33 0.0005 | 27 0.0005 | 28 0.0005 | 26 0.0005 | 214 0.0004 $N^{0.014\ 109}$ |

*Table 48:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK FNIR(N, T= 0, R) | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 361 | SENSETIME-008 | [21] 0.0005 | [24] 0.0005 | [19] 0.0005 | [20] 0.0006 | [14] 0.0006 | [215] 0.0001 N^0.111 [116] | [22] 0.0004 | [26] 0.0005 | [23] 0.0005 | [21] 0.0005 | [24] 0.0005 | [161] 0.0003 N^0.022 [167] |
| 362 | SENSETIME-009 | [23] 0.0005 | [17] 0.0005 | [16] 0.0005 | [12] 0.0006 | [12] 0.0006 | [264] 0.0001 N^0.093 [69] | [26] 0.0004 | [27] 0.0005 | [34] 0.0005 | [26] 0.0005 | [23] 0.0005 | [195] 0.0004 N^0.016 [130] |
| 363 | SENSETIME-010 | [13] 0.0005 | [10] 0.0005 | [8] 0.0005 | [6] 0.0005 | [9] 0.0006 | [214] 0.0001 N^0.108 [106] | [15] 0.0004 | [17] 0.0004 | [16] 0.0004 | [13] 0.0004 | [14] 0.0004 | [216] 0.0004 N^0.008 [70] |
| 364 | SENSETIME-011 | [24] 0.0005 | [19] 0.0005 | [13] 0.0005 | [13] 0.0006 | [11] 0.0006 | [265] 0.0001 N^0.093 [70] | [25] 0.0004 | [30] 0.0005 | [28] 0.0005 | [24] 0.0005 | [21] 0.0005 | [193] 0.0004 N^0.016 [129] |
| 365 | SENSETIME-012 | [12] 0.0004 | [15] 0.0005 | [10] 0.0005 | [4] 0.0005 | [4] 0.0006 | [251] 0.0001 N^0.095 [79] | [17] 0.0004 | [18] 0.0004 | [14] 0.0004 | [10] 0.0004 | [9] 0.0004 | [265] 0.0004 N^0.000 [5] |
| 366 | SERENDIPITY-000 | [218] 0.0010 | [232] 0.0012 | [226] 0.0013 | [227] 0.0015 | [231] 0.0018 | [183] 0.0001 N^0.184 [285] | [216] 0.0008 | [218] 0.0008 | [215] 0.0009 | [213] 0.0009 | [207] 0.0009 | [235] 0.0004 N^0.050 [272] |
| 367 | SEVENTHSENSE-001 | [239] 0.0011 | [309] 0.0016 | [373] 0.0031 | [391] 0.0058 | [414] 0.0117 | [10] 0.0000 N^0.832 [458] | [266] 0.0009 | [393] 0.0015 | [422] 0.0030 | [431] 0.0056 | [443] 0.0115 | [9] 0.0000 N^0.880 [458] |
| 368 | SHAMAN-007 | [559] 0.0368 | [550] 0.0393 | [471] 0.0413 | [460] 0.0439 | [453] 0.0470 | [461] 0.0120 N^0.083 [49] | [576] 0.0306 | [569] 0.0311 | [474] 0.0316 | [461] 0.0323 | [454] 0.0334 | [468] 0.0205 N^0.029 [205] |
| 369 | SIAT-001 | [314] 0.0015 | [308] 0.0016 | [296] 0.0018 | [289] 0.0021 | [292] 0.0026 | [250] 0.0001 N^0.185 [286] | [271] 0.0009 | [282] 0.0010 | [279] 0.0010 | [284] 0.0011 | [282] 0.0012 | [130] 0.0003 N^0.091 [346] |
| 370 | SIAT-002 | [304] 0.0014 | [312] 0.0017 | [297] 0.0018 | [287] 0.0021 | [282] 0.0025 | [257] 0.0001 N^0.182 [279] | [301] 0.0010 | [311] 0.0011 | [305] 0.0011 | [298] 0.0012 | [289] 0.0012 | [254] 0.0004 N^0.064 [309] |
| 371 | SJTU-001 | [22] 0.0005 | [25] 0.0005 | [23] 0.0005 | [29] 0.0006 | [22] 0.0007 | [163] 0.0001 N^0.133 [172] | [20] 0.0004 | [20] 0.0004 | [20] 0.0004 | [18] 0.0005 | [17] 0.0005 | [179] 0.0004 N^0.016 [133] |
| 372 | SPARKTEK-000 | [129] 0.0007 | [116] 0.0008 | [133] 0.0009 | [119] 0.0010 | [134] 0.0011 | [175] 0.0001 N^0.161 [226] | [121] 0.0006 | [121] 0.0006 | [114] 0.0006 | [106] 0.0006 | [102] 0.0006 | [237] 0.0004 N^0.026 [184] |
| 373 | SPARKTEK-001 | [59] 0.0006 | [54] 0.0006 | [52] 0.0006 | [45] 0.0007 | [49] 0.0008 | [211] 0.0001 N^0.124 [150] | [86] 0.0005 | [83] 0.0005 | [71] 0.0005 | [64] 0.0005 | [63] 0.0005 | [287] 0.0005 N^0.007 [58] |
| 374 | SPARSH-001 | [98] 0.0007 | [103] 0.0007 | [104] 0.0008 | [105] 0.0009 | [106] 0.0011 | [171] 0.0001 N^0.157 [217] | [117] 0.0006 | [108] 0.0006 | [98] 0.0006 | [93] 0.0006 | [85] 0.0006 | [300] 0.0005 N^0.009 [78] |
| 375 | SQISOFT-001 | [391] 0.0025 | [412] 0.0038 | [409] 0.0055 | [410] 0.0081 | [415] 0.0118 | [14] 0.0000 N^0.537 [455] | [141] 0.0006 | [167] 0.0007 | [182] 0.0008 | [198] 0.0008 | [221] 0.0010 | [52] 0.0001 N^0.160 [392] |
| 376 | SQISOFT-002 | [151] 0.0008 | [177] 0.0009 | [199] 0.0012 | [226] 0.0015 | [246] 0.0019 | [20] 0.0000 N^0.301 [433] | [108] 0.0006 | [105] 0.0006 | [112] 0.0006 | [110] 0.0006 | [119] 0.0007 | [122] 0.0002 N^0.060 [296] |
| 377 | STCON-001 | [46] 0.0005 | [46] 0.0005 | [47] 0.0006 | [46] 0.0007 | [39] 0.0007 | [193] 0.0001 N^0.129 [163] | [56] 0.0005 | [46] 0.0005 | [45] 0.0005 | [36] 0.0005 | [42] 0.0005 | [230] 0.0004 N^0.013 [103] |
| 378 | STCON-002 | [35] 0.0005 | [36] 0.0005 | [40] 0.0006 | [30] 0.0006 | [30] 0.0007 | [206] 0.0001 N^0.119 [132] | [53] 0.0005 | [58] 0.0005 | [40] 0.0005 | [50] 0.0005 | [46] 0.0005 | [202] 0.0004 N^0.018 [145] |
| 379 | STCON-003 | [36] 0.0005 | [38] 0.0005 | [36] 0.0006 | [28] 0.0006 | [27] 0.0007 | [218] 0.0001 N^0.114 [121] | [62] 0.0005 | [53] 0.0005 | [46] 0.0005 | [47] 0.0005 | [53] 0.0005 | [232] 0.0004 N^0.013 [100] |
| 380 | STCON-004 | [32] 0.0005 | [34] 0.0005 | [28] 0.0005 | [16] 0.0006 | [19] 0.0007 | [239] 0.0001 N^0.105 [96] | [45] 0.0005 | [38] 0.0005 | [47] 0.0005 | [38] 0.0005 | [32] 0.0005 | [212] 0.0004 N^0.015 [125] |
| 381 | STCON-005 | [30] 0.0005 | [31] 0.0005 | [29] 0.0005 | [18] 0.0006 | [17] 0.0007 | [262] 0.0001 N^0.097 [82] | [42] 0.0005 | [39] 0.0005 | [41] 0.0005 | [35] 0.0005 | [29] 0.0005 | [211] 0.0004 N^0.015 [126] |
| 382 | STELLARTECS-001 | [331] 0.0016 | [323] 0.0018 | [309] 0.0020 | [295] 0.0022 | [290] 0.0026 | [315] 0.0002 N^0.158 [219] | [378] 0.0013 | [370] 0.0014 | [337] 0.0014 | [329] 0.0014 | [315] 0.0014 | [410] 0.0010 N^0.023 [172] |
| 383 | SYNESIS-003 | [528] 0.0157 | [515] 0.0159 | [454] 0.0160 | [440] 0.0162 | [437] 0.0250 | [452] 0.0026 N^0.129 [161] | [563] 0.0156 | [556] 0.0156 | [470] 0.0156 | [459] 0.0156 | [452] 0.0242 | [392] 0.0009 N^0.196 [408] |
| 384 | SYNESIS-003 | [584] 0.1453 | [583] 0.1697 | [477] 0.1873 | [465] 0.2085 | [458] 0.2315 | [463] 0.0176 N^0.158 [221] | [586] 0.0825 | [585] 0.0866 | [477] 0.0917 | [465] 0.0995 | [456] 0.1101 | [465] 0.0216 N^0.098 [355] |
| 385 | SYNESIS-005 | [489] 0.0081 | [471] 0.0082 | [431] 0.0082 | [413] 0.0083 | [397] 0.0084 | [460] 0.0069 N^0.012 [3] | [538] 0.0081 | [529] 0.0081 | [461] 0.0081 | [444] 0.0081 | [431] 0.0081 | [462] 0.0081 N^0.000 [9] |
| 386 | TECH5-001 | [403] 0.0028 | [408] 0.0036 | [397] 0.0044 | [387] 0.0053 | [384] 0.0067 | [93] 0.0001 N^0.294 [425] | [365] 0.0012 | [361] 0.0013 | [354] 0.0014 | [349] 0.0016 | [355] 0.0019 | [89] 0.0002 N^0.148 [381] |
| 387 | TECH5-002 | [337] 0.0017 | [364] 0.0023 | [359] 0.0028 | [352] 0.0034 | [359] 0.0044 | [28] 0.0000 N^0.324 [440] | [105] 0.0006 | [154] 0.0007 | [170] 0.0007 | [188] 0.0008 | [224] 0.0010 | [36] 0.0000 N^0.190 [406] |
| 388 | TECH5-003 | [100] 0.0007 | [104] 0.0007 | [99] 0.0008 | [107] 0.0009 | [107] 0.0011 | [154] 0.0001 N^0.161 [227] | [111] 0.0006 | [104] 0.0006 | [95] 0.0006 | [93] 0.0006 | [88] 0.0006 | [284] 0.0005 N^0.013 [102] |
| 389 | TECHAINER-000 | [345] 0.0017 | [336] 0.0019 | [312] 0.0020 | [306] 0.0023 | [286] 0.0025 | [375] 0.0003 N^0.128 [158] | [392] 0.0014 | [383] 0.0014 | [357] 0.0015 | [340] 0.0015 | [330] 0.0016 | [396] 0.0009 N^0.032 [221] |
| 390 | TEVIAN-005 | [457] 0.0055 | [461] 0.0071 | [432] 0.0082 | [424] 0.0103 | [419] 0.0128 | [249] 0.0001 N^0.285 [417] | [438] 0.0019 | [437] 0.0022 | [414] 0.0025 | [402] 0.0027 | [396] 0.0033 | [90] 0.0002 N^0.181 [402] |
| 391 | TEVIAN-006 | [359] 0.0019 | [347] 0.0021 | [333] 0.0023 | [313] 0.0024 | [302] 0.0027 | [393] 0.0004 N^0.123 [147] | [367] 0.0012 | [354] 0.0013 | [335] 0.0013 | [324] 0.0014 | [318] 0.0014 | [334] 0.0006 N^0.052 [276] |
| 392 | TEVIAN-007 | [286] 0.0014 | [277] 0.0014 | [255] 0.0015 | [248] 0.0016 | [230] 0.0018 | [404] 0.0004 N^0.090 [64] | [258] 0.0009 | [255] 0.0009 | [245] 0.0009 | [237] 0.0010 | [232] 0.0010 | [319] 0.0006 N^0.035 [235] |
| 393 | THAKAAIT-000 | [195] 0.0010 | [211] 0.0011 | [217] 0.0012 | [220] 0.0015 | [223] 0.0017 | [114] 0.0001 N^0.203 [315] | [174] 0.0007 | [169] 0.0007 | [163] 0.0007 | [158] 0.0007 | [160] 0.0008 | [198] 0.0004 N^0.045 [263] |
| 394 | TIGER-002 | [437] 0.0043 | [441] 0.0054 | [418] 0.0066 | [414] 0.0084 | [408] 0.0102 | [148] 0.0001 N^0.302 [434] | [362] 0.0012 | [381] 0.0014 | [378] 0.0016 | [379] 0.0020 | [389] 0.0026 | [31] 0.0000 N^0.256 [420] |
| 395 | TIGER-004 | [357] 0.0019 | [360] 0.0022 | [342] 0.0024 | [326] 0.0027 | [323] 0.0031 | [338] 0.0002 N^0.160 [224] | [419] 0.0017 | [414] 0.0018 | [394] 0.0019 | [381] 0.0020 | [375] 0.0022 | [315] 0.0006 N^0.083 [333] |
| 396 | TOSHIBA-000 | [417] 0.0033 | [420] 0.0043 | [404] 0.0050 | [390] 0.0058 | [423] 0.0151 | [15] 0.0000 N^0.462 [452] | [394] 0.0014 | [403] 0.0016 | [384] 0.0017 | [378] 0.0019 | [441] 0.0103 | [12] 0.0000 N^0.568 [455] |
| 397 | TOSHIBA-002 | [172] 0.0009 | [176] 0.0009 | [157] 0.0010 | [135] 0.0010 | [125] 0.0011 | [403] 0.0004 N^0.062 [21] | [267] 0.0009 | [250] 0.0009 | [231] 0.0009 | [218] 0.0009 | [205] 0.0009 | [394] 0.0009 N^0.002 [15] |
| 398 | TOSHIBA-003 | [138] 0.0008 | [121] 0.0008 | [101] 0.0008 | [90] 0.0009 | [80] 0.0010 | [350] 0.0002 N^0.080 [44] | [195] 0.0007 | [187] 0.0007 | [173] 0.0007 | [159] 0.0007 | [146] 0.0007 | [354] 0.0007 N^0.004 [26] |
| 399 | TOSS-000 | [199] 0.0010 | [187] 0.0010 | [173] 0.0010 | [160] 0.0011 | [140] 0.0012 | [400] 0.0004 N^0.069 [28] | [272] 0.0009 | [256] 0.0009 | [232] 0.0009 | [223] 0.0009 | [208] 0.0009 | [391] 0.0009 N^0.005 [44] |
| 400 | TRUEFACE-000 | [400] 0.0027 | [391] 0.0029 | [374] 0.0032 | [359] 0.0035 | [348] 0.0039 | [414] 0.0005 N^0.128 [157] | [453] 0.0022 | [441] 0.0022 | [408] 0.0023 | [396] 0.0024 | [385] 0.0024 | [422] 0.0012 N^0.044 [261] |
| 401 | TSIGHT-000 | [2] 0.0004 | [3] 0.0004 | [3] 0.0005 | [505] 1.0000 | [546] 1.0000 | [1] 0.0000 N^3.173 [467] | [4] 0.0004 | [1] 0.0004 | [3] 0.0004 | [590] 1.0000 | [593] 1.0000 | [1] 0.0000 N^3.187 [467] |
| 402 | UNISSEY-003 | [410] 0.0029 | [398] 0.0031 | [376] 0.0032 | [358] 0.0034 | [344] 0.0038 | [437] 0.0009 N^0.083 [48] | [471] 0.0026 | [455] 0.0026 | [416] 0.0026 | [401] 0.0027 | [392] 0.0027 | [445] 0.0022 N^0.014 [108] |
| 403 | UNIVERSEAI-000 | [165] 0.0008 | [95] 0.0007 | [91] 0.0007 | [94] 0.0009 | [126] 0.0011 | [298] 0.0002 N^0.108 [107] | [206] 0.0008 | [124] 0.0006 | [117] 0.0006 | [122] 0.0006 | [146] 0.0007 | [353] 0.0007 N^0.003 [1] |
| 404 | USEB-001 | [164] 0.0008 | [156] 0.0009 | [149] 0.0009 | [151] 0.0011 | [149] 0.0012 | [269] 0.0001 N^0.132 [169] | [191] 0.0007 | [181] 0.0007 | [168] 0.0007 | [151] 0.0007 | [149] 0.0007 | [344] 0.0007 N^0.007 [60] |
| 405 | USEB-002 | [219] 0.0010 | [206] 0.0011 | [191] 0.0011 | [189] 0.0013 | [181] 0.0014 | [339] 0.0002 N^0.110 [111] | [276] 0.0009 | [254] 0.0009 | [236] 0.0009 | [222] 0.0009 | [211] 0.0009 | [388] 0.0009 N^0.006 [51] |
| 406 | USEB-003 | [159] 0.0008 | [155] 0.0009 | [144] 0.0009 | [141] 0.0010 | [148] 0.0012 | [272] 0.0001 N^0.129 [164] | [183] 0.0007 | [175] 0.0007 | [172] 0.0007 | [156] 0.0007 | [152] 0.0008 | [318] 0.0006 N^0.018 [144] |
| 407 | USEB-004 | [191] 0.0009 | [179] 0.0010 | [164] 0.0010 | [163] 0.0011 | [155] 0.0012 | [351] 0.0002 N^0.096 [80] | [251] 0.0009 | [237] 0.0009 | [222] 0.0009 | [209] 0.0009 | [199] 0.0009 | [386] 0.0008 N^0.004 [33] |
| 408 | VD-001 | [541] 0.0227 | [539] 0.0272 | [467] 0.0312 | [457] 0.0360 | [449] 0.0414 | [445] 0.0014 N^0.206 [320] | [555] 0.0117 | [547] 0.0127 | [469] 0.0137 | [458] 0.0151 | [450] 0.0167 | [447] 0.0022 N^0.123 [368] |
| 409 | VDFSOFTWARE-002 | [287] 0.0014 | [319] 0.0018 | [332] 0.0023 | [340] 0.0029 | [347] 0.0038 | [19] 0.0000 N^0.358 [449] | [96] 0.0005 | [132] 0.0006 | [154] 0.0007 | [167] 0.0008 | [195] 0.0009 | [43] 0.0001 N^0.168 [396] |
| 410 | VERIDAS-001 | [358] 0.0019 | [370] 0.0024 | [361] 0.0028 | [357] 0.0034 | [355] 0.0041 | [104] 0.0001 N^0.261 [387] | [316] 0.0010 | [321] 0.0011 | [317] 0.0012 | [314] 0.0013 | [317] 0.0014 | [119] 0.0002 N^0.107 [358] |
| 411 | VERIDAS-002 | [356] 0.0019 | [369] 0.0024 | [360] 0.0028 | [356] 0.0034 | [354] 0.0041 | [100] 0.0001 N^0.261 [388] | [310] 0.0010 | [319] 0.0011 | [315] 0.0012 | [312] 0.0013 | [316] 0.0014 | [117] 0.0002 N^0.108 [360] |
| 412 | VERIDAS-003 | [274] 0.0013 | [279] 0.0015 | [270] 0.0016 | [266] 0.0019 | [268] 0.0022 | [231] 0.0001 N^0.181 [277] | [259] 0.0009 | [267] 0.0009 | [253] 0.0010 | [246] 0.0010 | [254] 0.0011 | [236] 0.0004 N^0.059 [294] |
| 413 | VERIDAS-004 | [200] 0.0010 | [190] 0.0010 | [175] 0.0011 | [144] 0.0011 | [136] 0.0012 | [409] 0.0004 N^0.058 [16] | [237] 0.0008 | [233] 0.0009 | [219] 0.0009 | [205] 0.0009 | [197] 0.0009 | [360] 0.0007 N^0.012 [95] |
| 414 | VERIGRAM-000 | [375] 0.0021 | [356] 0.0022 | [323] 0.0022 | [297] 0.0022 | [273] 0.0023 | [446] 0.0015 N^0.025 [5] | [448] 0.0021 | [433] 0.0021 | [402] 0.0021 | [389] 0.0021 | [370] 0.0021 | [443] 0.0020 N^0.002 [17] |
| 415 | VERIGRAM-001 | [372] 0.0021 | [353] 0.0021 | [322] 0.0022 | [305] 0.0023 | [275] 0.0023 | [442] 0.0012 N^0.042 [10] | [447] 0.0020 | [431] 0.0020 | [401] 0.0020 | [385] 0.0020 | [366] 0.0020 | [442] 0.0019 N^0.003 [20] |
| 416 | VIANTE-000 | [177] 0.0009 | [167] 0.0009 | [156] 0.0009 | [139] 0.0010 | [132] 0.0011 | [357] 0.0003 N^0.088 [61] | [236] 0.0008 | [225] 0.0008 | [210] 0.0008 | [202] 0.0009 | [187] 0.0009 | [377] 0.0008 N^0.005 [46] |
| 417 | VIANTE-001 | [172] 0.0009 | [159] 0.0009 | [147] 0.0009 | [129] 0.0010 | [113] 0.0011 | [367] 0.0003 N^0.079 [43] | [234] 0.0008 | [219] 0.0008 | [212] 0.0008 | [200] 0.0008 | [183] 0.0008 | [380] 0.0008 N^0.003 [21] |
| 418 | VIANTE-002 | [45] 0.0005 | [47] 0.0005 | [43] 0.0006 | [37] 0.0006 | [33] 0.0007 | [234] 0.0001 N^0.111 [117] | [67] 0.0005 | [69] 0.0005 | [63] 0.0005 | [61] 0.0005 | [56] 0.0005 | [275] 0.0004 N^0.008 [65] |
| 419 | VIANTE-003 | [47] 0.0005 | [42] 0.0005 | [41] 0.0006 | [38] 0.0006 | [32] 0.0007 | [248] 0.0001 N^0.107 [100] | [65] 0.0005 | [73] 0.0005 | [61] 0.0005 | [60] 0.0005 | [55] 0.0005 | [274] 0.0004 N^0.008 [66] |
| 420 | VIETTELAI-000 | [433] 0.0040 | [418] 0.0041 | [394] 0.0042 | [375] 0.0043 | [363] 0.0045 | [450] 0.0024 N^0.039 [9] | [492] 0.0038 | [483] 0.0038 | [436] 0.0038 | [417] 0.0039 | [405] 0.0039 | [452] 0.0031 N^0.013 [106] |
| 421 | VIETTELAI-001 | [333] 0.0016 | [314] 0.0017 | [287] 0.0018 | [270] 0.0019 | [264] 0.0021 | [413] 0.0005 N^0.093 [68] | [397] 0.0015 | [389] 0.0015 | [361] 0.0015 | [341] 0.0015 | [327] 0.0015 | [425] 0.0013 N^0.012 [98] |
| 422 | VIETTELAI-002 | [332] 0.0016 | [313] 0.0017 | [286] 0.0018 | [269] 0.0019 | [263] 0.0021 | [410] 0.0004 N^0.094 [72] | [398] 0.0015 | [388] 0.0015 | [362] 0.0015 | [342] 0.0015 | [329] 0.0015 | [426] 0.0013 N^0.012 [97] |
| 423 | VIETTELAI-003 | [72] 0.0006 | [75] 0.0006 | [84] 0.0007 | [82] 0.0008 | [79] 0.0010 | [83] 0.0001 N^0.179 [268] | [77] 0.0005 | [72] 0.0005 | [67] 0.0005 | [70] 0.0005 | [74] 0.0006 | [167] 0.0003 N^0.030 [206] |
| 424 | VIGILANTSOLUTIONS-008 | [374] 0.0021 | [379] 0.0026 | [371] 0.0031 | [366] 0.0037 | [358] 0.0044 | [144] 0.0001 N^0.250 [376] | [215] 0.0008 | [259] 0.0009 | [286] 0.0011 | [299] 0.0012 | [304] 0.0013 | [55] 0.0001 N^0.175 [399] |
| 425 | VINBIGDATA-001 | [135] 0.0007 | [150] 0.0009 | [163] 0.0010 | [170] 0.0011 | [183] 0.0014 | [47] 0.0000 N^0.223 [340] | [76] 0.0005 | [86] 0.0005 | [82] 0.0005 | [83] 0.0006 | [86] 0.0006 | [135] 0.0003 N^0.043 [259] |
| 426 | VINBIGDATA-002 | [60] 0.0006 | [61] 0.0006 | [61] 0.0006 | [59] 0.0008 | [78] 0.0009 | [68] 0.0000 N^0.183 [282] | [58] 0.0005 | [44] 0.0005 | [54] 0.0005 | [46] 0.0005 | [44] 0.0005 | [225] 0.0004 N^0.015 [120] |
| 427 | VINBIGDATA-003 | [50] 0.0005 | [51] 0.0006 | [64] 0.0007 | [60] 0.0008 | [73] 0.0009 | [57] 0.0000 N^0.190 [297] | [59] 0.0005 | [55] 0.0005 | [56] 0.0005 | [53] 0.0005 | [45] 0.0005 | [218] 0.0004 N^0.016 [134] |
| 428 | VINBIGDATA-004 | [56] 0.0006 | [50] 0.0006 | [56] 0.0006 | [52] 0.0007 | [55] 0.0008 | [146] 0.0001 N^0.146 [198] | [54] 0.0005 | [49] 0.0005 | [39] 0.0005 | [51] 0.0005 | [49] 0.0005 | [182] 0.0004 N^0.020 [153] |
| 429 | VISIONBOX-000 | [292] 0.0014 | [296] 0.0016 | [295] 0.0018 | [493] 1.0000 | [464] 0.9526 | [2] 0.0000 N^2.661 [465] | [252] 0.0009 | [253] 0.0009 | [258] 0.0010 | [472] 1.0000 | [464] 0.9525 | [2] 0.0000 N^2.857 [466] |
| 430 | VISIONBOX-001 | [390] 0.0025 | [380] 0.0026 | [358] 0.0027 | [344] 0.0030 | [327] 0.0033 | [427] 0.0007 N^0.094 [71] | [457] 0.0023 | [446] 0.0023 | [409] 0.0023 | [395] 0.0023 | [381] 0.0024 | [441] 0.0019 N^0.015 [122] |
| 431 | VISIONBOX-002 | [289] 0.0014 | [280] 0.0015 | [266] 0.0016 | [261] 0.0018 | [260] 0.0021 | [319] 0.0002 N^0.142 [193] | [348] 0.0011 | [342] 0.0012 | [324] 0.0012 | [305] 0.0012 | [292] 0.0012 | [367] 0.0008 N^0.031 [215] |
| 432 | VISIONLABS-004 | [373] 0.0021 | [381] 0.0026 | [372] 0.0031 | [373] 0.0043 | [385] 0.0068 | [17] 0.0000 N^0.395 [451] | [360] 0.0012 | [365] 0.0013 | [376] 0.0016 | [398] 0.0024 | [411] 0.0044 | [17] 0.0000 N^0.440 [450] |

*Table 49:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES AT GIVEN RANK | | ENROL MOST RECENT | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T= 0, R) | | RANK 1 | | | | | | RANK 50 | | | | | |
| # | ALGORITHM | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ | N=0.64M | N=1.6M | N=3.0M | N=6.0M | N=12.0M | $aN^b$ |
| 433 | VISIONLABS-005 | 354 0.0019 | 361 0.0023 | 357 0.0027 | 361 0.0035 | 367 0.0049 | 27 0.0000 N^0.328 443 | 354 0.0012 | 346 0.0013 | 365 0.0015 | 373 0.0019 | 393 0.0029 | 22 0.0000 N^0.303 438 |
| 434 | VISIONLABS-006 | 320 0.0015 | 320 0.0018 | 314 0.0021 | 331 0.0027 | 349 0.0040 | 24 0.0000 N^0.324 441 | 351 0.0011 | 344 0.0012 | 351 0.0014 | 369 0.0018 | 390 0.0026 | 25 0.0000 N^0.281 432 |
| 435 | VISIONLABS-007 | 315 0.0015 | 317 0.0017 | 303 0.0019 | 302 0.0023 | 328 0.0033 | 70 0.0000 N^0.256 382 | 342 0.0011 | 338 0.0012 | 323 0.0012 | 313 0.0013 | 356 0.0019 | 76 0.0001 N^0.156 388 |
| 436 | VISIONLABS-008 | 313 0.0015 | 304 0.0016 | 285 0.0018 | 292 0.0022 | 296 0.0027 | 205 0.0001 N^0.199 309 | 375 0.0013 | 367 0.0013 | 343 0.0014 | 352 0.0016 | 361 0.0020 | 102 0.0002 N^0.140 379 |
| 437 | VISIONLABS-009 | 130 0.0007 | 128 0.0008 | 128 0.0009 | 147 0.0011 | 179 0.0014 | 45 0.0000 N^0.219 336 | 145 0.0006 | 147 0.0006 | 151 0.0007 | 172 0.0008 | 238 0.0010 | 51 0.0001 N^0.158 390 |
| 438 | VISIONLABS-010 | 211 0.0010 | 201 0.0010 | 201 0.0012 | 202 0.0013 | 222 0.0017 | 192 0.0001 N^0.176 260 | 260 0.0009 | 246 0.0009 | 255 0.0010 | 266 0.0011 | 301 0.0013 | 92 0.0002 N^0.121 367 |
| 439 | VISIONLABS-011 | 136 0.0008 | 139 0.0008 | 141 0.0009 | 153 0.0011 | 190 0.0015 | 48 0.0000 N^0.219 338 | 172 0.0007 | 172 0.0007 | 183 0.0008 | 196 0.0008 | 268 0.0011 | 54 0.0001 N^0.162 393 |
| 440 | VISTEAM-000 | 420 0.0034 | 432 0.0048 | 413 0.0060 | 407 0.0077 | 405 0.0097 | 33 0.0000 N^0.361 450 | 202 0.0008 | 286 0.0010 | 328 0.0012 | 334 0.0014 | 343 0.0018 | 19 0.0000 N^0.288 435 |
| 441 | VIXVIZION-009 | 326 0.0016 | 338 0.0019 | 331 0.0023 | 336 0.0028 | 330 0.0033 | 76 0.0000 N^0.262 389 | 179 0.0007 | 222 0.0008 | 237 0.0009 | 245 0.0010 | 260 0.0011 | 68 0.0001 N^0.148 382 |
| 442 | VNPAY-001 | 229 0.0011 | 216 0.0011 | 206 0.0012 | 199 0.0013 | 191 0.0015 | 324 0.0002 N^0.118 129 | 262 0.0009 | 249 0.0009 | 238 0.0009 | 236 0.0010 | 226 0.0010 | 329 0.0006 N^0.030 209 |
| 443 | VNPT-001 | 341 0.0017 | 325 0.0018 | 305 0.0019 | 286 0.0021 | 279 0.0024 | 381 0.0003 N^0.119 131 | 389 0.0014 | 384 0.0014 | 355 0.0015 | 337 0.0015 | 321 0.0015 | 413 0.0010 N^0.022 164 |
| 444 | VNPT-002 | 307 0.0015 | 288 0.0015 | 268 0.0016 | 258 0.0017 | 247 0.0019 | 412 0.0005 N^0.087 58 | 384 0.0014 | 371 0.0014 | 347 0.0014 | 328 0.0014 | 314 0.0014 | 420 0.0012 N^0.012 94 |
| 445 | VNPT-003 | 265 0.0012 | 245 0.0013 | 238 0.0014 | 221 0.0015 | 217 0.0016 | 372 0.0003 N^0.100 84 | 343 0.0011 | 327 0.0011 | 309 0.0011 | 289 0.0011 | 277 0.0012 | 398 0.0009 N^0.015 121 |
| 446 | VNPT-004 | 263 0.0012 | 246 0.0013 | 229 0.0013 | 213 0.0014 | 213 0.0016 | 384 0.0003 N^0.092 67 | 345 0.0011 | 331 0.0011 | 310 0.0011 | 290 0.0011 | 273 0.0012 | 408 0.0010 N^0.011 91 |
| 447 | VNPT-005 | 264 0.0012 | 249 0.0013 | 228 0.0013 | 209 0.0014 | 203 0.0016 | 402 0.0004 N^0.077 38 | 338 0.0011 | 322 0.0011 | 306 0.0011 | 283 0.0011 | 271 0.0011 | 399 0.0009 N^0.014 107 |
| 448 | VOCORD-005 | 458 0.0057 | 458 0.0068 | 428 0.0079 | 422 0.0095 | 413 0.0114 | 342 0.0002 N^0.238 362 | 482 0.0031 | 476 0.0033 | 432 0.0034 | 416 0.0038 | 409 0.0041 | 395 0.0009 N^0.092 347 |
| 449 | VOGIC-000 | 378 0.0022 | 368 0.0024 | 350 0.0026 | 334 0.0028 | 326 0.0032 | 402 0.0004 N^0.129 162 | 427 0.0018 | 416 0.0018 | 390 0.0019 | 374 0.0019 | 362 0.0020 | 413 0.0011 N^0.036 237 |
| 450 | VTS-001 | 209 0.0010 | 235 0.0012 | 234 0.0013 | 243 0.0016 | 249 0.0019 | 79 0.0000 N^0.226 344 | 146 0.0006 | 151 0.0007 | 155 0.0007 | 149 0.0007 | 164 0.0008 | 110 0.0002 N^0.077 323 |
| 451 | VTS-002 | 278 0.0013 | 297 0.0016 | 301 0.0019 | 303 0.0023 | 306 0.0028 | 60 0.0000 N^0.256 381 | 182 0.0007 | 196 0.0008 | 207 0.0008 | 214 0.0009 | 231 0.0010 | 86 0.0002 N^0.113 362 |
| 452 | VTS-003 | 125 0.0007 | 133 0.0008 | 125 0.0009 | 131 0.0010 | 143 0.0012 | 131 0.0001 N^0.173 252 | 80 0.0005 | 87 0.0005 | 86 0.0005 | 81 0.0006 | 89 0.0006 | 138 0.0003 N^0.043 256 |
| 453 | VTS-004 | 77 0.0006 | 82 0.0006 | 86 0.0007 | 81 0.0008 | 83 0.0010 | 105 0.0001 N^0.170 241 | 39 0.0005 | 56 0.0005 | 55 0.0005 | 59 0.0005 | 58 0.0005 | 128 0.0003 N^0.041 249 |
| 454 | VTS-005 | 65 0.0006 | 67 0.0006 | 79 0.0007 | 75 0.0008 | 77 0.0009 | 81 0.0001 N^0.177 265 | 55 0.0005 | 48 0.0005 | 42 0.0005 | 43 0.0005 | 41 0.0005 | 221 0.0004 N^0.015 118 |
| 455 | WINK-001 | 369 0.0021 | 376 0.0026 | 365 0.0030 | 360 0.0035 | 356 0.0043 | 147 0.0001 N^0.249 374 | 330 0.0011 | 352 0.0011 | 326 0.0012 | 318 0.0013 | 322 0.0015 | 112 0.0002 N^0.114 364 |
| 456 | WINK-002 | 167 0.0008 | 181 0.0010 | 196 0.0011 | 204 0.0013 | 216 0.0016 | 53 0.0000 N^0.227 346 | 95 0.0005 | 109 0.0006 | 108 0.0006 | 107 0.0006 | 120 0.0007 | 105 0.0002 N^0.069 317 |
| 457 | WINK-003 | 363 0.0020 | 343 0.0020 | 317 0.0021 | 298 0.0022 | 278 0.0024 | 432 0.0008 N^0.067 26 | 434 0.0018 | 418 0.0018 | 389 0.0019 | 372 0.0019 | 358 0.0019 | 438 0.0015 N^0.015 123 |
| 458 | XAILIENT-000 | 362 0.0019 | 371 0.0024 | 364 0.0030 | 363 0.0036 | 365 0.0046 | 44 0.0000 N^0.298 428 | 254 0.0009 | 275 0.0010 | 277 0.0010 | 301 0.0012 | 298 0.0013 | 84 0.0001 N^0.132 373 |
| 459 | XFORWARDAI-000 | 348 0.0018 | 332 0.0019 | 311 0.0020 | 308 0.0023 | 291 0.0026 | 371 0.0003 N^0.130 165 | 408 0.0016 | 397 0.0016 | 371 0.0016 | 353 0.0016 | 338 0.0017 | 417 0.0011 N^0.022 165 |
| 460 | XFORWARDAI-001 | 338 0.0017 | 315 0.0017 | 281 0.0017 | 264 0.0018 | 252 0.0020 | 423 0.0007 N^0.066 25 | 406 0.0015 | 396 0.0015 | 370 0.0015 | 344 0.0015 | 331 0.0016 | 436 0.0014 N^0.005 42 |
| 461 | XFORWARDAI-002 | 328 0.0016 | 300 0.0016 | 275 0.0017 | 256 0.0017 | 237 0.0018 | 433 0.0008 N^0.047 13 | 402 0.0015 | 394 0.0015 | 368 0.0015 | 343 0.0015 | 326 0.0015 | 435 0.0014 N^0.004 35 |
| 462 | YITU-002 | 271 0.0013 | 287 0.0015 | 292 0.0018 | 288 0.0021 | 293 0.0026 | 71 0.0000 N^0.247 373 | 169 0.0007 | 184 0.0007 | 188 0.0008 | 184 0.0008 | 198 0.0009 | 103 0.0002 N^0.095 349 |
| 463 | YITU-003 | 383 0.0023 | 377 0.0026 | 362 0.0028 | 346 0.0031 | 340 0.0036 | 365 0.0003 N^0.154 213 | 426 0.0018 | 415 0.0018 | 392 0.0019 | 380 0.0020 | 372 0.0021 | 379 0.0008 N^0.059 293 |
| 464 | YITU-004 | 170 0.0009 | 191 0.0010 | 221 0.0012 | 218 0.0015 | 361 0.0044 | 13 0.0000 N^0.500 454 | 137 0.0006 | 136 0.0006 | 130 0.0006 | 117 0.0006 | 398 0.0033 | 14 0.0000 N^0.470 451 |
| 465 | YITU-005 | 361 0.0019 | 348 0.0021 | 328 0.0022 | 314 0.0024 | 305 0.0028 | 401 0.0004 N^0.120 137 | 421 0.0017 | 408 0.0017 | 383 0.0017 | 365 0.0017 | 344 0.0018 | 434 0.0014 N^0.014 114 |
| 466 | YOUVERSE-001 | 159 0.0008 | 161 0.0009 | 161 0.0010 | 155 0.0011 | 167 0.0013 | 219 0.0001 N^0.150 205 | 155 0.0006 | 145 0.0006 | 143 0.0007 | 148 0.0007 | 144 0.0007 | 169 0.0003 N^0.047 266 |
| 467 | YUAN-000 | 285 0.0014 | 266 0.0014 | 250 0.0014 | 232 0.0015 | 515 1.0000 | 5 0.0000 N^1.807 463 | 382 0.0013 | 364 0.0013 | 336 0.0013 | 320 0.0013 | 465 1.0000 | 5 0.0000 N^1.793 463 |

*Table 50:* **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 1 | 20FACE-000 | 619 - | 619 - | 561 0.0549 | 553 0.0265 | 549 0.0194 | 543 0.0143 | 533 0.0095 | 555 1.272 |
| 2 | 3DIVI-003 | 619 - | 619 - | 571 0.0830 | 565 0.0441 | 564 0.0346 | 560 0.0267 | 558 0.0188 | 567 1.444 |
| 3 | 3DIVI-004 | 619 - | 619 - | 519 0.0172 | 505 0.0088 | 502 0.0072 | 496 0.0058 | 491 0.0046 | 510 1.089 |
| 4 | 3DIVI-005 | 619 - | 619 - | 520 0.0173 | 506 0.0088 | 501 0.0071 | 494 0.0058 | 492 0.0046 | 511 1.090 |
| 5 | 3DIVI-006 | 619 - | 619 - | 531 0.0237 | 539 0.0168 | 545 0.0157 | 546 0.0151 | 552 0.0145 | 537 1.160 |
| 6 | AADI-000 | 619 - | 619 - | 615 0.9991 | 615 0.9987 | 615 0.9984 | 615 0.9979 | 615 0.9969 | 615 9.988 |
| 7 | AADI-001 | 619 - | 619 - | 611 0.7887 | 611 0.7331 | 611 0.7098 | 612 0.6856 | 612 0.6528 | 611 7.655 |
| 8 | AADI-002 | 619 - | 619 - | 610 0.6994 | 610 0.6287 | 610 0.5993 | 608 0.5688 | 608 0.5255 | 610 6.734 |
| 9 | ACER-000 | 619 - | 619 - | 489 0.0102 | 466 0.0048 | 460 0.0037 | 454 0.0030 | 442 0.0022 | 468 1.050 |
| 10 | ACER-001 | 619 - | 619 - | 431 0.0047 | 433 0.0029 | 429 0.0024 | 425 0.0021 | 420 0.0019 | 429 1.028 |
| 11 | ADIANCE-001 | 619 - | 619 - | 618 1.0000 | 617 1.0000 | 618 1.0000 | 618 1.0000 | 616 1.0000 | 617 10.000 |
| 12 | ADIANCE-002 | 619 - | 619 - | 237 0.0012 | 237 0.0010 | 225 0.0009 | 224 0.0009 | 220 0.0008 | 235 1.009 |
| 13 | ADVANCE-000 | 619 - | 619 - | 301 0.0016 | 314 0.0013 | 314 0.0012 | 308 0.0011 | 310 0.0011 | 305 1.012 |
| 14 | AFISBIOMETRICS-000 | 619 - | 619 - | 244 0.0012 | 155 0.0008 | 135 0.0007 | 133 0.0006 | 100 0.0006 | 174 1.007 |
| 15 | AFRENGINE-000 | 619 - | 619 - | 389 0.0029 | 375 0.0019 | 374 0.0017 | 378 0.0015 | 373 0.0014 | 379 1.018 |
| 16 | AFRENGINE-001 | 619 - | 619 - | 262 0.0013 | 309 0.0013 | 325 0.0013 | 336 0.0013 | 348 0.0013 | 297 1.012 |
| 17 | AFRENGINE-002 | 619 - | 619 - | 261 0.0013 | 308 0.0013 | 322 0.0013 | 337 0.0013 | 349 0.0013 | 296 1.012 |
| 18 | AFRENGINE-003 | 619 - | 619 - | 366 0.0023 | 404 0.0022 | 416 0.0022 | 427 0.0022 | 439 0.0022 | 396 1.020 |
| 19 | AIGROUP-000 | 619 - | 619 - | 215 0.0011 | 186 0.0008 | 177 0.0008 | 181 0.0008 | 188 0.0007 | 191 1.008 |
| 20 | AIZE-001 | 619 - | 619 - | 437 0.0053 | 437 0.0033 | 442 0.0030 | 443 0.0027 | 448 0.0023 | 437 1.032 |
| 21 | AIZE-002 | 619 - | 619 - | 397 0.0031 | 381 0.0019 | 381 0.0017 | 383 0.0016 | 386 0.0014 | 385 1.019 |
| 22 | AIZE-006 | 619 - | 619 - | 407 0.0036 | 398 0.0021 | 389 0.0018 | 382 0.0016 | 368 0.0013 | 401 1.021 |
| 23 | AKUVOX-001 | 619 - | 619 - | 268 0.0014 | 281 0.0012 | 287 0.0011 | 294 0.0011 | 305 0.0011 | 273 1.011 |
| 24 | ALCHERA-000 | 619 - | 619 - | 516 0.0159 | 522 0.0122 | 527 0.0114 | 532 0.0108 | 538 0.0102 | 521 1.114 |
| 25 | ALCHERA-001 | 619 - | 619 - | 614 0.9869 | 614 0.9782 | 614 0.9735 | 614 0.9679 | 613 0.9589 | 614 9.811 |
| 26 | ALCHERA-002 | 619 - | 619 - | 574 0.0946 | 572 0.0552 | 570 0.0440 | 568 0.0351 | 564 0.0251 | 572 1.542 |
| 27 | ALCHERA-003 | 619 - | 619 - | 488 0.0102 | 472 0.0051 | 471 0.0043 | 469 0.0036 | 468 0.0030 | 474 1.053 |
| 28 | ALCHERA-004 | 619 - | 619 - | 493 0.0106 | 464 0.0045 | 455 0.0035 | 447 0.0028 | 438 0.0022 | 466 1.048 |
| 29 | ALCHERA-005 | 619 - | 619 - | 460 0.0070 | 434 0.0030 | 425 0.0023 | 412 0.0019 | 391 0.0015 | 436 1.032 |
| 30 | ALCHERA-006 | 619 - | 619 - | 160 0.0009 | 168 0.0008 | 179 0.0008 | 184 0.0008 | 183 0.0007 | 167 1.007 |
| 31 | ALCHERA-007 | 619 - | 619 - | 154 0.0009 | 157 0.0008 | 161 0.0007 | 164 0.0007 | 185 0.0007 | 156 1.007 |
| 32 | ALCHERA-008 | 619 - | 619 - | 182 0.0010 | 209 0.0009 | 218 0.0009 | 228 0.0009 | 239 0.0009 | 202 1.008 |
| 33 | ALGORITHMSINC-000 | 619 - | 619 - | 293 0.0016 | 319 0.0013 | 332 0.0013 | 343 0.0013 | 363 0.0013 | 315 1.012 |
| 34 | ALLGOVISION-000 | 619 - | 619 - | 496 0.0111 | 499 0.0080 | 505 0.0074 | 506 0.0069 | 511 0.0064 | 499 1.076 |
| 35 | ALLGOVISION-001 | 619 - | 619 - | 474 0.0086 | 462 0.0045 | 458 0.0036 | 453 0.0030 | 449 0.0023 | 461 1.045 |
| 36 | ALLGOVISION-002 | 619 - | 619 - | 410 0.0037 | 405 0.0022 | 391 0.0018 | 384 0.0016 | 362 0.0013 | 404 1.022 |
| 37 | ALLGOVISION-003 | 619 - | 619 - | 321 0.0018 | 315 0.0013 | 316 0.0012 | 331 0.0012 | 335 0.0011 | 319 1.012 |
| 38 | ALLGOVISION-004 | 619 - | 619 - | 485 0.0101 | 461 0.0044 | 445 0.0033 | 435 0.0025 | 425 0.0019 | 465 1.046 |
| 39 | ALLGOVISION-005 | 619 - | 619 - | 316 0.0017 | 322 0.0013 | 317 0.0012 | 319 0.0012 | 326 0.0011 | 317 1.012 |
| 40 | ALTTEKGLOBAL-000 | 619 - | 619 - | 596 0.2740 | 599 0.2740 | 602 0.2740 | 603 0.2740 | 605 0.2740 | 599 3.466 |
| 41 | ANKE-000 | 619 - | 619 - | 507 0.0129 | 489 0.0070 | 483 0.0057 | 484 0.0047 | 481 0.0037 | 494 1.070 |
| 42 | ANKE-001 | 619 - | 619 - | 508 0.0129 | 490 0.0070 | 485 0.0058 | 485 0.0047 | 482 0.0037 | 495 1.070 |
| 43 | ANKE-002 | 619 - | 619 - | 374 0.0024 | 363 0.0016 | 358 0.0015 | 361 0.0014 | 356 0.0013 | 364 1.016 |
| 44 | ANTHEUS-000 | 619 - | 619 - | 459 0.0069 | 480 0.0060 | 488 0.0059 | 495 0.0058 | 505 0.0057 | 475 1.055 |
| 45 | ARATEK-001 | 619 - | 619 - | 131 0.0008 | 120 0.0007 | 122 0.0006 | 123 0.0006 | 125 0.0006 | 123 1.006 |
| 46 | ARMATURA-000 | 619 - | 619 - | 290 0.0015 | 343 0.0015 | 356 0.0015 | 370 0.0015 | 387 0.0015 | 339 1.013 |
| 47 | AWARE-003 | 619 - | 619 - | 543 0.0303 | 537 0.0158 | 532 0.0124 | 526 0.0097 | 521 0.0072 | 538 1.160 |
| 48 | AWARE-004 | 619 - | 619 - | 566 0.0676 | 561 0.0345 | 558 0.0271 | 555 0.0205 | 551 0.0142 | 561 1.351 |
| 49 | AWARE-005 | 619 - | 619 - | 544 0.0308 | 538 0.0164 | 534 0.0130 | 529 0.0104 | 526 0.0079 | 540 1.164 |
| 50 | AWARE-006 | 619 - | 619 - | 568 0.0694 | 563 0.0366 | 559 0.0284 | 557 0.0220 | 555 0.0155 | 563 1.368 |
| 51 | AWARE-007 | 619 - | 619 - | 352 0.0021 | 301 0.0012 | 278 0.0011 | 253 0.0010 | 227 0.0009 | 313 1.012 |
| 52 | AWIROS-000 | 619 - | 619 - | 463 0.0073 | 467 0.0048 | 472 0.0044 | 479 0.0041 | 485 0.0039 | 463 1.046 |
| 53 | AYONIX-000 | 619 - | 619 - | 603 0.4503 | 604 0.3538 | 604 0.3174 | 604 0.2831 | 603 0.2379 | 604 4.286 |
| 54 | AYONIX-001 | 619 - | 619 - | 597 0.3412 | 596 0.2335 | 594 0.1974 | 591 0.1649 | 591 0.1271 | 596 3.223 |
| 55 | AYONIX-002 | 619 - | 619 - | 598 0.3412 | 597 0.2336 | 595 0.1975 | 590 0.1649 | 592 0.1271 | 597 3.224 |
| 56 | BIOIDTECHSWISS-001 | 619 - | 619 - | 134 0.0008 | 141 0.0007 | 148 0.0007 | 144 0.0007 | 143 0.0006 | 137 1.006 |
| 57 | BIOMETRIK-001 | 619 - | 619 - | 423 0.0044 | 405 0.0023 | 401 0.0019 | 391 0.0016 | 379 0.0014 | 411 1.023 |
| 58 | CAMVI-003 | 619 - | 619 - | 560 0.0517 | 571 0.0514 | 572 0.0514 | 575 0.0514 | 576 0.0514 | 569 1.463 |
| 59 | CAMVI-004 | 619 - | 619 - | 558 0.0466 | 568 0.0463 | 571 0.0463 | 572 0.0462 | 575 0.0462 | 565 1.417 |
| 60 | CAMVI-005 | 619 - | 619 - | 565 0.0650 | 573 0.0646 | 577 0.0646 | 578 0.0645 | 581 0.0645 | 573 1.582 |
| 61 | CANON-001 | 619 - | 619 - | 106 0.0007 | 112 0.0006 | 105 0.0006 | 104 0.0006 | 98 0.0006 | 112 1.006 |
| 62 | CANON-002 | 619 - | 619 - | 138 0.0008 | 106 0.0006 | 101 0.0006 | 93 0.0006 | 94 0.0006 | 107 1.006 |
| 63 | CANON-003 | 619 - | 619 - | 53 0.0006 | 46 0.0005 | 33 0.0005 | 34 0.0005 | 42 0.0005 | 47 1.005 |
| 64 | CIB-000 | 619 - | 619 - | 233 0.0012 | 227 0.0010 | 215 0.0009 | 210 0.0008 | 209 0.0008 | 229 1.009 |
| 65 | CLEARVIEWAI-000 | 619 - | 619 - | 107 0.0007 | 110 0.0006 | 112 0.0006 | 100 0.0006 | 95 0.0006 | 110 1.006 |
| 66 | CLEARVIEWAI-001 | 619 - | 619 - | 110 0.0007 | 128 0.0007 | 137 0.0007 | 146 0.0007 | 157 0.0007 | 131 1.006 |
| 67 | CLEARVIEWAI-002 | 619 - | 619 - | 119 0.0008 | 129 0.0007 | 138 0.0007 | 143 0.0007 | 149 0.0007 | 130 1.006 |
| 68 | CLEARVIEWAI-003 | 619 - | 619 - | 113 0.0008 | 131 0.0007 | 132 0.0007 | 138 0.0007 | 156 0.0007 | 129 1.006 |
| 69 | CLOUDWALK-HR-000 | 619 - | 619 - | 223 0.0011 | 267 0.0011 | 279 0.0011 | 290 0.0011 | 299 0.0010 | 260 1.010 |
| 70 | CLOUDWALK-MT-000 | 619 - | 619 - | 273 0.0014 | 334 0.0014 | 340 0.0014 | 355 0.0014 | 376 0.0014 | 325 1.013 |
| 71 | CLOUDWALK-MT-001 | 619 - | 619 - | 271 0.0014 | 336 0.0014 | 341 0.0014 | 354 0.0014 | 377 0.0014 | 323 1.013 |
| 72 | CLOUDWALK-MT-002 | 619 - | 619 - | 272 0.0014 | 333 0.0014 | 345 0.0014 | 359 0.0014 | 374 0.0014 | 324 1.013 |

*Table 51:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 73 | COGENT-000 | 619- | 619- | 486 0.0101 | 512 0.0093 | 515 0.0092 | 448 0.0028 | 435 0.0021 | 506 1.085 |
| 74 | COGENT-001 | 619- | 619- | 487 0.0101 | 511 0.0093 | 516 0.0092 | 449 0.0028 | 434 0.0021 | 505 1.085 |
| 75 | COGENT-002 | 619- | 619- | 401 0.0033 | 380 0.0019 | 375 0.0017 | 369 0.0015 | 345 0.0013 | 384 1.019 |
| 76 | COGENT-003 | 619- | 619- | 403 0.0035 | 396 0.0021 | 385 0.0018 | 386 0.0016 | 372 0.0014 | 400 1.020 |
| 77 | COGENT-004 | 619- | 619- | 303 0.0016 | 296 0.0012 | 291 0.0011 | 301 0.0011 | 297 0.0010 | 293 1.011 |
| 78 | COGENT-005 | 619- | 619- | 258 0.0013 | 261 0.0011 | 259 0.0010 | 271 0.0010 | 280 0.0010 | 261 1.010 |
| 79 | COGENT-006 | 619- | 619- | 158 0.0009 | 170 0.0008 | 160 0.0007 | 174 0.0007 | 182 0.0007 | 162 1.007 |
| 80 | COGENT-007 | 619- | 619- | 109 0.0007 | 115 0.0006 | 119 0.0006 | 126 0.0006 | 127 0.0006 | 119 1.006 |
| 81 | COGNITEC-000 | 619- | 619- | 533 0.0249 | 528 0.0133 | 524 0.0104 | 518 0.0082 | 509 0.0062 | 529 1.133 |
| 82 | COGNITEC-001 | 619- | 619- | 498 0.0114 | 478 0.0058 | 477 0.0047 | 477 0.0039 | 470 0.0031 | 479 1.059 |
| 83 | COGNITEC-002 | 619- | 619- | 439 0.0053 | 439 0.0033 | 436 0.0028 | 436 0.0026 | 447 0.0023 | 438 1.032 |
| 84 | COGNITEC-003 | 619- | 619- | 444 0.0059 | 445 0.0037 | 446 0.0033 | 455 0.0030 | 459 0.0028 | 447 1.036 |
| 85 | COGNITEC-004 | 619- | 619- | 388 0.0028 | 365 0.0017 | 347 0.0014 | 322 0.0012 | 303 0.0010 | 369 1.016 |
| 86 | COGNITEC-005 | 619- | 619- | 254 0.0013 | 214 0.0009 | 208 0.0009 | 199 0.0008 | 200 0.0008 | 220 1.009 |
| 87 | COGNITEC-006 | 619- | 619- | 238 0.0012 | 210 0.0009 | 203 0.0008 | 198 0.0008 | 195 0.0008 | 209 1.008 |
| 88 | COGNITEC-007 | 619- | 619- | 197 0.0010 | 161 0.0008 | 158 0.0007 | 155 0.0007 | 153 0.0007 | 169 1.007 |
| 89 | COGNITEC-008 | 619- | 619- | 183 0.0010 | 142 0.0007 | 136 0.0007 | 139 0.0007 | 140 0.0006 | 147 1.007 |
| 90 | COGNITEC-009 | 619- | 619- | 229 0.0012 | 208 0.0009 | 210 0.0009 | 214 0.0008 | 213 0.0008 | 210 1.008 |
| 91 | COMPAL-000 | 619- | 619- | 270 0.0014 | 295 0.0012 | 310 0.0012 | 318 0.0012 | 329 0.0011 | 288 1.011 |
| 92 | COMPAL-001 | 619- | 619- | 456 0.0067 | 487 0.0066 | 495 0.0066 | 502 0.0065 | 514 0.0065 | 482 1.059 |
| 93 | COMPAL-002 | 619- | 619- | 455 0.0067 | 486 0.0066 | 494 0.0065 | 501 0.0065 | 515 0.0065 | 481 1.059 |
| 94 | CORSIGHT-000 | 619- | 619- | 114 0.0008 | 114 0.0006 | 117 0.0006 | 124 0.0006 | 139 0.0006 | 114 1.006 |
| 95 | CORSIGHT-001 | 619- | 619- | 118 0.0008 | 127 0.0007 | 134 0.0007 | 131 0.0006 | 135 0.0006 | 127 1.006 |
| 96 | CUBOX-000 | 619- | 619- | 204 0.0011 | 248 0.0010 | 262 0.0010 | 273 0.0010 | 284 0.0010 | 240 1.009 |
| 97 | CUBOX-001 | 619- | 619- | 166 0.0009 | 188 0.0008 | 195 0.0008 | 205 0.0008 | 214 0.0008 | 183 1.008 |
| 98 | CUBOX-002 | 619- | 619- | 226 0.0011 | 260 0.0011 | 277 0.0011 | 289 0.0011 | 307 0.0011 | 256 1.010 |
| 99 | CUBOX-003 | 619- | 619- | 242 0.0012 | 283 0.0012 | 302 0.0012 | 315 0.0011 | 325 0.0011 | 275 1.011 |
| 100 | CYBERLINK-000 | 619- | 619- | 405 0.0036 | 416 0.0024 | 421 0.0022 | 424 0.0021 | 419 0.0019 | 415 1.023 |
| 101 | CYBERLINK-001 | 619- | 619- | 399 0.0032 | 386 0.0020 | 378 0.0017 | 368 0.0015 | 366 0.0013 | 391 1.019 |
| 102 | CYBERLINK-002 | 619- | 619- | 362 0.0023 | 382 0.0019 | 393 0.0019 | 399 0.0018 | 407 0.0017 | 378 1.018 |
| 103 | CYBERLINK-003 | 619- | 619- | 240 0.0012 | 224 0.0010 | 223 0.0009 | 212 0.0008 | 204 0.0008 | 232 1.009 |
| 104 | CYBERLINK-004 | 619- | 619- | 256 0.0013 | 277 0.0011 | 289 0.0011 | 292 0.0011 | 304 0.0011 | 270 1.011 |
| 105 | CYBERLINK-005 | 619- | 619- | 278 0.0015 | 297 0.0012 | 304 0.0012 | 306 0.0011 | 313 0.0011 | 290 1.011 |
| 106 | CYBERLINK-006 | 619- | 619- | 130 0.0008 | 149 0.0007 | 146 0.0007 | 149 0.0007 | 162 0.0007 | 142 1.007 |
| 107 | DAHUA-000 | 619- | 619- | 479 0.0091 | 483 0.0064 | 487 0.0059 | 492 0.0056 | 499 0.0052 | 485 1.061 |
| 108 | DAHUA-001 | 619- | 619- | 450 0.0064 | 448 0.0039 | 453 0.0034 | 458 0.0031 | 460 0.0028 | 451 1.038 |
| 109 | DAHUA-002 | 619- | 619- | 281 0.0015 | 274 0.0011 | 283 0.0011 | 281 0.0010 | 279 0.0010 | 280 1.011 |
| 110 | DAHUA-003 | 619- | 619- | 144 0.0008 | 97 0.0006 | 89 0.0006 | 81 0.0005 | 77 0.0005 | 105 1.006 |
| 111 | DAHUA-004 | 619- | 619- | 102 0.0007 | 94 0.0006 | 94 0.0006 | 92 0.0006 | 91 0.0005 | 98 1.006 |
| 112 | DAON-000 | 619- | 619- | 409 0.0037 | 440 0.0034 | 449 0.0033 | 462 0.0033 | 475 0.0033 | 435 1.031 |
| 113 | DAON-001 | 619- | 619- | 534 0.0252 | 550 0.0249 | 555 0.0249 | 559 0.0248 | 563 0.0248 | 550 1.225 |
| 114 | DAON-002 | 619- | 619- | 170 0.0009 | 140 0.0007 | 133 0.0007 | 134 0.0006 | 137 0.0006 | 144 1.007 |
| 115 | DAON-003 | 619- | 619- | 99 0.0007 | 86 0.0006 | 84 0.0006 | 79 0.0005 | 82 0.0005 | 88 1.005 |
| 116 | DATECH-000 | 619- | 619- | 184 0.0010 | 159 0.0008 | 164 0.0007 | 172 0.0007 | 174 0.0007 | 166 1.007 |
| 117 | DATECH-001 | 619- | 619- | 117 0.0008 | 117 0.0006 | 114 0.0006 | 110 0.0006 | 112 0.0006 | 115 1.006 |
| 118 | DECATUR-000 | 619- | 619- | 318 0.0017 | 299 0.0012 | 296 0.0011 | 286 0.0011 | 268 0.0009 | 304 1.012 |
| 119 | DEEPGLINT-001 | 619- | 619- | 217 0.0011 | 245 0.0010 | 246 0.0010 | 252 0.0010 | 261 0.0009 | 239 1.009 |
| 120 | DEEPSEA-001 | 619- | 619- | 414 0.0040 | 376 0.0019 | 354 0.0015 | 332 0.0012 | 291 0.0010 | 392 1.019 |
| 121 | DERMALOG-003 | 619- | 619- | 580 0.1256 | 577 0.0741 | 575 0.0600 | 574 0.0477 | 572 0.0345 | 577 1.729 |
| 122 | DERMALOG-004 | 619- | 619- | 579 0.1248 | 576 0.0737 | 574 0.0595 | 573 0.0473 | 571 0.0341 | 576 1.725 |
| 123 | DERMALOG-005 | 619- | 619- | 512 0.0147 | 525 0.0127 | 531 0.0123 | 537 0.0121 | 546 0.0120 | 522 1.116 |
| 124 | DERMALOG-006 | 619- | 619- | 469 0.0079 | 488 0.0067 | 492 0.0064 | 498 0.0063 | 508 0.0062 | 487 1.062 |
| 125 | DERMALOG-007 | 619- | 619- | 476 0.0089 | 482 0.0063 | 482 0.0057 | 490 0.0054 | 496 0.0050 | 480 1.059 |
| 126 | DERMALOG-008 | 619- | 619- | 325 0.0025 | 362 0.0016 | 352 0.0014 | 341 0.0013 | 336 0.0012 | 363 1.016 |
| 127 | DERMALOG-009 | 619- | 619- | 323 0.0024 | 392 0.0020 | 403 0.0020 | 410 0.0019 | 421 0.0019 | 389 1.019 |
| 128 | DERMALOG-010 | 619- | 619- | 328 0.0018 | 370 0.0017 | 377 0.0017 | 394 0.0017 | 405 0.0017 | 362 1.016 |
| 129 | DERMALOG-011 | 619- | 619- | 243 0.0012 | 199 0.0009 | 193 0.0008 | 186 0.0008 | 176 0.0007 | 199 1.008 |
| 130 | DERMALOG-012 | 619- | 619- | 311 0.0017 | 302 0.0012 | 312 0.0012 | 313 0.0011 | 324 0.0011 | 302 1.012 |
| 131 | DERMALOG-013 | 619- | 619- | 294 0.0016 | 279 0.0012 | 286 0.0011 | 283 0.0010 | 290 0.0010 | 282 1.011 |
| 132 | DERMALOG-014 | 619- | 619- | 395 0.0030 | 426 0.0027 | 432 0.0026 | 442 0.0026 | 454 0.0026 | 421 1.025 |
| 133 | DERMALOG-015 | 619- | 619- | 241 0.0012 | 257 0.0011 | 272 0.0011 | 280 0.0010 | 295 0.0010 | 258 1.010 |
| 134 | DERMALOG-016 | 619- | 619- | 208 0.0011 | 250 0.0010 | 260 0.0010 | 272 0.0010 | 292 0.0010 | 242 1.009 |
| 135 | DERMALOG-017 | 619- | 619- | 214 0.0011 | 249 0.0010 | 261 0.0010 | 274 0.0010 | 290 0.0010 | 243 1.009 |
| 136 | DIGIDATA-000 | 619- | 619- | 607 0.5896 | 608 0.5890 | 608 0.5889 | 609 0.5889 | 610 0.5889 | 608 6.302 |
| 137 | DILUSENSE-000 | 619- | 619- | 330 0.0018 | 291 0.0012 | 271 0.0011 | 264 0.0010 | 241 0.0009 | 294 1.011 |
| 138 | DILUSENSE-001 | 619- | 619- | 218 0.0011 | 203 0.0009 | 200 0.0008 | 204 0.0008 | 202 0.0008 | 203 1.008 |
| 139 | EINETWORKSINDIA-000 | 619- | 619- | 276 0.0014 | 275 0.0011 | 288 0.0011 | 298 0.0011 | 308 0.0011 | 277 1.011 |
| 140 | EINETWORKSINDIA-001 | 619- | 619- | 265 0.0014 | 259 0.0011 | 268 0.0010 | 269 0.0010 | 274 0.0010 | 265 1.010 |
| 141 | EINETWORKSINDIA-002 | 619- | 619- | 126 0.0008 | 132 0.0007 | 125 0.0006 | 136 0.0006 | 146 0.0006 | 126 1.006 |
| 142 | EYEDEA-003 | 619- | 619- | 570 0.0797 | 566 0.0447 | 565 0.0359 | 561 0.0286 | 560 0.0208 | 568 1.445 |
| 143 | F8-001 | 619- | 619- | 502 0.0116 | 514 0.0101 | 519 0.0099 | 525 0.0097 | 532 0.0095 | 513 1.093 |
| 144 | FACEHAWK-001 | 619- | 619- | 542 0.0300 | 551 0.0257 | 556 0.0249 | 558 0.0244 | 562 0.0239 | 551 1.236 |

*Table 52:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 145 | FACEPHI-000 | 619 - | 619 - | 284 0.0015 | 337 0.0014 | 342 0.0014 | 352 0.0014 | 369 0.0014 | 330 1.013 |
| 146 | FASTLAHORE-000 | 619 - | 619 - | 537 0.0261 | 543 0.0178 | 542 0.0155 | 542 0.0138 | 543 0.0115 | 542 1.169 |
| 147 | FCI-000 | 619 - | 619 - | 285 0.0015 | 324 0.0013 | 330 0.0013 | 340 0.0013 | 351 0.0013 | 318 1.012 |
| 148 | FINCORE-000 | 619 - | 619 - | 492 0.0105 | 469 0.0049 | 461 0.0038 | 456 0.0030 | 445 0.0023 | 470 1.051 |
| 149 | FIRSTCREDITKZ-001 | 619 - | 619 - | 149 0.0008 | 184 0.0008 | 190 0.0008 | 201 0.0008 | 211 0.0008 | 177 1.007 |
| 150 | FISSIONSFR-001 | 619 - | 619 - | 575 0.0980 | 581 0.0976 | 582 0.0975 | 584 0.0975 | 586 0.0975 | 580 1.879 |
| 151 | FPT-000 | 619 - | 619 - | 292 0.0016 | 332 0.0014 | 339 0.0014 | 347 0.0013 | 359 0.0013 | 329 1.013 |
| 152 | FUJITSULAB-000 | 619 - | 619 - | 332 0.0018 | 311 0.0013 | 306 0.0012 | 297 0.0011 | 283 0.0010 | 311 1.012 |
| 153 | FUJITSULAB-001 | 619 - | 619 - | 283 0.0015 | 288 0.0012 | 294 0.0011 | 284 0.0011 | 293 0.0010 | 285 1.011 |
| 154 | GLORY-000 | 619 - | 619 - | 584 0.1778 | 586 0.1388 | 586 0.1263 | 586 0.1151 | 587 0.1004 | 585 2.295 |
| 155 | GLORY-001 | 619 - | 619 - | 581 0.1265 | 580 0.0964 | 580 0.0866 | 581 0.0775 | 582 0.0670 | 581 1.900 |
| 156 | GORILLA-001 | 619 - | 619 - | 562 0.0600 | 556 0.0300 | 553 0.0227 | 551 0.0171 | 542 0.0114 | 556 1.306 |
| 157 | GORILLA-002 | 619 - | 619 - | 526 0.0194 | 507 0.0090 | 497 0.0068 | 488 0.0052 | 487 0.0039 | 515 1.094 |
| 158 | GORILLA-003 | 619 - | 619 - | 547 0.0359 | 531 0.0144 | 523 0.0103 | 514 0.0076 | 498 0.0052 | 535 1.156 |
| 159 | GORILLA-004 | 619 - | 619 - | 445 0.0059 | 432 0.0028 | 423 0.0023 | 411 0.0019 | 385 0.0014 | 434 1.030 |
| 160 | GORILLA-005 | 619 - | 619 - | 387 0.0028 | 345 0.0015 | 334 0.0013 | 302 0.0011 | 260 0.0009 | 359 1.015 |
| 161 | GORILLA-006 | 619 - | 619 - | 269 0.0014 | 202 0.0009 | 199 0.0008 | 196 0.0008 | 191 0.0007 | 227 1.009 |
| 162 | GORILLA-007 | 619 - | 619 - | 260 0.0013 | 194 0.0009 | 188 0.0008 | 169 0.0007 | 170 0.0007 | 207 1.008 |
| 163 | GORILLA-008 | 619 - | 619 - | 225 0.0011 | 178 0.0008 | 173 0.0008 | 165 0.0007 | 171 0.0007 | 186 1.008 |
| 164 | GRIAULE-000 | 619 - | 619 - | 355 0.0021 | 357 0.0016 | 361 0.0015 | 362 0.0014 | 355 0.0013 | 355 1.015 |
| 165 | GRIAULE-001 | 619 - | 619 - | 143 0.0008 | 145 0.0007 | 150 0.0007 | 152 0.0007 | 164 0.0007 | 145 1.007 |
| 166 | GTELOTS-000 | 619 - | 619 - | 588 0.1940 | 592 0.1938 | 593 0.1938 | 596 0.1938 | 600 0.1938 | 592 2.745 |
| 167 | GTELOTS-001 | 619 - | 619 - | 514 0.0155 | 532 0.0151 | 540 0.0151 | 545 0.0150 | 553 0.0150 | 531 1.136 |
| 168 | HIK-003 | 619 - | 619 - | 499 0.0115 | 477 0.0058 | 475 0.0046 | 473 0.0038 | 465 0.0029 | 483 1.059 |
| 169 | HIK-004 | 619 - | 619 - | 497 0.0111 | 476 0.0057 | 474 0.0045 | 466 0.0035 | 461 0.0028 | 477 1.058 |
| 170 | HIK-005 | 619 - | 619 - | 421 0.0044 | 412 0.0024 | 397 0.0019 | 387 0.0016 | 379 0.0014 | 414 1.023 |
| 171 | HIK-006 | 619 - | 619 - | 422 0.0044 | 411 0.0024 | 398 0.0019 | 388 0.0016 | 378 0.0014 | 413 1.023 |
| 172 | HISIGN-001 | 619 - | 619 - | 152 0.0009 | 129 0.0007 | 128 0.0006 | 122 0.0006 | 130 0.0006 | 134 1.006 |
| 173 | HYPERVERGE-001 | 619 - | 619 - | 200 0.0010 | 222 0.0009 | 238 0.0009 | 245 0.0009 | 266 0.0009 | 219 1.009 |
| 174 | HYPERVERGE-002 | 619 - | 619 - | 195 0.0010 | 220 0.0009 | 237 0.0009 | 244 0.0009 | 252 0.0009 | 218 1.009 |
| 175 | HYPERVERGE-003 | 619 - | 619 - | 212 0.0011 | 232 0.0010 | 236 0.0009 | 246 0.0009 | 263 0.0009 | 224 1.009 |
| 176 | HYUNDAI-ROBOTICSLAB-000 | 619 - | 619 - | 367 0.0023 | 391 0.0020 | 405 0.0020 | 413 0.0019 | 423 0.0019 | 383 1.019 |
| 177 | HYUNDAI-ROBOTICSLAB-001 | 619 - | 619 - | 267 0.0014 | 266 0.0011 | 258 0.0010 | 261 0.0010 | 242 0.0009 | 267 1.010 |
| 178 | HYUNDAI-ROBOTICSLAB-002 | 619 - | 619 - | 100 0.0007 | 93 0.0006 | 99 0.0006 | 101 0.0006 | 97 0.0006 | 99 1.006 |
| 179 | HYUNDAI-ROBOTICSLAB-003 | 619 - | 619 - | 85 0.0007 | 71 0.0005 | 74 0.0005 | 74 0.0005 | 70 0.0005 | 77 1.005 |
| 180 | HZAILU-000 | 619 - | 619 - | 329 0.0018 | 304 0.0013 | 307 0.0012 | 303 0.0011 | 316 0.0011 | 308 1.012 |
| 181 | HZAILU-001 | 619 - | 619 - | 299 0.0016 | 326 0.0014 | 327 0.0013 | 333 0.0012 | 333 0.0011 | 322 1.012 |
| 182 | HZAILU-002 | 619 - | 619 - | 302 0.0016 | 323 0.0013 | 323 0.0013 | 325 0.0012 | 334 0.0011 | 321 1.012 |
| 183 | HZAILU-003 | 619 - | 619 - | 251 0.0013 | 256 0.0011 | 263 0.0010 | 270 0.0010 | 251 0.0009 | 257 1.010 |
| 184 | HZAILU-004 | 619 - | 619 - | 175 0.0009 | 201 0.0009 | 212 0.0009 | 222 0.0009 | 230 0.0009 | 195 1.008 |
| 185 | HZAILU-005 | 619 - | 619 - | 173 0.0009 | 197 0.0009 | 209 0.0009 | 217 0.0009 | 228 0.0009 | 194 1.008 |
| 186 | HZAILU-006 | 619 - | 619 - | 168 0.0009 | 196 0.0009 | 207 0.0009 | 218 0.0009 | 226 0.0009 | 193 1.008 |
| 187 | I2V-001 | 619 - | 619 - | 378 0.0026 | 417 0.0024 | 431 0.0024 | 431 0.0024 | 451 0.0024 | 406 1.022 |
| 188 | ICEHOUSE-000 | 619 - | 619 - | 424 0.0044 | 424 0.0026 | 419 0.0022 | 418 0.0020 | 409 0.0017 | 427 1.026 |
| 189 | ICEHOUSE-001 | 619 - | 619 - | 192 0.0010 | 185 0.0008 | 181 0.0008 | 187 0.0008 | 189 0.0007 | 185 1.008 |
| 190 | ICM-000 | 619 - | 619 - | 433 0.0049 | 459 0.0043 | 469 0.0042 | 480 0.0042 | 489 0.0041 | 454 1.040 |
| 191 | ICM-001 | 619 - | 619 - | 384 0.0028 | 402 0.0022 | 409 0.0021 | 417 0.0020 | 424 0.0019 | 397 1.020 |
| 192 | IDEMIA-003 | 619 - | 619 - | 454 0.0066 | 456 0.0041 | 457 0.0035 | 457 0.0031 | 452 0.0024 | 455 1.040 |
| 193 | IDEMIA-004 | 619 - | 619 - | 449 0.0063 | 443 0.0035 | 437 0.0029 | 432 0.0024 | 411 0.0018 | 442 1.035 |
| 194 | IDEMIA-005 | 619 - | 619 - | 468 0.0077 | 454 0.0041 | 447 0.0033 | 451 0.0029 | 458 0.0027 | 457 1.042 |
| 195 | IDEMIA-006 | 619 - | 619 - | 482 0.0092 | 468 0.0048 | 464 0.0039 | 467 0.0035 | 477 0.0034 | 467 1.049 |
| 196 | IDEMIA-007 | 619 - | 619 - | 359 0.0022 | 300 0.0012 | 276 0.0011 | 240 0.0009 | 216 0.0008 | 314 1.012 |
| 197 | IDEMIA-008 | 619 - | 619 - | 96 0.0007 | 89 0.0006 | 91 0.0006 | 97 0.0006 | 92 0.0005 | 90 1.005 |
| 198 | IDEMIA-009 | 619 - | 619 - | 69 0.0006 | 62 0.0005 | 66 0.0005 | 67 0.0005 | 64 0.0005 | 65 1.005 |
| 199 | IDEMIA-010 | 619 - | 619 - | 41 0.0005 | 53 0.0005 | 55 0.0005 | 62 0.0005 | 66 0.0005 | 48 1.005 |
| 200 | IDEMIA-011 | 619 - | 619 - | 91 0.0007 | 124 0.0007 | 139 0.0007 | 142 0.0007 | 150 0.0007 | 121 1.006 |
| 201 | IDEMIA-012 | 619 - | 619 - | 97 0.0007 | 123 0.0007 | 129 0.0007 | 135 0.0006 | 144 0.0006 | 120 1.006 |
| 202 | IDEMIA-013 | 619 - | 619 - | 2 0.0004 | 5 0.0004 | 5 0.0004 | 5 0.0004 | 6 0.0004 | 4 1.004 |
| 203 | IDENTY-000 | 619 - | 619 - | 619 1.0000 | 616 1.0000 | 616 1.0000 | 616 1.0000 | 617 1.0000 | 616 10.000 |
| 204 | IDENTY-001 | 619 - | 619 - | 193 0.0010 | 230 0.0010 | 229 0.0009 | 234 0.0009 | 248 0.0009 | 214 1.009 |
| 205 | IDENTY-002 | 619 - | 619 - | 90 0.0007 | 104 0.0006 | 110 0.0006 | 106 0.0006 | 110 0.0006 | 100 1.006 |
| 206 | IMAGUS-002 | 619 - | 619 - | 593 0.2200 | 585 0.1339 | 584 0.1087 | 582 0.0868 | 580 0.0629 | 586 2.305 |
| 207 | IMAGUS-003 | 619 - | 619 - | 600 0.3557 | 598 0.2489 | 598 0.2129 | 592 0.1788 | 593 0.1394 | 598 3.360 |
| 208 | IMAGUS-005 | 619 - | 619 - | 298 0.0016 | 303 0.0012 | 292 0.0011 | 279 0.0010 | 278 0.0010 | 295 1.012 |
| 209 | IMAGUS-006 | 619 - | 619 - | 305 0.0016 | 312 0.0013 | 309 0.0012 | 304 0.0011 | 315 0.0011 | 306 1.012 |
| 210 | IMAGUS-007 | 619 - | 619 - | 307 0.0016 | 278 0.0011 | 266 0.0010 | 247 0.0009 | 245 0.0009 | 283 1.011 |
| 211 | IMAGUS-008 | 619 - | 619 - | 573 0.0857 | 574 0.0698 | 576 0.0643 | 576 0.0587 | 577 0.0514 | 574 1.646 |
| 212 | IMDS-SOFTWARE-000 | 619 - | 619 - | 124 0.0008 | 102 0.0006 | 103 0.0006 | 102 0.0006 | 102 0.0006 | 109 1.006 |
| 213 | IMDS-SOFTWARE-001 | 619 - | 619 - | 127 0.0008 | 118 0.0006 | 116 0.0006 | 118 0.0006 | 115 0.0006 | 117 1.006 |
| 214 | IMDS-SOFTWARE-002 | 619 - | 619 - | 125 0.0008 | 108 0.0006 | 102 0.0006 | 103 0.0006 | 101 0.0006 | 108 1.006 |
| 215 | IMPERIAL-000 | 619 - | 619 - | 351 0.0021 | 346 0.0015 | 355 0.0015 | 360 0.0014 | 360 0.0013 | 350 1.014 |
| 216 | INCODE-000 | 619 - | 619 - | 559 0.0486 | 552 0.0258 | 550 0.0201 | 549 0.0157 | 541 0.0114 | 552 1.259 |

*Table 53:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 217 | INCODE-001 | 619 - | 619 - | 517 0.0164 | 501 0.0082 | 493 0.0065 | 489 0.0053 | 490 0.0041 | 503 1.084 |
| 218 | INCODE-002 | 619 - | 619 - | 521 0.0175 | 504 0.0088 | 496 0.0068 | 491 0.0054 | 488 0.0041 | 509 1.089 |
| 219 | INCODE-003 | 619 - | 619 - | 506 0.0127 | 481 0.0062 | 478 0.0049 | 474 0.0038 | 466 0.0029 | 489 1.064 |
| 220 | INCODE-004 | 619 - | 619 - | 400 0.0032 | 388 0.0020 | 387 0.0018 | 390 0.0016 | 390 0.0015 | 393 1.019 |
| 221 | INCODE-005 | 619 - | 619 - | 255 0.0013 | 246 0.0010 | 255 0.0010 | 249 0.0009 | 243 0.0009 | 250 1.010 |
| 222 | INCODE-006 | 619 - | 619 - | 101 0.0007 | 137 0.0007 | 152 0.0007 | 158 0.0007 | 168 0.0007 | 133 1.006 |
| 223 | INCODE-007 | 619 - | 619 - | 37 0.0005 | 37 0.0005 | 29 0.0005 | 29 0.0005 | 31 0.0005 | 35 1.004 |
| 224 | INNOV8TIF-000 | 619 - | 619 - | 572 0.0856 | 570 0.0479 | 566 0.0376 | 563 0.0296 | 561 0.0210 | 571 1.475 |
| 225 | INNOVATRICS-002 | 619 - | 619 - | 556 0.0448 | 559 0.0340 | 561 0.0319 | 564 0.0306 | 562 0.0295 | 560 1.319 |
| 226 | INNOVATRICS-003 | 619 - | 619 - | 536 0.0260 | 524 0.0124 | 517 0.0092 | 508 0.0072 | 497 0.0051 | 526 1.127 |
| 227 | INNOVATRICS-004 | 619 - | 619 - | 504 0.0120 | 479 0.0059 | 476 0.0047 | 472 0.0037 | 463 0.0029 | 486 1.061 |
| 228 | INNOVATRICS-005 | 619 - | 619 - | 350 0.0021 | 340 0.0015 | 333 0.0013 | 327 0.0012 | 314 0.0011 | 342 1.014 |
| 229 | INNOVATRICS-007 | 619 - | 619 - | 259 0.0013 | 254 0.0010 | 247 0.0010 | 232 0.0009 | 235 0.0009 | 253 1.010 |
| 230 | INNOVATRICS-008 | 619 - | 619 - | 174 0.0009 | 169 0.0008 | 167 0.0008 | 175 0.0007 | 177 0.0007 | 165 1.007 |
| 231 | INNOVATRICS-009 | 619 - | 619 - | 189 0.0010 | 207 0.0009 | 211 0.0009 | 221 0.0009 | 234 0.0009 | 204 1.008 |
| 232 | INNOVATRICS-010 | 619 - | 619 - | 203 0.0010 | 221 0.0009 | 226 0.0009 | 233 0.0009 | 236 0.0009 | 216 1.009 |
| 233 | INNOVATRICS-011 | 619 - | 619 - | 112 0.0008 | 135 0.0007 | 131 0.0007 | 127 0.0006 | 133 0.0006 | 132 1.006 |
| 234 | INNOVATRICS-012 | 619 - | 619 - | 111 0.0007 | 121 0.0007 | 127 0.0006 | 130 0.0006 | 128 0.0006 | 122 1.006 |
| 235 | INNOVATRICS-013 | 619 - | 619 - | 64 0.0006 | 52 0.0005 | 38 0.0005 | 36 0.0005 | 45 0.0005 | 55 1.005 |
| 236 | INNOVATRICS-014 | 619 - | 619 - | 49 0.0006 | 56 0.0005 | 63 0.0005 | 51 0.0005 | 37 0.0005 | 54 1.005 |
| 237 | INNOVATRICS-015 | 619 - | 619 - | 48 0.0006 | 48 0.0005 | 41 0.0005 | 33 0.0005 | 29 0.0005 | 44 1.005 |
| 238 | INSPUR-000 | 619 - | 619 - | 73 0.0006 | 51 0.0005 | 59 0.0005 | 59 0.0005 | 54 0.0005 | 59 1.005 |
| 239 | INSPUR-001 | 619 - | 619 - | 66 0.0006 | 60 0.0005 | 48 0.0005 | 52 0.0005 | 57 0.0005 | 61 1.005 |
| 240 | INSPUR-002 | 619 - | 619 - | 57 0.0006 | 50 0.0005 | 30 0.0005 | 32 0.0005 | 32 0.0005 | 51 1.005 |
| 241 | INTELIGENSIA-000 | 619 - | 619 - | 306 0.0016 | 272 0.0011 | 250 0.0010 | 229 0.0009 | 212 0.0008 | 272 1.011 |
| 242 | INTELIGENSIA-001 | 619 - | 619 - | 92 0.0007 | 92 0.0006 | 86 0.0006 | 91 0.0006 | 90 0.0005 | 93 1.005 |
| 243 | INTELLIVISION-001 | 619 - | 619 - | 548 0.0361 | 549 0.0195 | 543 0.0156 | 539 0.0123 | 531 0.0091 | 546 1.196 |
| 244 | INTELLIVISION-002 | 619 - | 619 - | 490 0.0103 | 471 0.0051 | 467 0.0040 | 463 0.0034 | 456 0.0027 | 472 1.052 |
| 245 | INTELRESEARCH-001 | 619 - | 619 - | 80 0.0006 | 72 0.0005 | 70 0.0005 | 70 0.0005 | 65 0.0005 | 70 1.005 |
| 246 | INTEMA-000 | 619 - | 619 - | 123 0.0008 | 146 0.0007 | 151 0.0007 | 154 0.0007 | 166 0.0007 | 139 1.006 |
| 247 | INTEMA-001 | 619 - | 619 - | 12 0.0005 | 17 0.0005 | 19 0.0005 | 19 0.0004 | 16 0.0004 | 17 1.004 |
| 248 | INTOZI-001 | 619 - | 619 - | 239 0.0012 | 195 0.0009 | 178 0.0008 | 159 0.0007 | 165 0.0007 | 197 1.008 |
| 249 | INTSYSMSU-000 | 619 - | 619 - | 582 0.1454 | 584 0.1317 | 587 0.1269 | 587 0.1222 | 590 0.1160 | 584 2.200 |
| 250 | IONETWORKS-001 | 619 - | 619 - | 402 0.0034 | 435 0.0032 | 444 0.0032 | 459 0.0031 | 471 0.0031 | 433 1.029 |
| 251 | IREX-000 | 619 - | 619 - | 417 0.0040 | 451 0.0039 | 465 0.0039 | 476 0.0039 | 486 0.0039 | 446 1.036 |
| 252 | IRISID-001 | 619 - | 619 - | 209 0.0011 | 154 0.0007 | 149 0.0007 | 140 0.0007 | 120 0.0006 | 159 1.007 |
| 253 | IRISID-002 | 619 - | 619 - | 165 0.0009 | 122 0.0007 | 123 0.0006 | 116 0.0006 | 103 0.0006 | 135 1.006 |
| 254 | ISTL-001 | 619 - | 619 - | 354 0.0021 | 378 0.0019 | 390 0.0018 | 401 0.0018 | 410 0.0018 | 374 1.017 |
| 255 | ISYSTEMS-002 | 619 - | 619 - | 448 0.0061 | 459 0.0041 | 459 0.0037 | 464 0.0034 | 474 0.0032 | 453 1.039 |
| 256 | ISYSTEMS-003 | 619 - | 619 - | 434 0.0049 | 444 0.0037 | 452 0.0034 | 461 0.0032 | 469 0.0031 | 441 1.034 |
| 257 | KAKAO-000 | 619 - | 619 - | 220 0.0011 | 162 0.0008 | 153 0.0007 | 137 0.0007 | 134 0.0006 | 170 1.007 |
| 258 | KAKAO-001 | 619 - | 619 - | 199 0.0010 | 235 0.0010 | 244 0.0010 | 257 0.0010 | 265 0.0009 | 226 1.009 |
| 259 | KAKAO-002 | 619 - | 619 - | 188 0.0010 | 233 0.0010 | 241 0.0010 | 243 0.0009 | 257 0.0009 | 223 1.009 |
| 260 | KEDACOM-001 | 619 - | 619 - | 464 0.0074 | 492 0.0070 | 498 0.0069 | 505 0.0069 | 517 0.0068 | 488 1.064 |
| 261 | KEYLESS-000 | 619 - | 619 - | 98 0.0007 | 90 0.0006 | 83 0.0006 | 87 0.0005 | 80 0.0005 | 89 1.005 |
| 262 | KLOUDSPOT-001 | 619 - | 619 - | 344 0.0020 | 318 0.0013 | 303 0.0012 | 312 0.0011 | 309 0.0011 | 326 1.013 |
| 263 | KNERON-000 | 619 - | 619 - | 442 0.0056 | 473 0.0056 | 481 0.0056 | 493 0.0056 | 501 0.0056 | 469 1.050 |
| 264 | KNERON-001 | 619 - | 619 - | 541 0.0292 | 555 0.0292 | 560 0.0292 | 562 0.0292 | 566 0.0291 | 553 1.262 |
| 265 | KNOWUTECH-000 | 619 - | 619 - | 120 0.0008 | 76 0.0006 | 73 0.0005 | 75 0.0005 | 74 0.0005 | 85 1.005 |
| 266 | KOGNIZA-000 | 619 - | 619 - | 247 0.0013 | 273 0.0011 | 290 0.0011 | 300 0.0011 | 318 0.0011 | 268 1.010 |
| 267 | LEBENTECH-000 | 619 - | 619 - | 605 0.5395 | 607 0.5376 | 607 0.5371 | 607 0.5367 | 609 0.5362 | 607 5.841 |
| 268 | LIGHTVISION-000 | 619 - | 619 - | 382 0.0027 | 397 0.0021 | 406 0.0020 | 415 0.0020 | 427 0.0019 | 394 1.020 |
| 269 | LINE-000 | 619 - | 619 - | 333 0.0019 | 290 0.0012 | 265 0.0010 | 230 0.0009 | 207 0.0008 | 292 1.011 |
| 270 | LINE-001 | 619 - | 619 - | 122 0.0008 | 111 0.0006 | 113 0.0006 | 105 0.0006 | 111 0.0006 | 113 1.006 |
| 271 | LINECLOVA-002 | 619 - | 619 - | 178 0.0009 | 192 0.0009 | 197 0.0008 | 208 0.0008 | 215 0.0008 | 190 1.008 |
| 272 | LINECLOVA-003 | 619 - | 619 - | 71 0.0006 | 57 0.0005 | 44 0.0005 | 55 0.0005 | 25 0.0005 | 60 1.005 |
| 273 | LOOKMAN-003 | 619 - | 619 - | 473 0.0084 | 496 0.0075 | 504 0.0073 | 510 0.0072 | 520 0.0070 | 492 1.068 |
| 274 | LOOKMAN-004 | 619 - | 619 - | 475 0.0087 | 497 0.0075 | 503 0.0073 | 509 0.0072 | 519 0.0070 | 493 1.069 |
| 275 | LOOKMAN-005 | 619 - | 619 - | 467 0.0077 | 493 0.0071 | 500 0.0070 | 507 0.0069 | 518 0.0069 | 490 1.065 |
| 276 | LSSYTECH-000 | 619 - | 619 - | 59 0.0006 | 80 0.0006 | 79 0.0005 | 84 0.0005 | 93 0.0005 | 71 1.005 |
| 277 | LUXAND-001 | 619 - | 619 - | 386 0.0028 | 385 0.0020 | 388 0.0018 | 393 0.0017 | 401 0.0016 | 387 1.019 |
| 278 | MANTRA-000 | 619 - | 619 - | 264 0.0014 | 239 0.0010 | 234 0.0009 | 225 0.0009 | 223 0.0008 | 247 1.009 |
| 279 | MAVERICKRESEARCH-000 | 619 - | 619 - | 491 0.0104 | 503 0.0085 | 506 0.0081 | 516 0.0080 | 525 0.0078 | 500 1.079 |
| 280 | MAXVISION-000 | 619 - | 619 - | 346 0.0021 | 328 0.0014 | 315 0.0012 | 309 0.0011 | 285 0.0010 | 336 1.013 |
| 281 | MAXVISION-001 | 619 - | 619 - | 148 0.0008 | 167 0.0008 | 172 0.0008 | 182 0.0008 | 179 0.0007 | 160 1.007 |
| 282 | MAXVISION-002 | 619 - | 619 - | 153 0.0009 | 171 0.0008 | 183 0.0008 | 192 0.0008 | 201 0.0008 | 164 1.007 |
| 283 | MAXVISION-003 | 619 - | 619 - | 164 0.0009 | 166 0.0008 | 175 0.0008 | 183 0.0008 | 180 0.0007 | 163 1.007 |
| 284 | MAXVISION-004 | 619 - | 619 - | 147 0.0008 | 177 0.0008 | 186 0.0008 | 197 0.0008 | 205 0.0008 | 171 1.007 |
| 285 | MEGVII-001 | 619 - | 619 - | 501 0.0116 | 509 0.0092 | 509 0.0086 | 520 0.0083 | 528 0.0080 | 507 1.085 |
| 286 | MEGVII-002 | 619 - | 619 - | 503 0.0117 | 510 0.0092 | 511 0.0087 | 519 0.0083 | 527 0.0079 | 508 1.086 |
| 287 | MEGVII-003 | 619 - | 619 - | 162 0.0009 | 187 0.0008 | 196 0.0008 | 206 0.0008 | 210 0.0008 | 180 1.008 |
| 288 | MEGVII-004 | 619 - | 619 - | 145 0.0008 | 181 0.0008 | 191 0.0008 | 203 0.0008 | 206 0.0008 | 176 1.007 |

*Table 54:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 289 | MEGVII-005 | 619 - | 619 - | 146 0.0008 | 182 0.0008 | 192 0.0008 | 202 0.0008 | 208 0.0008 | 175 1.007 |
| 290 | METSAKUURCOMPANY-000 | 619 - | 619 - | 129 0.0008 | 103 0.0006 | 88 0.0006 | 82 0.0005 | 84 0.0005 | 101 1.006 |
| 291 | METSAKUURCOMPANY-001 | 619 - | 619 - | 194 0.0010 | 179 0.0008 | 184 0.0008 | 190 0.0008 | 192 0.0008 | 187 1.008 |
| 292 | MIAXIS-001 | 619 - | 619 - | 252 0.0013 | 251 0.0010 | 256 0.0010 | 268 0.0010 | 277 0.0010 | 254 1.010 |
| 293 | MIAXIS-002 | 619 - | 619 - | 253 0.0013 | 271 0.0011 | 275 0.0011 | 275 0.0010 | 289 0.0010 | 266 1.010 |
| 294 | MICROFOCUS-003 | 619 - | 619 - | 609 0.5940 | 606 0.4690 | 606 0.4202 | 606 0.3722 | 607 0.3093 | 606 5.359 |
| 295 | MICROFOCUS-004 | 619 - | 619 - | 606 0.5762 | 605 0.4517 | 605 0.4024 | 605 0.3558 | 606 0.2954 | 605 5.198 |
| 296 | MICROFOCUS-005 | 619 - | 619 - | 601 0.4240 | 601 0.3025 | 599 0.2604 | 599 0.2206 | 595 0.1721 | 601 3.859 |
| 297 | MICROFOCUS-006 | 619 - | 619 - | 602 0.4266 | 602 0.3046 | 600 0.2621 | 601 0.2230 | 596 0.1743 | 602 3.878 |
| 298 | MICROSOFT-003 | 619 - | 619 - | 289 0.0015 | 216 0.0009 | 198 0.0008 | 176 0.0007 | 129 0.0006 | 234 1.009 |
| 299 | MICROSOFT-004 | 619 - | 619 - | 275 0.0014 | 204 0.0009 | 168 0.0008 | 150 0.0007 | 116 0.0006 | 215 1.009 |
| 300 | MICROSOFT-005 | 619 - | 619 - | 331 0.0018 | 212 0.0009 | 189 0.0008 | 162 0.0007 | 138 0.0006 | 248 1.009 |
| 301 | MICROSOFT-006 | 619 - | 619 - | 338 0.0019 | 270 0.0011 | 227 0.0009 | 188 0.0008 | 152 0.0007 | 281 1.011 |
| 302 | MOBBL-000 | 619 - | 619 - | 365 0.0023 | 359 0.0016 | 350 0.0014 | 351 0.0014 | 350 0.0013 | 360 1.015 |
| 303 | MOBBL-001 | 619 - | 619 - | 500 0.0115 | 519 0.0109 | 525 0.0108 | 531 0.0108 | 539 0.0107 | 516 1.099 |
| 304 | MOBBL-002 | 619 - | 619 - | 591 0.2119 | 595 0.2119 | 597 0.2119 | 598 0.2119 | 602 0.2119 | 595 2.907 |
| 305 | MUKH-002 | 619 - | 619 - | 535 0.0255 | 529 0.0136 | 526 0.0109 | 521 0.0086 | 516 0.0066 | 530 1.136 |
| 306 | NAVERCLOUD-000 | 619 - | 619 - | 163 0.0009 | 165 0.0008 | 171 0.0008 | 180 0.0008 | 193 0.0008 | 161 1.007 |
| 307 | NEC-000 | 619 - | 619 - | 518 0.0167 | 502 0.0083 | 489 0.0063 | 486 0.0049 | 479 0.0035 | 504 1.084 |
| 308 | NEC-001 | 619 - | 619 - | 527 0.0206 | 530 0.0138 | 533 0.0125 | 535 0.0115 | 540 0.0109 | 528 1.132 |
| 309 | NEC-002 | 619 - | 619 - | 169 0.0009 | 174 0.0008 | 174 0.0008 | 179 0.0008 | 186 0.0007 | 172 1.007 |
| 310 | NEC-003 | 619 - | 619 - | 250 0.0013 | 289 0.0012 | 300 0.0012 | 316 0.0011 | 328 0.0011 | 279 1.011 |
| 311 | NEC-004 | 619 - | 619 - | 207 0.0011 | 242 0.0010 | 252 0.0010 | 250 0.0010 | 262 0.0009 | 238 1.009 |
| 312 | NEC-005 | 619 - | 619 - | 141 0.0008 | 164 0.0008 | 169 0.0008 | 173 0.0007 | 178 0.0007 | 155 1.007 |
| 313 | NEC-006 | 619 - | 619 - | 171 0.0009 | 191 0.0009 | 202 0.0008 | 200 0.0008 | 199 0.0008 | 188 1.008 |
| 314 | NEC-007 | 619 - | 619 - | 87 0.0007 | 105 0.0006 | 111 0.0006 | 115 0.0006 | 122 0.0006 | 102 1.006 |
| 315 | NEC-008 | 619 - | 619 - | 105 0.0007 | 126 0.0007 | 126 0.0006 | 132 0.0006 | 141 0.0006 | 124 1.006 |
| 316 | NEC-009 | 619 - | 619 - | 68 0.0006 | 88 0.0006 | 100 0.0006 | 108 0.0006 | 118 0.0006 | 87 1.005 |
| 317 | NEC-010 | 619 - | 619 - | 29 0.0005 | 34 0.0005 | 46 0.0005 | 42 0.0005 | 34 0.0005 | 36 1.004 |
| 318 | NEC-011 | 619 - | 619 - | 14 0.0005 | 15 0.0004 | 12 0.0004 | 8 0.0004 | 13 0.0004 | 14 1.004 |
| 319 | NEC-012 | 619 - | 619 - | 16 0.0005 | 10 0.0004 | 13 0.0004 | 10 0.0004 | 9 0.0004 | 12 1.004 |
| 320 | NEUROTECHNOLOGY-003 | 619 - | 619 - | 529 0.0222 | 523 0.0123 | 518 0.0096 | 513 0.0075 | 500 0.0054 | 525 1.122 |
| 321 | NEUROTECHNOLOGY-004 | 619 - | 619 - | 436 0.0052 | 438 0.0033 | 441 0.0029 | 441 0.0026 | 440 0.0022 | 439 1.032 |
| 322 | NEUROTECHNOLOGY-005 | 619 - | 619 - | 415 0.0040 | 423 0.0026 | 428 0.0024 | 428 0.0022 | 432 0.0021 | 425 1.025 |
| 323 | NEUROTECHNOLOGY-006 | 619 - | 619 - | 522 0.0178 | 498 0.0078 | 484 0.0058 | 482 0.0045 | 473 0.0032 | 502 1.082 |
| 324 | NEUROTECHNOLOGY-007 | 619 - | 619 - | 404 0.0036 | 408 0.0023 | 413 0.0021 | 414 0.0019 | 412 0.0018 | 407 1.022 |
| 325 | NEUROTECHNOLOGY-008 | 619 - | 619 - | 326 0.0018 | 280 0.0012 | 281 0.0011 | 276 0.0010 | 271 0.0010 | 291 1.011 |
| 326 | NEUROTECHNOLOGY-009 | 619 - | 619 - | 213 0.0011 | 193 0.0009 | 194 0.0008 | 193 0.0008 | 194 0.0008 | 196 1.008 |
| 327 | NEUROTECHNOLOGY-010 | 619 - | 619 - | 157 0.0009 | 138 0.0007 | 145 0.0007 | 145 0.0007 | 161 0.0007 | 143 1.007 |
| 328 | NEUROTECHNOLOGY-012 | 619 - | 619 - | 84 0.0007 | 102 0.0006 | 106 0.0006 | 109 0.0006 | 119 0.0006 | 97 1.006 |
| 329 | NEUROTECHNOLOGY-013 | 619 - | 619 - | 86 0.0007 | 99 0.0006 | 104 0.0006 | 113 0.0006 | 126 0.0006 | 95 1.005 |
| 330 | NEUROTECHNOLOGY-014 | 619 - | 619 - | 50 0.0006 | 55 0.0005 | 62 0.0005 | 64 0.0005 | 71 0.0005 | 53 1.005 |
| 331 | NEUROTECHNOLOGY-015 | 619 - | 619 - | 43 0.0005 | 27 0.0005 | 36 0.0005 | 45 0.0005 | 52 0.0005 | 34 1.004 |
| 332 | NEUROTECHNOLOGY-016 | 619 - | 619 - | 33 0.0005 | 40 0.0005 | 53 0.0005 | 60 0.0005 | 51 0.0005 | 39 1.004 |
| 333 | NEWLAND-002 | 619 - | 619 - | 569 0.0783 | 569 0.0477 | 562 0.0394 | 562 0.0329 | 563 0.0261 | 570 1.466 |
| 334 | NEXTMETALABS-001 | 619 - | 619 - | 545 0.0312 | 535 0.0156 | 529 0.0116 | 522 0.0087 | 507 0.0061 | 536 1.159 |
| 335 | NEXTMETALABS-002 | 619 - | 619 - | 72 0.0006 | 63 0.0005 | 60 0.0005 | 54 0.0005 | 47 0.0005 | 66 1.005 |
| 336 | NEXTMETALABS-003 | 619 - | 619 - | 70 0.0006 | 65 0.0005 | 57 0.0005 | 50 0.0005 | 50 0.0005 | 67 1.005 |
| 337 | NOBLIS-001 | 619 - | 619 - | 595 0.2489 | 589 0.1769 | 589 0.1539 | 589 0.1336 | 588 0.1109 | 591 2.676 |
| 338 | NOBLIS-002 | 619 - | 619 - | 585 0.1791 | 582 0.1104 | 581 0.0900 | 579 0.0719 | 579 0.0532 | 582 2.074 |
| 339 | NOMINDER-000 | 619 - | 619 - | 89 0.0007 | 70 0.0005 | 72 0.0005 | 72 0.0005 | 63 0.0005 | 78 1.005 |
| 340 | NOMINDER-001 | 619 - | 619 - | 228 0.0011 | 258 0.0011 | 273 0.0011 | 282 0.0010 | 302 0.0010 | 253 1.010 |
| 341 | NOMINDER-002 | 619 - | 619 - | 230 0.0012 | 262 0.0011 | 274 0.0011 | 285 0.0011 | 301 0.0010 | 259 1.010 |
| 342 | NOMINDER-003 | 619 - | 619 - | 231 0.0012 | 268 0.0011 | 282 0.0011 | 291 0.0011 | 312 0.0011 | 262 1.010 |
| 343 | NOMINDER-004 | 619 - | 619 - | 210 0.0011 | 241 0.0010 | 251 0.0010 | 259 0.0010 | 276 0.0010 | 236 1.009 |
| 344 | NOMINDER-005 | 619 - | 619 - | 221 0.0011 | 253 0.0010 | 264 0.0010 | 277 0.0010 | 294 0.0010 | 246 1.009 |
| 345 | NOTIONTAG-000 | 619 - | 619 - | 349 0.0021 | 371 0.0017 | 376 0.0017 | 392 0.0016 | 399 0.0016 | 366 1.016 |
| 346 | NTECHLAB-003 | 619 - | 619 - | 447 0.0060 | 428 0.0028 | 418 0.0022 | 403 0.0018 | 392 0.0015 | 432 1.029 |
| 347 | NTECHLAB-004 | 619 - | 619 - | 429 0.0046 | 403 0.0022 | 384 0.0018 | 375 0.0015 | 345 0.0012 | 408 1.023 |
| 348 | NTECHLAB-005 | 619 - | 619 - | 427 0.0046 | 399 0.0021 | 372 0.0016 | 335 0.0013 | 288 0.0010 | 405 1.022 |
| 349 | NTECHLAB-006 | 619 - | 619 - | 416 0.0040 | 372 0.0018 | 349 0.0014 | 305 0.0011 | 231 0.0009 | 388 1.019 |
| 350 | NTECHLAB-007 | 619 - | 619 - | 363 0.0023 | 316 0.0013 | 280 0.0011 | 263 0.0010 | 224 0.0008 | 328 1.013 |
| 351 | NTECHLAB-008 | 619 - | 619 - | 263 0.0014 | 190 0.0008 | 185 0.0008 | 178 0.0007 | 163 0.0007 | 208 1.008 |
| 352 | NTECHLAB-009 | 619 - | 619 - | 172 0.0009 | 147 0.0007 | 144 0.0007 | 129 0.0006 | 123 0.0006 | 149 1.007 |
| 353 | NTECHLAB-010 | 619 - | 619 - | 115 0.0008 | 130 0.0007 | 140 0.0007 | 141 0.0007 | 159 0.0007 | 128 1.006 |
| 354 | NTECHLAB-011 | 619 - | 619 - | 88 0.0007 | 87 0.0006 | 87 0.0006 | 85 0.0005 | 89 0.0005 | 86 1.005 |
| 355 | OMNIGARDE-000 | 619 - | 619 - | 93 0.0007 | 75 0.0005 | 71 0.0005 | 68 0.0005 | 62 0.0005 | 80 1.005 |
| 356 | OMNIGARDE-001 | 619 - | 619 - | 45 0.0005 | 41 0.0005 | 45 0.0005 | 31 0.0005 | 23 0.0004 | 40 1.004 |
| 357 | OMNIGARDE-002 | 619 - | 619 - | 60 0.0006 | 49 0.0005 | 43 0.0005 | 37 0.0005 | 19 0.0004 | 52 1.005 |
| 358 | ONFIDO-000 | 619 - | 619 - | 322 0.0018 | 265 0.0011 | 242 0.0010 | 216 0.0009 | 173 0.0007 | 269 1.010 |
| 359 | ONFIDO-001 | 619 - | 619 - | 257 0.0013 | 163 0.0008 | 141 0.0007 | 125 0.0006 | 107 0.0006 | 178 1.008 |
| 360 | OPTIEXACTA-000 | 619 - | 619 - | 39 0.0005 | 38 0.0005 | 42 0.0005 | 48 0.0005 | 59 0.0005 | 38 1.004 |

*Table 55:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 361 | PANGIAM-000 | 619 - | 619 - | 142 0.0008 | 148 0.0007 | 154 0.0007 | 153 0.0007 | 160 0.0007 | 148 1.007 |
| 362 | PANGIAM-001 | 619 - | 619 - | 451 0.0065 | 484 0.0064 | 491 0.0064 | 499 0.0064 | 512 0.0064 | 478 1.058 |
| 363 | PAPSAV1923-000 | 619 - | 619 - | 132 0.0008 | 152 0.0007 | 156 0.0007 | 168 0.0007 | 190 0.0007 | 151 1.007 |
| 364 | PARAVISION-000 | 619 - | 619 - | 523 0.0187 | 541 0.0170 | 547 0.0166 | 550 0.0164 | 557 0.0163 | 534 1.155 |
| 365 | PARAVISION-001 | 619 - | 619 - | 406 0.0036 | 409 0.0023 | 414 0.0021 | 416 0.0020 | 413 0.0018 | 409 1.023 |
| 366 | PARAVISION-002 | 619 - | 619 - | 413 0.0039 | 418 0.0025 | 417 0.0022 | 419 0.0020 | 417 0.0018 | 416 1.024 |
| 367 | PARAVISION-003 | 619 - | 619 - | 396 0.0031 | 401 0.0021 | 404 0.0020 | 405 0.0018 | 404 0.0017 | 398 1.020 |
| 368 | PARAVISION-004 | 619 - | 619 - | 248 0.0013 | 252 0.0010 | 254 0.0010 | 260 0.0010 | 269 0.0009 | 252 1.010 |
| 369 | PARAVISION-005 | 619 - | 619 - | 234 0.0012 | 247 0.0010 | 253 0.0010 | 262 0.0010 | 272 0.0010 | 244 1.009 |
| 370 | PARAVISION-007 | 619 - | 619 - | 137 0.0008 | 150 0.0007 | 147 0.0007 | 151 0.0007 | 155 0.0007 | 141 1.006 |
| 371 | PARAVISION-009 | 619 - | 619 - | 83 0.0006 | 95 0.0006 | 107 0.0006 | 112 0.0006 | 112 0.0006 | 92 1.005 |
| 372 | PARAVISION-012 | 619 - | 619 - | 62 0.0006 | 84 0.0006 | 93 0.0006 | 94 0.0006 | 99 0.0006 | 82 1.005 |
| 373 | PARAVISION-014 | 619 - | 619 - | 56 0.0006 | 82 0.0006 | 85 0.0006 | 90 0.0006 | 96 0.0006 | 76 1.005 |
| 374 | PARAVISION-016 | 619 - | 619 - | 63 0.0006 | 83 0.0006 | 92 0.0006 | 96 0.0006 | 106 0.0006 | 81 1.005 |
| 375 | PARAVISION-019 | 619 - | 619 - | 27 0.0005 | 32 0.0005 | 35 0.0005 | 38 0.0005 | 36 0.0005 | 30 1.004 |
| 376 | PARAVISION-021 | 619 - | 619 - | 13 0.0005 | 12 0.0004 | 14 0.0004 | 18 0.0004 | 21 0.0004 | 11 1.004 |
| 377 | PENTACOMP-000 | 619 - | 619 - | 613 0.9751 | 612 0.8897 | 612 0.7909 | 611 0.5980 | 358 0.0013 | 612 9.021 |
| 378 | PENTACOMP-001 | 619 - | 619 - | 274 0.0014 | 264 0.0011 | 257 0.0010 | 254 0.0010 | 240 0.0009 | 264 1.010 |
| 379 | PIXELALL-002 | 619 - | 619 - | 419 0.0041 | 420 0.0025 | 415 0.0021 | 400 0.0018 | 398 0.0016 | 420 1.025 |
| 380 | PIXELALL-003 | 619 - | 619 - | 324 0.0018 | 307 0.0013 | 305 0.0012 | 296 0.0011 | 306 0.0011 | 309 1.012 |
| 381 | PIXELALL-004 | 619 - | 619 - | 310 0.0017 | 287 0.0012 | 285 0.0011 | 288 0.0011 | 281 0.0010 | 289 1.011 |
| 382 | PIXELALL-005 | 619 - | 619 - | 288 0.0015 | 317 0.0013 | 320 0.0012 | 334 0.0012 | 343 0.0012 | 307 1.012 |
| 383 | PRIVID-000 | 619 - | 619 - | 528 0.0208 | 548 0.0207 | 551 0.0207 | 556 0.0207 | 559 0.0207 | 544 1.186 |
| 384 | PROTENET-000 | 619 - | 619 - | 564 0.0632 | 567 0.0461 | 569 0.0426 | 570 0.0400 | 573 0.0377 | 566 1.436 |
| 385 | PROTENET-001 | 619 - | 619 - | 341 0.0020 | 361 0.0016 | 369 0.0016 | 374 0.0015 | 382 0.0014 | 356 1.015 |
| 386 | PSL-000 | 619 - | 619 - | 26 0.0005 | 13 0.0004 | 11 0.0004 | 15 0.0004 | 7 0.0004 | 16 1.004 |
| 387 | PSL-001 | 619 - | 619 - | 18 0.0005 | 20 0.0005 | 17 0.0004 | 14 0.0004 | 14 0.0004 | 19 1.004 |
| 388 | PSL-002 | 619 - | 619 - | 6 0.0004 | 7 0.0004 | 6 0.0004 | 3 0.0004 | 5 0.0004 | 6 1.004 |
| 389 | PSL-003 | 619 - | 619 - | 9 0.0004 | 6 0.0004 | 9 0.0004 | 11 0.0004 | 10 0.0004 | 9 1.004 |
| 390 | PSL-004 | 619 - | 619 - | 7 0.0004 | 8 0.0004 | 7 0.0004 | 9 0.0004 | 11 0.0004 | 8 1.004 |
| 391 | PSL-005 | 619 - | 619 - | 11 0.0005 | 16 0.0004 | 16 0.0004 | 13 0.0004 | 12 0.0004 | 15 1.004 |
| 392 | PSL-006 | 619 - | 619 - | 8 0.0004 | 9 0.0004 | 8 0.0004 | 7 0.0004 | 8 0.0004 | 7 1.004 |
| 393 | PTAKURATSATU-000 | 619 - | 619 - | 383 0.0027 | 369 0.0017 | 368 0.0016 | 358 0.0014 | 352 0.0013 | 371 1.017 |
| 394 | PYRAMID-000 | 619 - | 619 - | 108 0.0007 | 116 0.0006 | 120 0.0006 | 121 0.0006 | 131 0.0006 | 118 1.006 |
| 395 | PYRAMID-001 | 619 - | 619 - | 140 0.0008 | 144 0.0007 | 142 0.0007 | 148 0.0007 | 158 0.0007 | 140 1.006 |
| 396 | QAZSMARTVISIONAI-000 | 619 - | 619 - | 21 0.0005 | 21 0.0005 | 22 0.0005 | 20 0.0004 | 15 0.0004 | 23 1.004 |
| 397 | QAZSMARTVISIONAI-001 | 619 - | 619 - | 5 0.0004 | 3 0.0004 | 4 0.0004 | 6 0.0004 | 4 0.0004 | 5 1.004 |
| 398 | QAZSMARTVISIONAI-002 | 619 - | 619 - | 1 0.0004 | 2 0.0004 | 3 0.0004 | 1 0.0004 | 2 0.0004 | 2 1.004 |
| 399 | QAZSMARTVISIONAI-003 | 619 - | 619 - | 4 0.0004 | 4 0.0004 | 2 0.0004 | 4 0.0004 | 3 0.0004 | 3 1.004 |
| 400 | QNAP-000 | 619 - | 619 - | 465 0.0074 | 452 0.0040 | 450 0.0033 | 452 0.0029 | 453 0.0025 | 456 1.040 |
| 401 | QNAP-001 | 619 - | 619 - | 411 0.0038 | 421 0.0025 | 424 0.0023 | 426 0.0021 | 429 0.0020 | 418 1.024 |
| 402 | QNAP-002 | 619 - | 619 - | 426 0.0045 | 453 0.0040 | 466 0.0040 | 478 0.0039 | 484 0.0038 | 448 1.037 |
| 403 | QNAP-003 | 619 - | 619 - | 372 0.0024 | 366 0.0017 | 360 0.0015 | 345 0.0013 | 341 0.0012 | 367 1.016 |
| 404 | QNAP-004 | 619 - | 619 - | 291 0.0016 | 292 0.0012 | 293 0.0011 | 278 0.0010 | 287 0.0010 | 286 1.011 |
| 405 | QNAP-005 | 619 - | 619 - | 390 0.0029 | 427 0.0027 | 434 0.0027 | 445 0.0027 | 457 0.0027 | 422 1.025 |
| 406 | QNAP-006 | 619 - | 619 - | 224 0.0011 | 215 0.0009 | 214 0.0009 | 215 0.0009 | 217 0.0008 | 212 1.009 |
| 407 | QUANTASOFT-001 | 619 - | 619 - | 592 0.2174 | 588 0.1641 | 588 0.1465 | 588 0.1309 | 589 0.1113 | 588 2.537 |
| 408 | RANKONE-002 | 619 - | 619 - | 524 0.0191 | 517 0.0108 | 512 0.0089 | 511 0.0074 | 504 0.0057 | 519 1.108 |
| 409 | RANKONE-003 | 619 - | 619 - | 525 0.0191 | 516 0.0108 | 513 0.0089 | 512 0.0074 | 502 0.0057 | 518 1.108 |
| 410 | RANKONE-004 | 619 - | 619 - | 554 0.0412 | 549 0.0222 | 548 0.0174 | 541 0.0138 | 534 0.0098 | 549 1.222 |
| 411 | RANKONE-005 | 619 - | 619 - | 478 0.0090 | 470 0.0051 | 470 0.0043 | 465 0.0035 | 462 0.0028 | 471 1.051 |
| 412 | RANKONE-006 | 619 - | 619 - | 430 0.0047 | 425 0.0027 | 426 0.0023 | 420 0.0020 | 406 0.0017 | 428 1.027 |
| 413 | RANKONE-007 | 619 - | 619 - | 392 0.0030 | 384 0.0020 | 379 0.0017 | 371 0.0015 | 353 0.0013 | 386 1.019 |
| 414 | RANKONE-009 | 619 - | 619 - | 342 0.0020 | 310 0.0013 | 308 0.0012 | 310 0.0011 | 300 0.0010 | 316 1.012 |
| 415 | RANKONE-010 | 619 - | 619 - | 334 0.0019 | 335 0.0014 | 326 0.0013 | 323 0.0012 | 320 0.0011 | 335 1.013 |
| 416 | RANKONE-011 | 619 - | 619 - | 227 0.0011 | 200 0.0009 | 206 0.0009 | 207 0.0008 | 203 0.0008 | 205 1.008 |
| 417 | RANKONE-012 | 619 - | 619 - | 196 0.0010 | 183 0.0008 | 180 0.0008 | 191 0.0008 | 197 0.0008 | 181 1.008 |
| 418 | RANKONE-013 | 619 - | 619 - | 94 0.0007 | 85 0.0006 | 82 0.0005 | 80 0.0005 | 78 0.0005 | 84 1.005 |
| 419 | RANKONE-014 | 619 - | 619 - | 74 0.0006 | 74 0.0005 | 77 0.0005 | 76 0.0005 | 85 0.0005 | 72 1.005 |
| 420 | RANKONE-015 | 619 - | 619 - | 77 0.0006 | 77 0.0006 | 81 0.0005 | 88 0.0005 | 88 0.0005 | 83 1.005 |
| 421 | REALNETWORKS-000 | 619 - | 619 - | 553 0.0399 | 545 0.0193 | 539 0.0147 | 534 0.0109 | 523 0.0075 | 547 1.199 |
| 422 | REALNETWORKS-001 | 619 - | 619 - | 552 0.0399 | 546 0.0193 | 538 0.0147 | 533 0.0109 | 524 0.0075 | 548 1.199 |
| 423 | REALNETWORKS-002 | 619 - | 619 - | 549 0.0391 | 544 0.0187 | 536 0.0140 | 530 0.0106 | 522 0.0075 | 545 1.193 |
| 424 | REALNETWORKS-003 | 619 - | 619 - | 532 0.0238 | 520 0.0113 | 510 0.0086 | 504 0.0067 | 495 0.0050 | 523 1.117 |
| 425 | REALNETWORKS-004 | 619 - | 619 - | 530 0.0233 | 518 0.0109 | 508 0.0084 | 500 0.0065 | 493 0.0047 | 520 1.113 |
| 426 | REALNETWORKS-005 | 619 - | 619 - | 340 0.0019 | 294 0.0012 | 267 0.0010 | 266 0.0010 | 229 0.0009 | 300 1.012 |
| 427 | REALNETWORKS-006 | 619 - | 619 - | 202 0.0010 | 180 0.0008 | 176 0.0008 | 161 0.0007 | 148 0.0007 | 182 1.008 |
| 428 | REALNETWORKS-007 | 619 - | 619 - | 185 0.0010 | 172 0.0008 | 163 0.0007 | 156 0.0007 | 142 0.0006 | 173 1.007 |
| 429 | REALNETWORKS-008 | 619 - | 619 - | 135 0.0008 | 96 0.0006 | 97 0.0006 | 99 0.0006 | 76 0.0005 | 103 1.006 |
| 430 | REALSENSE-001 | 619 - | 619 - | 78 0.0006 | 73 0.0005 | 76 0.0005 | 77 0.0005 | 68 0.0005 | 75 1.005 |
| 431 | RECOGNITO-000 | 619 - | 619 - | 58 0.0006 | 43 0.0005 | 51 0.0005 | 44 0.0005 | 43 0.0005 | 49 1.005 |
| 432 | RECOGNITO-001 | 619 - | 619 - | 52 0.0006 | 67 0.0005 | 68 0.0005 | 71 0.0005 | 75 0.0005 | 64 1.005 |

*Table 56:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 433 | RECOGNITO-002 | 619 - | 619 - | 40 0.0005 | 36 0.0005 | 40 0.0005 | 49 0.0005 | 60 0.0005 | 37 1.004 |
| 434 | REMARKAI-000 | 619 - | 619 - | 394 0.0030 | 367 0.0017 | 359 0.0015 | 344 0.0013 | 339 0.0012 | 373 1.017 |
| 435 | REMARKAI-000 | 619 - | 619 - | 472 0.0083 | 457 0.0042 | 451 0.0034 | 450 0.0029 | 450 0.0024 | 459 1.043 |
| 436 | REMARKAI-002 | 619 - | 619 - | 470 0.0079 | 447 0.0037 | 438 0.0029 | 434 0.0024 | 422 0.0019 | 452 1.039 |
| 437 | RENDIP-000 | 619 - | 619 - | 236 0.0012 | 226 0.0010 | 219 0.0009 | 211 0.0008 | 221 0.0008 | 228 1.009 |
| 438 | REVEALMEDIA-000 | 619 - | 619 - | 282 0.0015 | 236 0.0010 | 240 0.0009 | 231 0.0009 | 232 0.0009 | 251 1.010 |
| 439 | RNIT-000 | 619 - | 619 - | 222 0.0011 | 240 0.0010 | 243 0.0010 | 258 0.0010 | 264 0.0009 | 237 1.009 |
| 440 | ROC-016 | 619 - | 619 - | 79 0.0006 | 81 0.0006 | 80 0.0005 | 89 0.0005 | 79 0.0005 | 79 1.005 |
| 441 | ROC-017 | 619 - | 619 - | 65 0.0006 | 79 0.0006 | 78 0.0005 | 78 0.0005 | 81 0.0005 | 73 1.005 |
| 442 | ROC-018 | 619 - | 619 - | 44 0.0005 | 45 0.0005 | 54 0.0005 | 58 0.0005 | 61 0.0005 | 45 1.005 |
| 443 | ROC-019 | 619 - | 619 - | 28 0.0005 | 30 0.0005 | 28 0.0005 | 28 0.0005 | 41 0.0005 | 28 1.004 |
| 444 | ROC-020 | 619 - | 619 - | 35 0.0005 | 31 0.0005 | 34 0.0005 | 26 0.0005 | 40 0.0005 | 31 1.004 |
| 445 | ROC-021 | 619 - | 619 - | 22 0.0005 | 23 0.0005 | 21 0.0005 | 24 0.0005 | 28 0.0005 | 21 1.004 |
| 446 | S1-000 | 619 - | 619 - | 345 0.0021 | 331 0.0014 | 328 0.0013 | 326 0.0012 | 330 0.0011 | 340 1.013 |
| 447 | S1-001 | 619 - | 619 - | 385 0.0028 | 400 0.0021 | 411 0.0021 | 421 0.0020 | 428 0.0020 | 395 1.020 |
| 448 | S1-002 | 619 - | 619 - | 205 0.0011 | 238 0.0010 | 245 0.0010 | 255 0.0010 | 273 0.0010 | 231 1.009 |
| 449 | S1-003 | 619 - | 619 - | 219 0.0011 | 234 0.0010 | 235 0.0009 | 237 0.0009 | 247 0.0009 | 230 1.009 |
| 450 | S1-004 | 619 - | 619 - | 186 0.0010 | 219 0.0009 | 230 0.0009 | 238 0.0009 | 244 0.0009 | 213 1.009 |
| 451 | S1-005 | 619 - | 619 - | 198 0.0010 | 223 0.0009 | 232 0.0009 | 242 0.0009 | 258 0.0009 | 217 1.009 |
| 452 | S1-006 | 619 - | 619 - | 180 0.0010 | 211 0.0009 | 220 0.0009 | 227 0.0009 | 238 0.0009 | 201 1.008 |
| 453 | S1-007 | 619 - | 619 - | 151 0.0009 | 173 0.0008 | 182 0.0008 | 189 0.0008 | 198 0.0008 | 168 1.007 |
| 454 | SANSAP-001 | 619 - | 619 - | 136 0.0008 | 134 0.0007 | 109 0.0006 | 111 0.0006 | 114 0.0006 | 125 1.006 |
| 455 | SCANOVATE-000 | 619 - | 619 - | 428 0.0046 | 406 0.0023 | 394 0.0019 | 366 0.0015 | 340 0.0012 | 412 1.023 |
| 456 | SCANOVATE-001 | 619 - | 619 - | 435 0.0050 | 410 0.0023 | 396 0.0019 | 372 0.0015 | 337 0.0012 | 419 1.024 |
| 457 | SCYLLA-001 | 619 - | 619 - | 327 0.0018 | 313 0.0013 | 298 0.0012 | 293 0.0011 | 298 0.0010 | 310 1.012 |
| 458 | SENSETIME-000 | 619 - | 619 - | 357 0.0022 | 379 0.0019 | 386 0.0018 | 395 0.0017 | 400 0.0016 | 375 1.017 |
| 459 | SENSETIME-001 | 619 - | 619 - | 358 0.0022 | 374 0.0018 | 383 0.0018 | 389 0.0016 | 395 0.0015 | 372 1.017 |
| 460 | SENSETIME-002 | 619 - | 619 - | 509 0.0133 | 527 0.0132 | 535 0.0132 | 540 0.0132 | 548 0.0132 | 524 1.119 |
| 461 | SENSETIME-003 | 619 - | 619 - | 81 0.0006 | 98 0.0006 | 108 0.0006 | 107 0.0006 | 113 0.0006 | 91 1.005 |
| 462 | SENSETIME-004 | 619 - | 619 - | 76 0.0006 | 69 0.0005 | 69 0.0005 | 63 0.0005 | 67 0.0005 | 69 1.005 |
| 463 | SENSETIME-005 | 619 - | 619 - | 32 0.0005 | 28 0.0005 | 23 0.0005 | 23 0.0005 | 24 0.0005 | 27 1.004 |
| 464 | SENSETIME-006 | 619 - | 619 - | 30 0.0005 | 29 0.0005 | 31 0.0005 | 35 0.0005 | 35 0.0005 | 29 1.004 |
| 465 | SENSETIME-007 | 619 - | 619 - | 23 0.0005 | 26 0.0005 | 27 0.0005 | 25 0.0005 | 33 0.0005 | 26 1.004 |
| 466 | SENSETIME-008 | 619 - | 619 - | 24 0.0005 | 22 0.0005 | 24 0.0005 | 22 0.0005 | 26 0.0005 | 22 1.004 |
| 467 | SENSETIME-009 | 619 - | 619 - | 17 0.0005 | 24 0.0005 | 25 0.0005 | 30 0.0005 | 27 0.0005 | 25 1.004 |
| 468 | SENSETIME-010 | 619 - | 619 - | 10 0.0005 | 11 0.0004 | 10 0.0004 | 16 0.0004 | 17 0.0004 | 10 1.004 |
| 469 | SENSETIME-011 | 619 - | 619 - | 19 0.0005 | 25 0.0005 | 26 0.0005 | 27 0.0005 | 30 0.0005 | 24 1.004 |
| 470 | SENSETIME-012 | 619 - | 619 - | 15 0.0005 | 14 0.0004 | 15 0.0004 | 12 0.0004 | 18 0.0004 | 13 1.004 |
| 471 | SERENDIPITY-000 | 619 - | 619 - | 232 0.0012 | 229 0.0010 | 221 0.0009 | 219 0.0009 | 218 0.0008 | 225 1.009 |
| 472 | SEVENTHSENSE-001 | 619 - | 619 - | 309 0.0016 | 355 0.0016 | 370 0.0016 | 381 0.0016 | 393 0.0015 | 349 1.014 |
| 473 | SHAMAN-003 | 619 - | 619 - | 577 0.1240 | 578 0.0819 | 578 0.0704 | 577 0.0613 | 578 0.0515 | 578 1.786 |
| 474 | SHAMAN-004 | 619 - | 619 - | 594 0.2218 | 587 0.1470 | 585 0.1238 | 585 0.1046 | 584 0.0822 | 587 2.409 |
| 475 | SHAMAN-006 | 619 - | 619 - | 551 0.0395 | 560 0.0341 | 563 0.0329 | 566 0.0320 | 570 0.0312 | 558 1.313 |
| 476 | SHAMAN-007 | 619 - | 619 - | 550 0.0393 | 558 0.0339 | 562 0.0328 | 565 0.0319 | 569 0.0311 | 557 1.312 |
| 477 | SHENNONA-000 | 619 - | 619 - | 505 0.0120 | 521 0.0118 | 530 0.0118 | 536 0.0117 | 544 0.0117 | 517 1.106 |
| 478 | SHUFTI-001 | 619 - | 619 - | 494 0.0107 | 515 0.0103 | 522 0.0103 | 528 0.0102 | 537 0.0102 | 514 1.093 |
| 479 | SIAT-001 | 619 - | 619 - | 308 0.0016 | 306 0.0013 | 299 0.0012 | 287 0.0011 | 282 0.0010 | 301 1.012 |
| 480 | SIAT-002 | 619 - | 619 - | 312 0.0017 | 298 0.0012 | 311 0.0012 | 317 0.0012 | 311 0.0011 | 303 1.012 |
| 481 | SJTU-001 | 619 - | 619 - | 25 0.0005 | 18 0.0005 | 20 0.0005 | 21 0.0005 | 20 0.0004 | 20 1.004 |
| 482 | SJTU-002 | 619 - | 619 - | 20 0.0005 | 19 0.0005 | 18 0.0005 | 17 0.0004 | 22 0.0004 | 18 1.004 |
| 483 | SMILART-004 | 619 - | 619 - | 612 0.9648 | 613 0.9641 | 613 0.9640 | 613 0.9639 | 614 0.9637 | 613 9.678 |
| 484 | SMILART-005 | 619 - | 619 - | | | | | | 618 10.000 |
| 485 | SPARKTEK-000 | 619 - | 619 - | 116 0.0008 | 119 0.0006 | 115 0.0006 | 114 0.0006 | 121 0.0006 | 116 1.006 |
| 486 | SPARKTEK-001 | 619 - | 619 - | 54 0.0006 | 64 0.0005 | 67 0.0005 | 73 0.0005 | 83 0.0005 | 63 1.005 |
| 487 | SPARSH-001 | 619 - | 619 - | 103 0.0007 | 101 0.0006 | 95 0.0006 | 98 0.0006 | 108 0.0006 | 96 1.006 |
| 488 | SQISOFT-001 | 619 - | 619 - | 412 0.0038 | 269 0.0011 | 222 0.0009 | 195 0.0008 | 167 0.0007 | 331 1.013 |
| 489 | SQISOFT-002 | 619 - | 619 - | 177 0.0009 | 133 0.0007 | 124 0.0006 | 117 0.0006 | 105 0.0006 | 138 1.006 |
| 490 | STAQU-000 | 619 - | 619 - | 457 0.0068 | 475 0.0057 | 429 0.0054 | 487 0.0052 | 494 0.0050 | 473 1.052 |
| 491 | STCON-000 | 619 - | 619 - | 587 0.1826 | 591 0.1826 | 591 0.1826 | 594 0.1826 | 598 0.1825 | 590 2.643 |
| 492 | STCON-001 | 619 - | 619 - | 46 0.0005 | 42 0.0005 | 39 0.0005 | 43 0.0005 | 46 0.0005 | 41 1.004 |
| 493 | STCON-002 | 619 - | 619 - | 36 0.0005 | 47 0.0005 | 49 0.0005 | 39 0.0005 | 50 0.0005 | 42 1.004 |
| 494 | STCON-003 | 619 - | 619 - | 38 0.0005 | 44 0.0005 | 52 0.0005 | 47 0.0005 | 53 0.0005 | 43 1.005 |
| 495 | STCON-004 | 619 - | 619 - | 34 0.0005 | 33 0.0005 | 37 0.0005 | 46 0.0005 | 38 0.0005 | 32 1.004 |
| 496 | STCON-005 | 619 - | 619 - | 31 0.0005 | 35 0.0005 | 32 0.0005 | 40 0.0005 | 39 0.0005 | 33 1.004 |
| 497 | STELLARTECS-001 | 619 - | 619 - | 323 0.0018 | 339 0.0014 | 346 0.0014 | 353 0.0014 | 370 0.0014 | 337 1.013 |
| 498 | SUMSUB-000 | 619 - | 619 - | 557 0.0453 | 554 0.0270 | 552 0.0221 | 553 0.0182 | 549 0.0136 | 554 1.264 |
| 499 | SYNESIS-003 | 619 - | 619 - | 515 0.0159 | 536 0.0156 | 544 0.0156 | 548 0.0156 | 556 0.0156 | 533 1.141 |
| 500 | SYNESIS-003 | 619 - | 619 - | 583 0.1697 | 583 0.1169 | 583 0.1045 | 583 0.0950 | 585 0.0866 | 583 2.118 |
| 501 | SYNESIS-005 | 619 - | 619 - | 471 0.0082 | 500 0.0081 | 507 0.0081 | 517 0.0081 | 529 0.0081 | 497 1.073 |
| 502 | T4ISB-000 | 619 - | 619 - | 484 0.0100 | 513 0.0100 | 521 0.0100 | 527 0.0100 | 535 0.0100 | 512 1.090 |
| 503 | TCC-000 | 619 - | 619 - | 555 0.0448 | 564 0.0413 | 568 0.0405 | 569 0.0399 | 574 0.0393 | 564 1.376 |
| 504 | TCC-001 | 619 - | 619 - | 443 0.0059 | 449 0.0039 | 454 0.0034 | 460 0.0032 | 467 0.0030 | 449 1.037 |

*Table 57:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 505 | TECH5-001 | 619 - | 619 - | 408 0.0036 | 390 0.0020 | 380 0.0017 | 373 0.0015 | 361 0.0013 | 399 1.020 |
| 506 | TECH5-002 | 619 - | 619 - | 364 0.0023 | 263 0.0011 | 216 0.0009 | 185 0.0008 | 154 0.0007 | 287 1.011 |
| 507 | TECH5-003 | 619 - | 619 - | 104 0.0007 | 91 0.0006 | 98 0.0006 | 95 0.0006 | 104 0.0006 | 94 1.005 |
| 508 | TECHAINER-000 | 619 - | 619 - | 336 0.0019 | 353 0.0016 | 362 0.0015 | 367 0.0015 | 383 0.0014 | 352 1.015 |
| 509 | TEVIAN-003 | 619 - | 619 - | 511 0.0145 | 494 0.0072 | 486 0.0058 | 483 0.0046 | 480 0.0036 | 498 1.074 |
| 510 | TEVIAN-004 | 619 - | 619 - | 495 0.0111 | 474 0.0056 | 473 0.0045 | 471 0.0036 | 464 0.0029 | 476 1.057 |
| 511 | TEVIAN-005 | 619 - | 619 - | 461 0.0071 | 446 0.0037 | 443 0.0030 | 439 0.0026 | 437 0.0022 | 450 1.037 |
| 512 | TEVIAN-006 | 619 - | 619 - | 347 0.0021 | 341 0.0015 | 344 0.0014 | 350 0.0013 | 354 0.0013 | 343 1.014 |
| 513 | TEVIAN-007 | 619 - | 619 - | 277 0.0014 | 244 0.0010 | 248 0.0010 | 256 0.0010 | 255 0.0009 | 249 1.010 |
| 514 | THAKAAIT-000 | 619 - | 619 - | 211 0.0011 | 175 0.0008 | 162 0.0007 | 171 0.0007 | 169 0.0007 | 179 1.008 |
| 515 | TIGER-000 | 619 - | 619 - | 563 0.0613 | 557 0.0308 | 554 0.0234 | 552 0.0176 | 545 0.0118 | 559 1.313 |
| 516 | TIGER-002 | 619 - | 619 - | 441 0.0054 | 430 0.0028 | 422 0.0023 | 397 0.0018 | 381 0.0014 | 430 1.029 |
| 517 | TIGER-003 | 619 - | 619 - | 440 0.0054 | 431 0.0028 | 420 0.0022 | 398 0.0018 | 380 0.0014 | 431 1.029 |
| 518 | TIGER-004 | 619 - | 619 - | 360 0.0022 | 377 0.0019 | 395 0.0019 | 406 0.0018 | 414 0.0018 | 377 1.017 |
| 519 | TONGYITRANS-000 | 619 - | 619 - | 453 0.0066 | 441 0.0035 | 440 0.0029 | 439 0.0026 | 444 0.0023 | 443 1.035 |
| 520 | TONGYITRANS-001 | 619 - | 619 - | 452 0.0066 | 442 0.0035 | 439 0.0029 | 438 0.0026 | 443 0.0023 | 444 1.035 |
| 521 | TOSHIBA-000 | 619 - | 619 - | 420 0.0043 | 414 0.0024 | 408 0.0020 | 402 0.0018 | 403 0.0016 | 417 1.024 |
| 522 | TOSHIBA-001 | 619 - | 619 - | 425 0.0045 | 422 0.0025 | 412 0.0021 | 404 0.0018 | 402 0.0016 | 424 1.025 |
| 523 | TOSHIBA-002 | 619 - | 619 - | 176 0.0009 | 213 0.0009 | 224 0.0009 | 236 0.0009 | 250 0.0009 | 206 1.008 |
| 524 | TOSHIBA-003 | 619 - | 619 - | 121 0.0008 | 151 0.0007 | 159 0.0007 | 177 0.0007 | 187 0.0007 | 150 1.007 |
| 525 | TOSS-000 | 619 - | 619 - | 187 0.0010 | 217 0.0009 | 228 0.0009 | 241 0.0009 | 256 0.0009 | 211 1.009 |
| 526 | TRUEFACE-000 | 619 - | 619 - | 391 0.0029 | 419 0.0025 | 430 0.0024 | 429 0.0023 | 441 0.0022 | 410 1.023 |
| 527 | TSIGHT-000 | 619 - | 619 - | 3 0.0004 | 1 0.0004 | 1 0.0004 | 2 0.0004 | 1 0.0004 | 1 1.004 |
| 528 | TTIGROUP-000 | 619 - | 619 - | 295 0.0016 | 329 0.0014 | 338 0.0014 | 346 0.0013 | 357 0.0013 | 327 1.013 |
| 529 | TURINGTECHVIP-001 | 619 - | 619 - | 480 0.0091 | 508 0.0090 | 514 0.0090 | 524 0.0089 | 530 0.0089 | 501 1.081 |
| 530 | UNISSEY-001 | 619 - | 619 - | 590 0.2098 | 594 0.2097 | 596 0.2097 | 597 0.2097 | 601 0.2097 | 594 2.887 |
| 531 | UNISSEY-002 | 619 - | 619 - | 586 0.1808 | 590 0.1805 | 590 0.1805 | 593 0.1805 | 597 0.1804 | 589 2.625 |
| 532 | UNISSEY-003 | 619 - | 619 - | 398 0.0031 | 429 0.0028 | 433 0.0027 | 444 0.0027 | 455 0.0026 | 426 1.025 |
| 533 | UNIVERSEAI-000 | 619 - | 619 - | 95 0.0007 | 113 0.0006 | 118 0.0006 | 120 0.0006 | 124 0.0006 | 111 1.006 |
| 534 | USEB-000 | 619 - | 619 - | 546 0.0342 | 534 0.0153 | 528 0.0115 | 523 0.0089 | 506 0.0060 | 539 1.162 |
| 535 | USEB-001 | 619 - | 619 - | 156 0.0009 | 156 0.0008 | 165 0.0007 | 166 0.0007 | 181 0.0007 | 157 1.007 |
| 536 | USEB-002 | 619 - | 619 - | 206 0.0011 | 225 0.0010 | 239 0.0009 | 239 0.0009 | 254 0.0009 | 222 1.009 |
| 537 | USEB-003 | 619 - | 619 - | 155 0.0009 | 160 0.0008 | 166 0.0007 | 170 0.0007 | 175 0.0007 | 158 1.007 |
| 538 | USEB-004 | 619 - | 619 - | 179 0.0010 | 205 0.0009 | 213 0.0009 | 226 0.0009 | 237 0.0009 | 198 1.008 |
| 539 | VD-000 | 619 - | 619 - | 604 0.4735 | 603 0.3202 | 601 0.2693 | 600 0.2212 | 594 0.1675 | 603 4.056 |
| 540 | VD-001 | 619 - | 619 - | 539 0.0272 | 542 0.0178 | 546 0.0158 | 544 0.0143 | 547 0.0127 | 543 1.171 |
| 541 | VD-002 | 619 - | 619 - | 481 0.0092 | 495 0.0073 | 499 0.0070 | 503 0.0067 | 513 0.0064 | 491 1.068 |
| 542 | VD-003 | 619 - | 619 - | 462 0.0073 | 485 0.0065 | 490 0.0064 | 497 0.0063 | 510 0.0062 | 484 1.059 |
| 543 | VDFSOFTWARE-001 | 619 - | 619 - | 589 0.2055 | 593 0.1958 | 592 0.1926 | 593 0.1893 | 599 0.1853 | 593 2.773 |
| 544 | VDFSOFTWARE-002 | 619 - | 619 - | 319 0.0018 | 231 0.0010 | 187 0.0008 | 162 0.0007 | 132 0.0006 | 245 1.009 |
| 545 | VERIDAS-001 | 619 - | 619 - | 370 0.0024 | 352 0.0016 | 336 0.0013 | 324 0.0012 | 321 0.0011 | 354 1.015 |
| 546 | VERIDAS-002 | 619 - | 619 - | 369 0.0024 | 351 0.0016 | 331 0.0013 | 321 0.0012 | 319 0.0011 | 353 1.015 |
| 547 | VERIDAS-003 | 619 - | 619 - | 279 0.0015 | 276 0.0011 | 270 0.0011 | 265 0.0010 | 267 0.0009 | 271 1.011 |
| 548 | VERIDAS-004 | 619 - | 619 - | 190 0.0010 | 206 0.0009 | 217 0.0009 | 223 0.0009 | 233 0.0009 | 200 1.008 |
| 549 | VERIGRAM-000 | 619 - | 619 - | 356 0.0022 | 394 0.0021 | 410 0.0021 | 423 0.0021 | 433 0.0021 | 390 1.019 |
| 550 | VERIGRAM-001 | 619 - | 619 - | 353 0.0021 | 393 0.0020 | 407 0.0020 | 422 0.0020 | 431 0.0020 | 380 1.018 |
| 551 | VERIJELAS-000 | 619 - | 619 - | 599 0.3545 | 600 0.2973 | 603 0.2802 | 602 0.2652 | 604 0.2486 | 600 3.742 |
| 552 | VIANTE-000 | 619 - | 619 - | 167 0.0009 | 198 0.0009 | 204 0.0009 | 209 0.0008 | 225 0.0008 | 192 1.008 |
| 553 | VIANTE-001 | 619 - | 619 - | 159 0.0009 | 189 0.0008 | 201 0.0008 | 213 0.0008 | 219 0.0008 | 184 1.008 |
| 554 | VIANTE-002 | 619 - | 619 - | 47 0.0005 | 58 0.0005 | 64 0.0005 | 66 0.0005 | 69 0.0005 | 56 1.005 |
| 555 | VIANTE-003 | 619 - | 619 - | 42 0.0005 | 54 0.0005 | 61 0.0005 | 65 0.0005 | 73 0.0005 | 50 1.005 |
| 556 | VIETTELAI-000 | 619 - | 619 - | 418 0.0041 | 450 0.0039 | 462 0.0039 | 475 0.0039 | 483 0.0038 | 445 1.036 |
| 557 | VIETTELAI-001 | 619 - | 619 - | 314 0.0017 | 350 0.0016 | 364 0.0015 | 376 0.0015 | 389 0.0015 | 347 1.014 |
| 558 | VIETTELAI-002 | 619 - | 619 - | 313 0.0017 | 349 0.0016 | 365 0.0015 | 377 0.0015 | 388 0.0015 | 346 1.014 |
| 559 | VIETTELAI-003 | 619 - | 619 - | 75 0.0006 | 78 0.0006 | 75 0.0005 | 69 0.0005 | 72 0.0005 | 74 1.005 |
| 560 | VIGILANTSOLUTIONS-003 | 619 - | 619 - | 567 0.0691 | 562 0.0346 | 557 0.0259 | 554 0.0198 | 550 0.0137 | 562 1.352 |
| 561 | VIGILANTSOLUTIONS-004 | 619 - | 619 - | 578 0.1246 | 575 0.0702 | 573 0.0553 | 571 0.0431 | 568 0.0302 | 575 1.696 |
| 562 | VIGILANTSOLUTIONS-005 | 619 - | 619 - | 477 0.0089 | 458 0.0042 | 448 0.0033 | 440 0.0026 | 426 0.0019 | 460 1.043 |
| 563 | VIGILANTSOLUTIONS-006 | 619 - | 619 - | 483 0.0096 | 463 0.0045 | 456 0.0035 | 446 0.0028 | 430 0.0020 | 464 1.046 |
| 564 | VIGILANTSOLUTIONS-007 | 619 - | 619 - | 393 0.0030 | 356 0.0016 | 335 0.0013 | 320 0.0012 | 270 0.0010 | 365 1.016 |
| 565 | VIGILANTSOLUTIONS-008 | 619 - | 619 - | 379 0.0026 | 344 0.0015 | 319 0.0012 | 299 0.0011 | 259 0.0009 | 351 1.014 |
| 566 | VINBIGDATA-001 | 619 - | 619 - | 150 0.0009 | 100 0.0006 | 90 0.0006 | 86 0.0005 | 86 0.0005 | 104 1.006 |
| 567 | VINBIGDATA-002 | 619 - | 619 - | 61 0.0006 | 39 0.0005 | 50 0.0005 | 56 0.0005 | 44 0.0005 | 46 1.005 |
| 568 | VINBIGDATA-003 | 619 - | 619 - | 51 0.0006 | 61 0.0005 | 58 0.0005 | 57 0.0005 | 55 0.0005 | 57 1.005 |
| 569 | VINBIGDATA-004 | 619 - | 619 - | 50 0.0006 | 59 0.0005 | 56 0.0005 | 41 0.0005 | 49 0.0005 | 58 1.005 |
| 570 | VIRTISHA-000 | 619 - | 619 - | 513 0.0154 | 533 0.0153 | 541 0.0152 | 547 0.0152 | 554 0.0152 | 532 1.138 |
| 571 | VISIONBOX-000 | 619 - | 619 - | 296 0.0016 | 282 0.0012 | 284 0.0011 | 267 0.0010 | 253 0.0009 | 284 1.011 |
| 572 | VISIONBOX-001 | 619 - | 619 - | 380 0.0026 | 413 0.0024 | 427 0.0023 | 430 0.0023 | 446 0.0023 | 402 1.022 |
| 573 | VISIONBOX-002 | 619 - | 619 - | 280 0.0015 | 305 0.0013 | 318 0.0012 | 330 0.0012 | 342 0.0012 | 299 1.012 |
| 574 | VISIONLABS-004 | 619 - | 619 - | 381 0.0026 | 368 0.0017 | 367 0.0016 | 364 0.0015 | 365 0.0013 | 368 1.016 |
| 575 | VISIONLABS-005 | 619 - | 619 - | 361 0.0023 | 358 0.0016 | 353 0.0015 | 348 0.0013 | 346 0.0013 | 357 1.015 |
| 576 | VISIONLABS-006 | 619 - | 619 - | 320 0.0018 | 330 0.0014 | 329 0.0013 | 338 0.0013 | 344 0.0012 | 333 1.013 |

*Table 58:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES OUTSIDE RANK R | | RESOURCE USAGE | | ENROL MOST RECENT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FNIR(N, T=0, R) | | TEMPLATE | | FRVT 2018 MUGSHOTS | | | | | |
| # | ALGORITHM | BYTES | MSEC | R=1 | R=5 | R=10 | R=20 | R=50 | WORK-10 |
| 577 | VISIONLABS-007 | 619 - | 619 - | 317 0.0017 | 320 0.0013 | 321 0.0013 | 329 0.0012 | 338 0.0012 | 320 1.012 |
| 578 | VISIONLABS-008 | 619 - | 619 - | 304 0.0016 | 342 0.0015 | 351 0.0014 | 356 0.0014 | 367 0.0013 | 338 1.013 |
| 579 | VISIONLABS-009 | 619 - | 619 - | 125 0.0008 | 143 0.0007 | 143 0.0007 | 147 0.0007 | 147 0.0006 | 136 1.006 |
| 580 | VISIONLABS-010 | 619 - | 619 - | 201 0.0010 | 228 0.0010 | 233 0.0009 | 235 0.0009 | 246 0.0009 | 221 1.009 |
| 581 | VISIONLABS-011 | 619 - | 619 - | 139 0.0008 | 158 0.0008 | 170 0.0008 | 163 0.0007 | 172 0.0007 | 154 1.007 |
| 582 | VISTEAM-000 | 619 - | 619 - | 432 0.0048 | 395 0.0021 | 371 0.0016 | 342 0.0013 | 286 0.0010 | 403 1.022 |
| 583 | VISTEAM-001 | 619 - | 619 - | 438 0.0053 | 415 0.0024 | 392 0.0018 | 363 0.0015 | 317 0.0011 | 423 1.025 |
| 584 | VIXVIZION-009 | 619 - | 619 - | 335 0.0019 | 293 0.0012 | 269 0.0011 | 251 0.0010 | 222 0.0008 | 298 1.012 |
| 585 | VNIS-000 | 619 - | 619 - | 576 0.1054 | 570 0.0880 | 579 0.0825 | 580 0.0771 | 583 0.0707 | 579 1.811 |
| 586 | VNPAY-001 | 619 - | 619 - | 216 0.0011 | 243 0.0010 | 249 0.0010 | 248 0.0009 | 249 0.0009 | 233 1.009 |
| 587 | VNPT-001 | 619 - | 619 - | 325 0.0018 | 347 0.0015 | 357 0.0015 | 365 0.0015 | 384 0.0014 | 345 1.014 |
| 588 | VNPT-002 | 619 - | 619 - | 286 0.0015 | 338 0.0014 | 348 0.0014 | 357 0.0014 | 371 0.0014 | 332 1.013 |
| 589 | VNPT-003 | 619 - | 619 - | 245 0.0013 | 284 0.0012 | 301 0.0012 | 311 0.0011 | 327 0.0011 | 278 1.011 |
| 590 | VNPT-004 | 619 - | 619 - | 246 0.0013 | 286 0.0012 | 297 0.0011 | 314 0.0011 | 331 0.0011 | 276 1.011 |
| 591 | VNPT-005 | 619 - | 619 - | 249 0.0013 | 285 0.0012 | 295 0.0011 | 307 0.0011 | 322 0.0011 | 274 1.011 |
| 592 | VOCORD-003 | 619 - | 619 - | 446 0.0060 | 436 0.0032 | 435 0.0027 | 433 0.0024 | 436 0.0021 | 440 1.032 |
| 593 | VOCORD-004 | 619 - | 619 - | 466 0.0077 | 465 0.0046 | 468 0.0040 | 470 0.0036 | 472 0.0032 | 462 1.045 |
| 594 | VOCORD-005 | 619 - | 619 - | 458 0.0068 | 460 0.0044 | 463 0.0039 | 468 0.0036 | 476 0.0033 | 458 1.042 |
| 595 | VOCORD-006 | 619 - | 619 - | 616 1.0000 | 619 1.0000 | 617 1.0000 | 617 1.0000 | 618 1.0000 | 619 10.000 |
| 596 | VOGIC-000 | 619 - | 619 - | 368 0.0024 | 389 0.0020 | 402 0.0020 | 409 0.0019 | 416 0.0018 | 381 1.019 |
| 597 | VSOFT-001 | 619 - | 619 - | 510 0.0143 | 491 0.0070 | 480 0.0055 | 481 0.0045 | 478 0.0034 | 496 1.072 |
| 598 | VSOFT-002 | 619 - | 619 - | 339 0.0019 | 325 0.0013 | 324 0.0013 | 328 0.0012 | 323 0.0011 | 334 1.013 |
| 599 | VTS-000 | 619 - | 619 - | 608 0.5936 | 609 0.5935 | 609 0.5935 | 610 0.5935 | 611 0.5935 | 609 6.341 |
| 600 | VTS-001 | 619 - | 619 - | 230 0.0012 | 176 0.0008 | 157 0.0007 | 160 0.0007 | 151 0.0007 | 189 1.008 |
| 601 | VTS-002 | 619 - | 619 - | 297 0.0016 | 253 0.0010 | 231 0.0009 | 220 0.0009 | 196 0.0008 | 263 1.010 |
| 602 | VTS-003 | 619 - | 619 - | 133 0.0008 | 109 0.0006 | 96 0.0006 | 83 0.0005 | 87 0.0005 | 106 1.006 |
| 603 | VTS-004 | 619 - | 619 - | 82 0.0006 | 66 0.0005 | 65 0.0005 | 61 0.0005 | 56 0.0005 | 68 1.005 |
| 604 | VTS-005 | 619 - | 619 - | 67 0.0006 | 68 0.0005 | 47 0.0005 | 53 0.0005 | 48 0.0005 | 62 1.005 |
| 605 | WINK-000 | 619 - | 619 - | 540 0.0291 | 540 0.0170 | 537 0.0142 | 538 0.0121 | 536 0.0100 | 541 1.168 |
| 606 | WINK-001 | 619 - | 619 - | 376 0.0026 | 360 0.0016 | 343 0.0014 | 339 0.0013 | 332 0.0011 | 361 1.016 |
| 607 | WINK-002 | 619 - | 619 - | 181 0.0010 | 136 0.0007 | 121 0.0006 | 119 0.0006 | 109 0.0006 | 146 1.007 |
| 608 | WINK-003 | 619 - | 619 - | 343 0.0020 | 383 0.0019 | 399 0.0019 | 408 0.0019 | 418 0.0018 | 376 1.017 |
| 609 | XAILIENT-000 | 619 - | 619 - | 371 0.0024 | 327 0.0014 | 313 0.0012 | 295 0.0011 | 275 0.0010 | 341 1.014 |
| 610 | XFORWARDAI-000 | 619 - | 619 - | 337 0.0019 | 364 0.0017 | 373 0.0016 | 385 0.0016 | 397 0.0016 | 358 1.015 |
| 611 | XFORWARDAI-001 | 619 - | 619 - | 315 0.0017 | 354 0.0016 | 366 0.0016 | 380 0.0015 | 396 0.0015 | 348 1.014 |
| 612 | XFORWARDAI-002 | 619 - | 619 - | 300 0.0016 | 348 0.0015 | 363 0.0015 | 379 0.0015 | 394 0.0015 | 344 1.014 |
| 613 | YISHENG-001 | 619 - | 619 - | 538 0.0262 | 526 0.0128 | 520 0.0099 | 515 0.0077 | 503 0.0057 | 527 1.131 |
| 614 | YITU-002 | 619 - | 619 - | 287 0.0015 | 218 0.0009 | 205 0.0009 | 194 0.0008 | 184 0.0007 | 241 1.009 |
| 615 | YITU-003 | 619 - | 619 - | 377 0.0026 | 387 0.0020 | 400 0.0019 | 407 0.0018 | 415 0.0018 | 382 1.019 |
| 616 | YITU-004 | 619 - | 619 - | 191 0.0010 | 139 0.0007 | 130 0.0007 | 128 0.0006 | 136 0.0006 | 152 1.007 |
| 617 | YITU-005 | 619 - | 619 - | 348 0.0021 | 373 0.0018 | 382 0.0018 | 396 0.0017 | 408 0.0017 | 370 1.017 |
| 618 | YOUVERSE-001 | 619 - | 619 - | 161 0.0009 | 153 0.0007 | 155 0.0007 | 157 0.0007 | 145 0.0006 | 153 1.007 |
| 619 | YUAN-000 | 619 - | 619 - | 266 0.0014 | 321 0.0013 | 337 0.0013 | 349 0.0013 | 364 0.0013 | 312 1.012 |

*Table 59:* **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 1 | 20FACE-000 | [543]0.462 | [560]0.347 | [567]0.229 | [533]0.763 | [519]0.450 | [539]0.301 | [513]1.000 | [535]1.000 | [532]1.000 | [466]0.424 | [472]0.255 | [311]0.772 | [344]0.599 | [466]0.938 | [487]0.836 |
| 2 | 3DIVI-003 | [546]0.481 | [569]0.400 | [575]0.281 | [522]0.685 | [542]0.626 | [562]0.497 | | | | [487]0.605 | [494]0.445 | | | [431]0.821 | [472]0.717 |
| 3 | 3DIVI-004 | [505]0.255 | [527]0.168 | [532]0.093 | [471]0.400 | [504]0.343 | [528]0.237 | | | | [443]0.277 | [460]0.172 | | | [387]0.607 | [434]0.485 |
| 4 | 3DIVI-005 | [504]0.254 | [524]0.166 | [531]0.092 | [470]0.395 | [502]0.339 | [527]0.234 | [392]0.998 | [410]0.996 | [442]0.990 | [507]0.864 | [517]0.846 | | | [385]0.597 | [433]0.484 |
| 5 | 3DIVI-006 | [502]0.252 | [526]0.167 | [534]0.095 | [474]0.403 | [503]0.342 | [529]0.238 | | | | [444]0.283 | [461]0.174 | | | [391]0.615 | [435]0.490 |
| 6 | AADI-000 | [611]1.000 | [611]1.000 | [613]1.000 | [609]1.000 | [619]1.000 | [610]1.000 | [590]1.000 | [607]1.000 | [558]1.000 | [546]1.000 | [558]1.000 | [471]1.000 | [413]1.000 | [589]1.000 | [557]1.000 |
| 7 | AADI-001 | | | | | | | | [492]1.000 | [501]0.998 | | | | | | [524]1.000 |
| 8 | AADI-002 | [593]0.952 | [599]0.939 | [603]0.899 | [581]0.995 | [594]0.992 | [600]0.980 | [444]1.000 | [453]0.999 | [484]0.997 | [525]0.995 | [526]0.986 | [358]0.999 | [383]0.997 | [497]0.998 | [518]0.996 |
| 9 | ACER-000 | [482]0.207 | [514]0.145 | [519]0.073 | [446]0.300 | [473]0.246 | [496]0.157 | [305]0.987 | [334]0.981 | [392]0.955 | [433]0.201 | [451]0.114 | | | [361]0.490 | [409]0.363 |
| 10 | ACER-001 | [412]0.108 | [445]0.056 | [451]0.025 | [361]0.136 | [387]0.109 | [408]0.069 | [442]1.000 | [464]0.999 | [500]0.998 | [368]0.068 | [384]0.036 | [277]0.406 | [314]0.250 | [359]0.479 | [333]0.206 |
| 11 | ADIANCE-001 | [616]1.000 | [613]1.000 | [617]1.000 | [594]1.000 | [600]1.000 | [604]1.000 | [617]1.000 | [582]1.000 | [579]1.000 | [503]0.833 | [514]0.828 | [369]1.000 | [396]1.000 | [463]0.931 | [489]0.852 |
| 12 | ADIANCE-002 | [283]0.032 | [281]0.012 | [277]0.005 | [578]0.991 | [585]0.955 | [574]0.634 | [461]1.000 | [476]1.000 | [227]0.333 | [517]0.974 | [510]0.692 | [415]1.000 | [509]1.000 | [493]0.996 | [495]0.902 |
| 13 | ADVANCE-000 | [304]0.038 | [338]0.020 | [332]0.008 | [465]0.371 | [487]0.270 | [470]0.112 | [378]0.997 | [316]0.973 | [264]0.470 | [419]0.163 | [399]0.047 | [323]0.884 | [349]0.667 | [309]0.326 | [340]0.209 |
| 14 | AFISBIOMETRICS-000 | [350]0.053 | [390]0.029 | [370]0.010 | [319]0.104 | [343]0.081 | [344]0.044 | [166]0.619 | [187]0.511 | [232]0.340 | [314]0.034 | [313]0.015 | [206]0.154 | [238]0.080 | [232]0.172 | [253]0.127 |
| 15 | AFRENGINE-000 | [285]0.033 | [334]0.019 | [354]0.009 | [464]0.364 | [488]0.271 | [488]0.149 | [154]0.565 | [149]0.360 | [188]0.237 | [435]0.206 | [450]0.114 | [305]0.743 | [280]0.135 | [300]0.308 | [355]0.231 |
| 16 | AFRENGINE-001 | [126]0.008 | [127]0.003 | [183]0.002 | [186]0.041 | [198]0.031 | [215]0.020 | [150]0.550 | [115]0.227 | [114]0.119 | [246]0.018 | [297]0.013 | [174]0.113 | [197]0.051 | [167]0.120 | [195]0.097 |
| 17 | AFRENGINE-002 | [102]0.006 | [116]0.003 | [169]0.002 | [205]0.046 | [168]0.026 | [182]0.018 | [229]0.901 | [144]0.344 | [91]0.106 | [179]0.008 | [172]0.004 | [331]0.926 | [174]0.039 | [140]0.102 | [117]0.075 |
| 18 | AFRENGINE-003 | [133]0.008 | [174]0.004 | [236]0.003 | [197]0.045 | [211]0.033 | [224]0.022 | [108]0.333 | [85]0.183 | [101]0.113 | [204]0.011 | [192]0.005 | [166]0.097 | [173]0.039 | [157]0.111 | [148]0.083 |
| 19 | AIGROUP-000 | [302]0.037 | [258]0.010 | [253]0.003 | [585]0.996 | [593]0.991 | [595]0.935 | [406]0.999 | [416]0.997 | [204]0.276 | [497]0.693 | [498]0.522 | [500]1.000 | [431]1.000 | [461]0.929 | [490]0.858 |
| 20 | AIZE-001 | [426]0.126 | [468]0.076 | [469]0.033 | [397]0.187 | [419]0.143 | [436]0.087 | [357]0.995 | [384]0.994 | [426]0.983 | [390]0.101 | [407]0.052 | [266]0.364 | [307]0.216 | [325]0.387 | [383]0.289 |
| 21 | AIZE-002 | [574]0.787 | [431]0.051 | [415]0.019 | [597]1.000 | [590]0.981 | [401]0.065 | [480]1.000 | [523]1.000 | [537]1.000 | [456]0.355 | [380]0.036 | [384]1.000 | [319]0.272 | [519]1.000 | [609]1.000 |
| 22 | AIZE-006 | [417]0.116 | [420]0.045 | [422]0.020 | [593]1.000 | [596]0.996 | [555]0.413 | | [434]0.998 | [315]0.708 | [450]0.304 | [409]0.055 | [561]1.000 | [584]1.000 | [445]0.885 | [443]0.544 |
| 23 | AKUVOX-001 | [262]0.026 | [288]0.013 | [282]0.005 | [259]0.067 | [269]0.050 | [281]0.031 | [276]0.979 | [300]0.964 | [341]0.844 | [254]0.019 | [254]0.009 | [302]0.737 | [320]0.275 | [194]0.141 | [216]0.110 |
| 24 | ALCHERA-000 | [493]0.228 | [511]0.138 | [513]0.070 | [433]0.259 | [457]0.216 | [486]0.146 | [413]0.999 | [439]0.999 | [477]0.996 | [425]0.176 | [448]0.111 | | | [427]0.803 | [427]0.456 |
| 25 | ALCHERA-001 | [607]1.000 | [607]0.999 | [610]0.999 | [600]1.000 | [603]1.000 | [609]1.000 | | | | [608]1.000 | [588]1.000 | | | [539]1.000 | [591]1.000 |
| 26 | ALCHERA-002 | [577]0.806 | [578]0.485 | [579]0.301 | [521]0.685 | [537]0.591 | [557]0.442 | [469]1.000 | [472]1.000 | [509]0.999 | [502]0.827 | [512]0.770 | | | [428]0.811 | [467]0.705 |
| 27 | ALCHERA-003 | [538]0.449 | [517]0.155 | [514]0.070 | [447]0.304 | [469]0.239 | [493]0.152 | [459]1.000 | [455]0.999 | [483]0.997 | [424]0.172 | [440]0.097 | | | [352]0.464 | [408]0.362 |
| 28 | ALCHERA-004 | [551]0.519 | [568]0.394 | [564]0.211 | [514]0.642 | [532]0.529 | [545]0.327 | [358]0.995 | [371]0.991 | [334]0.811 | [467]0.424 | [470]0.232 | [297]0.708 | [336]0.515 | [376]0.546 | [421]0.398 |
| 29 | ALCHERA-005 | [536]0.432 | [556]0.314 | [552]0.145 | [497]0.534 | [517]0.439 | [531]0.258 | [325]0.991 | [348]0.985 | [374]0.935 | [451]0.305 | [457]0.144 | [327]0.908 | [335]0.502 | [348]0.455 | [393]0.309 |
| 30 | ALCHERA-006 | [187]0.012 | [184]0.005 | [180]0.002 | [153]0.033 | [155]0.025 | [168]0.017 | [316]0.989 | [337]0.982 | [361]0.913 | [115]0.006 | [113]0.003 | [286]0.493 | [221]0.068 | [113]0.094 | [100]0.072 |
| 31 | ALCHERA-007 | [145]0.009 | [143]0.004 | [149]0.002 | [132]0.029 | [137]0.023 | [153]0.016 | [326]0.991 | [340]0.982 | [353]0.885 | [96]0.005 | [98]0.003 | [306]0.743 | [253]0.086 | [89]0.087 | [90]0.069 |
| 32 | ALCHERA-008 | [121]0.008 | [134]0.003 | [138]0.002 | [134]0.029 | [125]0.022 | [136]0.015 | [84]0.272 | [80]0.177 | [88]0.105 | [200]0.010 | [81]0.002 | [323]1.000 | [374]0.975 | [247]0.186 | [177]0.090 |
| 33 | ALGORITHMSINC-000 | [289]0.034 | [317]0.016 | [327]0.007 | [469]0.394 | [408]0.130 | [300]0.033 | [194]0.760 | [185]0.507 | [243]0.367 | [392]0.108 | [306]0.014 | [389]1.000 | [384]0.997 | [342]0.433 | [291]0.153 |
| 34 | ALLGOVISION-000 | [437]0.138 | [482]0.088 | [493]0.045 | [412]0.202 | [436]0.166 | [466]0.106 | [334]0.993 | [367]0.990 | [424]0.982 | [395]0.117 | [420]0.066 | | | [320]0.526 | [420]0.396 |
| 35 | ALLGOVISION-001 | [453]0.154 | [489]0.101 | [500]0.052 | [439]0.275 | [461]0.221 | [485]0.141 | [342]0.993 | [350]0.986 | [371]0.932 | [413]0.150 | [433]0.081 | | | [362]0.491 | [418]0.389 |
| 36 | ALLGOVISION-002 | [432]0.132 | [459]0.065 | [457]0.028 | [531]0.753 | [531]0.520 | [471]0.113 | [288]0.983 | [288]0.950 | [347]0.868 | [437]0.232 | [403]0.049 | [566]1.000 | [402]1.000 | [410]0.716 | [411]0.366 |
| 37 | ALLGOVISION-003 | [282]0.032 | [321]0.017 | [322]0.007 | [276]0.075 | [298]0.059 | [317]0.037 | [176]0.656 | [206]0.566 | [252]0.423 | [267]0.021 | [271]0.010 | [196]0.147 | [250]0.082 | [358]0.478 | [264]0.133 |
| 38 | ALLGOVISION-004 | [459]0.164 | [494]0.114 | [508]0.059 | [420]0.229 | [449]0.190 | [475]0.120 | [208]0.804 | [239]0.747 | [298]0.627 | [396]0.121 | [419]0.066 | [279]0.418 | [318]0.271 | [335]0.412 | [371]0.262 |
| 39 | ALLGOVISION-005 | [295]0.035 | [343]0.020 | [344]0.008 | [293]0.086 | [307]0.064 | [321]0.038 | [128]0.468 | [157]0.393 | [198]0.262 | [333]0.043 | [292]0.012 | [376]1.000 | [368]0.943 | [308]0.324 | [286]0.149 |
| 40 | ALTTEKGLOBAL-000 | [513]0.282 | [550]0.278 | [573]0.275 | [438]0.272 | [484]0.267 | [532]0.259 | [172]0.646 | [213]0.617 | [284]0.571 | [461]0.383 | [486]0.381 | [290]0.536 | [331]0.434 | [396]0.641 | [459]0.636 |
| 41 | ANKE-000 | [467]0.183 | [496]0.117 | [510]0.062 | [432]0.256 | [459]0.220 | [491]0.151 | [354]0.995 | [383]0.994 | [439]0.990 | [619]1.000 | [579]1.000 | | | [558]1.000 | [576]1.000 |
| 42 | ANKE-001 | [466]0.183 | [499]0.119 | [511]0.062 | [431]0.256 | [460]0.220 | [492]0.151 | [359]0.995 | [391]0.994 | [454]0.992 | [586]1.000 | [610]1.000 | | | [512]1.000 | [615]1.000 |
| 43 | ANKE-002 | [364]0.062 | [395]0.031 | [394]0.013 | [317]0.103 | [339]0.079 | [363]0.050 | [265]0.974 | [287]0.947 | [329]0.793 | [316]0.034 | [333]0.018 | | | [275]0.245 | [327]0.190 |
| 44 | ANTHEUS-000 | [375]0.066 | [410]0.041 | [424]0.020 | [538]0.805 | [549]0.664 | [444]0.093 | [237]0.927 | [246]0.793 | [290]0.598 | [506]0.862 | [499]0.548 | [597]1.000 | [415]1.000 | [479]0.972 | [493]0.900 |
| 45 | ARATEK-001 | [444]0.141 | [239]0.008 | [216]0.003 | [561]0.958 | [266]0.050 | [211]0.020 | [511]1.000 | [536]1.000 | [510]0.999 | [383]0.086 | [206]0.005 | [341]0.971 | [186]0.045 | [580]1.000 | [521]0.999 |
| 46 | ARMATURA-000 | [138]0.008 | [160]0.004 | [192]0.002 | [148]0.031 | [158]0.025 | [190]0.018 | [88]0.283 | [105]0.214 | [119]0.124 | [118]0.006 | [124]0.003 | [96]0.048 | [92]0.019 | [102]0.090 | [108]0.073 |

*Table 60:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 47 | AWARE-003 | 463 0.173 | 505 0.128 | 523 0.082 | 460 0.351 | 496 0.298 | 518 0.204 | 301 0.986 | 345 0.984 | 414 0.977 | 468 0.428 | 485 0.378 | | | 371 0.530 | 426 0.443 |
| 48 | AWARE-004 | 525 0.354 | 548 0.268 | 557 0.174 | 510 0.619 | 528 0.509 | 550 0.375 | 464 1.000 | 478 1.000 | 512 0.999 | 462 0.397 | 475 0.279 | | | 429 0.816 | 456 0.631 |
| 49 | AWARE-005 | 558 0.606 | 563 0.363 | 526 0.085 | 454 0.342 | 476 0.253 | 499 0.163 | 456 1.000 | 484 1.000 | 514 0.999 | 441 0.255 | 454 0.122 | | | 453 0.916 | 470 0.714 |
| 50 | AWARE-006 | 544 0.470 | 549 0.275 | 558 0.175 | 490 0.466 | 510 0.398 | 535 0.283 | 437 1.000 | 467 0.999 | 504 0.999 | 458 0.368 | 471 0.254 | | | 415 0.749 | 453 0.623 |
| 51 | AWARE-007 | 273 0.029 | 316 0.016 | 324 0.007 | 405 0.197 | 320 0.070 | 318 0.038 | 387 0.998 | 375 0.992 | 312 0.694 | 337 0.044 | 346 0.021 | 344 0.982 | 324 0.338 | 278 0.250 | 313 0.177 |
| 52 | AWIROS-000 | 572 0.761 | 552 0.291 | 477 0.037 | 546 0.909 | 524 0.471 | 468 0.109 | 522 1.000 | 504 1.000 | 536 1.000 | 518 0.976 | 482 0.320 | 418 1.000 | 387 0.998 | 502 0.999 | 496 0.924 |
| 53 | AYONIX-000 | 580 0.846 | 592 0.811 | 599 0.724 | 559 0.956 | 583 0.939 | 592 0.892 | 393 0.998 | 425 0.998 | 475 0.995 | 514 0.954 | 520 0.891 | | | 482 0.982 | 502 0.959 |
| 54 | AYONIX-001 | 584 0.874 | 594 0.824 | 597 0.701 | 553 0.946 | 577 0.920 | 588 0.845 | 453 1.000 | 462 0.999 | 479 0.996 | 528 0.999 | 529 0.998 | | | 476 0.969 | 498 0.926 |
| 55 | AYONIX-002 | 585 0.874 | 593 0.824 | 598 0.701 | 554 0.946 | 578 0.920 | 587 0.845 | 455 1.000 | 463 0.999 | 478 0.996 | 510 0.915 | 513 0.821 | | | 477 0.969 | 497 0.926 |
| 56 | BIOIDTECHSWISS-001 | 149 0.009 | 154 0.004 | 140 0.002 | 176 0.038 | 194 0.030 | 240 0.023 | 375 0.997 | 355 0.987 | 181 0.207 | 197 0.010 | 229 0.007 | 136 0.067 | 167 0.036 | 241 0.179 | 307 0.172 |
| 57 | BIOMETRIK-001 | 416 0.115 | 433 0.052 | 440 0.023 | 350 0.129 | 374 0.098 | 385 0.058 | 494 1.000 | 479 1.000 | 482 0.997 | 351 0.054 | 367 0.027 | 263 0.341 | 301 0.183 | 392 0.618 | 324 0.188 |
| 58 | CAMVI-003 | 399 0.094 | 463 0.071 | 505 0.057 | 373 0.152 | 410 0.132 | 467 0.108 | 277 0.979 | 308 0.970 | 378 0.940 | 394 0.114 | 441 0.100 | | | 330 0.402 | 415 0.377 |
| 59 | CAMVI-004 | 410 0.106 | 464 0.071 | 503 0.054 | 425 0.240 | 412 0.136 | 453 0.100 | 439 1.000 | 459 0.999 | 488 0.998 | 389 0.100 | 432 0.081 | | | 423 0.787 | 437 0.507 |
| 60 | CAMVI-005 | 438 0.138 | 487 0.099 | 522 0.076 | 486 0.451 | 446 0.179 | 480 0.132 | 446 1.000 | 473 1.000 | 497 0.998 | 416 0.156 | 449 0.112 | | | 498 0.999 | 511 0.983 |
| 61 | CANON-001 | 177 0.012 | 193 0.005 | 190 0.002 | 141 0.031 | 146 0.023 | 137 0.015 | 168 0.629 | 147 0.357 | 182 0.208 | 162 0.008 | 170 0.004 | 140 0.068 | 158 0.034 | 189 0.139 | 182 0.092 |
| 62 | CANON-002 | 151 0.009 | 170 0.004 | 177 0.002 | 118 0.027 | 114 0.020 | 104 0.013 | 132 0.481 | 159 0.400 | 193 0.245 | 224 0.013 | 176 0.004 | 149 0.075 | 187 0.046 | 250 0.188 | 208 0.106 |
| 63 | CANON-003 | 80 0.005 | 91 0.002 | 76 0.001 | 60 0.015 | 69 0.012 | 72 0.010 | 85 0.274 | 95 0.199 | 123 0.127 | 114 0.006 | 188 0.004 | 51 0.030 | 66 0.014 | 52 0.076 | 58 0.063 |
| 64 | CIB-000 | 320 0.044 | 277 0.012 | 269 0.005 | 278 0.077 | 257 0.045 | 252 0.025 | 497 1.000 | 497 1.000 | 527 1.000 | 244 0.017 | 238 0.008 | 188 0.141 | 222 0.068 | 448 0.894 | 439 0.521 |
| 65 | CLEARVIEWAI-000 | 192 0.013 | 202 0.005 | 186 0.002 | 172 0.036 | 162 0.025 | 151 0.016 | 417 0.999 | 317 0.974 | 137 0.139 | 163 0.008 | 159 0.004 | 118 0.057 | 129 0.027 | 288 0.268 | 137 0.080 |
| 66 | CLEARVIEWAI-001 | 120 0.007 | 140 0.003 | 115 0.001 | 110 0.026 | 113 0.020 | 102 0.013 | 56 0.234 | 70 0.167 | 76 0.093 | 105 0.005 | 78 0.002 | 86 0.042 | 82 0.016 | 68 0.079 | 74 0.066 |
| 67 | CLEARVIEWAI-002 | 132 0.008 | 152 0.004 | 128 0.002 | 128 0.029 | 123 0.022 | 112 0.013 | 67 0.250 | 76 0.170 | 78 0.095 | 106 0.006 | 93 0.002 | 85 0.041 | 84 0.017 | 62 0.078 | 69 0.066 |
| 68 | CLEARVIEWAI-003 | 81 0.005 | 89 0.002 | 84 0.001 | 90 0.019 | 88 0.015 | 78 0.010 | 27 0.172 | 46 0.128 | 45 0.074 | 90 0.005 | 72 0.002 | 73 0.038 | 71 0.015 | 65 0.079 | 67 0.065 |
| 69 | CLOUDWALK-HR-000 | 65 0.004 | 78 0.002 | 105 0.001 | 62 0.015 | 77 0.013 | 92 0.012 | 31 0.179 | 43 0.123 | 61 0.085 | 91 0.005 | 139 0.003 | 59 0.033 | 89 0.018 | 130 0.099 | 118 0.075 |
| 70 | CLOUDWALK-MT-000 | 46 0.003 | 76 0.002 | 120 0.001 | 61 0.015 | 76 0.013 | 96 0.012 | 22 0.160 | 28 0.099 | 39 0.066 | 49 0.002 | 63 0.002 | 18 0.018 | 31 0.009 | 45 0.072 | 59 0.063 |
| 71 | CLOUDWALK-MT-001 | 39 0.003 | 71 0.002 | 113 0.001 | 47 0.013 | 61 0.012 | 91 0.011 | 4 0.093 | 6 0.059 | 12 0.049 | 12 0.001 | 16 0.001 | 14 0.015 | 14 0.006 | 12 0.056 | 16 0.049 |
| 72 | CLOUDWALK-MT-002 | 37 0.003 | 68 0.002 | 112 0.001 | 41 0.012 | 57 0.011 | 88 0.011 | 1 0.070 | 3 0.053 | 8 0.045 | 8 0.001 | 7 0.001 | 7 0.013 | 8 0.005 | 10 0.055 | 9 0.047 |
| 73 | COGENT-000 | 447 0.143 | 436 0.052 | 458 0.029 | 388 0.175 | 415 0.140 | 456 0.100 | 365 0.996 | 396 0.995 | 450 0.991 | | | | | | |
| 74 | COGENT-001 | 446 0.143 | 435 0.052 | 459 0.029 | 389 0.175 | 416 0.140 | 457 0.100 | 366 0.996 | 397 0.995 | 449 0.991 | | | | | | |
| 75 | COGENT-002 | 456 0.157 | 417 0.044 | 410 0.017 | 345 0.124 | 372 0.098 | 398 0.063 | 403 0.998 | 422 0.998 | 461 0.994 | | | | | | |
| 76 | COGENT-003 | 480 0.199 | 421 0.045 | 401 0.015 | 343 0.121 | 366 0.095 | 394 0.061 | 404 0.999 | 427 0.998 | 471 0.995 | | | | | | |
| 77 | COGENT-004 | 485 0.209 | 396 0.033 | 293 0.006 | 257 0.067 | 270 0.051 | 285 0.031 | 391 0.998 | 420 0.997 | 474 0.995 | 271 0.022 | 289 0.012 | 180 0.126 | 226 0.072 | 349 0.456 | 316 0.178 |
| 78 | COGENT-005 | 339 0.050 | 248 0.009 | 255 0.003 | 218 0.050 | 230 0.037 | 242 0.023 | 362 0.996 | 360 0.989 | 224 0.315 | 206 0.011 | 218 0.006 | 156 0.082 | 182 0.043 | 451 0.905 | 329 0.202 |
| 79 | COGENT-006 | 163 0.010 | 163 0.004 | 163 0.002 | 157 0.033 | 147 0.023 | 143 0.015 | 118 0.376 | 118 0.229 | 131 0.135 | 131 0.006 | 145 0.003 | 280 0.422 | 137 0.028 | 180 0.130 | 234 0.120 |
| 80 | COGENT-007 | 127 0.008 | 135 0.003 | 129 0.002 | 121 0.027 | 121 0.021 | 133 0.014 | 62 0.241 | 86 0.185 | 108 0.116 | 135 0.006 | 141 0.003 | 94 0.047 | 102 0.022 | 200 0.144 | 256 0.129 |
| 81 | COGNITEC-000 | 491 0.225 | 521 0.160 | 533 0.095 | 482 0.439 | 497 0.303 | 516 0.200 | 361 0.995 | 373 0.992 | 402 0.971 | | | | | | |
| 82 | COGNITEC-001 | 478 0.191 | 488 0.101 | 501 0.052 | 586 0.997 | 465 0.230 | 482 0.135 | 562 1.000 | 557 1.000 | 397 0.965 | | | | | | |
| 83 | COGNITEC-002 | 422 0.120 | 438 0.053 | 446 0.024 | 527 0.990 | 445 0.178 | 460 0.101 | 576 1.000 | 499 1.000 | 393 0.956 | | | | | | |
| 84 | COGNITEC-003 | 404 0.099 | 434 0.052 | 448 0.025 | 419 0.222 | 434 0.162 | 454 0.100 | 579 1.000 | 509 1.000 | 381 0.945 | | | | | | |
| 85 | COGNITEC-004 | 352 0.055 | 394 0.031 | 395 0.013 | 348 0.127 | 370 0.097 | 386 0.058 | 356 0.995 | 364 0.990 | 365 0.918 | 366 0.068 | 385 0.038 | 257 0.316 | 306 0.196 | 293 0.288 | 347 0.218 |
| 86 | COGNITEC-005 | 354 0.055 | 254 0.009 | 243 0.003 | 239 0.058 | 245 0.041 | 234 0.022 | 591 1.000 | 606 1.000 | 350 0.877 | 331 0.041 | 368 0.028 | 209 0.157 | 258 0.092 | 240 0.179 | 282 0.145 |
| 87 | COGNITEC-006 | 272 0.029 | 233 0.007 | 226 0.003 | 250 0.065 | 241 0.040 | 228 0.022 | 577 1.000 | 618 1.000 | 505 0.999 | 301 0.030 | 302 0.013 | 216 0.171 | 243 0.081 | 404 0.681 | 344 0.214 |
| 88 | COGNITEC-007 | 197 0.013 | 200 0.005 | 196 0.002 | 221 0.051 | 216 0.034 | 204 0.019 | 189 0.735 | 216 0.621 | 266 0.483 | 284 0.026 | 287 0.012 | 233 0.210 | 252 0.085 | 220 0.160 | 247 0.124 |
| 89 | COGNITEC-008 | 202 0.014 | 206 0.006 | 202 0.002 | 213 0.048 | 221 0.035 | 214 0.020 | 174 0.653 | 205 0.564 | 250 0.414 | 285 0.026 | 283 0.011 | 191 0.142 | 220 0.068 | 212 0.155 | 244 0.122 |
| 90 | COGNITEC-009 | 204 0.014 | 210 0.006 | 214 0.003 | 211 0.048 | 222 0.035 | 213 0.020 | 169 0.633 | 204 0.554 | 251 0.415 | 280 0.025 | 284 0.011 | 178 0.124 | 206 0.059 | 214 0.156 | 250 0.126 |
| 91 | COMPAL-000 | 520 0.323 | 236 0.008 | 228 0.003 | 204 0.046 | 237 0.038 | 269 0.028 | | | 541 1.000 | 370 0.072 | 423 0.068 | 169 0.104 | 242 0.081 | 202 0.147 | 269 0.135 |
| 92 | COMPAL-001 | 528 0.366 | 282 0.012 | 334 0.008 | 203 0.046 | 210 0.033 | 235 0.023 | 338 0.993 | 165 0.422 | 133 0.136 | 221 0.013 | 252 0.009 | 160 0.090 | 153 0.033 | 162 0.117 | 155 0.084 |

*Table 61:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*



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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 93 | COMPAL-002 | 529 0.368 | 283 0.012 | 333 0.008 | 208 0.046 | 206 0.033 | 237 0.023 | 337 0.993 | 164 0.419 | 132 0.135 | 223 0.013 | 251 0.009 | 159 0.090 | 151 0.033 | 164 0.118 | 157 0.084 |
| 94 | CORSIGHT-000 | 213 0.016 | 211 0.006 | 203 0.002 | 177 0.038 | 186 0.029 | 183 0.018 | 260 0.971 | 189 0.517 | 163 0.173 | 188 0.009 | 182 0.004 | 137 0.067 | 154 0.033 | 384 0.594 | 169 0.088 |
| 95 | CORSIGHT-001 | 209 0.015 | 219 0.006 | 201 0.002 | 194 0.043 | 212 0.034 | 209 0.020 | 230 0.903 | 176 0.460 | 162 0.171 | 173 0.008 | 175 0.004 | 124 0.062 | 144 0.030 | 188 0.138 | 143 0.082 |
| 96 | CUBOX-000 | 72 0.004 | 104 0.002 | 126 0.002 | 100 0.022 | 104 0.019 | 120 0.014 | 79 0.267 | 64 0.158 | 77 0.093 | 81 0.004 | 103 0.003 | 47 0.028 | 67 0.014 | 46 0.073 | 56 0.062 |
| 97 | CUBOX-001 | 82 0.005 | 111 0.003 | 117 0.001 | 112 0.026 | 118 0.021 | 122 0.014 | 98 0.311 | 104 0.212 | 112 0.118 | 141 0.007 | 135 0.003 | 142 0.068 | 108 0.023 | 80 0.085 | 76 0.067 |
| 98 | CUBOX-002 | 92 0.006 | 125 0.003 | 133 0.002 | 119 0.027 | 132 0.022 | 163 0.016 | 82 0.270 | 81 0.181 | 85 0.103 | 84 0.004 | 77 0.002 | 75 0.038 | 78 0.016 | 58 0.077 | 53 0.061 |
| 99 | CUBOX-003 | 94 0.006 | 131 0.003 | 168 0.002 | 120 0.027 | 138 0.023 | 166 0.017 | 105 0.321 | 108 0.219 | 127 0.131 | 119 0.006 | 133 0.003 | 105 0.051 | 101 0.021 | 75 0.082 | 71 0.066 |
| 100 | CYBERLINK-000 | 436 0.137 | 446 0.056 | 436 0.022 | 381 0.162 | 391 0.116 | 411 0.070 | 382 0.997 | 400 0.995 | 423 0.981 | 363 0.063 | 377 0.032 | | | 314 0.339 | 357 0.232 |
| 101 | CYBERLINK-001 | 402 0.096 | 439 0.053 | 434 0.022 | 365 0.138 | 388 0.109 | 404 0.067 | 381 0.997 | 395 0.995 | 428 0.983 | 360 0.062 | 372 0.031 | | | 399 0.652 | 359 0.239 |
| 102 | CYBERLINK-002 | 305 0.038 | 304 0.014 | 303 0.006 | 263 0.068 | 284 0.053 | 291 0.032 | 346 0.994 | 359 0.988 | 394 0.957 | 278 0.024 | 293 0.013 | | | 294 0.288 | 296 0.157 |
| 103 | CYBERLINK-003 | 324 0.045 | 234 0.007 | 239 0.003 | 200 0.045 | 220 0.035 | 220 0.021 | 348 0.994 | 314 0.972 | 340 0.843 | 213 0.012 | 231 0.007 | 168 0.100 | 196 0.051 | 320 0.368 | 236 0.120 |
| 104 | CYBERLINK-004 | 475 0.188 | 226 0.007 | 223 0.003 | 248 0.063 | 224 0.036 | 232 0.022 | 533 1.000 | 540 1.000 | 516 0.999 | 219 0.013 | 228 0.007 | 170 0.109 | 192 0.050 | 472 0.954 | 386 0.291 |
| 105 | CYBERLINK-005 | 483 0.208 | 260 0.010 | 258 0.004 | 227 0.054 | 242 0.041 | 258 0.026 | 473 1.000 | 482 1.000 | 354 0.887 | 228 0.014 | 234 0.007 | 158 0.089 | 179 0.043 | 460 0.926 | 374 0.266 |
| 106 | CYBERLINK-006 | 134 0.008 | 102 0.002 | 92 0.001 | 99 0.022 | 100 0.018 | 97 0.012 | 333 0.992 | 351 0.986 | 231 0.339 | 85 0.004 | 91 0.002 | 147 0.073 | 72 0.015 | 246 0.186 | 192 0.096 |
| 107 | DAHUA-000 | 427 0.127 | 478 0.086 | 490 0.044 | 392 0.179 | 411 0.135 | 434 0.083 | | | | | | | | | |
| 108 | DAHUA-001 | 409 0.105 | 466 0.072 | 473 0.036 | 372 0.151 | 399 0.122 | 424 0.075 | 303 0.987 | 330 0.980 | 372 0.933 | | | | | | |
| 109 | DAHUA-002 | 260 0.026 | 305 0.015 | 300 0.006 | 243 0.060 | 261 0.046 | 271 0.029 | 183 0.678 | 217 0.634 | 275 0.517 | 241 0.017 | 244 0.008 | | | 218 0.159 | 248 0.125 |
| 110 | DAHUA-003 | 256 0.025 | 294 0.014 | 280 0.005 | 226 0.054 | 244 0.041 | 245 0.024 | 170 0.643 | 208 0.574 | 257 0.441 | 216 0.013 | 225 0.006 | 154 0.081 | 180 0.043 | 183 0.134 | 214 0.109 |
| 111 | DAHUA-004 | 198 0.014 | 224 0.007 | 215 0.003 | 160 0.033 | 166 0.026 | 156 0.016 | 149 0.547 | 181 0.479 | 229 0.337 | 175 0.008 | 181 0.004 | 106 0.051 | 126 0.027 | 161 0.113 | 186 0.094 |
| 112 | DAON-000 | 434 0.135 | 355 0.022 | 355 0.009 | 283 0.079 | 302 0.061 | 322 0.039 | 466 1.000 | 480 1.000 | 496 0.998 | 281 0.025 | 294 0.013 | 218 0.173 | 257 0.091 | 437 0.846 | 306 0.172 |
| 113 | DAON-001 | 369 0.063 | 412 0.043 | 462 0.031 | 387 0.173 | 414 0.140 | 449 0.096 | 252 0.953 | 289 0.952 | 386 0.951 | 364 0.064 | 402 0.048 | 205 0.154 | 264 0.098 | 222 0.161 | 258 0.129 |
| 114 | DAON-002 | 221 0.018 | 227 0.007 | 233 0.003 | 366 0.139 | 254 0.044 | 249 0.024 | | | 552 1.000 | 233 0.015 | 223 0.006 | 186 0.141 | 185 0.044 | 527 1.000 | 419 0.395 |
| 115 | DAON-003 | 122 0.008 | 130 0.003 | 121 0.002 | 149 0.031 | 142 0.023 | 140 0.015 | 304 0.987 | 139 0.319 | 140 0.140 | 133 0.006 | 137 0.003 | 138 0.067 | 136 0.028 | 121 0.096 | 129 0.079 |
| 116 | DATECH-000 | 312 0.041 | 235 0.007 | 230 0.003 | 318 0.104 | 235 0.038 | 208 0.020 | 388 0.998 | 409 0.996 | 425 0.982 | 237 0.016 | 210 0.005 | 330 0.926 | 347 0.632 | 302 0.311 | 219 0.111 |
| 117 | DATECH-001 | 303 0.037 | 220 0.006 | 206 0.002 | 384 0.168 | 303 0.062 | 185 0.018 | 371 0.996 | 380 0.993 | 403 0.971 | 327 0.039 | 235 0.007 | 357 0.998 | 364 0.875 | 318 0.360 | 322 0.184 |
| 118 | DECATUR-000 | 316 0.043 | 358 0.023 | 361 0.010 | 291 0.085 | 311 0.066 | 330 0.040 | 193 0.754 | 223 0.671 | 270 0.503 | 289 0.027 | 307 0.014 | 217 0.173 | 265 0.098 | 271 0.239 | 294 0.156 |
| 119 | DEEPGLINT-001 | 157 0.010 | 124 0.003 | 142 0.002 | 83 0.018 | 79 0.014 | 73 0.010 | 492 1.000 | 469 1.000 | 268 0.498 | 136 0.006 | 179 0.004 | | | 217 0.159 | 193 0.097 |
| 120 | DEEPSEA-001 | 382 0.073 | 423 0.046 | 432 0.021 | 352 0.129 | 376 0.101 | 389 0.059 | 311 0.988 | 347 0.984 | 408 0.973 | 377 0.077 | 391 0.041 | | | 310 0.326 | 365 0.251 |
| 121 | DERMALOG-003 | 555 0.549 | 577 0.482 | 582 0.360 | 528 0.715 | 546 0.655 | 566 0.526 | | | | 494 0.677 | 500 0.554 | | | 442 0.870 | 482 0.791 |
| 122 | DERMALOG-004 | 556 0.553 | 576 0.479 | 581 0.358 | 526 0.711 | 547 0.657 | 564 0.526 | 373 0.997 | 401 0.995 | 453 0.991 | 485 0.603 | 496 0.458 | | | 439 0.856 | 477 0.751 |
| 123 | DERMALOG-005 | 476 0.189 | 481 0.088 | 485 0.043 | 408 0.201 | 426 0.154 | 450 0.096 | 367 0.996 | 366 0.990 | 385 0.949 | 449 0.300 | 473 0.267 | | | 390 0.614 | 428 0.459 |
| 124 | DERMALOG-006 | 401 0.095 | 432 0.051 | 450 0.025 | 363 0.137 | 380 0.105 | 403 0.067 | 314 0.989 | 333 0.981 | 373 0.933 | 358 0.059 | 374 0.031 | | | 307 0.318 | 354 0.230 |
| 125 | DERMALOG-007 | 474 0.188 | 479 0.086 | 482 0.040 | 407 0.200 | 424 0.152 | 446 0.093 | 368 0.996 | 365 0.990 | 384 0.949 | 388 0.099 | 405 0.052 | | | 329 0.557 | 390 0.299 |
| 126 | DERMALOG-008 | 507 0.267 | 419 0.045 | 409 0.017 | 422 0.231 | 363 0.094 | 377 0.054 | 491 1.000 | 516 1.000 | 529 1.000 | 354 0.057 | 362 0.025 | 272 0.382 | 295 0.158 | 467 0.940 | 464 0.678 |
| 127 | DERMALOG-009 | 310 0.041 | 341 0.020 | 351 0.009 | 294 0.086 | 312 0.066 | 333 0.040 | 476 1.000 | 500 1.000 | 523 1.000 | 303 0.031 | 319 0.016 | 361 0.999 | 388 0.999 | 434 0.840 | 348 0.222 |
| 128 | DERMALOG-010 | 226 0.018 | 223 0.007 | 238 0.003 | 359 0.134 | 288 0.055 | 239 0.023 | 440 1.000 | 448 0.999 | 388 0.951 | 385 0.089 | 393 0.041 | 368 1.000 | 373 0.971 | 368 0.522 | 343 0.211 |
| 129 | DERMALOG-011 | 325 0.046 | 348 0.021 | 323 0.007 | 402 0.193 | 351 0.087 | 325 0.039 | 416 0.999 | 433 0.998 | 429 0.985 | 402 0.129 | 299 0.013 | 348 0.991 | 341 0.567 | 418 0.764 | 353 0.228 |
| 130 | DERMALOG-012 | 313 0.041 | 347 0.021 | 329 0.007 | 297 0.089 | 314 0.067 | 328 0.040 | 241 0.932 | 183 0.505 | 216 0.294 | 273 0.023 | 277 0.011 | 197 0.147 | 231 0.076 | 334 0.409 | 245 0.123 |
| 131 | DERMALOG-013 | 311 0.041 | 346 0.021 | 326 0.007 | 303 0.095 | 317 0.068 | 327 0.040 | 350 0.995 | 278 0.908 | 221 0.306 | 274 0.024 | 280 0.011 | 204 0.153 | 232 0.076 | 323 0.541 | 254 0.128 |
| 132 | DERMALOG-014 | 314 0.042 | 350 0.021 | 346 0.008 | 304 0.095 | 318 0.069 | 335 0.041 | 351 0.995 | 277 0.907 | 223 0.312 | 293 0.028 | 323 0.016 | 208 0.156 | 239 0.080 | 341 0.432 | 280 0.144 |
| 133 | DERMALOG-015 | 167 0.011 | 186 0.005 | 198 0.002 | 179 0.039 | 199 0.031 | 218 0.021 | 209 0.809 | 136 0.313 | 144 0.143 | 182 0.009 | 193 0.005 | 130 0.065 | 155 0.033 | 216 0.158 | 149 0.083 |
| 134 | DERMALOG-016 | 89 0.006 | 84 0.002 | 108 0.001 | 115 0.027 | 111 0.020 | 121 0.014 | 251 0.953 | 131 0.288 | 64 0.088 | 76 0.004 | 102 0.003 | 67 0.035 | 86 0.017 | 374 0.544 | 104 0.073 |
| 135 | DERMALOG-017 | 103 0.006 | 120 0.003 | 134 0.002 | 139 0.030 | 143 0.023 | 171 0.017 | 164 0.608 | 110 0.220 | 92 0.106 | 117 0.006 | 131 0.003 | 81 0.041 | 99 0.020 | 263 0.220 | 106 0.073 |
| 136 | DIGIDATA-000 | 559 0.619 | 583 0.610 | 594 0.598 | 506 0.588 | 536 0.577 | 569 0.560 | 420 0.999 | 388 0.994 | 380 0.941 | 490 0.646 | 504 0.643 | 314 0.789 | 354 0.722 | 432 0.824 | 485 0.816 |
| 137 | DILUSENSE-000 | 345 0.053 | 392 0.030 | 388 0.012 | 313 0.100 | 338 0.078 | 355 0.047 | 216 0.850 | 219 0.651 | 265 0.482 | 325 0.039 | 352 0.022 | 296 0.664 | 311 0.242 | 257 0.203 | 293 0.154 |
| 138 | DILUSENSE-001 | 223 0.018 | 242 0.008 | 242 0.003 | 206 0.046 | 217 0.034 | 226 0.022 | 199 0.769 | 171 0.450 | 226 0.332 | 234 0.015 | 266 0.010 | 288 0.511 | 269 0.109 | 187 0.137 | 220 0.112 |

*Table 62:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 139 | EINETWORKSINDIA-000 | 353 0.055 | 309 0.015 | 299 0.006 | 245 0.063 | 255 0.044 | 260 0.026 | 401 0.998 | 419 0.997 | 447 0.991 | 258 0.020 | 259 0.010 | 284 0.455 | 204 0.057 | 219 0.159 | 261 0.132 |
| 140 | EINETWORKSINDIA-001 | 403 0.096 | 315 0.016 | 295 0.006 | 285 0.080 | 263 0.048 | 259 0.026 | 398 0.998 | 417 0.997 | 443 0.990 | 371 0.072 | 256 0.009 | 339 0.957 | 234 0.077 | 422 0.782 | 315 0.178 |
| 141 | EINETWORKSINDIA-002 | 130 0.008 | 149 0.004 | 135 0.002 | 123 0.028 | 128 0.022 | 115 0.014 | 63 0.243 | 89 0.188 | 105 0.114 | 160 0.007 | 118 0.003 | 337 0.953 | 353 0.701 | 175 0.127 | 112 0.074 |
| 142 | EYEDEA-003 | 549 0.507 | 566 0.387 | 572 0.265 | 512 0.625 | 534 0.543 | 553 0.404 | 372 0.997 | 392 0.994 | 438 0.989 | 482 0.570 | 488 0.392 | | | 425 0.792 | 461 0.658 |
| 143 | F8-001 | 541 0.458 | 523 0.166 | 470 0.035 | | | | 415 0.999 | 437 0.998 | 476 0.995 | | | | | | |
| 144 | FACEHAWK-001 | 460 0.168 | 502 0.123 | 520 0.073 | 534 0.771 | 545 0.654 | 537 0.289 | 232 0.913 | 258 0.851 | 314 0.700 | 504 0.844 | 497 0.502 | 551 1.000 | 410 1.000 | 487 0.990 | 492 0.900 |
| 145 | FACEPHI-000 | 183 0.012 | 218 0.006 | 229 0.003 | 189 0.042 | 218 0.034 | 248 0.024 | 133 0.487 | 150 0.367 | 187 0.236 | 194 0.009 | 219 0.006 | 133 0.066 | 143 0.030 | 184 0.135 | 241 0.121 |
| 146 | FASTLAHORE-000 | 540 0.457 | 538 0.214 | 536 0.098 | 608 1.000 | 614 1.000 | 605 1.000 | 507 1.000 | 508 1.000 | 522 1.000 | 542 1.000 | 566 1.000 | 477 1.000 | 441 1.000 | 567 1.000 | 569 1.000 |
| 147 | FCI-000 | 284 0.033 | 214 0.006 | 231 0.003 | 155 0.033 | 169 0.026 | 196 0.019 | 278 0.979 | 292 0.957 | 336 0.820 | 180 0.008 | 173 0.004 | 312 0.777 | 338 0.538 | 142 0.103 | 133 0.079 |
| 148 | FINCORE-000 | 473 0.187 | 510 0.133 | 517 0.071 | 437 0.267 | 458 0.217 | 483 0.140 | 457 1.000 | 488 1.000 | 466 0.995 | 427 0.187 | 447 0.108 | 292 0.598 | 329 0.418 | 350 0.458 | 402 0.349 |
| 149 | FIRSTCREDITKZ-001 | 97 0.006 | 108 0.003 | 90 0.001 | 107 0.025 | 105 0.019 | 109 0.013 | 117 0.372 | 129 0.282 | 158 0.167 | 156 0.007 | 169 0.004 | 123 0.061 | 138 0.028 | 124 0.097 | 132 0.079 |
| 150 | FISSIONSFR-001 | 443 0.141 | 500 0.121 | 539 0.105 | 403 0.194 | 444 0.177 | 495 0.153 | 226 0.893 | 260 0.859 | 327 0.789 | 430 0.193 | 459 0.169 | 335 0.950 | 356 0.748 | 360 0.490 | 425 0.441 |
| 151 | FPT-000 | 431 0.129 | 252 0.009 | 256 0.004 | 515 0.643 | 506 0.372 | 313 0.036 | 418 0.999 | 438 0.999 | 446 0.991 | 457 0.360 | 442 0.100 | 589 1.000 | 399 1.000 | 464 0.935 | 463 0.668 |
| 152 | FUJITSULAB-000 | 498 0.246 | 342 0.020 | 342 0.008 | 269 0.070 | 291 0.056 | 306 0.035 | | | | 275 0.024 | 301 0.013 | 220 0.177 | 262 0.093 | 272 0.240 | 295 0.156 |
| 153 | FUJITSULAB-001 | 563 0.655 | 329 0.018 | 312 0.006 | 330 0.112 | 293 0.058 | 293 0.033 | 363 0.996 | 377 0.992 | 379 0.940 | 277 0.024 | 285 0.011 | 303 0.739 | 323 0.310 | 277 0.247 | 284 0.146 |
| 154 | GLORY-000 | 537 0.440 | 564 0.367 | 578 0.294 | 505 0.586 | 535 0.547 | 560 0.470 | 352 0.995 | 394 0.995 | 458 0.993 | 472 0.453 | 487 0.381 | | | 433 0.839 | 483 0.795 |
| 155 | GLORY-001 | 524 0.354 | 553 0.305 | 569 0.236 | 503 0.582 | 533 0.537 | 558 0.448 | 343 0.994 | 381 0.993 | 445 0.991 | 464 0.408 | 483 0.336 | | | 430 0.819 | 478 0.753 |
| 156 | GORILLA-001 | 571 0.743 | 570 0.406 | 570 0.246 | 507 0.590 | 521 0.453 | 542 0.314 | 516 1.000 | 534 1.000 | 548 1.000 | 473 0.468 | 479 0.299 | | | 598 1.000 | 469 0.710 |
| 157 | GORILLA-002 | 506 0.265 | 532 0.188 | 540 0.105 | 456 0.342 | 485 0.268 | 505 0.170 | 504 1.000 | 542 1.000 | 459 0.993 | 440 0.250 | 455 0.137 | | | 507 1.000 | 432 0.466 |
| 158 | GORILLA-003 | 568 0.690 | 557 0.317 | 554 0.157 | 520 0.684 | 516 0.434 | 530 0.247 | 560 1.000 | 558 1.000 | 540 1.000 | 463 0.407 | 465 0.213 | | | 547 1.000 | 447 0.562 |
| 159 | GORILLA-004 | 433 0.135 | 485 0.089 | 486 0.043 | 409 0.202 | 432 0.160 | 458 0.101 | 261 0.972 | 293 0.958 | 356 0.902 | 407 0.135 | 426 0.072 | | | 344 0.438 | 392 0.309 |
| 160 | GORILLA-005 | 395 0.086 | 451 0.058 | 452 0.026 | 391 0.179 | 418 0.142 | 438 0.088 | 198 0.768 | 228 0.697 | 281 0.548 | 384 0.088 | 389 0.040 | | | 305 0.315 | 349 0.223 |
| 161 | GORILLA-006 | 330 0.046 | 378 0.027 | 371 0.010 | 338 0.118 | 355 0.089 | 373 0.053 | 161 0.597 | 197 0.525 | 237 0.361 | 291 0.028 | 296 0.013 | 214 0.166 | 261 0.093 | 262 0.218 | 292 0.154 |
| 162 | GORILLA-007 | 326 0.046 | 375 0.027 | 366 0.010 | 314 0.101 | 337 0.077 | 350 0.045 | 167 0.622 | 199 0.529 | 238 0.361 | 283 0.026 | 286 0.012 | 244 0.264 | 268 0.108 | 238 0.178 | 273 0.138 |
| 163 | GORILLA-008 | 318 0.043 | 359 0.024 | 345 0.008 | 328 0.111 | 346 0.083 | 358 0.048 | 147 0.536 | 173 0.457 | 211 0.286 | 300 0.030 | 282 0.011 | 260 0.319 | 256 0.090 | 236 0.178 | 262 0.132 |
| 164 | GRIAULE-000 | 319 0.043 | 339 0.020 | 349 0.008 | 287 0.082 | 306 0.063 | 320 0.038 | 384 0.997 | 398 0.995 | 387 0.951 | 310 0.033 | 343 0.020 | 225 0.185 | 267 0.107 | 254 0.198 | 300 0.166 |
| 165 | GRIAULE-001 | 190 0.012 | 171 0.004 | 160 0.002 | 220 0.051 | 180 0.028 | 160 0.016 | 238 0.927 | 261 0.863 | 294 0.621 | 150 0.007 | 136 0.003 | 351 0.995 | 346 0.610 | 128 0.099 | 141 0.081 |
| 166 | GTELOTS-000 | 487 0.217 | 536 0.206 | 561 0.198 | 448 0.307 | 478 0.256 | 526 0.232 | 200 0.770 | 232 0.727 | 305 0.651 | 447 0.292 | 476 0.282 | 499 1.000 | 429 1.000 | 386 0.603 | 451 0.577 |
| 167 | GTELOTS-001 | 379 0.069 | 424 0.046 | 453 0.026 | 334 0.115 | 354 0.088 | 370 0.052 | 224 0.883 | 227 0.680 | 254 0.432 | 372 0.072 | 404 0.051 | 251 0.287 | 290 0.150 | 297 0.294 | 360 0.239 |
| 168 | HIK-003 | 455 0.156 | 490 0.102 | 504 0.057 | 401 0.190 | 428 0.158 | 463 0.105 | 280 0.980 | 306 0.968 | 367 0.924 | 410 0.142 | 431 0.080 | | | 347 0.445 | 406 0.359 |
| 169 | HIK-004 | 454 0.155 | 486 0.098 | 502 0.054 | 395 0.182 | 425 0.153 | 459 0.101 | 291 0.983 | 318 0.976 | 382 0.946 | 408 0.137 | 430 0.077 | | | 343 0.434 | 404 0.353 |
| 170 | HIK-005 | 406 0.102 | 414 0.043 | 416 0.019 | 308 0.098 | 336 0.077 | 359 0.048 | 443 1.000 | 466 0.999 | 491 0.998 | 365 0.068 | 382 0.036 | | | 372 0.541 | 368 0.258 |
| 171 | HIK-006 | 445 0.141 | 425 0.047 | 420 0.020 | 327 0.111 | 350 0.086 | 371 0.052 | 518 1.000 | 532 1.000 | 519 0.999 | | | | | | |
| 172 | HISIGN-001 | 212 0.015 | 231 0.007 | 232 0.003 | 240 0.058 | 229 0.037 | 230 0.022 | 409 0.999 | 361 0.990 | 248 0.397 | 225 0.013 | 227 0.007 | 319 0.824 | 305 0.194 | 199 0.143 | 215 0.109 |
| 173 | HYPERVERGE-001 | 148 0.009 | 165 0.004 | 181 0.002 | 180 0.039 | 196 0.031 | 212 0.020 | 78 0.267 | 102 0.210 | 134 0.136 | 152 0.007 | 168 0.004 | 111 0.053 | 131 0.027 | 135 0.101 | 150 0.083 |
| 174 | HYPERVERGE-002 | 131 0.008 | 138 0.003 | 147 0.002 | 163 0.034 | 174 0.027 | 187 0.018 | 83 0.270 | 96 0.201 | 115 0.120 | 113 0.006 | 123 0.003 | 98 0.048 | 111 0.023 | 112 0.093 | 125 0.077 |
| 175 | HYPERVERGE-003 | 210 0.015 | 230 0.007 | 227 0.003 | 225 0.053 | 247 0.042 | 264 0.026 | 112 0.346 | 125 0.274 | 166 0.174 | 208 0.012 | 212 0.005 | 129 0.065 | 152 0.033 | 146 0.106 | 158 0.085 |
| 176 | HYUNDAI-ROBOTICSLAB-000 | 252 0.025 | 292 0.013 | 311 0.006 | 517 0.673 | 520 0.452 | 413 0.070 | 223 0.876 | 224 0.672 | 255 0.439 | 476 0.492 | 464 0.197 | 380 1.000 | 391 0.999 | 441 0.870 | 460 0.654 |
| 177 | HYUNDAI-ROBOTICSLAB-001 | 264 0.027 | 284 0.013 | 290 0.005 | 548 0.921 | 559 0.774 | 543 0.322 | 284 0.981 | 256 0.839 | 259 0.444 | 498 0.699 | 490 0.404 | 404 1.000 | 411 1.000 | 465 0.937 | 484 0.807 |
| 178 | HYUNDAI-ROBOTICSLAB-002 | 173 0.011 | 196 0.005 | 165 0.002 | 489 0.463 | 448 0.189 | 217 0.021 | 110 0.344 | 117 0.228 | 125 0.129 | 367 0.068 | 245 0.008 | 381 1.000 | 326 0.981 | 346 0.440 | 305 0.171 |
| 179 | HYUNDAI-ROBOTICSLAB-003 | 168 0.011 | 192 0.005 | 155 0.002 | 260 0.067 | 215 0.034 | 170 0.017 | 101 0.311 | 112 0.221 | 124 0.127 | 253 0.019 | 186 0.004 | 354 0.995 | 357 0.781 | 270 0.236 | 204 0.102 |
| 180 | HZAILU-000 | 292 0.035 | 337 0.019 | 350 0.008 | 249 0.064 | 271 0.051 | 284 0.031 | 289 0.983 | 303 0.967 | 335 0.811 | 262 0.020 | 262 0.010 | 256 0.316 | 235 0.077 | 207 0.153 | 239 0.120 |
| 181 | HZAILU-001 | 215 0.016 | 243 0.008 | 263 0.004 | 475 0.414 | 447 0.183 | 251 0.024 | 397 0.998 | 354 0.986 | 203 0.273 | 432 0.196 | 348 0.021 | 530 1.000 | 385 0.997 | 403 0.679 | 407 0.360 |
| 182 | HZAILU-002 | 205 0.015 | 238 0.008 | 260 0.004 | 219 0.050 | 238 0.039 | 250 0.024 | 318 0.990 | 229 0.701 | 213 0.291 | 209 0.012 | 215 0.006 | 135 0.066 | 160 0.034 | 312 0.330 | 217 0.110 |
| 183 | HZAILU-003 | 206 0.015 | 240 0.008 | 261 0.004 | 222 0.051 | 234 0.038 | 233 0.022 | 187 0.711 | 201 0.545 | 206 0.278 | 193 0.009 | 189 0.004 | 126 0.062 | 149 0.031 | 166 0.119 | 191 0.096 |
| 184 | HZAILU-004 | 147 0.009 | 164 0.004 | 167 0.002 | 191 0.042 | 200 0.031 | 195 0.019 | 145 0.528 | 156 0.390 | 191 0.241 | 108 0.006 | 108 0.003 | 91 0.045 | 100 0.021 | 139 0.102 | 164 0.087 |

*Table 63:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 185 | HZAILU-005 | 107 0.007 | 132 0.003 | 125 0.002 | 154 0.033 | 154 0.025 | 150 0.015 | 127 0.463 | 141 0.329 | 175 0.199 | 72 0.004 | 69 0.002 | 58 0.033 | 70 0.015 | 115 0.094 | 128 0.078 |
| 186 | HZAILU-006 | 116 0.007 | 129 0.003 | 119 0.001 | 124 0.028 | 129 0.022 | 134 0.015 | 138 0.497 | 140 0.325 | 174 0.198 | 69 0.004 | 65 0.002 | 52 0.030 | 64 0.014 | 106 0.091 | 124 0.077 |
| 187 | I2V-001 | 582 0.851 | 428 0.049 | 268 0.004 | 580 0.994 | 586 0.966 | 414 0.071 | 468 1.000 | 427 1.000 | 416 0.977 | 509 0.905 | 310 0.014 | 459 1.000 | 350 0.683 | 483 0.983 | 398 0.328 |
| 188 | ICEHOUSE-000 | 517 0.298 | 484 0.089 | 468 0.033 | 547 0.917 | 571 0.852 | 525 0.223 | 452 1.000 | 481 1.000 | 506 0.999 | 393 0.111 | 406 0.052 | 582 1.000 | 409 1.000 | 355 0.470 | 400 0.346 |
| 189 | ICEHOUSE-001 | 230 0.019 | 255 0.009 | 248 0.003 | 235 0.056 | 246 0.041 | 254 0.025 | 141 0.510 | 146 0.348 | 177 0.204 | 220 0.013 | 226 0.007 | 167 0.098 | 189 0.046 | 242 0.179 | 187 0.094 |
| 190 | ICM-000 | 358 0.058 | 368 0.026 | 391 0.012 | 331 0.113 | 362 0.092 | 397 0.062 | 298 0.984 | 301 0.965 | 396 0.960 | 320 0.036 | 354 0.022 | 241 0.241 | 281 0.136 | 273 0.242 | 339 0.208 |
| 191 | ICM-001 | 535 0.421 | 389 0.029 | 363 0.010 | 551 0.942 | 474 0.251 | 366 0.051 | 515 1.000 | 494 1.000 | 456 0.993 | 452 0.307 | 314 0.015 | 386 1.000 | 283 0.145 | 443 0.878 | 309 0.173 |
| 192 | IDEMIA-003 | 554 0.541 | 426 0.047 | 427 0.021 | 612 1.000 | 435 0.165 | 428 0.079 | | | 586 1.000 | 400 0.123 | 412 0.061 | | | 420 0.766 | 454 0.630 |
| 193 | IDEMIA-004 | 351 0.054 | 407 0.037 | 426 0.021 | 368 0.144 | 395 0.118 | 427 0.079 | 270 0.975 | 315 0.973 | 398 0.967 | 399 0.123 | 413 0.061 | | | 419 0.766 | 455 0.630 |
| 194 | IDEMIA-005 | 373 0.065 | 416 0.044 | 449 0.025 | 394 0.181 | 423 0.150 | 461 0.102 | 279 0.979 | 323 0.977 | 406 0.973 | 403 0.130 | 424 0.070 | | | 444 0.879 | 474 0.743 |
| 195 | IDEMIA-006 | 370 0.064 | 413 0.043 | 447 0.025 | 436 0.266 | 463 0.226 | 498 0.161 | 294 0.984 | 338 0.982 | 420 0.980 | 411 0.144 | 437 0.090 | | | 412 0.733 | 441 0.531 |
| 196 | IDEMIA-007 | 294 0.035 | 328 0.018 | 336 0.008 | 273 0.073 | 287 0.055 | 294 0.033 | 601 1.000 | 597 1.000 | 582 1.000 | 346 0.052 | 355 0.022 | 223 0.182 | 270 0.109 | 599 1.000 | 509 0.982 |
| 197 | IDEMIA-008 | 53 0.003 | 70 0.002 | 74 0.001 | 75 0.016 | 74 0.013 | 68 0.009 | 80 0.267 | 93 0.195 | 122 0.126 | 89 0.005 | 111 0.003 | 68 0.036 | 95 0.019 | 147 0.106 | 180 0.092 |
| 198 | IDEMIA-009 | 51 0.003 | 52 0.001 | 51 0.001 | 42 0.012 | 48 0.011 | 48 0.008 | 40 0.193 | 49 0.131 | 66 0.088 | 55 0.003 | 62 0.002 | 39 0.027 | 58 0.013 | 49 0.074 | 63 0.064 |
| 199 | IDEMIA-010 | 21 0.002 | 10 0.001 | 18 0.001 | 21 0.009 | 21 0.008 | 32 0.007 | 48 0.219 | 38 0.120 | 40 0.068 | 37 0.002 | 42 0.001 | 42 0.028 | 30 0.009 | 40 0.070 | 44 0.058 |
| 200 | IDEMIA-011 | 17 0.002 | 19 0.001 | 49 0.001 | 29 0.010 | 41 0.009 | 54 0.008 | 38 0.192 | 11 0.070 | 14 0.050 | 20 0.002 | 29 0.001 | 33 0.025 | 24 0.008 | 16 0.061 | 21 0.052 |
| 201 | IDEMIA-012 | 12 0.002 | 17 0.001 | 40 0.001 | 25 0.010 | 28 0.009 | 46 0.008 | 8 0.116 | 9 0.069 | 10 0.048 | 35 0.002 | 39 0.001 | 17 0.016 | 16 0.007 | 19 0.061 | 19 0.051 |
| 202 | IDEMIA-013 | 1 0.001 | 1 0.000 | 1 0.000 | 1 0.006 | 1 0.006 | 2 0.005 | 157 0.569 | 128 0.282 | 7 0.044 | 5 0.001 | 6 0.001 | 5 0.010 | 3 0.004 | 9 0.055 | 11 0.048 |
| 203 | IDENTY-000 | 612 1.000 | 615 1.000 | 619 1.000 | 592 0.999 | 598 0.999 | 603 0.999 | 592 1.000 | 605 1.000 | 564 1.000 | 521 0.991 | 528 0.991 | 370 1.000 | 400 1.000 | 490 0.992 | 516 0.992 |
| 204 | IDENTY-001 | 112 0.007 | 113 0.003 | 123 0.002 | 104 0.024 | 110 0.020 | 131 0.014 | 76 0.264 | 67 0.162 | 83 0.103 | 88 0.005 | 85 0.002 | 99 0.048 | 113 0.023 | 119 0.094 | 145 0.082 |
| 205 | IDENTY-002 | 109 0.007 | 109 0.003 | 100 0.001 | 94 0.021 | 94 0.016 | 87 0.011 | 59 0.240 | 63 0.152 | 73 0.092 | 86 0.005 | 79 0.002 | 93 0.047 | 105 0.022 | 70 0.080 | 70 0.066 |
| 206 | IMAGUS-002 | 589 0.907 | 588 0.749 | 592 0.564 | 552 0.944 | 565 0.816 | 576 0.645 | 510 1.000 | 537 1.000 | 533 1.000 | | | | | | |
| 207 | IMAGUS-003 | 588 0.897 | 591 0.807 | 595 0.669 | 558 0.954 | 575 0.909 | 586 0.809 | 519 1.000 | 506 1.000 | 530 1.000 | | | | | | |
| 208 | IMAGUS-005 | 285 0.034 | 332 0.018 | 337 0.008 | 295 0.088 | 310 0.066 | 331 0.040 | 236 0.925 | 255 0.836 | 302 0.642 | 294 0.029 | 321 0.016 | 213 0.161 | 263 0.094 | 267 0.231 | 325 0.189 |
| 209 | IMAGUS-006 | 306 0.039 | 336 0.019 | 339 0.008 | 300 0.093 | 319 0.069 | 340 0.042 | 282 0.980 | 273 0.896 | 292 0.617 | 292 0.028 | 311 0.015 | 212 0.161 | 259 0.092 | 282 0.260 | 319 0.181 |
| 210 | IMAGUS-007 | 317 0.043 | 357 0.022 | 360 0.009 | 312 0.100 | 328 0.073 | 347 0.045 | 263 0.973 | 271 0.892 | 303 0.647 | 304 0.031 | 316 0.016 | 215 0.169 | 266 0.098 | 287 0.265 | 318 0.181 |
| 211 | IMAGUS-008 | 600 0.995 | 601 0.974 | 590 0.523 | 562 0.958 | 560 0.774 | 536 0.285 | 449 1.000 | 404 0.996 | 313 0.697 | 479 0.520 | 425 0.071 | 321 1.000 | 339 0.540 | 362 0.518 | 363 0.246 |
| 212 | IMDS-SOFTWARE-000 | 225 0.018 | 217 0.006 | 212 0.003 | 468 0.389 | 209 0.033 | 198 0.019 | 487 1.000 | 518 1.000 | 549 1.000 | 217 0.013 | 200 0.005 | 236 0.216 | 176 0.041 | 528 1.000 | 597 1.000 |
| 213 | IMDS-SOFTWARE-001 | 254 0.025 | 221 0.006 | 210 0.003 | 540 0.834 | 213 0.034 | 186 0.018 | 483 1.000 | 521 1.000 | 538 1.000 | 256 0.019 | 195 0.005 | 281 0.424 | 203 0.055 | 509 1.000 | 618 1.000 |
| 214 | IMDS-SOFTWARE-002 | 224 0.018 | 216 0.006 | 209 0.003 | 467 0.389 | 208 0.033 | 197 0.019 | 486 1.000 | 519 1.000 | 550 1.000 | 218 0.013 | 199 0.005 | 235 0.216 | 175 0.041 | 529 1.000 | 598 1.000 |
| 215 | IMPERIAL-000 | 450 0.154 | 366 0.026 | 357 0.009 | 298 0.089 | 315 0.068 | 337 0.041 | 505 1.000 | 443 0.999 | 465 0.995 | 332 0.042 | 344 0.020 | | | 276 0.245 | 301 0.168 |
| 216 | INCODE-000 | 534 0.421 | 554 0.310 | 562 0.199 | 495 0.486 | 514 0.420 | 541 0.304 | 450 1.000 | 430 0.998 | 463 0.994 | | | | | | |
| 217 | INCODE-001 | 519 0.317 | 537 0.211 | 542 0.111 | 459 0.348 | 492 0.296 | 514 0.198 | 523 1.000 | 531 1.000 | 525 1.000 | | | | | | |
| 218 | INCODE-002 | 515 0.285 | 529 0.184 | 538 0.100 | 452 0.333 | 486 0.269 | 508 0.176 | 386 0.998 | 382 0.993 | 413 0.975 | | | | | | |
| 219 | INCODE-003 | 514 0.282 | 525 0.167 | 525 0.083 | 466 0.372 | 481 0.264 | 501 0.164 | 462 1.000 | 463 0.999 | 480 0.996 | | | | | | |
| 220 | INCODE-004 | 405 0.099 | 442 0.053 | 439 0.023 | 383 0.167 | 398 0.120 | 412 0.070 | 383 0.997 | 393 0.994 | 369 0.928 | 362 0.063 | 373 0.031 | | | 303 0.313 | 351 0.226 |
| 221 | INCODE-005 | 233 0.020 | 271 0.011 | 266 0.004 | 234 0.055 | 252 0.043 | 262 0.026 | 165 0.609 | 193 0.522 | 239 0.365 | 243 0.017 | 248 0.009 | 194 0.145 | 227 0.073 | 213 0.155 | 227 0.116 |
| 222 | INCODE-006 | 49 0.003 | 55 0.001 | 62 0.001 | 44 0.013 | 50 0.011 | 63 0.009 | 52 0.230 | 65 0.159 | 75 0.093 | 63 0.003 | 83 0.002 | 35 0.026 | 56 0.013 | 67 0.079 | 78 0.067 |
| 223 | INCODE-007 | 30 0.003 | 49 0.001 | 31 0.001 | 38 0.012 | 29 0.009 | 19 0.006 | 45 0.205 | 59 0.146 | 72 0.092 | 32 0.002 | 33 0.001 | 48 0.028 | 45 0.011 | 98 0.089 | 62 0.064 |
| 224 | INNOV8TIF-000 | 549 0.487 | 572 0.414 | 576 0.286 | 504 0.584 | 529 0.517 | 551 0.378 | 345 0.994 | 372 0.991 | 427 0.983 | 483 0.581 | 491 0.421 | 329 0.922 | 359 0.825 | 424 0.792 | 466 0.702 |
| 225 | INNOVATRICS-002 | 531 0.374 | 540 0.233 | 551 0.139 | 473 0.403 | 498 0.310 | 521 0.209 | 509 1.000 | 538 1.000 | 520 0.999 | | | | | | |
| 226 | INNOVATRICS-003 | 516 0.296 | 541 0.221 | 547 0.131 | 461 0.351 | 493 0.297 | 517 0.203 | 463 1.000 | 474 1.000 | 494 0.998 | | | | | | |
| 227 | INNOVATRICS-004 | 469 0.183 | 507 0.131 | 521 0.073 | 434 0.262 | 462 0.222 | 489 0.149 | 293 0.983 | 328 0.980 | 405 0.972 | | | | | | |
| 228 | INNOVATRICS-005 | 356 0.056 | 397 0.034 | 397 0.014 | 333 0.114 | 356 0.089 | 372 0.052 | 225 0.889 | 257 0.845 | 316 0.720 | 341 0.047 | 353 0.022 | | | 279 0.251 | 320 0.182 |
| 229 | INNOVATRICS-007 | 243 0.023 | 280 0.012 | 281 0.005 | 251 0.065 | 274 0.051 | 287 0.032 | 207 0.803 | 235 0.740 | 282 0.562 | 242 0.017 | 258 0.009 | 163 0.093 | 202 0.053 | 209 0.154 | 238 0.120 |
| 230 | INNOVATRICS-008 | 250 0.024 | 185 0.005 | 184 0.002 | 364 0.138 | 225 0.036 | 184 0.018 | 498 1.000 | 513 1.000 | 473 0.995 | 222 0.013 | 233 0.007 | 324 0.886 | 330 0.430 | 190 0.139 | 212 0.108 |

*Table 64:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 231 | INNOVATRICS-009 | 106 0.007 | 114 0.003 | 146 0.002 | 144 0.031 | 156 0.025 | 180 0.018 | 129 0.475 | 142 0.338 | 173 0.197 | 148 0.007 | 180 0.004 | 104 0.050 | 122 0.025 | 148 0.106 | 178 0.090 |
| 232 | INNOVATRICS-010 | 108 0.007 | 126 0.003 | 161 0.002 | 152 0.032 | 157 0.025 | 179 0.018 | 140 0.500 | 145 0.347 | 180 0.206 | 155 0.007 | 177 0.004 | 114 0.054 | 132 0.027 | 156 0.110 | 184 0.093 |
| 233 | INNOVATRICS-011 | 79 0.005 | 81 0.002 | 88 0.001 | 81 0.017 | 83 0.014 | 83 0.010 | 184 0.684 | 194 0.523 | 178 0.204 | 25 0.004 | 97 0.003 | 27 0.039 | 87 0.017 | 88 0.087 | 92 0.069 |
| 234 | INNOVATRICS-012 | 61 0.004 | 72 0.002 | 81 0.001 | 51 0.014 | 52 0.011 | 55 0.008 | 65 0.247 | 73 0.168 | 87 0.104 | 57 0.003 | 60 0.002 | 38 0.027 | 54 0.012 | 77 0.083 | 68 0.066 |
| 235 | INNOVATRICS-013 | 55 0.004 | 58 0.001 | 59 0.001 | 95 0.022 | 91 0.015 | 67 0.009 | 53 0.230 | 61 0.147 | 65 0.088 | 123 0.006 | 84 0.002 | 100 0.048 | 83 0.016 | 95 0.088 | 60 0.063 |
| 236 | INNOVATRICS-014 | 32 0.002 | 36 0.001 | 45 0.001 | 17 0.009 | 17 0.008 | 11 0.006 | 61 0.241 | 33 0.106 | 38 0.066 | 27 0.002 | 31 0.001 | 15 0.015 | 22 0.007 | 28 0.066 | 34 0.056 |
| 237 | INNOVATRICS-015 | 29 0.002 | 24 0.001 | 38 0.001 | 14 0.008 | 14 0.007 | 9 0.006 | 163 0.604 | 45 0.126 | 36 0.065 | 21 0.002 | 14 0.001 | 13 0.015 | 17 0.007 | 63 0.078 | 33 0.056 |
| 238 | INSPUR-000 | 208 0.015 | 213 0.006 | 194 0.002 | 563 0.963 | 576 0.915 | 585 0.740 | 467 1.000 | 487 1.000 | 503 0.999 | 513 0.936 | 501 0.584 | 467 1.000 | 460 1.000 | 496 0.998 | 505 0.966 |
| 239 | INSPUR-001 | 175 0.011 | 191 0.005 | 170 0.002 | 192 0.043 | 172 0.027 | 152 0.016 | 488 1.000 | 501 1.000 | 515 0.999 | 172 0.008 | 166 0.004 | 119 0.057 | 130 0.027 | 462 0.931 | 242 0.121 |
| 240 | INSPUR-002 | 182 0.012 | 198 0.005 | 164 0.002 | 193 0.043 | 189 0.029 | 164 0.016 | 517 1.000 | 451 0.999 | 421 0.980 | 166 0.008 | 158 0.004 | 110 0.052 | 120 0.024 | 474 0.957 | 190 0.095 |
| 241 | INTELIGENSIA-000 | 343 0.050 | 360 0.024 | 369 0.010 | 306 0.097 | 335 0.077 | 362 0.049 | 285 0.981 | 245 0.784 | 269 0.501 | 350 0.053 | 381 0.036 | 239 0.235 | 285 0.145 | 280 0.255 | 337 0.208 |
| 242 | INTELIGENSIA-001 | 185 0.012 | 197 0.005 | 174 0.002 | 174 0.037 | 178 0.028 | 169 0.017 | 203 0.787 | 172 0.451 | 129 0.132 | 165 0.008 | 178 0.004 | 139 0.067 | 150 0.031 | 179 0.129 | 224 0.115 |
| 243 | INTELLIVISION-001 | 550 0.508 | 551 0.279 | 555 0.158 | 488 0.459 | 511 0.404 | 540 0.302 | 478 1.000 | 485 1.000 | 517 0.999 | 454 0.328 | 467 0.219 | 307 0.749 | 343 0.598 | 406 0.685 | 448 0.562 |
| 244 | INTELLIVISION-002 | 512 0.281 | 516 0.154 | 518 0.072 | 423 0.236 | 450 0.196 | 478 0.127 | 425 0.999 | 446 0.999 | 487 0.997 | 406 0.134 | 427 0.073 | 283 0.437 | 322 0.297 | 351 0.460 | 401 0.348 |
| 245 | INTELRESEARCH-001 | 70 0.004 | 66 0.001 | 60 0.001 | 62 0.015 | 67 0.012 | 64 0.009 | 121 0.396 | 126 0.276 | 128 0.132 | 93 0.005 | 117 0.003 | 53 0.030 | 74 0.015 | 72 0.080 | 77 0.067 |
| 246 | INTEMA-000 | 139 0.008 | 89 0.002 | 83 0.001 | 101 0.022 | 96 0.017 | 98 0.012 | | 520 1.000 | 68 0.090 | 92 0.005 | 86 0.002 | 252 0.288 | 178 0.042 | 73 0.081 | 80 0.067 |
| 247 | INTEMA-001 | 13 0.002 | 38 0.001 | 36 0.001 | 508 0.604 | 539 0.603 | 571 0.602 | 10 0.127 | 22 0.092 | 31 0.062 | 73 0.004 | 67 0.002 | 298 0.715 | 321 0.294 | 56 0.076 | 54 0.062 |
| 248 | INTOZI-001 | 280 0.031 | 310 0.015 | 291 0.006 | 566 0.968 | 582 0.938 | 570 0.575 | 502 1.000 | 510 1.000 | 539 1.000 | 526 0.998 | 519 0.884 | 509 1.000 | 608 1.000 | 552 1.000 | 523 0.999 |
| 249 | INTSYSMSU-000 | 603 0.999 | 605 0.998 | 607 0.990 | 595 1.000 | 599 1.000 | 601 0.998 | 458 1.000 | 475 1.000 | 490 0.998 | 527 0.999 | 527 0.989 | | | 500 0.999 | 514 0.988 |
| 250 | IONETWORKS-001 | 411 0.108 | 300 0.014 | 314 0.007 | 238 0.057 | 259 0.045 | 272 0.029 | 460 1.000 | 483 1.000 | 495 0.998 | 235 0.016 | 249 0.009 | 181 0.129 | 218 0.065 | 296 0.294 | 221 0.112 |
| 251 | IREX-000 | 327 0.068 | 384 0.028 | 335 0.008 | 311 0.099 | 301 0.060 | 289 0.032 | 310 0.988 | 291 0.956 | 309 0.676 | 336 0.044 | 276 0.011 | 255 0.302 | 212 0.062 | 231 0.170 | 268 0.135 |
| 252 | IRISID-001 | 268 0.028 | 293 0.013 | 283 0.005 | 254 0.066 | 272 0.051 | 277 0.030 | 142 0.511 | 166 0.422 | 212 0.287 | 282 0.025 | 278 0.011 | 198 0.148 | 228 0.075 | 225 0.162 | 263 0.132 |
| 253 | IRISID-002 | 234 0.021 | 266 0.010 | 265 0.004 | 242 0.059 | 258 0.045 | 268 0.027 | 130 0.479 | 160 0.407 | 201 0.271 | 261 0.020 | 265 0.010 | 183 0.133 | 219 0.067 | 227 0.164 | 260 0.130 |
| 254 | ISTL-001 | 239 0.023 | 286 0.013 | 296 0.006 | 428 0.249 | 327 0.073 | 314 0.036 | 131 0.481 | 152 0.378 | 194 0.255 | 343 0.048 | 261 0.010 | 367 1.000 | 370 0.952 | 333 0.407 | 271 0.138 |
| 255 | ISYSTEMS-002 | 452 0.154 | 470 0.078 | 465 0.032 | 378 0.161 | 403 0.126 | 430 0.080 | 396 0.998 | 421 0.998 | 457 0.993 | | | | | | |
| 256 | ISYSTEMS-003 | 481 0.203 | 452 0.058 | 442 0.024 | 360 0.135 | 386 0.107 | 406 0.068 | 427 1.000 | 486 1.000 | 486 0.997 | | | | | | |
| 257 | KAKAO-000 | 267 0.028 | 307 0.015 | 306 0.006 | 271 0.071 | 290 0.056 | 304 0.034 | 146 0.534 | 177 0.462 | 225 0.319 | 257 0.019 | 260 0.010 | 187 0.141 | 229 0.075 | 215 0.158 | 237 0.120 |
| 258 | KAKAO-001 | 90 0.006 | 97 0.002 | 111 0.001 | 97 0.022 | 99 0.017 | 107 0.013 | 47 0.218 | 62 0.150 | 70 0.090 | 80 0.004 | 71 0.002 | 87 0.042 | 80 0.016 | 48 0.074 | 57 0.063 |
| 259 | KAKAO-002 | 71 0.004 | 79 0.002 | 94 0.001 | 78 0.017 | 84 0.014 | 89 0.011 | 46 0.217 | 50 0.134 | 56 0.083 | 58 0.003 | 56 0.002 | 72 0.037 | 52 0.012 | 43 0.070 | 45 0.058 |
| 260 | KEDACOM-001 | 309 0.040 | 356 0.022 | 393 0.013 | 305 0.096 | 325 0.072 | 376 0.054 | 315 0.989 | 352 0.986 | 409 0.973 | 352 0.055 | 395 0.043 | | | 299 0.305 | 373 0.264 |
| 261 | KEYLESS-000 | 189 0.012 | 205 0.006 | 191 0.002 | 185 0.041 | 185 0.029 | 172 0.017 | 347 0.994 | 356 0.987 | 261 0.455 | 207 0.011 | 165 0.004 | 350 0.994 | 365 0.888 | 204 0.149 | 200 0.099 |
| 262 | KLOUDSPOT-001 | 378 0.069 | 361 0.024 | 362 0.010 | | | 479 0.128 | 192 0.752 | 202 0.550 | 246 0.381 | | 439 0.094 | | | | 422 0.406 |
| 263 | KNERON-000 | | | 467 0.033 | | | 452 0.099 | | | | | | | | | |
| 264 | KNERON-001 | | | 498 0.051 | | | | | | | | | | | | |
| 265 | KNOWUTECH-000 | 218 0.017 | 250 0.009 | 247 0.003 | 216 0.049 | 236 0.038 | 236 0.023 | 120 0.392 | 134 0.310 | 172 0.196 | 214 0.012 | 224 0.006 | 161 0.091 | 190 0.046 | 177 0.129 | 207 0.103 |
| 266 | KOGNIZA-000 | 181 0.012 | 209 0.006 | 213 0.003 | 170 0.036 | 179 0.028 | 194 0.019 | 185 0.692 | 170 0.444 | 195 0.255 | 101 0.005 | 106 0.003 | 103 0.050 | 94 0.019 | 90 0.087 | 87 0.069 |
| 267 | LEBENTECH-000 | 610 1.000 | 610 1.000 | 612 1.000 | 582 0.995 | 573 0.882 | 577 0.659 | 530 1.000 | 545 1.000 | 559 1.000 | 480 0.527 | 481 0.319 | 379 1.000 | 371 0.955 | 473 0.954 | 462 0.665 |
| 268 | LIGHTVISION-000 | 396 0.086 | 377 0.027 | 373 0.010 | 549 0.925 | 561 0.797 | 416 0.072 | 465 1.000 | 491 1.000 | 502 0.998 | 311 0.033 | 324 0.017 | 397 1.000 | 412 1.000 | 260 0.214 | 290 0.153 |
| 269 | LINE-000 | 365 0.062 | 393 0.030 | 389 0.012 | 356 0.132 | 367 0.095 | 378 0.054 | | | 544 1.000 | 339 0.046 | 349 0.021 | 247 0.278 | 292 0.151 | 619 1.000 | 375 0.268 |
| 270 | LINE-001 | 276 0.030 | 176 0.004 | 157 0.002 | 252 0.066 | 171 0.027 | 149 0.015 | 489 1.000 | 517 1.000 | 551 1.000 | 187 0.009 | 174 0.004 | 143 0.072 | 161 0.034 | 526 1.000 | 491 0.858 |
| 271 | LINECLOVA-002 | 154 0.009 | 146 0.004 | 143 0.002 | 496 0.508 | 405 0.130 | 125 0.014 | 332 0.992 | 332 0.980 | 285 0.572 | 329 0.040 | 152 0.004 | 377 1.000 | 351 0.690 | 408 0.700 | 179 0.091 |
| 272 | LINECLOVA-003 | 66 0.004 | 100 0.002 | 82 0.001 | 509 0.612 | 540 0.606 | 572 0.603 | | | 547 1.000 | 107 0.006 | 94 0.002 | 342 0.974 | 296 0.158 | 155 0.110 | 196 0.097 |
| 273 | LOOKMAN-003 | 371 0.065 | 415 0.043 | 445 0.024 | 355 0.131 | 390 0.112 | 431 0.082 | | | | 382 0.084 | 414 0.061 | | | 317 0.355 | 391 0.304 |
| 274 | LOOKMAN-004 | 383 0.073 | 418 0.044 | 441 0.024 | 344 0.123 | 383 0.105 | 422 0.075 | 275 0.978 | 320 0.977 | 411 0.974 | | | | | | |
| 275 | LOOKMAN-005 | 338 0.050 | 391 0.030 | 407 0.017 | 315 0.102 | 348 0.086 | 399 0.063 | 281 0.980 | 322 0.977 | 407 0.973 | 361 0.062 | 397 0.047 | | | 301 0.308 | 378 0.273 |
| 276 | LSSYTECH-000 | 47 0.003 | 48 0.001 | 48 0.001 | 33 0.011 | 38 0.009 | 40 0.008 | 54 0.231 | 71 0.167 | 93 0.106 | 51 0.003 | 53 0.001 | 34 0.026 | 39 0.011 | 69 0.080 | 93 0.069 |

*Table 65:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*



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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 277 | LUXAND-001 | 441 0.140 | 448 0.057 | 414 0.019 | 499 0.553 | 508 0.385 | 497 0.161 | 567 1.000 | 402 0.995 | 317 0.723 | 491 0.662 | 469 0.222 | 328 0.909 | 352 0.698 | 450 0.903 | 449 0.562 |
| 278 | MANTRA-000 | 374 0.066 | 261 0.010 | 245 0.003 | 244 0.063 | 243 0.041 | 231 0.022 | 556 1.000 | 561 1.000 | 513 0.999 | 296 0.029 | 305 0.014 | 201 0.152 | 244 0.081 | 506 1.000 | 288 0.151 |
| 279 | MAVERICKRESEARCH-000 | 439 0.138 | 474 0.084 | 483 0.040 | 426 0.245 | 442 0.176 | 464 0.105 | 377 0.997 | 374 0.992 | 326 0.785 | 397 0.122 | 411 0.058 | 273 0.394 | 308 0.220 | 375 0.546 | 379 0.279 |
| 280 | MAXVISION-000 | 333 0.048 | 385 0.028 | 390 0.012 | 491 0.468 | 466 0.237 | 379 0.054 | 213 0.825 | 242 0.764 | 297 0.626 | 412 0.149 | 350 0.022 | 355 0.997 | 363 0.872 | 378 0.557 | 362 0.245 |
| 281 | MAXVISION-001 | 152 0.009 | 158 0.004 | 152 0.002 | 196 0.044 | 160 0.025 | 146 0.015 | 86 0.274 | 101 0.210 | 121 0.126 | 144 0.007 | 128 0.003 | 336 0.951 | 334 0.485 | 134 0.100 | 127 0.078 |
| 282 | MAXVISION-002 | 129 0.008 | 141 0.004 | 154 0.002 | 130 0.029 | 131 0.022 | 126 0.014 | 64 0.246 | 88 0.188 | 109 0.116 | 140 0.007 | 130 0.003 | 325 0.895 | 194 0.050 | 144 0.104 | 162 0.086 |
| 283 | MAXVISION-003 | 179 0.012 | 204 0.006 | 195 0.002 | 223 0.051 | 239 0.039 | 229 0.022 | 97 0.304 | 119 0.237 | 143 0.142 | 205 0.011 | 204 0.005 | 264 0.348 | 286 0.145 | 158 0.111 | 163 0.086 |
| 284 | MAXVISION-004 | 83 0.005 | 90 0.002 | 85 0.001 | 108 0.025 | 109 0.020 | 128 0.014 | 25 0.171 | 40 0.122 | 50 0.078 | 71 0.004 | 75 0.002 | 346 0.990 | 361 0.846 | 76 0.082 | 98 0.071 |
| 285 | MEGVII-001 | 484 0.209 | 465 0.072 | 472 0.036 | 340 0.119 | 371 0.097 | 391 0.061 | | | | | | | | | |
| 286 | MEGVII-002 | 501 0.251 | 469 0.077 | 475 0.037 | 341 0.120 | 369 0.096 | 388 0.059 | 412 0.999 | 436 0.998 | 349 0.870 | | | | | | |
| 287 | MEGVII-003 | 87 0.006 | 95 0.002 | 103 0.001 | 48 0.013 | 55 0.011 | 65 0.009 | 77 0.265 | 58 0.145 | 69 0.090 | 62 0.003 | 64 0.002 | 41 0.027 | 63 0.014 | 61 0.078 | 86 0.069 |
| 288 | MEGVII-004 | 16 0.002 | 32 0.001 | 55 0.001 | 19 0.009 | 24 0.008 | 47 0.008 | 42 0.196 | 12 0.072 | 16 0.053 | 11 0.001 | 28 0.001 | 4 0.009 | 11 0.005 | 27 0.065 | 49 0.060 |
| 289 | MEGVII-005 | 100 0.006 | 33 0.001 | 56 0.001 | 80 0.017 | 40 0.009 | 49 0.008 | 158 0.580 | 82 0.181 | 17 0.053 | 111 0.006 | 26 0.001 | 388 1.000 | 304 0.190 | 163 0.117 | 47 0.058 |
| 290 | METSAKUURCOMPANY-000 | 217 0.017 | 229 0.007 | 222 0.003 | 289 0.084 | 223 0.036 | 192 0.018 | 423 0.999 | 449 0.999 | 470 0.995 | 379 0.081 | 268 0.010 | 353 0.995 | 367 0.915 | 405 0.683 | 372 0.264 |
| 291 | METSAKUURCOMPANY-001 | 451 0.154 | 251 0.009 | 241 0.003 | 241 0.059 | 197 0.031 | 191 0.018 | 490 1.000 | 470 1.000 | 489 0.998 | 198 0.010 | 202 0.005 | 145 0.073 | 166 0.036 | 436 0.841 | 423 0.416 |
| 292 | MIAXIS-001 | 211 0.015 | 237 0.008 | 240 0.003 | 188 0.042 | 205 0.033 | 221 0.021 | 268 0.975 | 290 0.954 | 328 0.792 | 230 0.014 | 230 0.007 | 203 0.153 | 181 0.043 | 235 0.176 | 274 0.139 |
| 293 | MIAXIS-002 | 216 0.016 | 241 0.008 | 250 0.003 | 209 0.046 | 226 0.036 | 247 0.024 | 179 0.666 | 200 0.536 | 242 0.367 | 240 0.017 | 240 0.008 | 226 0.192 | 200 0.052 | 226 0.163 | 270 0.135 |
| 294 | MICROFOCUS-003 | 595 0.958 | 598 0.931 | 602 0.866 | 576 0.988 | 589 0.979 | 597 0.948 | | | | 520 0.982 | 523 0.945 | | | 489 0.991 | 507 0.977 |
| 295 | MICROFOCUS-004 | 604 0.999 | 606 0.999 | 611 0.999 | 575 0.984 | 587 0.975 | 596 0.940 | | | | 516 0.974 | 522 0.935 | | | 485 0.989 | 506 0.976 |
| 296 | MICROFOCUS-005 | 587 0.882 | 595 0.835 | 600 0.736 | 556 0.951 | 580 0.928 | 590 0.865 | | | | 512 0.935 | 518 0.848 | | | 484 0.985 | 504 0.965 |
| 297 | MICROFOCUS-006 | 598 0.983 | 602 0.978 | 604 0.963 | 555 0.950 | 579 0.923 | 589 0.858 | | | | 511 0.923 | 516 0.843 | | | 478 0.971 | 499 0.939 |
| 298 | MICROSOFT-003 | 334 0.048 | 380 0.027 | 382 0.011 | 335 0.117 | 359 0.091 | 382 0.056 | | | | 321 0.036 | 340 0.019 | | | 269 0.233 | 311 0.176 |
| 299 | MICROSOFT-004 | 329 0.046 | 367 0.026 | 374 0.010 | 329 0.111 | 352 0.087 | 375 0.053 | | | | 312 0.033 | 334 0.018 | | | 264 0.222 | 304 0.170 |
| 300 | MICROSOFT-005 | 331 0.047 | 364 0.025 | 368 0.010 | 299 0.090 | 322 0.070 | 338 0.041 | 432 0.999 | 209 0.582 | 233 0.347 | 286 0.027 | 298 0.013 | | | 243 0.180 | 267 0.134 |
| 301 | MICROSOFT-006 | 251 0.025 | 276 0.012 | 304 0.006 | 212 0.048 | 227 0.037 | 246 0.024 | 124 0.446 | 153 0.379 | 202 0.272 | 307 0.032 | 312 0.015 | | | 237 0.178 | 272 0.138 |
| 302 | MOBBL-000 | 360 0.059 | 383 0.028 | 381 0.011 | 324 0.107 | 333 0.077 | 345 0.045 | 321 0.990 | 346 0.984 | 364 0.918 | 319 0.036 | 330 0.017 | 293 0.621 | 282 0.145 | 266 0.228 | 303 0.170 |
| 303 | MOBBL-001 | 368 0.063 | 399 0.035 | 423 0.020 | 325 0.109 | 342 0.080 | 360 0.049 | 320 0.990 | 342 0.983 | 359 0.911 | 357 0.058 | 390 0.040 | 294 0.622 | 297 0.160 | 291 0.275 | 350 0.225 |
| 304 | MOBBL-002 | 489 0.218 | 540 0.215 | 566 0.213 | 417 0.217 | 455 0.207 | 513 0.197 | 364 0.996 | 363 0.990 | 318 0.729 | 445 0.285 | 477 0.283 | 356 0.997 | 362 0.866 | 380 0.558 | 444 0.553 |
| 305 | MUKH-002 | 605 0.999 | 581 0.594 | 541 0.110 | 450 0.326 | 472 0.242 | 494 0.153 | 514 1.000 | 498 1.000 | 433 0.987 | 422 0.170 | 436 0.089 | 304 0.741 | 327 0.382 | 326 0.389 | 381 0.286 |
| 306 | NAVERCLOUD-000 | 114 0.007 | 139 0.003 | 127 0.002 | 113 0.026 | 115 0.020 | 108 0.013 | 306 0.987 | 311 0.971 | 230 0.337 | 95 0.005 | 112 0.003 | 76 0.039 | 93 0.019 | 114 0.094 | 136 0.079 |
| 307 | NEC-000 | 415 0.111 | 472 0.079 | 495 0.046 | 385 0.171 | 417 0.140 | 442 0.093 | 290 0.983 | 326 0.979 | 401 0.968 | | | | | 357 0.474 | 416 0.377 |
| 308 | NEC-001 | 449 0.148 | 492 0.105 | 509 0.060 | 424 0.238 | 452 0.197 | 481 0.133 | 327 0.991 | 353 0.986 | 404 0.972 | 405 0.133 | 434 0.082 | | | 354 0.468 | 417 0.378 |
| 309 | NEC-002 | 219 0.017 | 101 0.002 | 116 0.001 | 136 0.029 | 112 0.020 | 113 0.013 | 441 1.000 | 460 0.999 | 467 0.995 | 169 0.008 | 213 0.005 | | | 402 0.676 | 387 0.292 |
| 310 | NEC-003 | 69 0.004 | 96 0.002 | 141 0.002 | 92 0.021 | 98 0.017 | 111 0.013 | 228 0.901 | 251 0.822 | 296 0.623 | 174 0.008 | 216 0.006 | 69 0.036 | 107 0.023 | 401 0.668 | 369 0.261 |
| 311 | NEC-004 | 28 0.002 | 63 0.001 | 91 0.001 | 66 0.015 | 73 0.013 | 79 0.010 | 173 0.649 | 214 0.618 | 283 0.570 | 82 0.004 | 153 0.004 | 20 0.019 | 53 0.012 | 132 0.100 | 170 0.088 |
| 312 | NEC-005 | 101 0.006 | 57 0.001 | 72 0.001 | 56 0.014 | 62 0.012 | 69 0.009 | 227 0.900 | 222 0.669 | 159 0.168 | 65 0.003 | 92 0.002 | 19 0.019 | 38 0.011 | 127 0.099 | 167 0.087 |
| 313 | NEC-006 | 153 0.009 | 75 0.002 | 87 0.001 | 105 0.024 | 102 0.018 | 105 0.013 | 217 0.855 | 175 0.457 | 100 0.112 | 70 0.004 | 105 0.003 | 37 0.026 | 57 0.013 | 118 0.094 | 142 0.081 |
| 314 | NEC-007 | 14 0.002 | 35 0.001 | 52 0.001 | 31 0.010 | 31 0.009 | 36 0.007 | 55 0.232 | 54 0.137 | 51 0.078 | 29 0.002 | 44 0.001 | 6 0.010 | 13 0.006 | 26 0.065 | 39 0.057 |
| 315 | NEC-008 | 19 0.002 | 44 0.001 | 61 0.001 | 27 0.010 | 27 0.009 | 41 0.008 | 19 0.157 | 20 0.089 | 26 0.060 | 24 0.002 | 43 0.001 | 3 0.008 | 9 0.005 | 21 0.062 | 28 0.055 |
| 316 | NEC-009 | 18 0.002 | 14 0.001 | 33 0.001 | 9 0.007 | 10 0.007 | 22 0.007 | 24 0.165 | 10 0.070 | 9 0.048 | 9 0.001 | 23 0.001 | 2 0.005 | 2 0.003 | 7 0.052 | 10 0.048 |
| 317 | NEC-010 | 11 0.002 | 7 0.001 | 10 0.001 | 2 0.006 | 5 0.006 | 7 0.006 | 34 0.184 | 8 0.067 | 1 0.043 | 7 0.001 | 11 0.001 | 1 0.004 | 1 0.003 | 2 0.050 | 7 0.046 |
| 318 | NEC-011 | 9 0.001 | 6 0.001 | 7 0.001 | 7 0.007 | 7 0.006 | 5 0.005 | 119 0.380 | 55 0.140 | 4 0.043 | 2 0.001 | 2 0.001 | 313 0.789 | 326 0.354 | 5 0.052 | 1 0.043 |
| 319 | NEC-012 | 2 0.001 | 4 0.001 | 5 0.001 | 3 0.006 | 2 0.006 | 1 0.005 | 13 0.134 | 5 0.056 | 2 0.043 | 1 0.001 | 4 0.001 | 89 0.044 | 12 0.005 | 1 0.049 | 2 0.043 |
| 320 | NEUROTECHNOLOGY-003 | 603 0.999 | 585 0.631 | 537 0.098 | 535 0.773 | 483 0.266 | 500 0.164 | 539 1.000 | 576 1.000 | 607 1.000 | | | | | | |
| 321 | NEUROTECHNOLOGY-004 | 420 0.119 | 458 0.062 | 455 0.027 | 369 0.146 | 392 0.117 | 418 0.073 | 369 0.996 | 387 0.994 | 441 0.990 | | | | | | |
| 322 | NEUROTECHNOLOGY-005 | 418 0.116 | 443 0.054 | 433 0.022 | 429 0.252 | 406 0.130 | 421 0.074 | 410 0.999 | 426 0.998 | 436 0.988 | | | | | | |

*Table 66:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 323 | NEUROTECHNOLOGY-006 | 599 0.987 | 547 0.249 | 544 0.121 | 599 1.000 | 512 0.418 | 519 0.206 | | | | | | | | | |
| 324 | NEUROTECHNOLOGY-007 | 503 0.252 | 457 0.062 | 430 0.021 | 584 0.996 | 440 0.173 | 405 0.068 | 499 1.000 | 495 1.000 | 485 0.997 | 455 0.339 | 383 0.036 | | | 563 1.000 | 515 0.989 |
| 325 | NEUROTECHNOLOGY-008 | 576 0.797 | 437 0.053 | 386 0.012 | 326 0.110 | 341 0.080 | 354 0.047 | 500 1.000 | 512 1.000 | 554 1.000 | 318 0.035 | 329 0.017 | 254 0.293 | 289 0.149 | 256 0.203 | 289 0.152 |
| 326 | NEUROTECHNOLOGY-009 | 261 0.026 | 308 0.015 | 294 0.006 | 255 0.066 | 278 0.052 | 288 0.032 | 177 0.657 | 210 0.583 | 253 0.430 | 260 0.020 | 269 0.010 | 202 0.153 | 249 0.082 | 228 0.165 | 257 0.129 |
| 327 | NEUROTECHNOLOGY-010 | 522 0.346 | 259 0.010 | 235 0.003 | 210 0.047 | 231 0.037 | 243 0.023 | 116 0.370 | 123 0.269 | 154 0.160 | 199 0.010 | 194 0.005 | 151 0.075 | 172 0.039 | 171 0.126 | 197 0.097 |
| 328 | NEUROTECHNOLOGY-012 | 398 0.092 | 222 0.006 | 185 0.002 | 201 0.045 | 204 0.032 | 201 0.019 | 471 1.000 | 295 0.959 | 136 0.139 | 181 0.008 | 161 0.004 | 122 0.061 | 140 0.028 | 454 0.916 | 171 0.088 |
| 329 | NEUROTECHNOLOGY-013 | 359 0.058 | 159 0.004 | 110 0.001 | 140 0.030 | 140 0.023 | 132 0.014 | 266 0.974 | 137 0.316 | 95 0.107 | 134 0.006 | 116 0.003 | 92 0.046 | 104 0.022 | 395 0.641 | 153 0.083 |
| 330 | NEUROTECHNOLOGY-014 | 119 0.007 | 122 0.003 | 104 0.001 | 142 0.031 | 130 0.022 | 110 0.013 | 155 0.568 | 106 0.215 | 94 0.107 | 125 0.006 | 114 0.003 | 108 0.052 | 115 0.024 | 117 0.094 | 116 0.075 |
| 331 | NEUROTECHNOLOGY-015 | 111 0.007 | 128 0.003 | 90 0.001 | 150 0.032 | 150 0.024 | 127 0.014 | 70 0.254 | 75 0.170 | 86 0.104 | 126 0.006 | 107 0.003 | 102 0.049 | 106 0.023 | 87 0.086 | 82 0.068 |
| 332 | NEUROTECHNOLOGY-016 | 117 0.007 | 121 0.003 | 89 0.001 | 125 0.028 | 120 0.021 | 106 0.013 | 58 0.238 | 83 0.182 | 98 0.109 | 151 0.007 | 142 0.003 | 112 0.054 | 121 0.025 | 111 0.093 | 105 0.073 |
| 333 | NEWLAND-002 | 552 0.522 | 573 0.437 | 577 0.294 | 498 0.535 | 522 0.466 | 546 0.335 | 422 0.999 | 450 0.999 | 493 0.998 | | | | | | |
| 334 | NEXTMETALABS-001 | 518 0.305 | 543 0.233 | 553 0.145 | 602 1.000 | 601 1.000 | 594 0.933 | 495 1.000 | 514 1.000 | 553 1.000 | 612 1.000 | 530 0.999 | 534 1.000 | 607 1.000 | 559 1.000 | 575 1.000 |
| 335 | NEXTMETALABS-002 | 135 0.008 | 150 0.004 | 137 0.002 | 443 0.285 | 402 0.125 | 195 0.019 | 74 0.260 | 94 0.198 | 111 0.118 | 404 0.131 | 356 0.023 | 382 1.000 | 390 0.999 | 365 0.510 | 367 0.257 |
| 336 | NEXTMETALABS-003 | 136 0.008 | 151 0.004 | 136 0.002 | 133 0.029 | 133 0.022 | 123 0.014 | 72 0.257 | 92 0.194 | 110 0.117 | 124 0.006 | 127 0.003 | 109 0.052 | 117 0.024 | 150 0.107 | 163 0.087 |
| 337 | NOBLIS-001 | 608 1.000 | 608 1.000 | 608 0.991 | 618 1.000 | 610 1.000 | 615 1.000 | 528 1.000 | 528 1.000 | 545 1.000 | | | | | | |
| 338 | NOBLIS-002 | 606 1.000 | 604 0.997 | 587 0.488 | 619 1.000 | 608 1.000 | 618 1.000 | 528 1.000 | 526 1.000 | 546 1.000 | | | | | | |
| 339 | NOMINDER-000 | 184 0.012 | 177 0.004 | 159 0.002 | 493 0.481 | 489 0.280 | 273 0.029 | 286 0.982 | 284 0.929 | 258 0.441 | 459 0.370 | 435 0.088 | 400 1.000 | 401 1.000 | 426 0.799 | 445 0.557 |
| 340 | NOMINDER-001 | 188 0.012 | 190 0.005 | 197 0.002 | 376 0.156 | 349 0.086 | 210 0.020 | 222 0.875 | 225 0.675 | 207 0.282 | 167 0.008 | 138 0.003 | 378 1.000 | 369 0.948 | 208 0.154 | 146 0.082 |
| 341 | NOMINDER-002 | 176 0.011 | 188 0.005 | 189 0.002 | 199 0.045 | 188 0.029 | 193 0.019 | 103 0.317 | 113 0.223 | 116 0.121 | 137 0.006 | 125 0.003 | 322 0.867 | 211 0.062 | 100 0.089 | 91 0.069 |
| 342 | NOMINDER-003 | 161 0.010 | 181 0.005 | 178 0.002 | 181 0.039 | 176 0.027 | 177 0.018 | 95 0.299 | 98 0.208 | 104 0.114 | 121 0.006 | 115 0.003 | 318 0.816 | 159 0.034 | 83 0.085 | 66 0.065 |
| 343 | NOMINDER-004 | 164 0.010 | 175 0.004 | 179 0.002 | 167 0.035 | 165 0.026 | 176 0.017 | 92 0.296 | 99 0.208 | 107 0.116 | 129 0.006 | 119 0.003 | 261 0.331 | 128 0.027 | 99 0.089 | 85 0.068 |
| 344 | NOMINDER-005 | 156 0.010 | 118 0.003 | 130 0.002 | 195 0.044 | 124 0.022 | 147 0.015 | 93 0.297 | 74 0.169 | 71 0.091 | 112 0.006 | 110 0.003 | 265 0.356 | 139 0.028 | 97 0.088 | 79 0.067 |
| 345 | NOTIONTAG-000 | 281 0.032 | 323 0.017 | 320 0.007 | 277 0.076 | 296 0.059 | 310 0.036 | 180 0.667 | 212 0.606 | 263 0.460 | 268 0.021 | 281 0.011 | 200 0.150 | 251 0.084 | 234 0.176 | 275 0.140 |
| 346 | NTECHLAB-003 | 387 0.079 | 441 0.053 | 456 0.028 | 370 0.148 | 393 0.118 | 423 0.075 | 220 0.872 | 254 0.835 | 325 0.749 | | | | | | |
| 347 | NTECHLAB-004 | 366 0.062 | 409 0.040 | 428 0.021 | 354 0.131 | 381 0.105 | 402 0.065 | 219 0.866 | 253 0.831 | 323 0.743 | 349 0.053 | 371 0.030 | | | 284 0.263 | 346 0.214 |
| 348 | NTECHLAB-005 | 363 0.062 | 411 0.042 | 429 0.021 | 353 0.130 | 378 0.102 | 400 0.063 | 210 0.813 | 243 0.769 | 306 0.657 | 373 0.073 | 387 0.039 | | | 295 0.294 | 352 0.227 |
| 349 | NTECHLAB-006 | 355 0.056 | 403 0.036 | 412 0.018 | 342 0.121 | 364 0.094 | 382 0.059 | 206 0.799 | 241 0.751 | 300 0.631 | 355 0.057 | 376 0.032 | | | 283 0.260 | 335 0.207 |
| 350 | NTECHLAB-007 | 308 0.040 | 363 0.025 | 383 0.012 | 290 0.085 | 313 0.067 | 336 0.041 | 205 0.793 | 240 0.747 | 301 0.638 | 308 0.032 | 331 0.017 | | | 265 0.223 | 310 0.176 |
| 351 | NTECHLAB-008 | 248 0.024 | 298 0.014 | 313 0.006 | 237 0.057 | 260 0.045 | 274 0.029 | 160 0.596 | 195 0.524 | 247 0.384 | 313 0.033 | 335 0.018 | | | 244 0.183 | 276 0.140 |
| 352 | NTECHLAB-009 | 155 0.009 | 183 0.005 | 200 0.002 | 126 0.028 | 134 0.022 | 129 0.014 | 143 0.517 | 167 0.423 | 219 0.303 | 231 0.015 | 242 0.008 | 171 0.109 | 209 0.061 | 195 0.142 | 223 0.114 |
| 353 | NTECHLAB-010 | 76 0.005 | 94 0.002 | 86 0.001 | 86 0.018 | 86 0.015 | 85 0.011 | 107 0.327 | 121 0.243 | 152 0.159 | 142 0.007 | 167 0.004 | 121 0.059 | 147 0.031 | 126 0.098 | 122 0.077 |
| 354 | NTECHLAB-011 | 86 0.006 | 107 0.003 | 97 0.001 | 85 0.018 | 85 0.015 | 81 0.010 | 87 0.282 | 107 0.219 | 141 0.140 | 186 0.009 | 184 0.004 | 148 0.074 | 171 0.038 | 105 0.091 | 115 0.075 |
| 355 | OMNIGARDE-000 | 160 0.010 | 155 0.004 | 158 0.002 | 131 0.029 | 141 0.023 | 130 0.014 | 106 0.324 | 120 0.243 | 147 0.150 | 157 0.007 | 150 0.003 | 115 0.055 | 135 0.027 | 185 0.136 | 181 0.092 |
| 356 | OMNIGARDE-001 | 68 0.004 | 80 0.002 | 67 0.001 | 91 0.019 | 90 0.015 | 76 0.010 | 41 0.193 | 53 0.135 | 62 0.087 | 67 0.003 | 66 0.002 | 54 0.030 | 69 0.015 | 60 0.078 | 83 0.068 |
| 357 | OMNIGARDE-002 | 52 0.003 | 56 0.001 | 58 0.001 | 49 0.013 | 47 0.011 | 50 0.008 | 18 0.156 | 35 0.111 | 52 0.079 | 43 0.002 | 51 0.001 | 24 0.022 | 48 0.012 | 66 0.079 | 81 0.068 |
| 358 | ONFIDO-000 | 299 0.026 | 290 0.013 | 310 0.006 | 198 0.045 | 219 0.035 | 225 0.022 | 273 0.978 | 312 0.971 | 383 0.947 | 298 0.029 | 317 0.016 | 242 0.244 | 273 0.113 | 252 0.194 | 285 0.149 |
| 359 | ONFIDO-001 | 235 0.021 | 268 0.010 | 272 0.005 | 146 0.031 | 144 0.023 | 124 0.014 | 212 0.823 | 237 0.744 | 288 0.592 | 227 0.014 | 222 0.006 | 193 0.142 | 205 0.059 | 206 0.152 | 240 0.121 |
| 360 | OPTIEXACTA-000 | 26 0.002 | 29 0.001 | 27 0.001 | 22 0.010 | 23 0.008 | 26 0.007 | 36 0.186 | 52 0.135 | 59 0.085 | 59 0.003 | 70 0.002 | 61 0.033 | 73 0.015 | 42 0.070 | 48 0.059 |
| 361 | PANGIAM-000 | 195 0.013 | 208 0.006 | 193 0.002 | 183 0.039 | 191 0.030 | 189 0.018 | 264 0.974 | 135 0.310 | 156 0.165 | 192 0.009 | 190 0.005 | 185 0.136 | 157 0.033 | 145 0.105 | 152 0.083 |
| 362 | PANGIAM-001 | 238 0.023 | 272 0.011 | 338 0.008 | 178 0.039 | 190 0.030 | 209 0.020 | 171 0.646 | 151 0.376 | 160 0.171 | 185 0.009 | 164 0.004 | 321 0.860 | 241 0.081 | 193 0.141 | 159 0.085 |
| 363 | PAPSAV1923-000 | 110 0.007 | 117 0.003 | 98 0.001 | 116 0.027 | 117 0.021 | 117 0.014 | 96 0.300 | 114 0.226 | 139 0.140 | 195 0.009 | 220 0.006 | 101 0.049 | 114 0.024 | 116 0.094 | 121 0.076 |
| 364 | PARAVISION-000 | 509 0.271 | 483 0.088 | 492 0.045 | 484 0.447 | 437 0.170 | 455 0.100 | 485 1.000 | 454 0.999 | 481 0.997 | 474 0.470 | 493 0.443 | | | 459 0.926 | 481 0.779 |
| 365 | PARAVISION-001 | 440 0.139 | 427 0.049 | 425 0.020 | 414 0.207 | 404 0.128 | 420 0.074 | 529 1.000 | 440 0.999 | 462 0.994 | 471 0.444 | 492 0.428 | | | 413 0.739 | 450 0.573 |
| 366 | PARAVISION-002 | 394 0.085 | 429 0.050 | 435 0.022 | 374 0.152 | 396 0.119 | 425 0.076 | 331 0.992 | 341 0.983 | 324 0.746 | 378 0.080 | 396 0.043 | | | 363 0.497 | 376 0.268 |
| 367 | PARAVISION-003 | 367 0.063 | 398 0.035 | 402 0.015 | 347 0.124 | 368 0.096 | 390 0.060 | 380 0.997 | 389 0.994 | 319 0.730 | 356 0.058 | 379 0.034 | | | 298 0.296 | 356 0.232 |
| 368 | PARAVISION-004 | 253 0.025 | 262 0.010 | 264 0.004 | 215 0.049 | 252 0.038 | 244 0.024 | 508 1.000 | 539 1.000 | 332 0.794 | 249 0.018 | 274 0.011 | | | 452 0.908 | 342 0.211 |

*Table 67:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 369 | PARAVISION-005 | 201 0.014 | 153 0.004 | 176 0.002 | 145 0.031 | 151 0.024 | 162 0.016 | 376 0.997 | 329 0.980 | 161 0.171 | 205 0.011 | 243 0.008 | | | 181 0.132 | 235 0.120 |
| 370 | PARAVISION-007 | 332 0.047 | 147 0.004 | 122 0.002 | 501 0.560 | 159 0.025 | 145 0.015 | 501 1.000 | 511 1.000 | 556 1.000 | 183 0.009 | 221 0.006 | 173 0.113 | 116 0.024 | 556 1.000 | 580 1.000 |
| 371 | PARAVISION-009 | 105 0.007 | 99 0.002 | 71 0.001 | 109 0.026 | 106 0.019 | 93 0.012 | 201 0.775 | 234 0.732 | 280 0.545 | 60 0.003 | 58 0.002 | 60 0.033 | 75 0.015 | 47 0.073 | 52 0.061 |
| 372 | PARAVISION-012 | 62 0.004 | 50 0.001 | 39 0.001 | 71 0.016 | 68 0.012 | 61 0.009 | 148 0.541 | 178 0.469 | 200 0.263 | 50 0.002 | 54 0.002 | 32 0.025 | 50 0.012 | 37 0.068 | 38 0.057 |
| 373 | PARAVISION-014 | 45 0.003 | 41 0.001 | 37 0.001 | 55 0.014 | 56 0.011 | 53 0.008 | 159 0.595 | 198 0.527 | 185 0.235 | 34 0.002 | 36 0.001 | 62 0.033 | 55 0.012 | 20 0.061 | 20 0.052 |
| 374 | PARAVISION-016 | 35 0.002 | 31 0.001 | 32 0.001 | 59 0.015 | 53 0.011 | 45 0.008 | 17 0.156 | 36 0.112 | 41 0.068 | 22 0.002 | 17 0.001 | 43 0.028 | 42 0.011 | 23 0.065 | 26 0.055 |
| 375 | PARAVISION-019 | 41 0.003 | 28 0.001 | 17 0.001 | 53 0.014 | 43 0.010 | 34 0.007 | 50 0.226 | 79 0.176 | 97 0.108 | 19 0.002 | 15 0.001 | 49 0.028 | 40 0.011 | 18 0.061 | 14 0.049 |
| 376 | PARAVISION-021 | 34 0.002 | 26 0.001 | 15 0.001 | 54 0.014 | 46 0.011 | 30 0.007 | 20 0.159 | 44 0.124 | 43 0.074 | 16 0.001 | 13 0.001 | 46 0.028 | 41 0.011 | 17 0.061 | 13 0.049 |
| 377 | PENTACOMP-000 | 601 0.996 | 603 0.996 | 609 0.996 | 589 0.998 | 597 0.998 | 602 0.998 | 322 0.991 | 369 0.990 | 440 0.990 | 515 0.955 | 524 0.955 | 473 1.000 | 405 1.000 | 475 0.960 | 501 0.958 |
| 378 | PENTACOMP-001 | 290 0.034 | 319 0.016 | 331 0.008 | 481 0.436 | 438 0.172 | 311 0.036 | 181 0.667 | 188 0.517 | 244 0.374 | 376 0.076 | 300 0.013 | 596 1.000 | 386 0.998 | 340 0.430 | 312 0.177 |
| 379 | PIXELALL-002 | 564 0.664 | 491 0.105 | 460 0.029 | 571 0.974 | 509 0.388 | 433 0.083 | | 533 1.000 | 535 1.000 | 484 0.602 | 400 0.047 | | | 601 1.000 | 525 1.000 |
| 380 | PIXELALL-003 | 335 0.048 | 351 0.021 | 353 0.009 | 316 0.102 | 326 0.073 | 341 0.043 | | 471 1.000 | 492 0.998 | 324 0.037 | 341 0.020 | | | 377 0.554 | 366 0.255 |
| 381 | PIXELALL-004 | 421 0.120 | 331 0.018 | 318 0.007 | 536 0.783 | 340 0.079 | 316 0.037 | | 502 1.000 | 508 0.999 | 344 0.051 | 315 0.015 | | | 491 0.994 | 500 0.942 |
| 382 | PIXELALL-005 | 386 0.078 | 274 0.012 | 267 0.004 | 487 0.456 | 267 0.050 | 267 0.027 | | 525 1.000 | 521 0.999 | 287 0.027 | 325 0.017 | 228 0.203 | 224 0.071 | 505 1.000 | 510 0.983 |
| 383 | PRIVID-000 | 299 0.036 | 372 0.027 | 437 0.022 | 568 0.968 | 567 0.825 | 520 0.207 | 448 1.000 | 266 0.882 | 176 0.199 | 486 0.603 | 463 0.195 | 562 1.000 | 556 1.000 | 480 0.972 | 488 0.842 |
| 384 | PROTENET-000 | 586 0.875 | 589 0.794 | 591 0.540 | 544 0.884 | 568 0.834 | 581 0.682 | 309 0.988 | 327 0.980 | 370 0.930 | 501 0.804 | 502 0.587 | 519 1.000 | 408 1.000 | 446 0.890 | 475 0.747 |
| 385 | PROTENET-001 | 296 0.035 | 345 0.021 | 352 0.009 | 398 0.187 | 382 0.105 | 342 0.043 | 233 0.914 | 203 0.551 | 235 0.347 | 398 0.123 | 375 0.032 | 430 1.000 | 406 1.000 | 388 0.610 | 412 0.367 |
| 386 | PSL-000 | 93 0.006 | 98 0.002 | 69 0.001 | 135 0.029 | 119 0.021 | 99 0.012 | 91 0.295 | 92 0.208 | 106 0.115 | 102 0.005 | 82 0.002 | 113 0.054 | 124 0.026 | 125 0.097 | 110 0.074 |
| 387 | PSL-001 | 75 0.005 | 88 0.002 | 63 0.001 | 98 0.022 | 95 0.017 | 84 0.011 | 28 0.173 | 41 0.122 | 46 0.075 | 64 0.003 | 55 0.002 | 45 0.028 | 60 0.013 | 50 0.075 | 50 0.060 |
| 388 | PSL-002 | 48 0.003 | 39 0.001 | 21 0.001 | 64 0.015 | 54 0.011 | 39 0.008 | 16 0.155 | 25 0.095 | 24 0.059 | 39 0.002 | 25 0.001 | 29 0.023 | 37 0.010 | 64 0.079 | 51 0.060 |
| 389 | PSL-003 | 63 0.004 | 42 0.001 | 20 0.001 | 52 0.014 | 51 0.011 | 33 0.007 | 43 0.202 | 31 0.102 | 35 0.064 | 36 0.002 | 35 0.001 | 27 0.022 | 33 0.010 | 141 0.102 | 42 0.058 |
| 390 | PSL-004 | 50 0.003 | 30 0.001 | 12 0.001 | 39 0.012 | 36 0.009 | 21 0.007 | 29 0.174 | 30 0.101 | 37 0.065 | 23 0.002 | 18 0.001 | 28 0.022 | 36 0.010 | 55 0.076 | 40 0.058 |
| 391 | PSL-005 | 40 0.003 | 40 0.001 | 44 0.001 | 35 0.011 | 33 0.009 | 29 0.007 | 7 0.115 | 16 0.083 | 30 0.062 | 28 0.002 | 19 0.001 | 31 0.024 | 49 0.012 | 25 0.065 | 35 0.057 |
| 392 | PSL-006 | 33 0.002 | 20 0.001 | 11 0.001 | 32 0.011 | 22 0.008 | 15 0.006 | 9 0.123 | 19 0.085 | 19 0.058 | 15 0.001 | 9 0.001 | 21 0.019 | 28 0.009 | 30 0.067 | 25 0.054 |
| 393 | PTAKURATSATU-000 | 357 0.057 | 402 0.036 | 408 0.017 | 382 0.165 | 401 0.124 | 415 0.071 | 249 0.947 | 281 0.923 | 346 0.867 | 340 0.046 | 351 0.022 | 231 0.206 | 275 0.120 | 268 0.232 | 317 0.179 |
| 394 | PYRAMID-000 | 128 0.008 | 144 0.004 | 132 0.002 | 117 0.027 | 116 0.021 | 114 0.013 | 60 0.241 | 90 0.190 | 117 0.122 | 146 0.007 | 126 0.003 | 282 0.433 | 199 0.051 | 120 0.095 | 111 0.074 |
| 395 | PYRAMID-001 | 139 0.008 | 173 0.004 | 175 0.002 | 229 0.054 | 262 0.047 | 323 0.039 | 75 0.262 | 91 0.193 | 120 0.125 | 288 0.027 | 360 0.023 | 150 0.075 | 183 0.044 | 131 0.100 | 151 0.083 |
| 396 | QAZSMARTVISIONAI-000 | 24 0.002 | 43 0.001 | 35 0.001 | 45 0.013 | 44 0.010 | 44 0.008 | 11 0.130 | 24 0.094 | 34 0.064 | 48 0.002 | 48 0.001 | 82 0.041 | 46 0.011 | 31 0.067 | 30 0.056 |
| 397 | QAZSMARTVISIONAI-001 | 15 0.002 | 9 0.001 | 6 0.001 | 18 0.009 | 19 0.008 | 12 0.006 | 5 0.095 | 7 0.065 | 13 0.050 | 26 0.002 | 20 0.001 | 36 0.026 | 20 0.007 | 13 0.058 | 17 0.049 |
| 398 | QAZSMARTVISIONAI-002 | 5 0.001 | 3 0.001 | 3 0.000 | 6 0.007 | 6 0.006 | 6 0.005 | 6 0.107 | 2 0.052 | 5 0.043 | 6 0.001 | 5 0.001 | 9 0.014 | 5 0.004 | 8 0.052 | 6 0.045 |
| 399 | QAZSMARTVISIONAI-003 | 8 0.001 | 5 0.001 | 4 0.000 | 4 0.006 | 4 0.006 | 3 0.005 | 2 0.071 | 4 0.053 | 6 0.044 | 4 0.001 | 3 0.001 | 16 0.016 | 6 0.004 | 4 0.052 | 4 0.044 |
| 400 | QNAP-000 | 597 0.971 | 506 0.129 | 499 0.052 | 588 0.998 | 468 0.238 | 473 0.117 | 532 1.000 | 544 1.000 | 555 1.000 | 428 0.191 | 422 0.068 | 291 0.539 | 317 0.263 | 495 0.998 | 513 0.985 |
| 401 | QNAP-001 | 393 0.083 | 440 0.053 | 444 0.024 | 390 0.176 | 413 0.137 | 435 0.085 | 244 0.942 | 282 0.927 | 348 0.869 | 380 0.081 | 392 0.041 | 268 0.368 | 310 0.227 | 313 0.331 | 364 0.248 |
| 402 | QNAP-002 | 323 0.044 | 369 0.026 | 392 0.013 | 362 0.136 | 384 0.106 | 407 0.068 | 211 0.818 | 244 0.769 | 293 0.617 | 348 0.052 | 363 0.025 | 249 0.281 | 299 0.171 | 289 0.272 | 345 0.214 |
| 403 | QNAP-003 | 298 0.036 | 335 0.019 | 347 0.008 | 573 0.980 | 569 0.835 | 383 0.057 | 470 1.000 | 376 0.992 | 240 0.365 | 478 0.502 | 421 0.066 | 441 1.000 | 404 1.000 | 440 0.865 | 414 0.373 |
| 404 | QNAP-004 | 286 0.034 | 295 0.014 | 286 0.005 | 591 0.999 | 574 0.890 | 319 0.038 | 496 1.000 | 411 0.996 | 217 0.295 | 477 0.501 | 410 0.056 | 523 1.000 | 393 0.999 | 456 0.922 | 440 0.530 |
| 405 | QNAP-005 | 220 0.018 | 264 0.010 | 279 0.005 | 288 0.082 | 309 0.066 | 349 0.045 | 126 0.460 | 148 0.359 | 184 0.227 | 276 0.024 | 308 0.014 | 172 0.110 | 210 0.061 | 173 0.127 | 211 0.108 |
| 406 | QNAP-006 | 222 0.018 | 244 0.008 | 246 0.003 | 231 0.055 | 250 0.042 | 263 0.026 | 135 0.488 | 155 0.388 | 192 0.244 | 255 0.019 | 253 0.009 | 182 0.131 | 217 0.065 | 186 0.137 | 206 0.102 |
| 407 | QUANTASOFT-001 | 569 0.713 | 586 0.638 | 580 0.493 | | | | | | | | | | | | |
| 408 | RANKONE-002 | 465 0.182 | 498 0.117 | 515 0.070 | 449 0.308 | 429 0.261 | 512 0.190 | | | | | | | | | |
| 409 | RANKONE-003 | 464 0.182 | 497 0.117 | 516 0.070 | 445 0.300 | 477 0.255 | 510 0.187 | | | | | | | | | |
| 410 | RANKONE-004 | 500 0.250 | 533 0.193 | 546 0.124 | 494 0.482 | 515 0.426 | 544 0.324 | | | | | | | | | |
| 411 | RANKONE-005 | 400 0.095 | 454 0.059 | 466 0.033 | 415 0.212 | 441 0.173 | 474 0.119 | 429 0.999 | 428 0.998 | 464 0.994 | | | | | | |
| 412 | RANKONE-006 | 362 0.060 | 404 0.037 | 418 0.019 | | | | 302 0.987 | 321 0.977 | 375 0.936 | | | | | | |
| 413 | RANKONE-007 | 287 0.034 | 354 0.022 | 377 0.011 | 339 0.118 | 365 0.095 | 393 0.061 | 269 0.975 | 302 0.967 | 366 0.923 | | | | | | |
| 414 | RANKONE-009 | 277 0.030 | 326 0.017 | 341 0.008 | 307 0.098 | 331 0.076 | 346 0.045 | 287 0.982 | 307 0.969 | 344 0.857 | 359 0.062 | 370 0.029 | | | 311 0.328 | 334 0.206 |

*Table 68:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 415 | RANKONE-010 | 240 0.023 | 291 0.013 | 319 0.007 | 280 0.077 | 294 0.058 | 309 0.036 | 231 0.904 | 249 0.799 | 304 0.648 | 347 0.052 | 366 0.027 | 232 0.208 | 274 0.119 | 281 0.259 | 328 0.194 |
| 416 | RANKONE-011 | 413 0.109 | 245 0.008 | 259 0.004 | 284 0.079 | 264 0.048 | 275 0.029 | | | | 323 0.037 | 332 0.017 | 222 0.182 | 260 0.092 | 481 0.977 | 431 0.465 |
| 417 | RANKONE-012 | 232 0.020 | 232 0.007 | 249 0.003 | 272 0.072 | 285 0.053 | 278 0.030 | | | | 297 0.029 | 305 0.014 | 193 0.144 | 225 0.072 | 353 0.465 | 255 0.128 |
| 418 | RANKONE-013 | 162 0.010 | 166 0.004 | 171 0.002 | 207 0.046 | 214 0.034 | 206 0.020 | 394 0.998 | 407 0.996 | 179 0.205 | 250 0.018 | 239 0.008 | 189 0.141 | 195 0.050 | 196 0.142 | 198 0.097 |
| 419 | RANKONE-014 | 123 0.008 | 119 0.003 | 109 0.001 | 158 0.033 | 152 0.024 | 138 0.015 | | | | 189 0.009 | 171 0.004 | 155 0.081 | 123 0.026 | 582 1.000 | 147 0.083 |
| 420 | RANKONE-015 | 84 0.005 | 103 0.002 | 102 0.001 | 106 0.024 | 103 0.018 | 100 0.012 | | | | 145 0.007 | 144 0.003 | 88 0.042 | 97 0.020 | 154 0.109 | 168 0.088 |
| 421 | REALNETWORKS-000 | 526 0.365 | 544 0.233 | 550 0.137 | 480 0.433 | 501 0.319 | 523 0.209 | | | | | | | | | |
| 422 | REALNETWORKS-001 | 527 0.365 | 545 0.233 | 549 0.137 | 479 0.433 | 500 0.319 | 522 0.209 | | | | | | | | | |
| 423 | REALNETWORKS-002 | 530 0.369 | 542 0.231 | 548 0.137 | 476 0.416 | 499 0.315 | 524 0.209 | | | | | | | | | |
| 424 | REALNETWORKS-003 | 510 0.272 | 520 0.159 | 528 0.090 | 455 0.342 | 482 0.266 | 506 0.172 | 426 0.999 | 435 0.998 | 432 0.987 | 420 0.164 | 443 0.103 | | | 364 0.500 | 410 0.364 |
| 425 | REALNETWORKS-004 | 496 0.242 | 519 0.158 | 527 0.090 | 462 0.353 | 480 0.263 | 503 0.169 | 445 1.000 | 456 0.999 | 455 0.992 | 423 0.170 | 444 0.103 | | | 389 0.613 | 413 0.370 |
| 426 | REALNETWORKS-005 | 344 0.052 | 379 0.027 | 385 0.012 | 302 0.094 | 329 0.074 | 353 0.047 | 292 0.983 | 309 0.970 | 355 0.894 | 322 0.037 | 327 0.017 | 237 0.223 | 277 0.123 | 261 0.215 | 299 0.165 |
| 427 | REALNETWORKS-006 | 255 0.025 | 299 0.014 | 288 0.005 | 265 0.068 | 282 0.053 | 292 0.032 | 335 0.993 | 331 0.980 | 339 0.836 | 236 0.016 | 246 0.008 | 177 0.120 | 213 0.063 | 210 0.154 | 226 0.116 |
| 428 | REALNETWORKS-007 | 229 0.019 | 256 0.009 | 257 0.004 | 236 0.057 | 253 0.043 | 266 0.027 | 328 0.991 | 325 0.979 | 343 0.853 | 211 0.012 | 209 0.005 | 285 0.463 | 214 0.063 | 192 0.140 | 202 0.100 |
| 429 | REALNETWORKS-008 | 186 0.012 | 209 0.006 | 218 0.003 | 173 0.037 | 187 0.029 | 188 0.018 | 312 0.988 | 304 0.967 | 199 0.262 | 177 0.008 | 185 0.004 | 141 0.068 | 163 0.035 | 176 0.129 | 218 0.110 |
| 430 | REALSENSE-001 | 124 0.008 | 106 0.003 | 93 0.001 | 74 0.016 | 75 0.013 | 70 0.009 | 111 0.346 | 111 0.221 | 113 0.119 | 77 0.004 | 87 0.002 | 44 0.028 | 65 0.014 | 59 0.077 | 65 0.065 |
| 431 | RECOGNITO-000 | 58 0.004 | 74 0.002 | 70 0.001 | 63 0.015 | 59 0.012 | 62 0.009 | 71 0.257 | 78 0.175 | 102 0.113 | 158 0.007 | 149 0.003 | 301 0.730 | 332 0.437 | 108 0.092 | 97 0.070 |
| 432 | RECOGNITO-001 | 27 0.002 | 46 0.001 | 43 0.001 | 30 0.010 | 30 0.009 | 31 0.007 | 259 0.971 | 275 0.899 | 80 0.099 | 66 0.003 | 73 0.002 | 63 0.033 | 76 0.016 | 51 0.075 | 61 0.063 |
| 433 | RECOGNITO-002 | 25 0.002 | 27 0.001 | 28 0.001 | 24 0.010 | 25 0.008 | 27 0.007 | 37 0.186 | 51 0.135 | 58 0.085 | 54 0.003 | 57 0.002 | 287 0.510 | 313 0.248 | 35 0.068 | 29 0.055 |
| 434 | REMARKAI-000 | 425 0.124 | 444 0.055 | 438 0.022 | 386 0.173 | 397 0.120 | 410 0.070 | 435 0.999 | 452 0.999 | 468 0.995 | 369 0.069 | 378 0.033 | | | 411 0.717 | 395 0.315 |
| 435 | REMARKAI-000 | 479 0.196 | 504 0.127 | 507 0.059 | 435 0.263 | 453 0.203 | 477 0.123 | | | | | | | | | |
| 436 | REMARKAI-002 | 472 0.186 | 503 0.123 | 506 0.058 | 427 0.248 | 451 0.196 | 476 0.122 | 341 0.993 | 370 0.991 | 419 0.980 | | | | | | |
| 437 | RENDIP-000 | 241 0.023 | 273 0.012 | 276 0.005 | 400 0.189 | 295 0.059 | 302 0.034 | 248 0.945 | 272 0.892 | 322 0.741 | 269 0.022 | 295 0.013 | 224 0.185 | 255 0.089 | 229 0.167 | 259 0.130 |
| 438 | REVEALMEDIA-000 | 242 0.023 | 275 0.012 | 285 0.005 | 228 0.054 | 249 0.042 | 256 0.025 | 191 0.752 | 226 0.677 | 278 0.533 | 266 0.021 | 275 0.011 | 164 0.093 | 198 0.051 | 198 0.143 | 231 0.118 |
| 439 | RNIT-000 | 532 0.388 | 269 0.011 | 217 0.003 | 190 0.042 | 167 0.026 | 153 0.016 | 182 0.672 | 186 0.510 | 210 0.285 | 212 0.012 | 198 0.005 | 240 0.239 | 177 0.041 | 169 0.123 | 156 0.084 |
| 440 | ROC-016 | 88 0.006 | 93 0.002 | 80 0.001 | 96 0.022 | 97 0.017 | 95 0.012 | | | | 161 0.008 | 160 0.004 | 90 0.045 | 98 0.020 | 151 0.108 | 135 0.079 |
| 441 | ROC-017 | 99 0.006 | 112 0.003 | 101 0.001 | 127 0.028 | 127 0.022 | 118 0.014 | 69 0.253 | 77 0.174 | 90 0.105 | 139 0.006 | 122 0.003 | 78 0.039 | 77 0.016 | 107 0.092 | 103 0.073 |
| 442 | ROC-018 | 95 0.006 | 83 0.002 | 73 0.001 | 102 0.023 | 101 0.018 | 90 0.011 | 49 0.223 | 57 0.142 | 67 0.089 | 109 0.006 | 96 0.002 | 71 0.037 | 68 0.015 | 79 0.085 | 73 0.066 |
| 443 | ROC-019 | 57 0.004 | 47 0.001 | 29 0.001 | 36 0.011 | 34 0.009 | 24 0.007 | 44 0.203 | 27 0.098 | 29 0.062 | 43 0.002 | 40 0.001 | 40 0.027 | 34 0.010 | 41 0.070 | 36 0.057 |
| 444 | ROC-020 | 54 0.003 | 51 0.001 | 47 0.001 | 77 0.017 | 72 0.013 | 56 0.008 | 35 0.186 | 29 0.101 | 32 0.063 | 56 0.003 | 52 0.001 | 84 0.041 | 51 0.012 | 34 0.067 | 23 0.053 |
| 445 | ROC-021 | 30 0.002 | 15 0.001 | 22 0.001 | 20 0.009 | 16 0.007 | 10 0.006 | 12 0.132 | 15 0.083 | 18 0.054 | 25 0.002 | 24 0.001 | 146 0.073 | 61 0.013 | 15 0.060 | 15 0.049 |
| 446 | S1-000 | 435 0.136 | 382 0.028 | 375 0.010 | 351 0.129 | 347 0.085 | 357 0.048 | 534 1.000 | 546 1.000 | 287 0.591 | 342 0.047 | 336 0.018 | 449 1.000 | 276 0.123 | 581 1.000 | 457 0.632 |
| 447 | S1-001 | 349 0.053 | 312 0.015 | 315 0.007 | 253 0.066 | 277 0.052 | 297 0.033 | 329 0.992 | 349 0.985 | 389 0.952 | 252 0.019 | 267 0.010 | 184 0.136 | 230 0.075 | 203 0.148 | 232 0.119 |
| 448 | S1-002 | 361 0.060 | 199 0.005 | 188 0.002 | 292 0.085 | 195 0.031 | 181 0.018 | 235 0.923 | 87 0.186 | 60 0.085 | 153 0.007 | 140 0.003 | 315 0.792 | 293 0.151 | 435 0.841 | 279 0.144 |
| 449 | S1-003 | 340 0.050 | 246 0.008 | 237 0.003 | 224 0.052 | 228 0.037 | 227 0.022 | 481 1.000 | 493 1.000 | 437 0.989 | 229 0.014 | 217 0.006 | 274 0.396 | 169 0.037 | 518 1.000 | 341 0.209 |
| 450 | S1-004 | 245 0.024 | 148 0.004 | 153 0.002 | 161 0.034 | 153 0.025 | 161 0.016 | 425 1.000 | 415 0.997 | 167 0.182 | 128 0.006 | 120 0.003 | 128 0.064 | 118 0.024 | 544 1.000 | 160 0.086 |
| 451 | S1-005 | 249 0.024 | 161 0.004 | 162 0.002 | 143 0.031 | 135 0.022 | 148 0.015 | 433 0.999 | 390 0.994 | 135 0.138 | 116 0.006 | 88 0.002 | 132 0.065 | 110 0.023 | 449 0.895 | 120 0.076 |
| 452 | S1-006 | 214 0.016 | 86 0.002 | 95 0.001 | 93 0.021 | 93 0.016 | 94 0.012 | 531 1.000 | 458 0.999 | 84 0.103 | 97 0.005 | 89 0.002 | 80 0.040 | 91 0.019 | 339 0.423 | 95 0.070 |
| 453 | S1-007 | 78 0.005 | 73 0.002 | 79 0.001 | 84 0.018 | 81 0.014 | 82 0.010 | 115 0.367 | 69 0.165 | 63 0.088 | 79 0.004 | 76 0.002 | 64 0.034 | 79 0.016 | 57 0.077 | 55 0.062 |
| 454 | SANSAP-001 | 178 0.012 | 162 0.004 | 166 0.002 | 532 0.755 | 526 0.480 | 255 0.025 | 313 0.989 | 294 0.958 | 168 0.184 | 279 0.024 | 214 0.006 | 374 1.000 | 381 0.991 | 292 0.280 | 246 0.124 |
| 455 | SCANOVATE-000 | 407 0.102 | 462 0.066 | 461 0.030 | 444 0.296 | 471 0.240 | 490 0.150 | 240 0.930 | 270 0.892 | 333 0.800 | 436 0.215 | 453 0.118 | | | 328 0.400 | 389 0.299 |
| 456 | SCANOVATE-001 | 429 0.128 | 473 0.081 | 476 0.037 | 442 0.281 | 464 0.227 | 484 0.140 | 242 0.934 | 279 0.910 | 338 0.832 | 429 0.192 | 445 0.103 | | | 332 0.404 | 385 0.290 |
| 457 | SCYLLA-001 | 275 0.030 | 320 0.016 | 328 0.007 | 590 0.999 | 595 0.995 | 580 0.680 | 178 0.664 | 174 0.457 | 218 0.299 | 469 0.431 | 474 0.279 | | 559 1.000 | 470 0.953 | 471 0.714 |
| 458 | SENSETIME-000 | 297 0.035 | 344 0.020 | 356 0.009 | 281 0.078 | 305 0.063 | 332 0.040 | 548 1.000 | 568 1.000 | 434 0.987 | | | | | | |
| 459 | SENSETIME-001 | 300 0.036 | 352 0.022 | 359 0.009 | 286 0.080 | 308 0.064 | 339 0.041 | | | | | | | | | |
| 460 | SENSETIME-002 | 301 0.036 | 303 0.014 | 398 0.014 | 346 0.124 | 181 0.028 | 241 0.023 | 374 0.997 | 386 0.994 | 417 0.978 | 306 0.032 | 328 0.017 | | | 369 0.523 | 298 0.160 |

*Table 69:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 461 | SENSETIME-003 | [64] 0.004 | [65] 0.001 | [65] 0.001 | [58] 0.014 | [65] 0.012 | [66] 0.009 | [162] 0.602 | [129] 0.471 | [220] 0.303 | [170] 0.008 | [208] 0.005 | | | [182] 0.133 | [225] 0.115 |
| 462 | SENSETIME-004 | [44] 0.003 | [60] 0.001 | [64] 0.001 | [65] 0.015 | [70] 0.013 | [74] 0.010 | [90] 0.293 | [109] 0.220 | [138] 0.140 | [120] 0.006 | [157] 0.004 | | | [160] 0.113 | [201] 0.100 |
| 463 | SENSETIME-005 | [166] 0.010 | [83] 0.002 | [57] 0.001 | [87] 0.018 | [80] 0.014 | [71] 0.010 | [66] 0.250 | [68] 0.163 | [74] 0.092 | [143] 0.007 | [162] 0.004 | [107] 0.051 | [112] 0.023 | [143] 0.104 | [183] 0.093 |
| 464 | SENSETIME-006 | [67] 0.004 | [62] 0.001 | [50] 0.001 | [69] 0.016 | [66] 0.012 | [58] 0.009 | [407] 0.999 | [431] 0.998 | [308] 0.676 | [74] 0.004 | [68] 0.002 | [65] 0.034 | [81] 0.016 | [109] 0.093 | [134] 0.079 |
| 465 | SENSETIME-007 | [42] 0.003 | [37] 0.001 | [34] 0.001 | [37] 0.012 | [39] 0.009 | [35] 0.007 | [447] 1.000 | [461] 0.999 | [277] 0.533 | [53] 0.003 | [47] 0.001 | [30] 0.024 | [43] 0.011 | [81] 0.085 | [113] 0.074 |
| 466 | SENSETIME-008 | [22] 0.002 | [34] 0.001 | [25] 0.001 | [34] 0.011 | [32] 0.009 | [28] 0.007 | [317] 0.990 | [158] 0.398 | [47] 0.075 | [44] 0.002 | [38] 0.001 | [23] 0.021 | [29] 0.009 | [71] 0.080 | [109] 0.074 |
| 467 | SENSETIME-009 | [7] 0.001 | [11] 0.001 | [14] 0.001 | [13] 0.008 | [11] 0.007 | [14] 0.006 | [422] 0.999 | [262] 0.866 | [22] 0.058 | [18] 0.001 | [21] 0.001 | [11] 0.015 | [18] 0.007 | [85] 0.085 | [139] 0.081 |
| 468 | SENSETIME-010 | [10] 0.001 | [13] 0.001 | [8] 0.001 | [23] 0.010 | [20] 0.008 | [23] 0.007 | [30] 0.176 | [26] 0.096 | [20] 0.058 | [30] 0.002 | [30] 0.001 | [22] 0.020 | [25] 0.008 | [94] 0.087 | [144] 0.082 |
| 469 | SENSETIME-011 | [6] 0.001 | [12] 0.001 | [16] 0.001 | [12] 0.008 | [12] 0.007 | [13] 0.006 | [421] 0.999 | [263] 0.866 | [21] 0.058 | [17] 0.001 | [22] 0.001 | [12] 0.015 | [19] 0.007 | [84] 0.085 | [140] 0.081 |
| 470 | SENSETIME-012 | [4] 0.001 | [8] 0.001 | [9] 0.001 | [8] 0.007 | [8] 0.006 | [8] 0.006 | [14] 0.135 | [21] 0.090 | [27] 0.060 | [13] 0.001 | [8] 0.001 | [25] 0.022 | [35] 0.010 | [82] 0.085 | [138] 0.080 |
| 471 | SERENDIPITY-000 | [241] 0.023 | [279] 0.012 | [275] 0.005 | [492] 0.474 | [456] 0.213 | [312] 0.036 | [215] 0.842 | [238] 0.745 | [286] 0.575 | [247] 0.018 | [263] 0.010 | [390] 1.000 | [394] 0.999 | [197] 0.142 | [209] 0.106 |
| 472 | SEVENTHSENSE-001 | [140] 0.009 | [172] 0.004 | [211] 0.003 | [232] 0.055 | [251] 0.043 | [253] 0.025 | [113] 0.346 | [116] 0.228 | [126] 0.130 | [196] 0.010 | [196] 0.005 | [295] 0.643 | [325] 0.351 | [174] 0.127 | [188] 0.095 |
| 473 | SHAMAN-003 | [548] 0.506 | [575] 0.451 | [580] 0.346 | [516] 0.650 | [538] 0.597 | [561] 0.472 | | | | | | | | | |
| 474 | SHAMAN-004 | [566] 0.678 | [584] 0.615 | [586] 0.488 | [539] 0.812 | [557] 0.754 | [575] 0.639 | | | | | | | | | |
| 475 | SHAMAN-006 | [470] 0.184 | [512] 0.141 | [530] 0.092 | [440] 0.278 | [467] 0.237 | [502] 0.168 | [272] 0.977 | [313] 0.971 | [395] 0.959 | | | | | | |
| 476 | SHAMAN-007 | [468] 0.183 | [513] 0.141 | [529] 0.092 | [441] 0.280 | [470] 0.240 | [504] 0.169 | | | | | | | | | |
| 477 | SHENNONA-000 | [337] 0.049 | [353] 0.022 | [400] 0.015 | [418] 0.221 | [433] 0.160 | [462] 0.104 | [144] 0.524 | [154] 0.383 | [186] 0.235 | [418] 0.160 | [416] 0.064 | [517] 1.000 | [397] 1.000 | [381] 0.558 | [403] 0.350 |
| 478 | SHUFTI-001 | [325] 0.045 | [386] 0.028 | [406] 0.016 | | [379] 0.104 | [384] 0.058 | [520] 1.000 | [218] 0.650 | [260] 0.453 | [353] 0.056 | [394] 0.042 | [271] 0.382 | [284] 0.145 | [274] 0.243 | [338] 0.208 |
| 479 | SIAT-001 | [428] 0.127 | [325] 0.017 | [316] 0.007 | [513] 0.641 | [505] 0.365 | [547] 0.348 | | | | [302] 0.031 | [304] 0.014 | | | | |
| 480 | SIAT-002 | [533] 0.417 | [349] 0.021 | [321] 0.007 | [550] 0.942 | [525] 0.478 | [559] 0.460 | | | | [460] 0.372 | [484] 0.356 | | | [457] 0.923 | [302] 0.169 |
| 481 | SJTU-001 | [59] 0.004 | [64] 0.001 | [46] 0.001 | [68] 0.016 | [60] 0.012 | [52] 0.008 | [122] 0.435 | [133] 0.306 | [145] 0.144 | [110] 0.006 | [129] 0.003 | [175] 0.114 | [168] 0.037 | [138] 0.101 | [154] 0.084 |
| 482 | SJTU-002 | [20] 0.002 | [18] 0.001 | [13] 0.001 | [43] 0.012 | [35] 0.009 | [25] 0.007 | [254] 0.958 | [280] 0.911 | [103] 0.113 | [99] 0.005 | [121] 0.003 | [289] 0.516 | [201] 0.052 | [172] 0.126 | [161] 0.086 |
| 483 | SMILART-004 | [596] 0.970 | [600] 0.968 | [605] 0.965 | [572] 0.977 | [588] 0.976 | [599] 0.973 | | | | | | | | | |
| 484 | SMILART-005 | | | | | | | | | | | | | | | |
| 485 | SPARKTEK-000 | [113] 0.007 | [115] 0.003 | [118] 0.001 | [89] 0.018 | [87] 0.015 | [80] 0.010 | [134] 0.487 | [162] 0.414 | [215] 0.294 | [127] 0.006 | [148] 0.003 | [97] 0.048 | [103] 0.022 | [149] 0.106 | [166] 0.087 |
| 486 | SPARKTEK-001 | [77] 0.005 | [23] 0.001 | [30] 0.001 | [26] 0.010 | [26] 0.009 | [37] 0.007 | [100] 0.311 | [100] 0.210 | [118] 0.124 | [41] 0.002 | [49] 0.001 | [26] 0.022 | [27] 0.009 | [54] 0.076 | [75] 0.067 |
| 487 | SPARSH-001 | [150] 0.009 | [157] 0.004 | [131] 0.002 | [557] 0.952 | [558] 0.759 | [326] 0.039 | [330] 0.992 | [319] 0.976 | [155] 0.164 | [421] 0.166 | [342] 0.020 | [604] 1.000 | [392] 0.999 | [393] 0.633 | [399] 0.340 |
| 488 | SQISOFT-001 | [492] 0.226 | [508] 0.132 | [489] 0.044 | [453] 0.340 | [475] 0.252 | [469] 0.111 | [253] 0.955 | [247] 0.794 | [291] 0.604 | [328] 0.040 | [339] 0.019 | [259] 0.317 | [291] 0.150 | [337] 0.420 | [326] 0.189 |
| 489 | SQISOFT-002 | [384] 0.074 | [388] 0.029 | [298] 0.006 | | | [593] 0.908 | | [276] 0.903 | [196] 0.258 | [489] 0.621 | [429] 0.074 | | | [471] 0.953 | [424] 0.435 |
| 490 | STAQU-000 | [521] 0.334 | [455] 0.061 | [431] 0.021 | [541] 0.848 | [518] 0.443 | [392] 0.061 | [474] 1.000 | [489] 1.000 | [511] 0.999 | [481] 0.535 | [388] 0.039 | [340] 0.961 | [302] 0.183 | [541] 1.000 | [522] 0.999 |
| 491 | STCON-000 | [471] 0.185 | [530] 0.184 | [559] 0.183 | [519] 0.679 | [551] 0.678 | [578] 0.675 | [114] 0.355 | [138] 0.318 | [209] 0.284 | [493] 0.674 | [507] 0.674 | [333] 0.941 | [360] 0.837 | [409] 0.709 | [468] 0.709 |
| 492 | STCON-001 | [56] 0.004 | [61] 0.001 | [53] 0.001 | [72] 0.016 | [63] 0.012 | [59] 0.009 | [39] 0.192 | [48] 0.131 | [55] 0.082 | [52] 0.003 | [50] 0.001 | [269] 0.370 | [294] 0.158 | [38] 0.069 | [32] 0.056 |
| 493 | STCON-002 | [43] 0.003 | [53] 0.001 | [41] 0.001 | [50] 0.013 | [49] 0.011 | [43] 0.008 | [23] 0.162 | [37] 0.115 | [44] 0.074 | [46] 0.002 | [46] 0.001 | [278] 0.415 | [303] 0.185 | [33] 0.067 | [27] 0.055 |
| 494 | STCON-003 | [38] 0.003 | [45] 0.001 | [42] 0.001 | [40] 0.012 | [42] 0.010 | [38] 0.007 | [26] 0.171 | [34] 0.110 | [42] 0.071 | [40] 0.002 | [41] 0.001 | [55] 0.031 | [32] 0.009 | [24] 0.065 | [22] 0.053 |
| 495 | STCON-004 | [23] 0.002 | [21] 0.001 | [26] 0.001 | [16] 0.009 | [18] 0.008 | [20] 0.006 | [15] 0.136 | [18] 0.083 | [25] 0.060 | [14] 0.001 | [12] 0.001 | [10] 0.014 | [7] 0.005 | [11] 0.055 | [8] 0.047 |
| 496 | STCON-005 | [31] 0.002 | [22] 0.001 | [23] 0.001 | [15] 0.008 | [15] 0.007 | [16] 0.006 | [21] 0.159 | [17] 0.083 | [23] 0.058 | [10] 0.001 | [10] 0.001 | [50] 0.029 | [10] 0.005 | [6] 0.052 | [5] 0.045 |
| 497 | STELLARTECS-001 | [271] 0.028 | [301] 0.014 | [308] 0.006 | [264] 0.068 | [273] 0.051 | [279] 0.031 | [255] 0.961 | [283] 0.927 | [310] 0.686 | [264] 0.021 | [250] 0.009 | [308] 0.750 | [316] 0.263 | [211] 0.155 | [222] 0.112 |
| 498 | SUMSUB-000 | [565] 0.675 | [574] 0.445 | [556] 0.173 | [545] 0.888 | [555] 0.742 | [548] 0.348 | [493] 1.000 | [515] 1.000 | [557] 1.000 | [505] 0.852 | [495] 0.453 | [557] 1.000 | [580] 1.000 | [494] 0.996 | [494] 0.901 |
| 499 | SYNESIS-003 | [414] 0.111 | [460] 0.065 | [463] 0.031 | [325] 0.155 | [400] 0.123 | [426] 0.078 | [262] 0.973 | [296] 0.960 | [357] 0.910 | [375] 0.075 | [386] 0.039 | | | [304] 0.314 | [358] 0.235 |
| 500 | SYNESIS-003 | [561] 0.648 | [580] 0.582 | [585] 0.443 | [524] 0.708 | [544] 0.646 | [563] 0.524 | | | | | | | | | |
| 501 | SYNESIS-005 | [336] 0.049 | [362] 0.024 | [379] 0.011 | [296] 0.088 | [323] 0.072 | [343] 0.043 | [355] 0.995 | [343] 0.983 | [330] 0.793 | [309] 0.032 | [318] 0.016 | | | [259] 0.214 | [297] 0.158 |
| 502 | T4ISB-000 | [263] 0.027 | [313] 0.015 | [372] 0.010 | [262] 0.068 | [281] 0.053 | [303] 0.034 | [152] 0.561 | [184] 0.505 | [262] 0.457 | [263] 0.021 | [270] 0.010 | [309] 0.759 | [300] 0.177 | [223] 0.161 | [249] 0.125 |
| 503 | TCC-000 | [591] 0.943 | [596] 0.868 | [583] 0.372 | [583] 0.996 | [592] 0.990 | [598] 0.967 | [512] 1.000 | [507] 1.000 | [518] 0.999 | [523] 0.992 | [525] 0.973 | [455] 1.000 | [470] 1.000 | [503] 1.000 | [520] 0.998 |
| 504 | TCC-001 | [570] 0.734 | [522] 0.161 | [474] 0.037 | [587] 0.997 | [591] 0.987 | [568] 0.554 | [521] 1.000 | [505] 1.000 | [526] 1.000 | [522] 0.991 | [521] 0.915 | [429] 1.000 | [414] 1.000 | [504] 1.000 | [517] 0.994 |
| 505 | TECH5-001 | [578] 0.807 | [447] 0.057 | [411] 0.018 | [579] 0.994 | [581] 0.935 | [381] 0.055 | [569] 1.000 | [543] 1.000 | [531] 1.000 | [439] 0.244 | [369] 0.028 | | | [492] 0.994 | [486] 0.817 |
| 506 | TECH5-002 | [348] 0.053 | [373] 0.027 | [384] 0.012 | [301] 0.094 | [321] 0.070 | [334] 0.040 | [221] 0.872 | [250] 0.803 | [295] 0.622 | [326] 0.039 | [338] 0.019 | [229] 0.205 | [271] 0.111 | [345] 0.440 | [321] 0.182 |

*Table 70:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 507 | TECH5-003 | 143 0.009 | 145 0.004 | 124 0.002 | 564 0.964 | 562 0.801 | 361 0.049 | 344 0.994 | 339 0.982 | 153 0.160 | 426 0.179 | 399 0.023 | 503 1.000 | 403 1.000 | 400 0.656 | 405 0.358 |
| 508 | TECHAINER-000 | 258 0.026 | 289 0.013 | 297 0.006 | 275 0.074 | 283 0.053 | 299 0.033 | 202 0.780 | 182 0.479 | 197 0.259 | 245 0.018 | 255 0.009 | 207 0.155 | 216 0.064 | 201 0.145 | 213 0.109 |
| 509 | TEVIAN-003 | 495 0.238 | 528 0.177 | 535 0.096 | 458 0.346 | 494 0.298 | 515 0.198 | | | | | | | | | |
| 510 | TEVIAN-004 | 461 0.169 | 495 0.117 | 512 0.063 | 416 0.216 | 443 0.176 | 472 0.115 | | | | | | | | | |
| 511 | TEVIAN-005 | 430 0.128 | 480 0.087 | 491 0.044 | 393 0.180 | 420 0.144 | 439 0.089 | 307 0.987 | 299 0.961 | 331 0.793 | | | | | | |
| 512 | TEVIAN-006 | 247 0.024 | 257 0.009 | 274 0.005 | 187 0.041 | 202 0.032 | 223 0.021 | 151 0.556 | 163 0.418 | 208 0.282 | 238 0.016 | 247 0.009 | 162 0.093 | 193 0.050 | 469 0.951 | 230 0.117 |
| 513 | TEVIAN-007 | 170 0.011 | 189 0.005 | 204 0.002 | 122 0.028 | 126 0.022 | 135 0.015 | 139 0.498 | 132 0.293 | 165 0.174 | 190 0.009 | 197 0.005 | 131 0.065 | 156 0.033 | 168 0.122 | 205 0.102 |
| 514 | THAKAAIT-000 | 315 0.042 | 311 0.015 | 287 0.005 | 268 0.071 | 279 0.052 | 280 0.031 | 408 0.999 | 424 0.998 | 422 0.981 | 251 0.018 | 257 0.009 | 248 0.279 | 223 0.071 | 221 0.160 | 243 0.121 |
| 515 | TIGER-000 | 542 0.461 | 567 0.390 | 571 0.261 | 502 0.565 | 527 0.500 | 549 0.366 | | | | | | | | | |
| 516 | TIGER-002 | 458 0.157 | 477 0.086 | 481 0.039 | 410 0.202 | 430 0.158 | 448 0.095 | 436 0.999 | 445 0.999 | 412 0.975 | | | | | | |
| 517 | TIGER-003 | 457 0.157 | 476 0.086 | 480 0.039 | 411 0.202 | 429 0.158 | 447 0.095 | | | | | | | | | |
| 518 | TIGER-004 | 293 0.035 | 322 0.017 | 343 0.008 | 320 0.105 | 324 0.072 | 329 0.040 | 136 0.489 | 161 0.410 | 205 0.278 | 387 0.095 | 320 0.016 | 385 1.000 | 378 0.990 | 329 0.402 | 278 0.143 |
| 519 | TONGYITRANS-000 | 408 0.105 | 467 0.074 | 479 0.038 | 367 0.141 | 389 0.112 | 409 0.069 | | | | | | | | | |
| 520 | TONGYITRANS-001 | 424 0.123 | 461 0.066 | 464 0.032 | 349 0.128 | 377 0.101 | 396 0.062 | | | | | | | | | |
| 521 | TOSHIBA-000 | 423 0.122 | 456 0.062 | 454 0.026 | 371 0.150 | 394 0.118 | 419 0.074 | 329 0.997 | 399 0.995 | 435 0.988 | | | | | | |
| 522 | TOSHIBA-001 | 490 0.222 | 450 0.057 | 413 0.018 | 358 0.133 | 360 0.092 | 380 0.054 | | | | | | | | | |
| 523 | TOSHIBA-002 | 74 0.005 | 67 0.001 | 77 0.001 | 79 0.017 | 89 0.015 | 101 0.012 | 32 0.179 | 23 0.094 | 28 0.060 | 31 0.002 | 32 0.001 | 56 0.031 | 23 0.008 | 29 0.066 | 24 0.054 |
| 524 | TOSHIBA-003 | 115 0.007 | 59 0.001 | 66 0.001 | 57 0.014 | 64 0.012 | 75 0.010 | 104 0.320 | 42 0.123 | 33 0.064 | 38 0.002 | 34 0.001 | 219 0.176 | 44 0.011 | 86 0.085 | 31 0.056 |
| 525 | TOSS-000 | 60 0.004 | 77 0.002 | 96 0.001 | 82 0.017 | 82 0.014 | 86 0.011 | 33 0.182 | 39 0.121 | 49 0.077 | 68 0.003 | 59 0.002 | 70 0.037 | 59 0.013 | 39 0.069 | 46 0.058 |
| 526 | TRUEFACE-000 | 329 0.046 | 330 0.018 | 340 0.008 | 282 0.079 | 304 0.062 | 324 0.039 | 360 0.995 | 265 0.881 | 267 0.493 | 299 0.030 | 322 0.016 | 227 0.194 | 272 0.111 | 249 0.188 | 281 0.145 |
| 527 | TSIGHT-000 | 3 0.001 | 2 0.001 | 2 0.000 | 5 0.006 | 3 0.006 | 4 0.005 | 3 0.075 | 1 0.051 | 3 0.043 | 3 0.001 | 1 0.001 | 8 0.013 | 4 0.004 | 3 0.050 | 3 0.044 |
| 528 | TTIGROUP-000 | 419 0.116 | 249 0.009 | 244 0.003 | 500 0.553 | 495 0.298 | 286 0.032 | 414 0.999 | 432 0.998 | 430 0.986 | 446 0.286 | 417 0.065 | 387 1.000 | 398 1.000 | 438 0.855 | 442 0.540 |
| 529 | TURINGTECHVIP-001 | 583 0.865 | 558 0.345 | 376 0.011 | 565 0.967 | 570 0.850 | 507 0.173 | 434 0.999 | 378 0.993 | 171 0.196 | 519 0.978 | 511 0.754 | 619 1.000 | 407 1.000 | 501 0.999 | 512 0.984 |
| 530 | UNISSEY-001 | 488 0.217 | 539 0.214 | 565 0.211 | 525 0.709 | 553 0.700 | 582 0.695 | 197 0.767 | 215 0.619 | 276 0.532 | 496 0.693 | 509 0.691 | 363 0.999 | 380 0.991 | 414 0.748 | 473 0.742 |
| 531 | UNISSEY-002 | 477 0.190 | 531 0.187 | 560 0.183 | 523 0.703 | 552 0.685 | 579 0.677 | 204 0.792 | 207 0.571 | 256 0.441 | 500 0.724 | 508 0.678 | 364 1.000 | 382 0.992 | 447 0.891 | 479 0.769 |
| 532 | UNISSEY-003 | 274 0.029 | 302 0.014 | 325 0.007 | 247 0.063 | 268 0.050 | 298 0.033 | 218 0.857 | 196 0.524 | 234 0.347 | 295 0.029 | 345 0.020 | 230 0.205 | 233 0.077 | 233 0.173 | 228 0.116 |
| 533 | UNIVERSEAI-000 | 141 0.009 | 136 0.003 | 114 0.001 | 111 0.026 | 108 0.020 | 103 0.013 | 81 0.268 | 72 0.168 | 82 0.102 | 176 0.008 | 205 0.005 | 165 0.094 | 125 0.026 | 137 0.101 | 131 0.079 |
| 534 | USEB-000 | 575 0.789 | 587 0.715 | 589 0.509 | 537 0.803 | 554 0.722 | 567 0.531 | 283 0.980 | 305 0.968 | 362 0.914 | 475 0.490 | 480 0.301 | 316 0.804 | 348 0.651 | 397 0.648 | 436 0.506 |
| 535 | USEB-001 | 146 0.009 | 142 0.004 | 150 0.002 | 129 0.029 | 139 0.023 | 154 0.016 | 324 0.991 | 335 0.982 | 351 0.877 | 94 0.005 | 99 0.003 | 299 0.728 | 246 0.081 | 91 0.087 | 84 0.068 |
| 536 | USEB-002 | 228 0.019 | 156 0.004 | 173 0.002 | 138 0.030 | 148 0.023 | 159 0.016 | 323 0.991 | 336 0.982 | 352 0.877 | 104 0.005 | 100 0.003 | 300 0.728 | 245 0.081 | 96 0.088 | 88 0.069 |
| 537 | USEB-003 | 144 0.009 | 139 0.003 | 148 0.002 | 114 0.026 | 122 0.021 | 142 0.015 | 299 0.984 | 310 0.971 | 345 0.861 | 103 0.005 | 101 0.003 | 276 0.398 | 184 0.044 | 92 0.087 | 94 0.069 |
| 538 | USEB-004 | 118 0.007 | 133 0.003 | 139 0.002 | 147 0.031 | 136 0.022 | 139 0.015 | 89 0.284 | 84 0.182 | 89 0.105 | 201 0.010 | 80 0.002 | 372 1.000 | 375 0.975 | 248 0.186 | 176 0.090 |
| 539 | VD-000 | 592 0.949 | 597 0.917 | 601 0.827 | 567 0.968 | 584 0.946 | 591 0.871 | | | | | | | | | |
| 540 | VD-001 | 511 0.278 | 534 0.201 | 543 0.115 | 451 0.331 | 490 0.281 | 511 0.188 | | | | | | | | | |
| 541 | VD-002 | 448 0.144 | 471 0.078 | 471 0.036 | 399 0.188 | 422 0.148 | 440 0.092 | 390 0.998 | 403 0.995 | 431 0.987 | 386 0.095 | 401 0.048 | 267 0.367 | 309 0.220 | 322 0.372 | 380 0.280 |
| 542 | VD-003 | 494 0.234 | 422 0.045 | 419 0.019 | 357 0.133 | 375 0.100 | 395 0.061 | 430 0.999 | 447 0.999 | 460 0.994 | 345 0.051 | 365 0.027 | 243 0.244 | 278 0.133 | 306 0.315 | 330 0.203 |
| 543 | VDFSOFTWARE-001 | 609 1.000 | 609 1.000 | 606 0.990 | 530 0.746 | 530 0.518 | 538 0.301 | 527 1.000 | 527 1.000 | 534 1.000 | 495 0.680 | 456 0.144 | 365 1.000 | 377 0.989 | 468 0.948 | 452 0.608 |
| 544 | VDFSOFTWARE-002 | 376 0.068 | 408 0.038 | 399 0.015 | 421 0.229 | 385 0.107 | 369 0.052 | 256 0.969 | 230 0.713 | 249 0.398 | 374 0.075 | 361 0.024 | 383 1.000 | 379 0.990 | 338 0.423 | 336 0.207 |
| 545 | VERIDAS-001 | 388 0.080 | 405 0.037 | 404 0.016 | 323 0.106 | 345 0.082 | 365 0.051 | 340 0.993 | 358 0.987 | 376 0.937 | 334 0.044 | 357 0.023 | 245 0.266 | 287 0.146 | 286 0.264 | 331 0.204 |
| 546 | VERIDAS-002 | 387 0.080 | 406 0.037 | 403 0.016 | 322 0.106 | 344 0.082 | 364 0.051 | 339 0.993 | 357 0.987 | 377 0.937 | 335 0.044 | 358 0.023 | 246 0.266 | 288 0.146 | 285 0.264 | 332 0.204 |
| 547 | VERIDAS-003 | 380 0.072 | 318 0.016 | 302 0.006 | 269 0.071 | 289 0.055 | 295 0.033 | 400 0.998 | 412 0.997 | 368 0.927 | 259 0.020 | 273 0.011 | 199 0.150 | 236 0.078 | 239 0.178 | 277 0.142 |
| 548 | VERIDAS-004 | 257 0.026 | 201 0.005 | 199 0.002 | 162 0.034 | 161 0.025 | 167 0.017 | 319 0.990 | 324 0.978 | 337 0.820 | 168 0.008 | 191 0.005 | 120 0.058 | 141 0.029 | 165 0.118 | 210 0.107 |
| 549 | VERIGRAM-000 | 159 0.010 | 180 0.005 | 220 0.003 | 165 0.034 | 182 0.028 | 219 0.021 | 188 0.722 | 103 0.212 | 96 0.108 | 132 0.006 | 151 0.004 | 157 0.088 | 119 0.024 | 78 0.083 | 89 0.069 |
| 550 | VERIGRAM-001 | 194 0.013 | 182 0.005 | 219 0.003 | 477 0.419 | 240 0.039 | 216 0.021 | 297 0.984 | 211 0.584 | 99 0.110 | 130 0.006 | 146 0.003 | 345 0.988 | 146 0.030 | 74 0.082 | 72 0.066 |
| 551 | VERIJELAS-000 | 581 0.846 | 590 0.799 | 596 0.681 | 543 0.868 | 564 0.813 | 583 0.697 | 424 0.999 | 442 0.999 | 472 0.995 | 453 0.324 | 466 0.216 | 332 0.933 | 340 0.561 | 383 0.589 | 429 0.462 |
| 552 | VIANTE-000 | 165 0.010 | 105 0.003 | 106 0.001 | 73 0.016 | 78 0.014 | 77 0.010 | 99 0.311 | 60 0.147 | 81 0.102 | 138 0.006 | 132 0.003 | 74 0.038 | 88 0.018 | 93 0.087 | 99 0.071 |

*Table 71:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 553 | VIANTE-001 | [85]0.006 | [54]0.001 | [68]0.001 | [28]0.010 | [37]0.009 | [57]0.008 | [73]0.260 | [32]0.105 | [54]0.080 | [47]0.002 | [45]0.001 | [253]0.293 | [26]0.008 | [32]0.067 | [37]0.057 |
| 554 | VIANTE-002 | [73]0.005 | [25]0.001 | [24]0.001 | [11]0.008 | [13]0.007 | [17]0.006 | [94]0.299 | [14]0.078 | [15]0.051 | [42]0.002 | [37]0.001 | [234]0.211 | [21]0.007 | [22]0.063 | [18]0.050 |
| 555 | VIANTE-003 | [104]0.007 | [16]0.001 | [19]0.001 | [10]0.008 | [9]0.007 | [18]0.006 | [109]0.340 | [13]0.076 | [11]0.049 | [33]0.002 | [27]0.001 | [176]0.117 | [15]0.006 | [14]0.060 | [12]0.049 |
| 556 | VIETTELAI-000 | [573]0.784 | [453]0.058 | [301]0.006 | [570]0.972 | [556]0.753 | [315]0.036 | [524]1.000 | [530]1.000 | [524]1.000 | [470]0.435 | [337]0.018 | [353]0.995 | [345]0.601 | [596]1.000 | [519]0.997 |
| 557 | VIETTELAI-001 | | | | | | | | | | | | | | | |
| 558 | VIETTELAI-002 | [499]0.249 | [267]0.010 | [251]0.003 | [527]0.713 | [543]0.631 | [554]0.411 | [123]0.438 | [124]0.269 | [148]0.152 | [465]0.421 | [462]0.176 | [362]0.999 | [372]0.957 | [416]0.750 | [438]0.514 |
| 559 | VIETTELAI-003 | [203]0.014 | [169]0.004 | [151]0.002 | [156]0.033 | [149]0.023 | [144]0.015 | [274]0.978 | [252]0.823 | [279]0.543 | [122]0.006 | [104]0.003 | [211]0.160 | [134]0.027 | [104]0.090 | [101]0.072 |
| 560 | VIGILANTSOLUTIONS-003 | [545]0.481 | [571]0.408 | [524]0.281 | [529]0.730 | [548]0.660 | [565]0.526 | [428]0.999 | [441]0.999 | [469]0.995 | | | | | | |
| 561 | VIGILANTSOLUTIONS-004 | [560]0.623 | [579]0.549 | [584]0.422 | [542]0.858 | [566]0.817 | [584]0.709 | [395]0.998 | [406]0.996 | [451]0.991 | | | | | | |
| 562 | VIGILANTSOLUTIONS-005 | [590]0.934 | [565]0.386 | [484]0.042 | | | | [506]1.000 | [541]1.000 | [542]1.000 | | | | | | |
| 563 | VIGILANTSOLUTIONS-006 | [594]0.955 | [561]0.349 | [487]0.043 | | | | [525]1.000 | [529]1.000 | [543]1.000 | | | | | | |
| 564 | VIGILANTSOLUTIONS-007 | [385]0.075 | [382]0.028 | [328]0.011 | [332]0.113 | [353]0.088 | [374]0.053 | [385]0.997 | [405]0.996 | [452]0.991 | [381]0.081 | [398]0.047 | [270]0.371 | [312]0.242 | [327]0.391 | [388]0.295 |
| 565 | VIGILANTSOLUTIONS-008 | [341]0.050 | [340]0.020 | [358]0.009 | [321]0.105 | [334]0.077 | [351]0.046 | [438]1.000 | [444]0.999 | [448]0.991 | [391]0.104 | [408]0.054 | [275]0.398 | [315]0.259 | [366]0.511 | [396]0.316 |
| 566 | VINBIGDATA-001 | [267]0.028 | [314]0.016 | [278]0.005 | [413]0.204 | [299]0.059 | [276]0.029 | [353]0.995 | [285]0.931 | [214]0.293 | [248]0.018 | [236]0.007 | [334]0.944 | [208]0.061 | [290]0.274 | [189]0.095 |
| 567 | VINBIGDATA-002 | [96]0.006 | [82]0.002 | [78]0.001 | [70]0.016 | [58]0.012 | [51]0.008 | [57]0.237 | [66]0.160 | [79]0.095 | [87]0.005 | [90]0.002 | [258]0.317 | [96]0.019 | [53]0.076 | [64]0.064 |
| 568 | VINBIGDATA-003 | [125]0.008 | [92]0.002 | [75]0.001 | [76]0.016 | [71]0.013 | [60]0.009 | [51]0.226 | [56]0.141 | [53]0.080 | [78]0.004 | [74]0.002 | [79]0.039 | [62]0.014 | [44]0.070 | [41]0.058 |
| 569 | VINBIGDATA-004 | [91]0.006 | [69]0.002 | [54]0.001 | [46]0.013 | [45]0.010 | [42]0.008 | [68]0.251 | [47]0.130 | [48]0.076 | [61]0.003 | [61]0.002 | [57]0.032 | [47]0.012 | [36]0.068 | [43]0.058 |
| 570 | VIRTISHA-000 | [307]0.039 | [374]0.027 | [417]0.019 | [569]0.970 | [550]0.674 | [487]0.147 | [190]0.748 | [190]0.517 | [236]0.351 | [508]0.899 | [505]0.644 | [446]1.000 | [486]1.000 | [486]0.990 | [503]0.960 |
| 571 | VISIONBOX-000 | [381]0.073 | [327]0.017 | [317]0.007 | [270]0.071 | [292]0.057 | [307]0.035 | [349]0.994 | [368]0.990 | [410]0.973 | [272]0.023 | [290]0.012 | [195]0.146 | [240]0.081 | [224]0.162 | [251]0.126 |
| 572 | VISIONBOX-001 | [562]0.649 | [509]0.133 | [307]0.006 | [483]0.441 | [300]0.060 | [282]0.031 | [482]1.000 | [522]1.000 | [528]1.000 | [414]0.152 | [264]0.010 | [359]0.999 | [254]0.087 | [458]0.924 | [377]0.269 |
| 573 | VISIONBOX-002 | [237]0.022 | [270]0.011 | [271]0.005 | [230]0.055 | [248]0.042 | [257]0.025 | [295]0.984 | [233]0.730 | [228]0.336 | [232]0.015 | [237]0.007 | [179]0.125 | [215]0.063 | [191]0.140 | [199]0.099 |
| 574 | VISIONLABS-004 | [397]0.090 | [449]0.057 | [443]0.024 | [406]0.199 | [431]0.159 | [451]0.097 | [245]0.943 | [269]0.888 | [321]0.739 | | | | | | |
| 575 | VISIONLABS-005 | [390]0.080 | [430]0.050 | [421]0.020 | [396]0.183 | [421]0.147 | [432]0.087 | [247]0.945 | [268]0.886 | [320]0.733 | | | | | | |
| 576 | VISIONLABS-006 | [322]0.044 | [371]0.026 | [367]0.010 | [337]0.117 | [358]0.090 | [368]0.051 | [196]0.762 | [220]0.668 | [273]0.506 | | | | | | |
| 577 | VISIONLABS-007 | [321]0.044 | [370]0.026 | [363]0.010 | [336]0.117 | [357]0.090 | [362]0.051 | [195]0.762 | [221]0.668 | [272]0.505 | [305]0.031 | [309]0.014 | | | [245]0.185 | [283]0.145 |
| 578 | VISIONLABS-008 | [266]0.027 | [285]0.013 | [284]0.005 | [261]0.068 | [276]0.051 | [290]0.032 | [156]0.569 | [180]0.475 | [222]0.309 | [239]0.017 | [241]0.008 | | | [205]0.151 | [233]0.119 |
| 579 | VISIONLABS-009 | [180]0.012 | [167]0.004 | [145]0.002 | [151]0.032 | [163]0.025 | [173]0.017 | [239]0.929 | [248]0.797 | [170]0.186 | [178]0.008 | [183]0.004 | | | [159]0.113 | [185]0.093 |
| 580 | VISIONLABS-010 | [200]0.014 | [187]0.005 | [182]0.002 | [164]0.034 | [177]0.027 | [200]0.019 | | | [151]0.159 | [164]0.008 | [163]0.004 | [116]0.055 | [127]0.027 | [152]0.109 | [174]0.089 |
| 581 | VISIONLABS-011 | [172]0.011 | [123]0.003 | [107]0.001 | [103]0.024 | [107]0.020 | [119]0.014 | | | [169]0.184 | [83]0.004 | [95]0.002 | [66]0.034 | [85]0.017 | [103]0.090 | [130]0.079 |
| 582 | VISTEAM-000 | [442]0.140 | [475]0.085 | [478]0.038 | [518]0.674 | [454]0.205 | [443]0.093 | [258]0.970 | [267]0.886 | [307]0.666 | [401]0.128 | [415]0.062 | [375]1.000 | [389]0.999 | [316]0.345 | [361]0.240 |
| 583 | VISTEAM-001 | [462]0.172 | [493]0.109 | [494]0.046 | [604]1.000 | [523]0.469 | [465]0.105 | [246]0.944 | [264]0.878 | [311]0.691 | [415]0.154 | [428]0.074 | [526]1.000 | [395]1.000 | [323]0.374 | [370]0.262 |
| 584 | VIXVIZION-009 | [346]0.053 | [376]0.027 | [387]0.012 | [309]0.098 | [332]0.077 | [356]0.048 | [271]0.976 | [236]0.742 | [274]0.513 | [330]0.041 | [347]0.021 | [250]0.286 | [298]0.165 | [356]0.472 | [314]0.178 |
| 585 | VNIS-000 | [553]0.541 | [555]0.311 | [568]0.234 | [485]0.448 | [513]0.418 | [552]0.383 | [370]0.996 | [379]0.993 | [391]0.953 | [499]0.701 | [506]0.663 | [320]0.838 | [358]0.782 | [407]0.696 | [458]0.635 |
| 586 | VNPAY-001 | [191]0.013 | [205]0.006 | [205]0.002 | [166]0.035 | [170]0.026 | [158]0.016 | [137]0.491 | [127]0.280 | [149]0.156 | [210]0.012 | [232]0.007 | [134]0.066 | [162]0.034 | [153]0.109 | [194]0.097 |
| 587 | VNPT-001 | [265]0.027 | [297]0.014 | [305]0.006 | [377]0.158 | [316]0.068 | [308]0.036 | [234]0.921 | [231]0.715 | [241]0.365 | [317]0.035 | [279]0.011 | [347]0.990 | [337]0.537 | [319]0.362 | [266]0.134 |
| 588 | VNPT-002 | [193]0.013 | [215]0.006 | [234]0.003 | [184]0.040 | [201]0.032 | [222]0.021 | [153]0.563 | [130]0.284 | [146]0.144 | [159]0.007 | [155]0.004 | [144]0.072 | [148]0.031 | [122]0.096 | [119]0.075 |
| 589 | VNPT-003 | [196]0.013 | [212]0.006 | [224]0.003 | [171]0.036 | [183]0.028 | [178]0.018 | [308]0.988 | [297]0.960 | [164]0.173 | [154]0.007 | [156]0.004 | [125]0.062 | [142]0.030 | [123]0.096 | [123]0.077 |
| 590 | VNPT-004 | [171]0.011 | [194]0.005 | [208]0.002 | [159]0.033 | [164]0.026 | [165]0.016 | [257]0.969 | [274]0.899 | [157]0.166 | [149]0.007 | [147]0.003 | [117]0.057 | [133]0.027 | [110]0.093 | [114]0.075 |
| 591 | VNPT-005 | [142]0.009 | [168]0.004 | [187]0.002 | [137]0.030 | [145]0.023 | [141]0.015 | [243]0.936 | [259]0.853 | [142]0.141 | [100]0.005 | [109]0.003 | [95]0.047 | [109]0.023 | [101]0.090 | [102]0.072 |
| 592 | VOCORD-003 | [523]0.350 | [501]0.122 | [496]0.047 | [404]0.195 | [427]0.155 | [445]0.093 | [411]0.999 | [429]0.998 | [444]0.990 | [417]0.157 | [446]0.105 | | | [331]0.404 | [384]0.289 |
| 593 | VOCORD-004 | [579]0.825 | [562]0.353 | [497]0.051 | [472]0.401 | [439]0.173 | [441]0.093 | [503]1.000 | [496]1.000 | [507]0.999 | [431]0.193 | [418]0.065 | | | [488]0.991 | [480]0.776 |
| 594 | VOCORD-005 | [567]0.687 | [518]0.156 | [488]0.043 | [379]0.161 | [407]0.130 | [429]0.080 | [419]0.999 | [418]0.997 | [399]0.968 | [409]0.138 | [438]0.090 | | | [324]0.381 | [382]0.287 |
| 595 | VOCORD-006 | [618]1.000 | [619]1.000 | [615]1.000 | [596]1.000 | [606]1.000 | [606]1.000 | [535]1.000 | [580]1.000 | [604]1.000 | [579]1.000 | [619]1.000 | | | [513]1.000 | [611]1.000 |
| 596 | VOGIC-000 | [391]0.080 | [296]0.014 | [292]0.006 | [560]0.957 | [572]0.855 | [556]0.435 | [405]0.999 | [413]0.997 | [289]0.597 | [524]0.993 | [515]0.834 | [487]1.000 | [418]1.000 | [499]0.999 | [508]0.977 |
| 597 | VSOFT-001 | [486]0.213 | [515]0.153 | [524]0.083 | [463]0.355 | [491]0.288 | [509]0.185 | [336]0.993 | [362]0.990 | [400]0.968 | [434]0.202 | [452]0.118 | [360]0.999 | [366]0.915 | [382]0.573 | [430]0.462 |
| 598 | VSOFT-002 | [342]0.050 | [381]0.028 | [380]0.011 | [430]0.254 | [373]0.098 | [348]0.045 | [402]0.998 | [414]0.997 | [363]0.916 | [315]0.034 | [326]0.017 | [349]0.992 | [342]0.596 | [255]0.201 | [287]0.150 |

*Table 72:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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| MISSES BELOW THRESHOLD, T | | ENROL RECENT MUGSHOT, N = 1.6M | | | | | | | | | ENROL APPLICATION PORTRAIT, N = 1.6M | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: MUGSHOT | | | ENROL: VISA | | ENROL: BORDER | | ENROL: VISA | |
| | | PROBE: MUGSHOT | | | PROBE: WEBCAM | | | PROBE: PROFILE | | | PROBE: BORDER | | PROBE: BORDER 10+YR | | PROBE: KIOSK | |
| # | ALGORITHM | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.0003 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 | FPIR=0.001 | FPIR=0.01 |
| 599 | VTS-000 | 557 0.605 | 582 0.598 | 593 0.595 | 511 0.624 | 541 0.619 | 573 0.613 | 431 0.999 | 457 0.999 | 499 0.998 | 488 0.613 | 503 0.609 | 310 0.760 | 355 0.739 | 417 0.761 | 476 0.749 |
| 600 | VTS-001 | 291 0.034 | 287 0.013 | 289 0.005 | 258 0.067 | 275 0.051 | 283 0.031 | 389 0.998 | 385 0.994 | 271 0.504 | 270 0.022 | 291 0.012 | 190 0.141 | 237 0.079 | 251 0.192 | 252 0.126 |
| 601 | VTS-002 | 347 0.053 | 365 0.026 | 364 0.010 | 310 0.098 | 330 0.075 | 352 0.046 | 454 1.000 | 468 1.000 | 390 0.953 | 338 0.045 | 364 0.026 | 238 0.231 | 279 0.133 | 336 0.417 | 323 0.187 |
| 602 | VTS-003 | 209 0.015 | 228 0.007 | 221 0.003 | 214 0.048 | 207 0.033 | 203 0.019 | 484 1.000 | 503 1.000 | 299 0.628 | 226 0.014 | 201 0.005 | 338 0.954 | 207 0.060 | 394 0.635 | 175 0.089 |
| 603 | VTS-004 | 169 0.011 | 178 0.004 | 156 0.002 | 233 0.055 | 175 0.027 | 157 0.016 | 429 1.000 | 524 1.000 | 342 0.847 | 184 0.009 | 154 0.004 | 366 1.000 | 188 0.046 | 258 0.203 | 107 0.073 |
| 604 | VTS-005 | 174 0.011 | 195 0.005 | 144 0.002 | 202 0.045 | 193 0.030 | 174 0.017 | 451 1.000 | 423 0.998 | 189 0.238 | 147 0.007 | 134 0.003 | 317 0.811 | 170 0.037 | 133 0.100 | 96 0.070 |
| 605 | WINK-000 | 508 0.269 | 535 0.203 | 545 0.122 | 478 0.432 | 507 0.377 | 534 0.274 | 300 0.985 | 344 0.983 | 418 0.979 | 442 0.261 | 458 0.163 | 606 1.000 | 416 1.000 | 398 0.649 | 446 0.558 |
| 606 | WINK-001 | 392 0.082 | 401 0.036 | 396 0.014 | 457 0.346 | 409 0.131 | 432 0.083 | 399 0.998 | 408 0.996 | 415 0.977 | 438 0.237 | 468 0.220 | 326 0.903 | 328 0.412 | 321 0.372 | 394 0.312 |
| 607 | WINK-002 | 246 0.024 | 278 0.012 | 273 0.005 | 380 0.162 | 361 0.092 | 417 0.072 | 296 0.984 | 298 0.961 | 358 0.910 | 448 0.294 | 478 0.289 | 343 0.978 | 333 0.441 | 315 0.342 | 397 0.316 |
| 608 | WINK-003 | 497 0.242 | 249 0.009 | 254 0.003 | 217 0.049 | 233 0.038 | 261 0.026 | 472 1.000 | 490 1.000 | 498 0.998 | 202 0.010 | 207 0.005 | 152 0.077 | 165 0.036 | 421 0.774 | 308 0.173 |
| 609 | XAILIENT-000 | 372 0.065 | 400 0.035 | 405 0.016 | 279 0.077 | 297 0.059 | 301 0.034 | 175 0.653 | 192 0.521 | 245 0.376 | 290 0.027 | 288 0.012 | 221 0.181 | 248 0.082 | 253 0.196 | 229 0.117 |
| 610 | XFORWARDAI-000 | 270 0.028 | 306 0.015 | 309 0.006 | 266 0.070 | 286 0.053 | 305 0.034 | 186 0.694 | 168 0.434 | 190 0.241 | 265 0.021 | 272 0.011 | 210 0.159 | 247 0.082 | 230 0.169 | 265 0.134 |
| 611 | XFORWARDAI-001 | 158 0.010 | 179 0.005 | 207 0.002 | 169 0.036 | 184 0.028 | 205 0.020 | 214 0.836 | 169 0.441 | 130 0.133 | 171 0.008 | 203 0.005 | 127 0.062 | 145 0.030 | 120 0.123 | 203 0.102 |
| 612 | XFORWARDAI-002 | 98 0.006 | 110 0.003 | 172 0.002 | 88 0.018 | 92 0.016 | 116 0.014 | 267 0.974 | 191 0.519 | 57 0.084 | 98 0.005 | 143 0.003 | 83 0.041 | 90 0.018 | 129 0.099 | 173 0.089 |
| 613 | YISHENG-001 | 539 0.451 | 559 0.346 | 563 0.206 | 574 0.983 | 563 0.808 | 533 0.269 | | | | 492 0.666 | 489 0.396 | | | 455 0.919 | 465 0.695 |
| 614 | YITU-002 | 278 0.031 | 324 0.017 | 330 0.008 | 246 0.063 | 265 0.049 | 270 0.028 | | | | | | | | | |
| 615 | YITU-003 | 279 0.031 | 333 0.018 | 348 0.008 | 256 0.067 | 280 0.052 | 296 0.033 | | | | | | | | | |
| 616 | YITU-004 | 227 0.019 | 253 0.009 | 262 0.004 | 168 0.035 | 173 0.027 | 175 0.017 | 250 0.947 | 286 0.936 | 360 0.912 | | | | | | |
| 617 | YITU-005 | 236 0.021 | 263 0.010 | 270 0.005 | 182 0.039 | 203 0.032 | 238 0.023 | | | | | | | | | |
| 618 | YOUVERSE-001 | 199 0.014 | 225 0.007 | 225 0.003 | 175 0.038 | 192 0.030 | 202 0.019 | 102 0.311 | 122 0.253 | 150 0.159 | 191 0.009 | 187 0.004 | 153 0.081 | 164 0.036 | 136 0.101 | 126 0.078 |
| 619 | YUAN-000 | 231 0.019 | 265 0.010 | 252 0.003 | 274 0.073 | 256 0.045 | 265 0.027 | 125 0.454 | 143 0.342 | 183 0.210 | 215 0.012 | 211 0.005 | 262 0.333 | 191 0.048 | 178 0.129 | 172 0.088 |

*Table 73:* **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

# Appendices

## Appendix A Accuracy on large-population FRVT 2018 mugshots

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| | | | | | |
|---|---|---|---|---|---|
| 2026/09/04 | FNIR(N, R, T) = | False neg. identification rate | N = Num. enrolled subjects | T = Threshold | $T = 0 \rightarrow$ Investigation |
| 10:23:32 | FPIR(N, T) = | False pos. identification rate | R = Num. candidates examined | | $T > 0 \rightarrow$ Identification |

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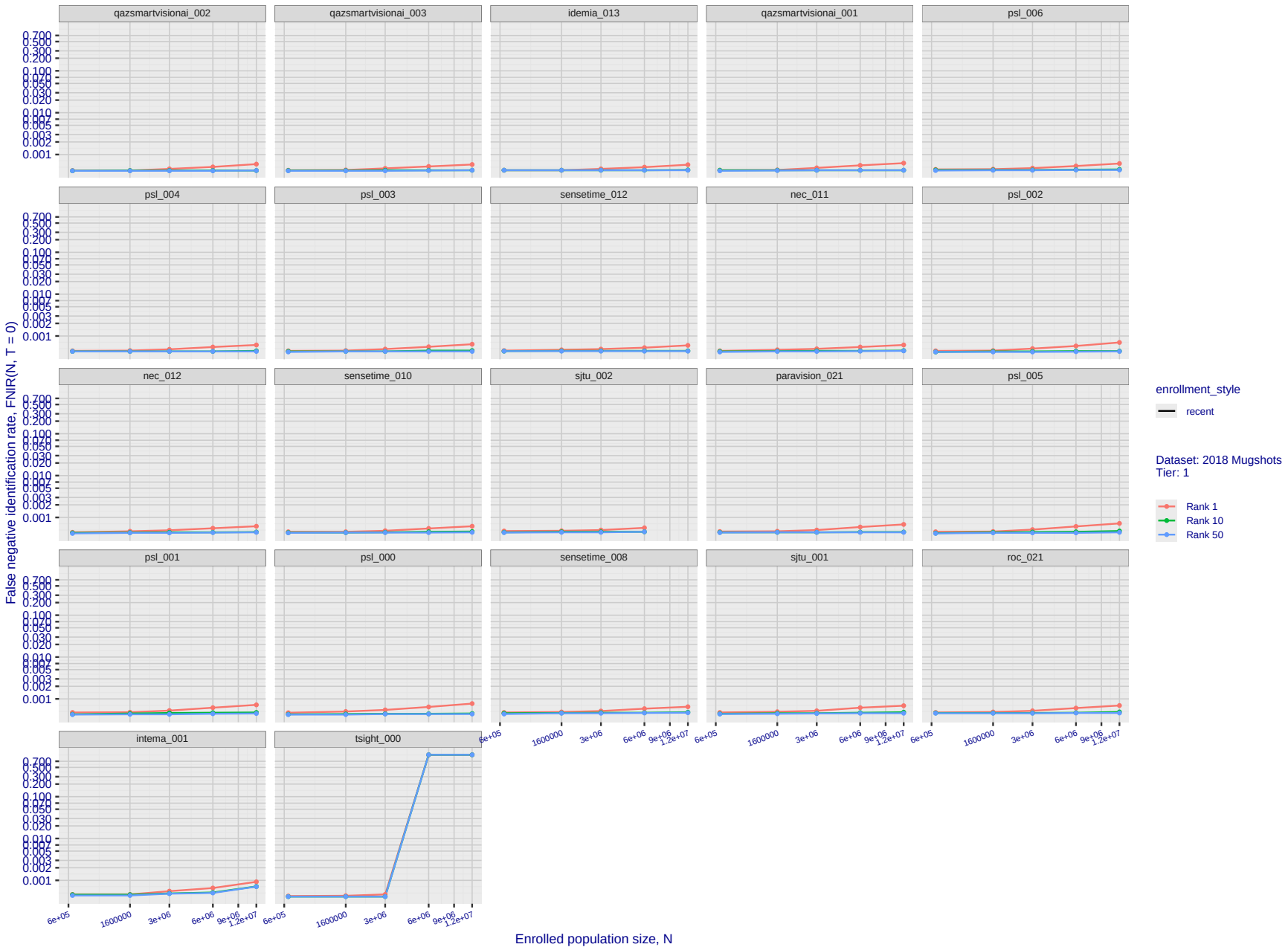


*Figure 20:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



2026/09/04
10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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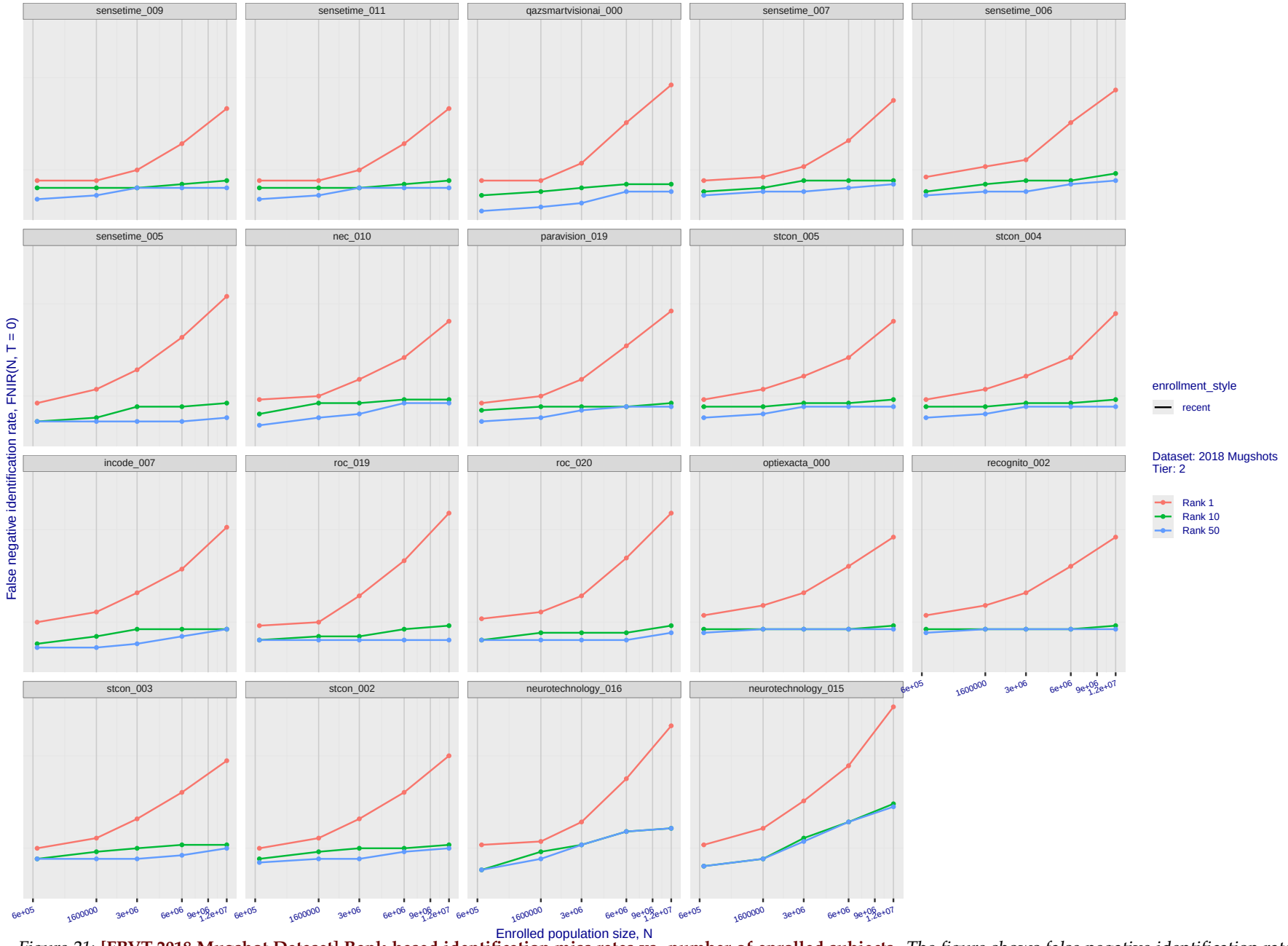


*Figure 21:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



2026/09/04
10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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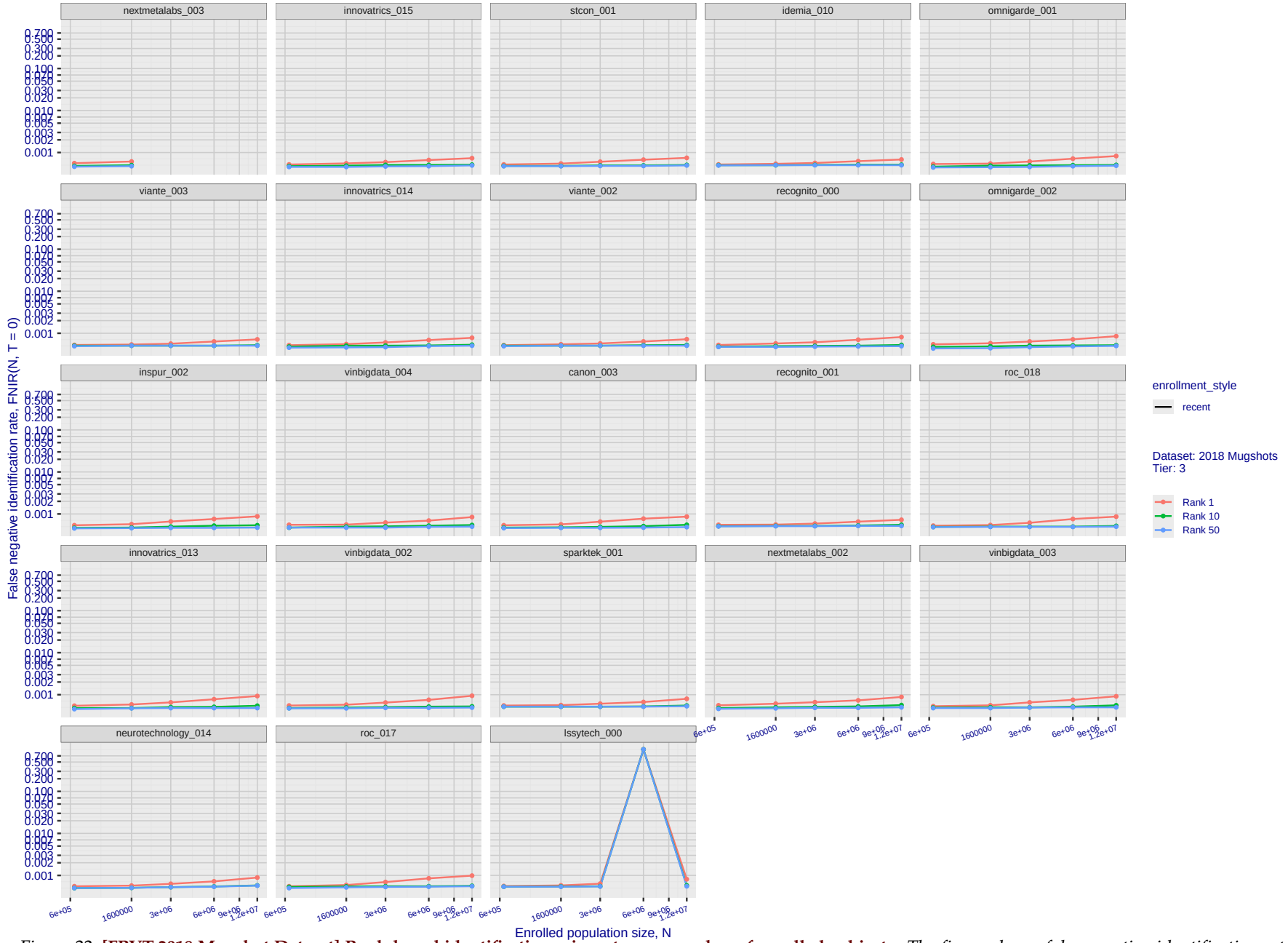


*Figure 22:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



2026/09/04
10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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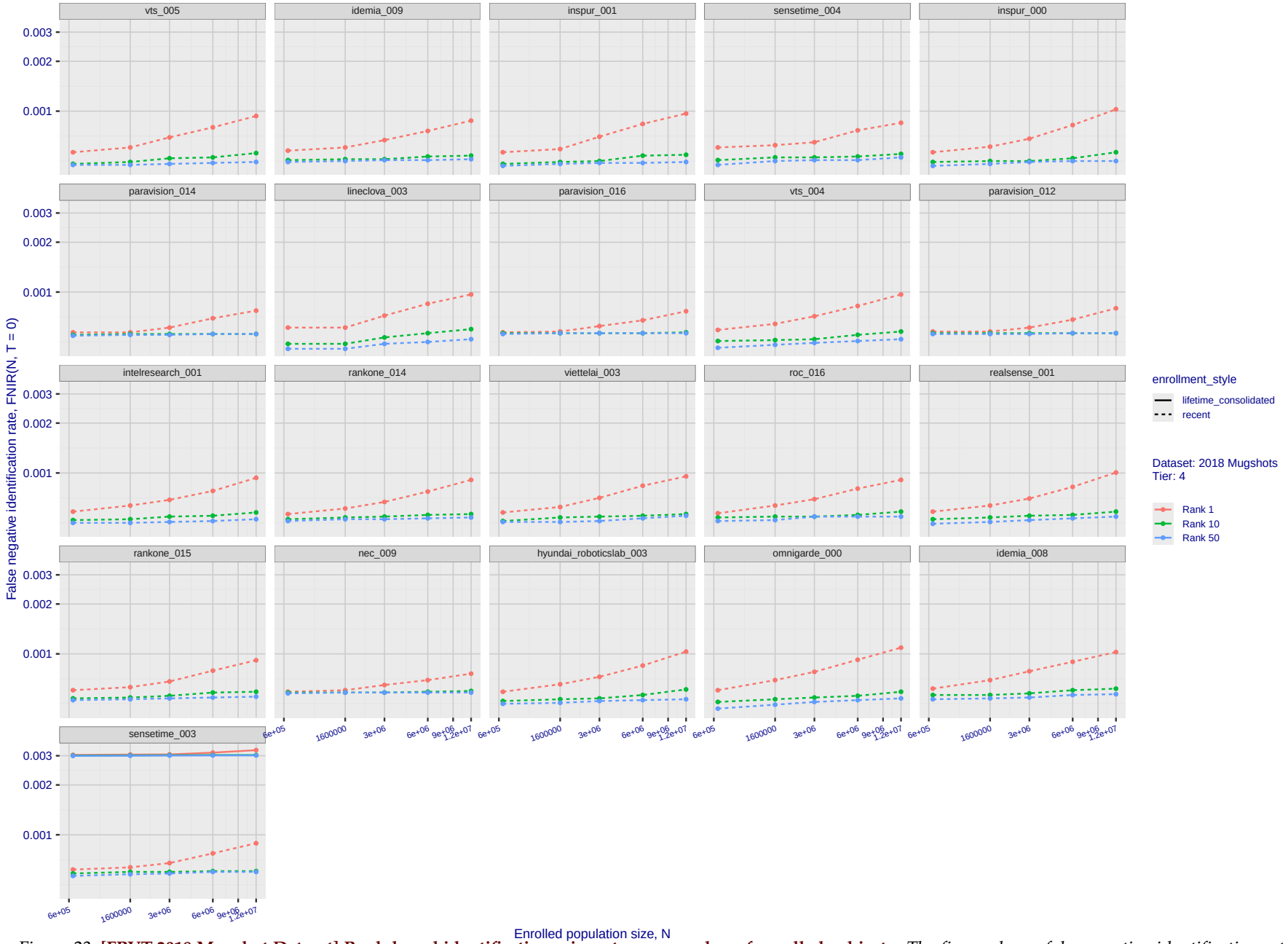


*Figure 23:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



2026/09/04 10:23:32

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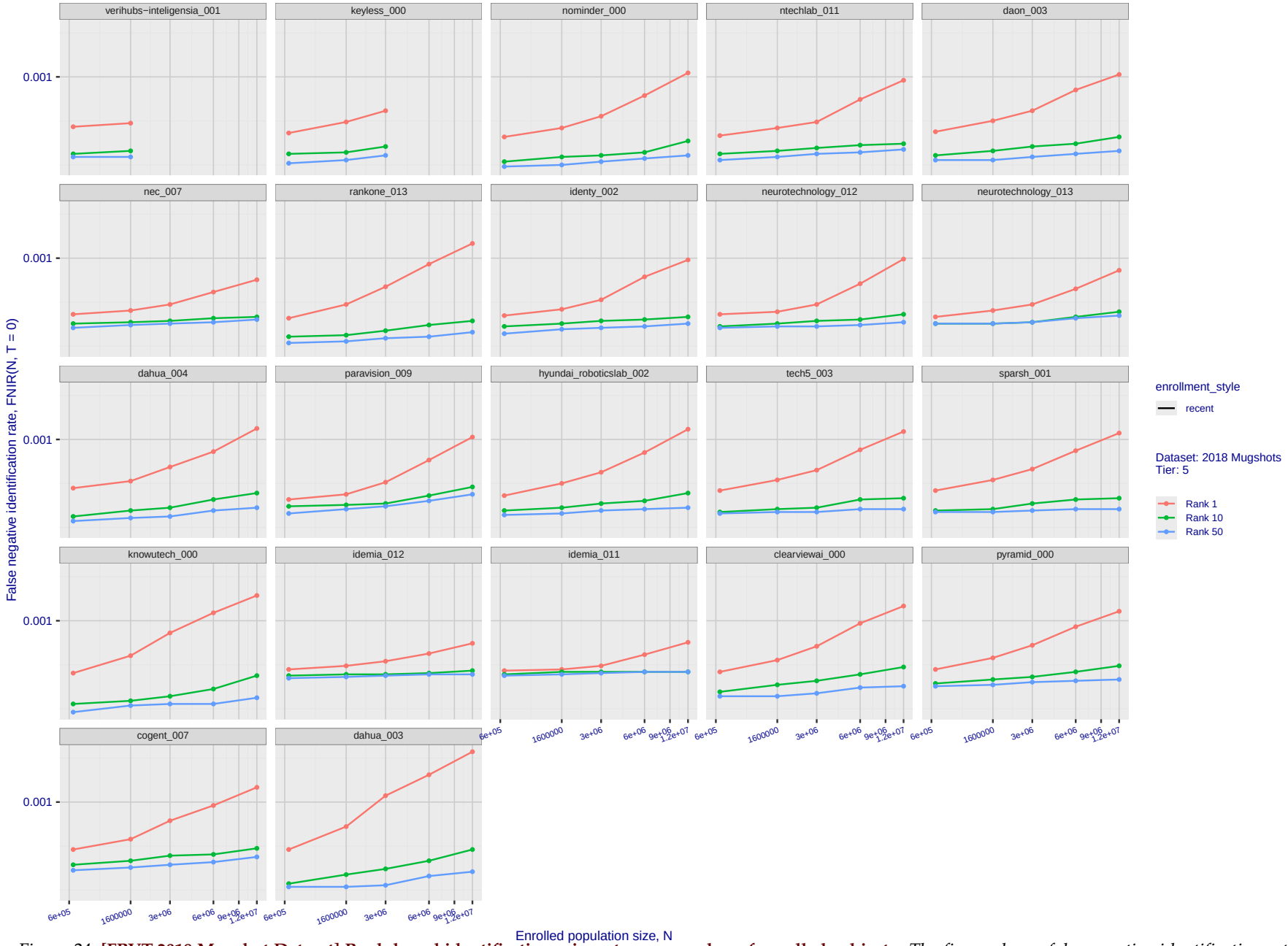


*Figure 24:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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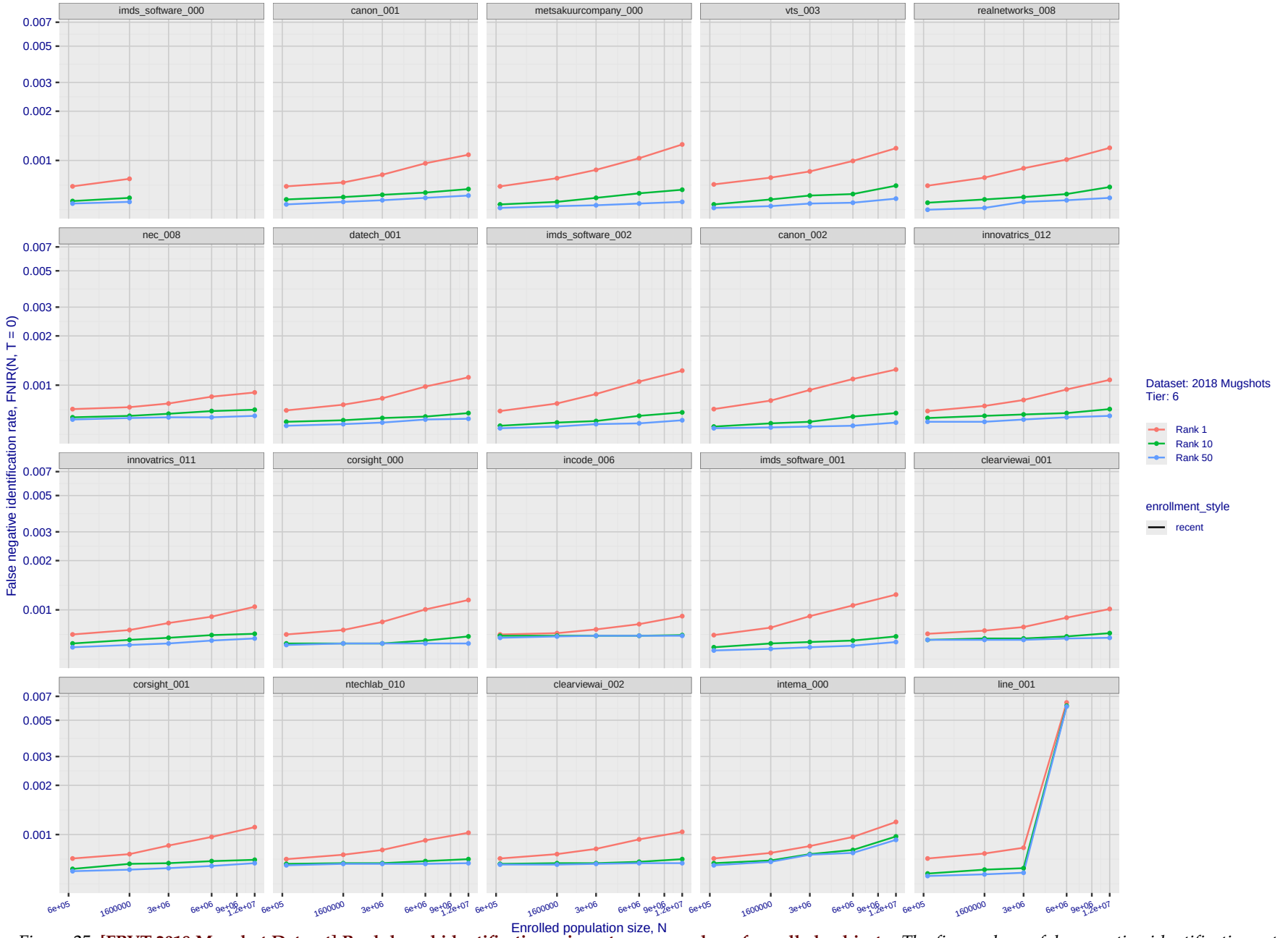


*Figure 25:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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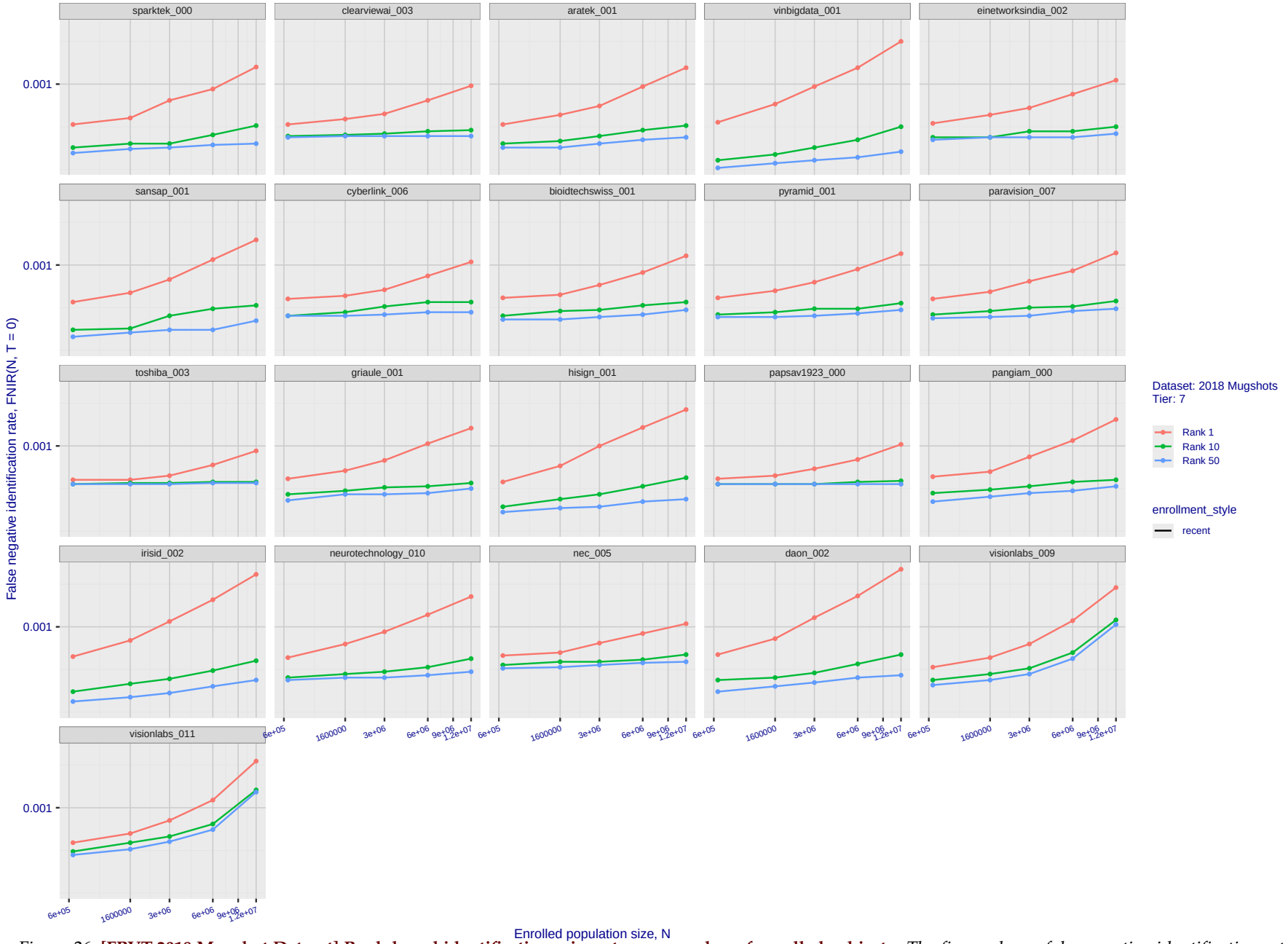


*Figure 26:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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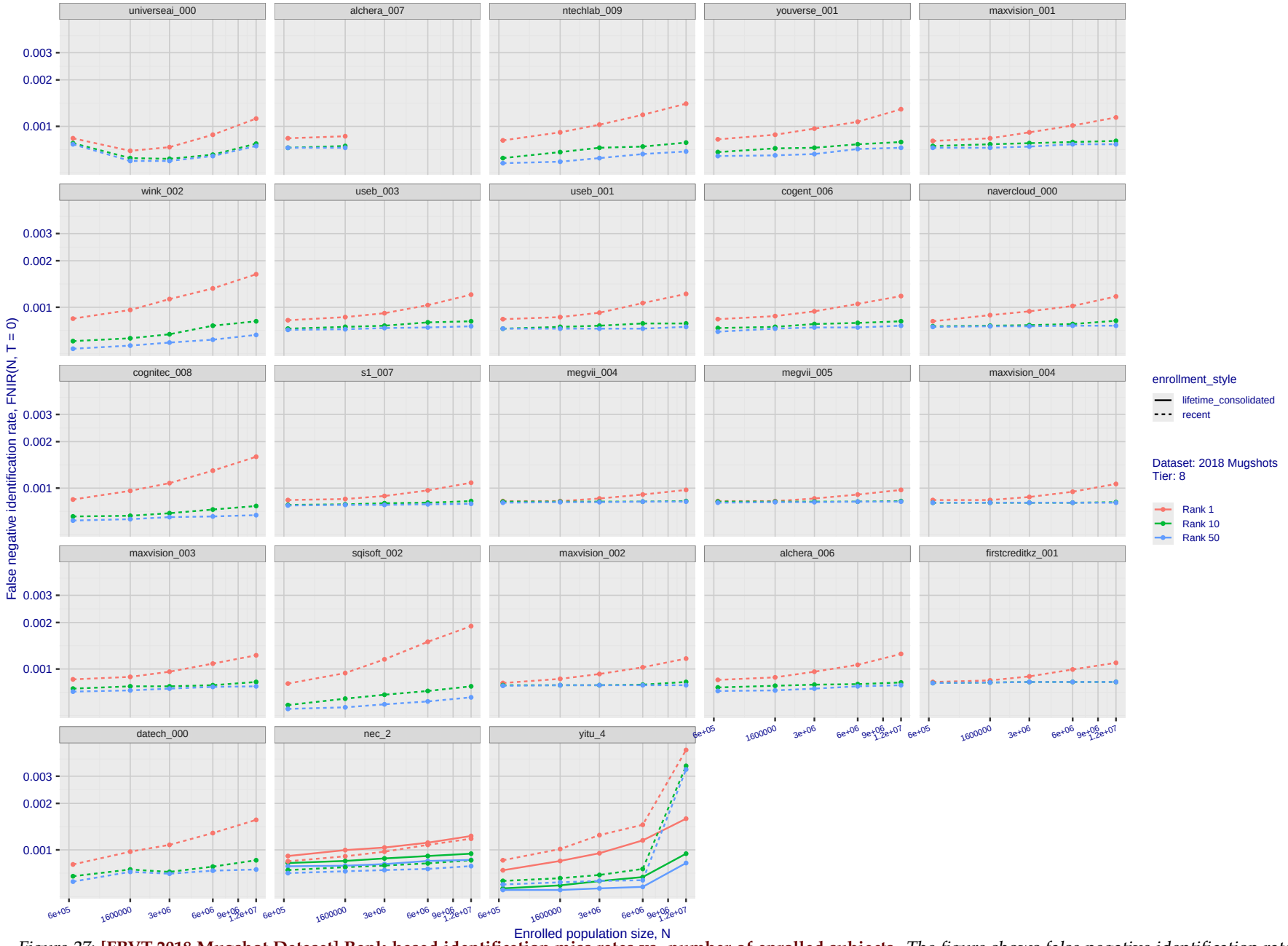


*Figure 27:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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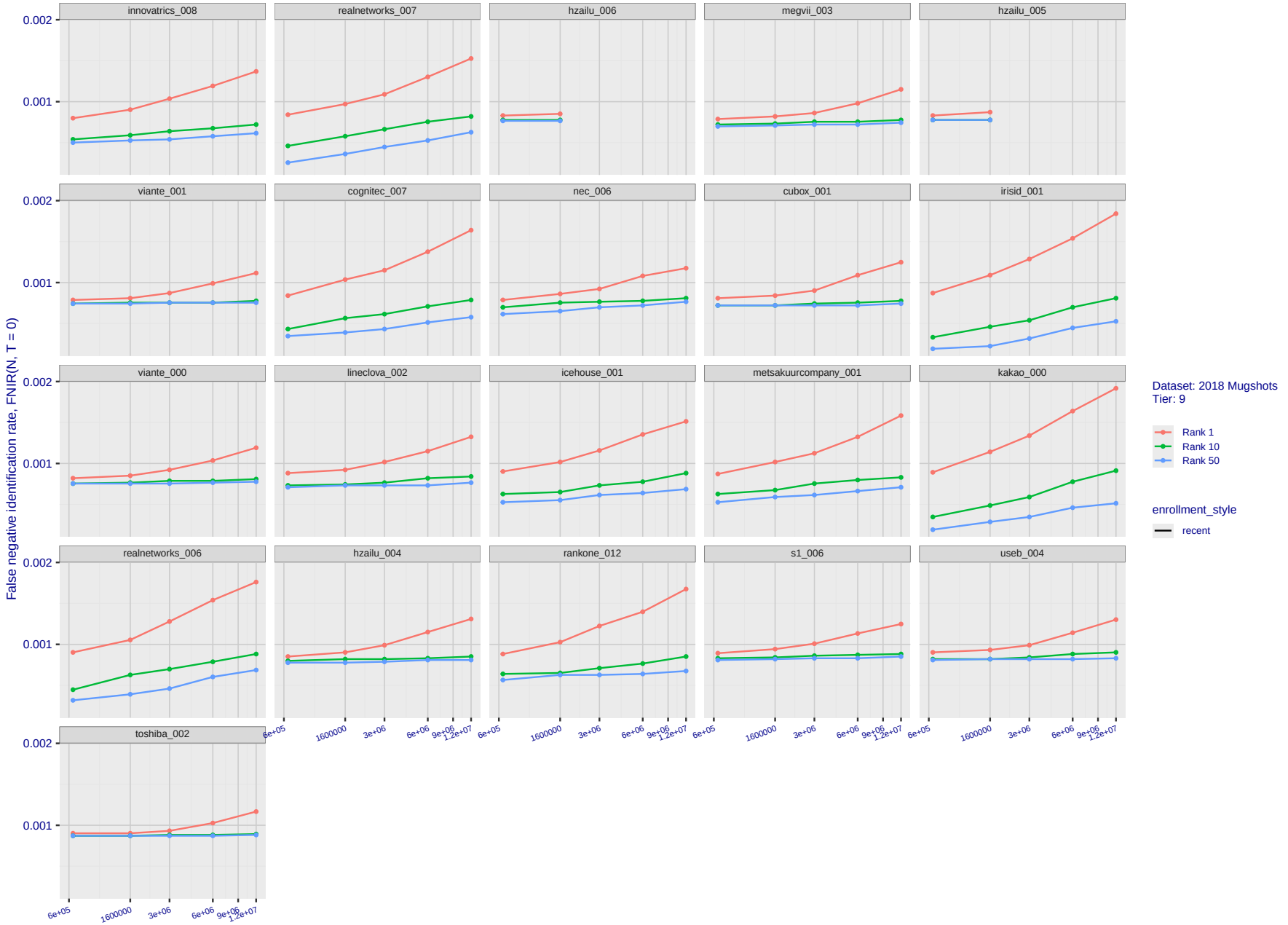


*Figure 28:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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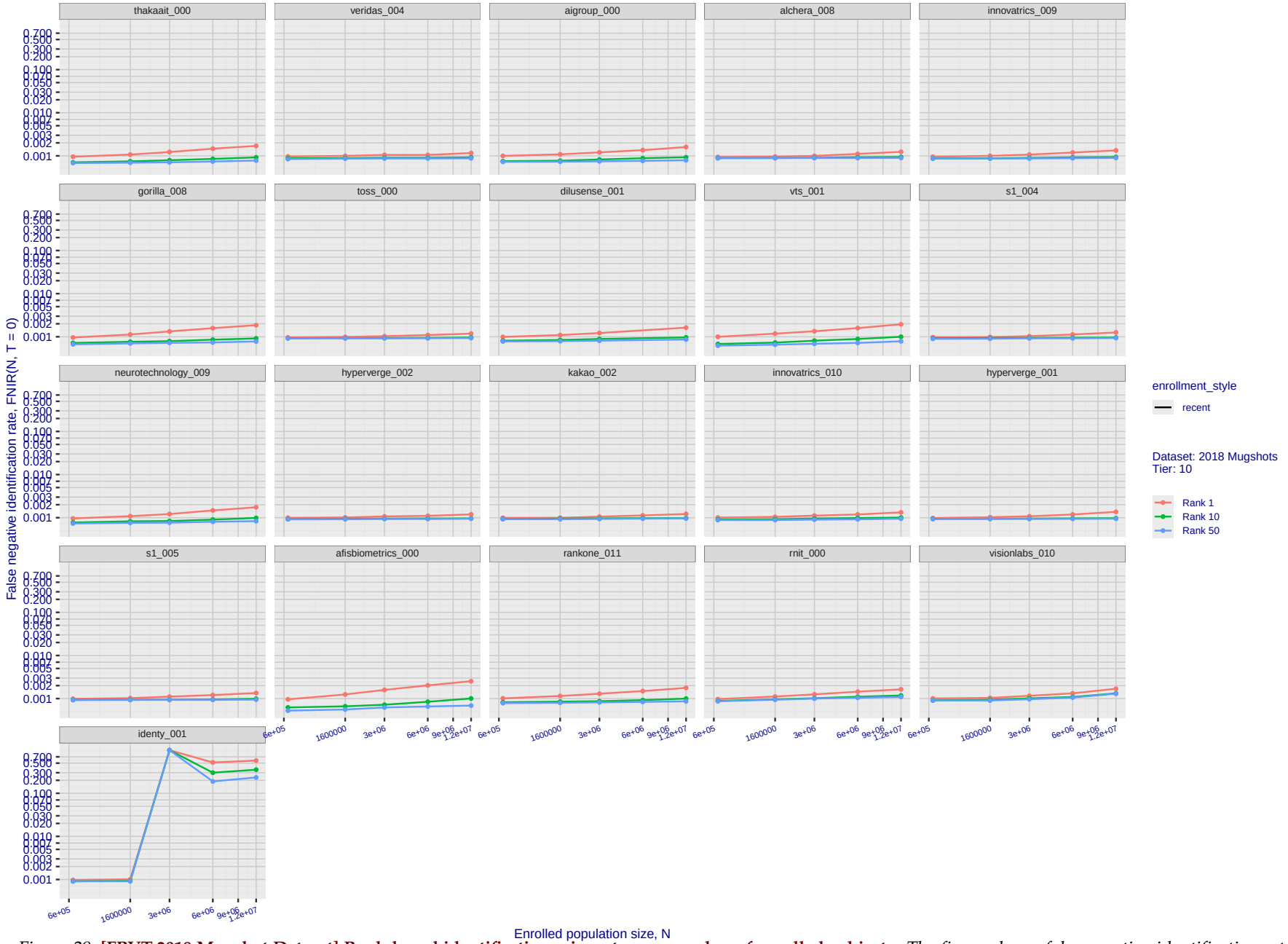


*Figure 29:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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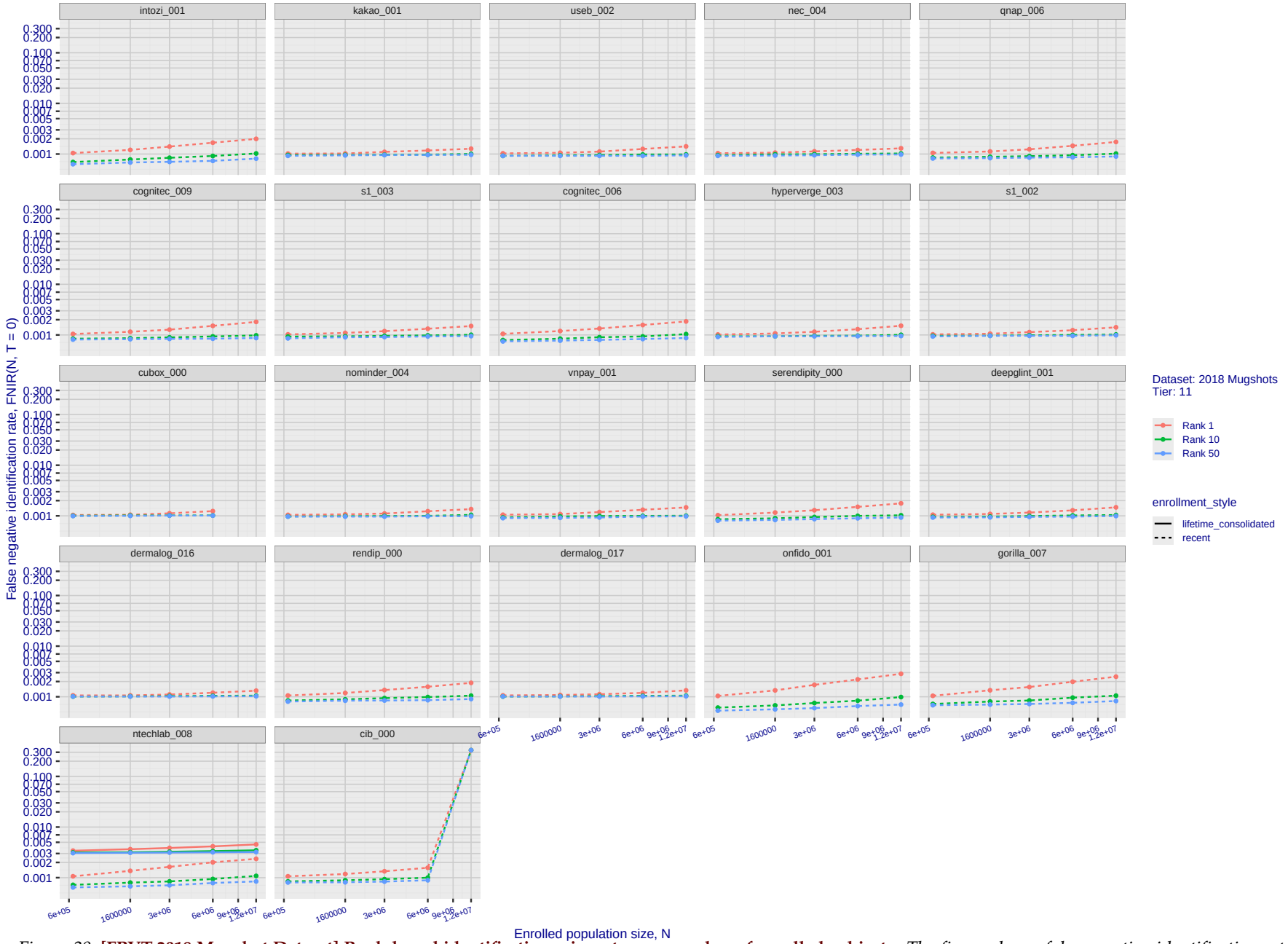


*Figure 30:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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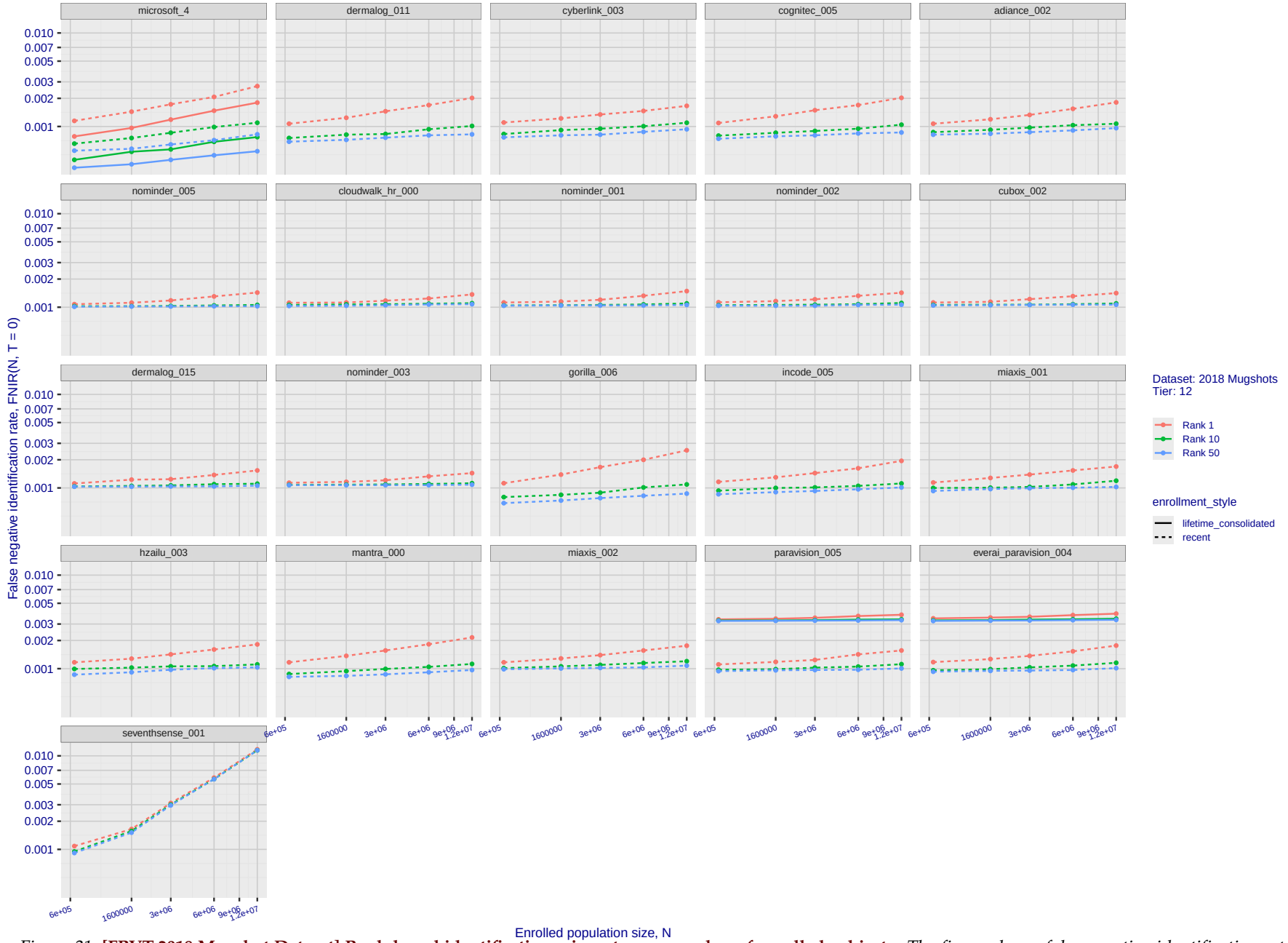


*Figure 31:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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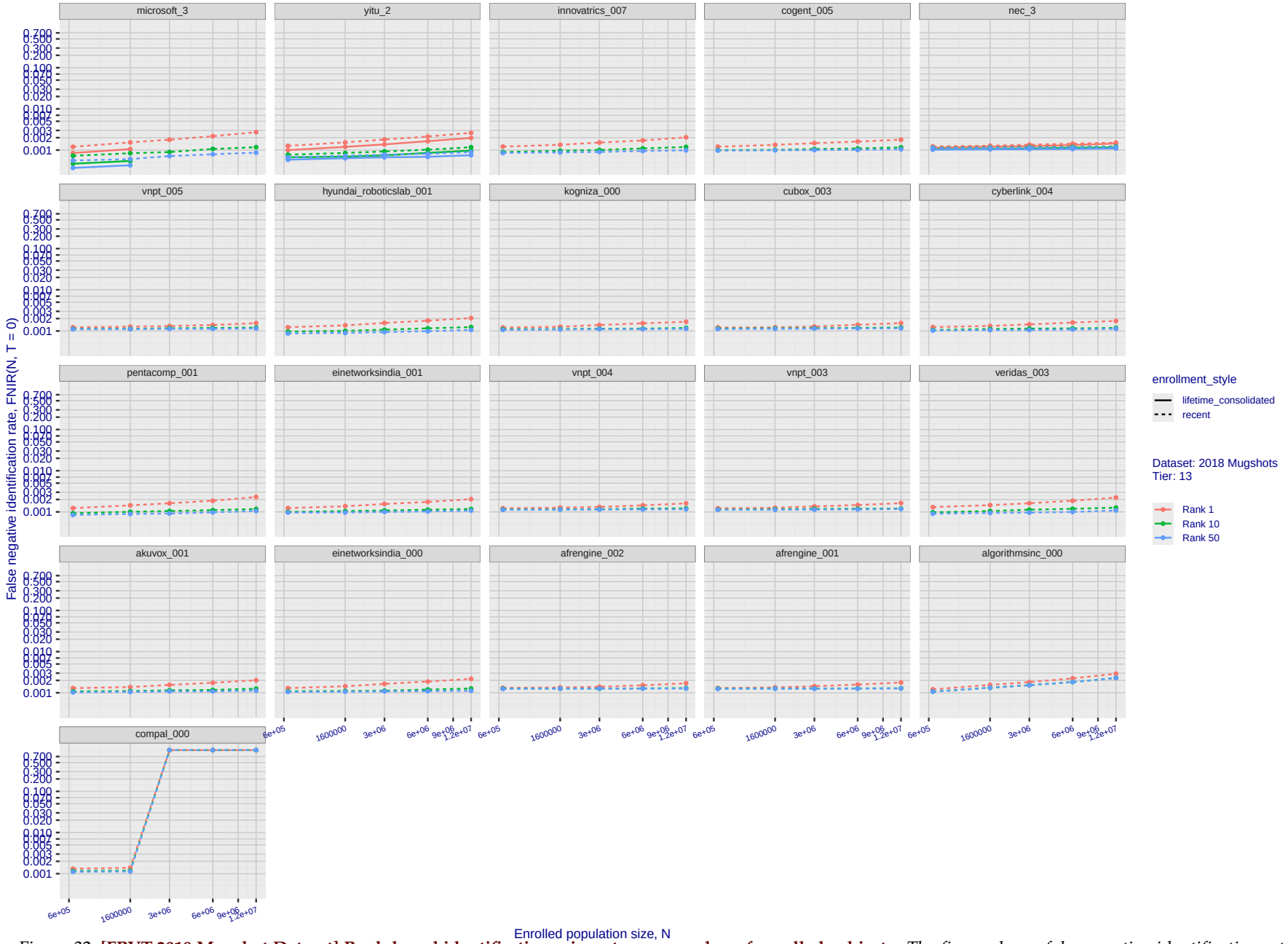


*Figure 32:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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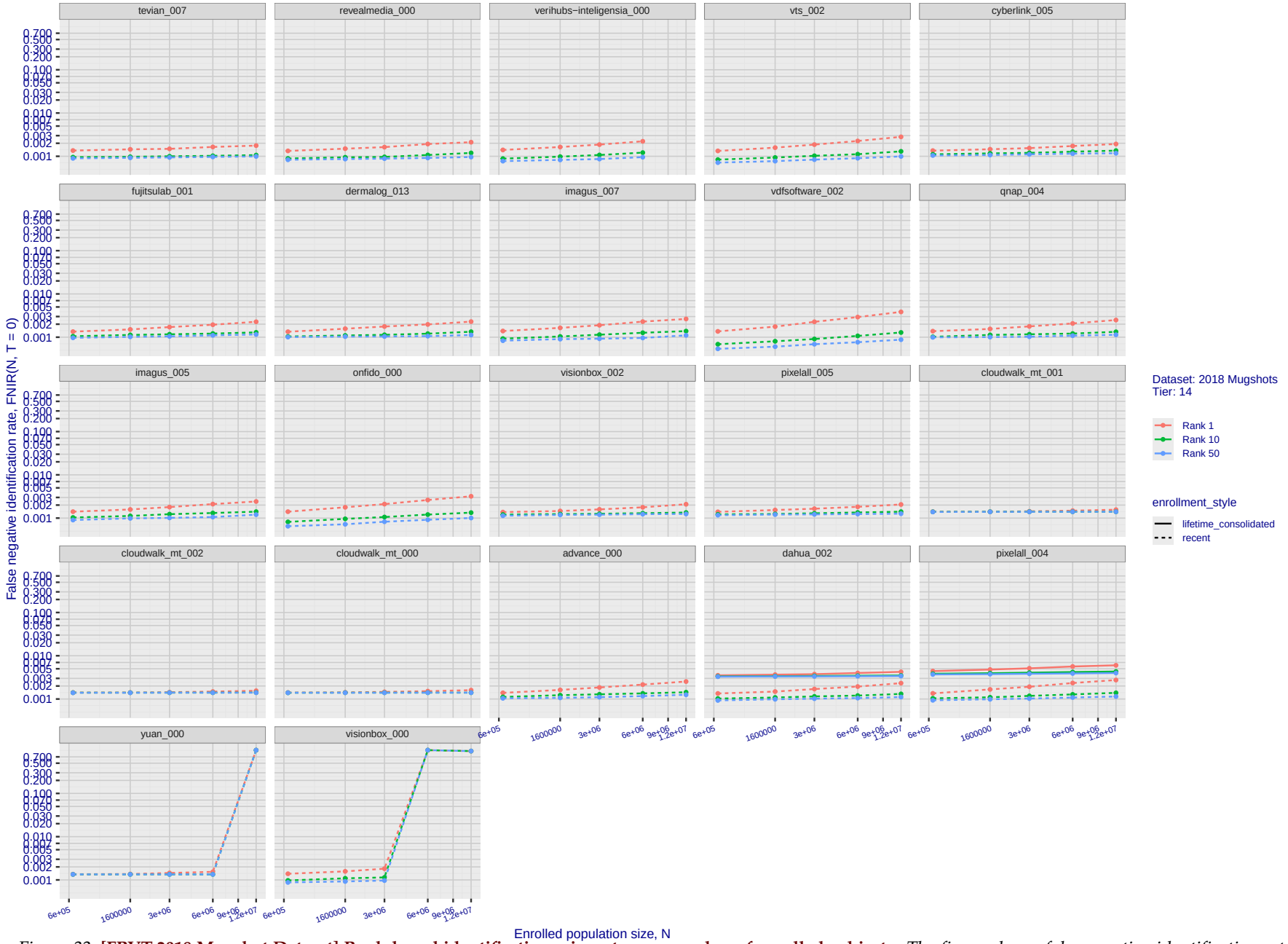


*Figure 33:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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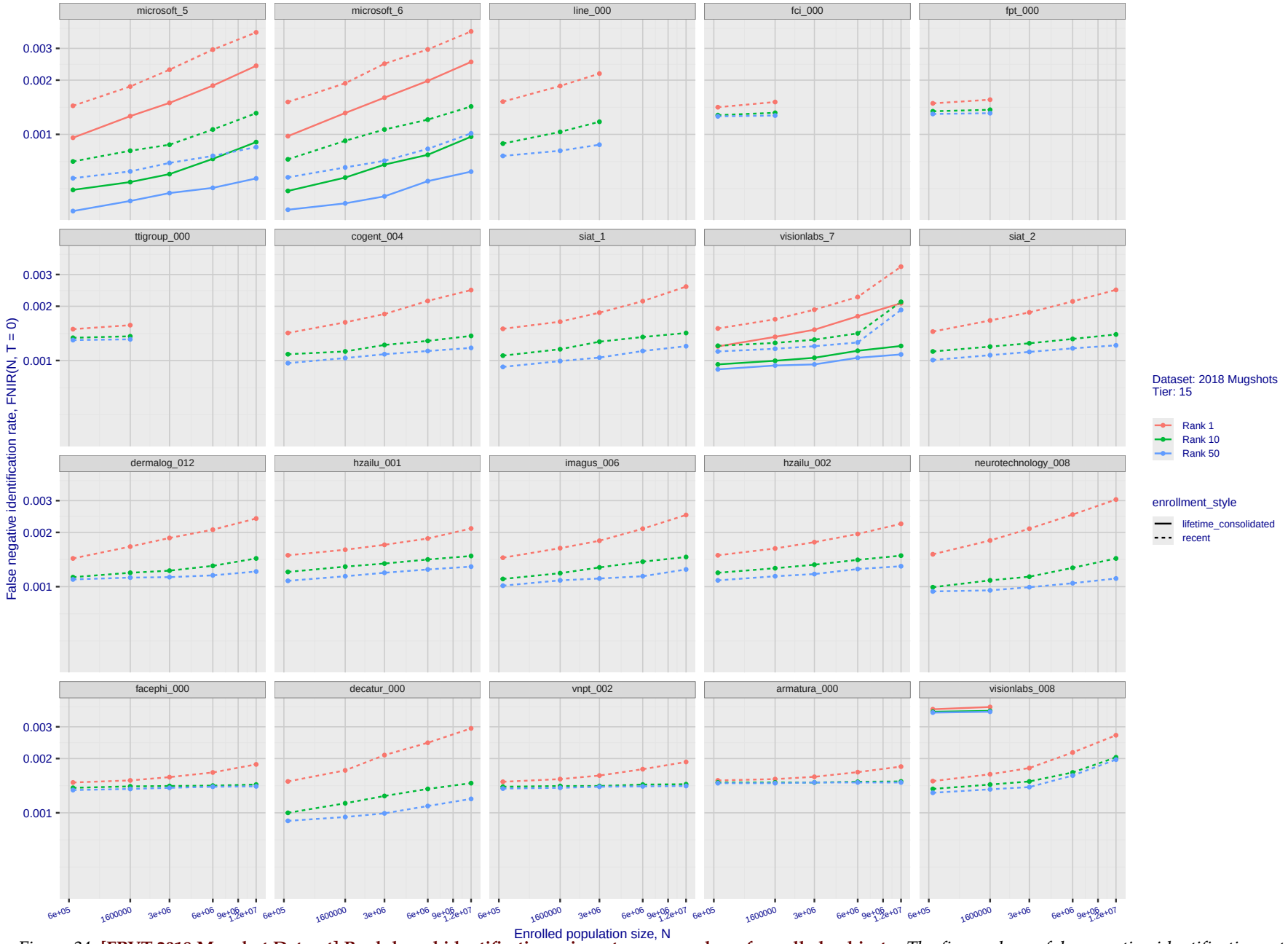


*Figure 34:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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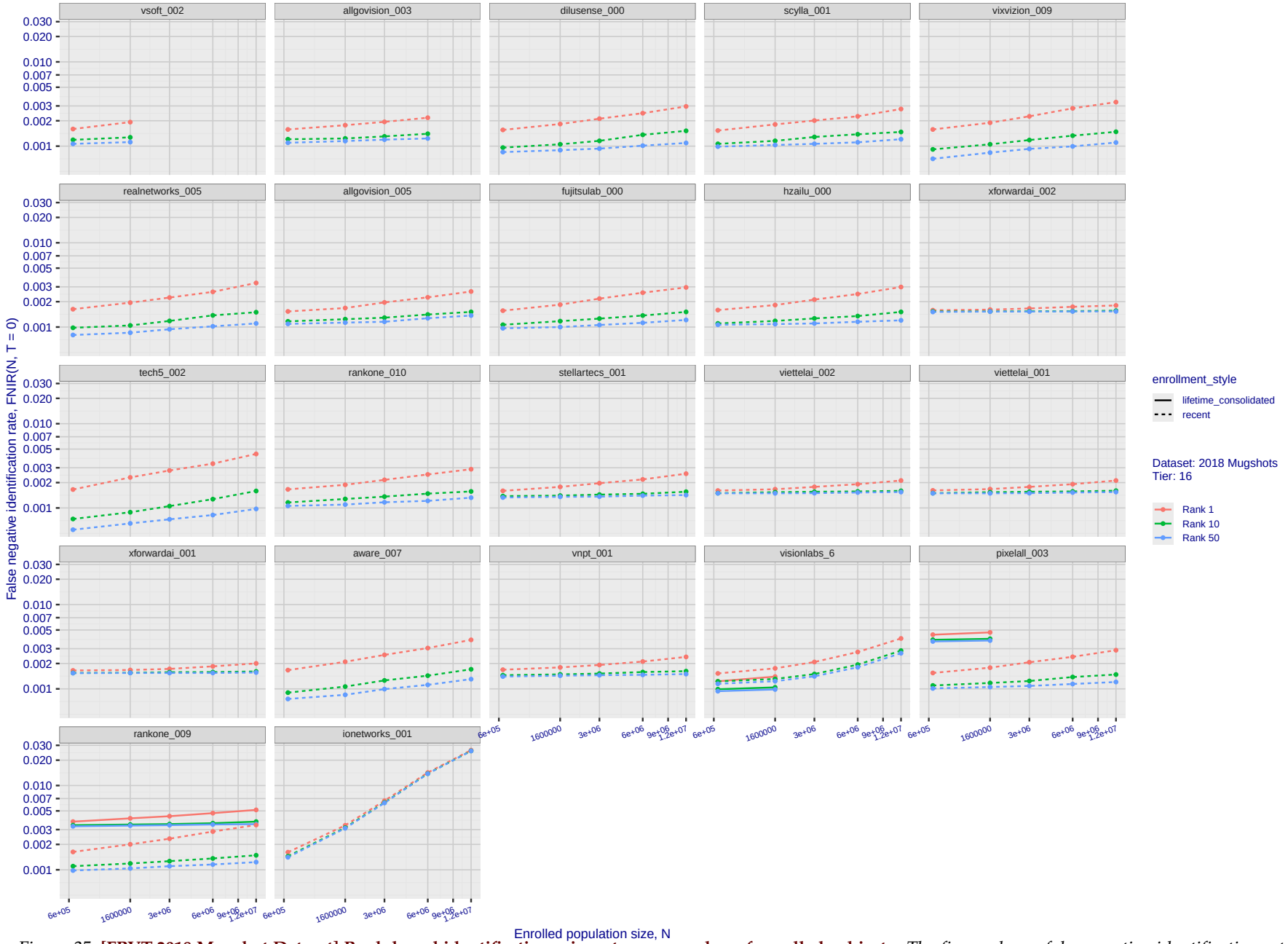


*Figure 35:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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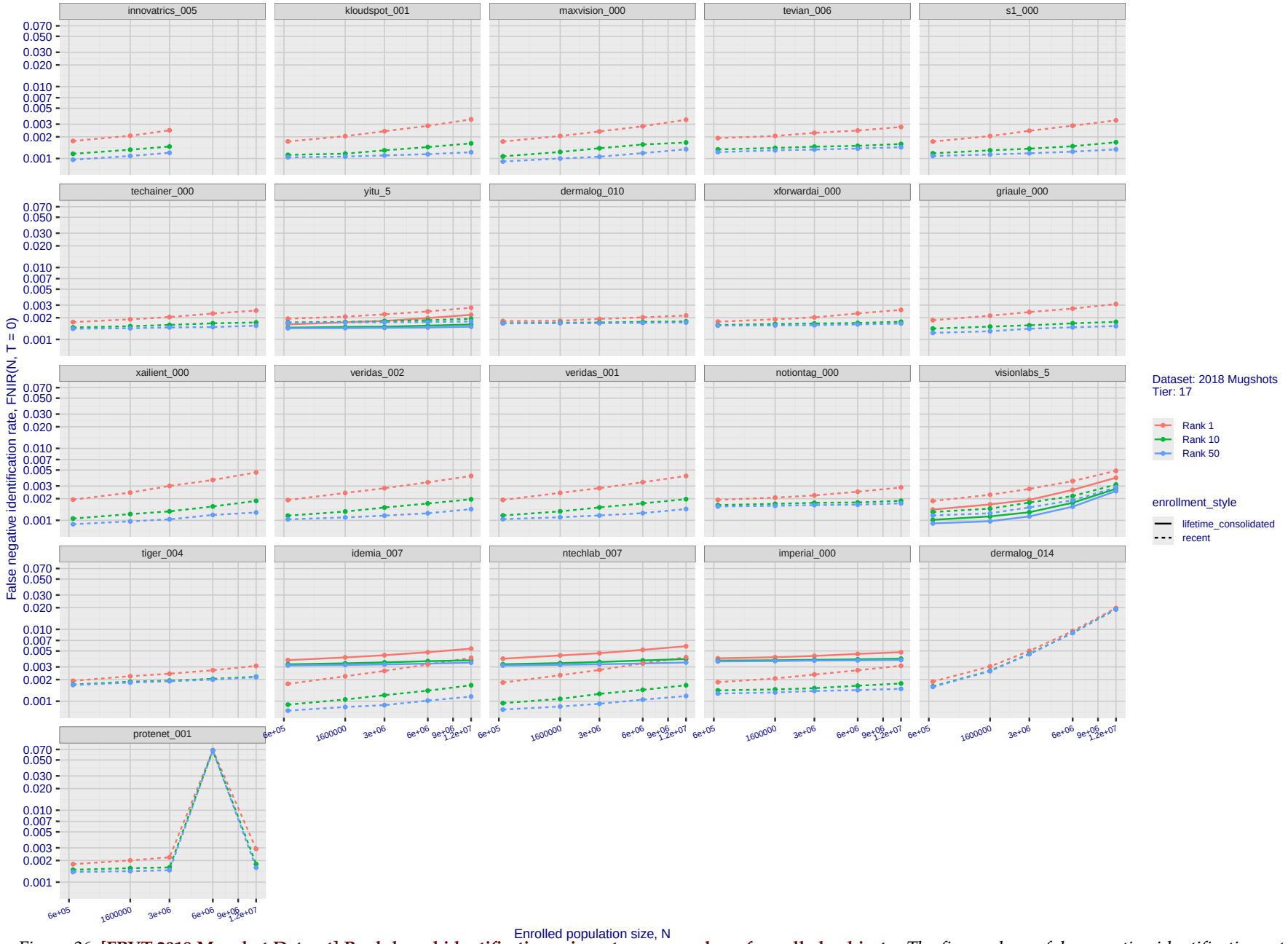


*Figure 36:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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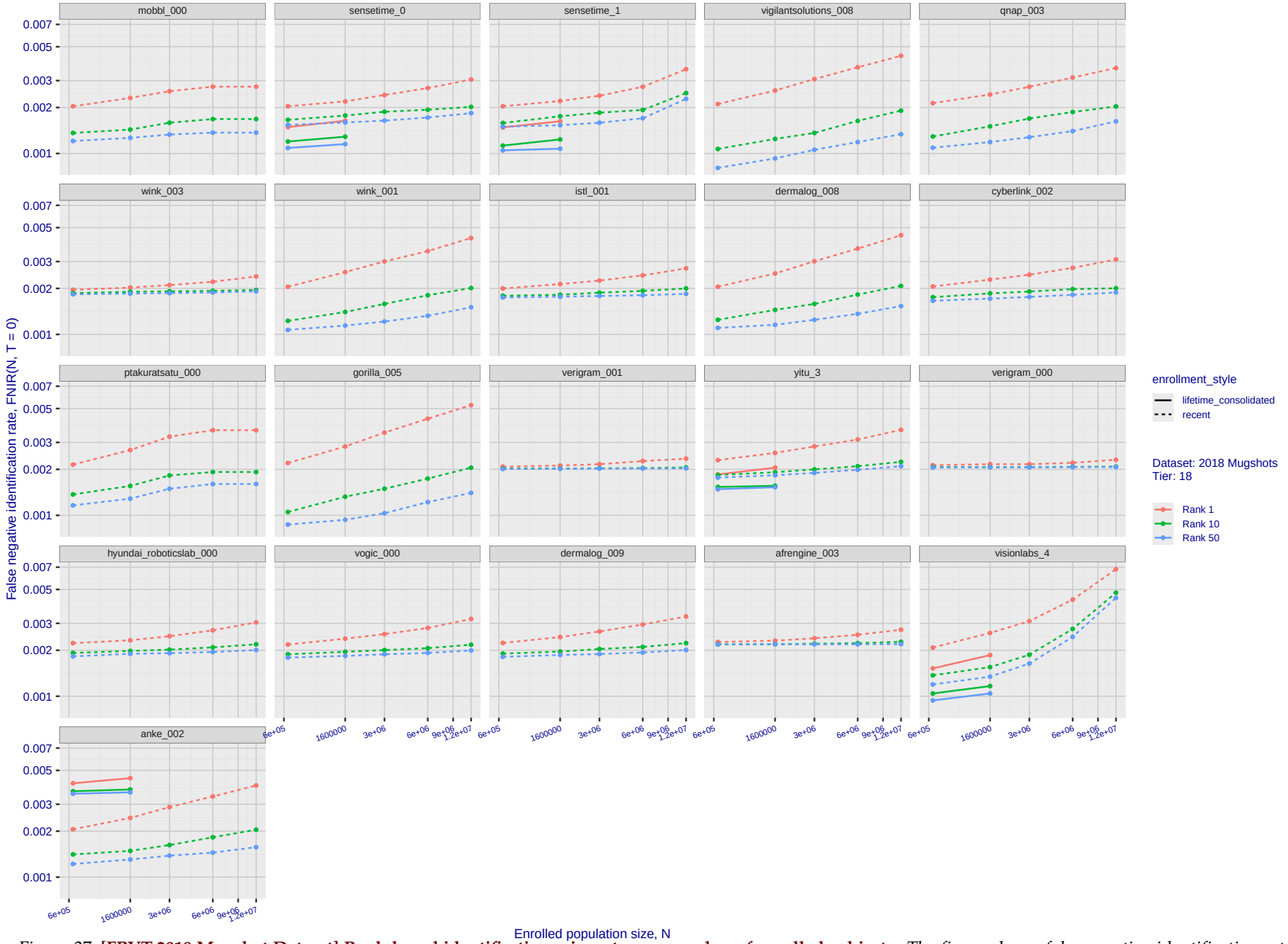


*Figure 37:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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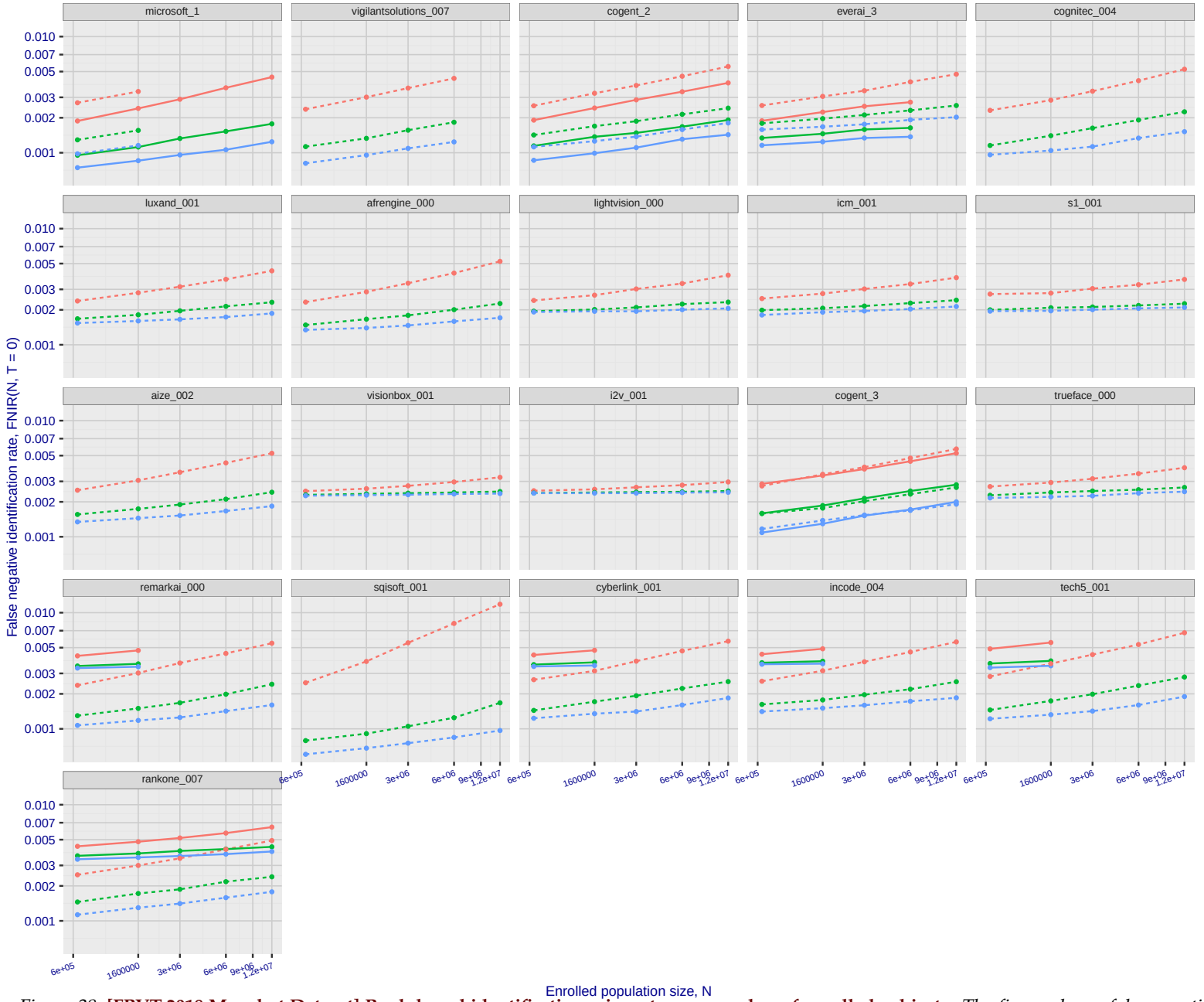


*Figure 38:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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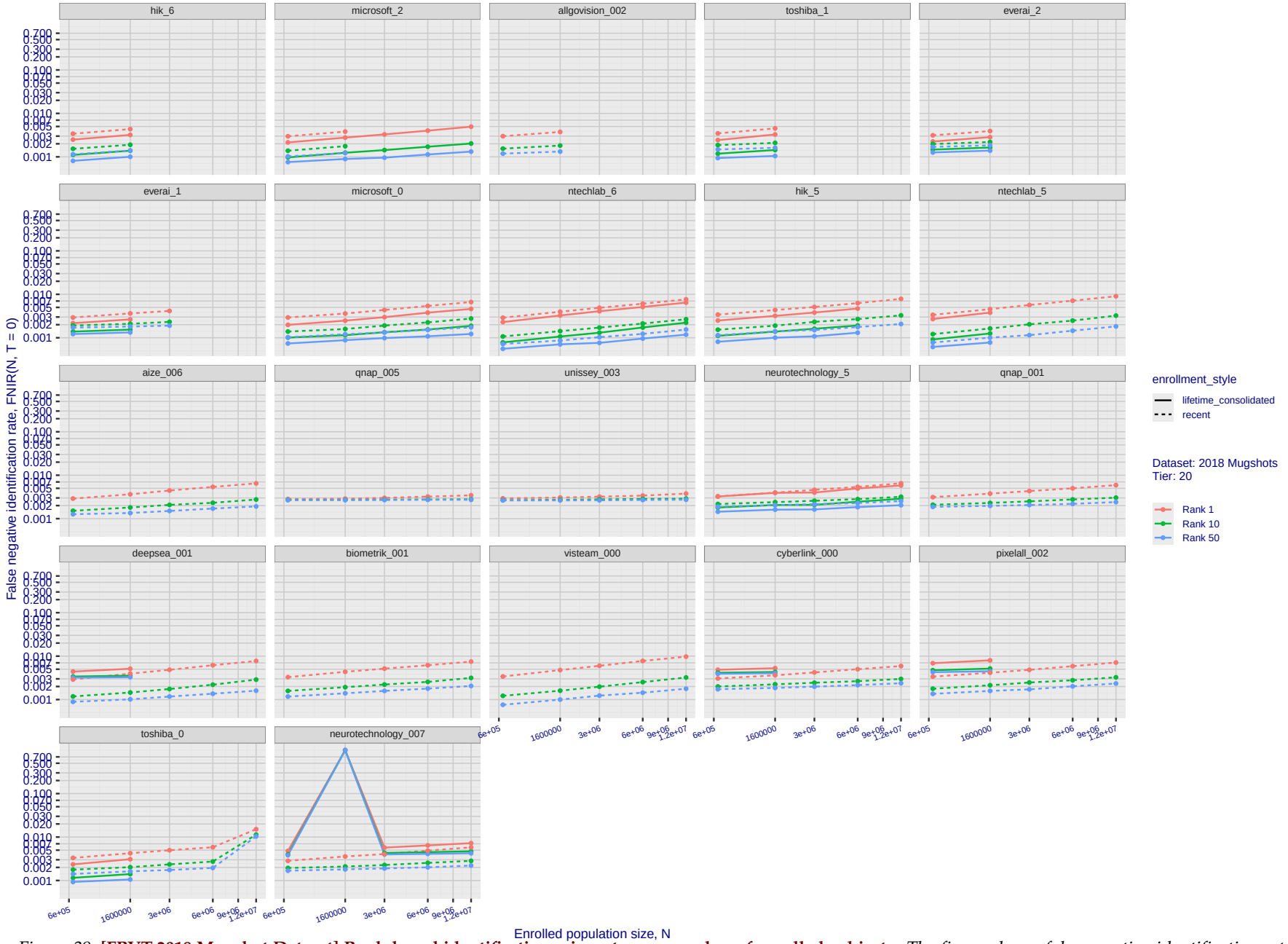


*Figure 39:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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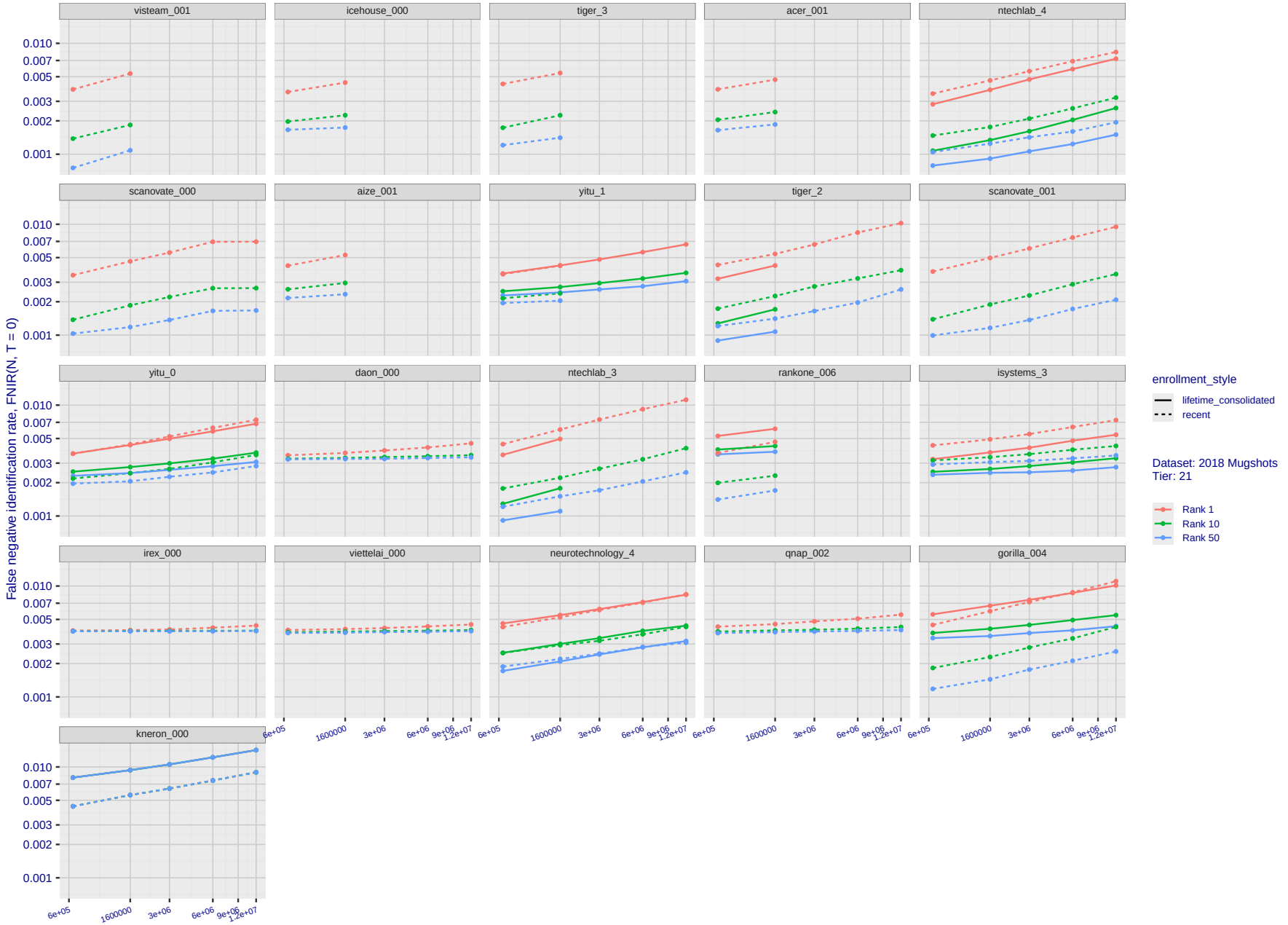


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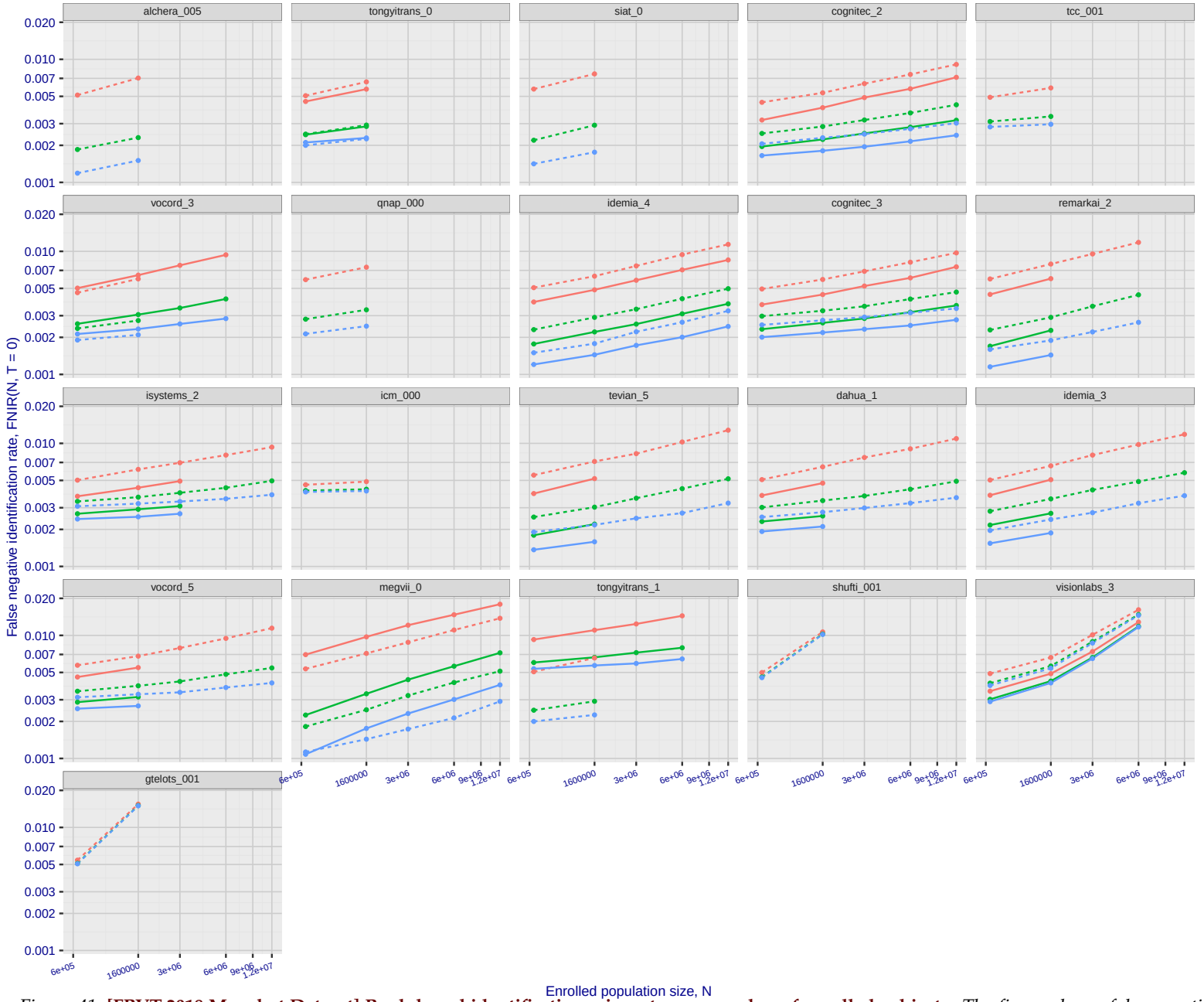


*Figure 41:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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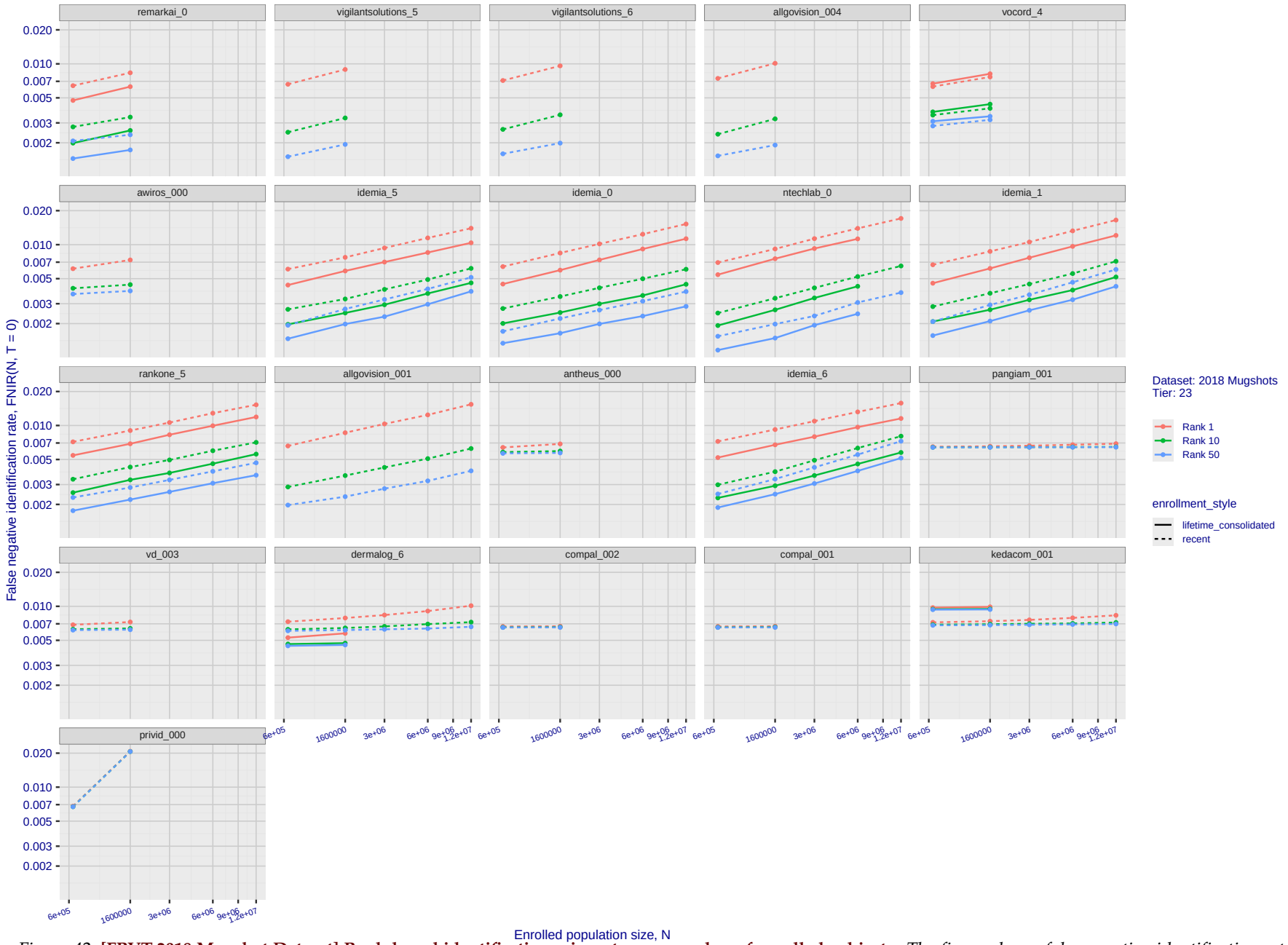


*Figure 42:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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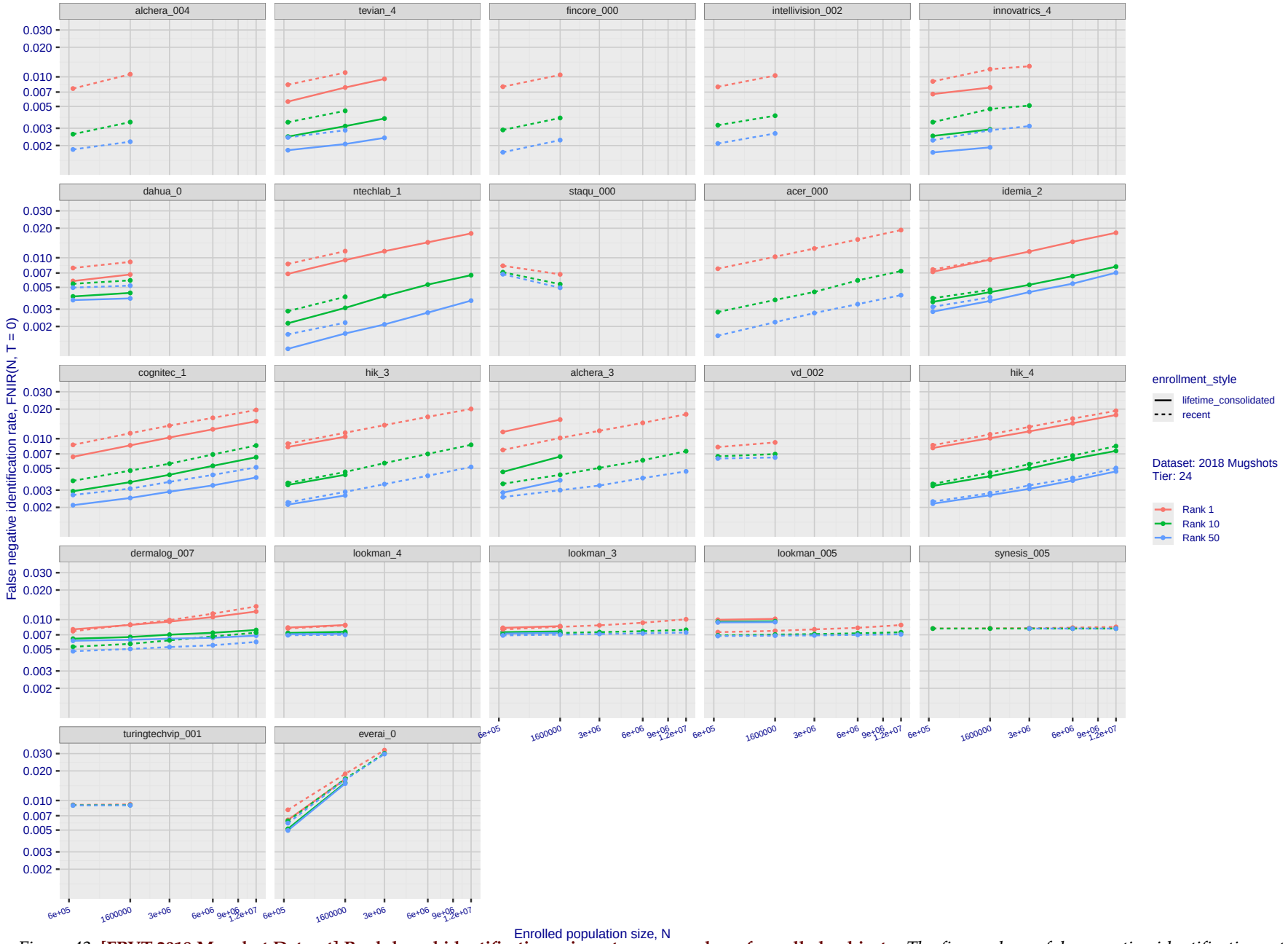


*Figure 43:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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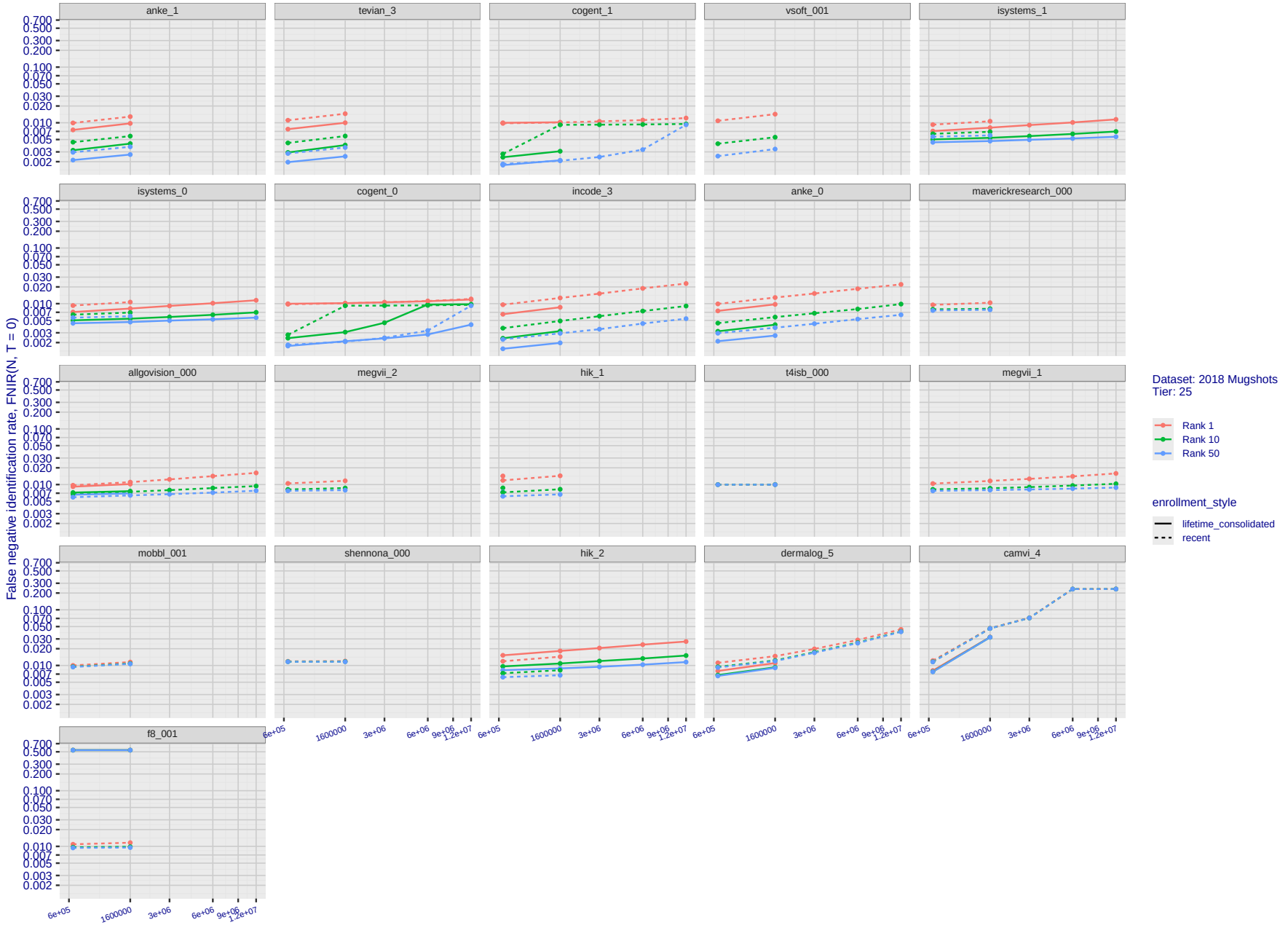


*Figure 44:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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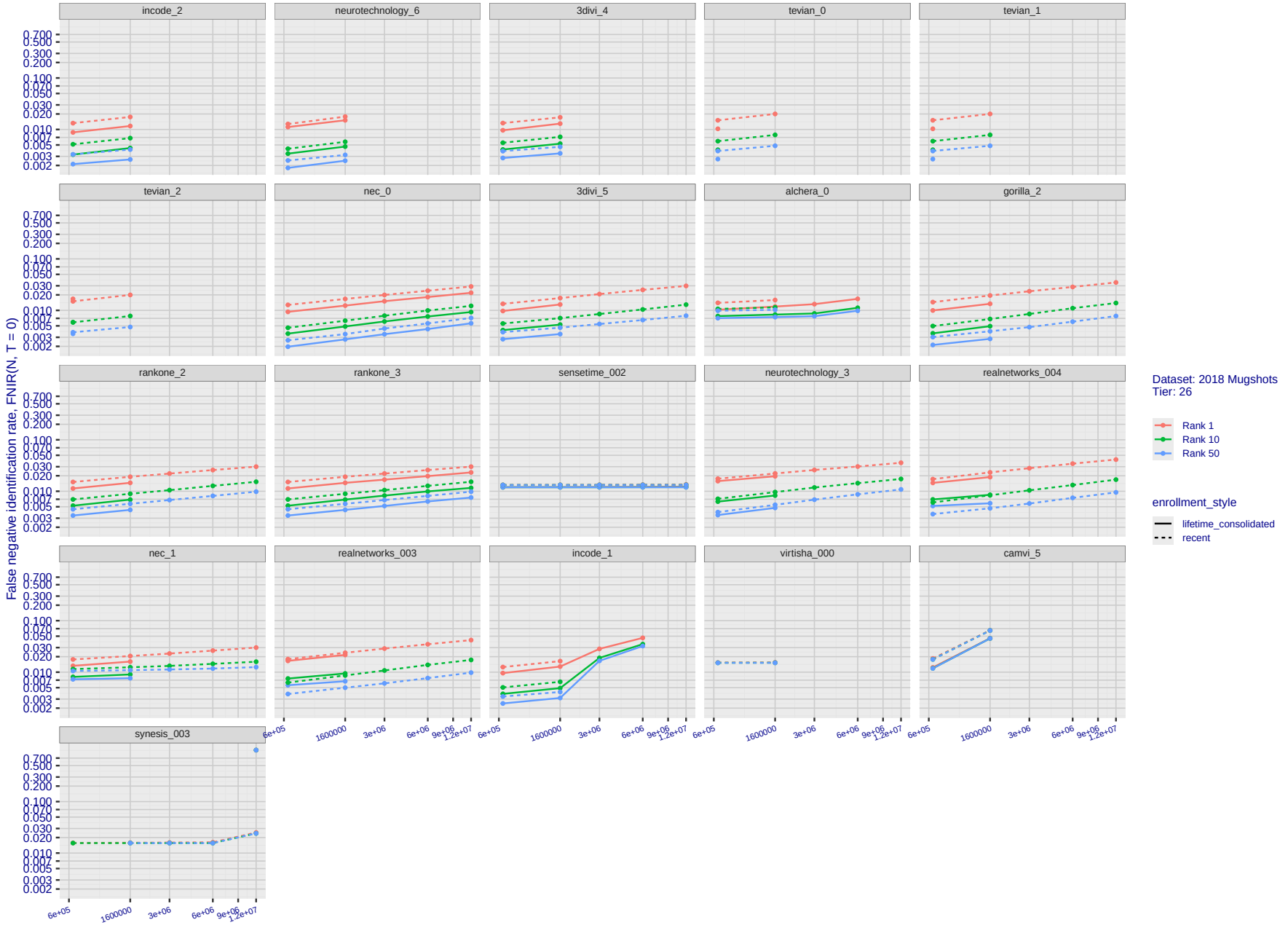


*Figure 45:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



2026/09/04 10:23:32
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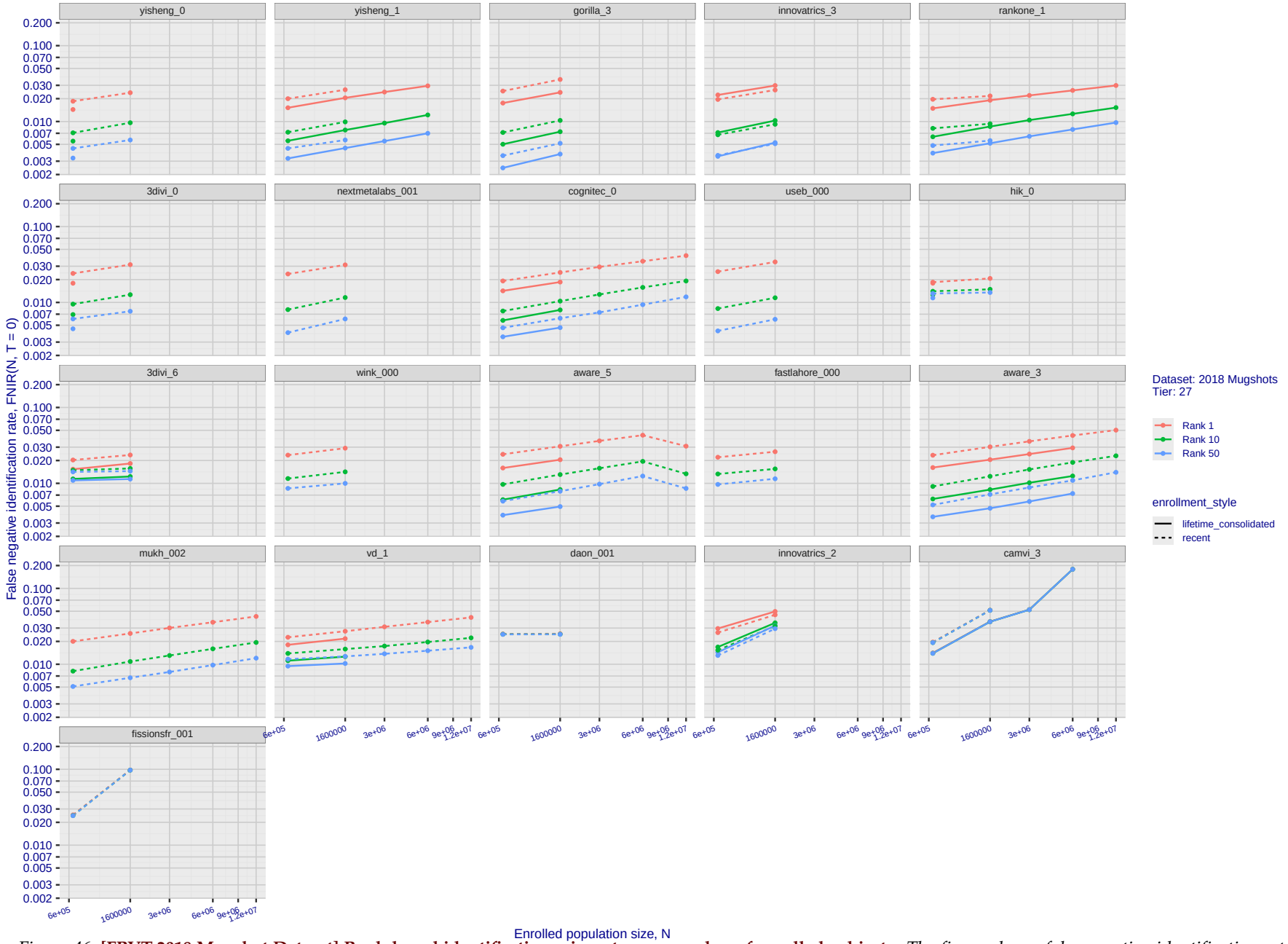


*Figure 46:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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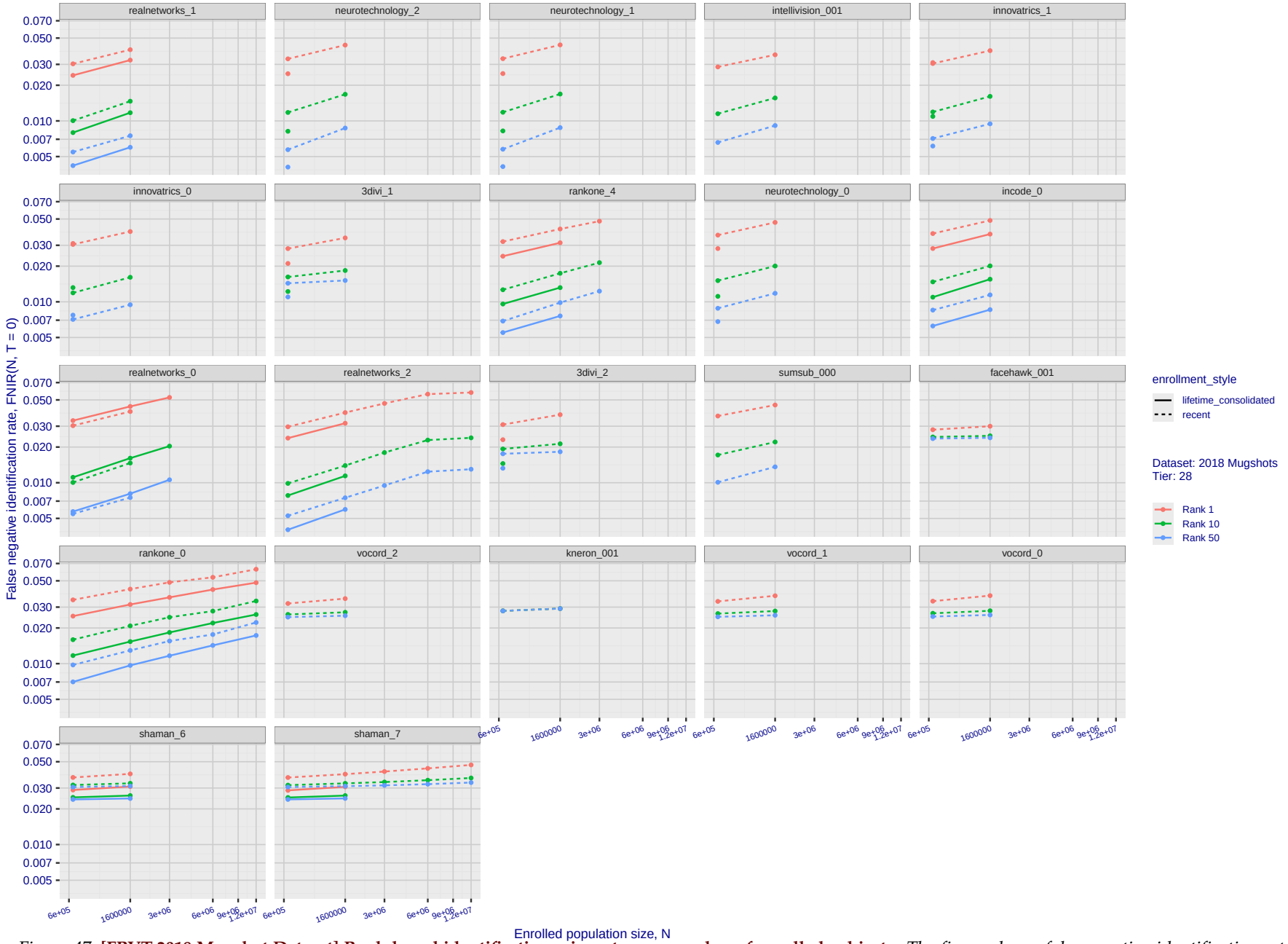


*Figure 47:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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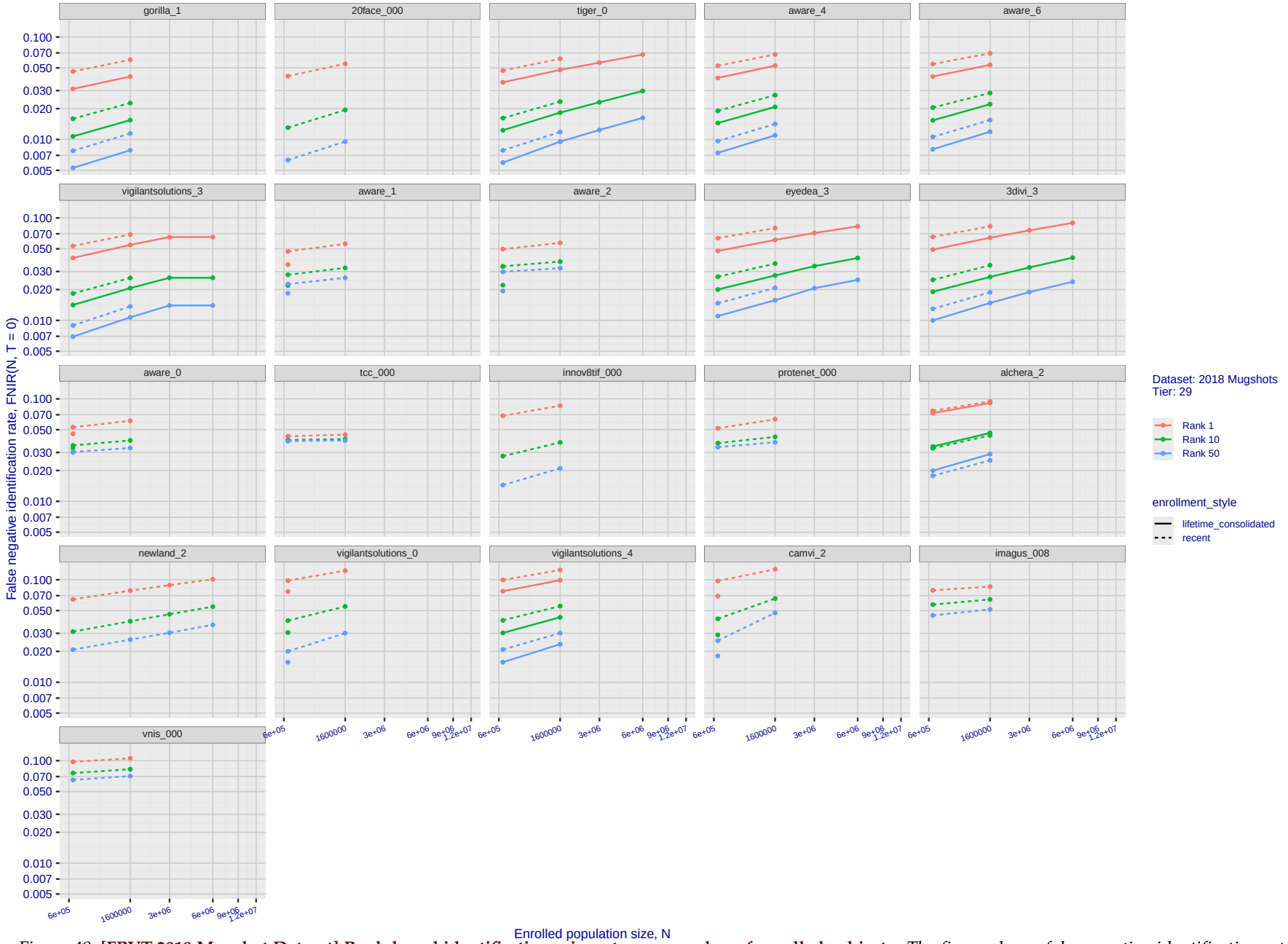


*Figure 48:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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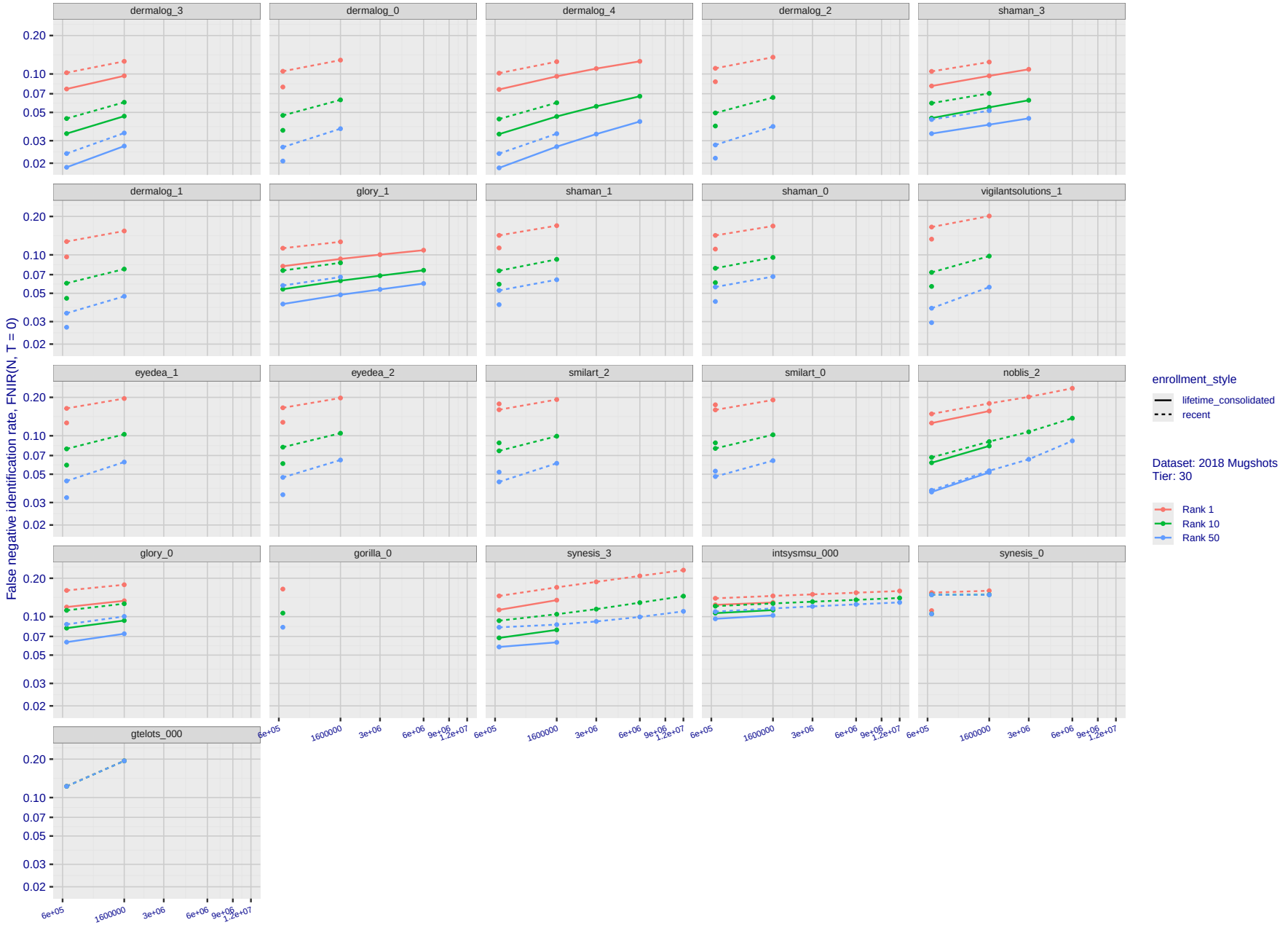


*Figure 49:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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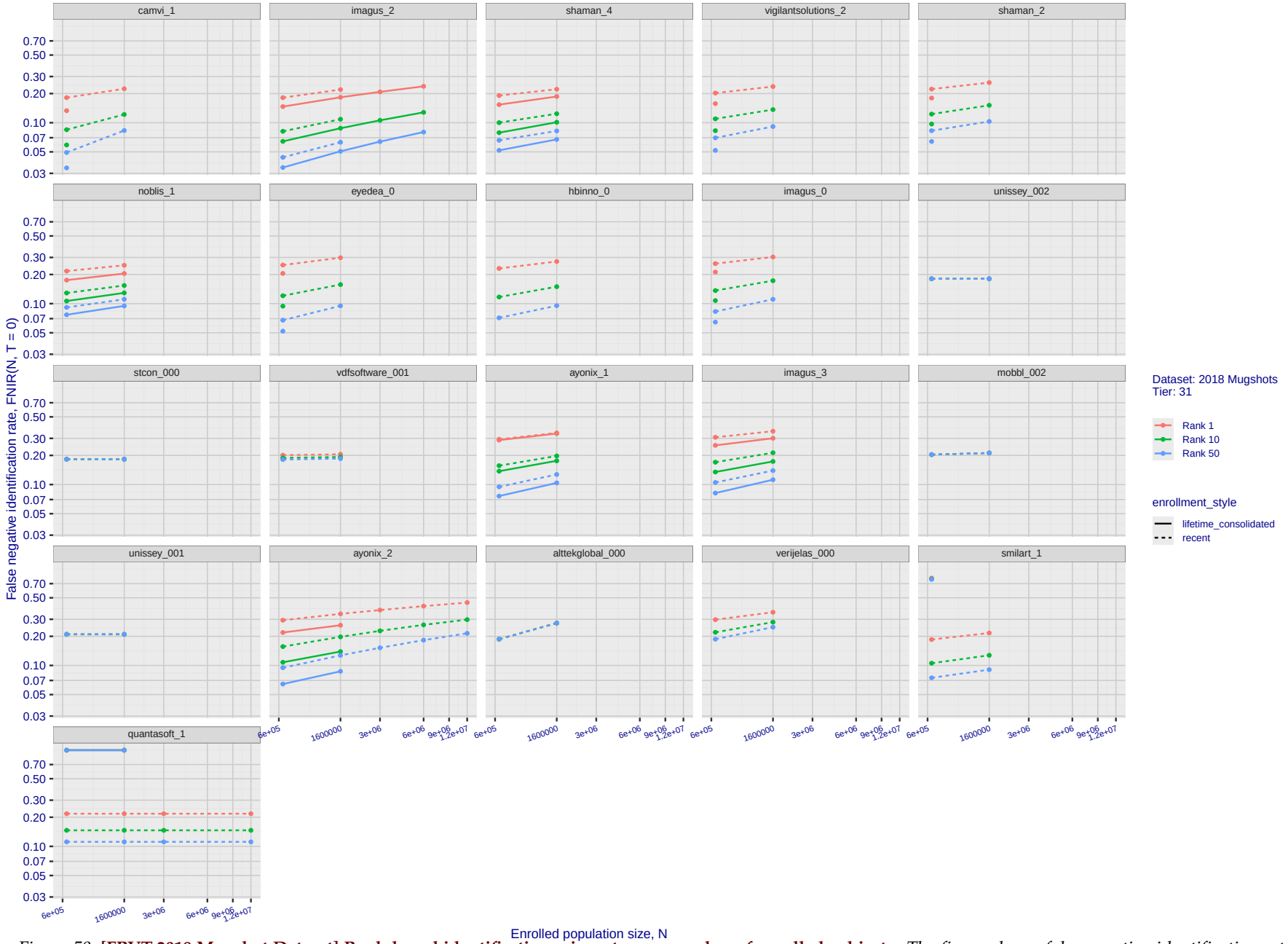


*Figure 50:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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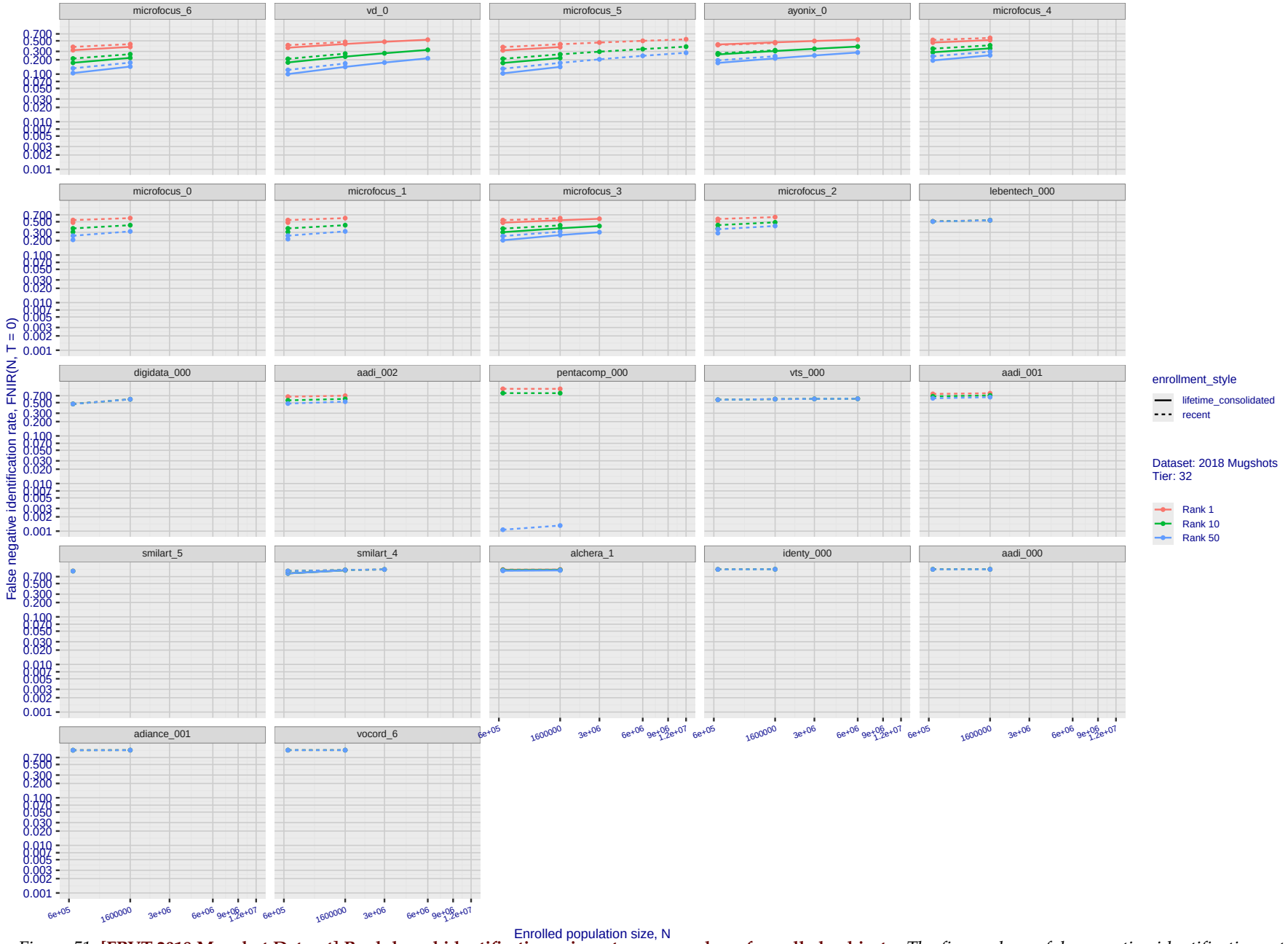


*Figure 51:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*



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| | | | | | |
|---|---|---|---|---|---|
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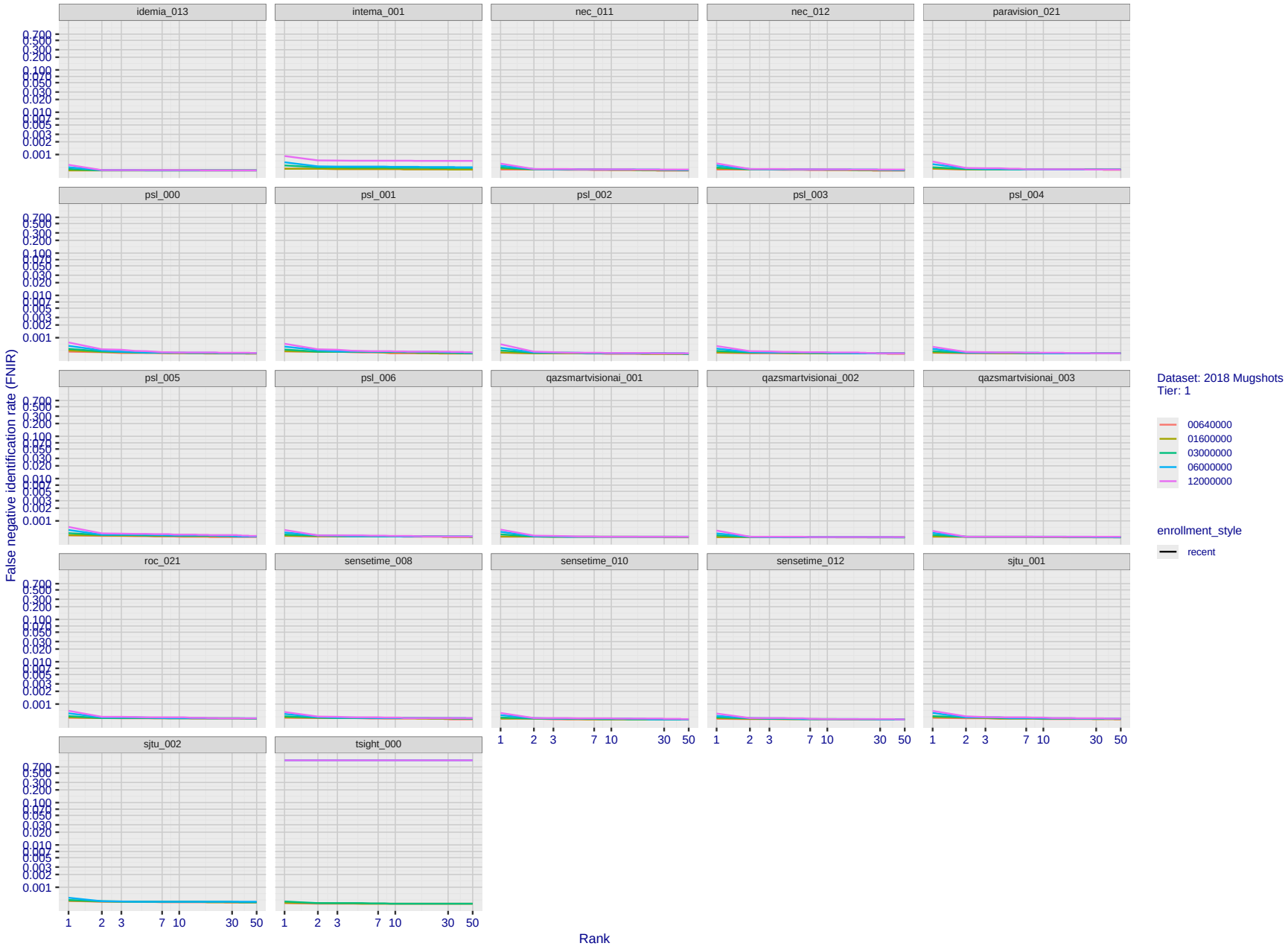


*Figure 52:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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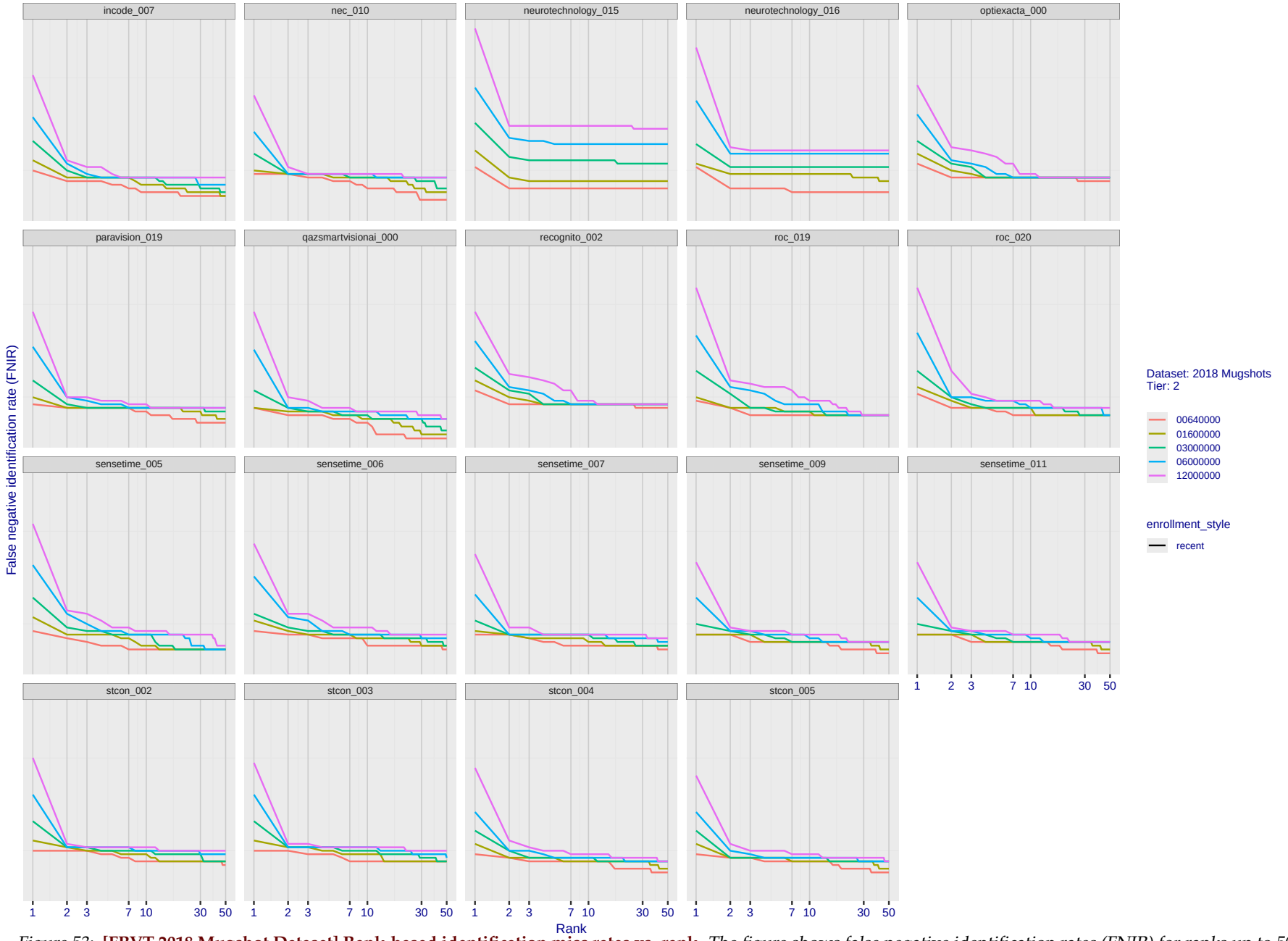


*Figure 53:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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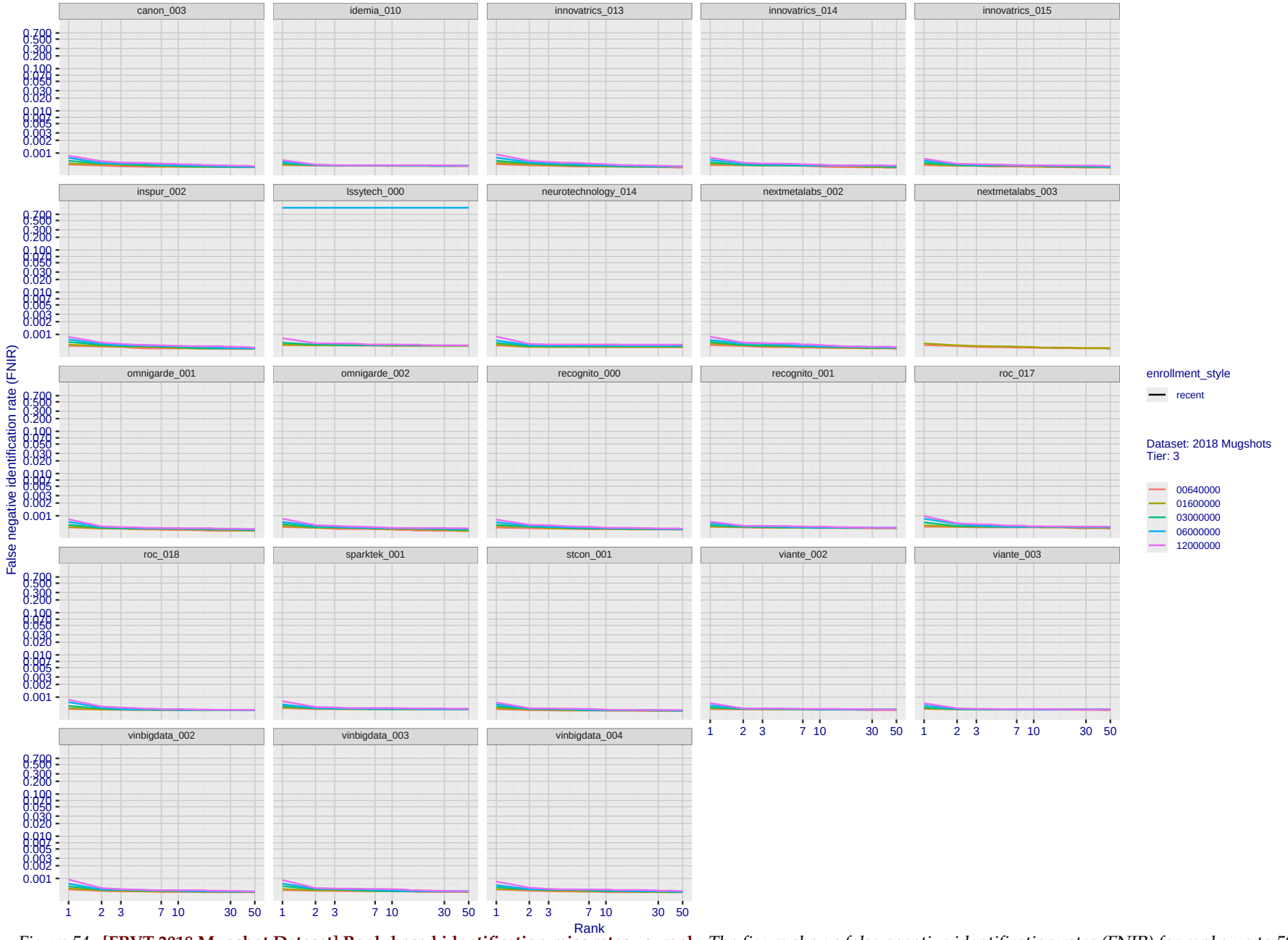


*Figure 54:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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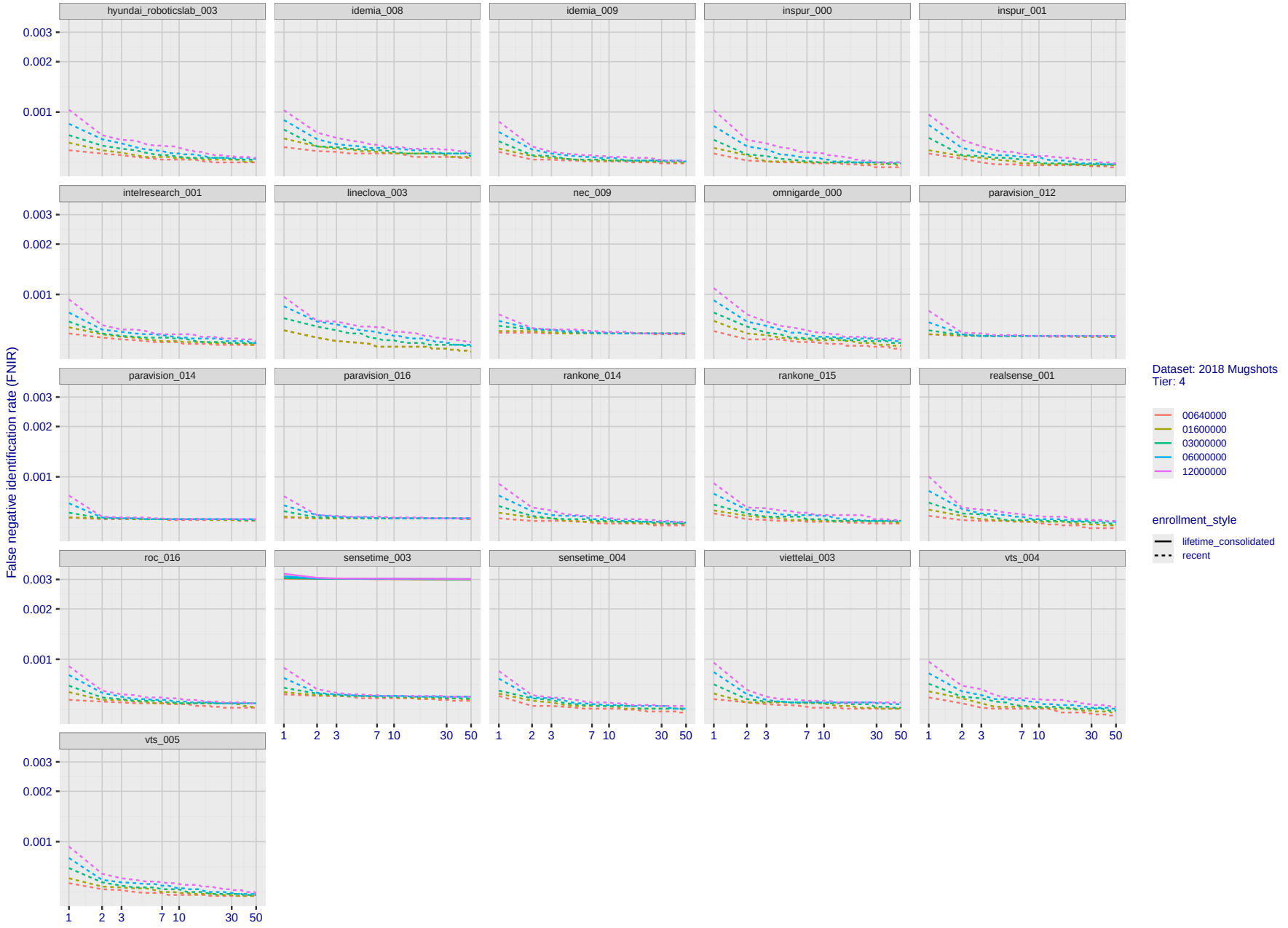


*Figure 55:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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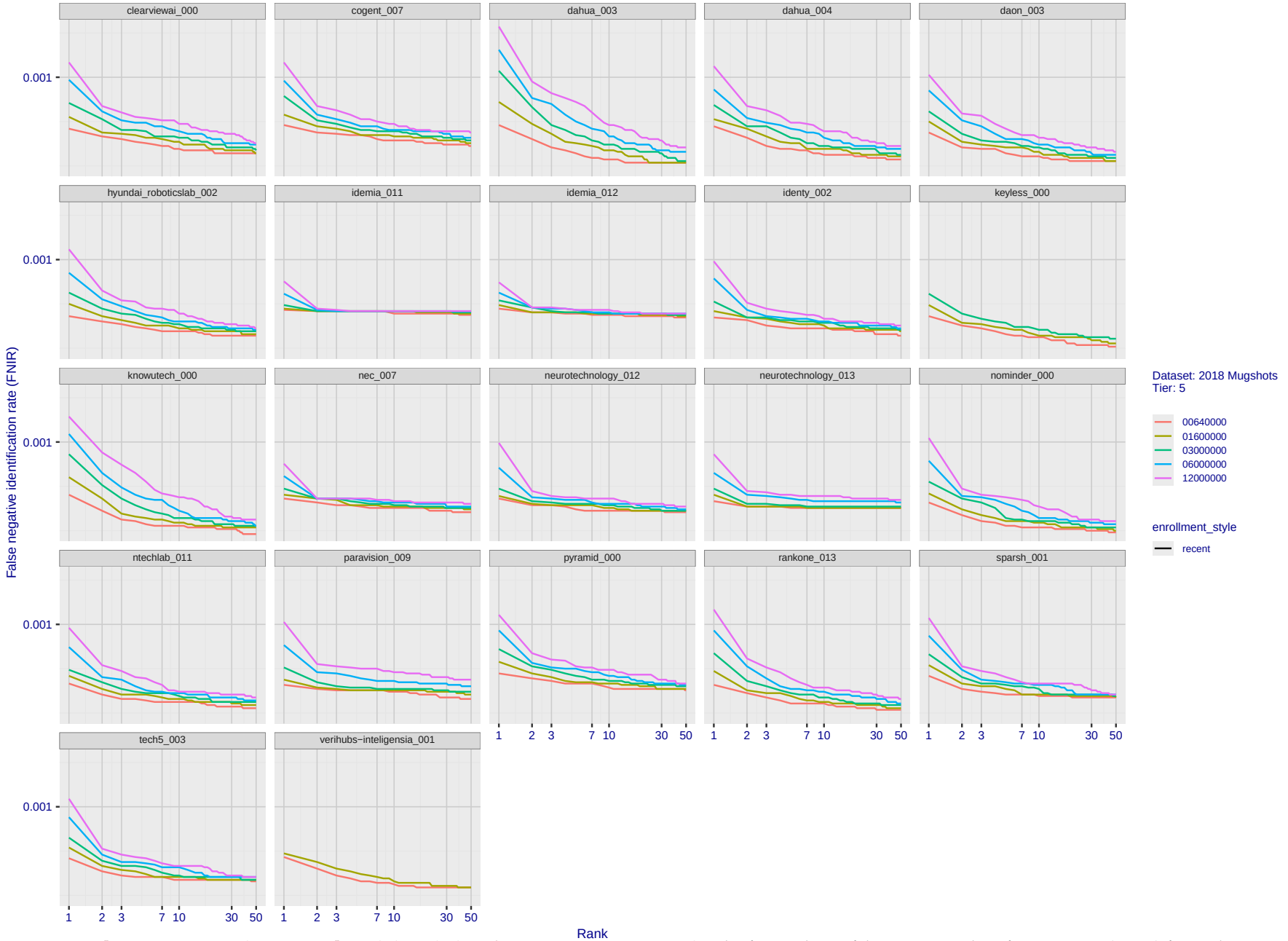


*Figure 56:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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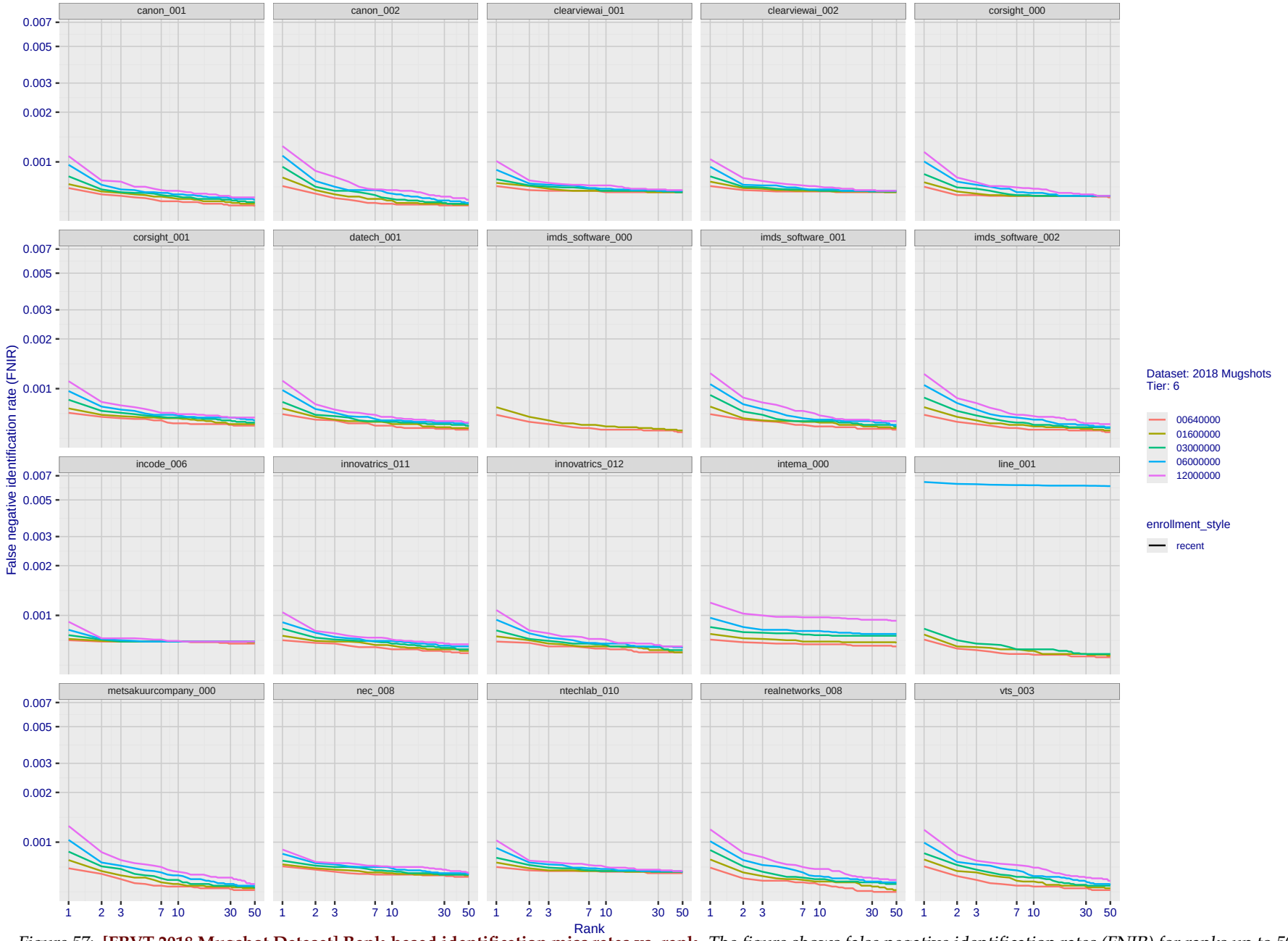


*Figure 57:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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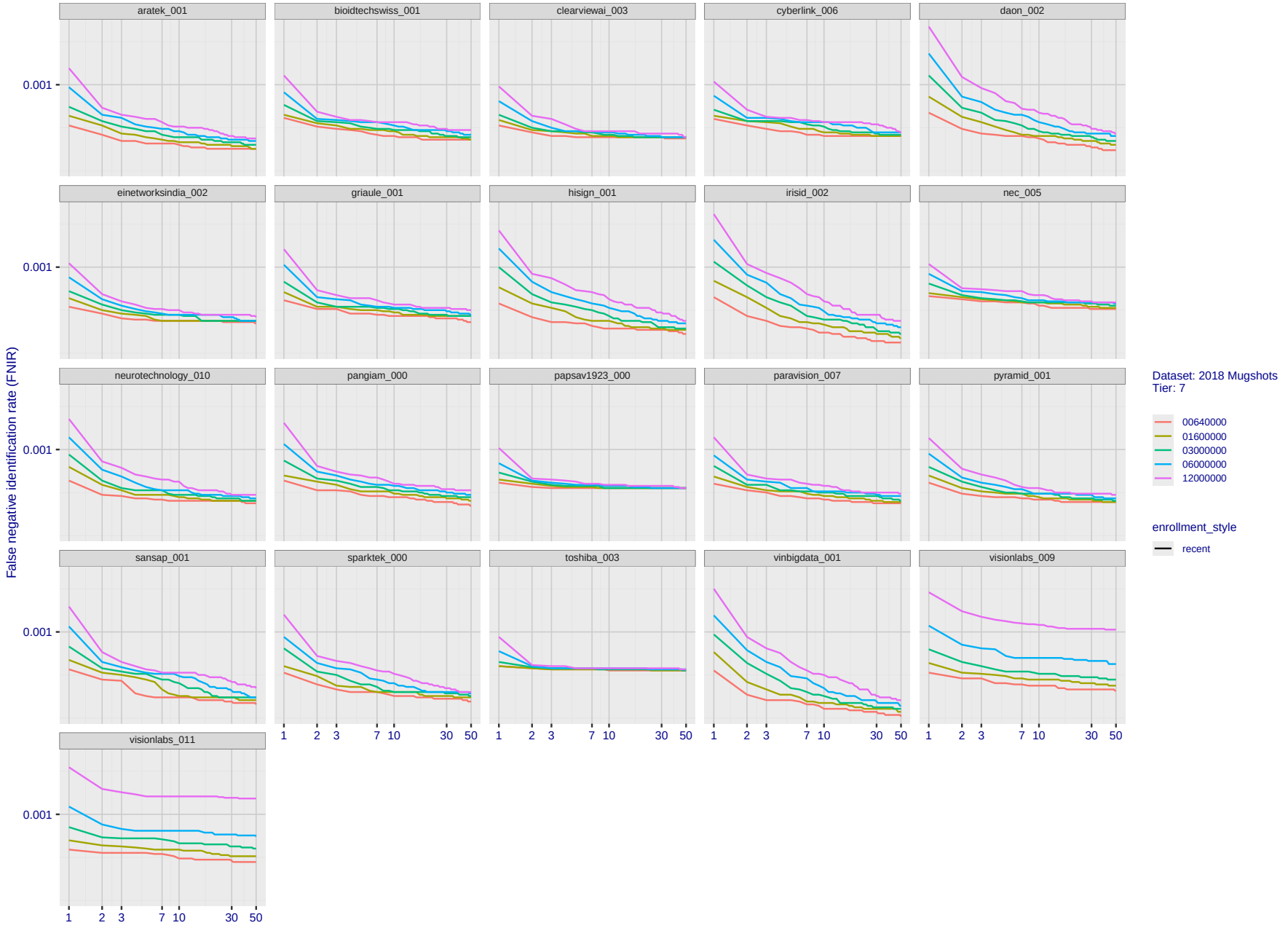


*Figure 58:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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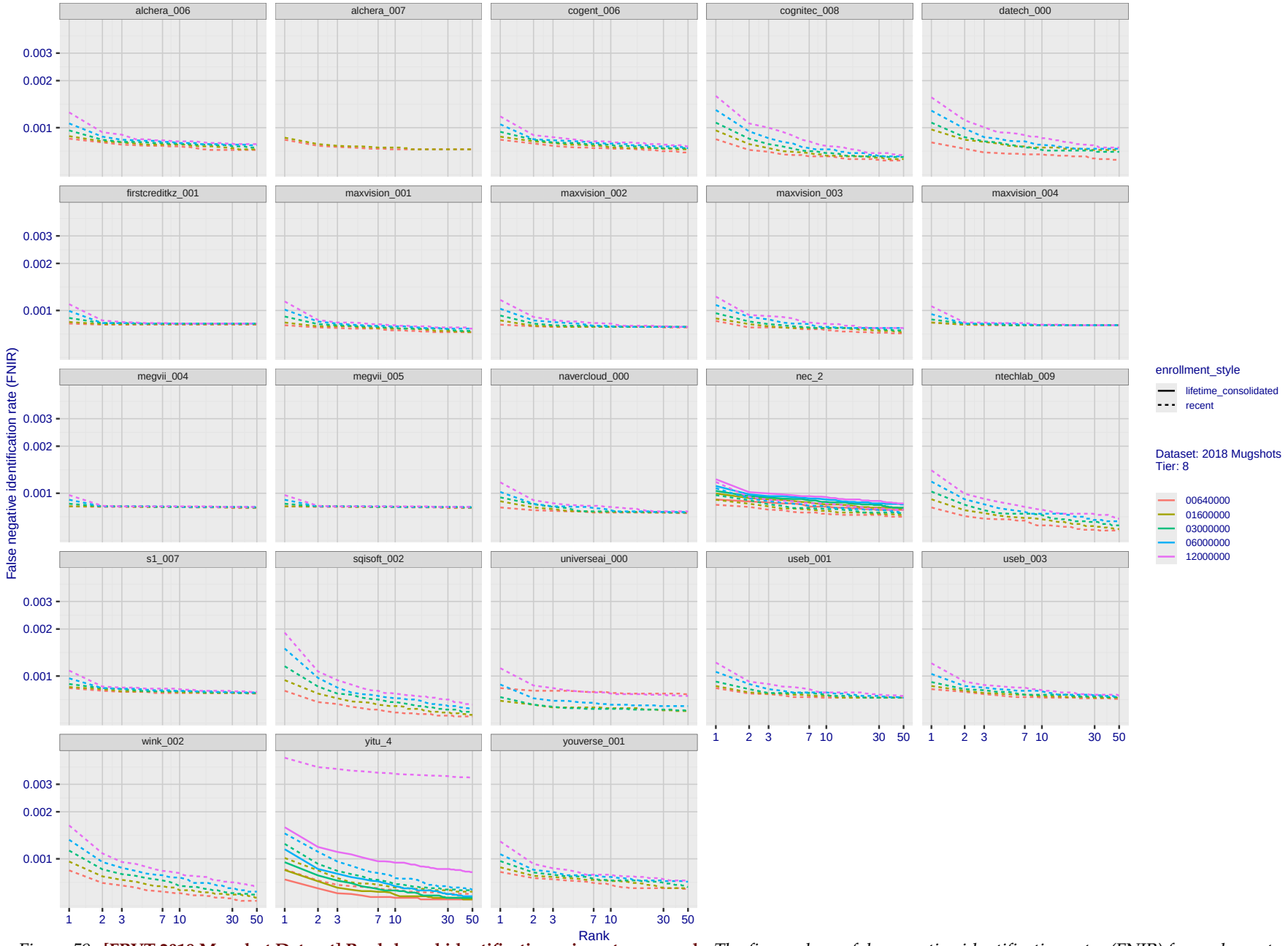


*Figure 59:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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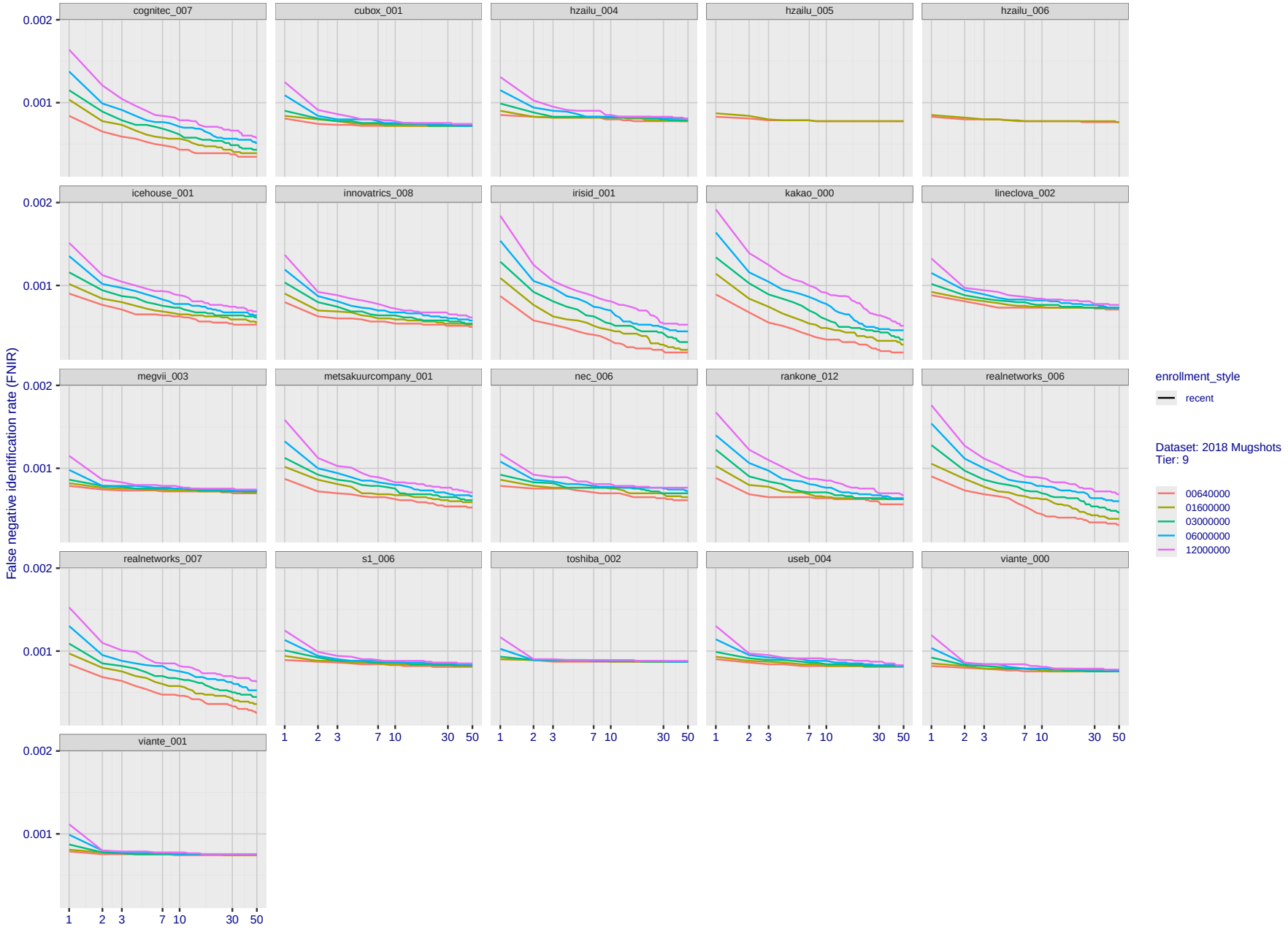


*Figure 60:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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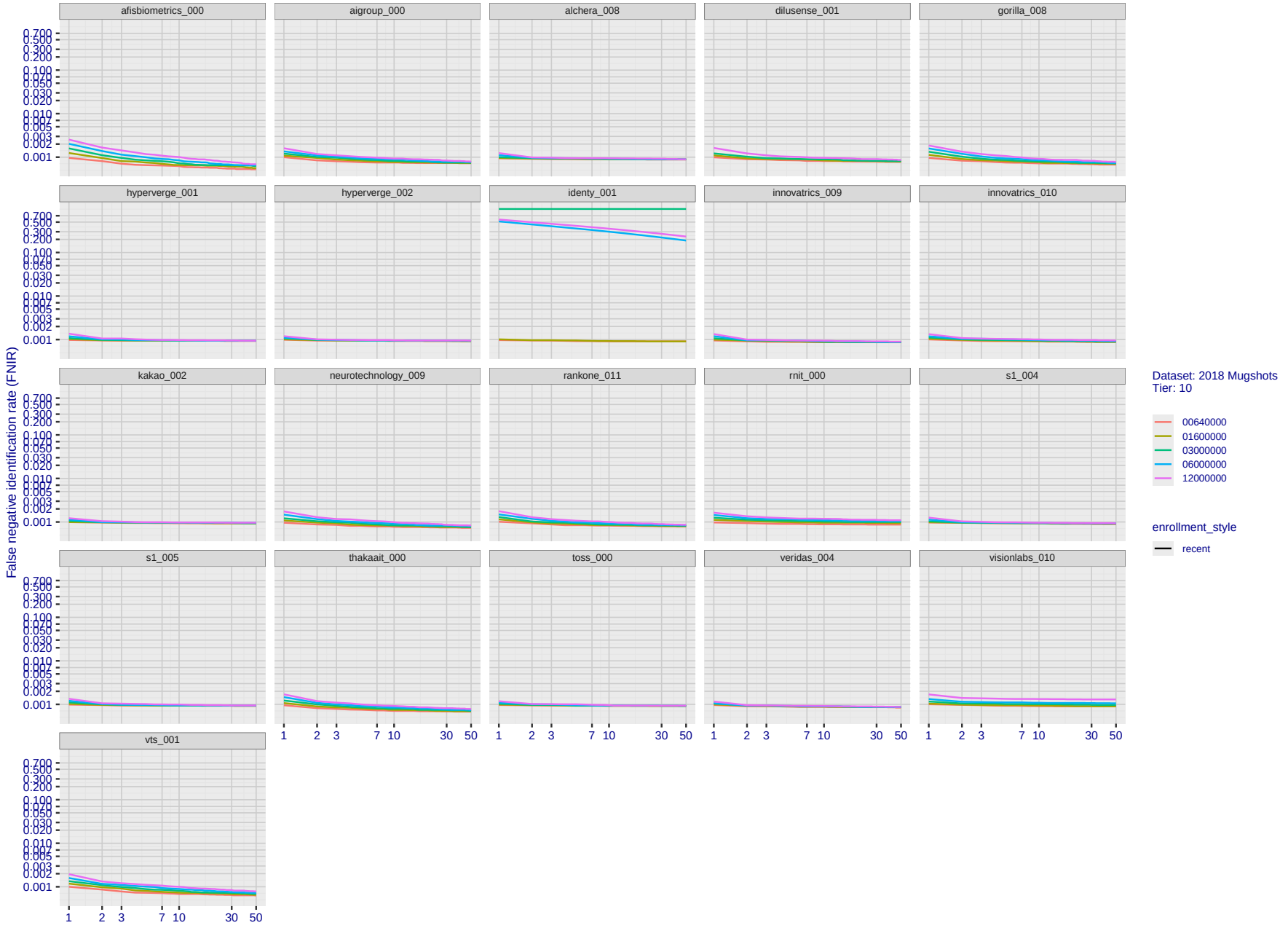


*Figure 61:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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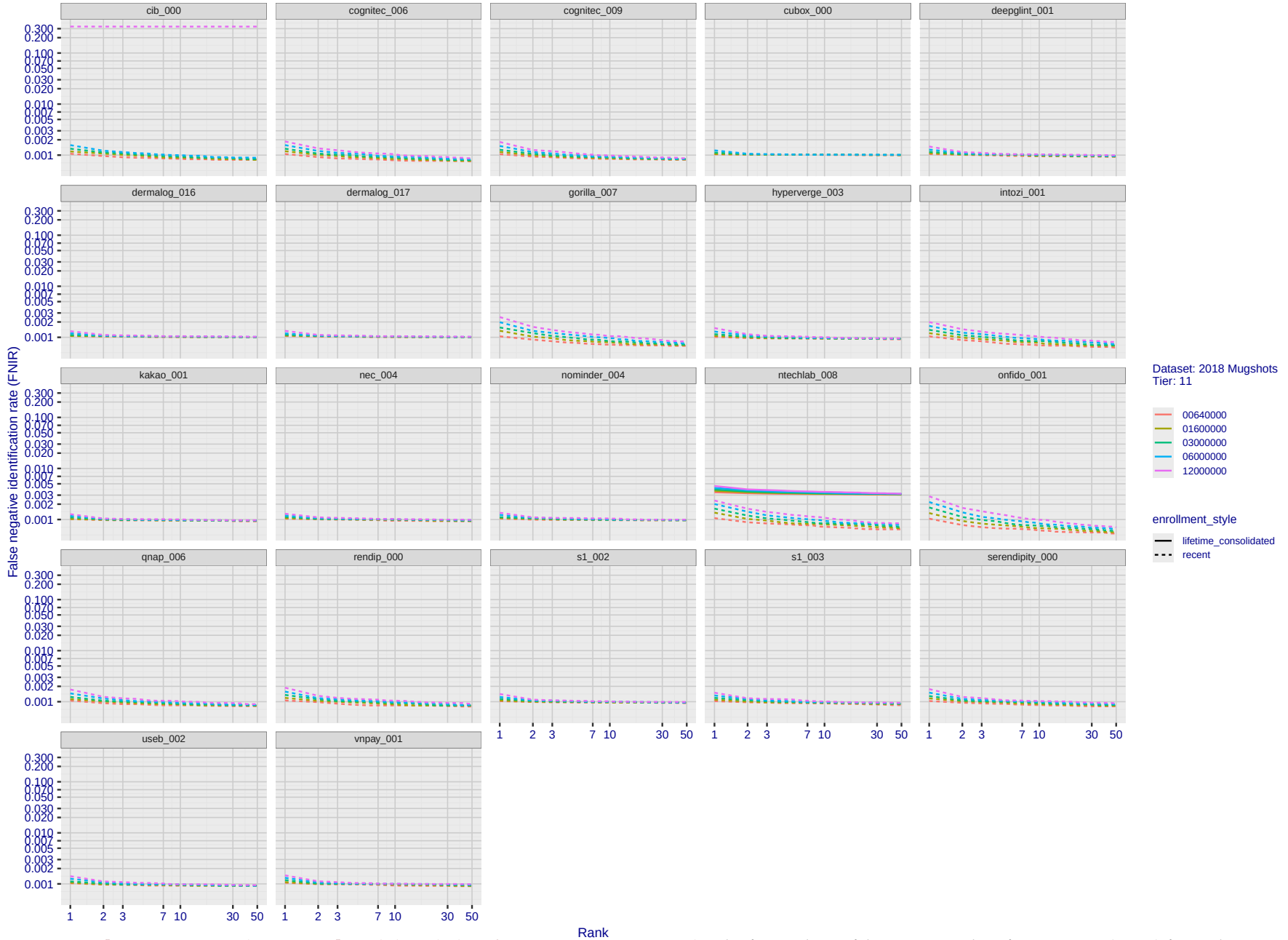


*Figure 62:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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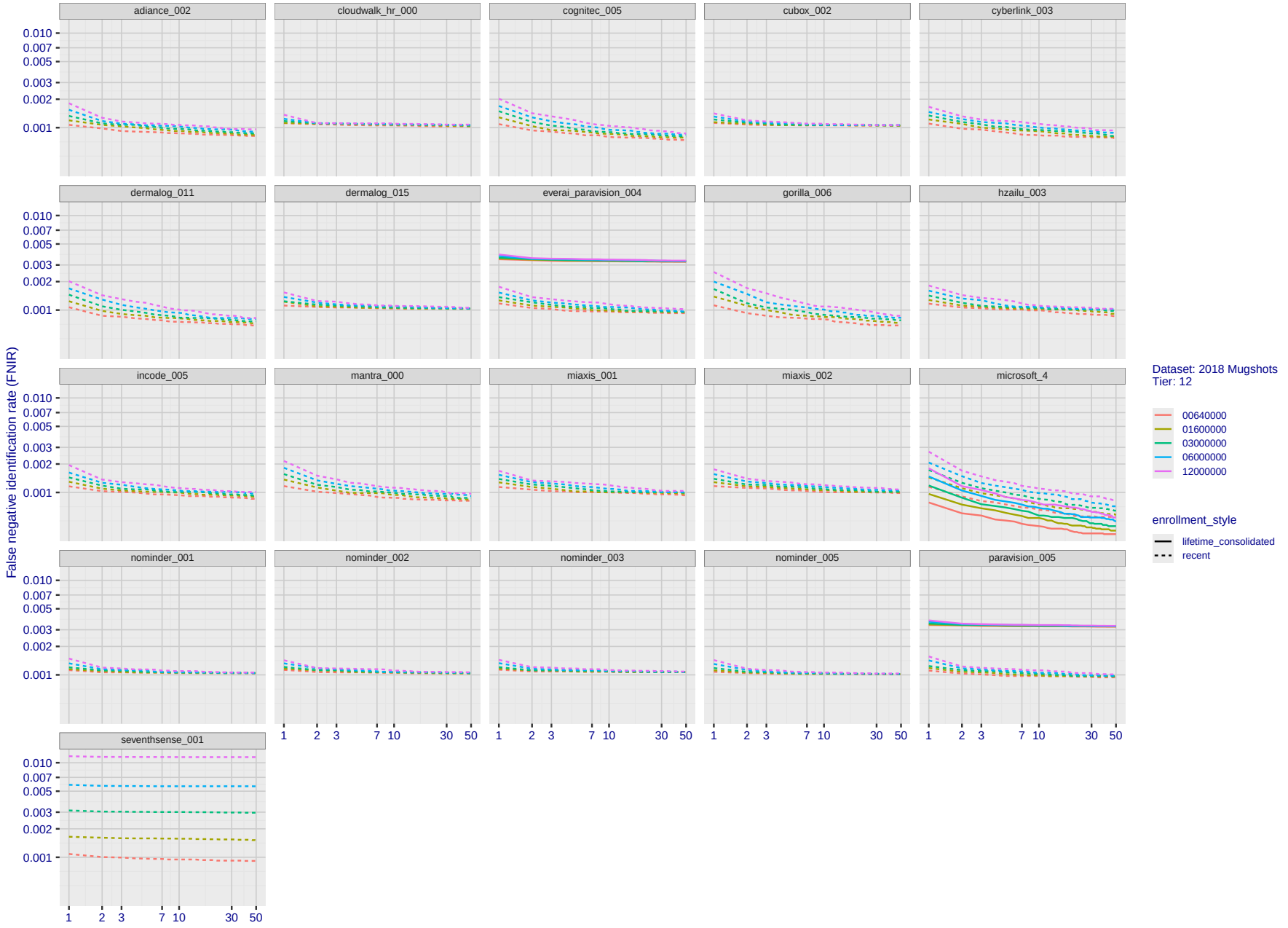


*Figure 63:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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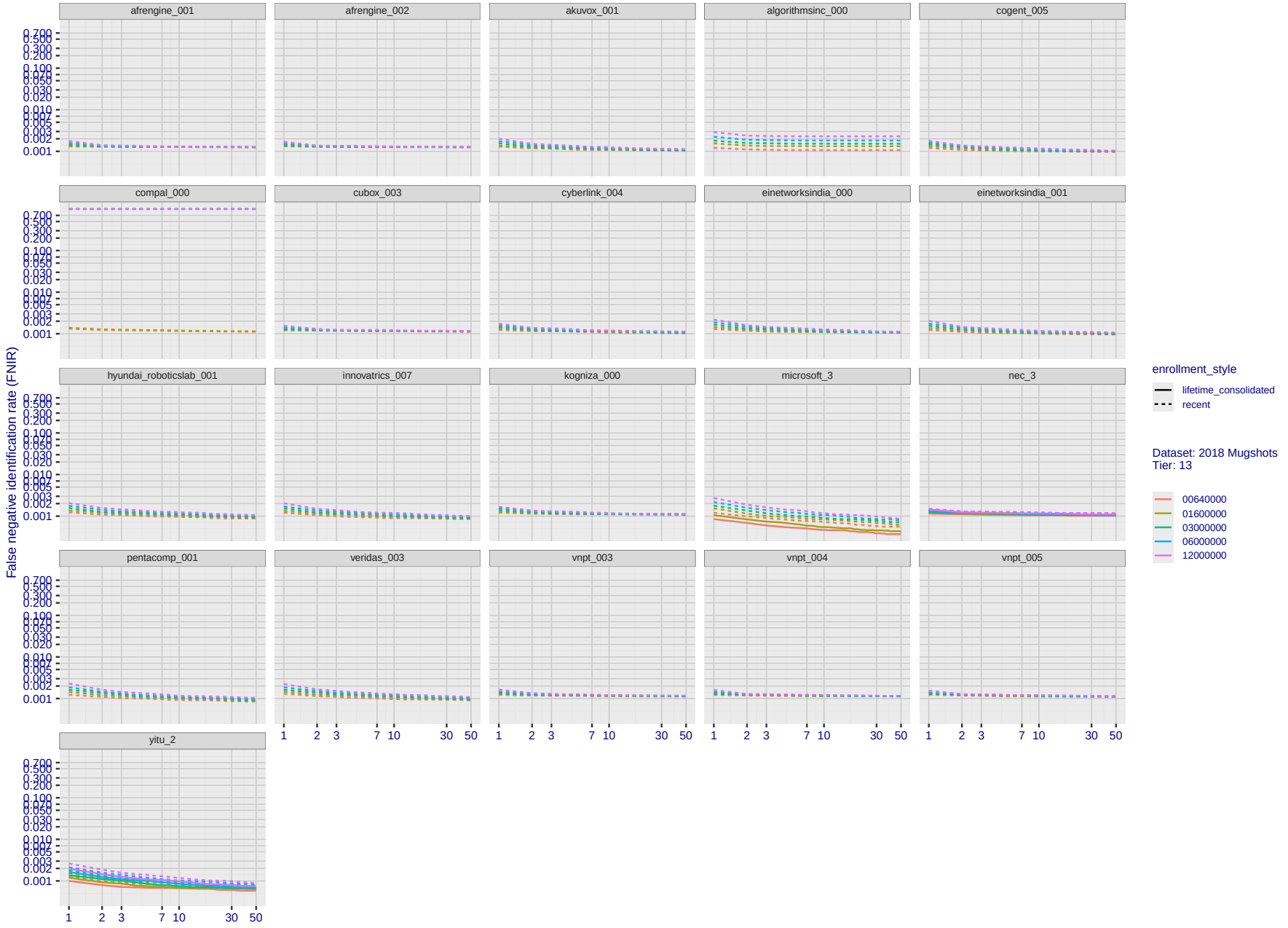


*Figure 64:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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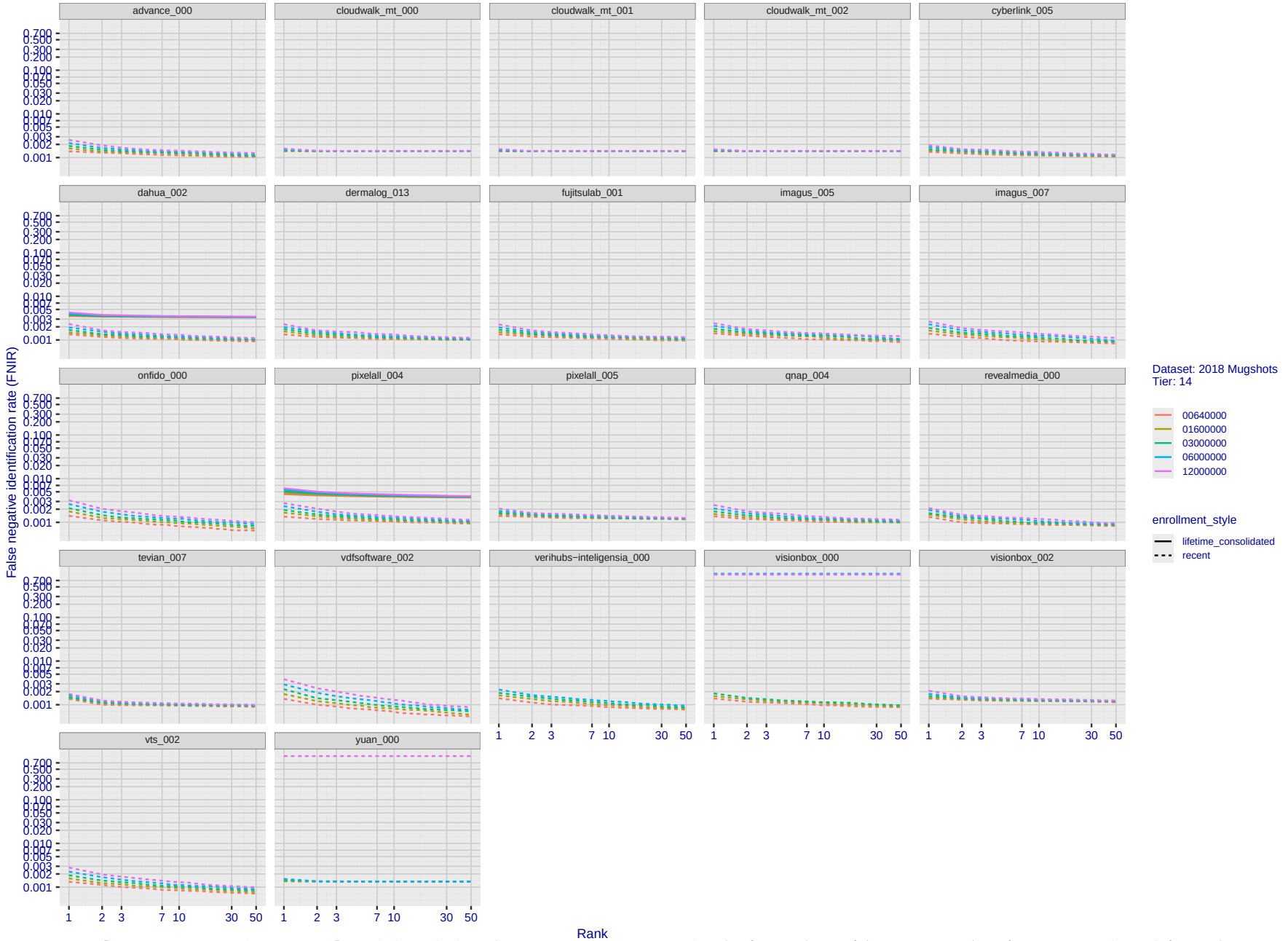


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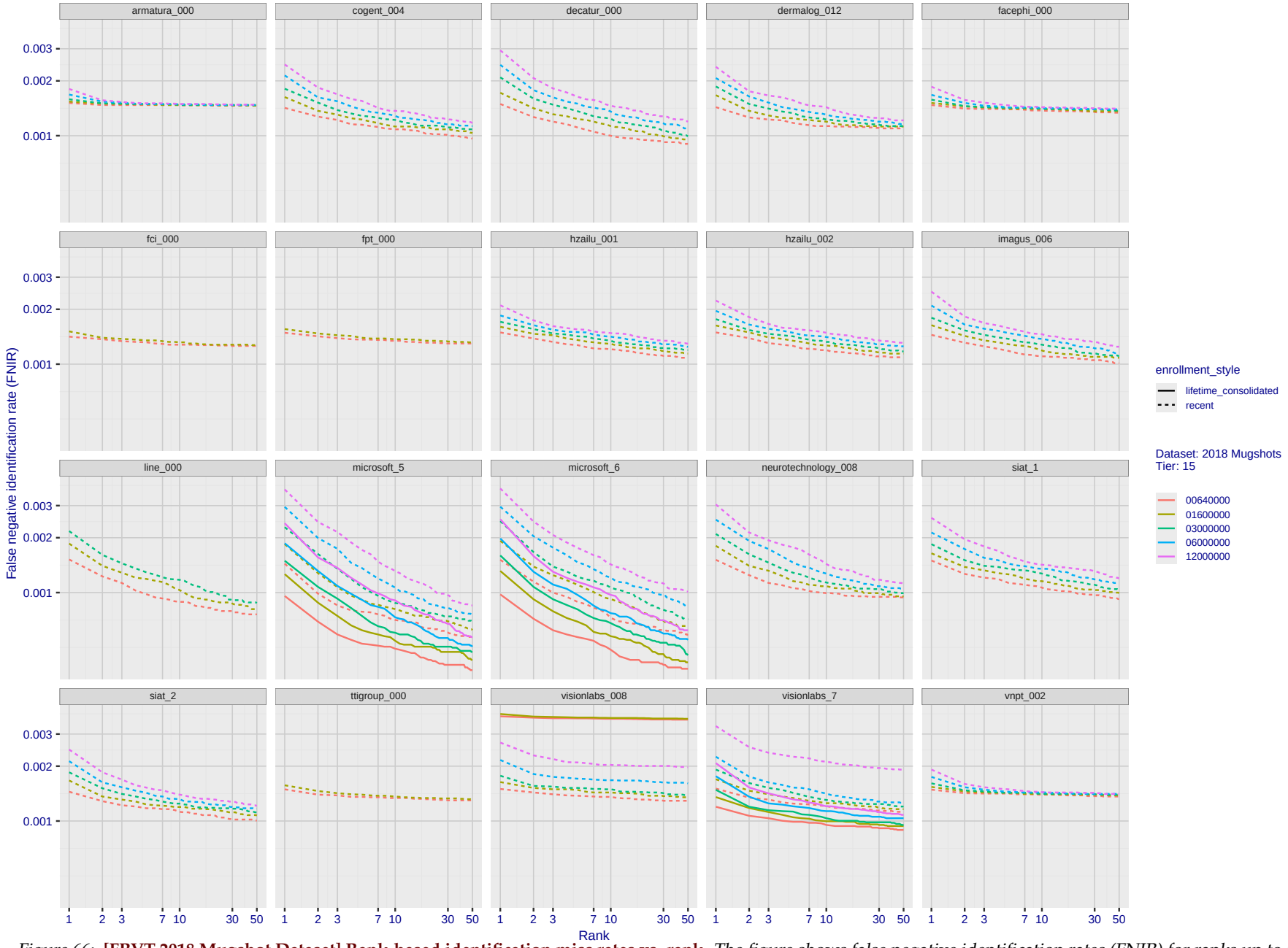


*Figure 66:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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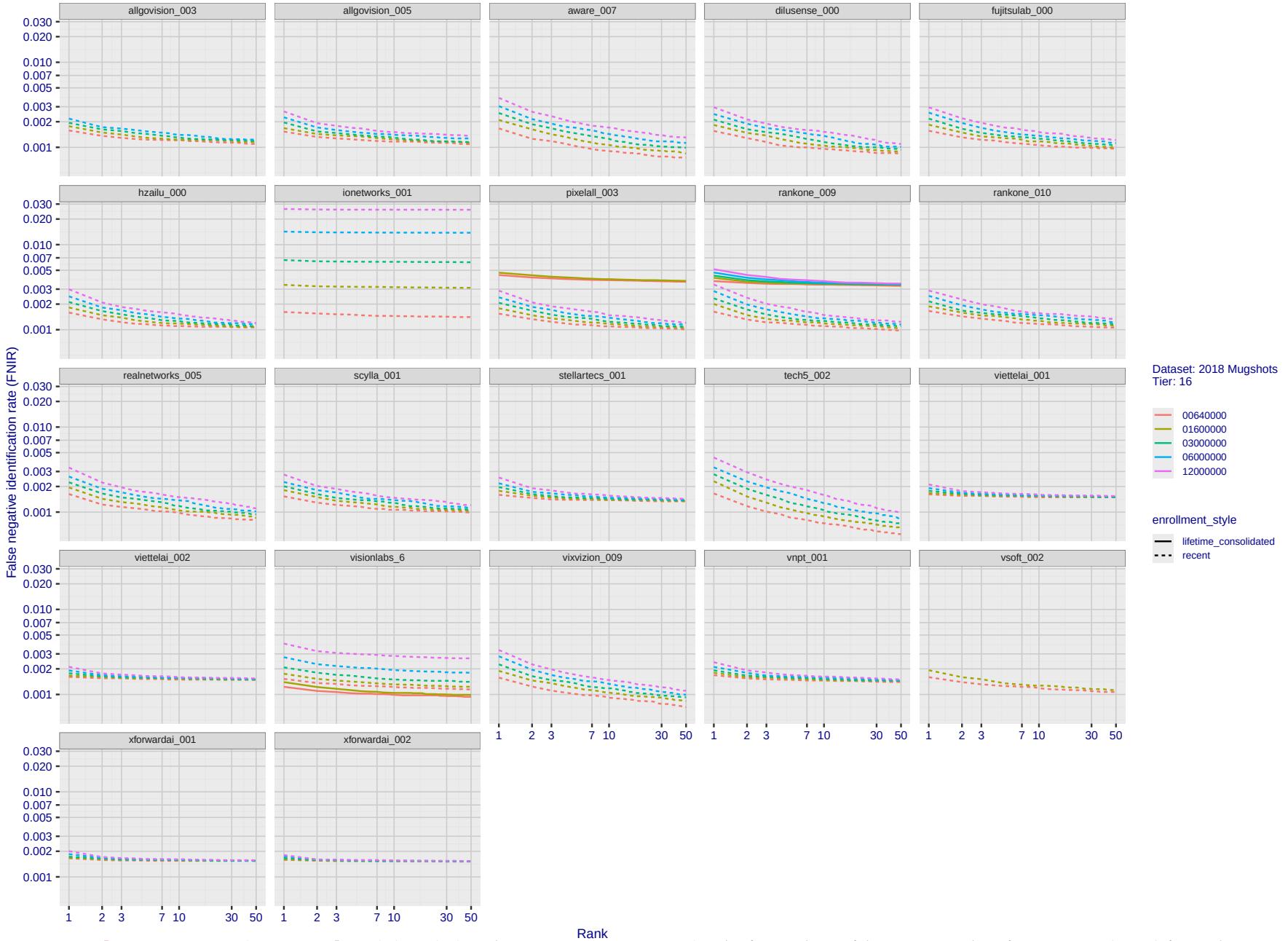


*Figure 67:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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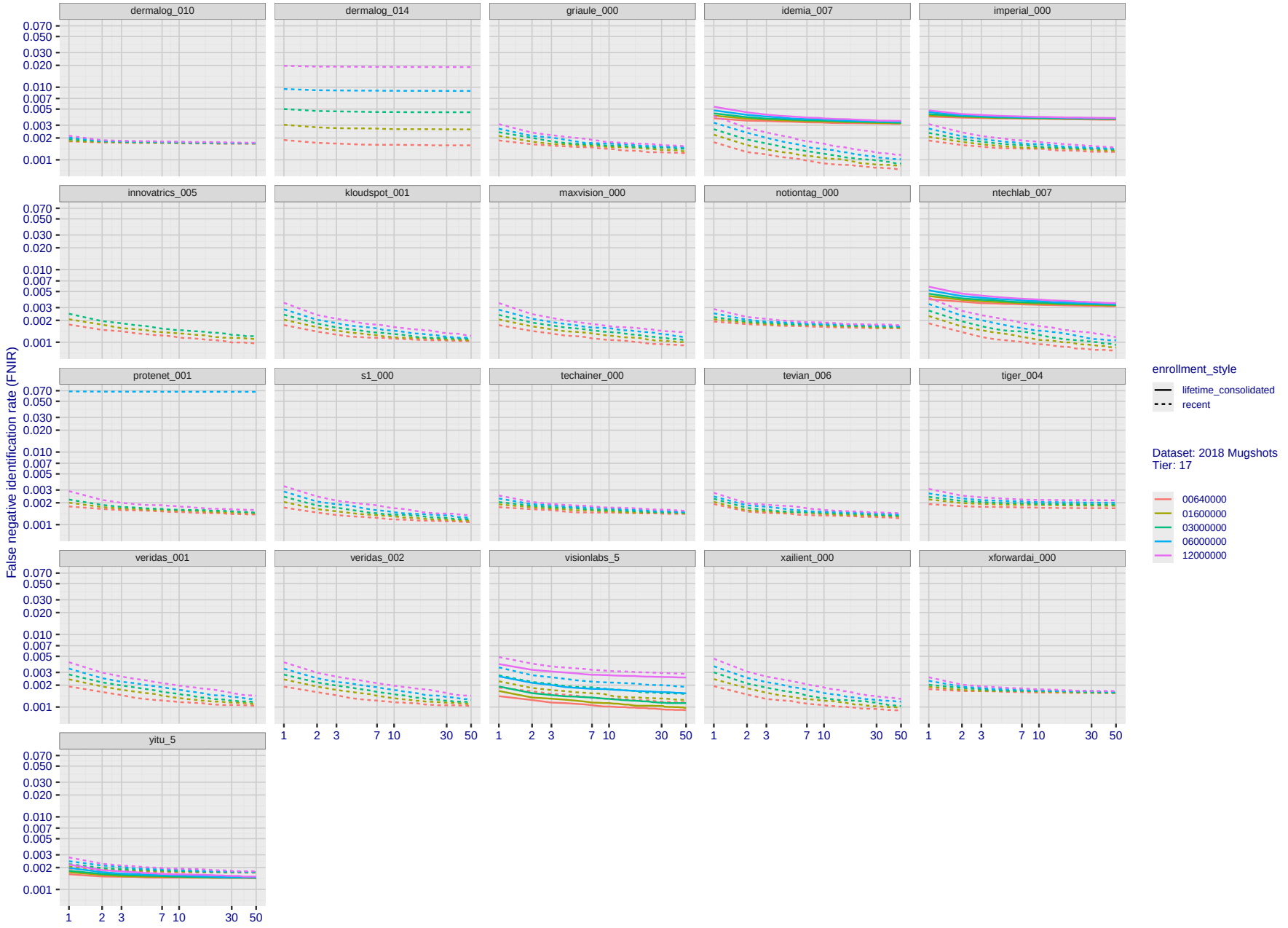


*Figure 68:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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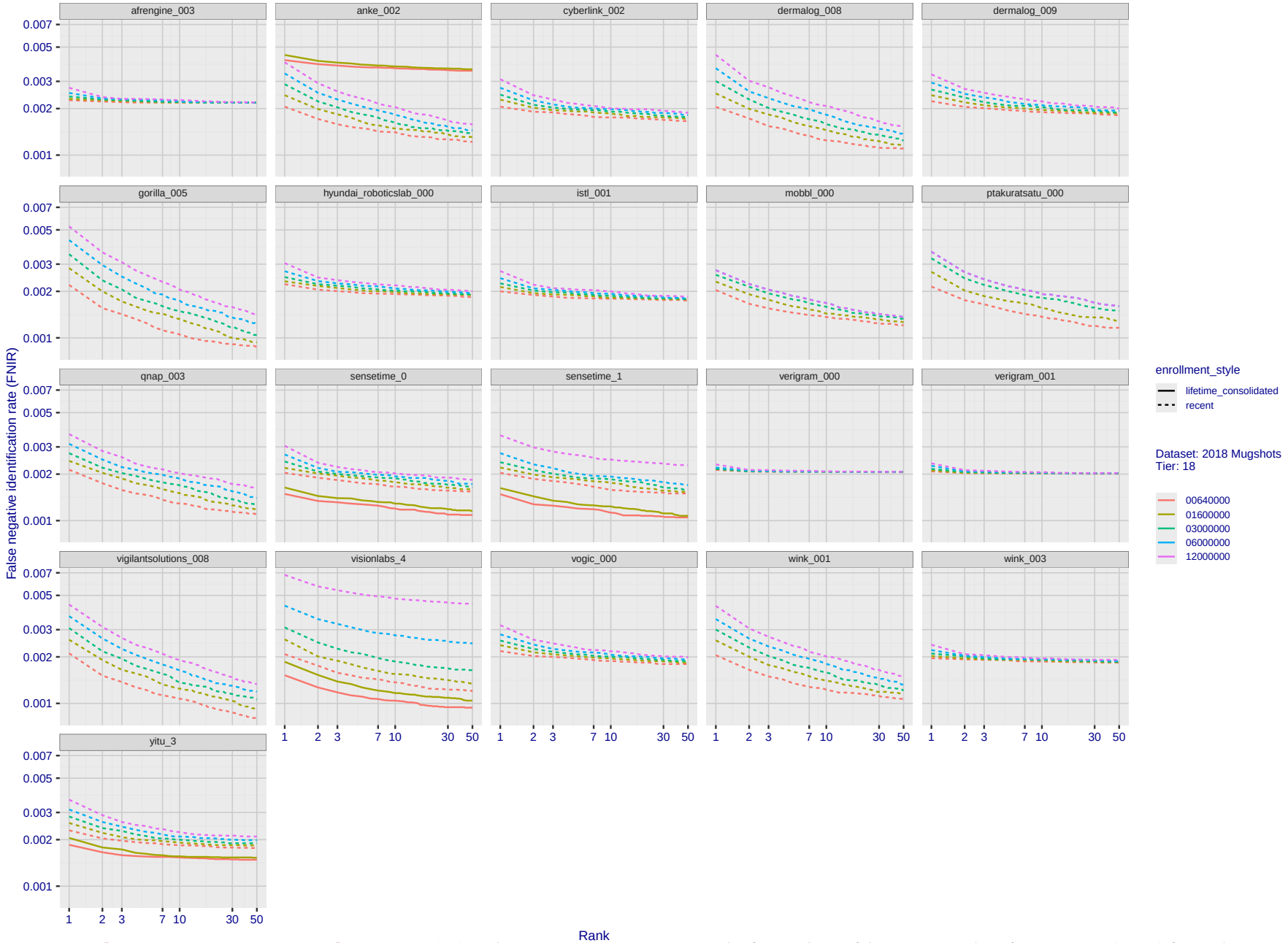


*Figure 69:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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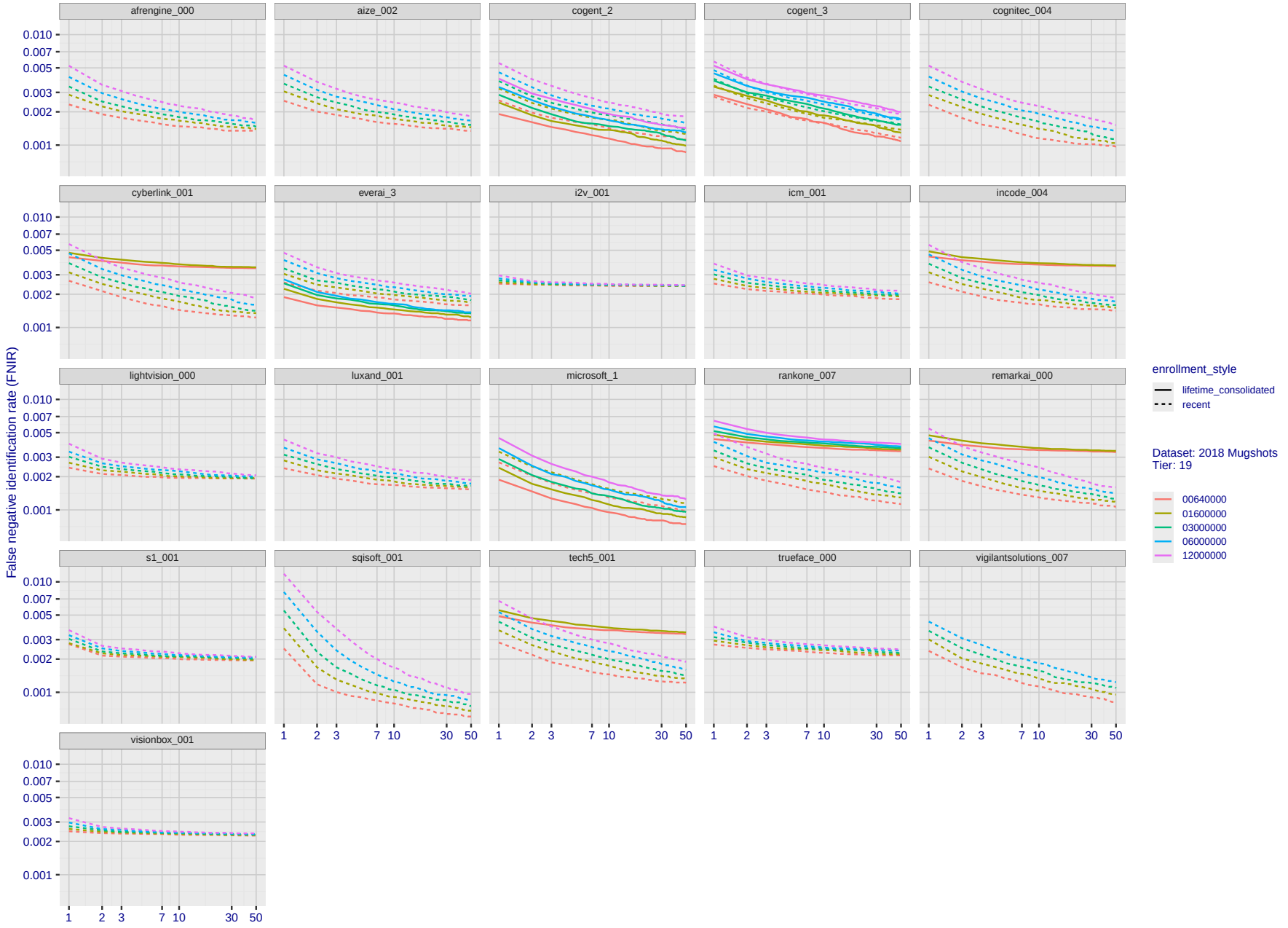


*Figure 70:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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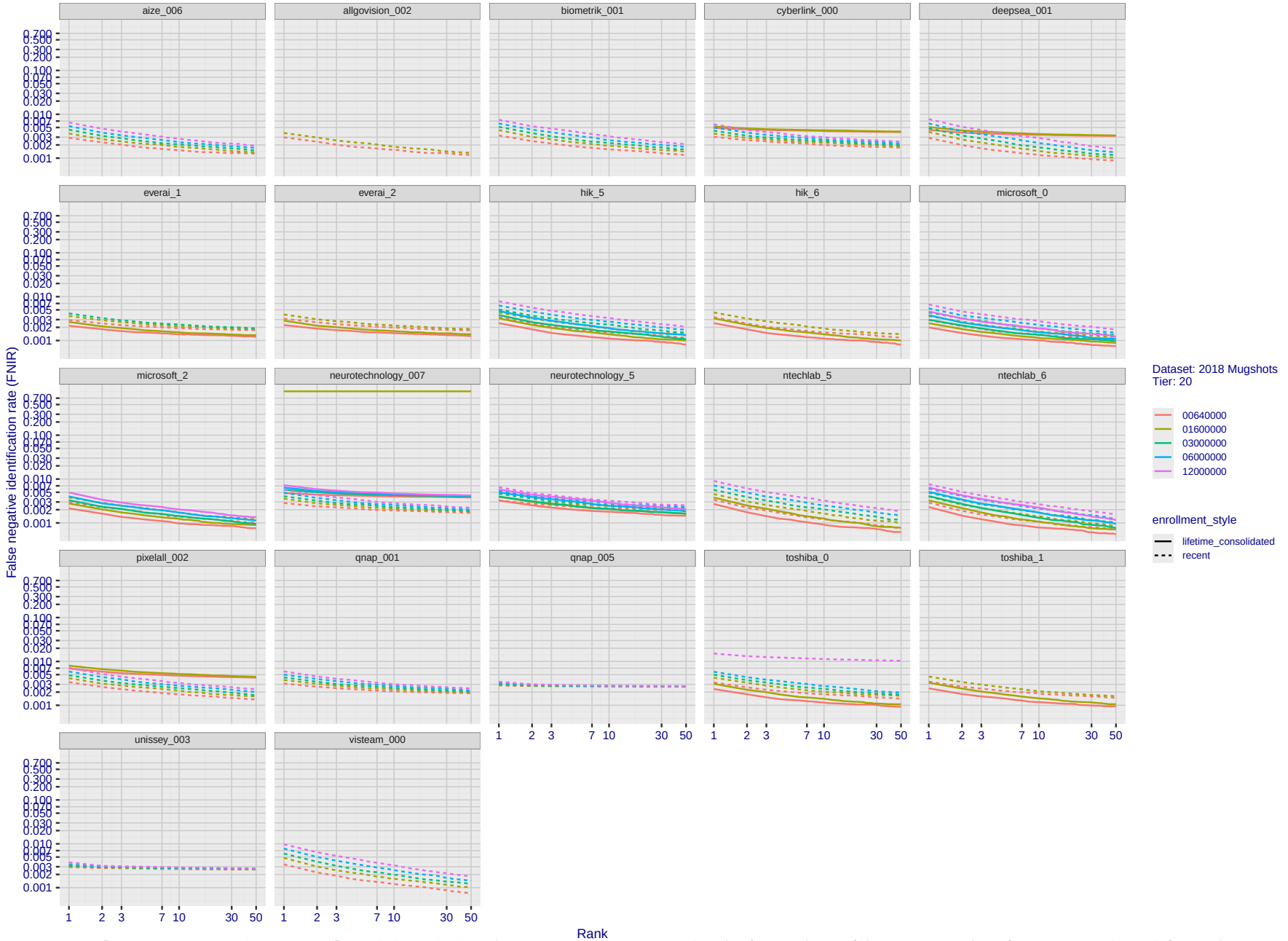


*Figure 71:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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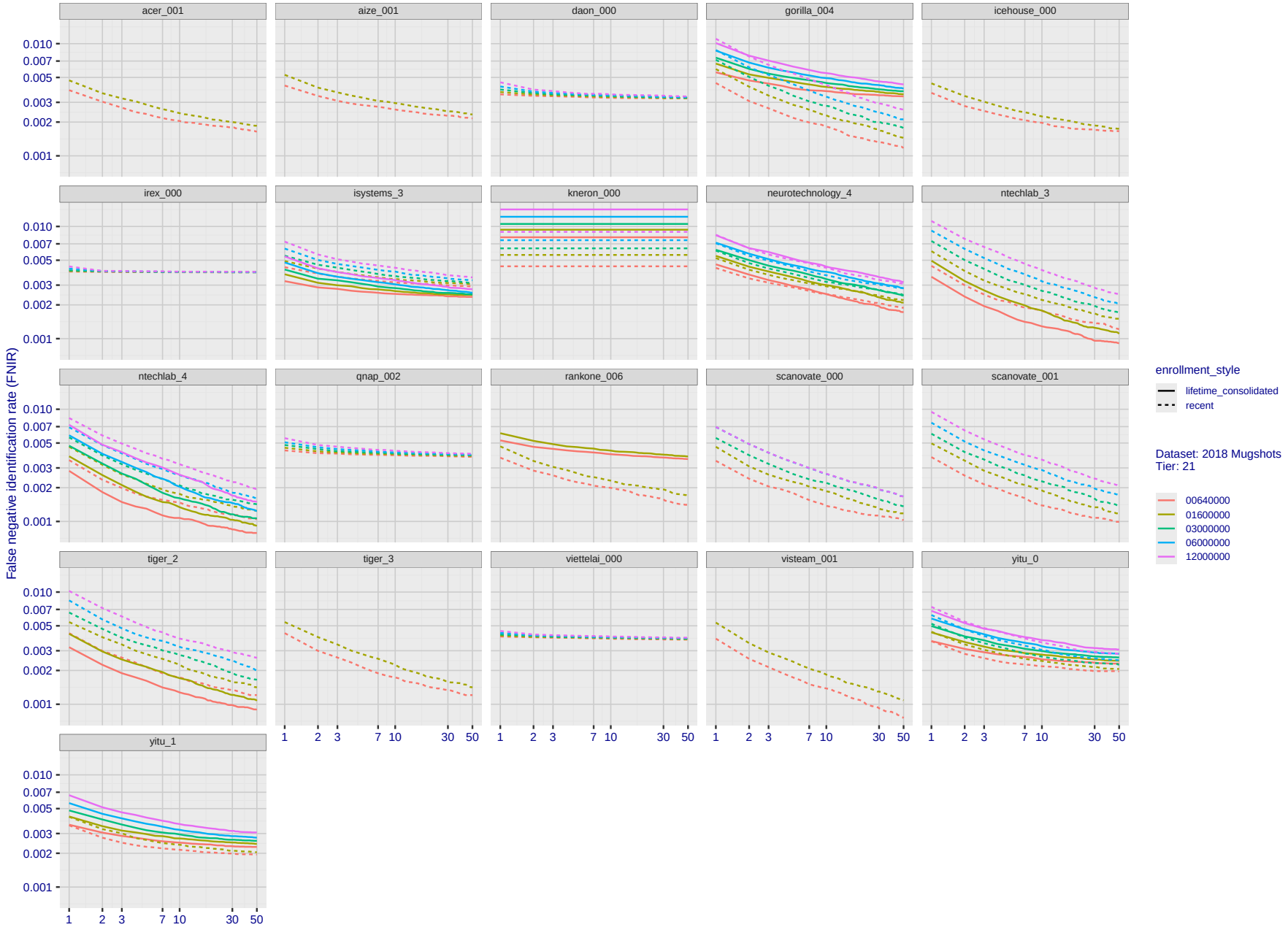


*Figure 72:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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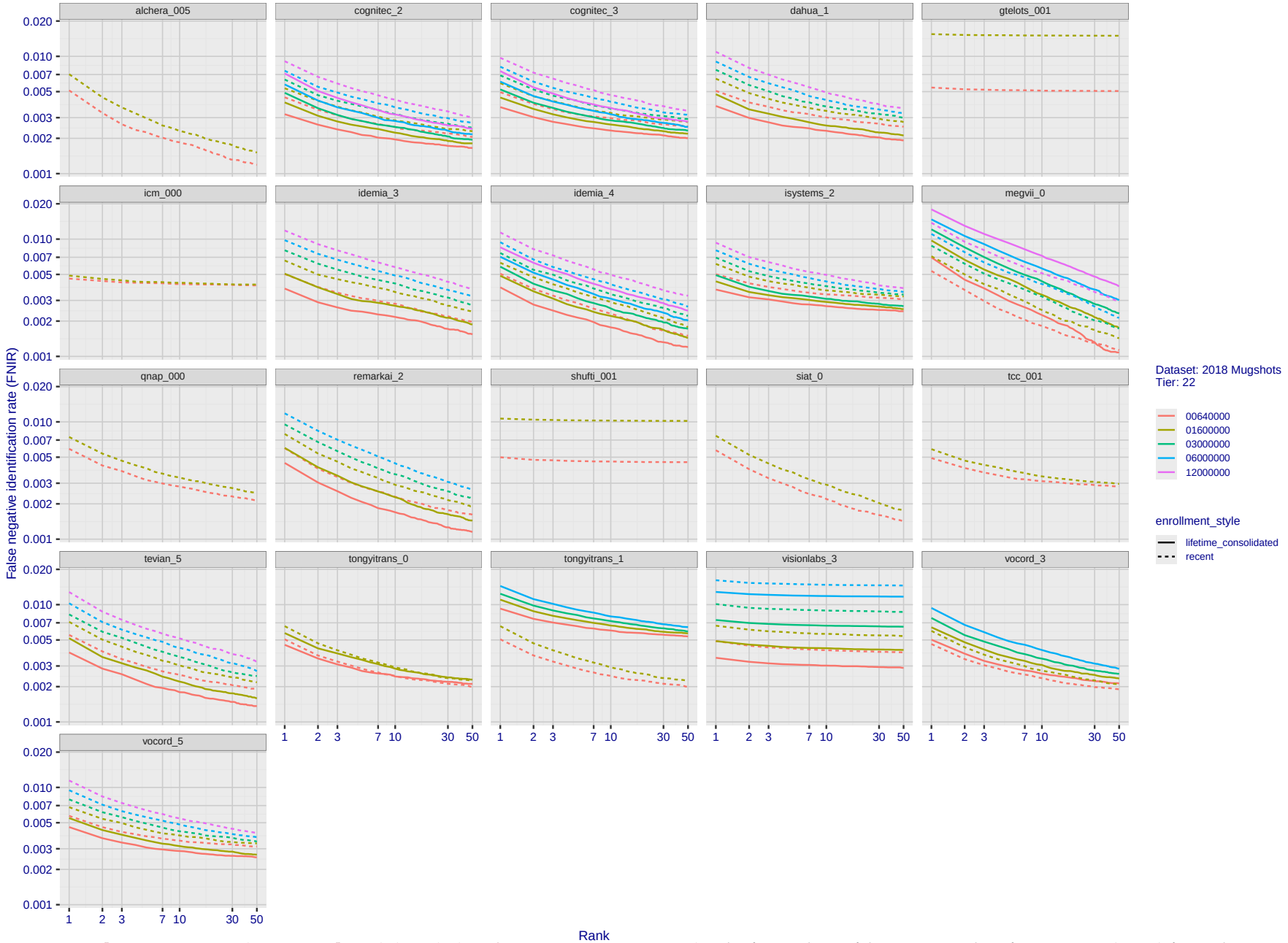


*Figure 73:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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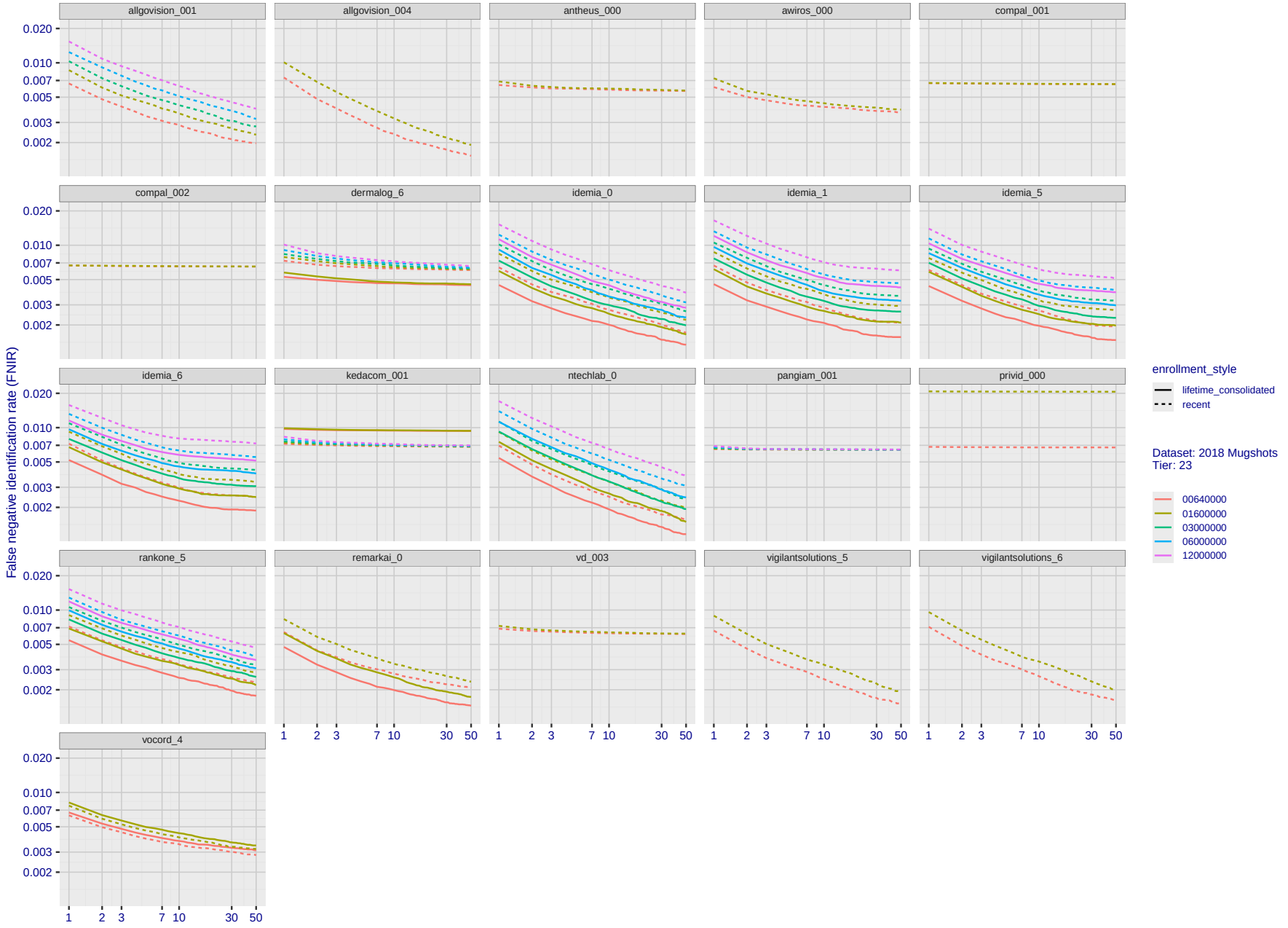


*Figure 74:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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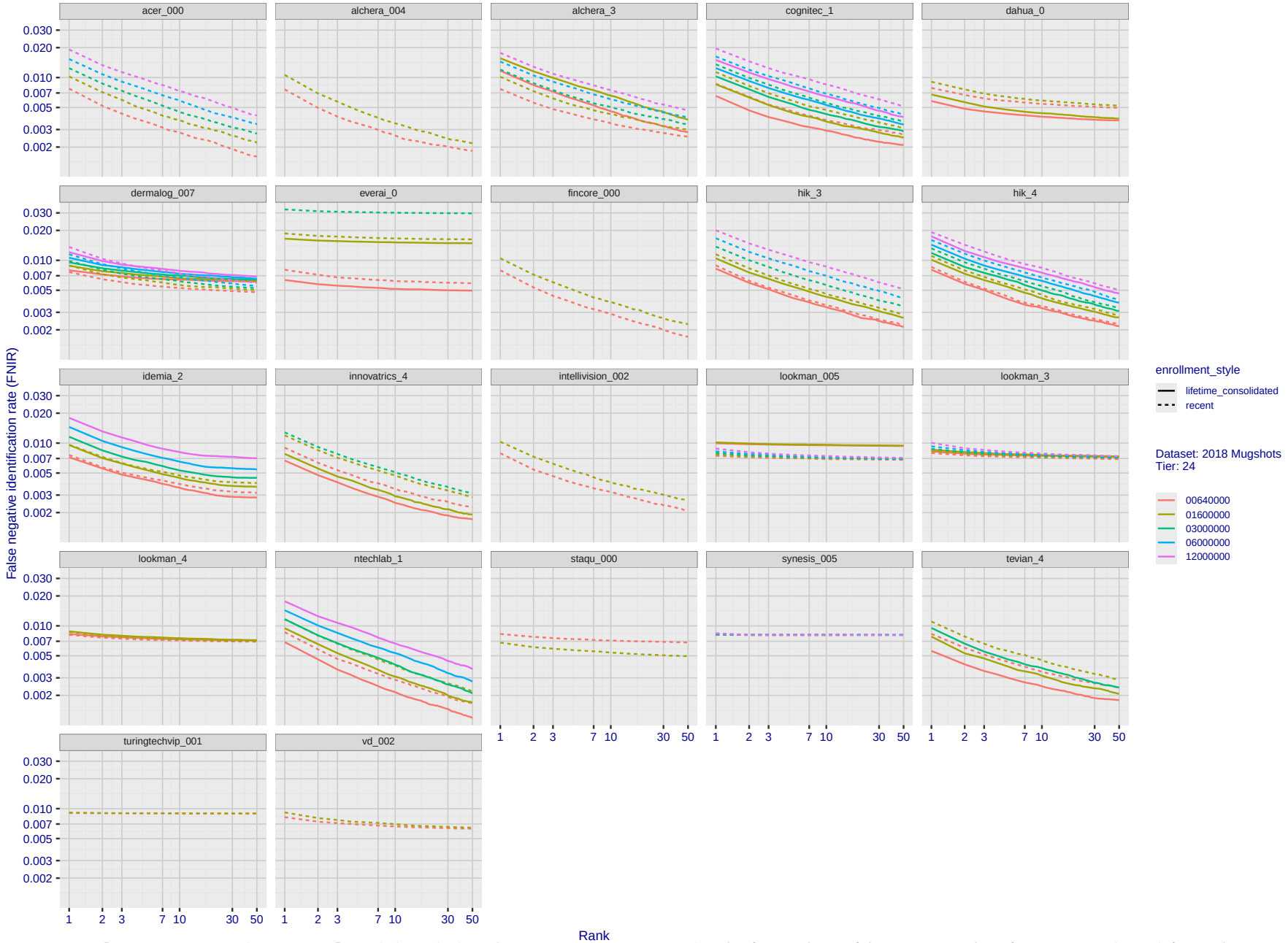


*Figure 75:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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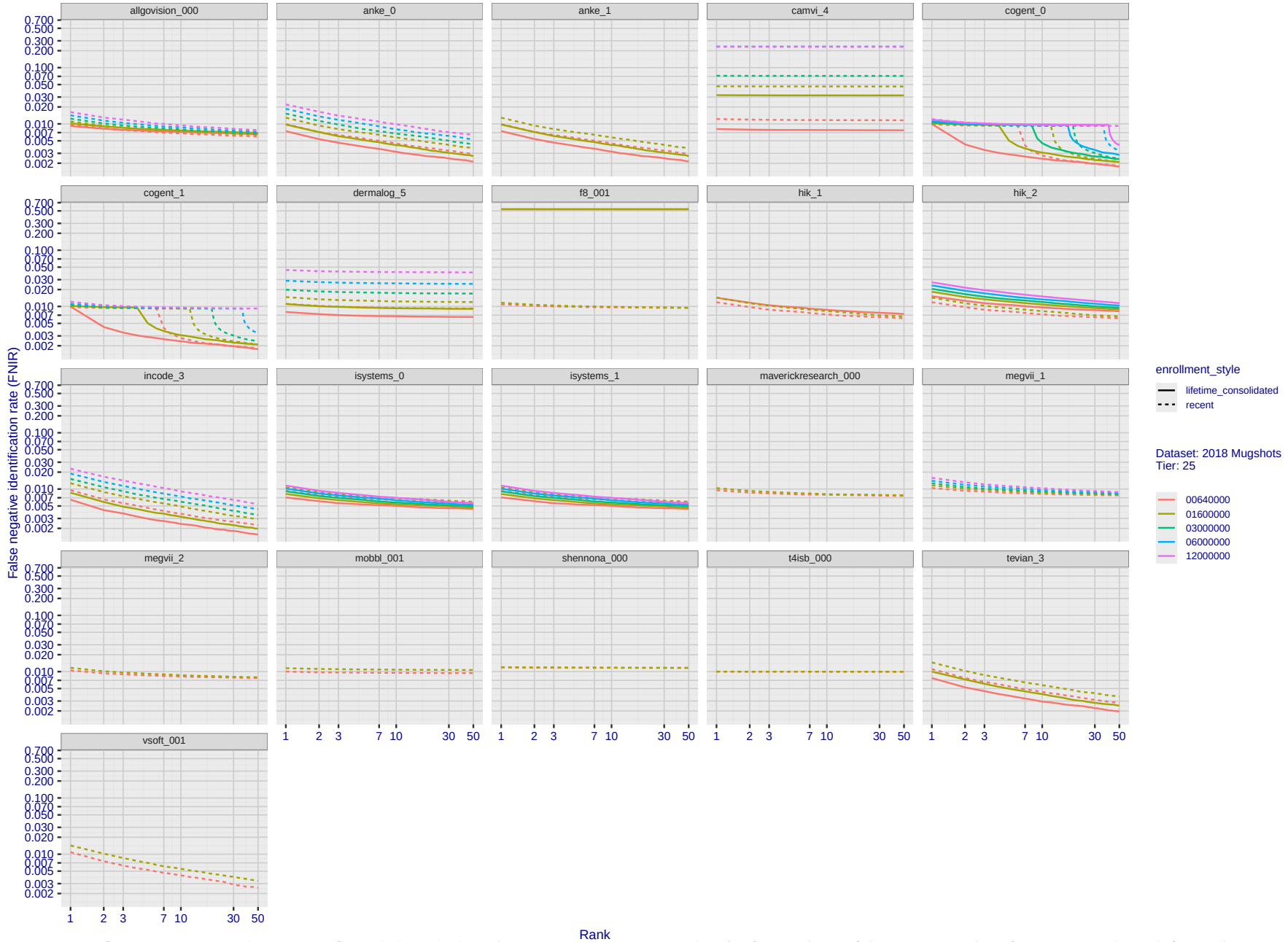


*Figure 76:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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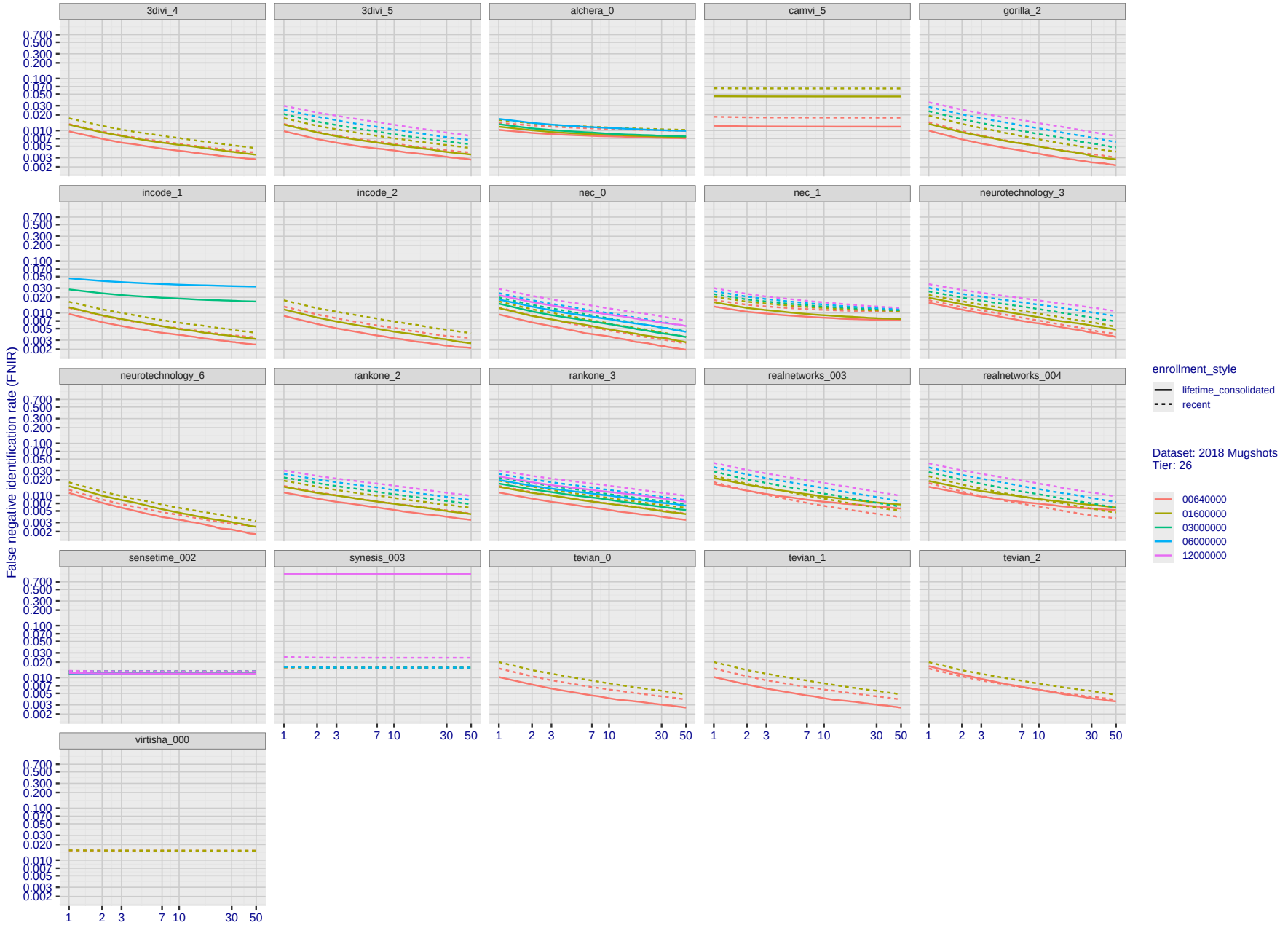


*Figure 77:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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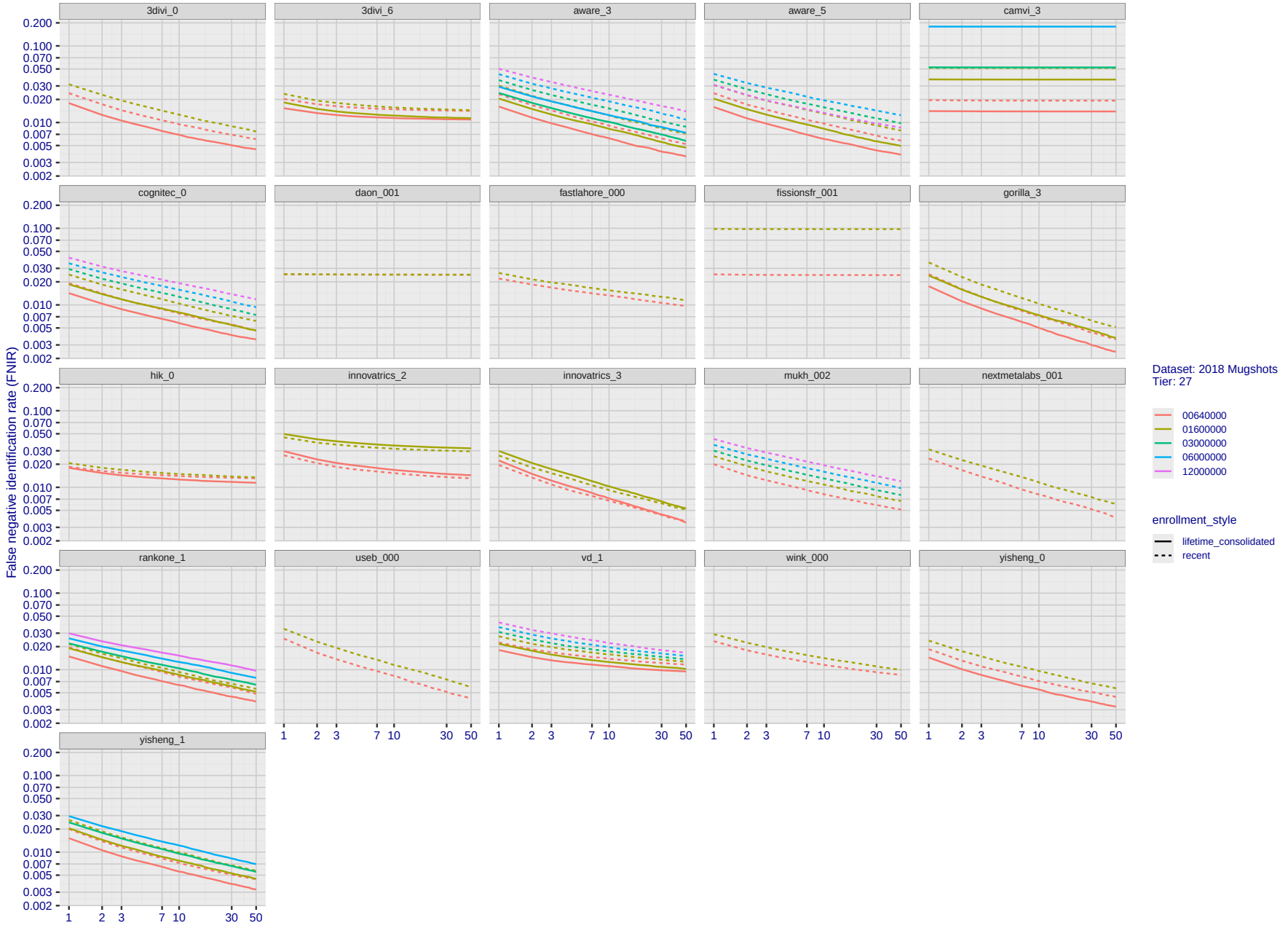


*Figure 78:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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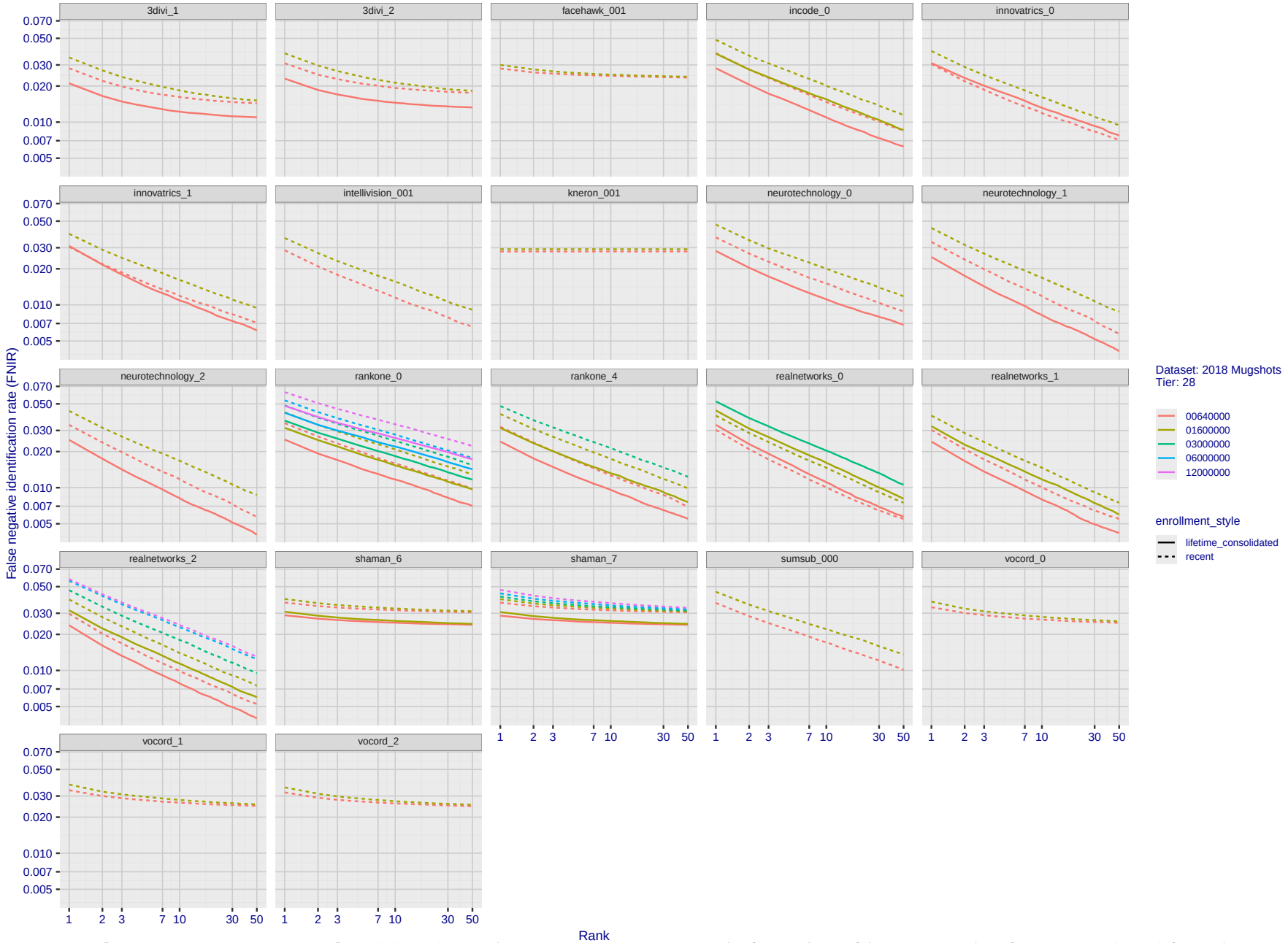


*Figure 79:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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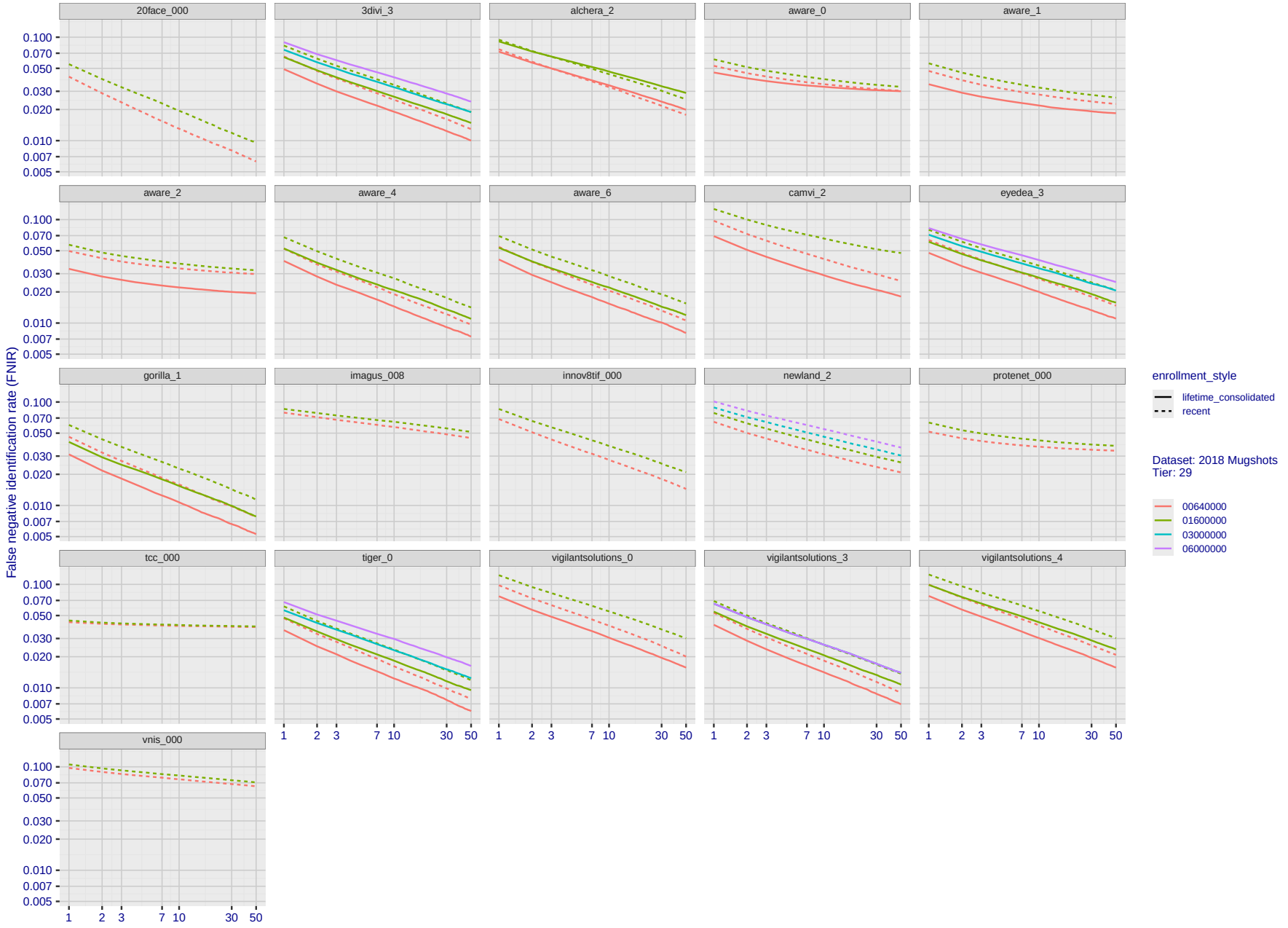


*Figure 80:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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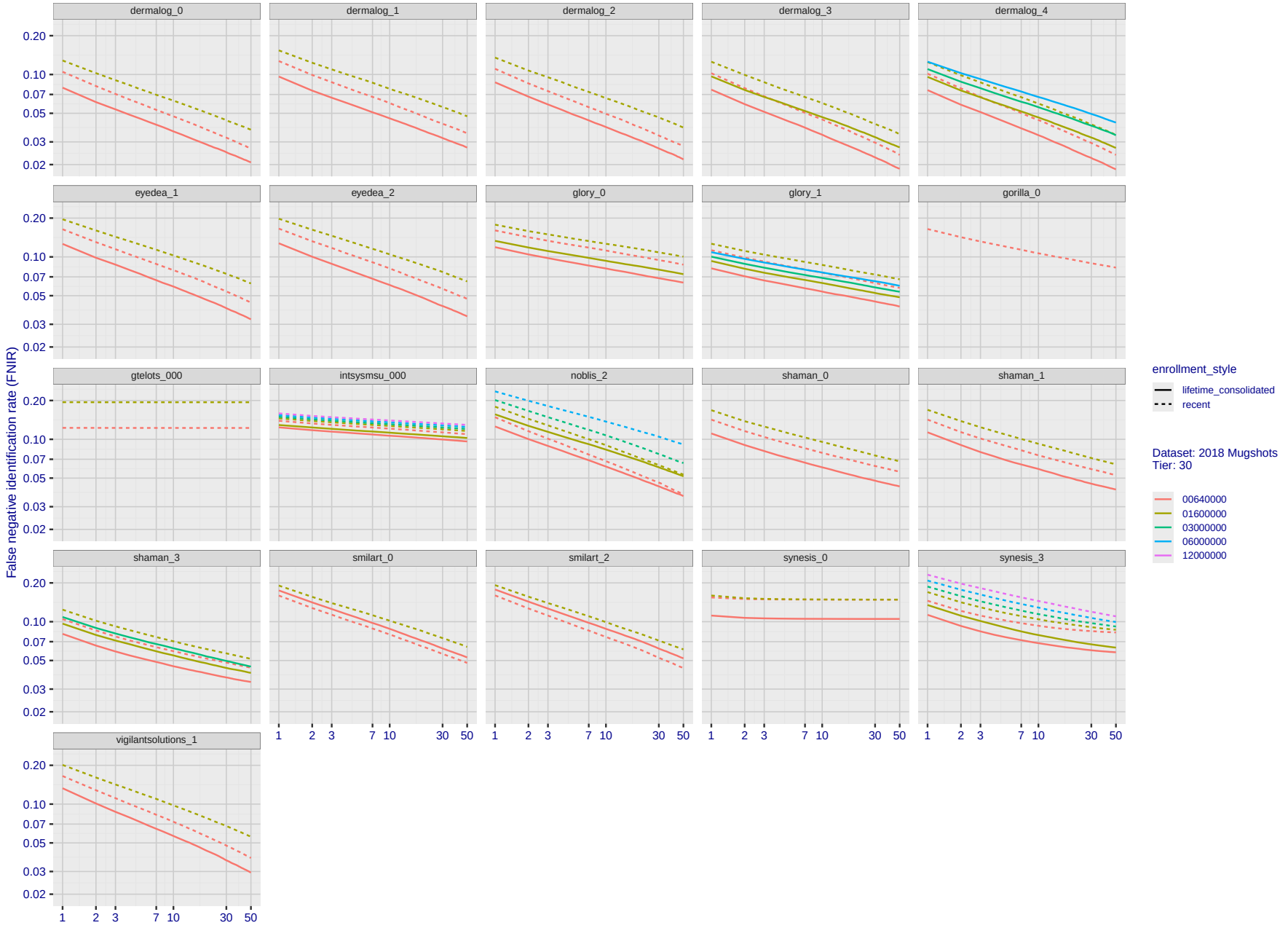


*Figure 81:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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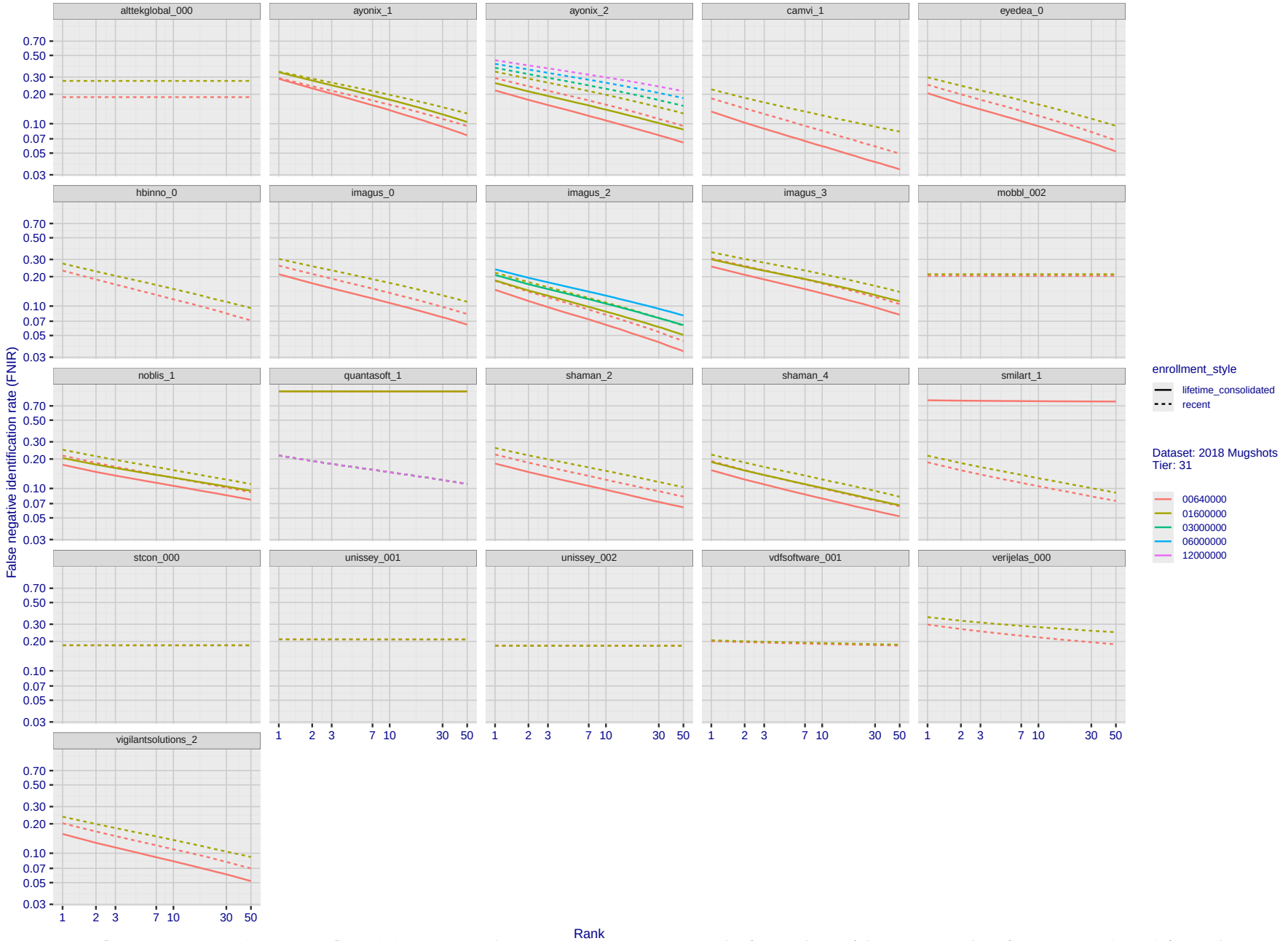


*Figure 82:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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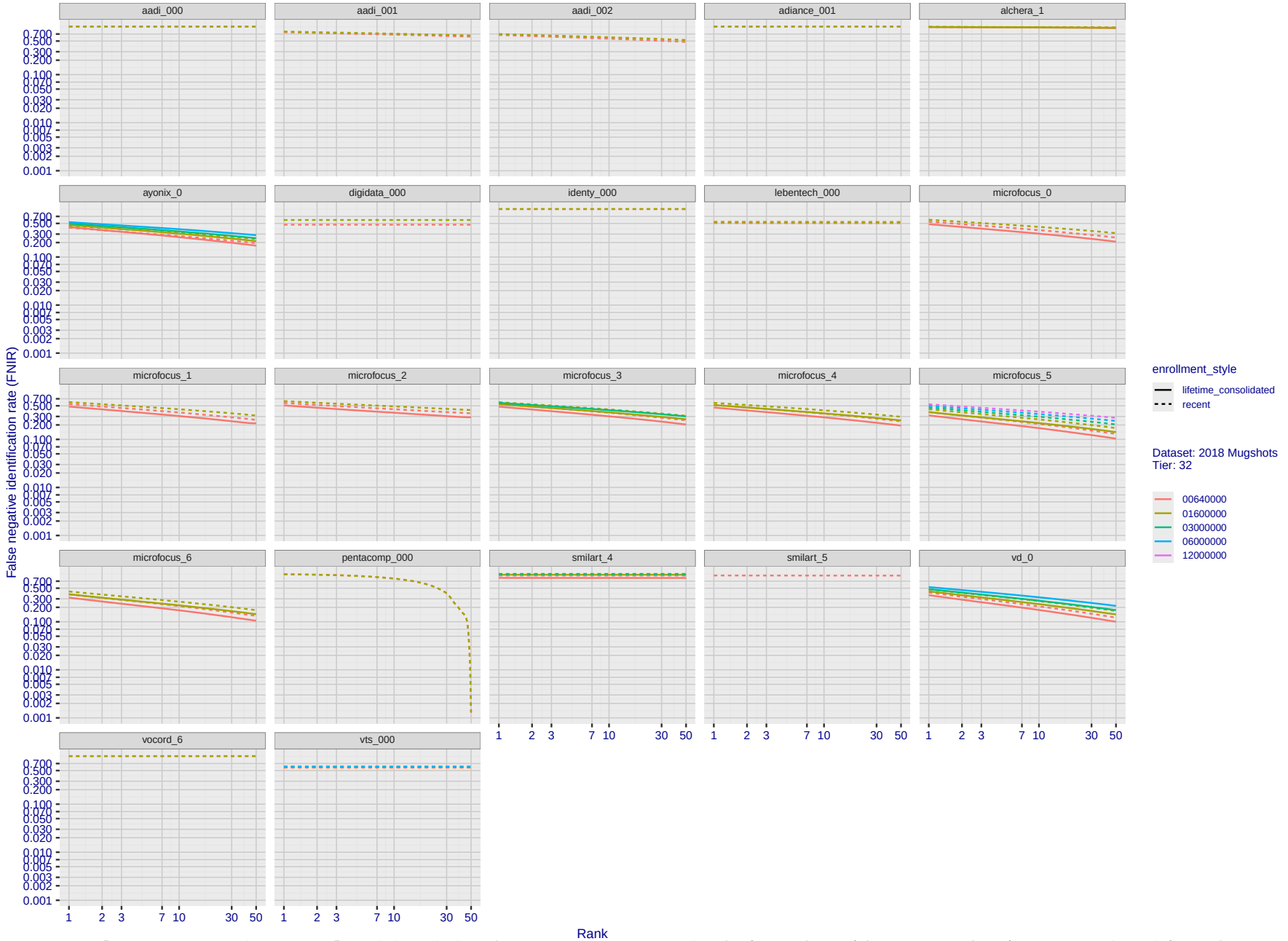


*Figure 83:* **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*



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| | | | | | |
|---|---|---|---|---|---|
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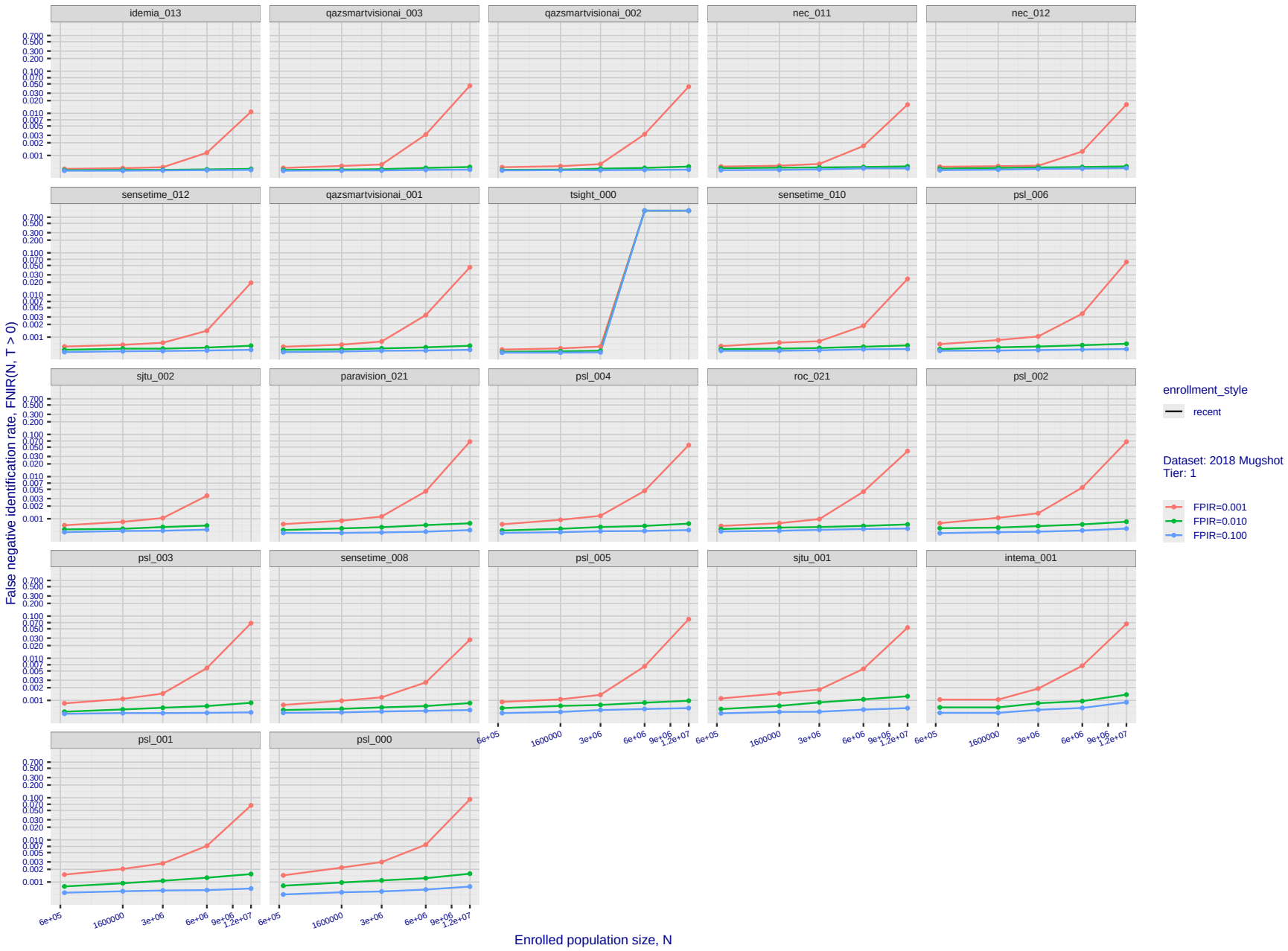


*Figure 84:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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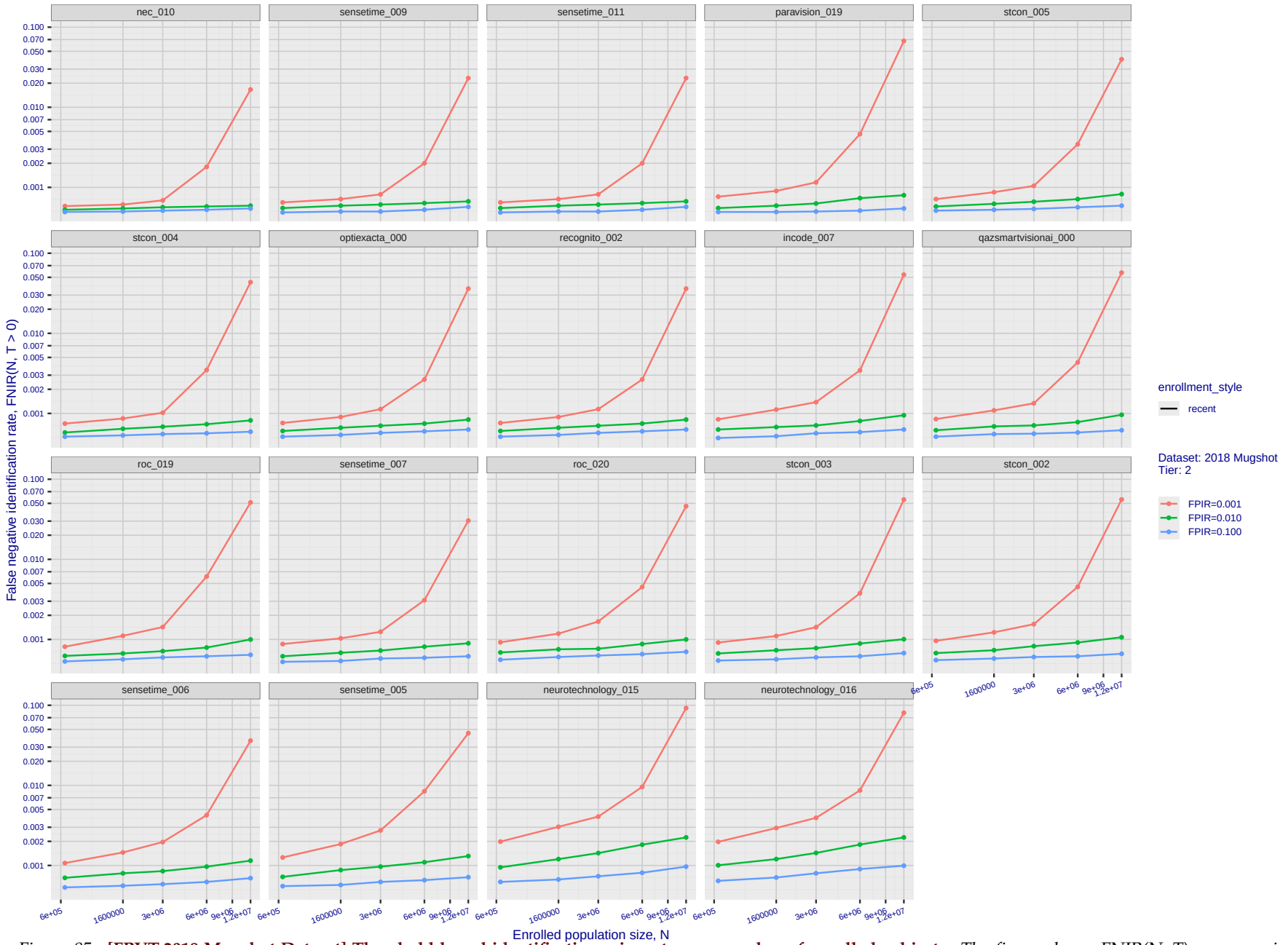


*Figure 85:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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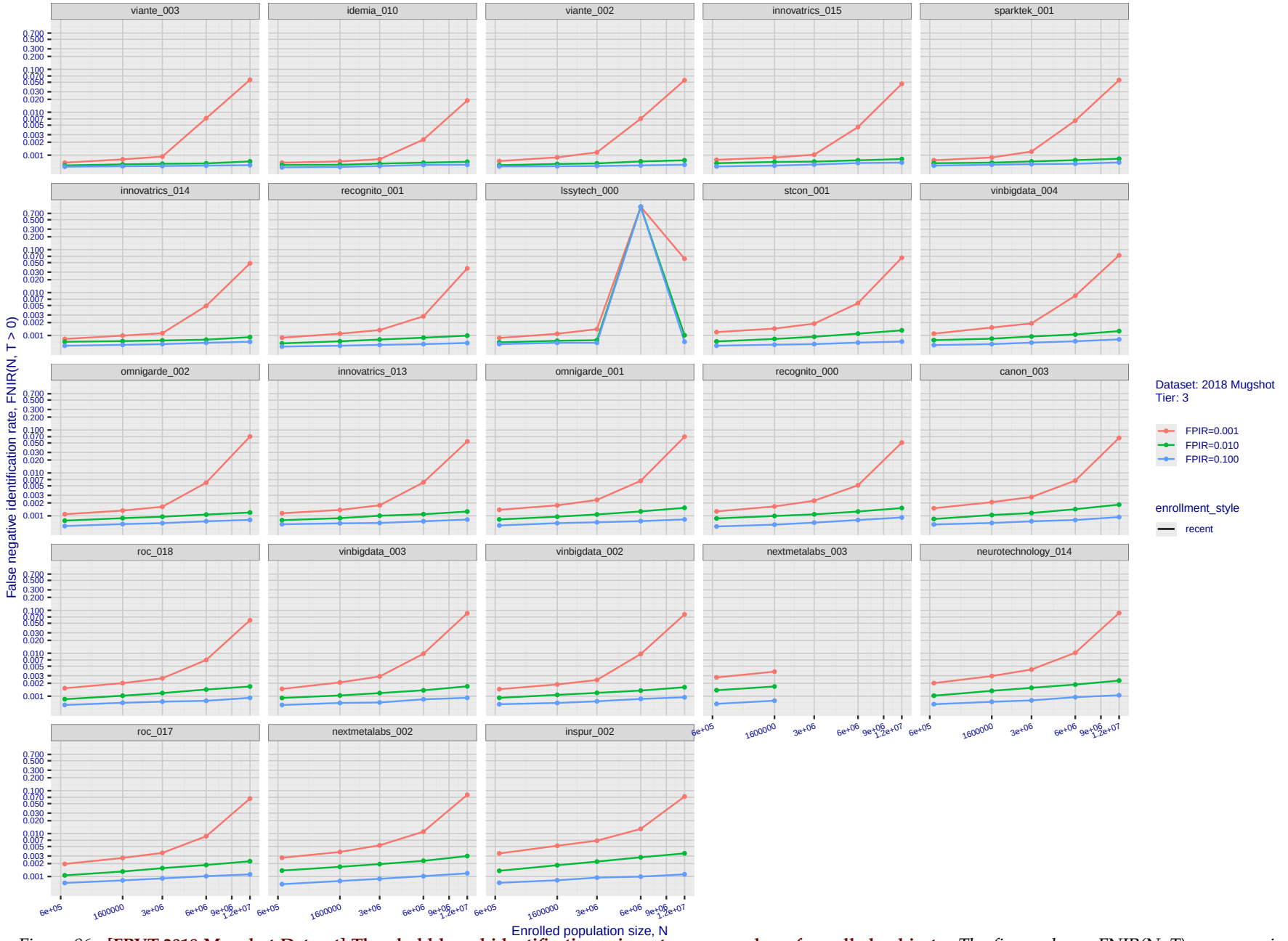


*Figure 86:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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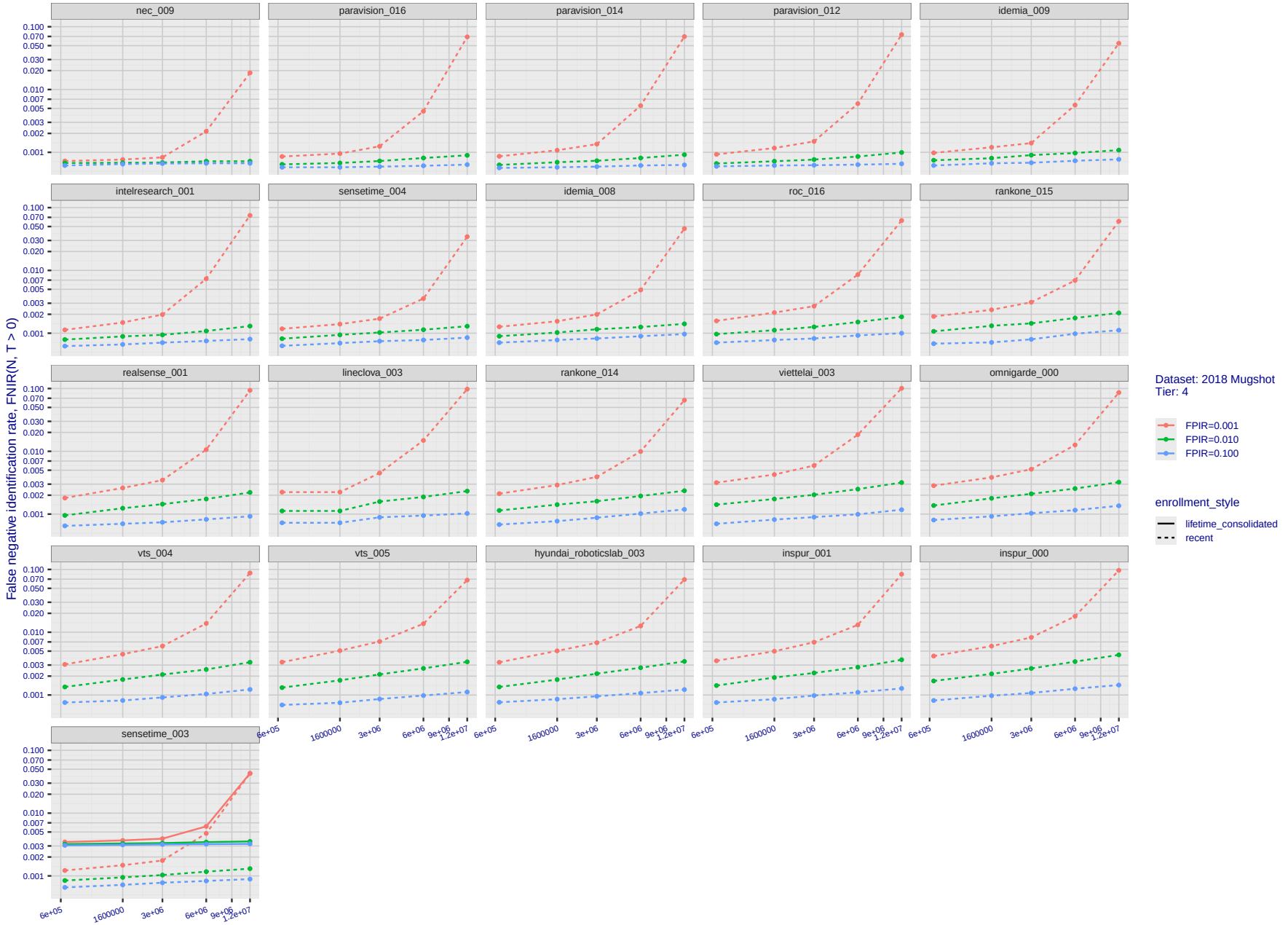


*Figure 87:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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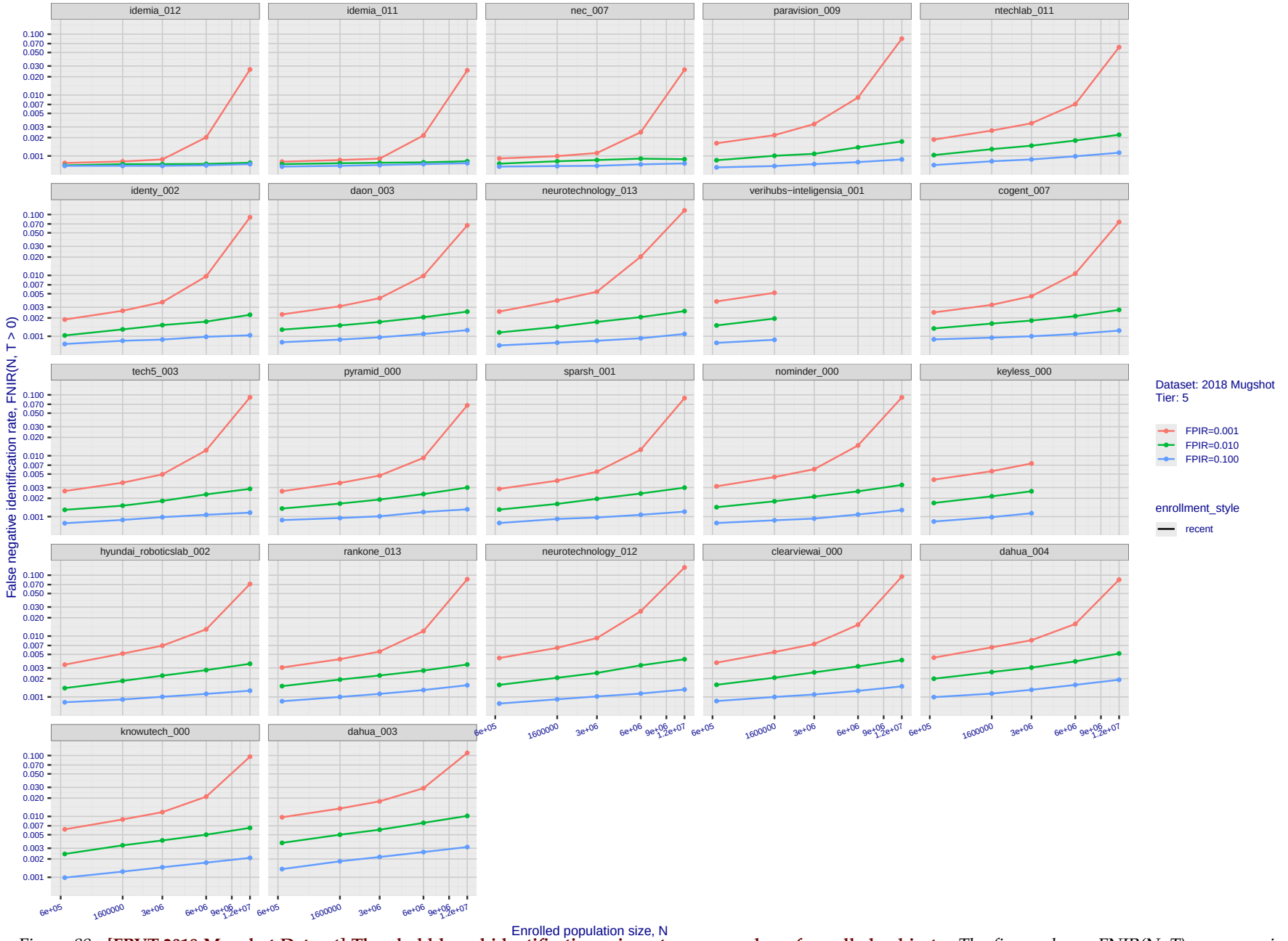


*Figure 88:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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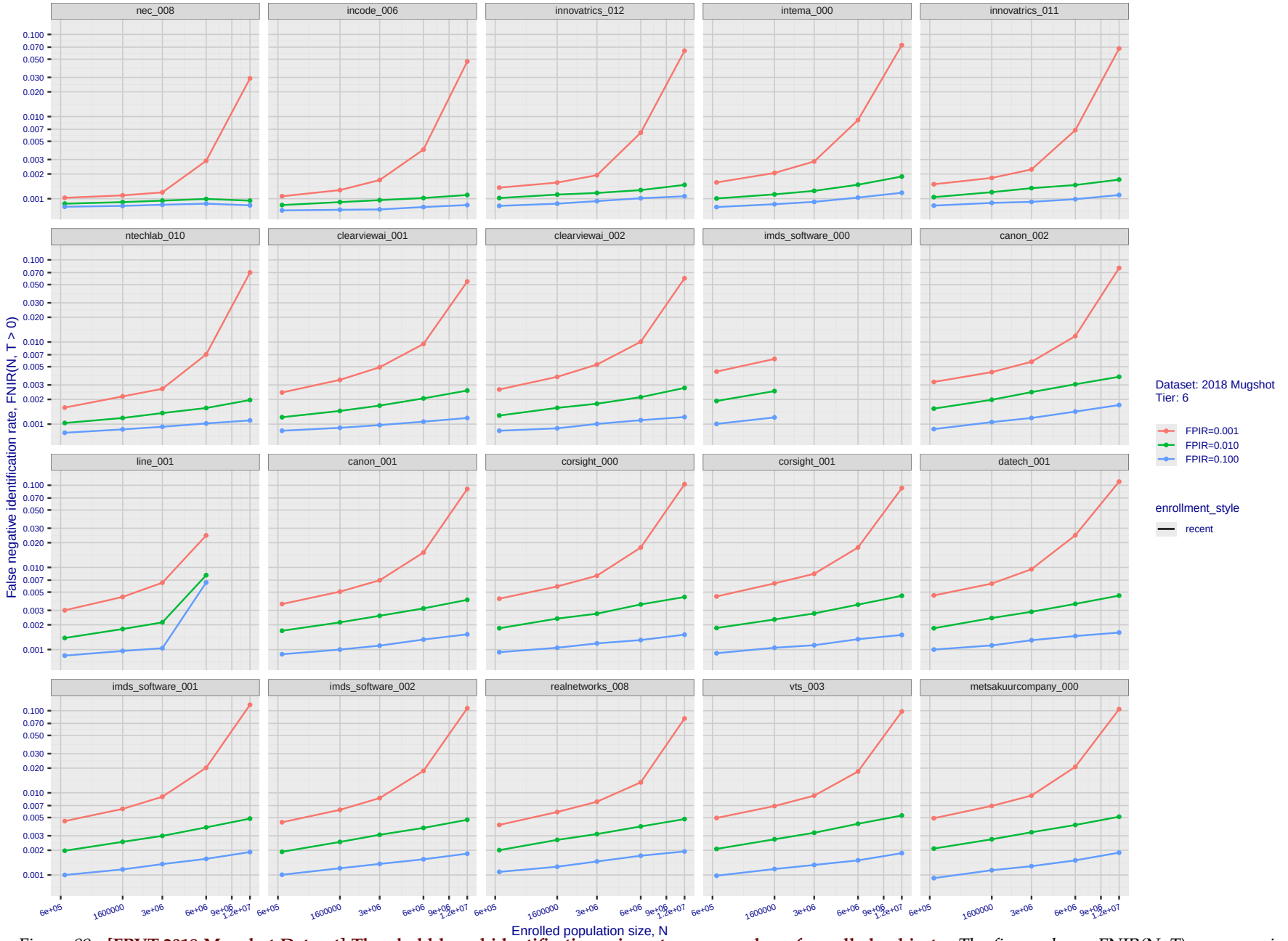


*Figure 89:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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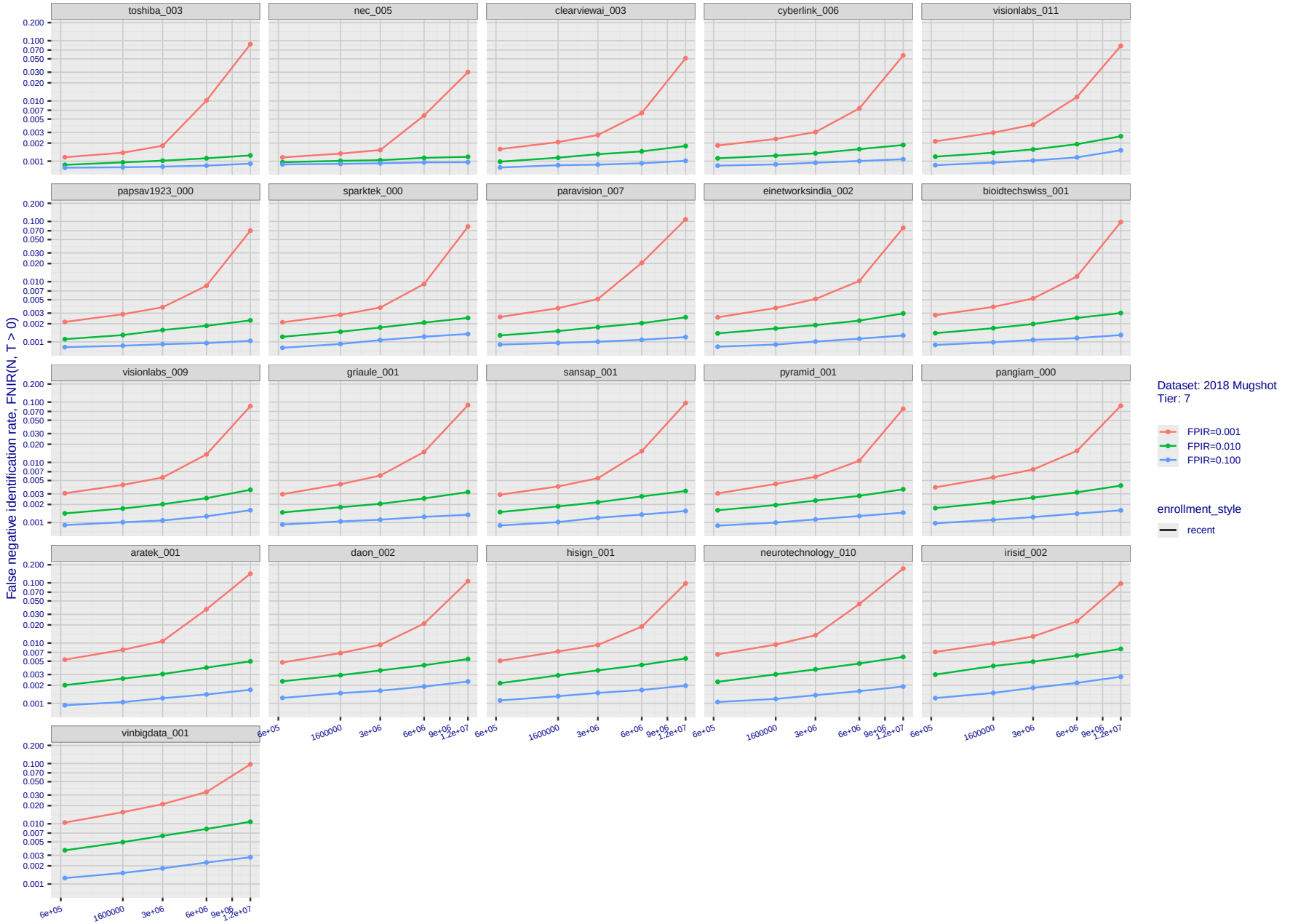


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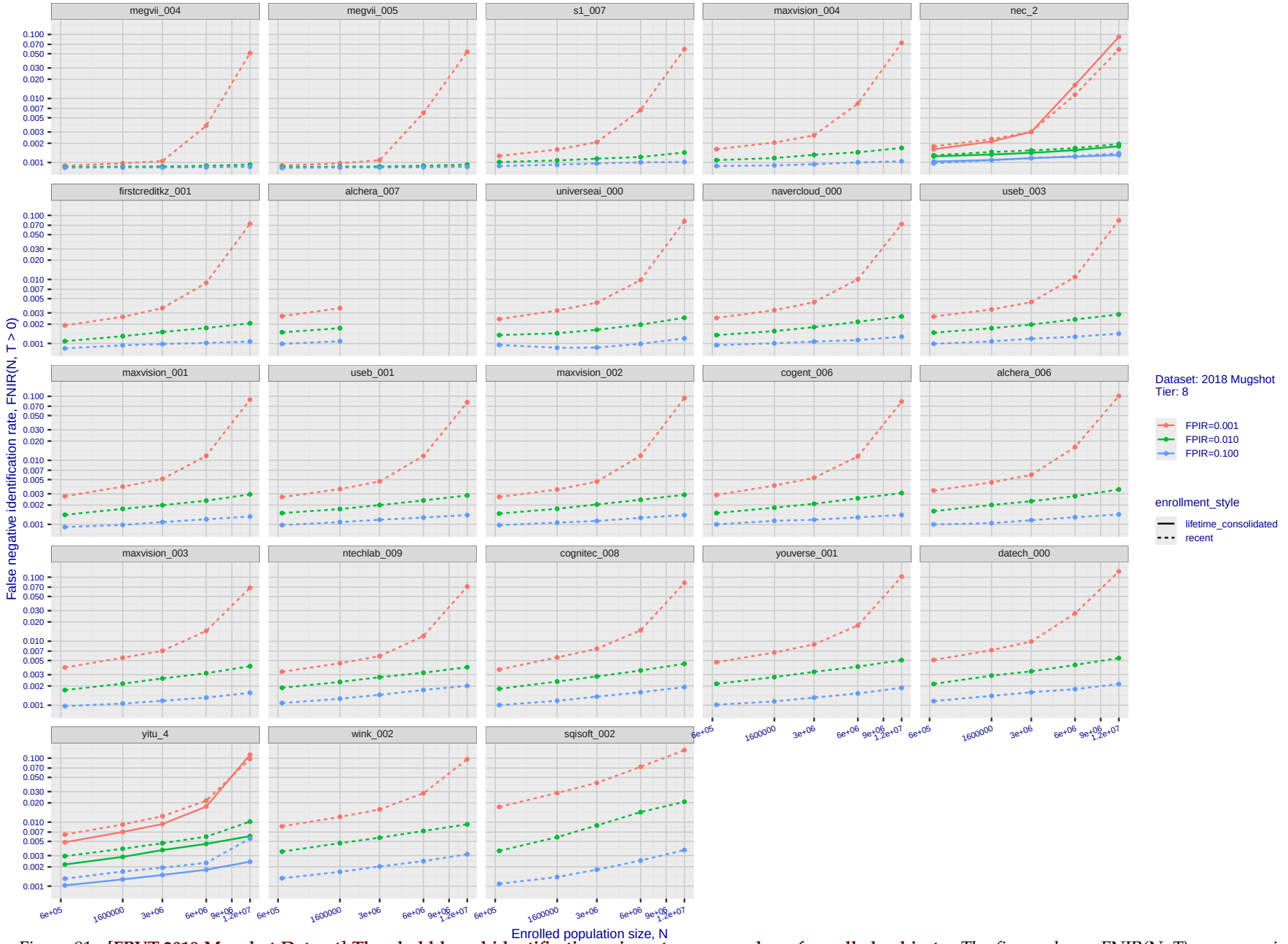


*Figure 91:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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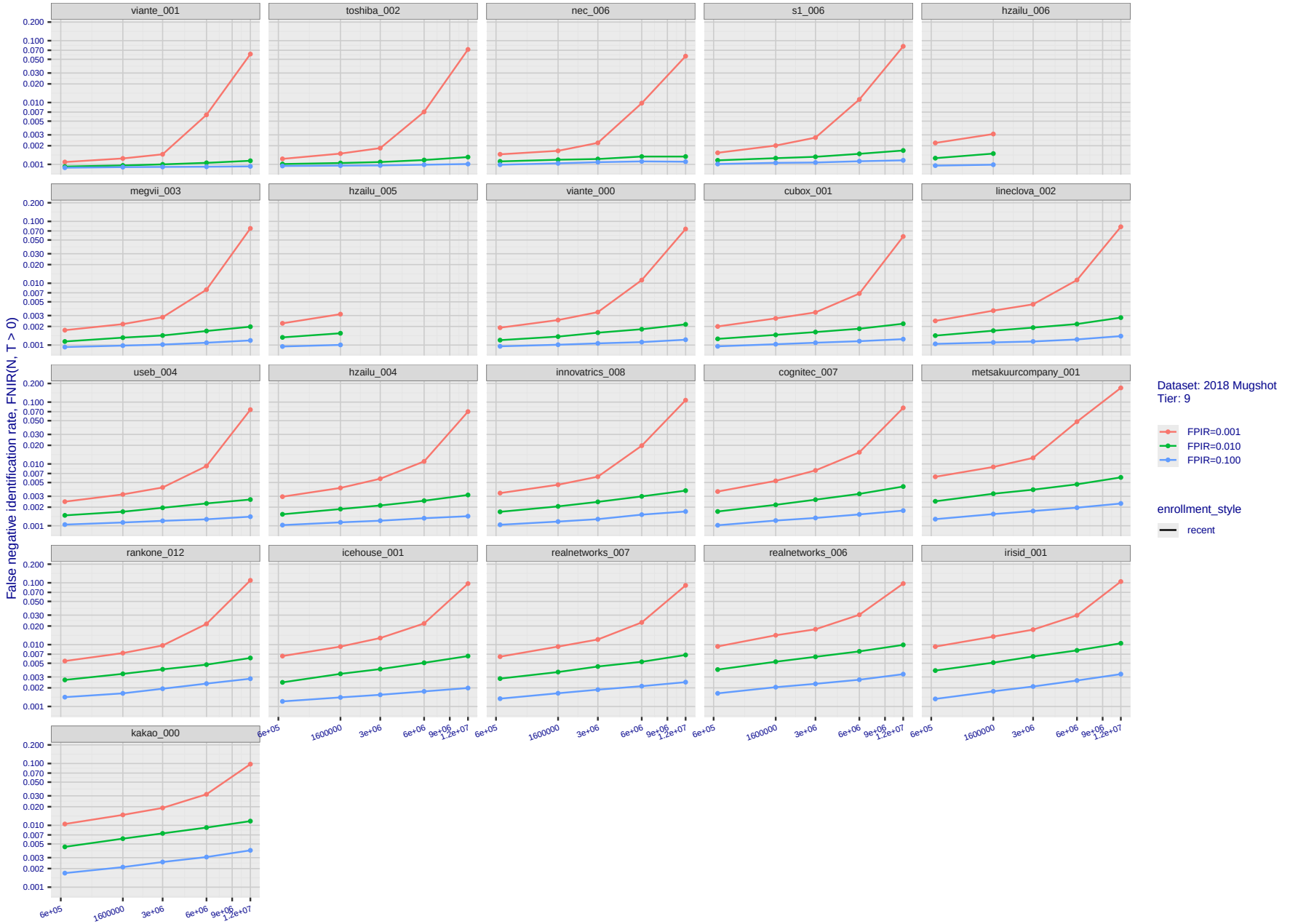


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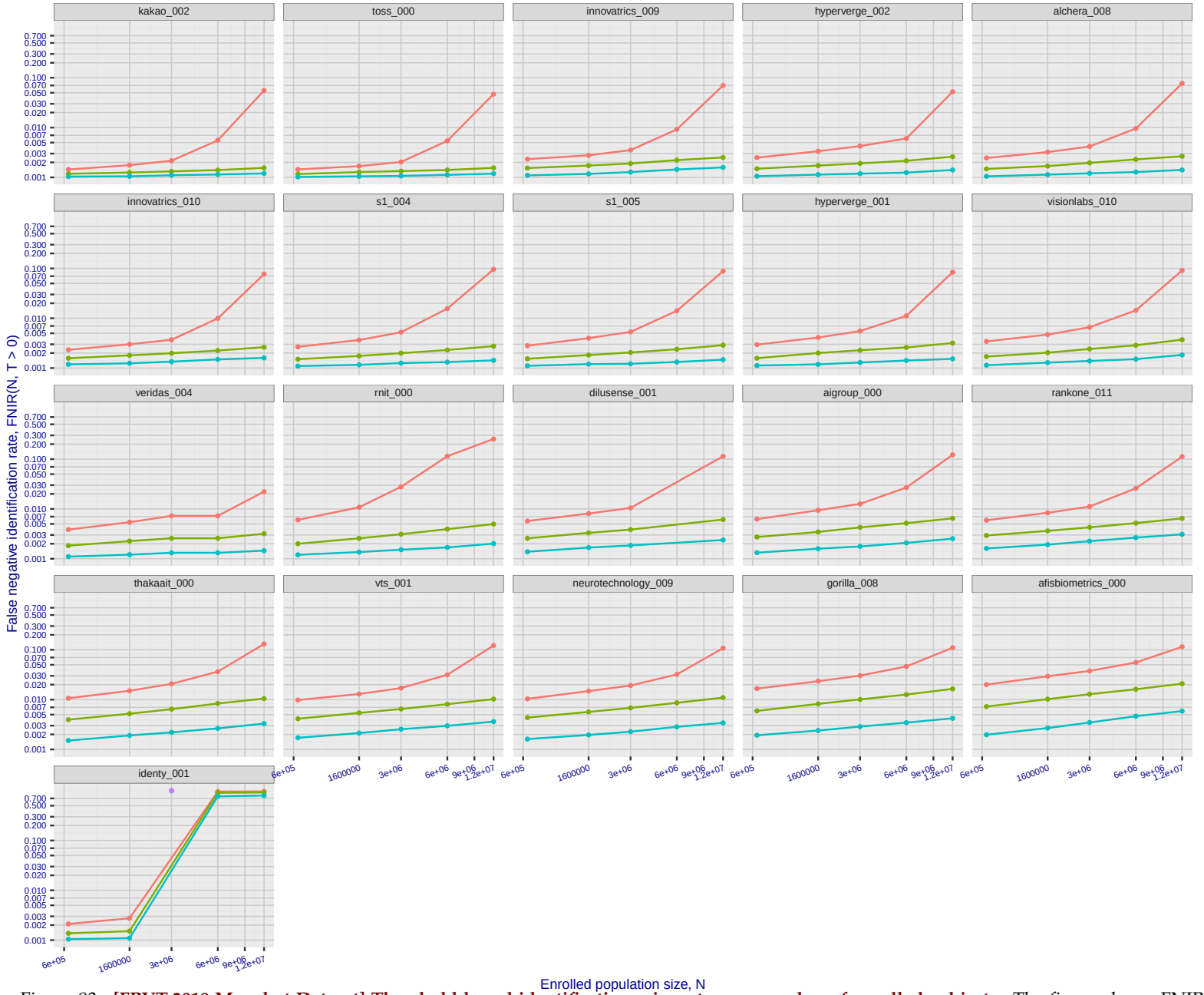


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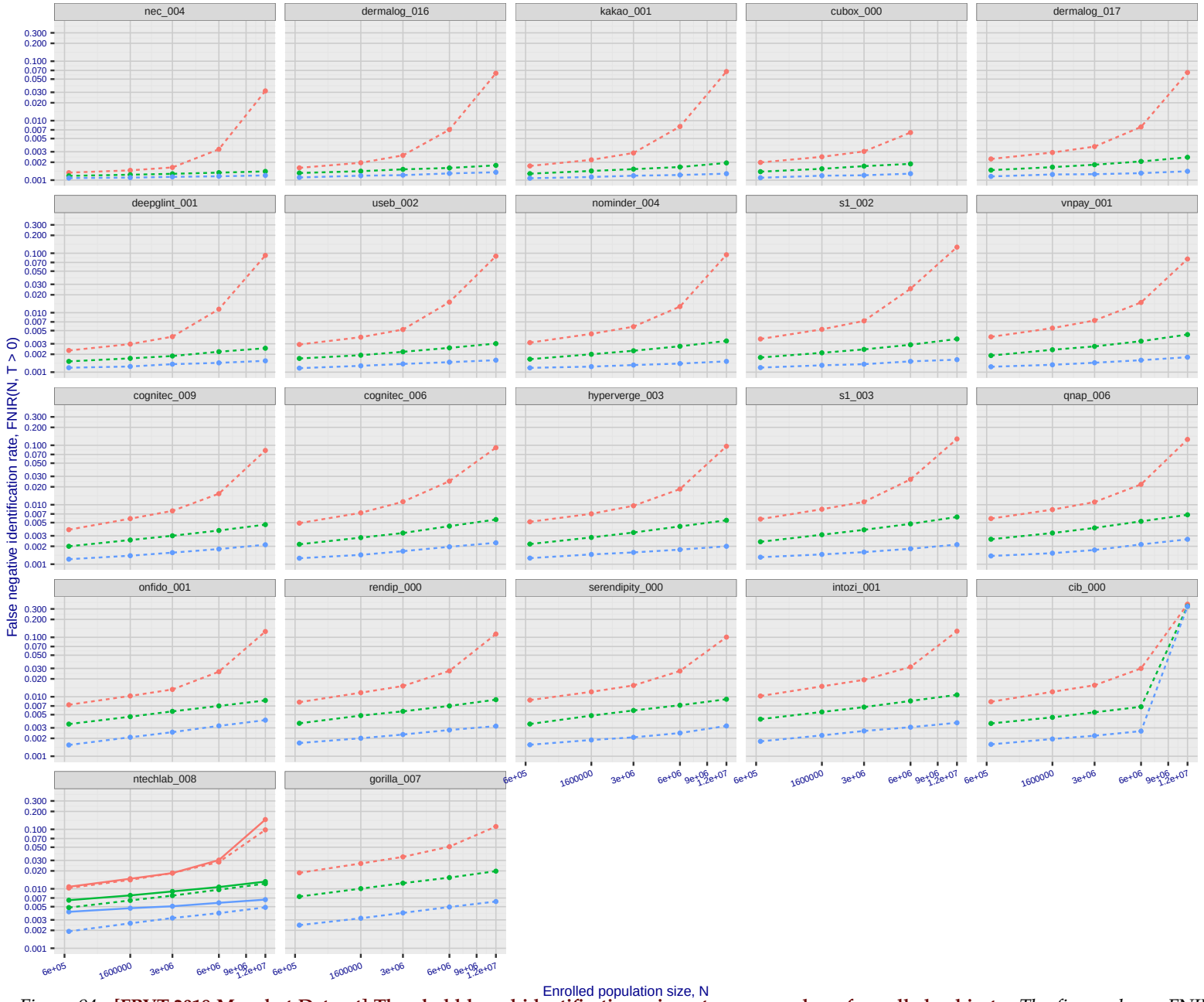


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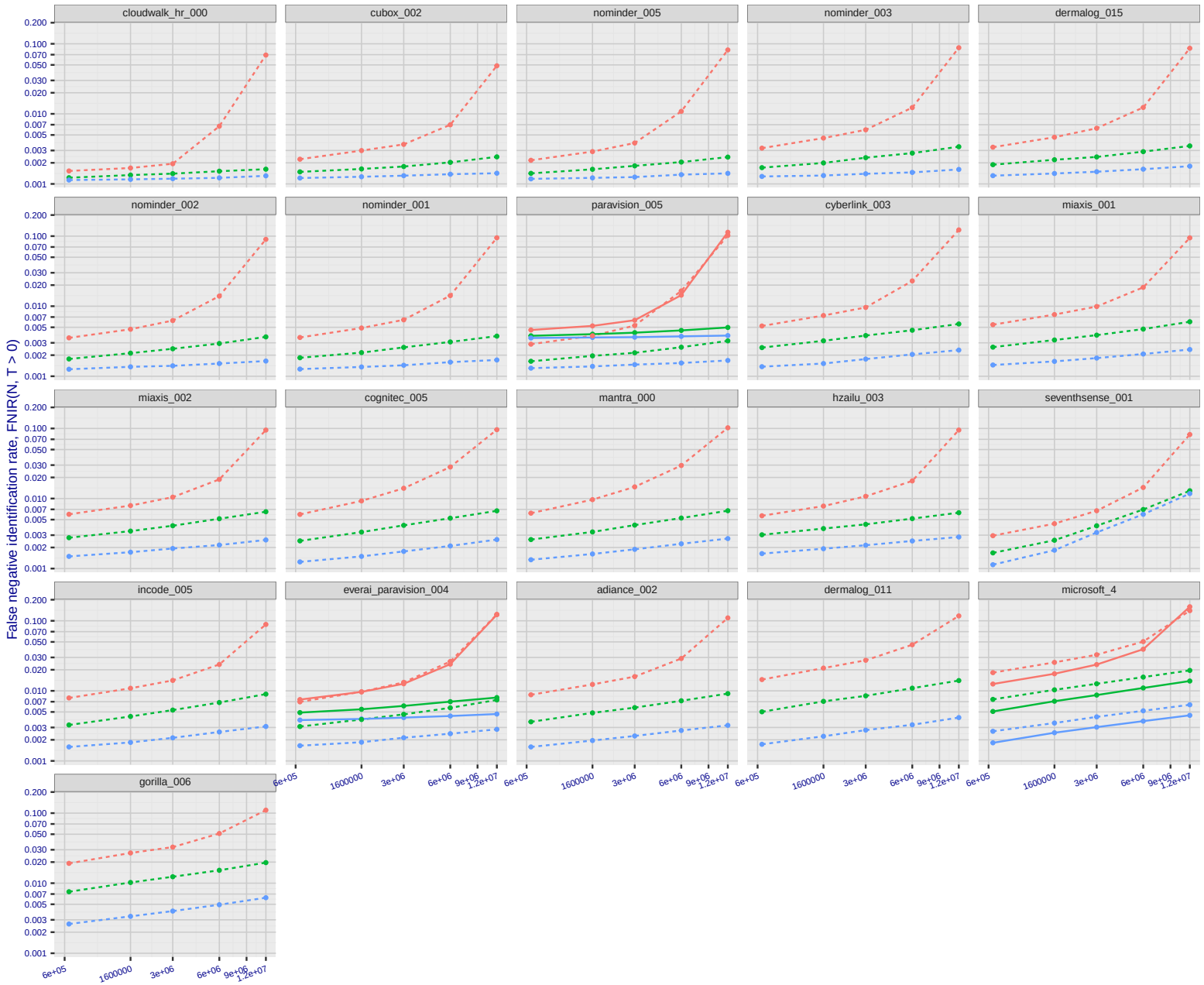


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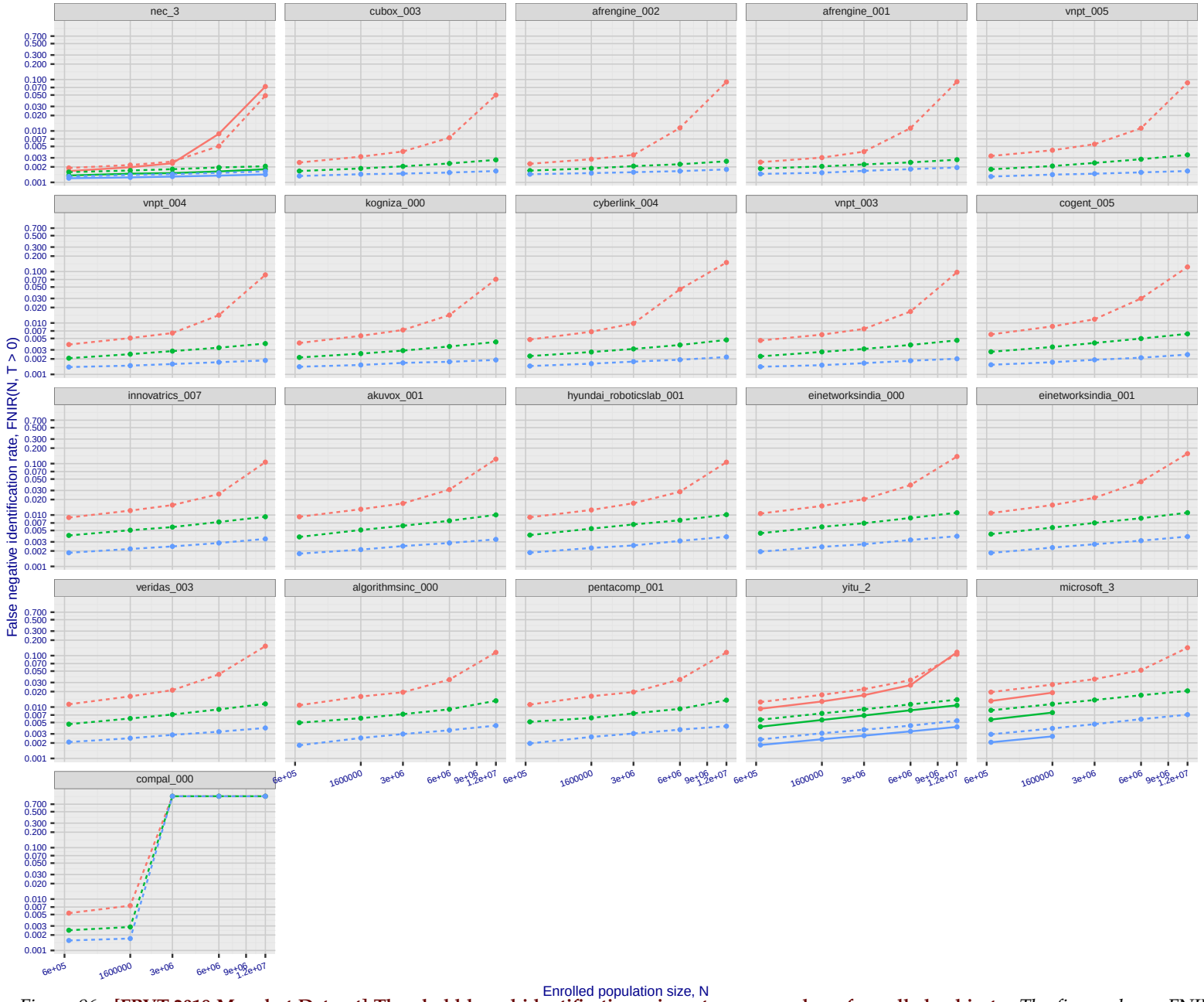


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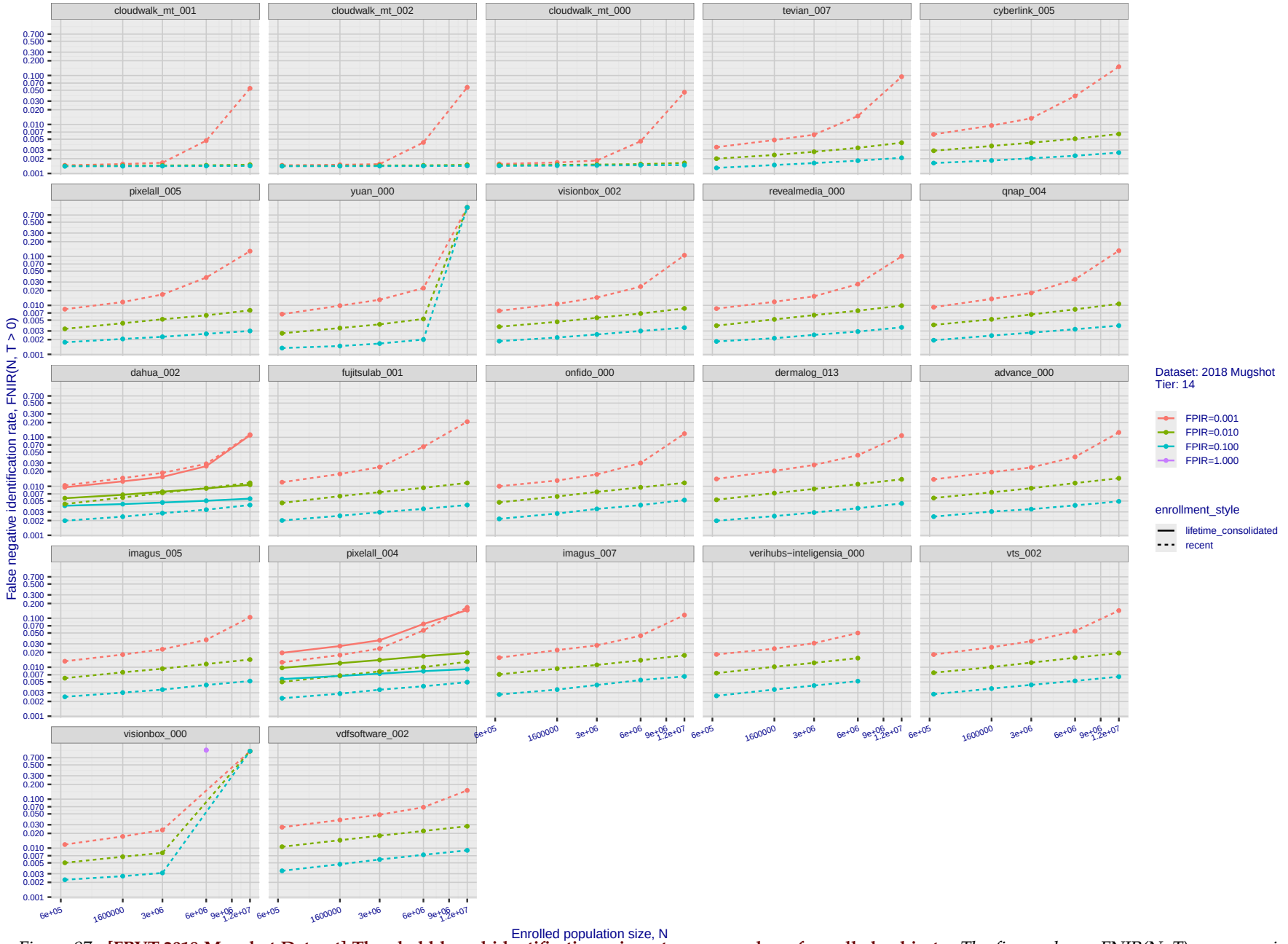


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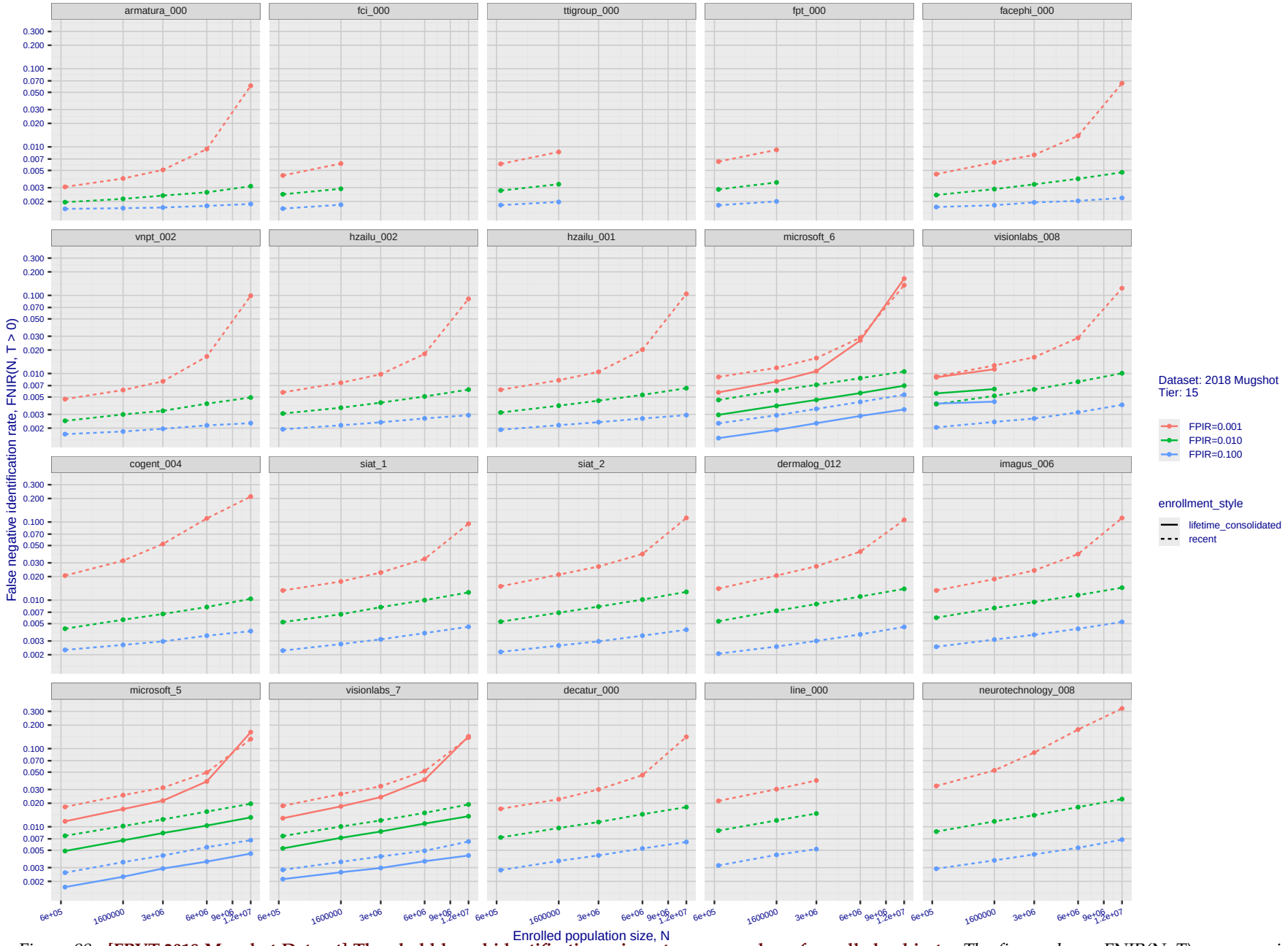


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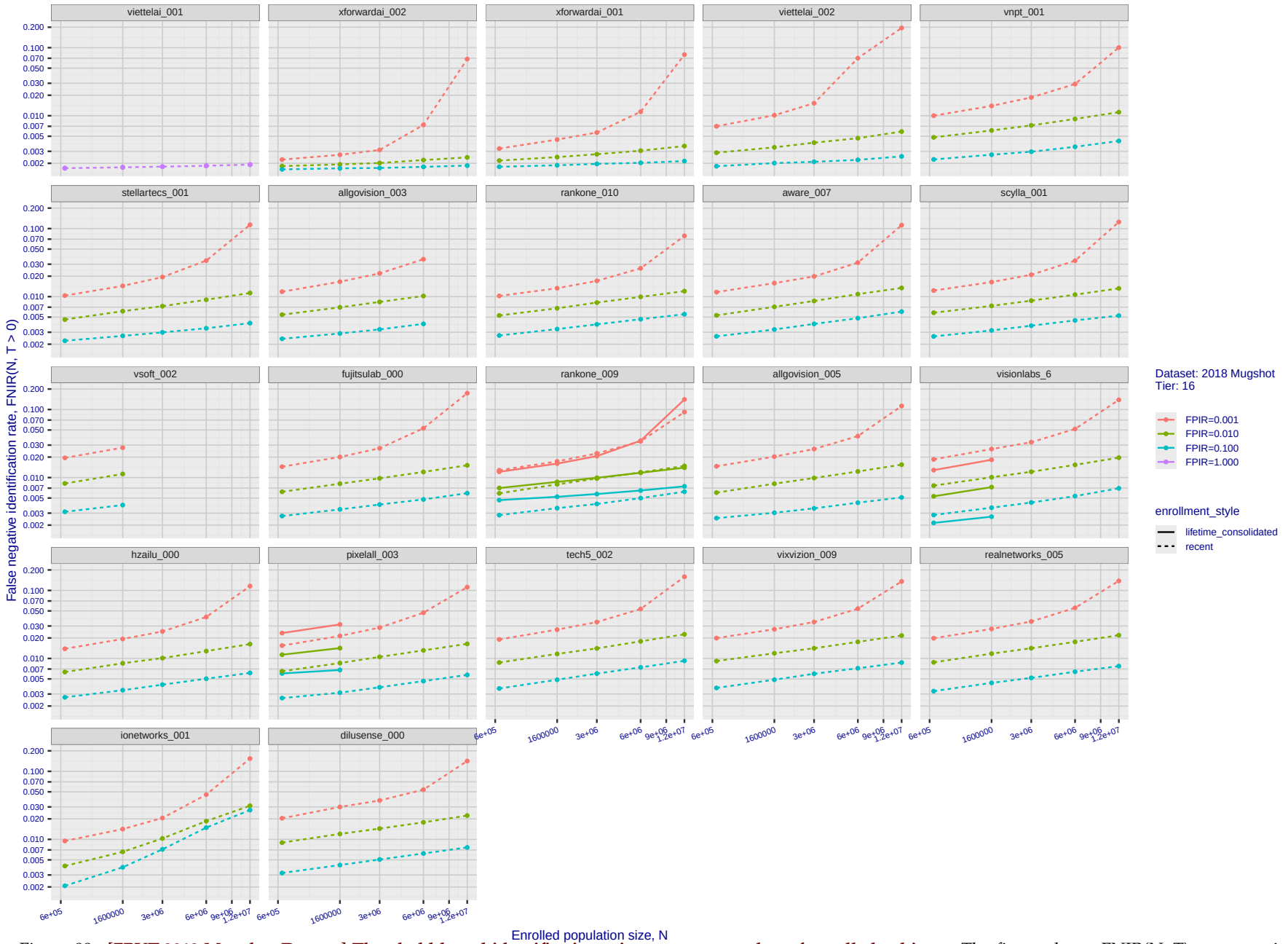


*Figure 99:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*

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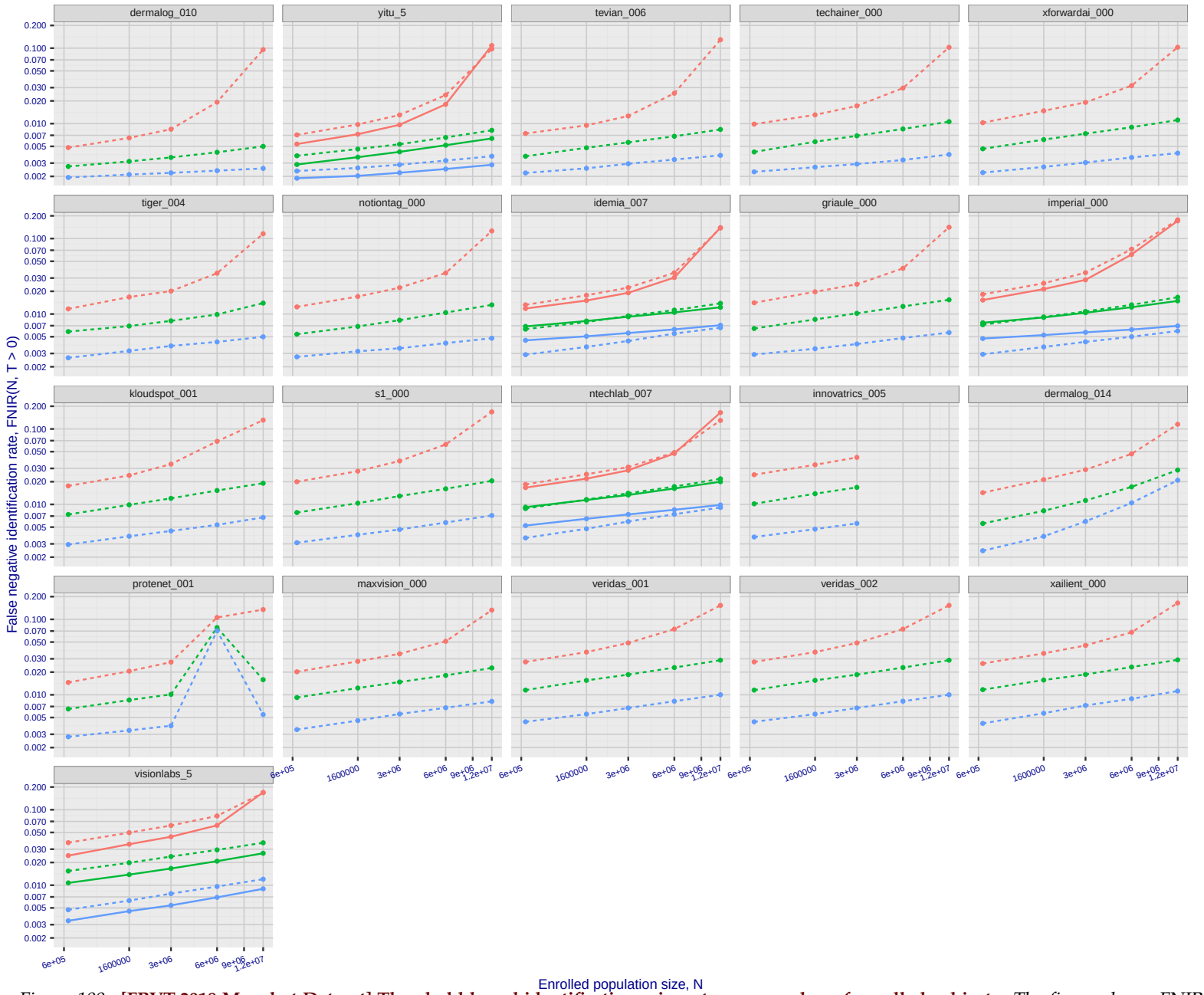


*Figure 100:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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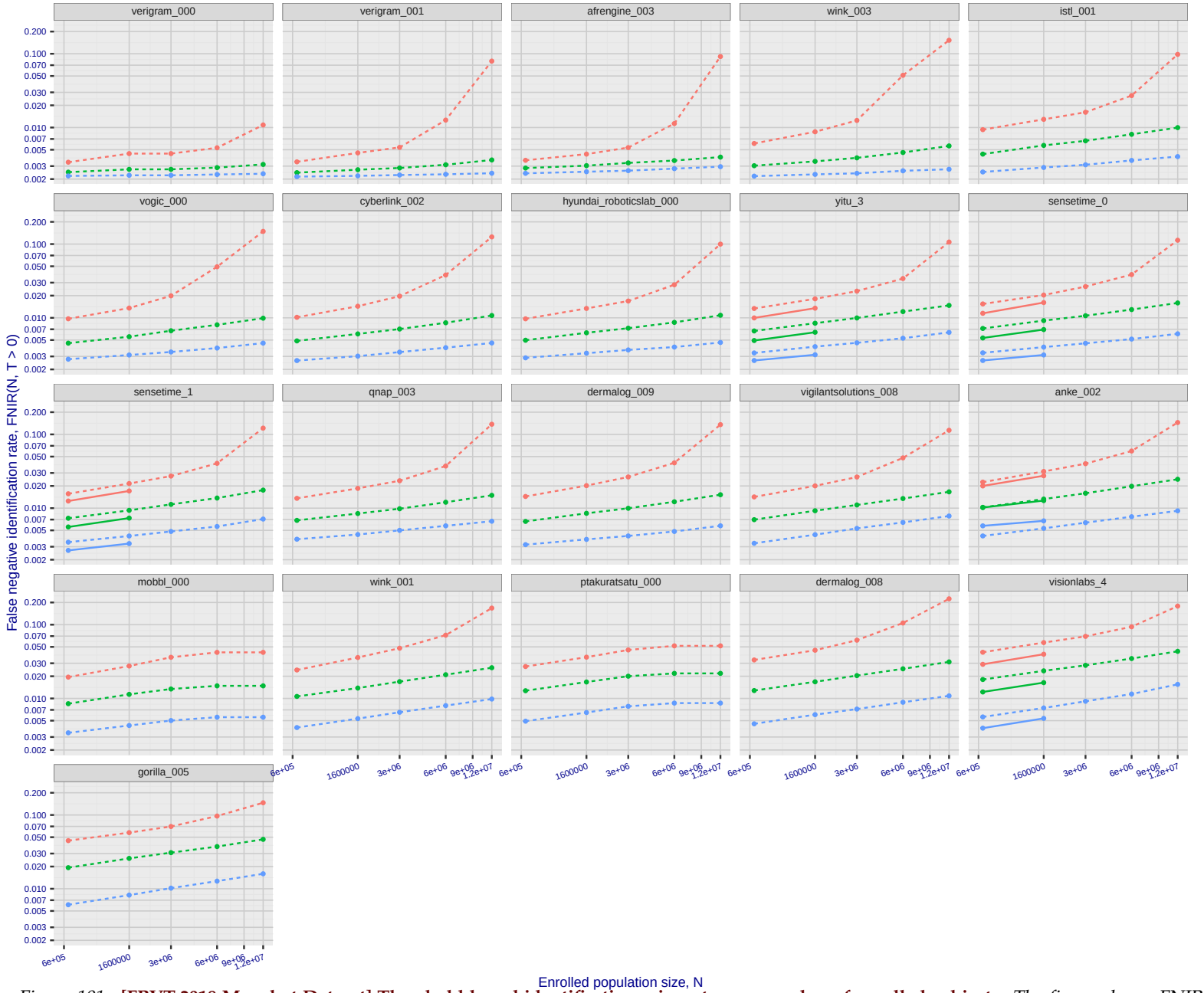


*Figure 101:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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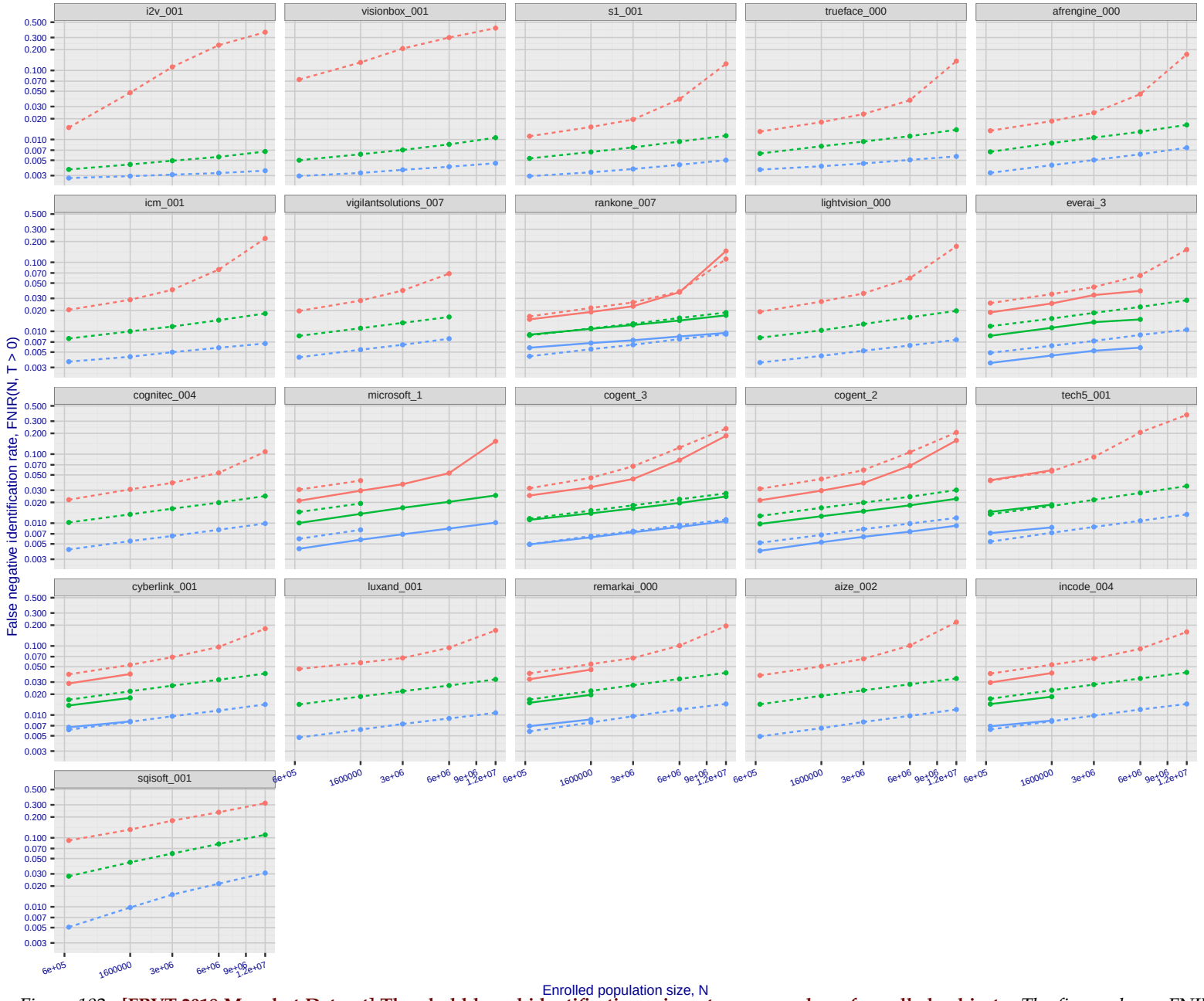


*Figure 102:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*

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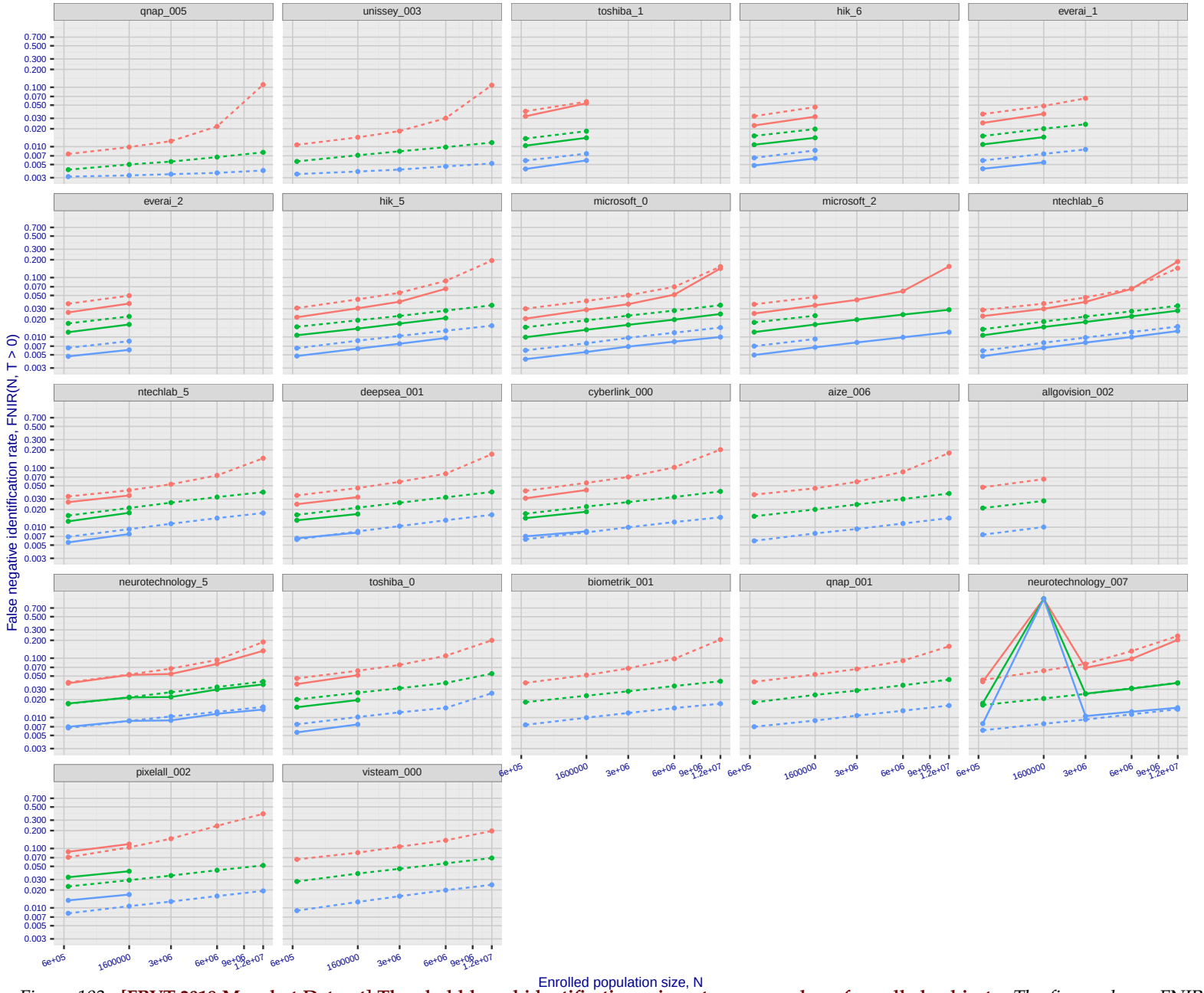


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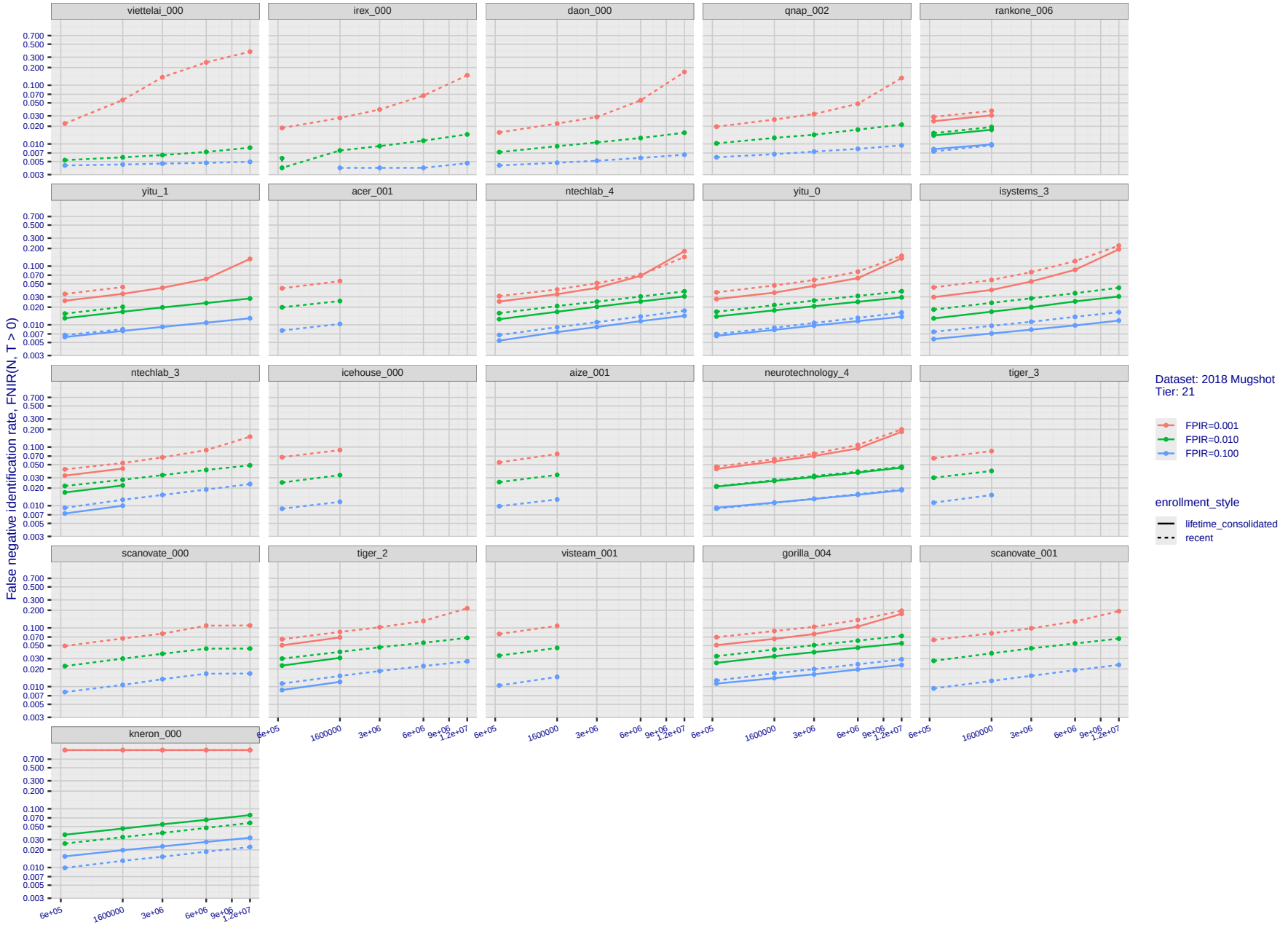


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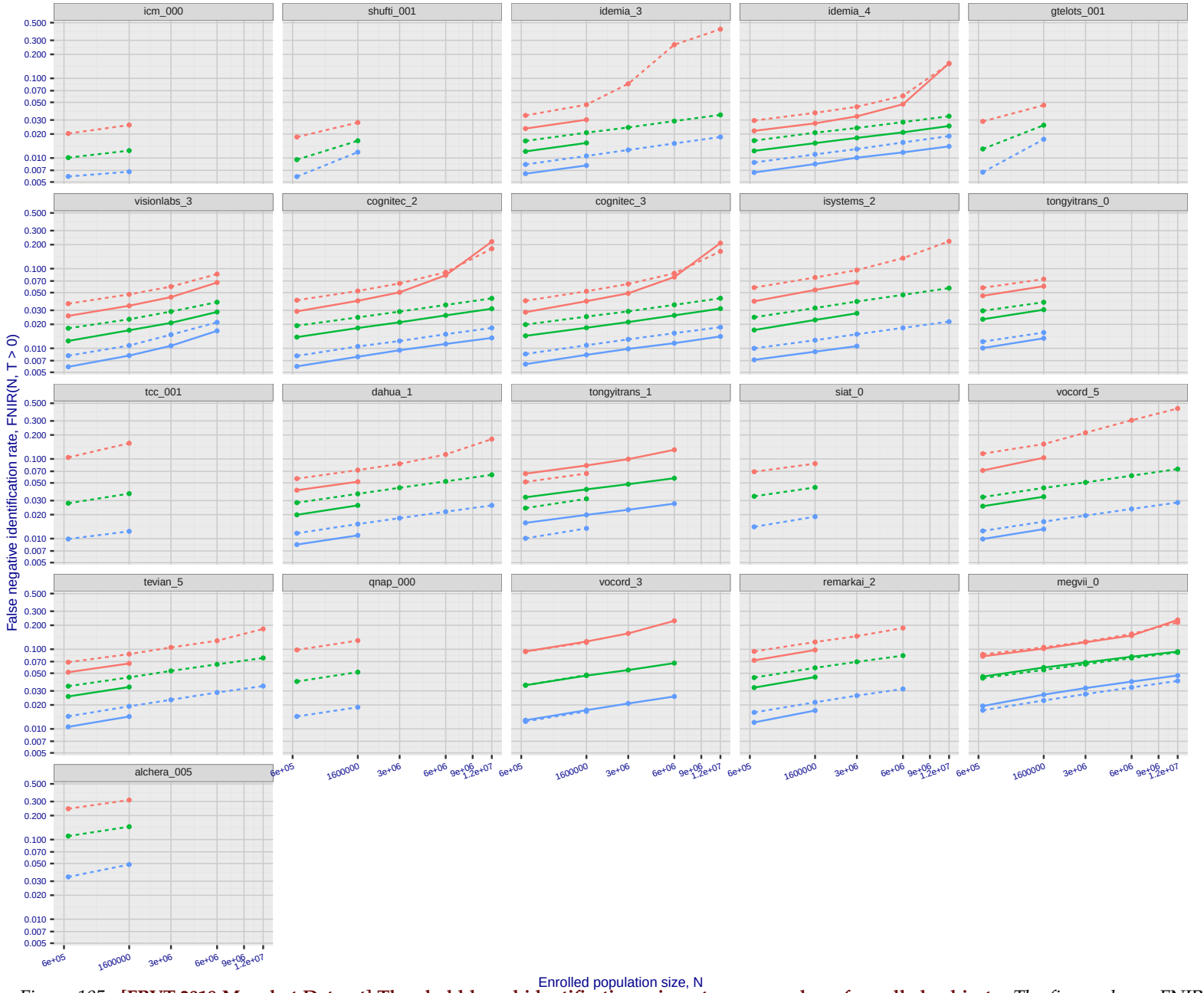


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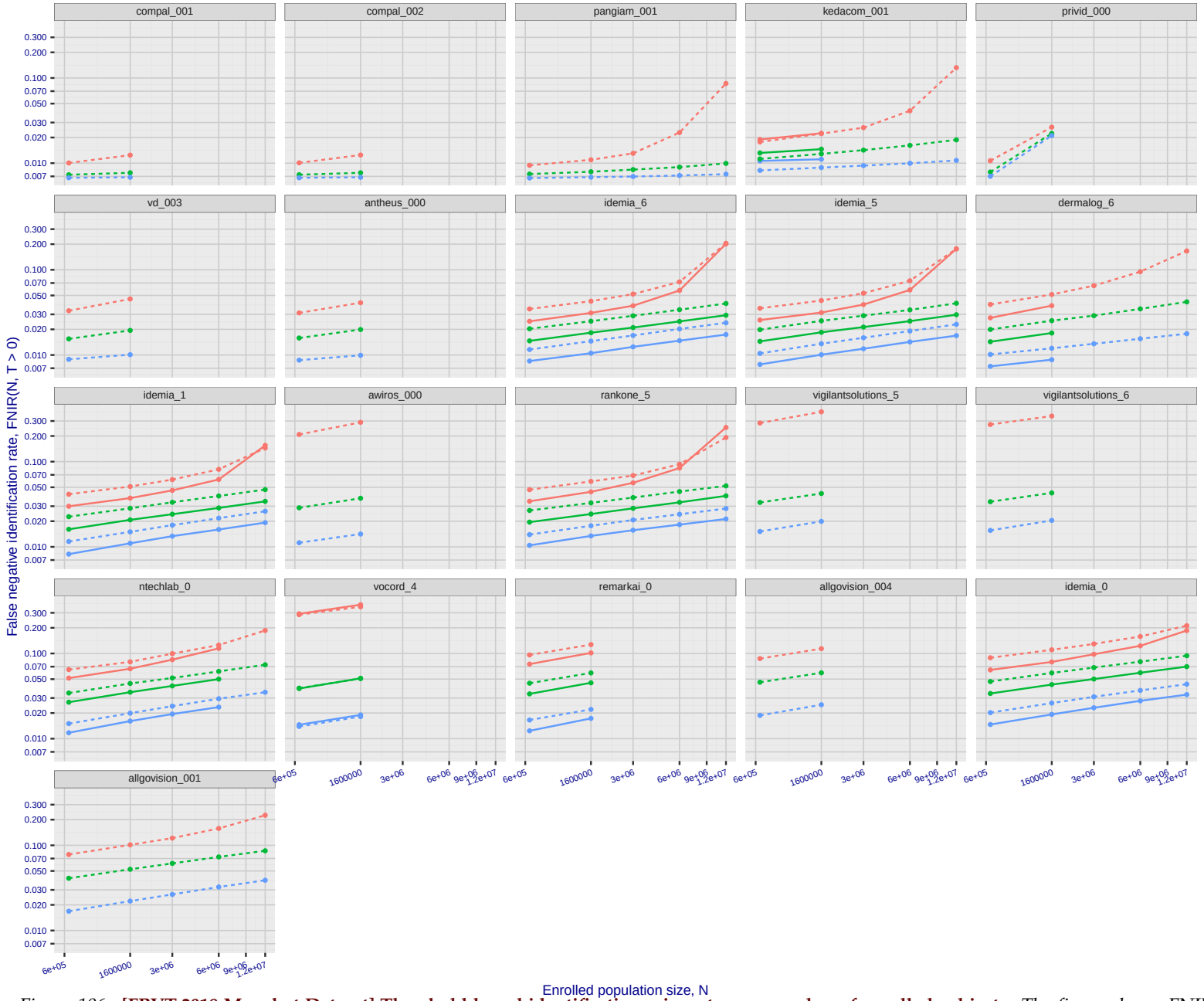


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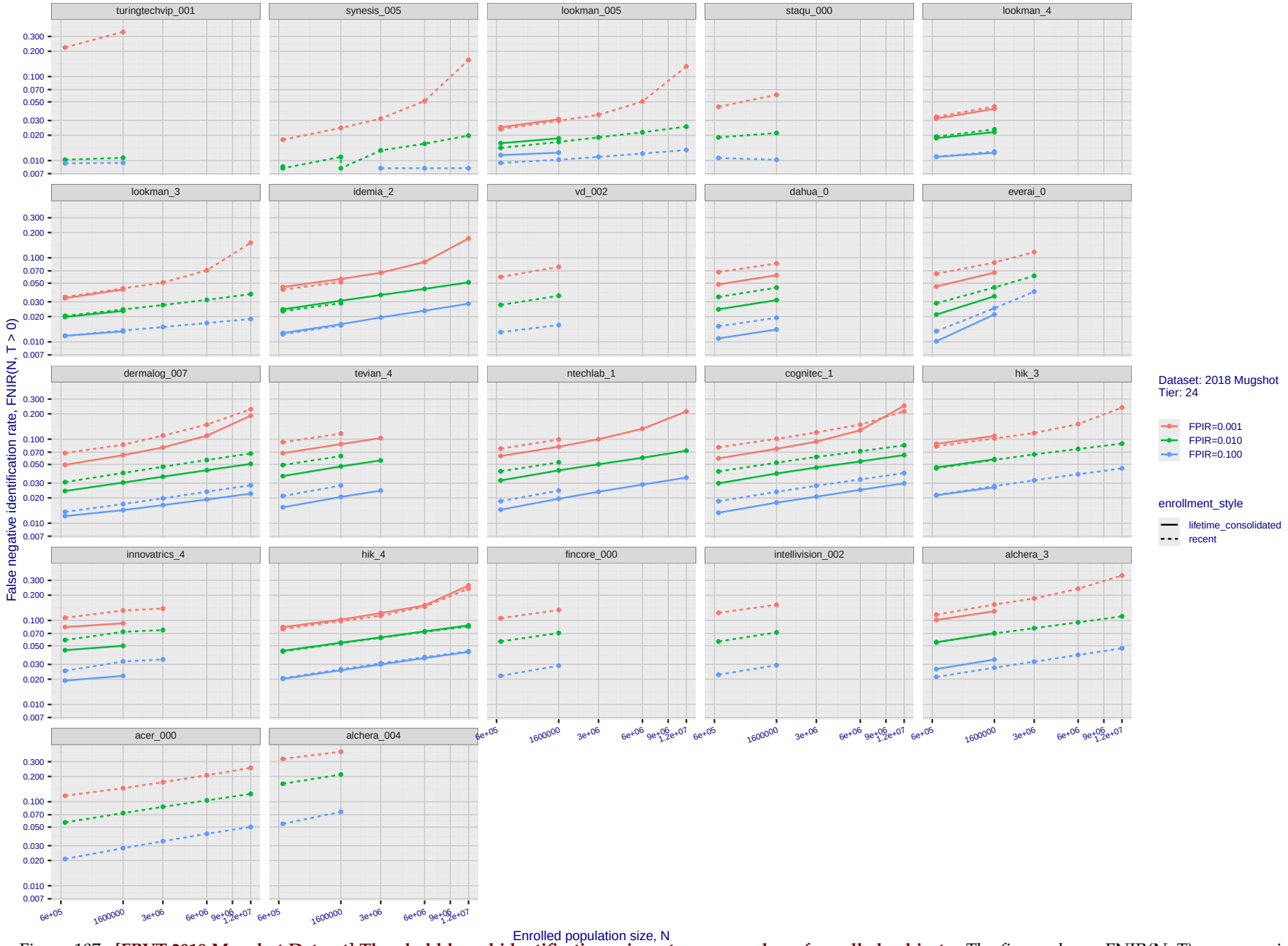


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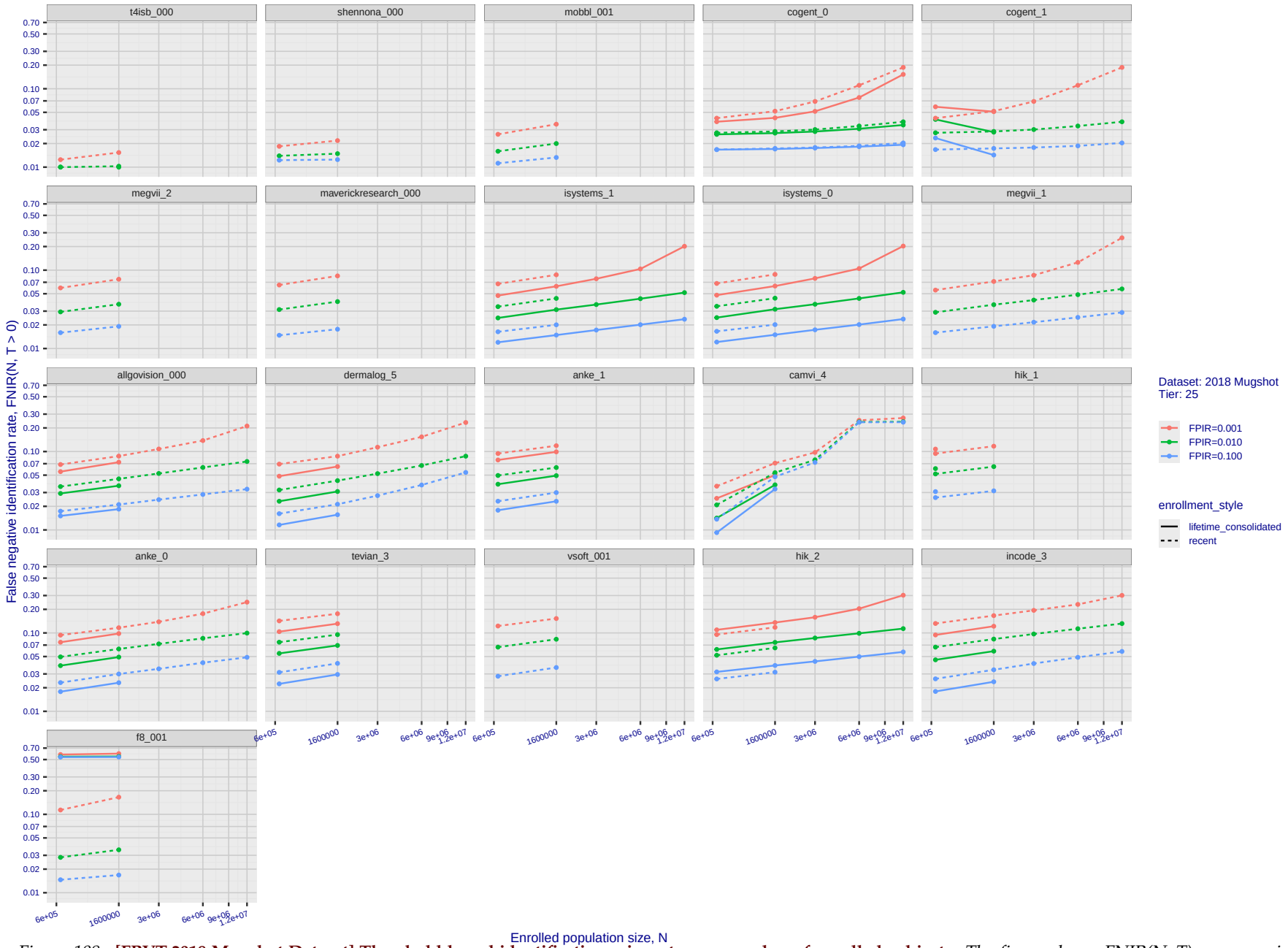


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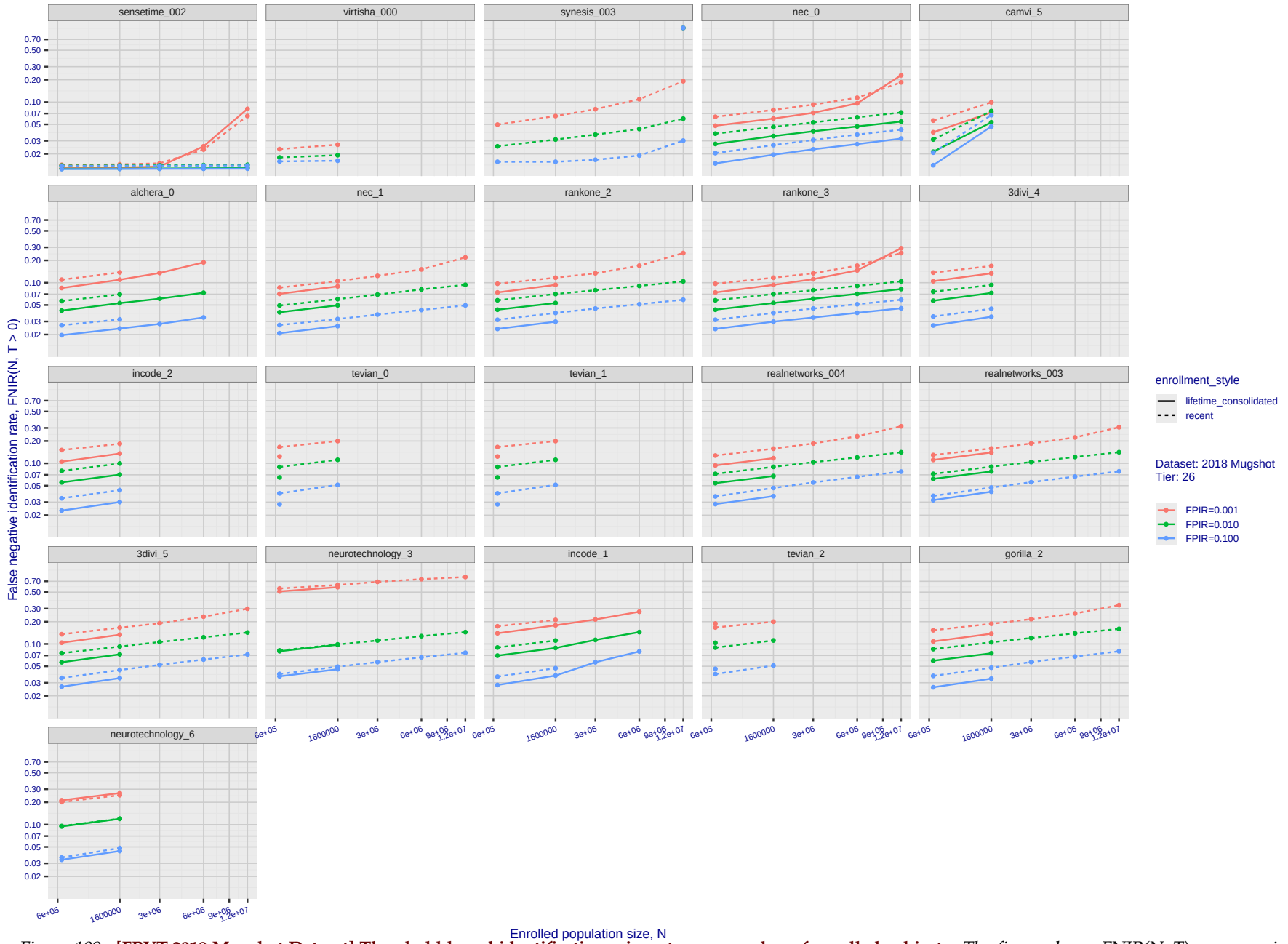


*Figure 109:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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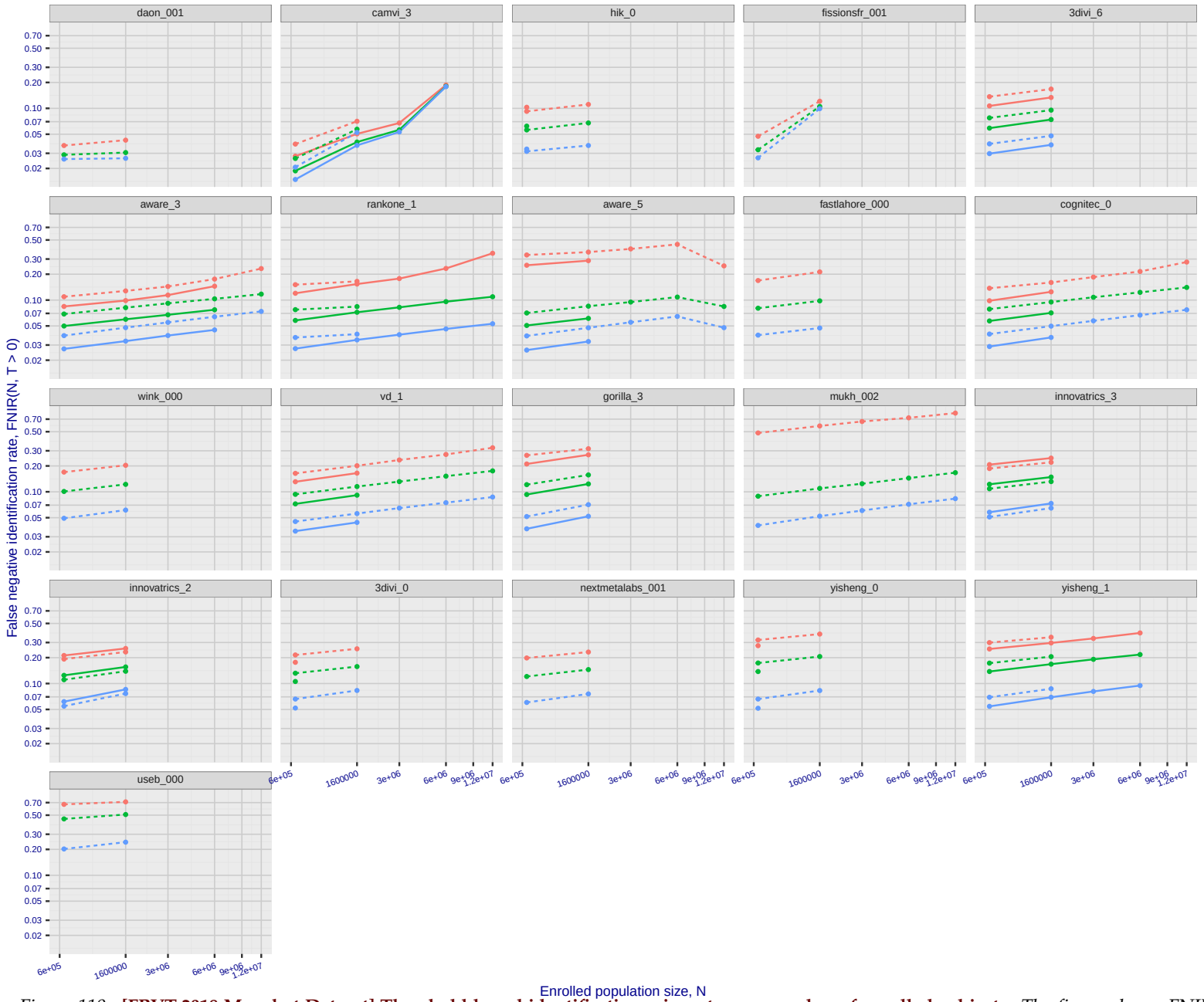


*Figure 110:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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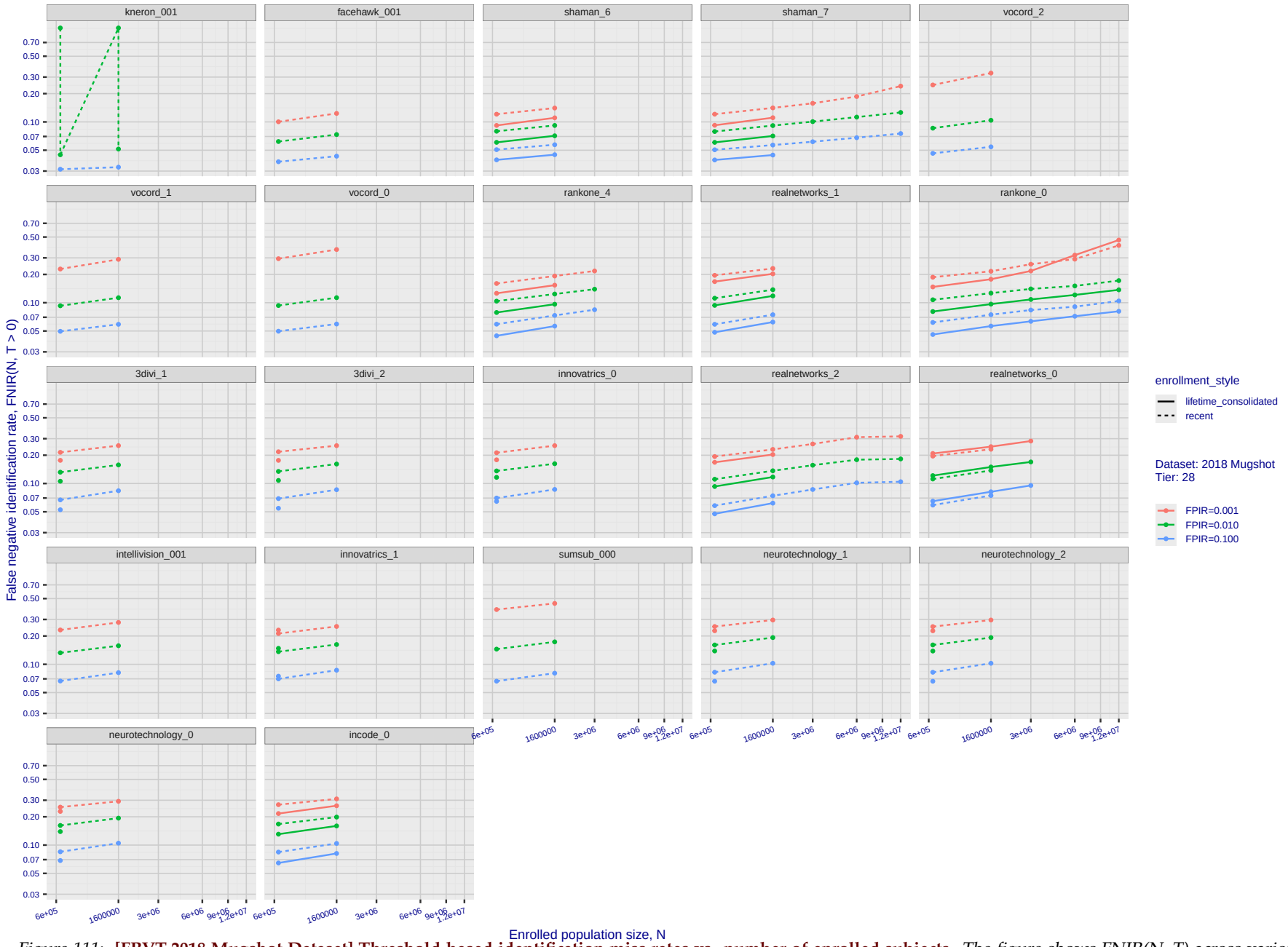


*Figure 111:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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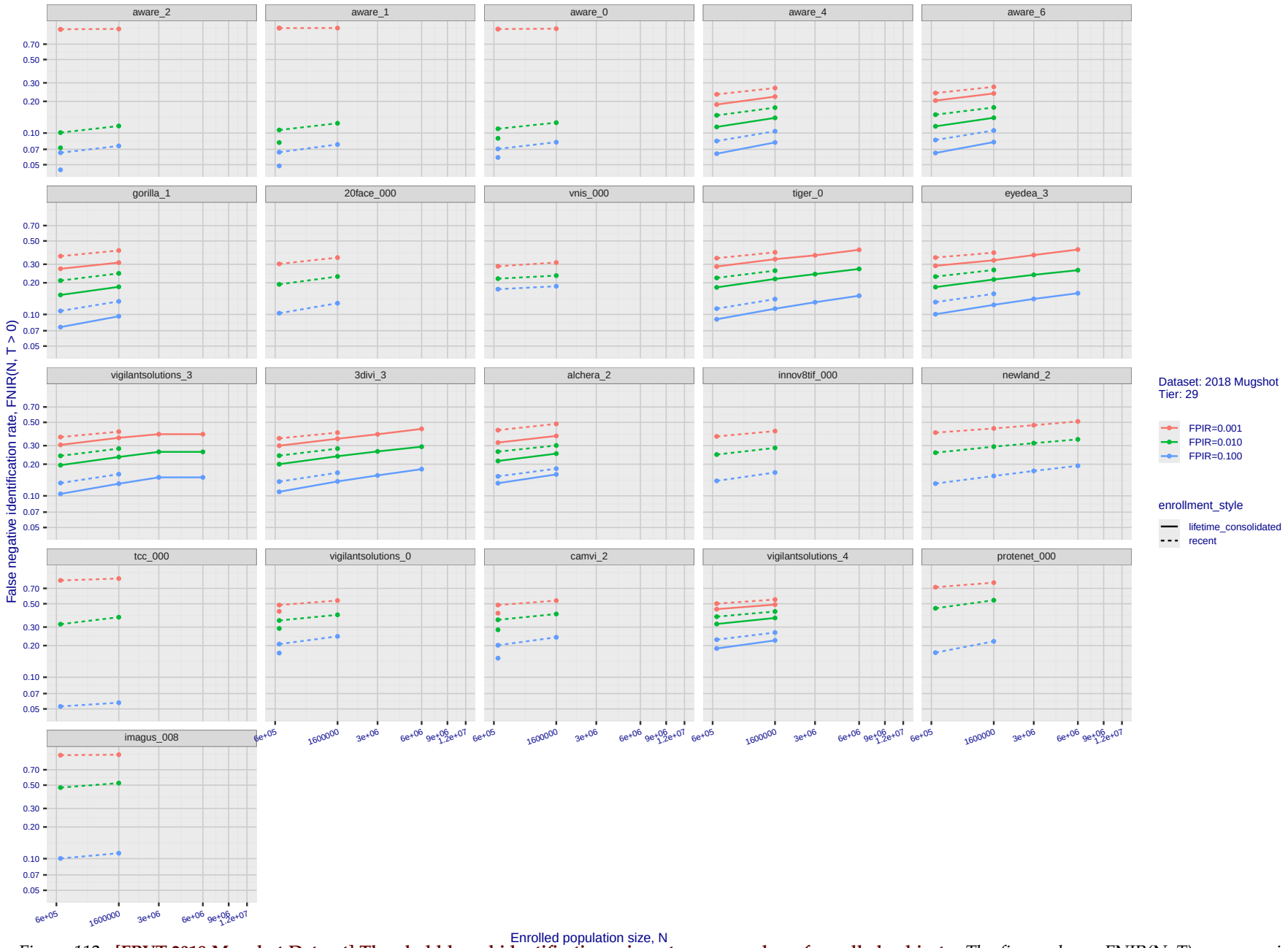


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*Figure 112:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*

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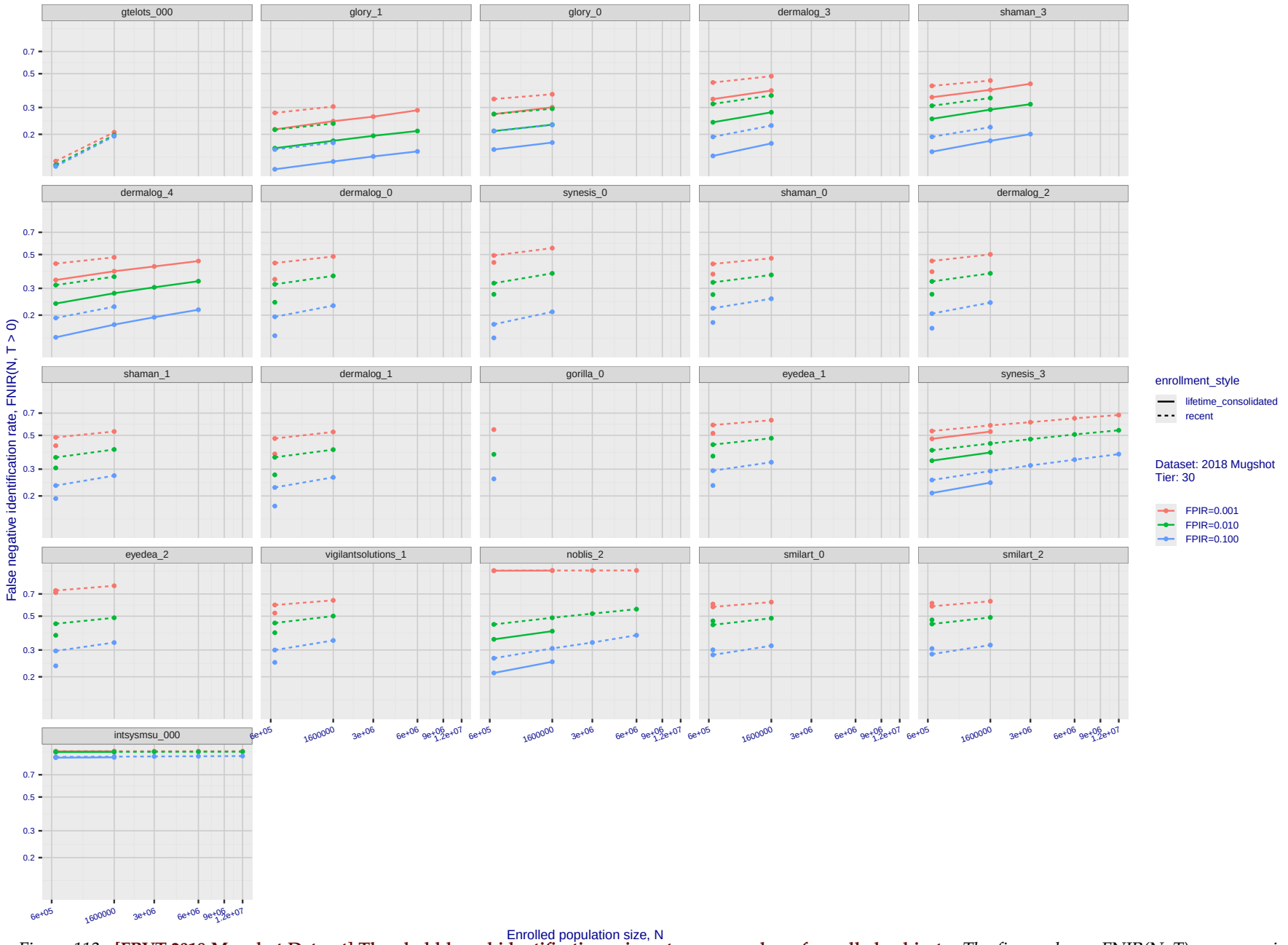


*Figure 113:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*

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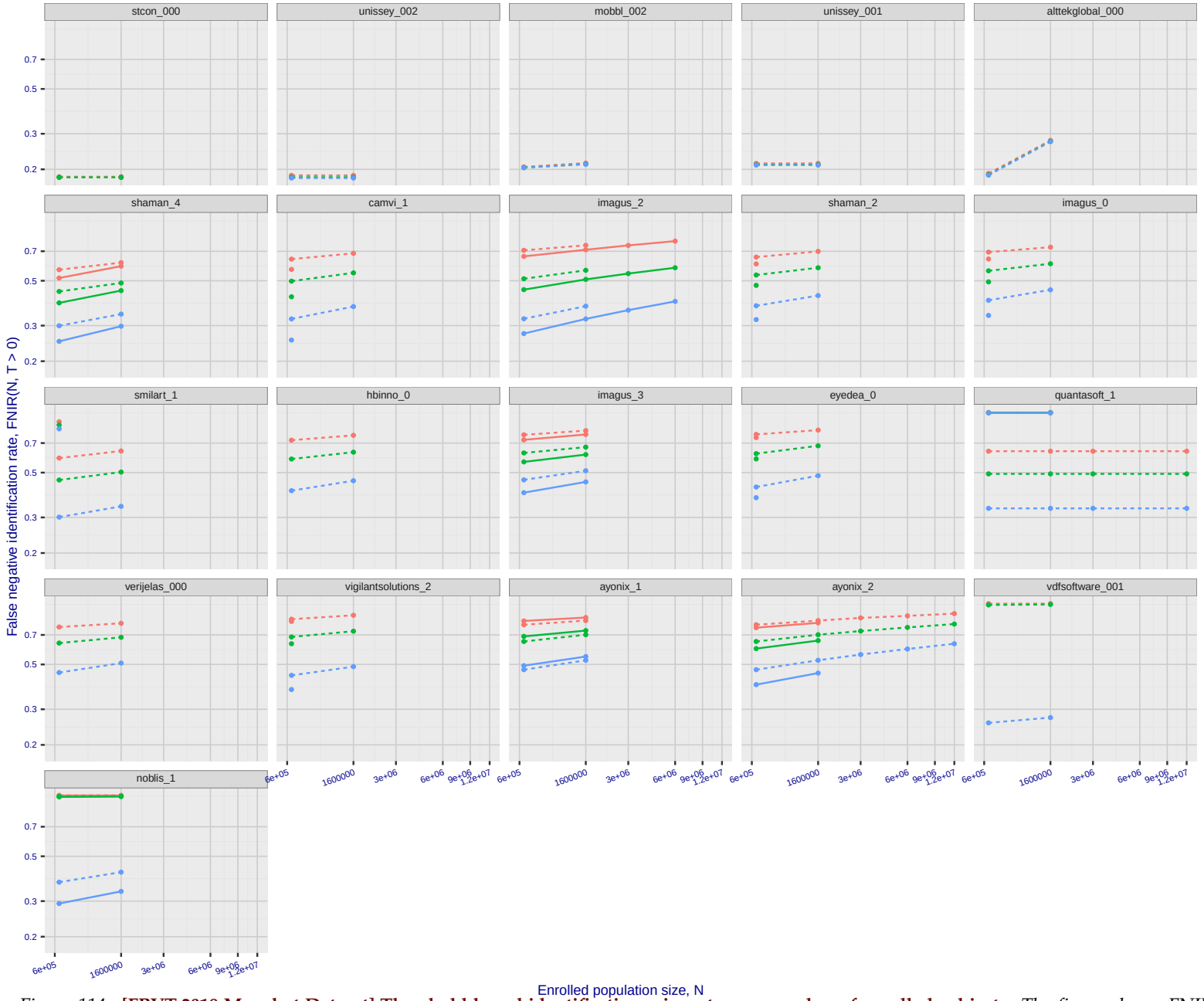


*Figure 114:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*



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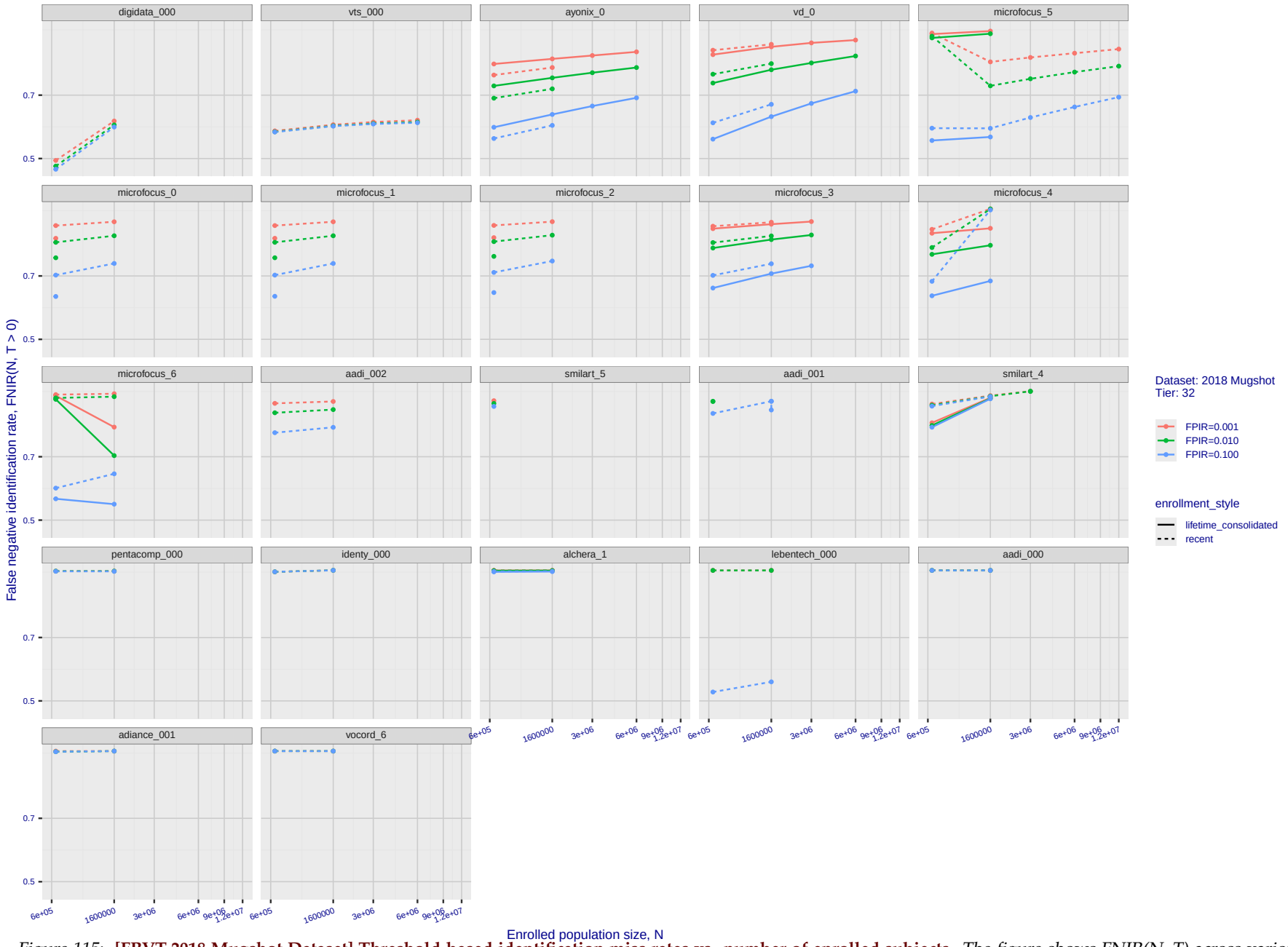


*Figure 115:* **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR($N_b$, 1, 0), then sorting by median FNIR($N_b$, T), $N_b$ = 640 000.*

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| | | | | | |
|---|---|---|---|---|---|
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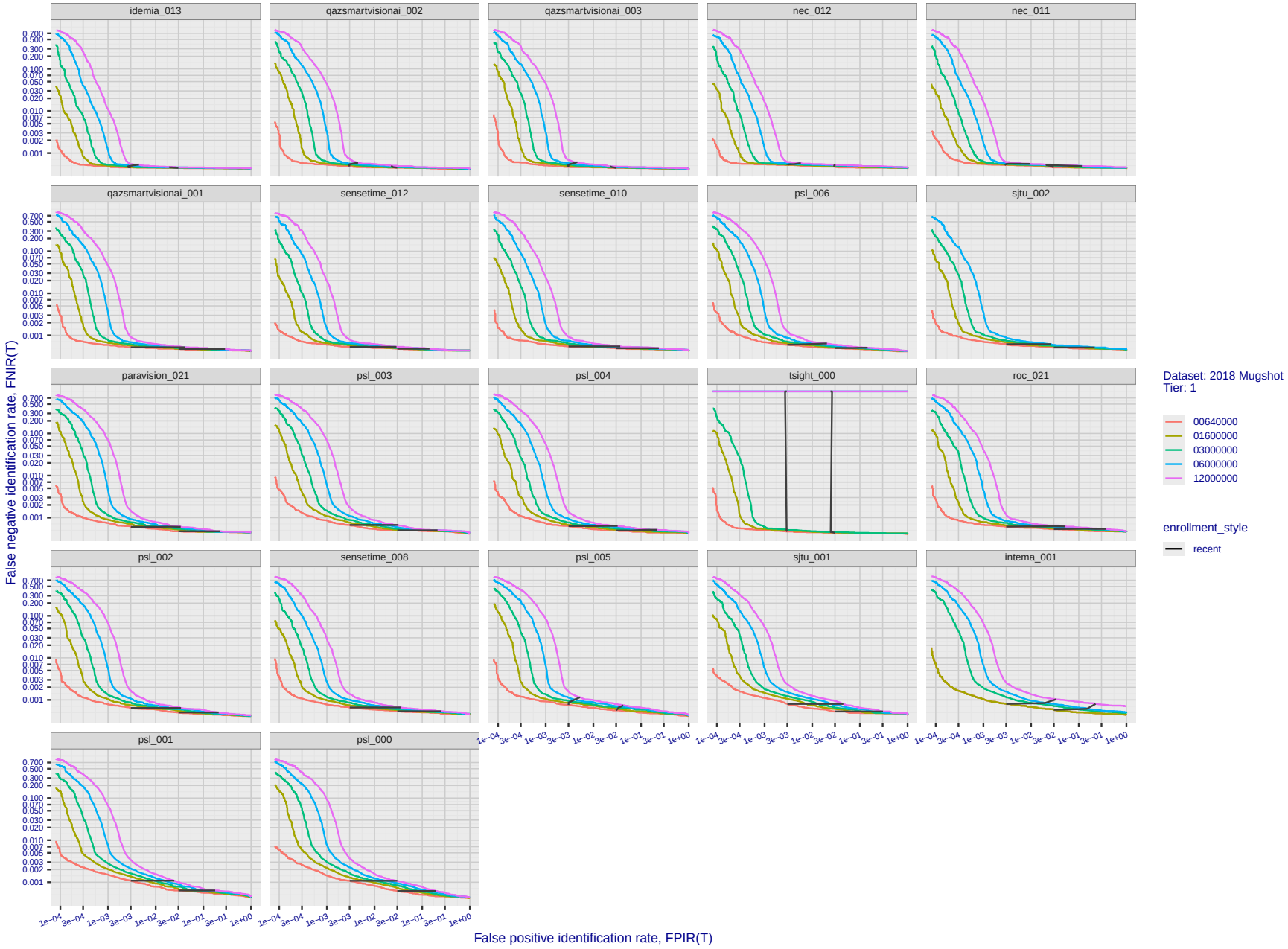


*Figure 116:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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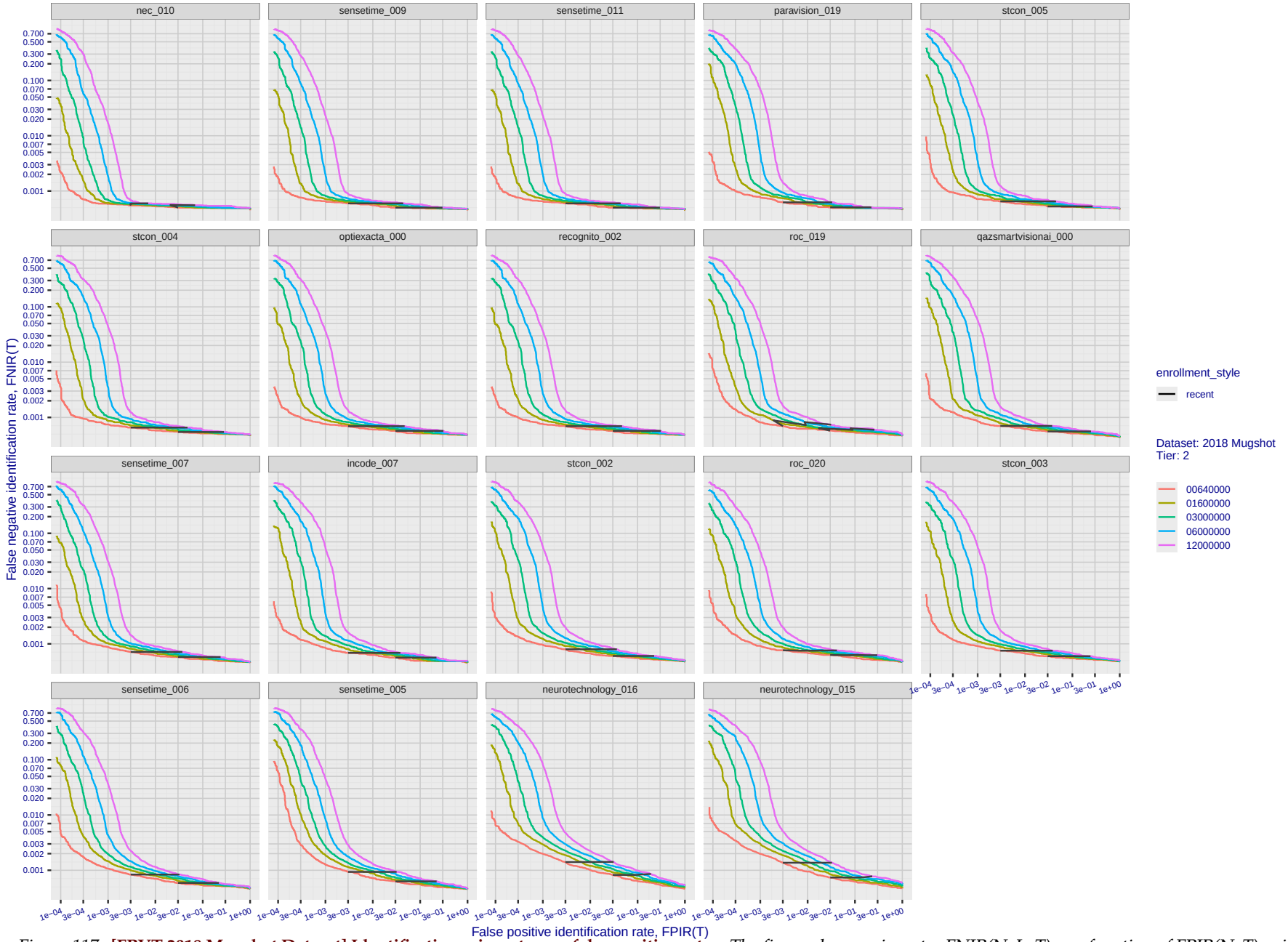


*Figure 117:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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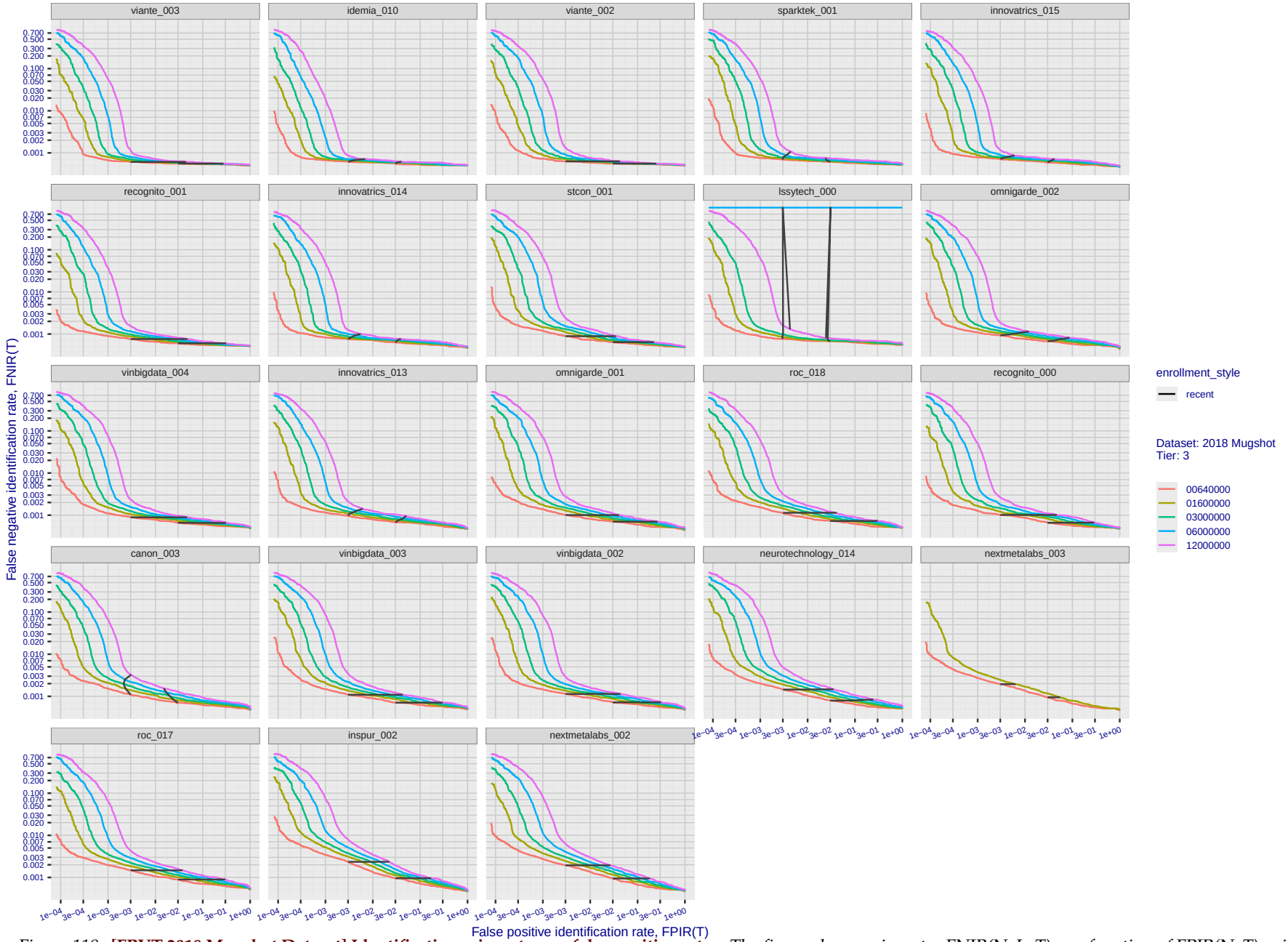


*Figure 118:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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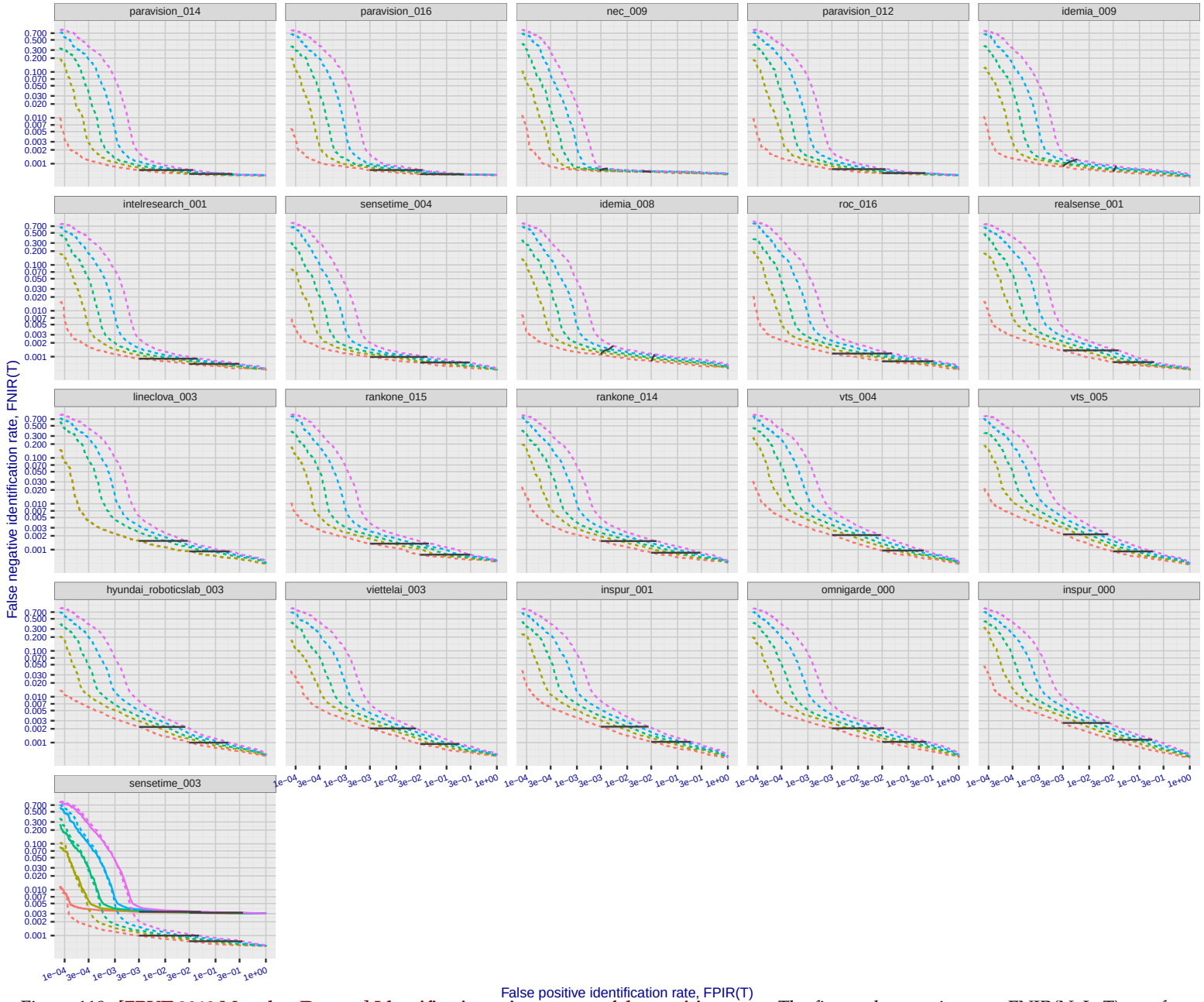


*Figure 119:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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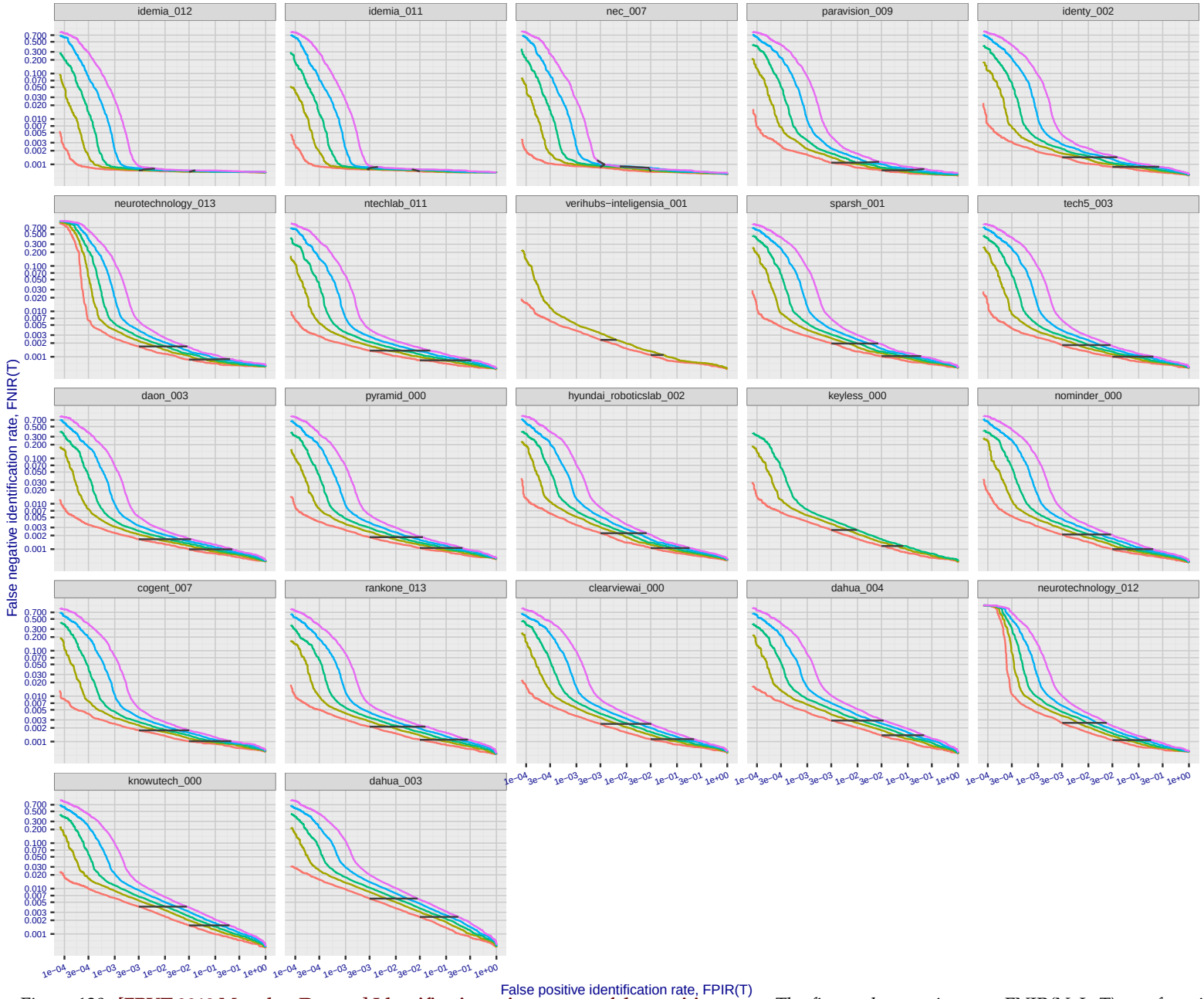


*Figure 120:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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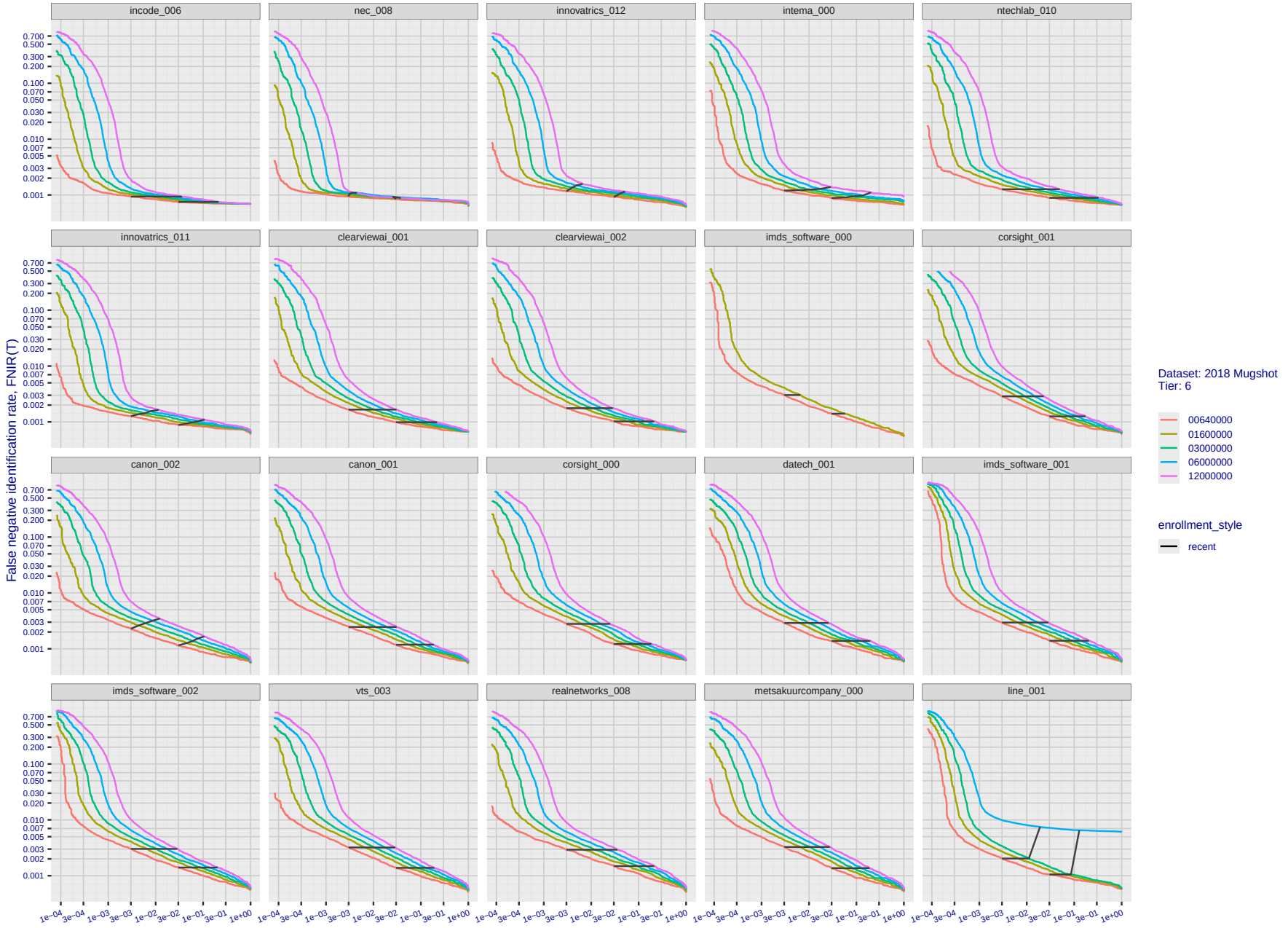


*Figure 121:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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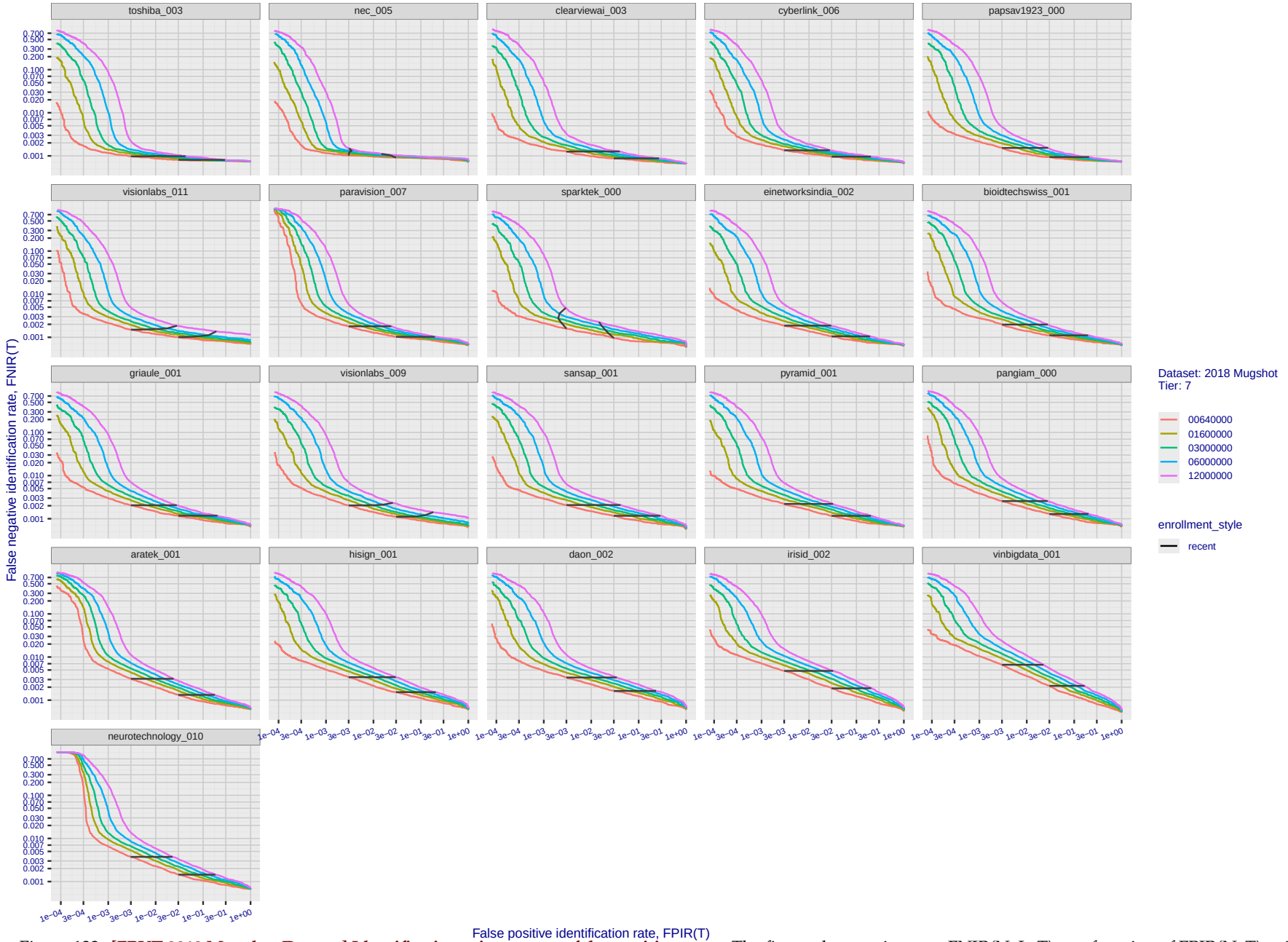


*Figure 122:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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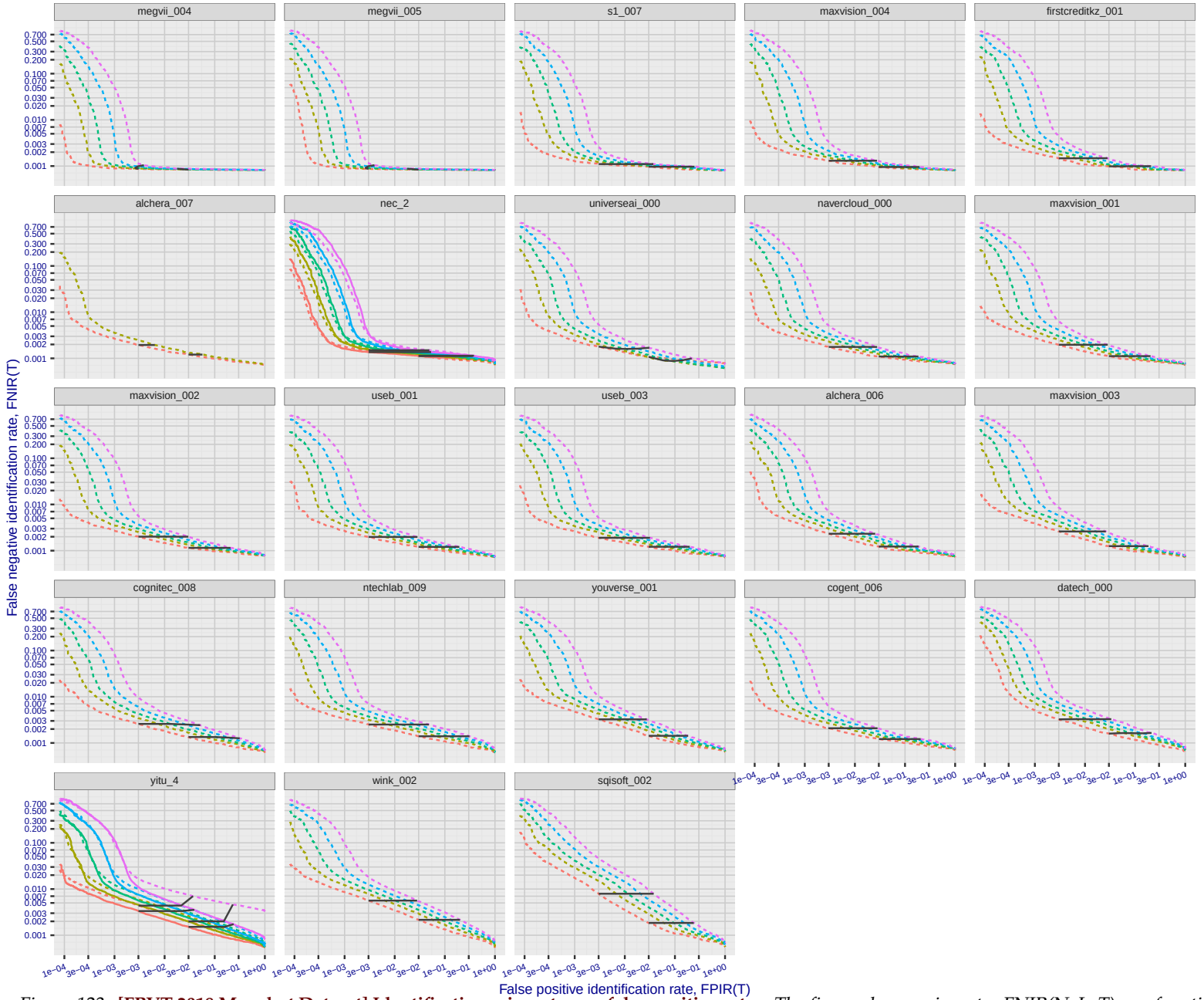


*Figure 123:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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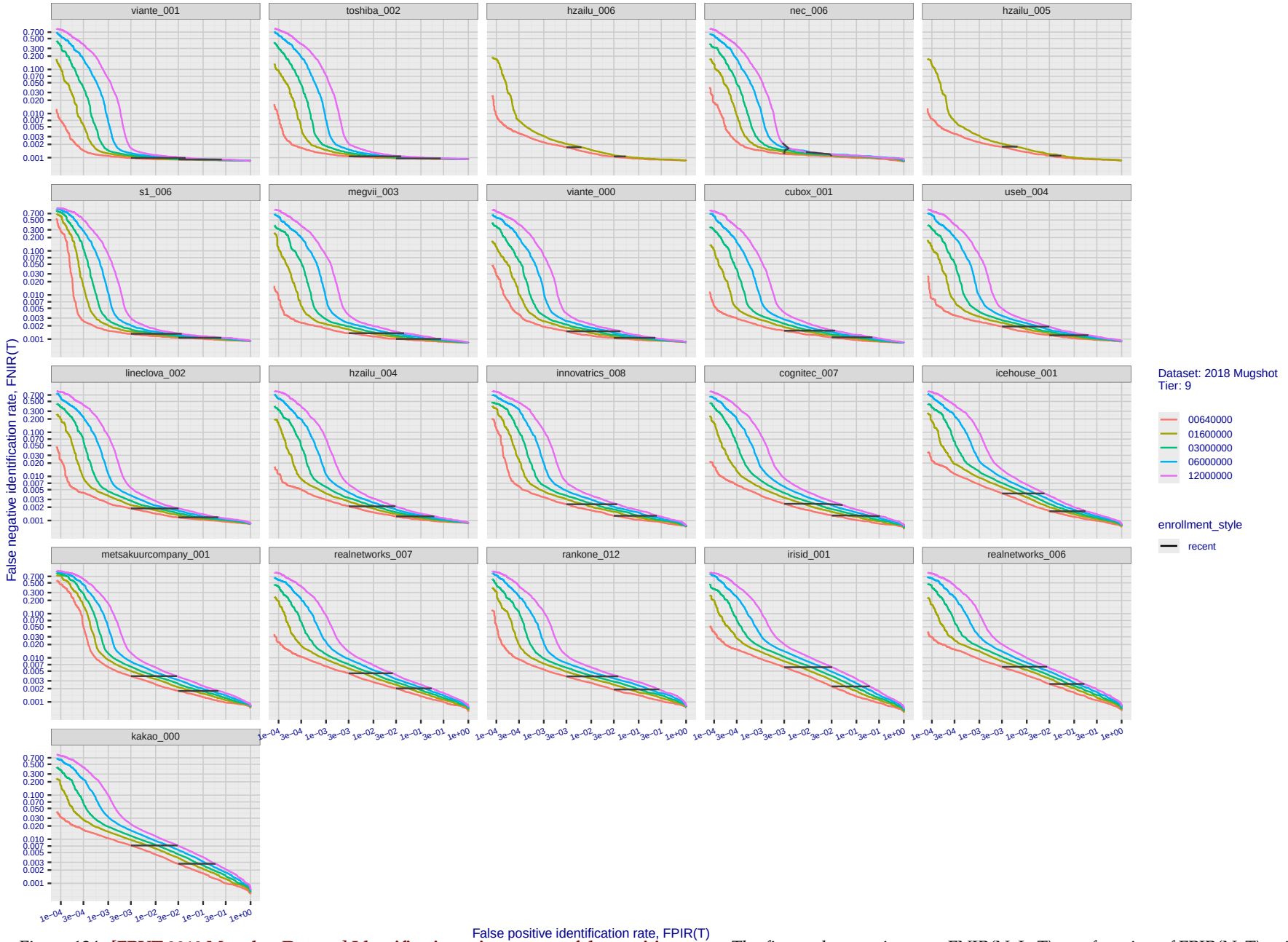


*Figure 124:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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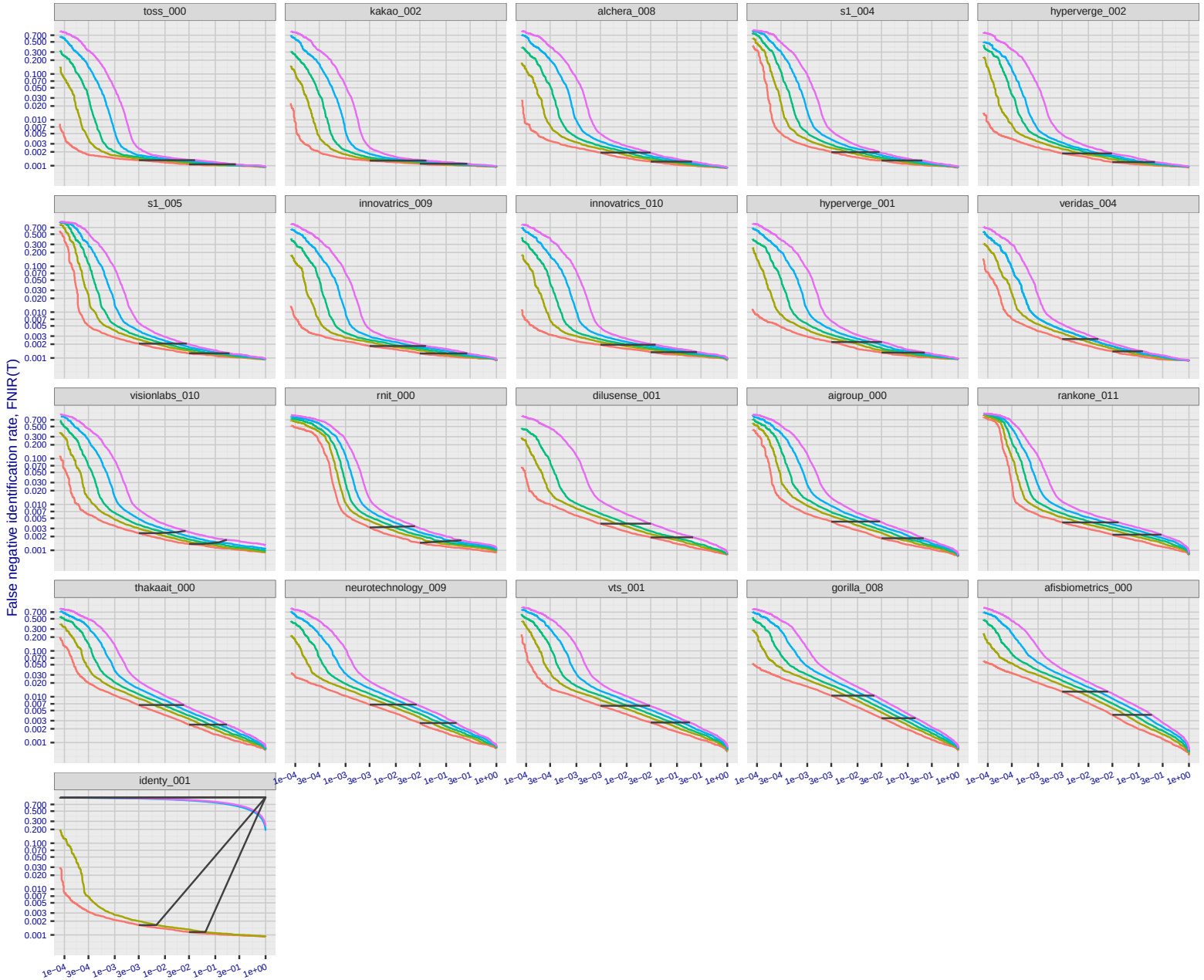


*Figure 125:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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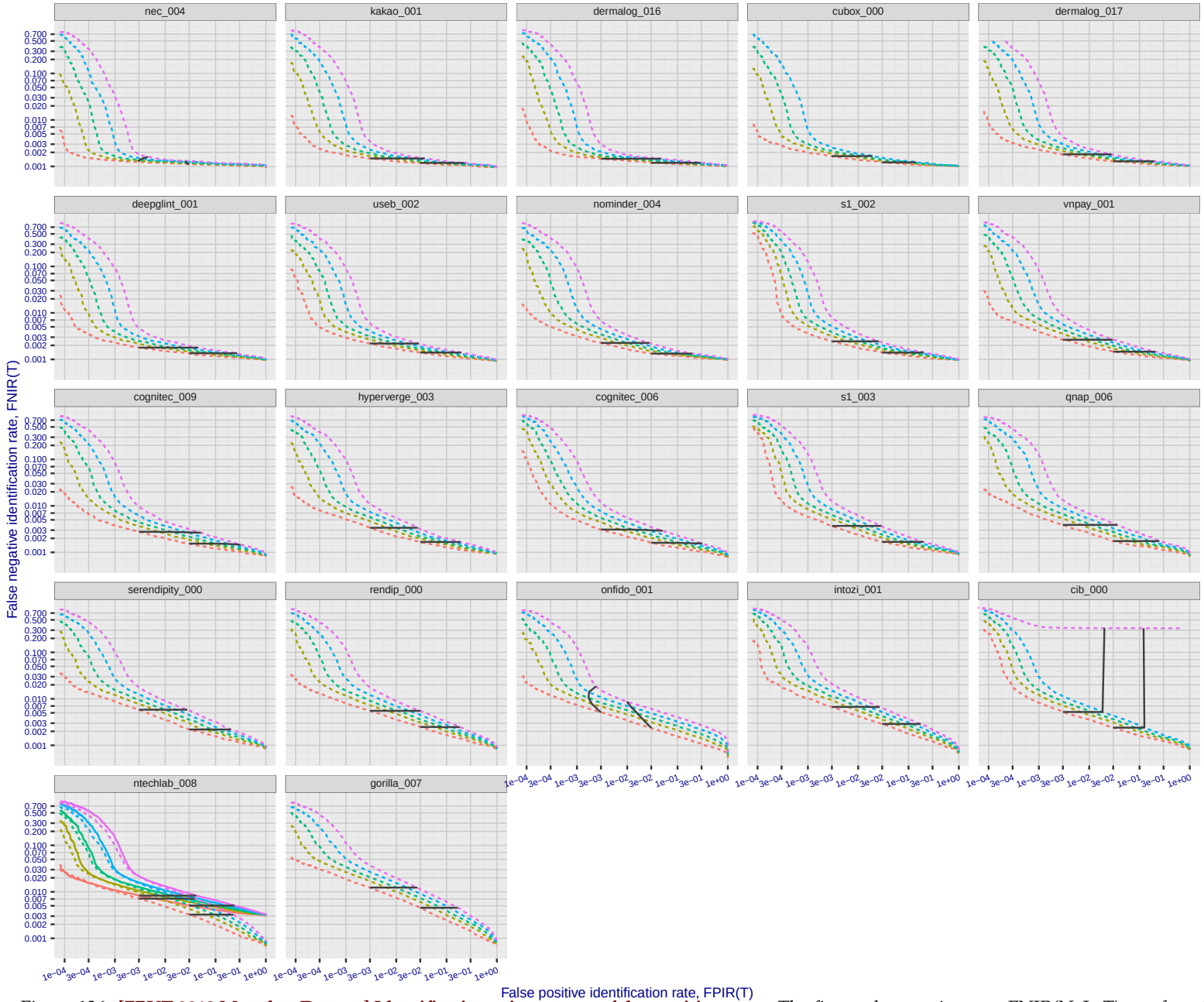


*Figure 126:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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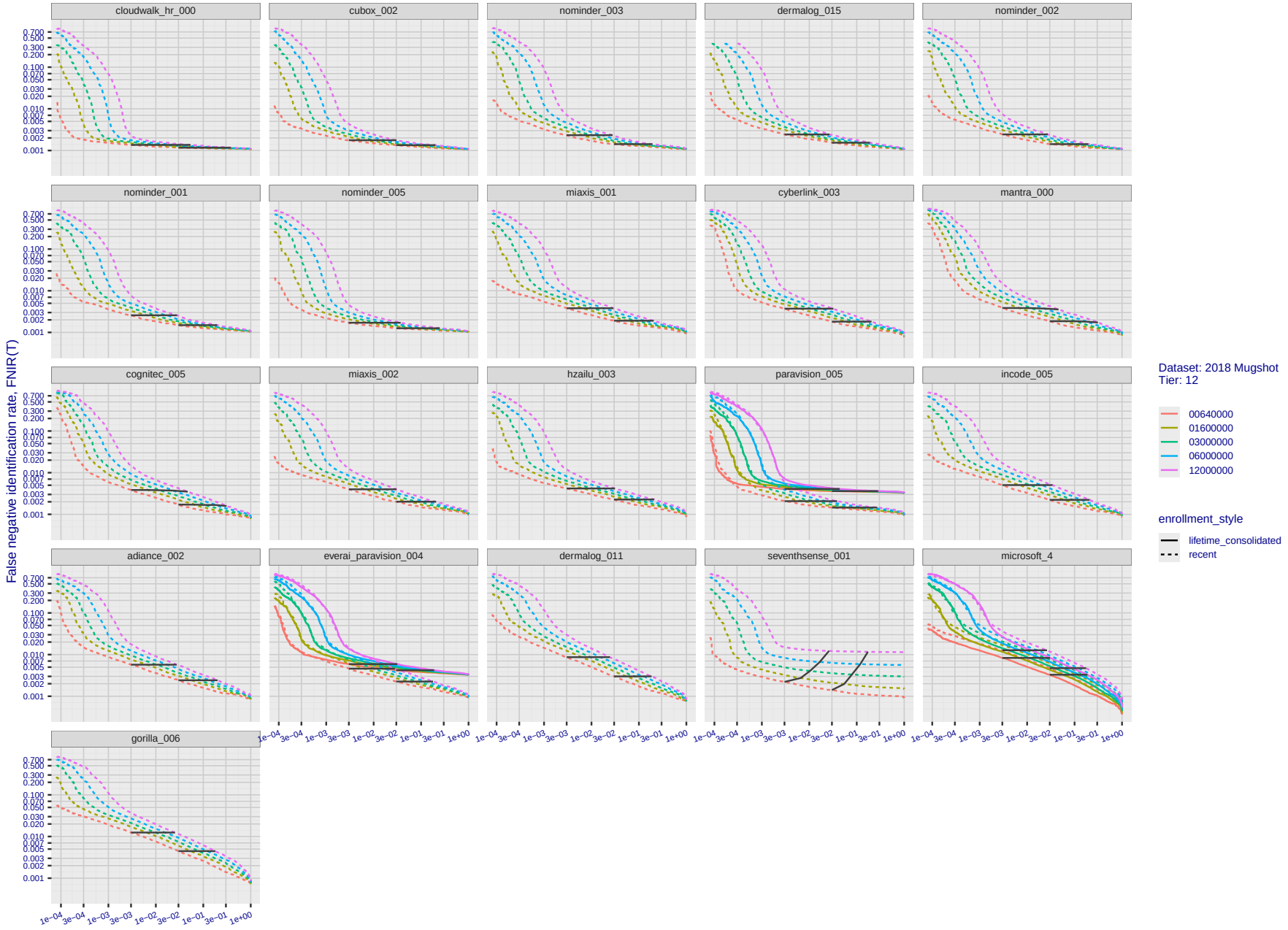


*Figure 127:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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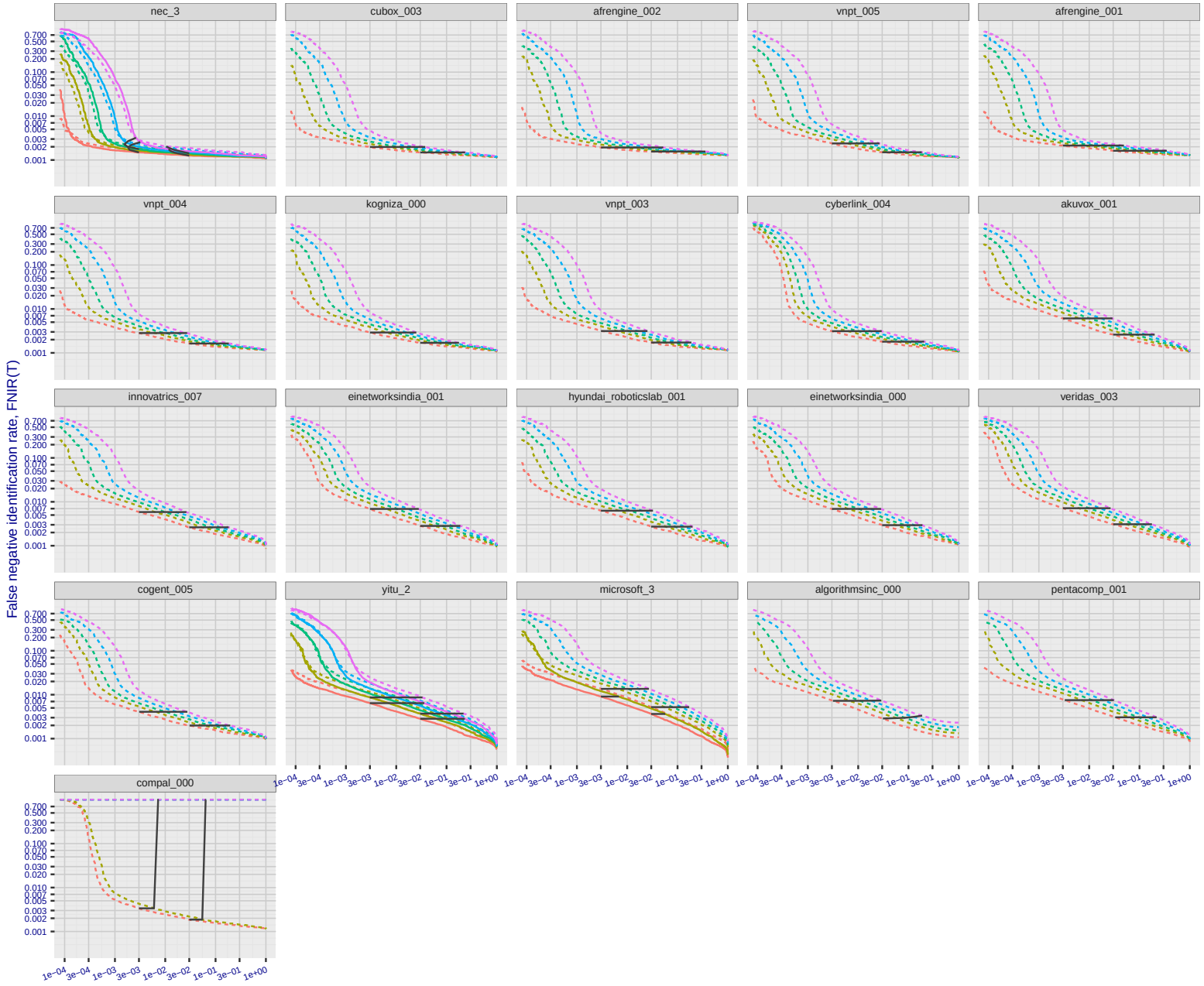


*Figure 128:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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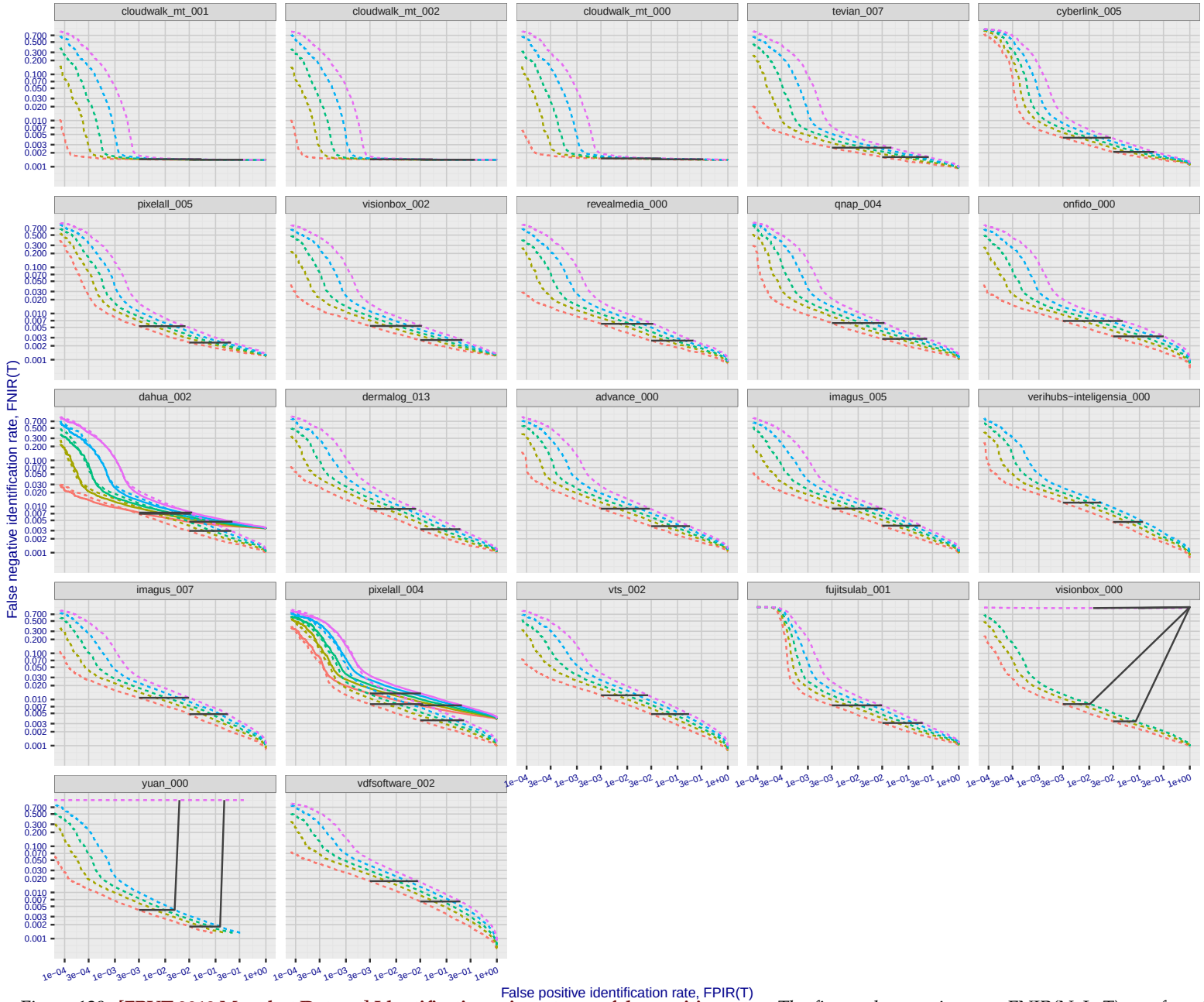


*Figure 129:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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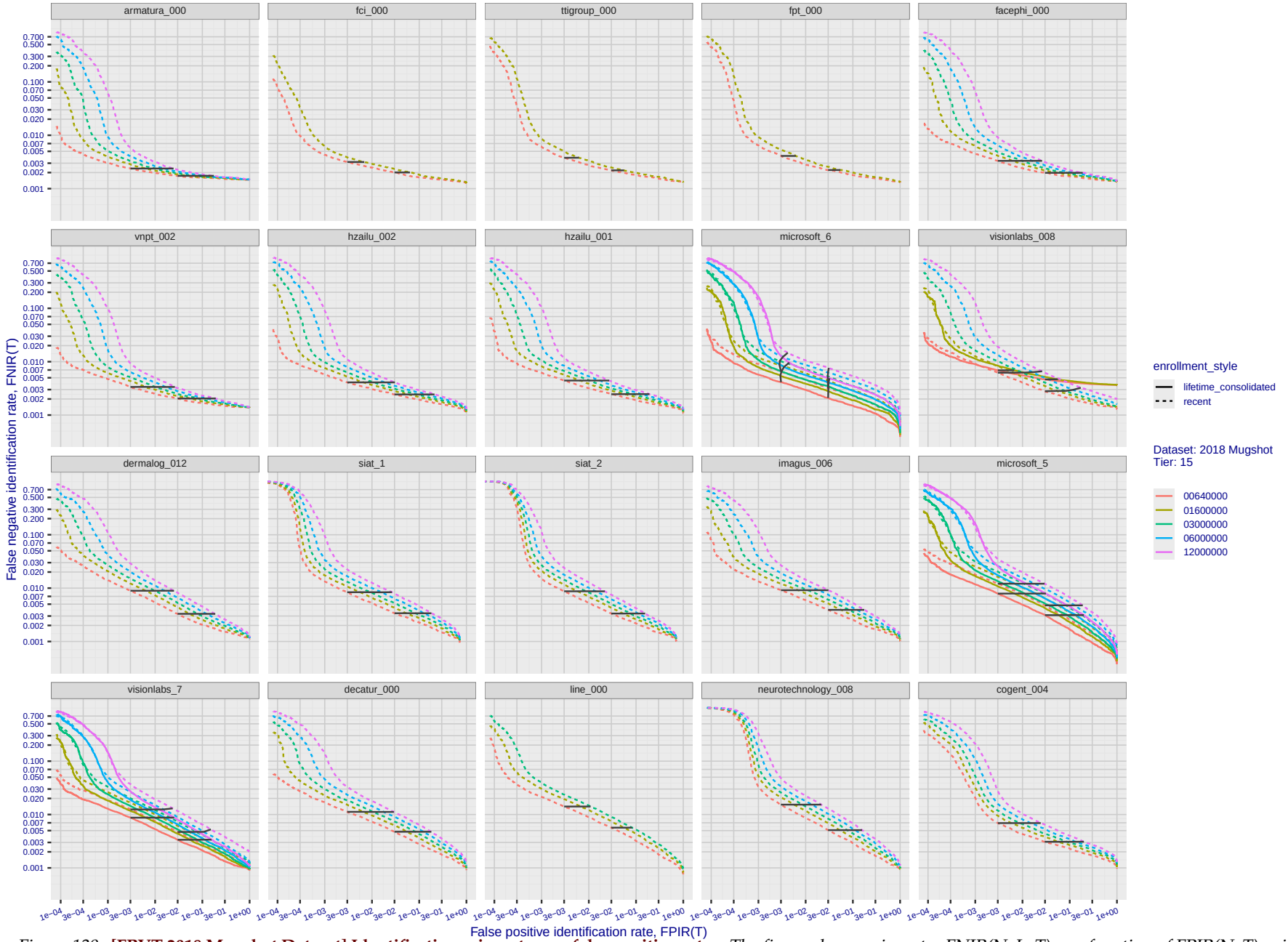


*Figure 130:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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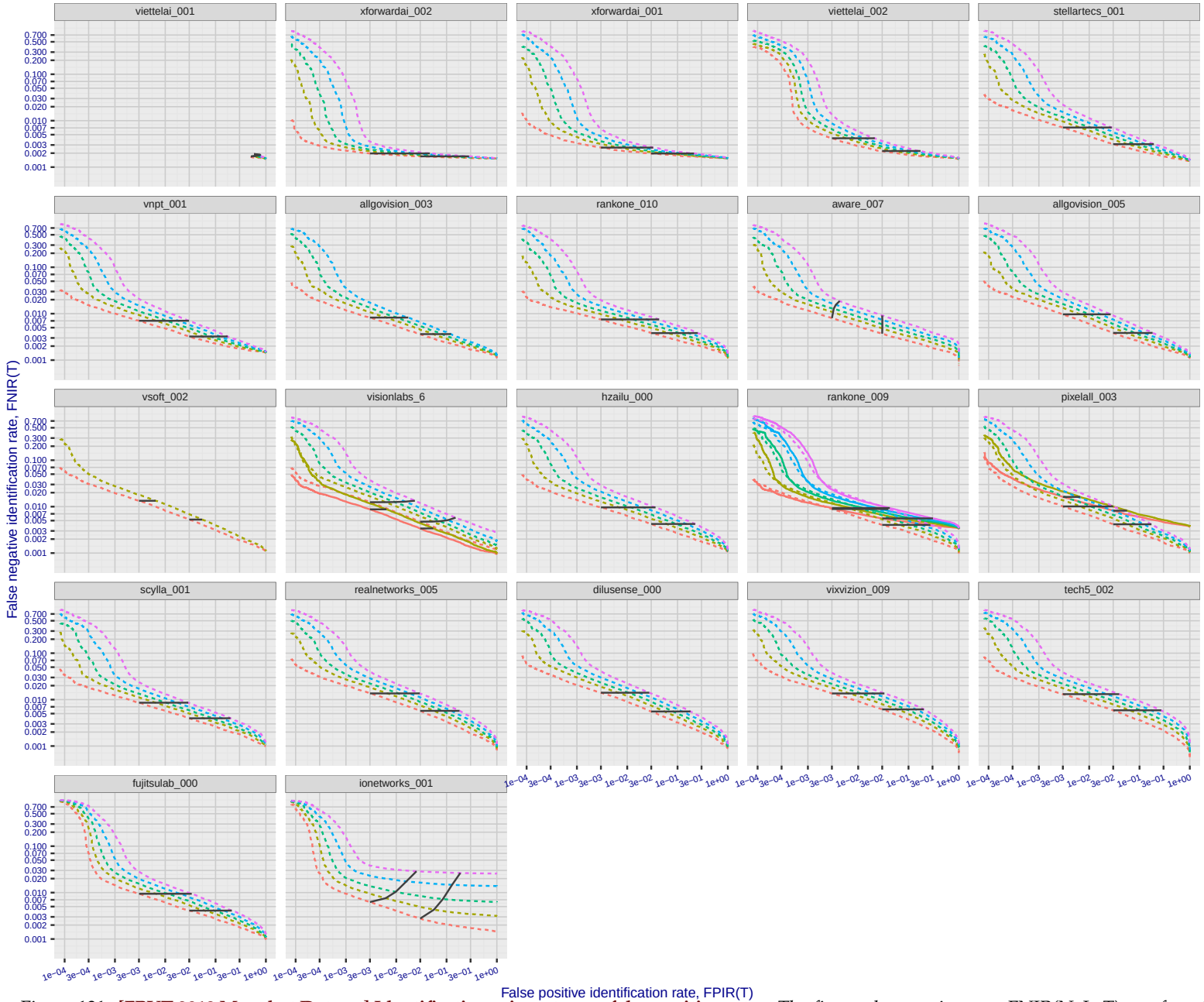


*Figure 131:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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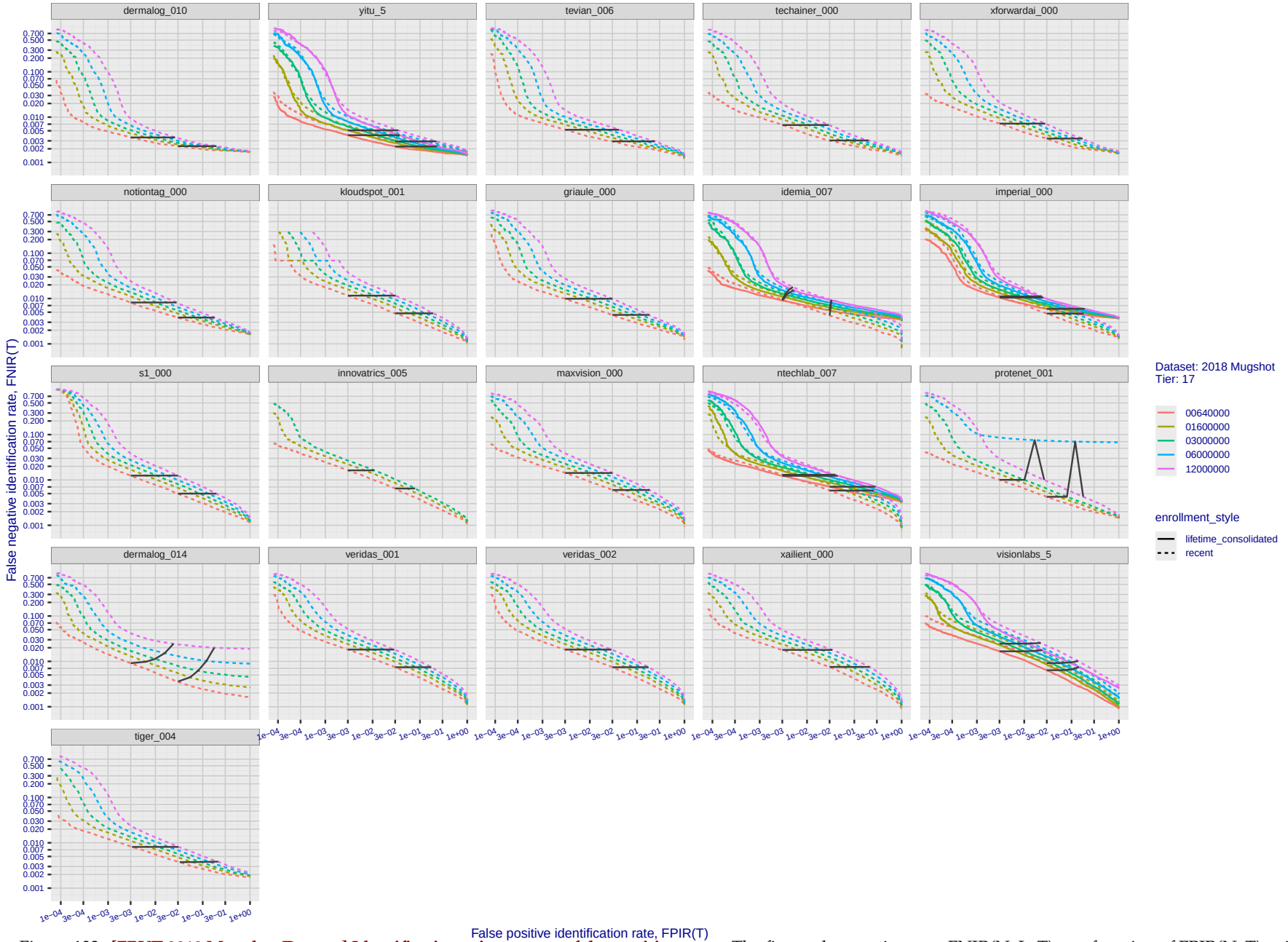


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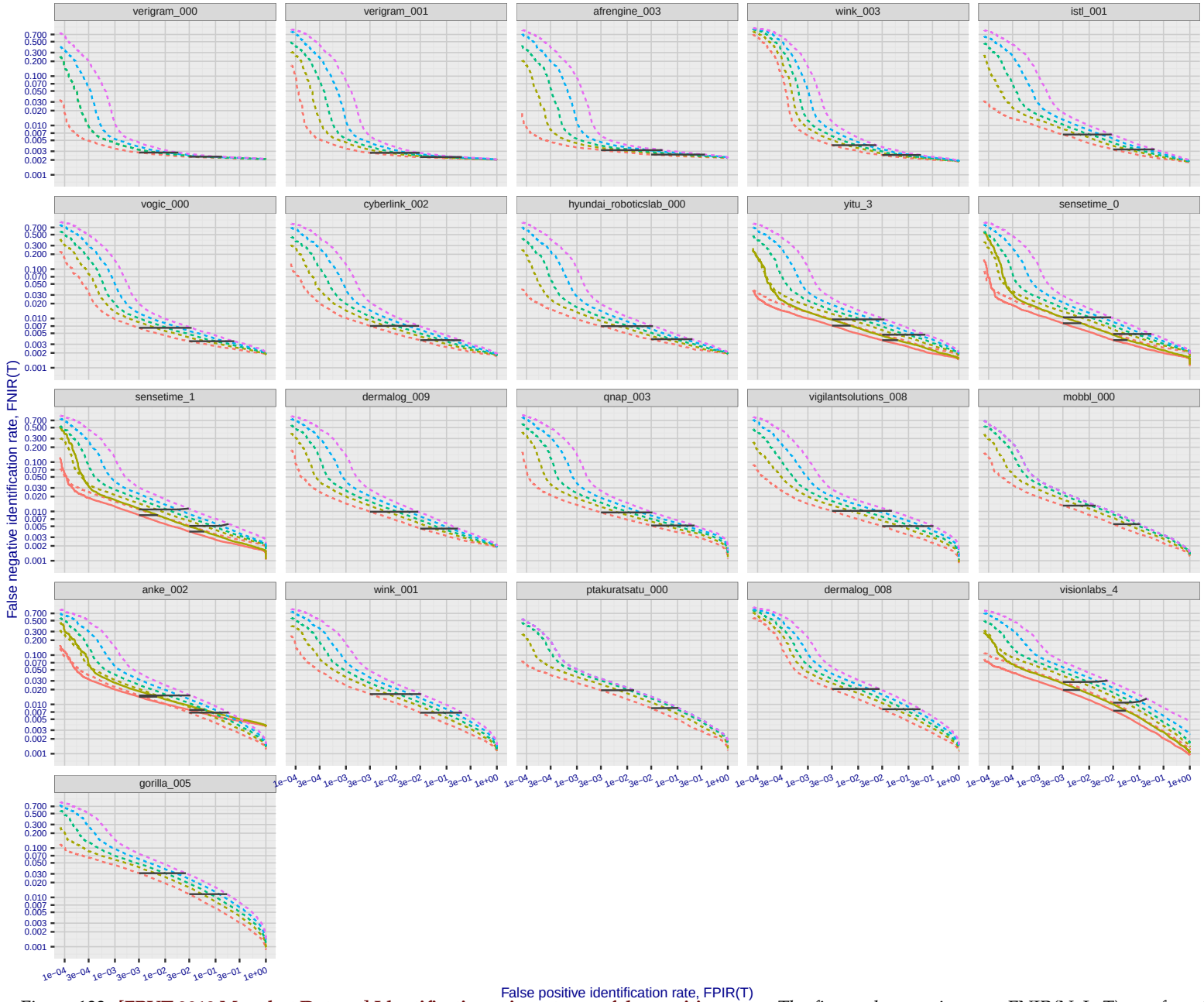


*Figure 133:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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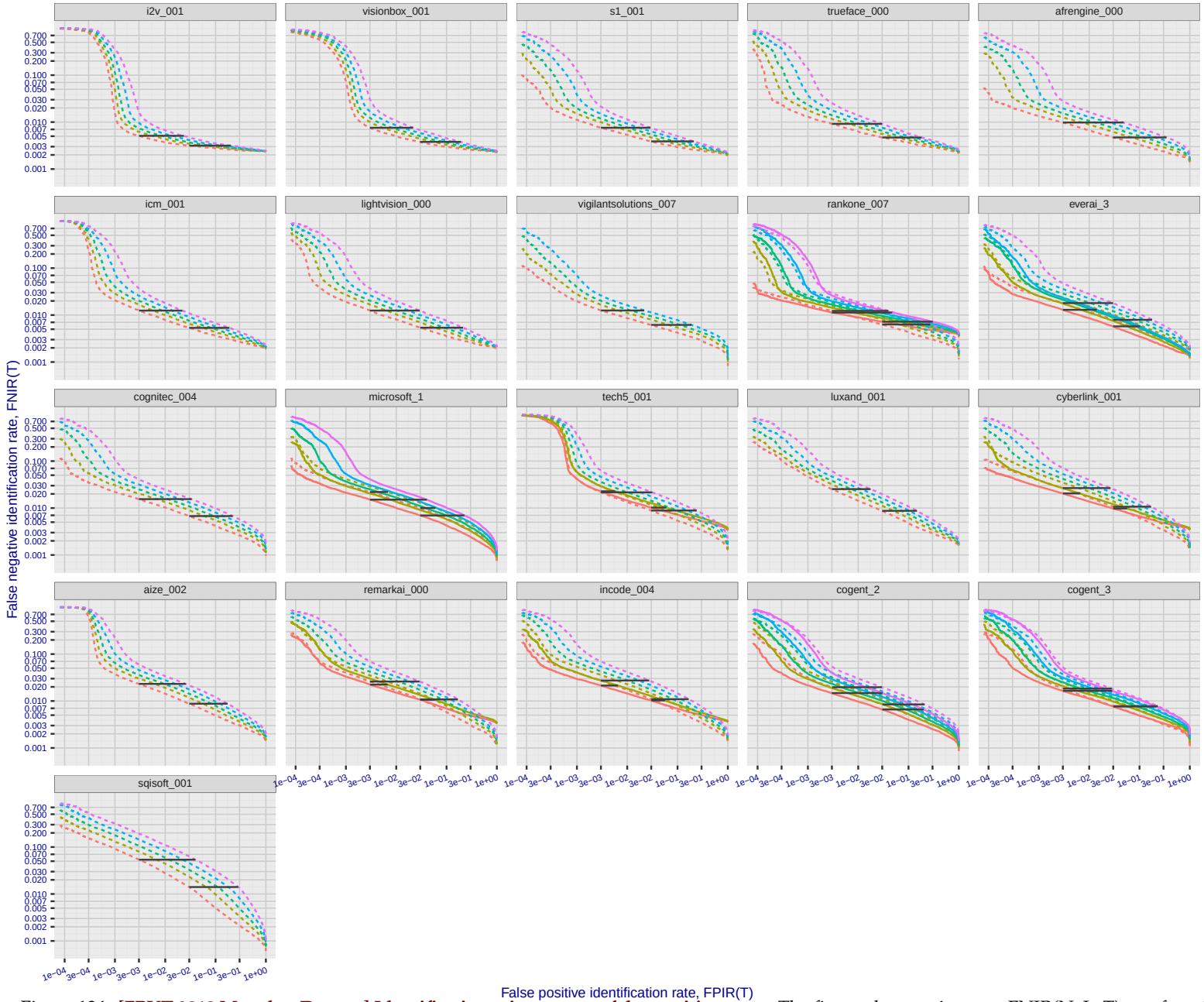


*Figure 134:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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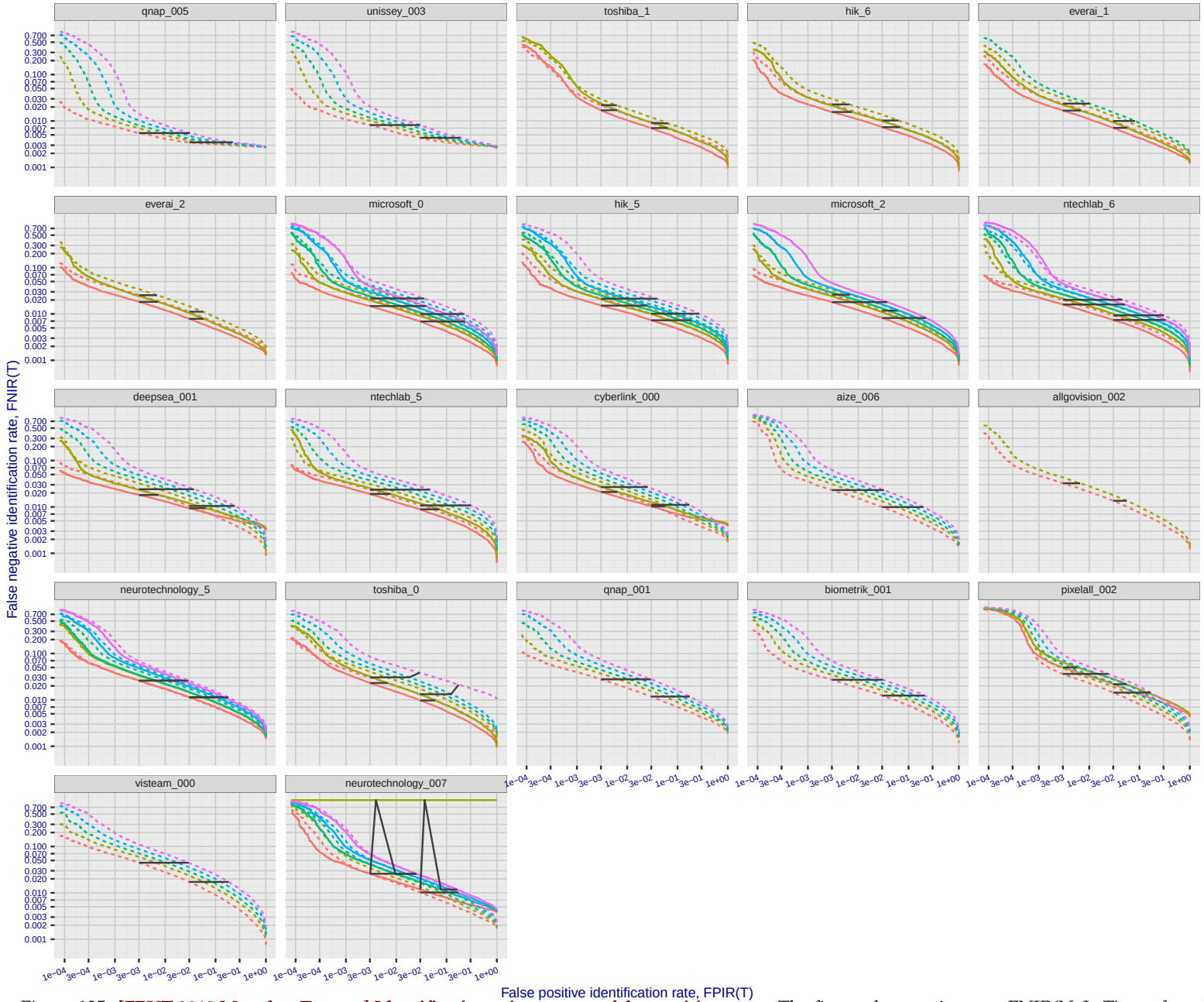


*Figure 135:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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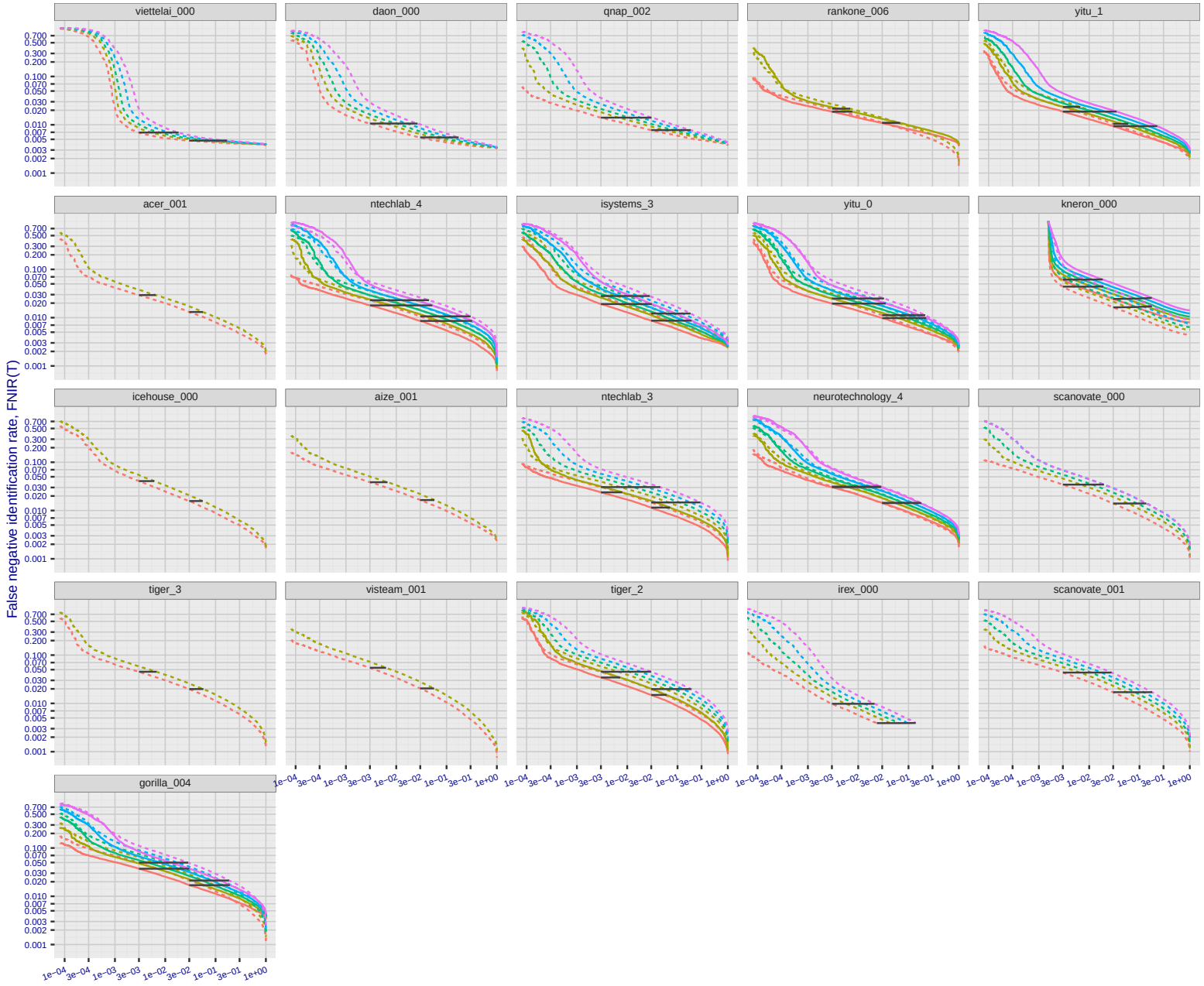


*Figure 136:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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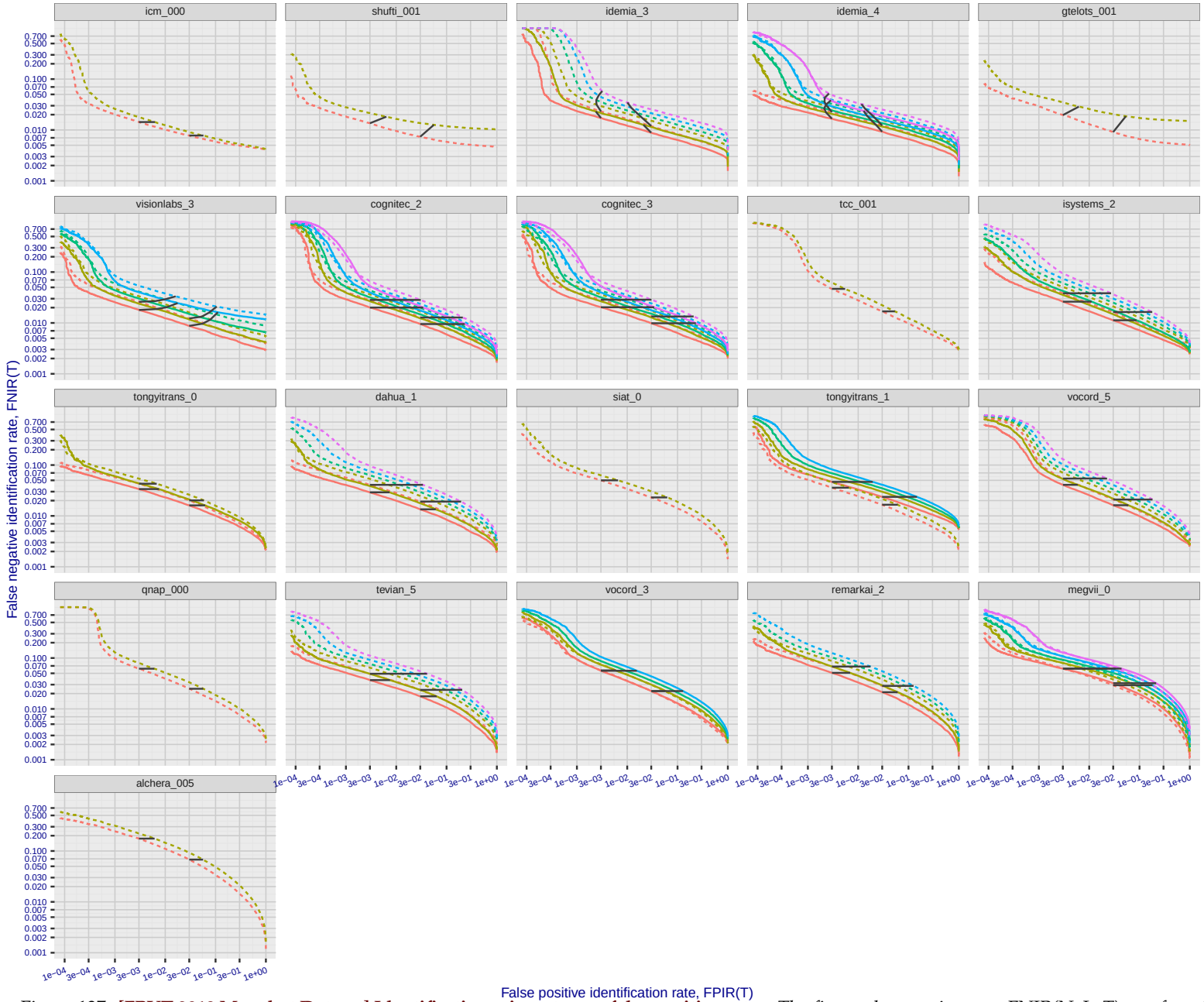


*Figure 137:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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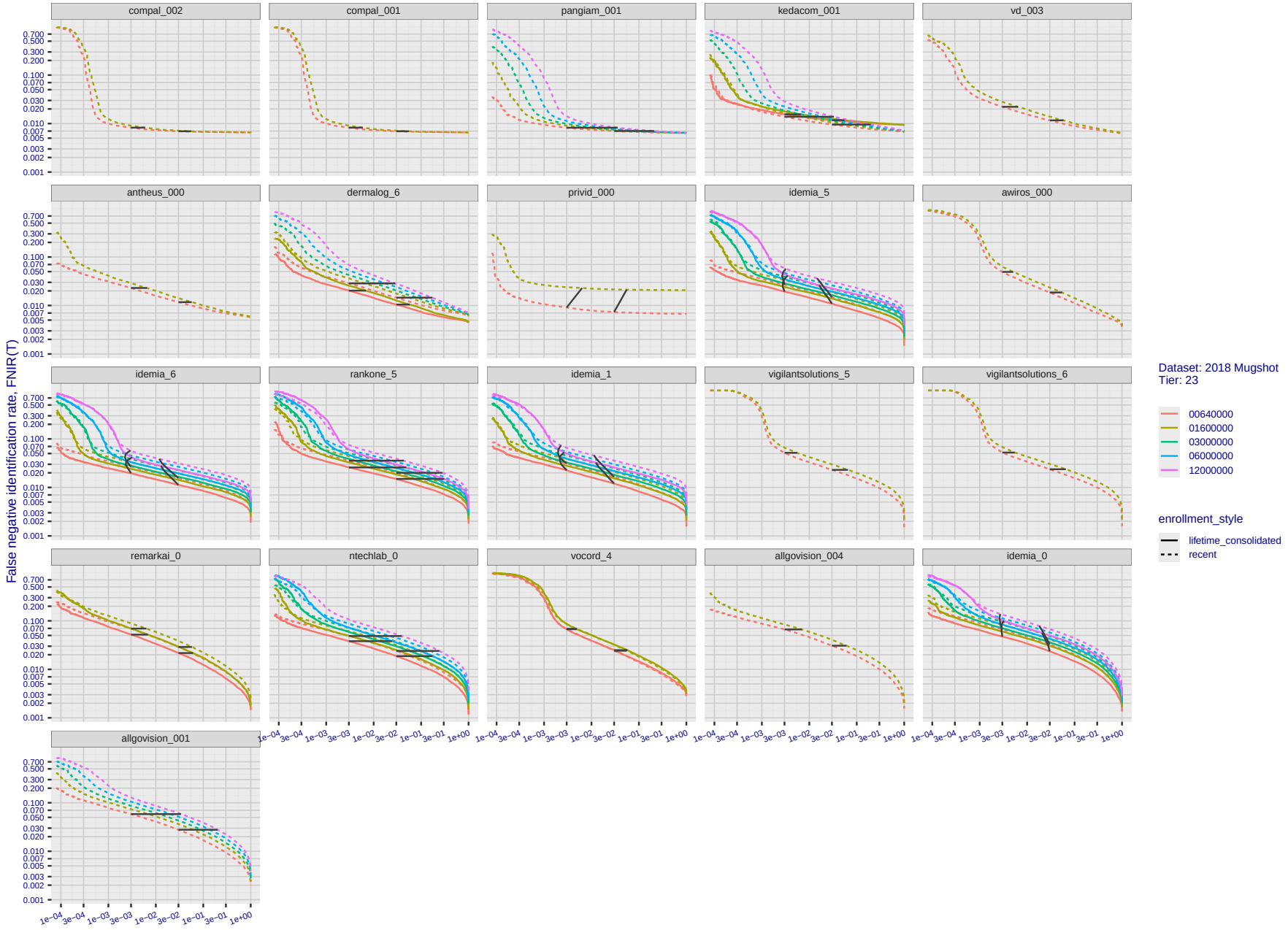


*Figure 138:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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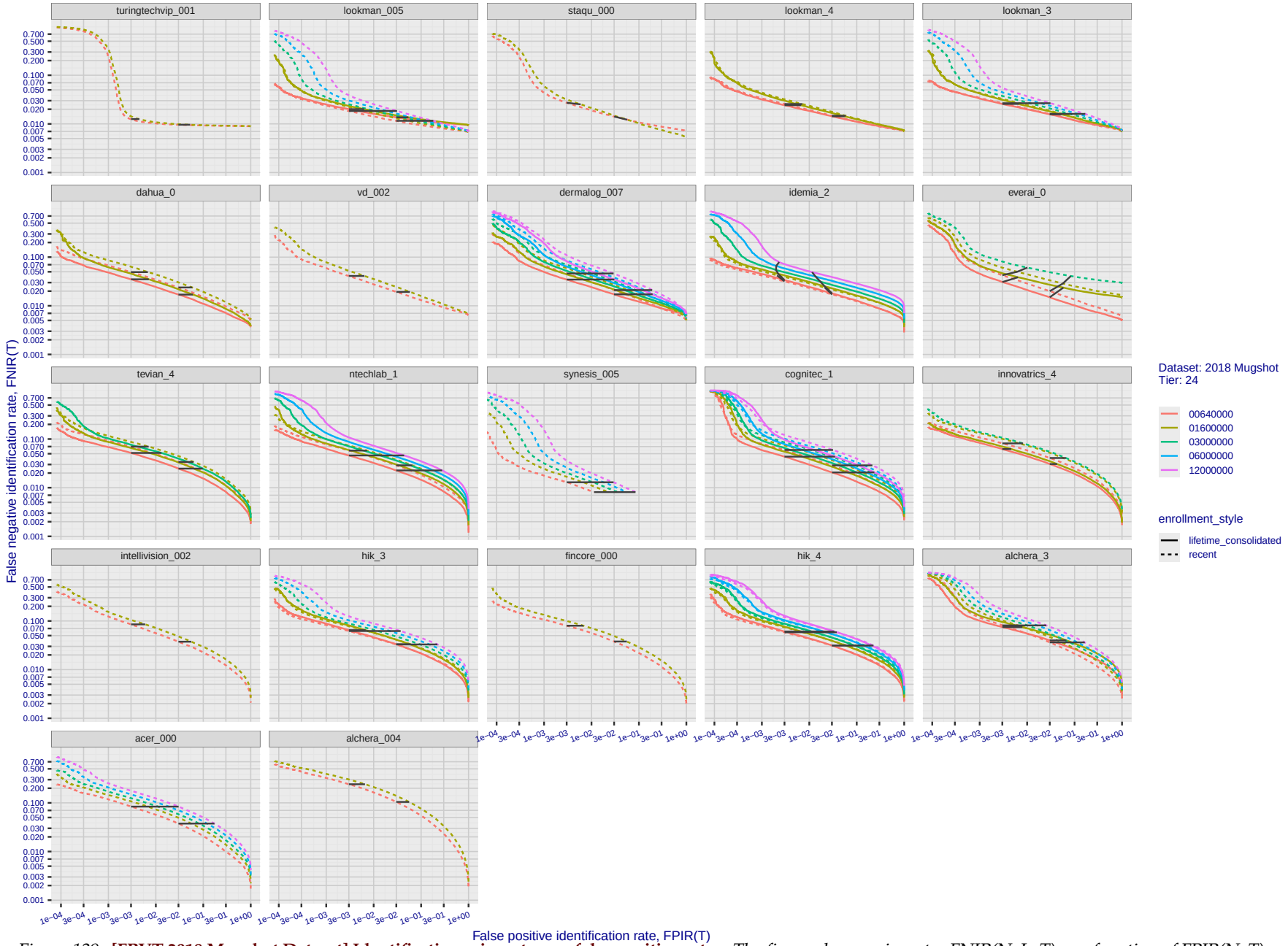


*Figure 139:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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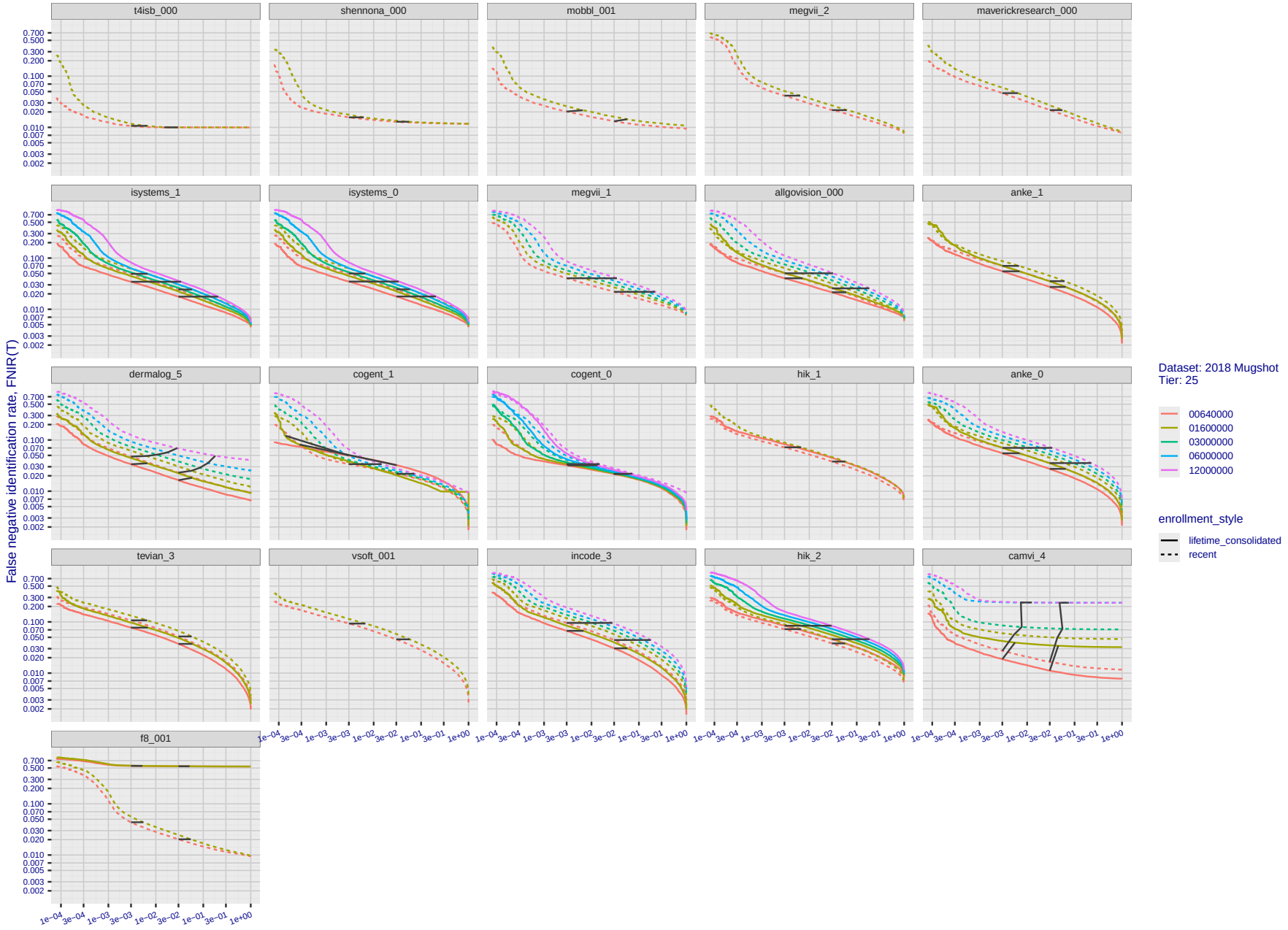


*Figure 140:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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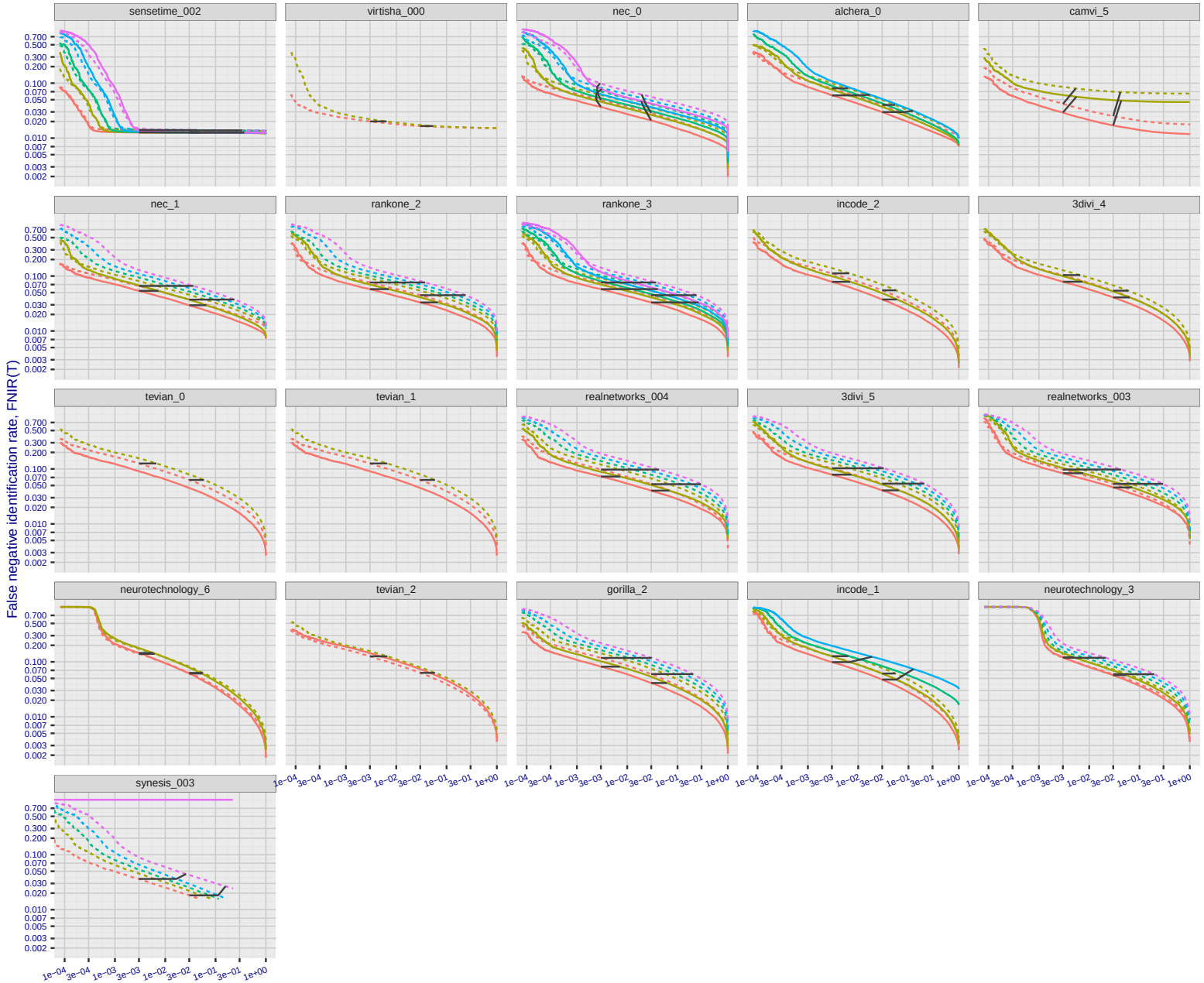


*Figure 141:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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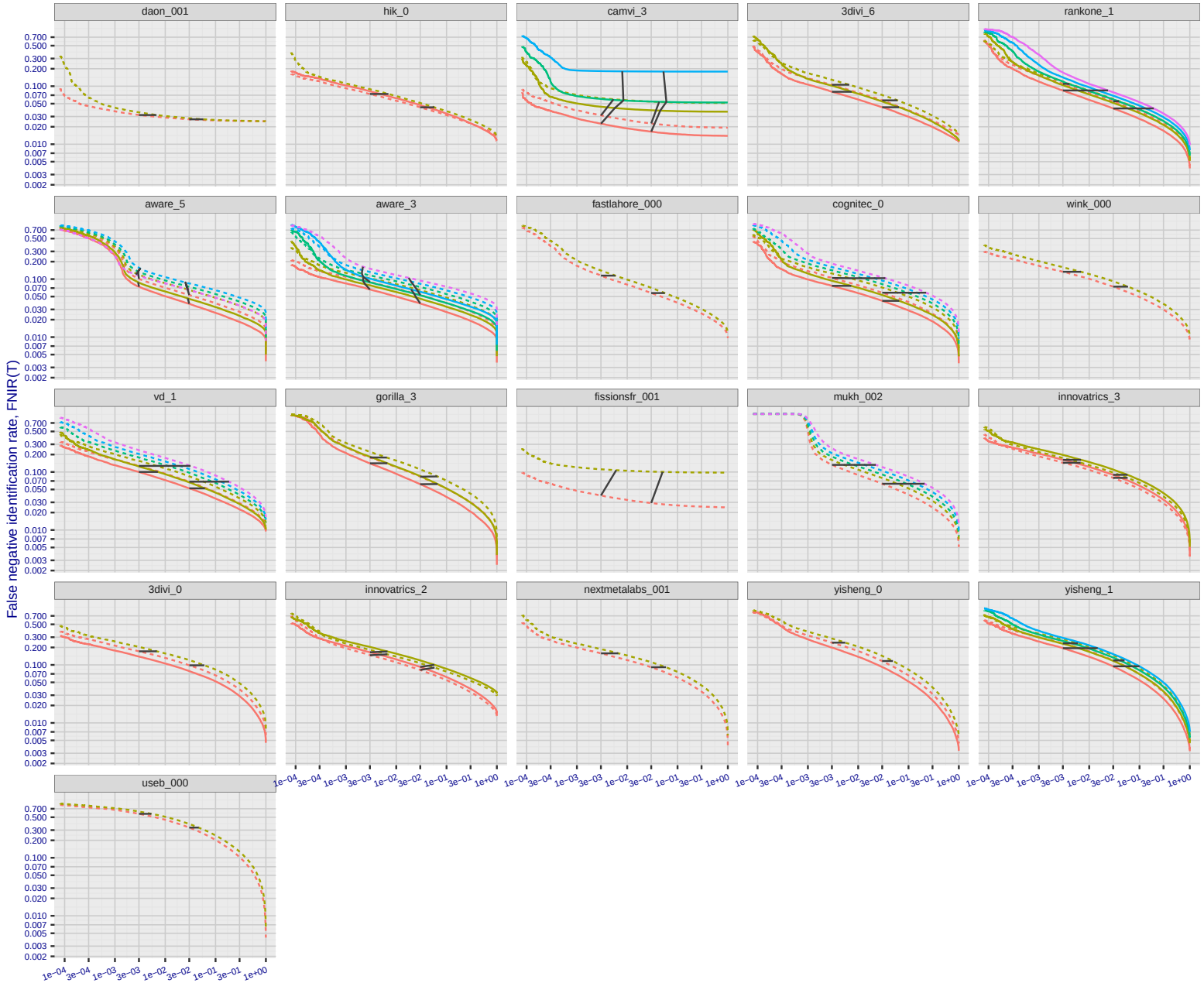


*Figure 142:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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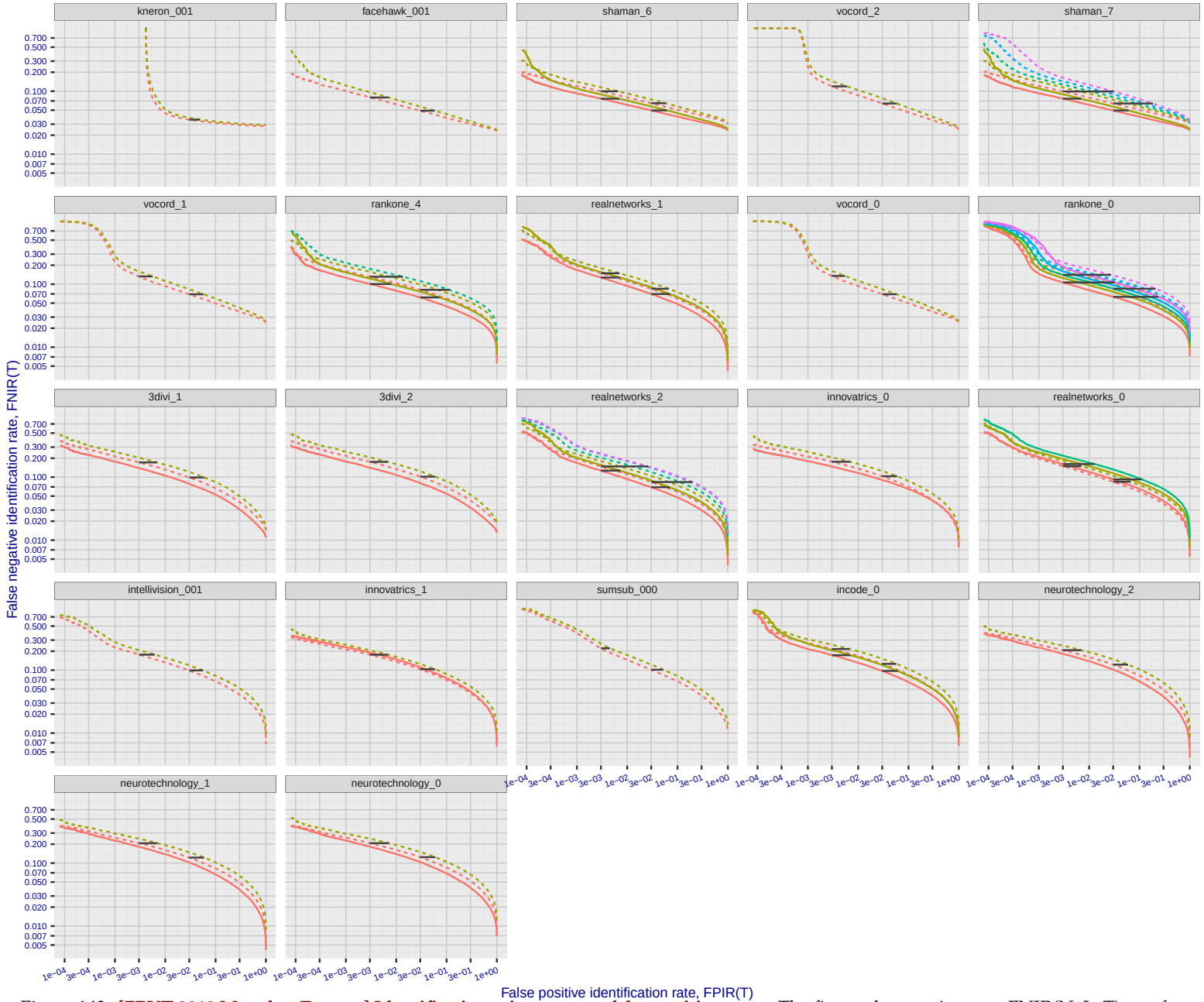


*Figure 143:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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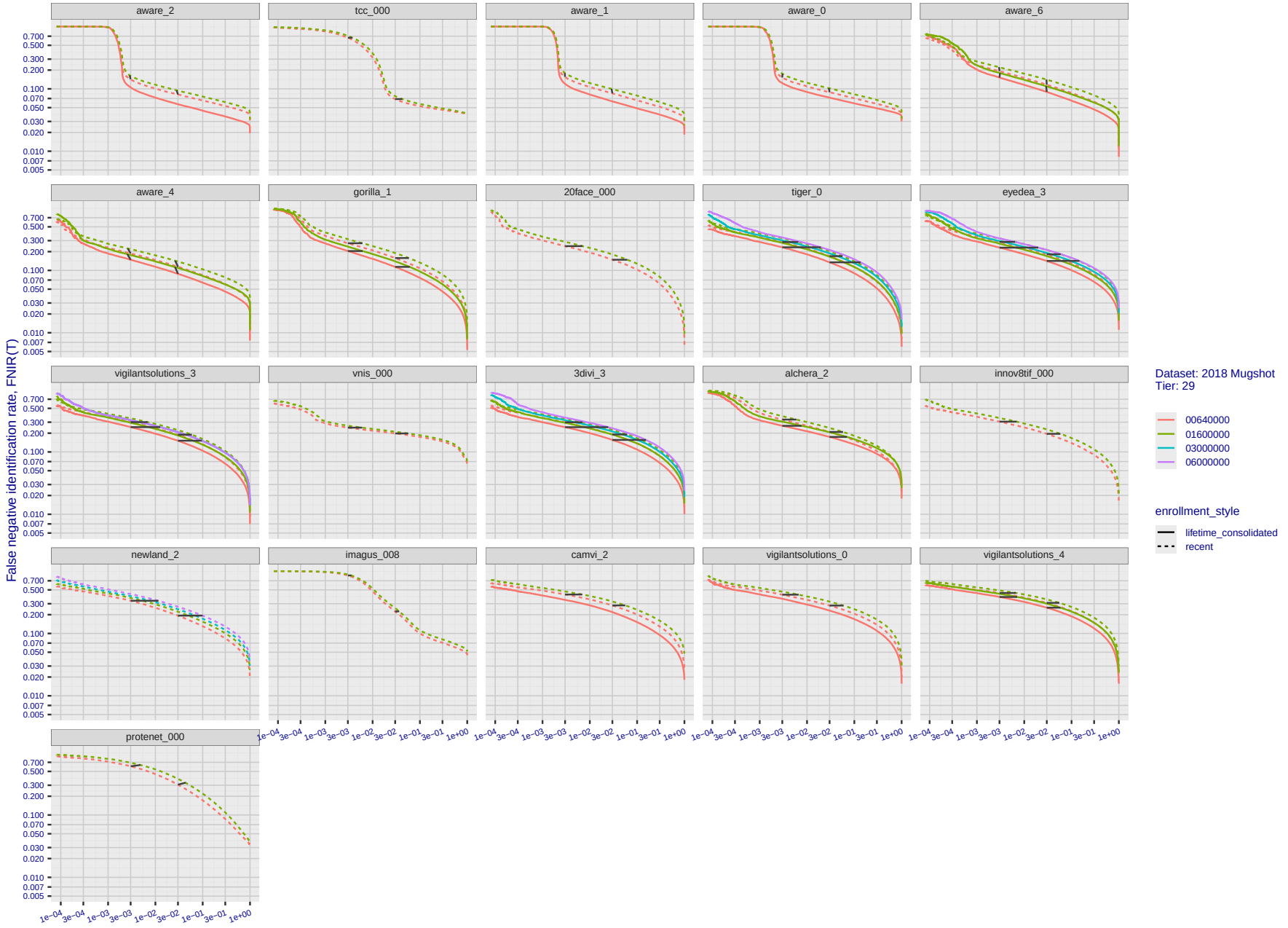


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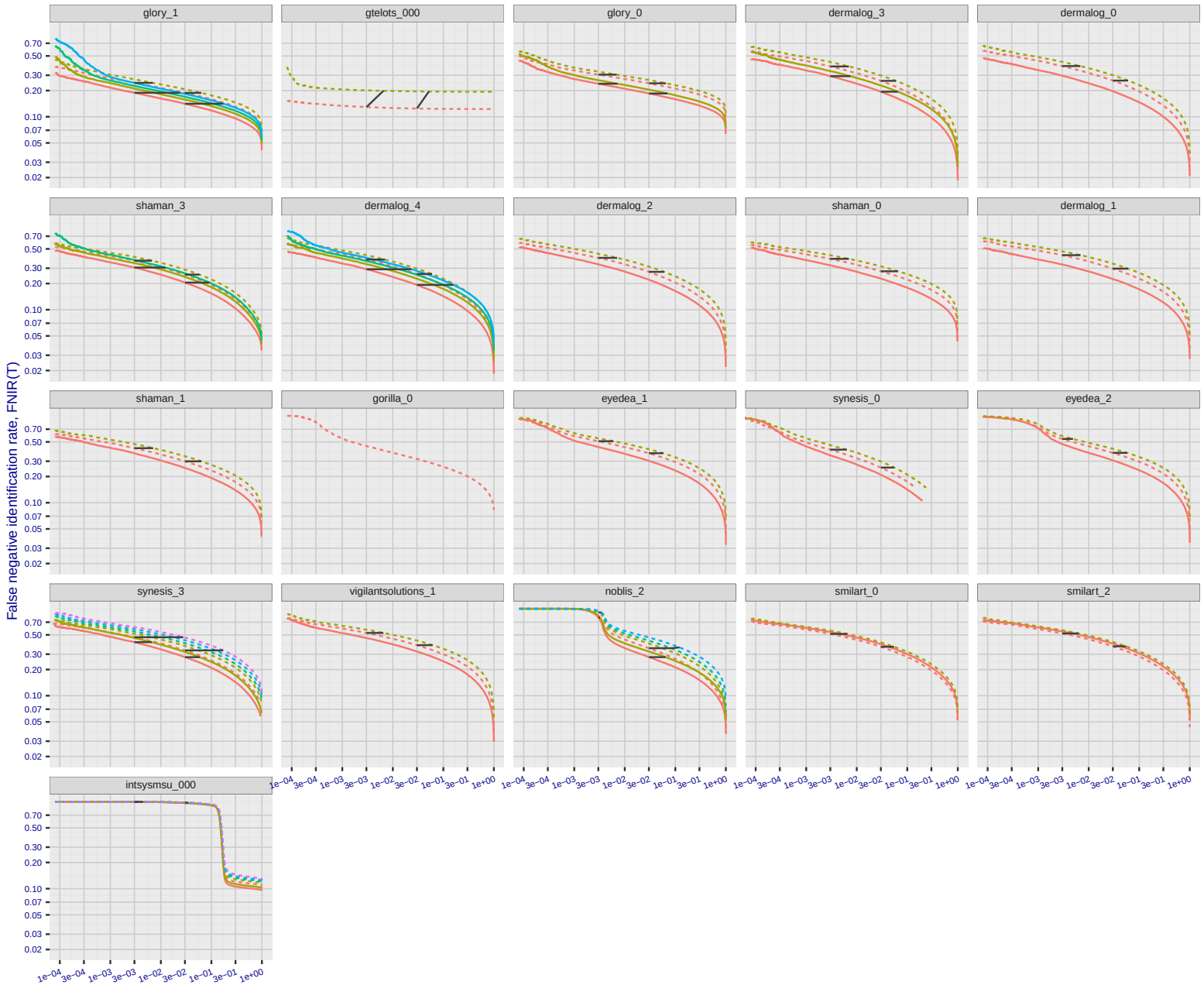


*Figure 145:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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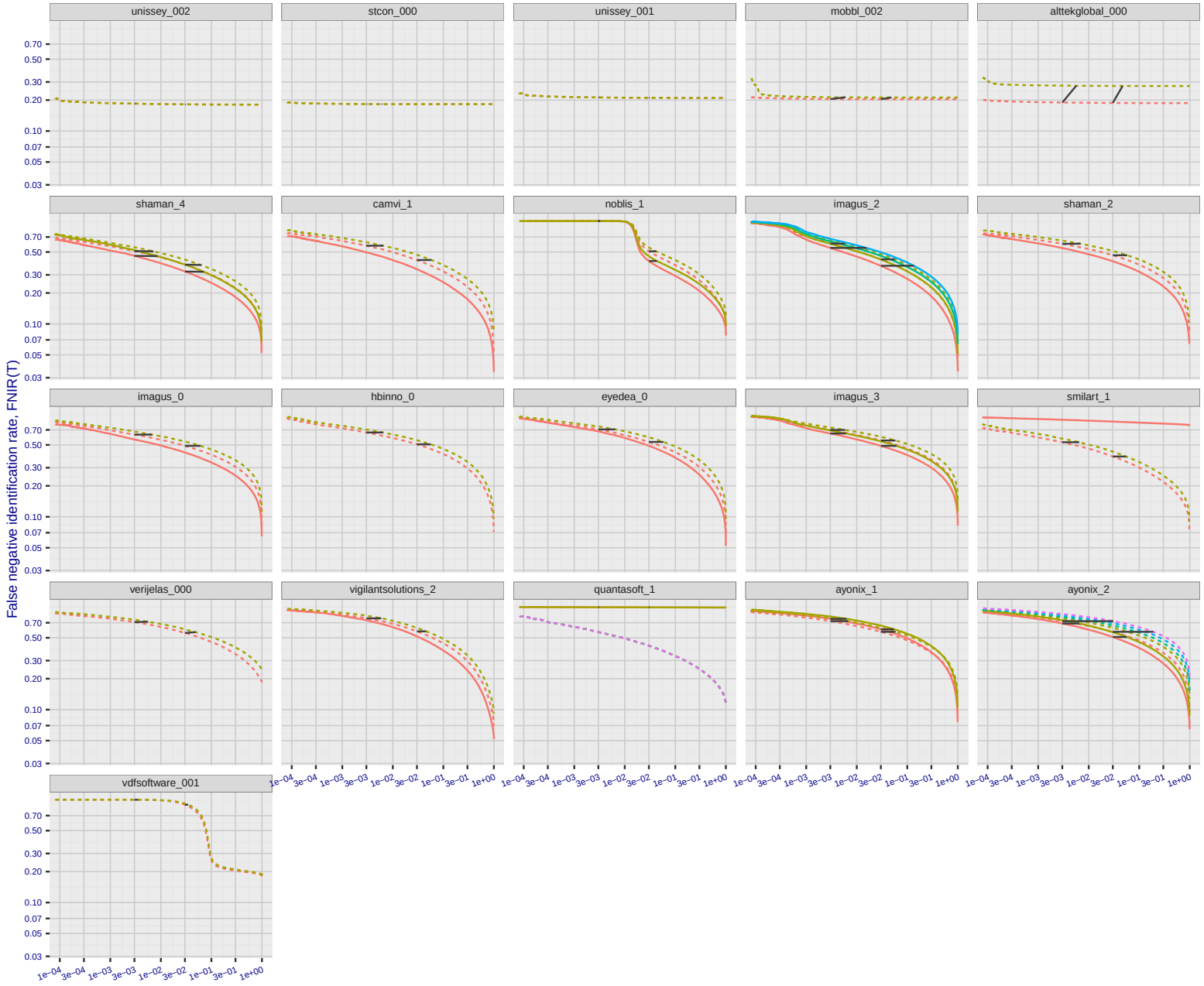


*Figure 146:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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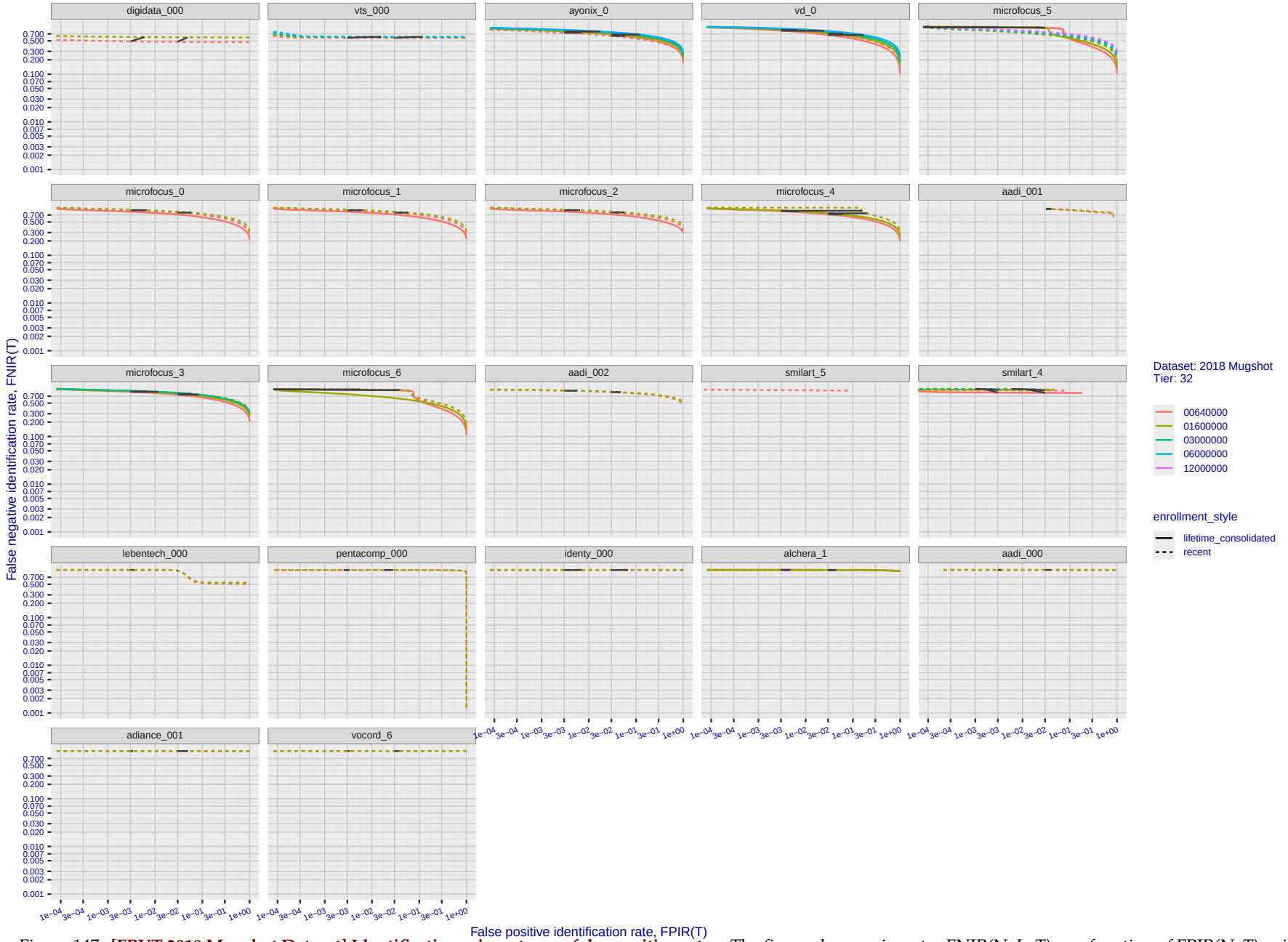


*Figure 147:* **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*



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# Appendix B Effect of time-lapse: Accuracy after face ageing

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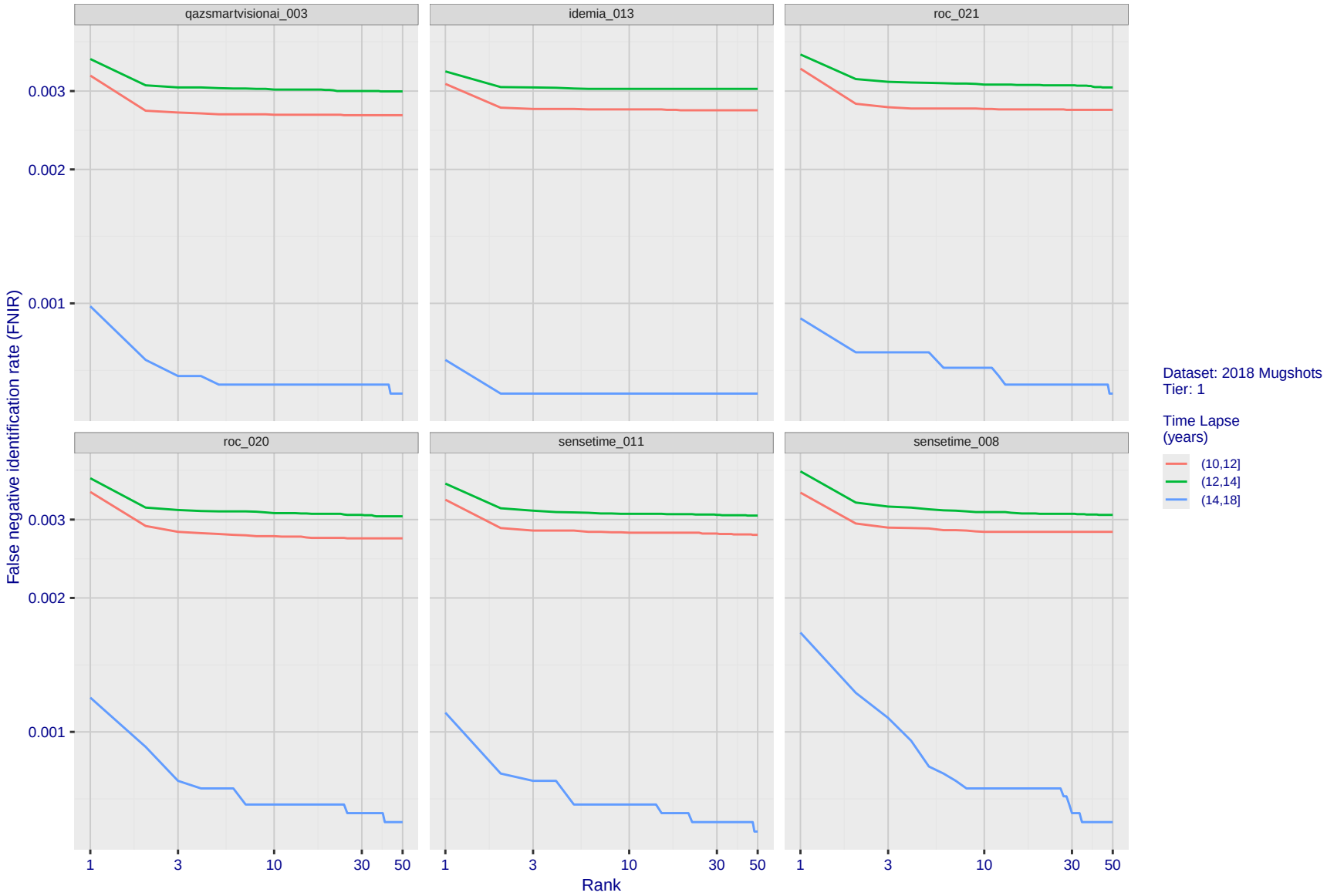


*Figure 148:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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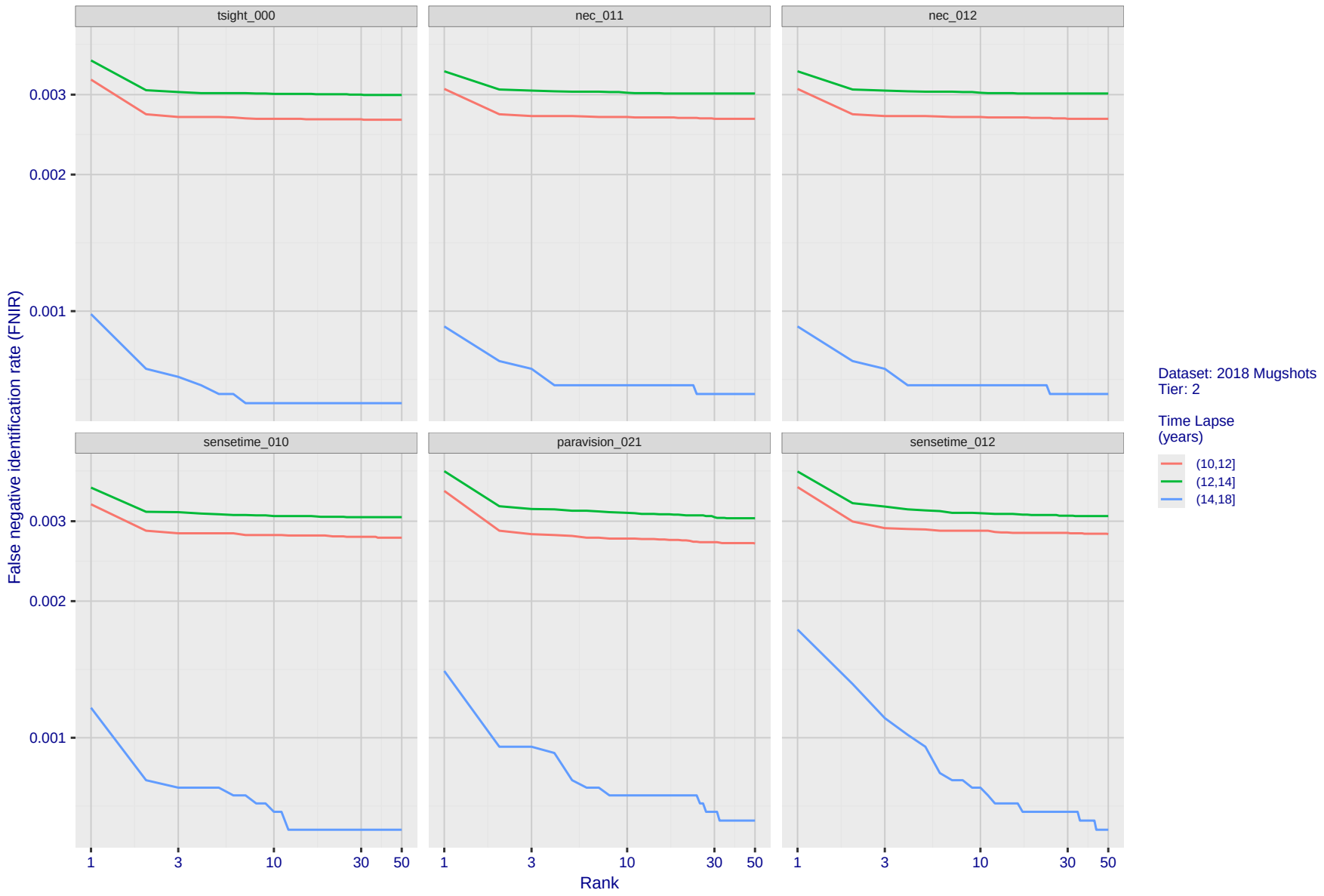


*Figure 149:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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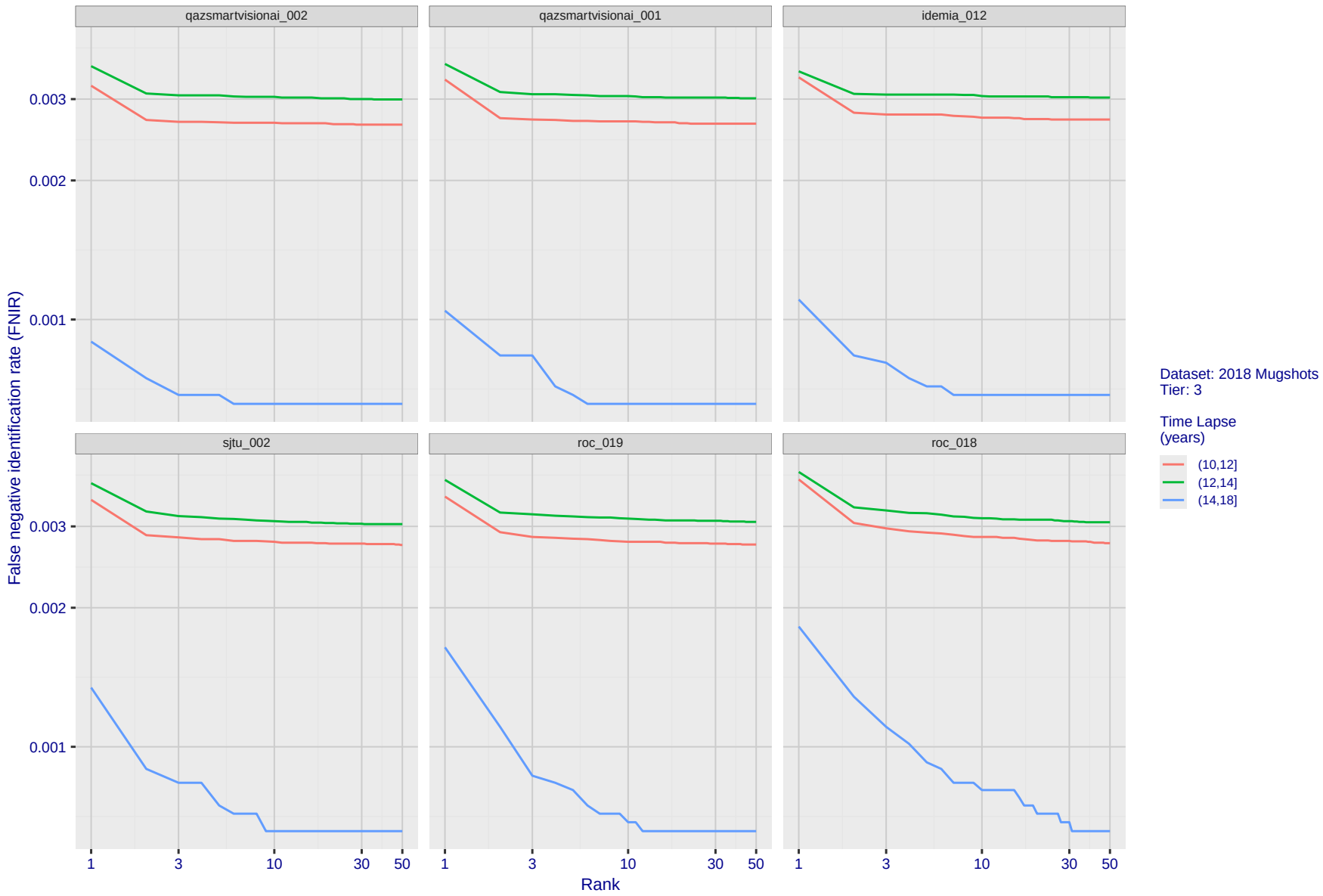


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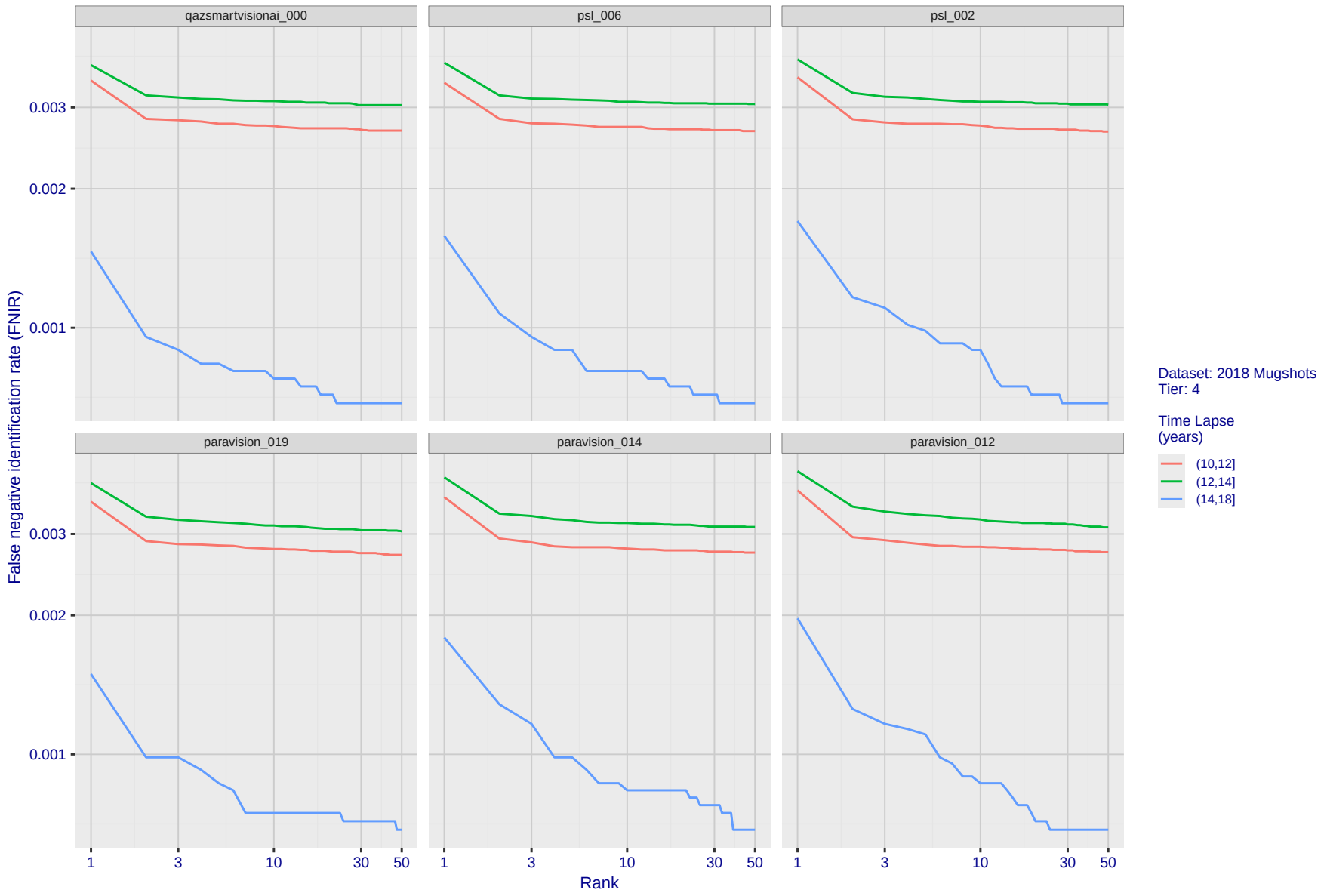


*Figure 151:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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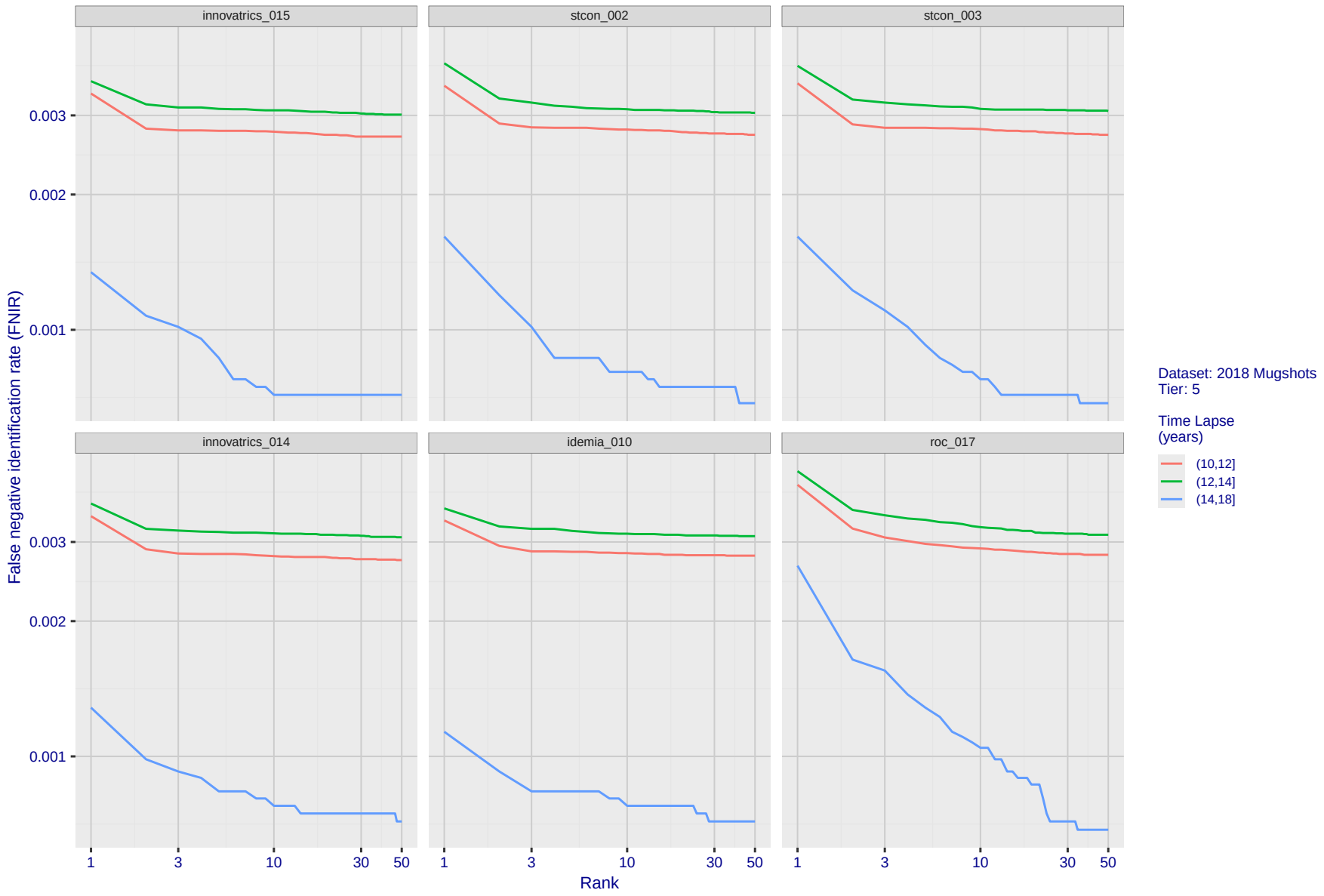


*Figure 152:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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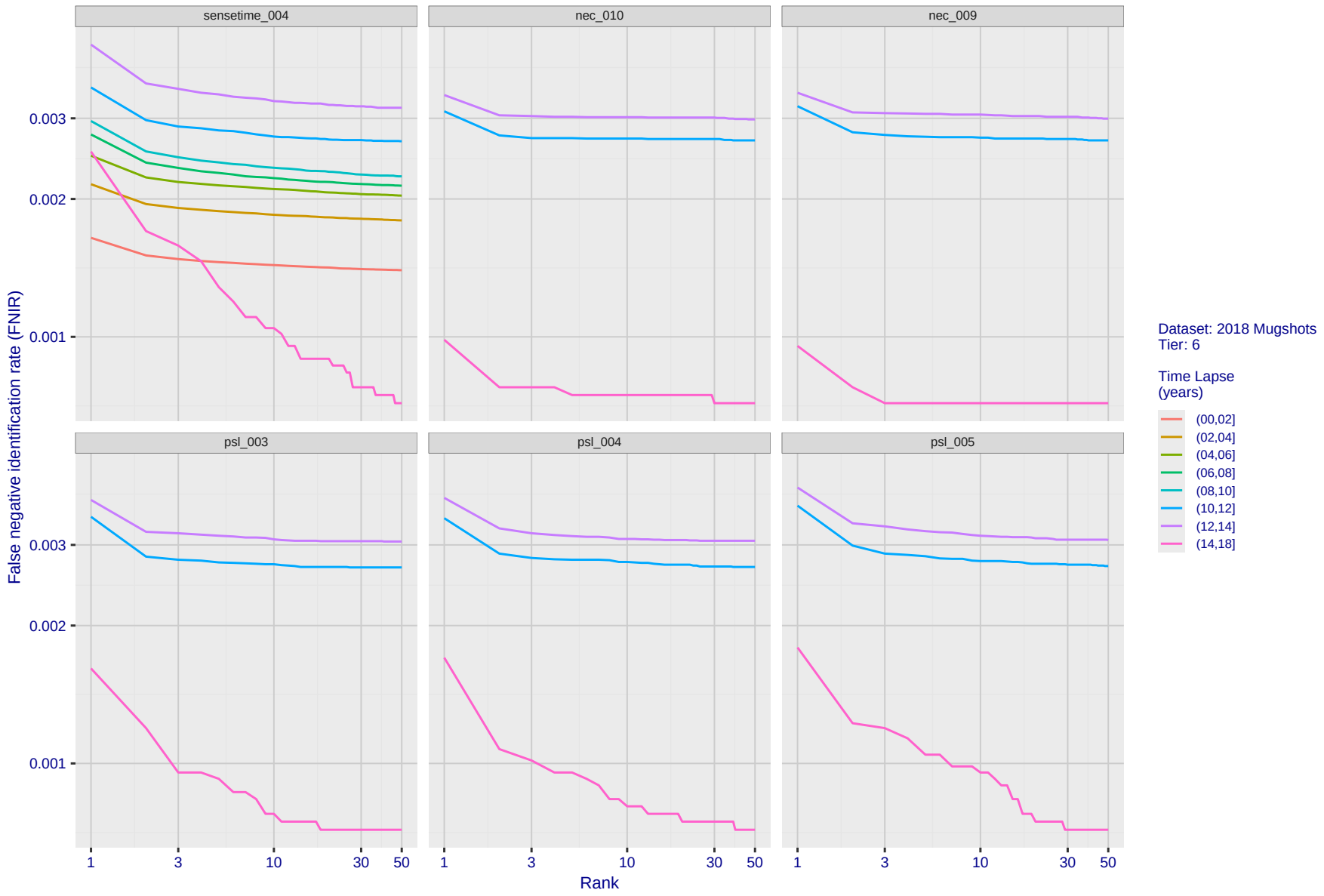


*Figure 153:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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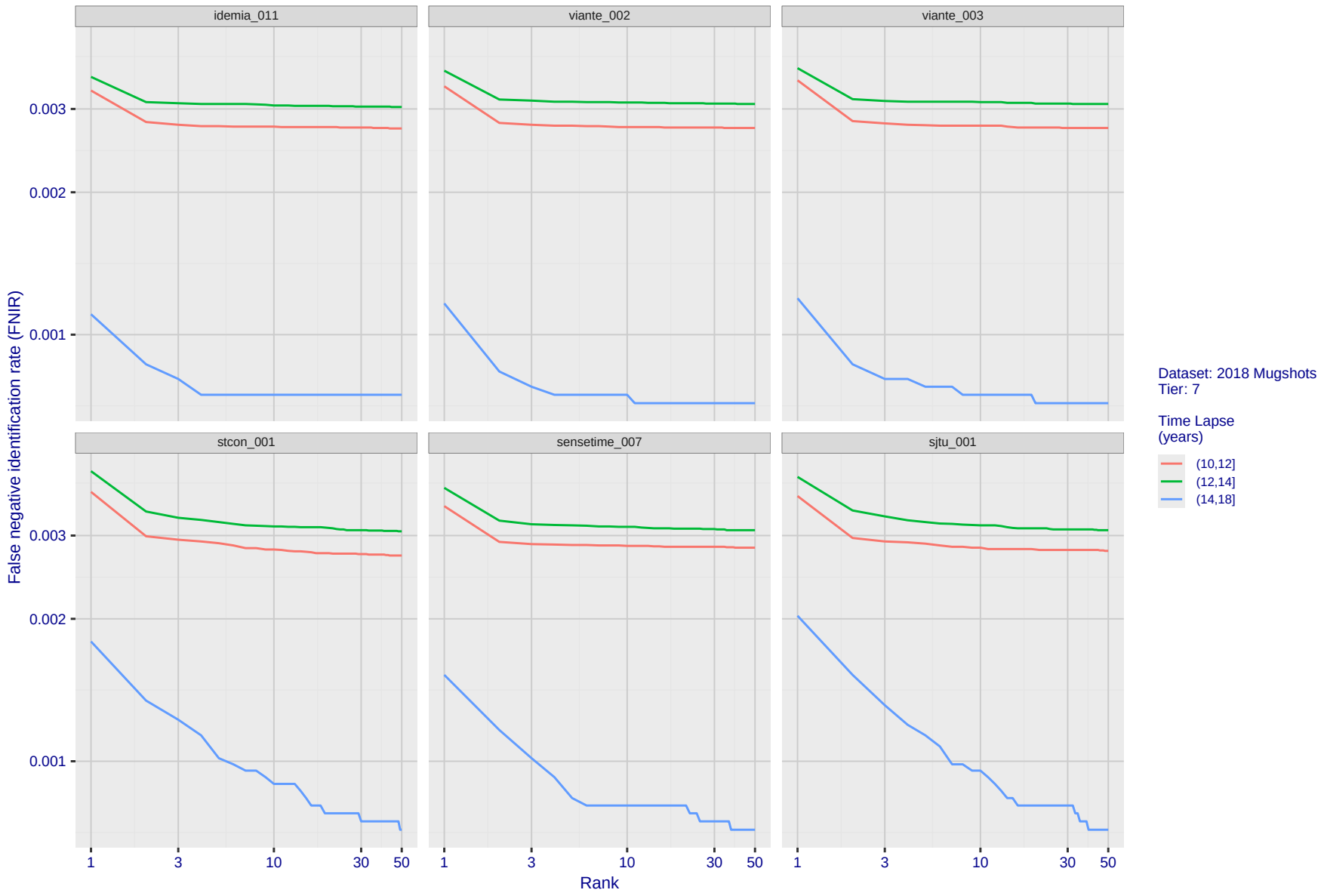


*Figure 154:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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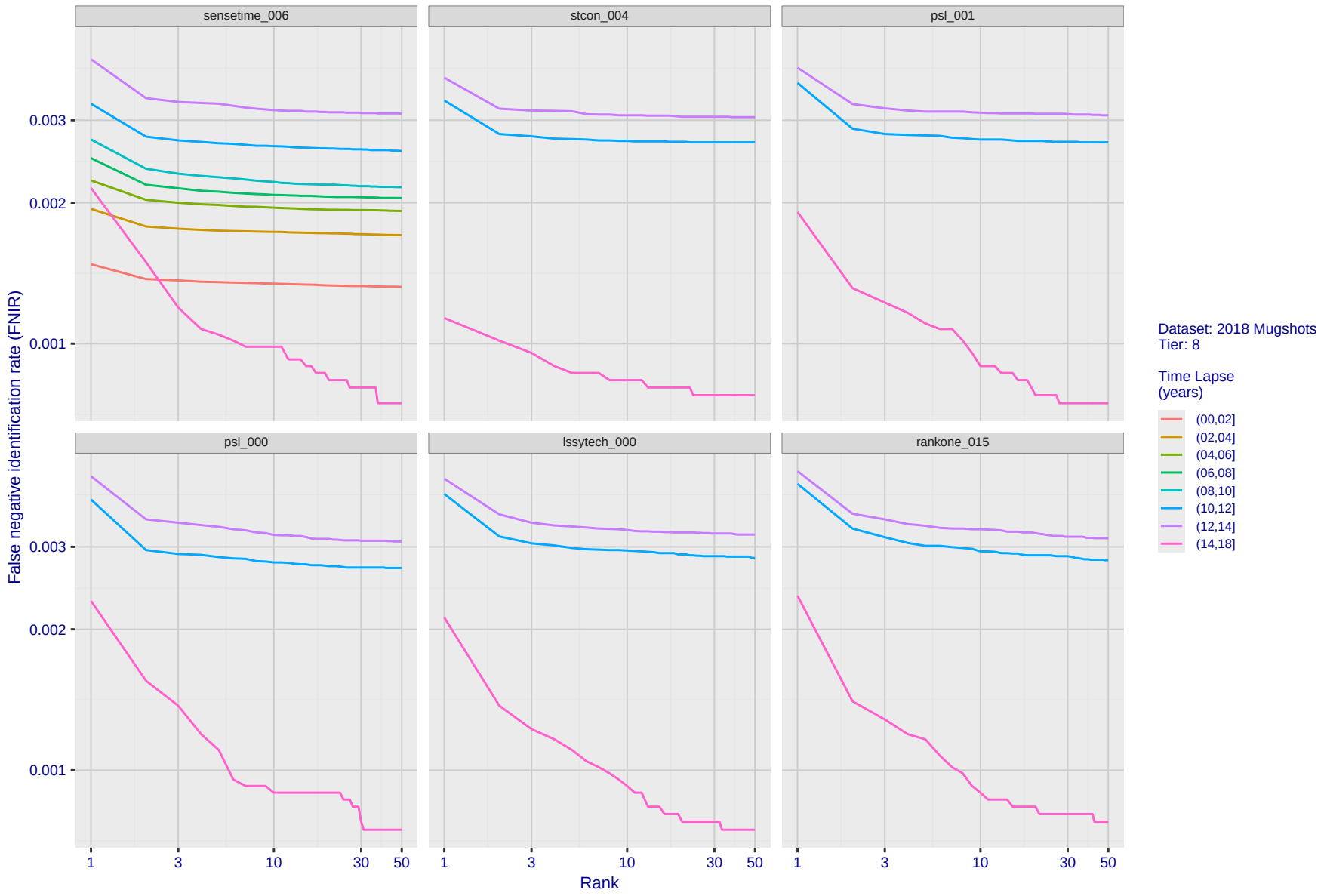


*Figure 155:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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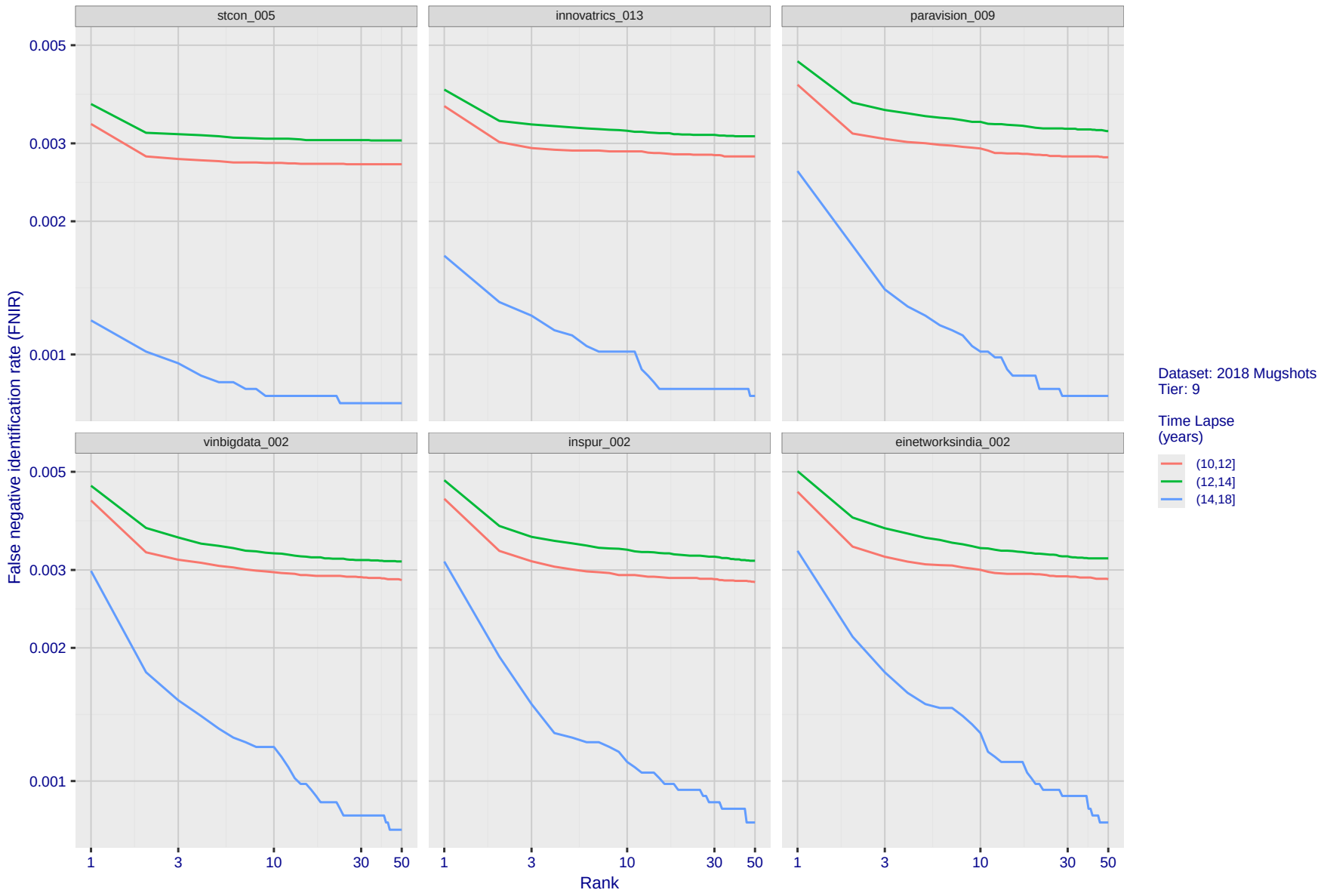


*Figure 156:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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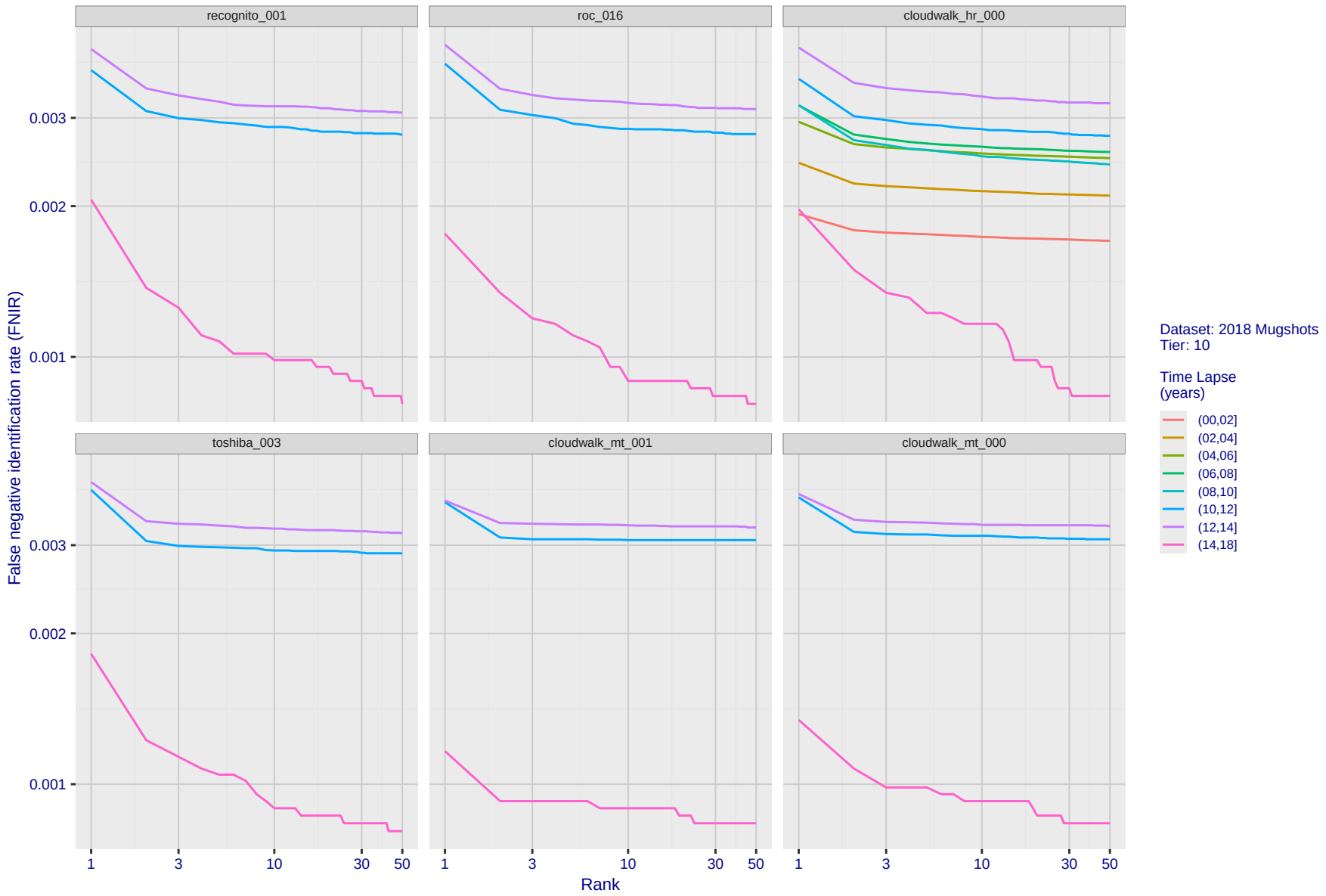


*Figure 157:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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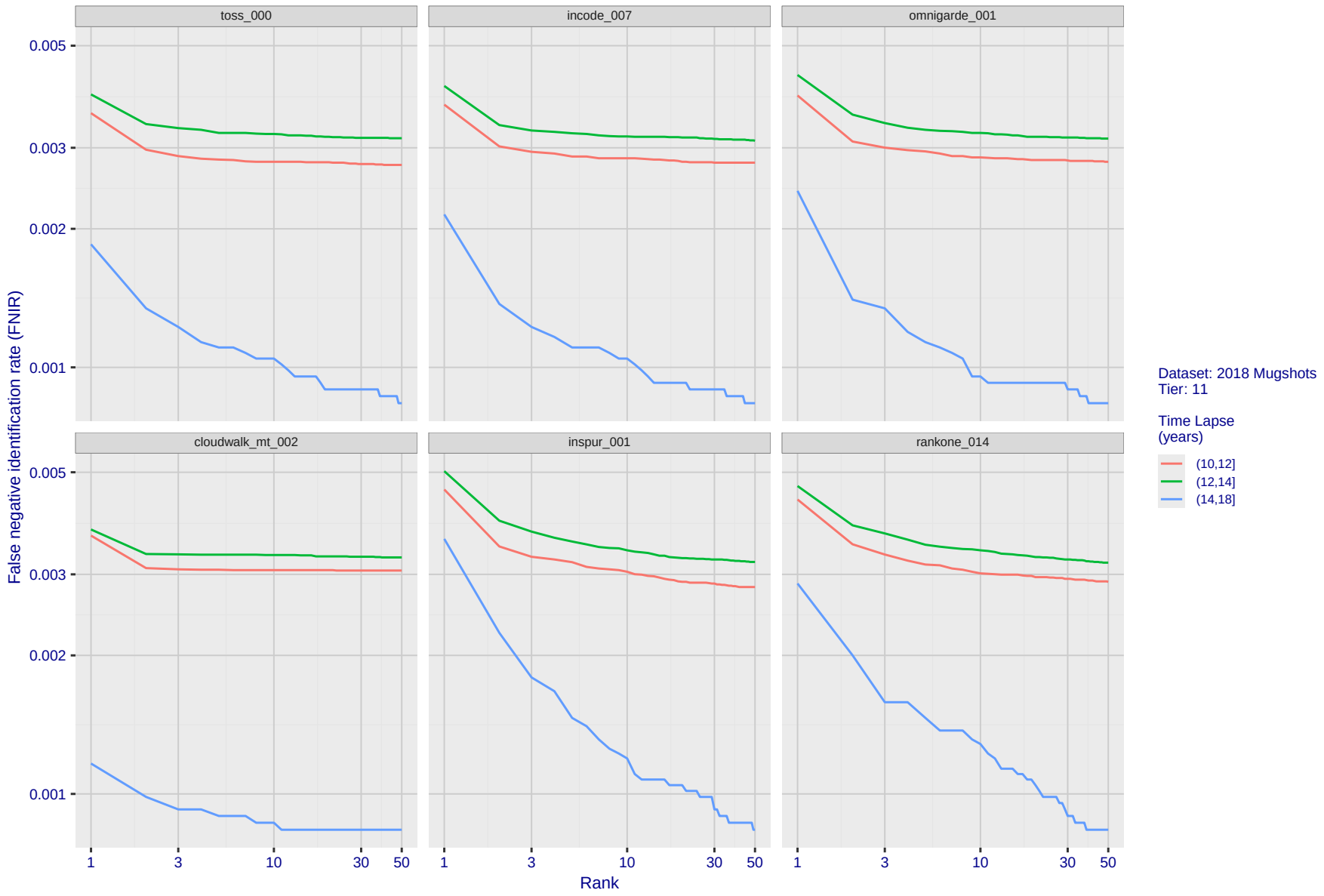


*Figure 158:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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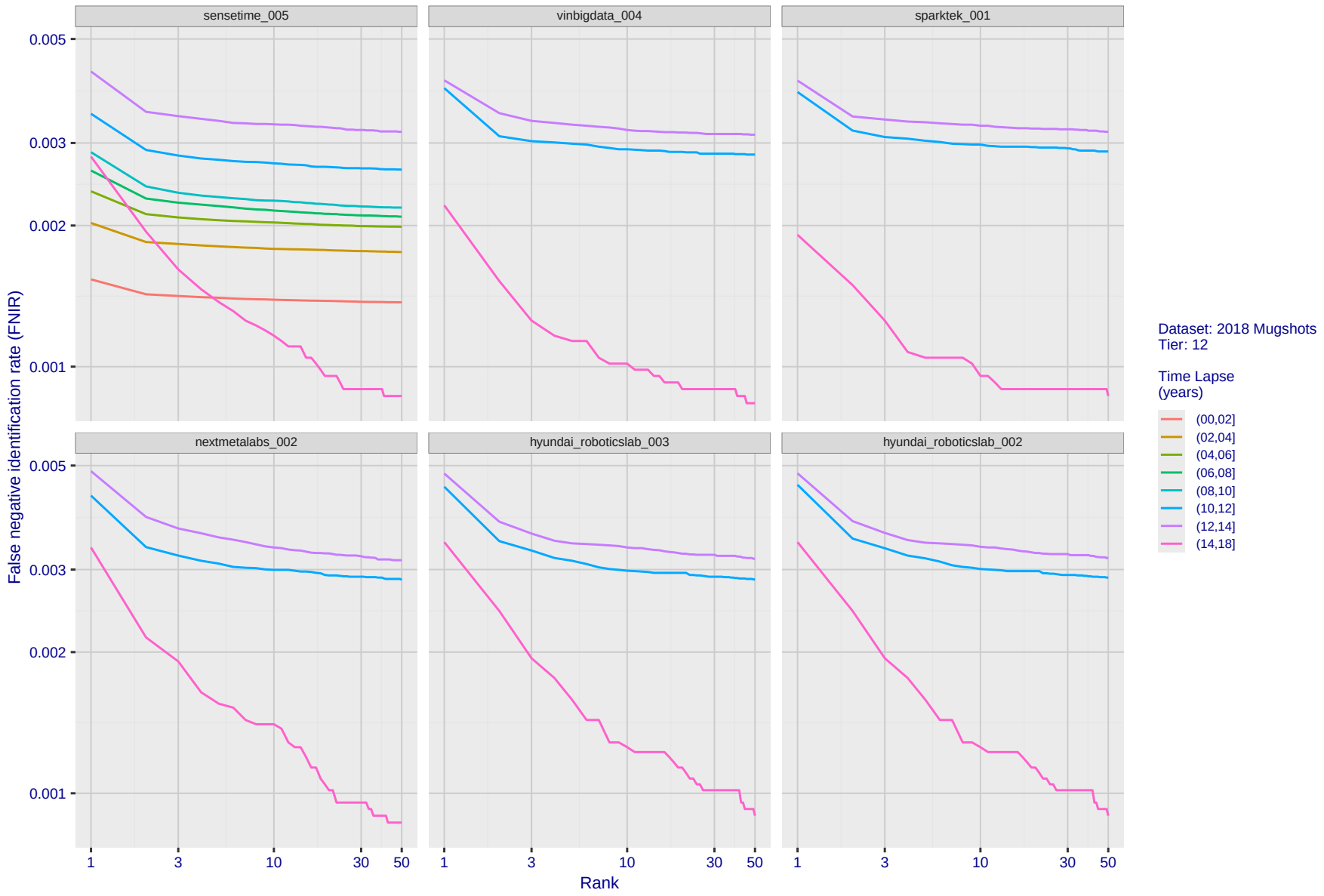


*Figure 159:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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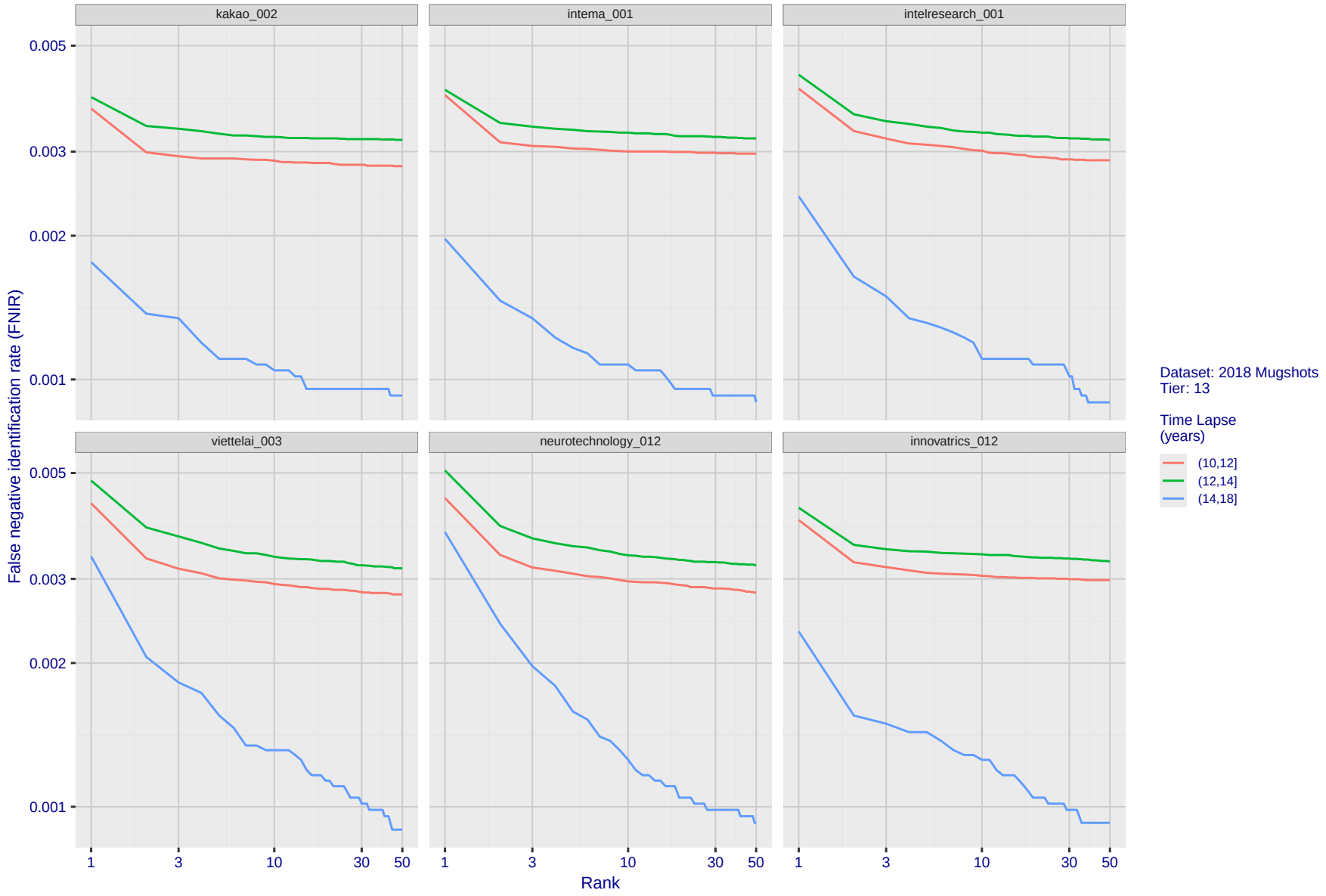


*Figure 160:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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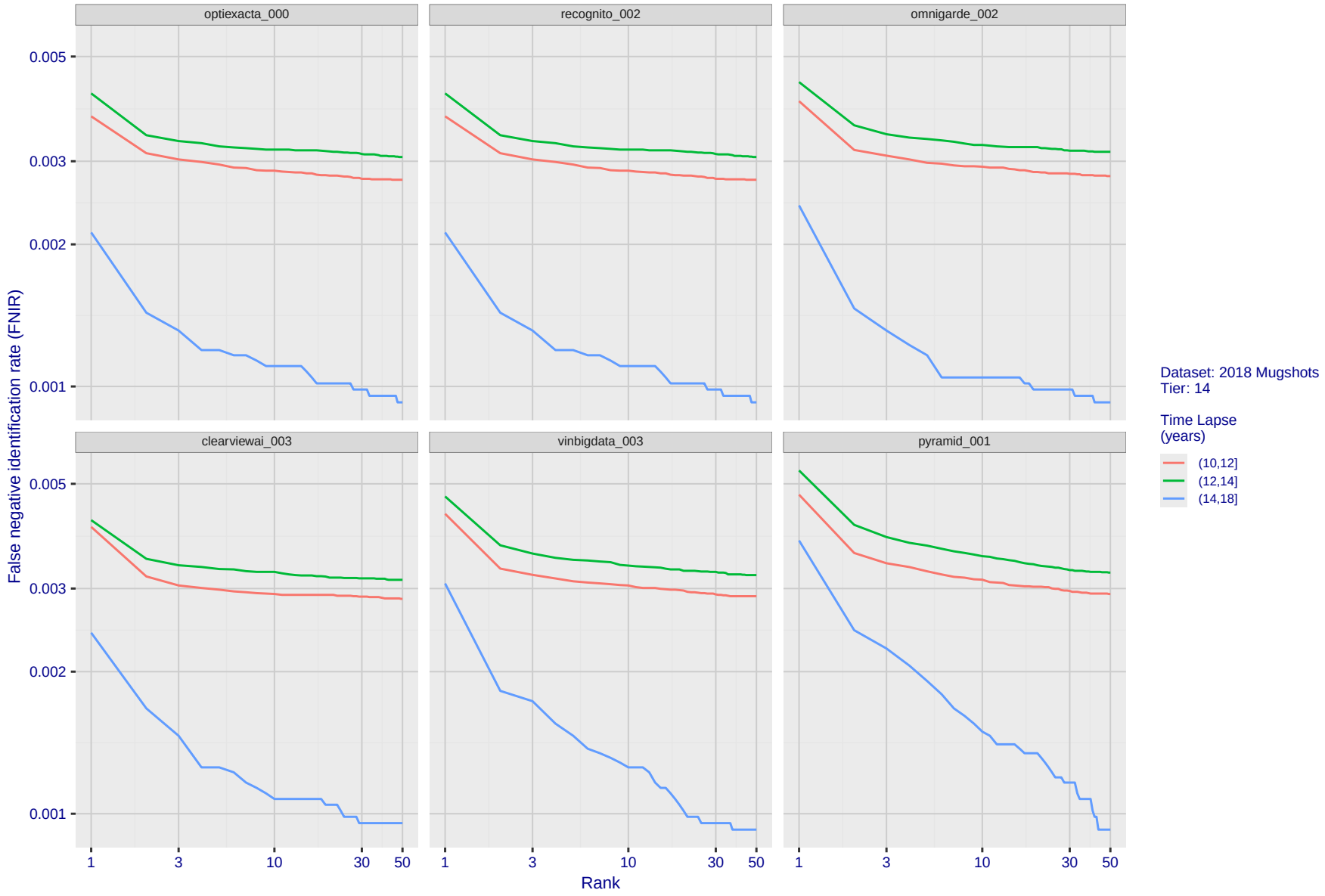


*Figure 161:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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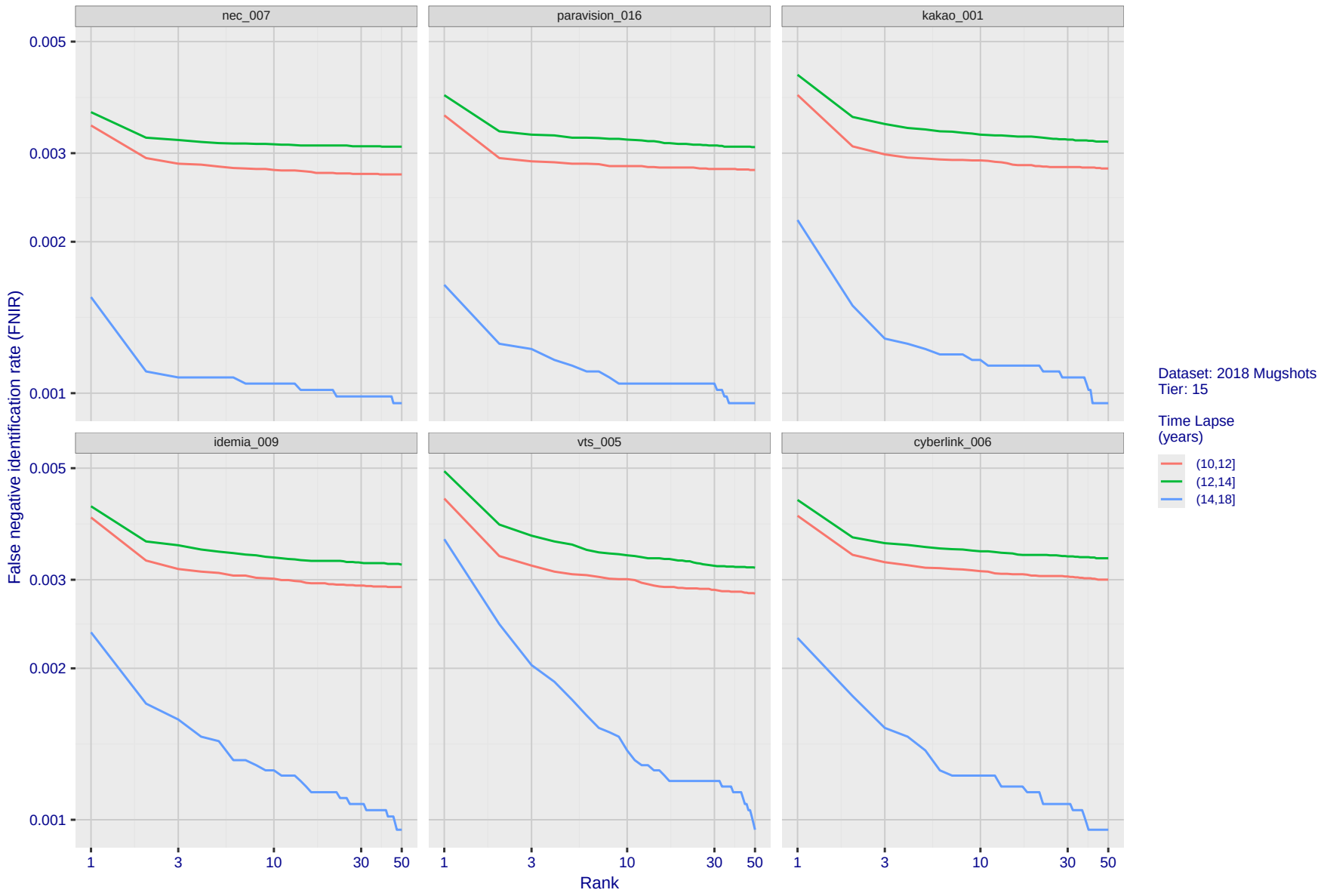


*Figure 162:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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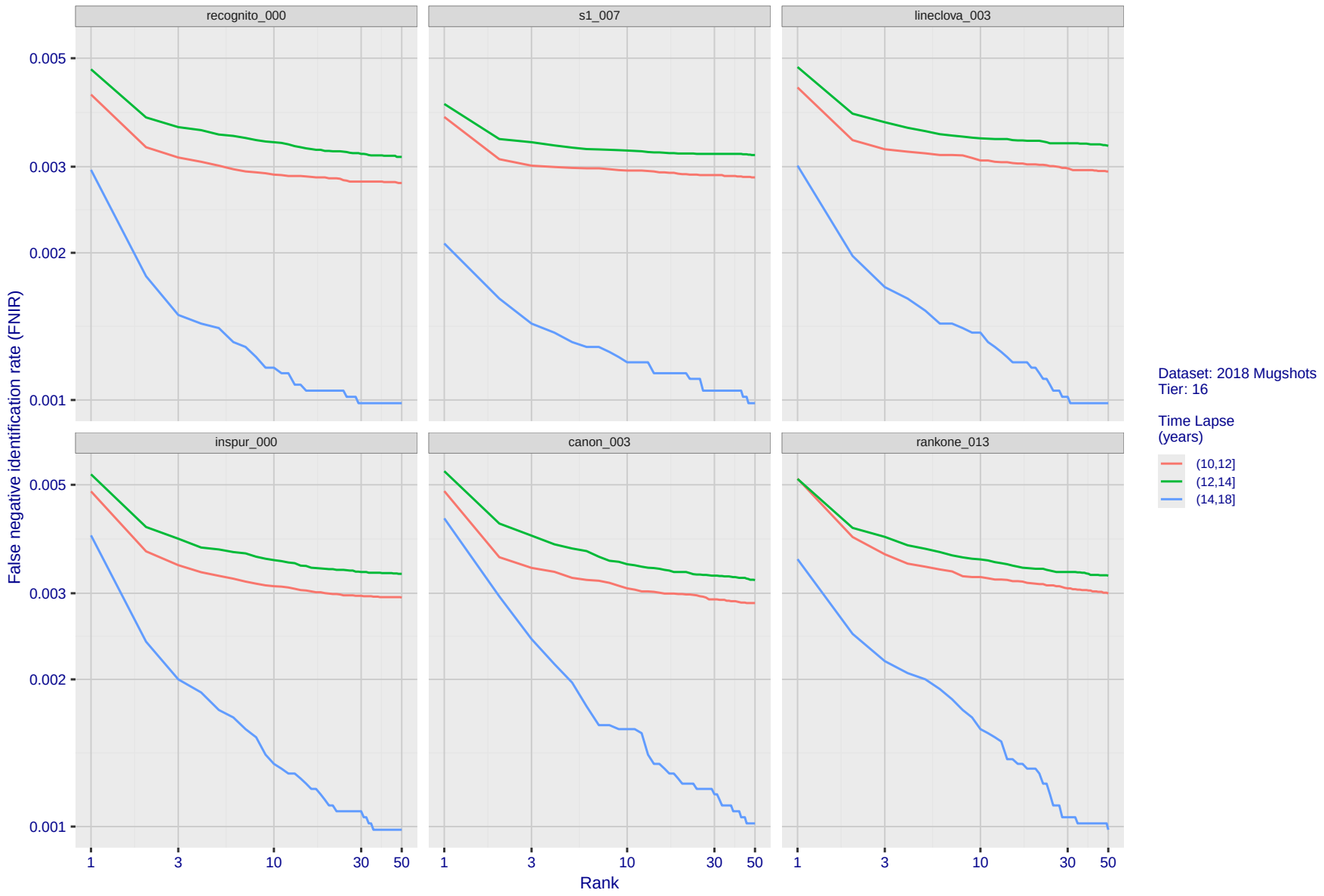


*Figure 163:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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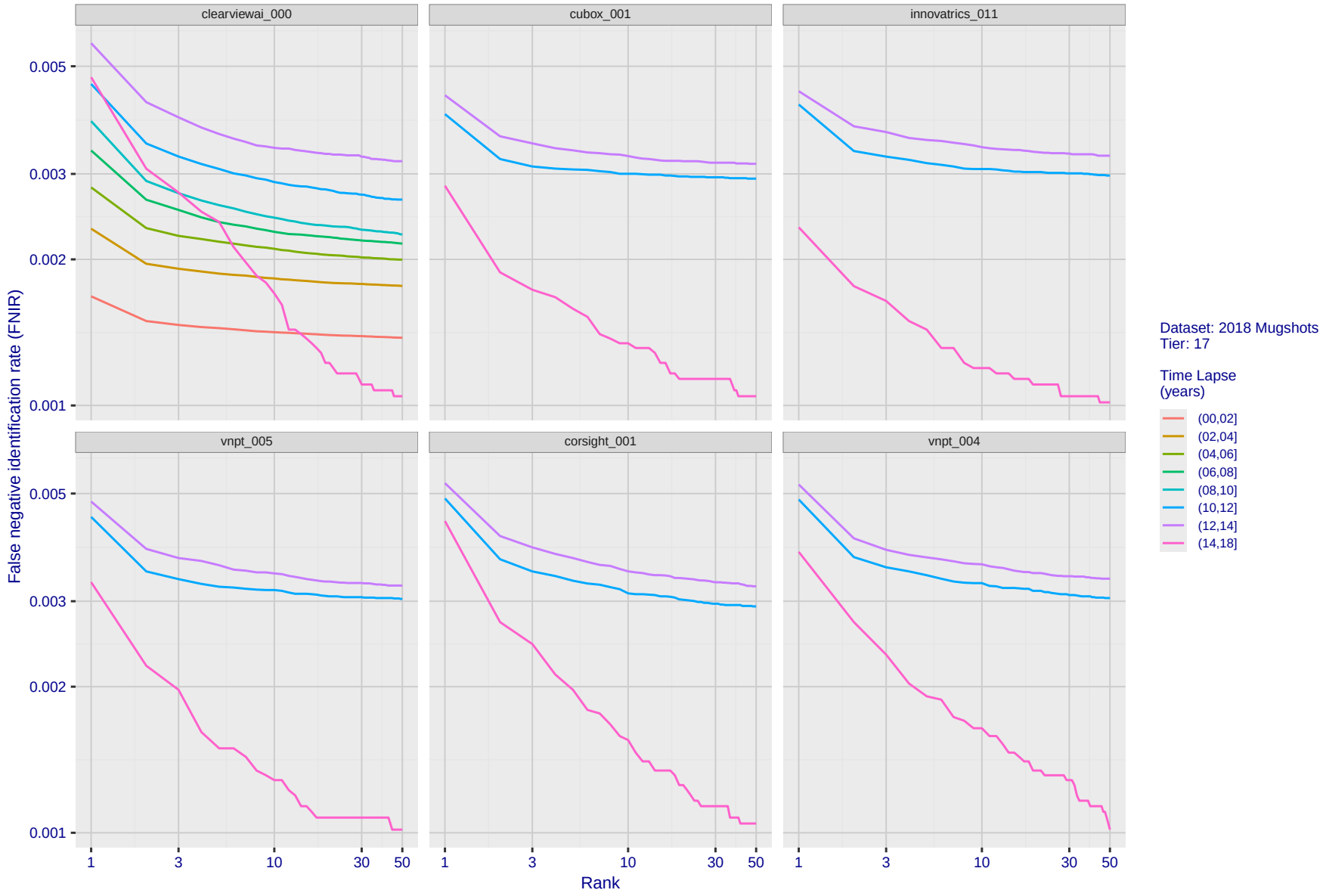


*Figure 164:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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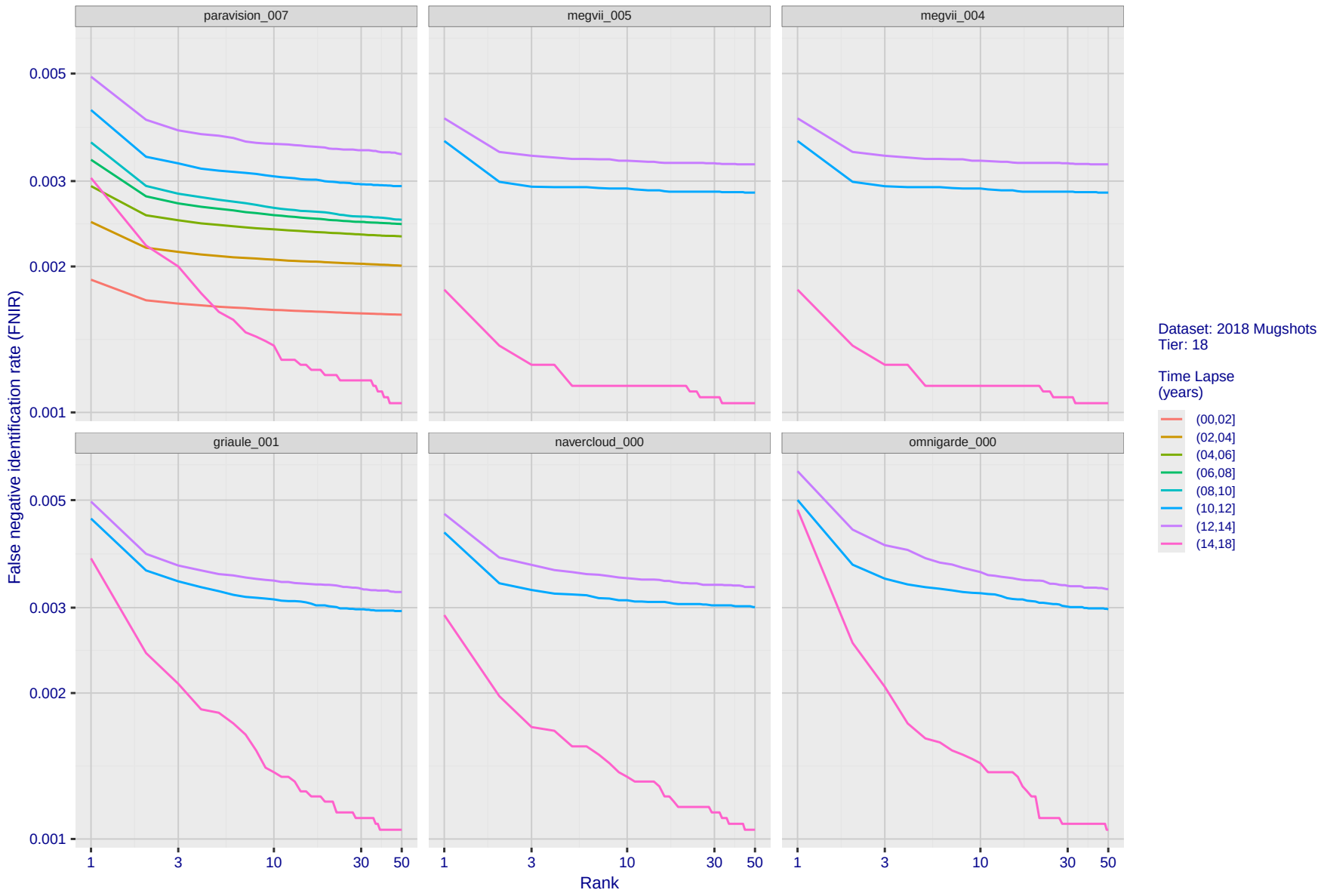


*Figure 165:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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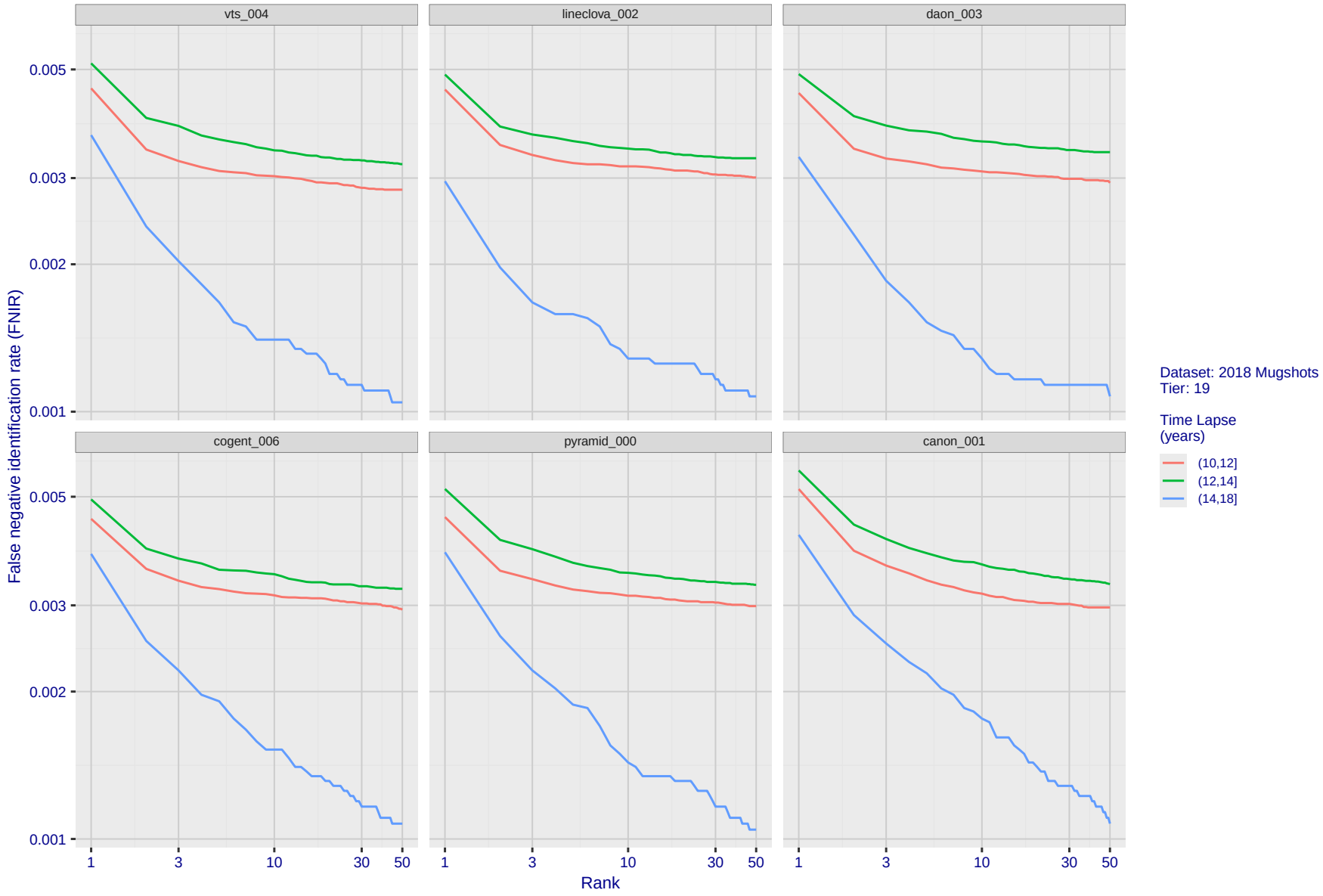


*Figure 166:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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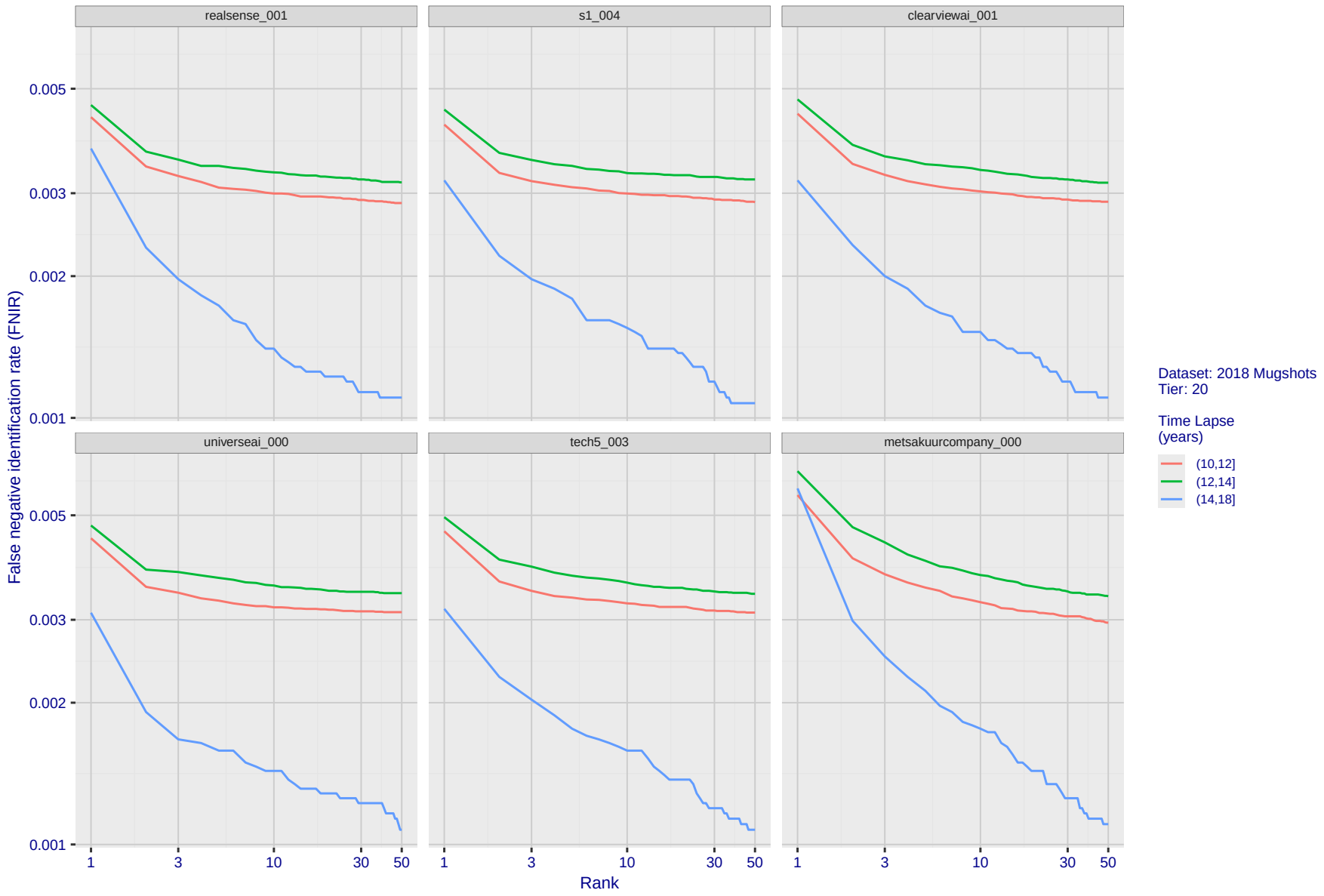


*Figure 167:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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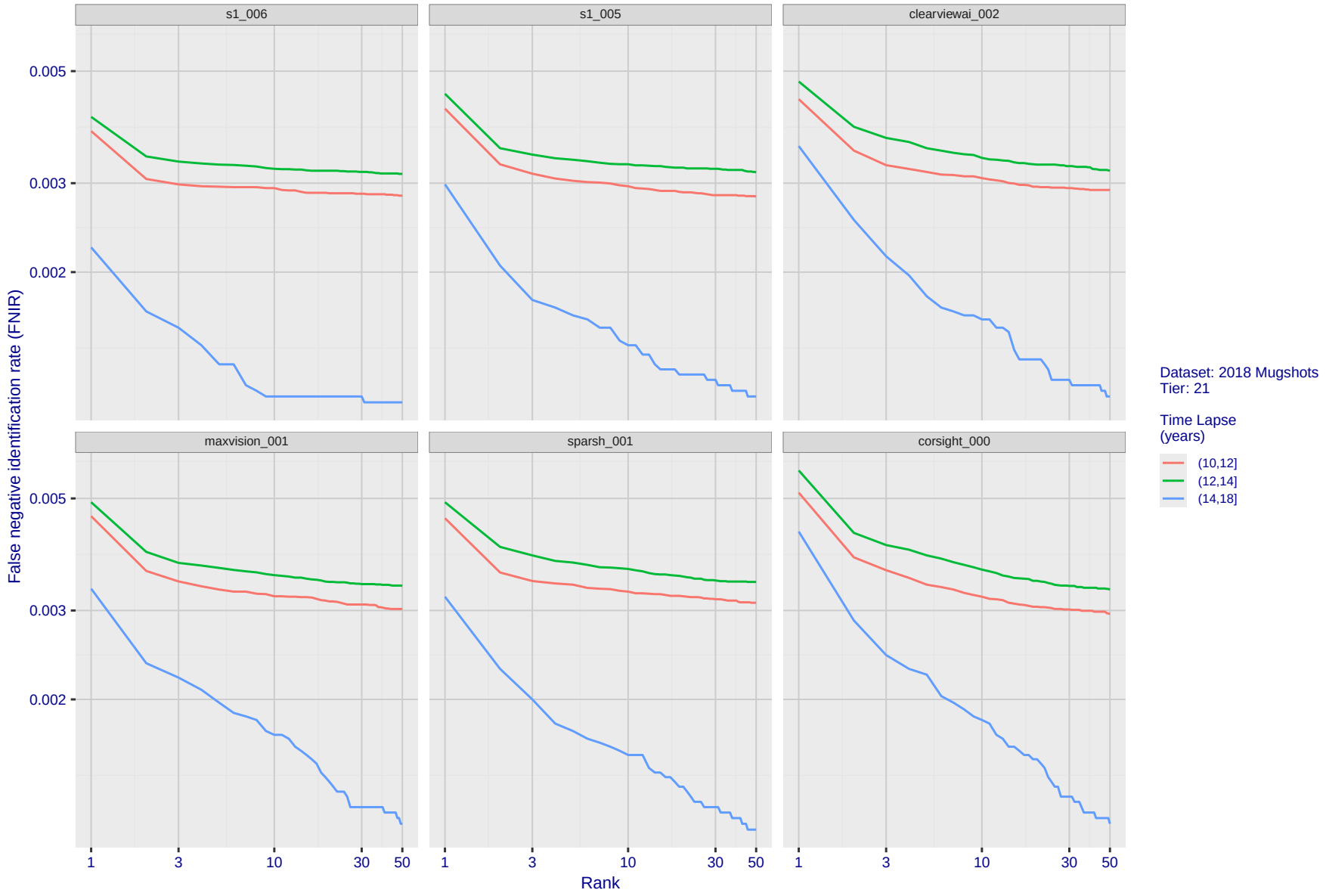


*Figure 168:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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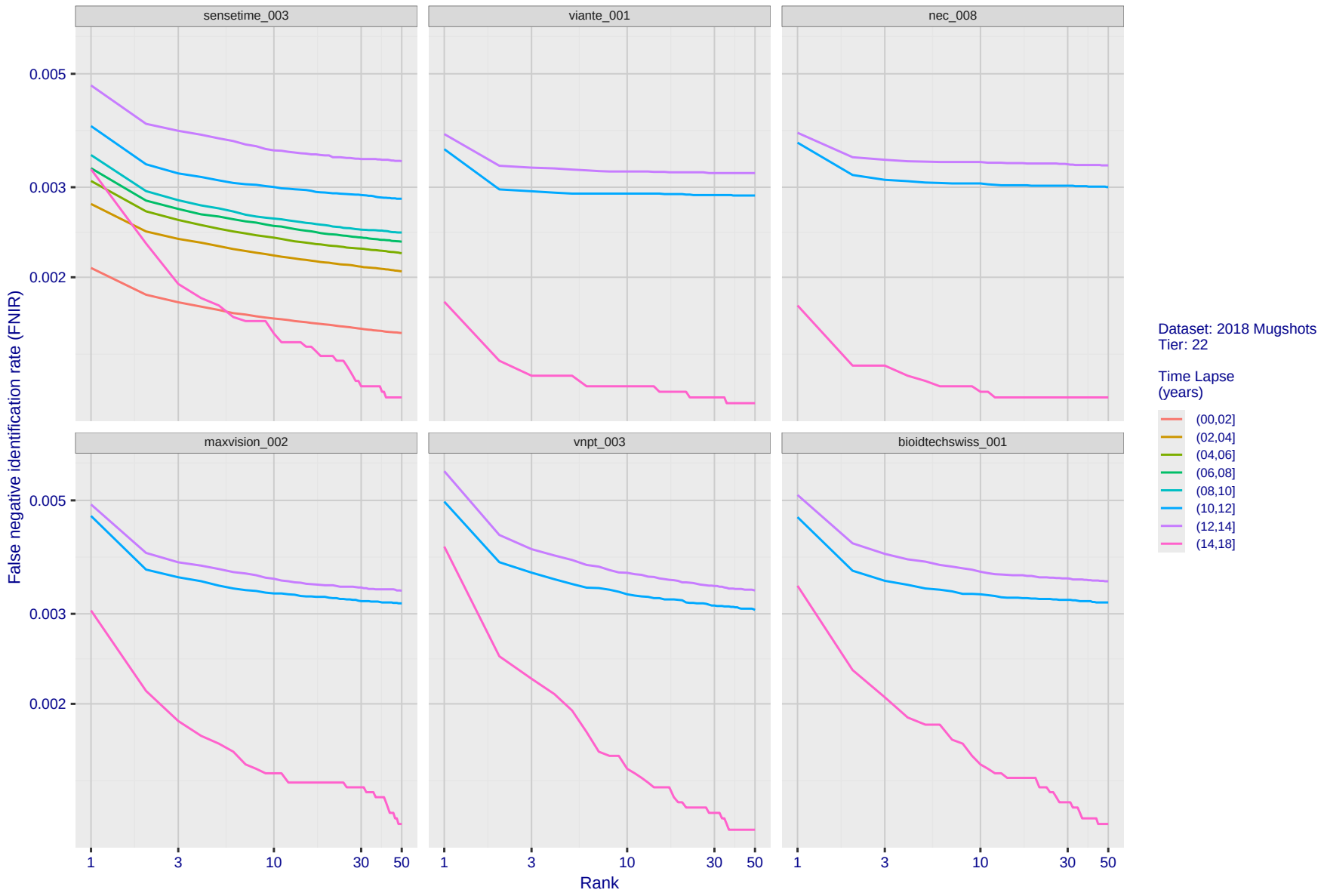


*Figure 169:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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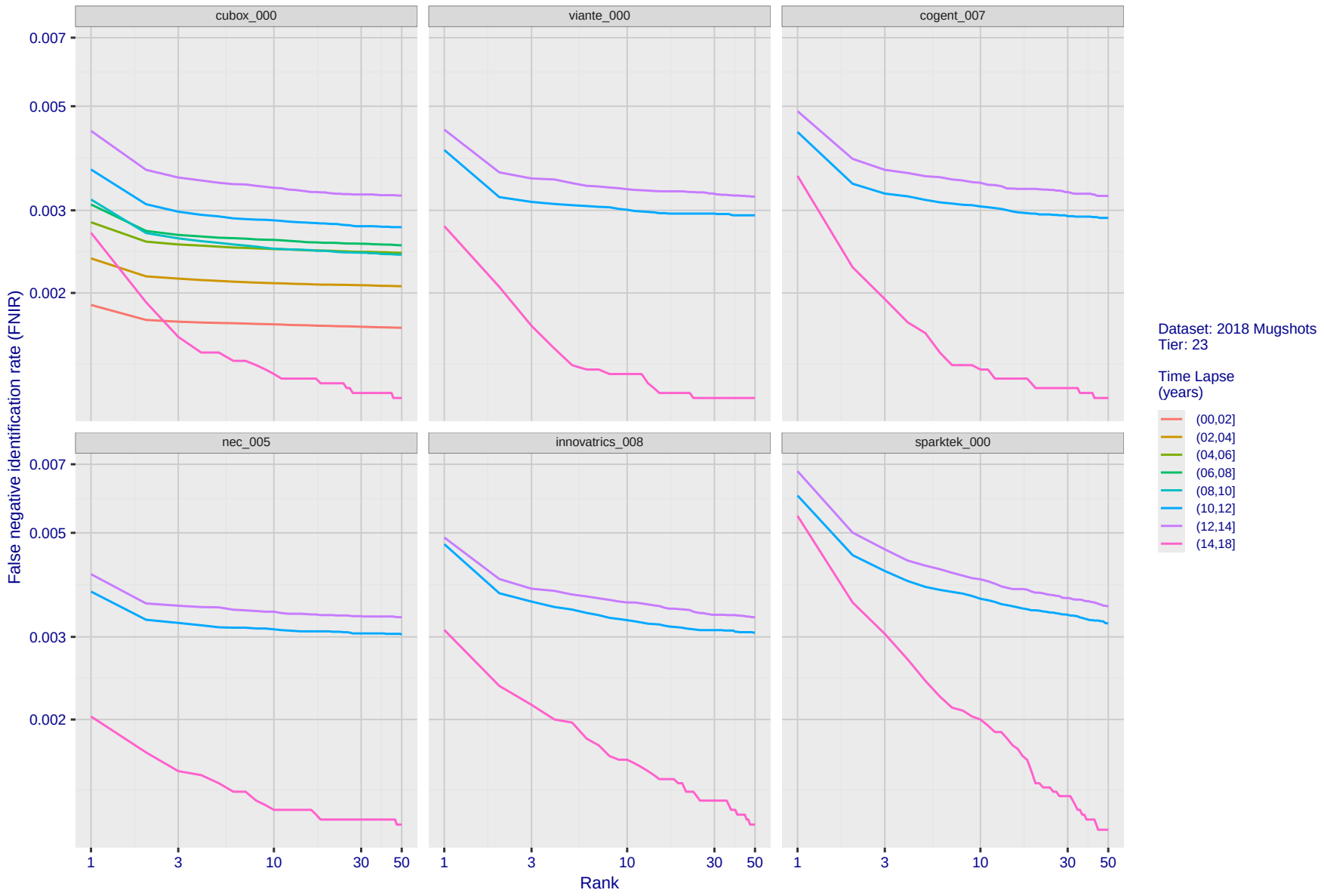


*Figure 170:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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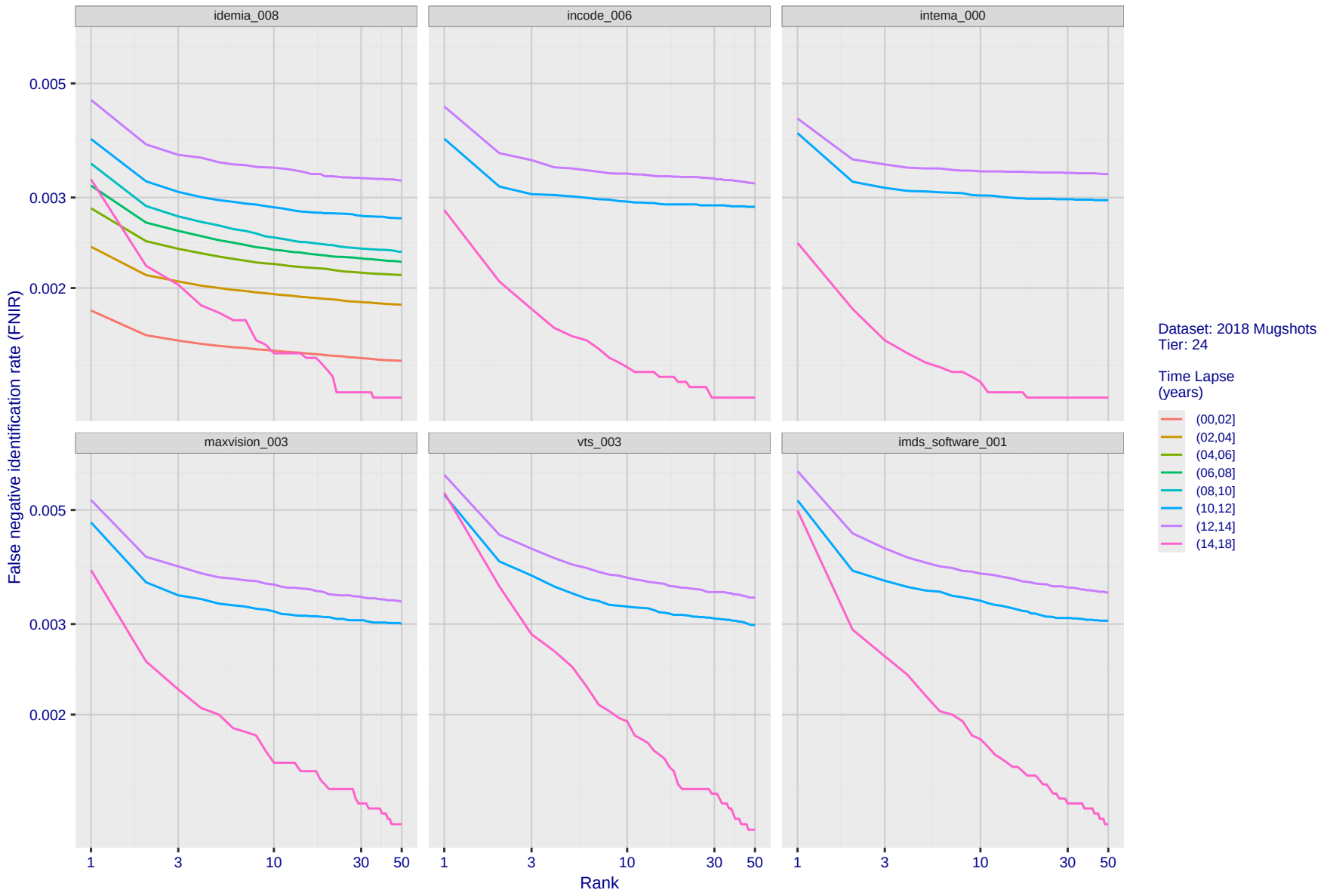


*Figure 171:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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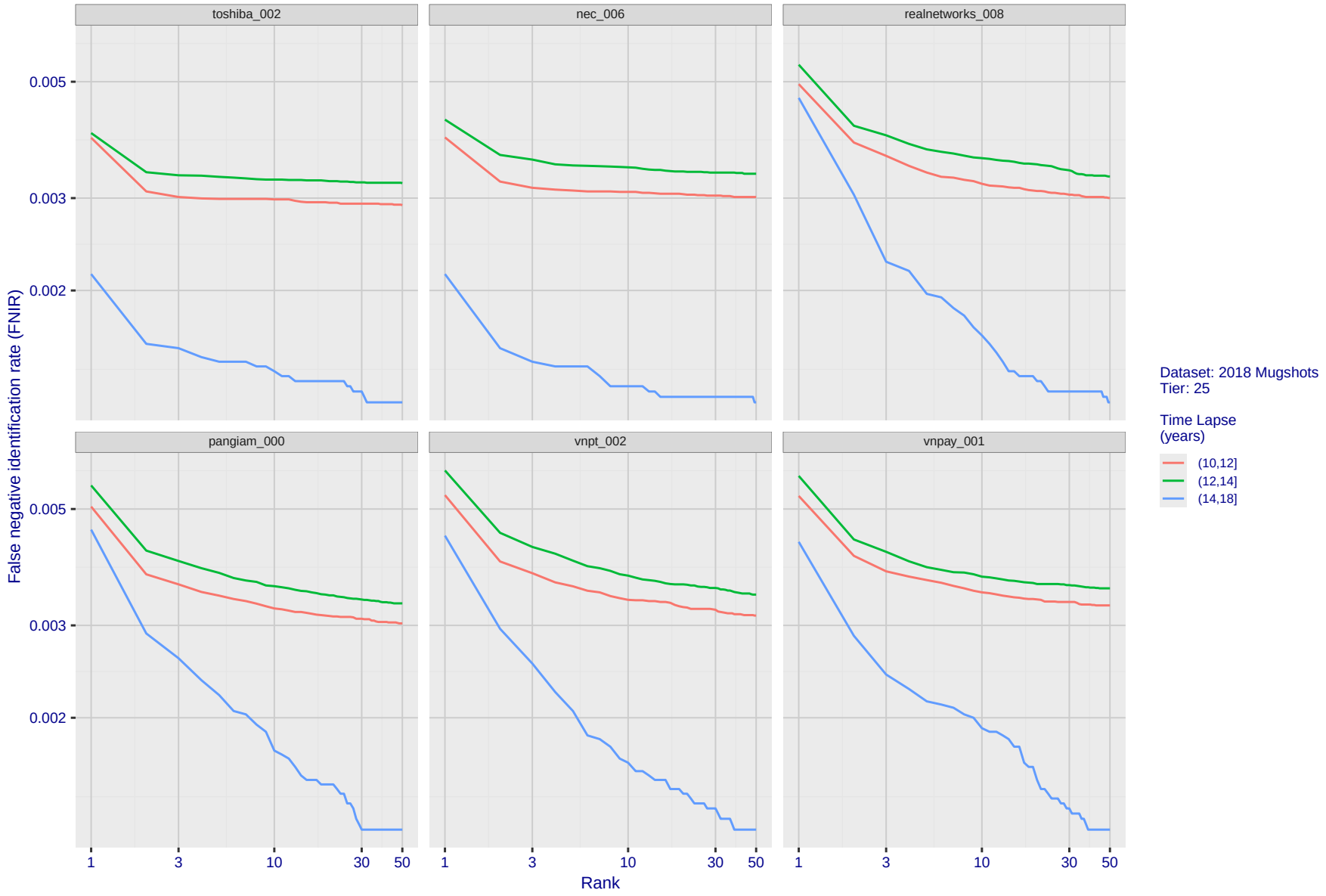


*Figure 172:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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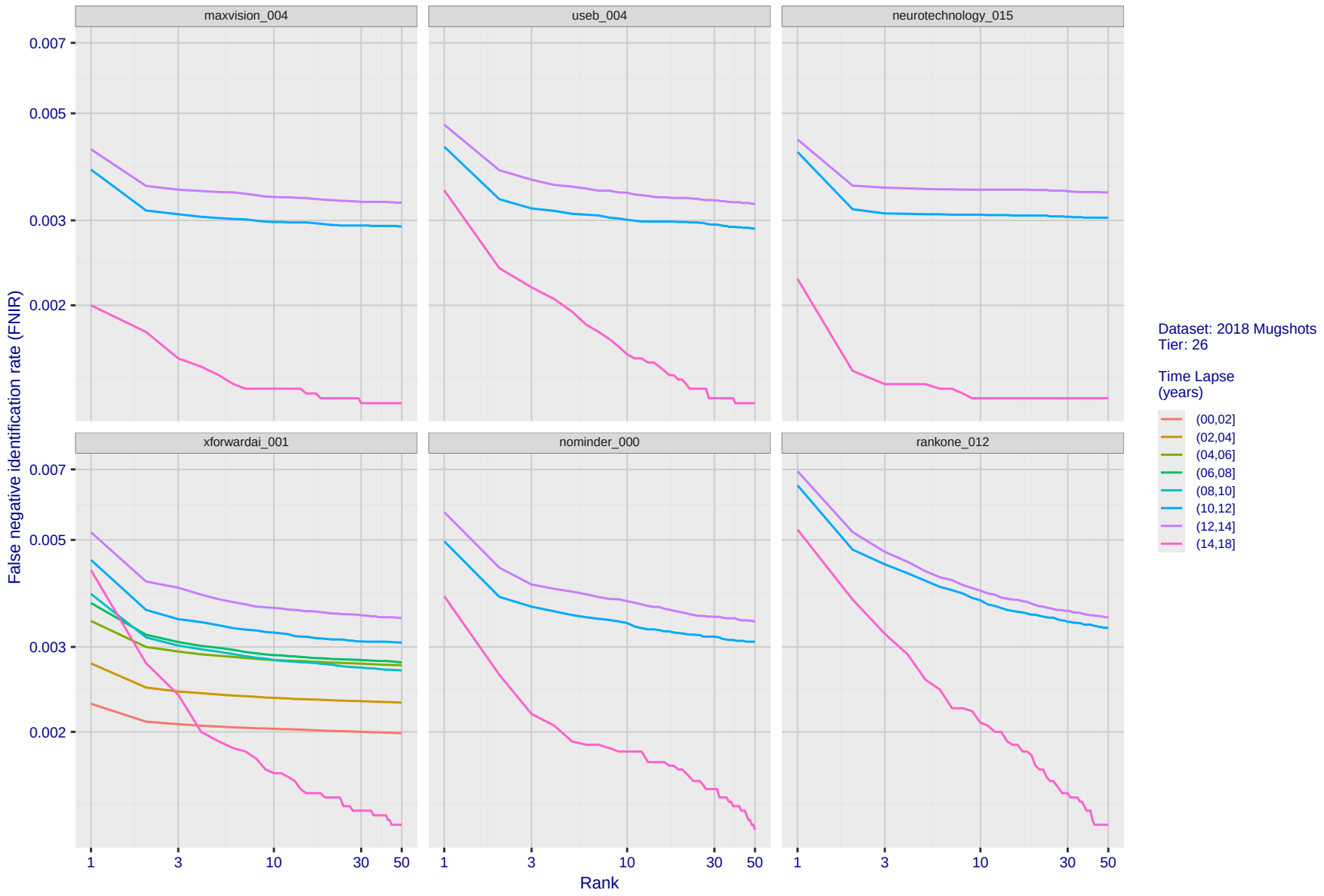


*Figure 173:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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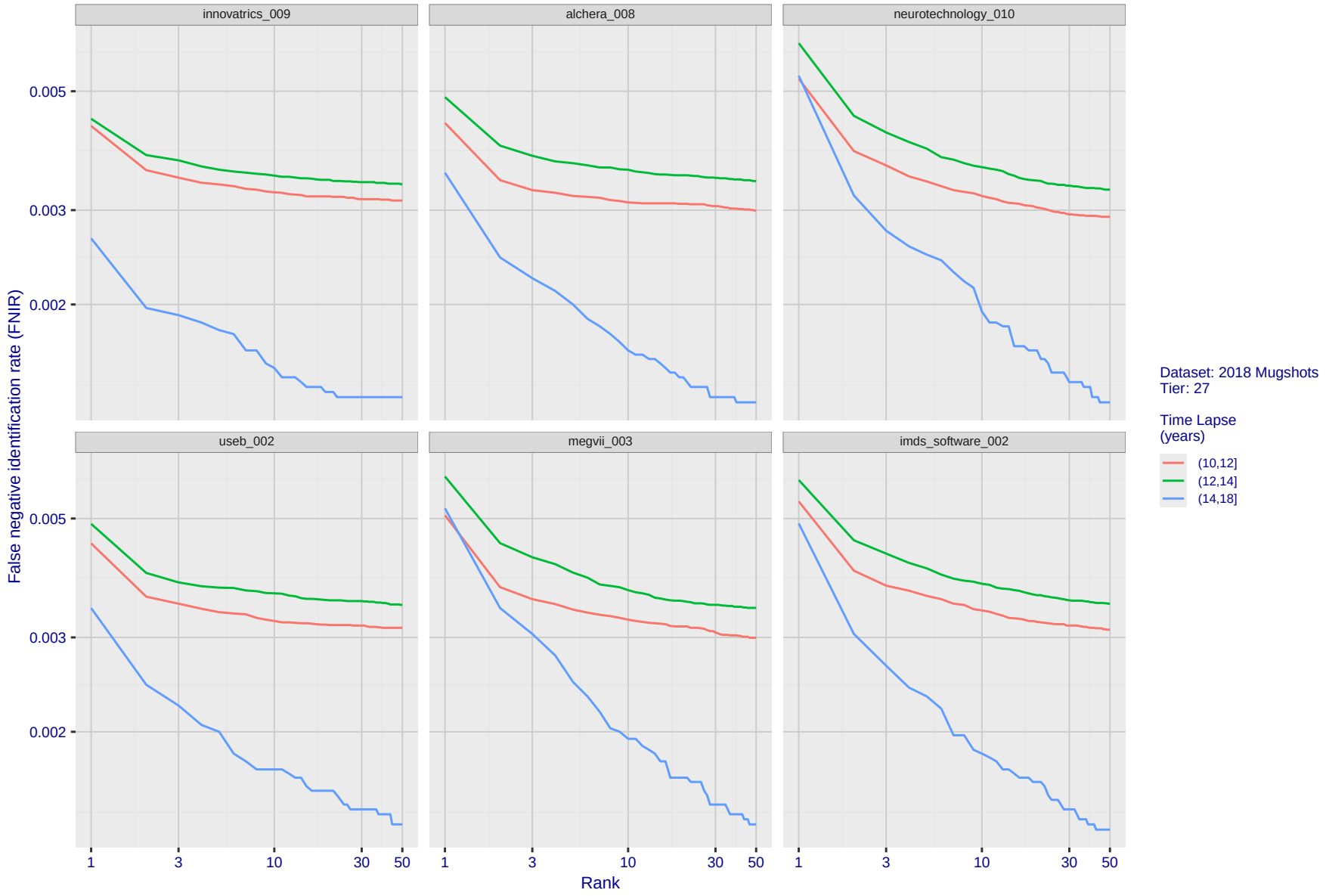


*Figure 174:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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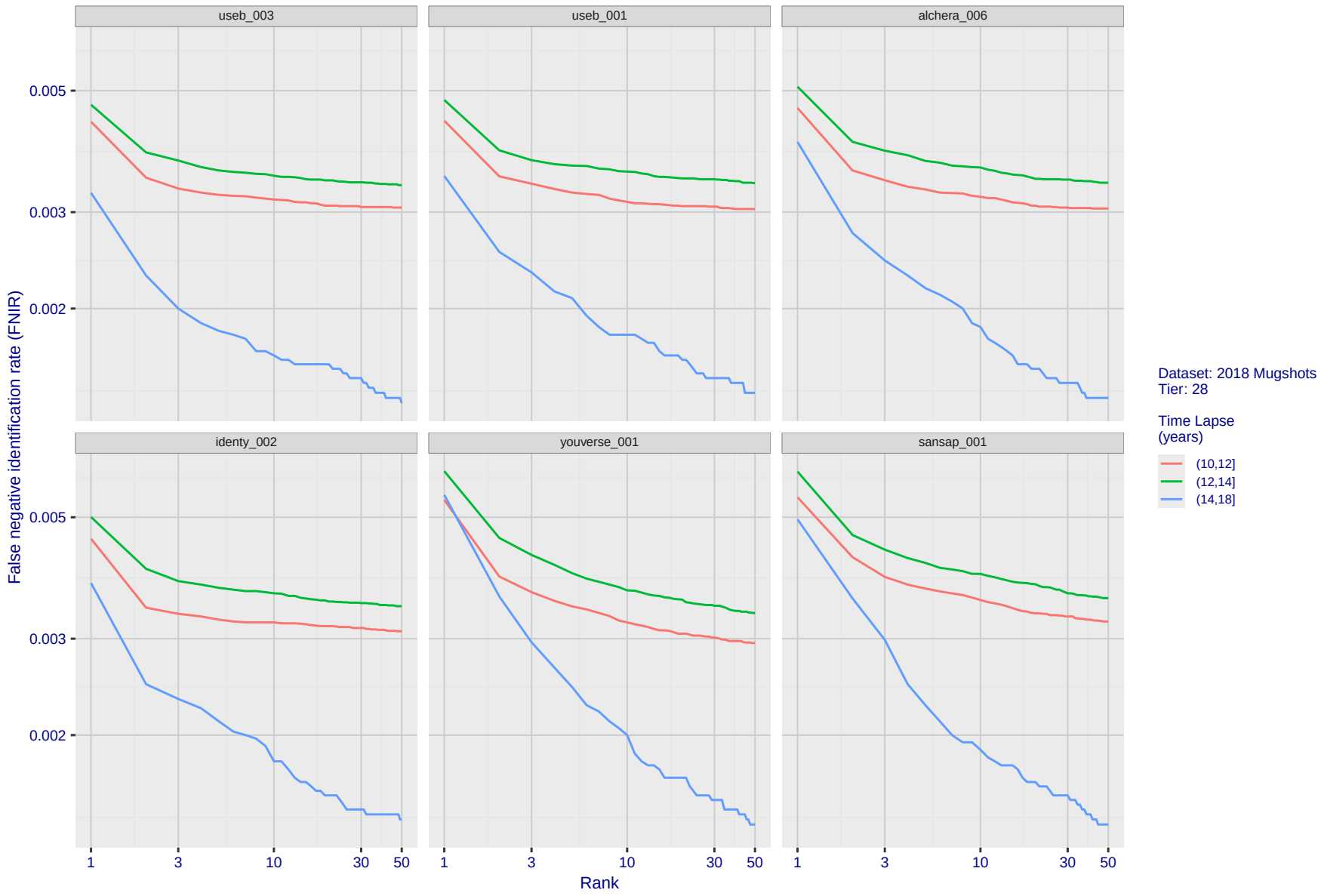


*Figure 175:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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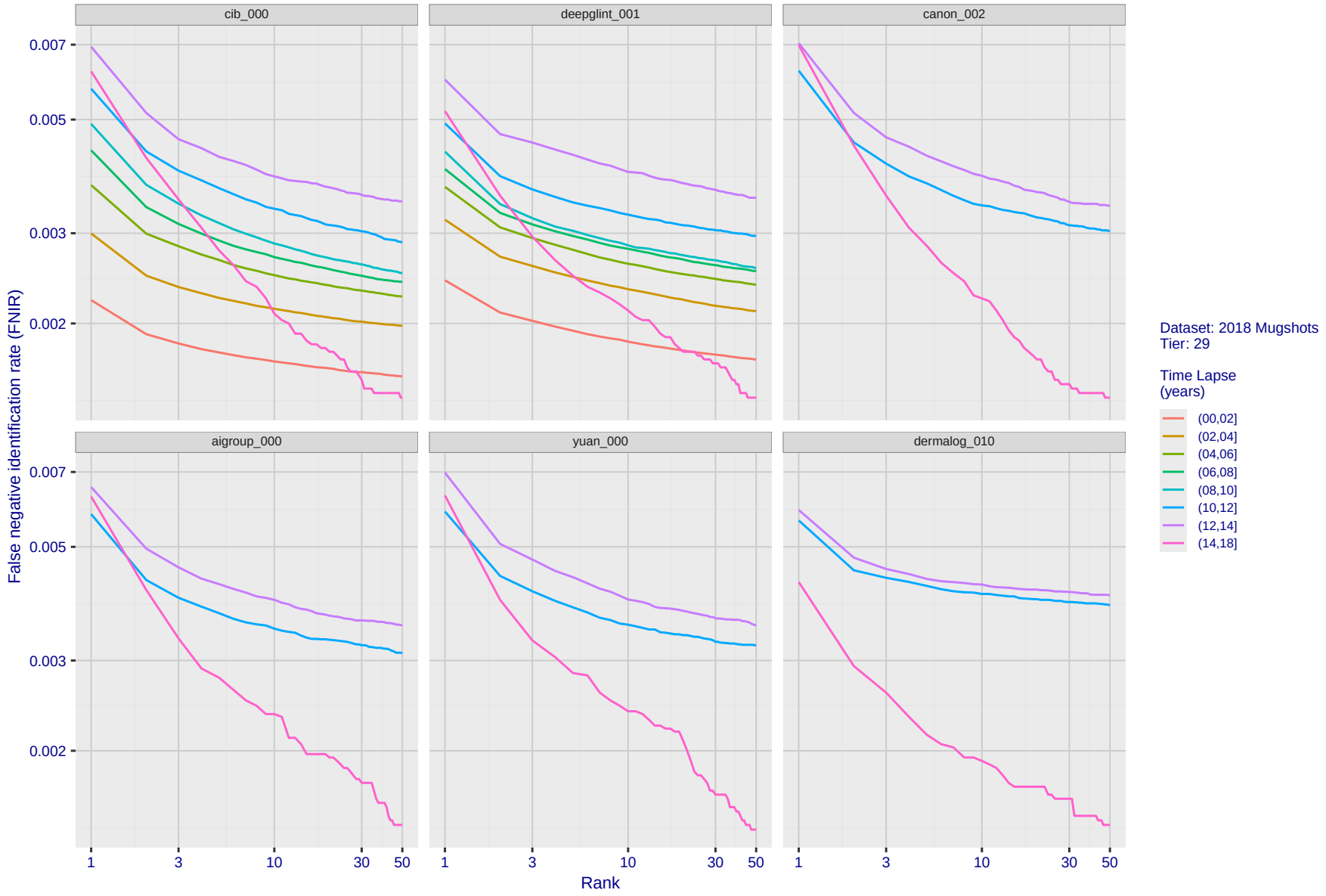


*Figure 176:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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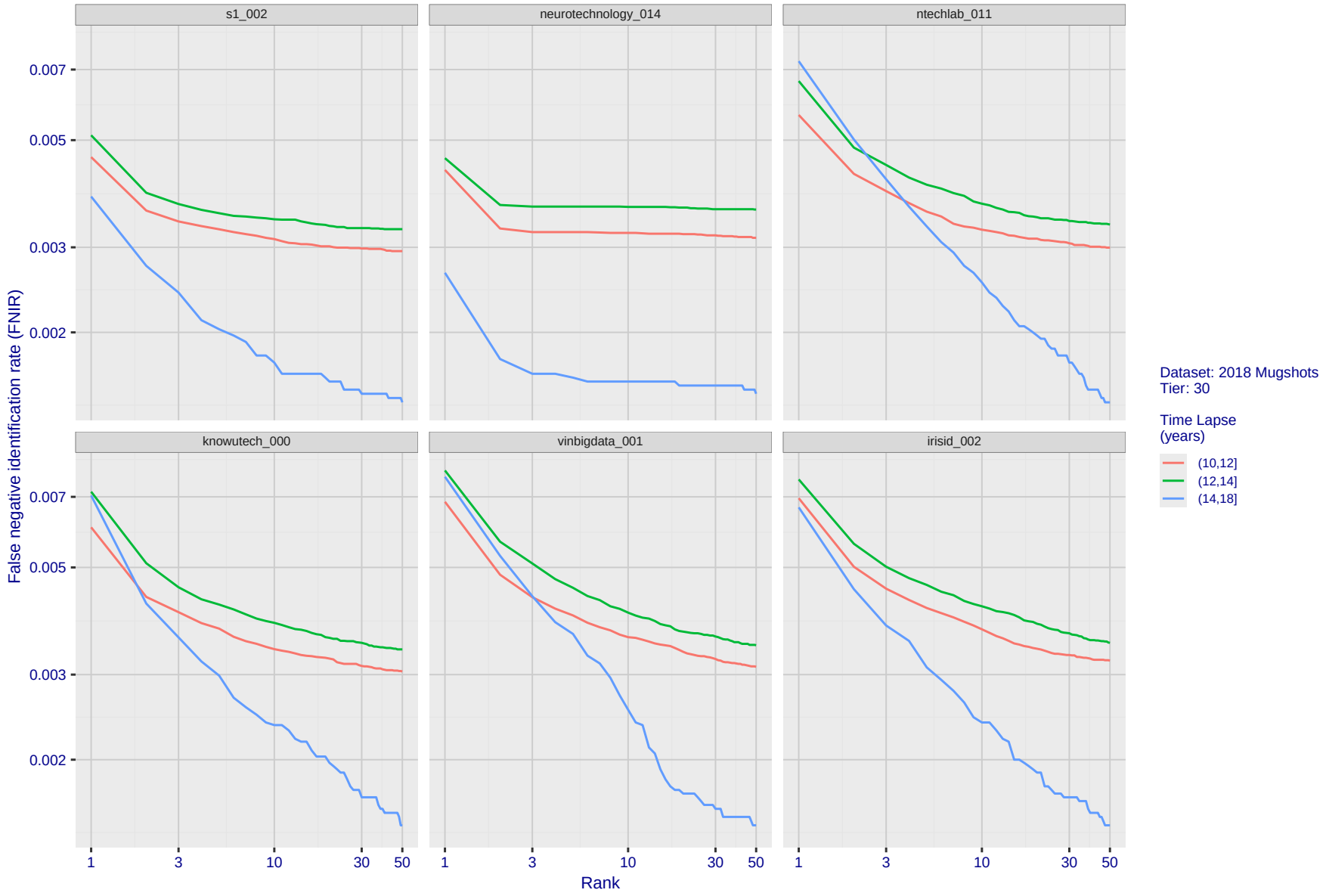


*Figure 177:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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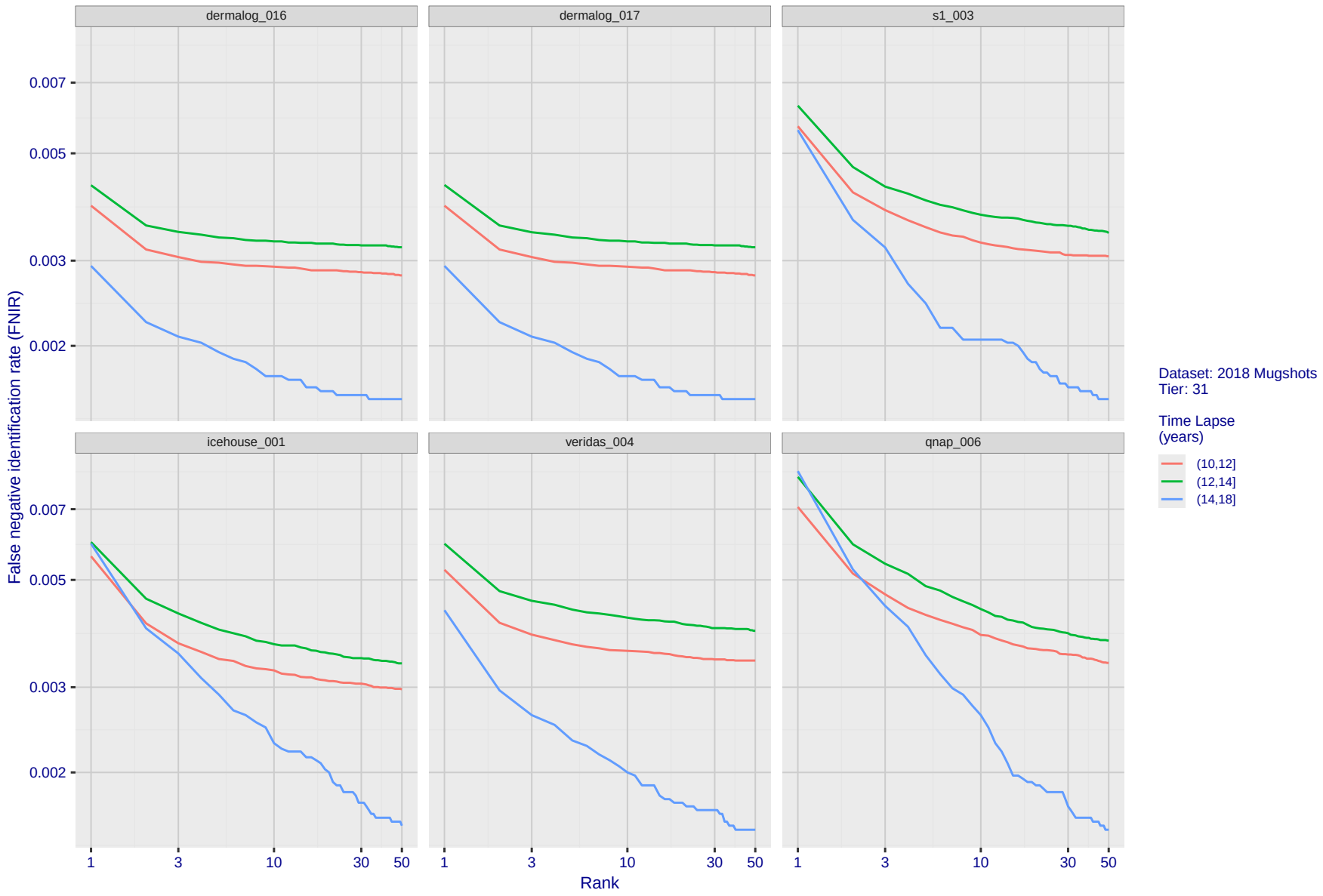


*Figure 178:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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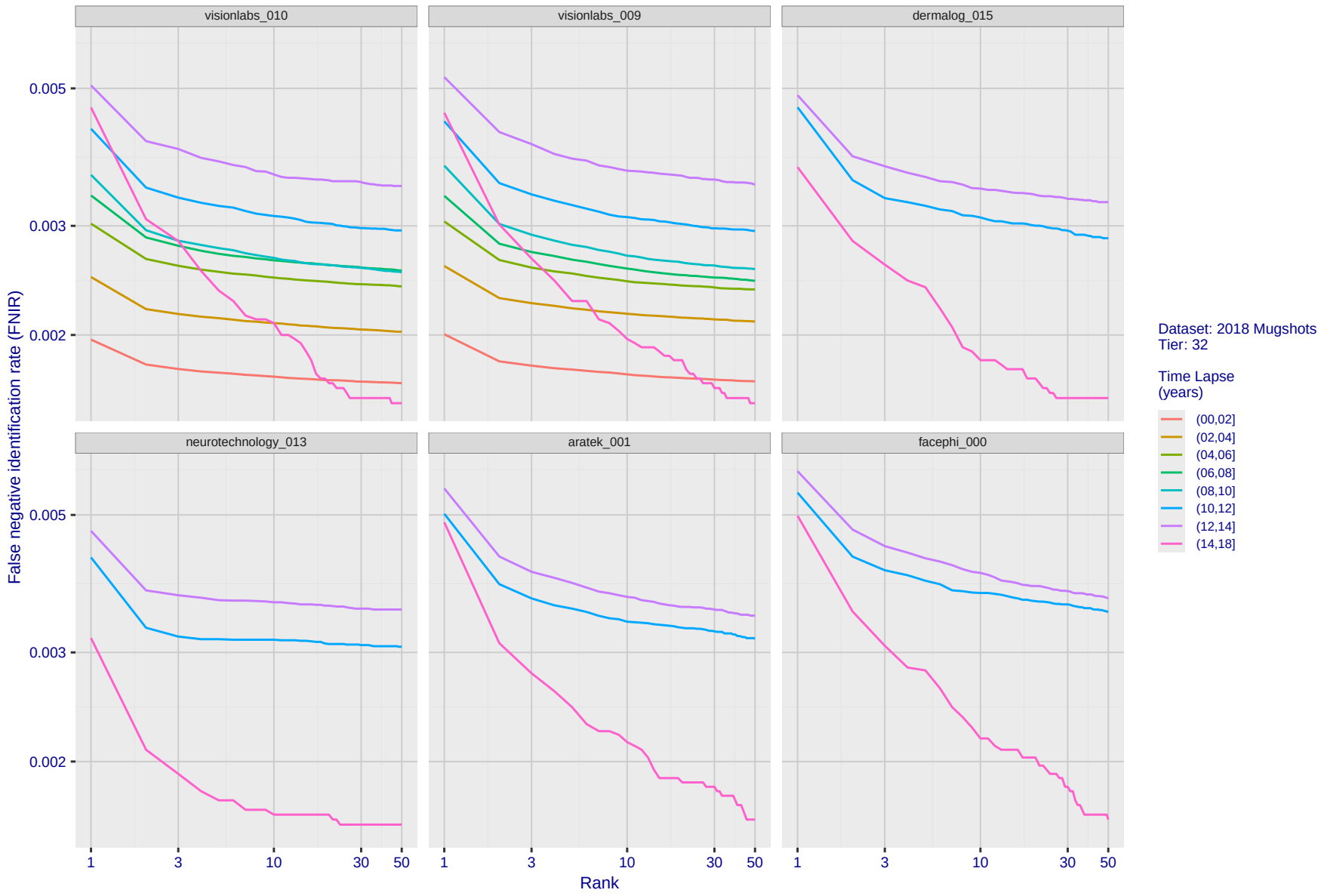


*Figure 179:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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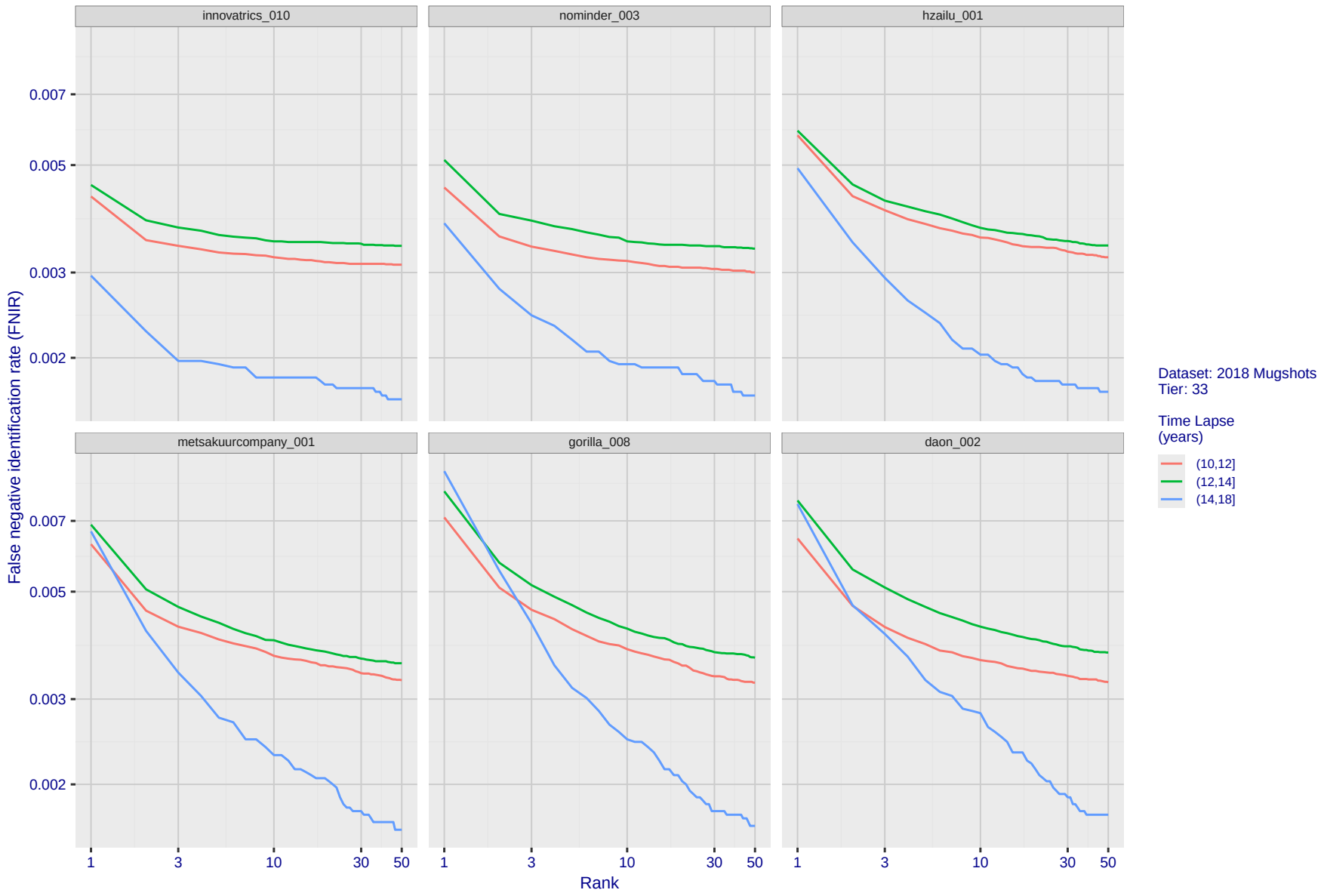


*Figure 180:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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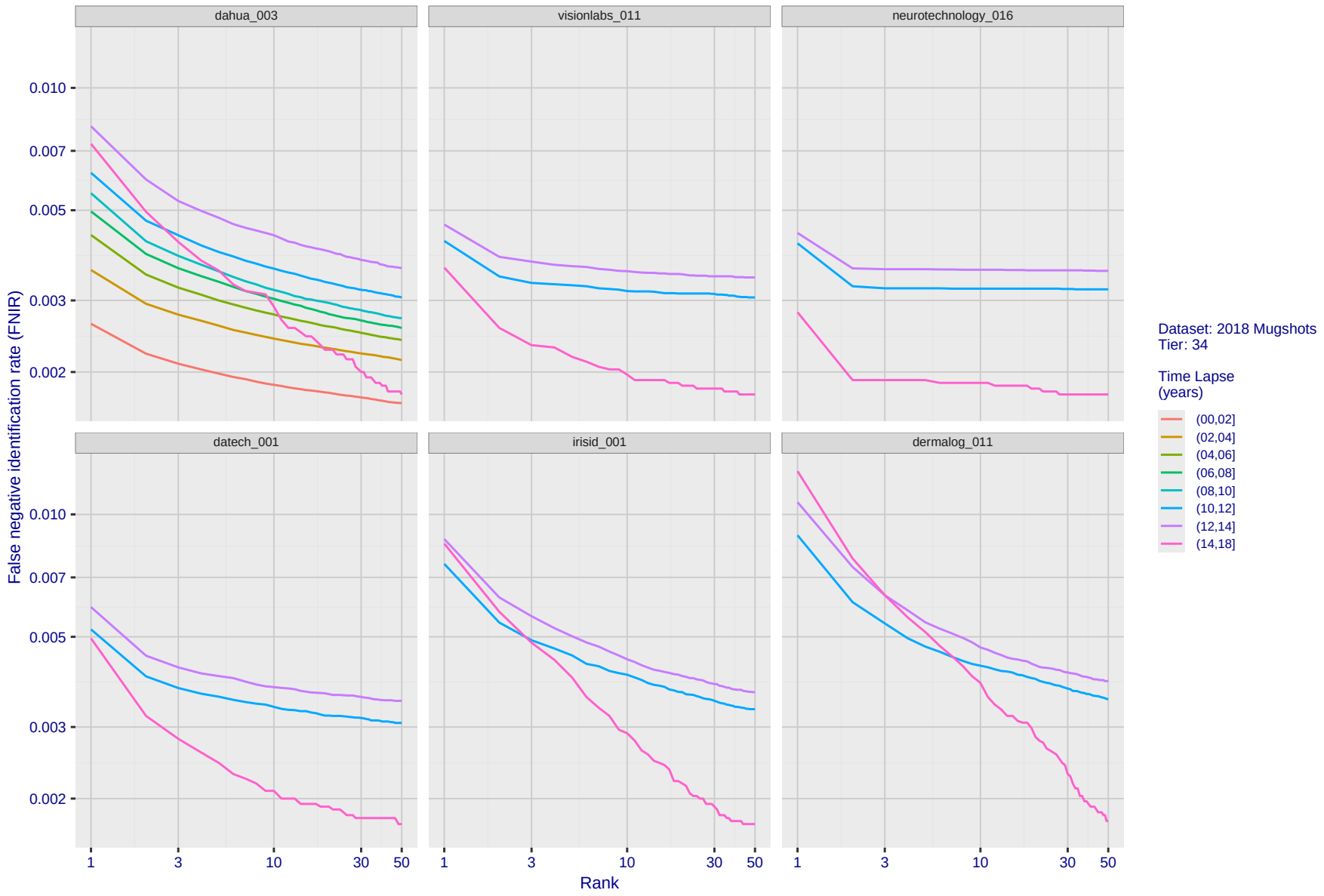


*Figure 181:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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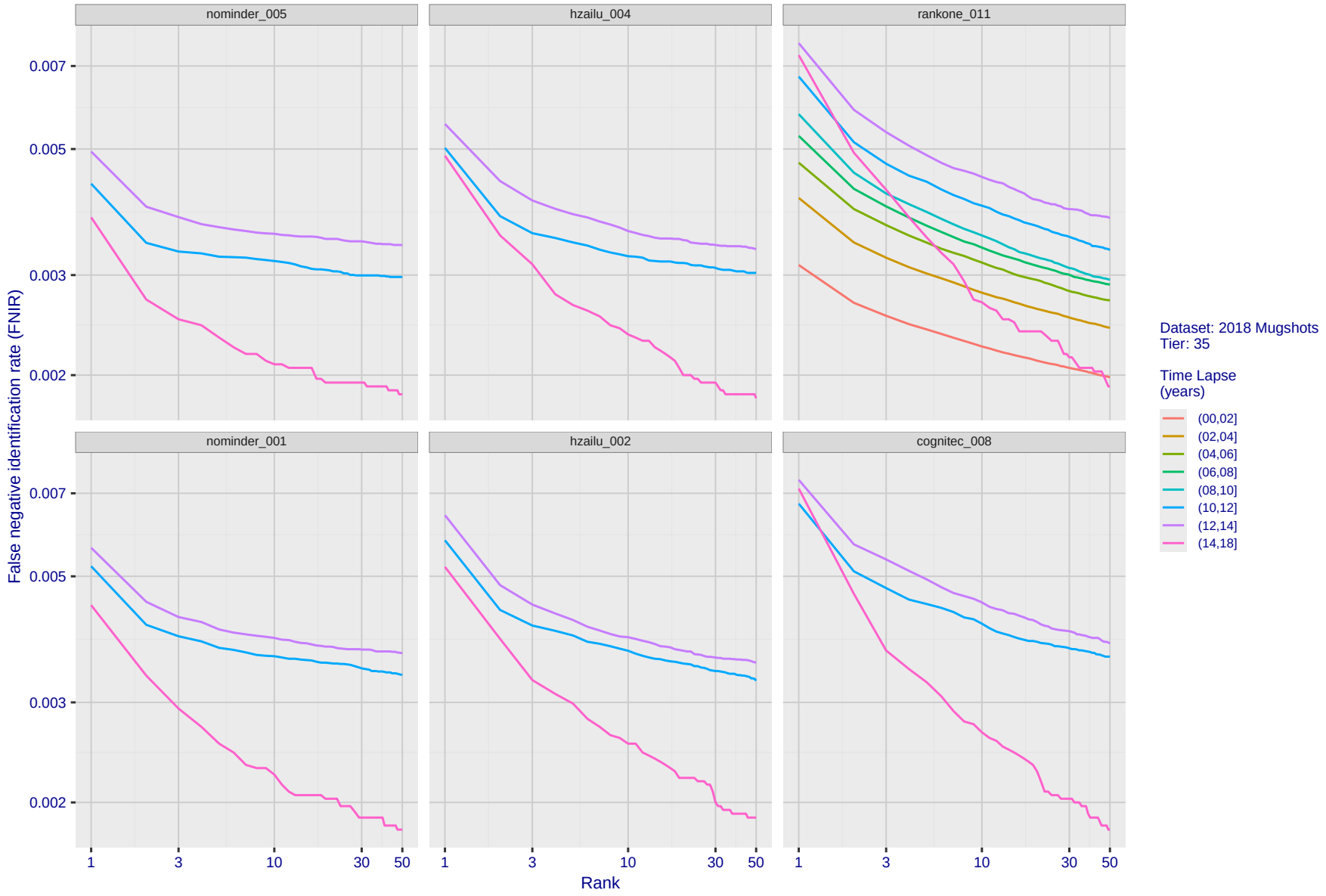


*Figure 182:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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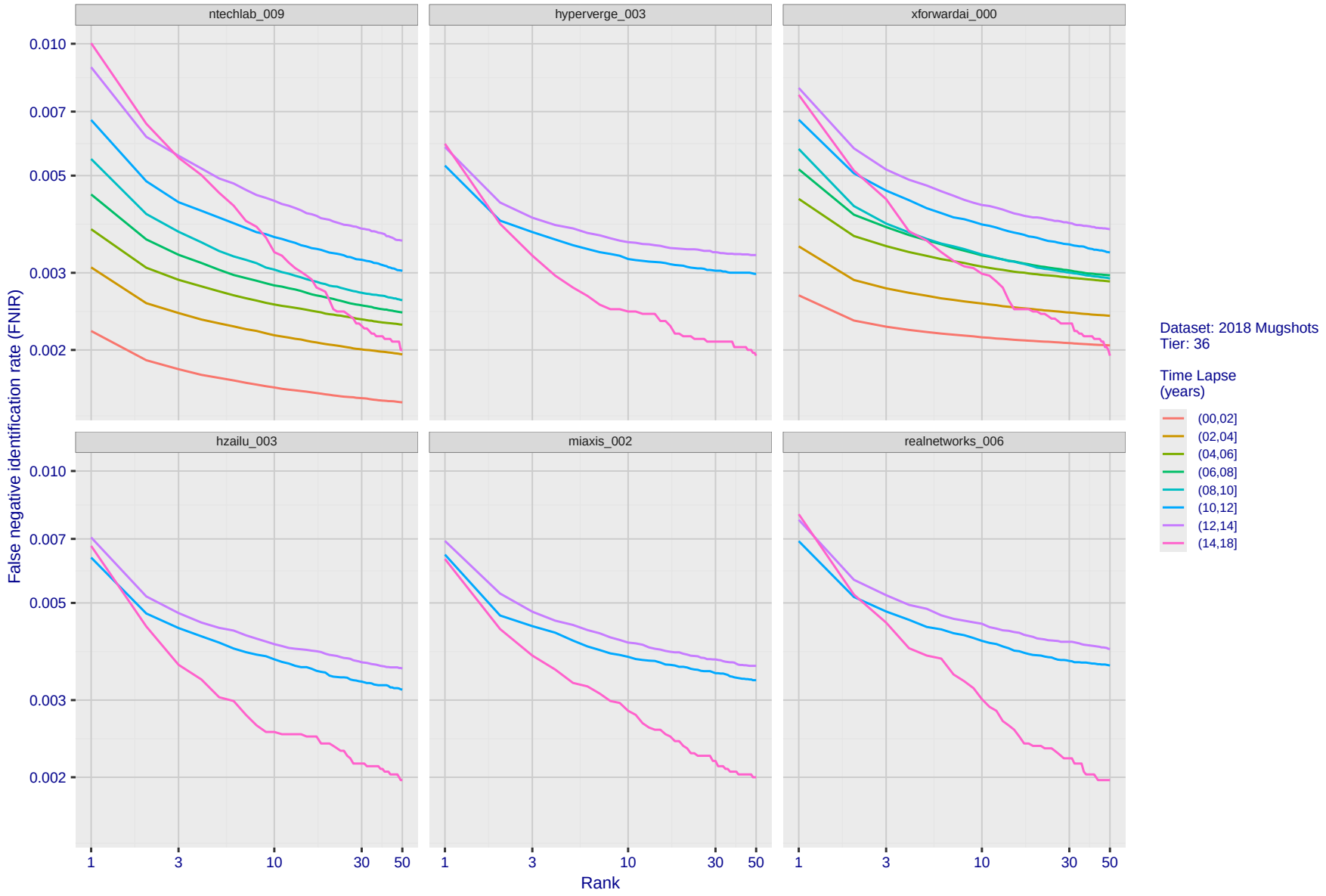


*Figure 183:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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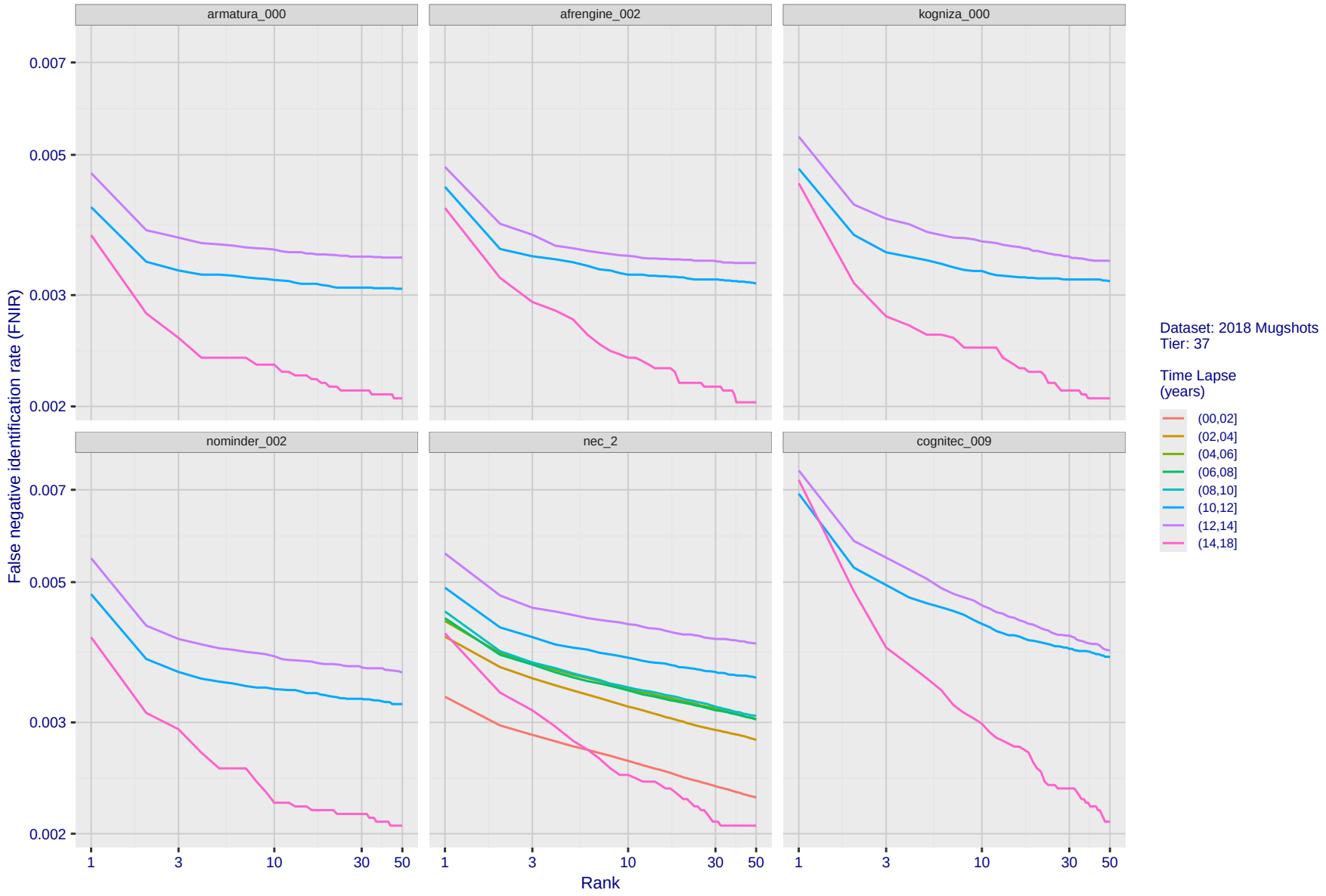


*Figure 184:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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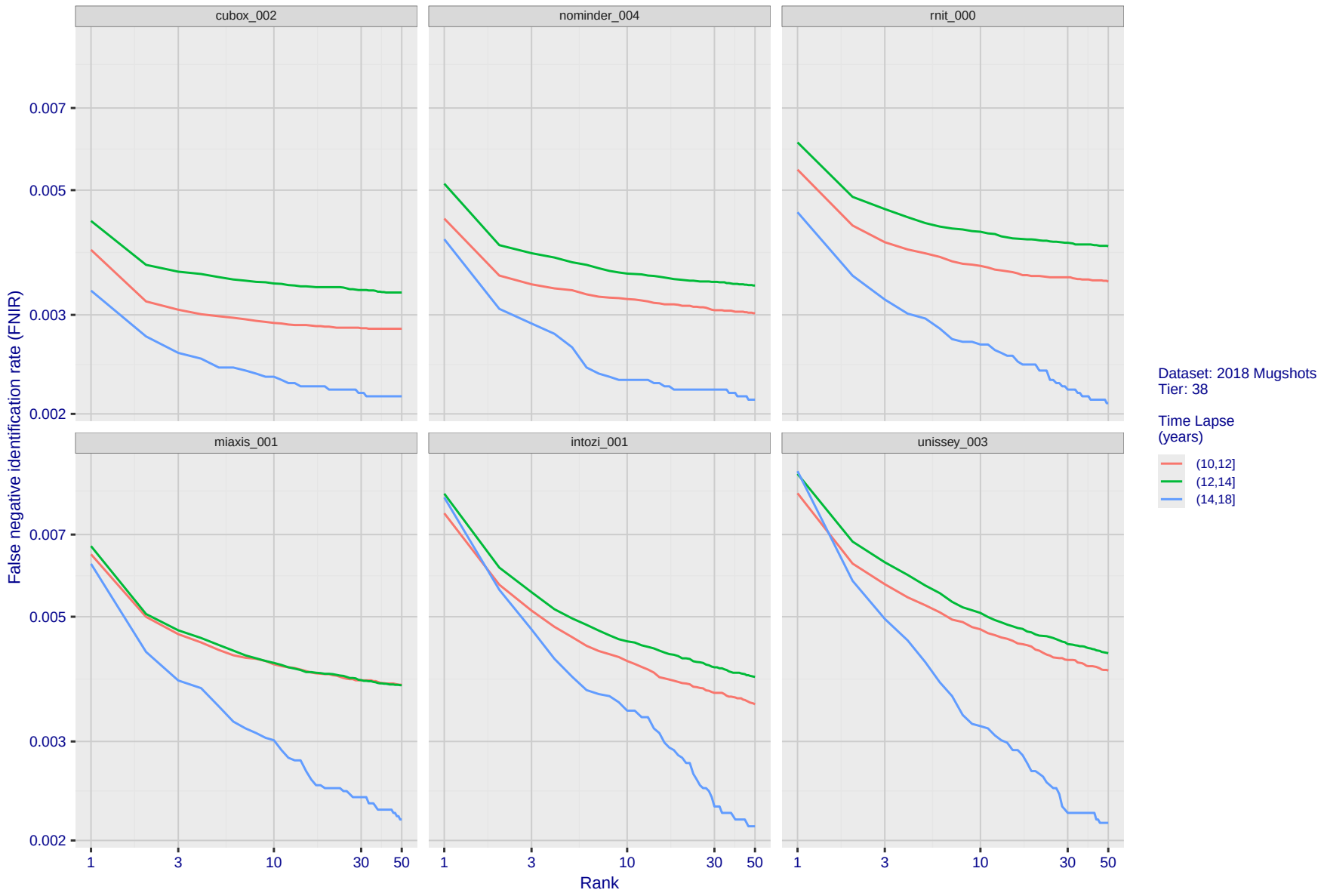


*Figure 185:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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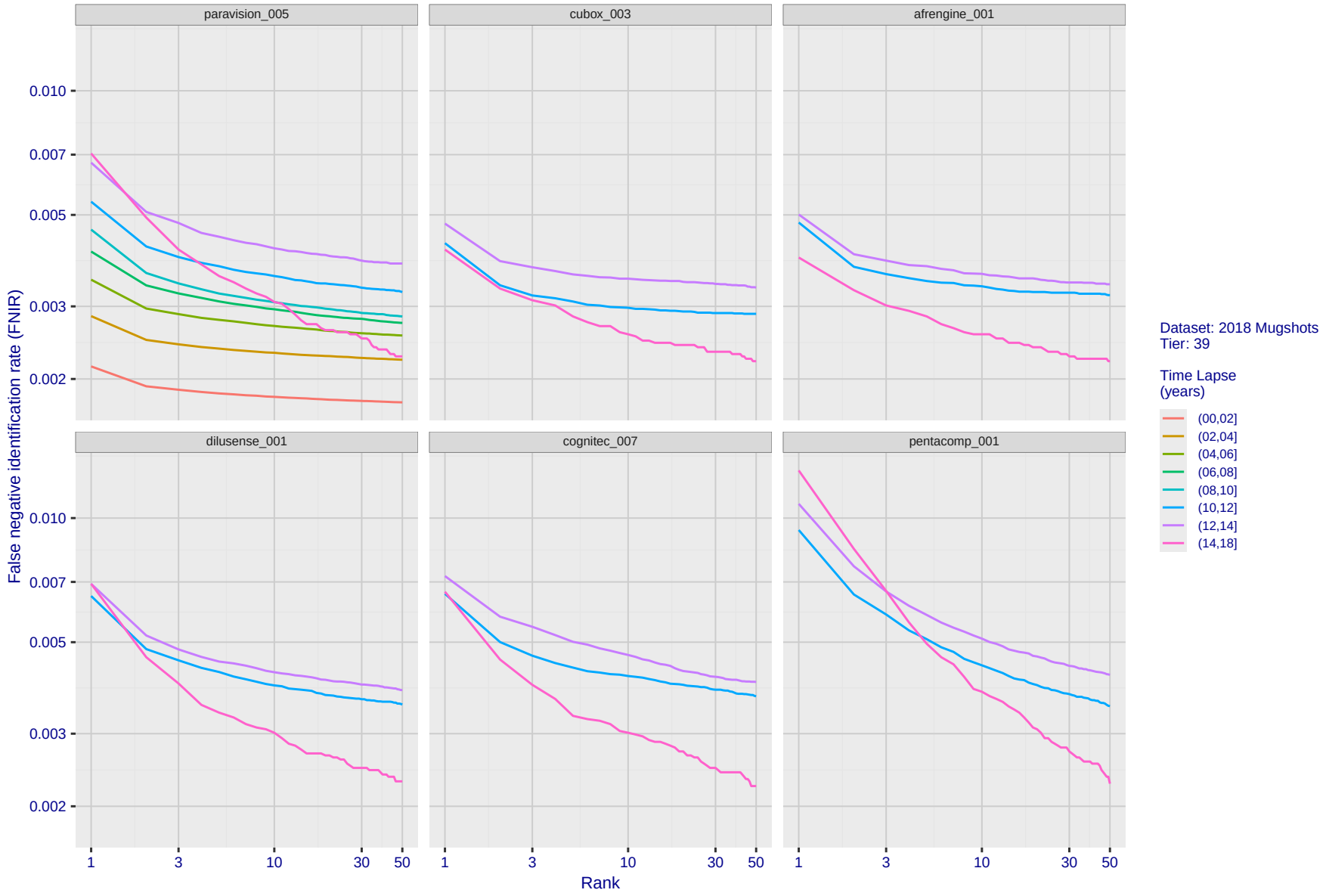


*Figure 186:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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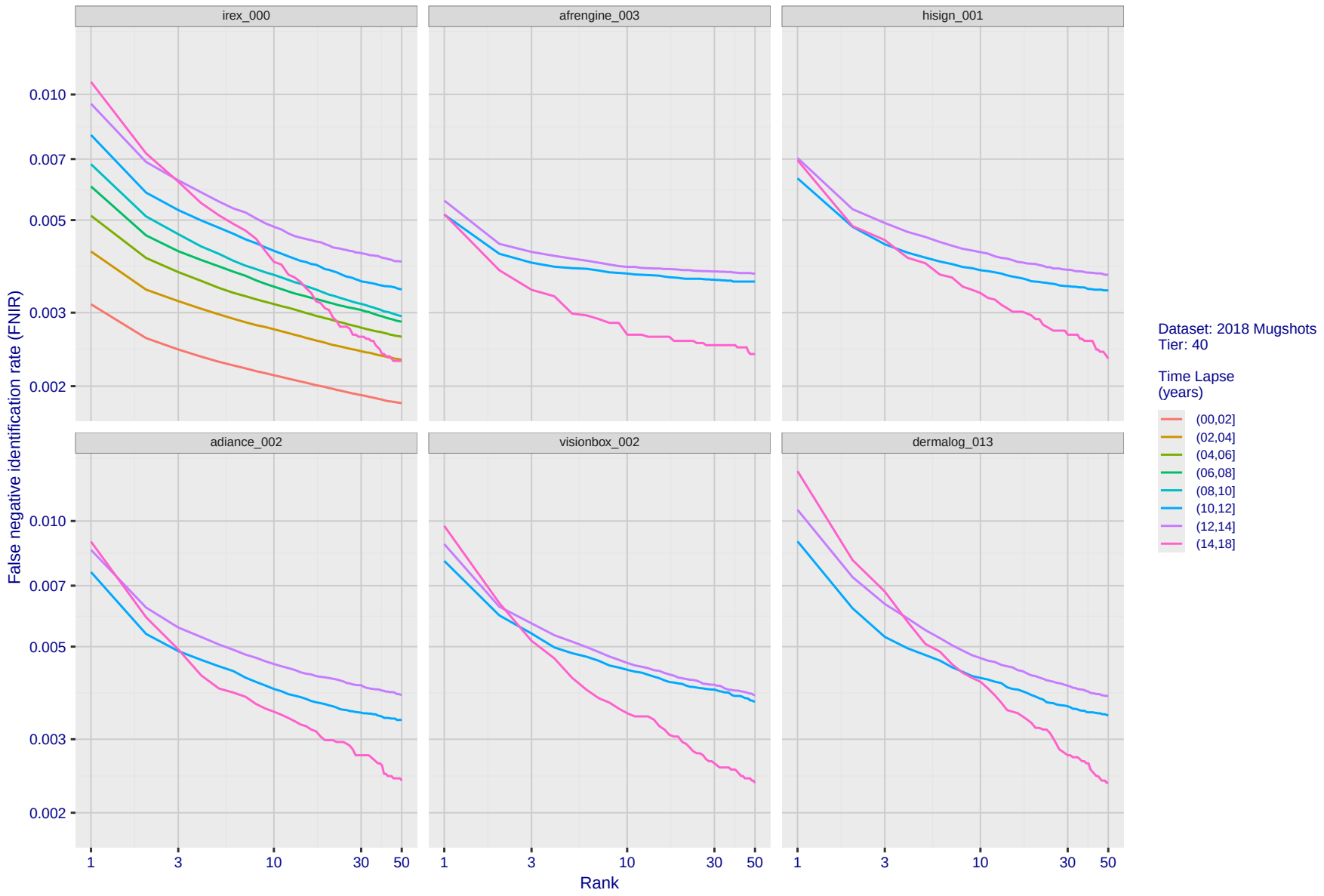


*Figure 187:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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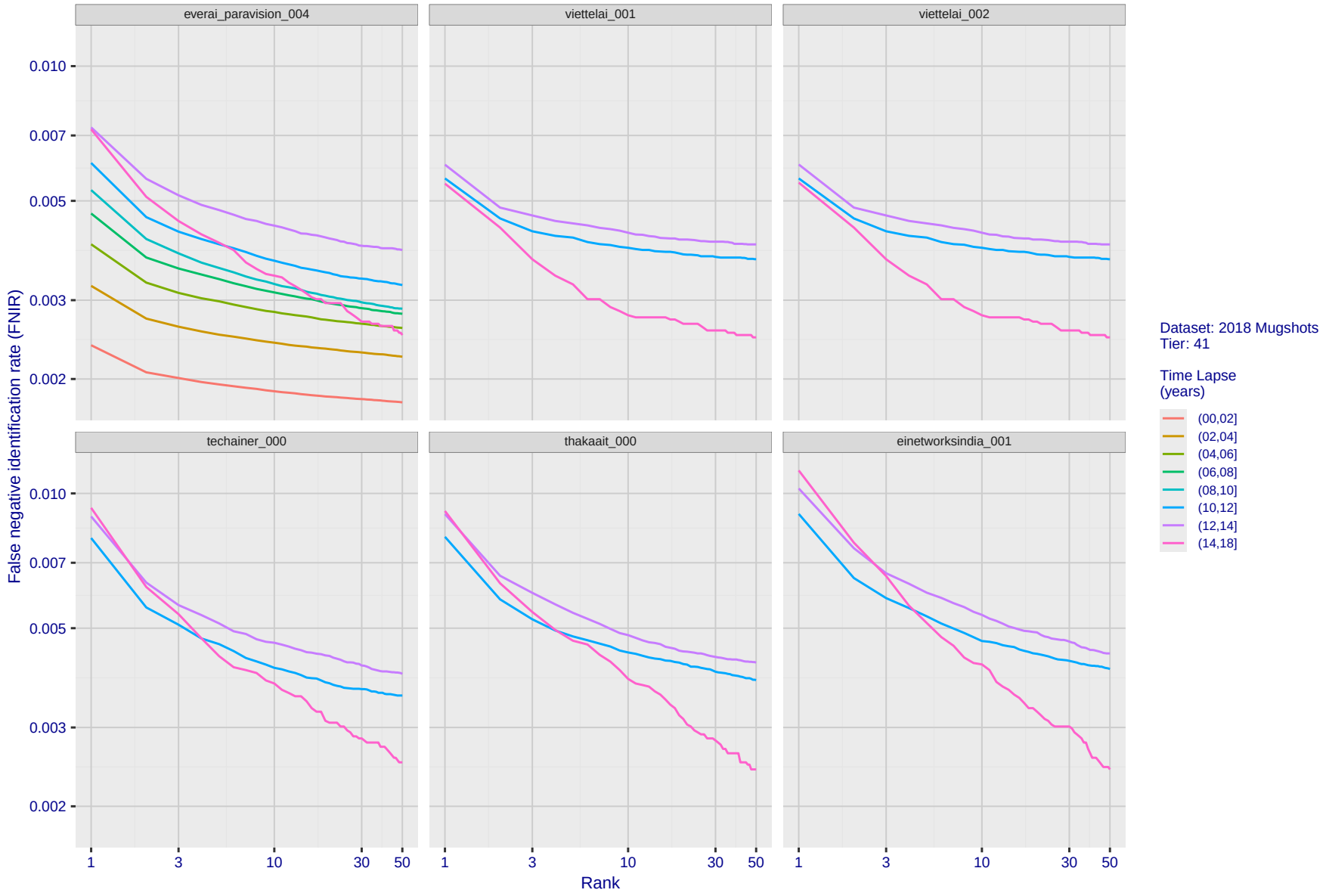


*Figure 188:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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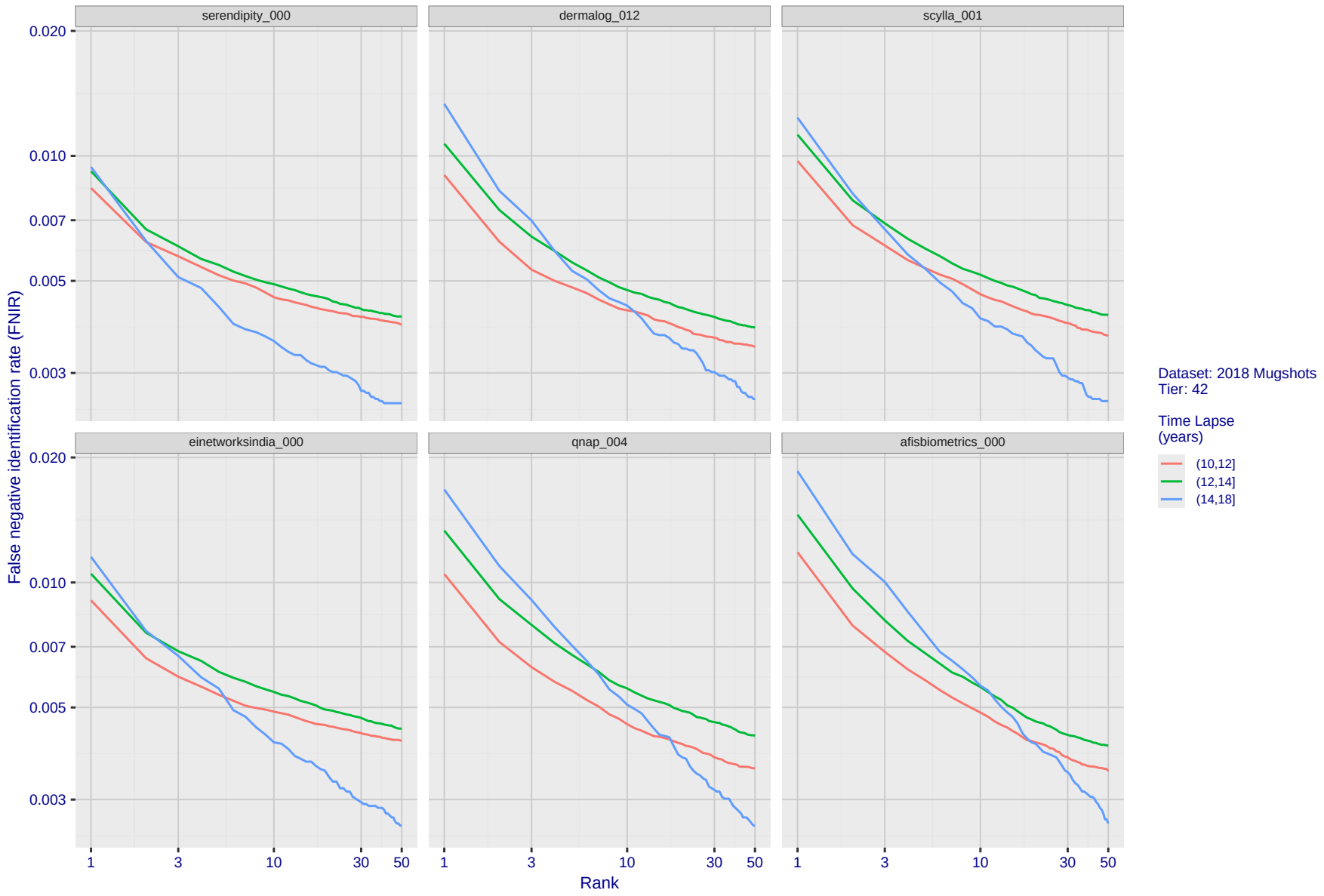


*Figure 189:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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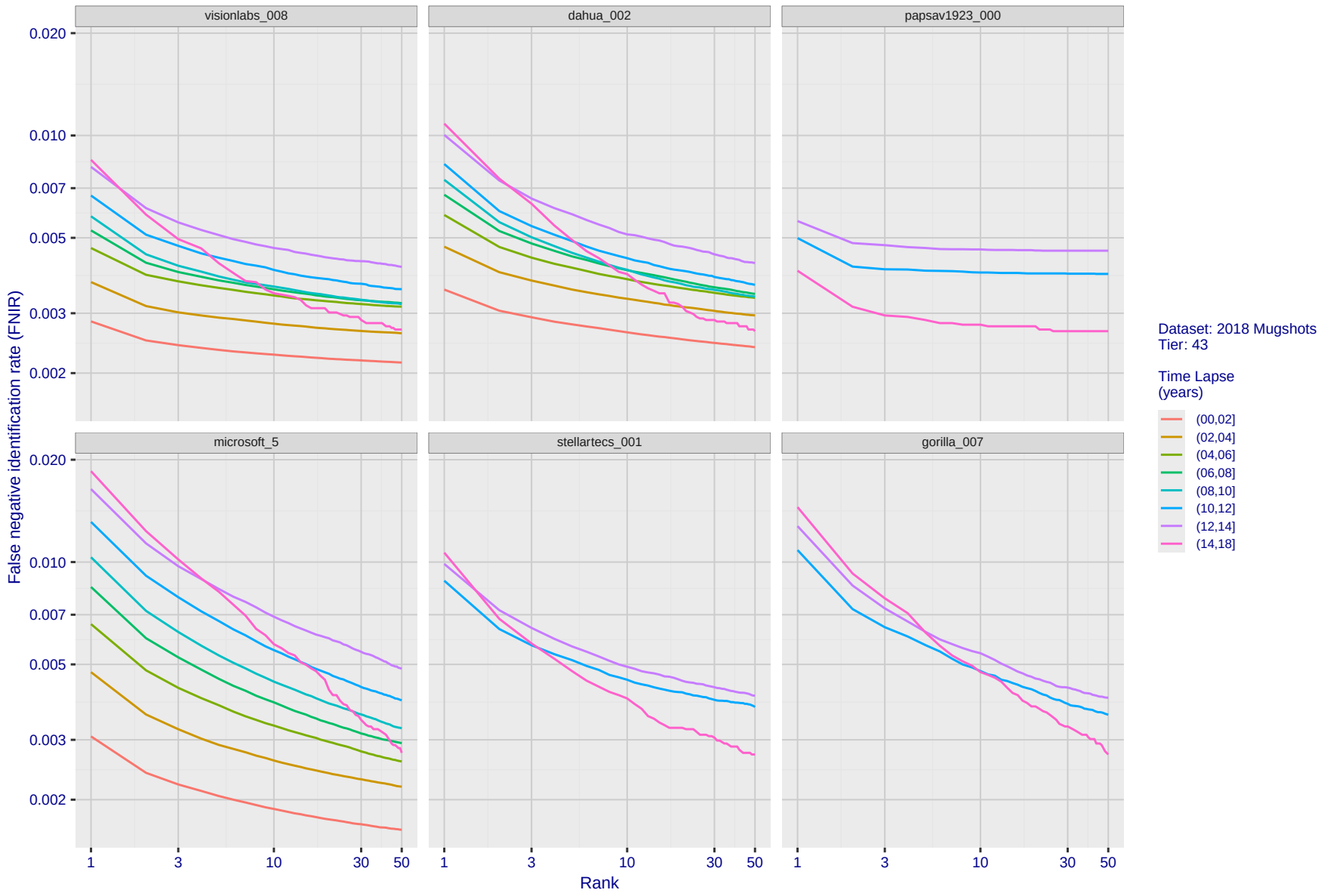


*Figure 190:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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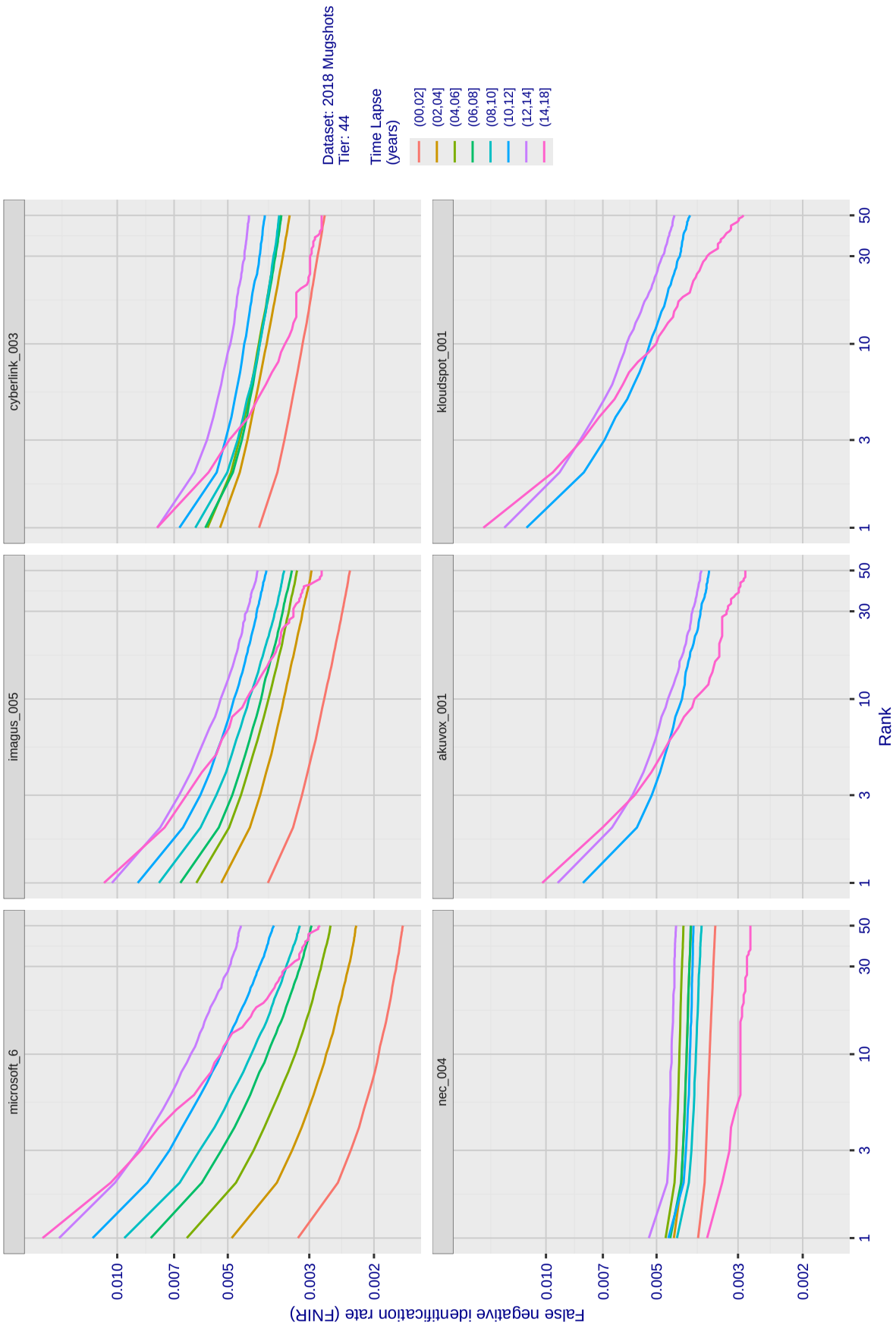


*Figure 191:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed.** *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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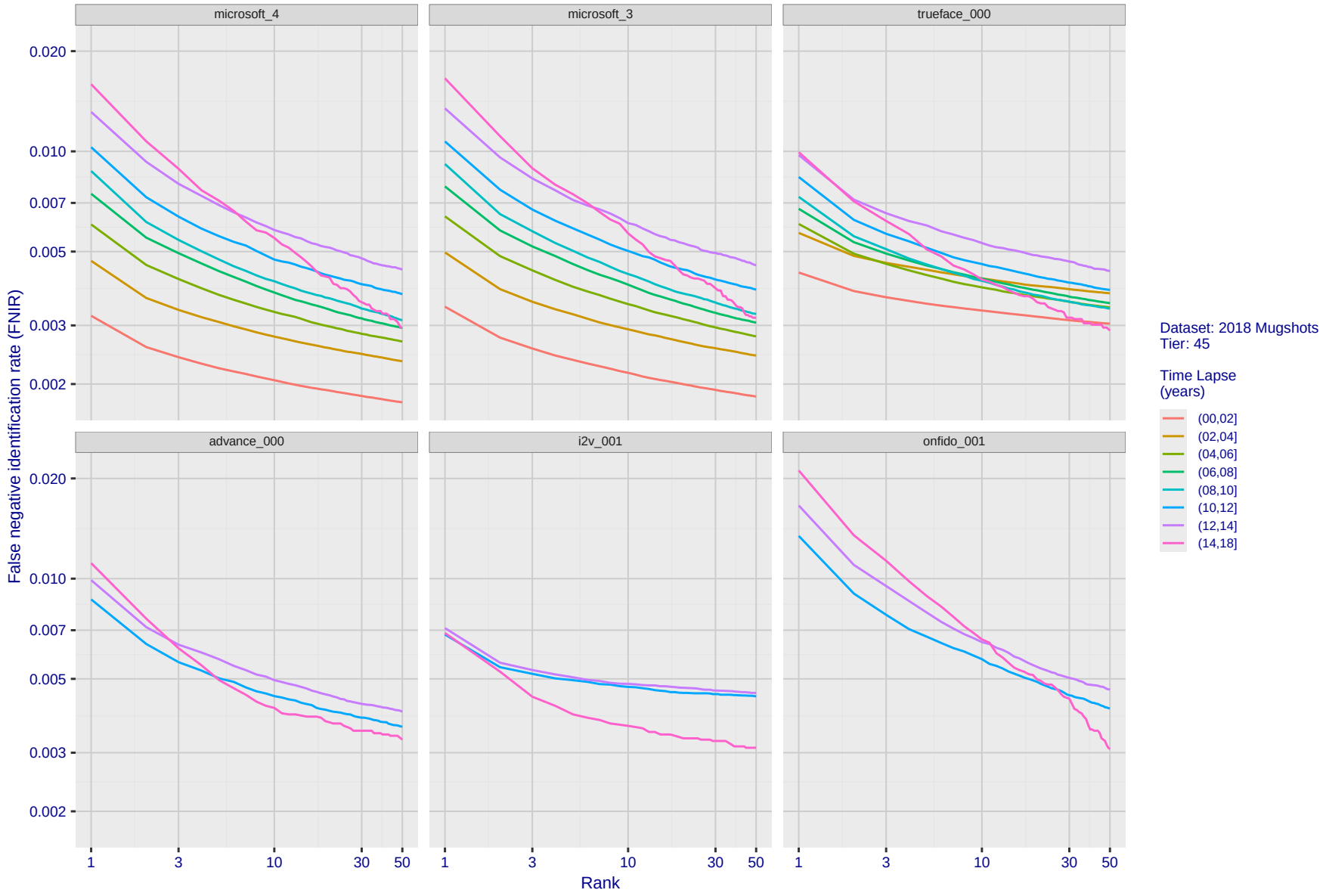


*Figure 192:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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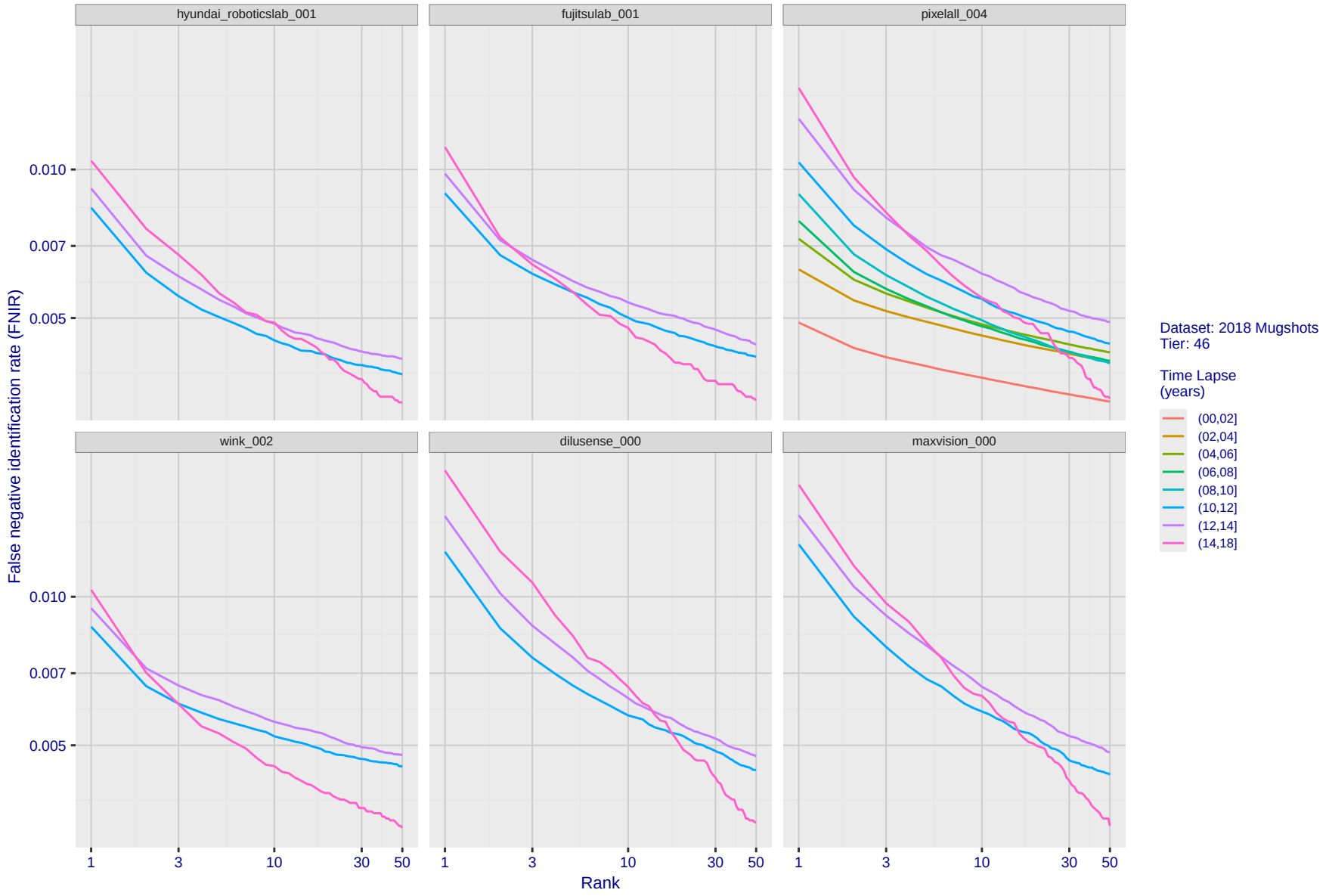


*Figure 193:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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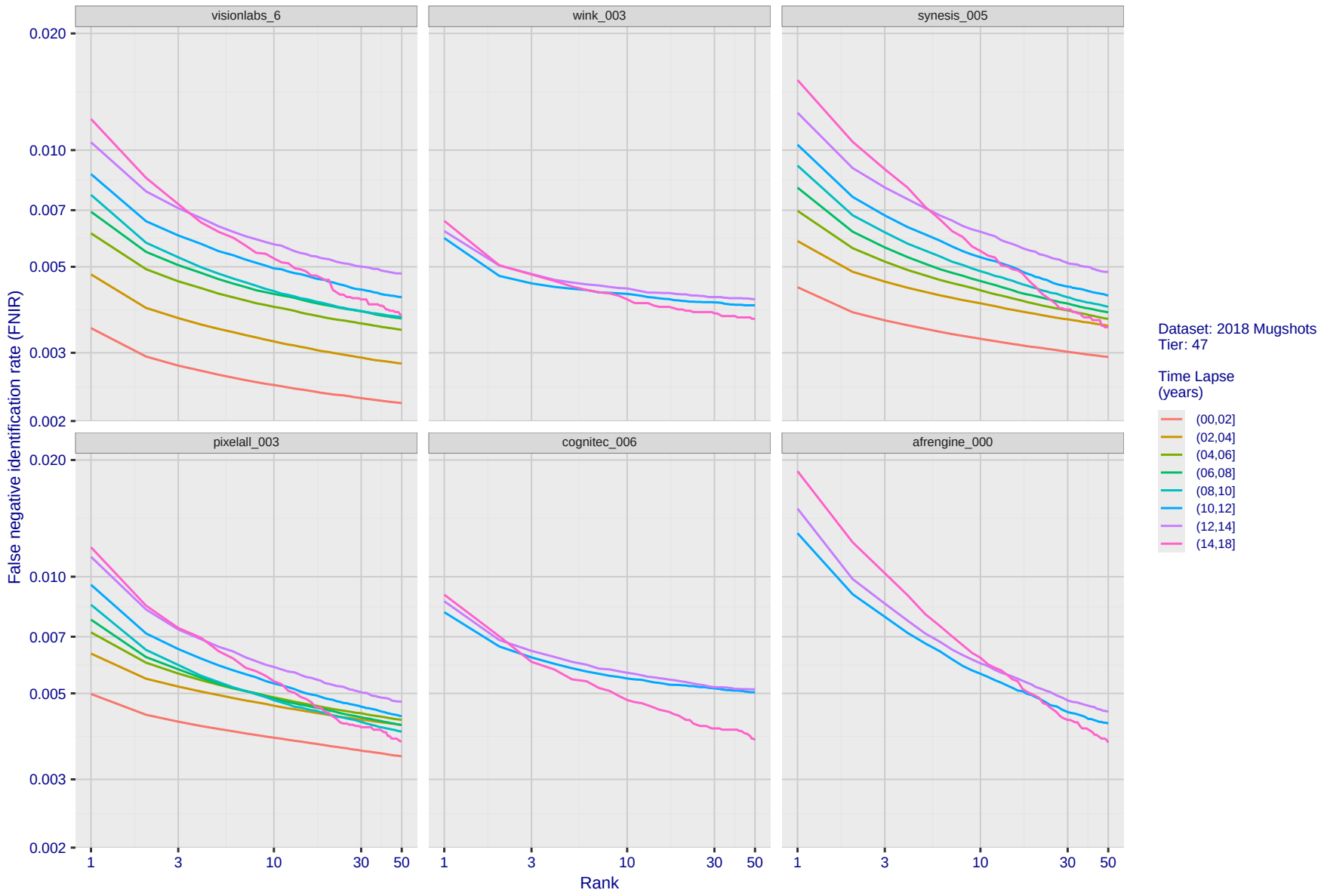


*Figure 194:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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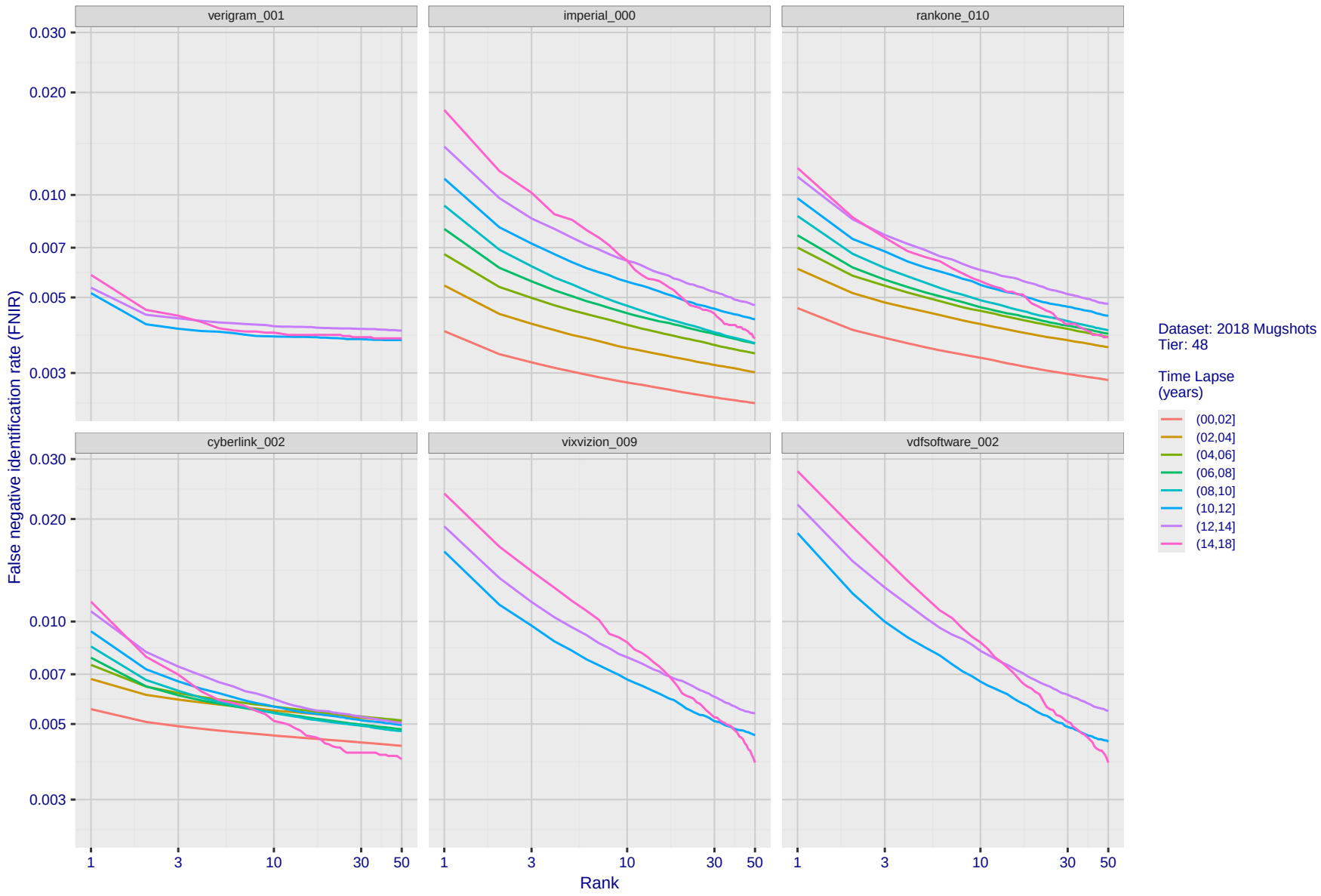


*Figure 195:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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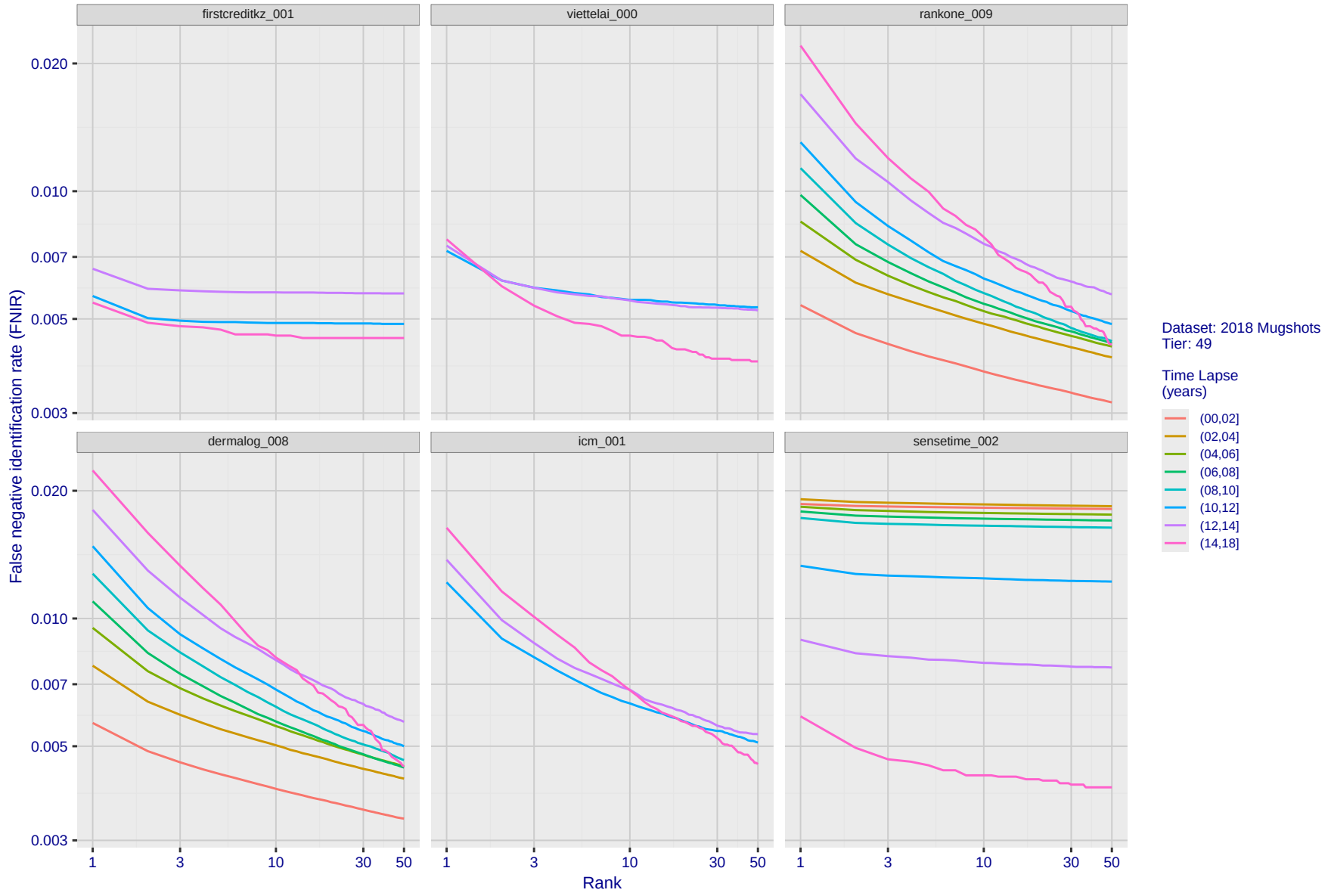


*Figure 196:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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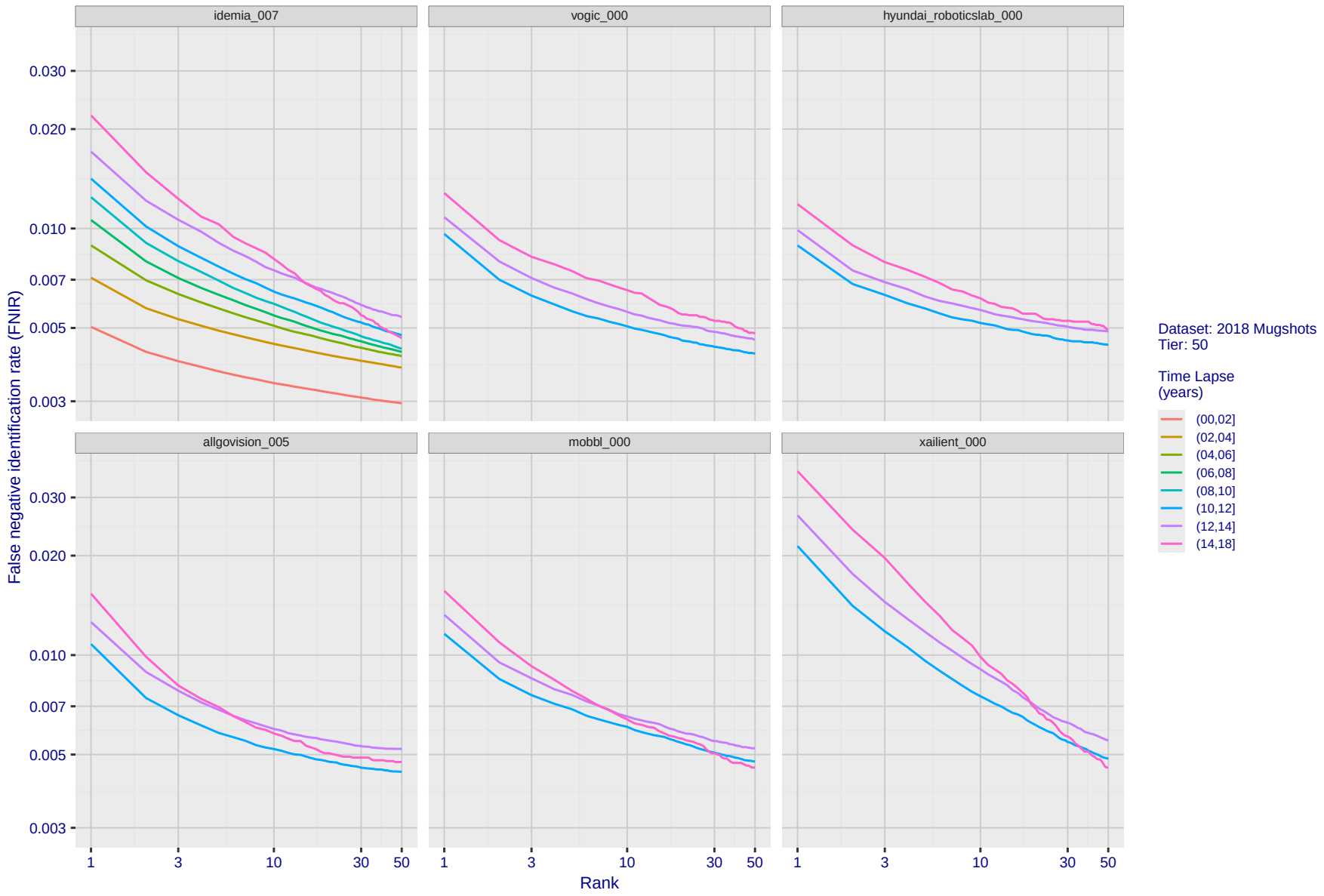


*Figure 197:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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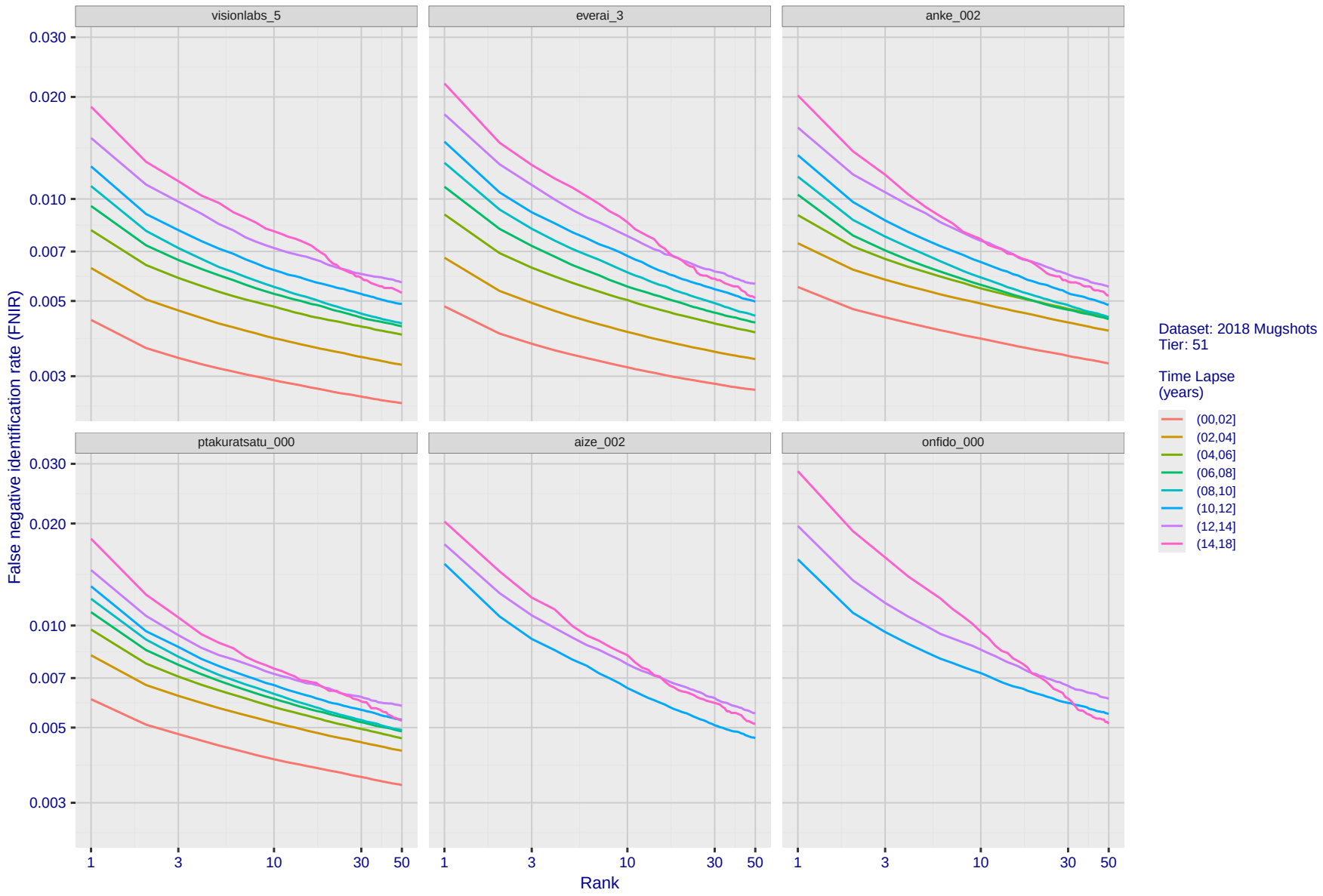


*Figure 198:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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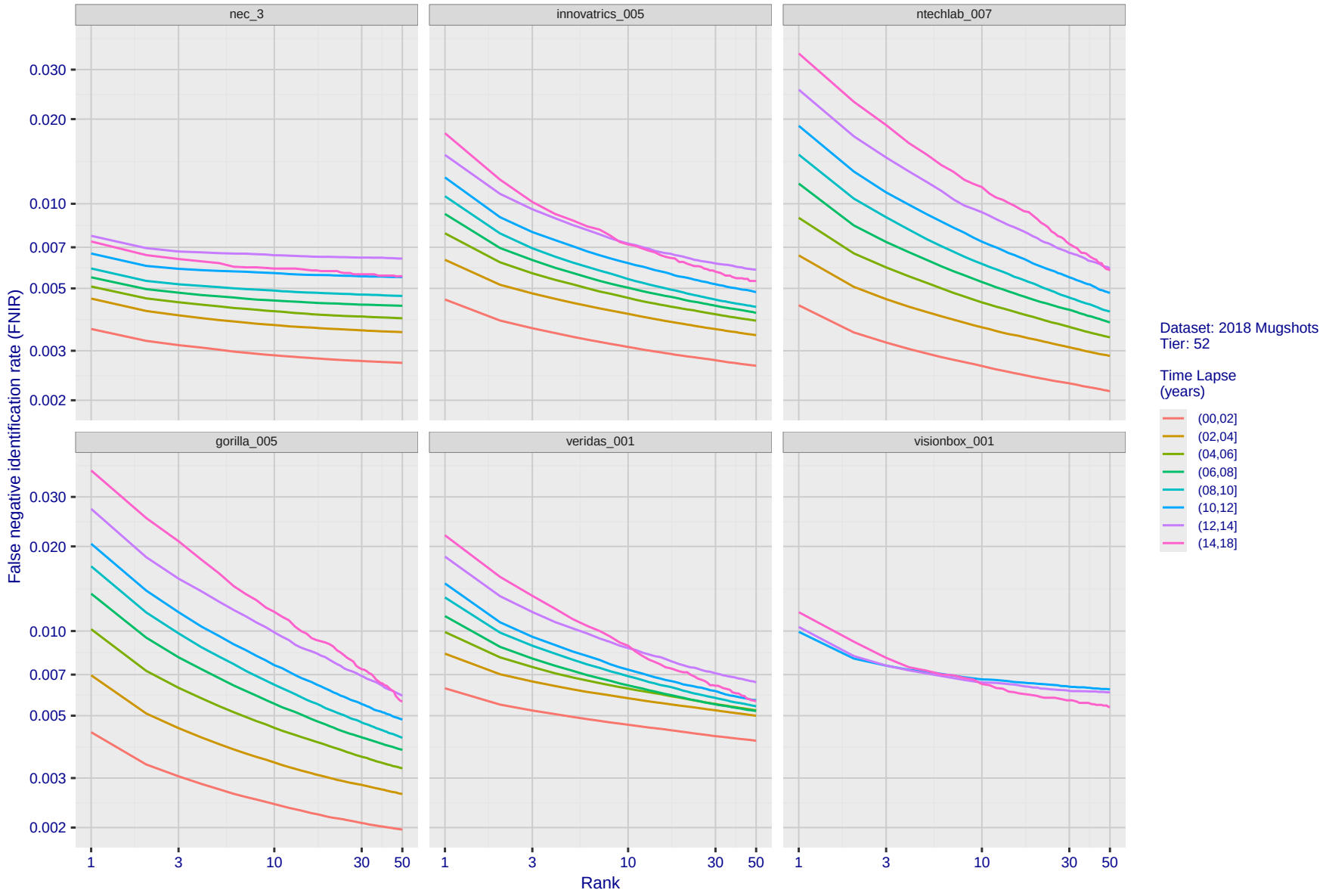


*Figure 199:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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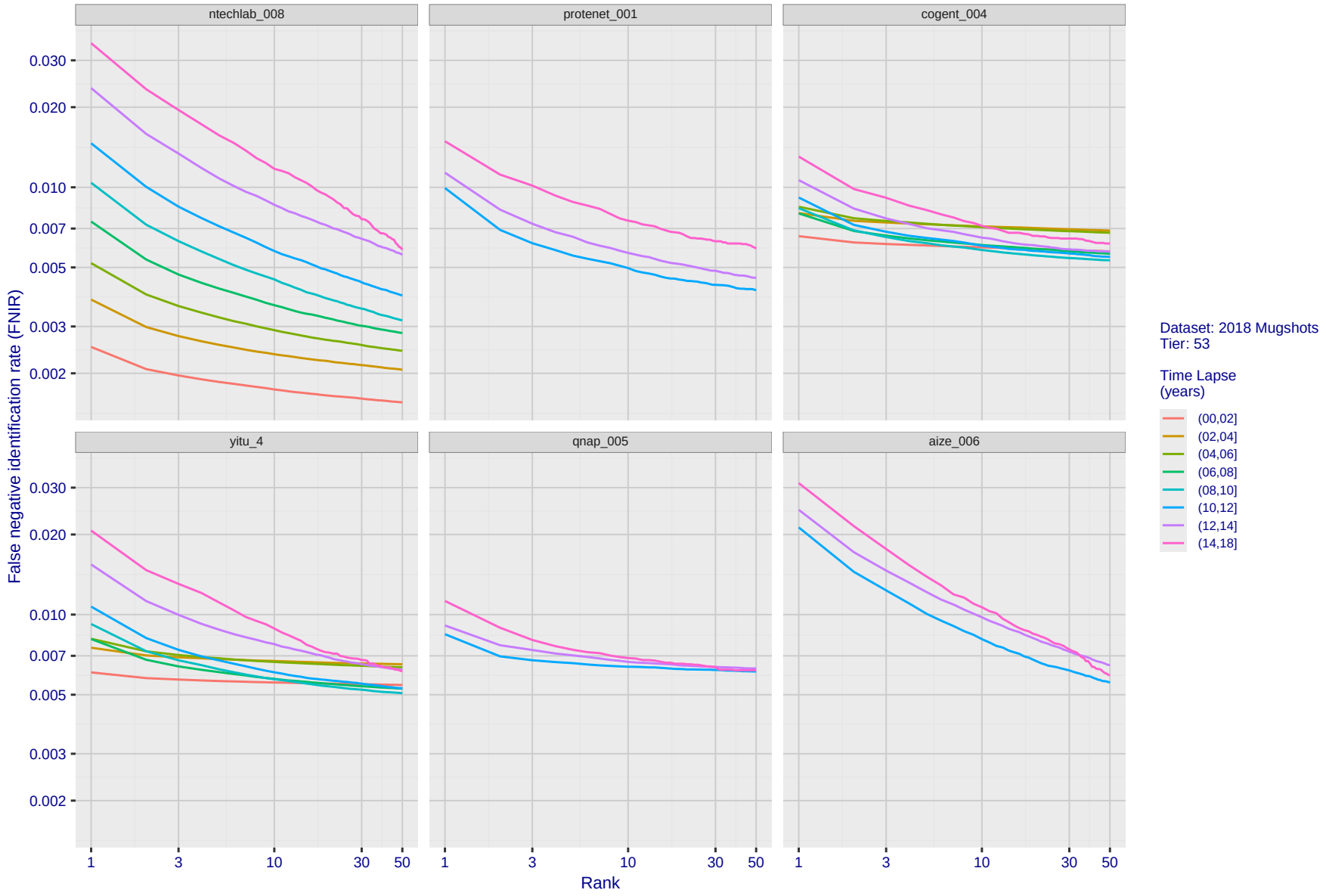


*Figure 200:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



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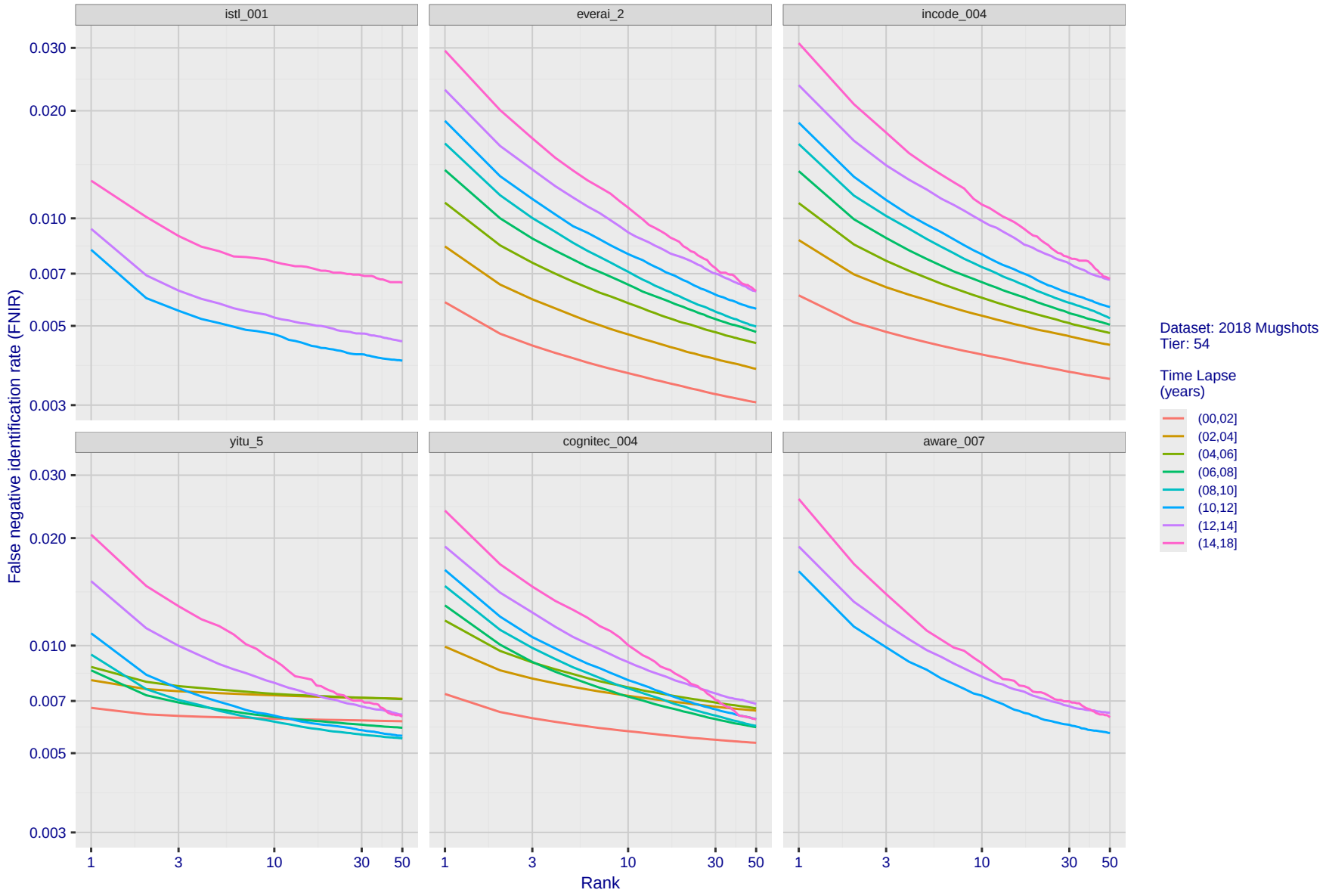


*Figure 201:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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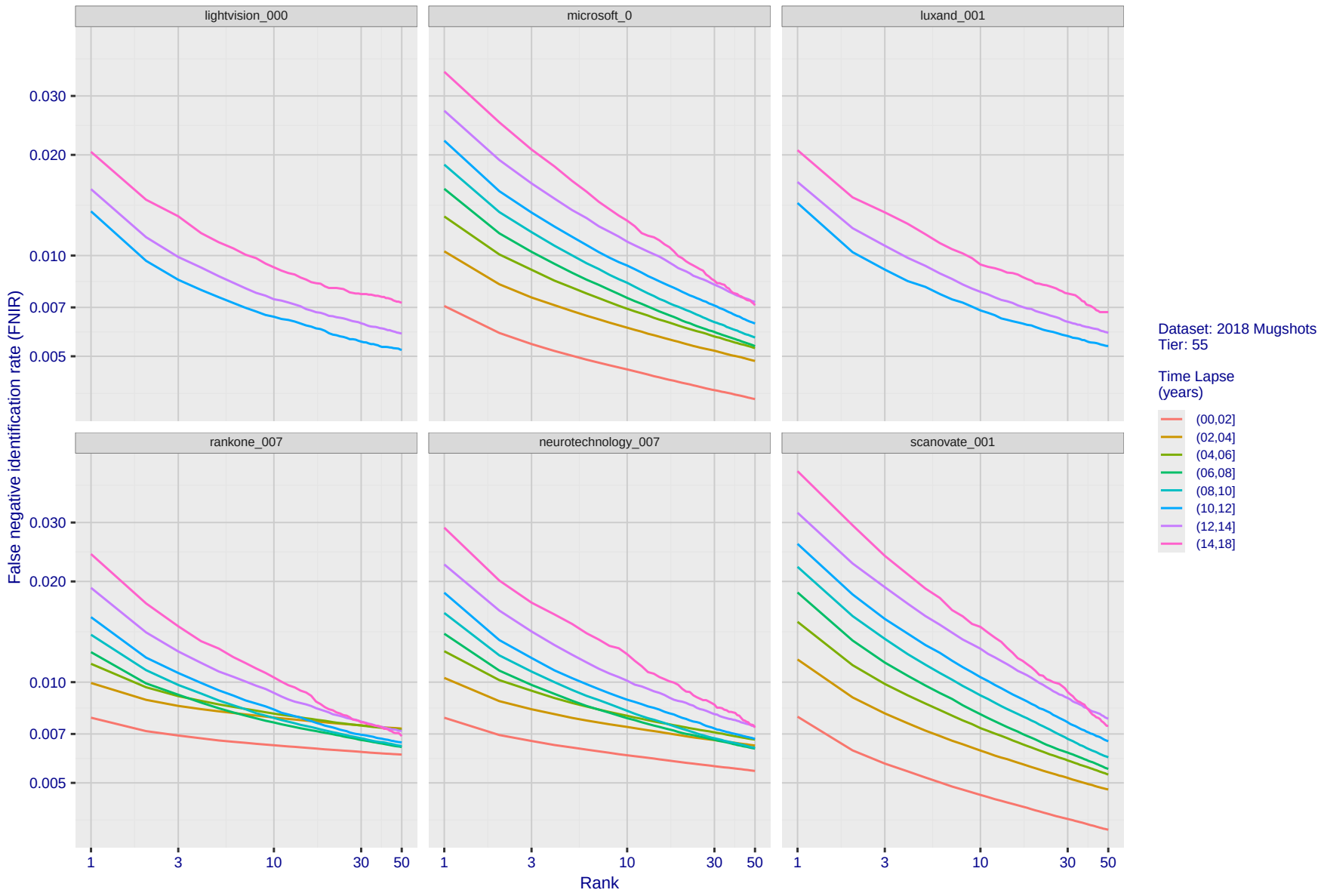


*Figure 202:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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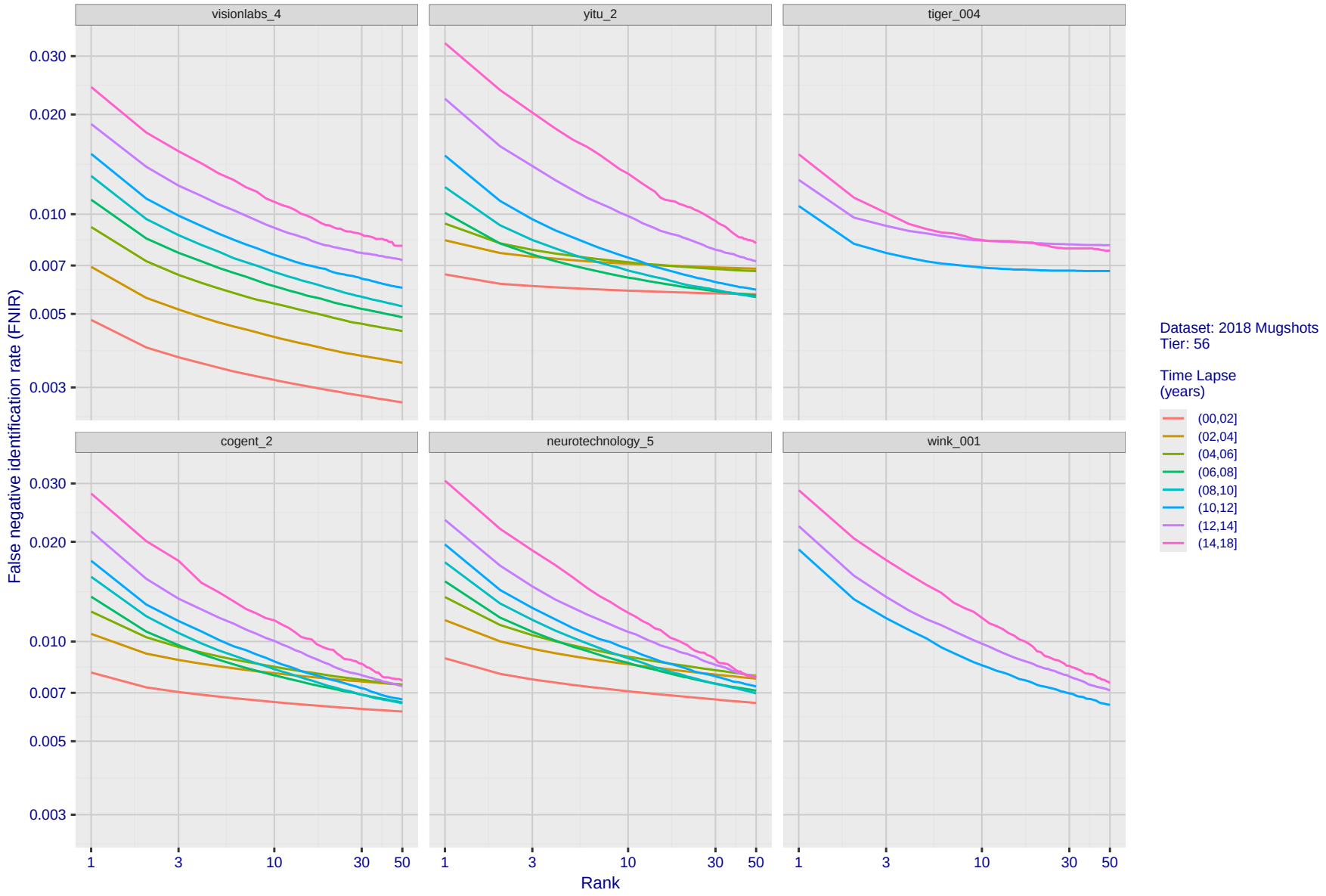


*Figure 203:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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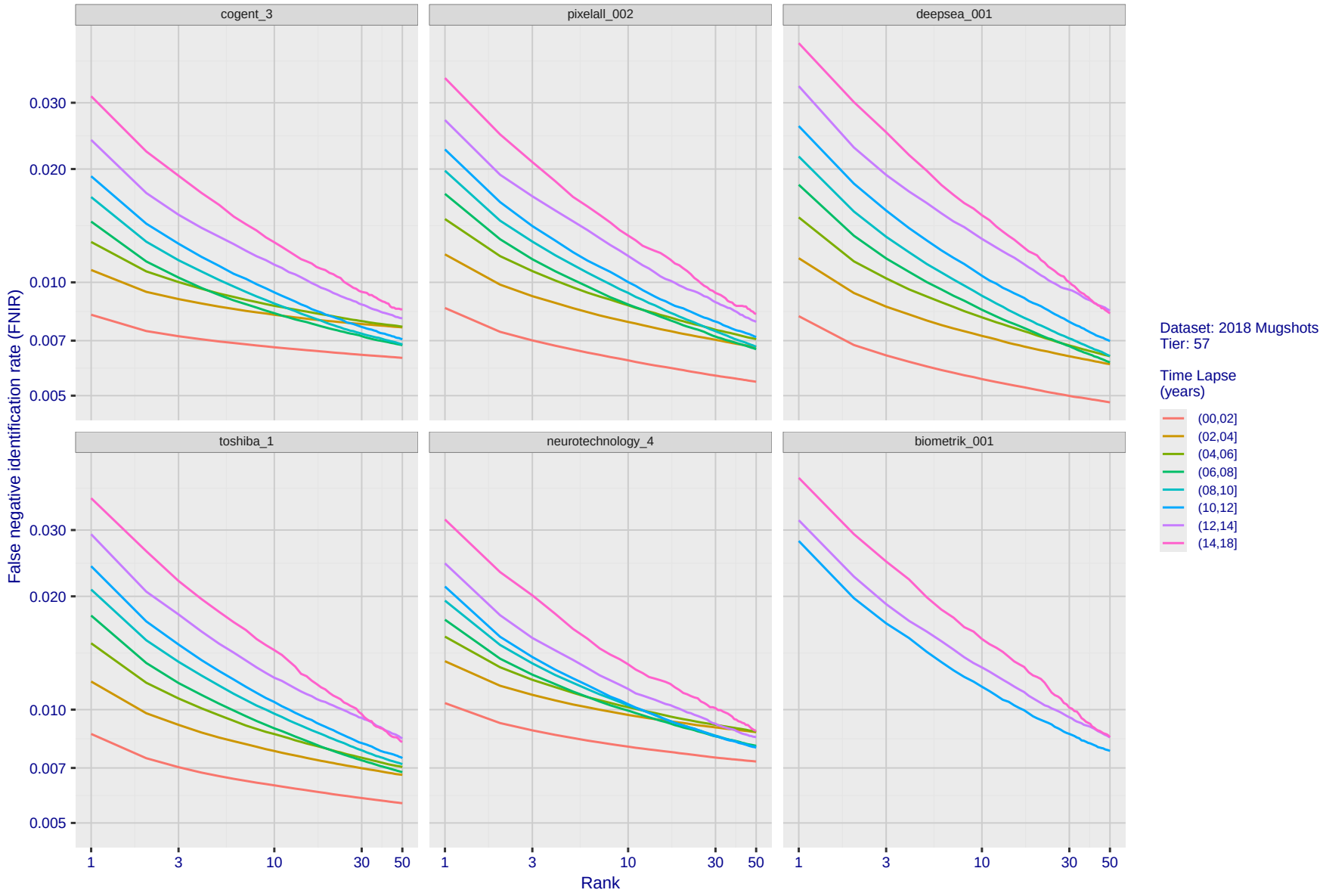


*Figure 204:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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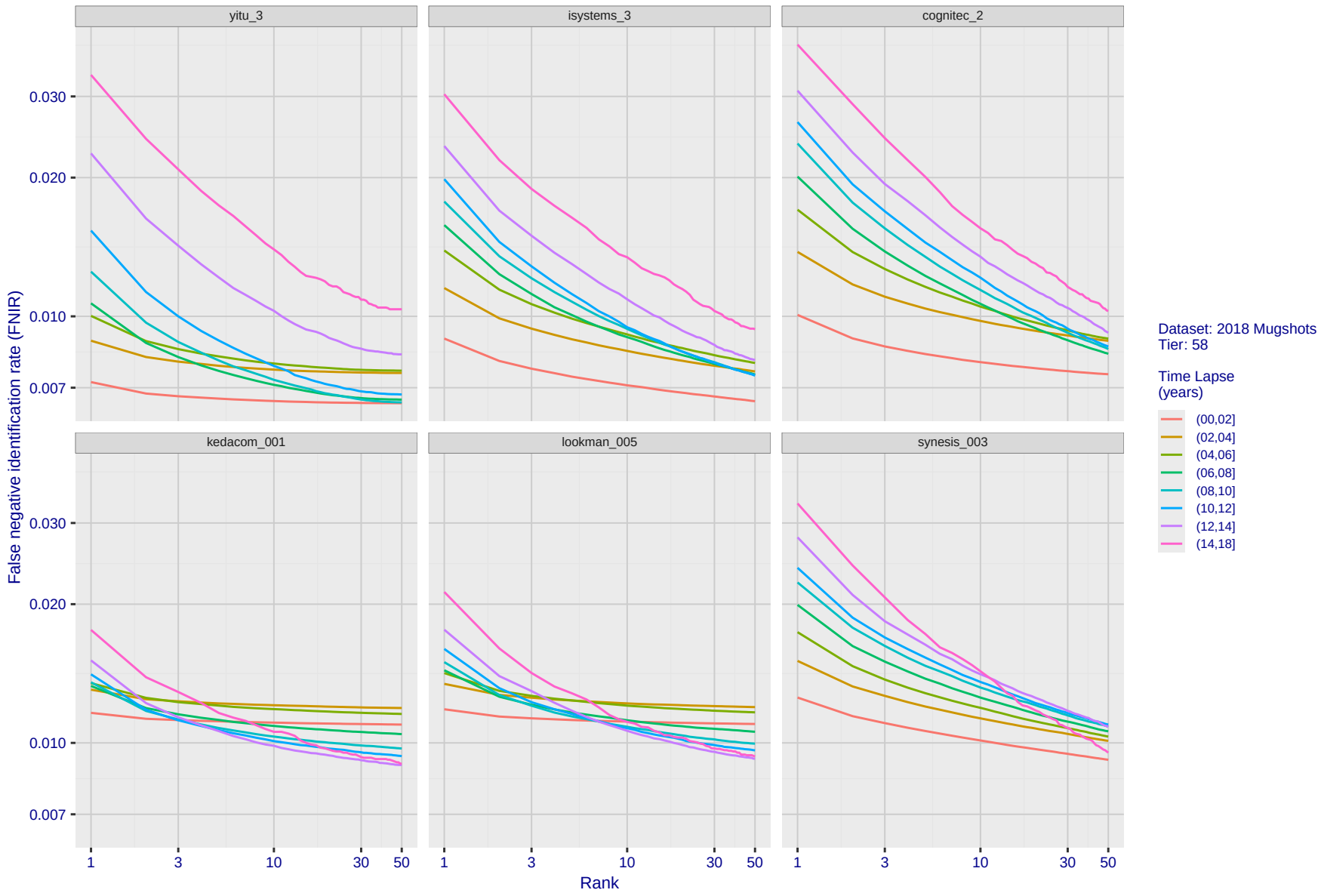


*Figure 205:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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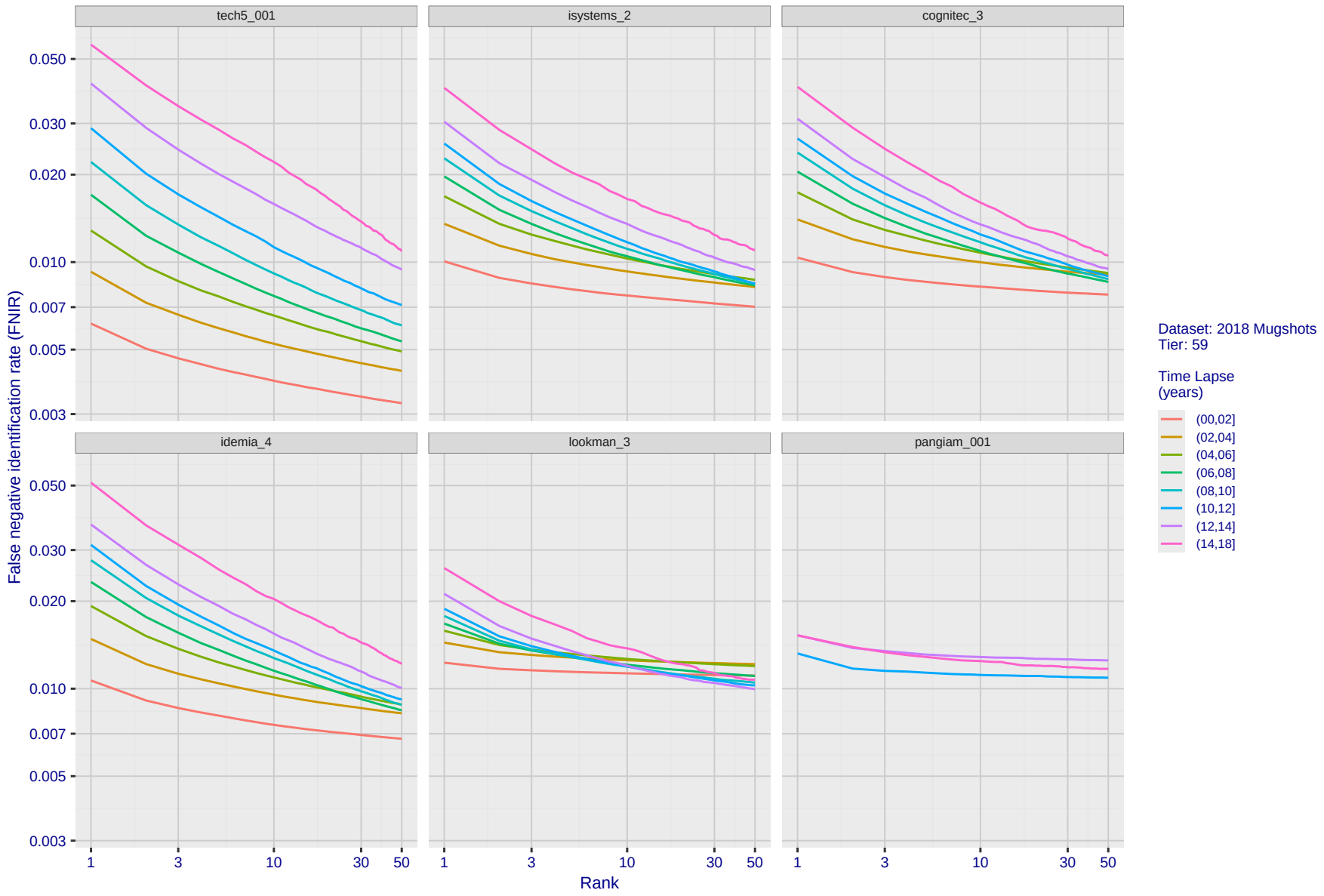


*Figure 206:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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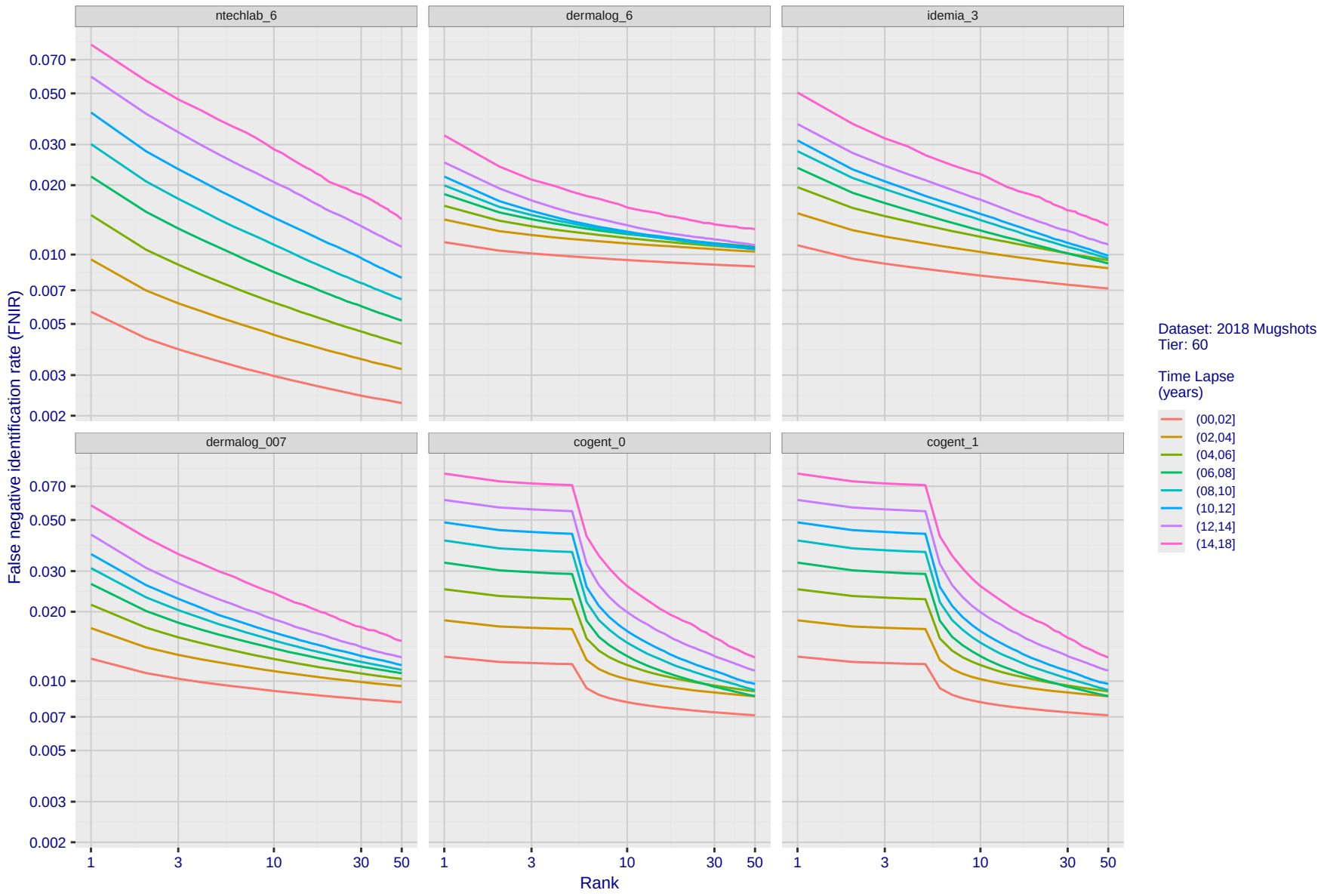


*Figure 207:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
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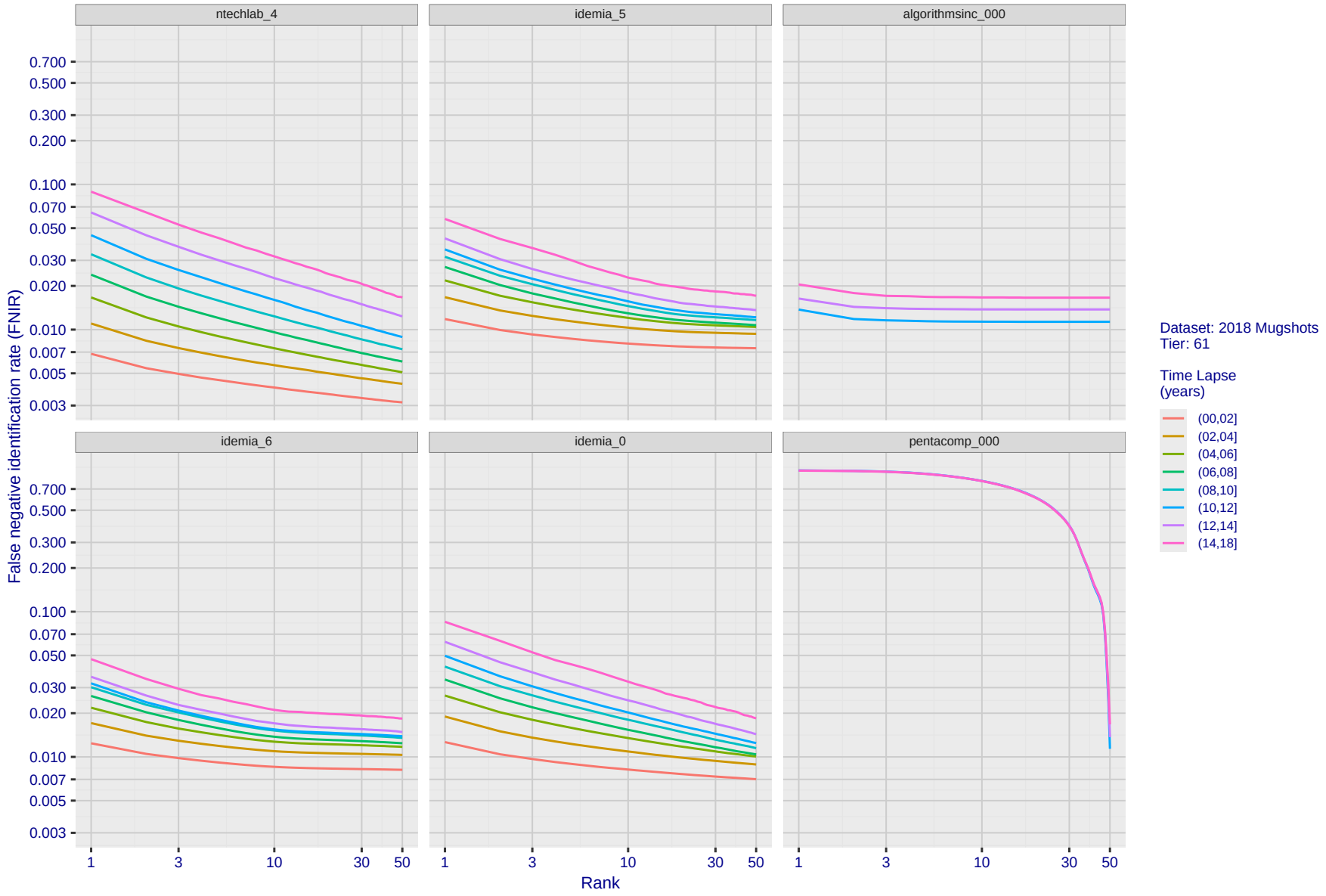


*Figure 208:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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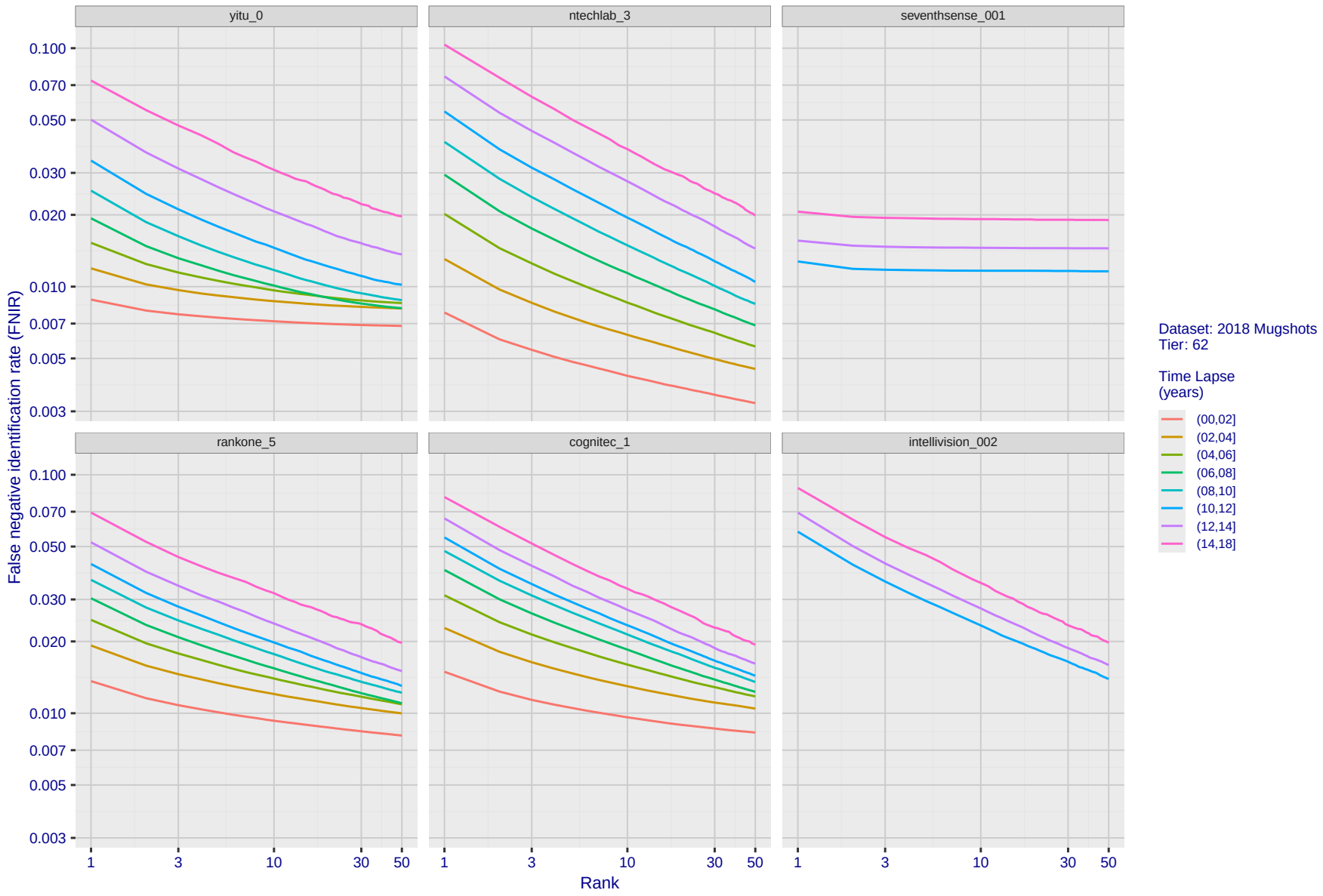


*Figure 209:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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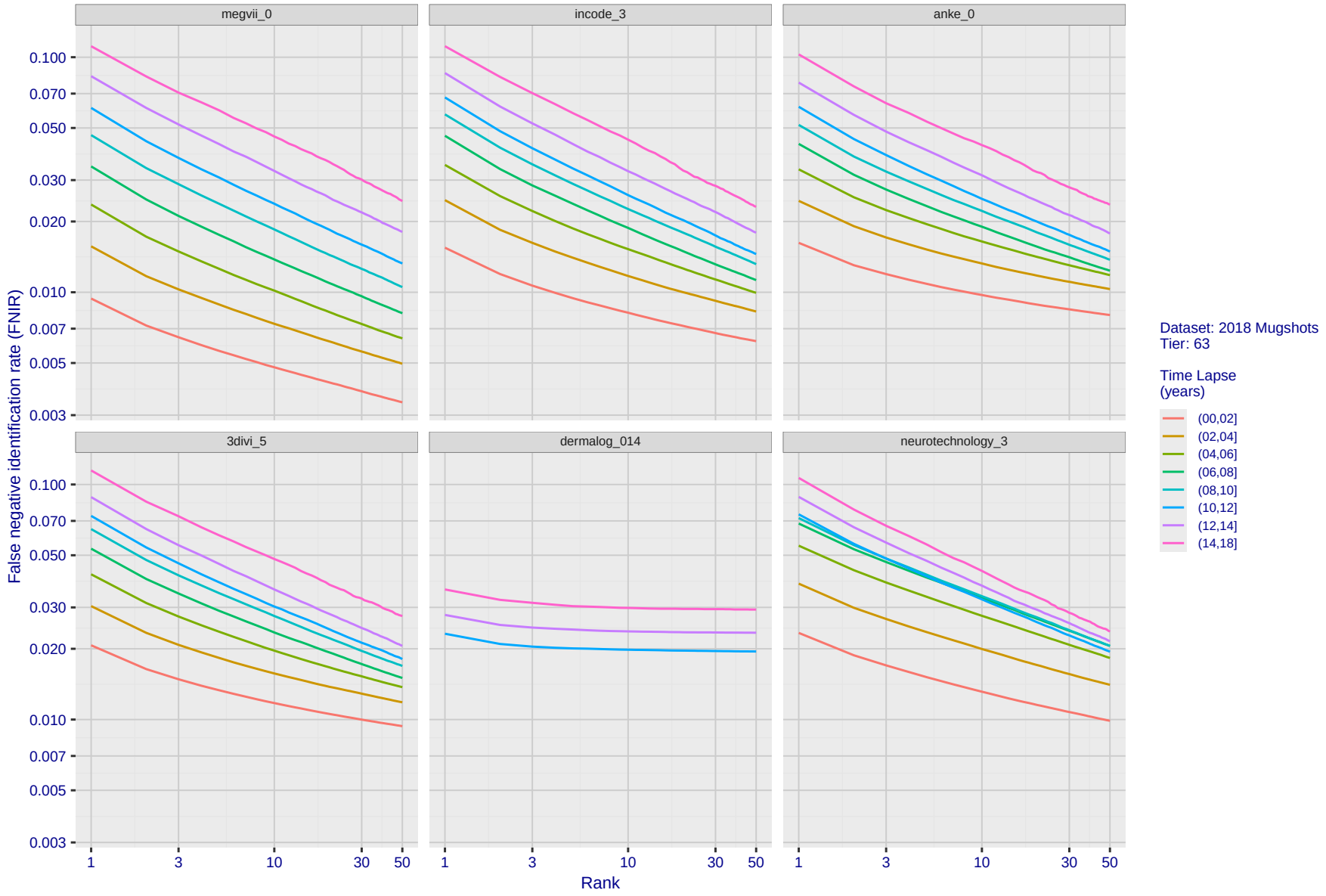


*Figure 210:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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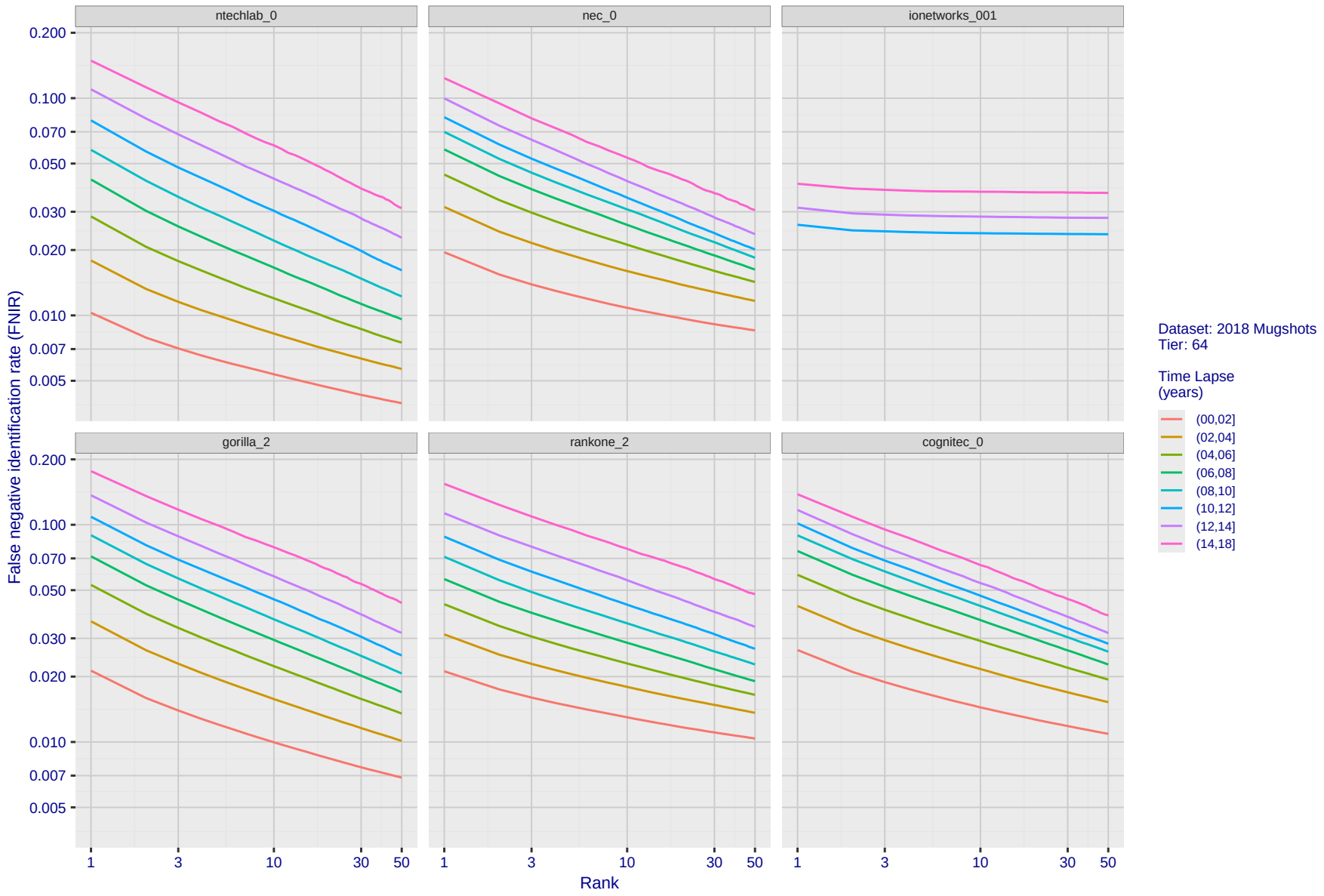


*Figure 211:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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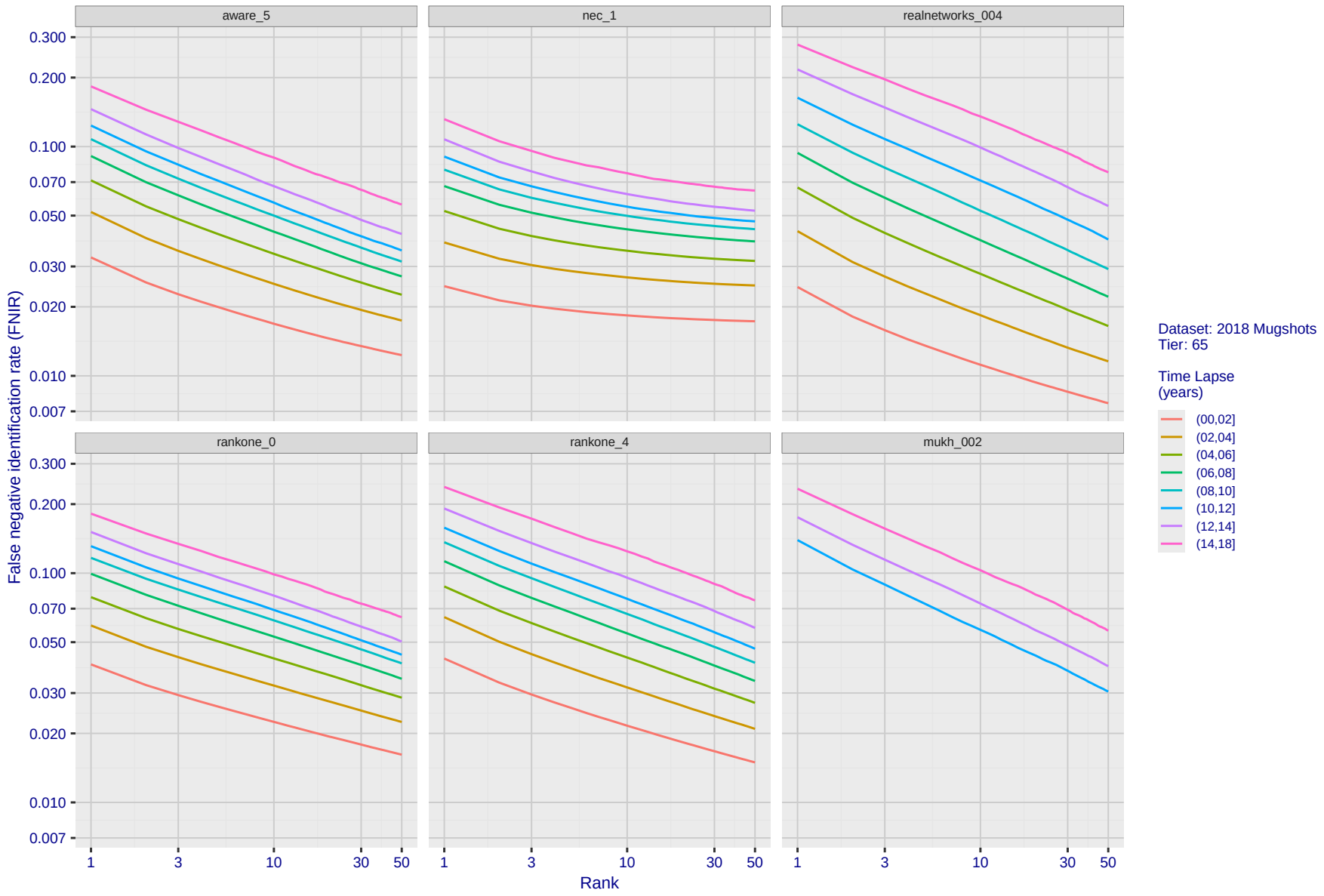


*Figure 212:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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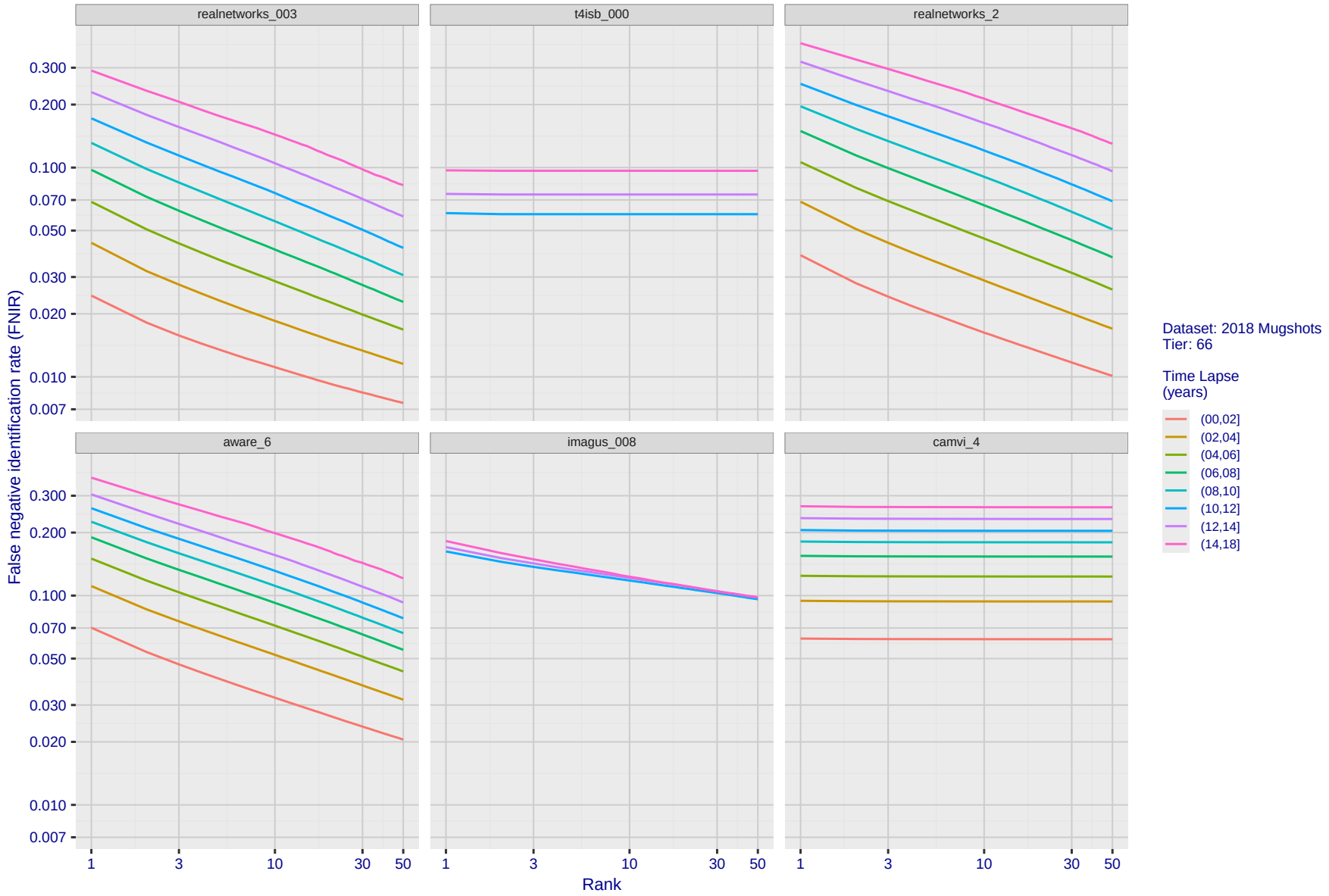


*Figure 213:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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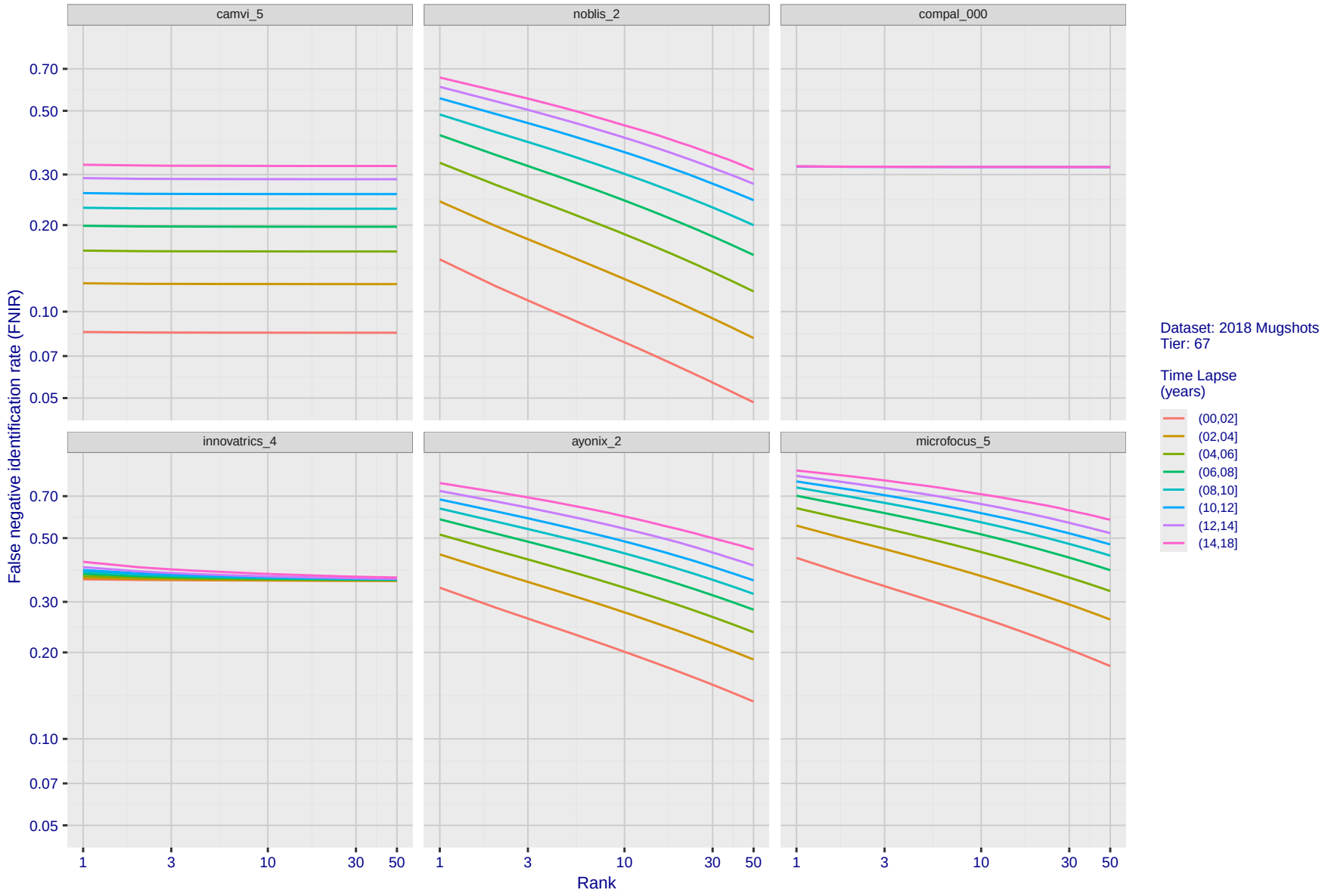


*Figure 214:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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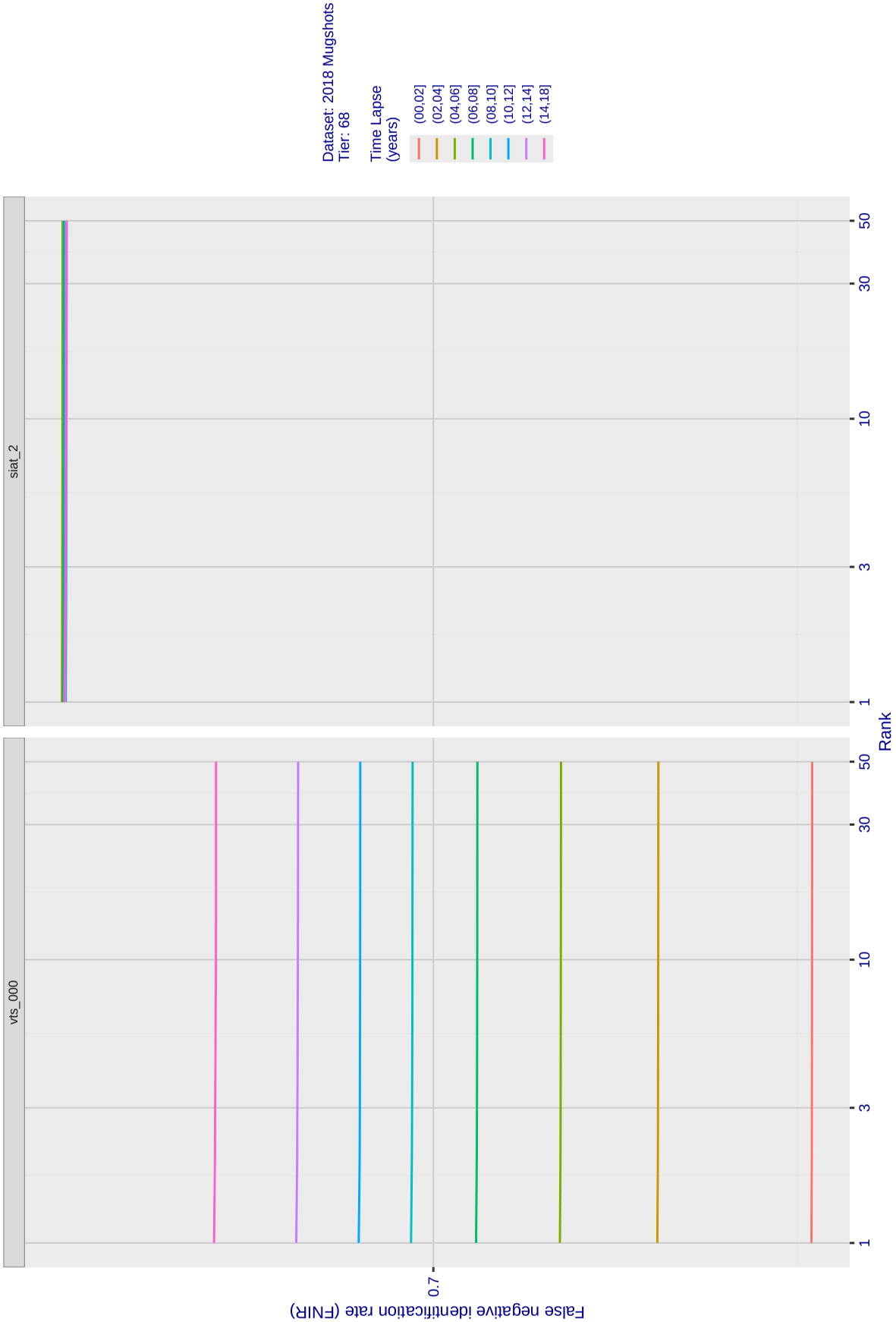


*Figure 215:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

| | | | | | |
|---|---|---|---|---|---|
| 2026/09/04 | FNIR(N, R, T) = | False neg. identification rate | N = Num. enrolled subjects | T = Threshold | $T = 0 \rightarrow$ Investigation |
| 10:23:32 | FPIR(N, T) = | False pos. identification rate | R = Num. candidates examined | | $T > 0 \rightarrow$ Identification |

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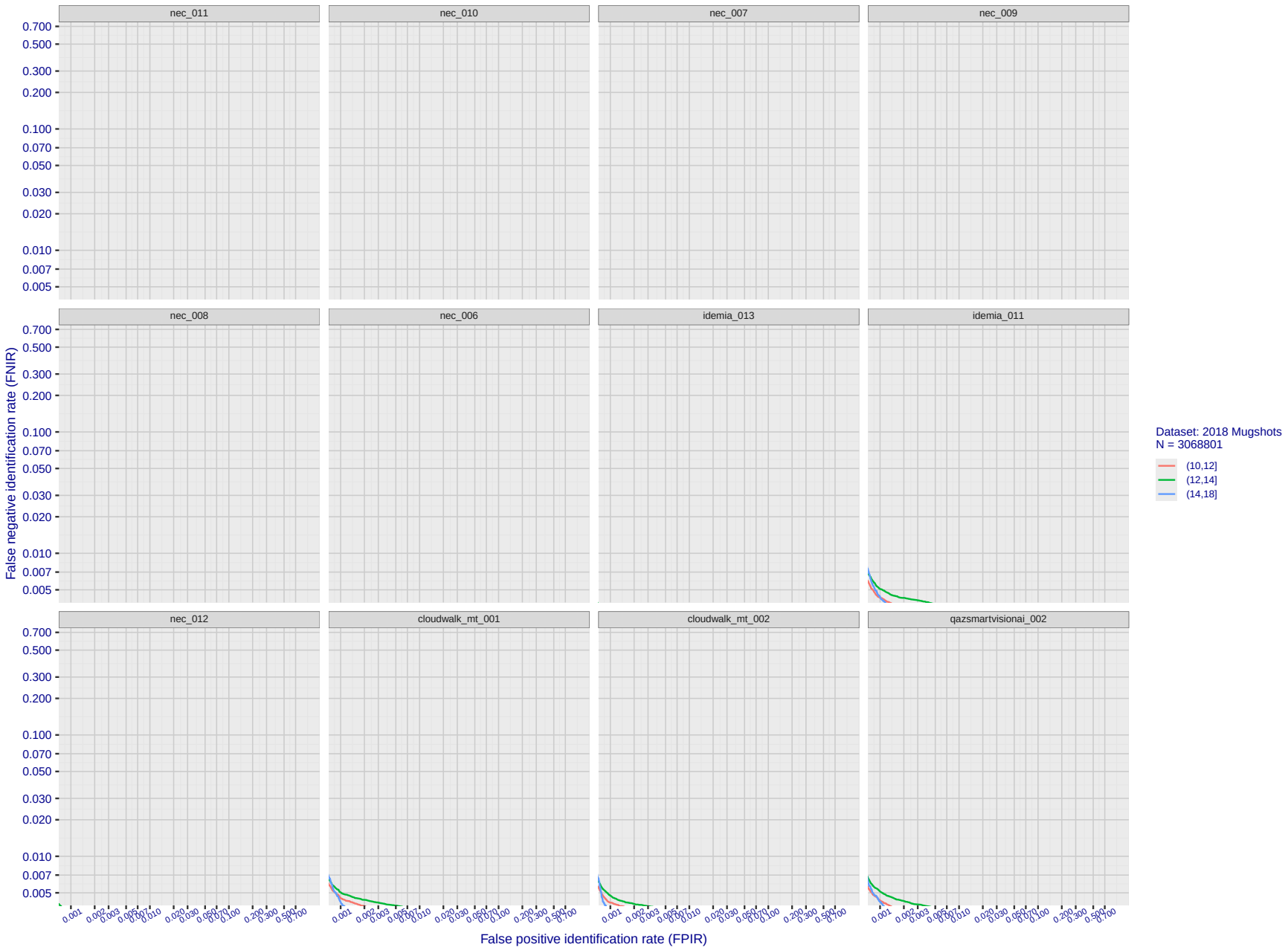


*Figure 216:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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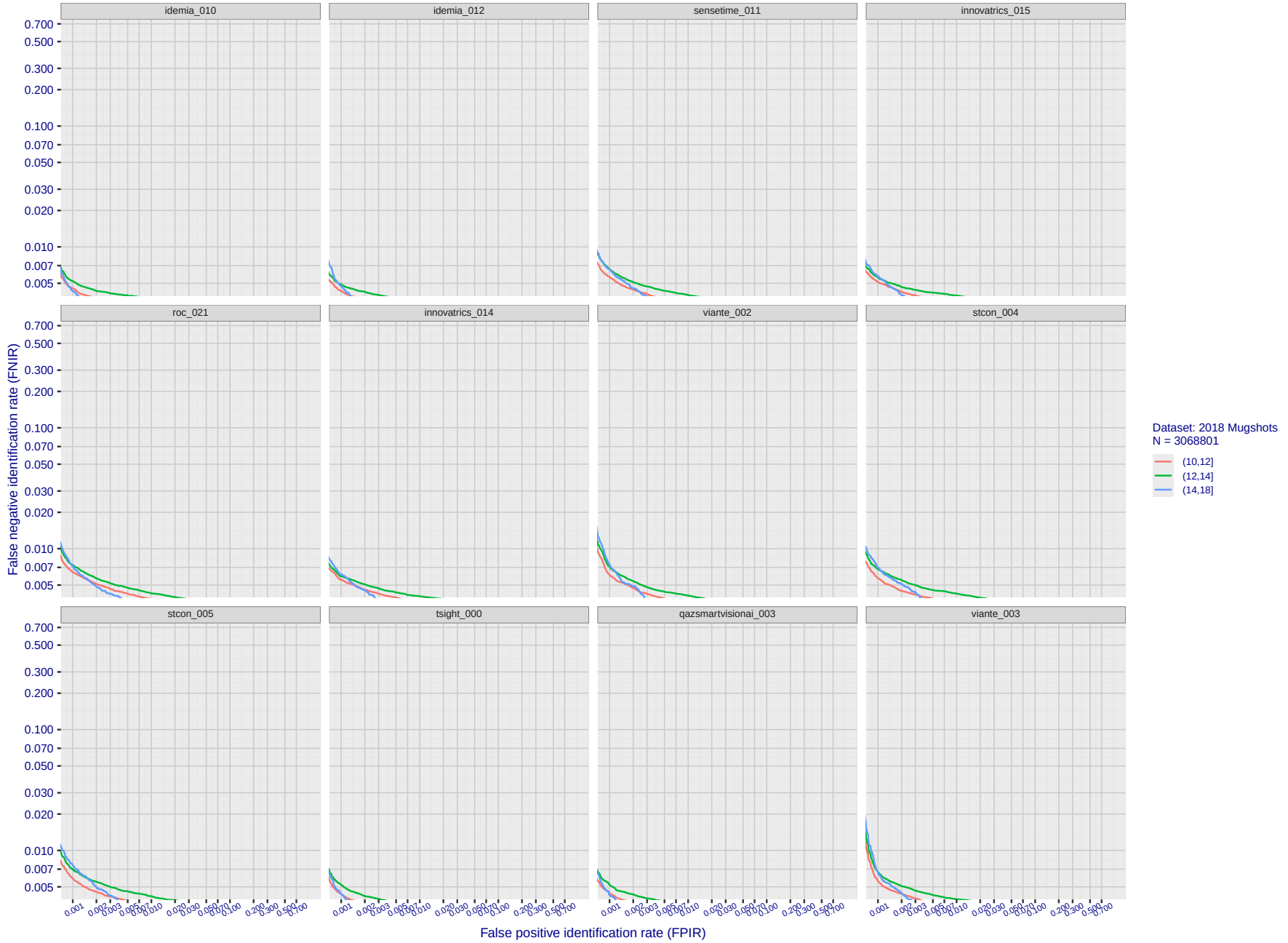


*Figure 217:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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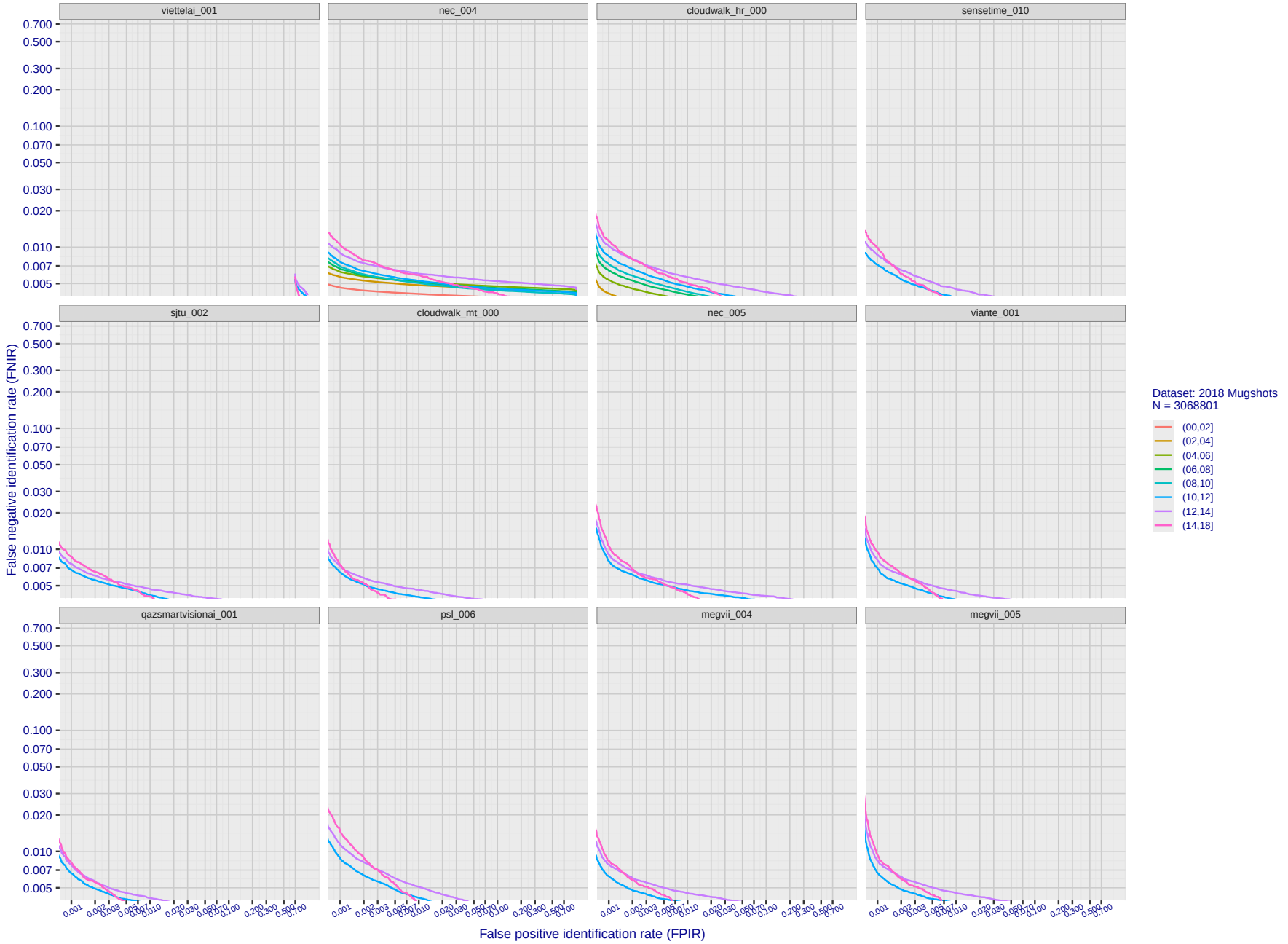


*Figure 218:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
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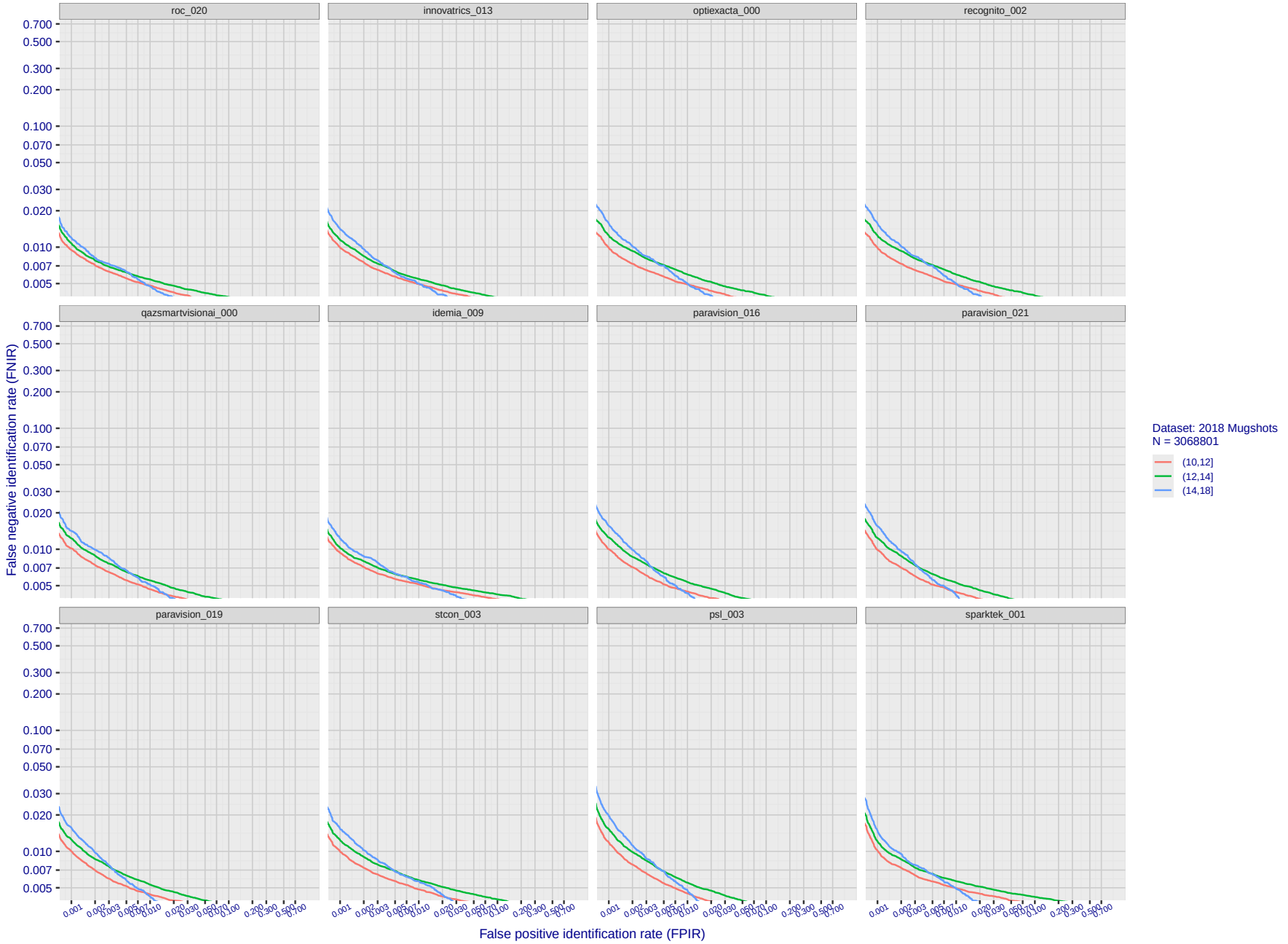


*Figure 219:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
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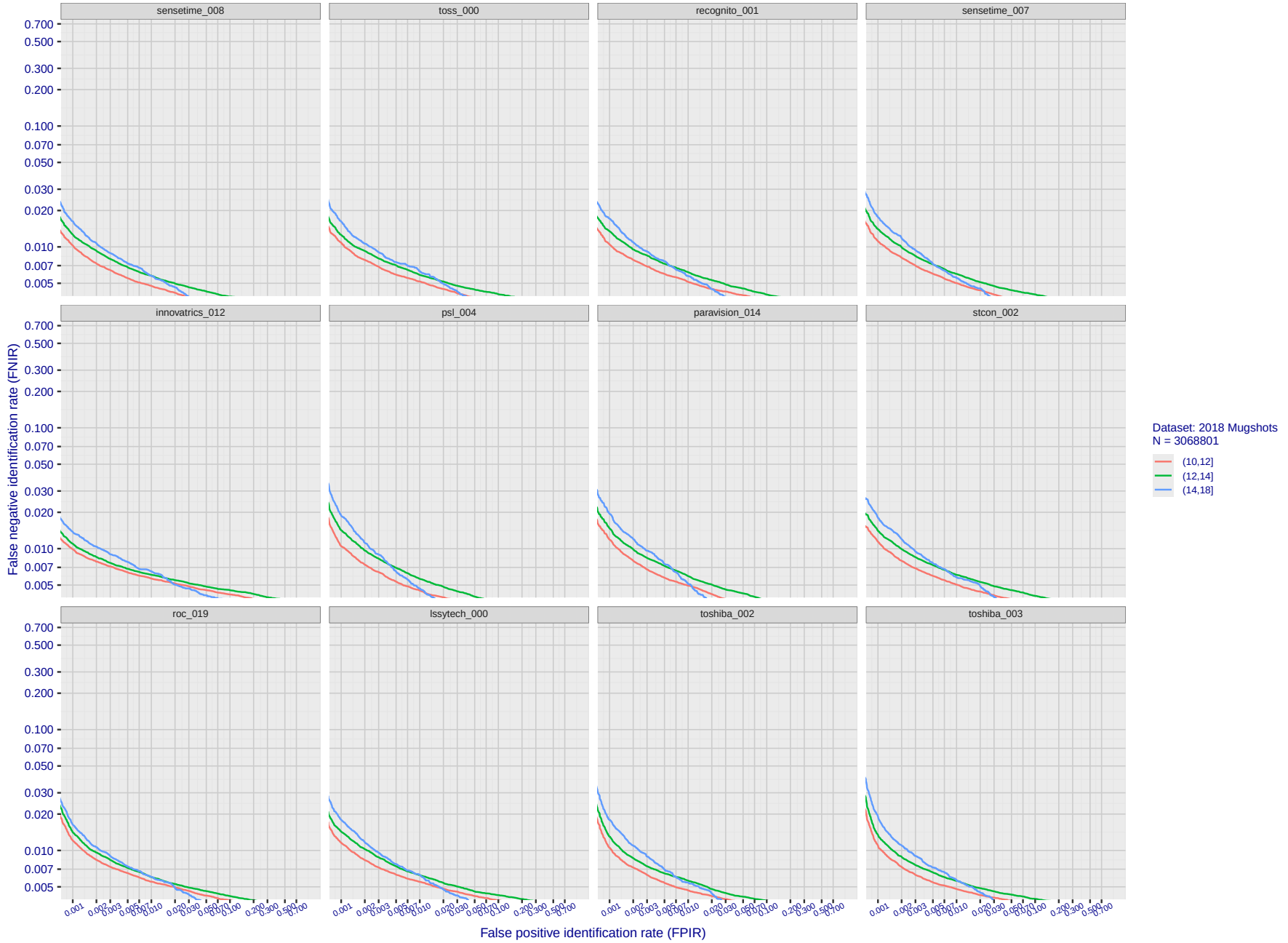


*Figure 220:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/09/04 10:23:32

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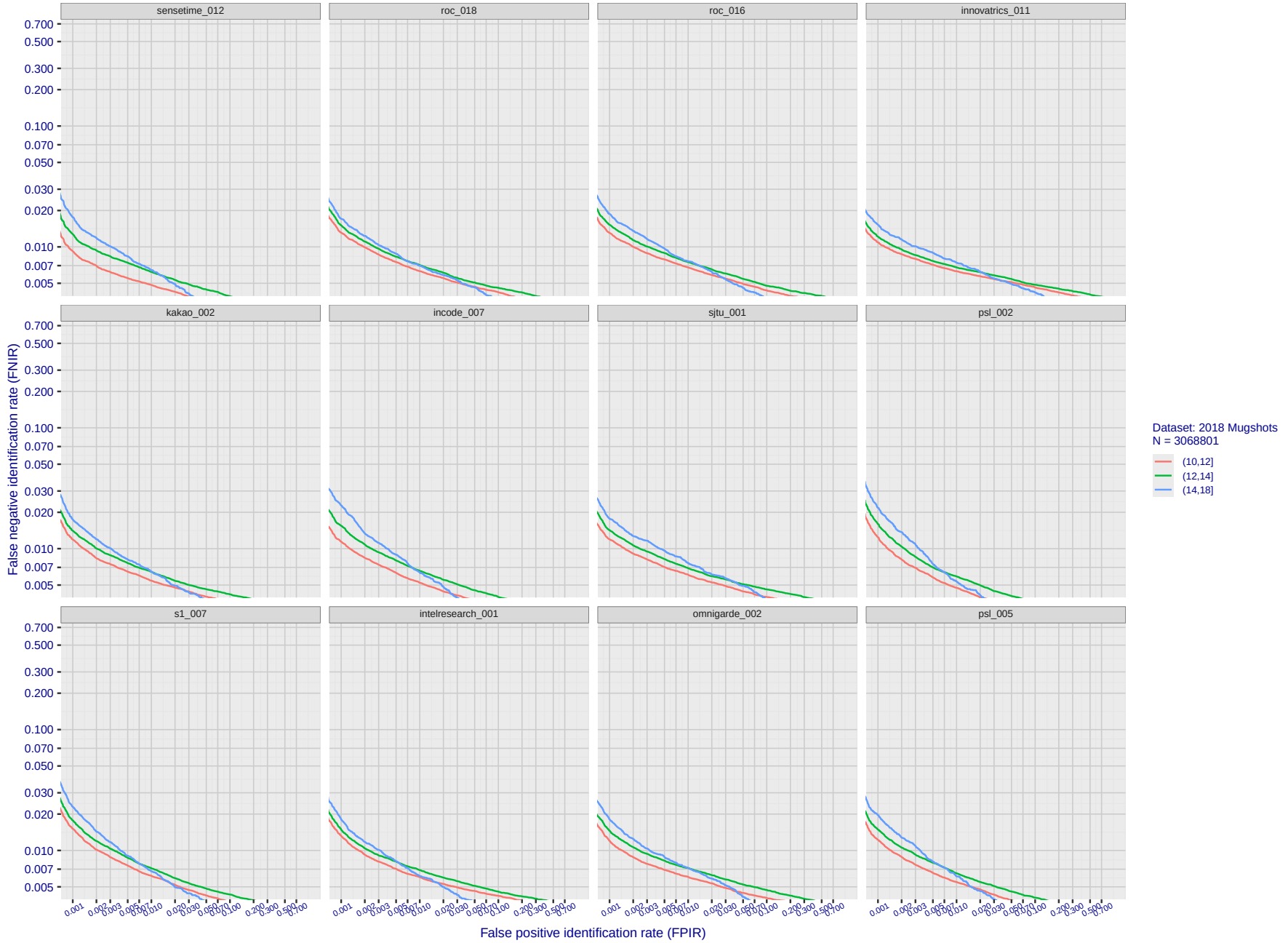


*Figure 221:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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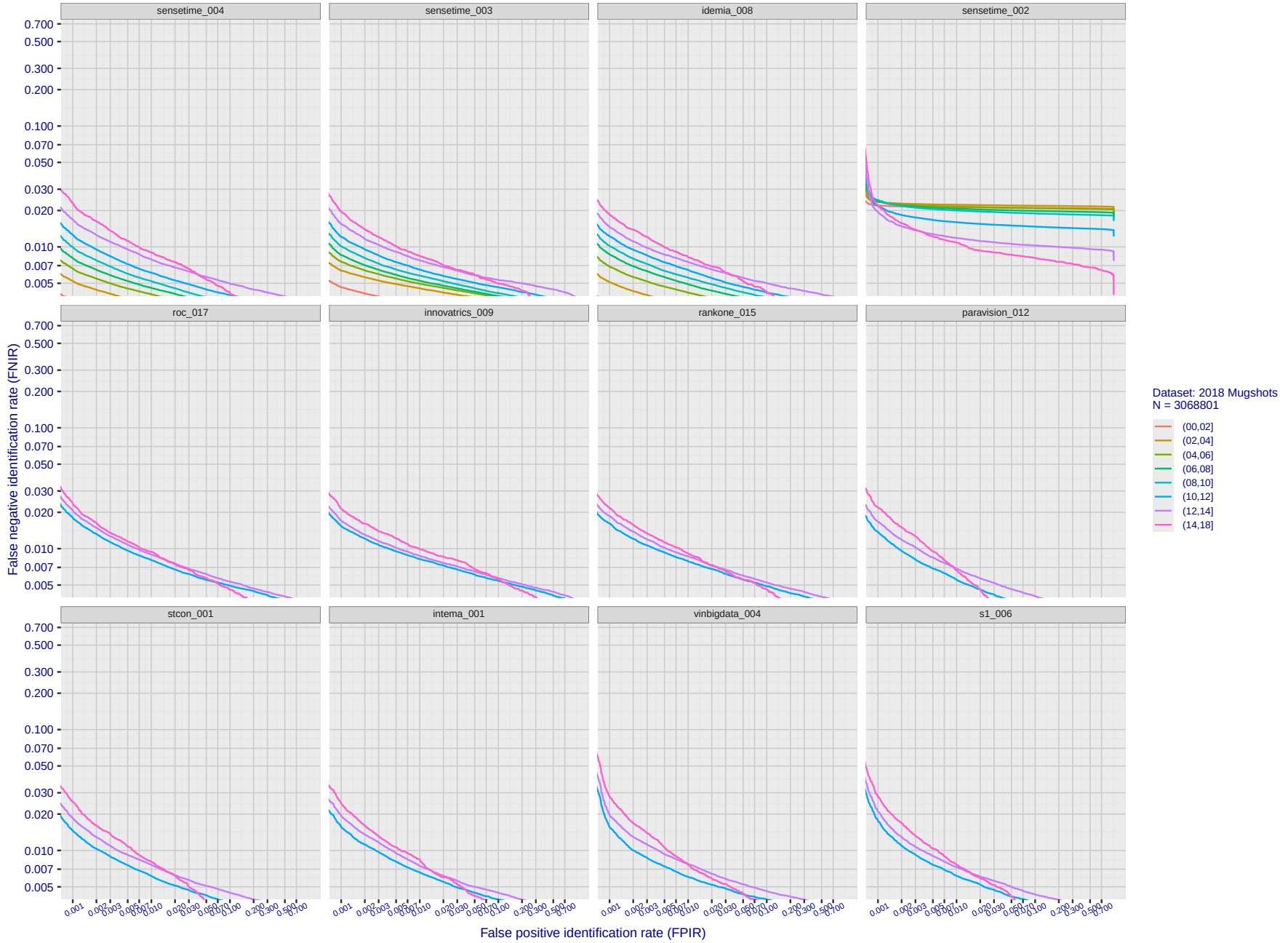


*Figure 222:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04
10:23:32
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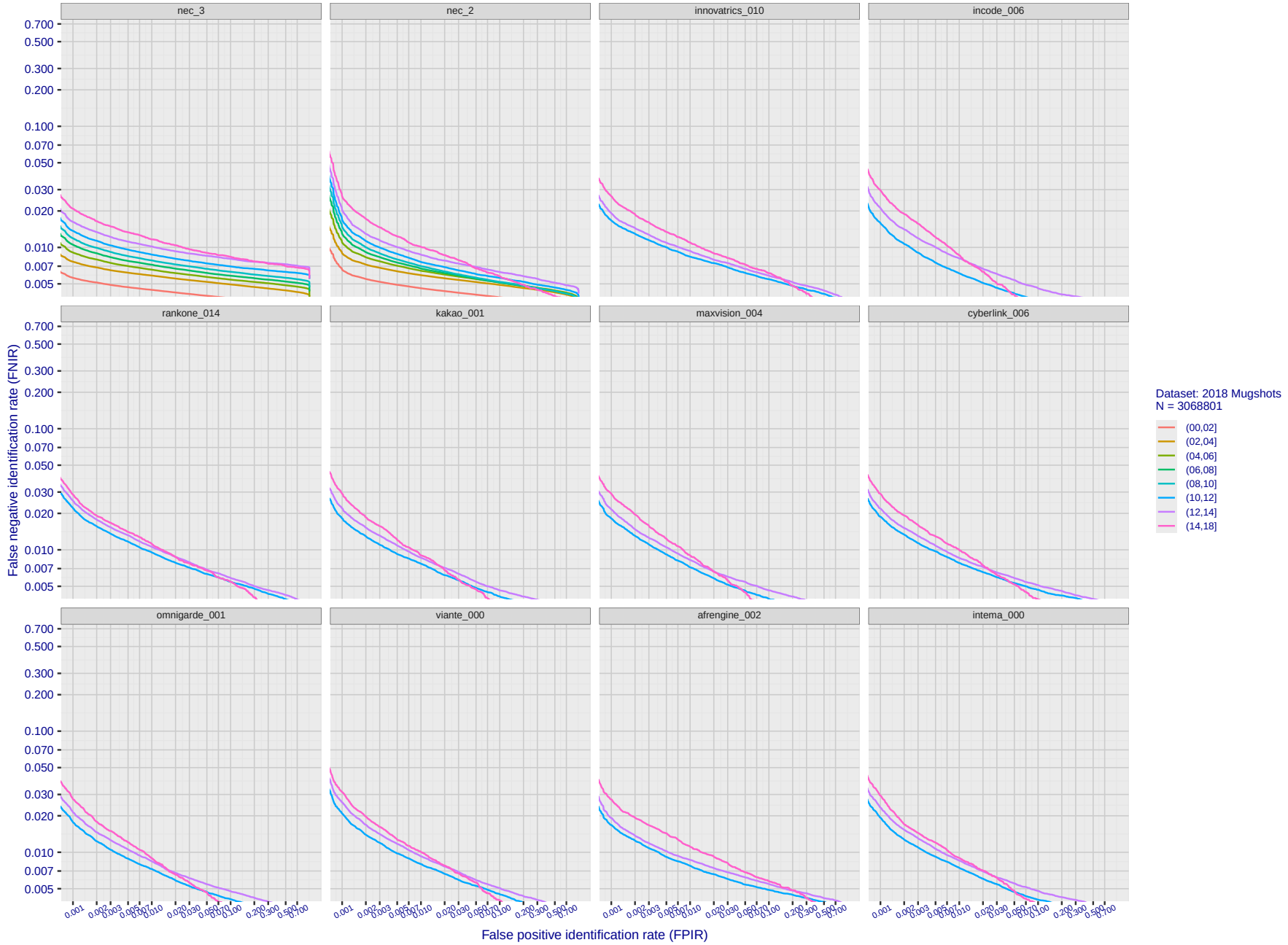


*Figure 223:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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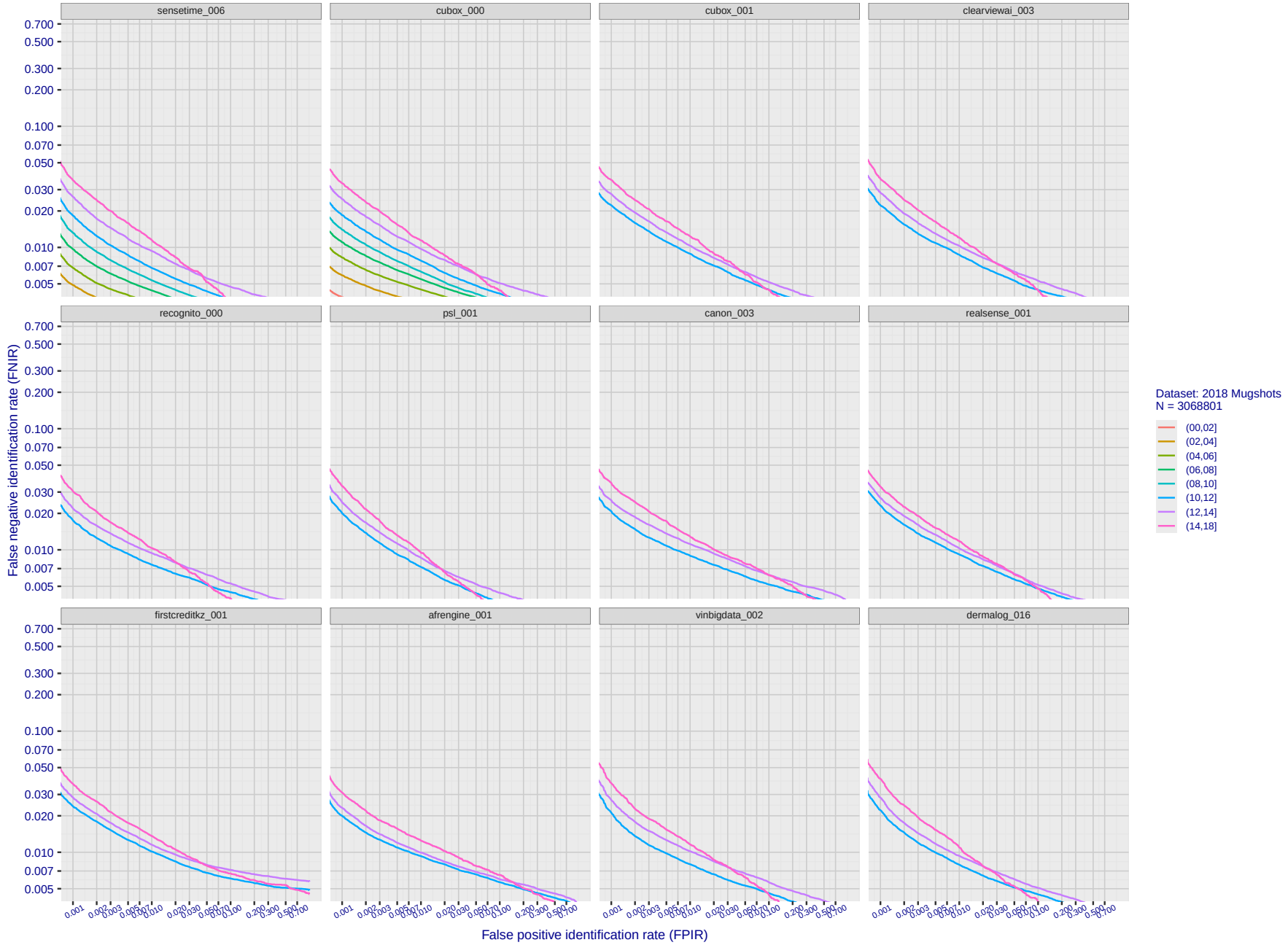


*Figure 224:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04 10:23:32
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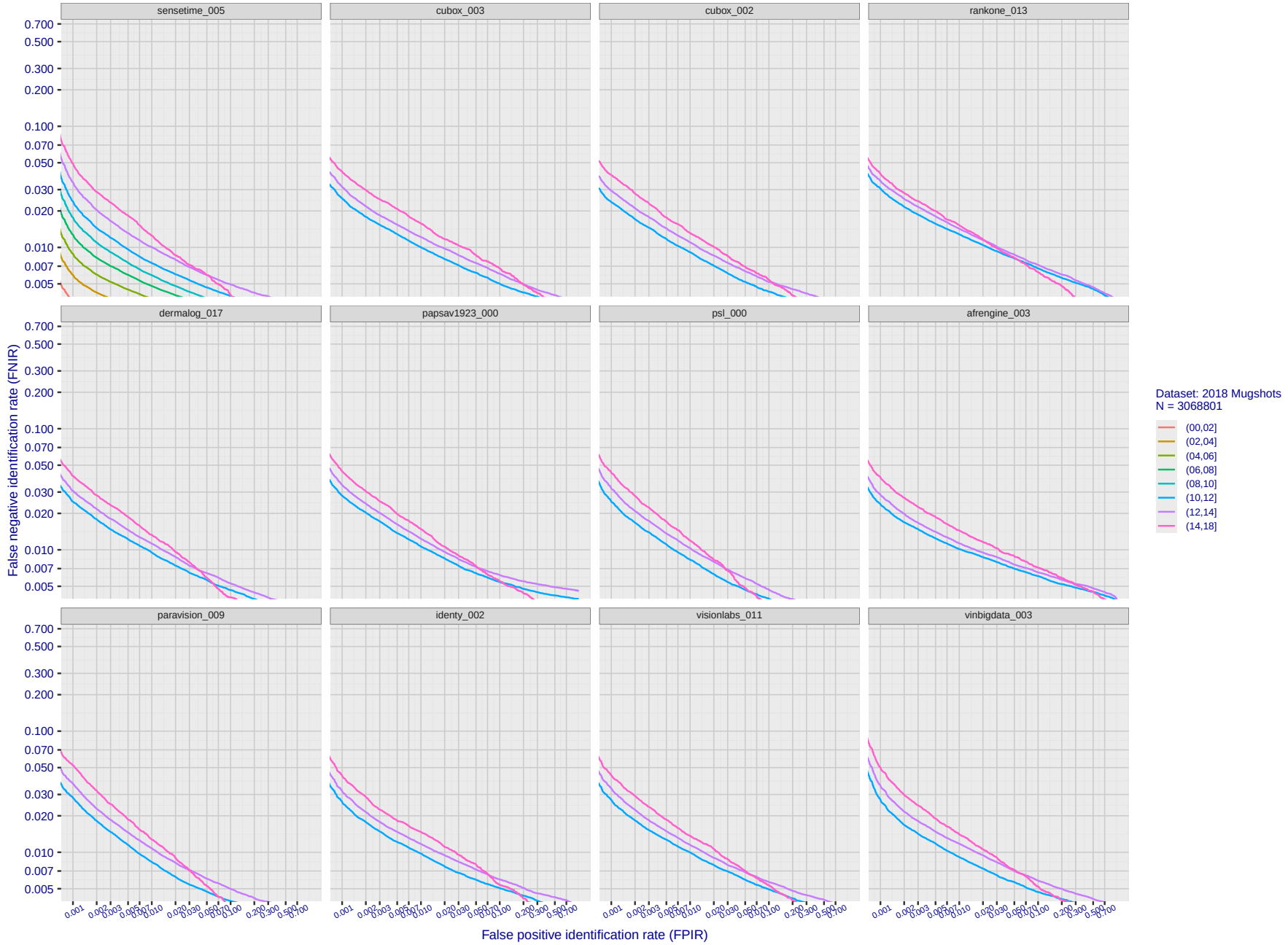


*Figure 225:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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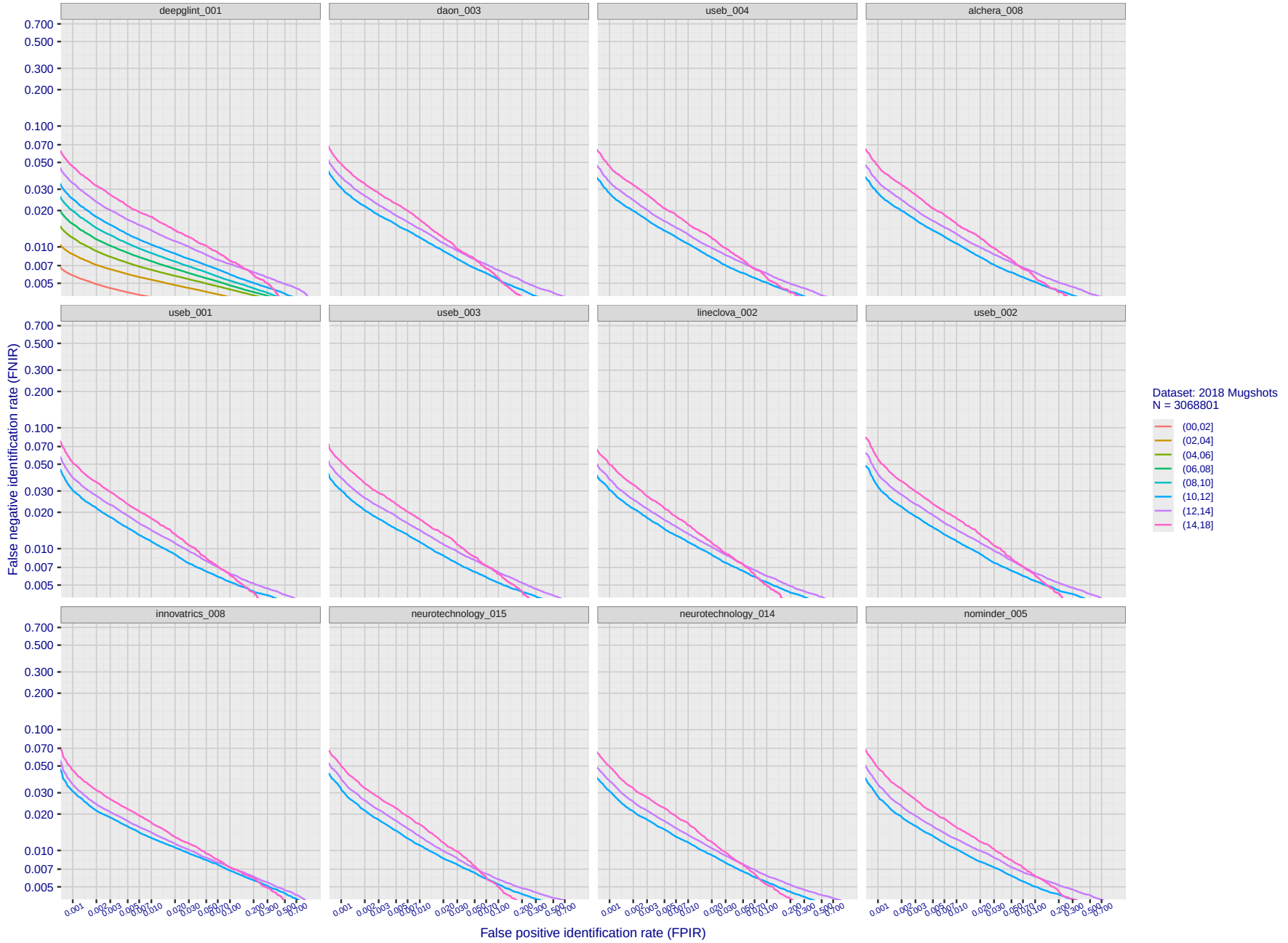


*Figure 226:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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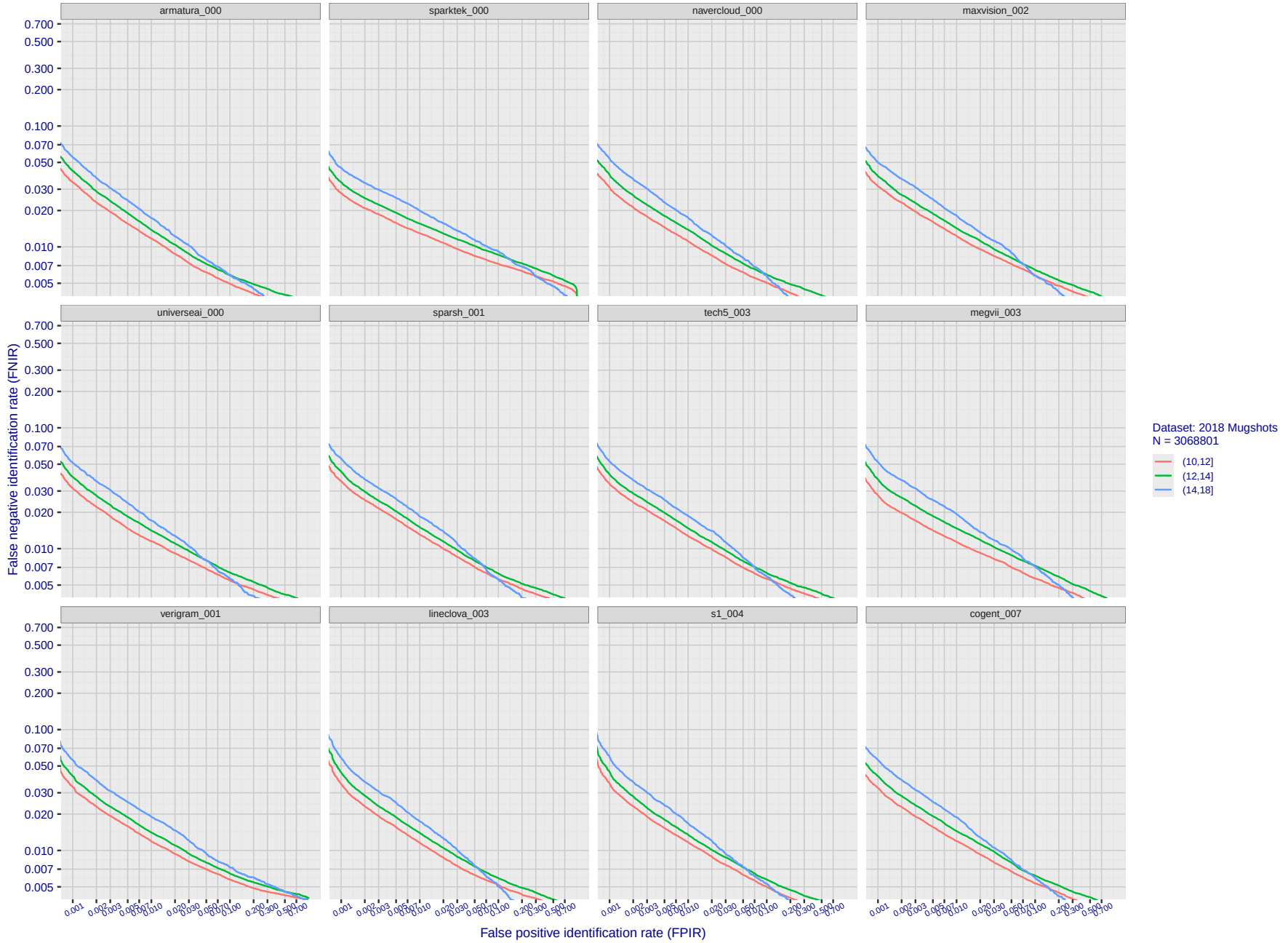


*Figure 227:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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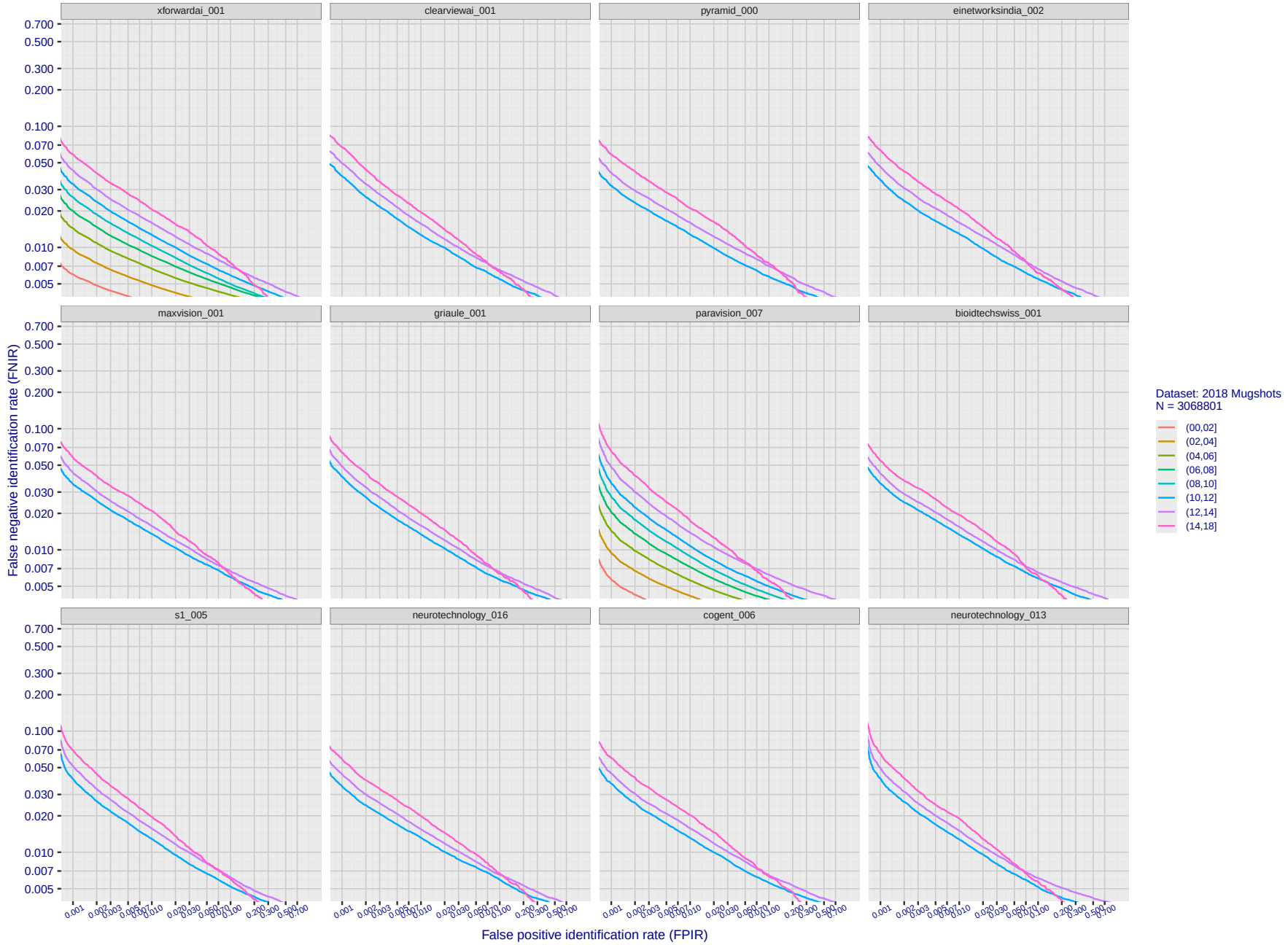


*Figure 228:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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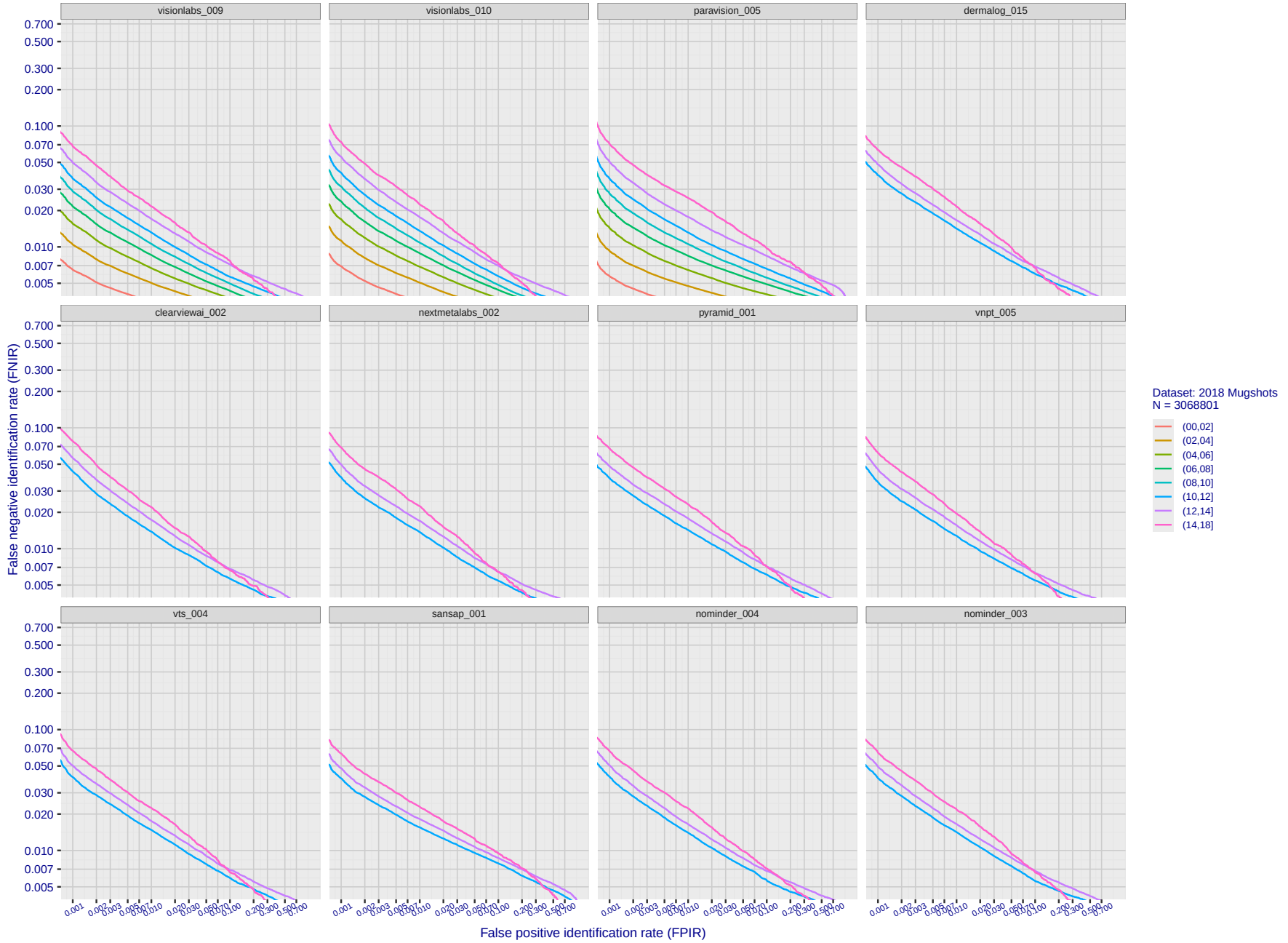


*Figure 229:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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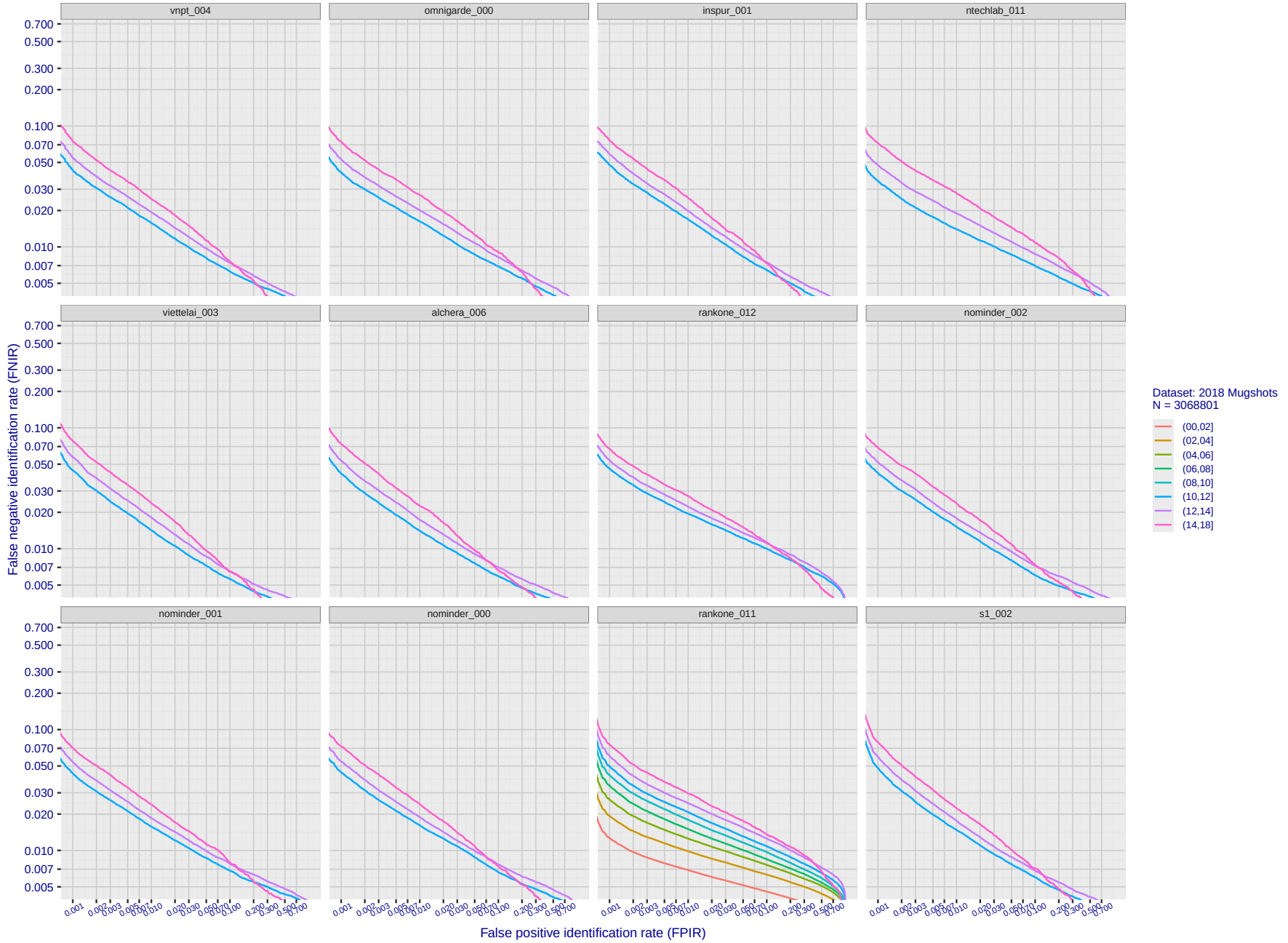


*Figure 230:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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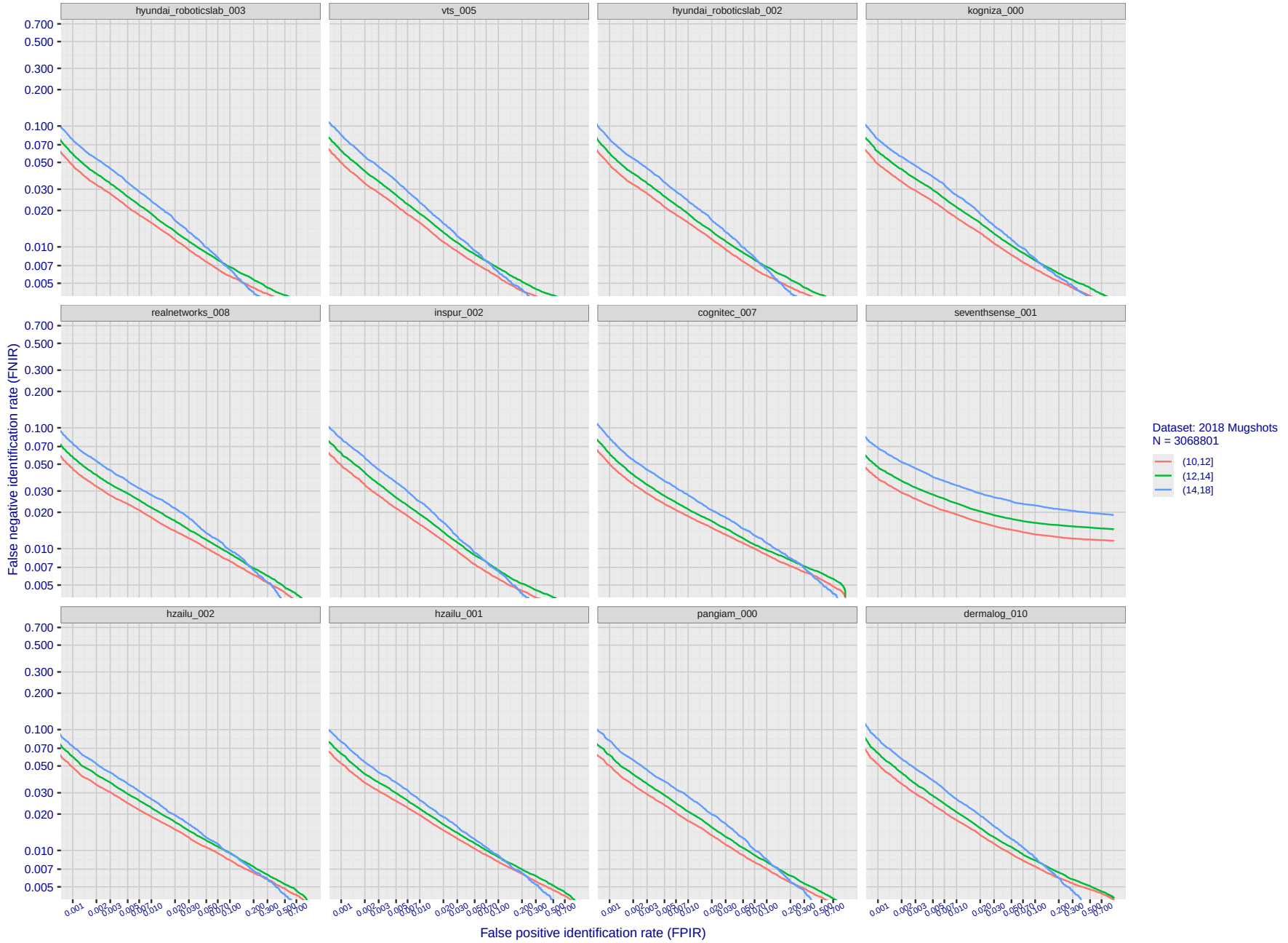


*Figure 231:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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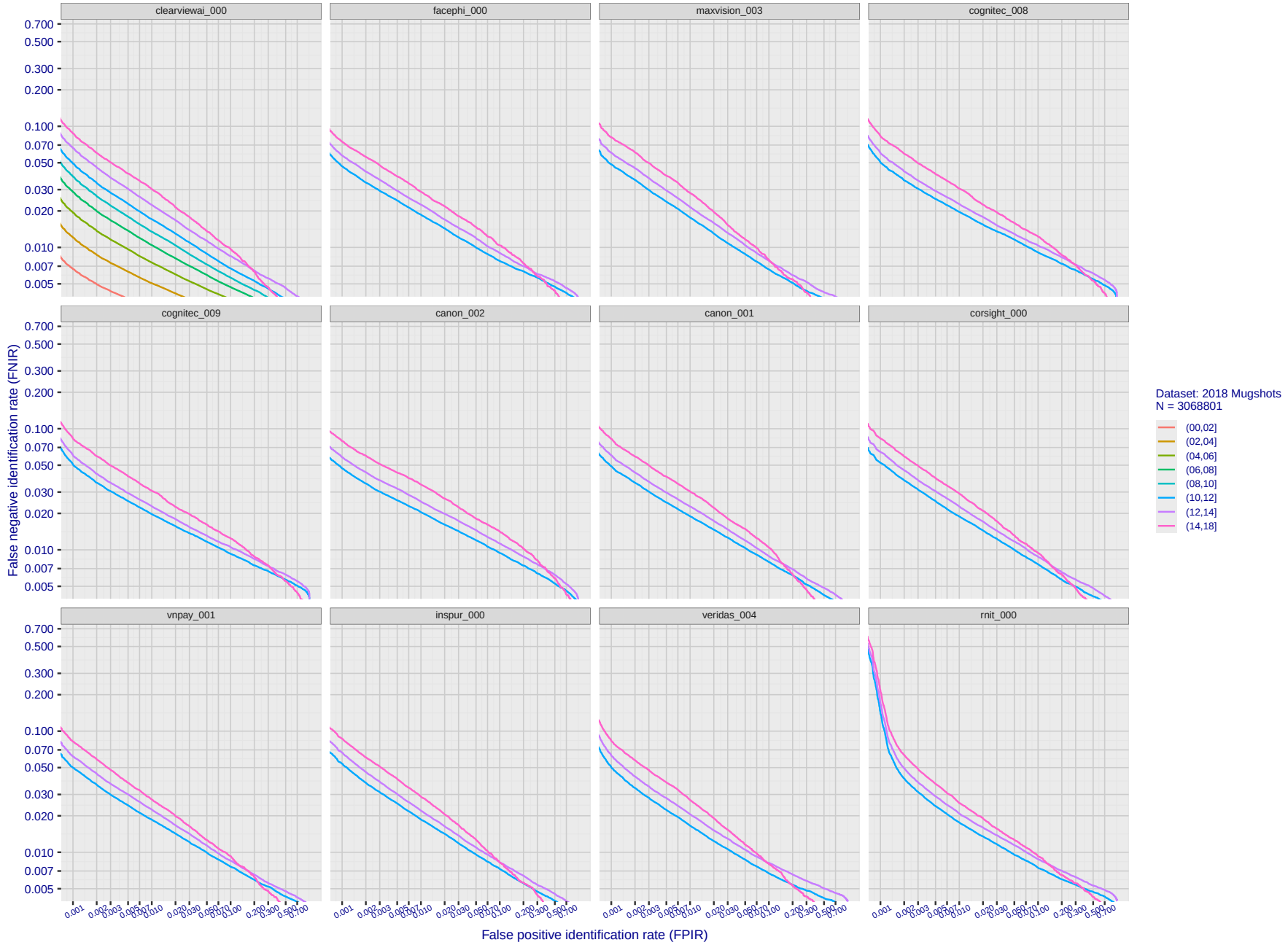


*Figure 232:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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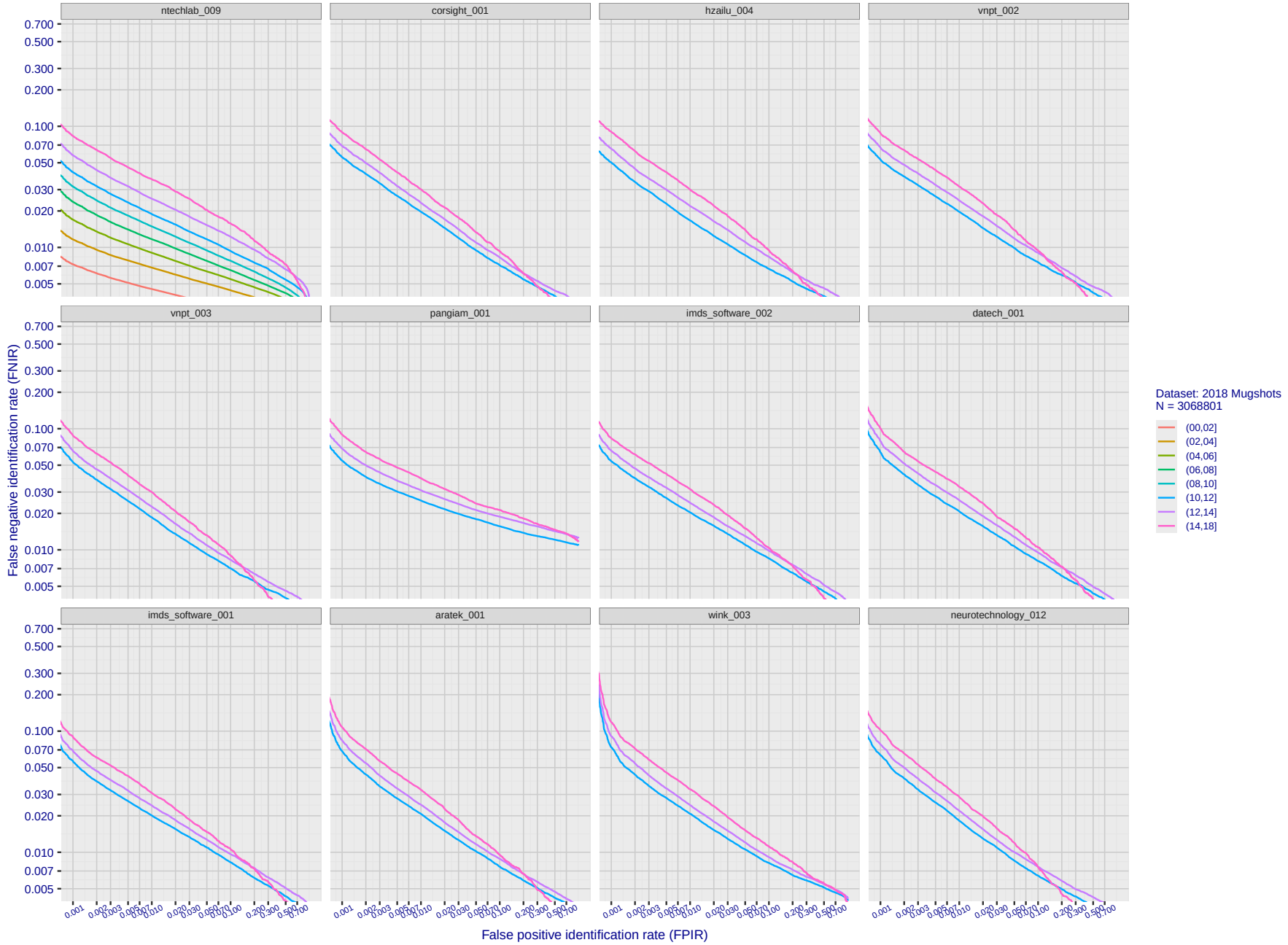


*Figure 233:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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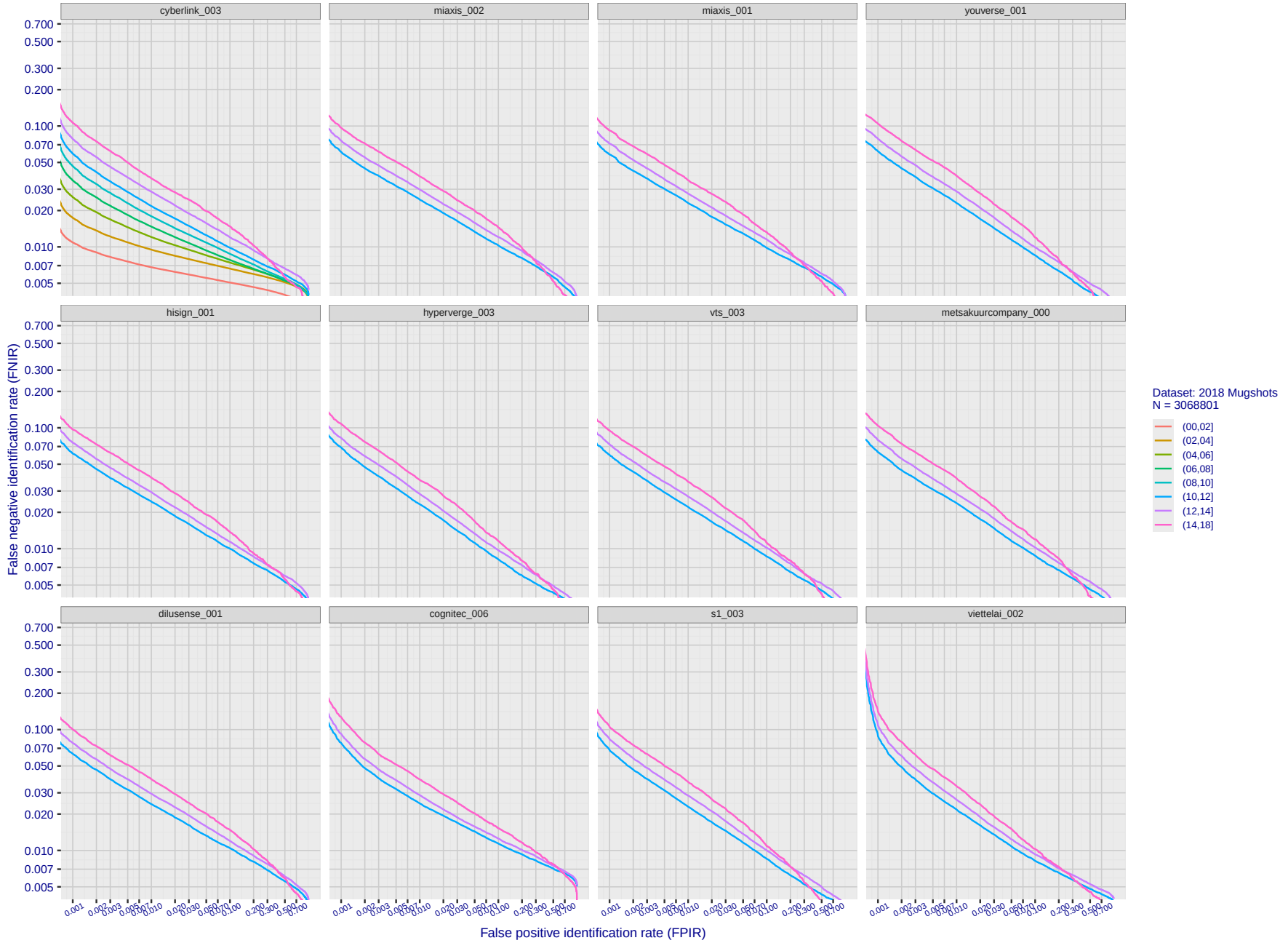


*Figure 234:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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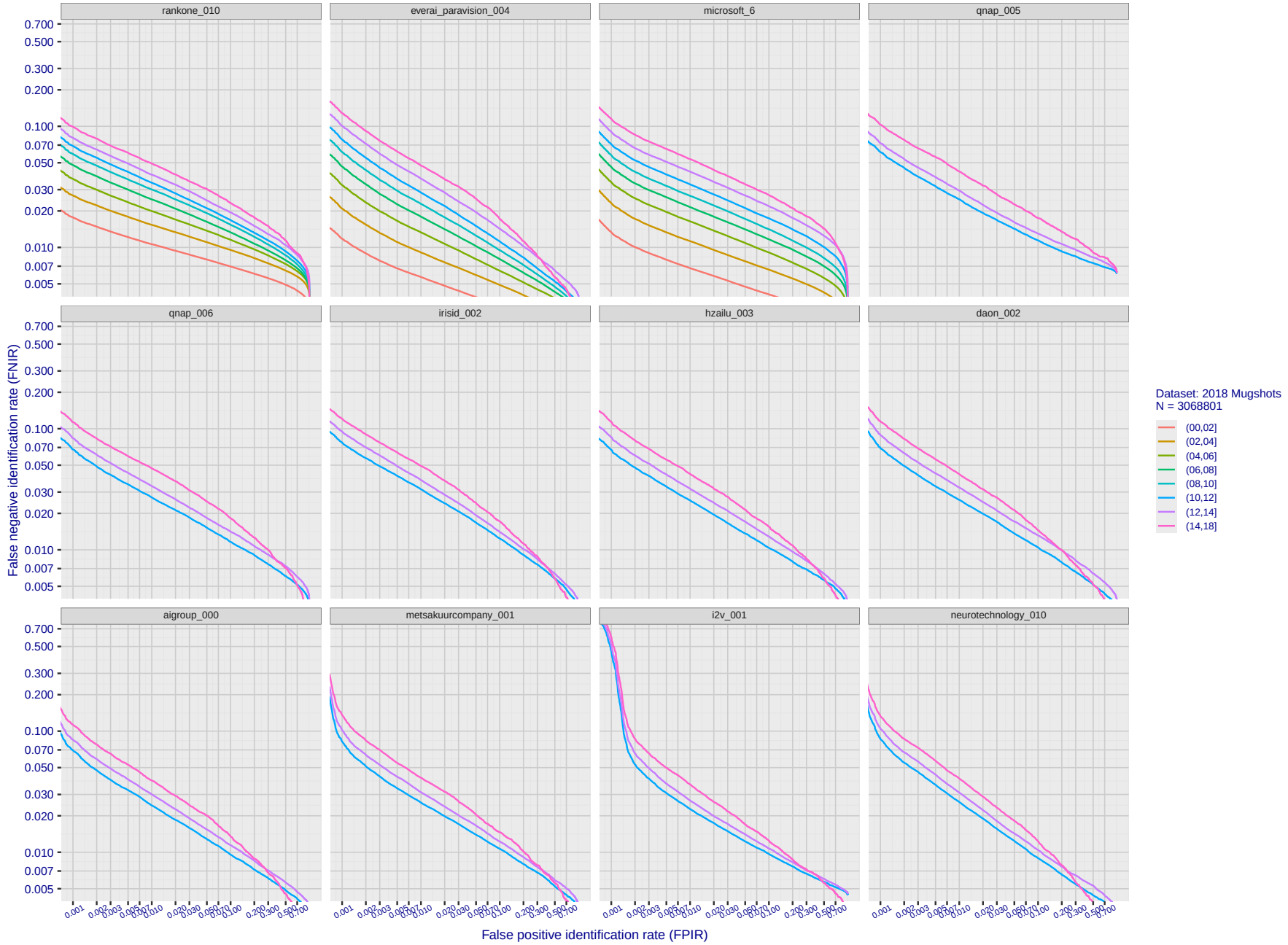


*Figure 235:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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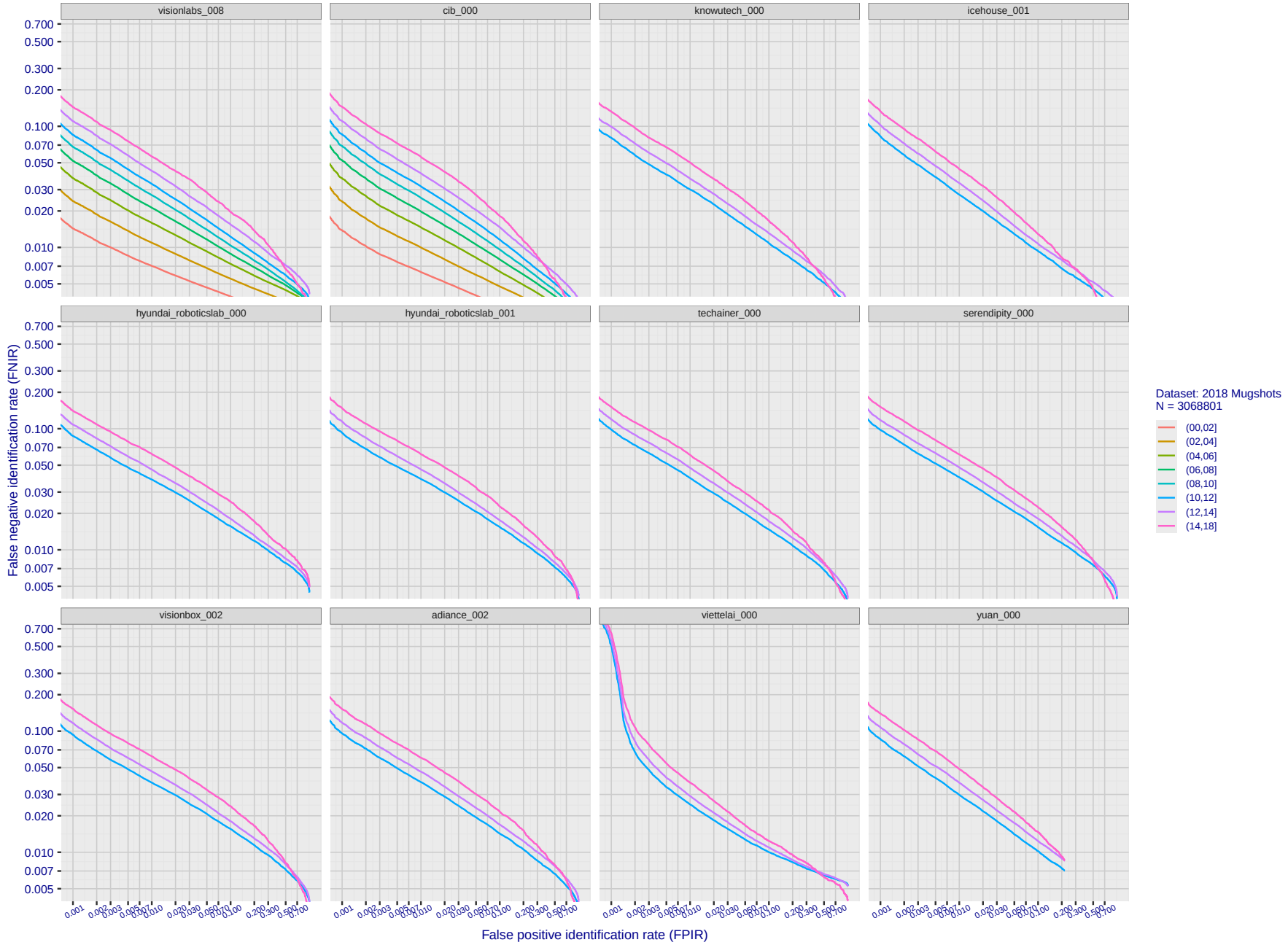


*Figure 236:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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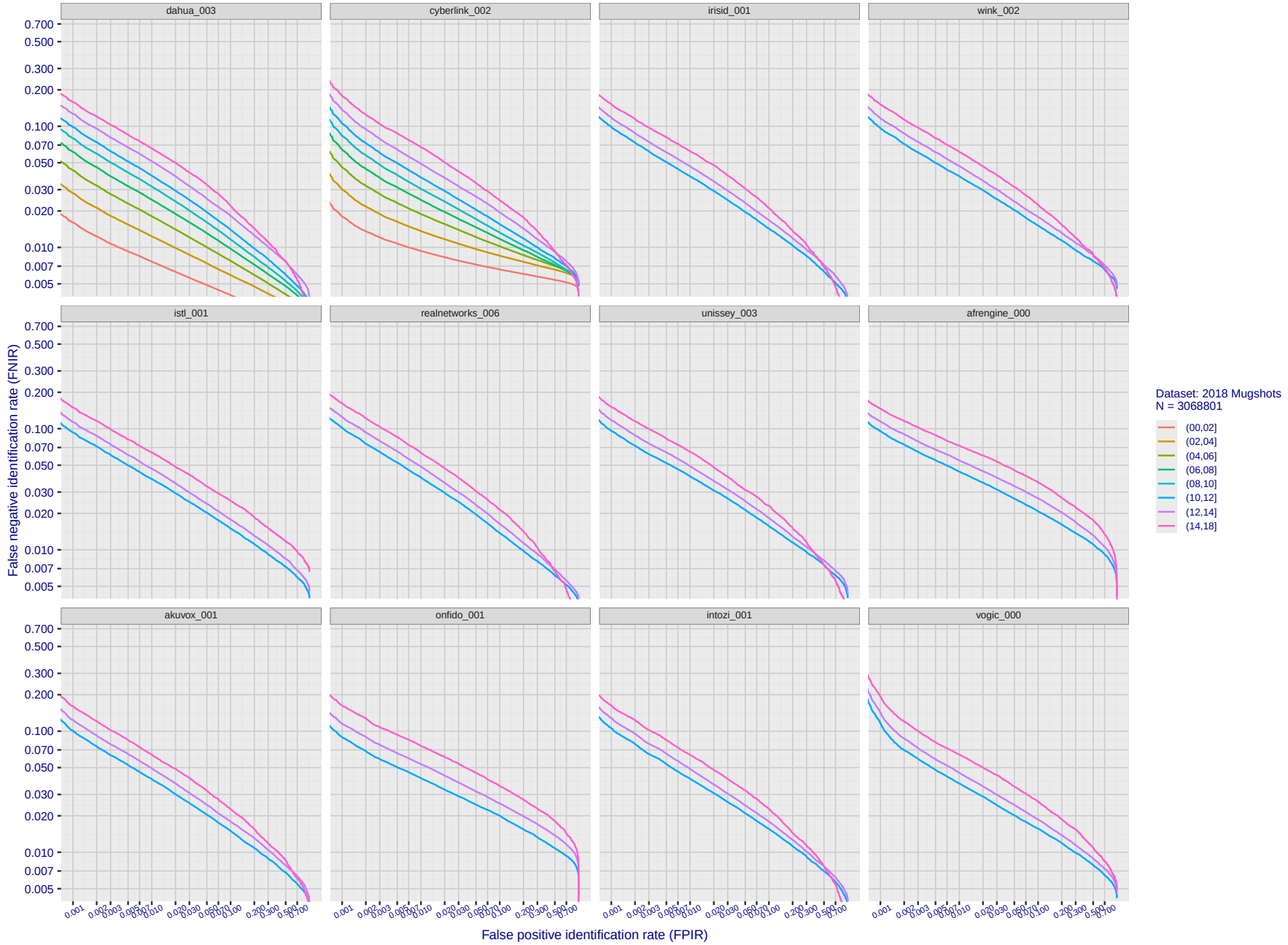


*Figure 237:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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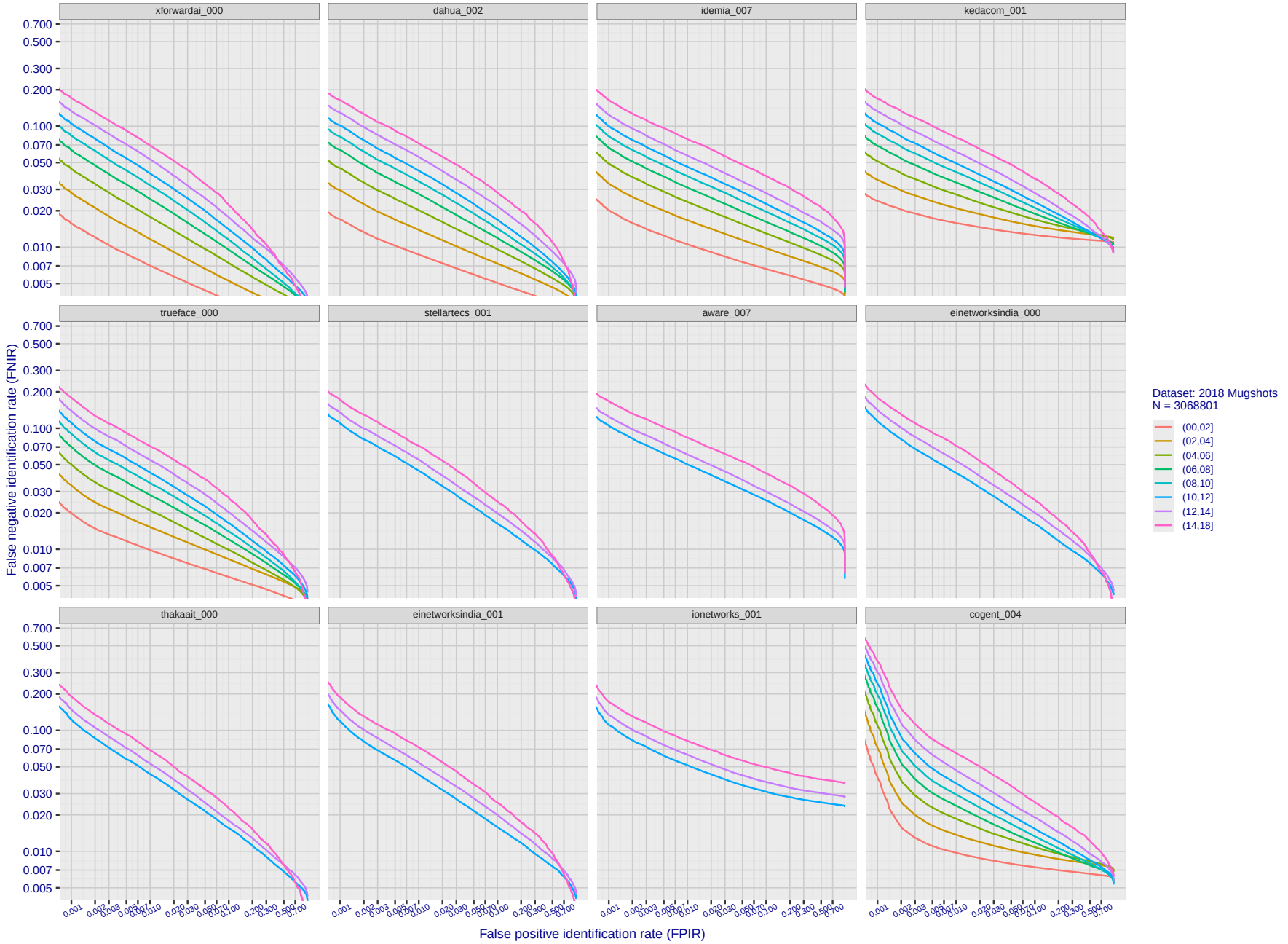


*Figure 238:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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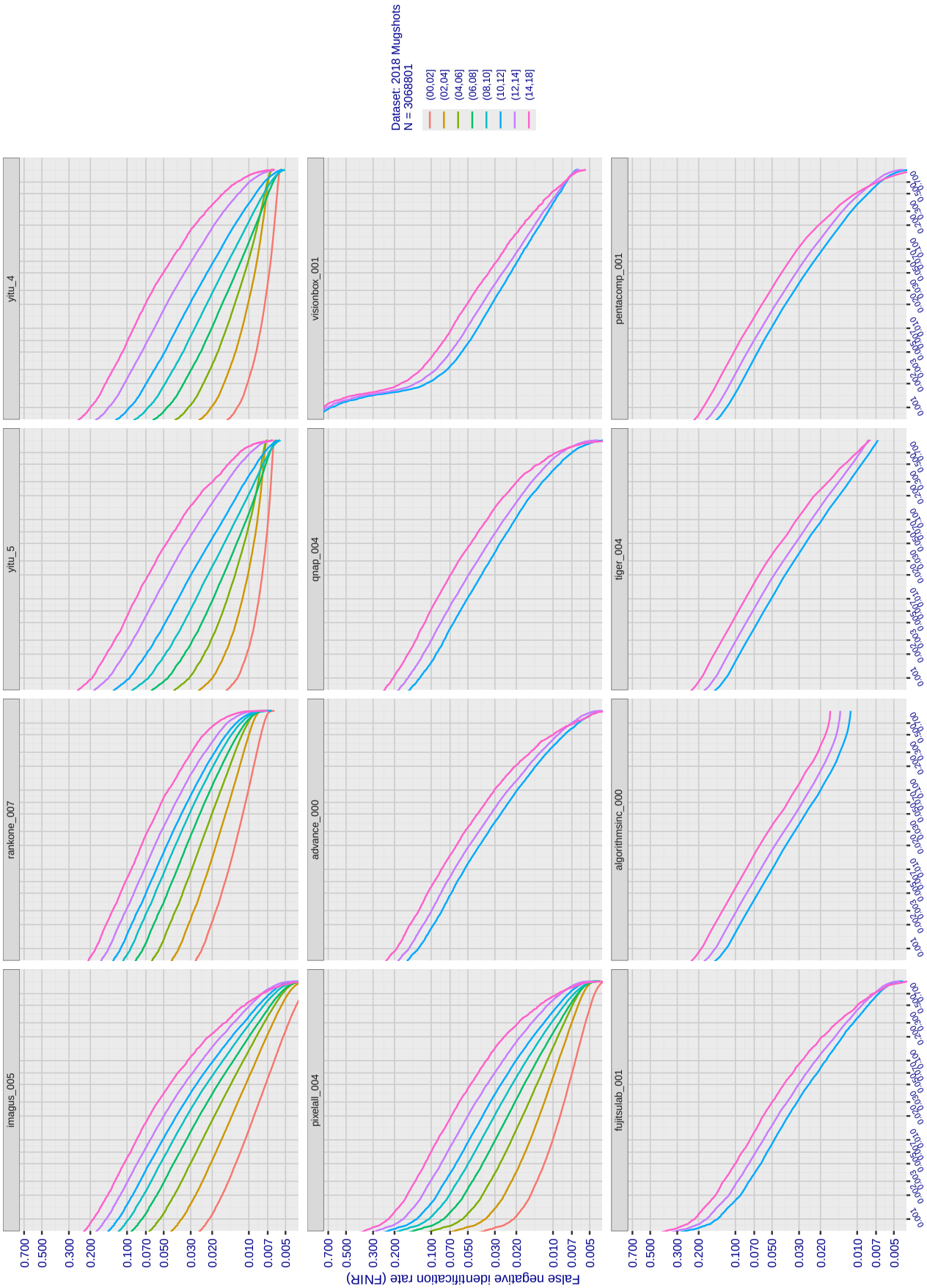


*Figure 239:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed.** *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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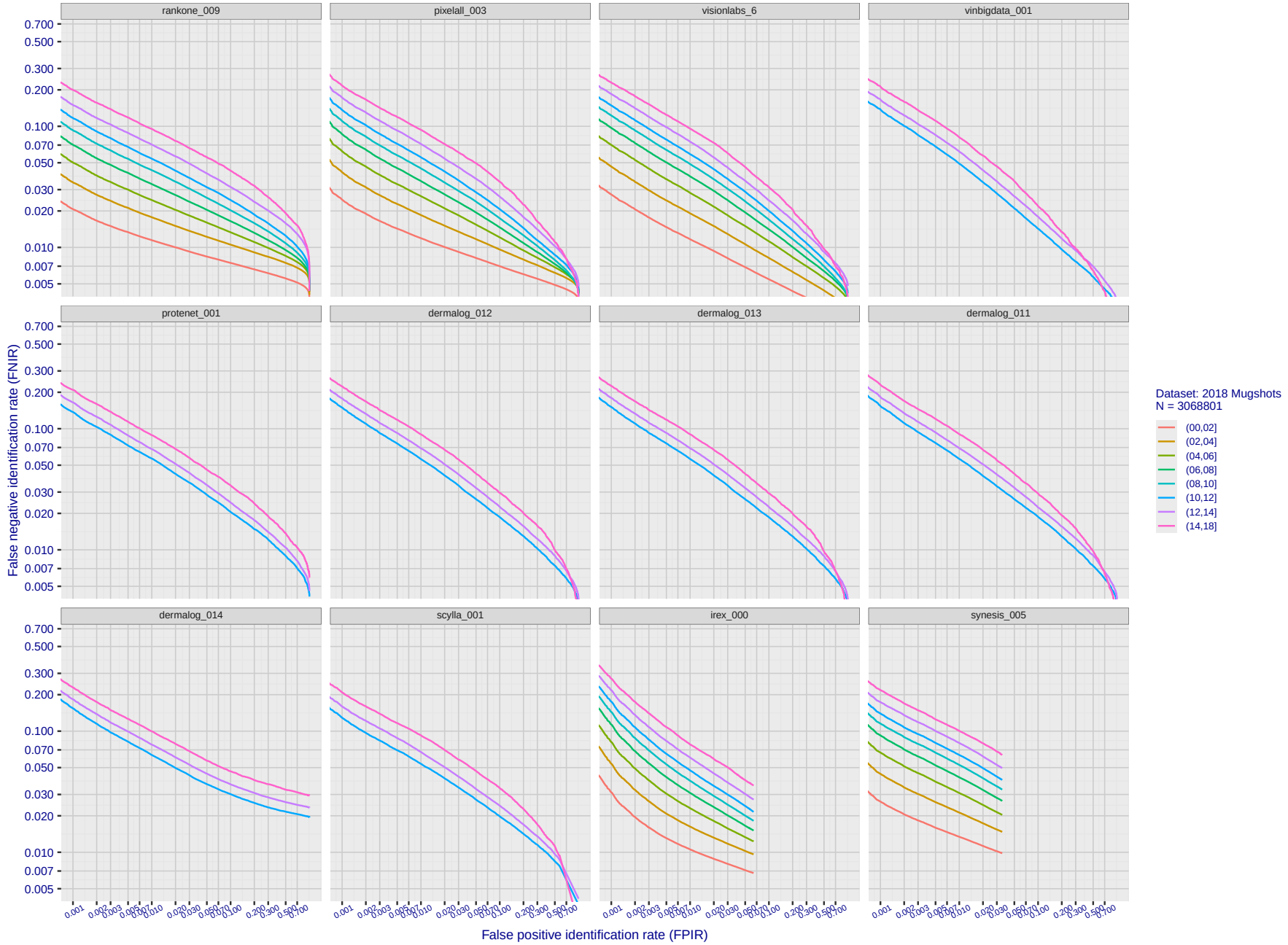


*Figure 240:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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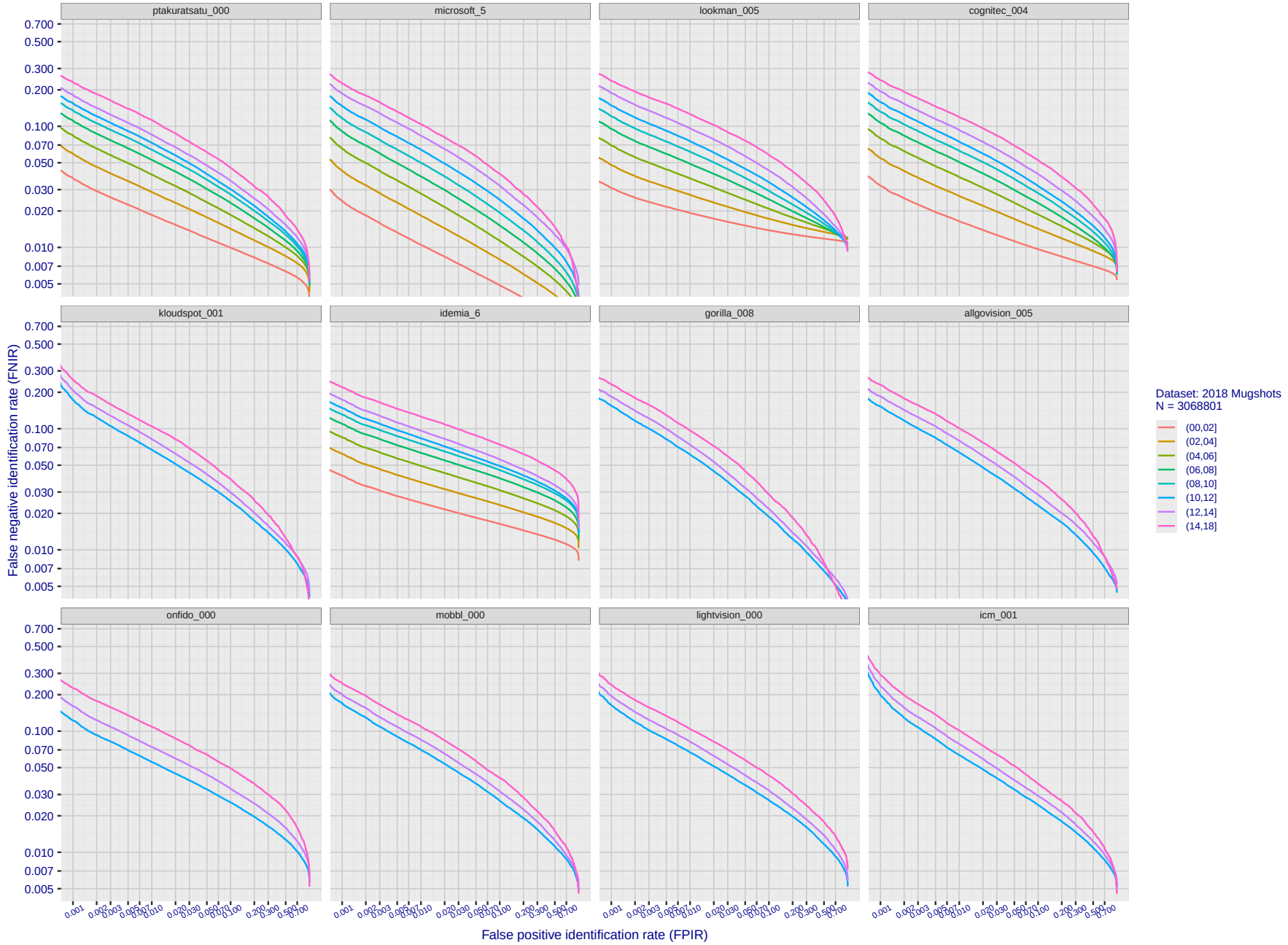


*Figure 241:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



2026/09/04
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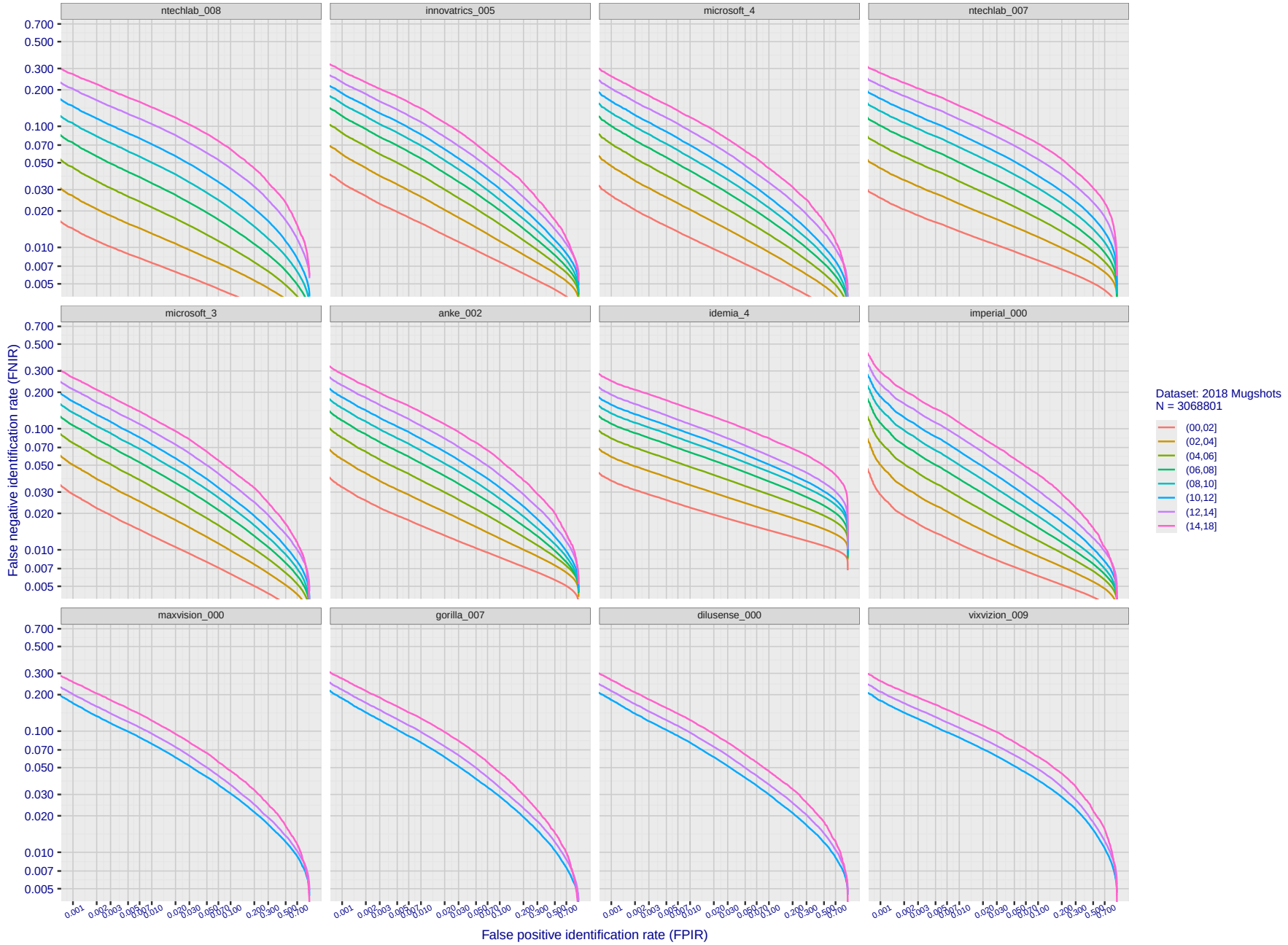


*Figure 242:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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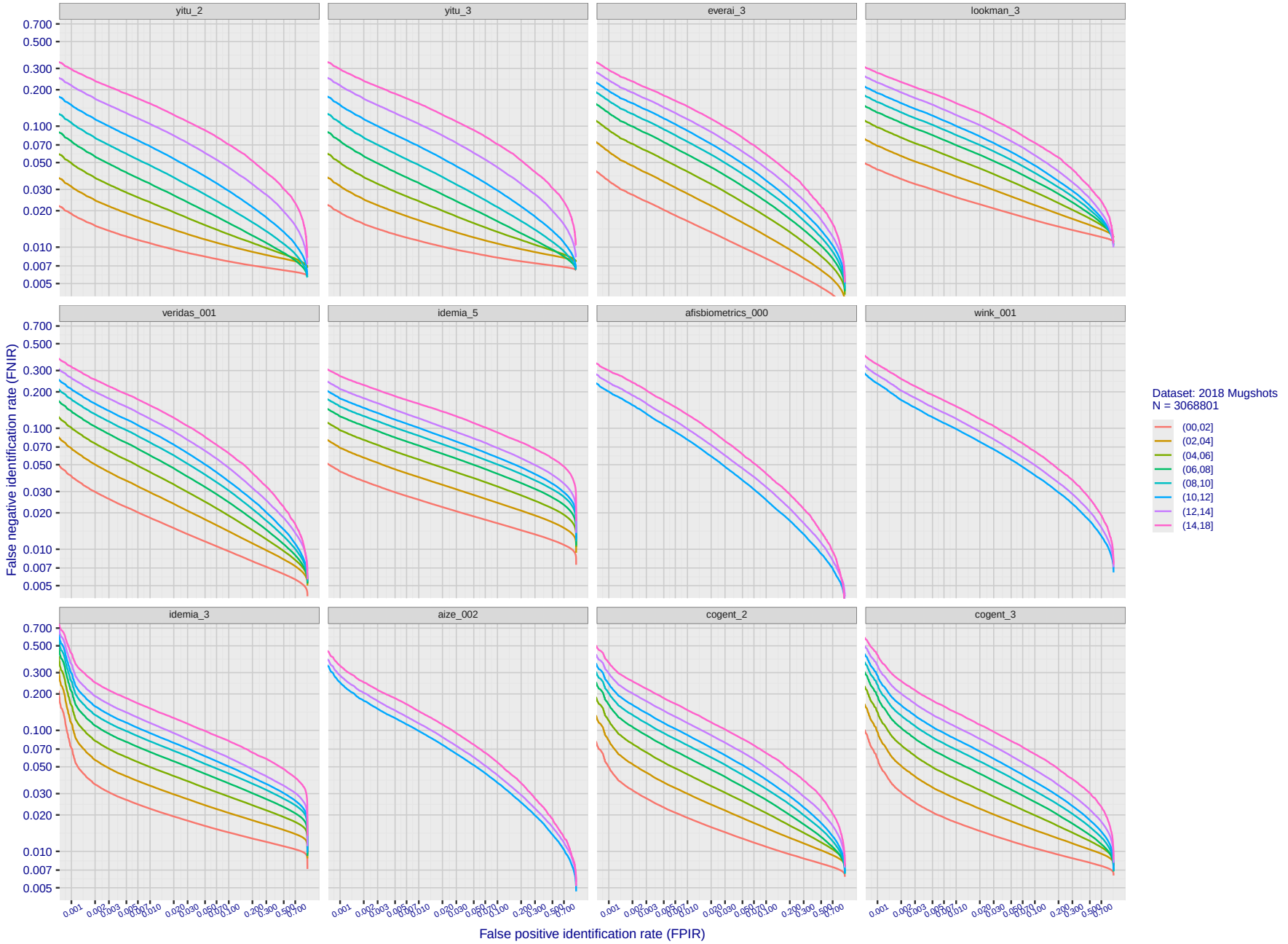


*Figure 243:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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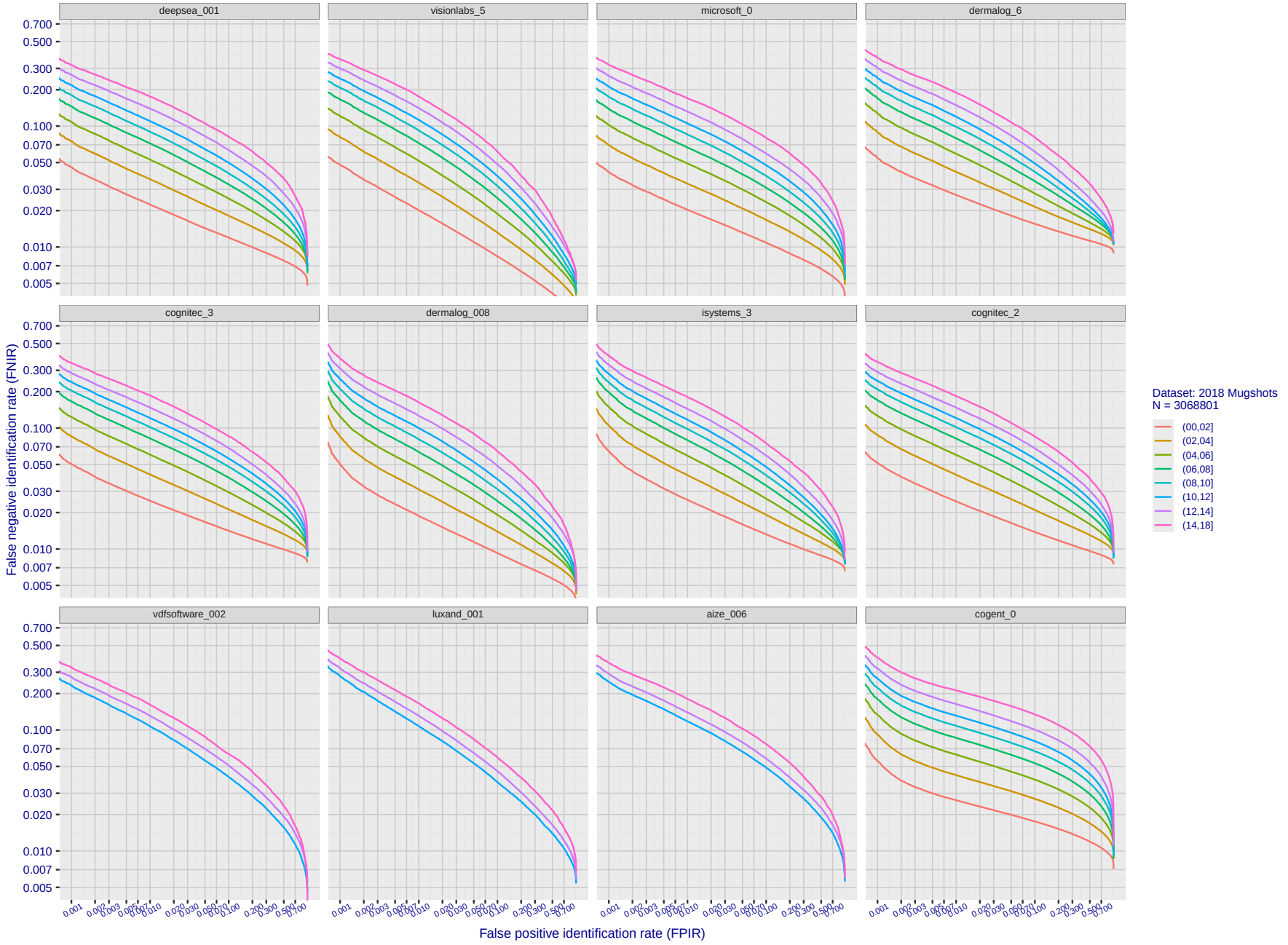


*Figure 244:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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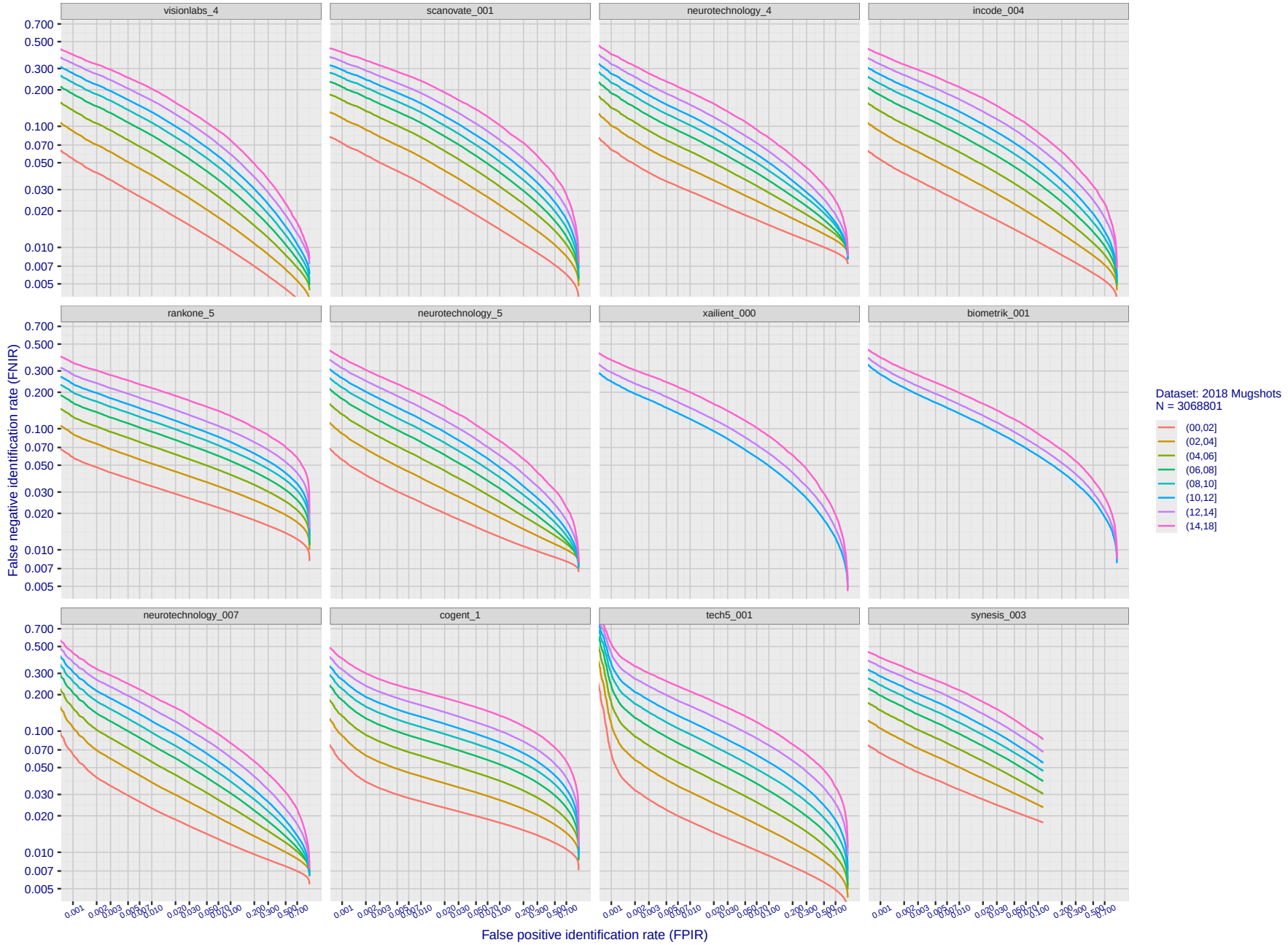


*Figure 245:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*



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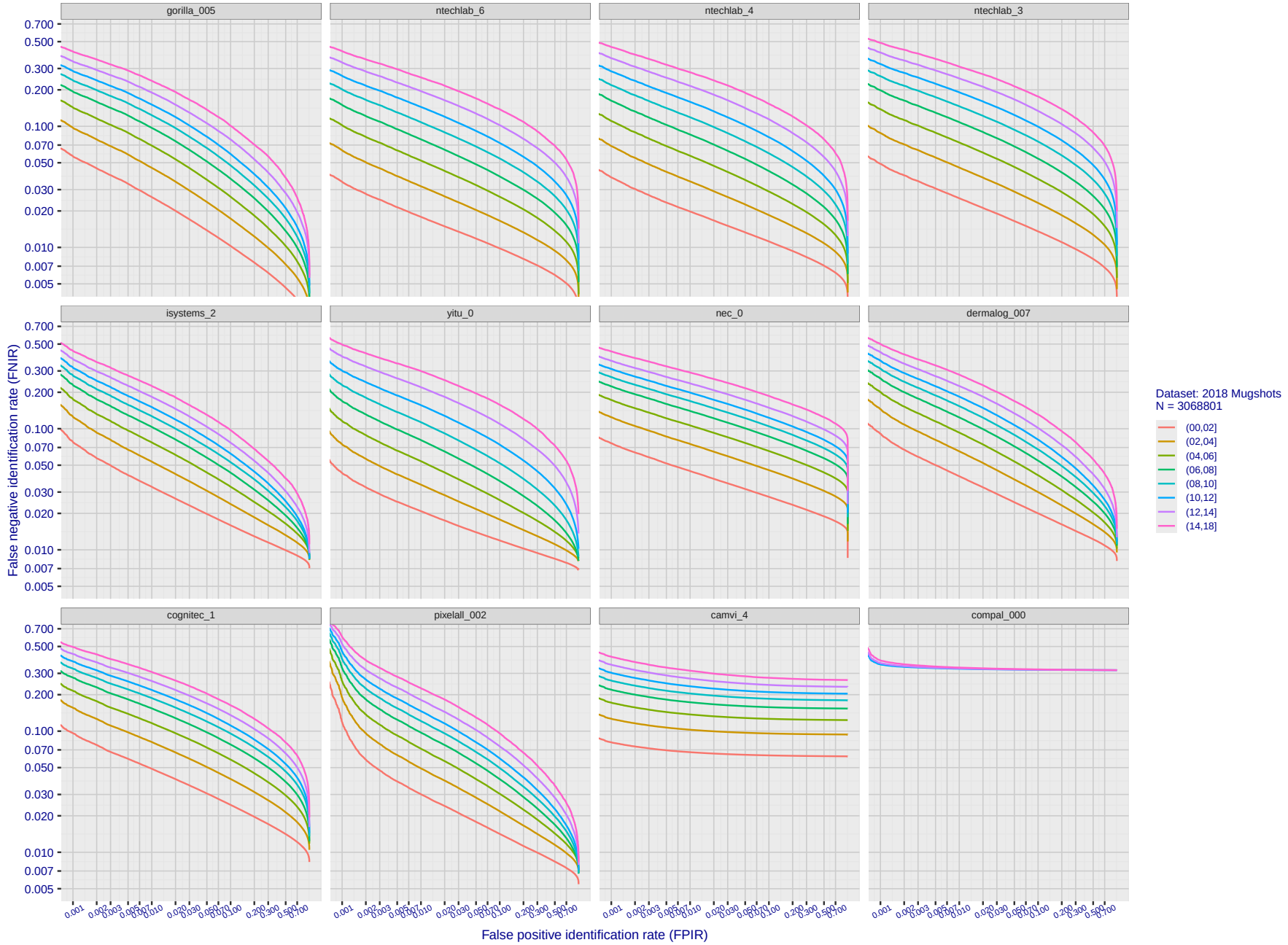


*Figure 246:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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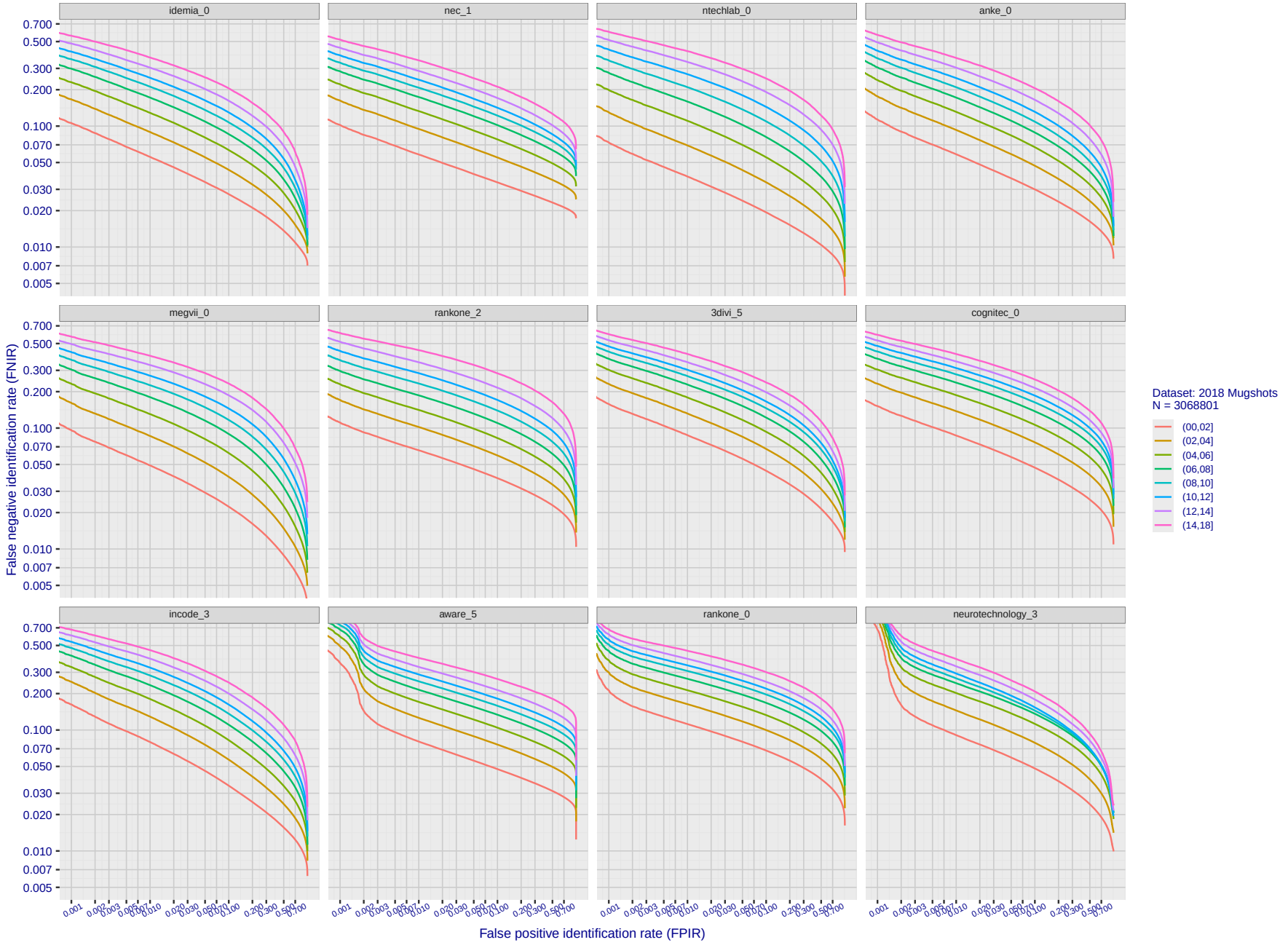


*Figure 247:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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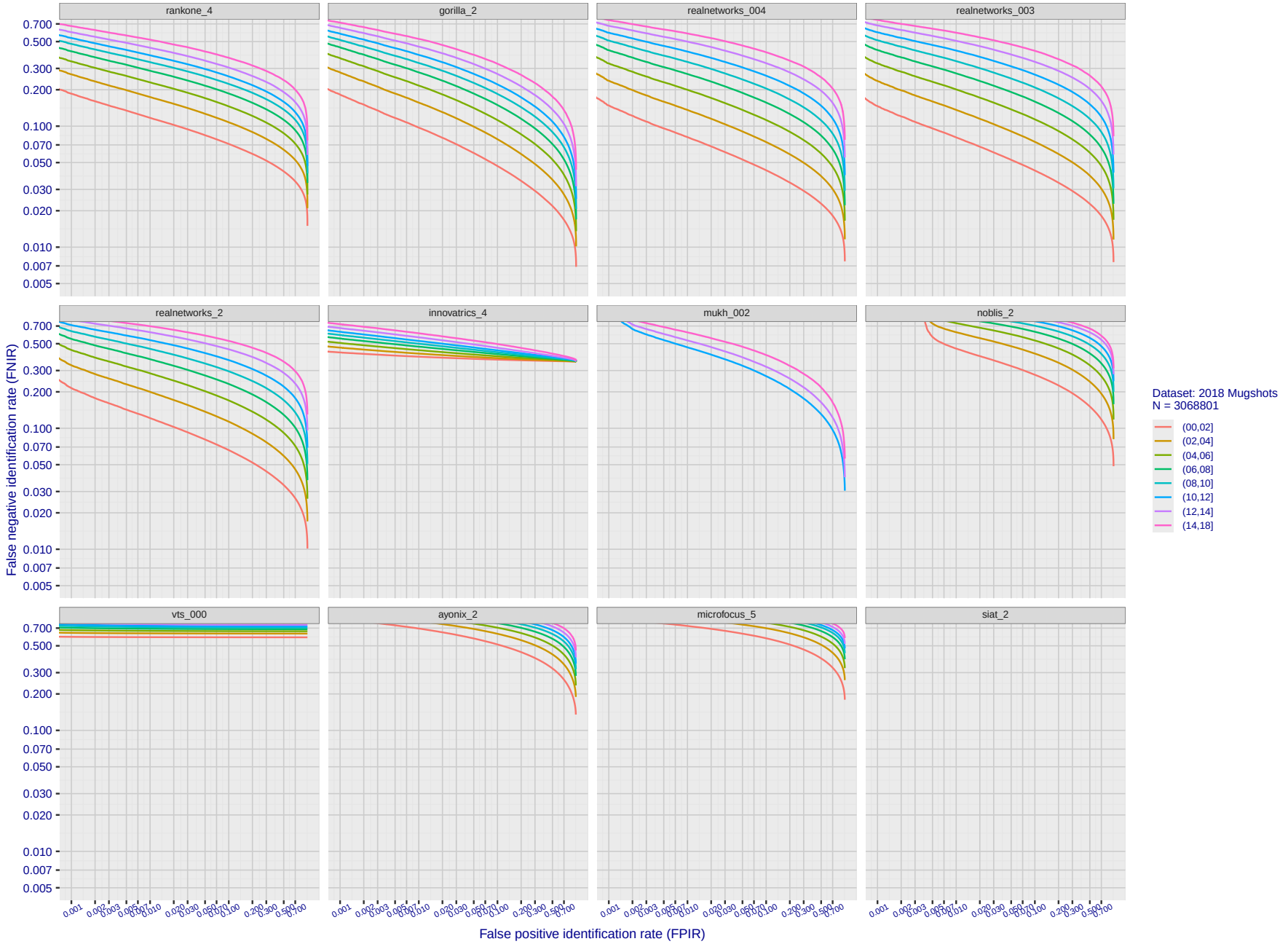


*Figure 248:* **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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| | | | | | |
|---|---|---|---|---|---|
| 2026/09/04 10:23:32 | FNIR(N, R, T) = FPIR(N, T) = | False neg. identification rate False pos. identification rate | N = Num. enrolled subjects R = Num. candidates examined | T = Threshold | $T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification |

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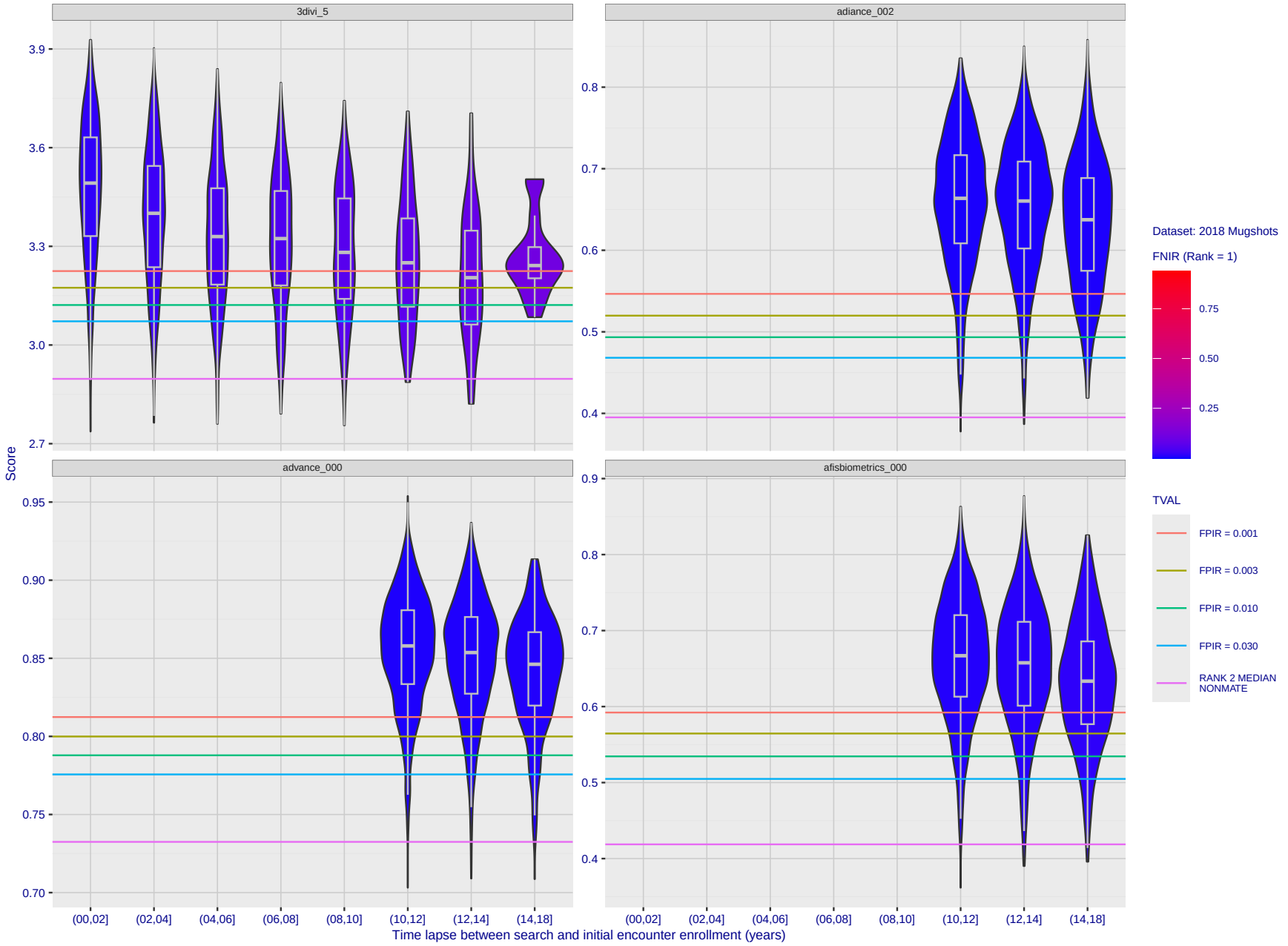


*Figure 249:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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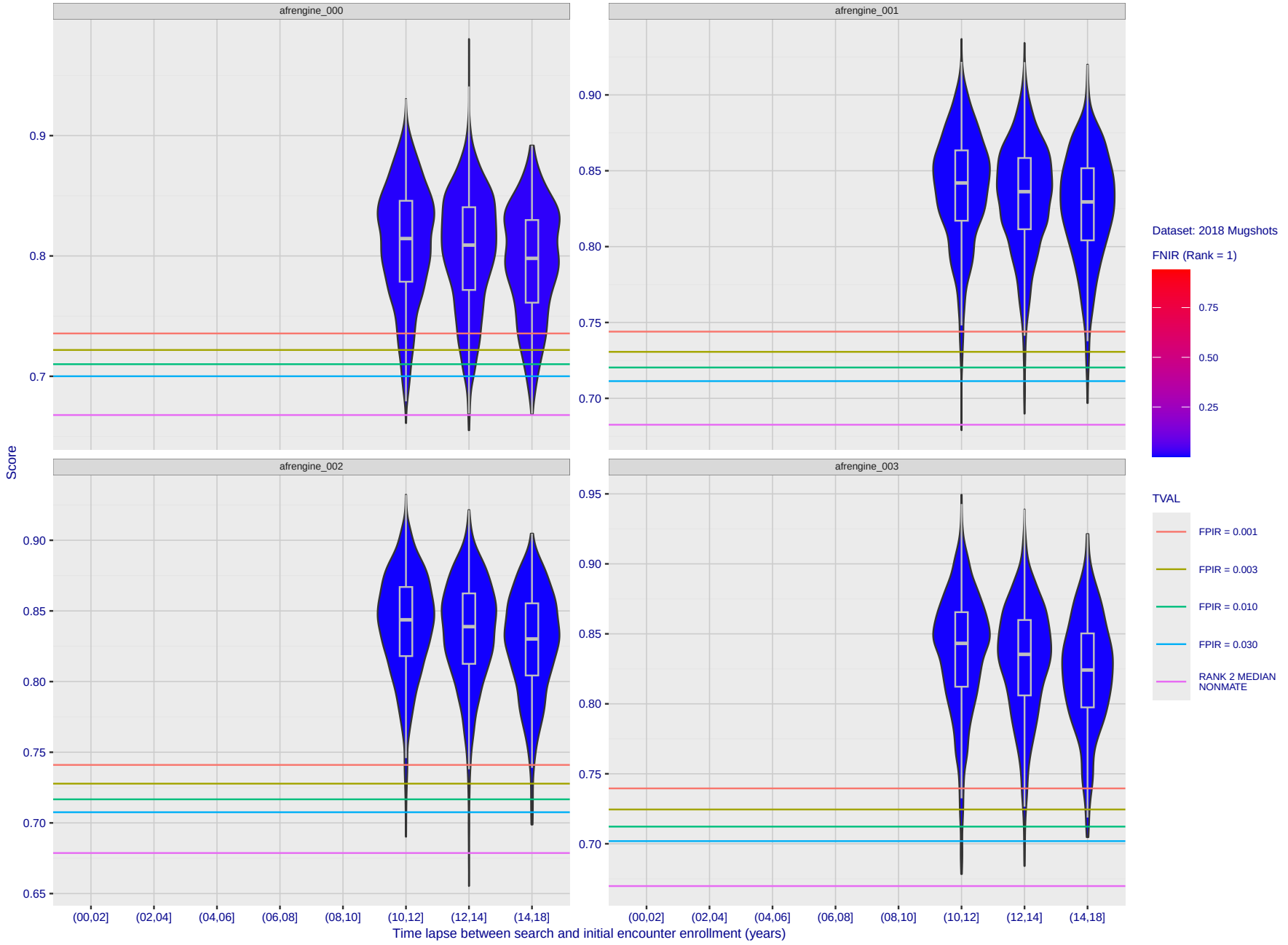


*Figure 250:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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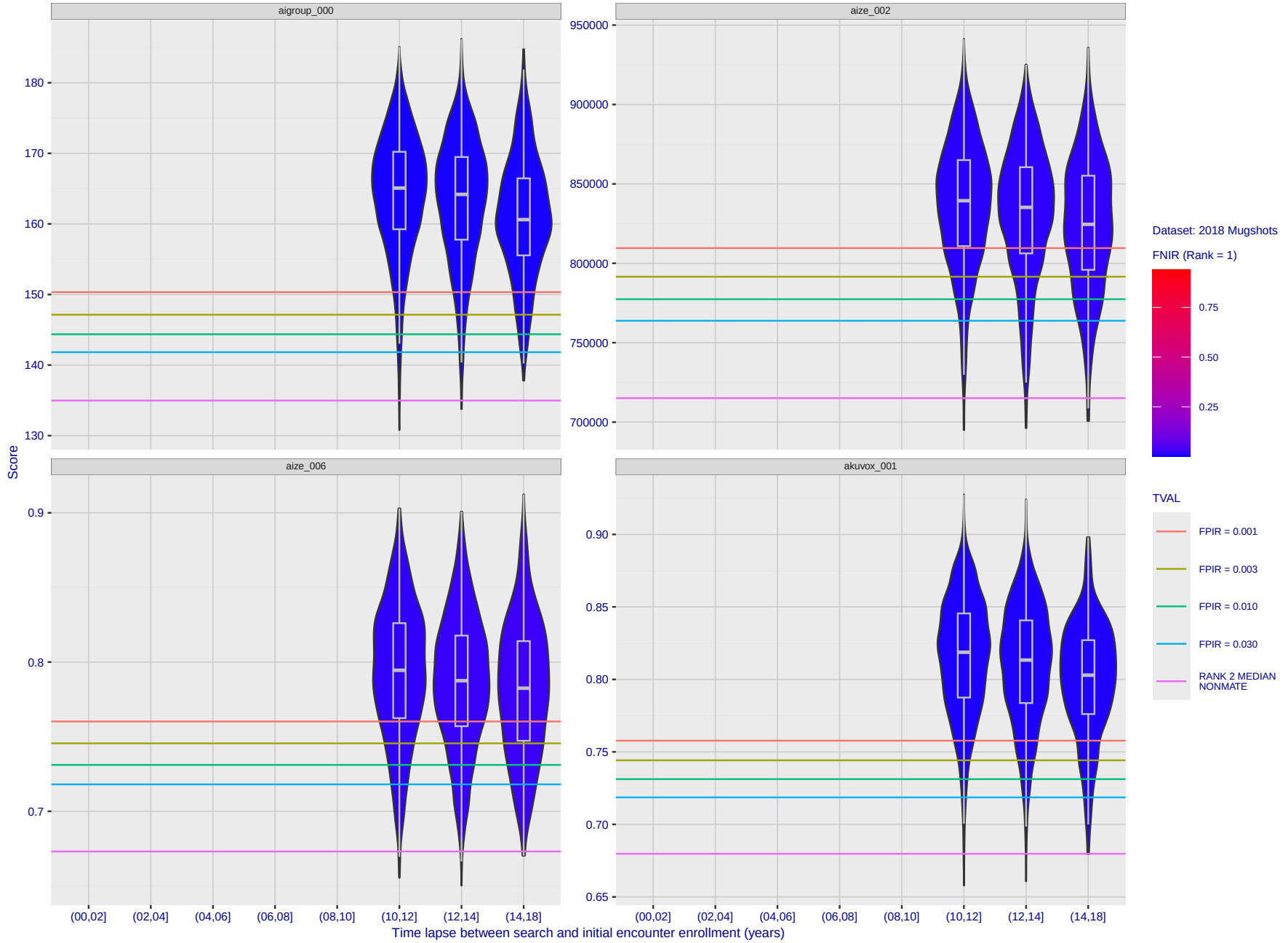


*Figure 251:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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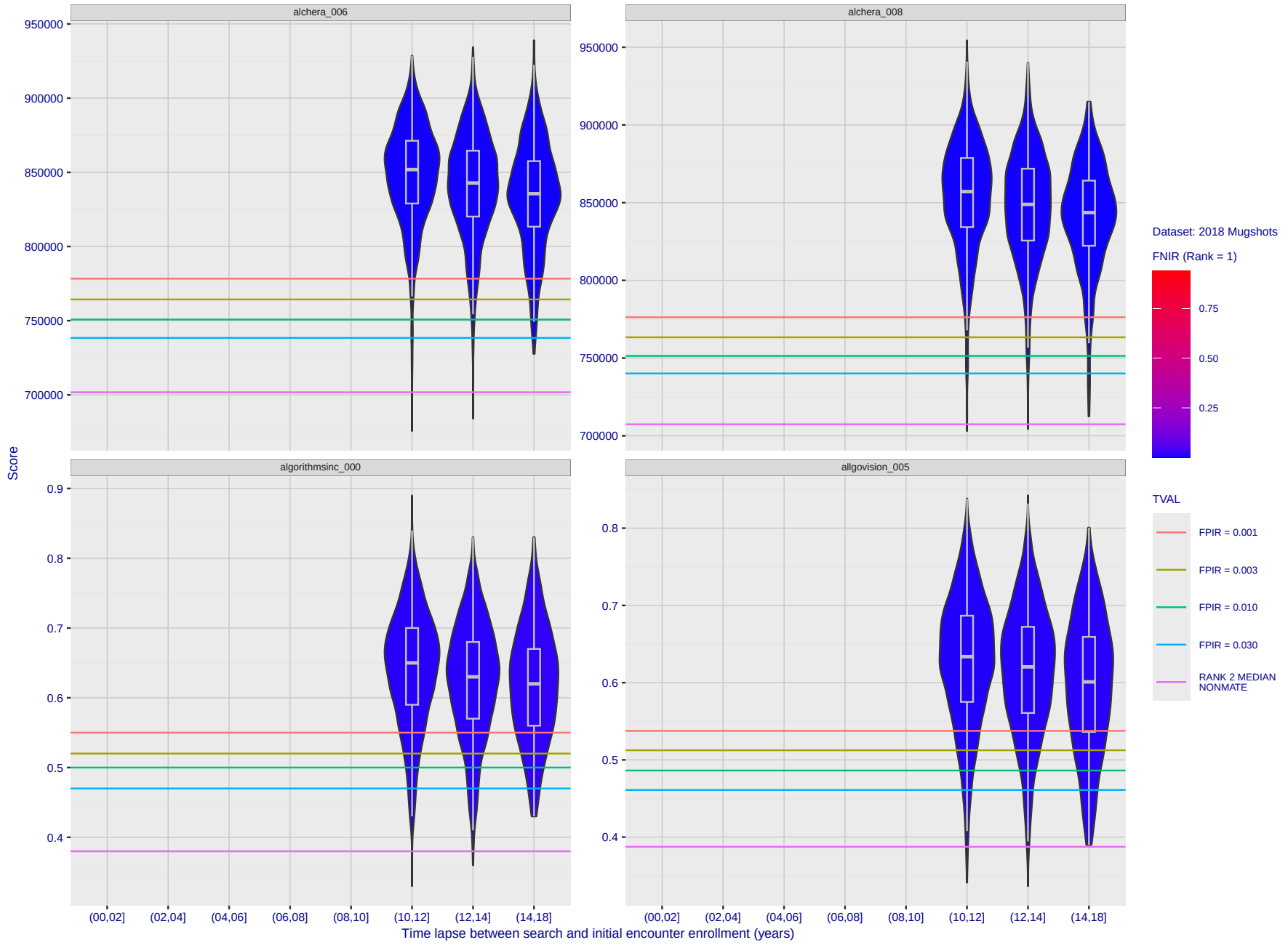


*Figure 252:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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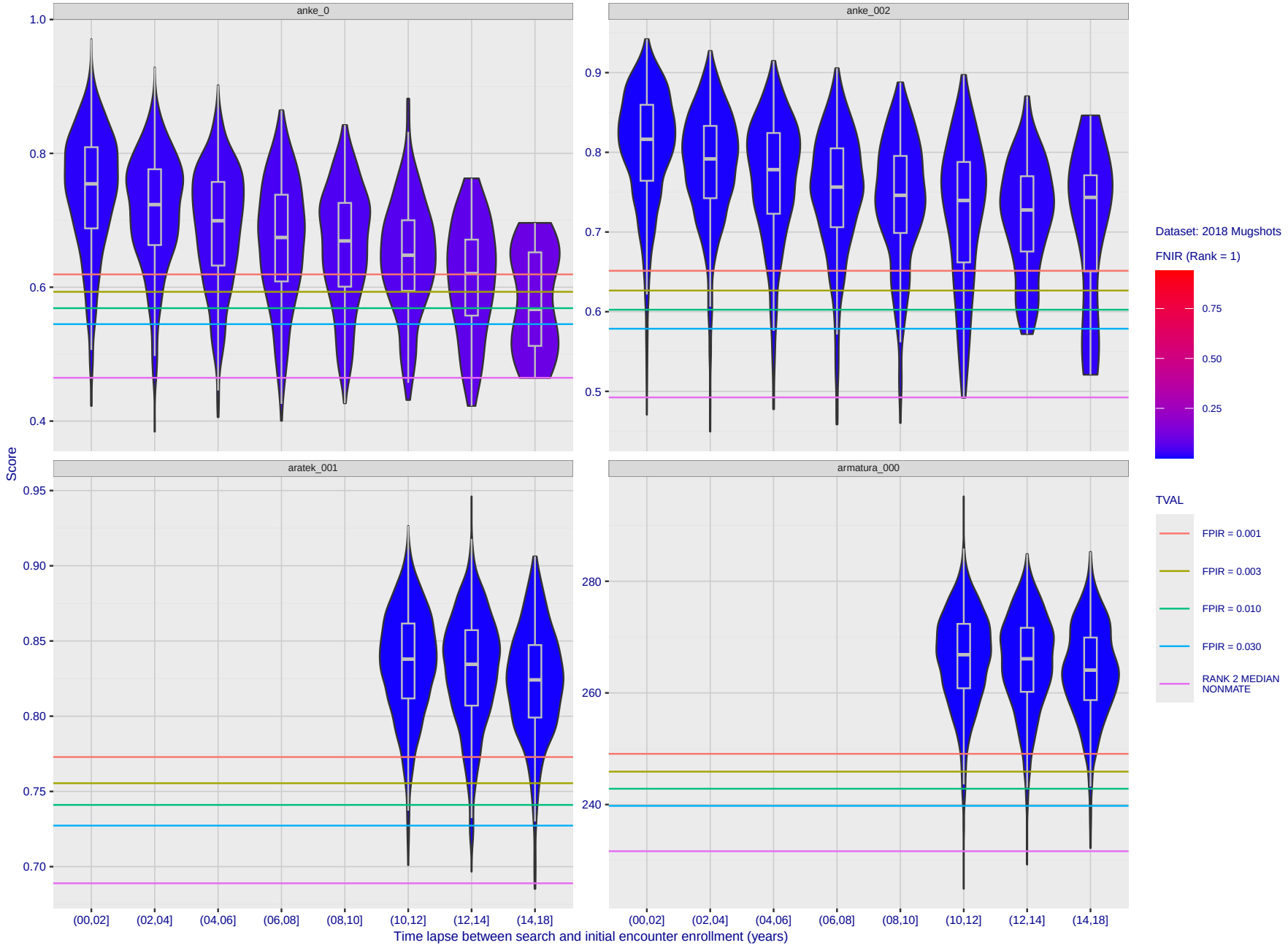


*Figure 253:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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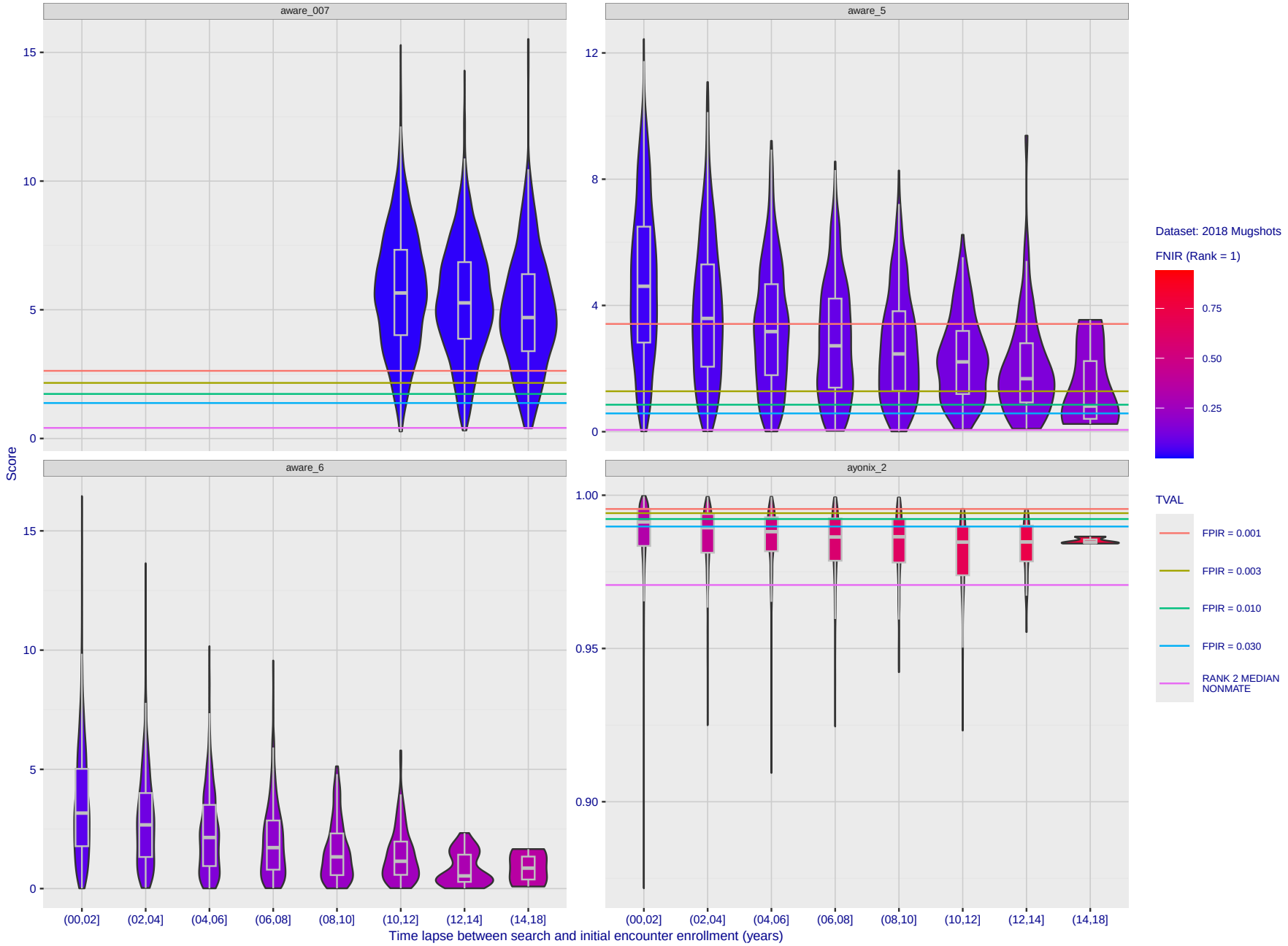


*Figure 254:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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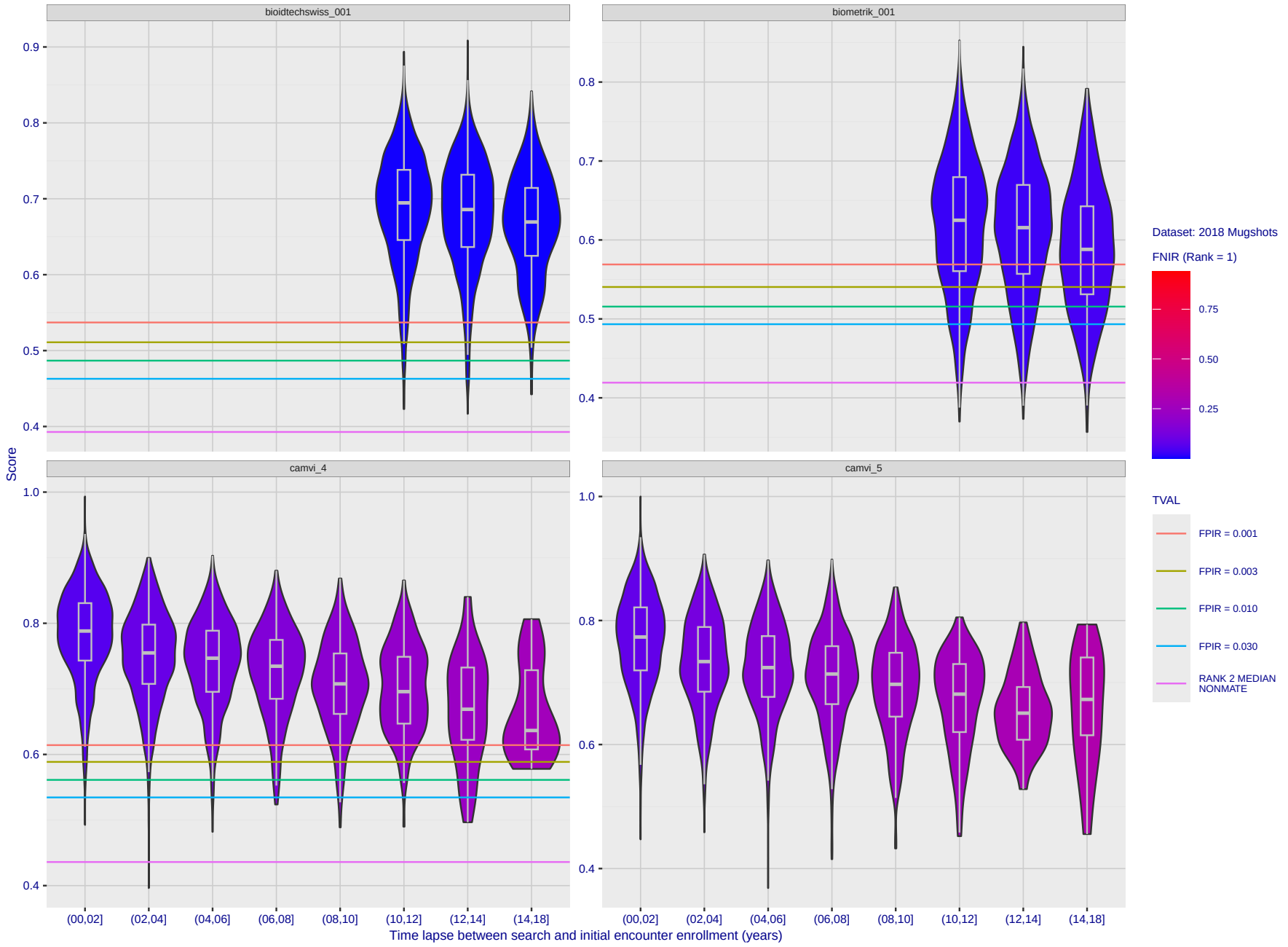


*Figure 255:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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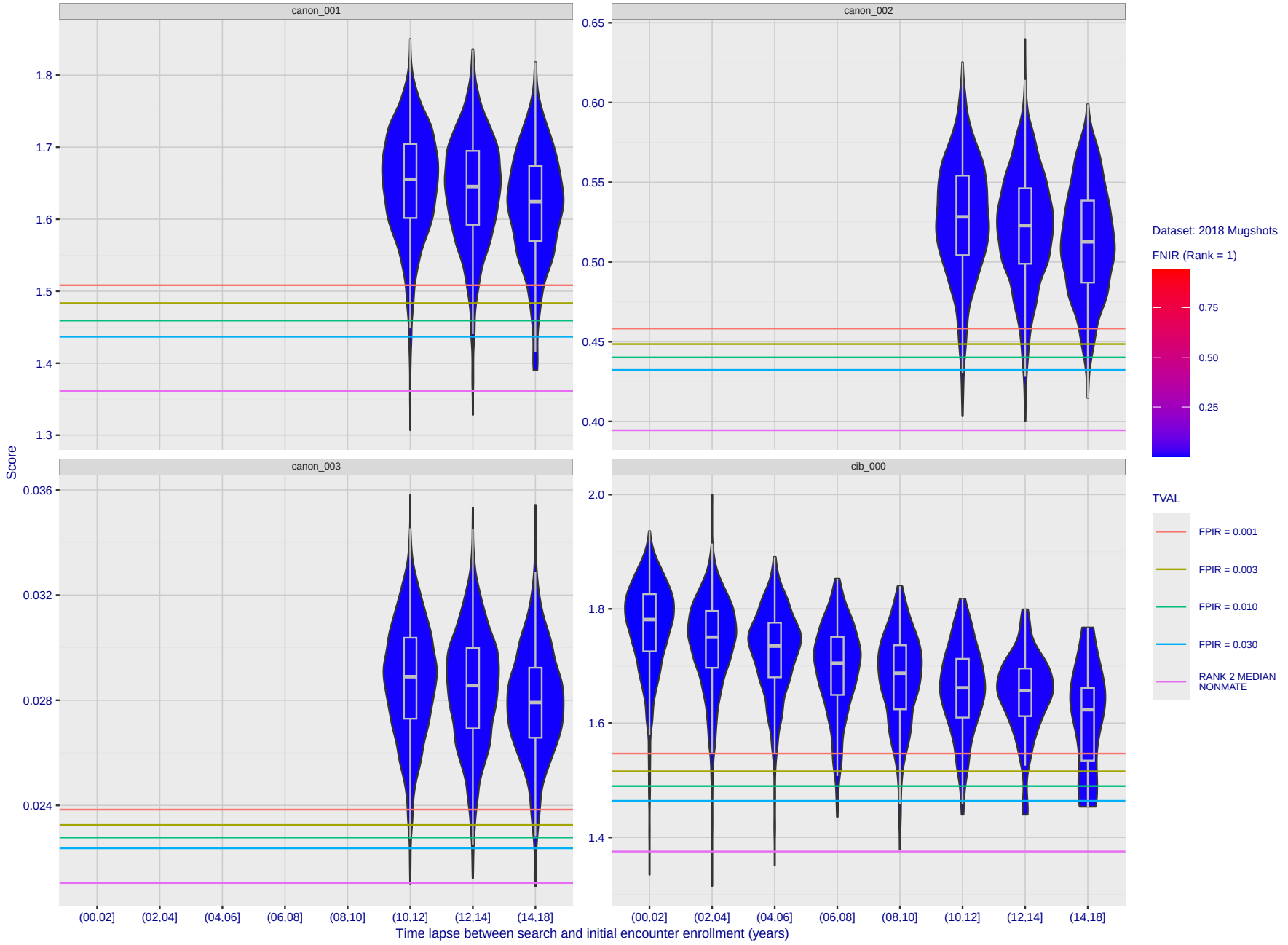


*Figure 256:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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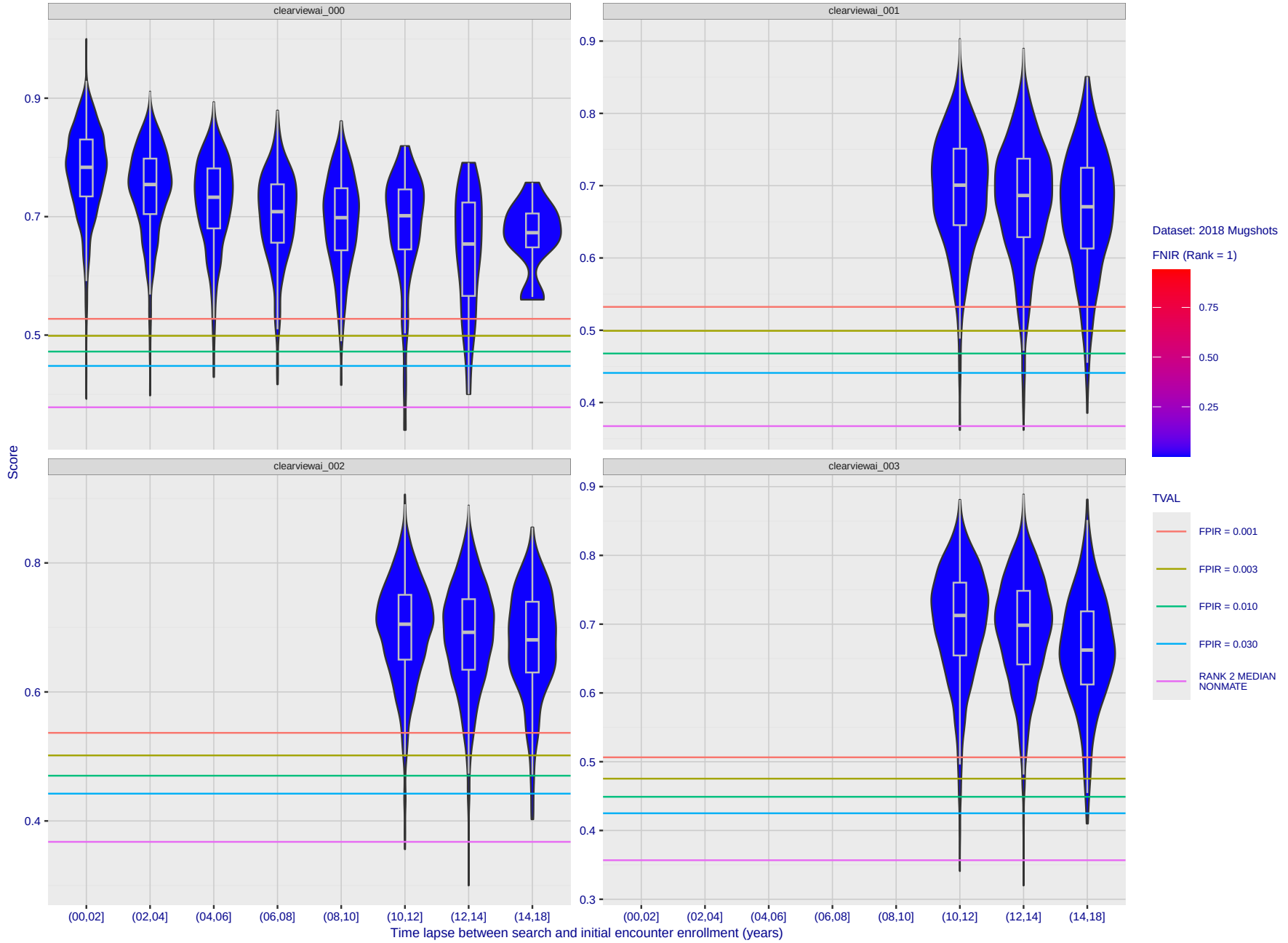


*Figure 257:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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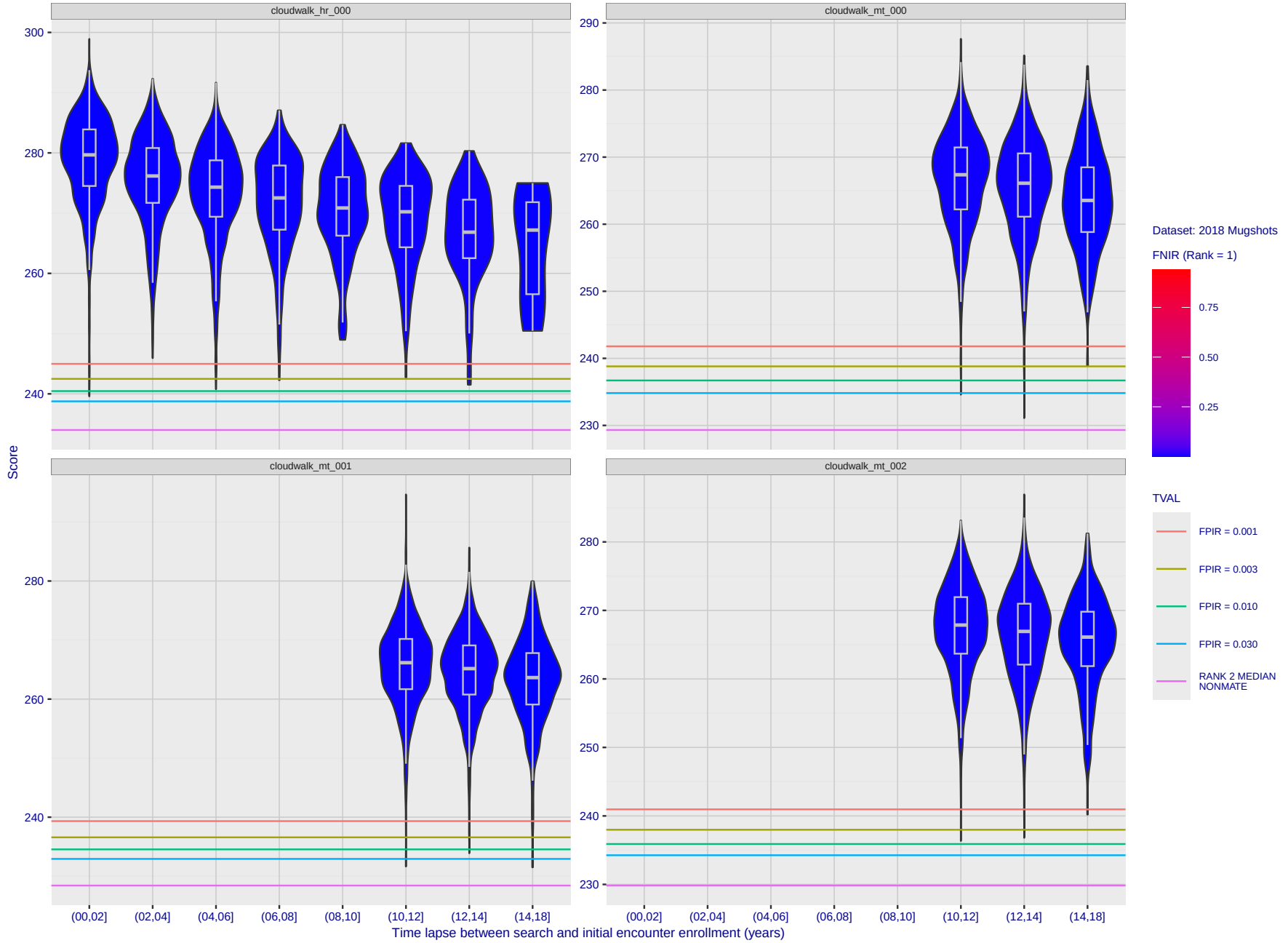


*Figure 258:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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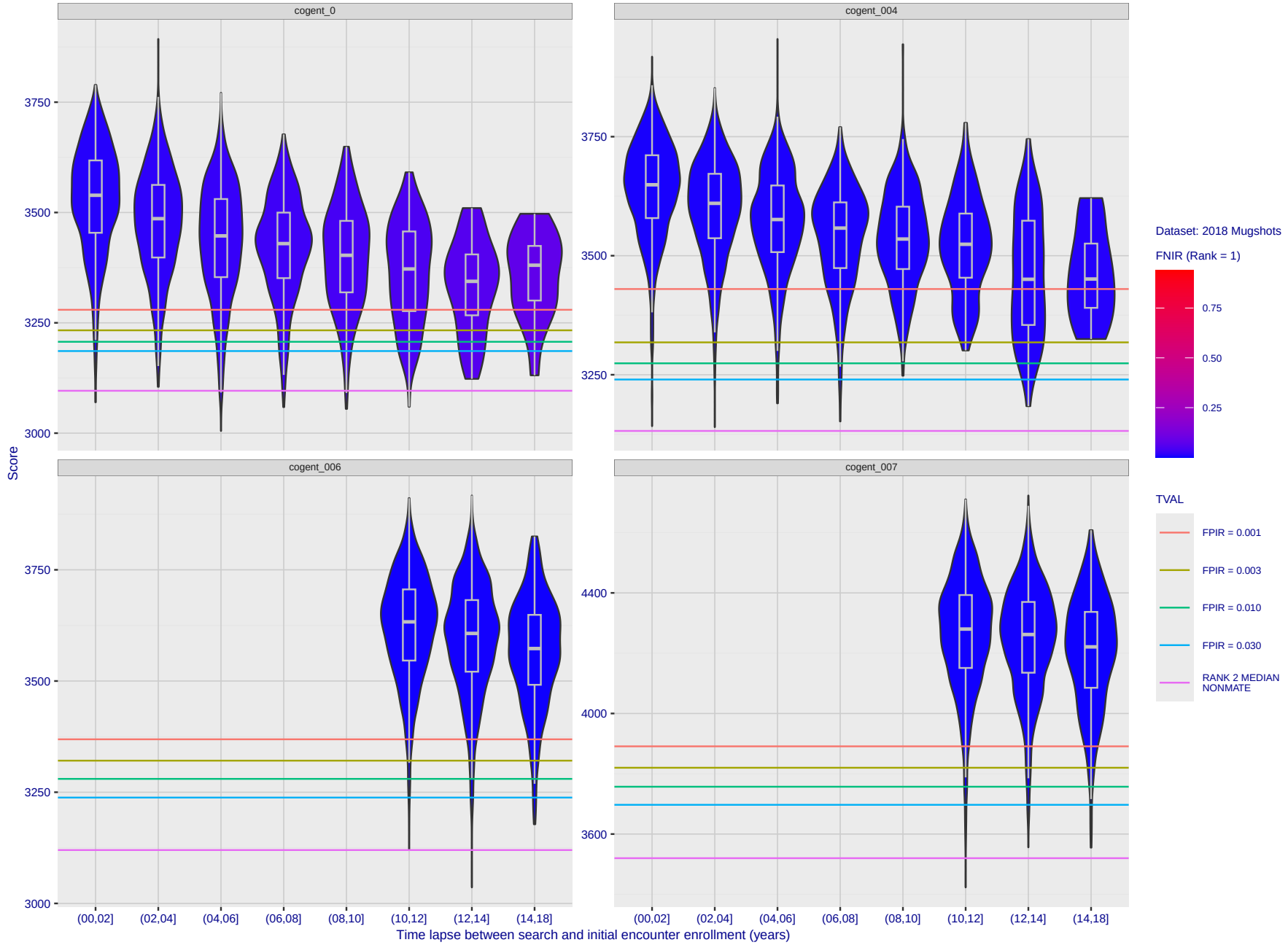


*Figure 259:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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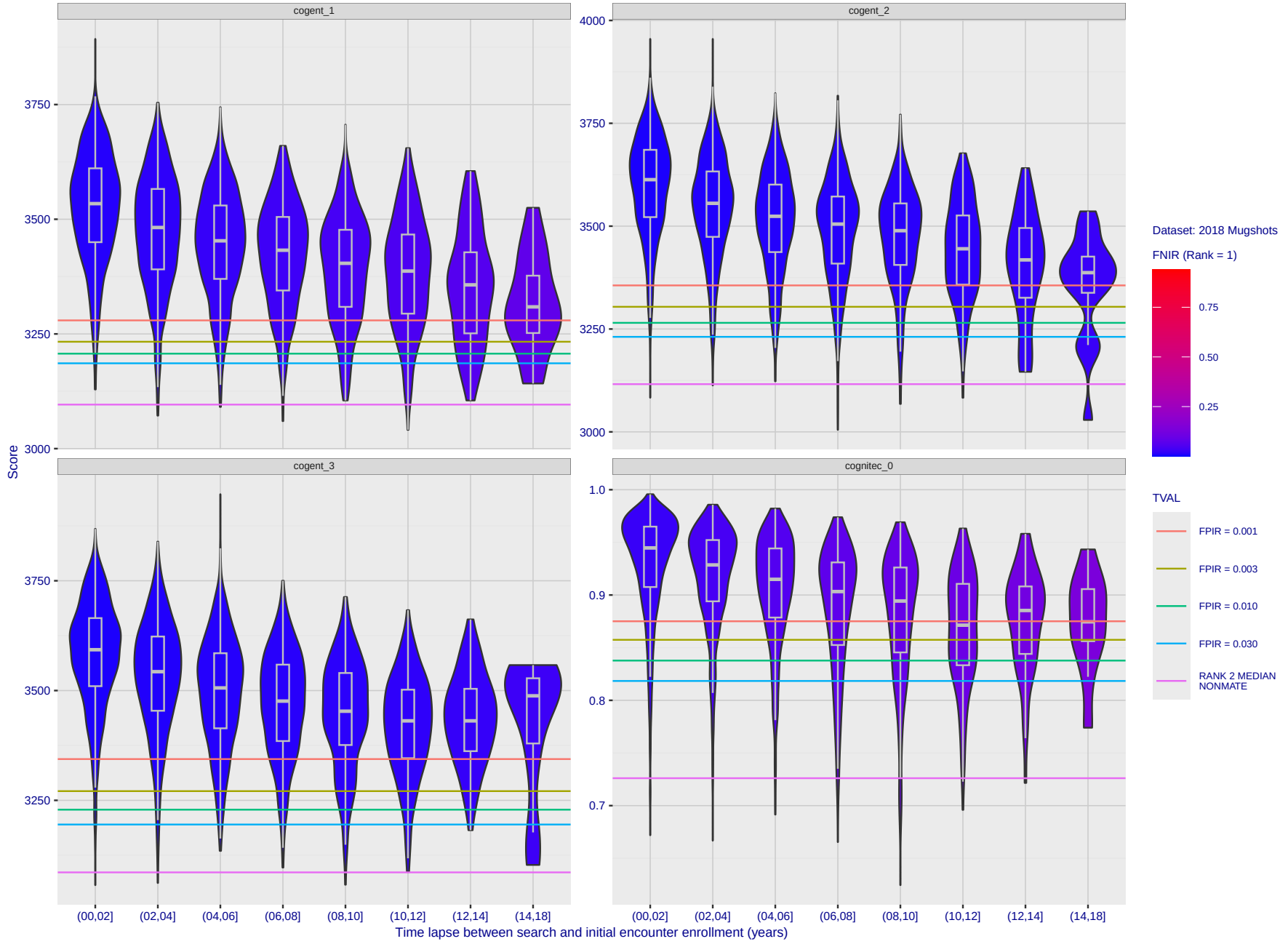


*Figure 260:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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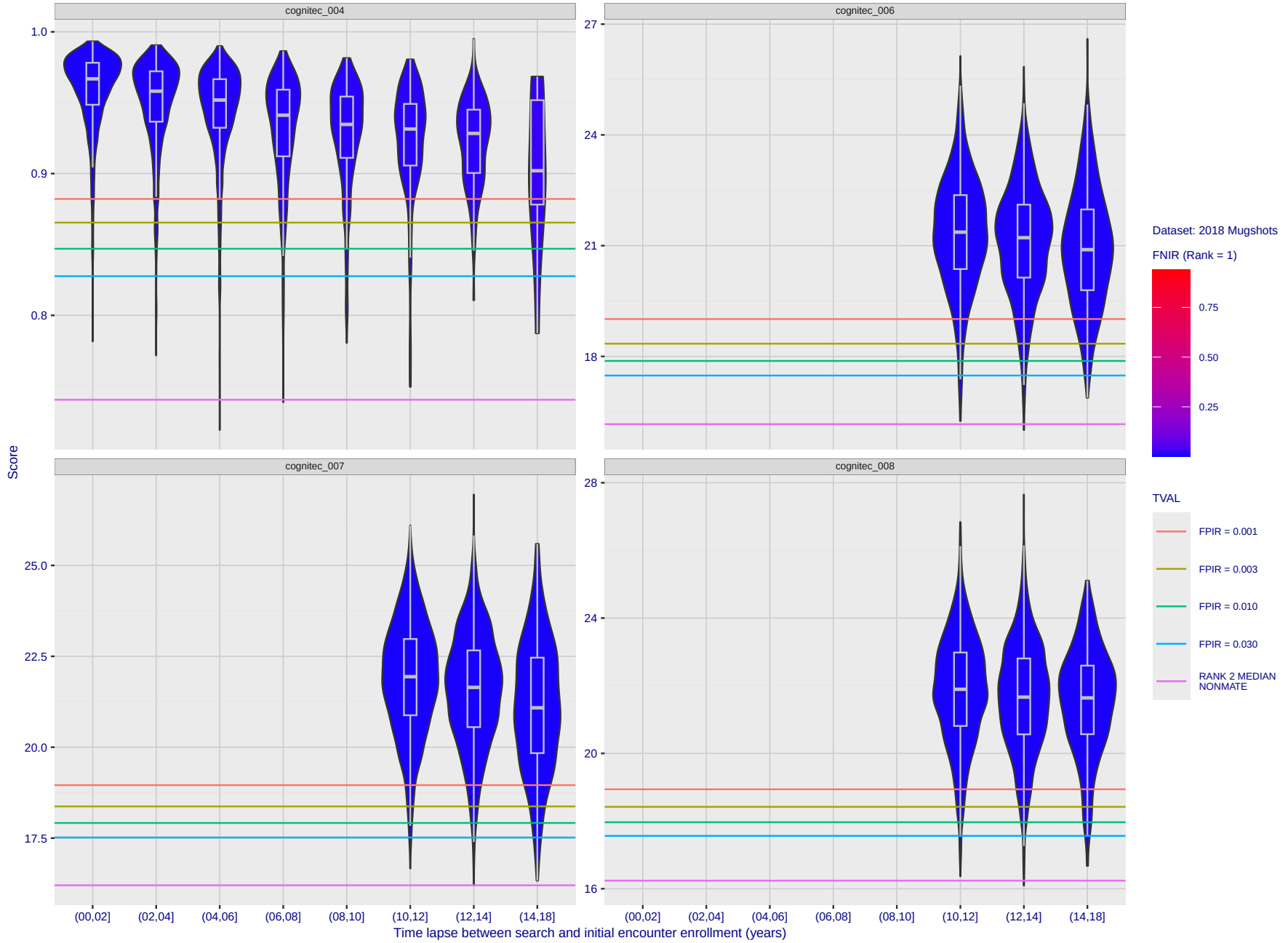


*Figure 261:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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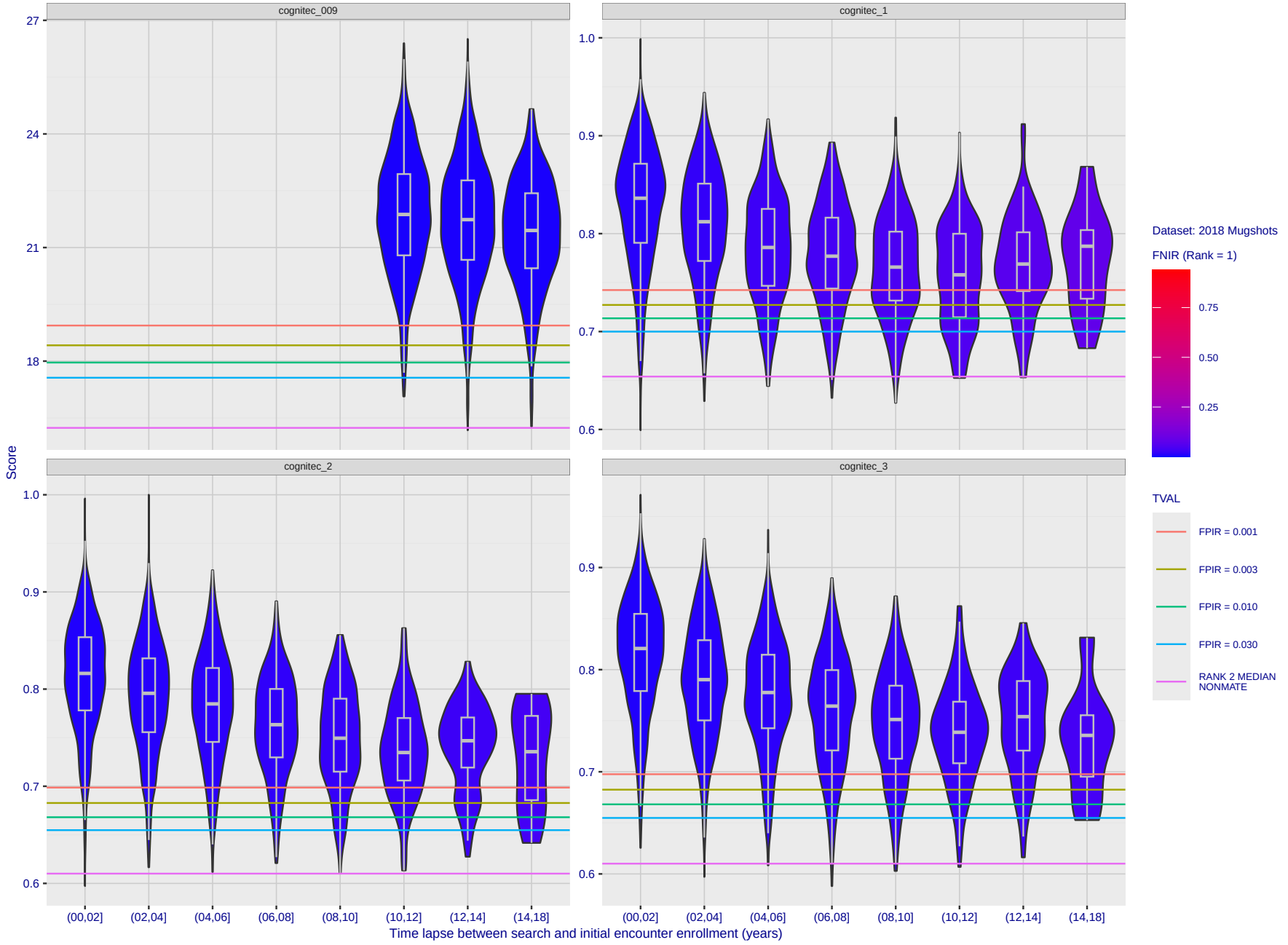


*Figure 262:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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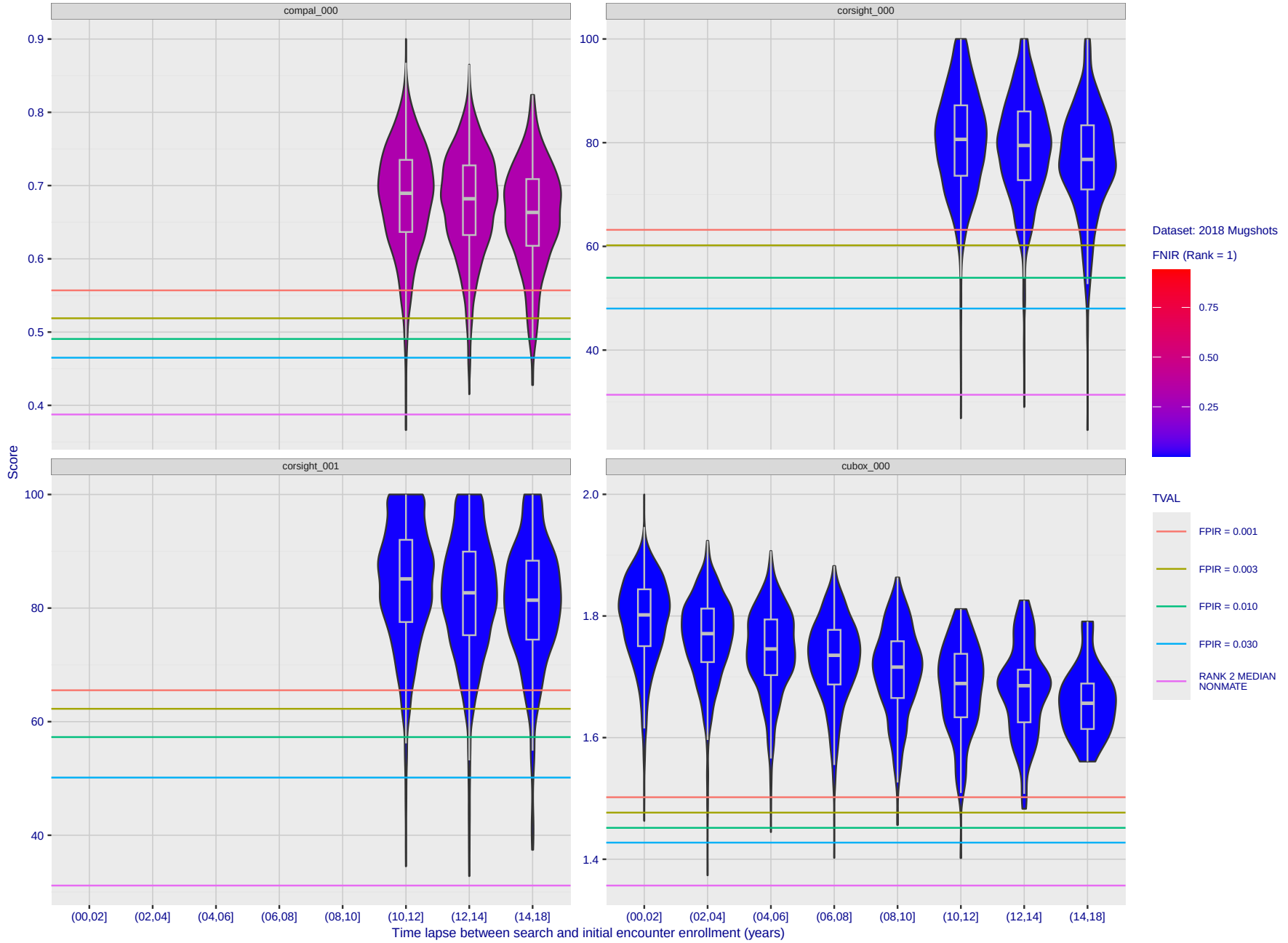


*Figure 263:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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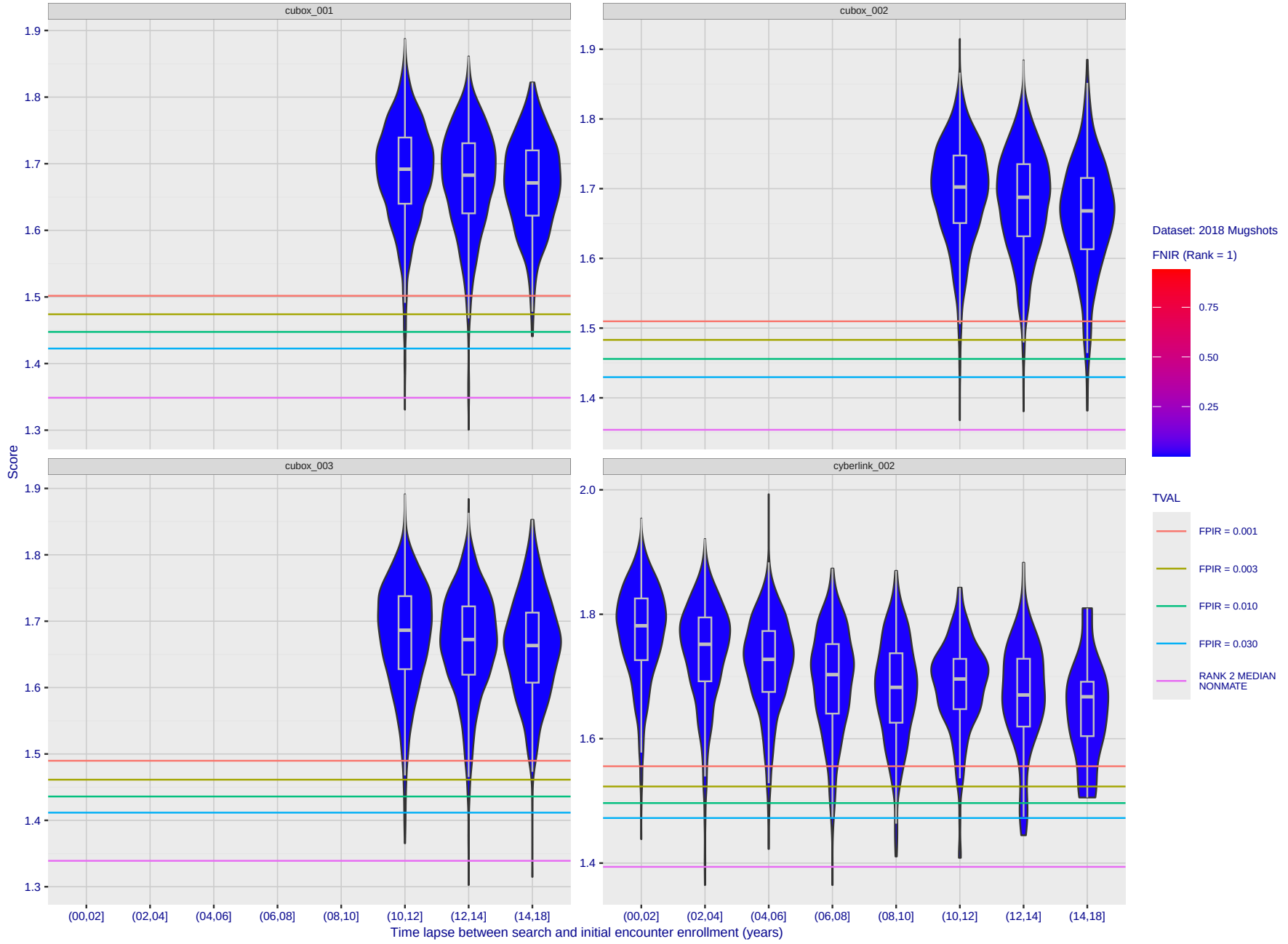


*Figure 264:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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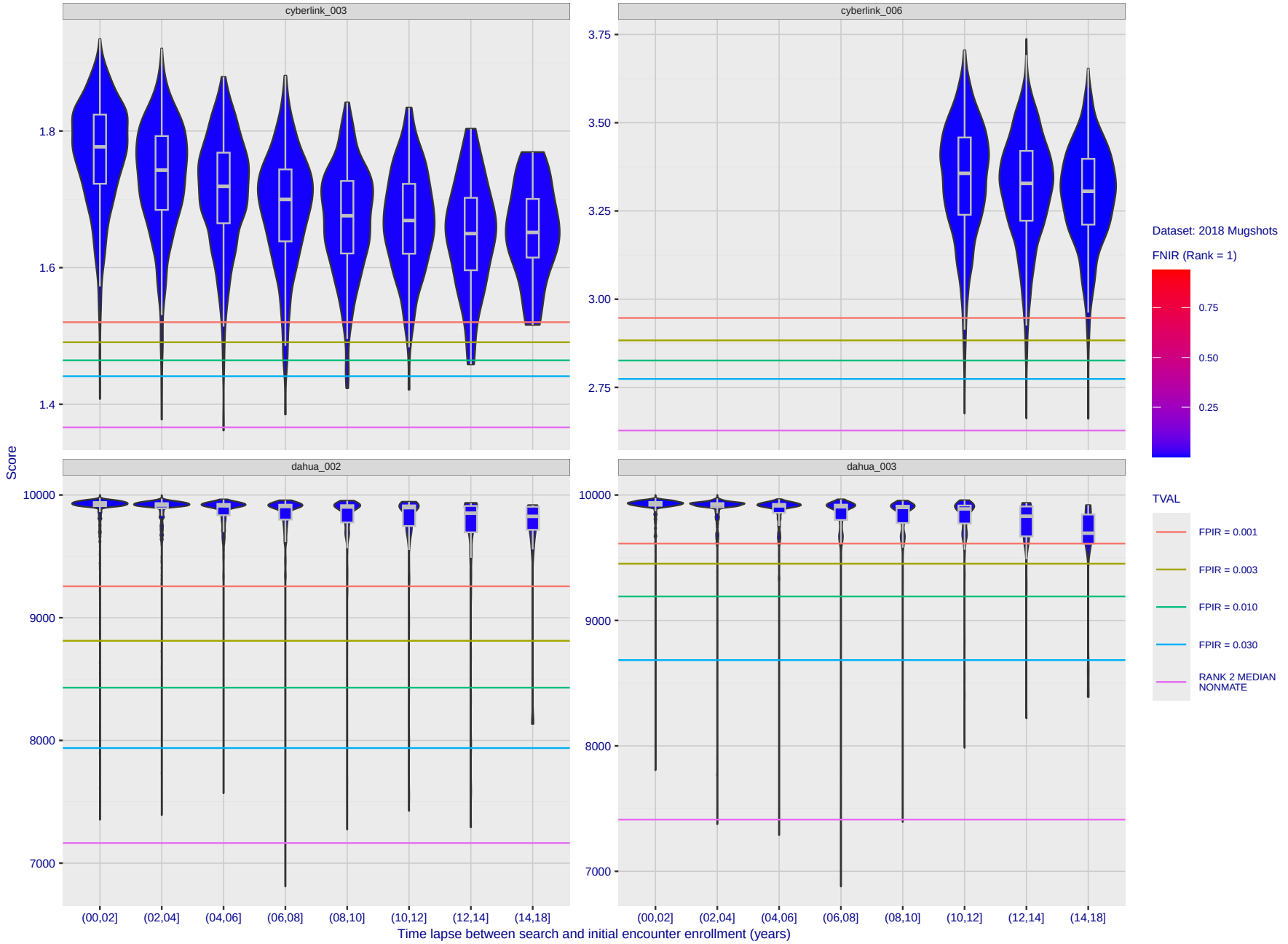


*Figure 265:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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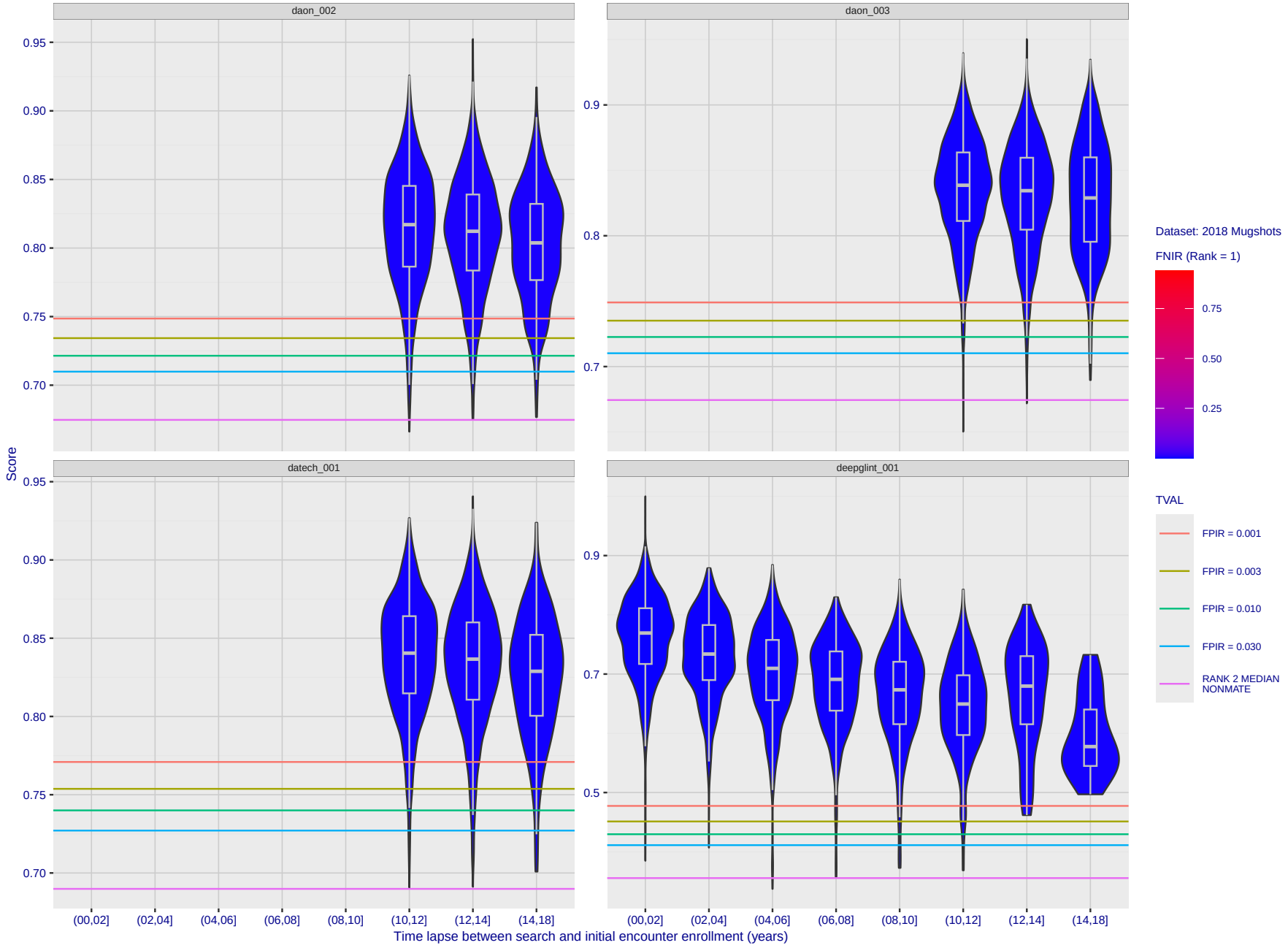


*Figure 266:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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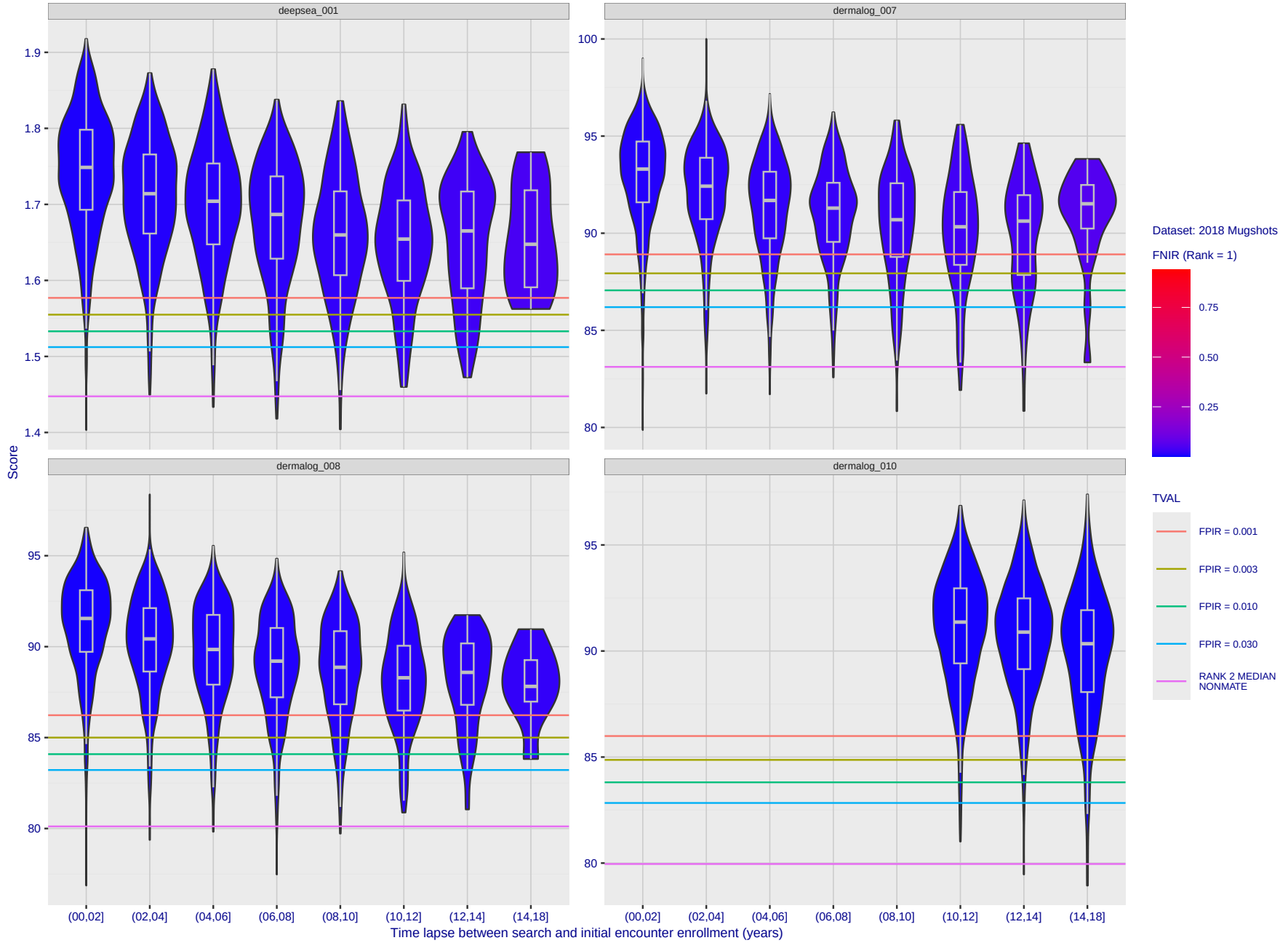


*Figure 267:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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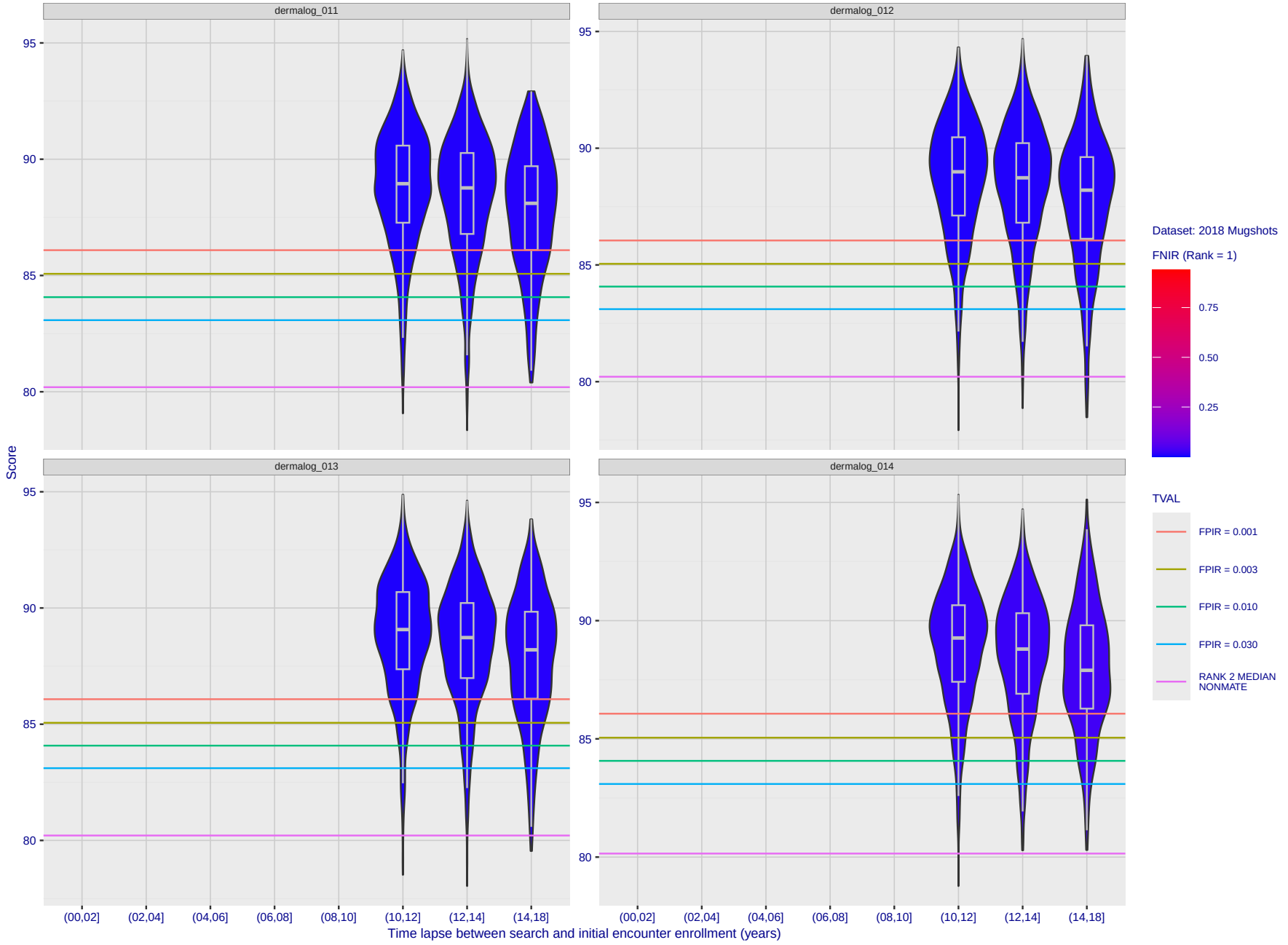


*Figure 268:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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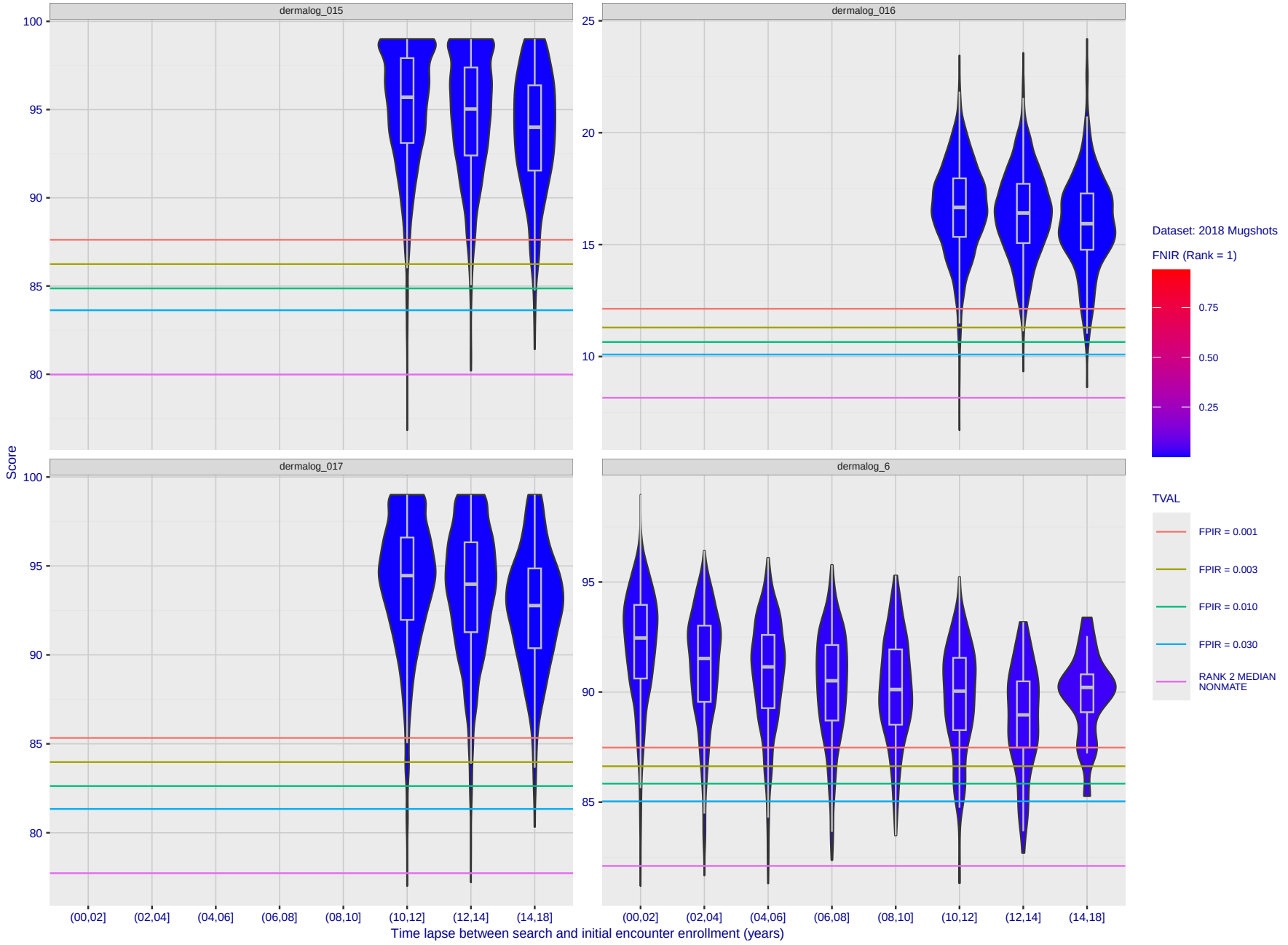


*Figure 269:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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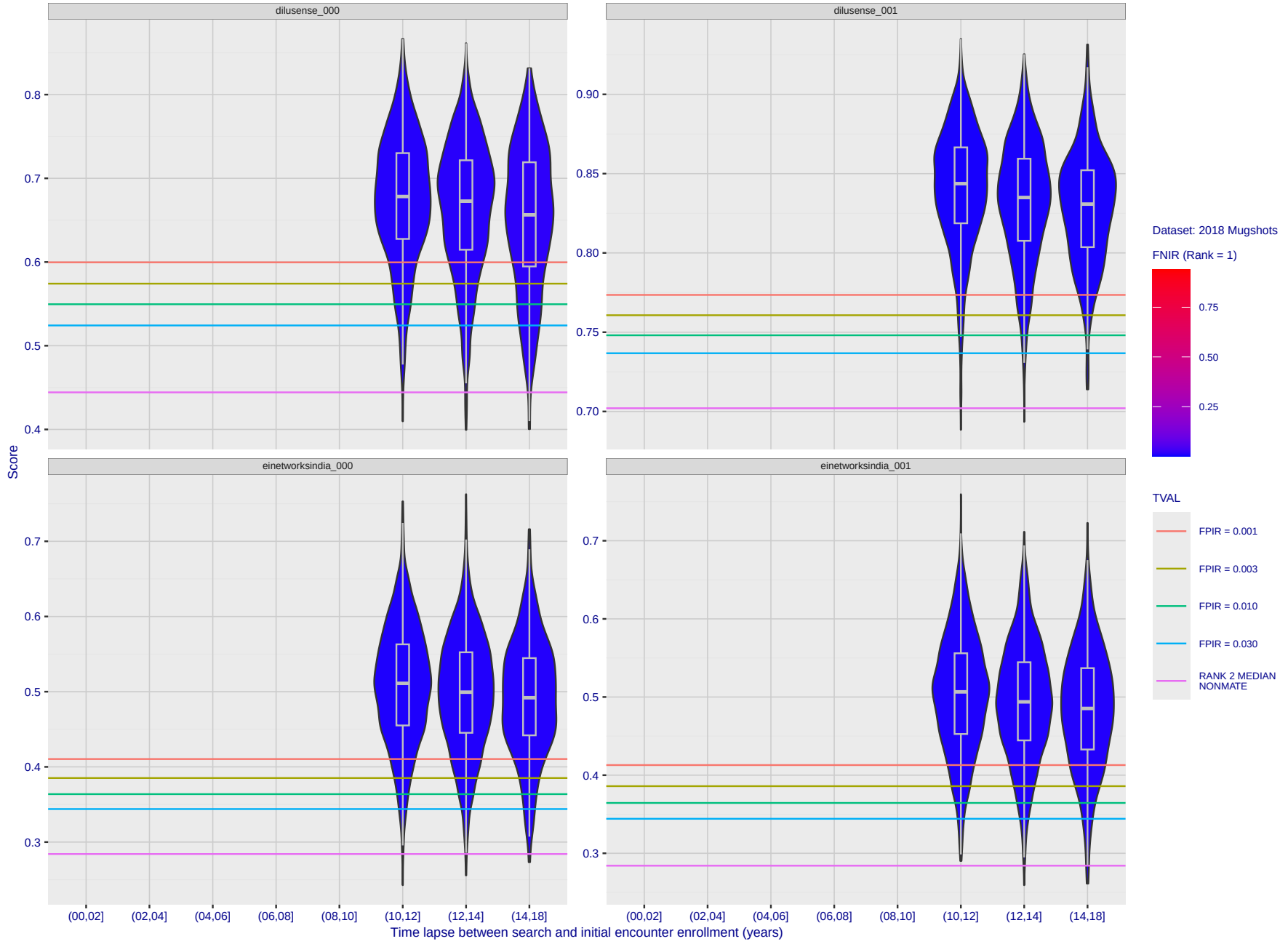


*Figure 270:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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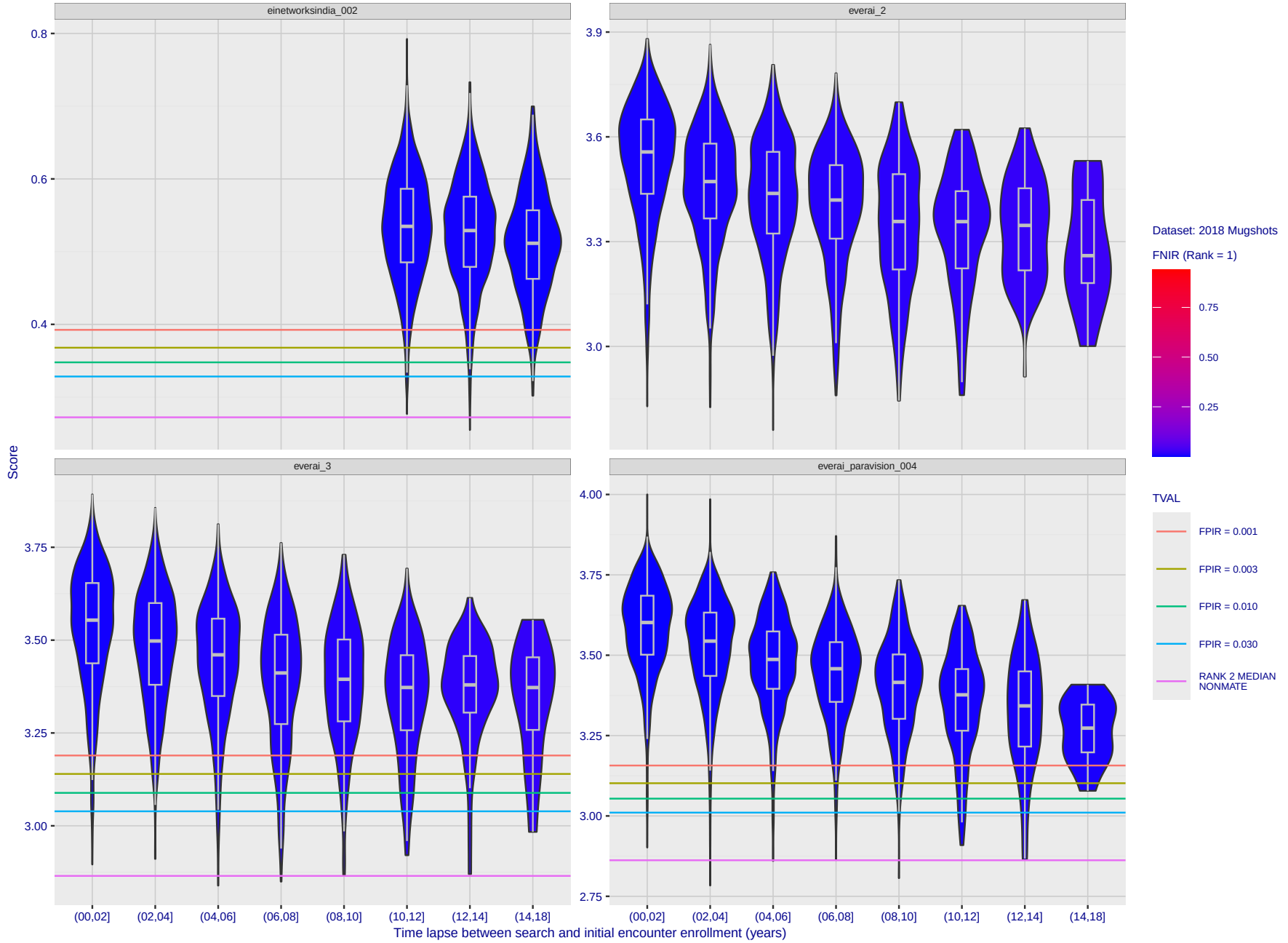


*Figure 271:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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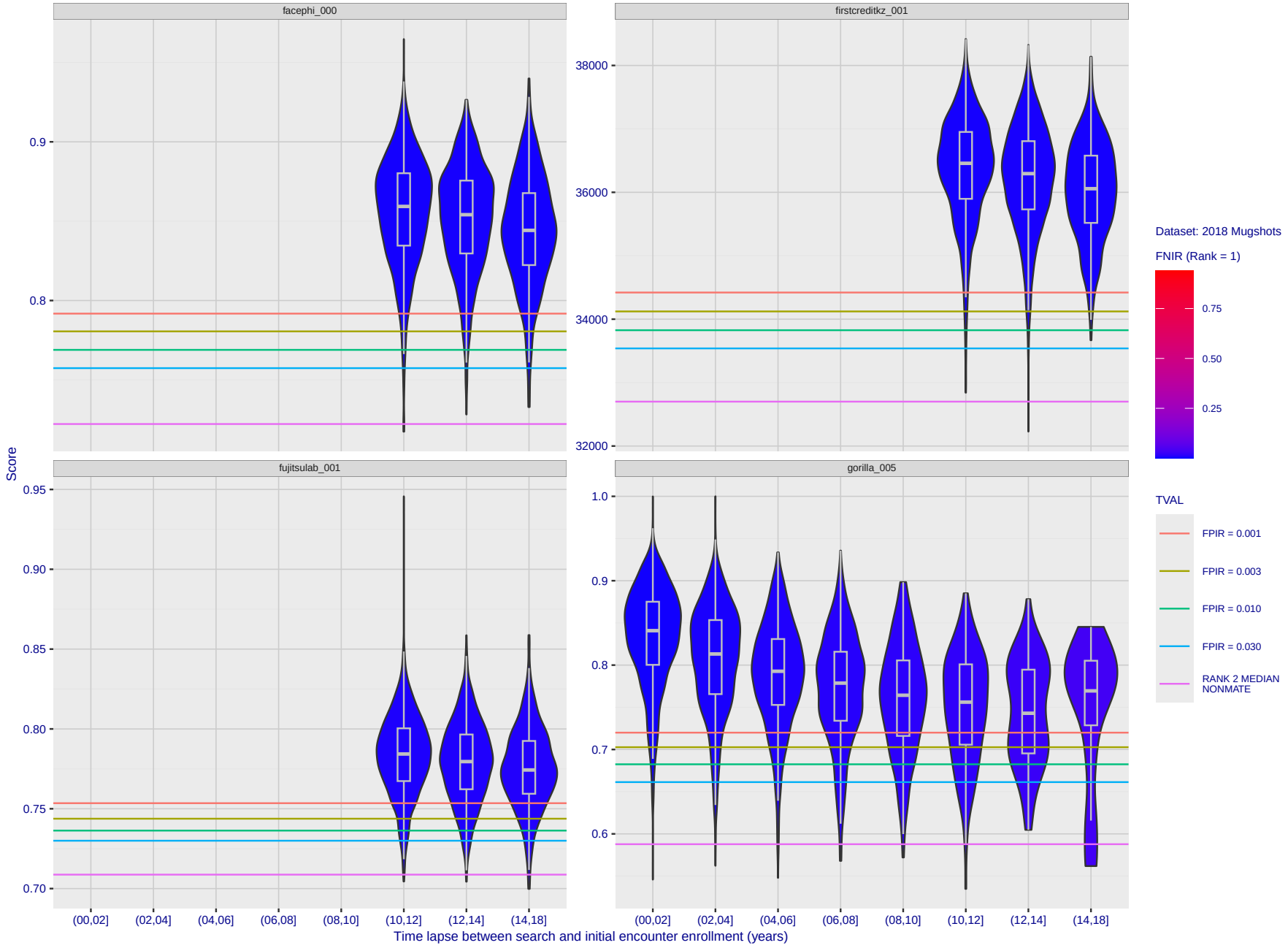


*Figure 272:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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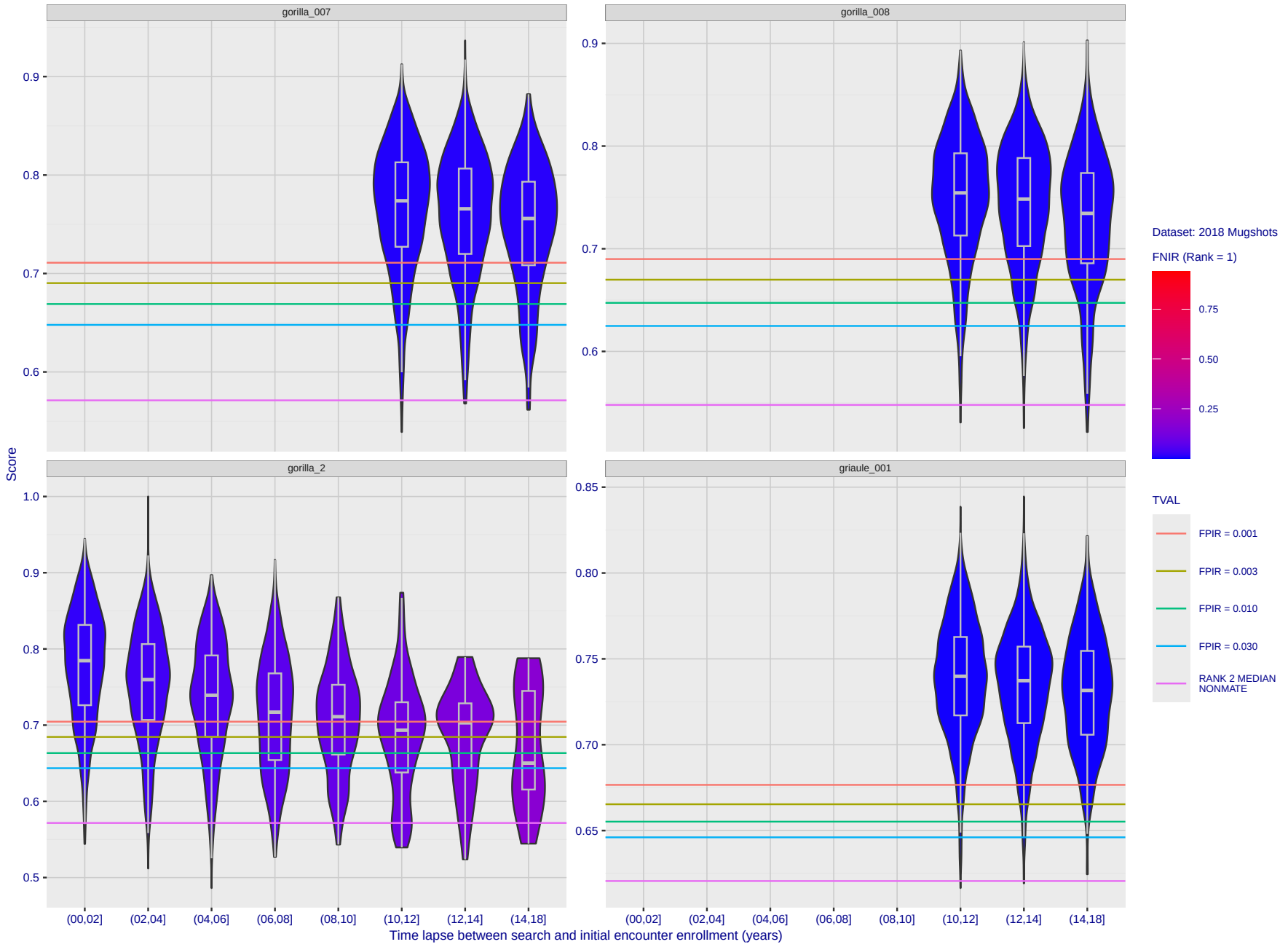


*Figure 273:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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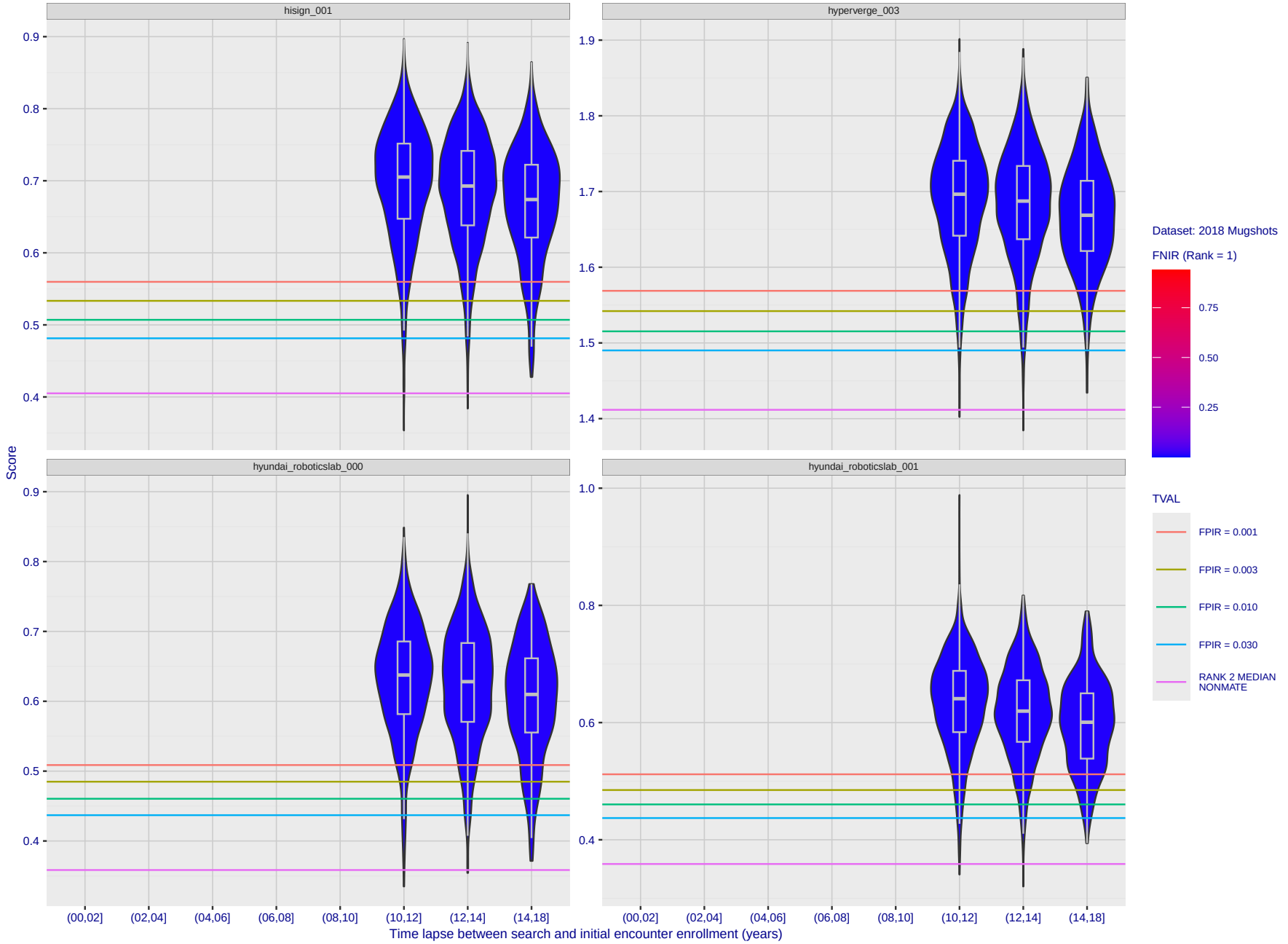


*Figure 274:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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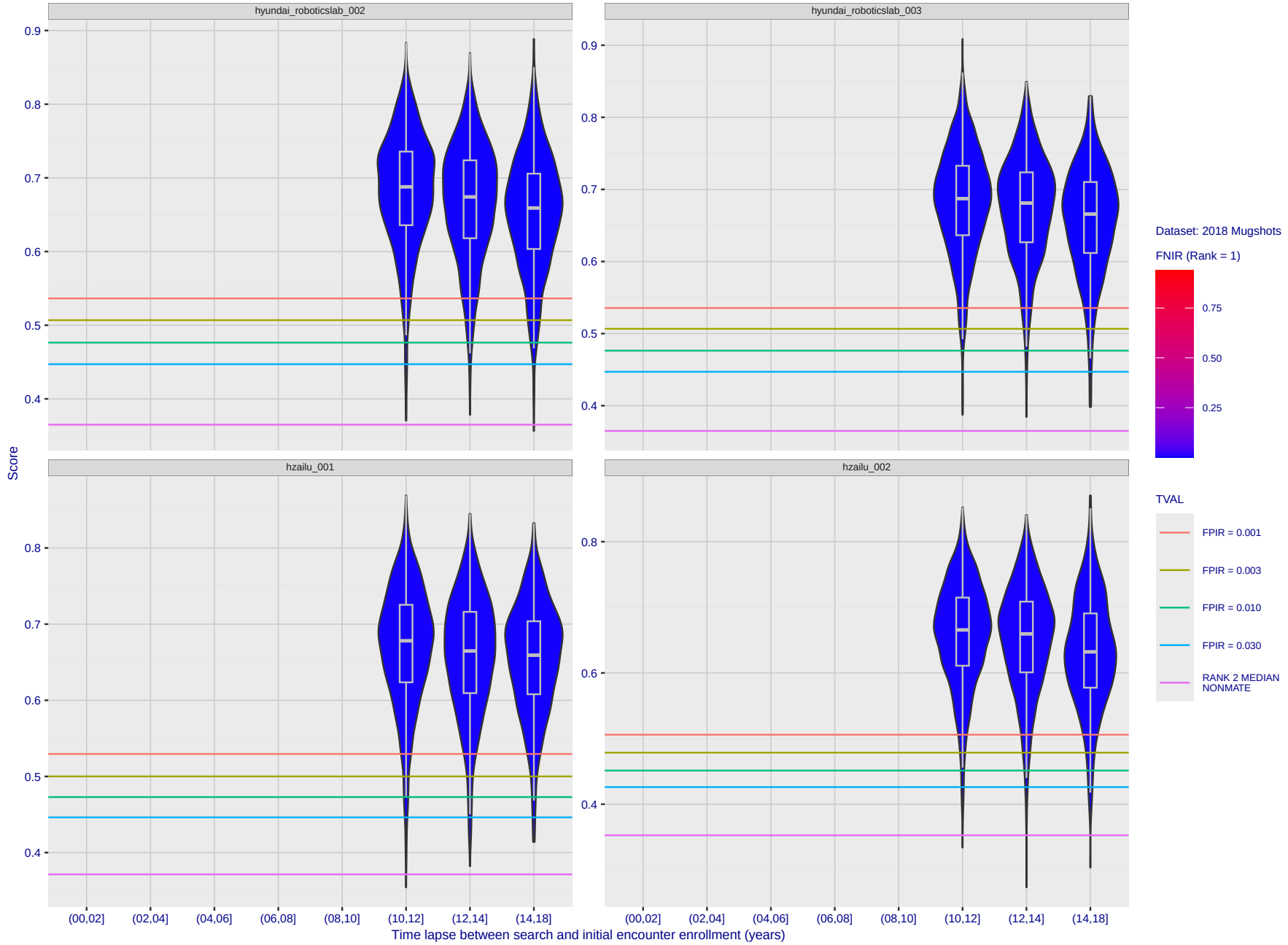


*Figure 275:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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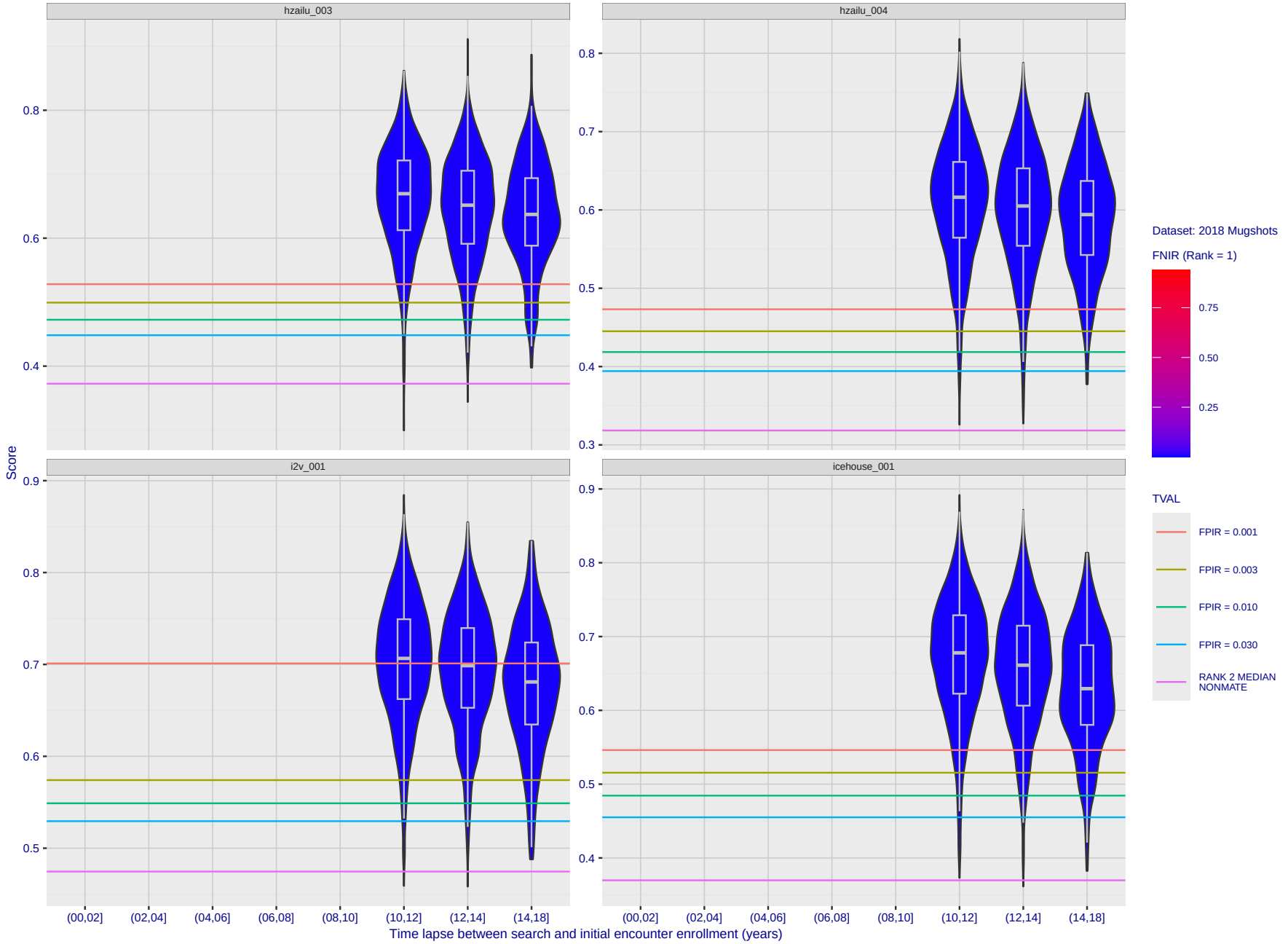


*Figure 276:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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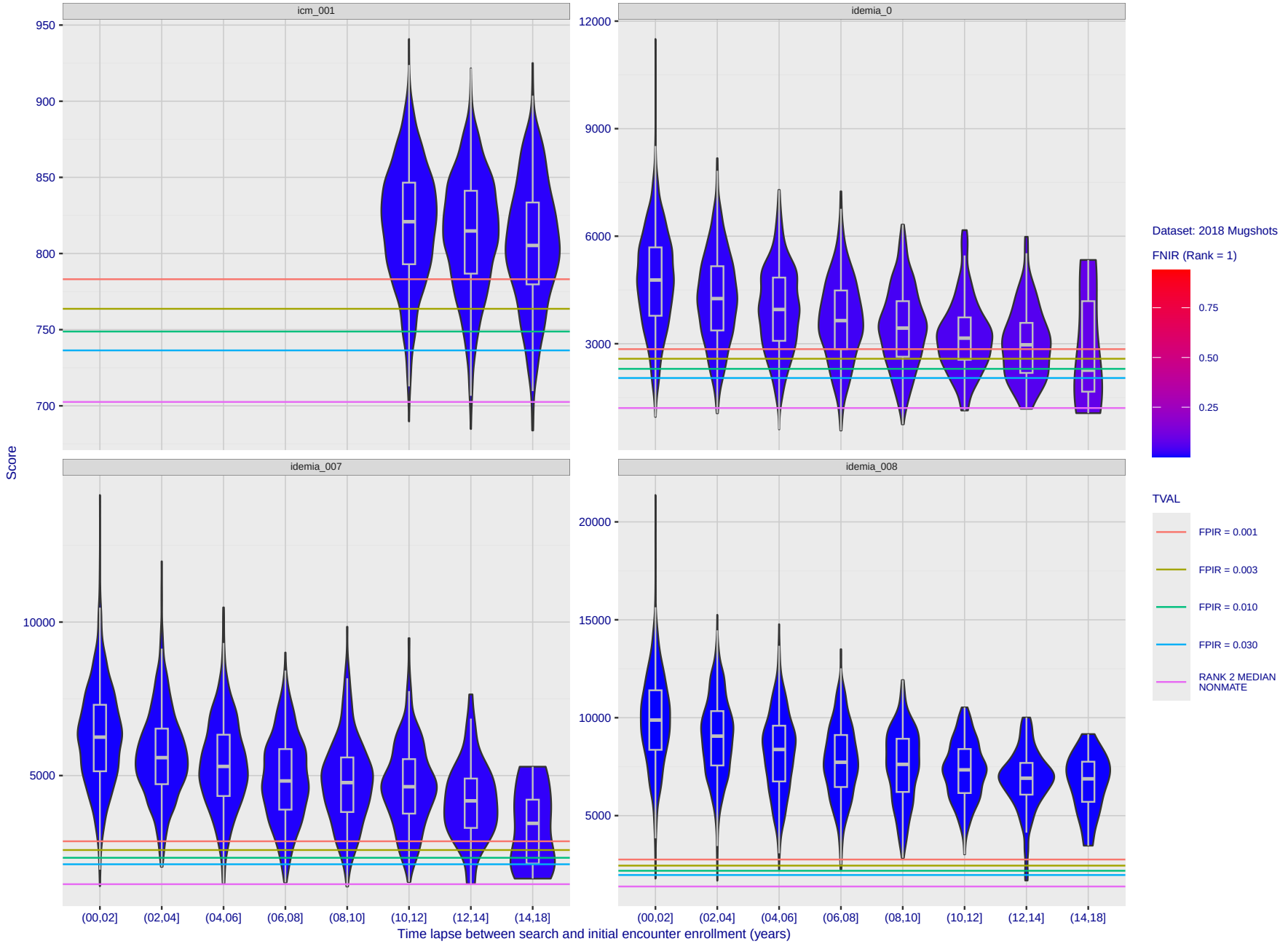


*Figure 277:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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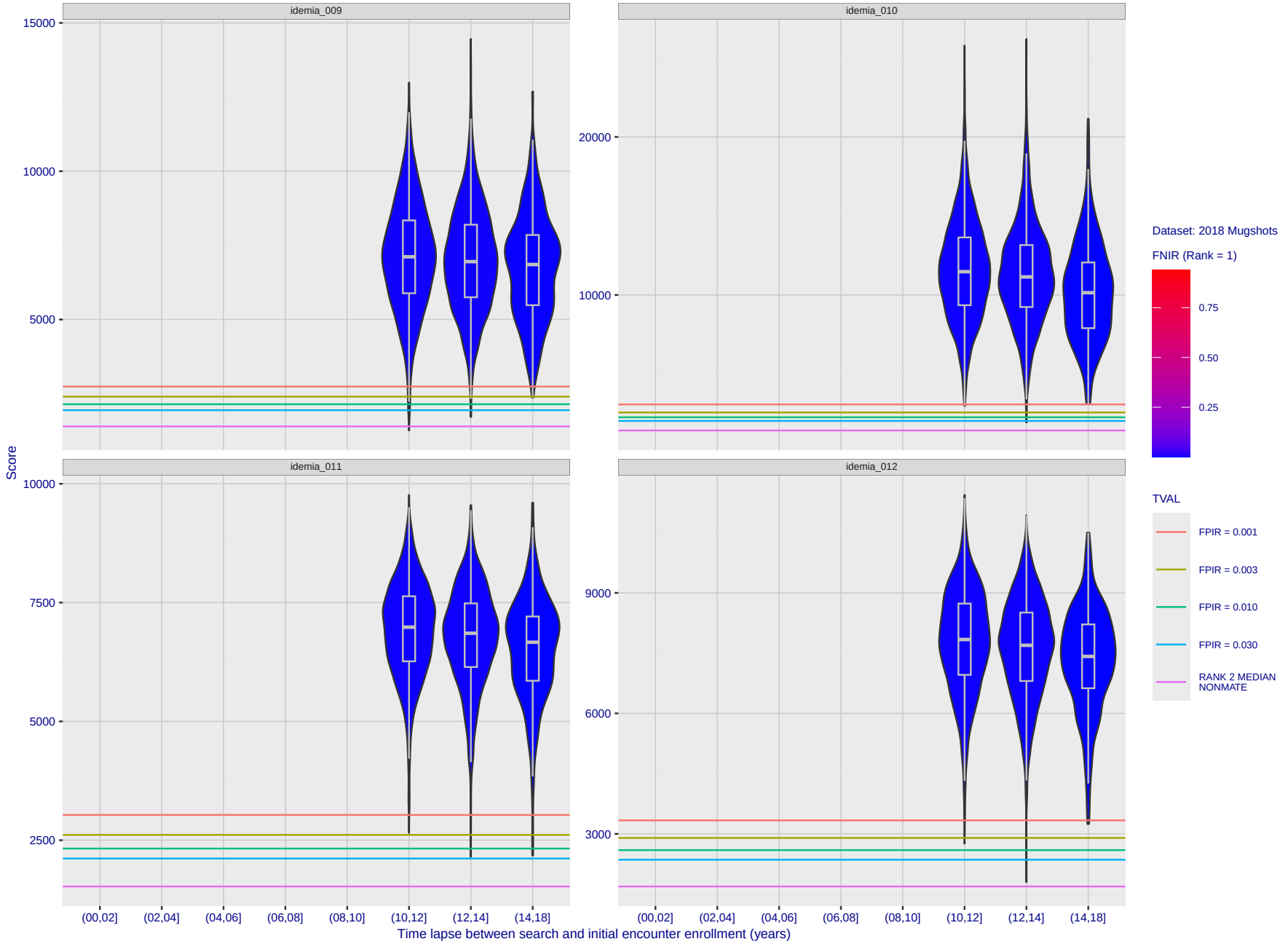


*Figure 278:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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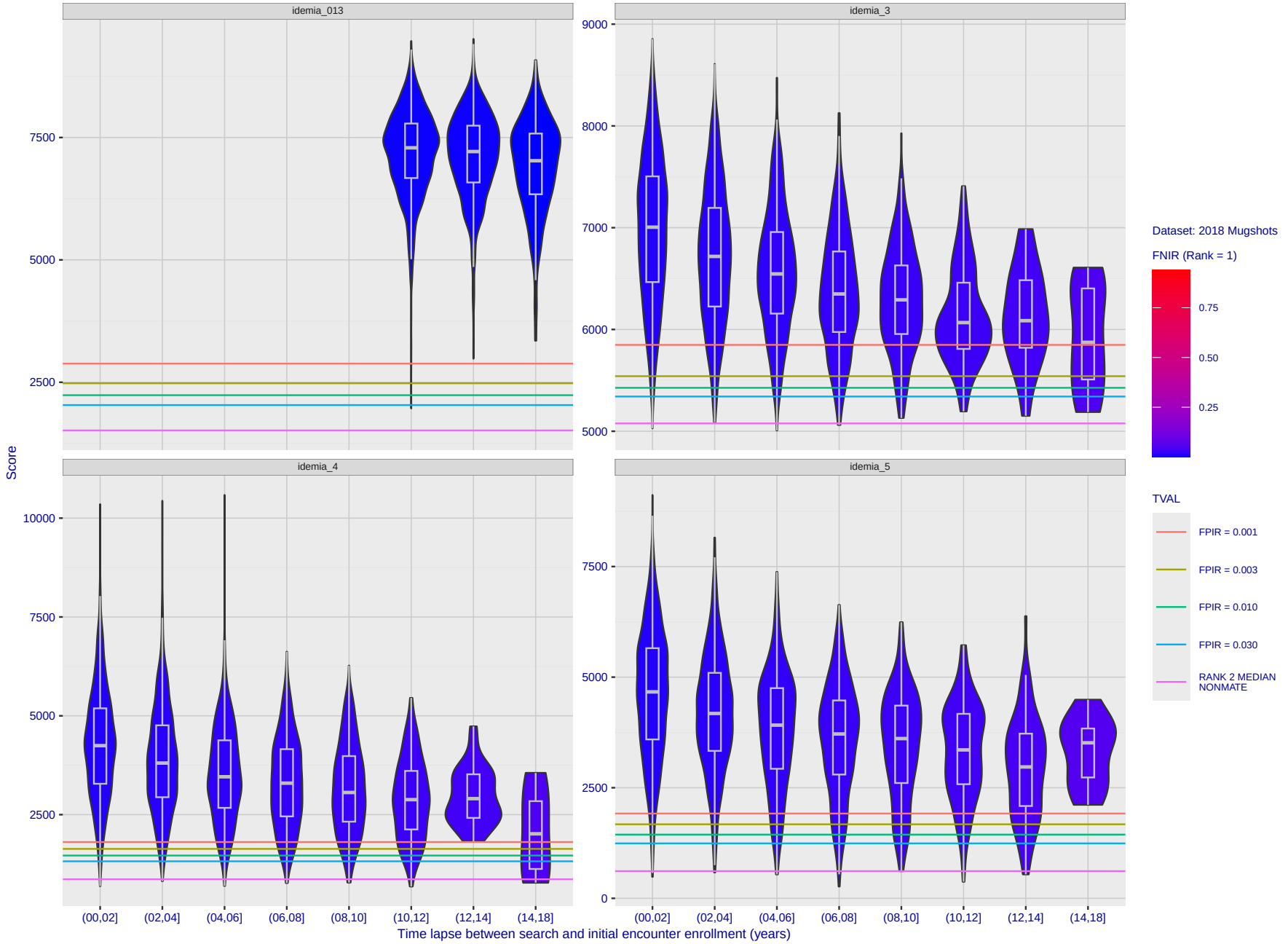


*Figure 279:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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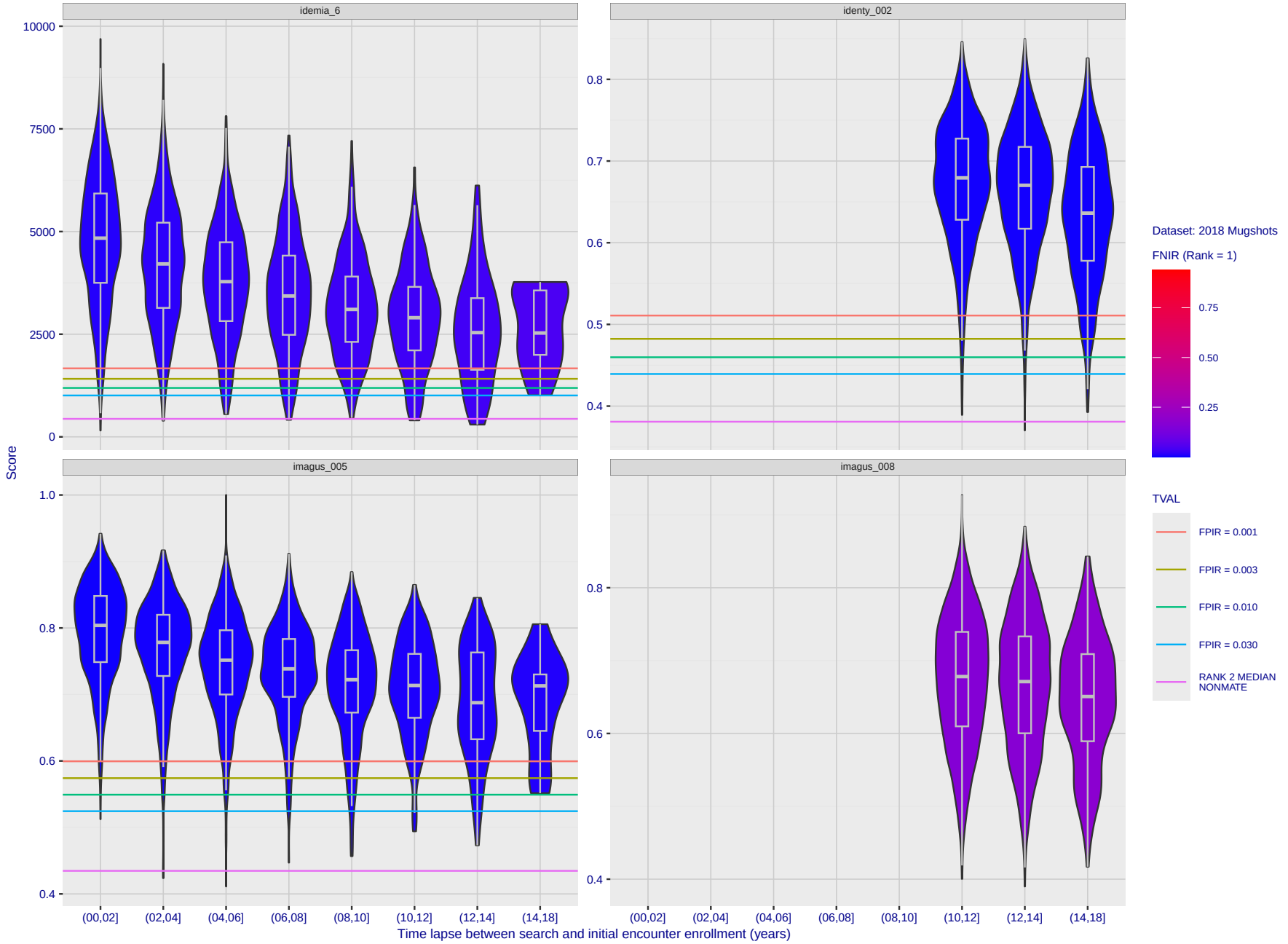


*Figure 280:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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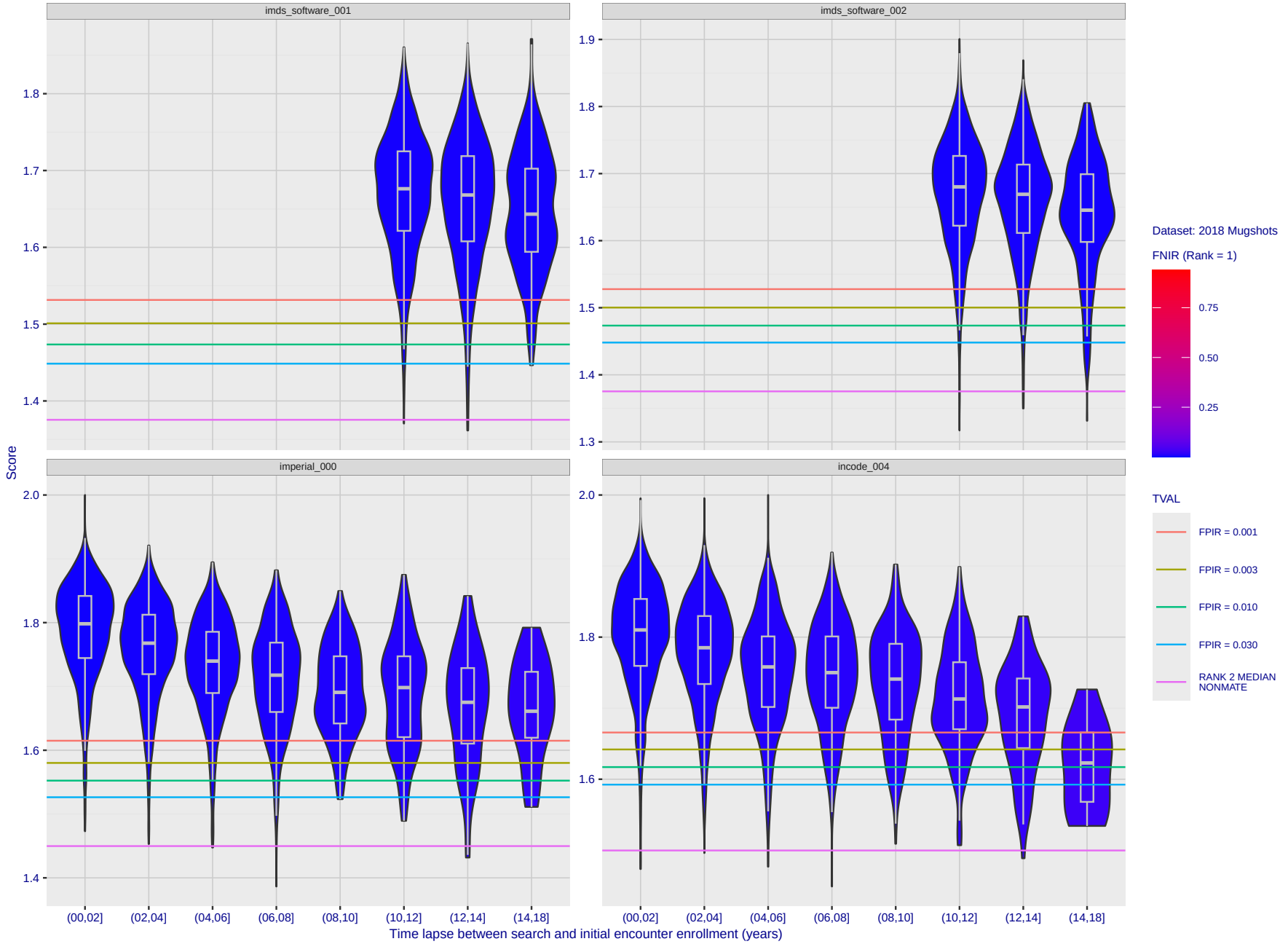


*Figure 281:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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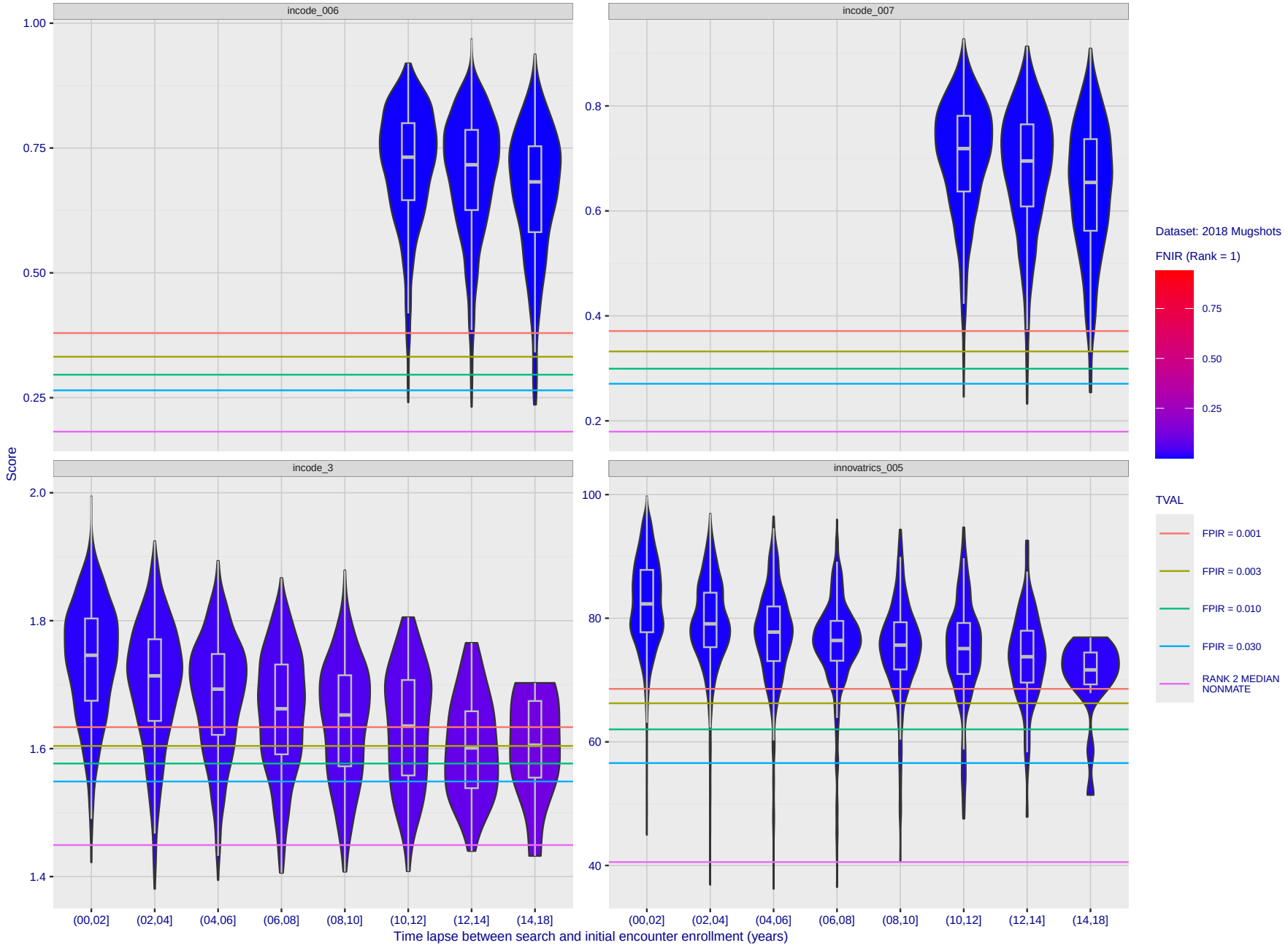


*Figure 282:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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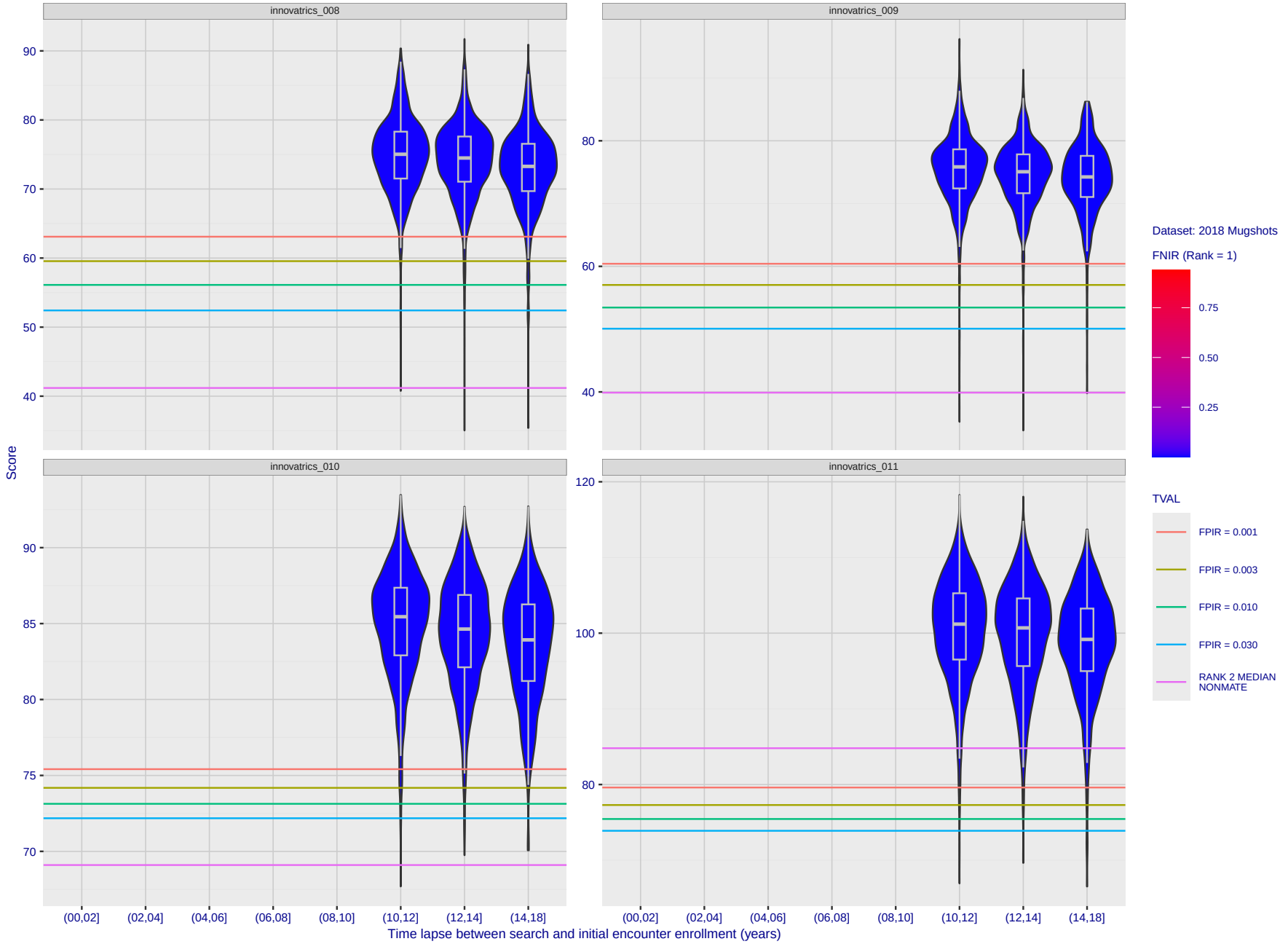


*Figure 283:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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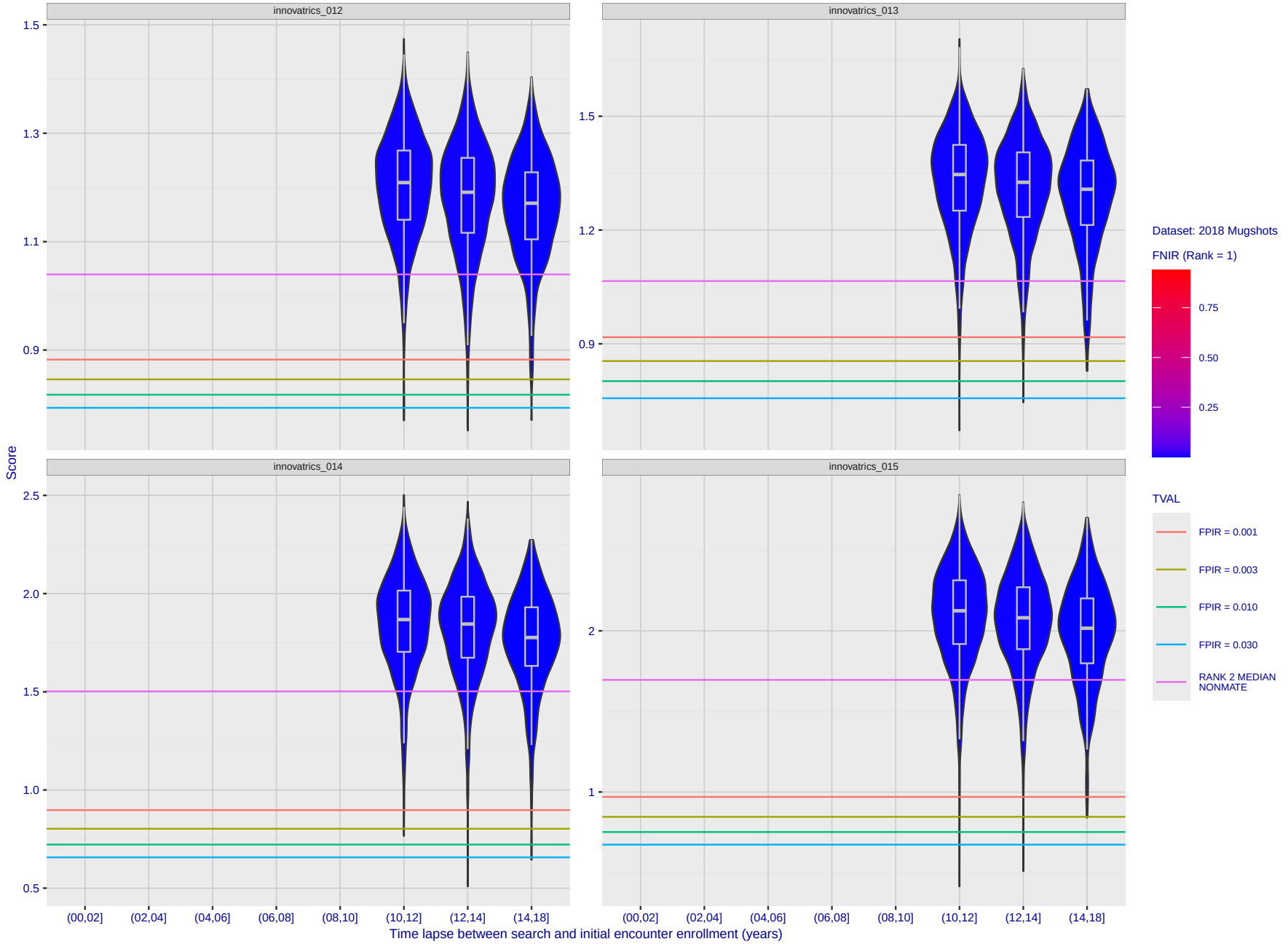


*Figure 284:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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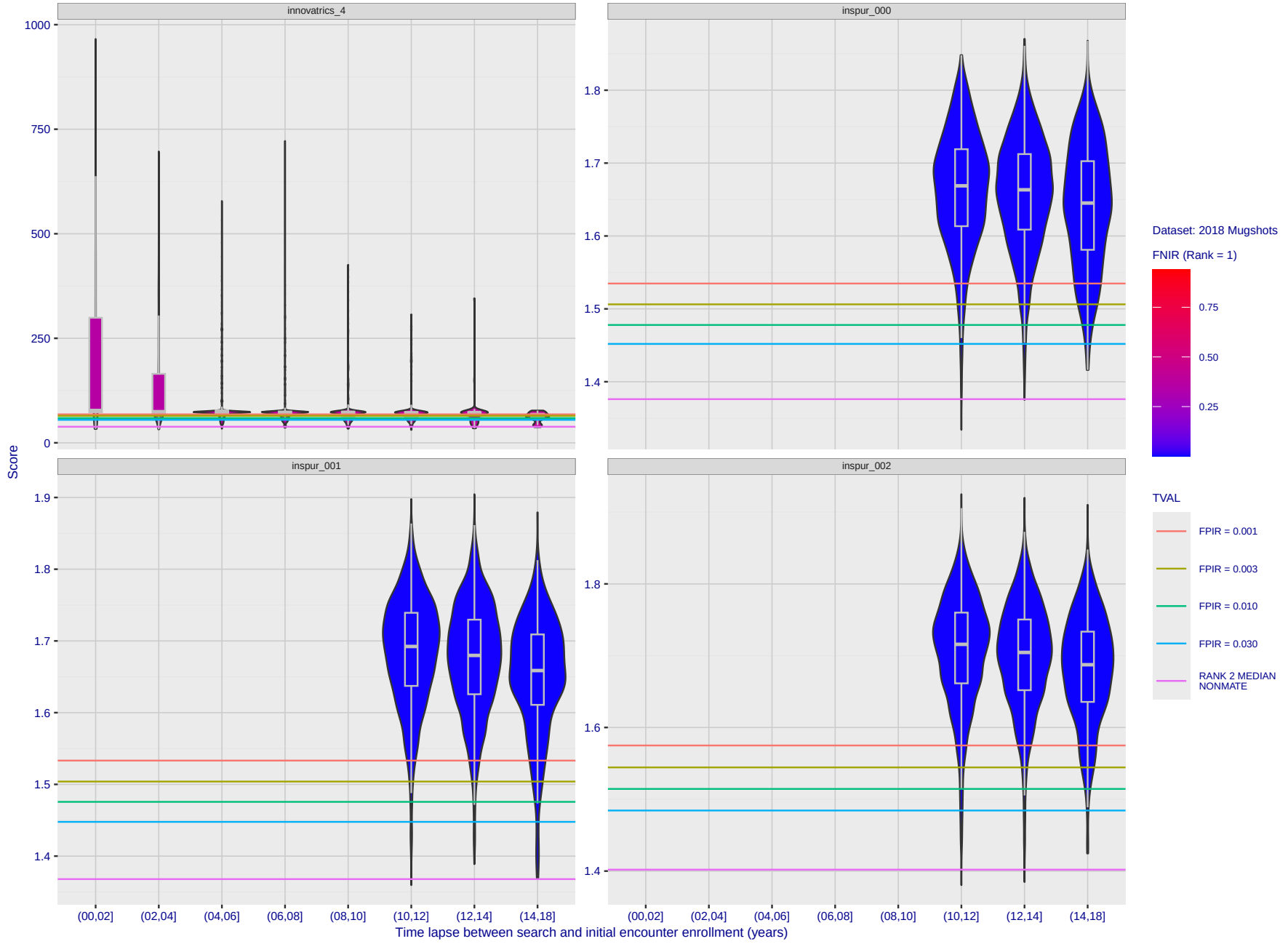


*Figure 285:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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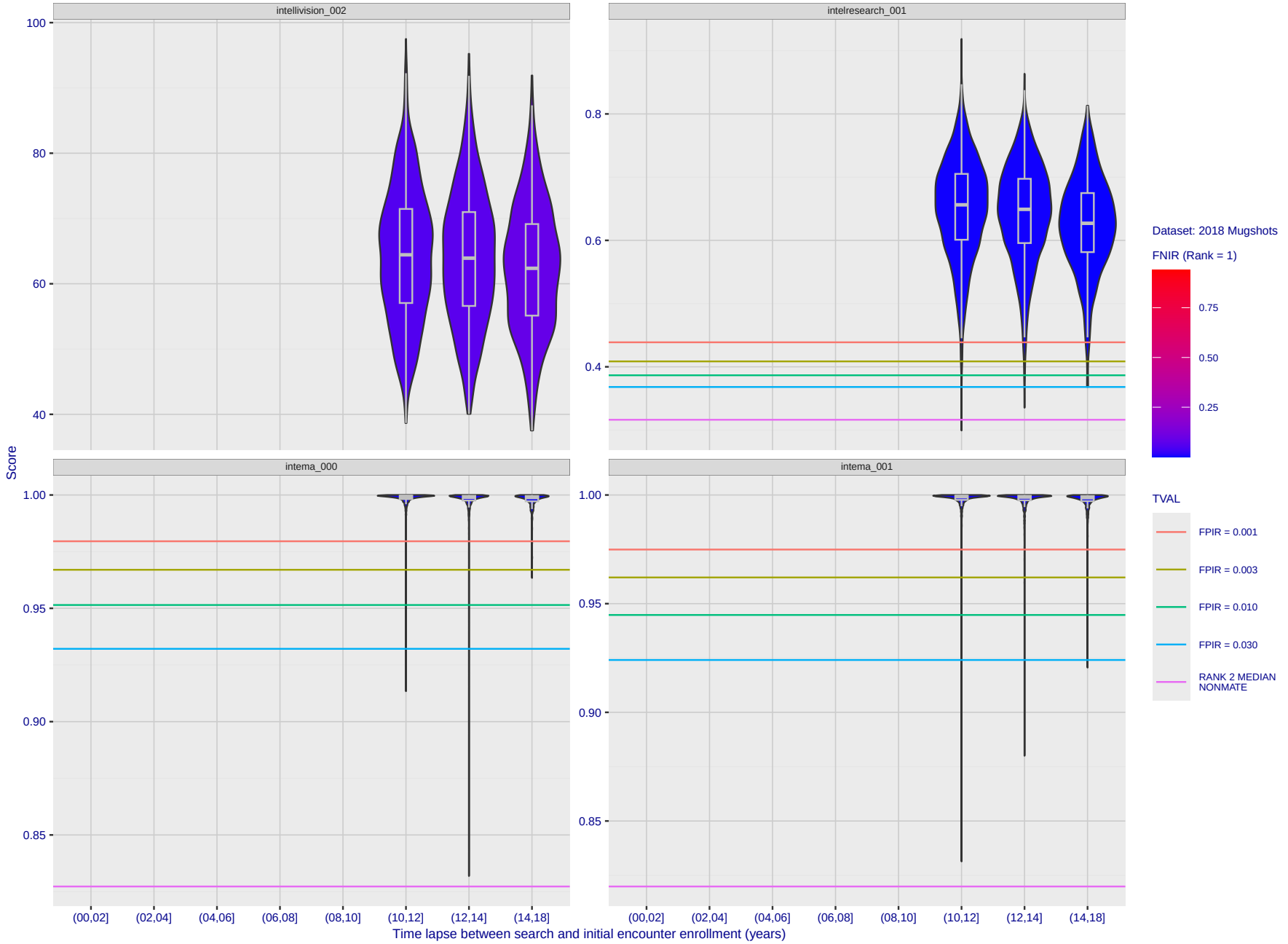


*Figure 286:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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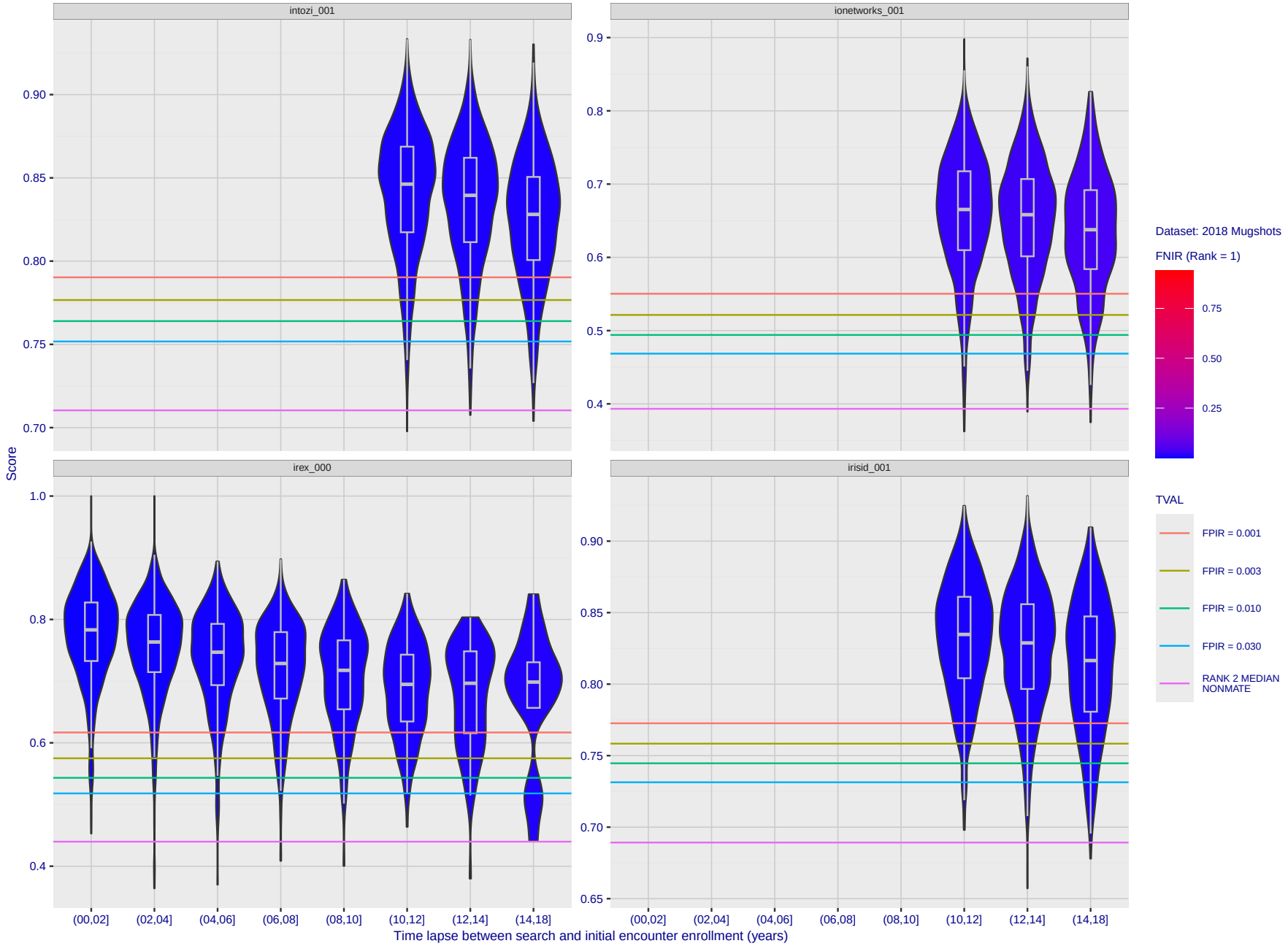


*Figure 287:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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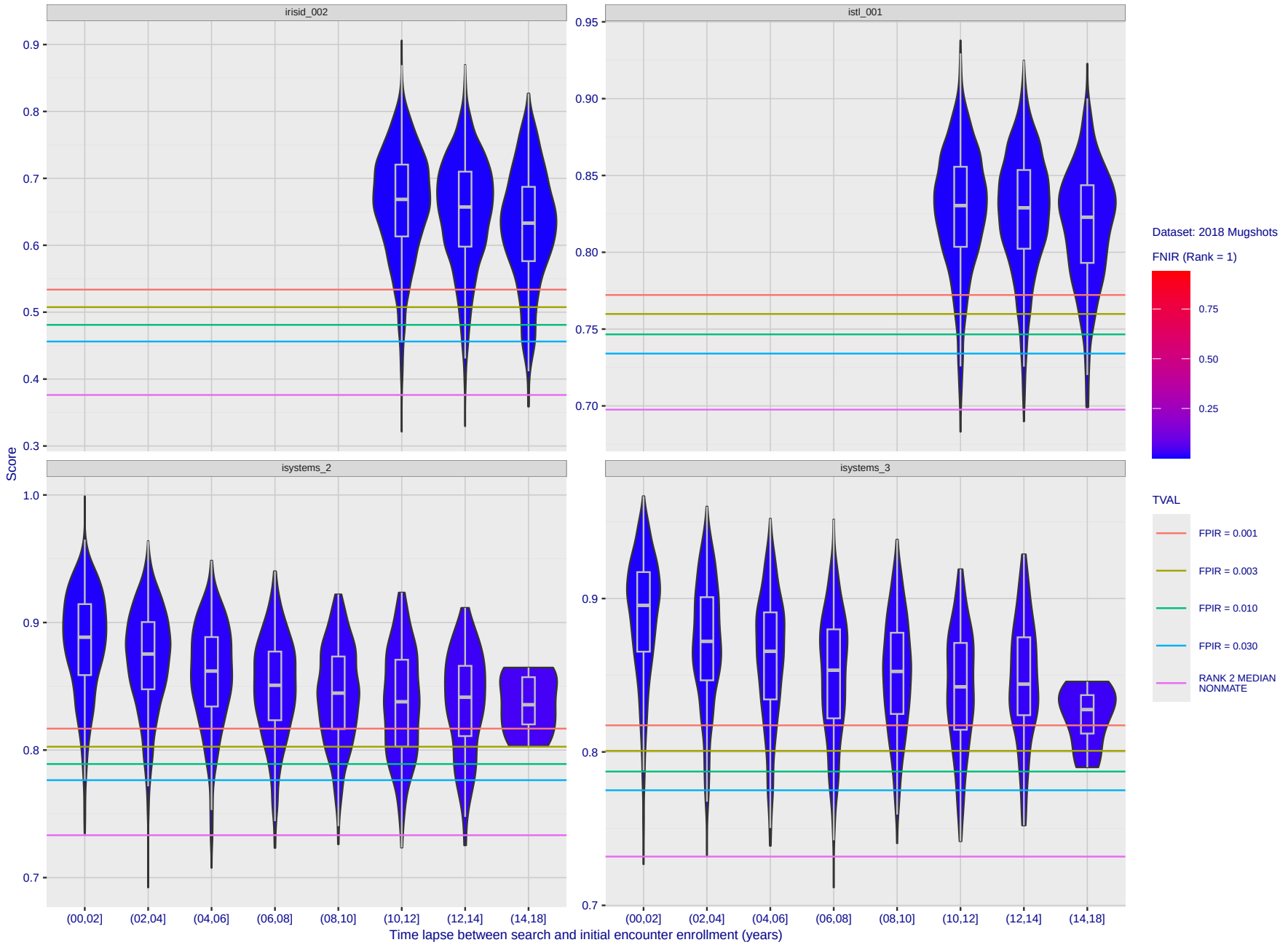


*Figure 288:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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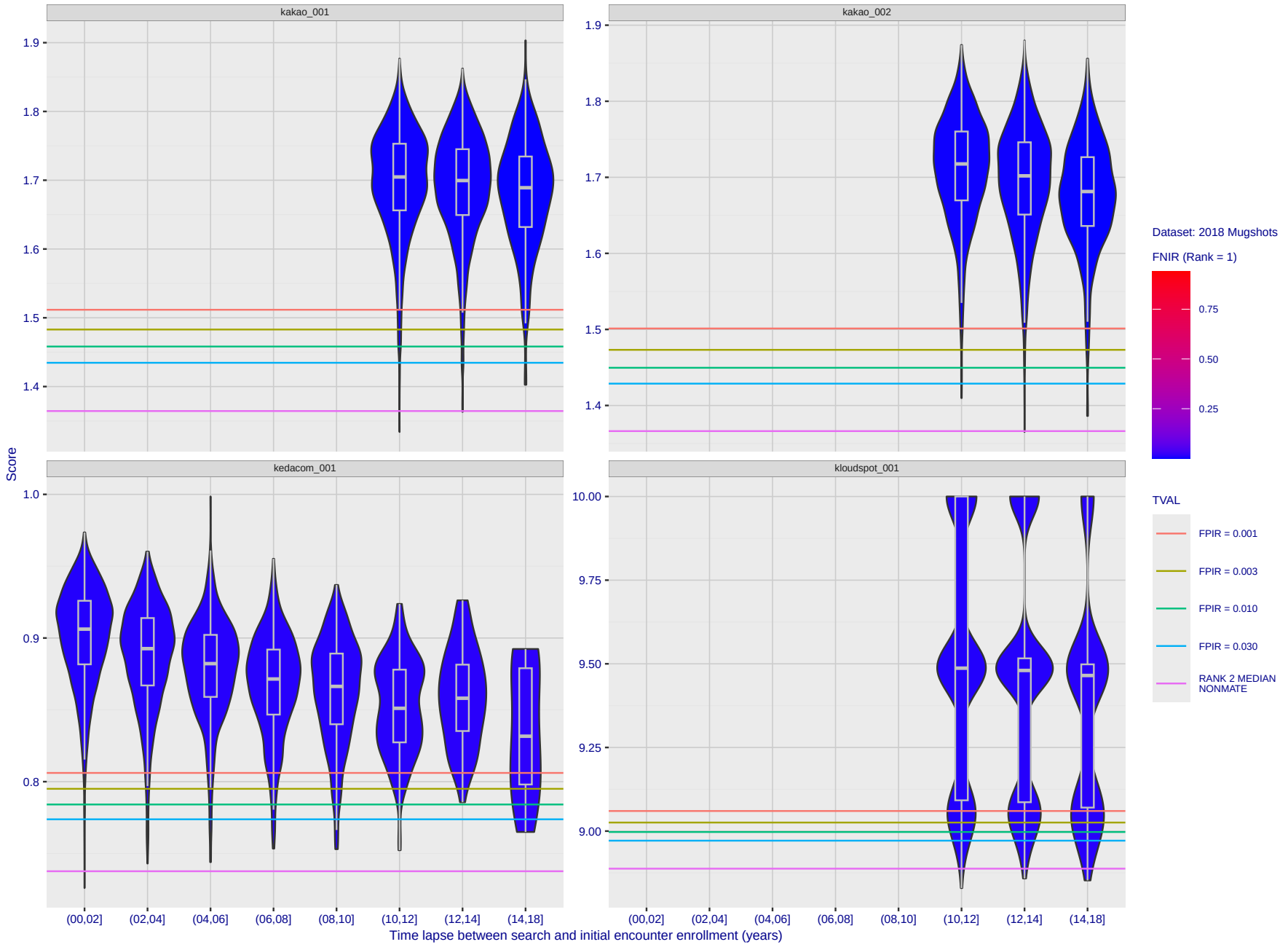


*Figure 289:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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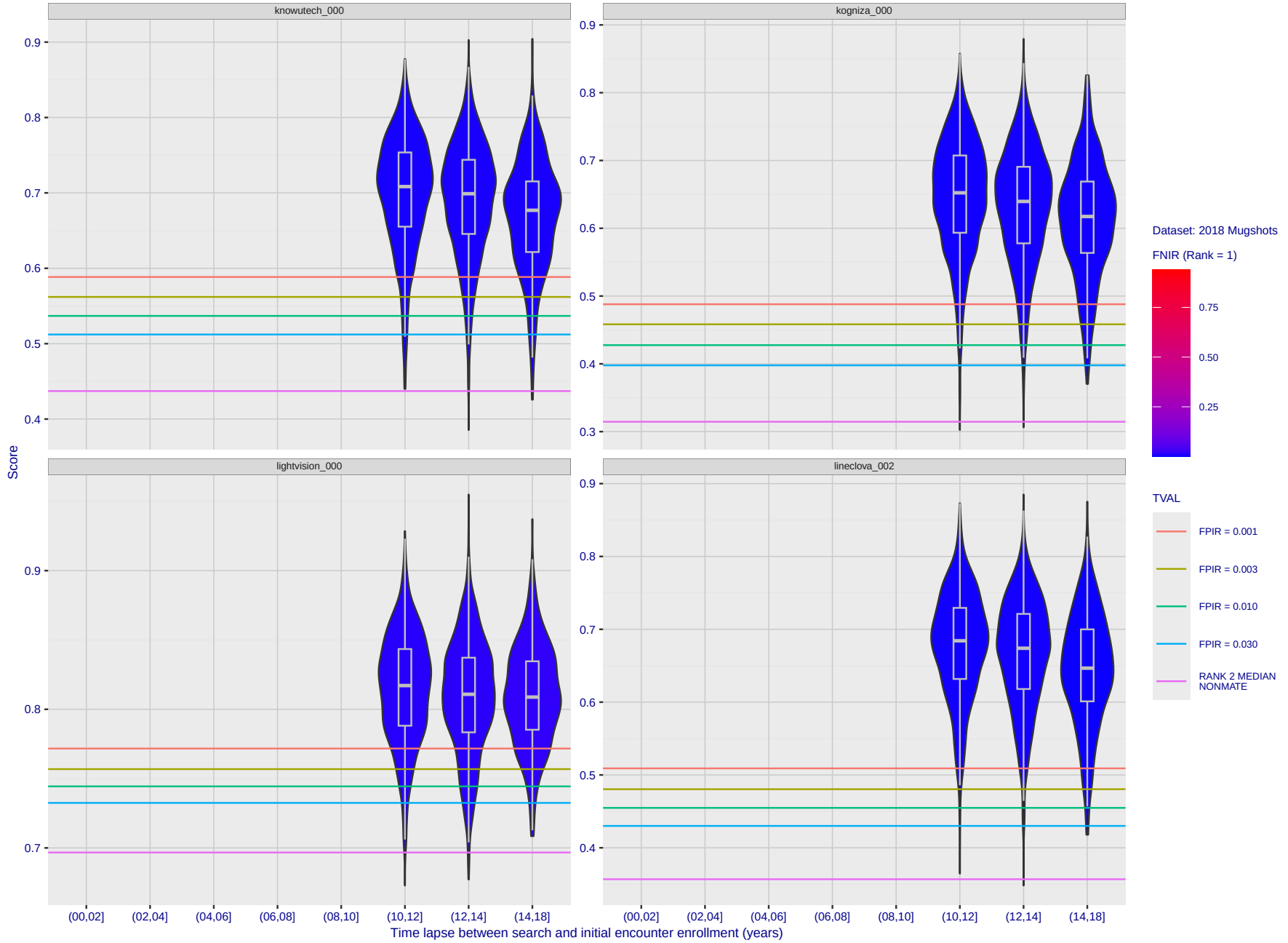


*Figure 290:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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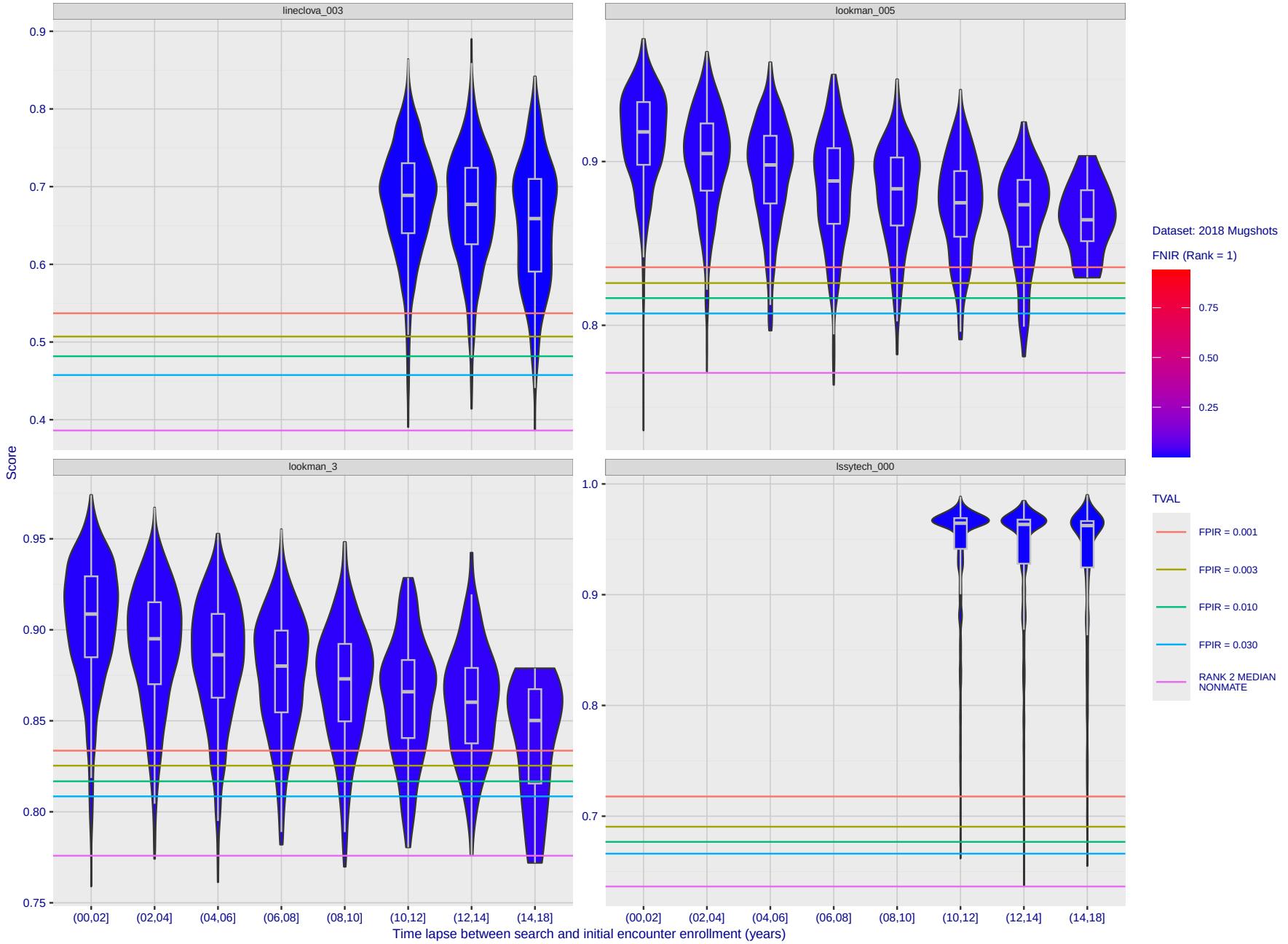


*Figure 291:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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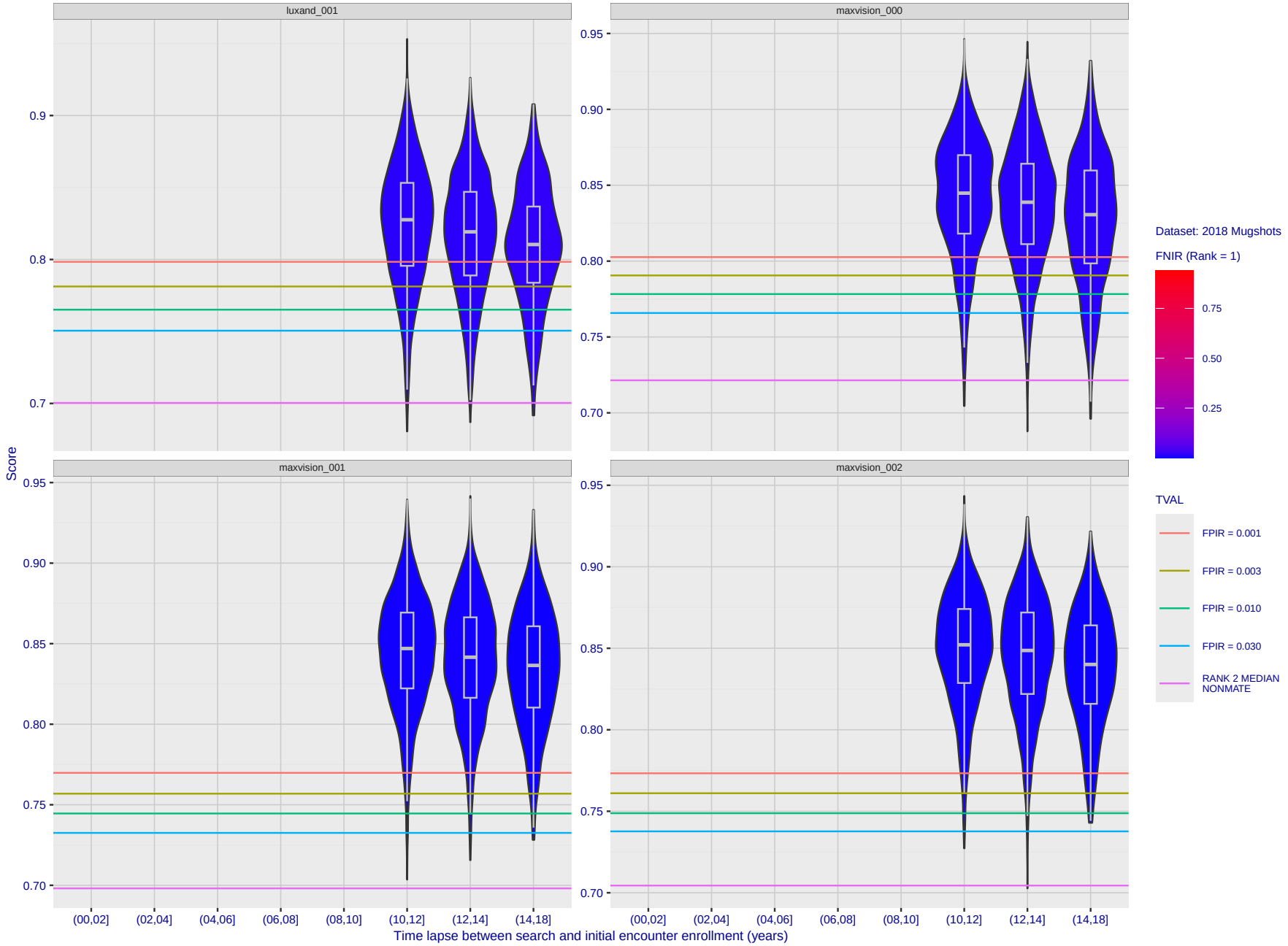


*Figure 292:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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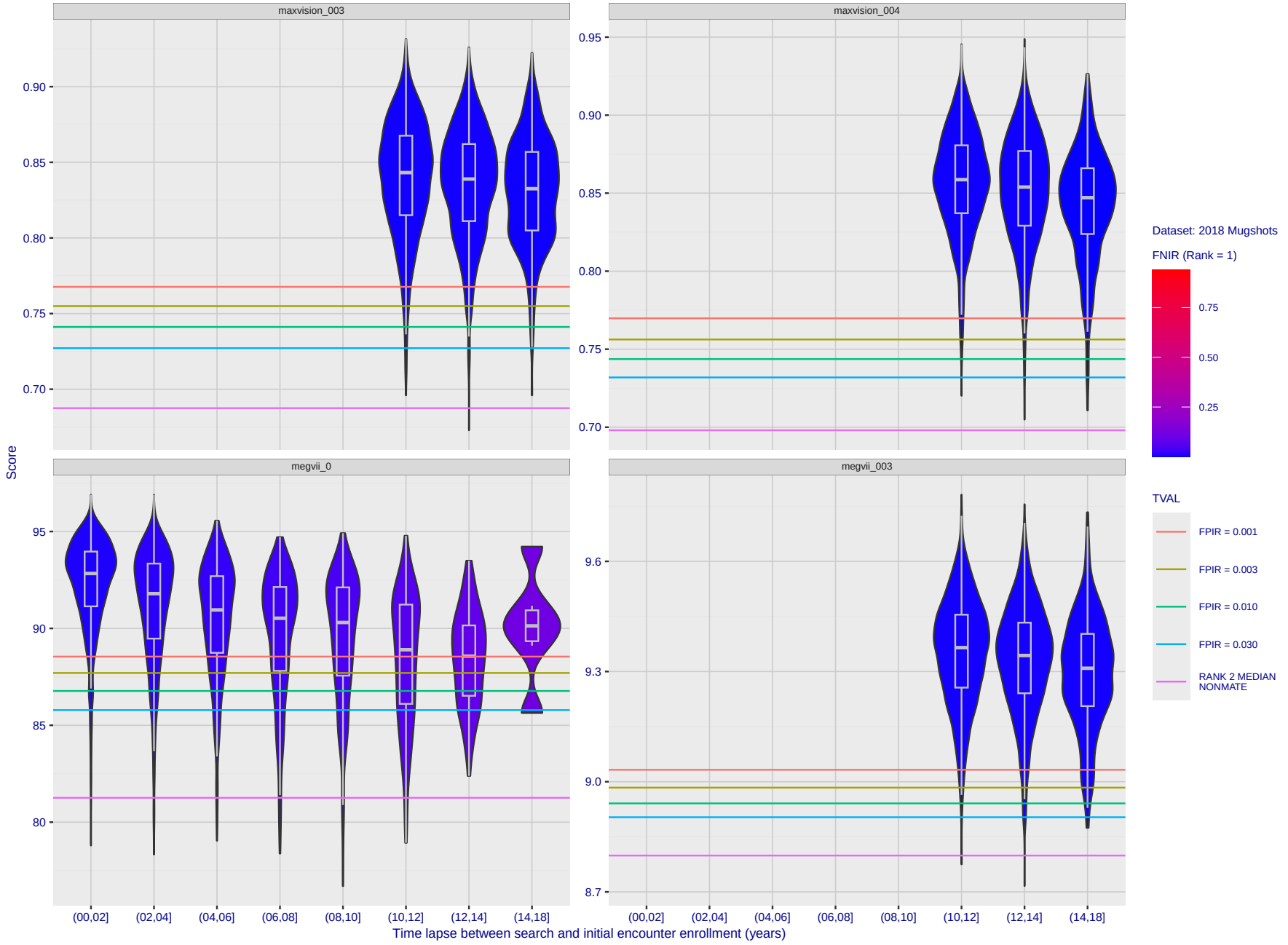


*Figure 293:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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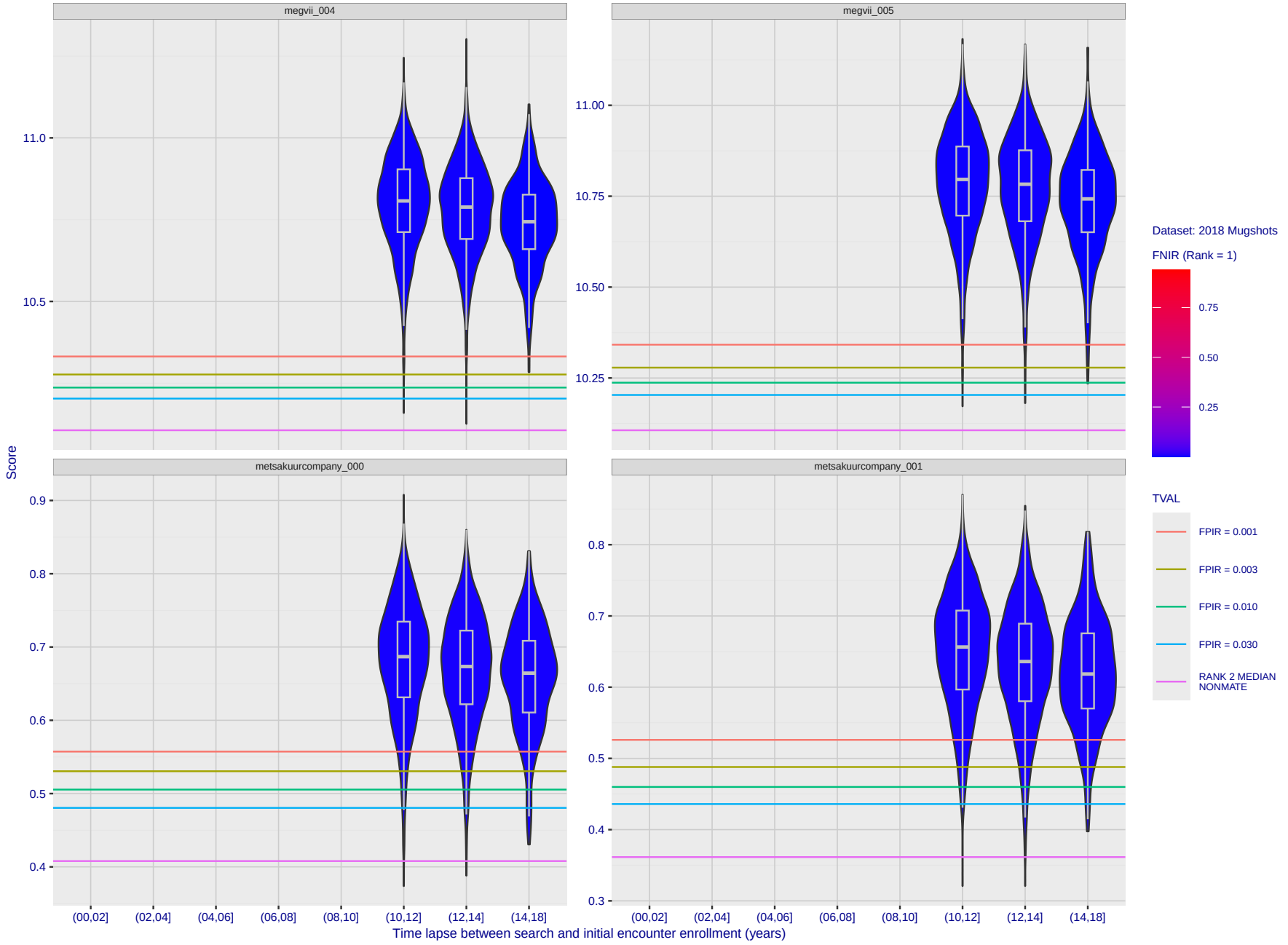


*Figure 294:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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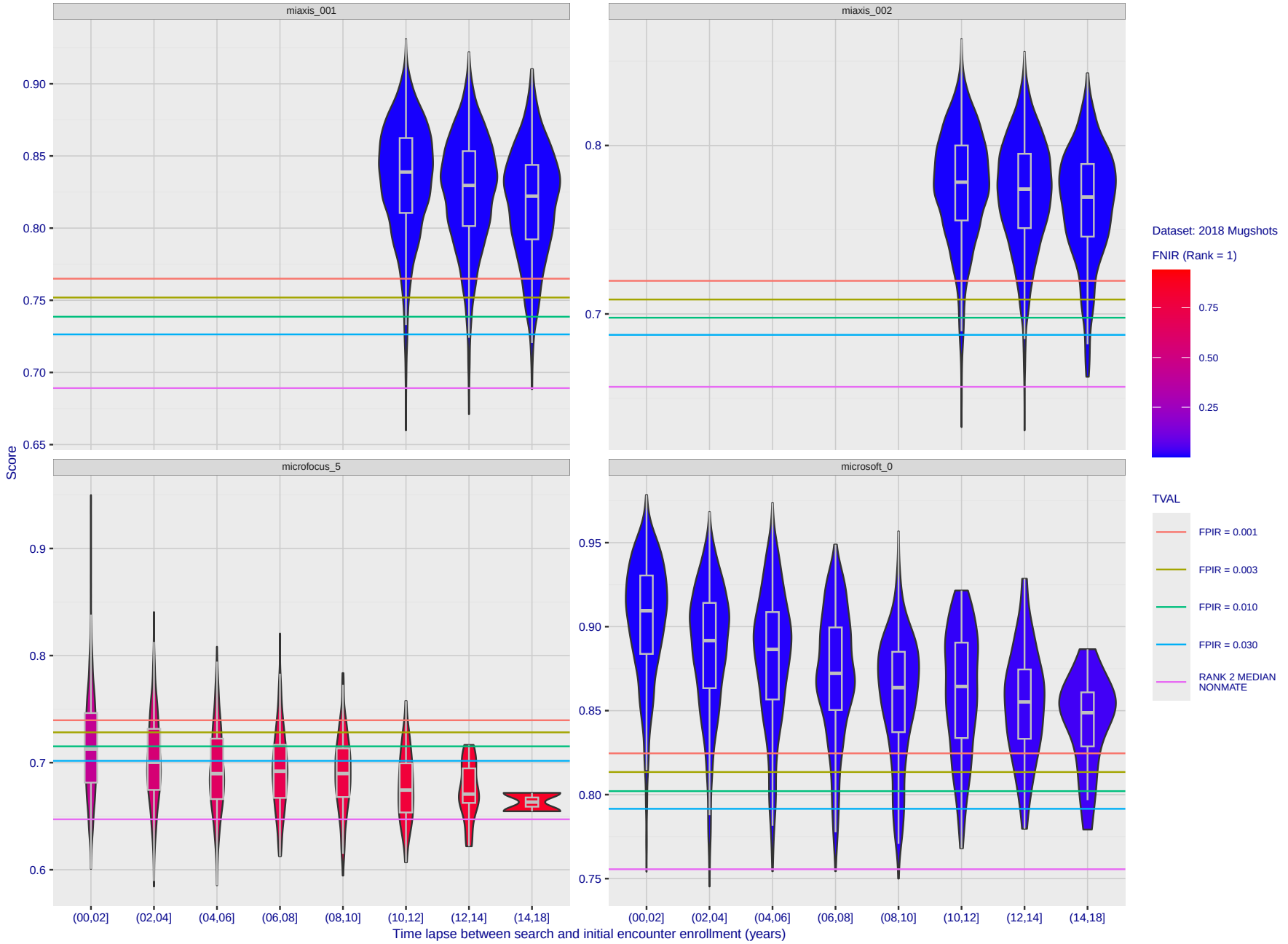


*Figure 295:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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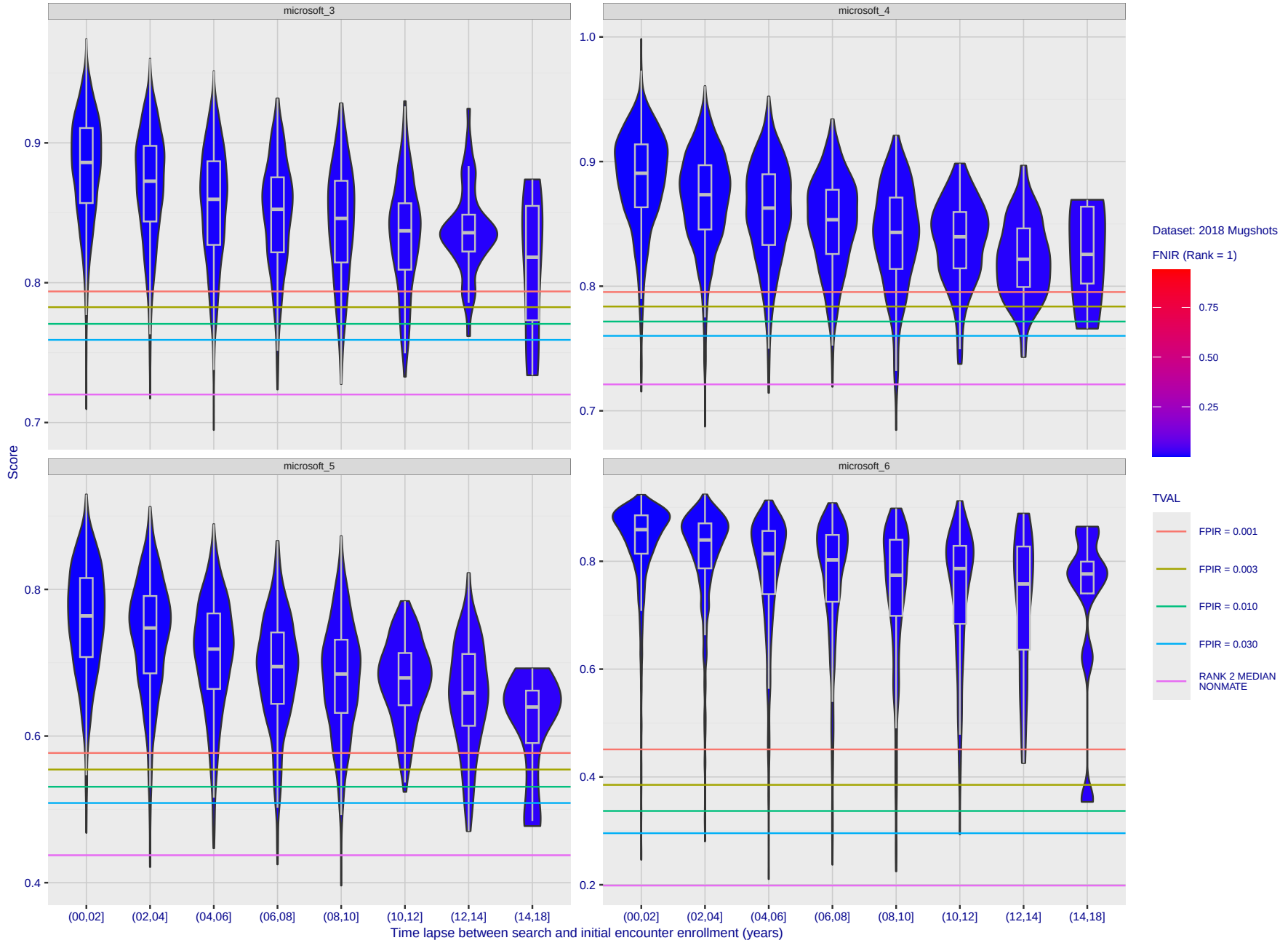


*Figure 296:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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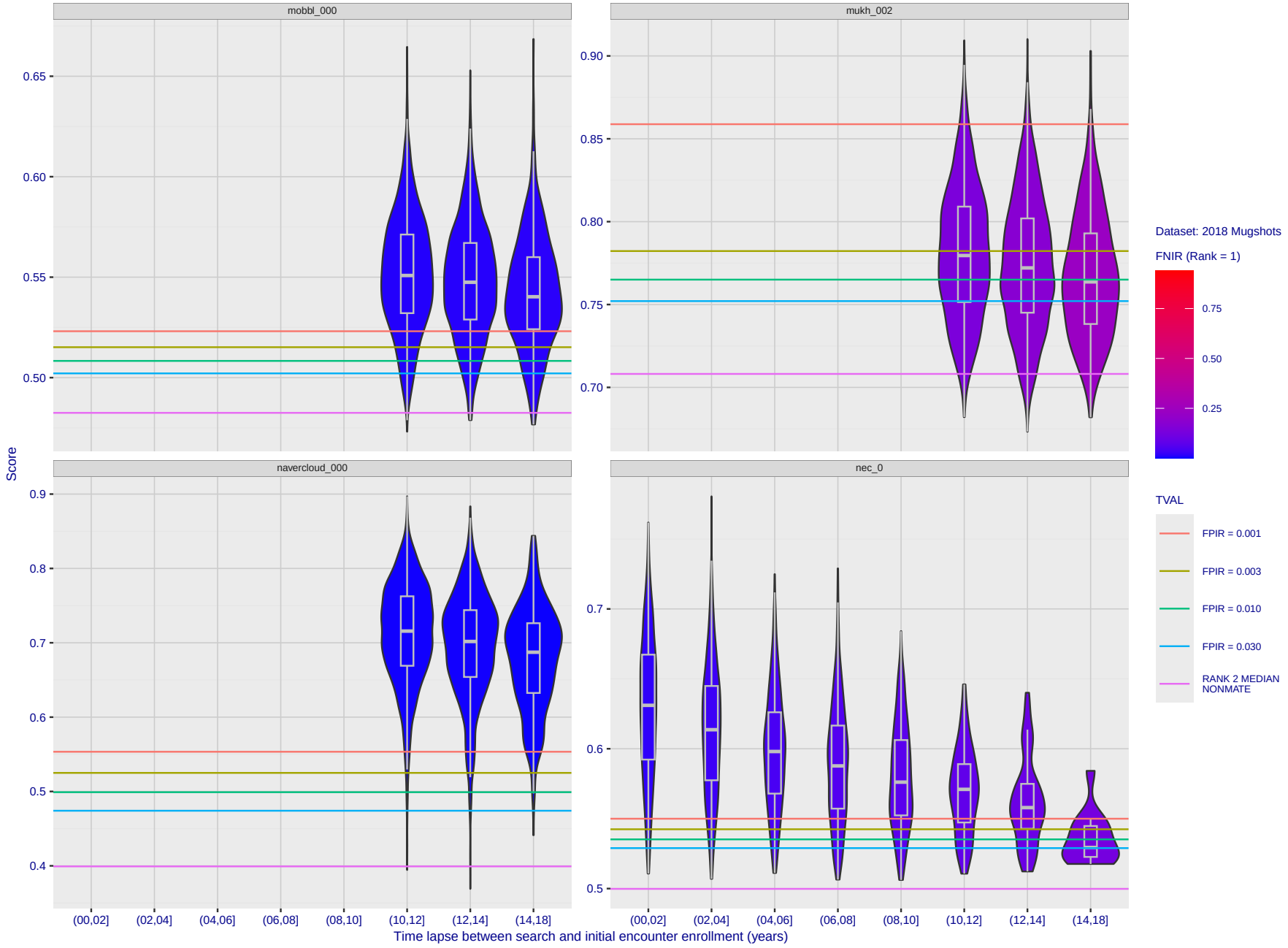


*Figure 297:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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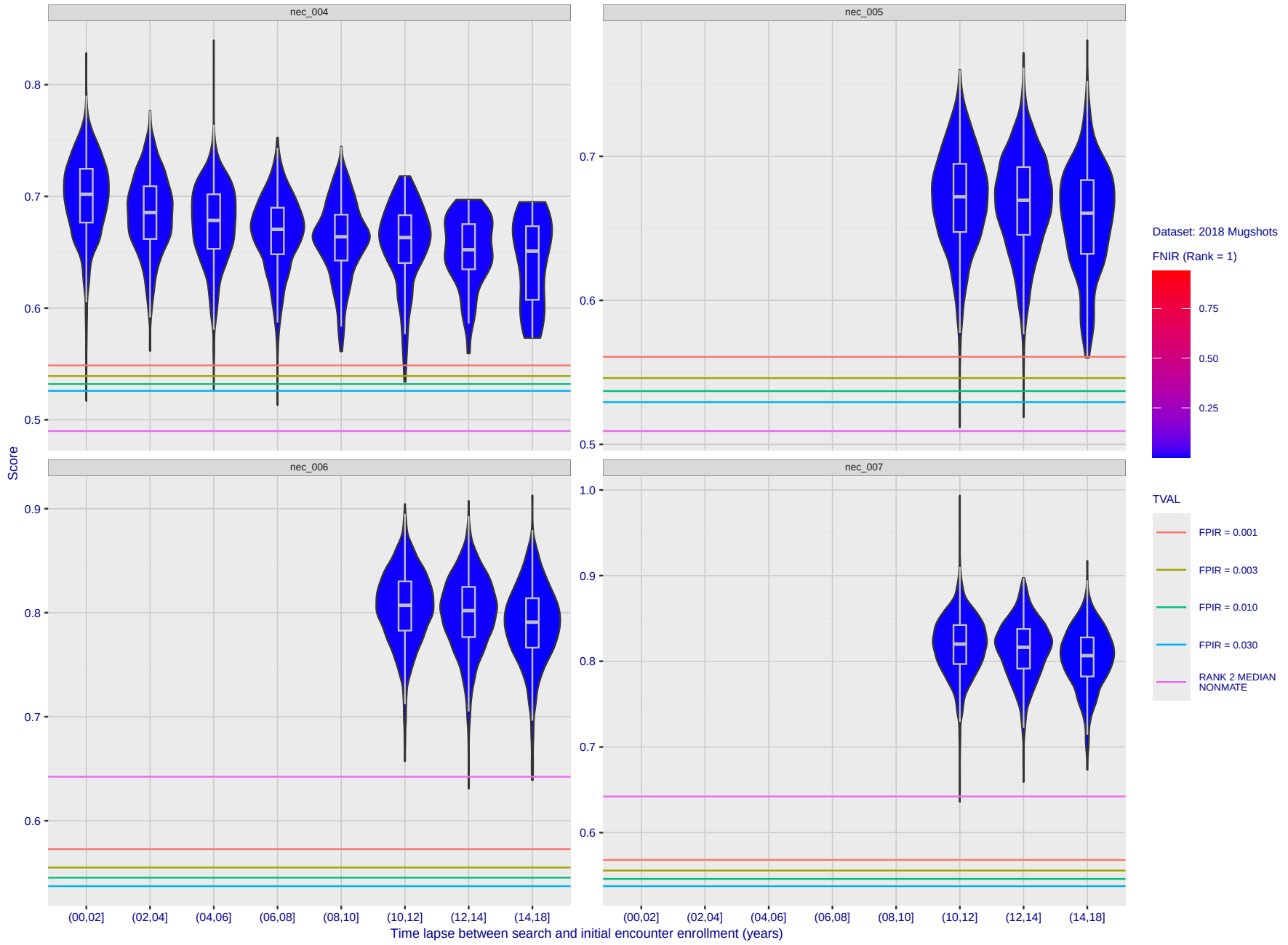


*Figure 298:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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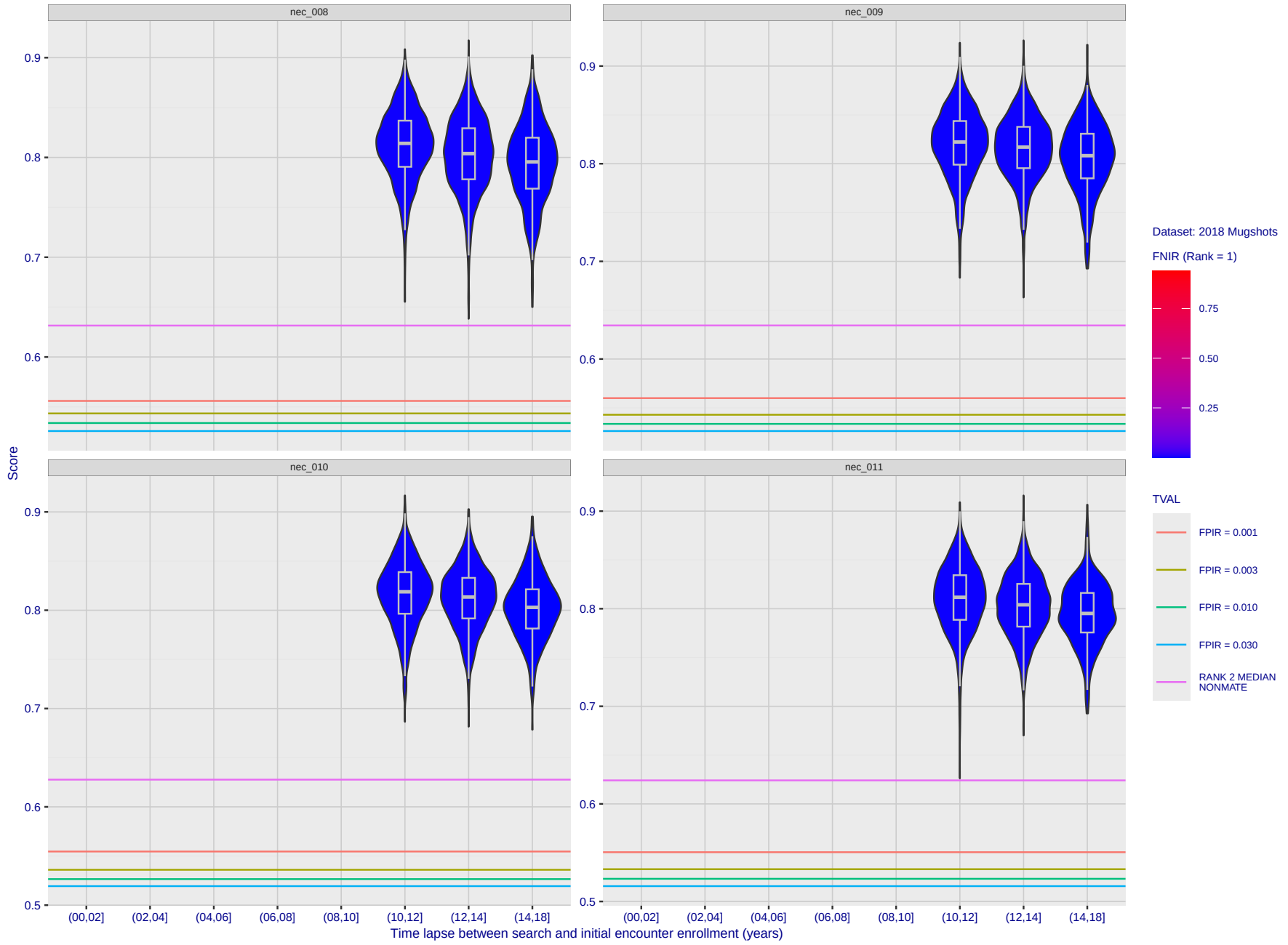


*Figure 299:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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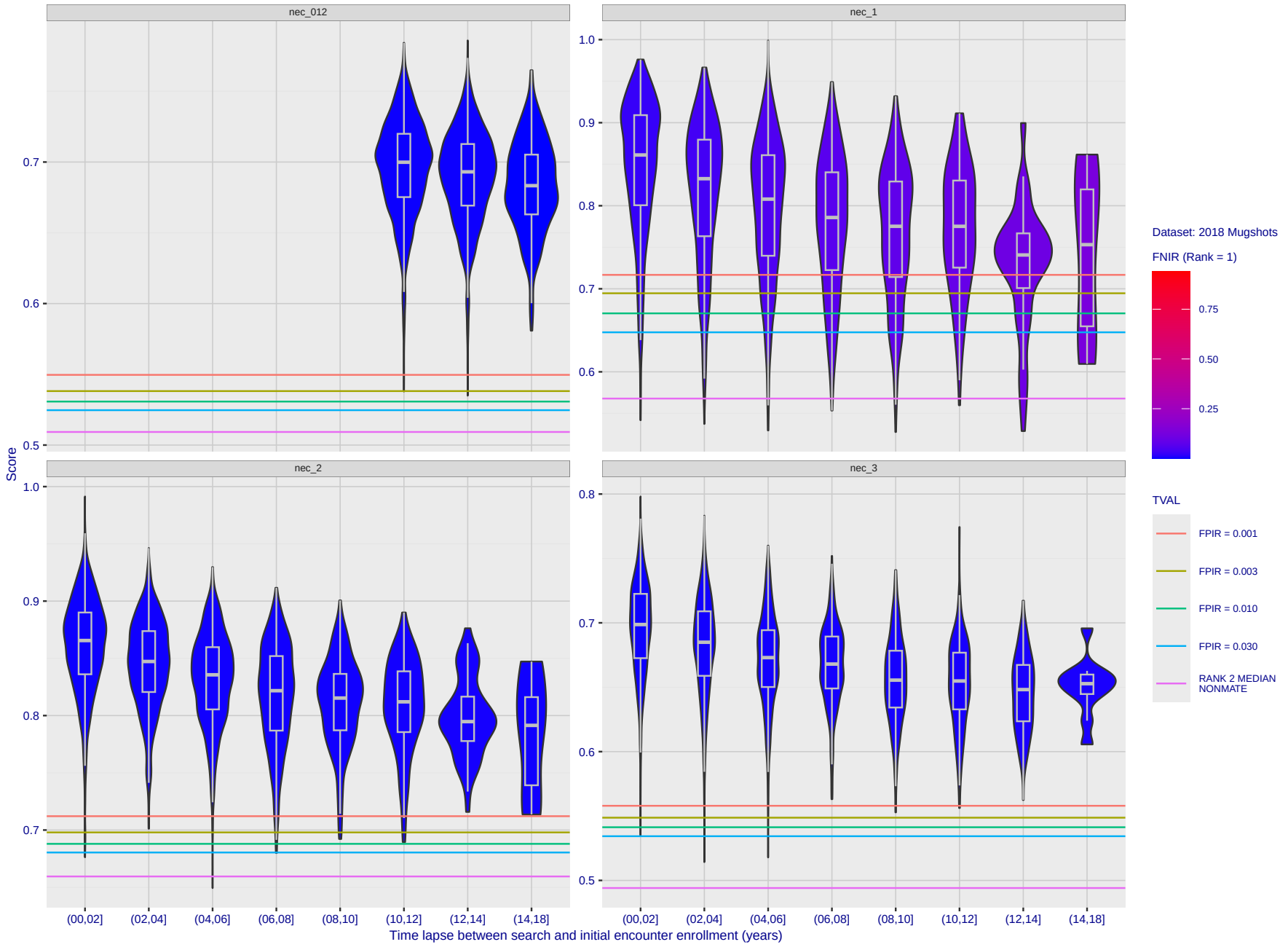


*Figure 300:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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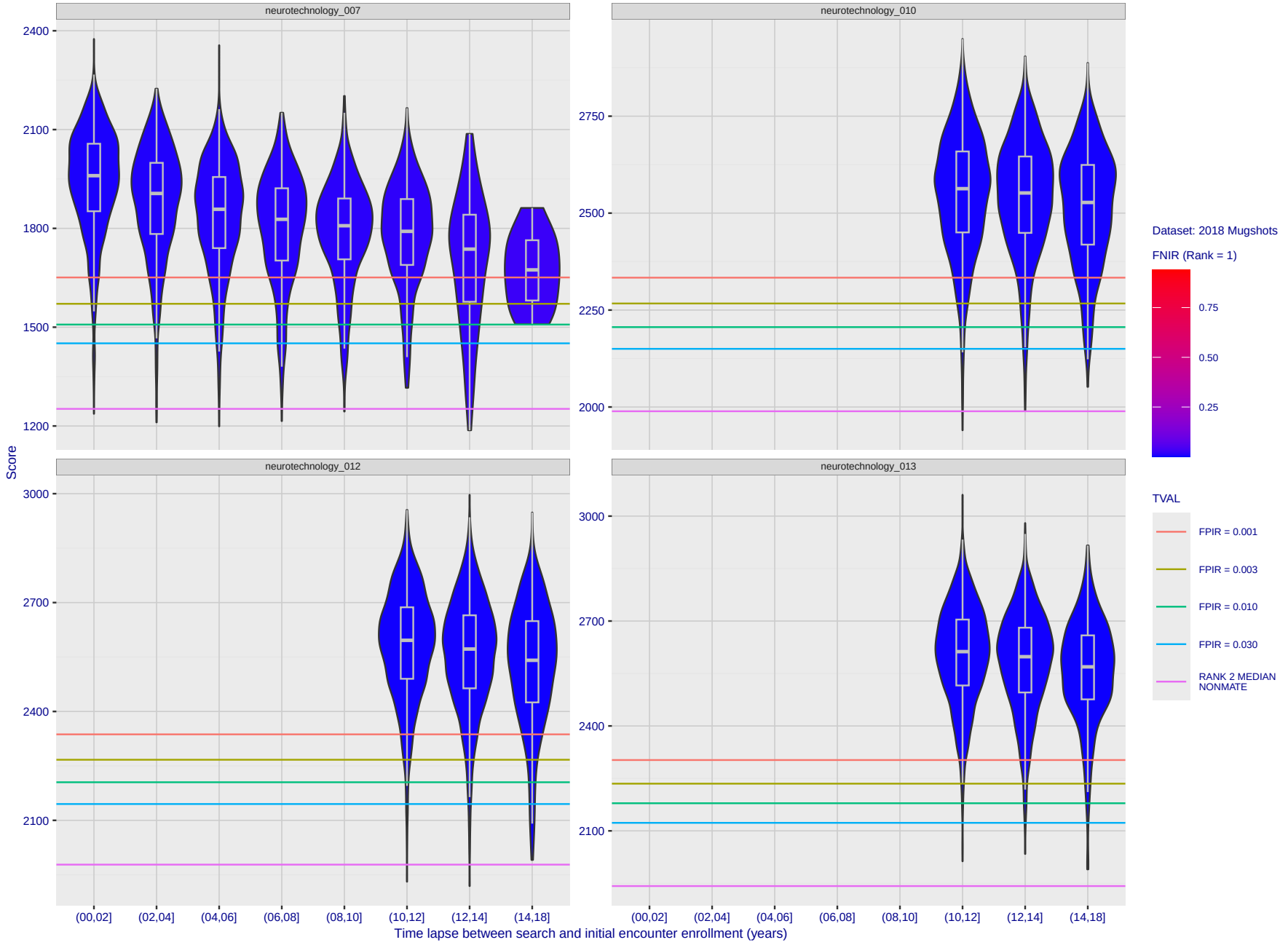


*Figure 301:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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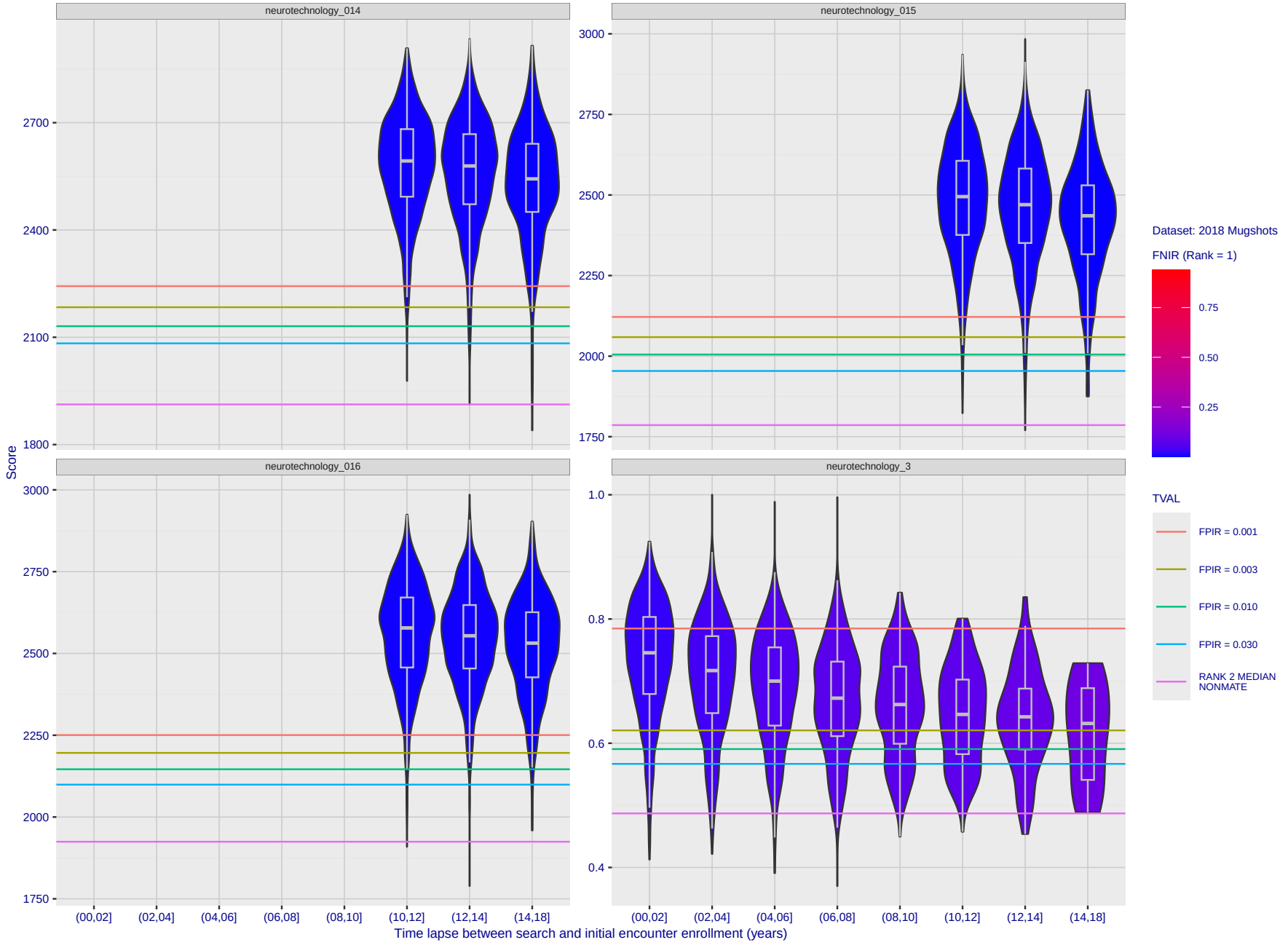


*Figure 302:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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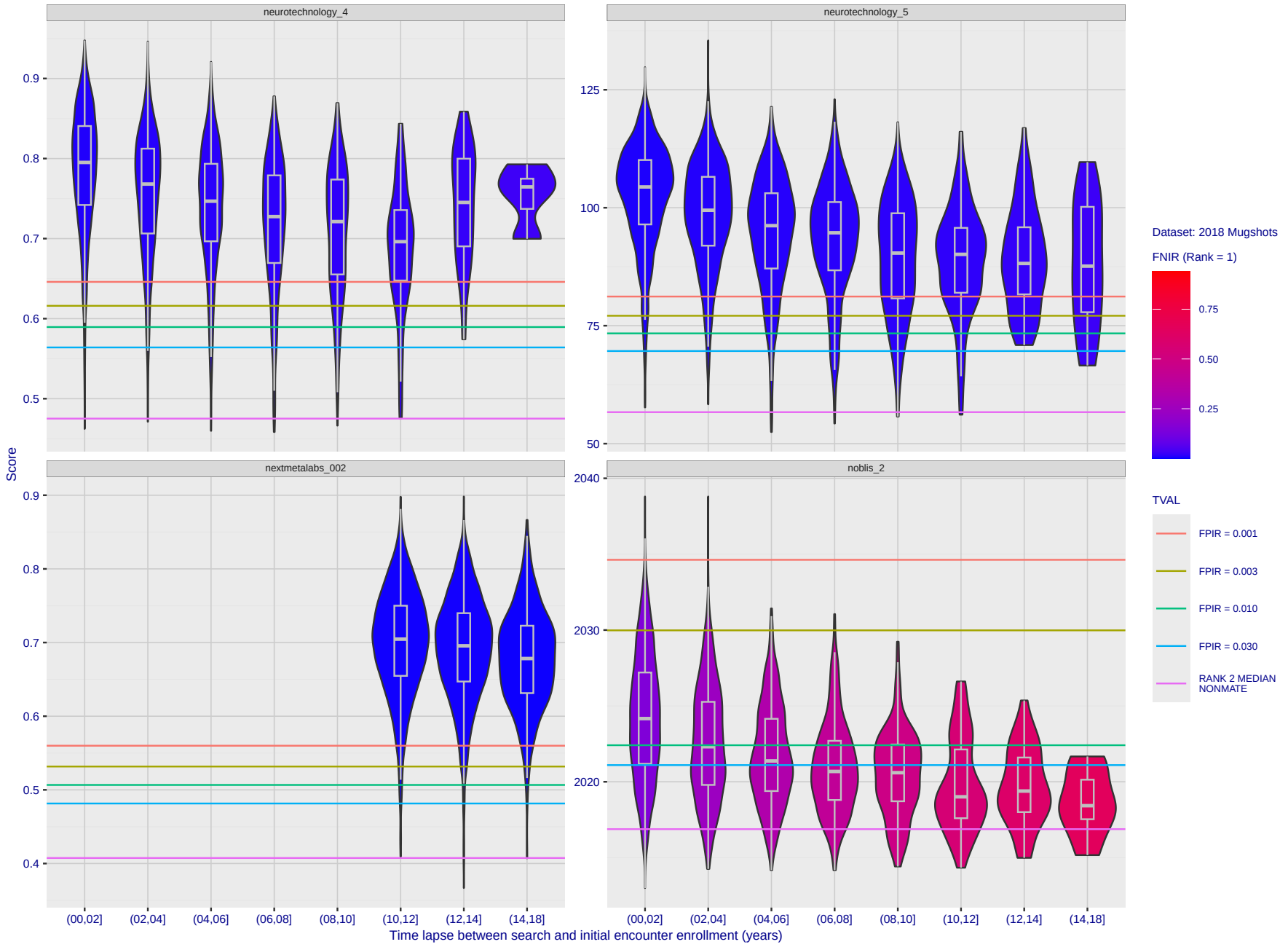


*Figure 303:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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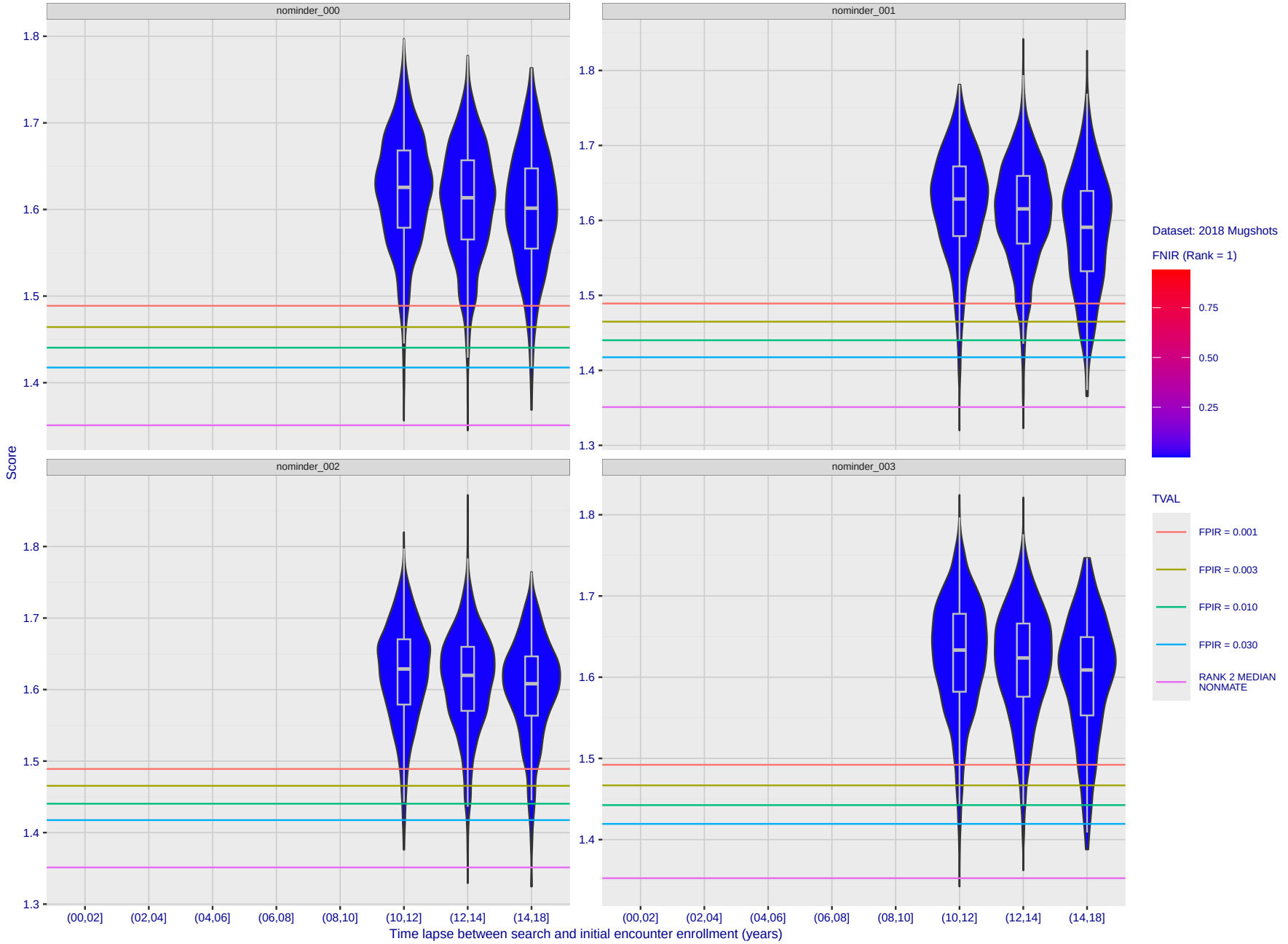


*Figure 304:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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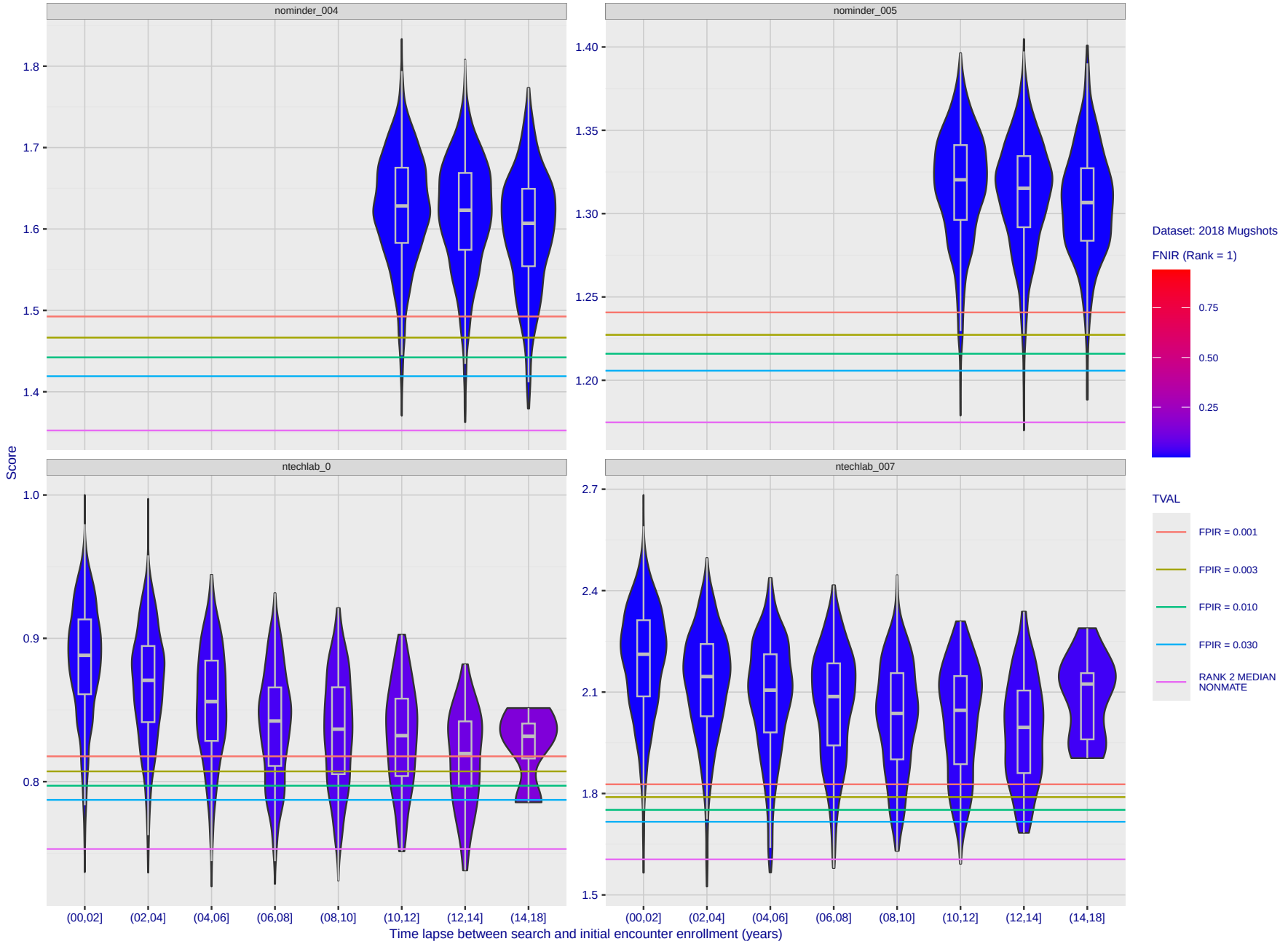


*Figure 305:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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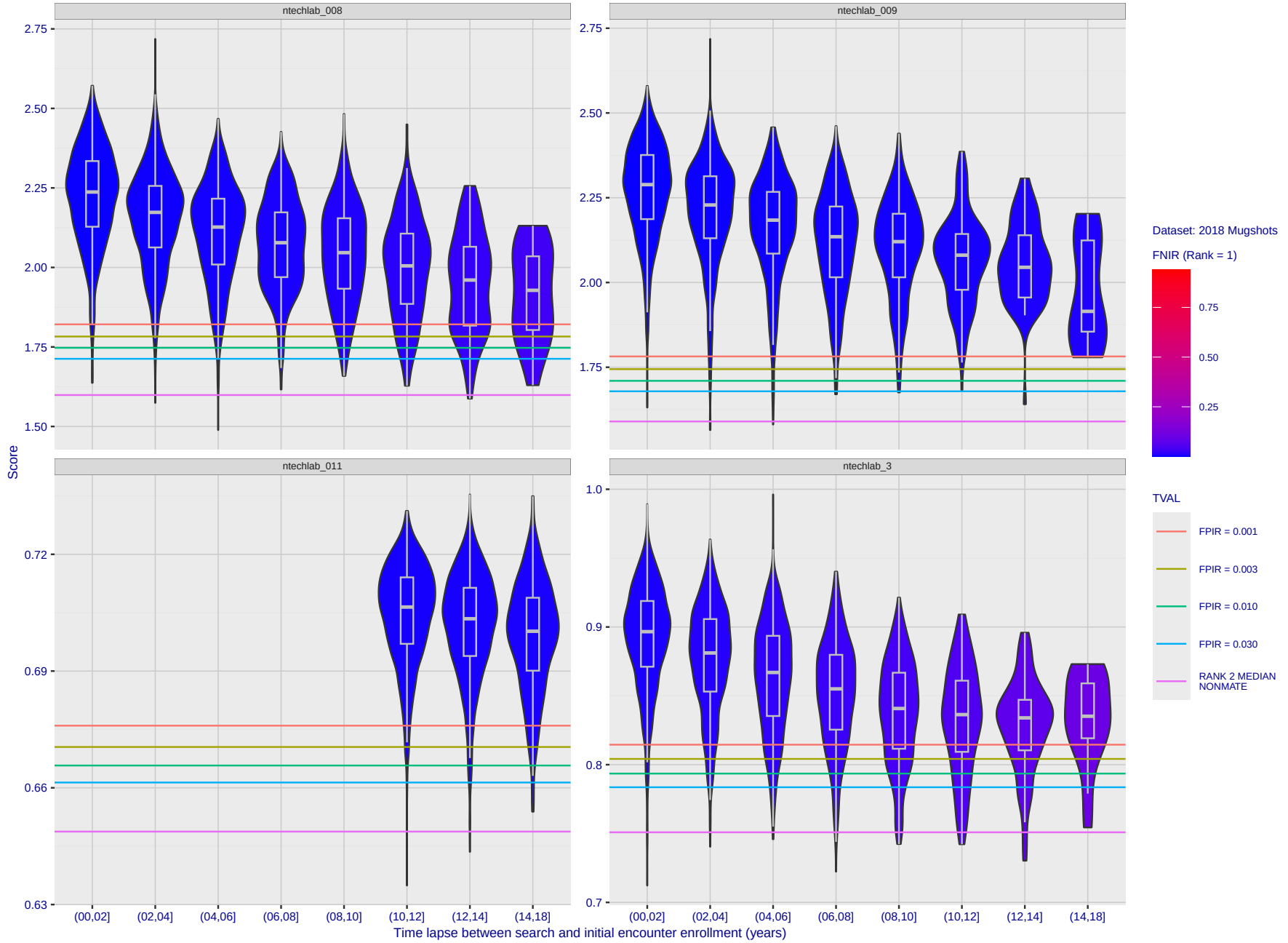


*Figure 306:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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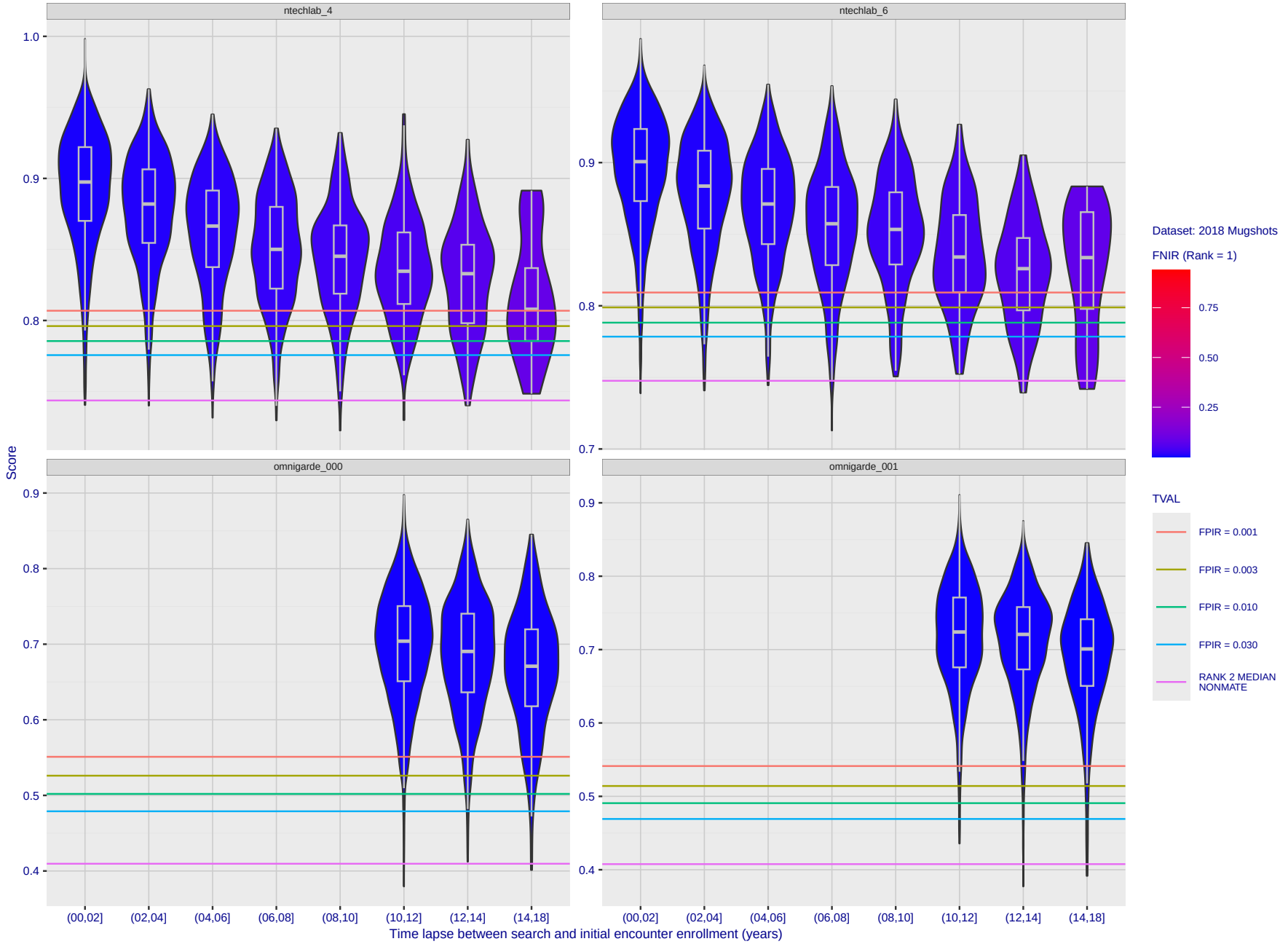


*Figure 307:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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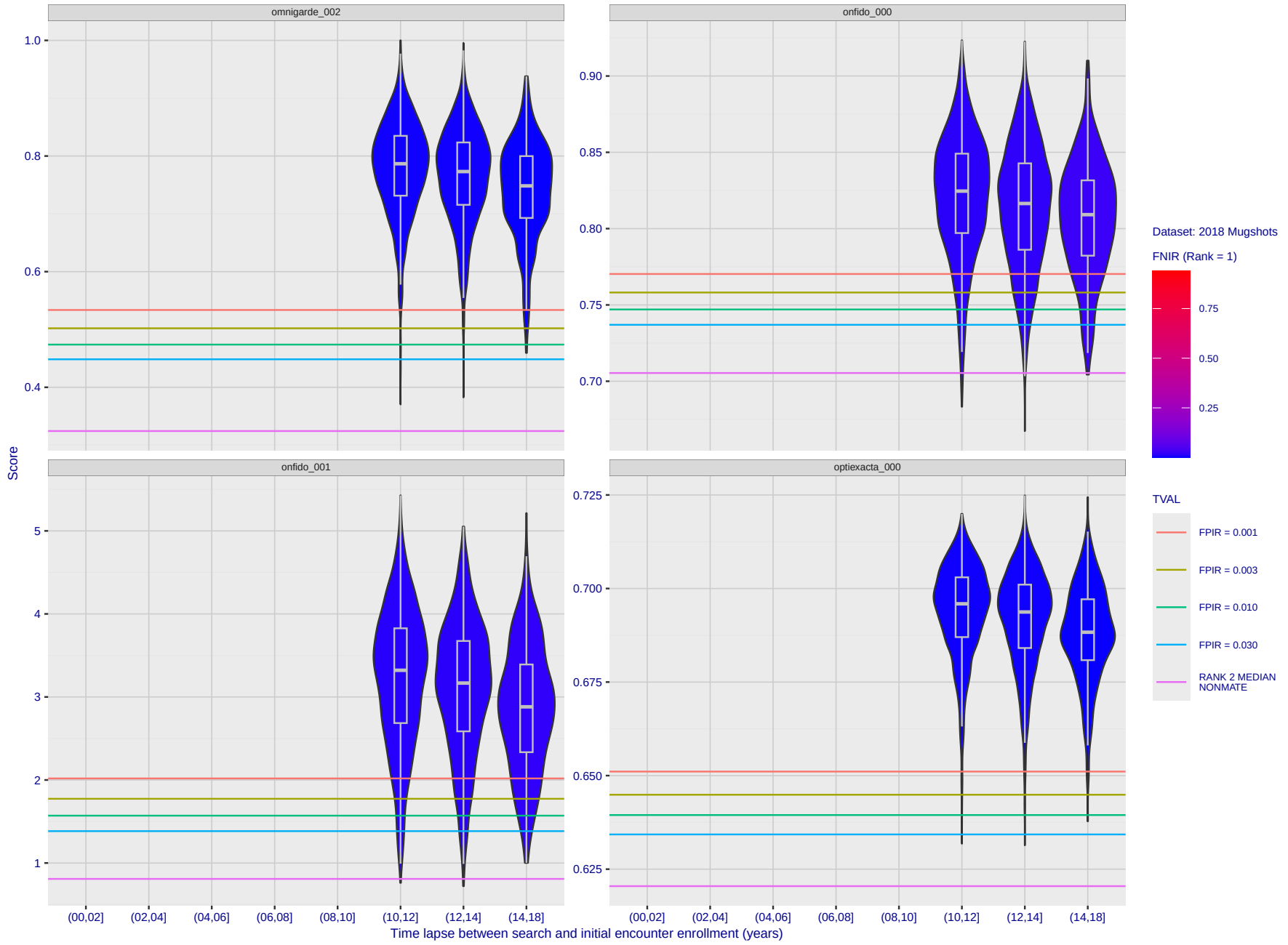


*Figure 308:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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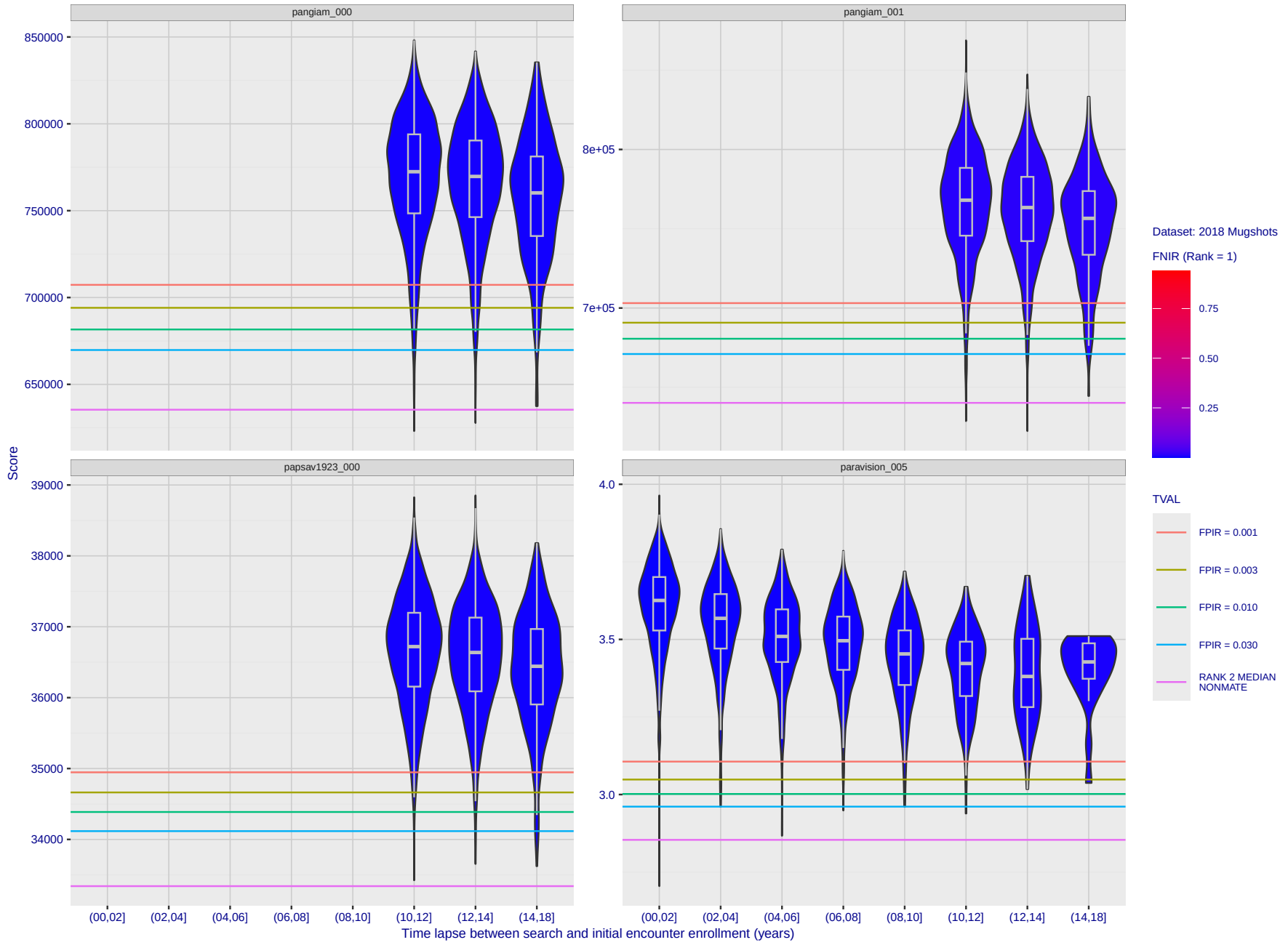


*Figure 309:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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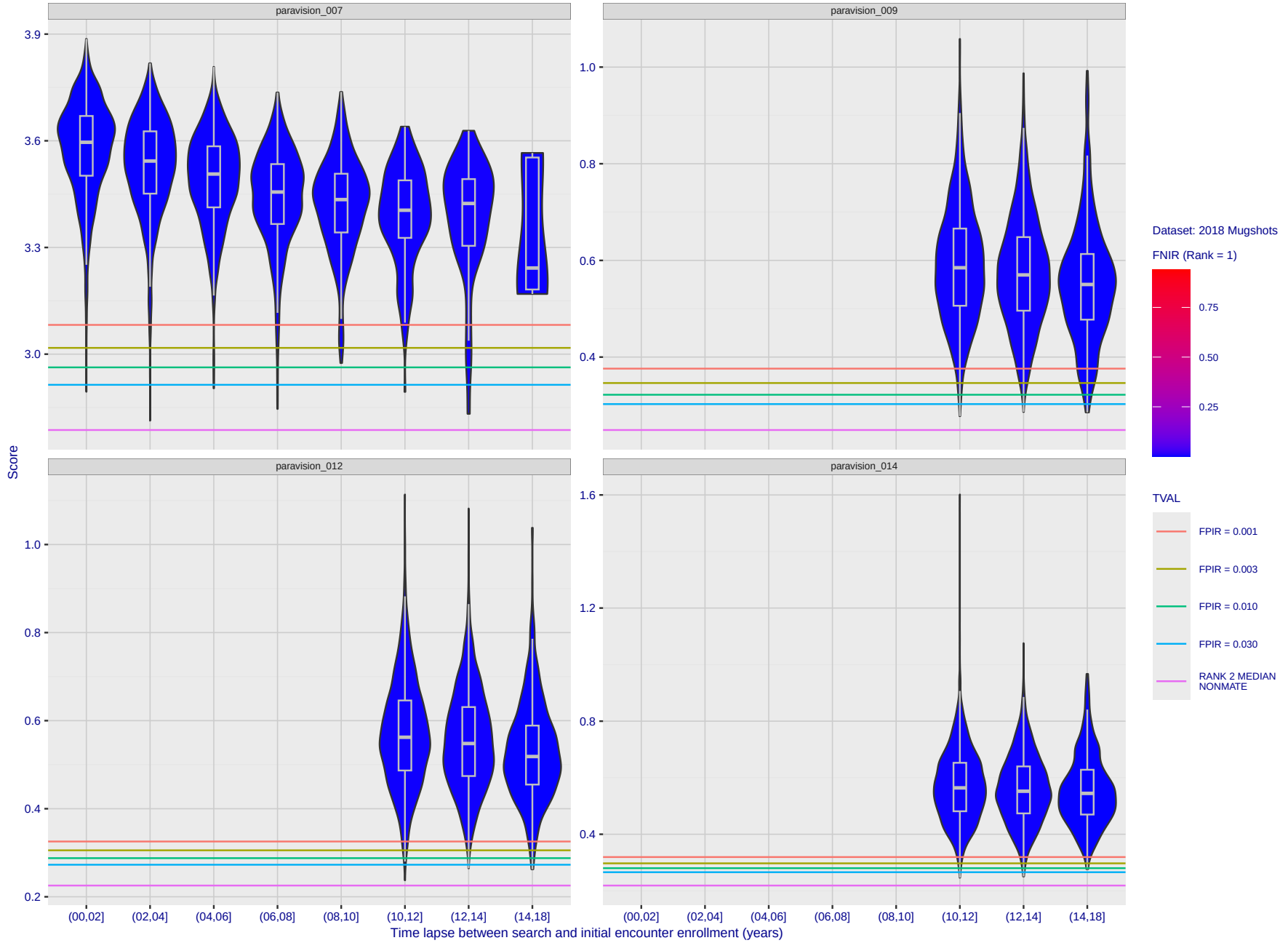


*Figure 310:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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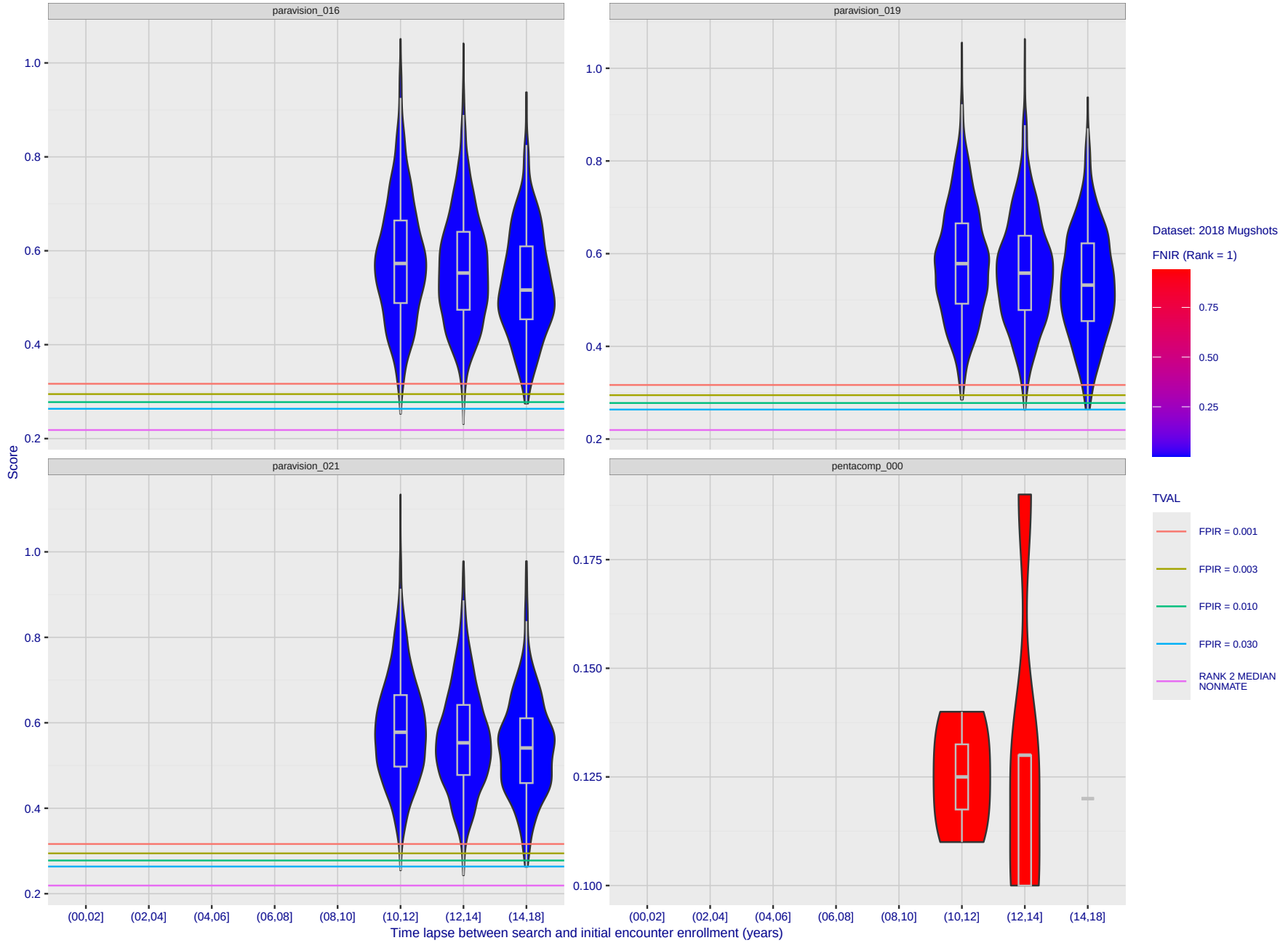


*Figure 311:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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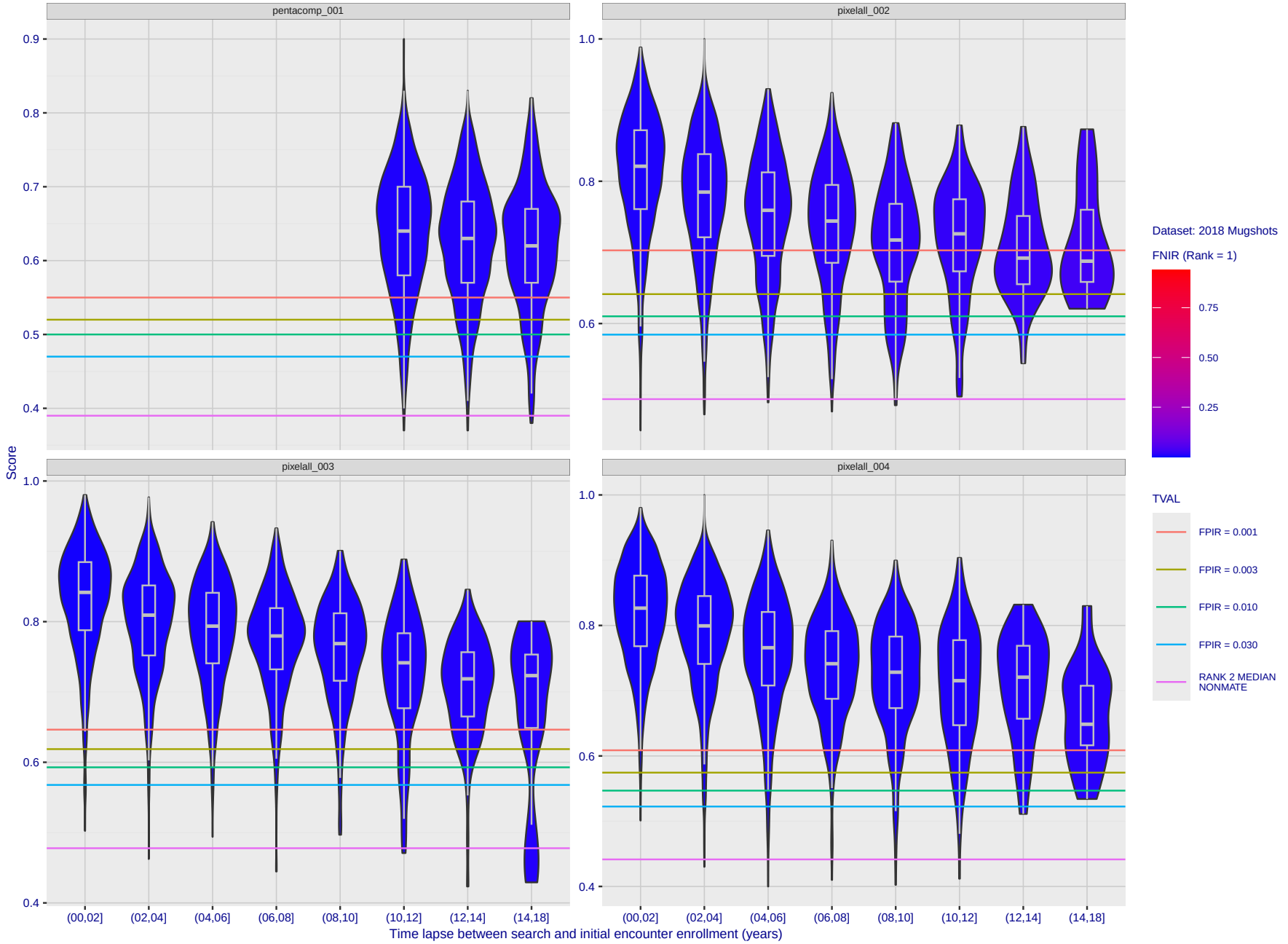


*Figure 312:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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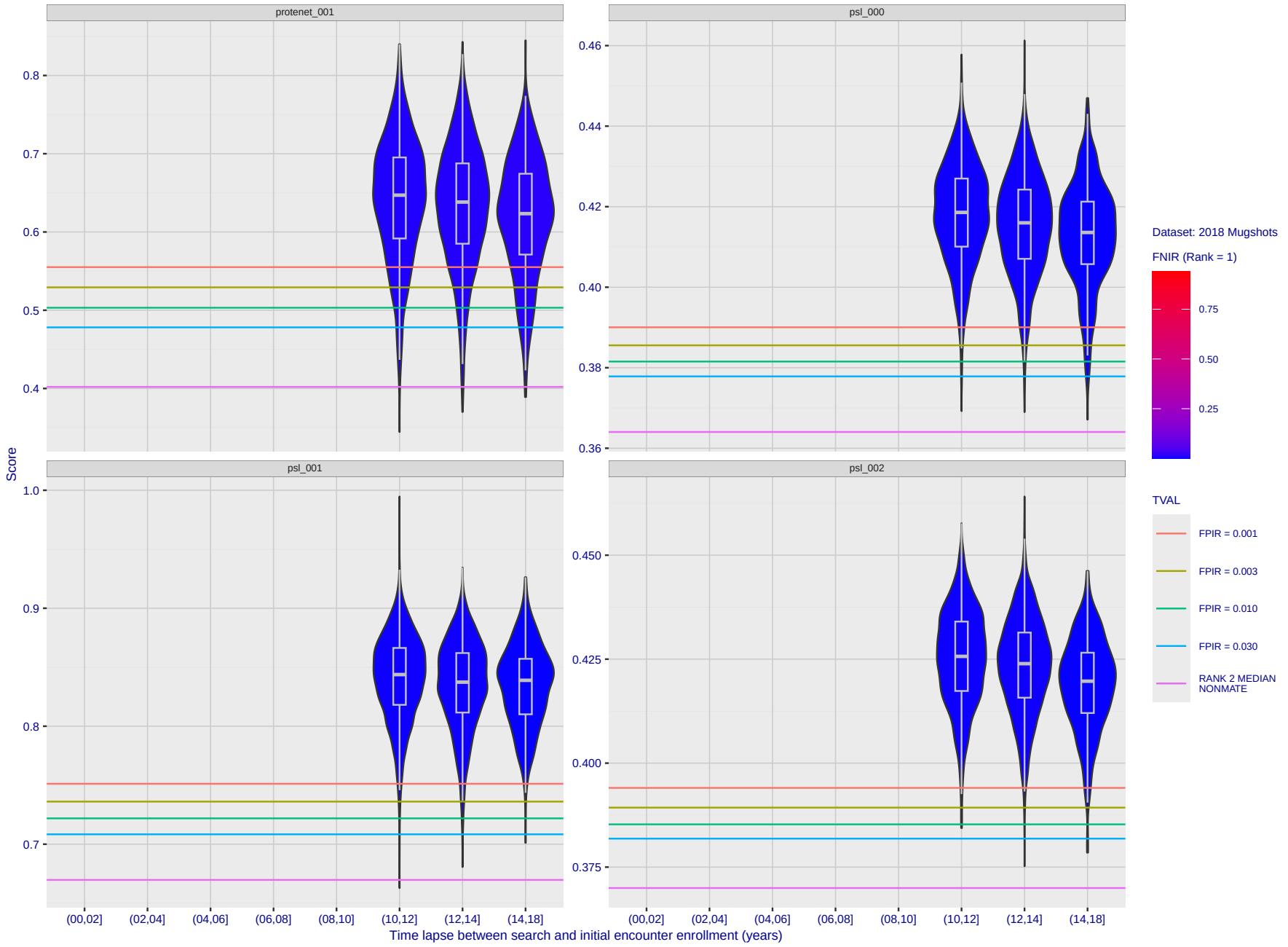


*Figure 313:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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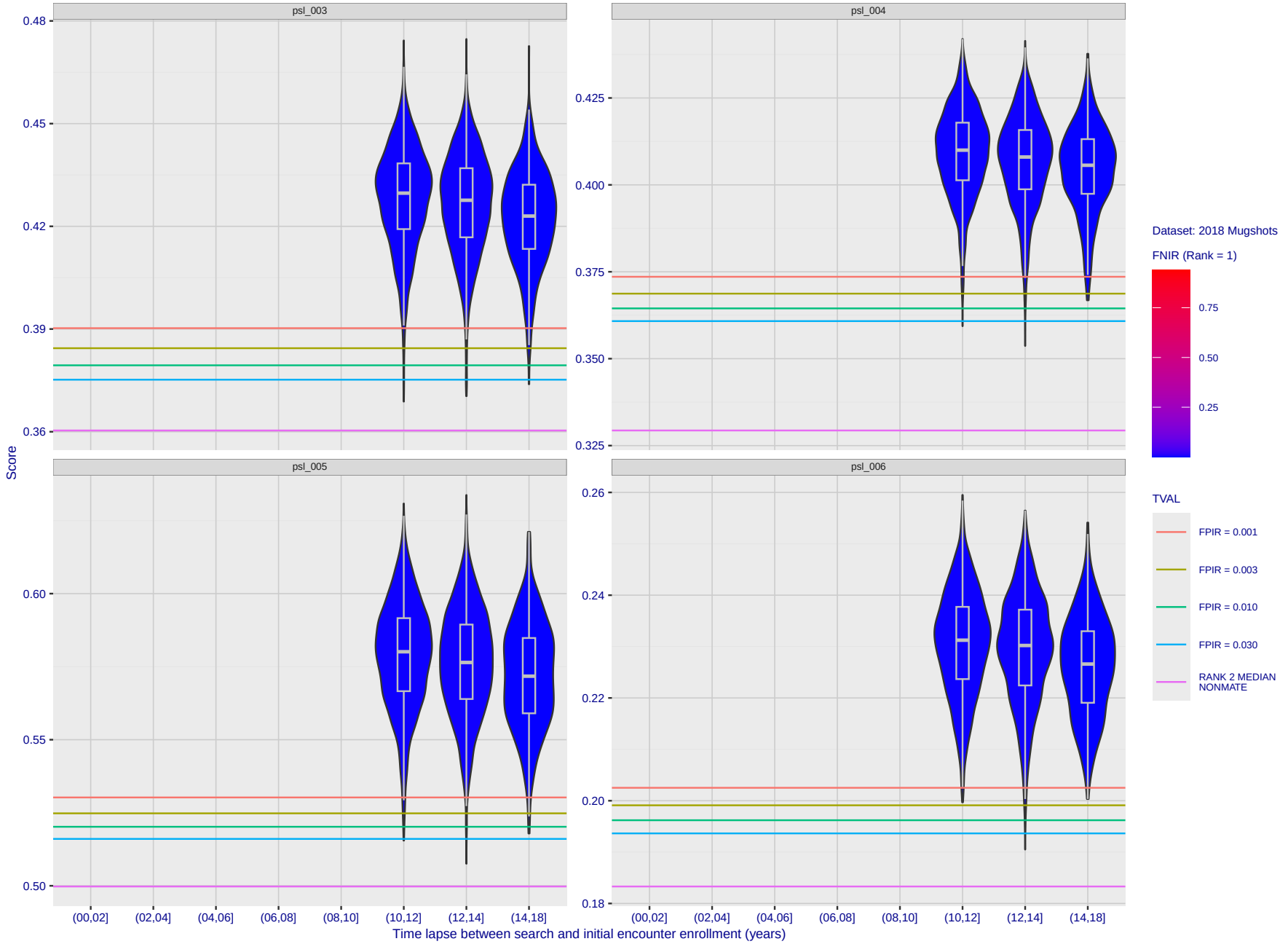


*Figure 314:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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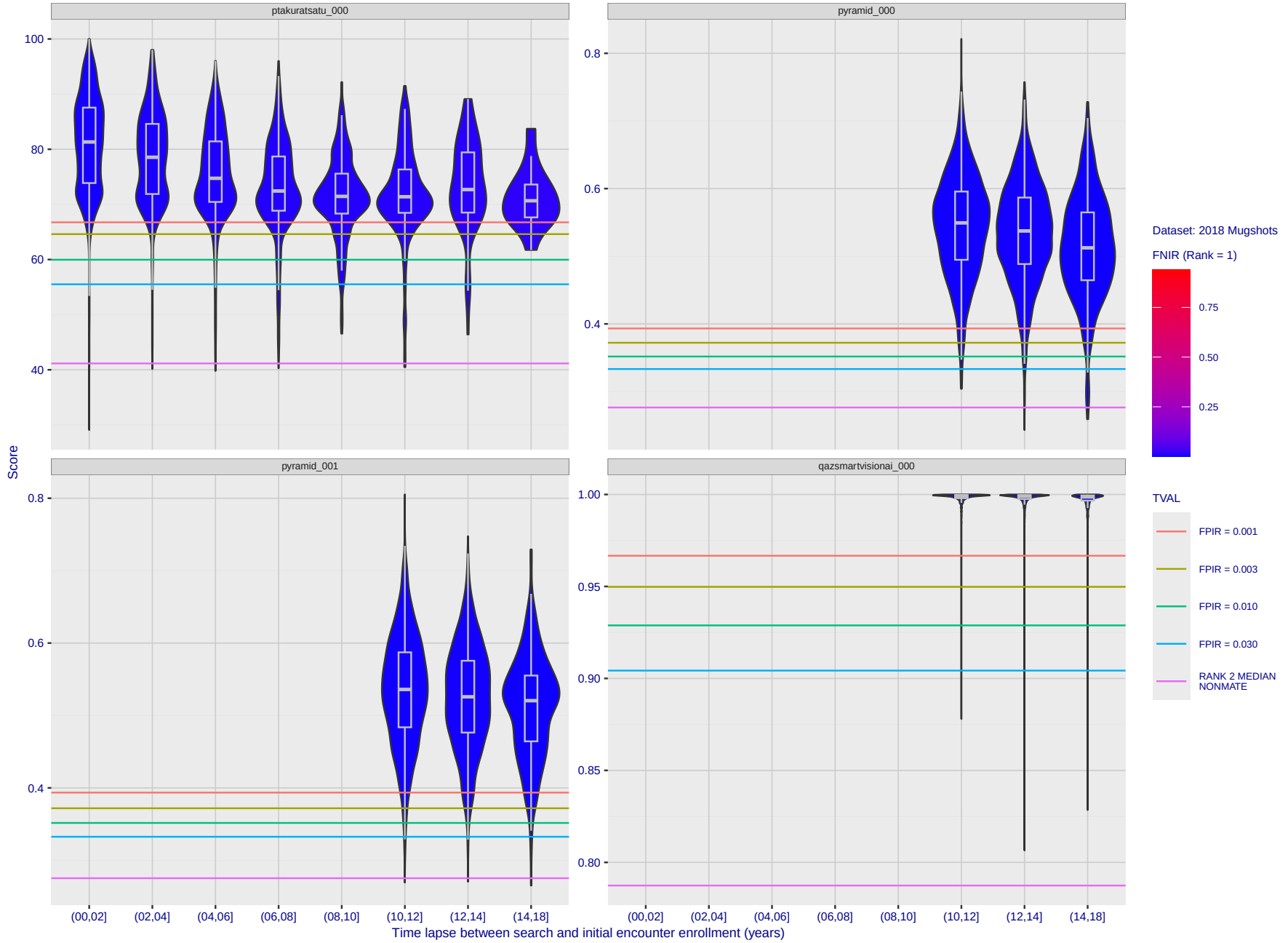


*Figure 315:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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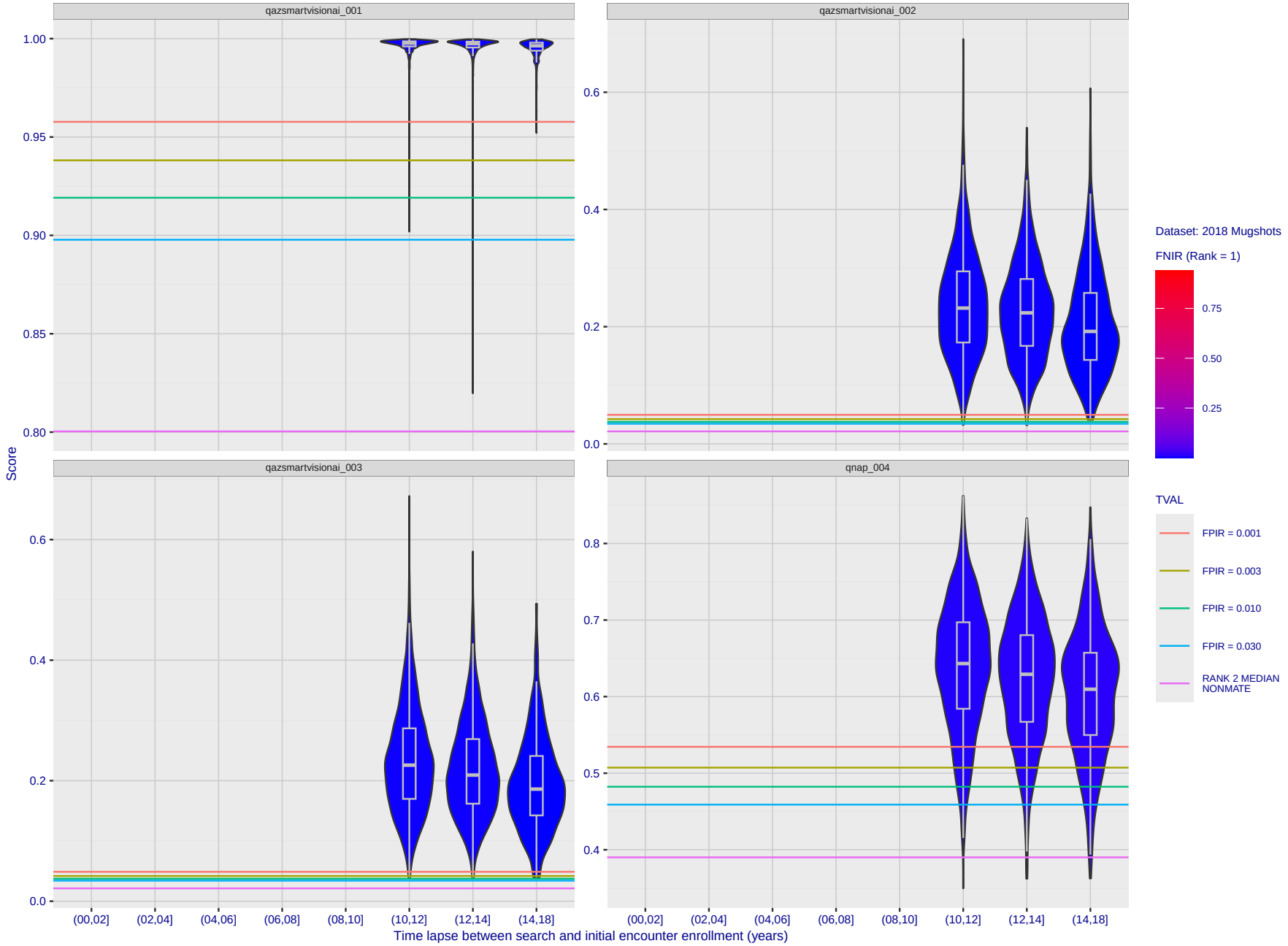


*Figure 316:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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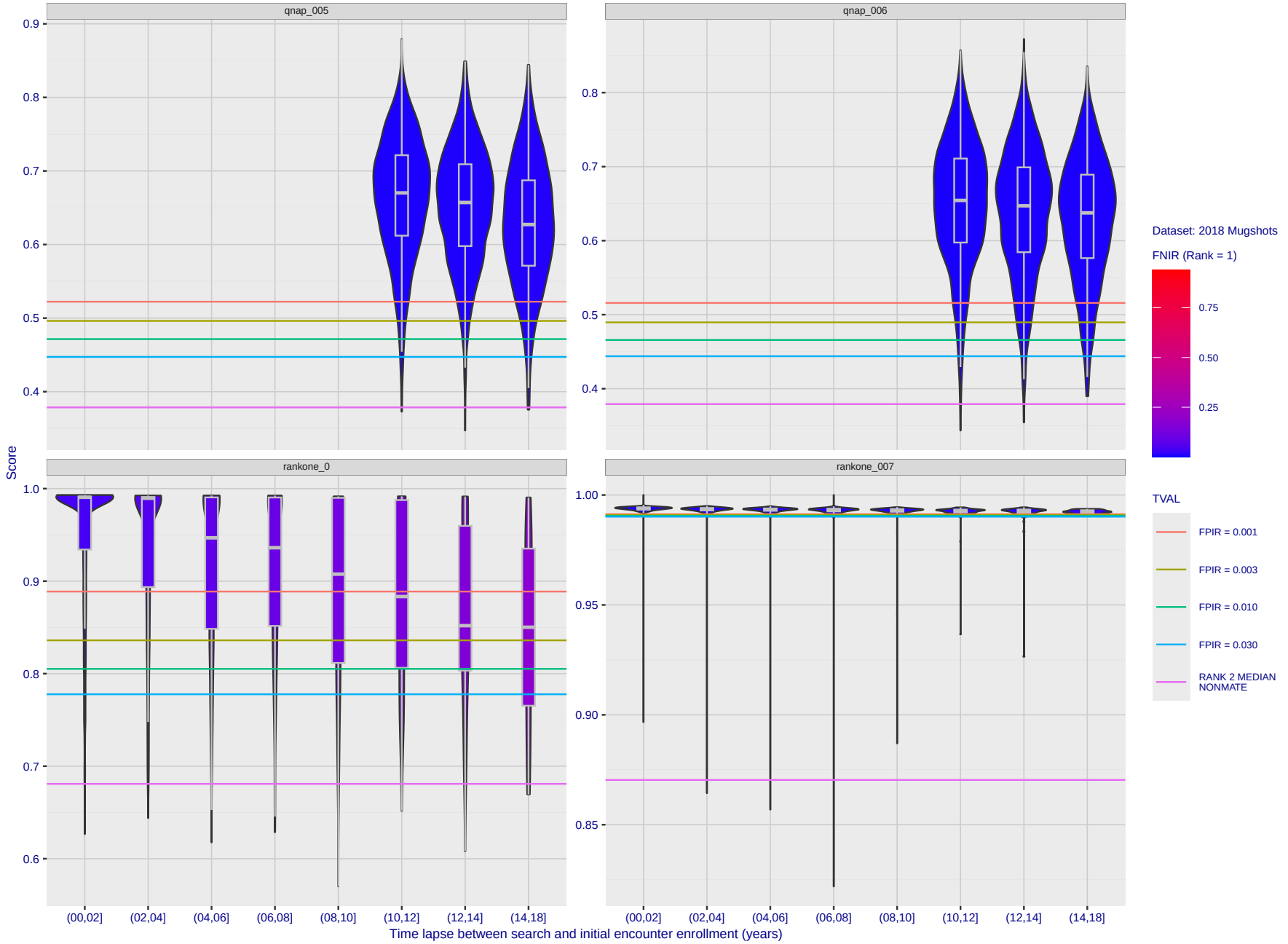


*Figure 317:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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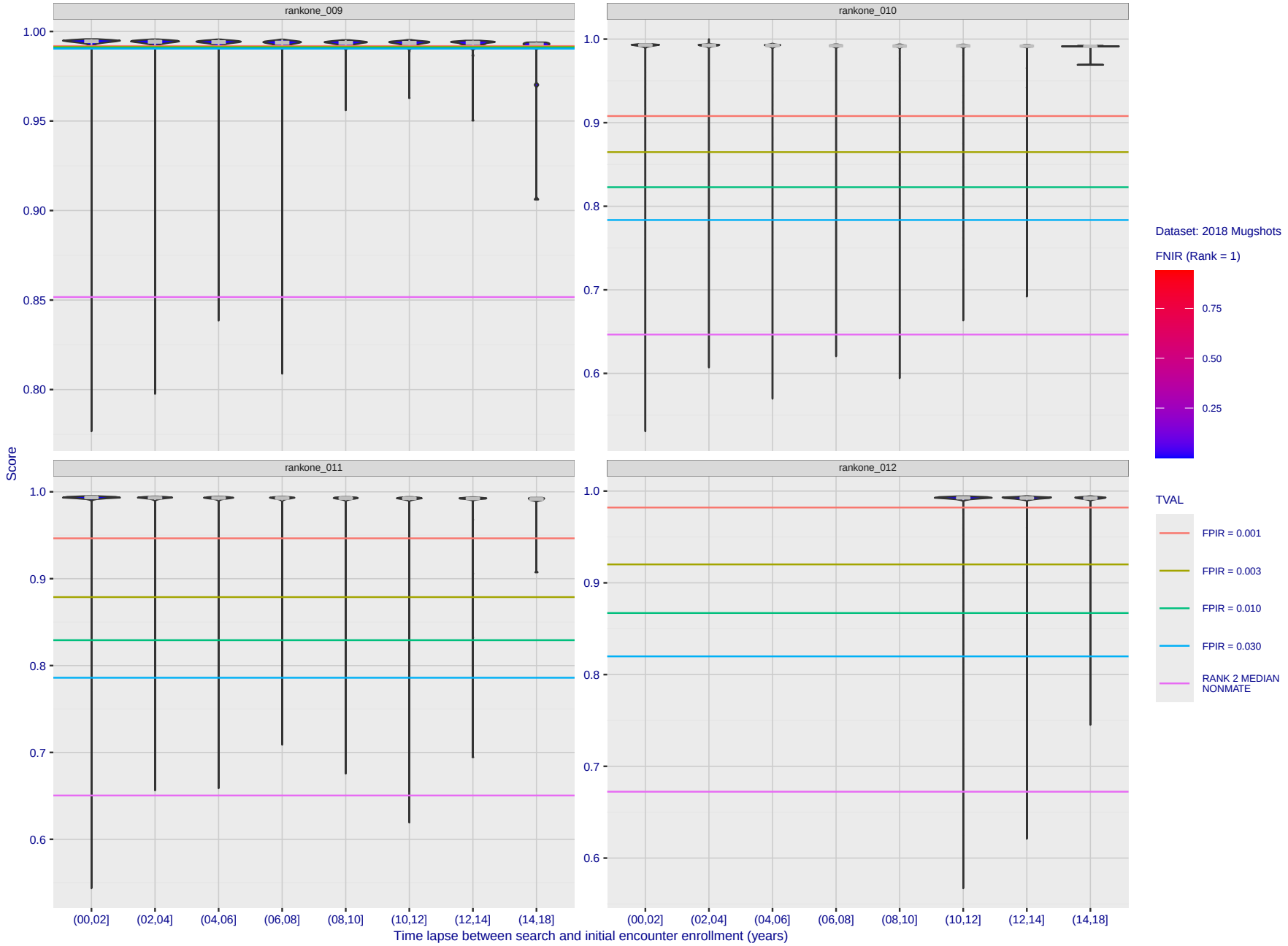


*Figure 318:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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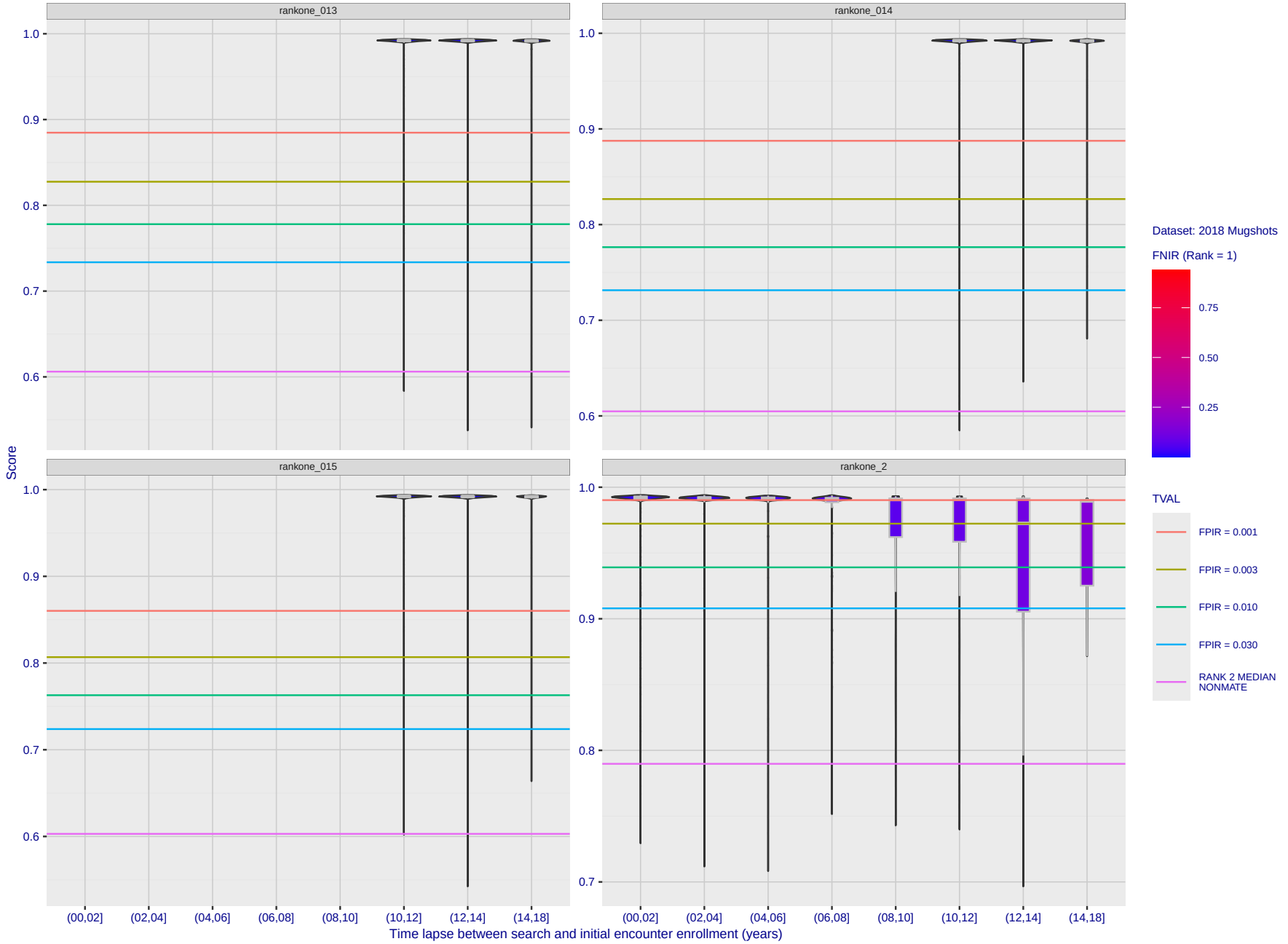


*Figure 319:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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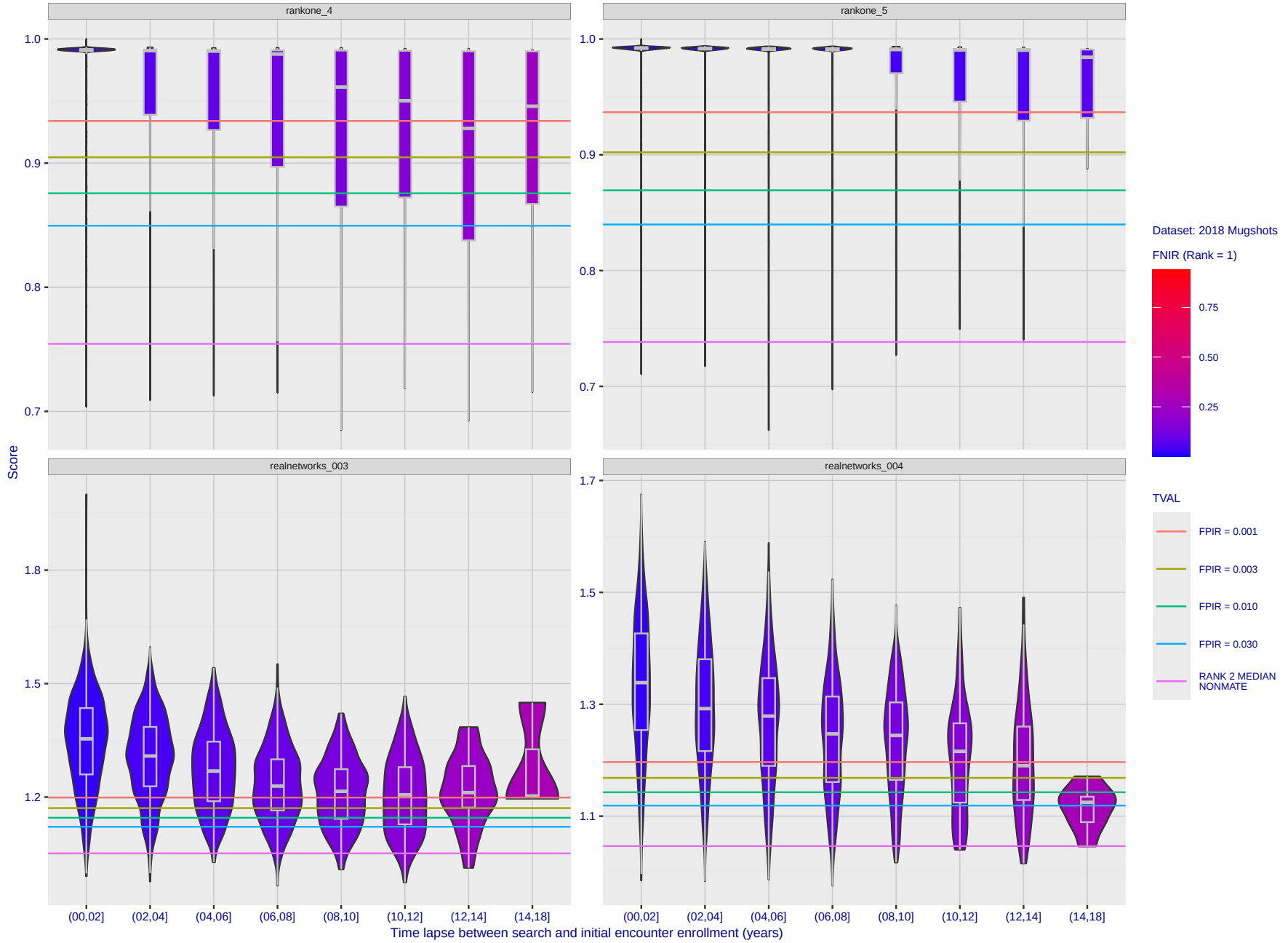


*Figure 320:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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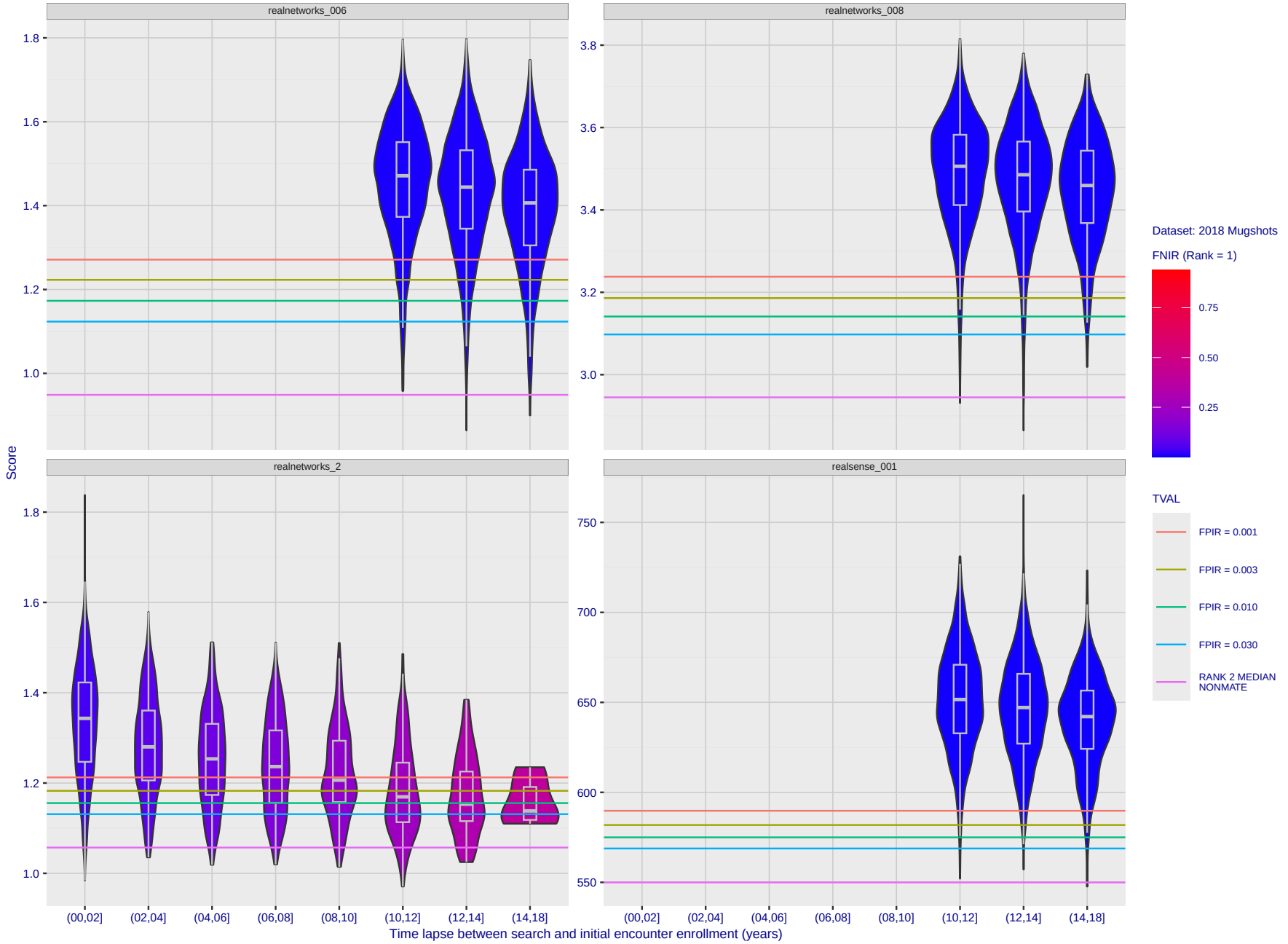


*Figure 321:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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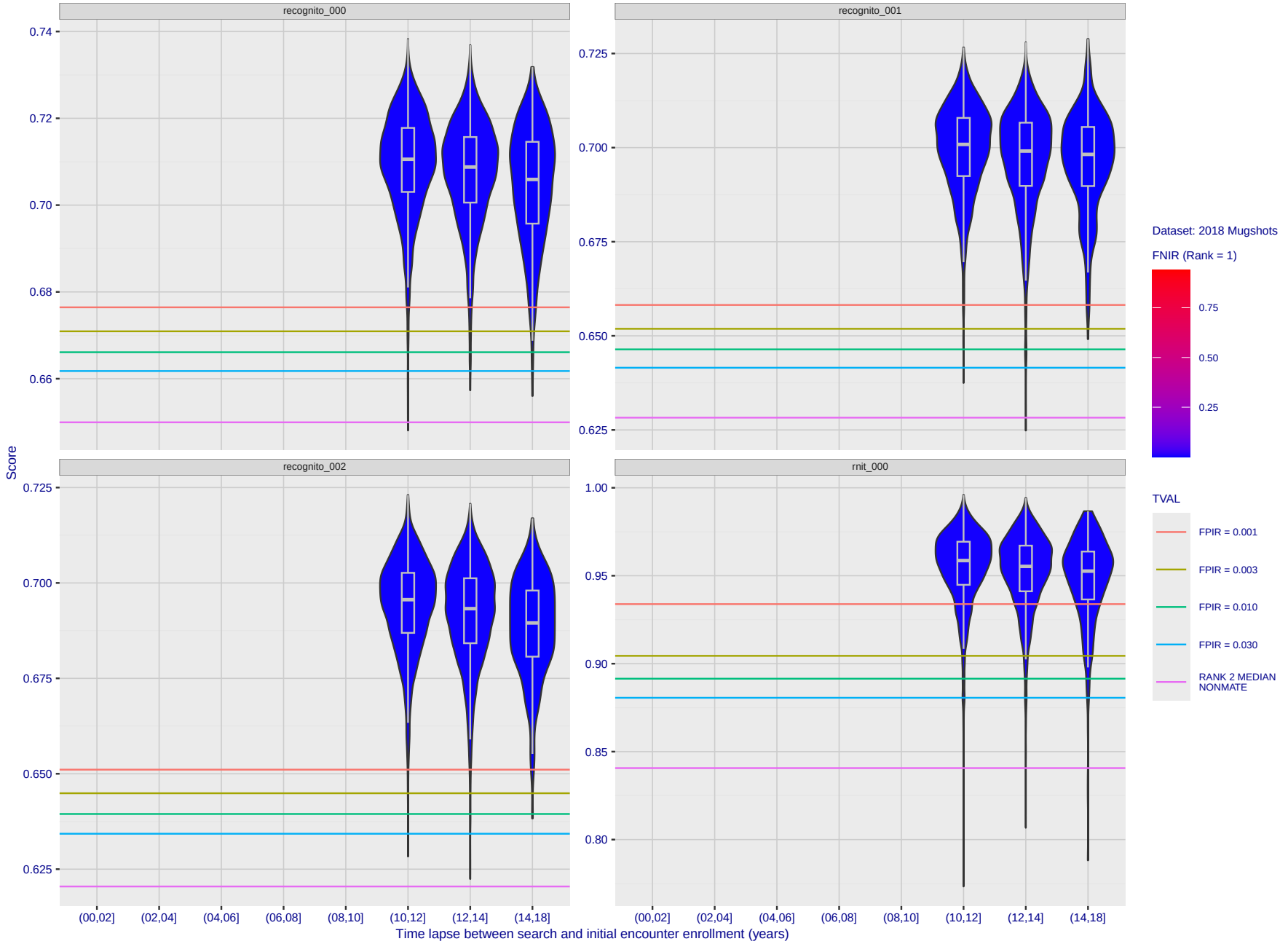


*Figure 322:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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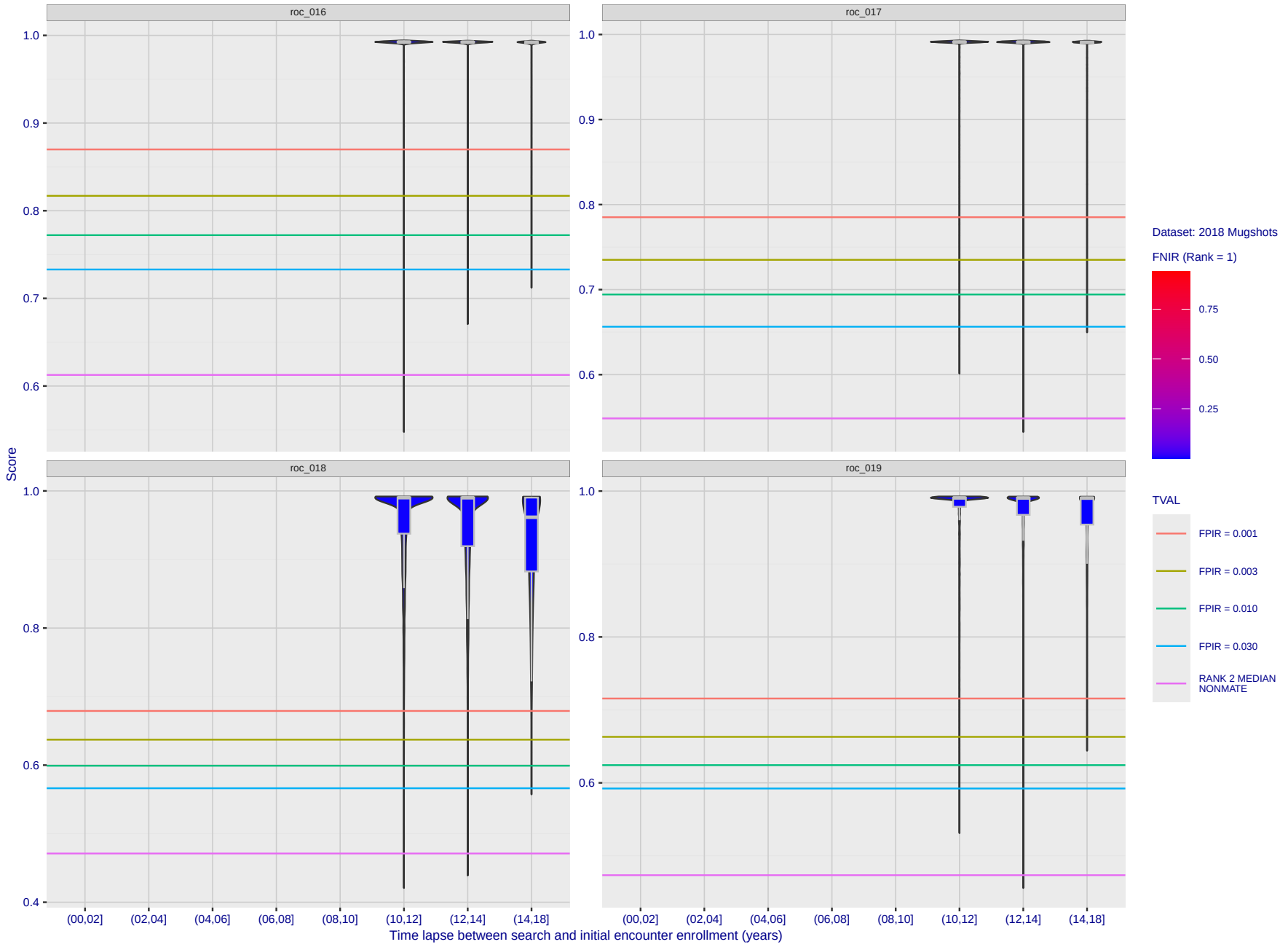


*Figure 323:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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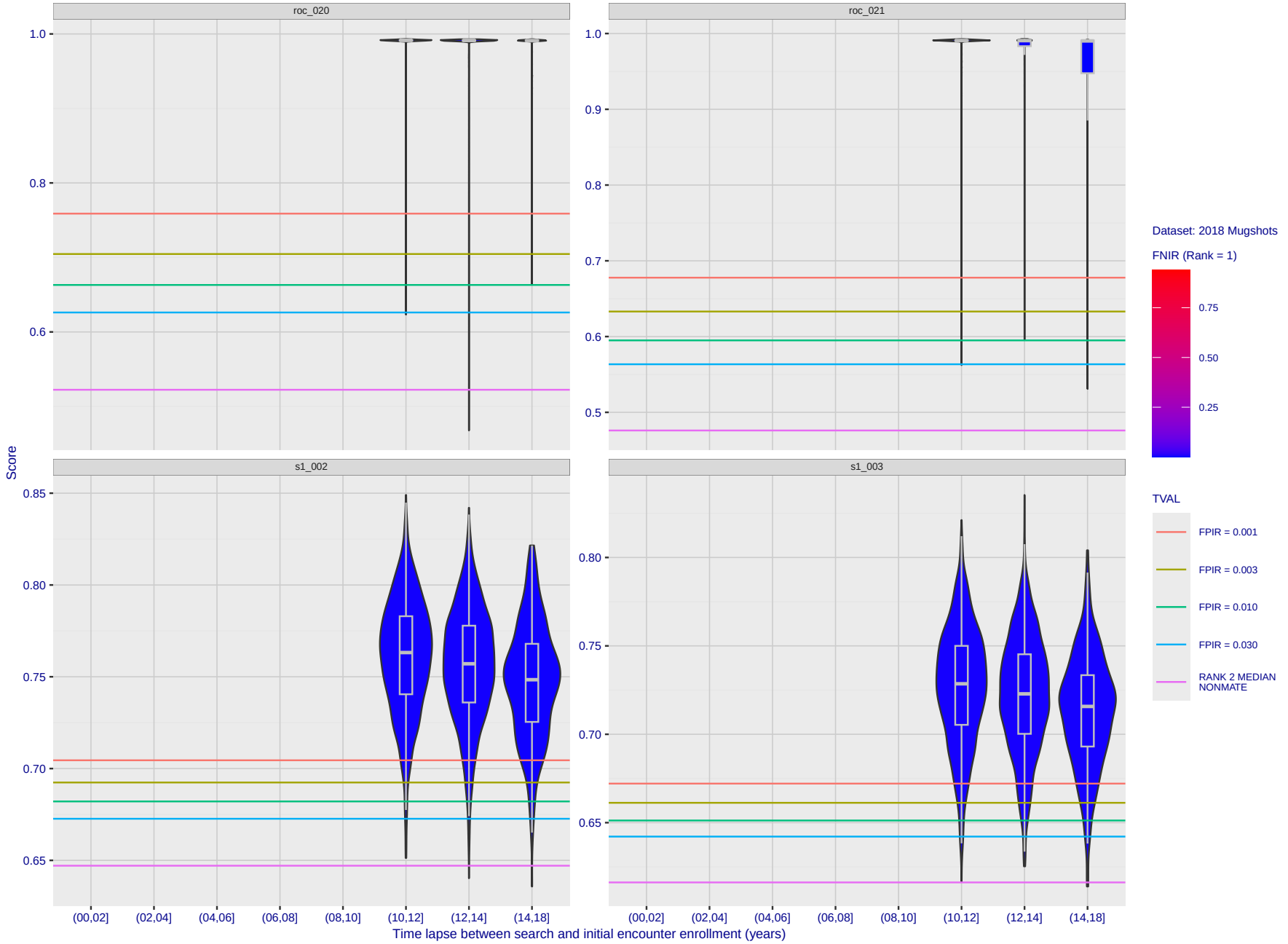


*Figure 324:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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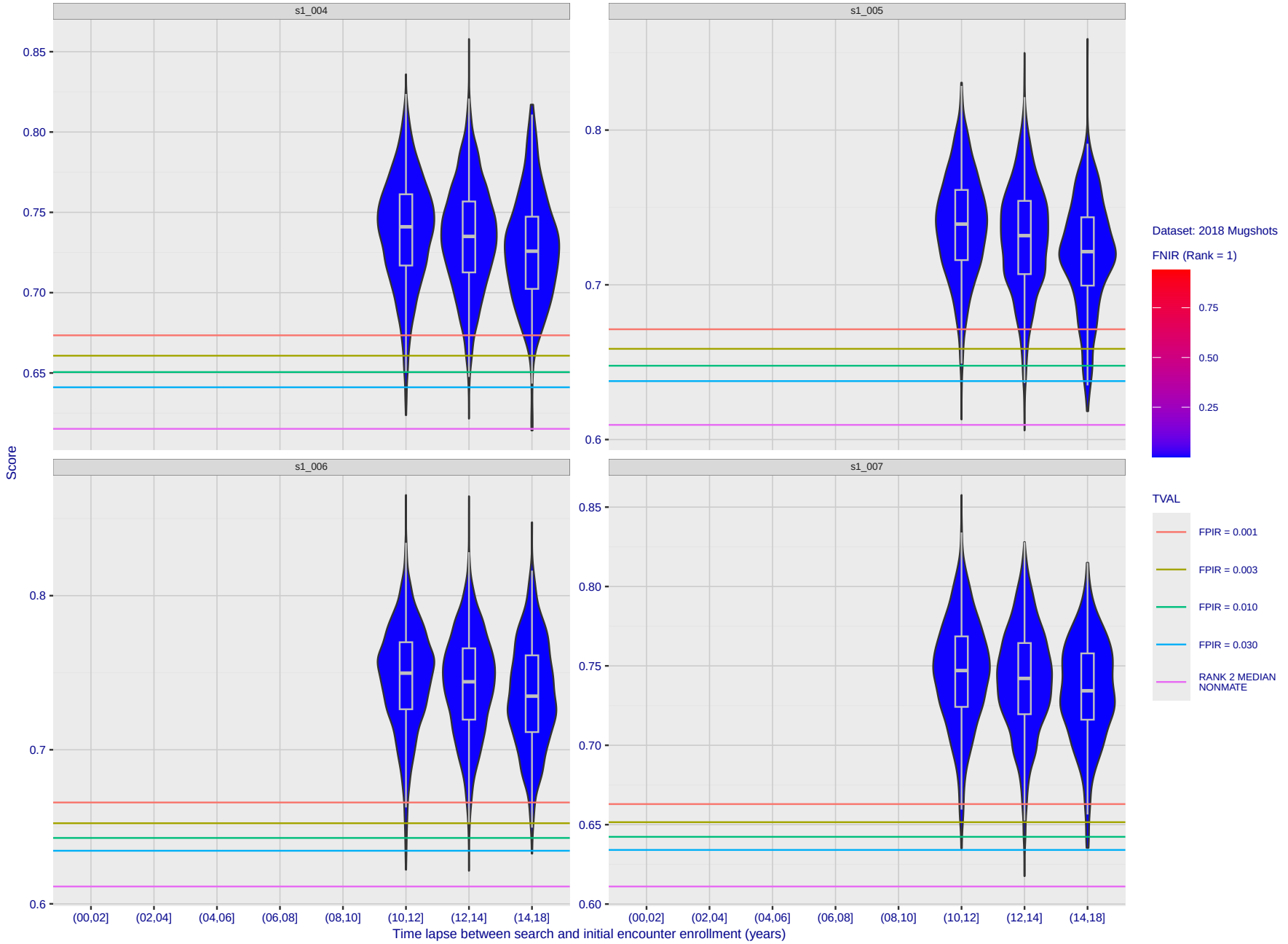


*Figure 325:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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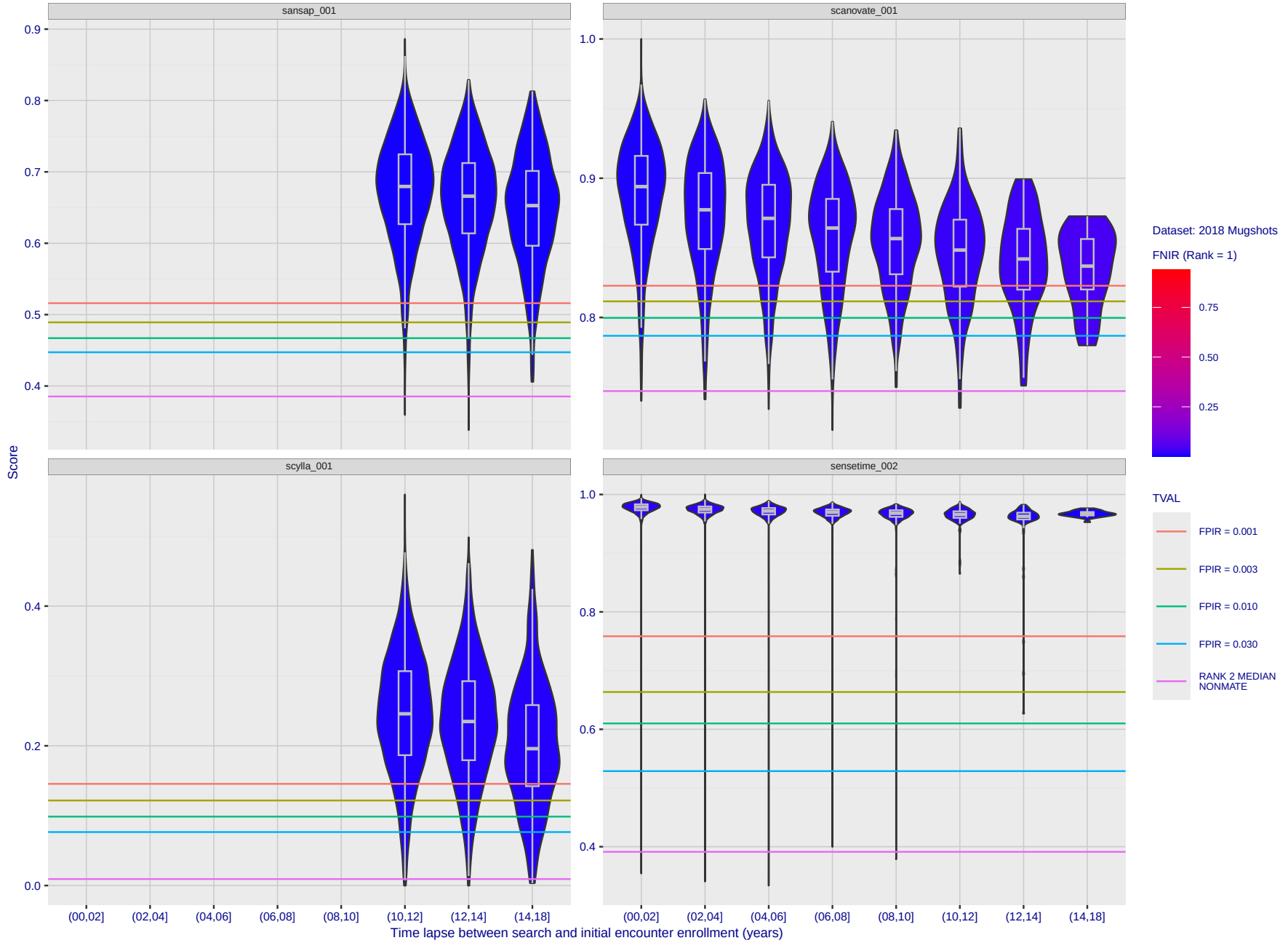


*Figure 326:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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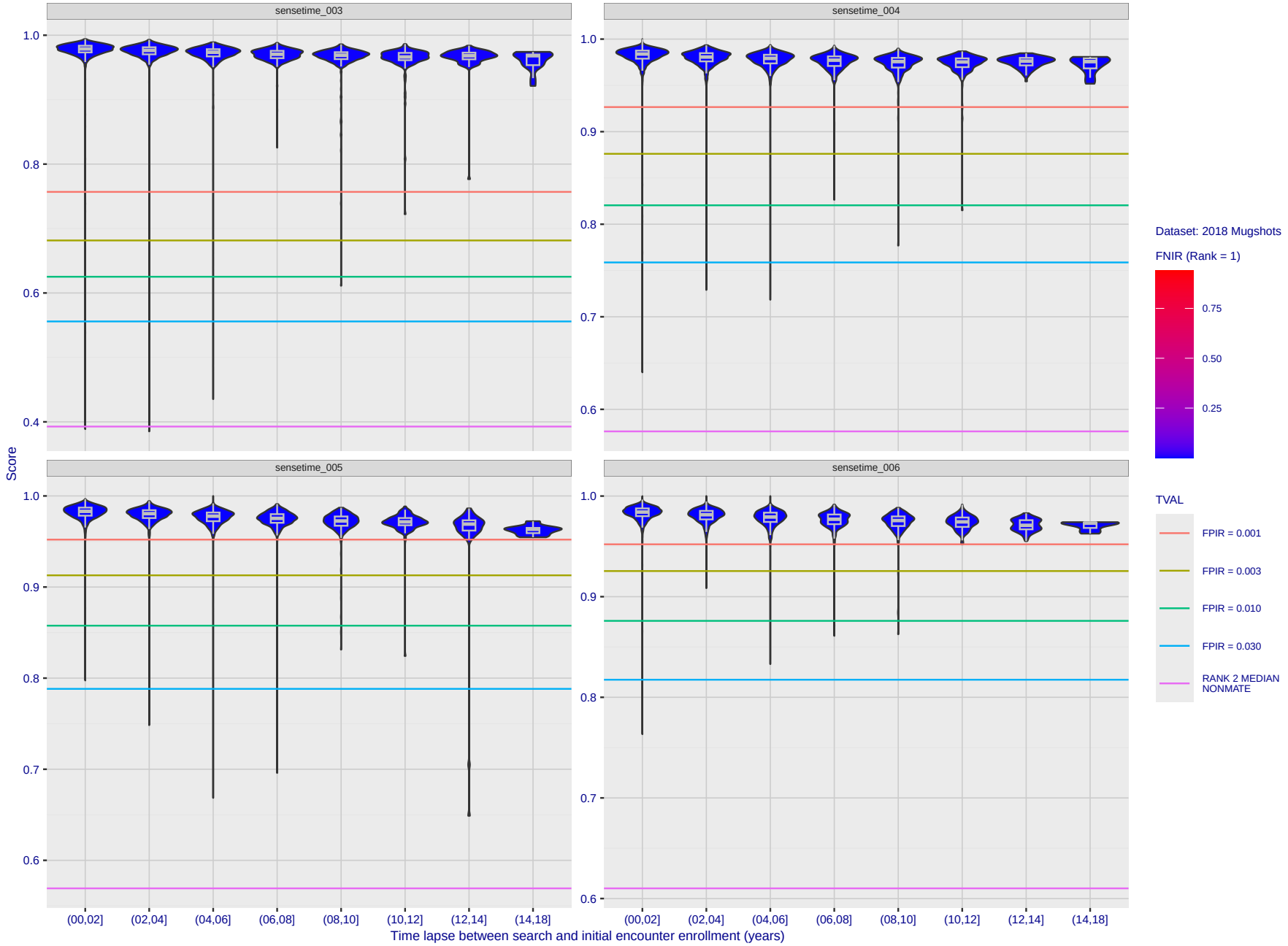


*Figure 327:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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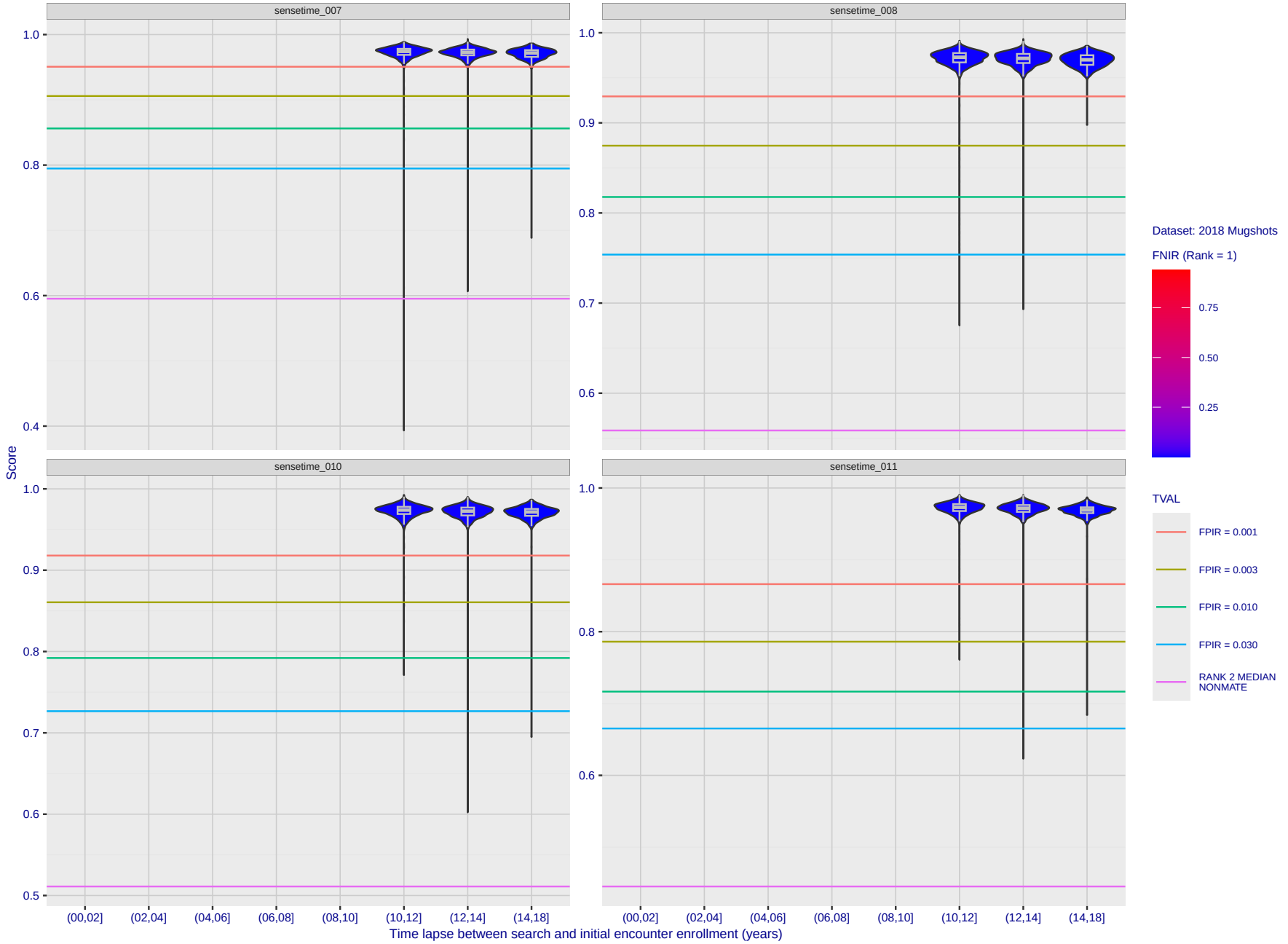


*Figure 328:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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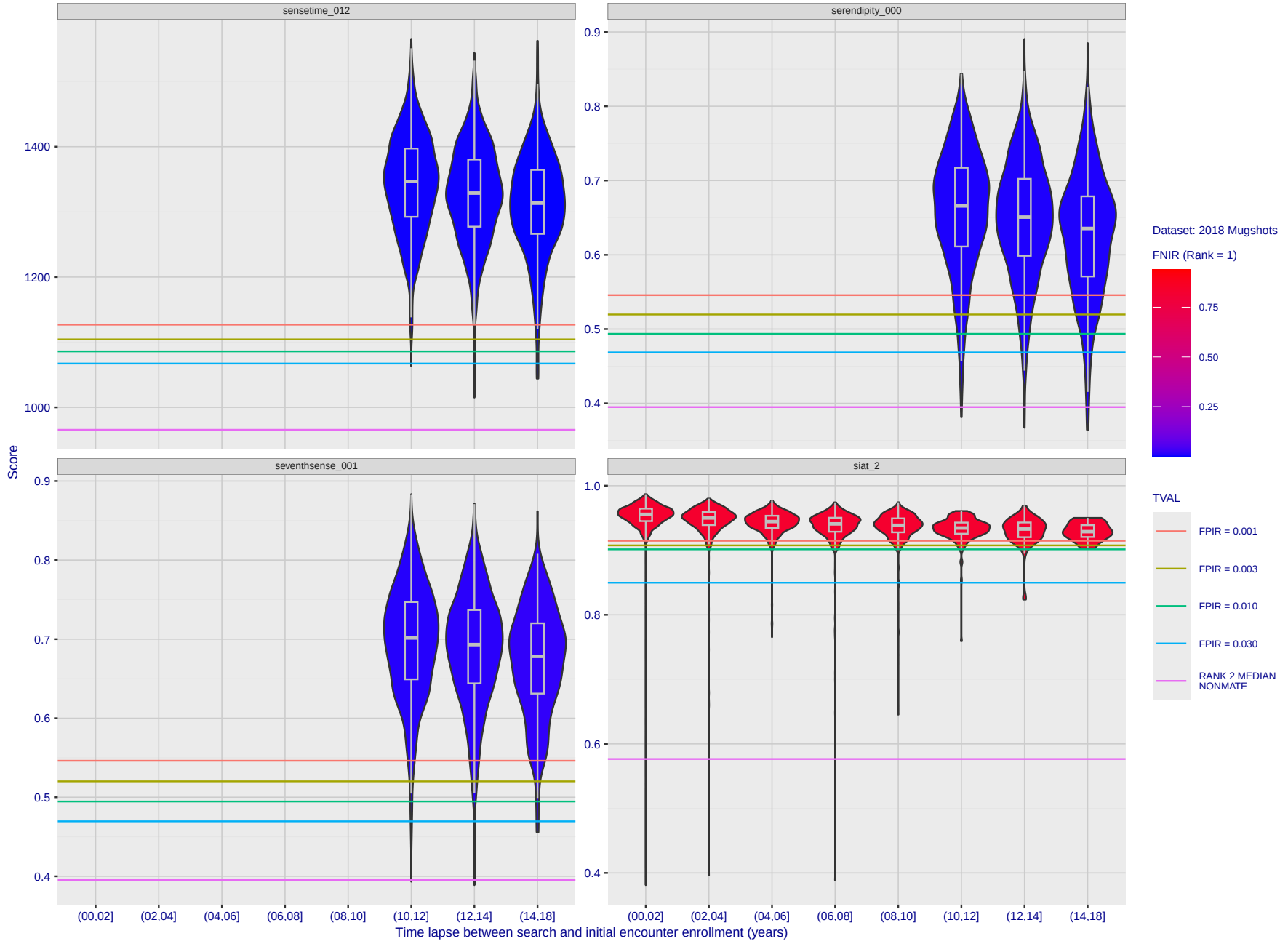


*Figure 329:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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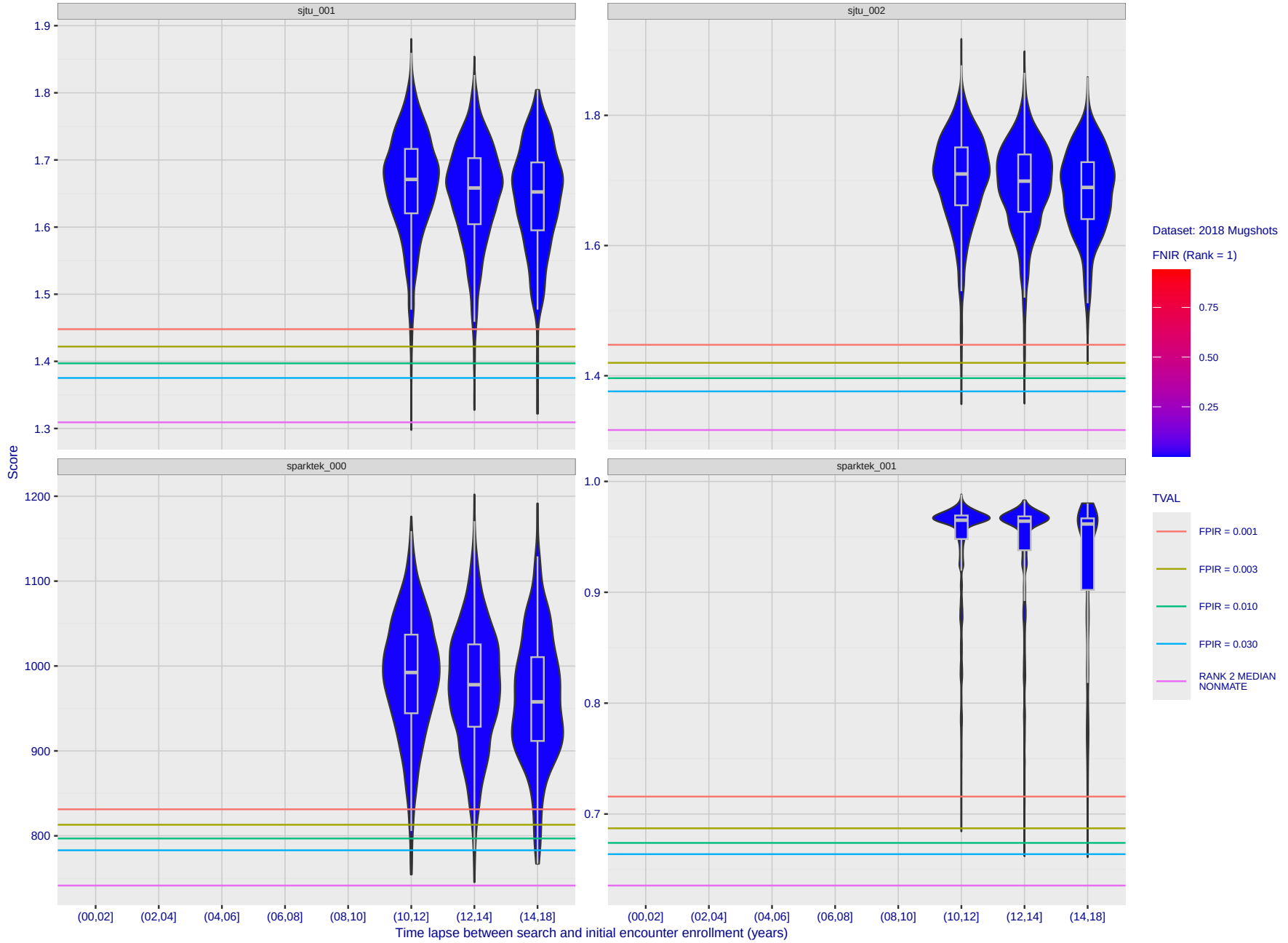


*Figure 330:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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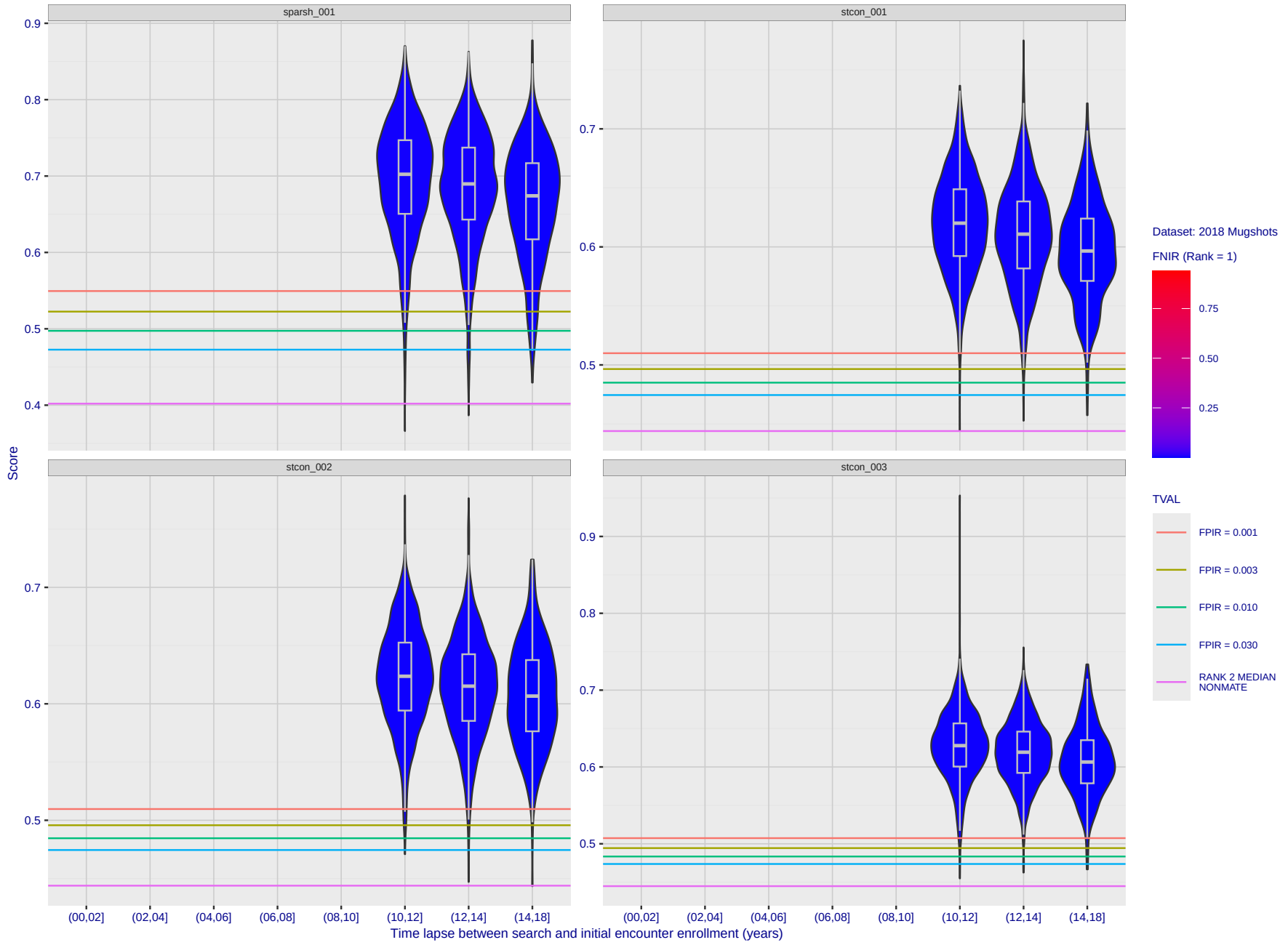


*Figure 331:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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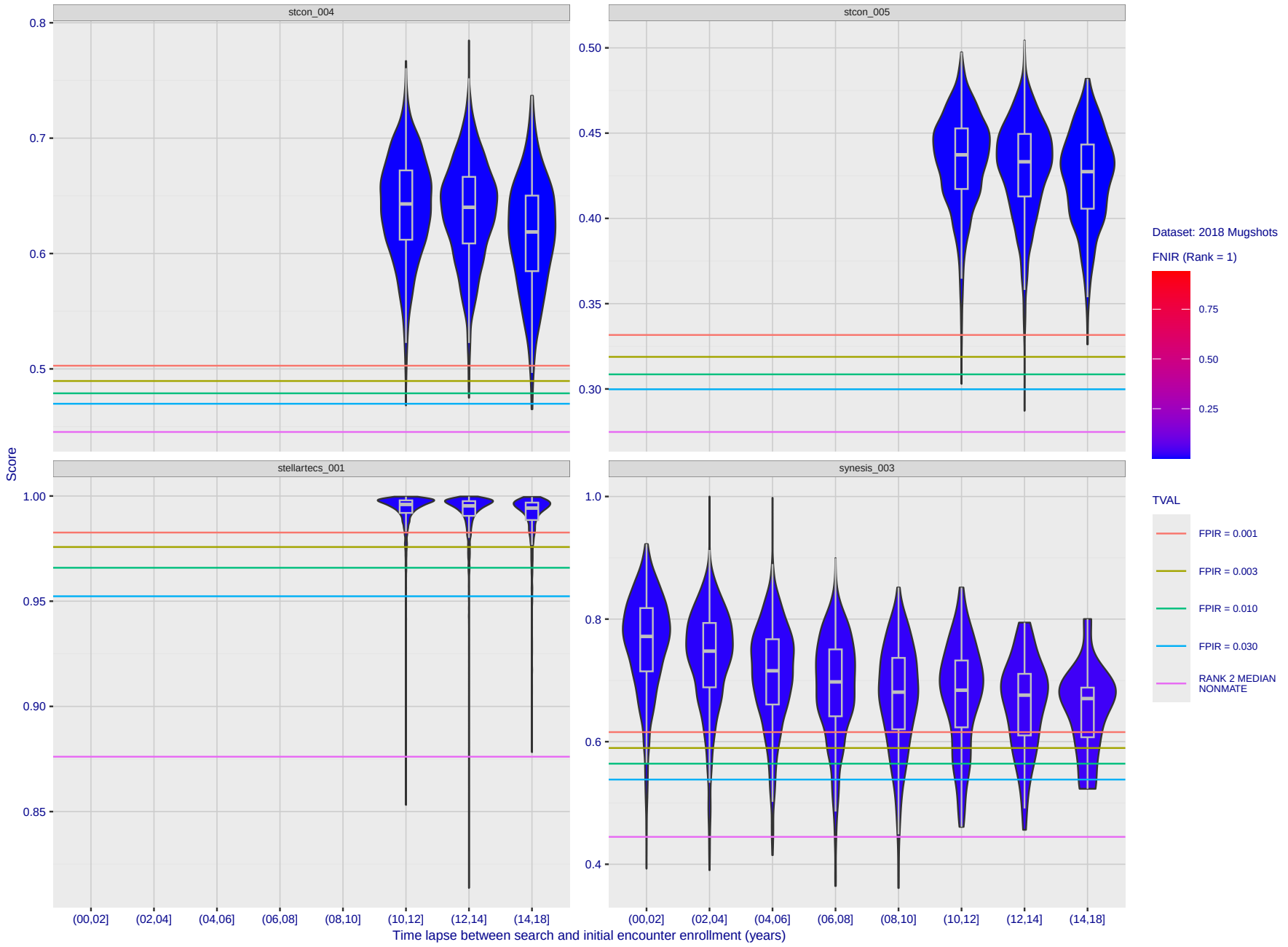


*Figure 332:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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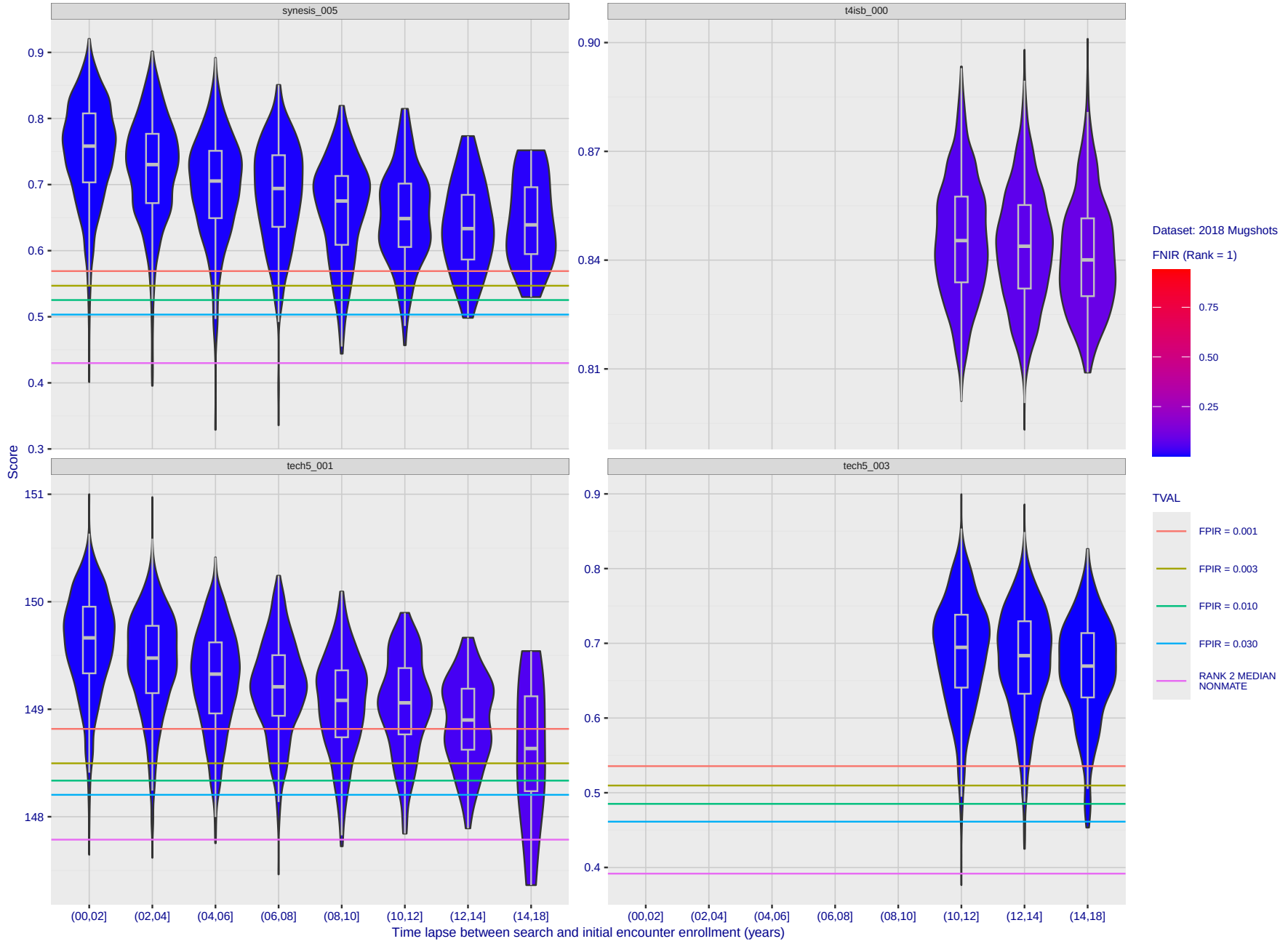


*Figure 333:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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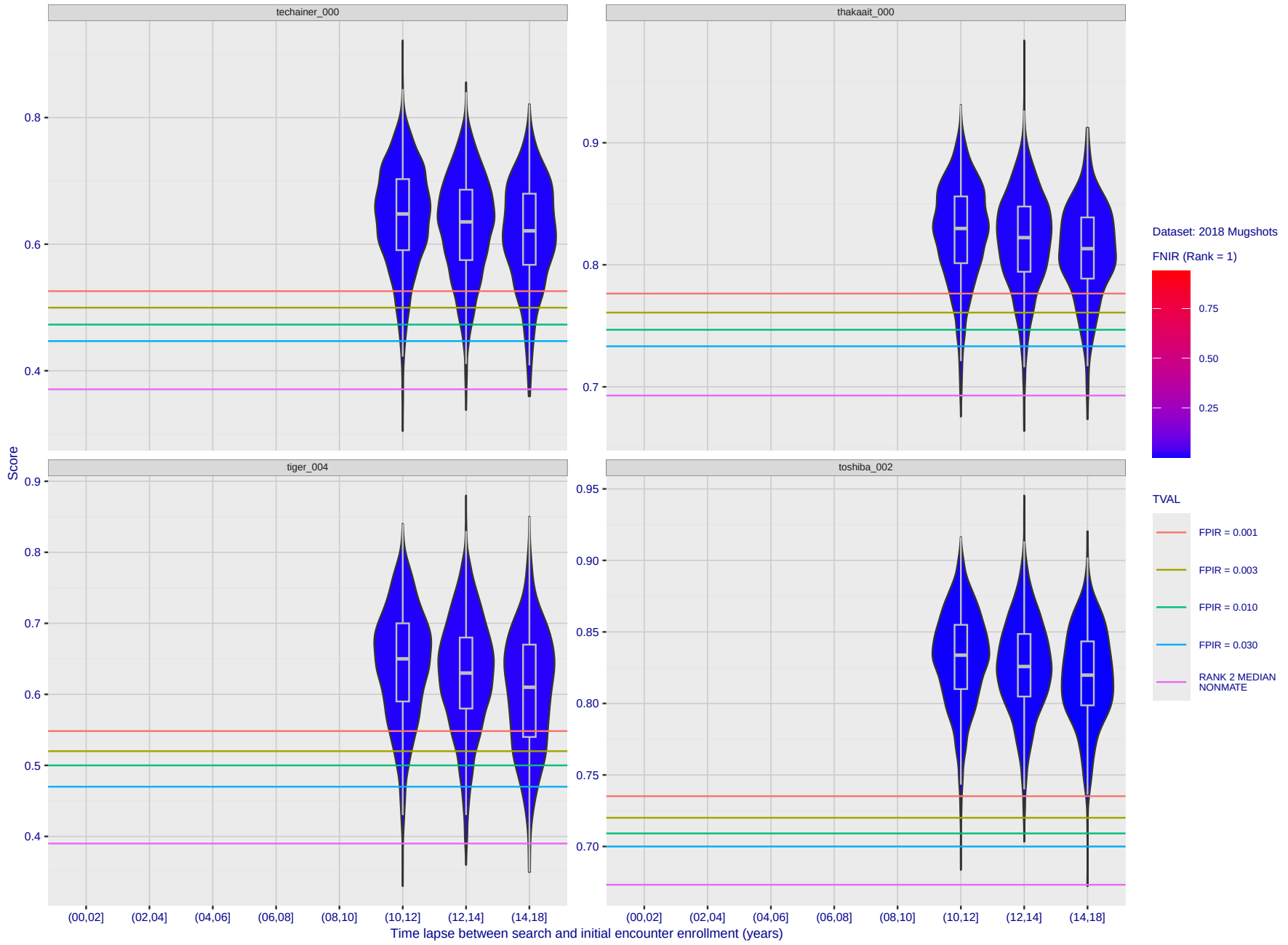


*Figure 334:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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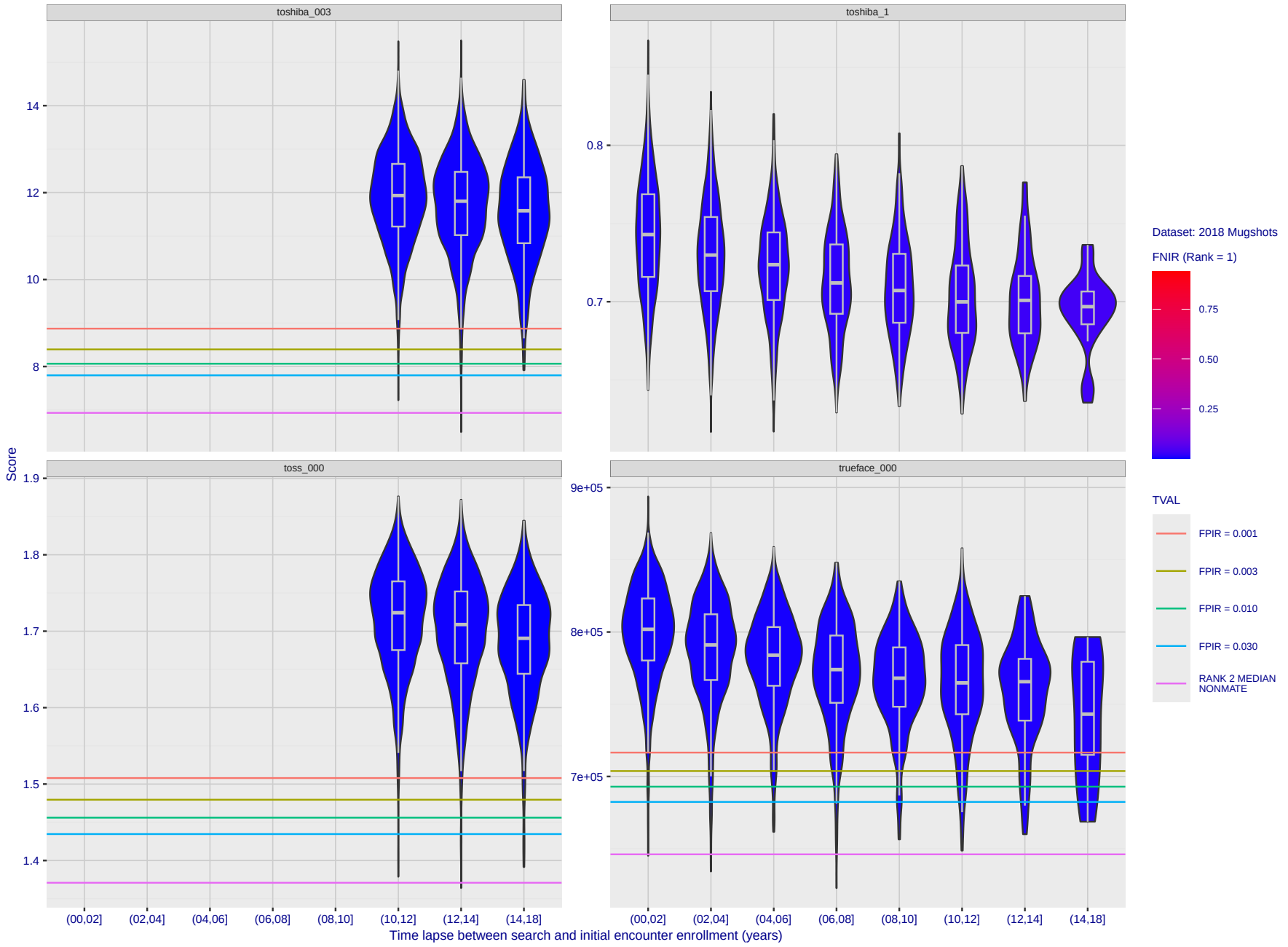


*Figure 335:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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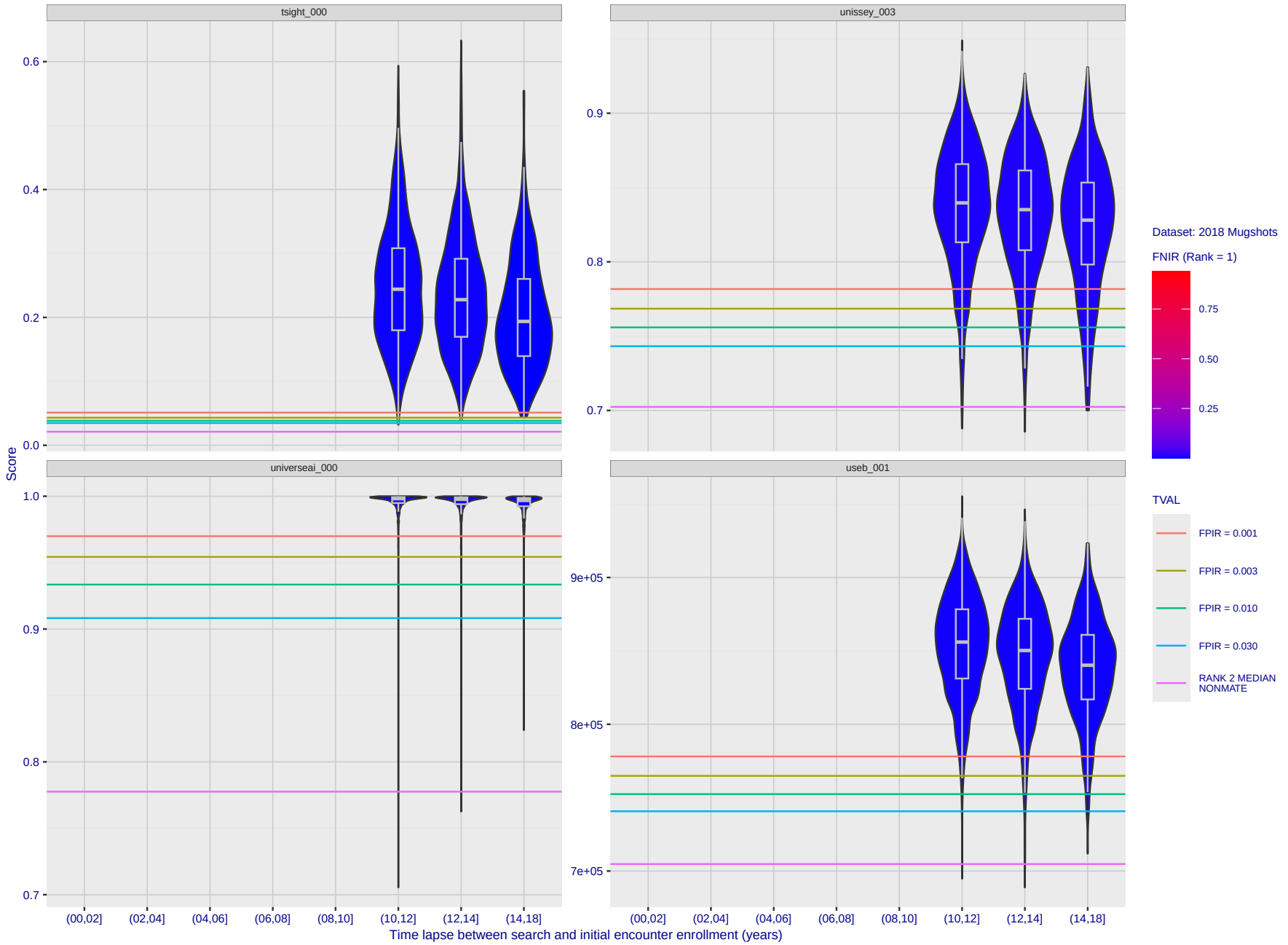


*Figure 336:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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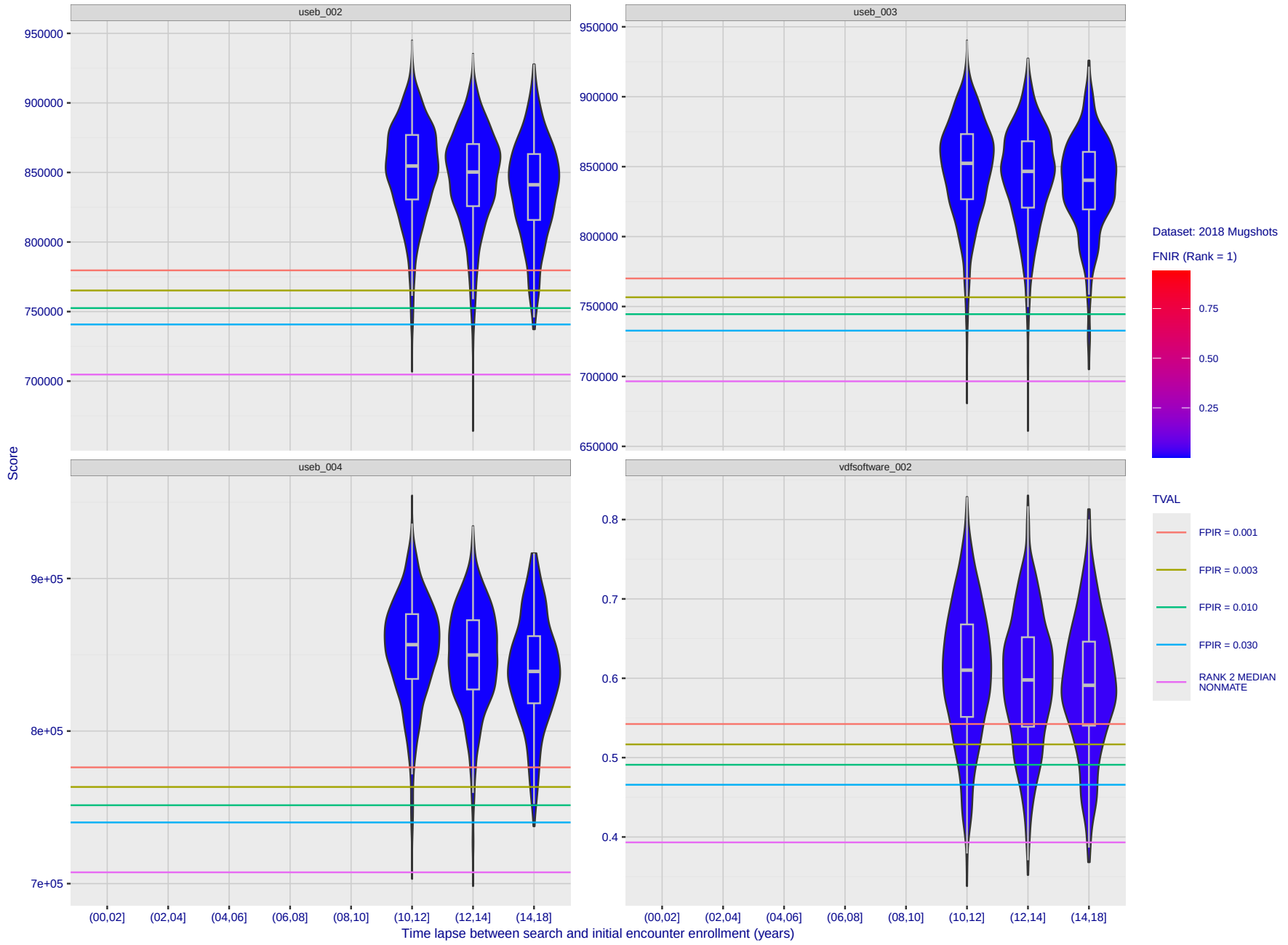


*Figure 337:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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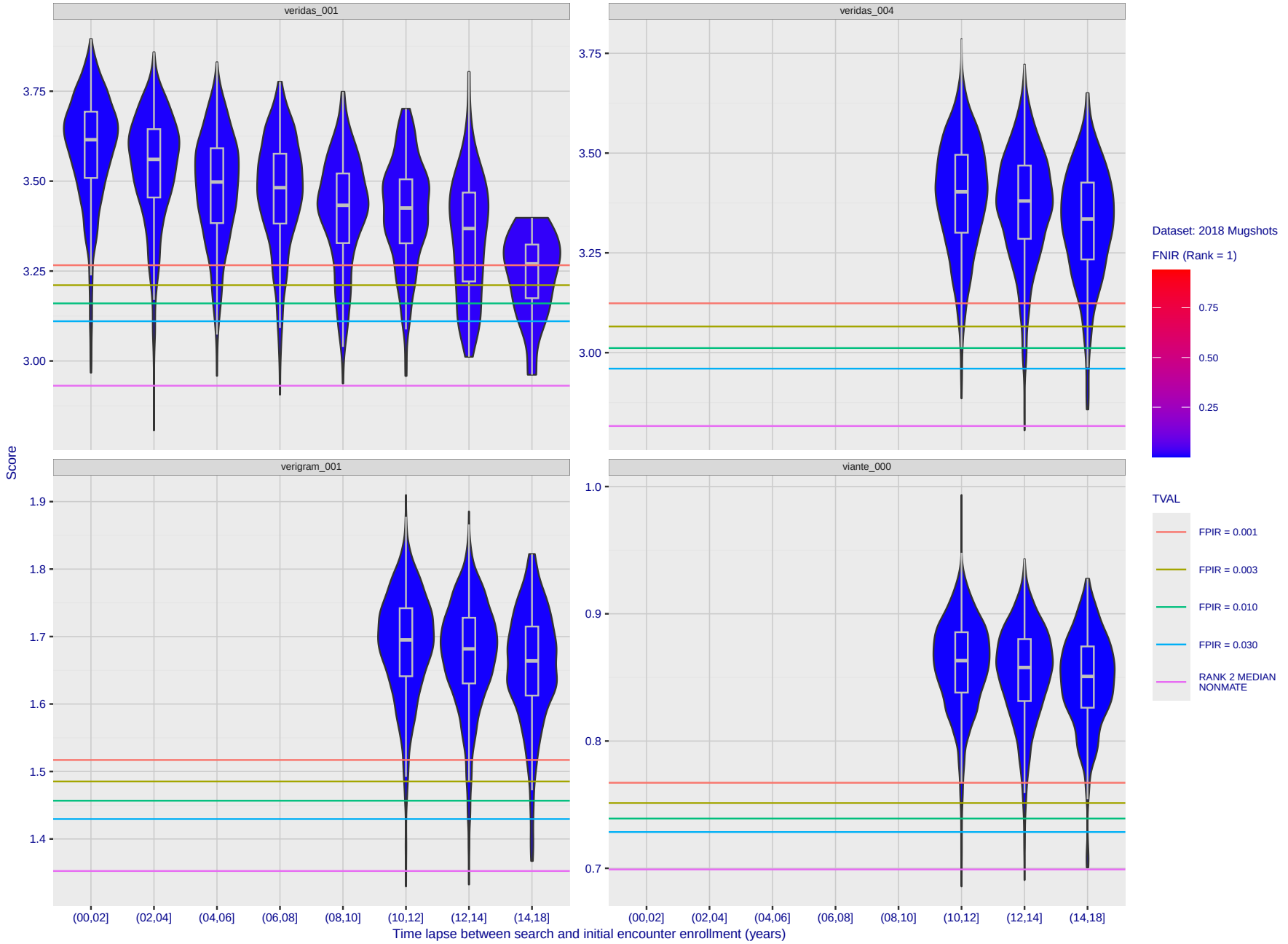


*Figure 338:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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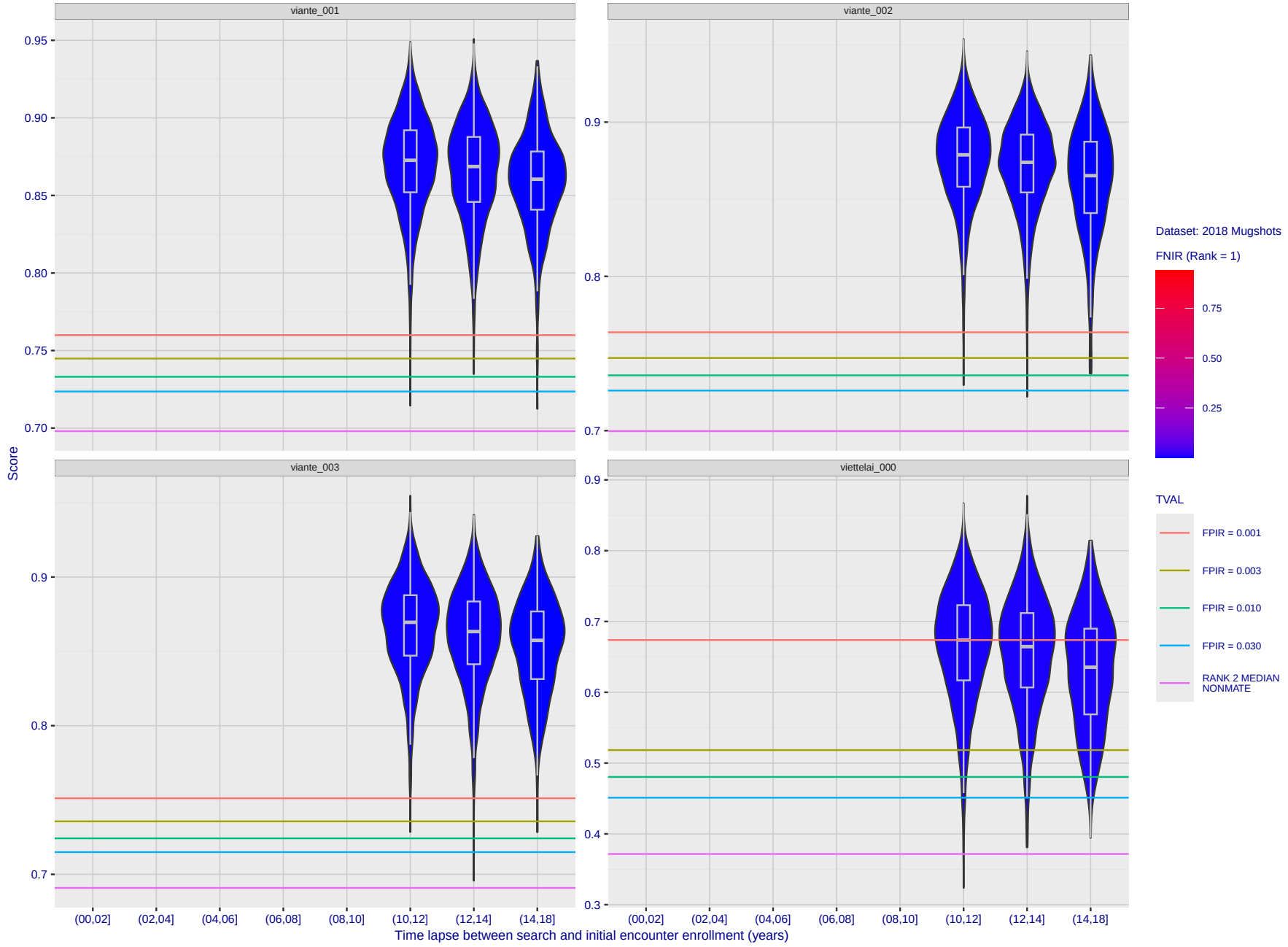


*Figure 339:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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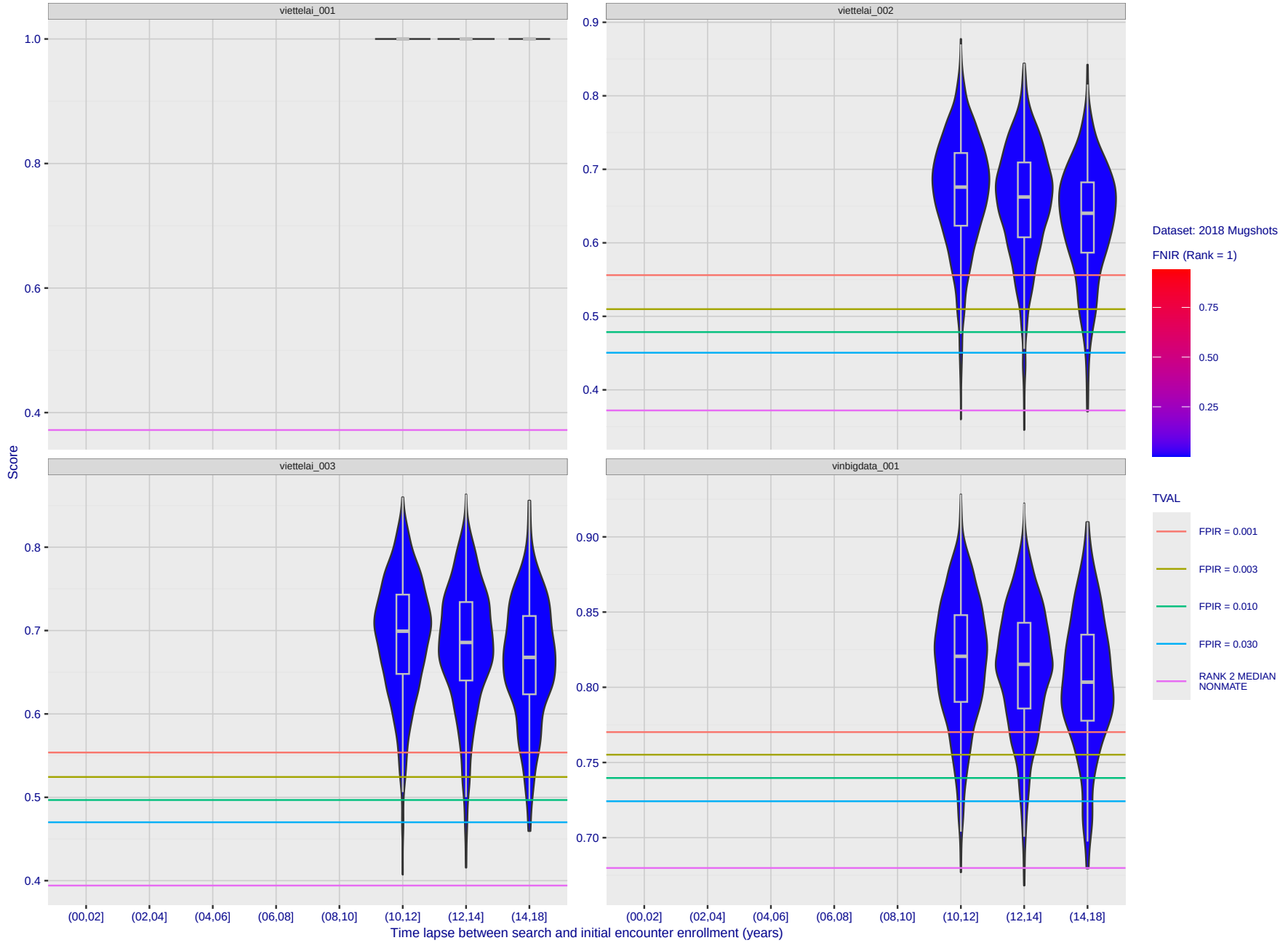


*Figure 340:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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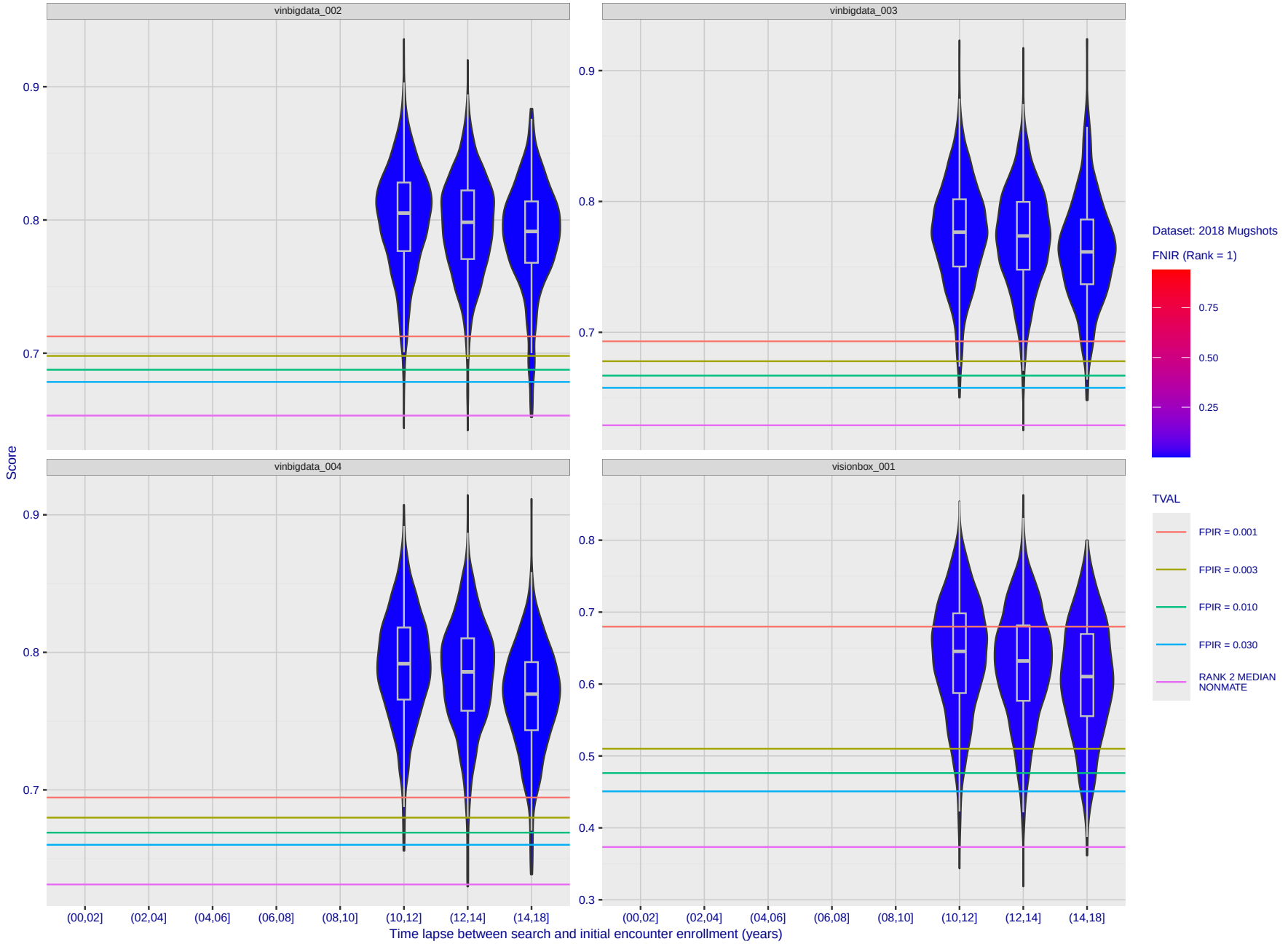


*Figure 341:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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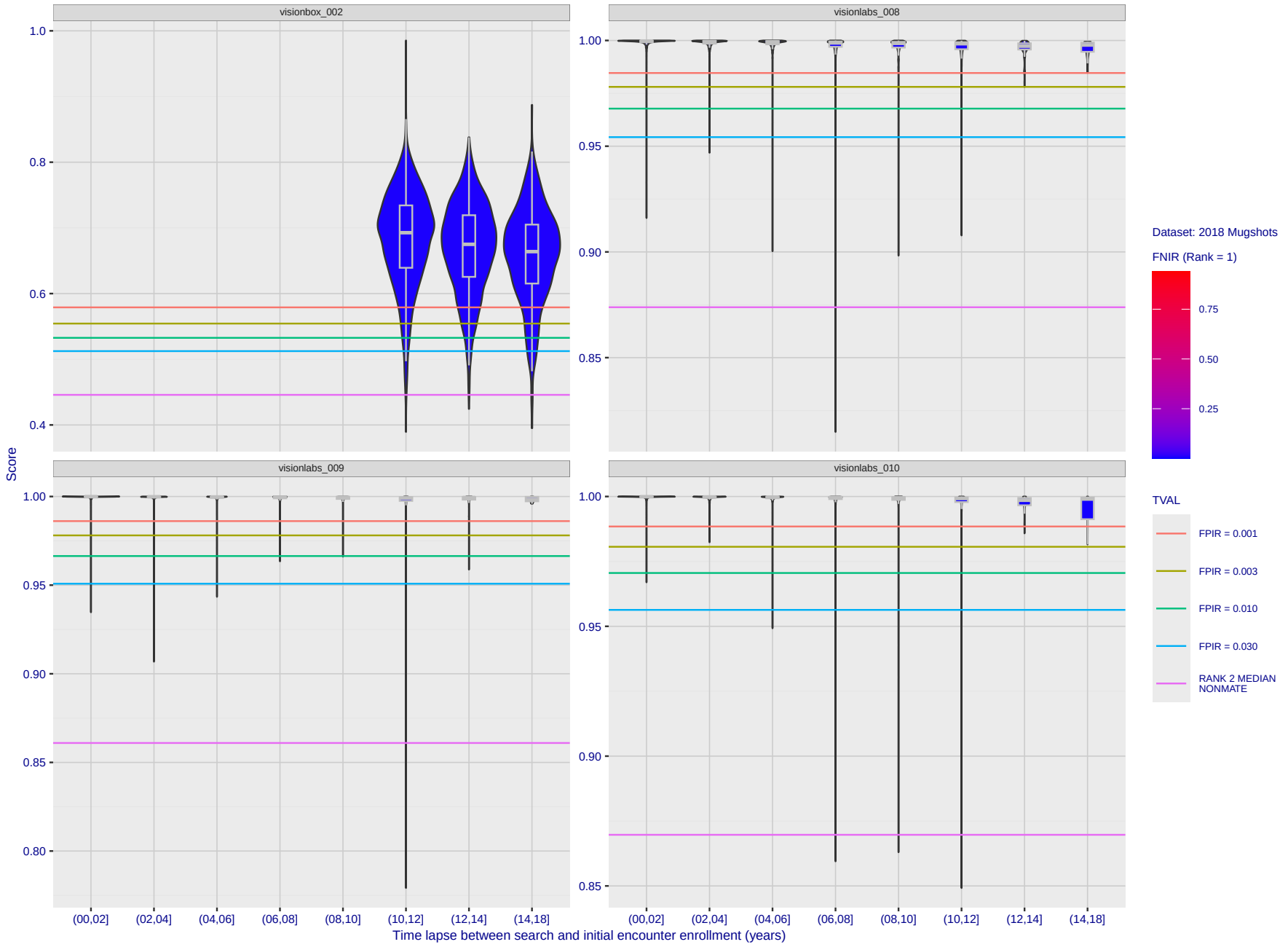


*Figure 342:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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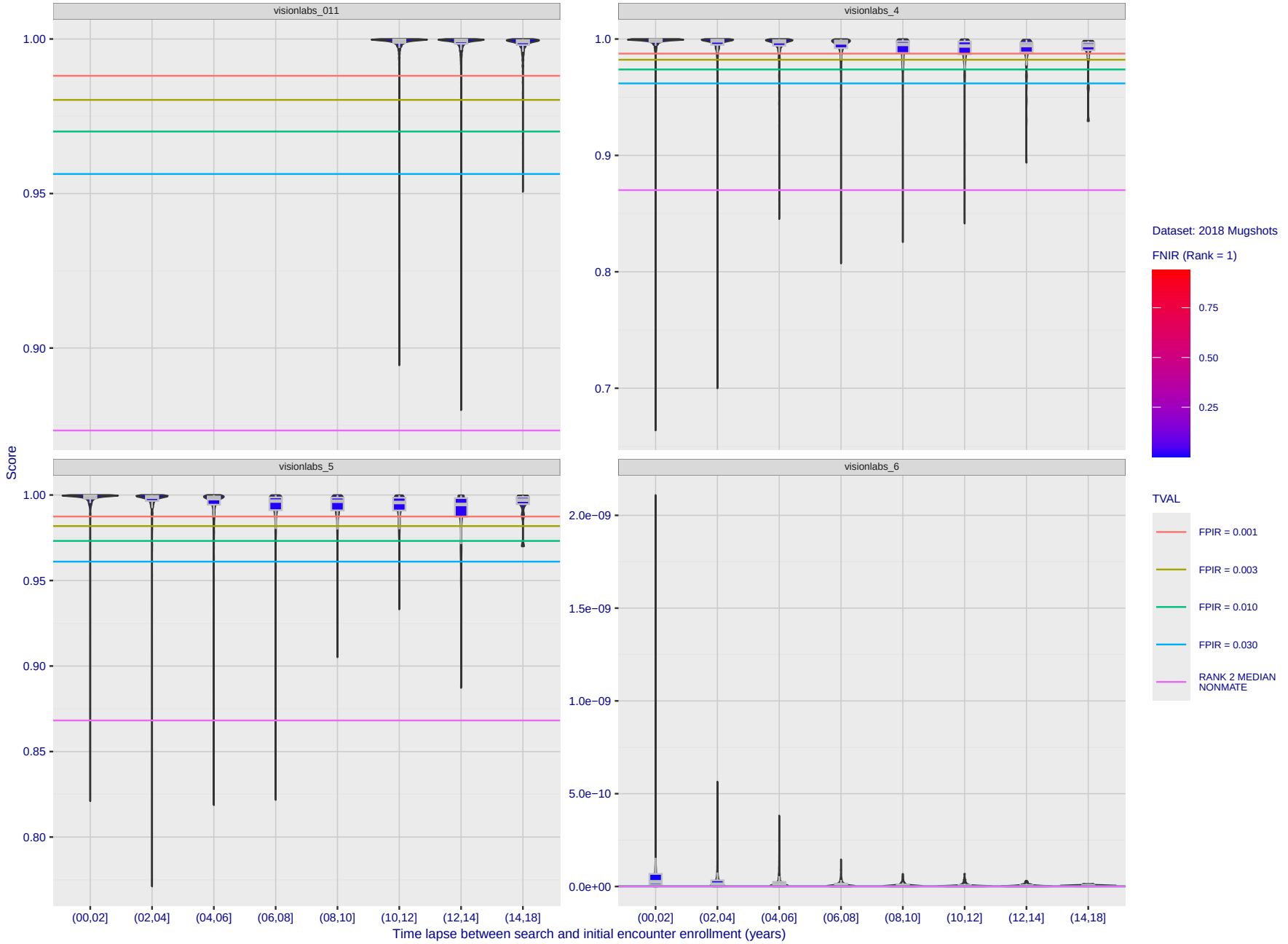


*Figure 343:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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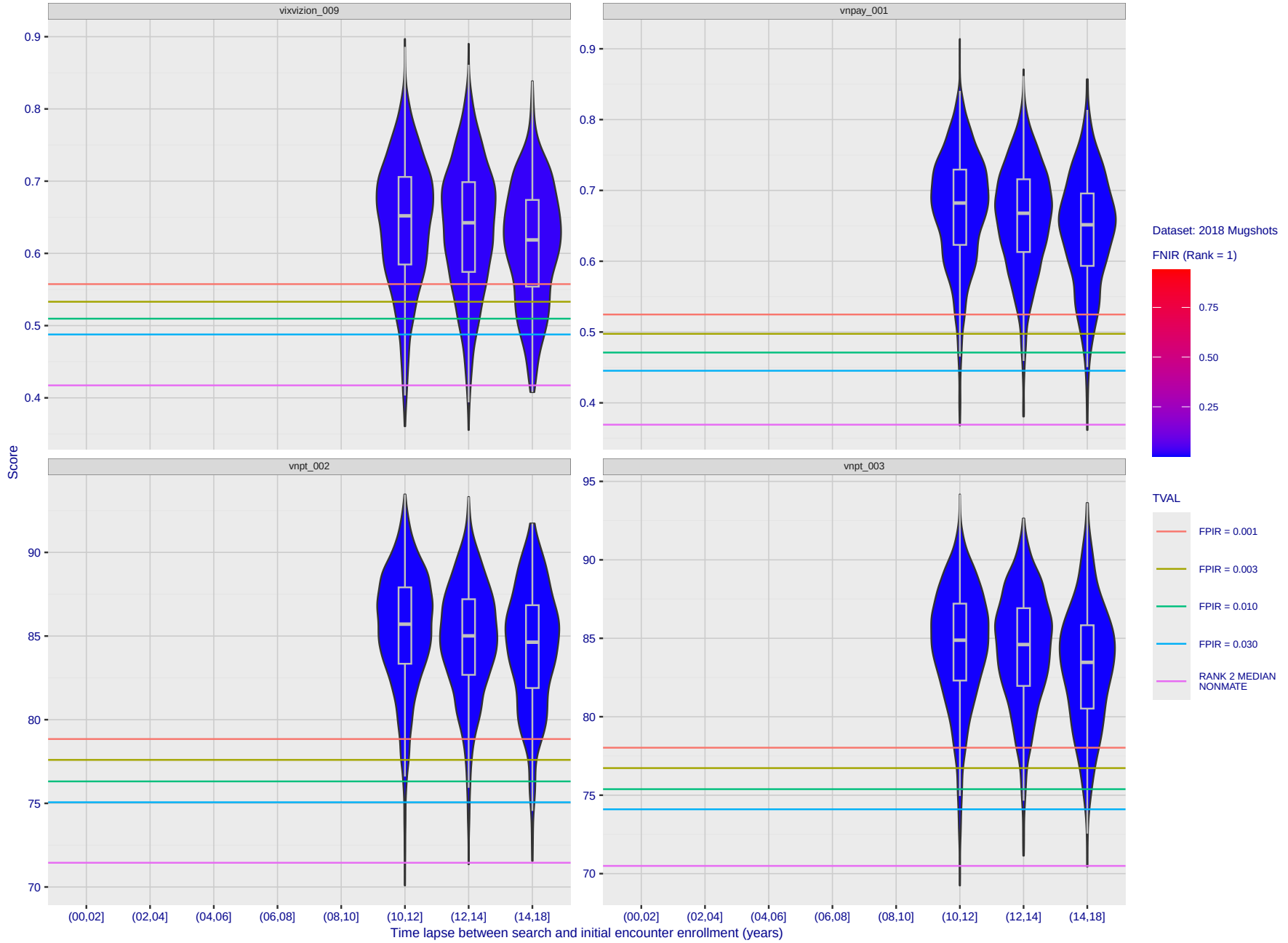


*Figure 344:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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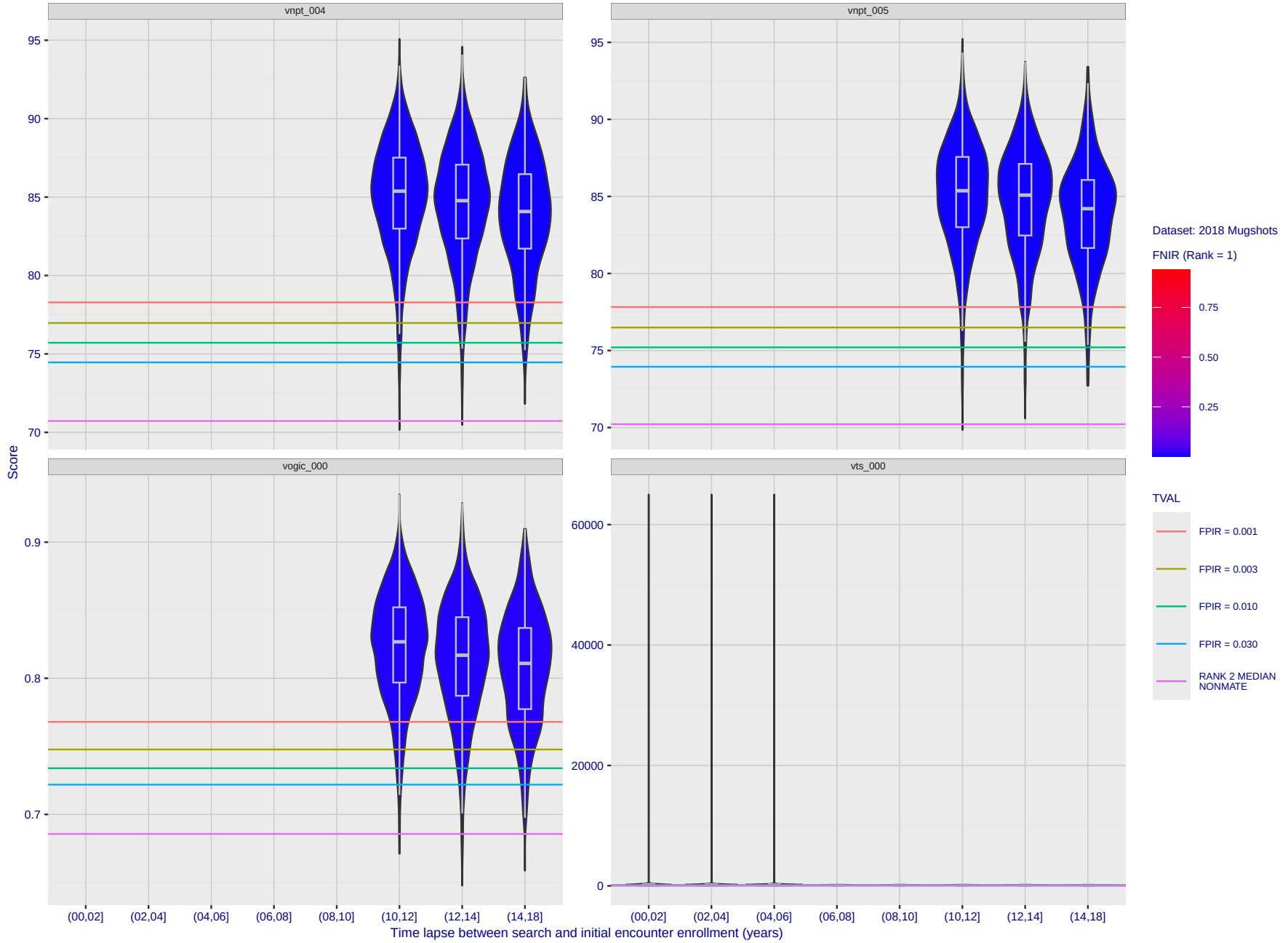


*Figure 345:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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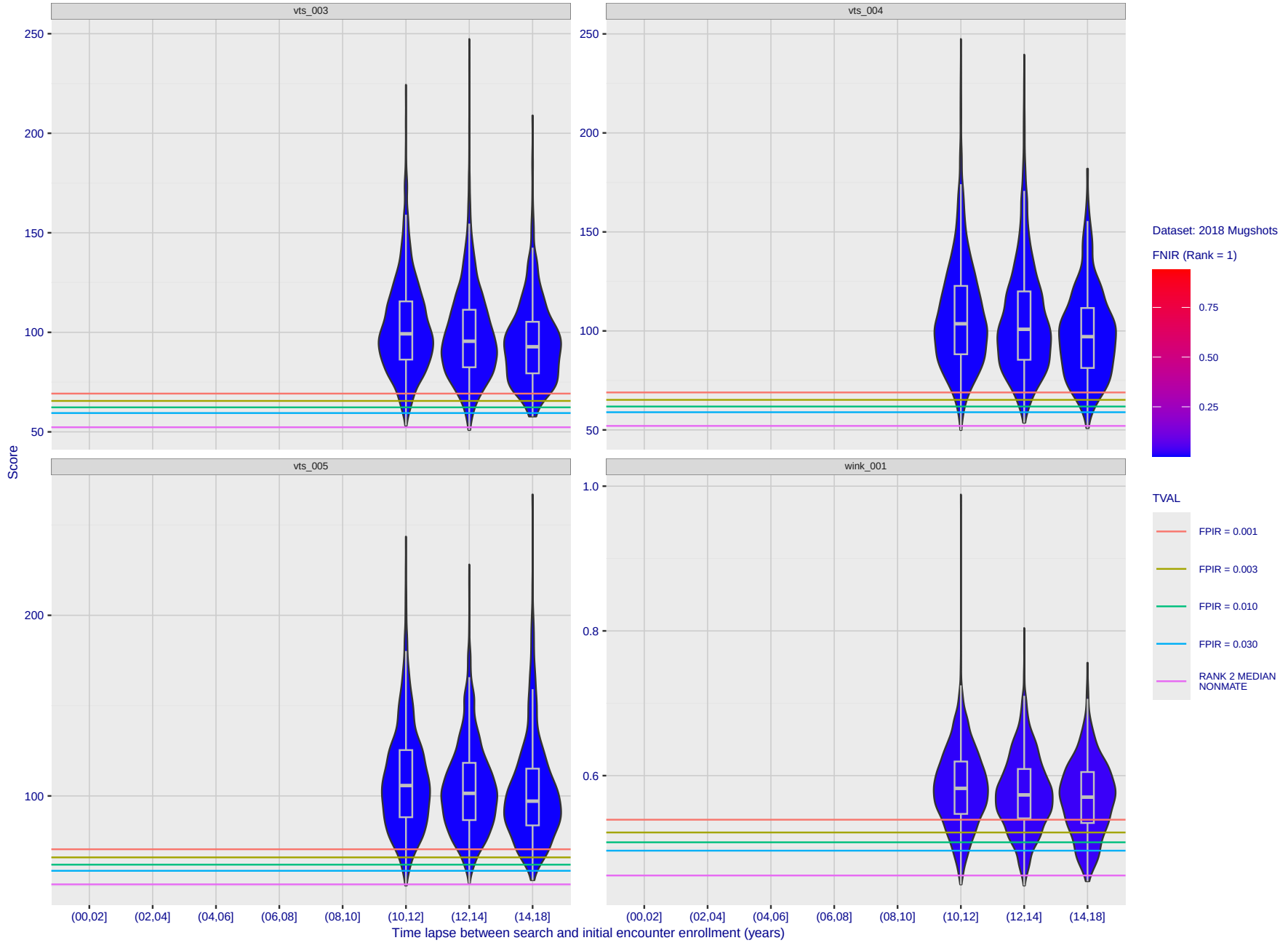


*Figure 346:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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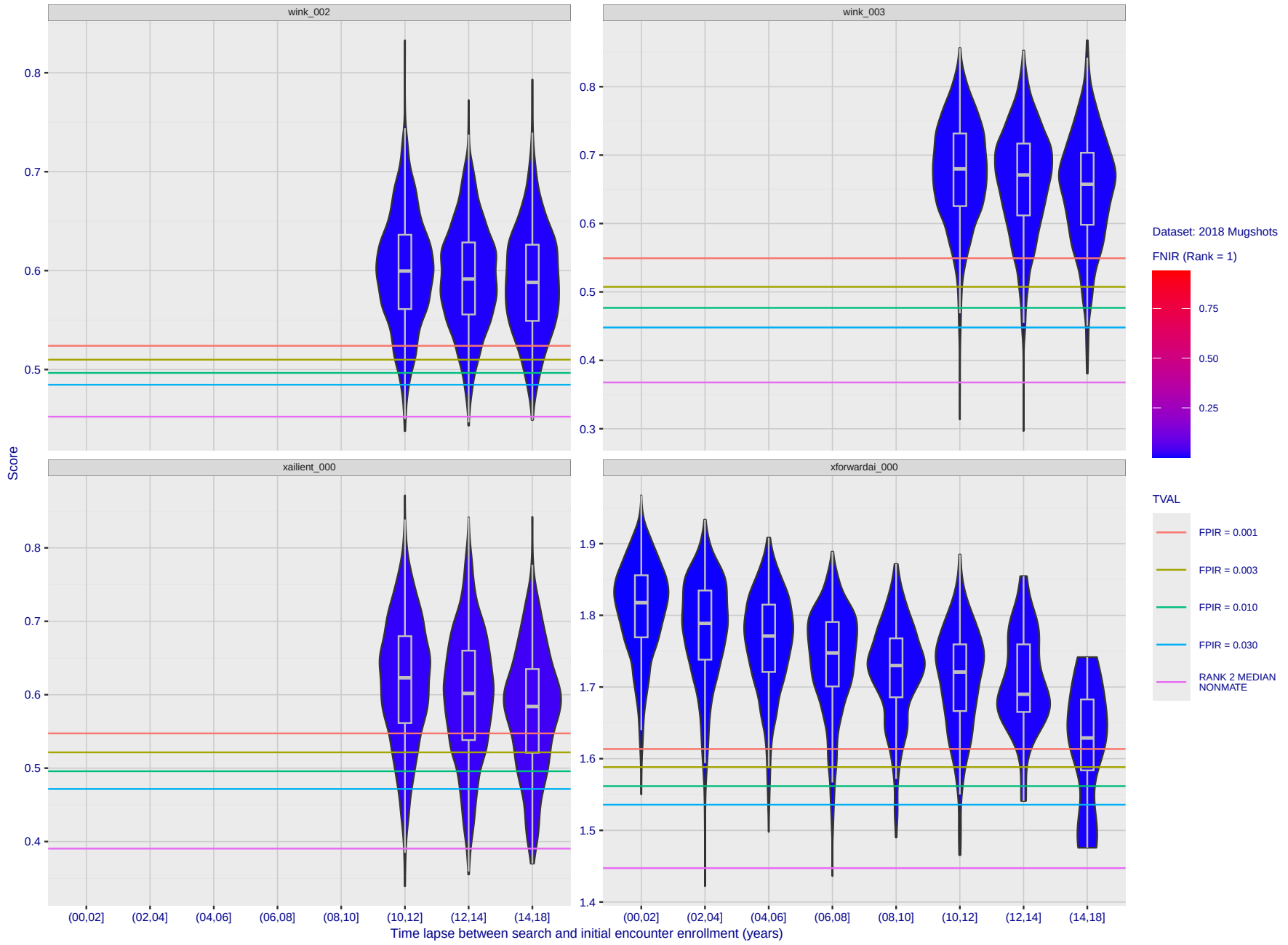


*Figure 347:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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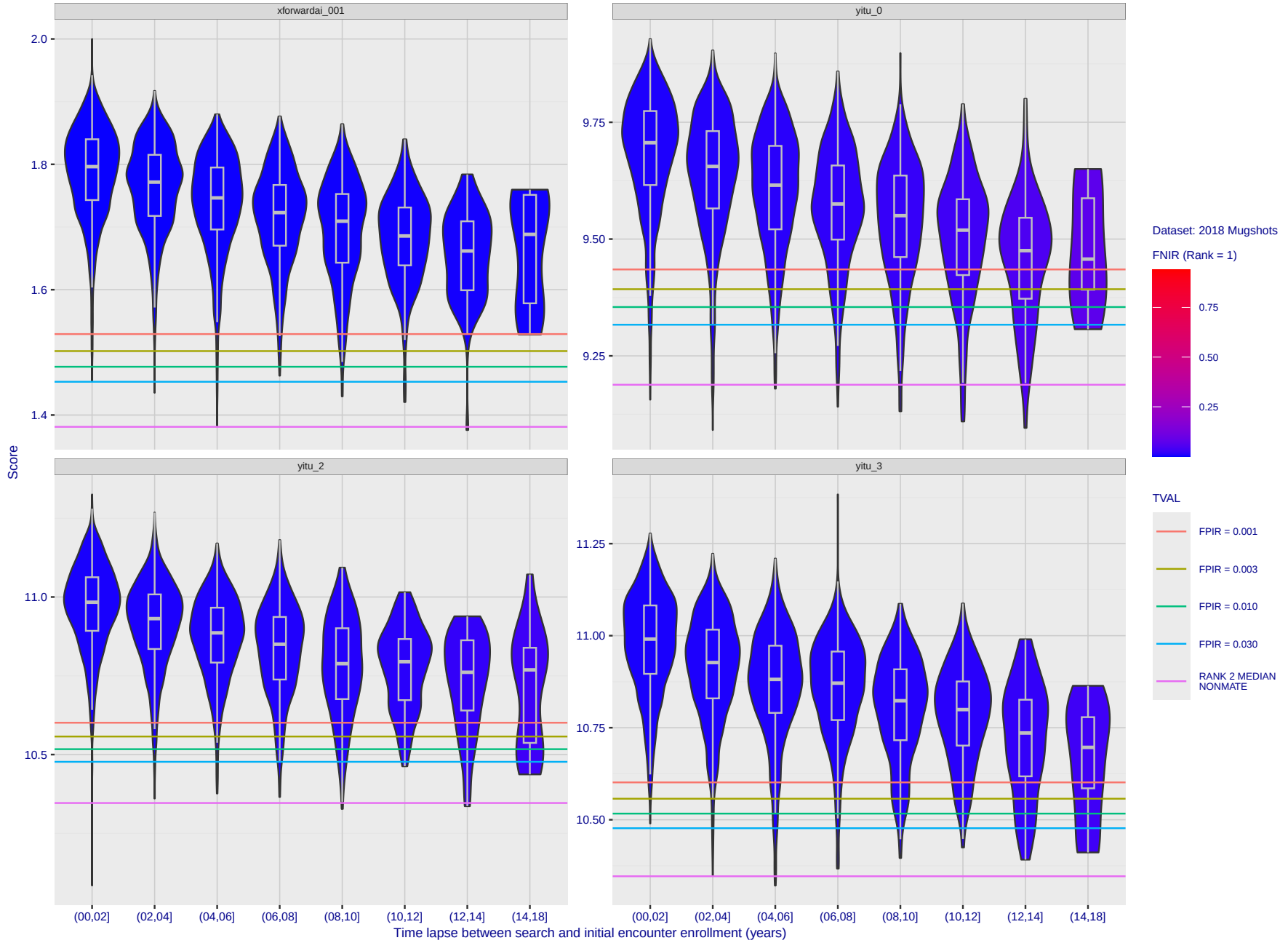


*Figure 348:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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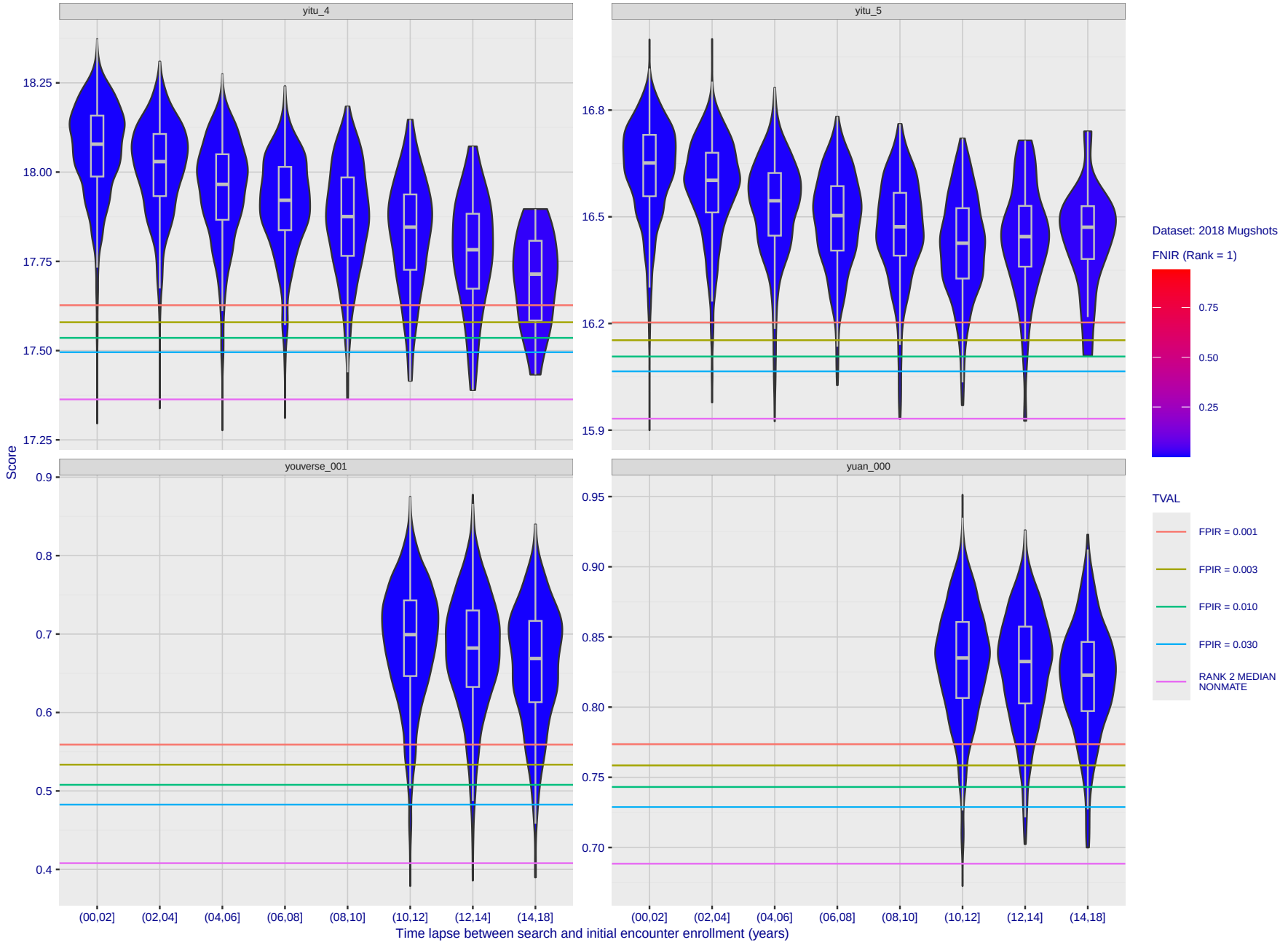


*Figure 349:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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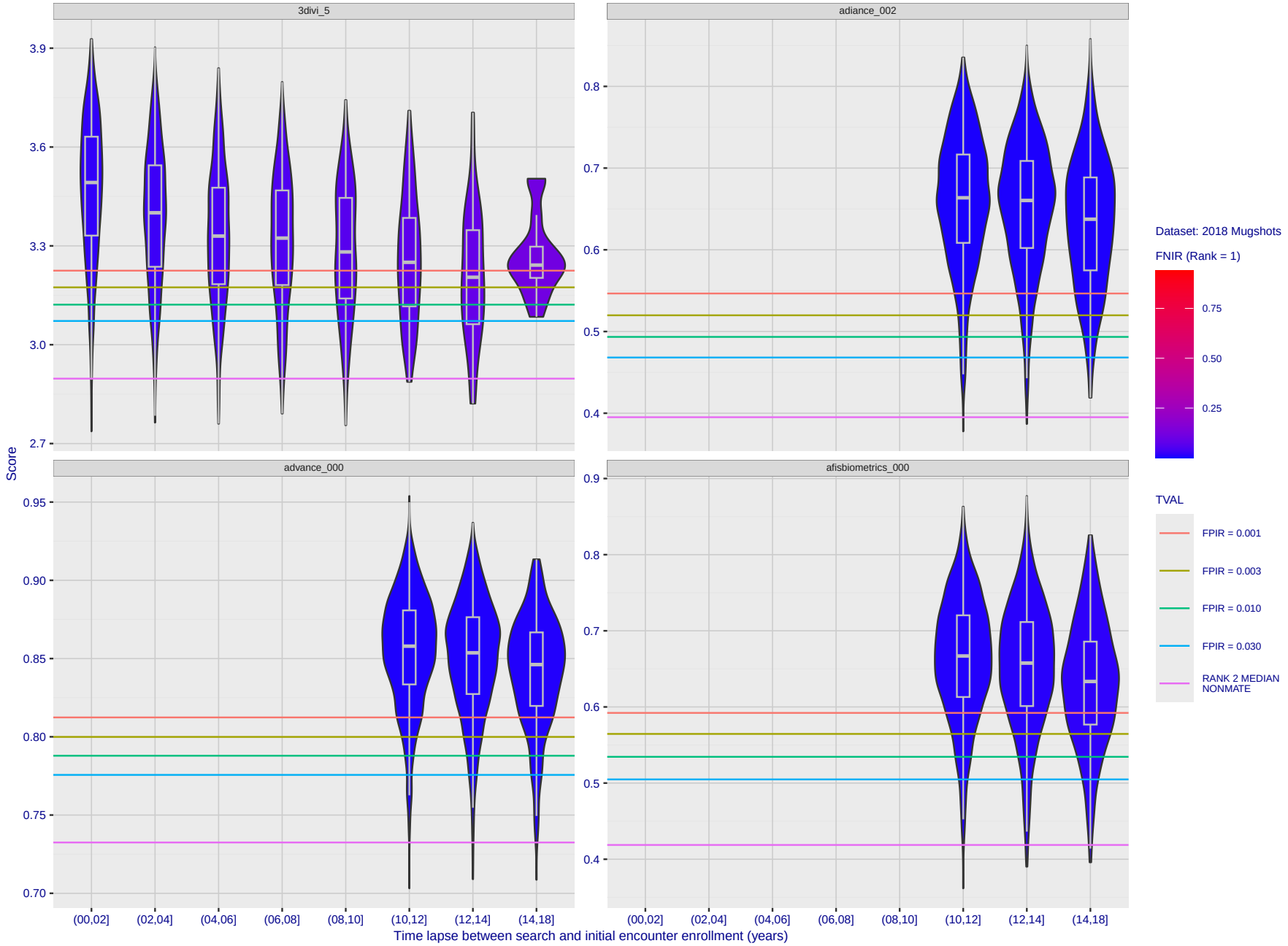


*Figure 350:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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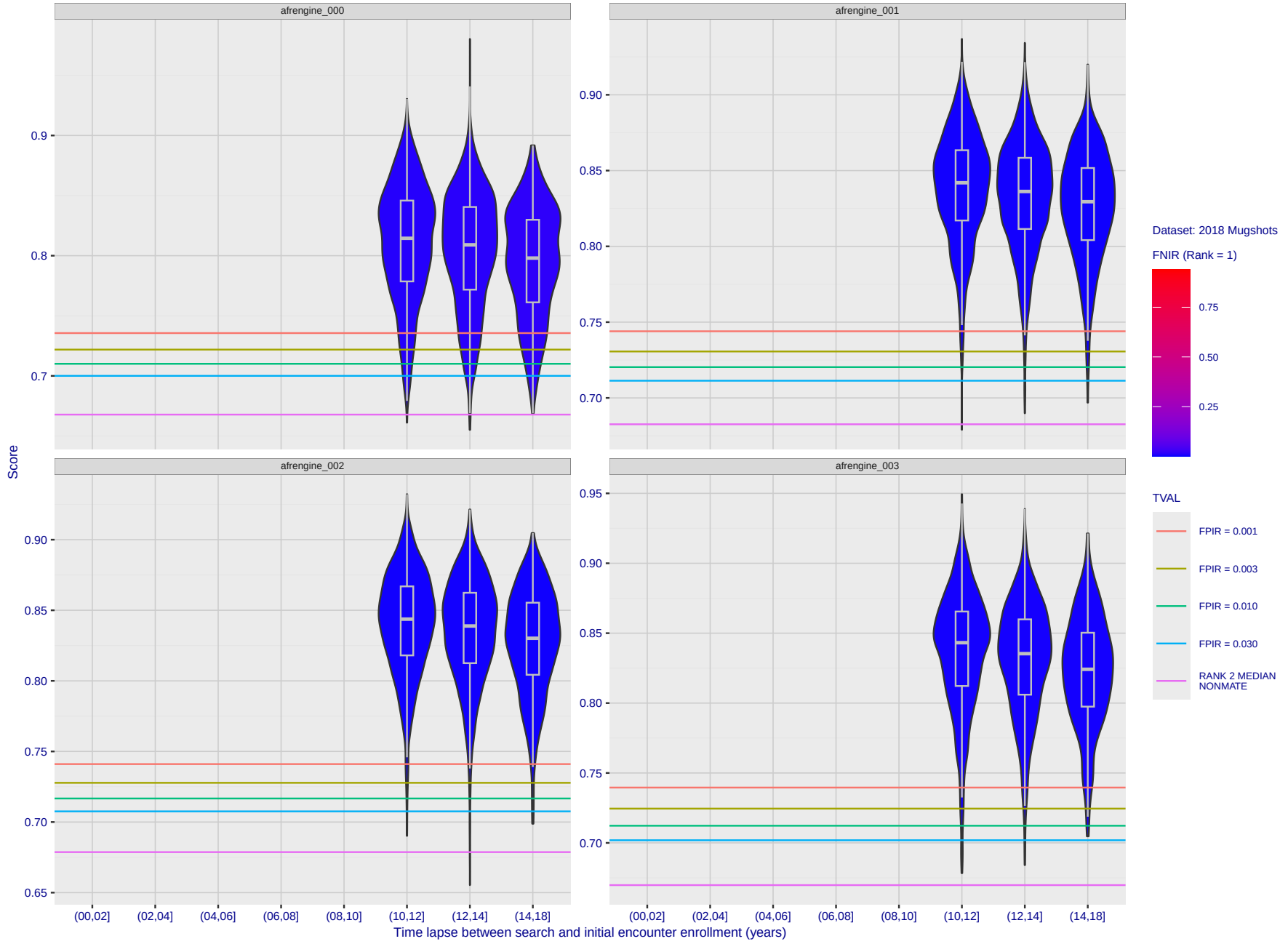


*Figure 351:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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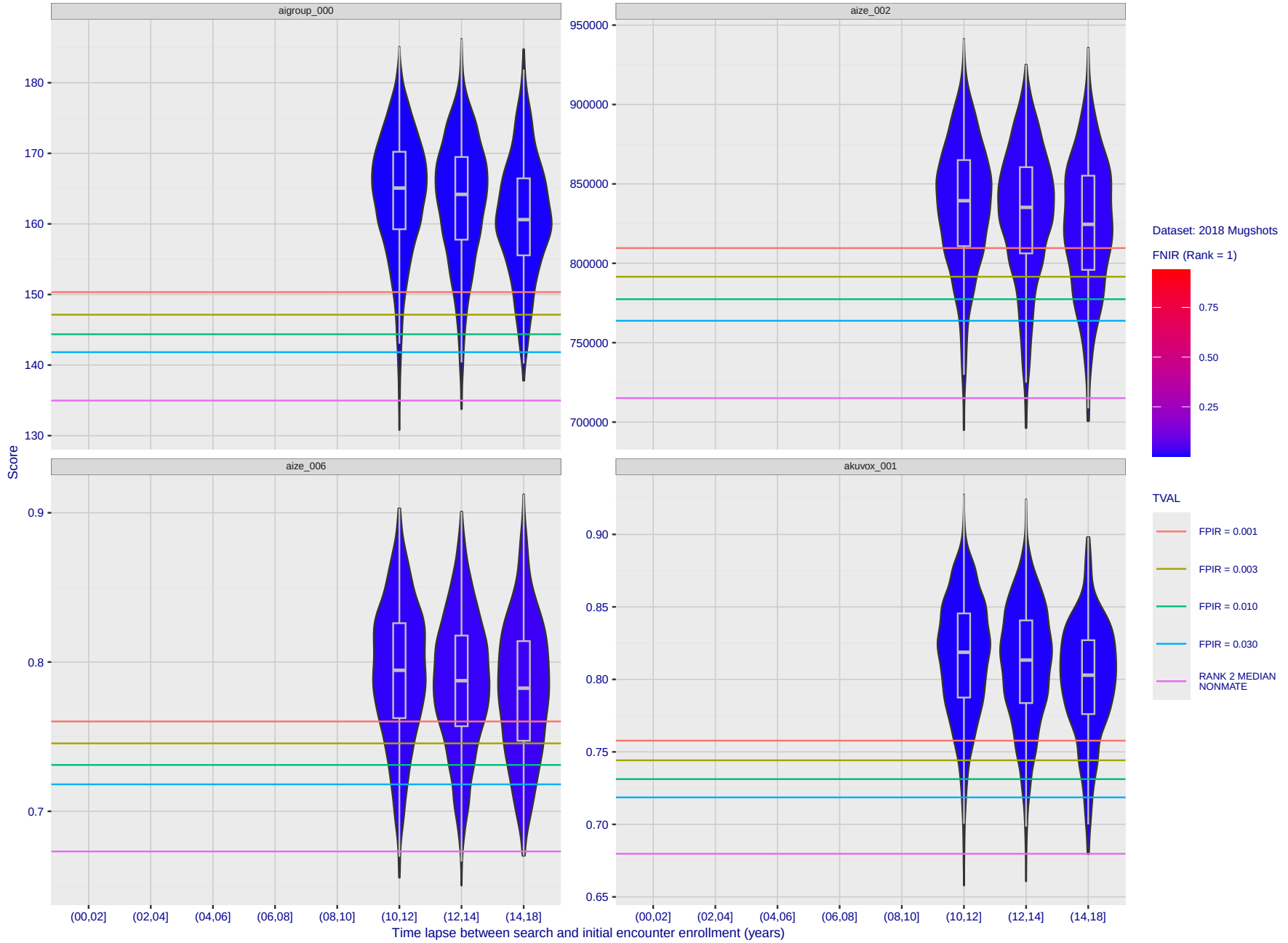


*Figure 352:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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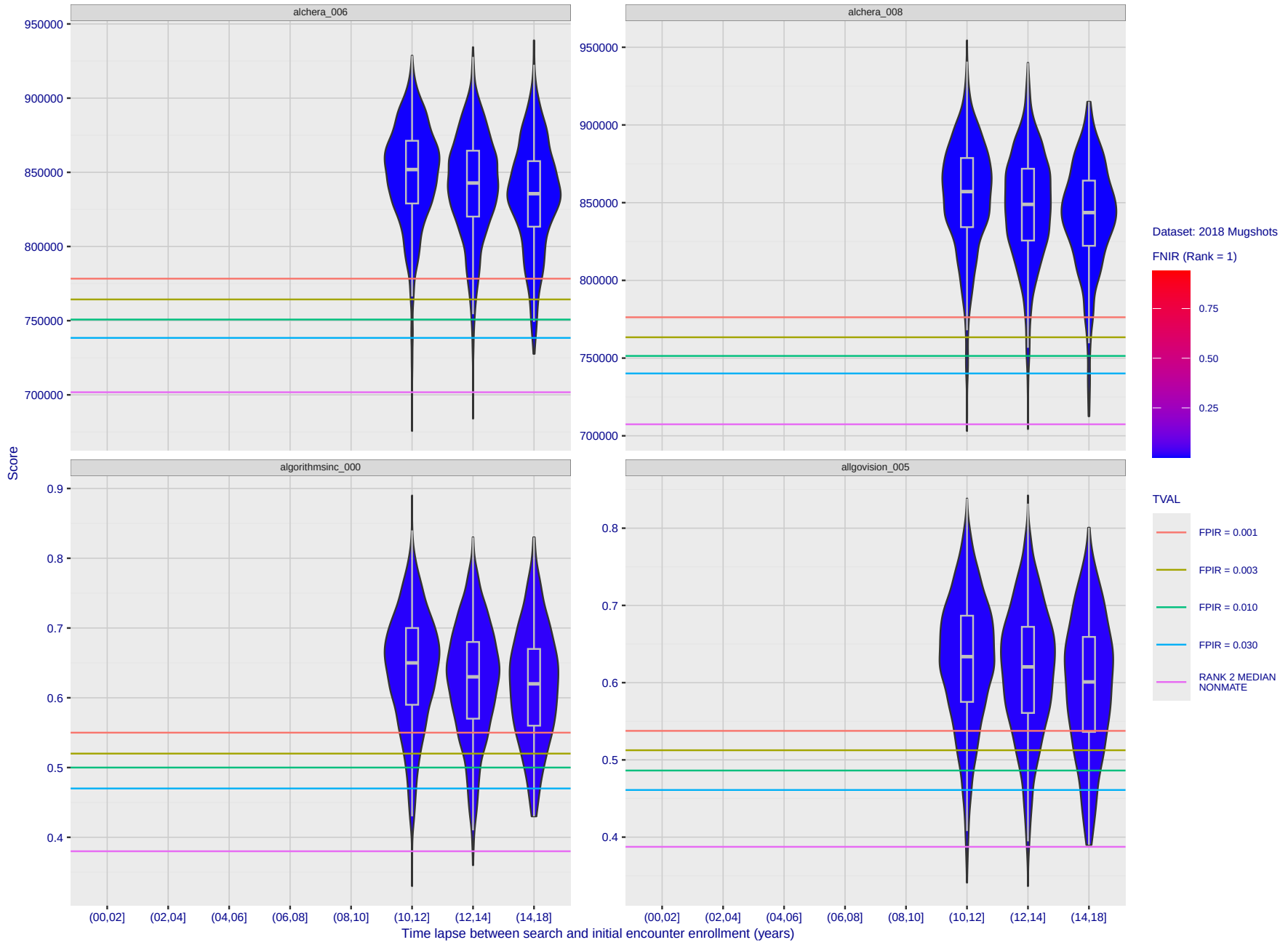


*Figure 353:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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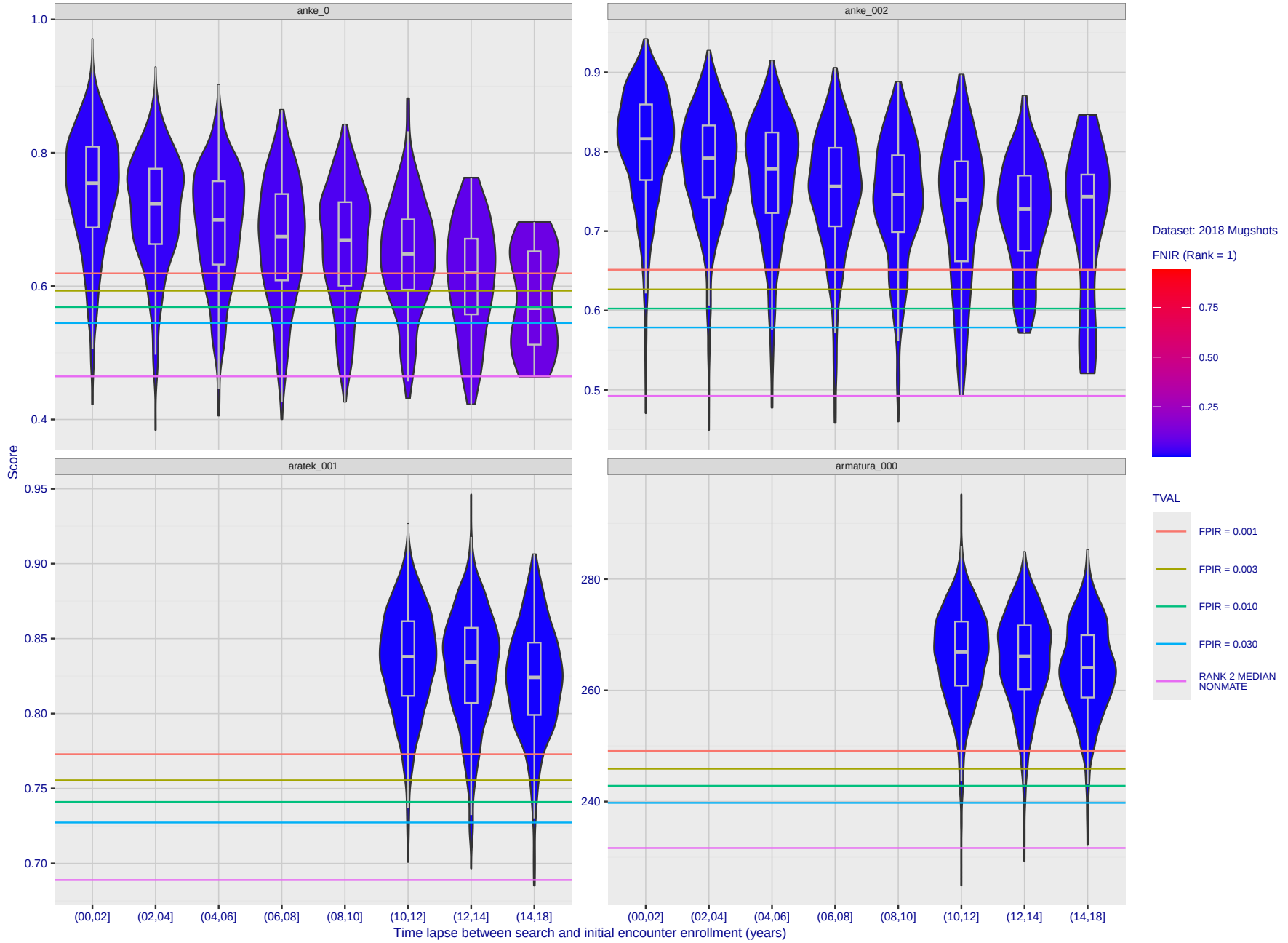


*Figure 354:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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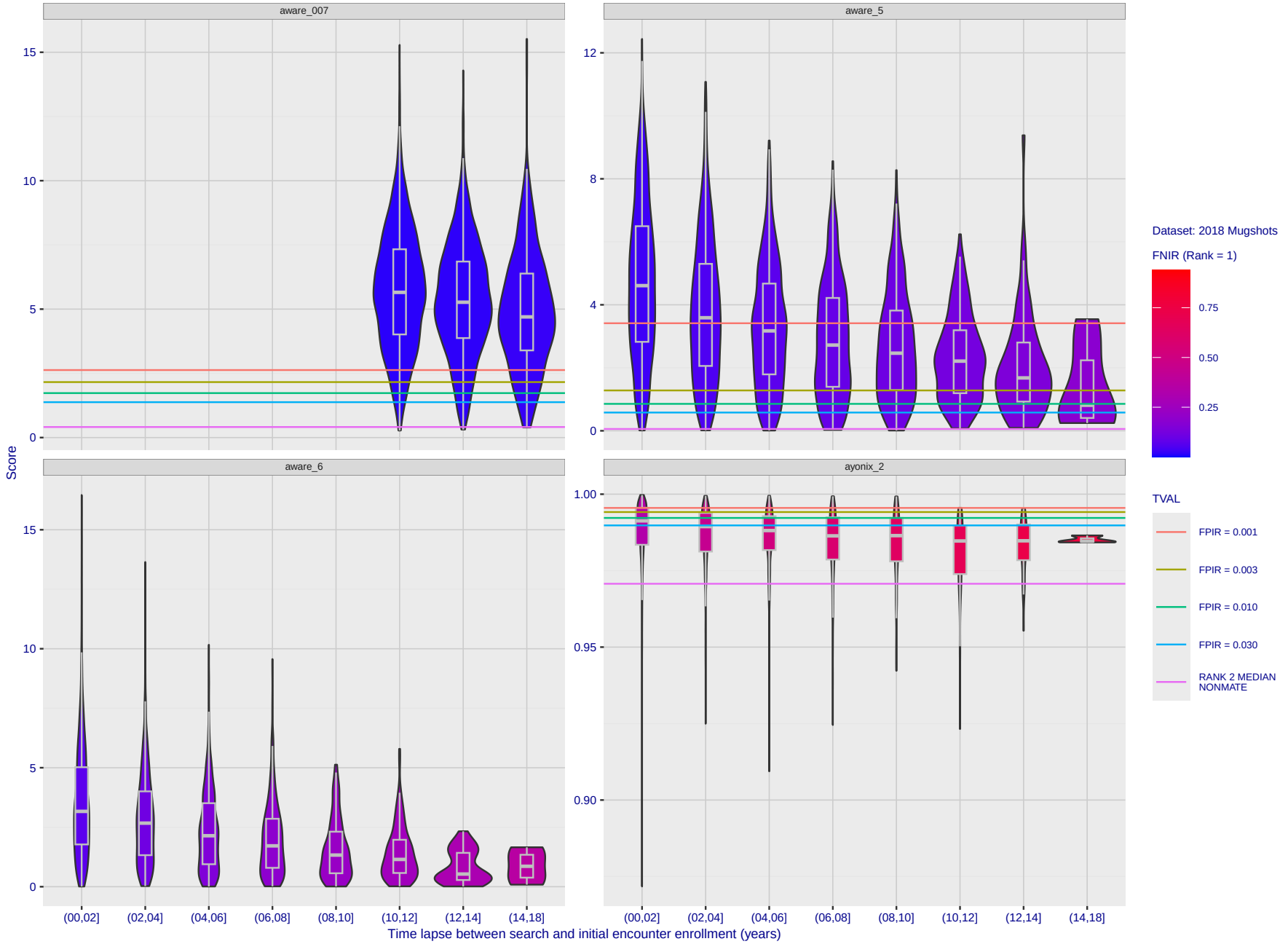


*Figure 355:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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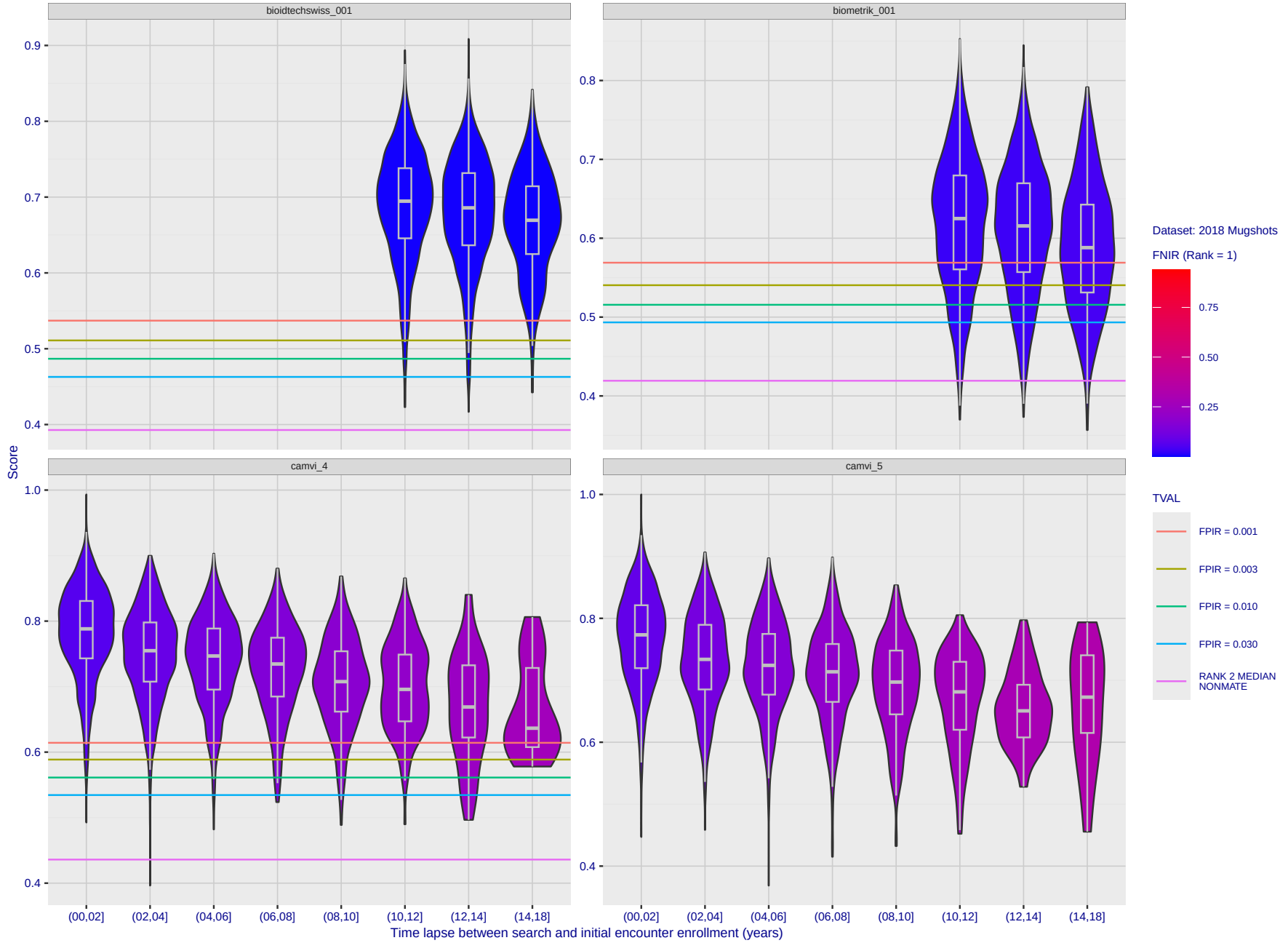


*Figure 356:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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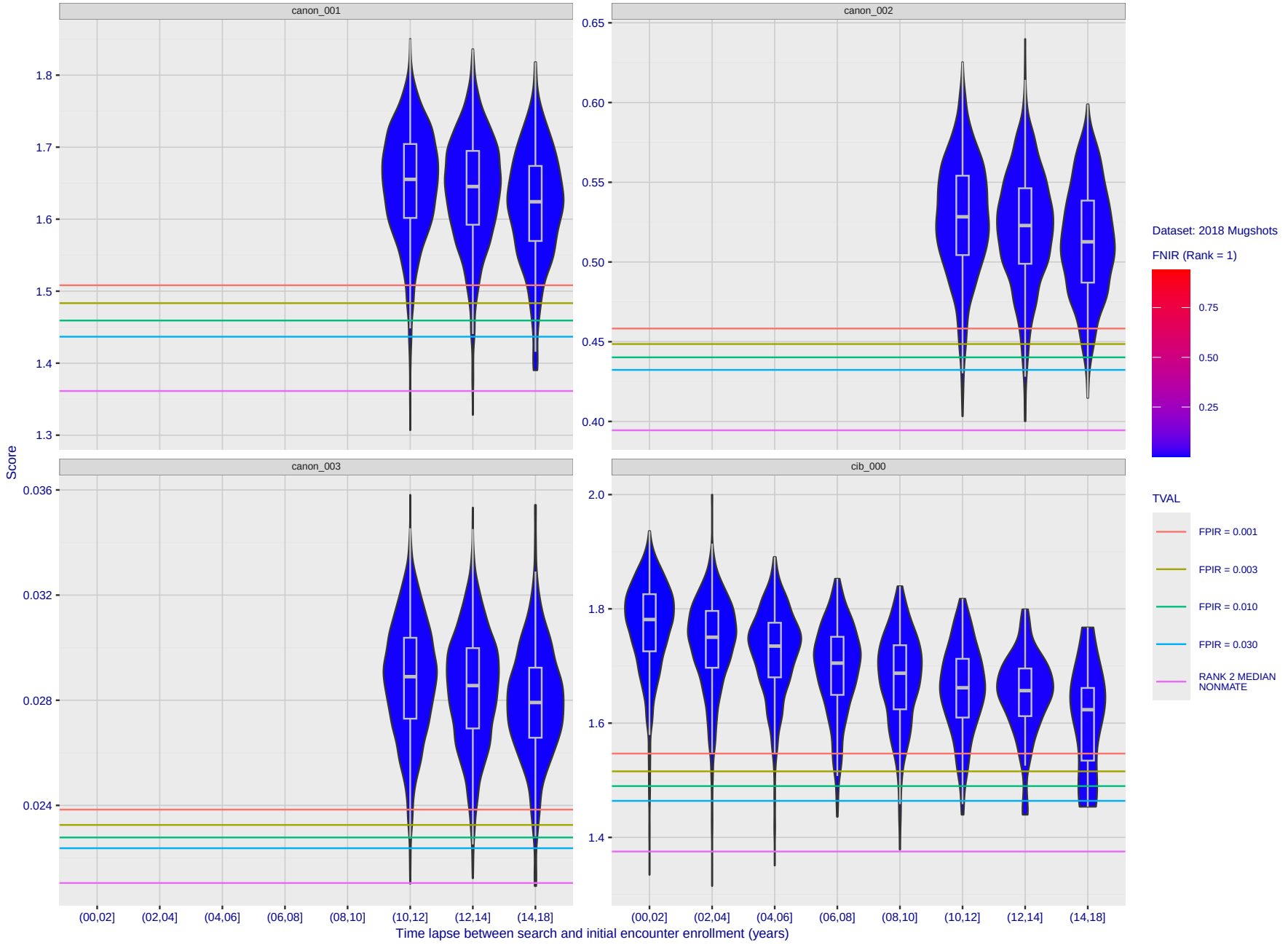


*Figure 357:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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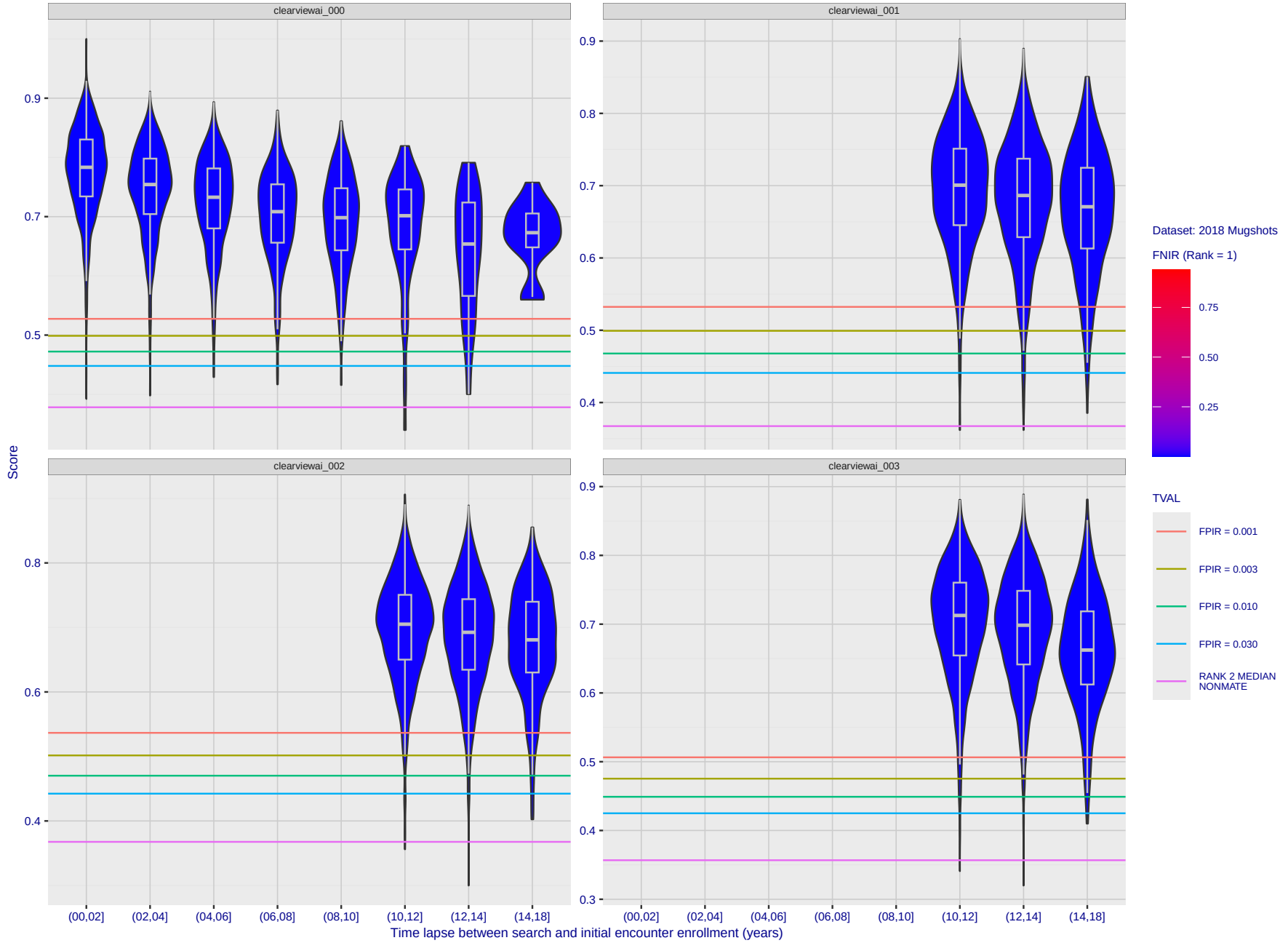


*Figure 358:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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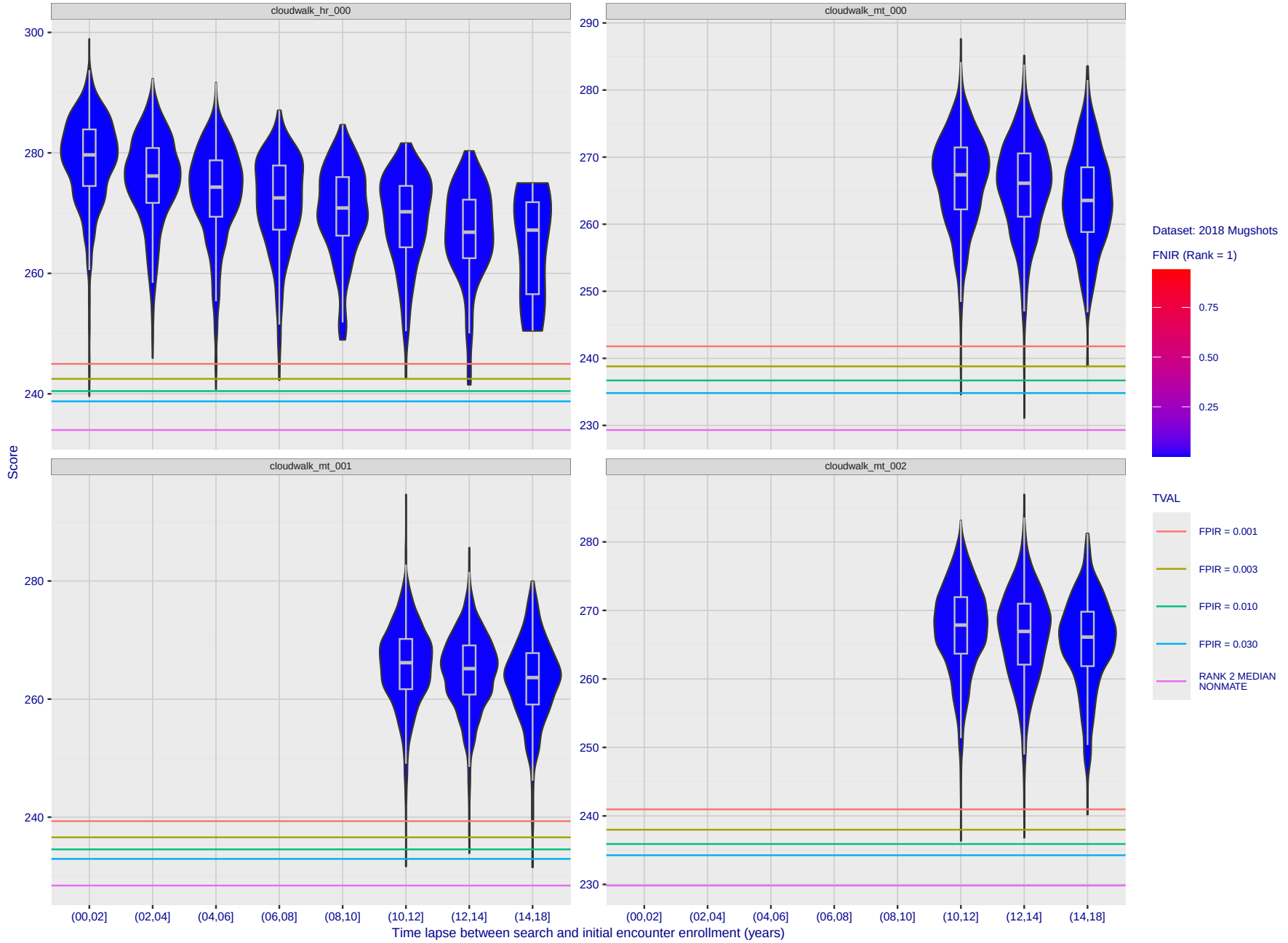


*Figure 359:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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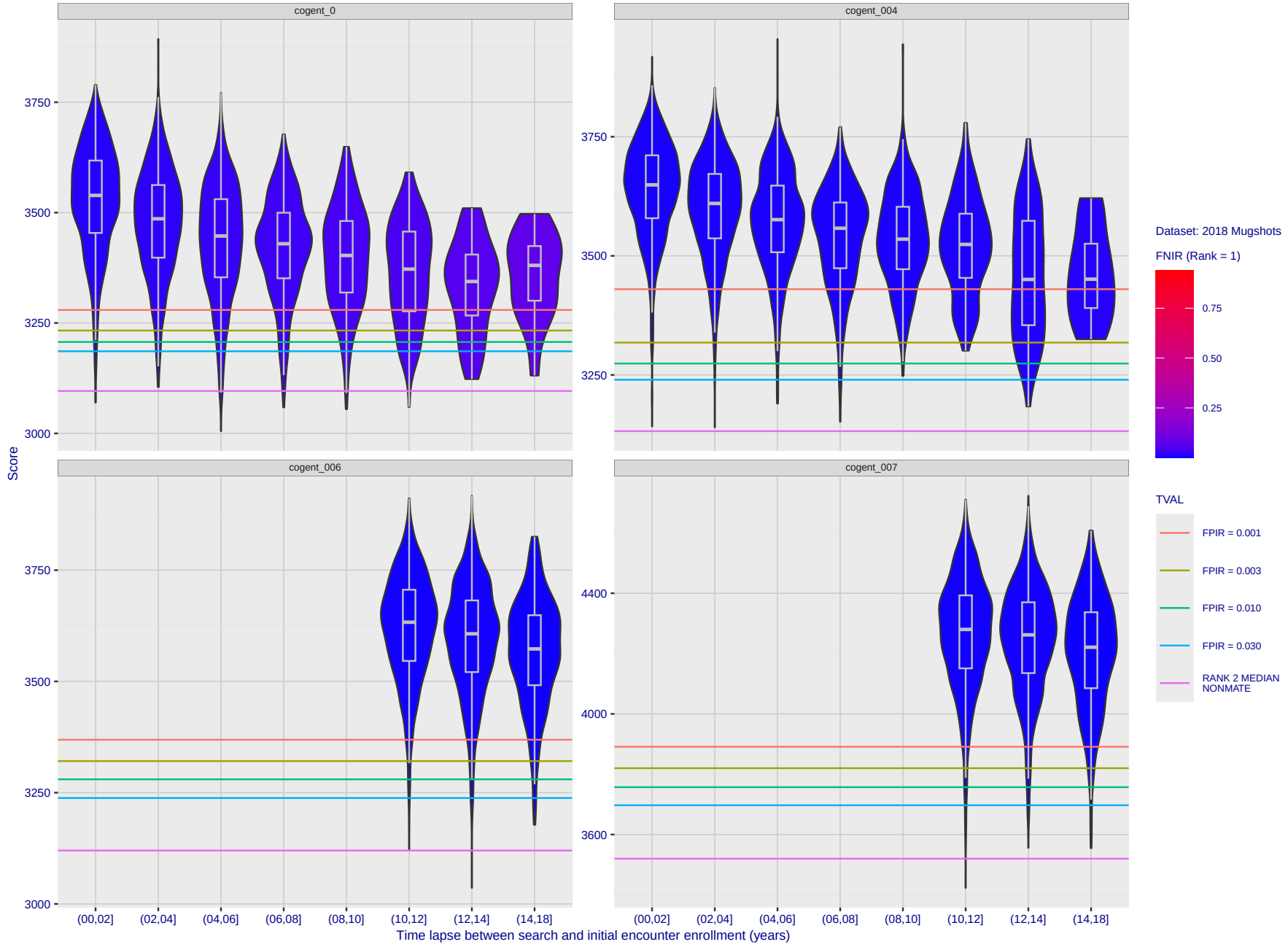


*Figure 360:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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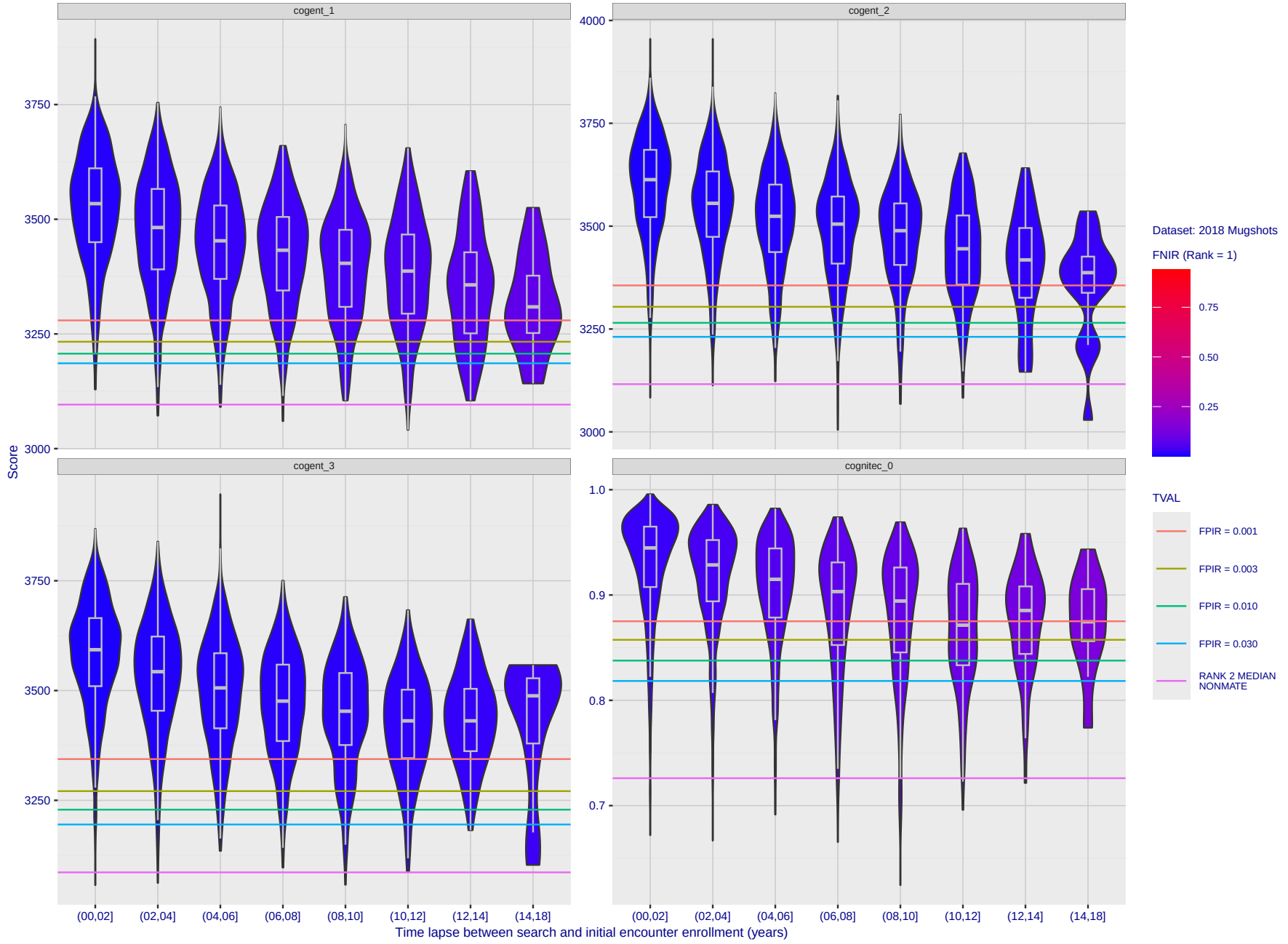


*Figure 361:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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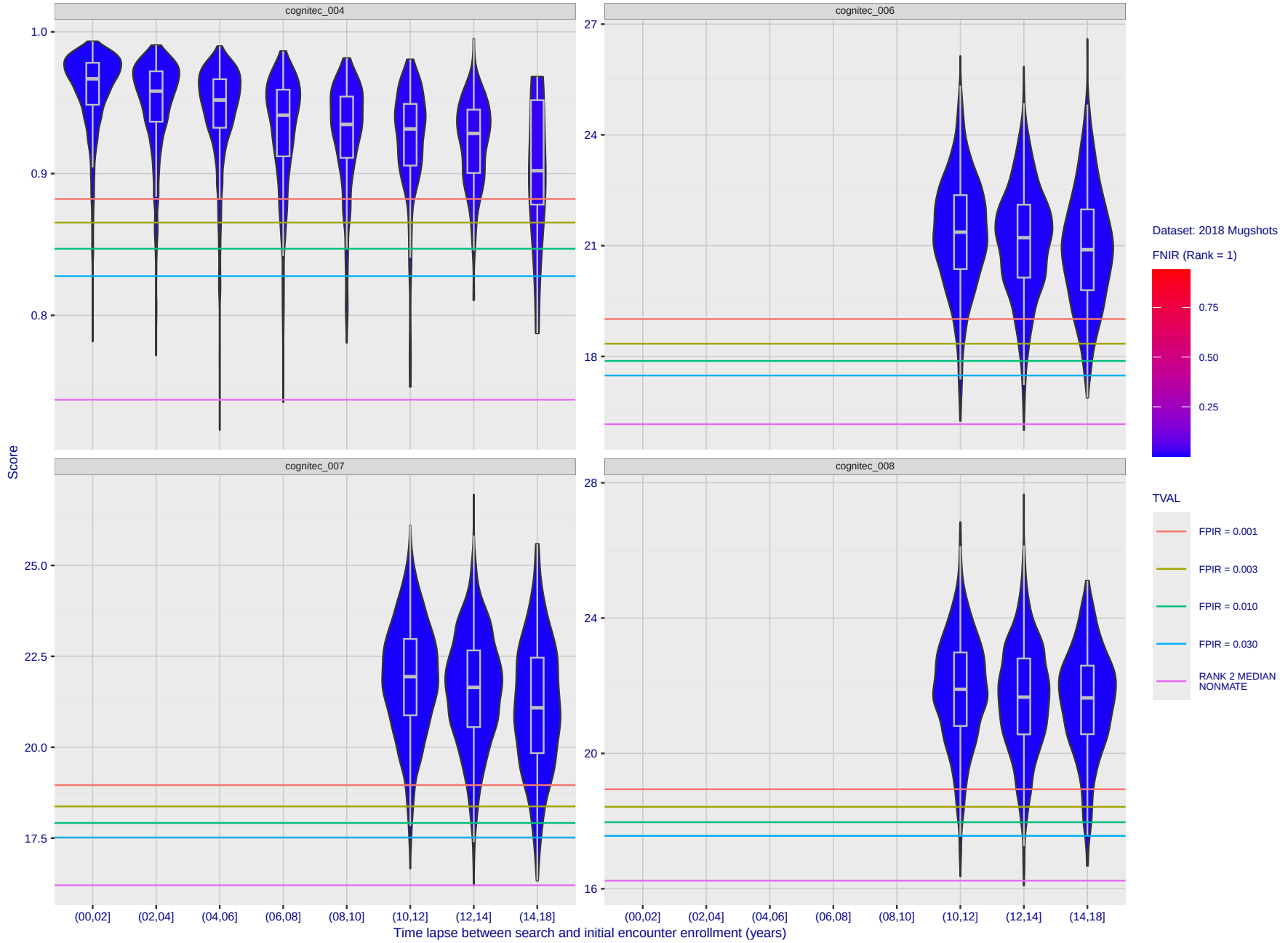


*Figure 362:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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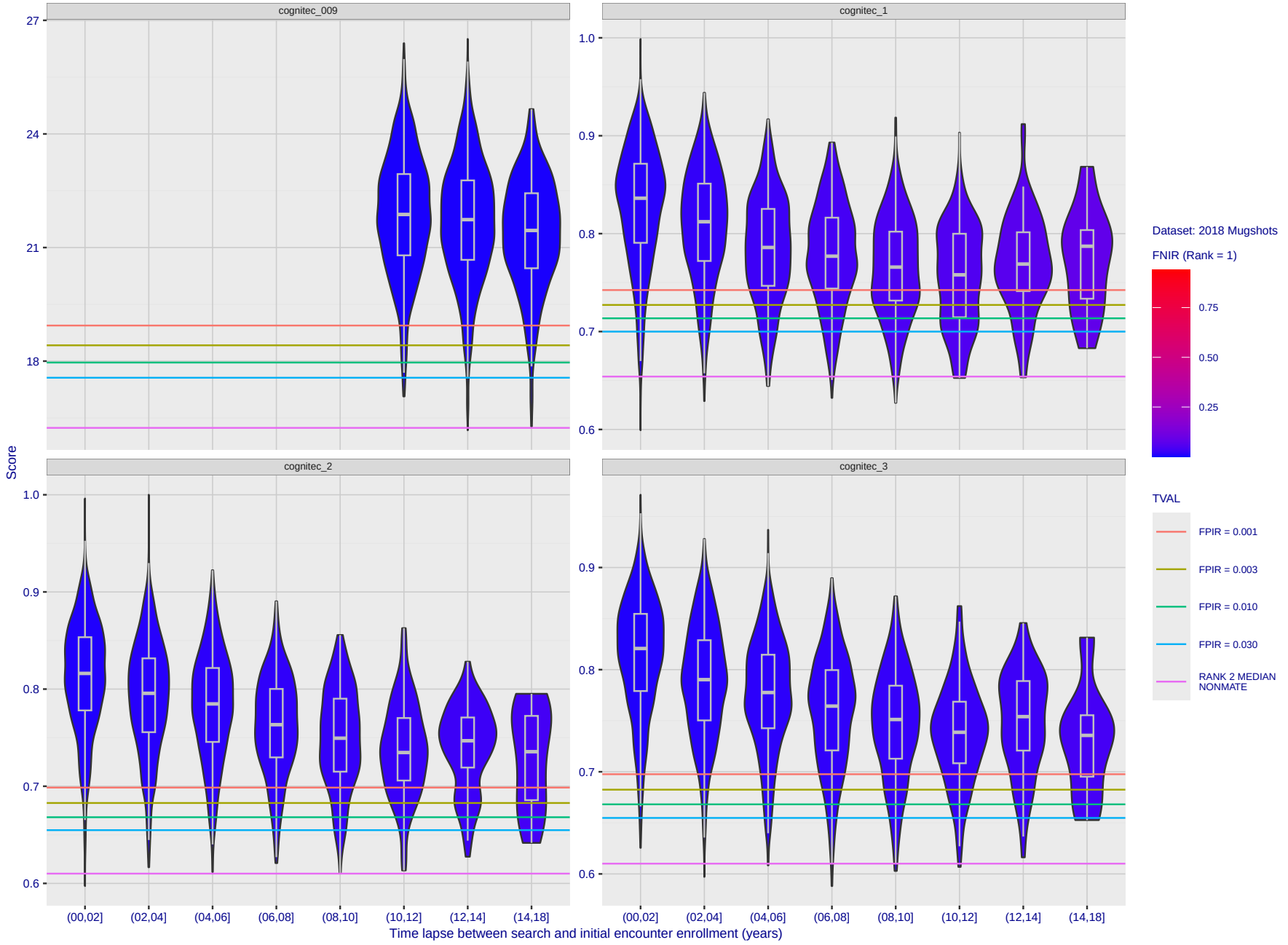


*Figure 363:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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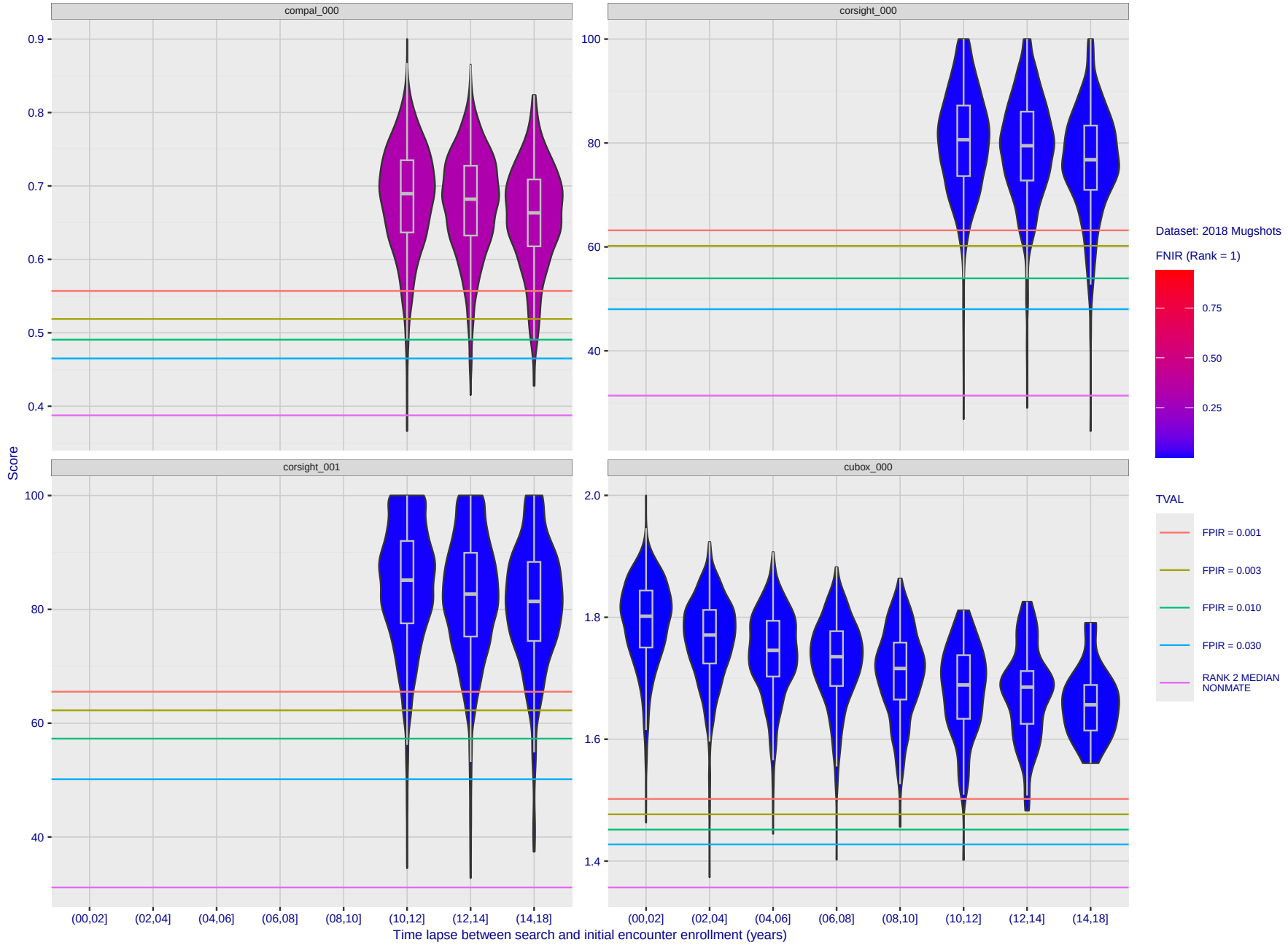


*Figure 364:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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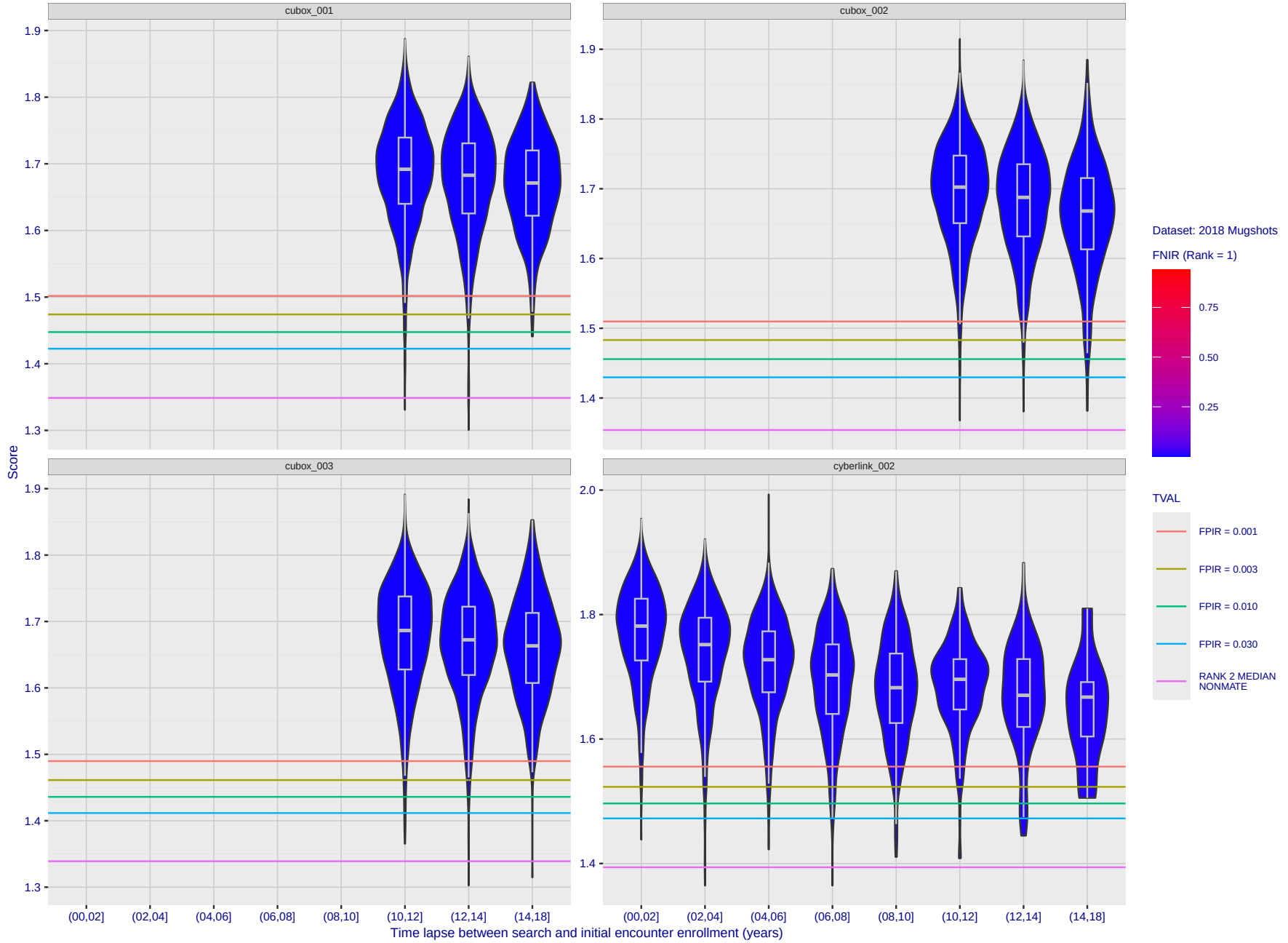


*Figure 365:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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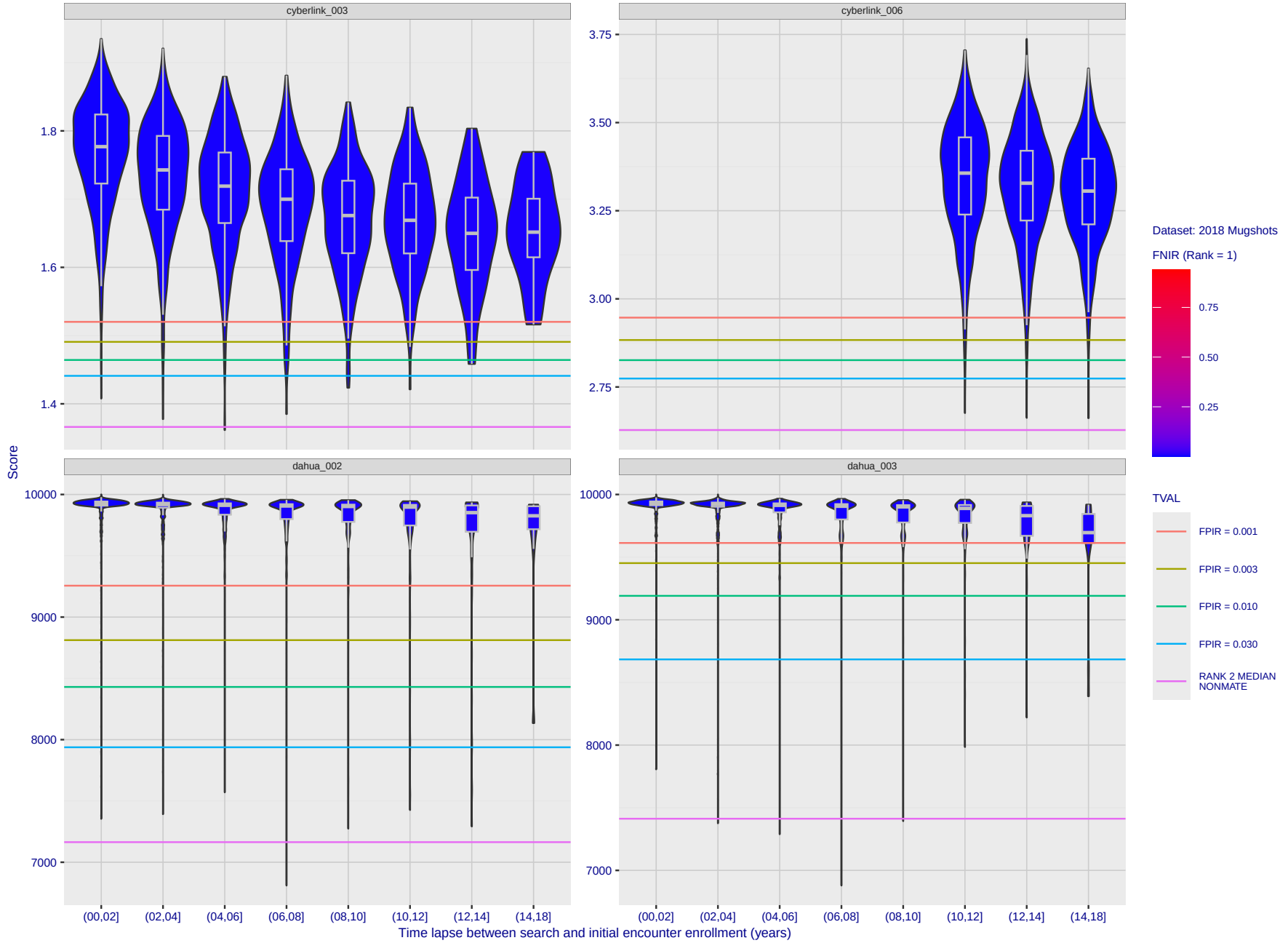


*Figure 366:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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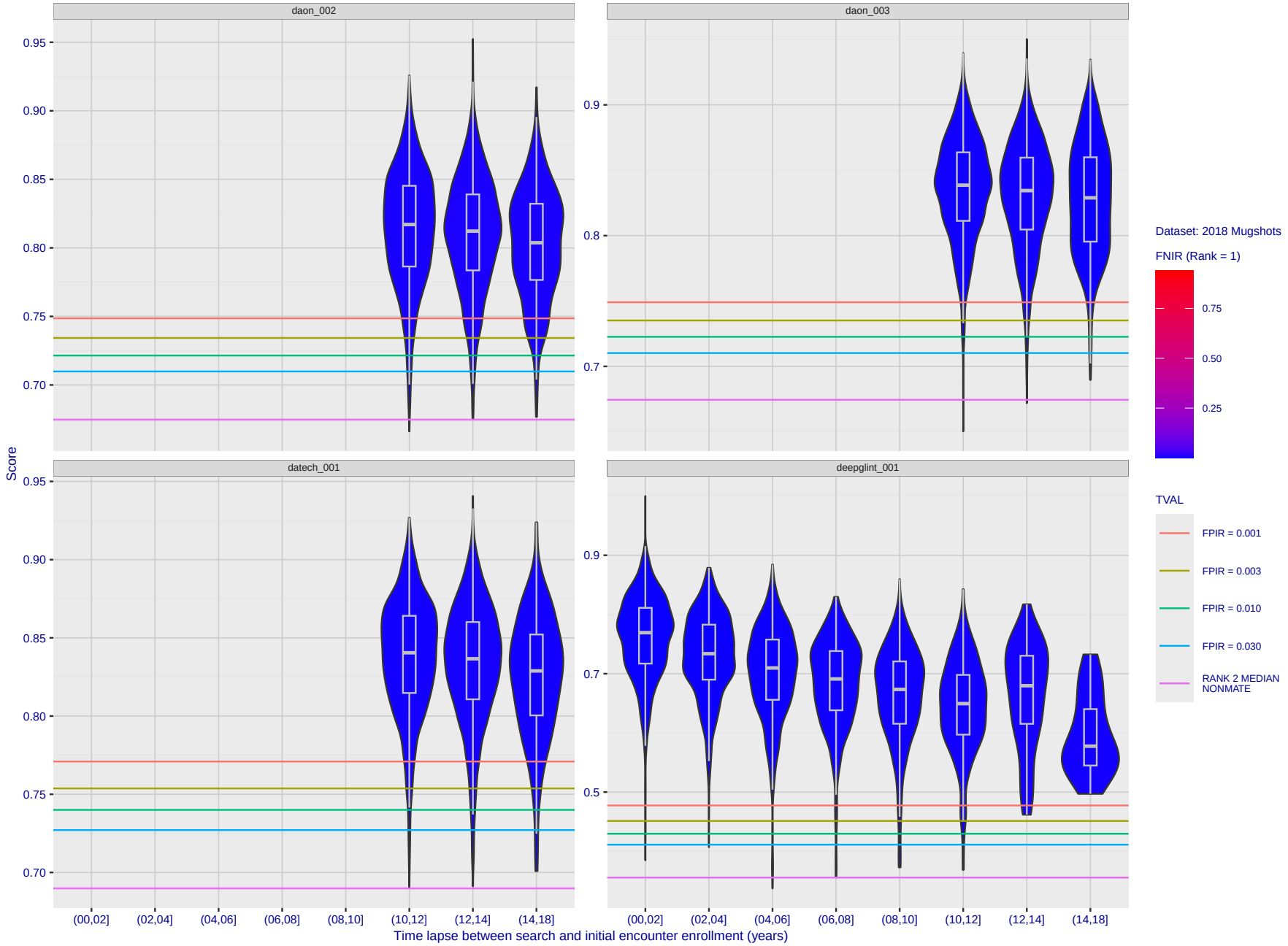


*Figure 367:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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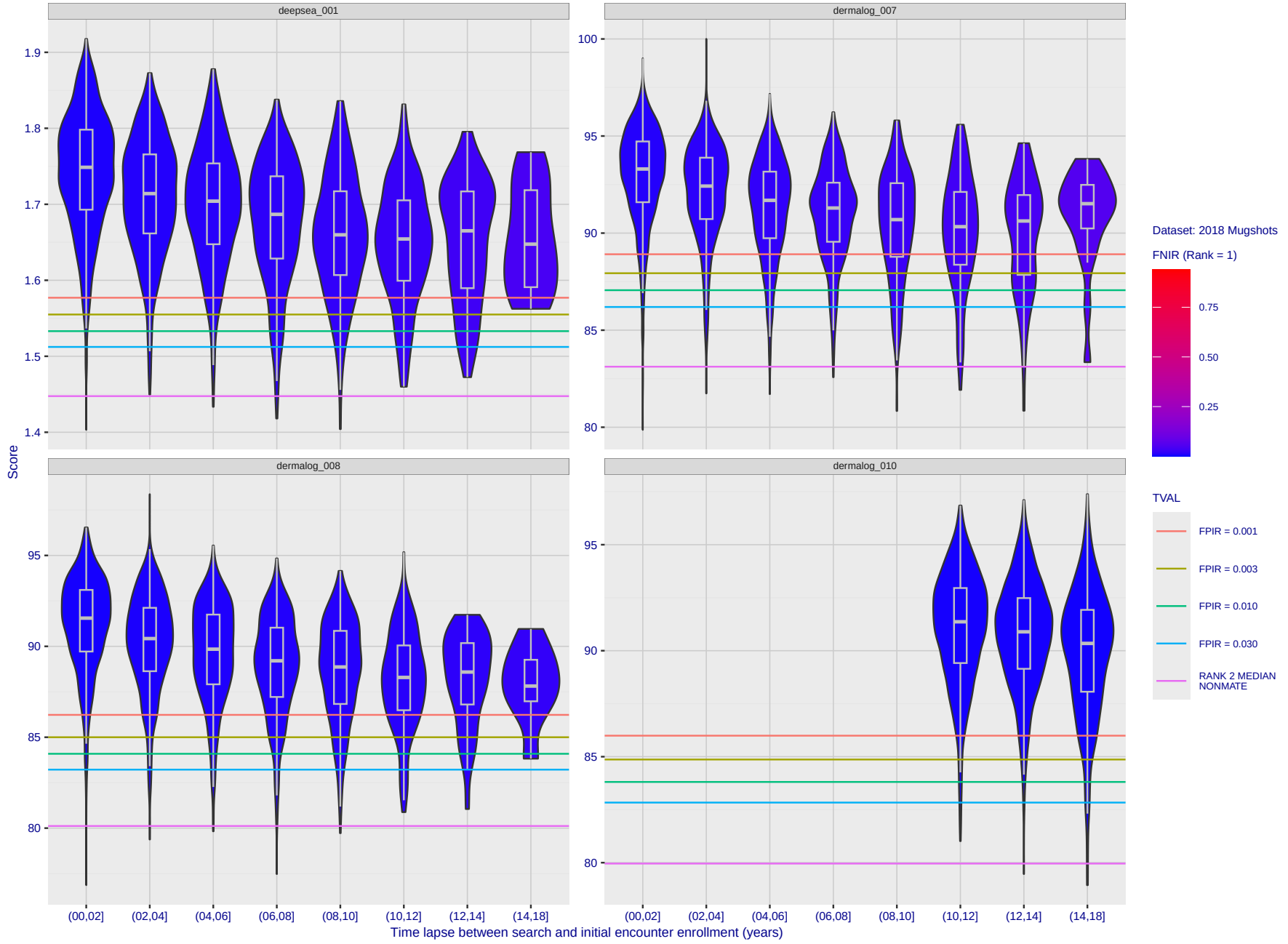


*Figure 368:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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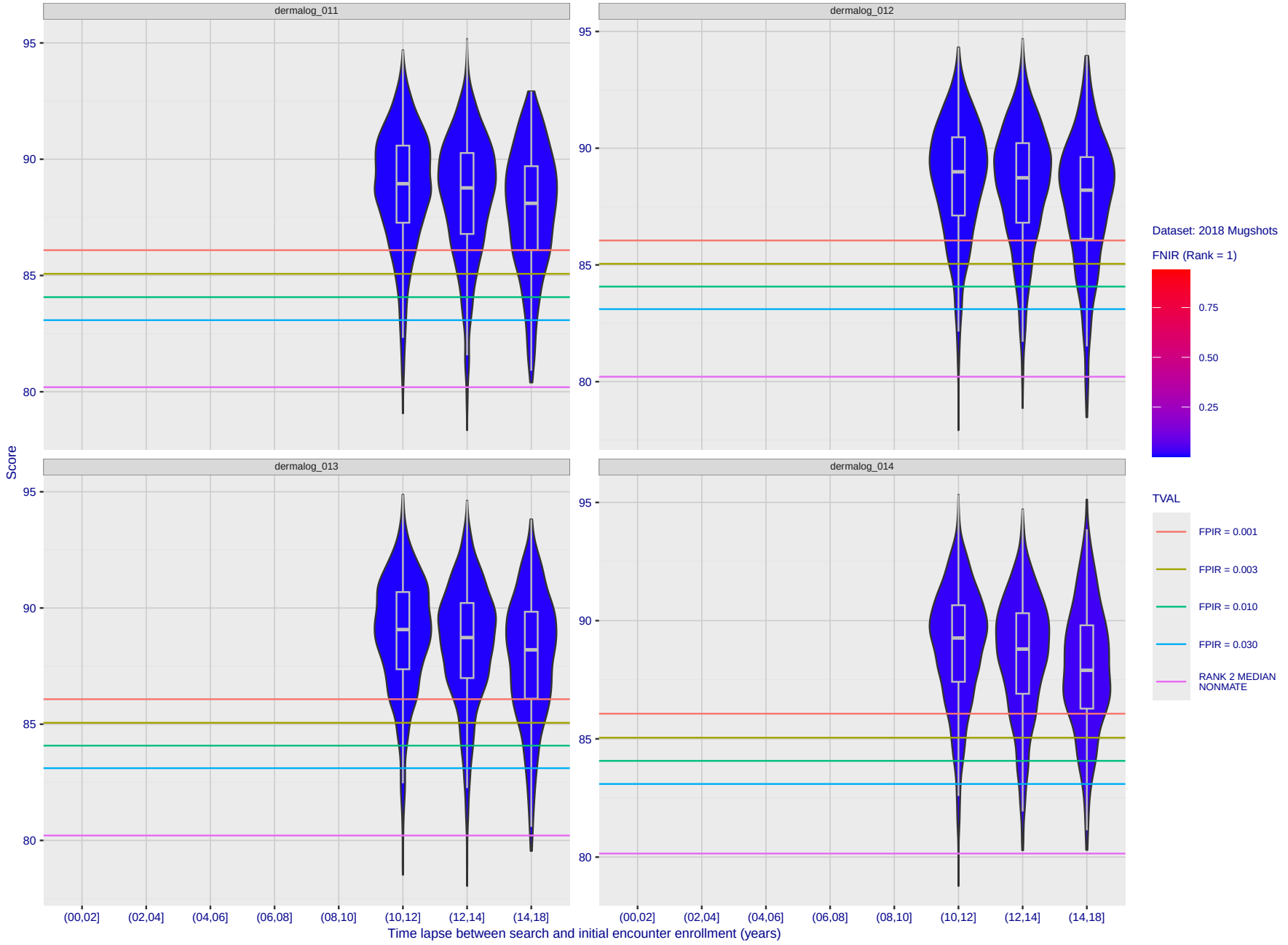


*Figure 369:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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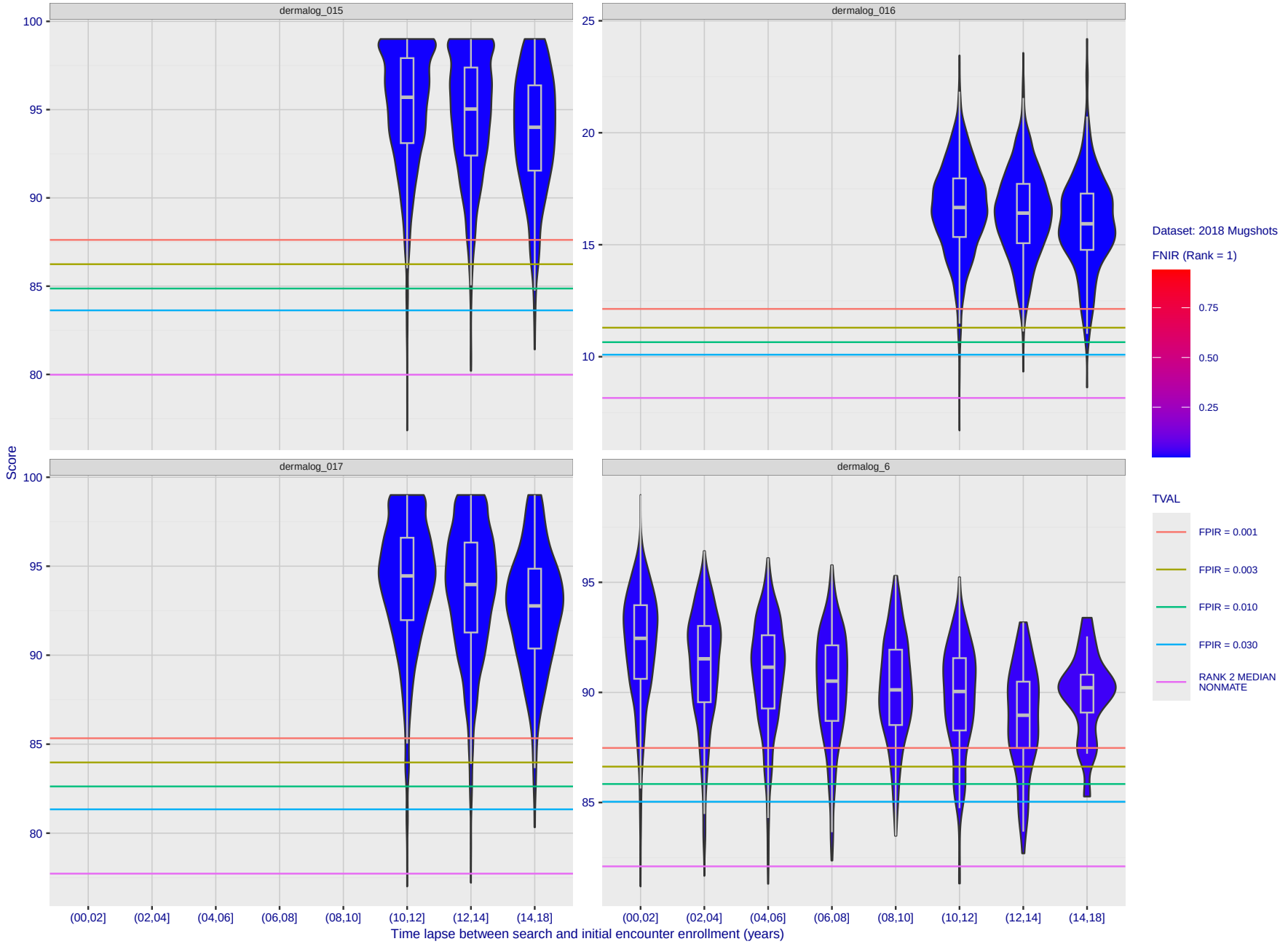


*Figure 370:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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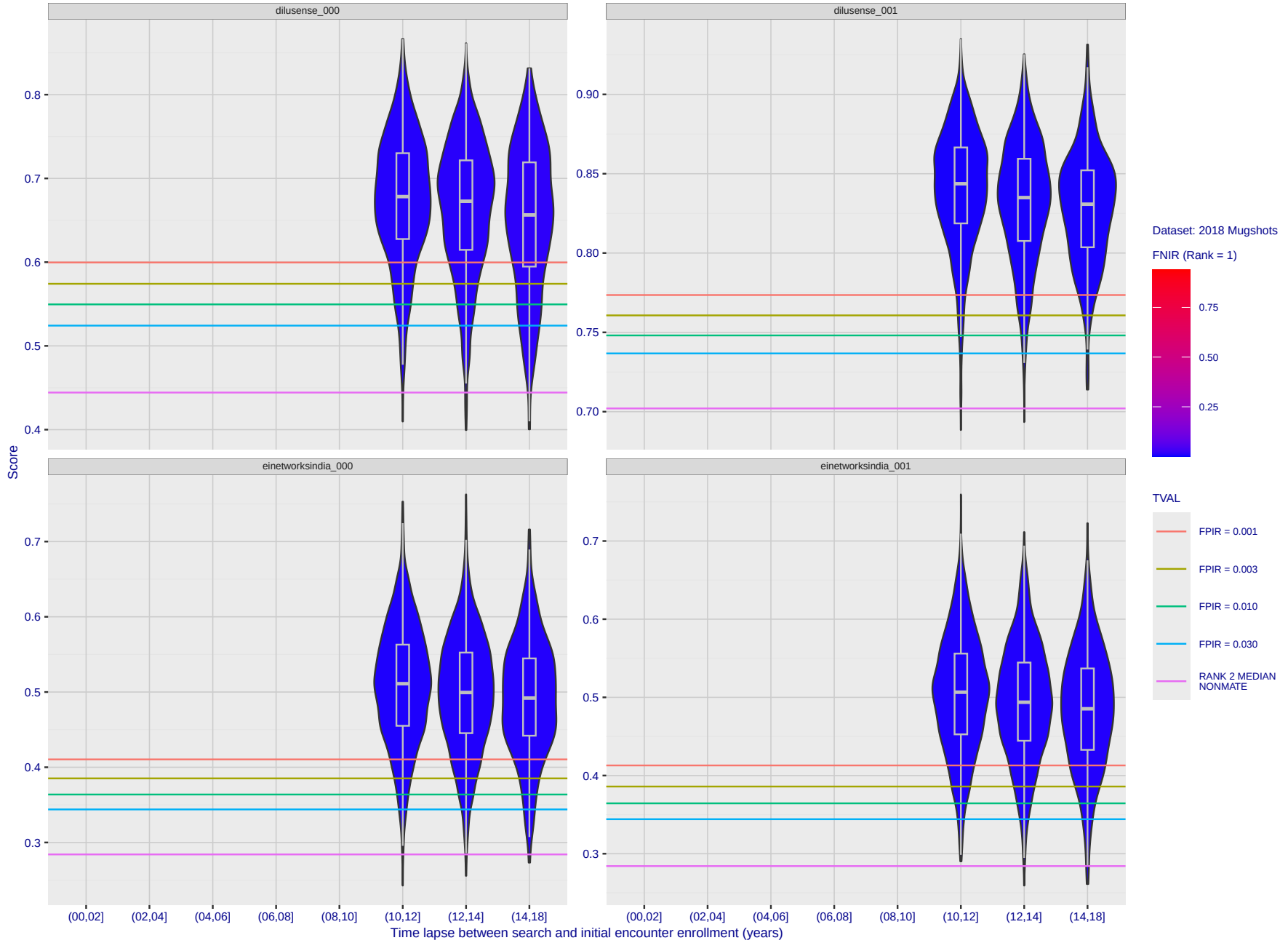


*Figure 371:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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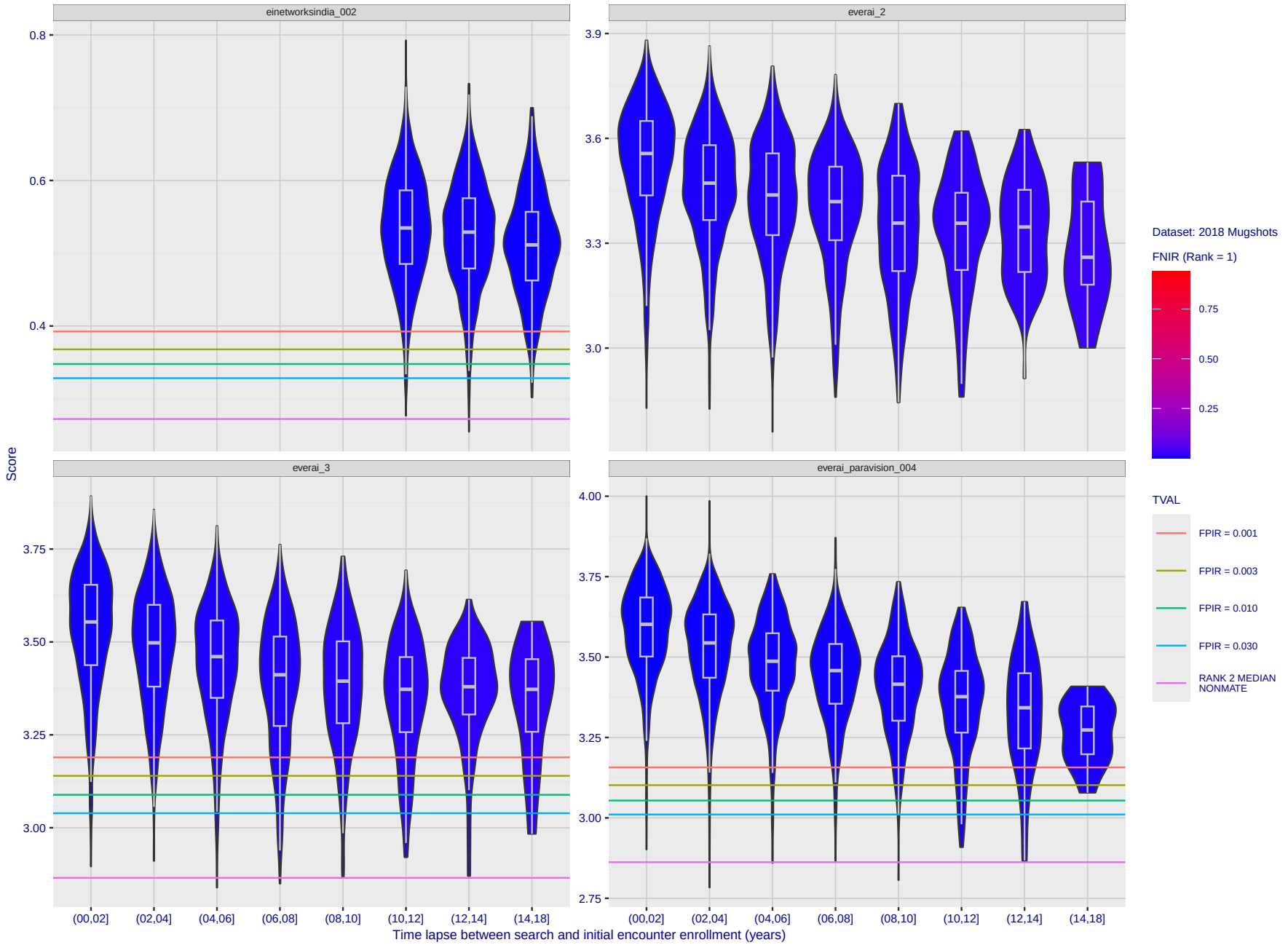


*Figure 372:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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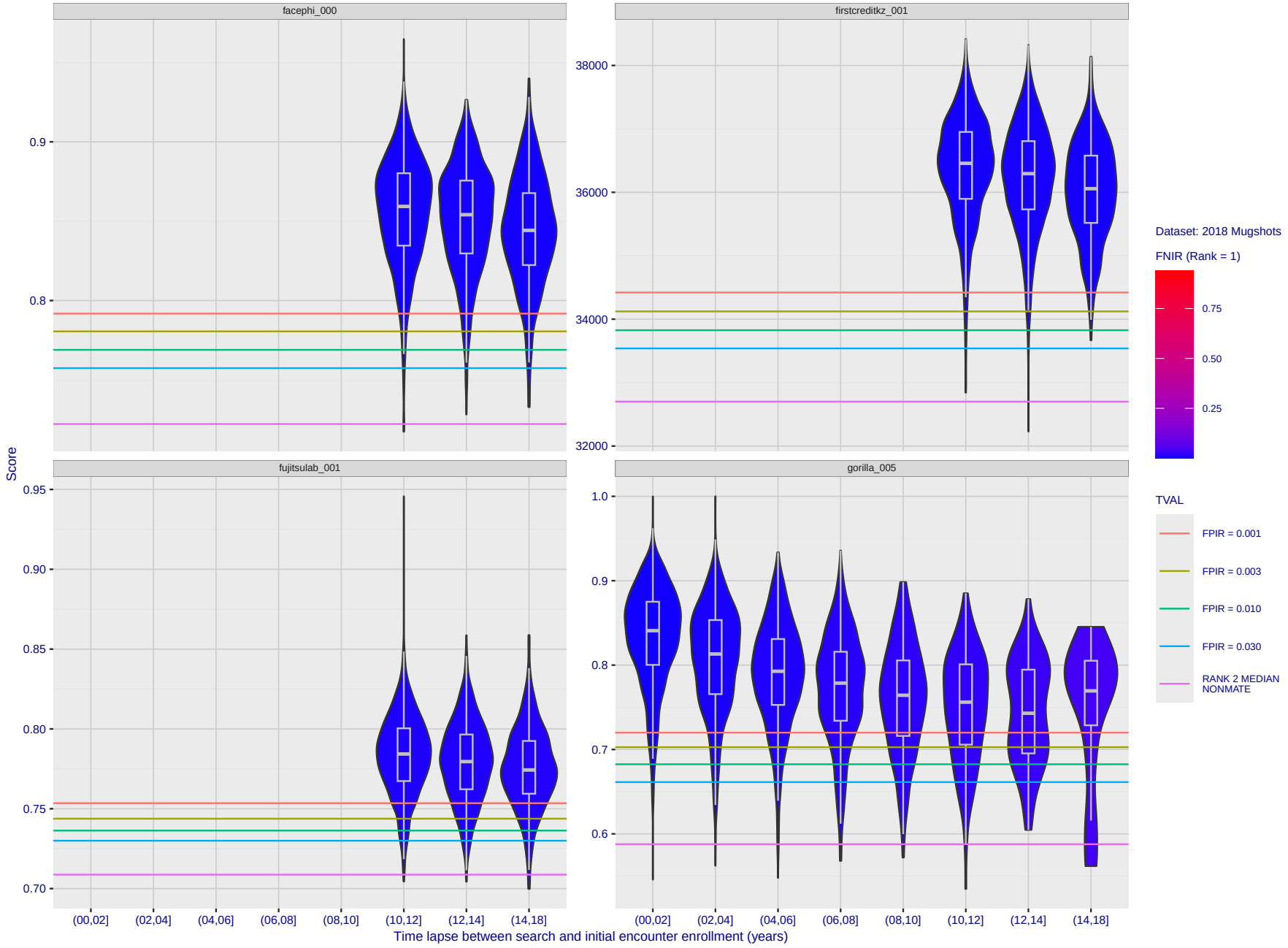


*Figure 373:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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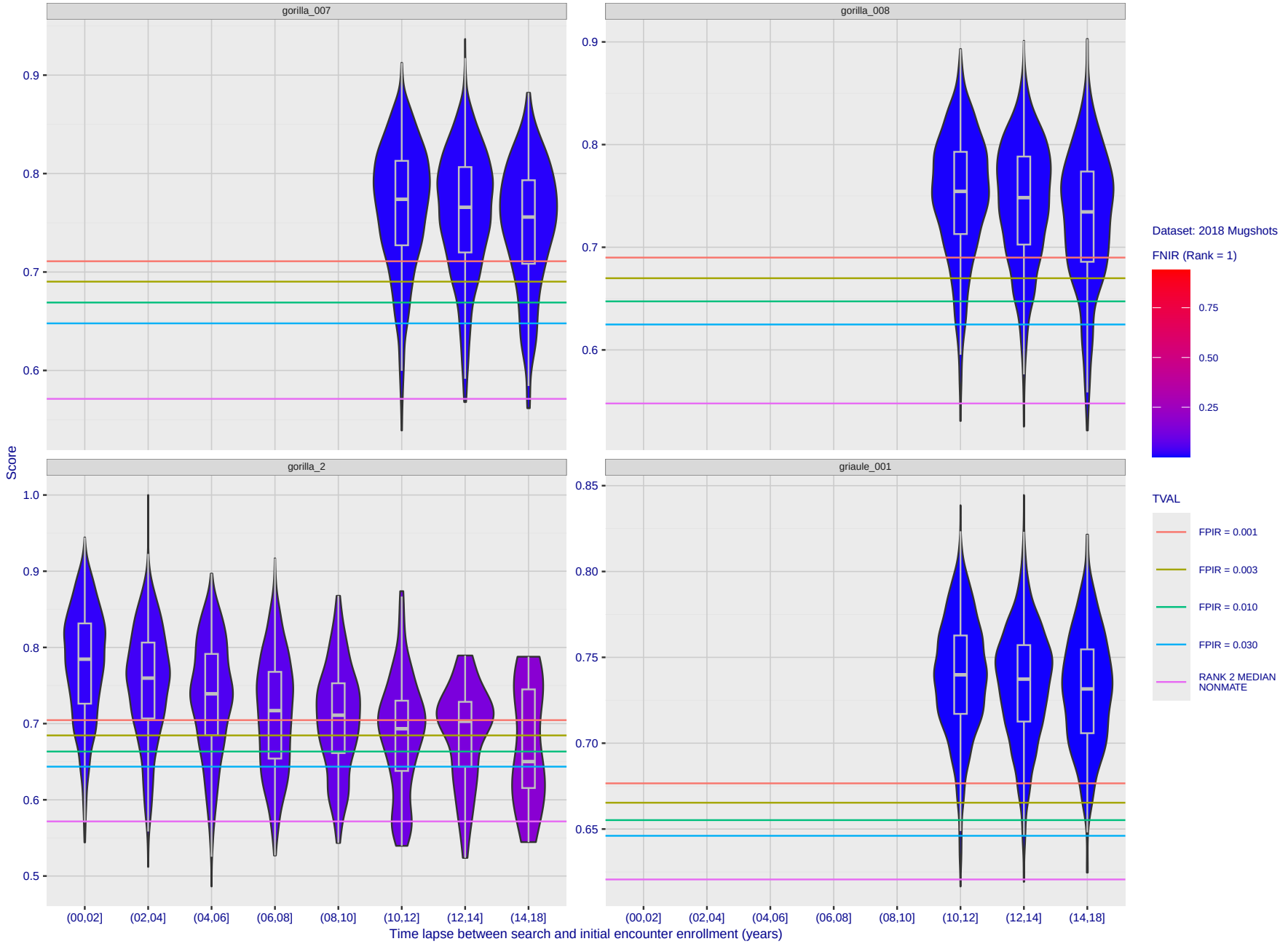


*Figure 374:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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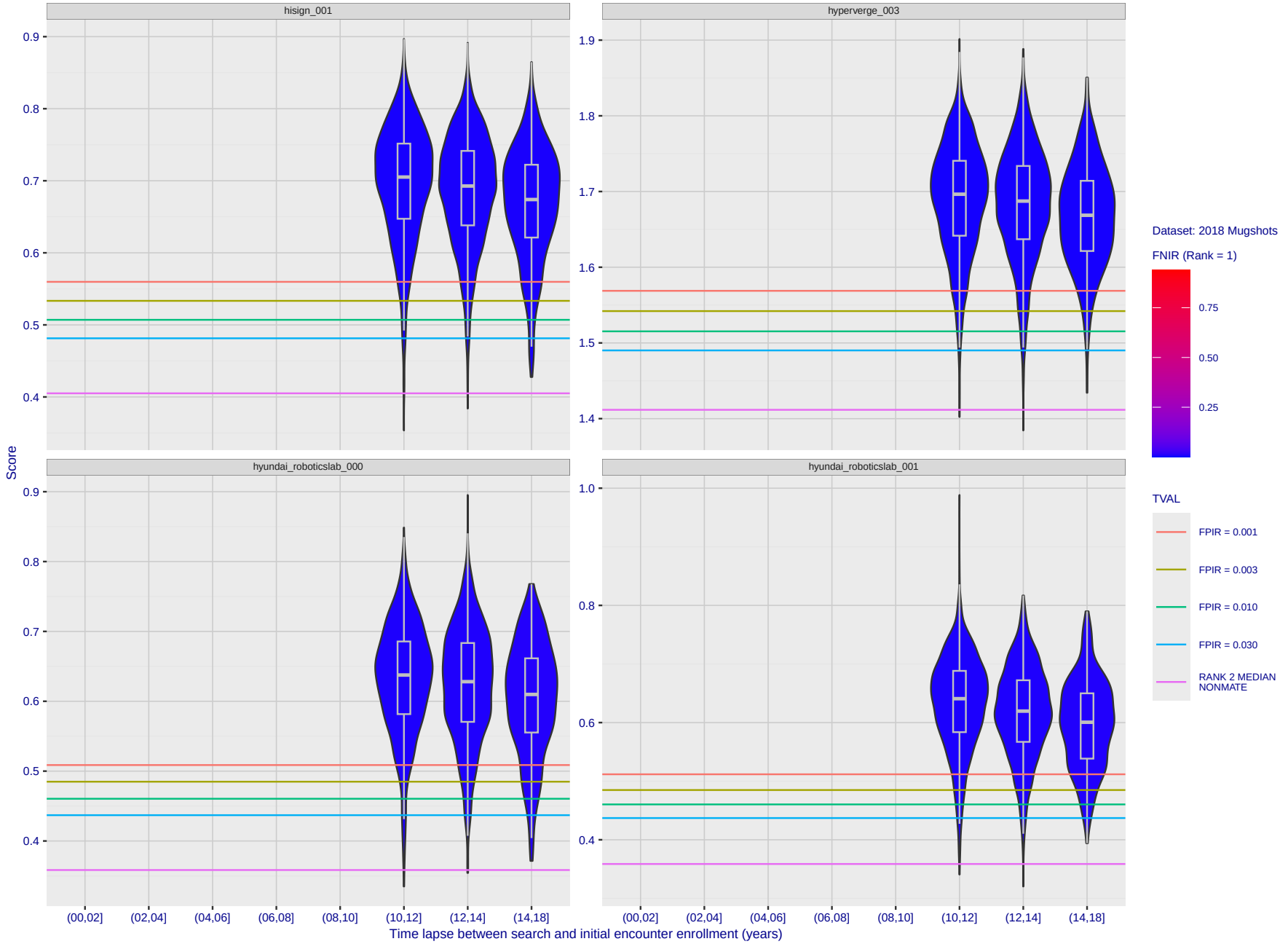


*Figure 375:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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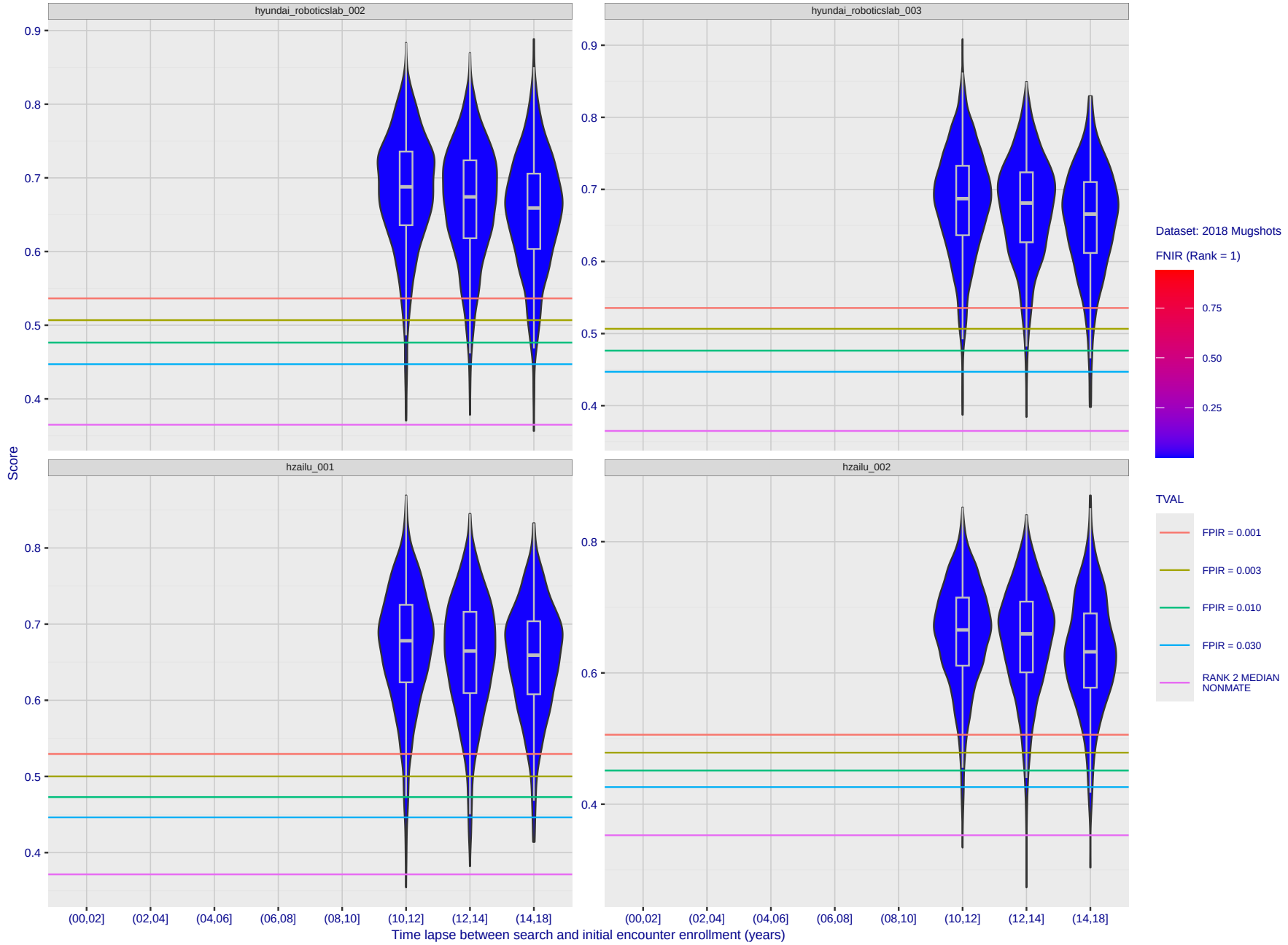


*Figure 376:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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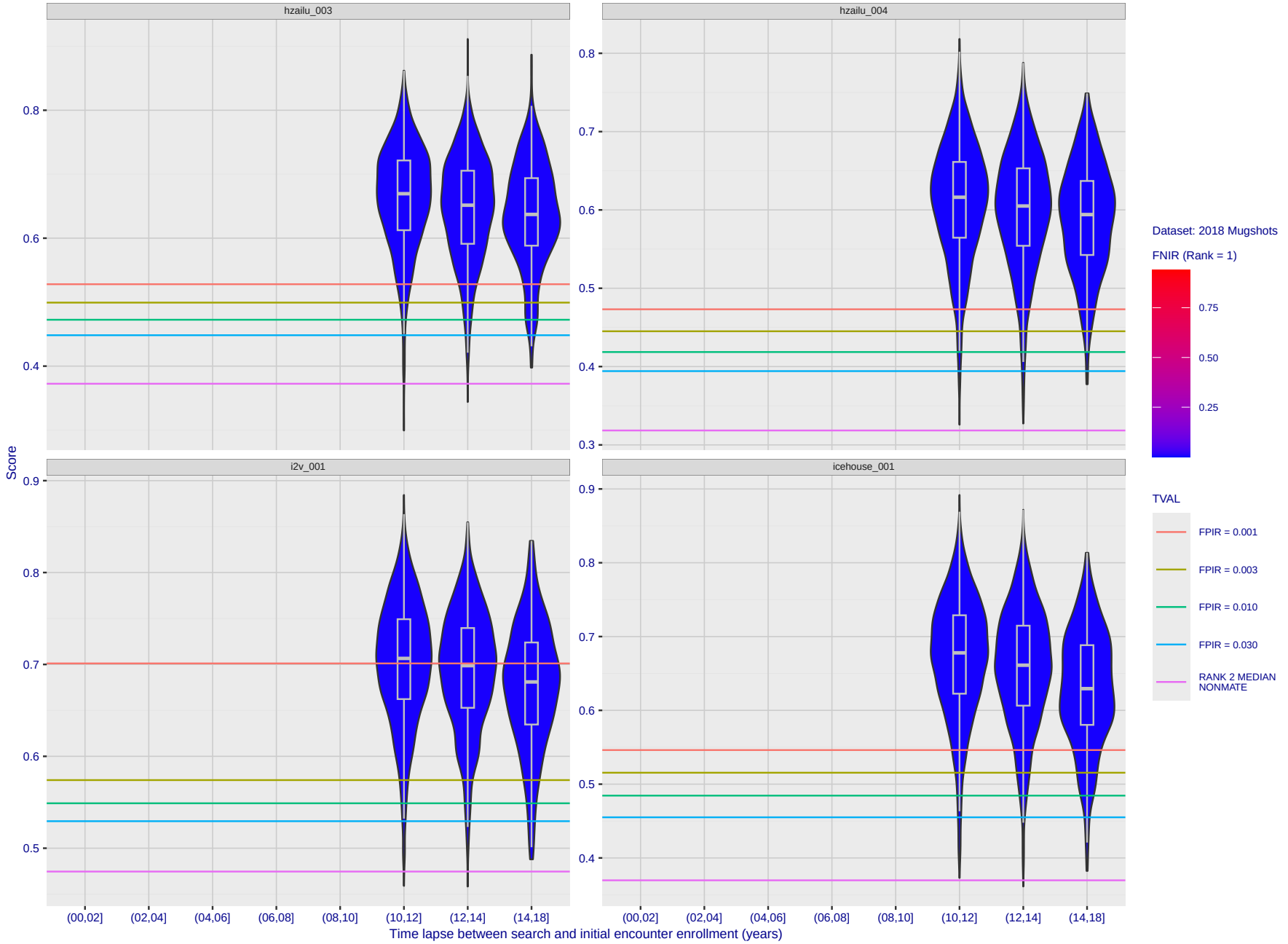


*Figure 377:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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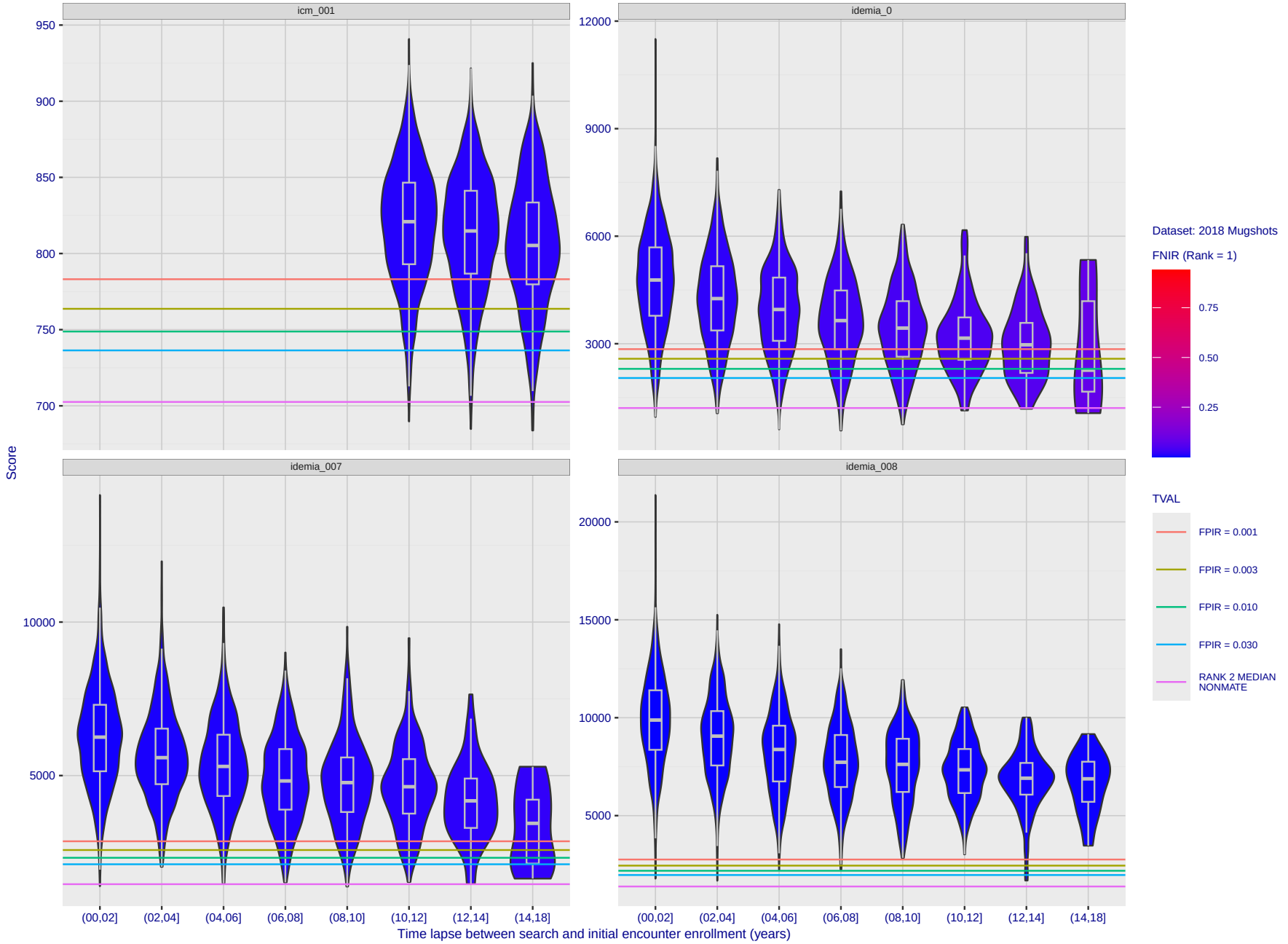


*Figure 378:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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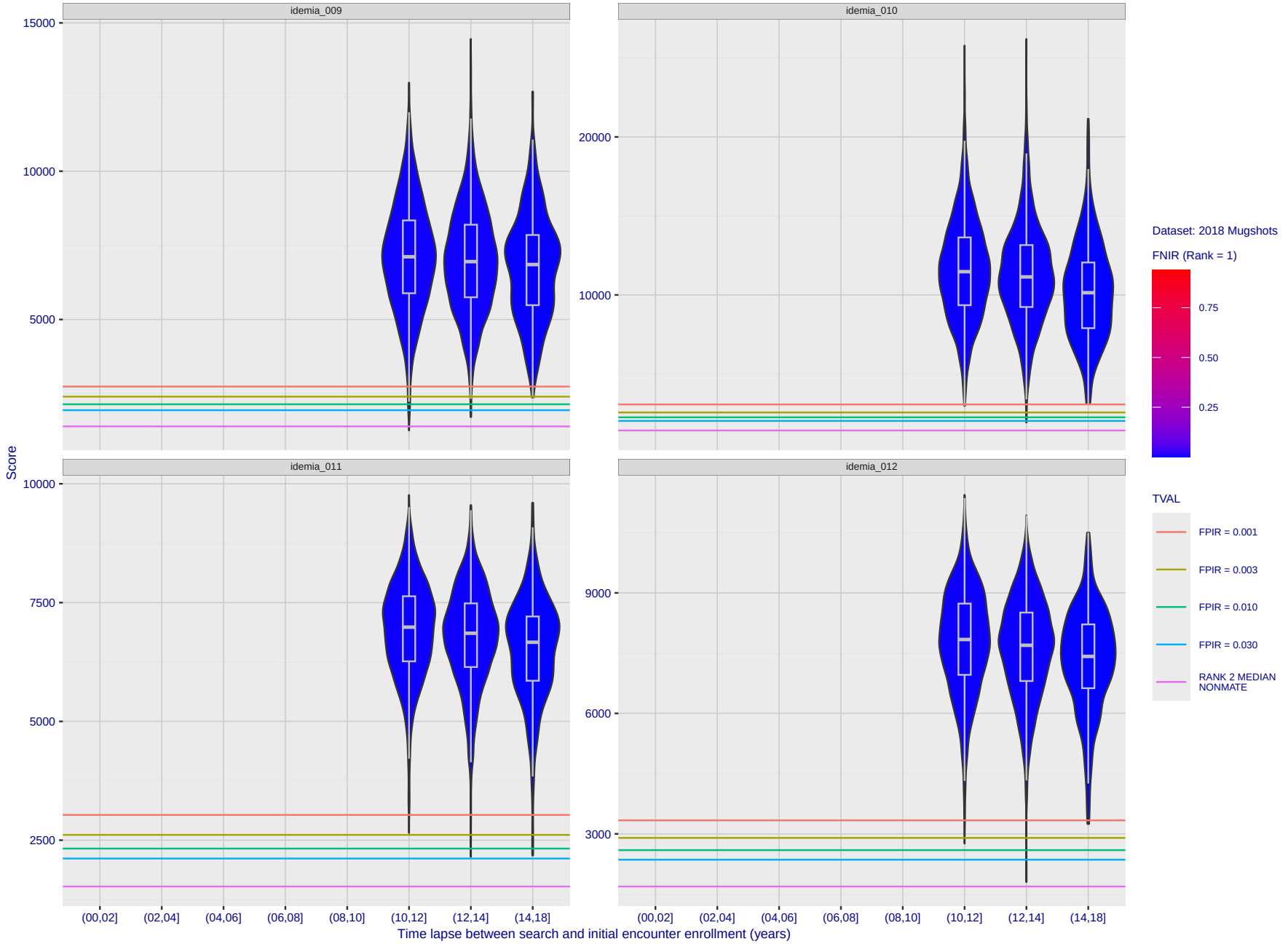


*Figure 379:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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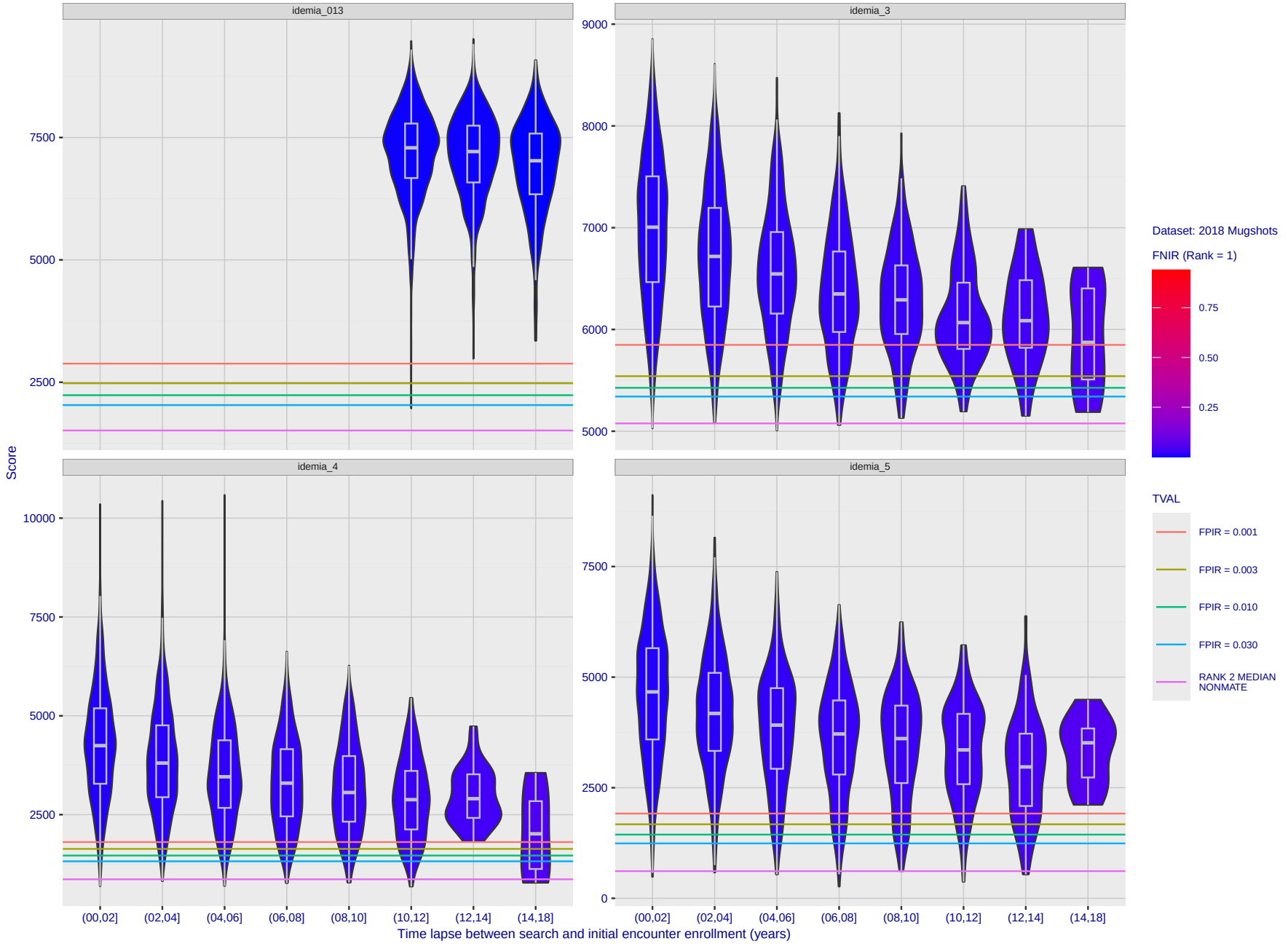


*Figure 380:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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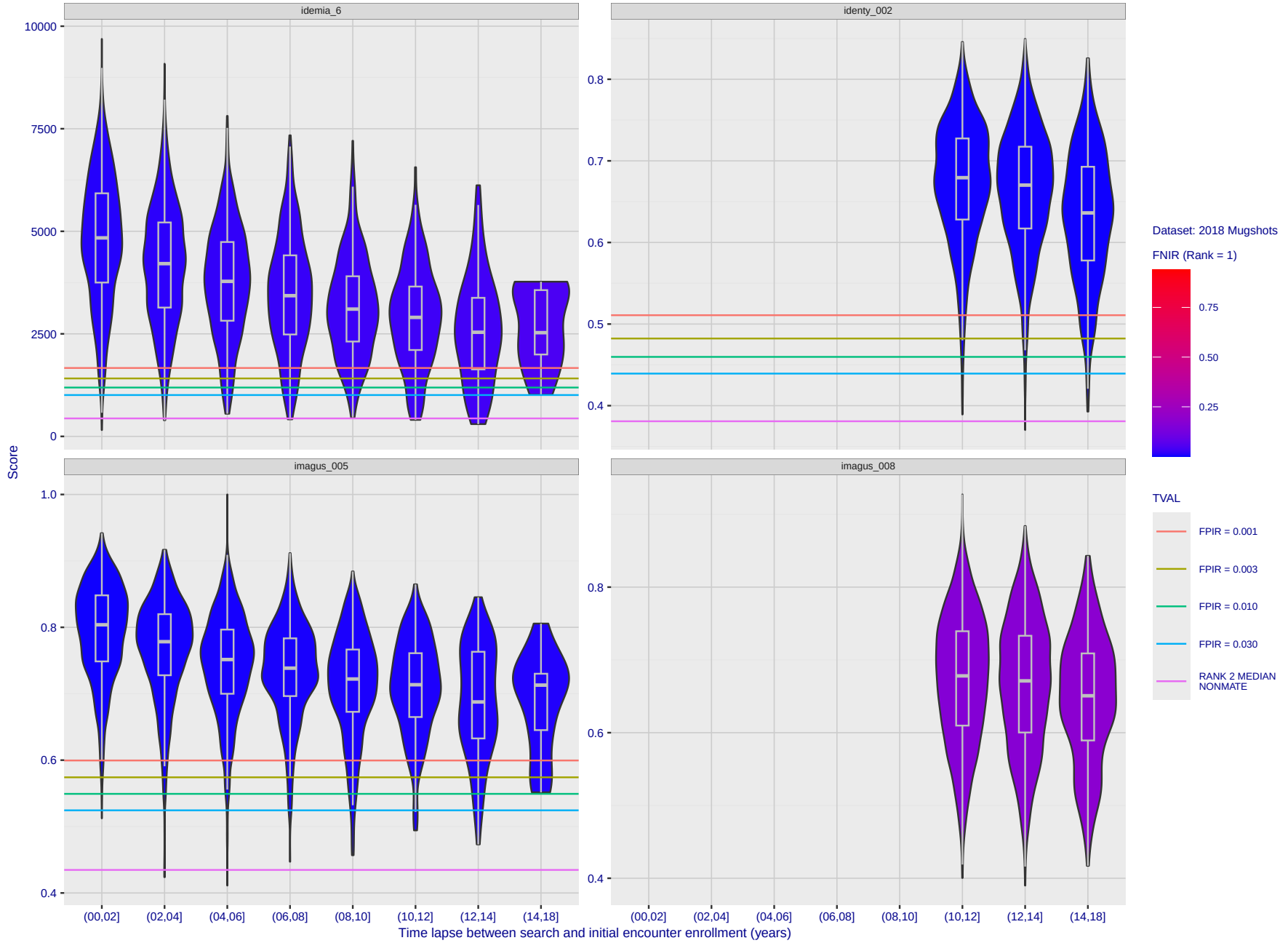


*Figure 381:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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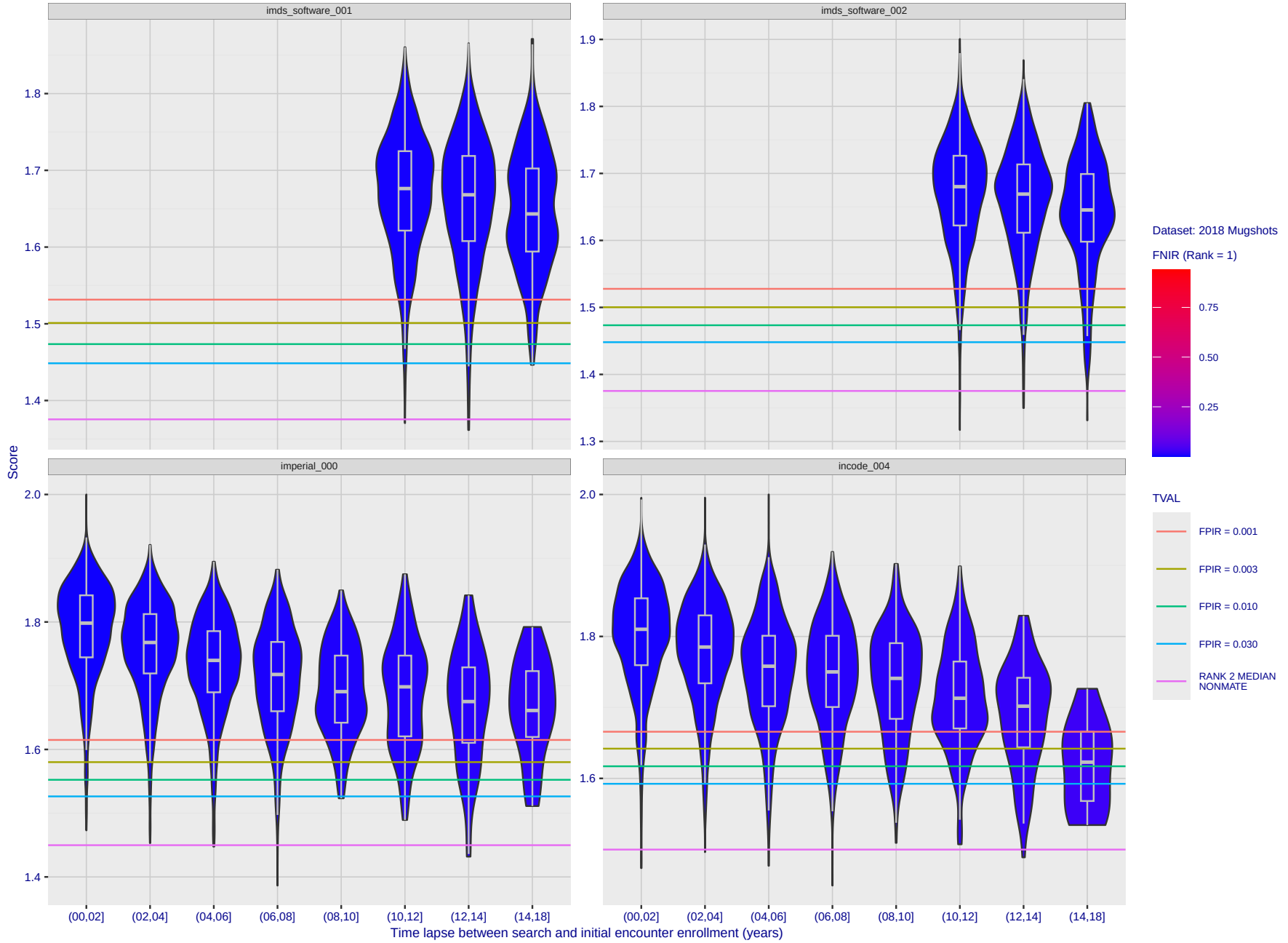


*Figure 382:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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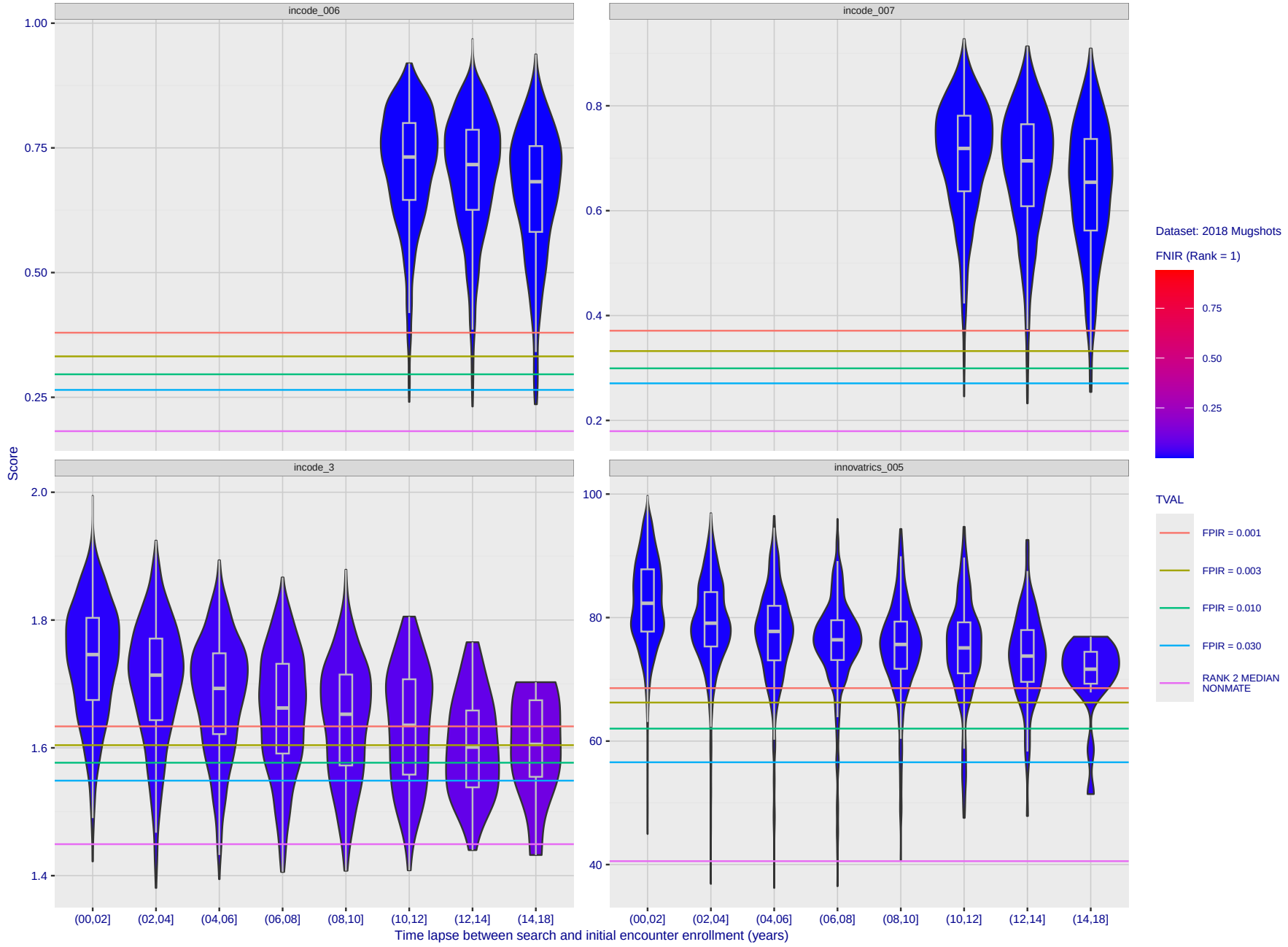


*Figure 383:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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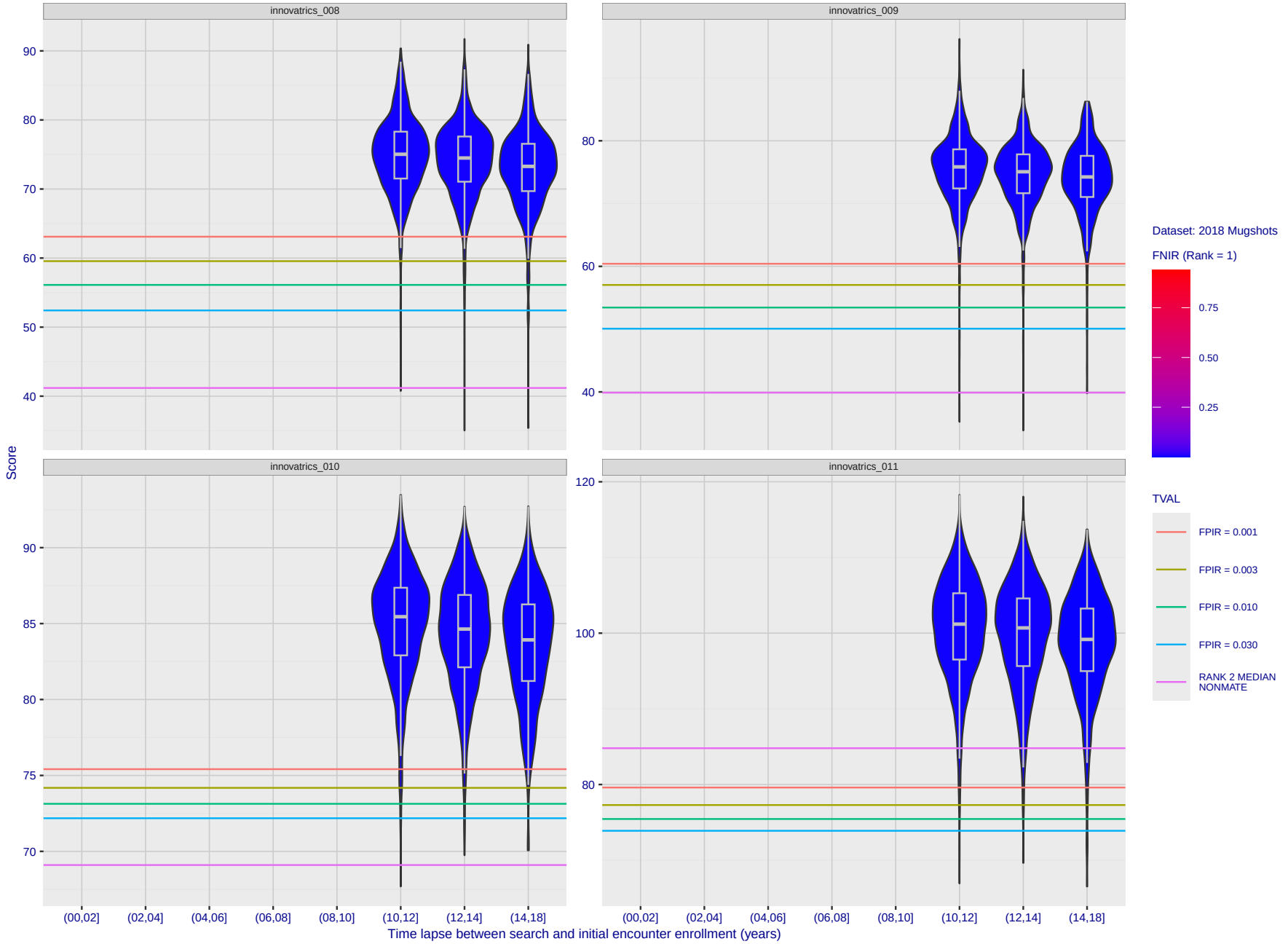


*Figure 384:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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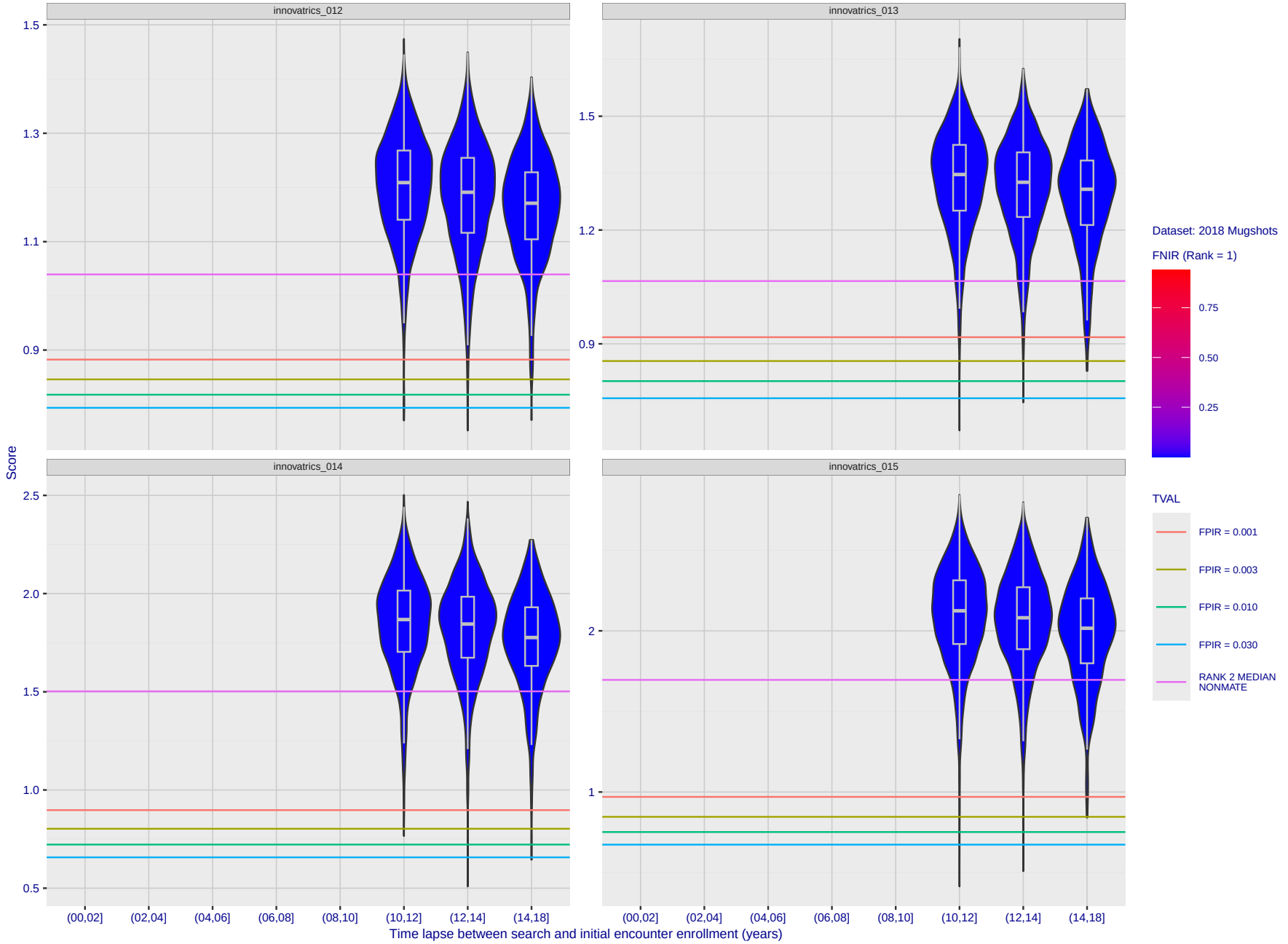


*Figure 385:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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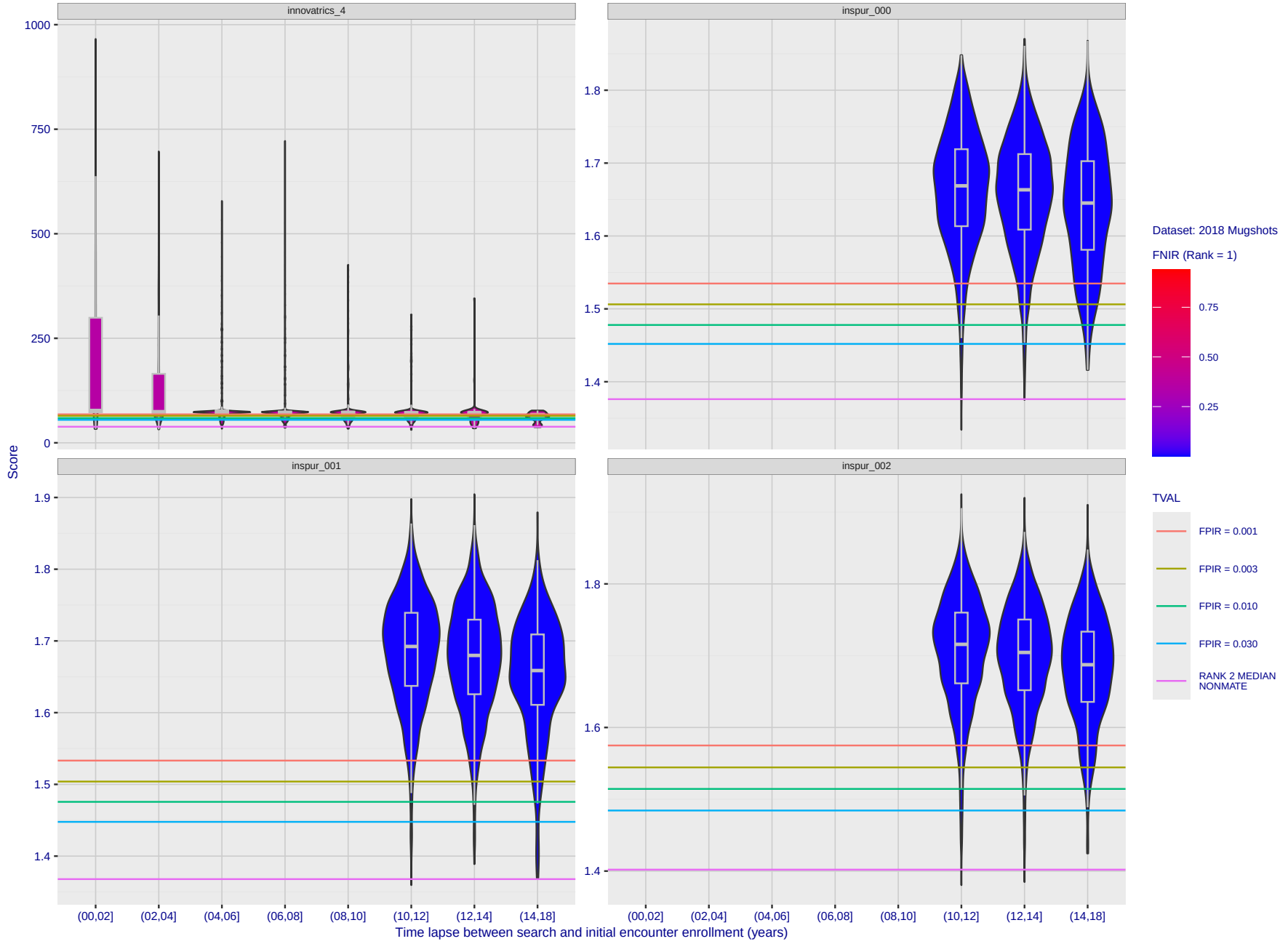


*Figure 386:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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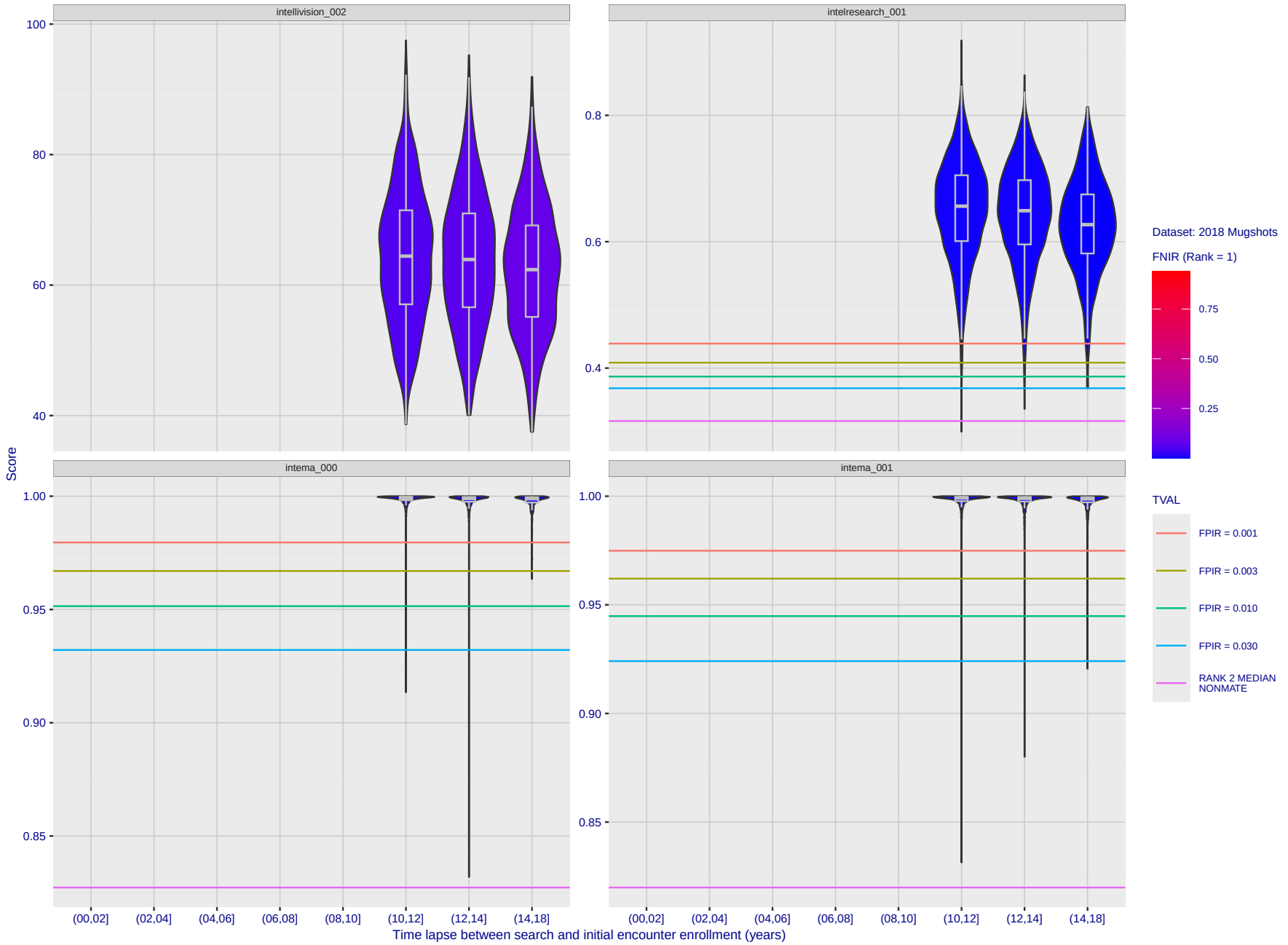


*Figure 387:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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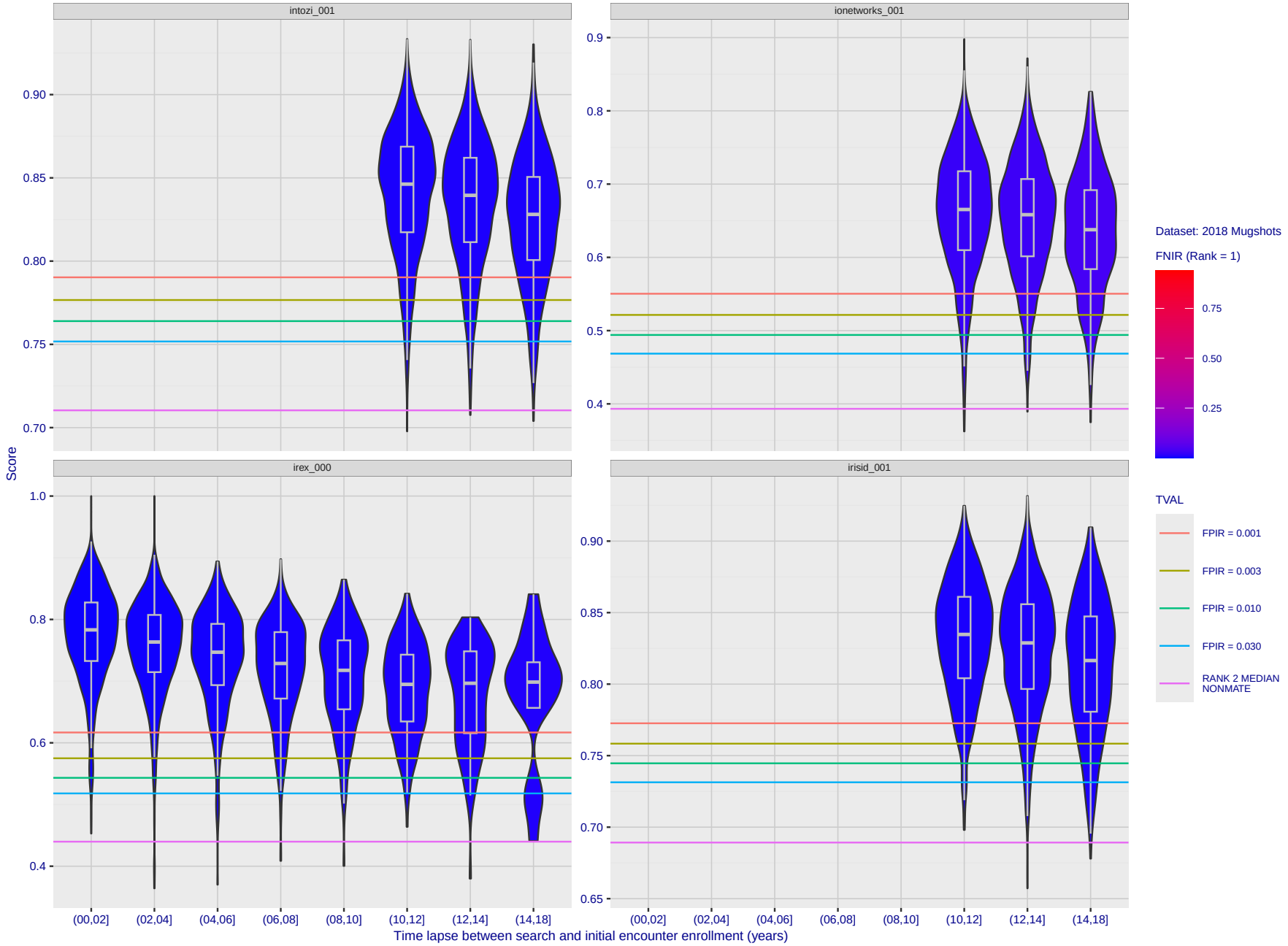


*Figure 388:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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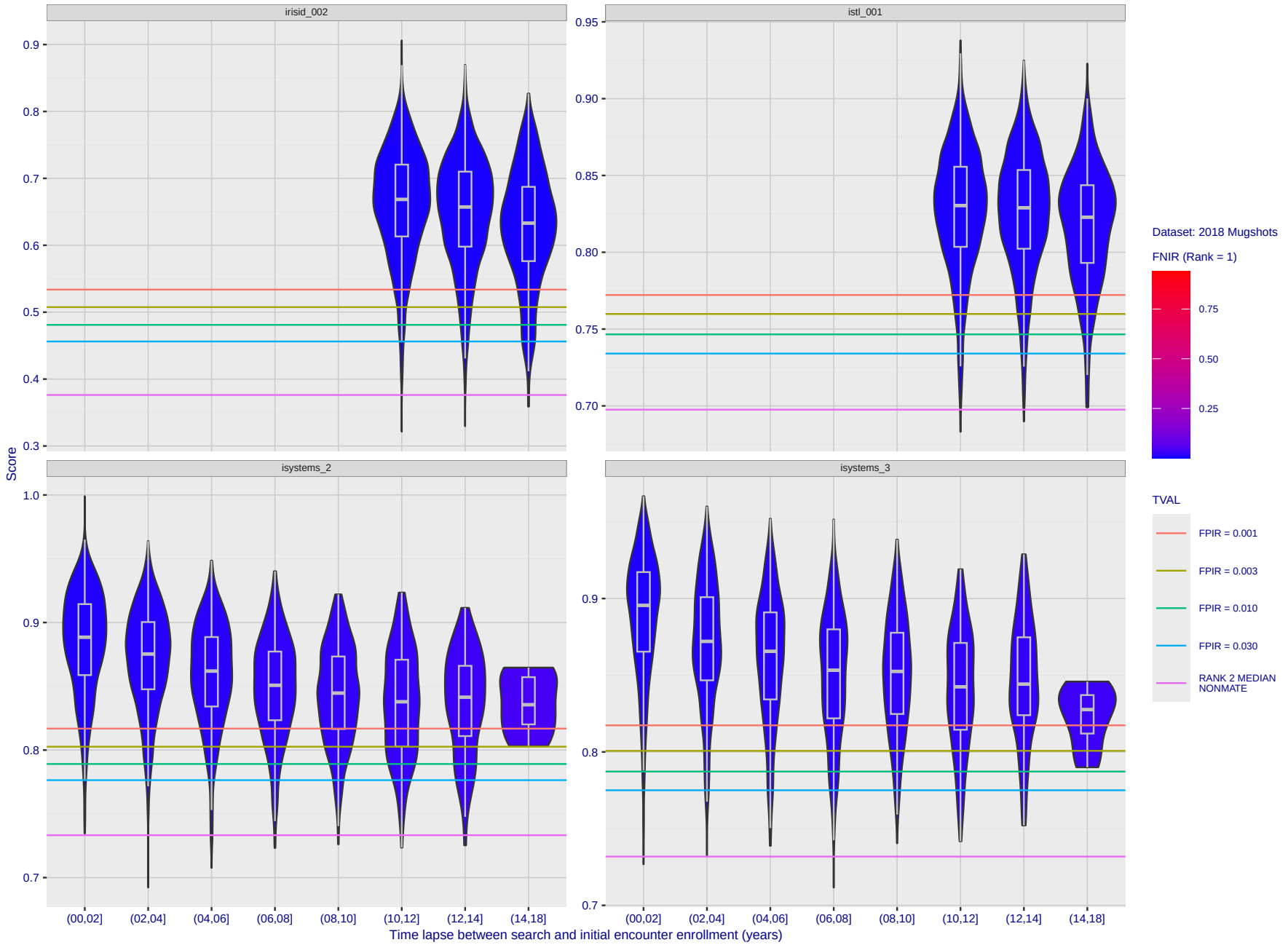


*Figure 389:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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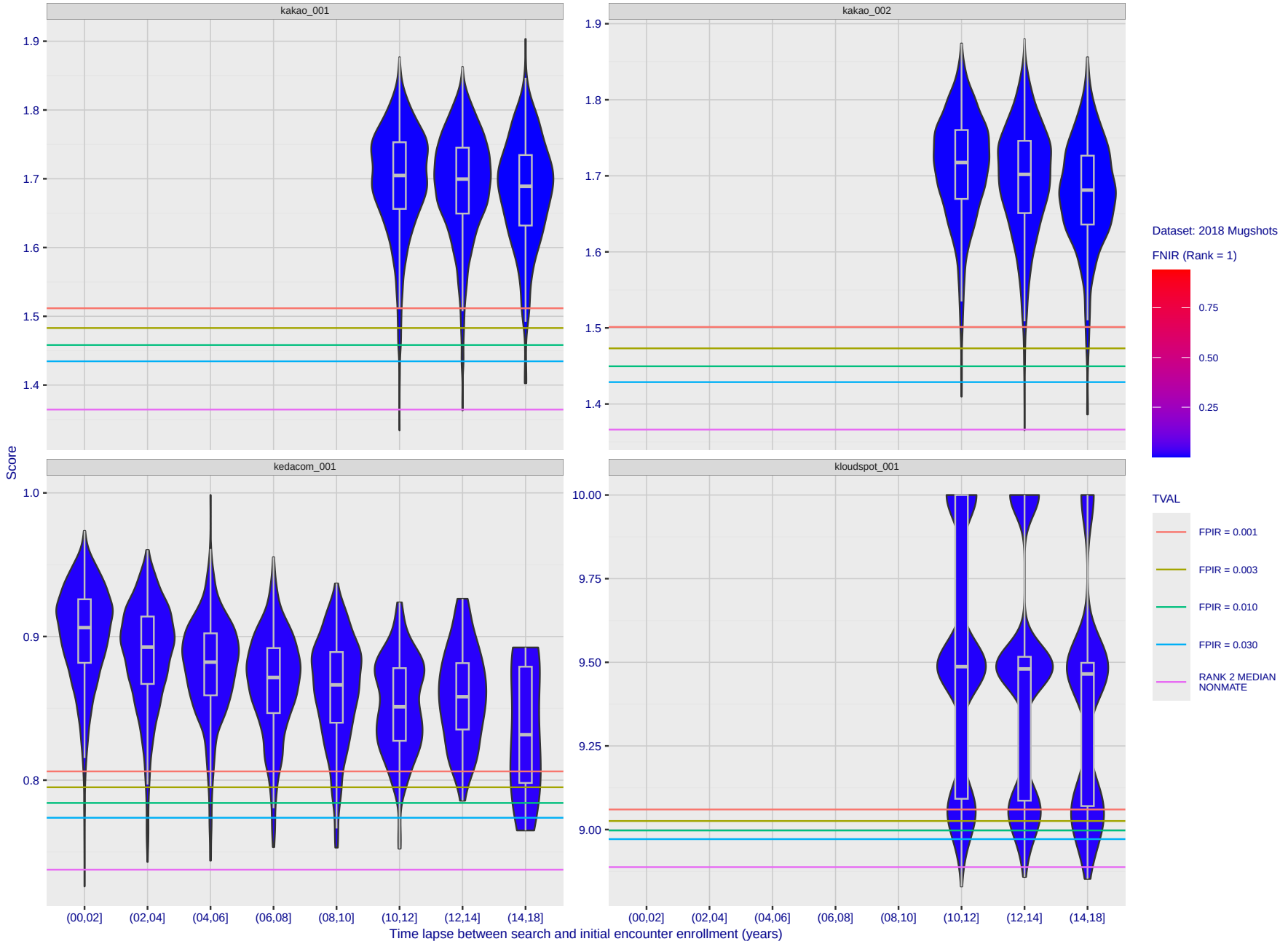


*Figure 390:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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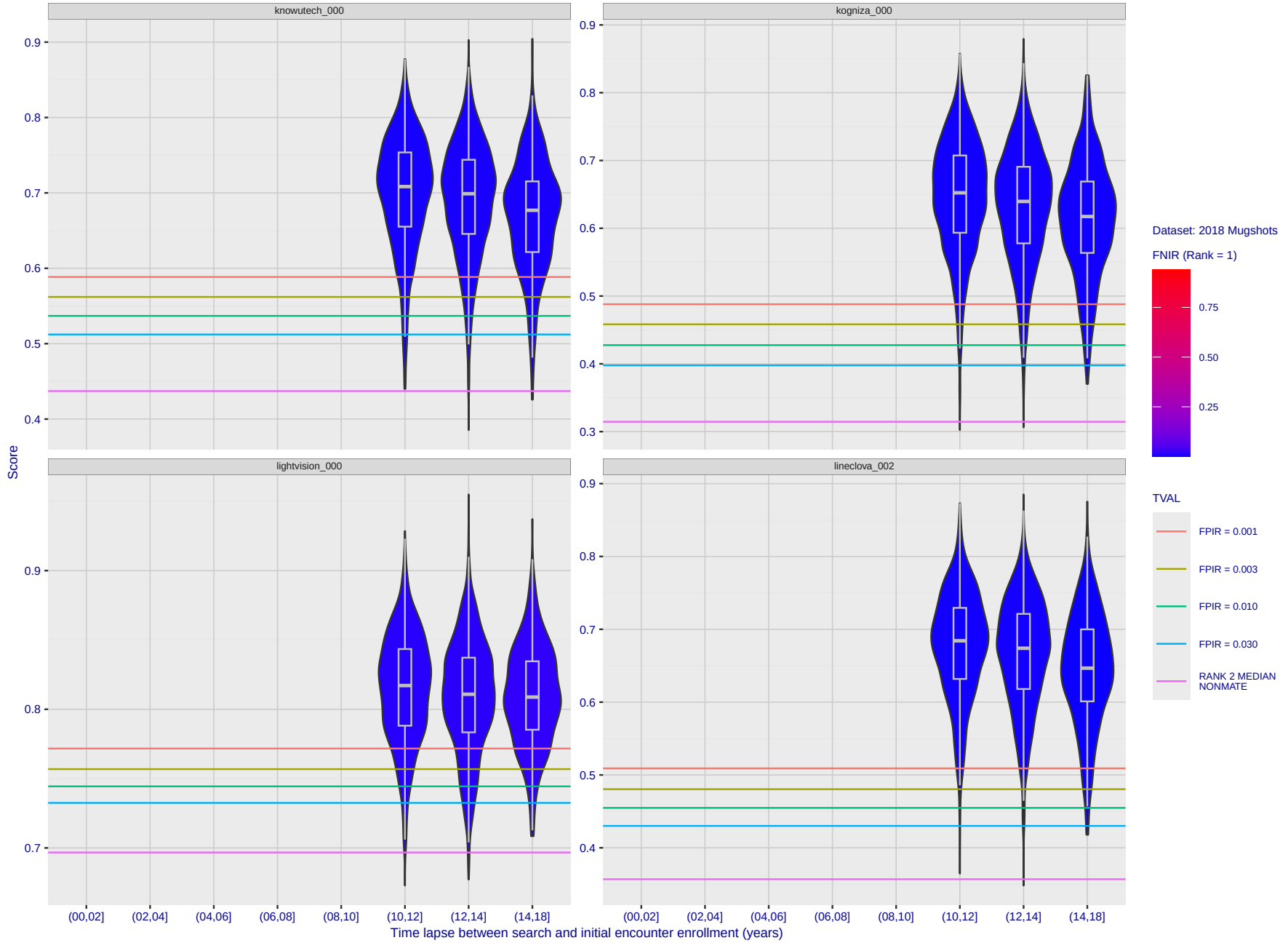


*Figure 391:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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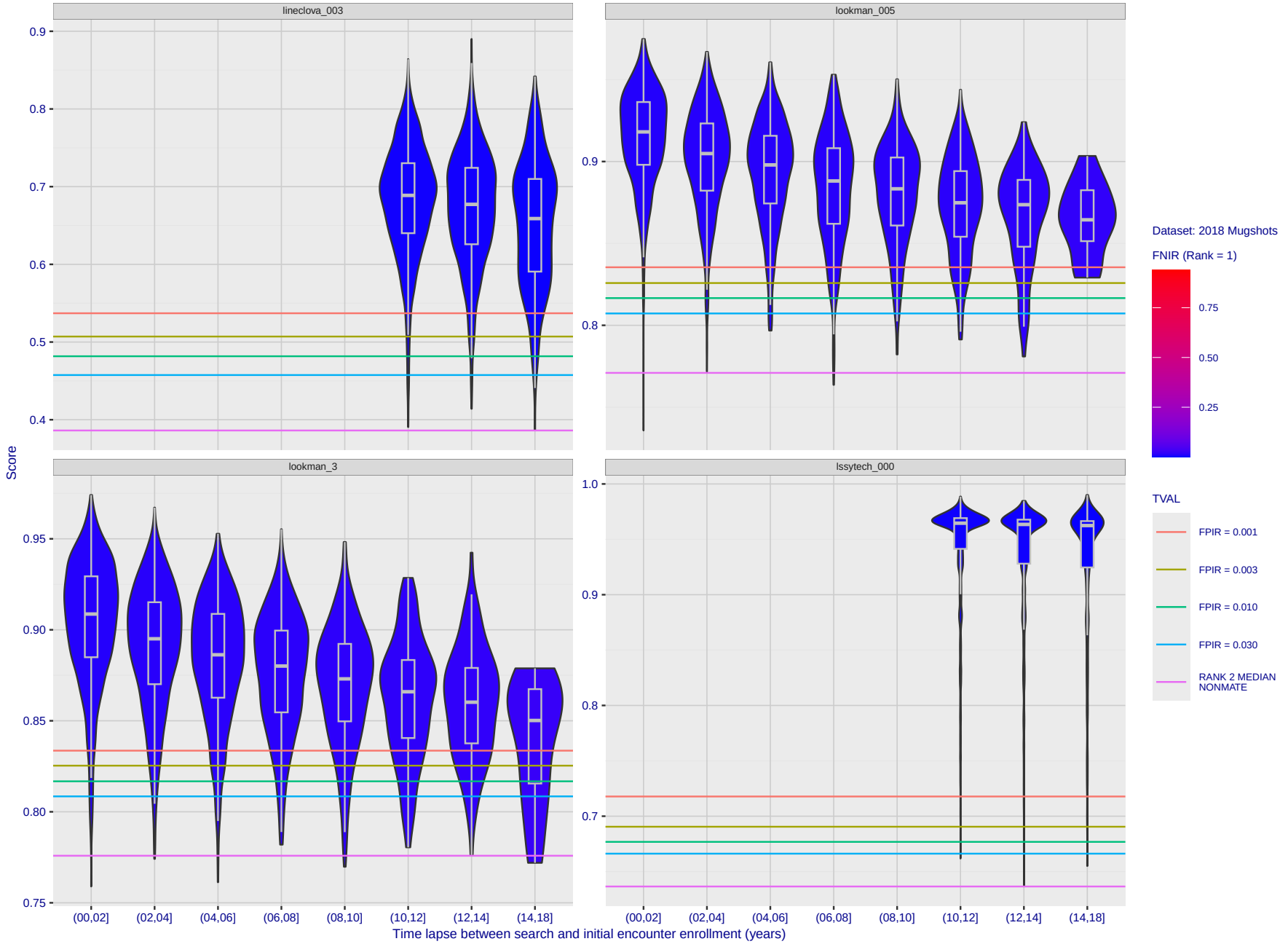


*Figure 392:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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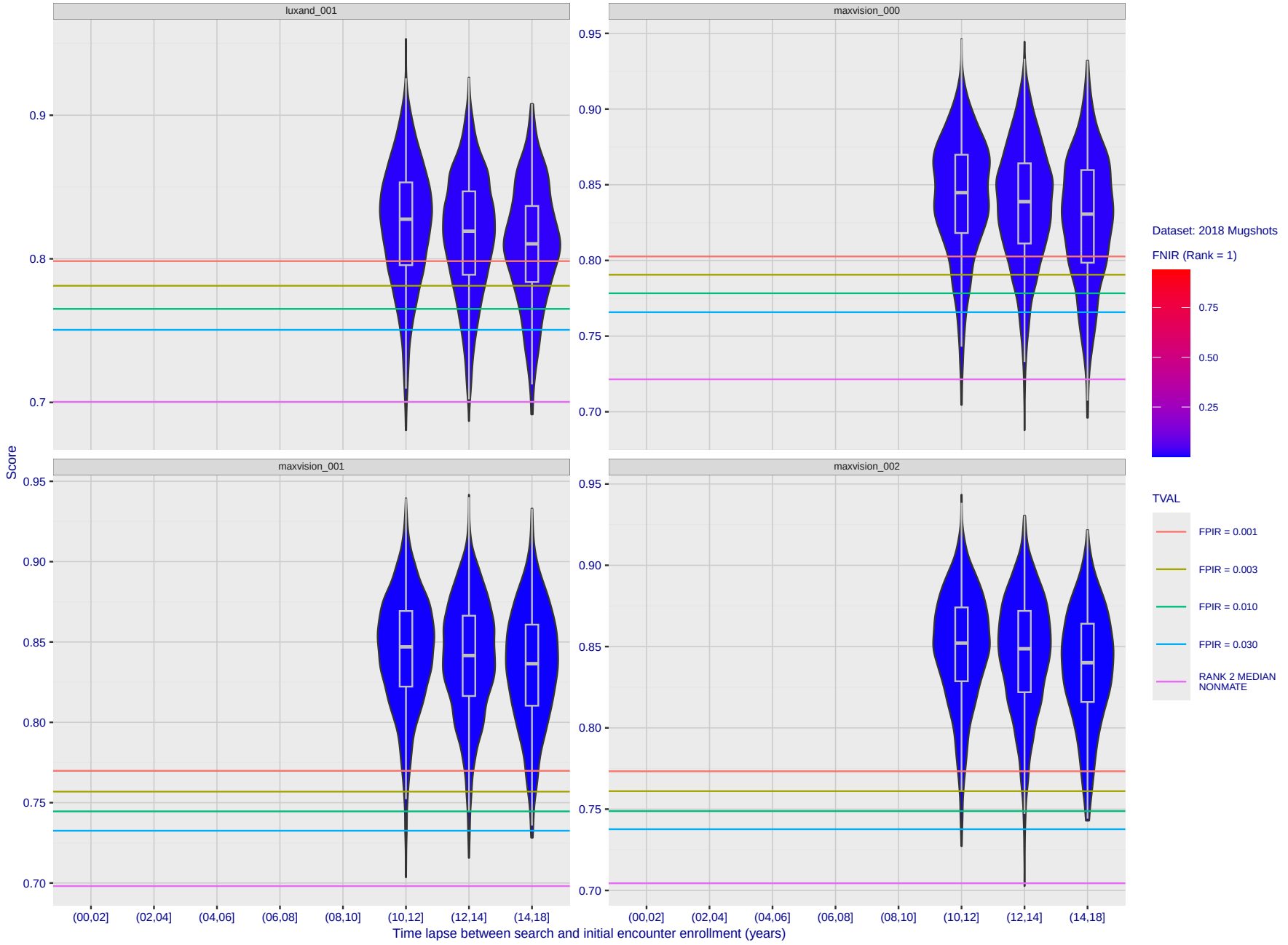


*Figure 393:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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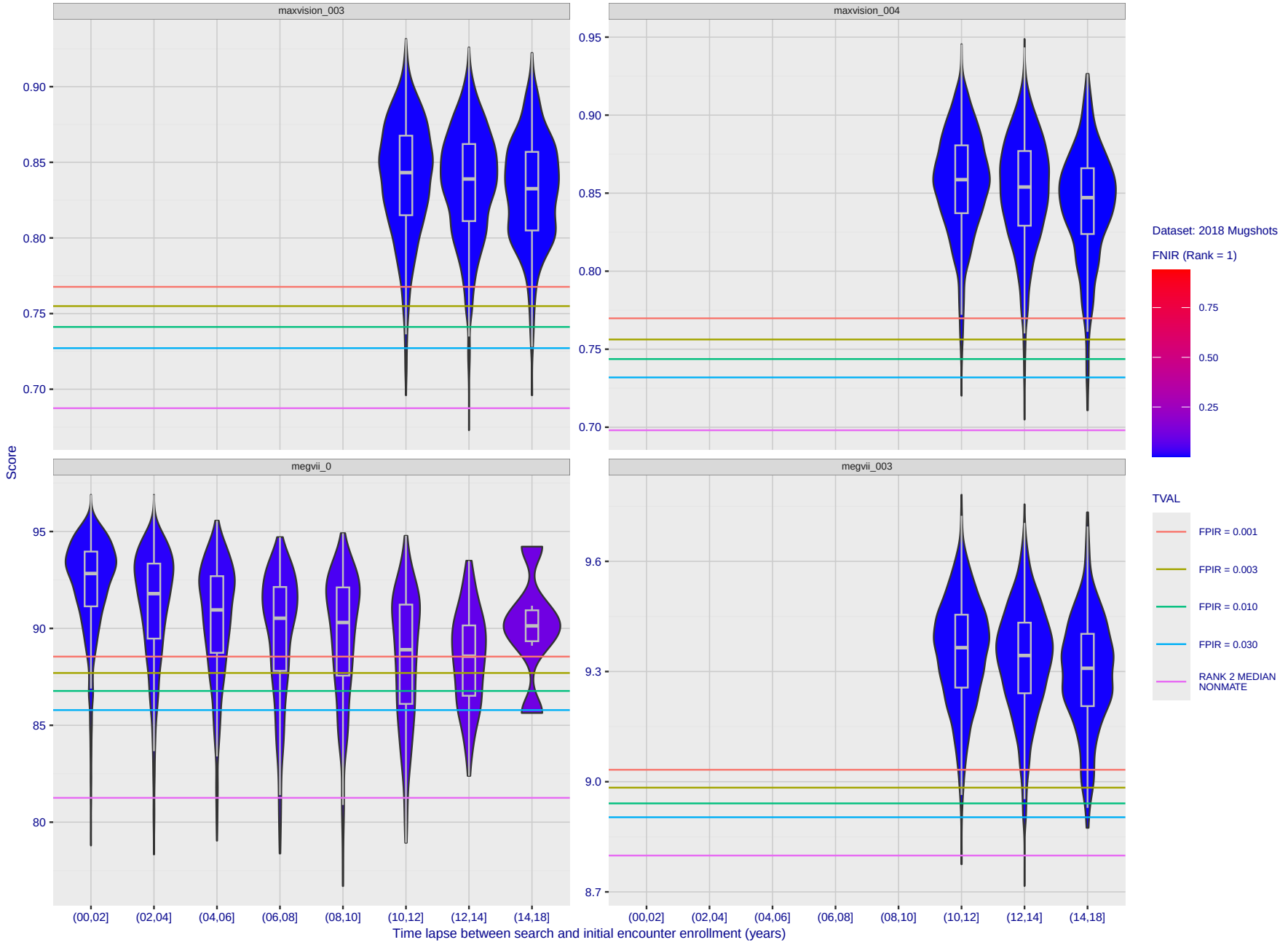


*Figure 394:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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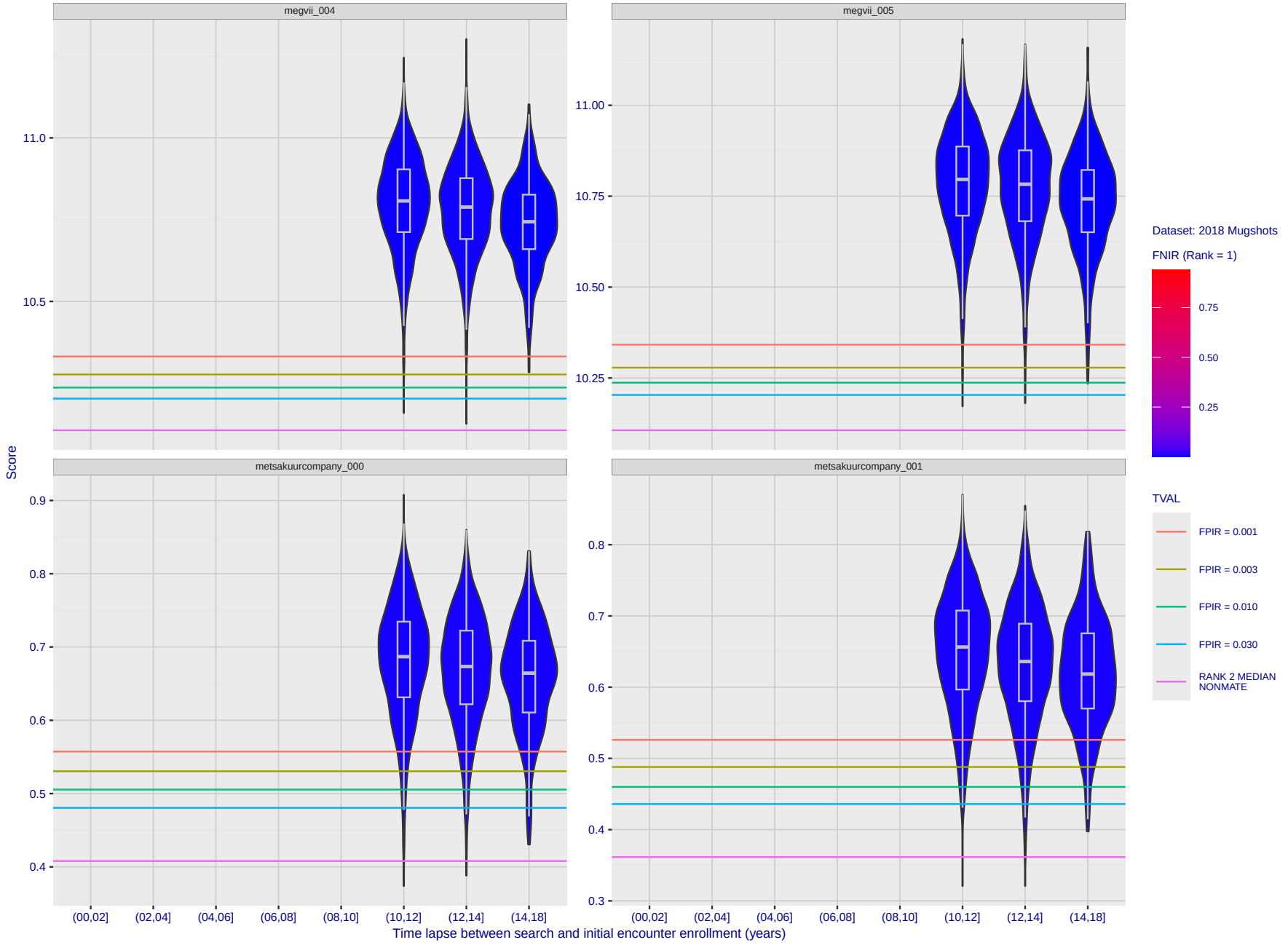


*Figure 395:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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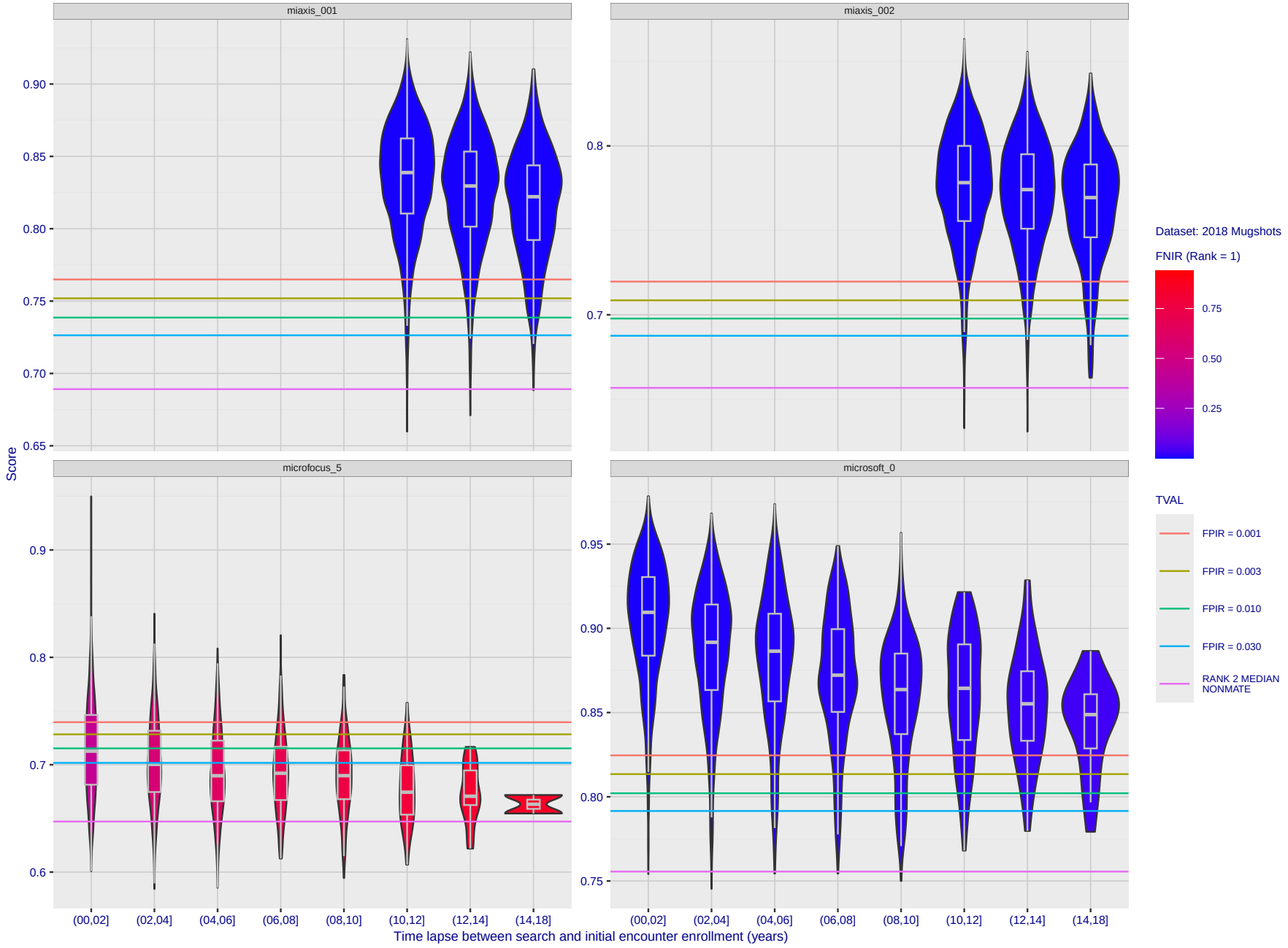


*Figure 396:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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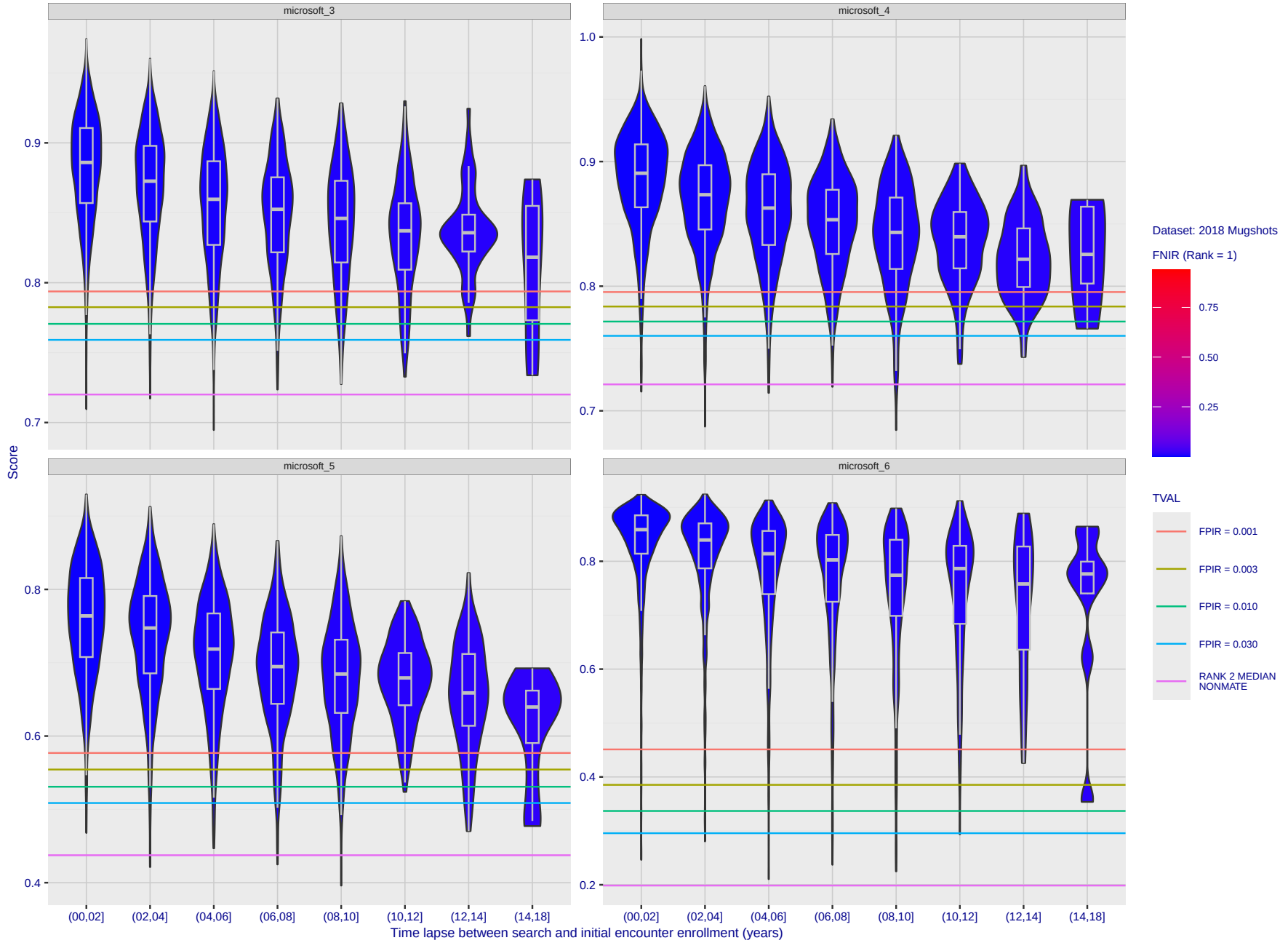


*Figure 397:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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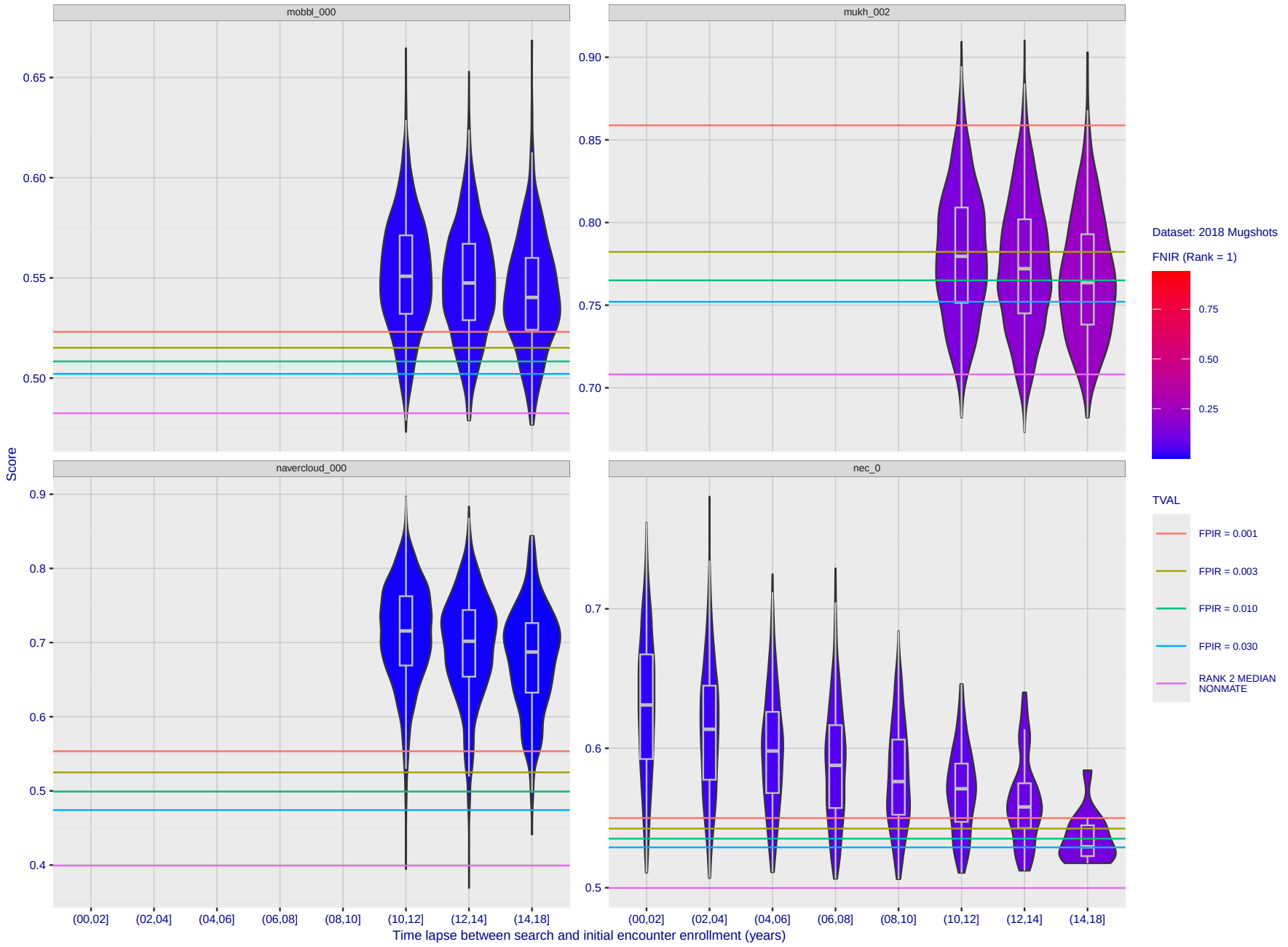


*Figure 398:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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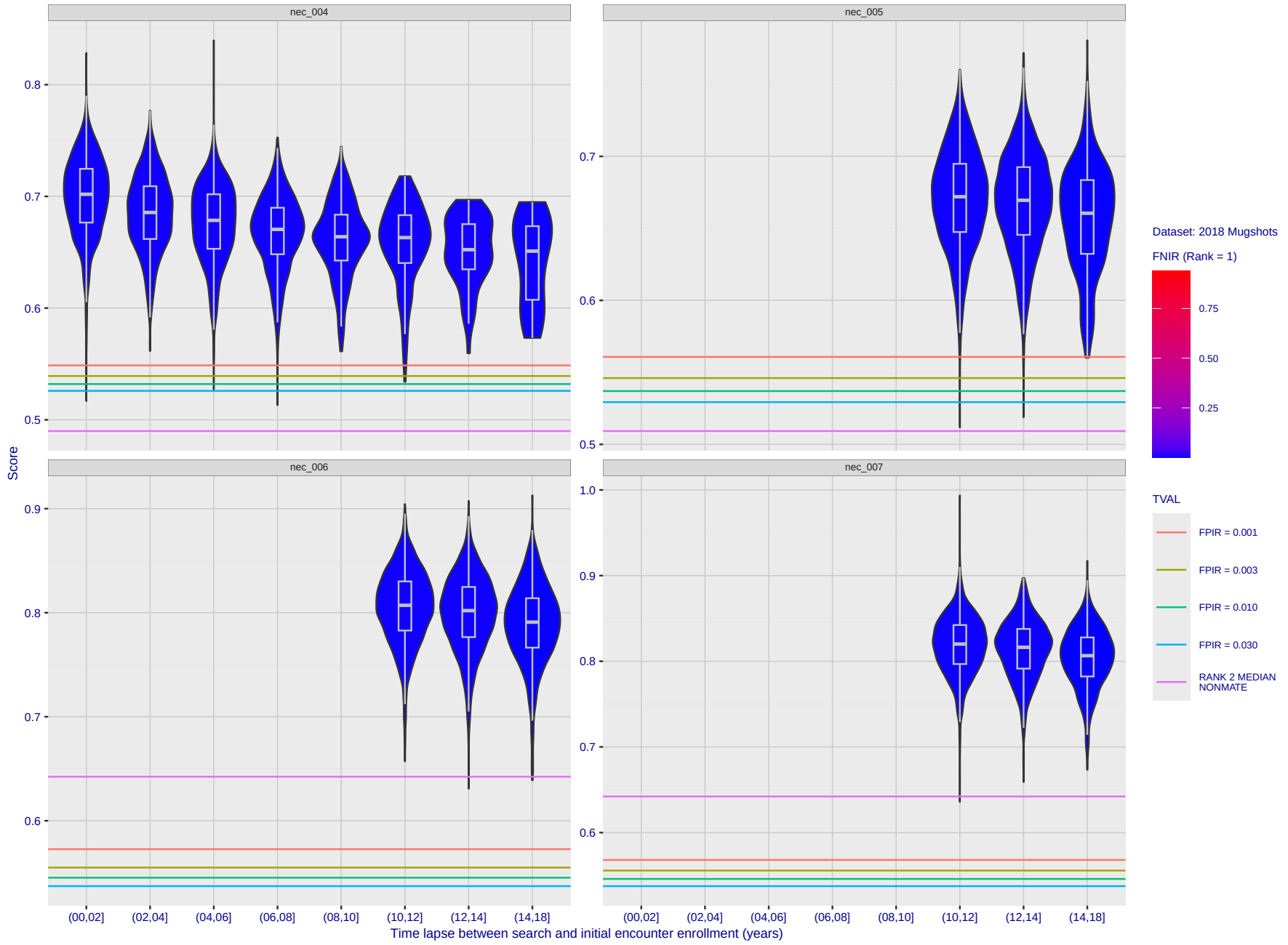


*Figure 399:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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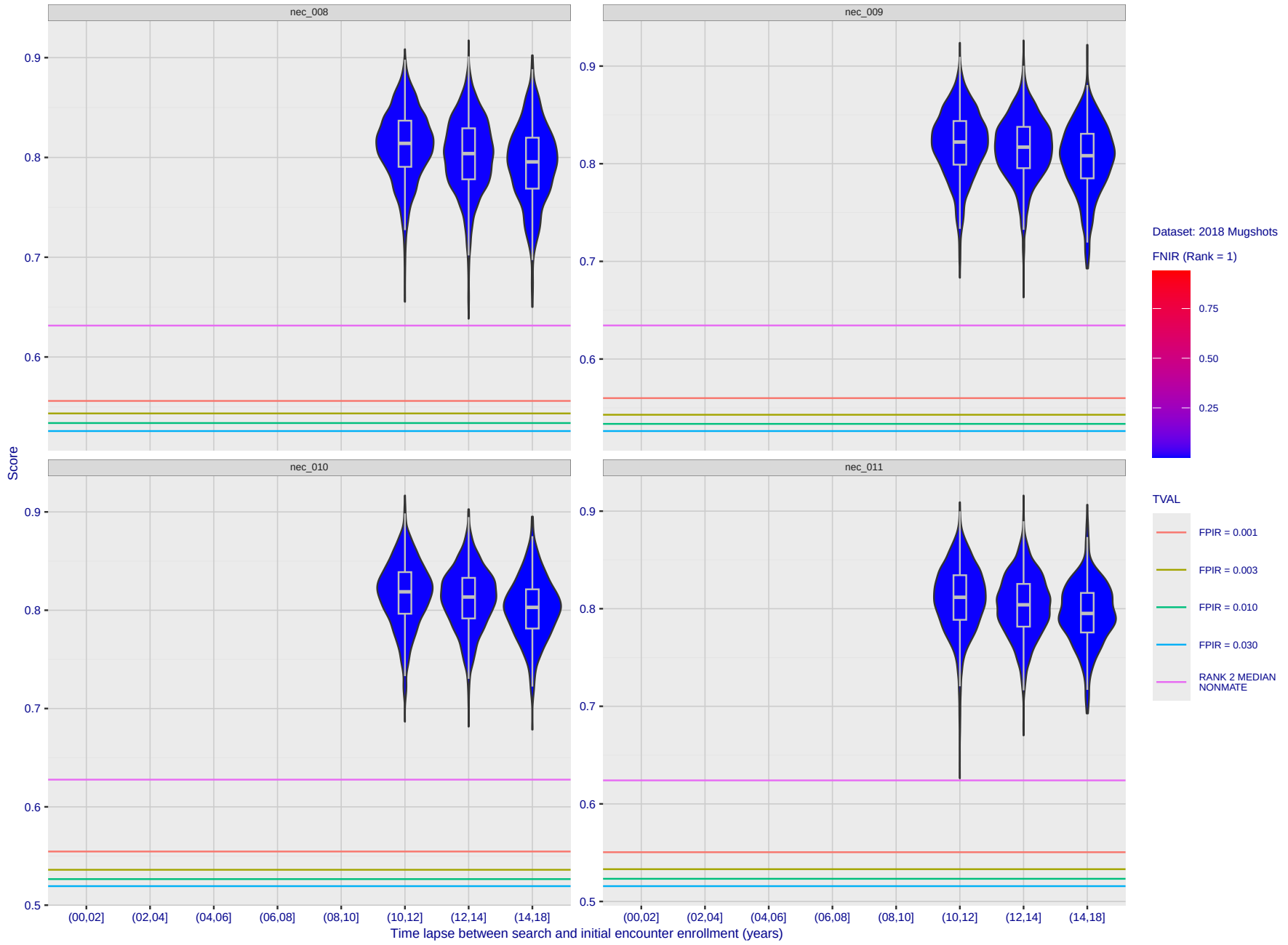


*Figure 400:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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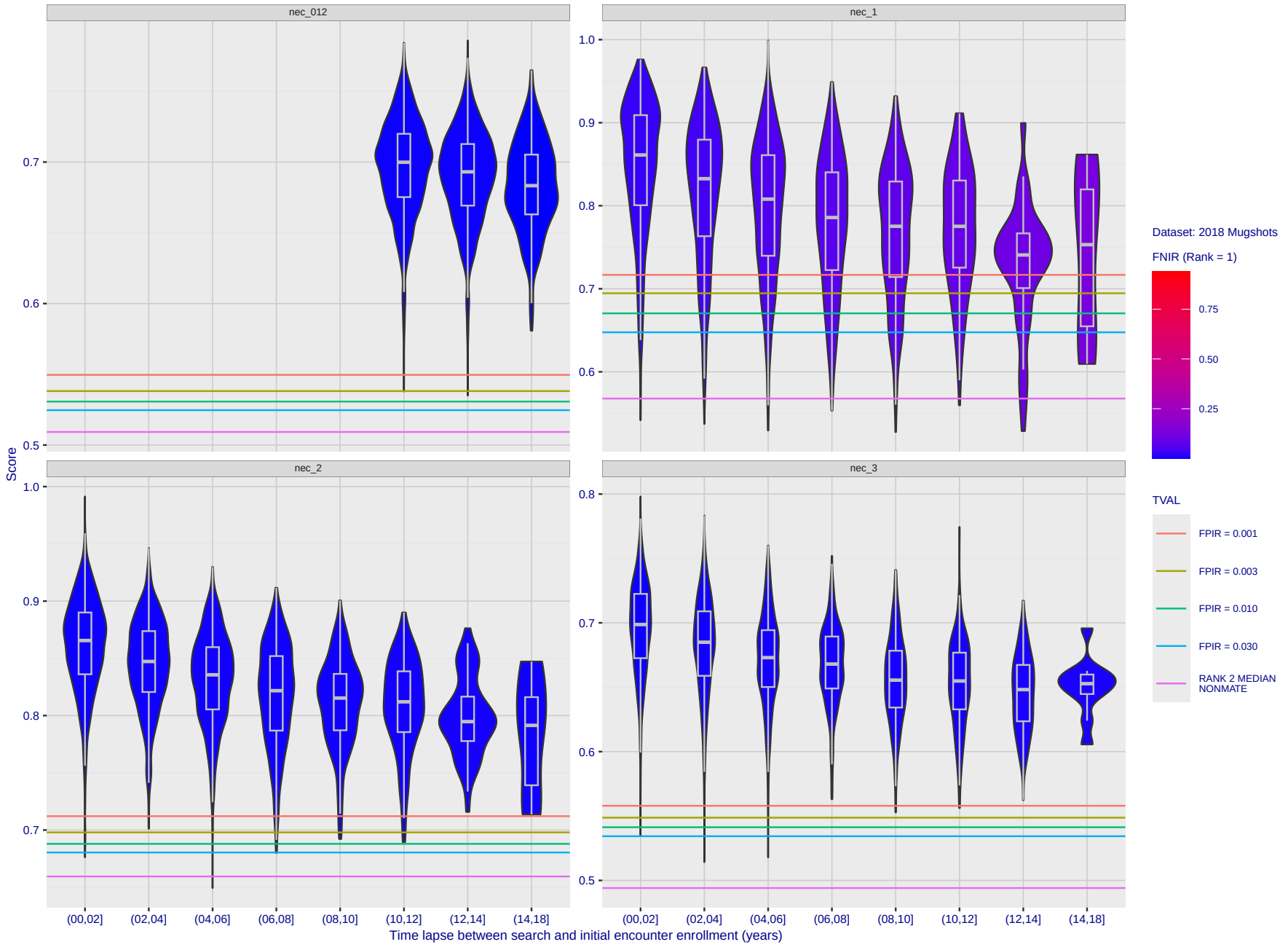


*Figure 401:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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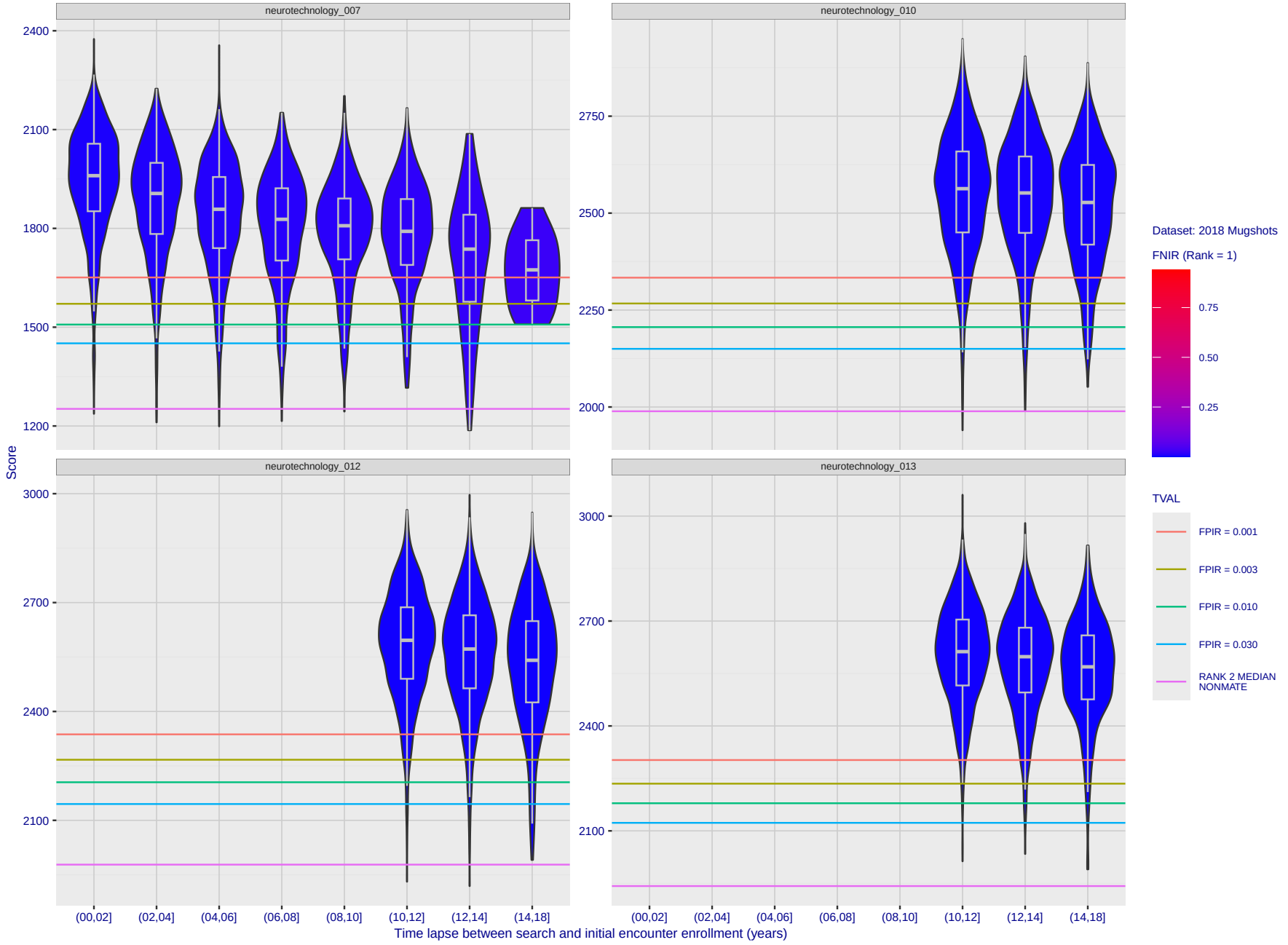


*Figure 402:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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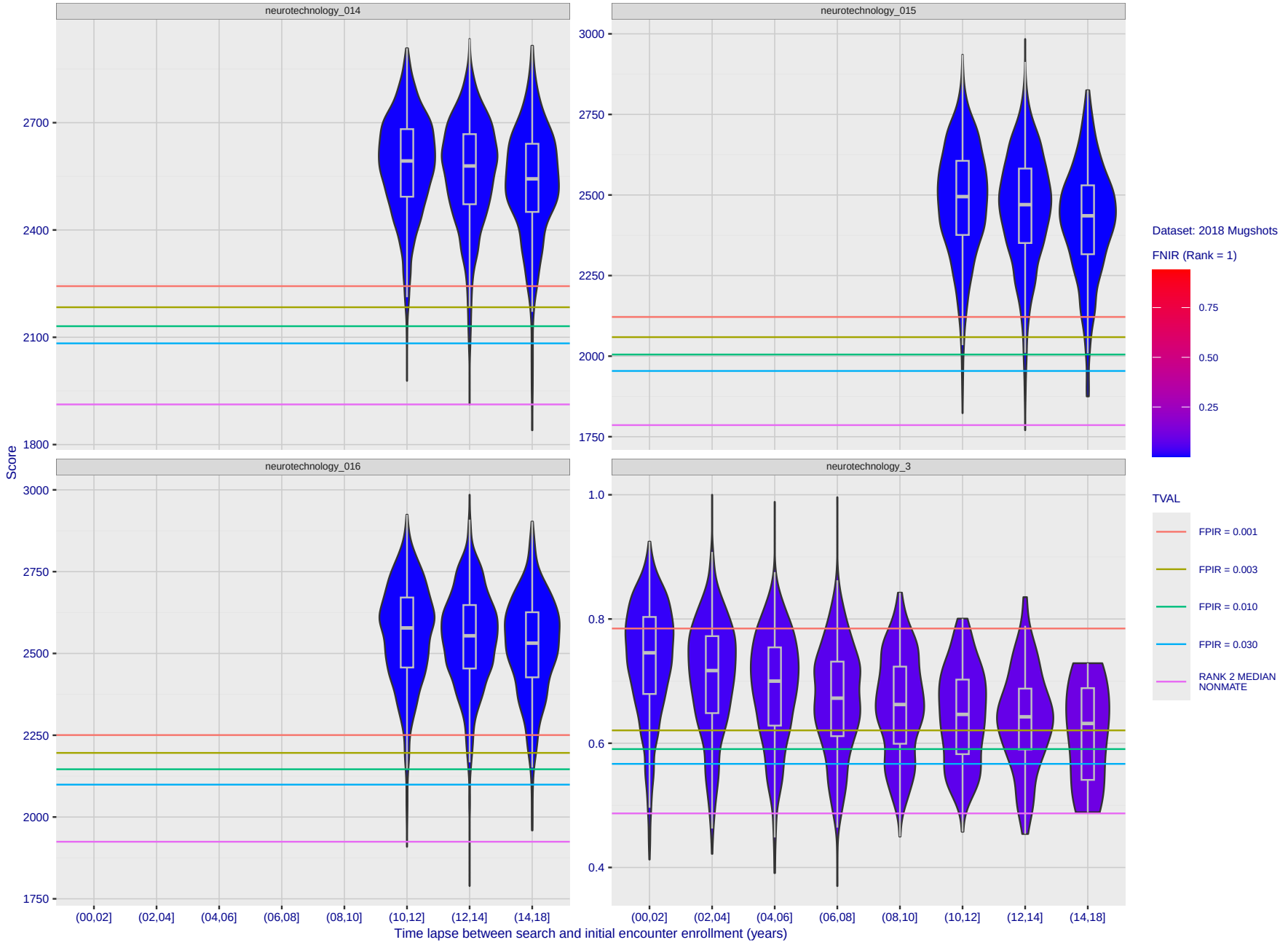


*Figure 403:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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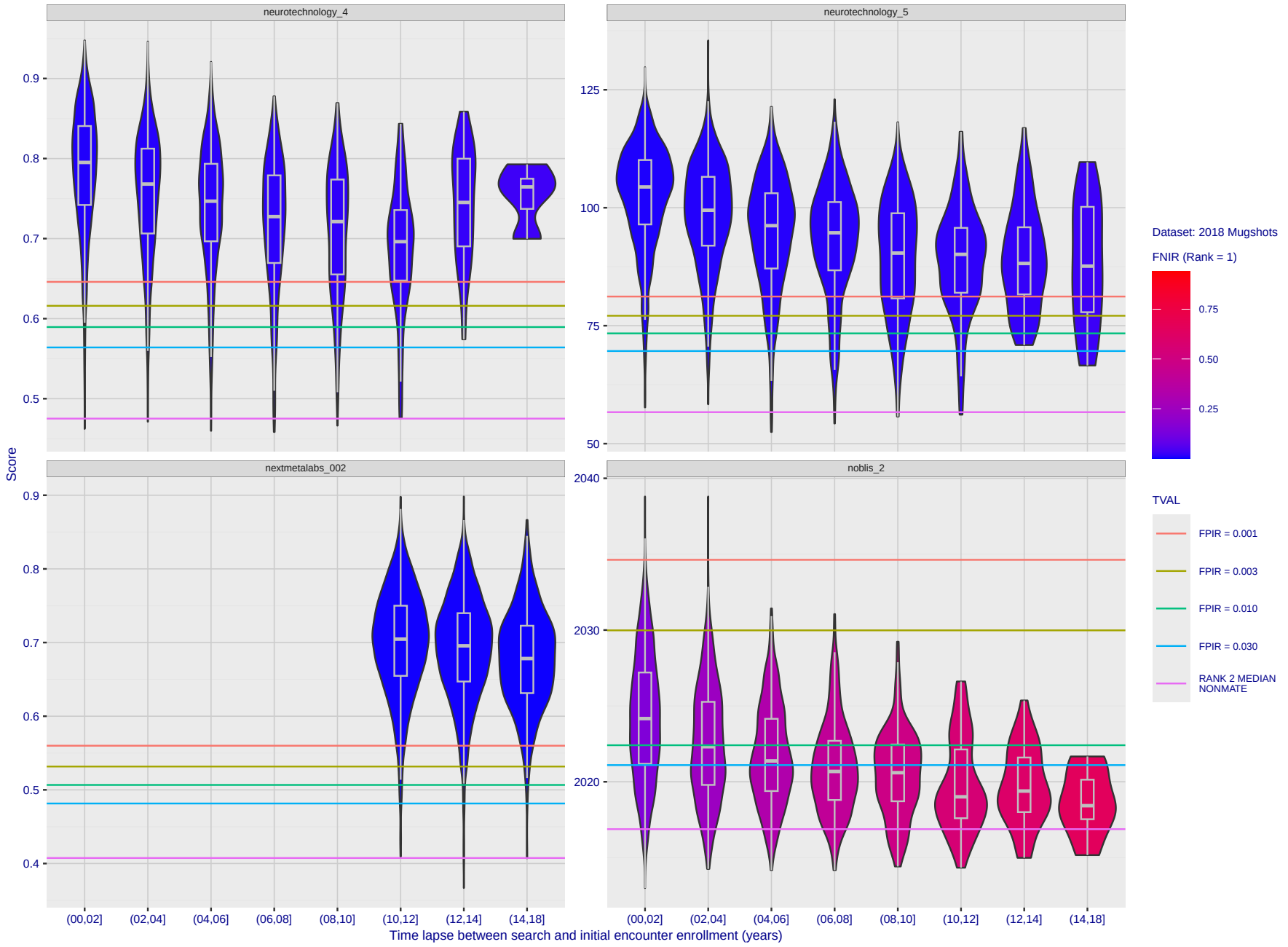


*Figure 404:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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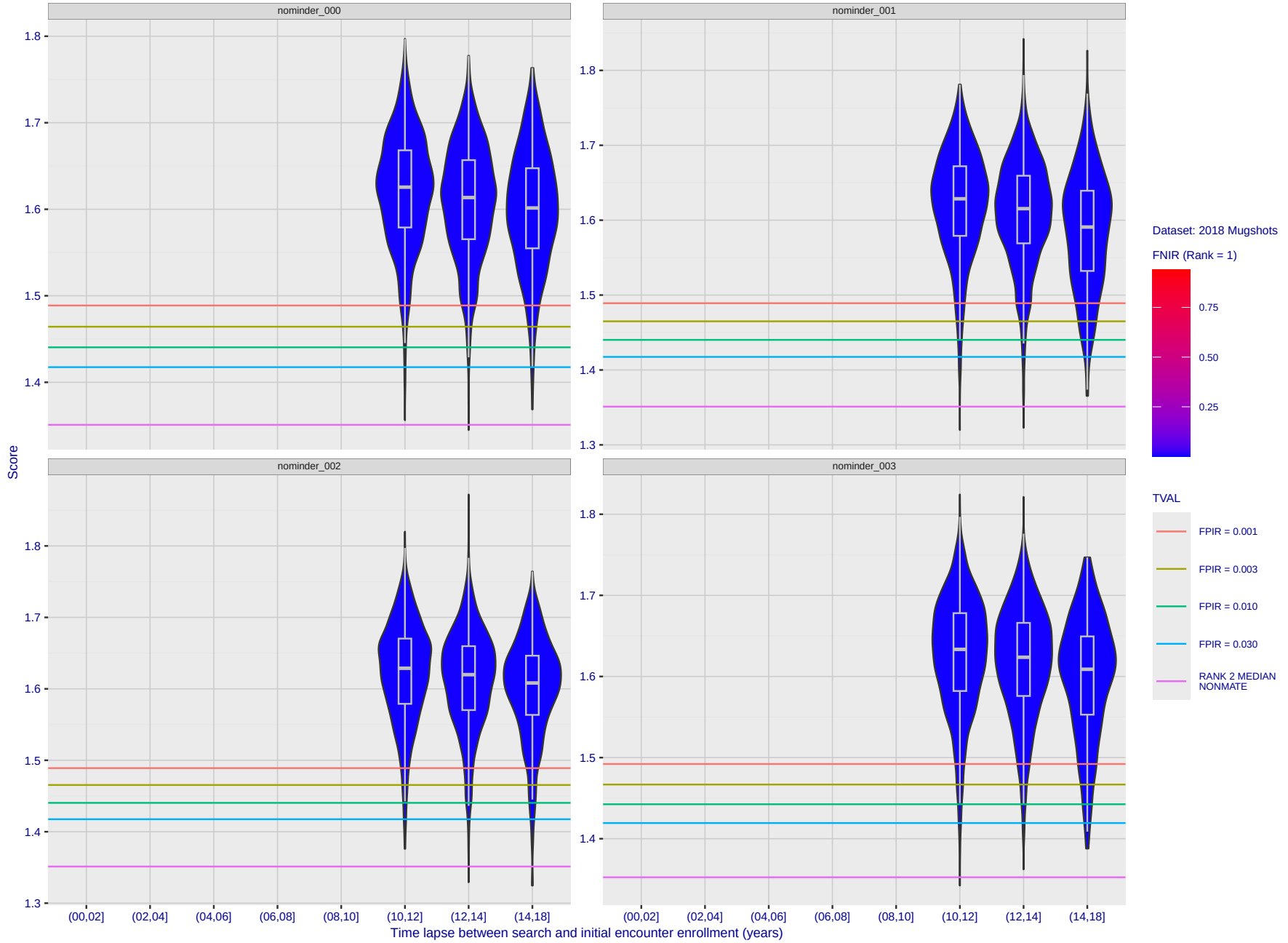


*Figure 405:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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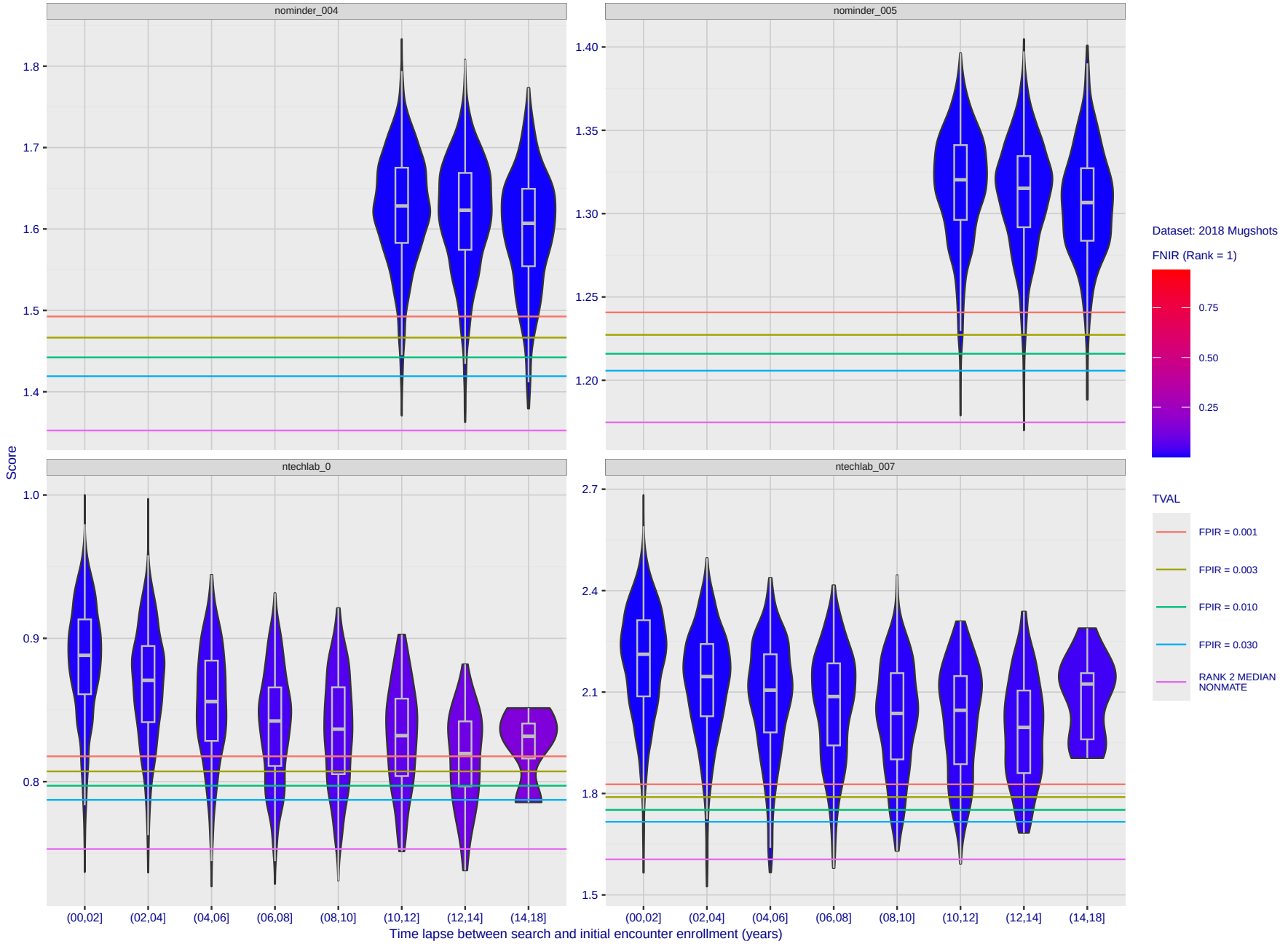


*Figure 406:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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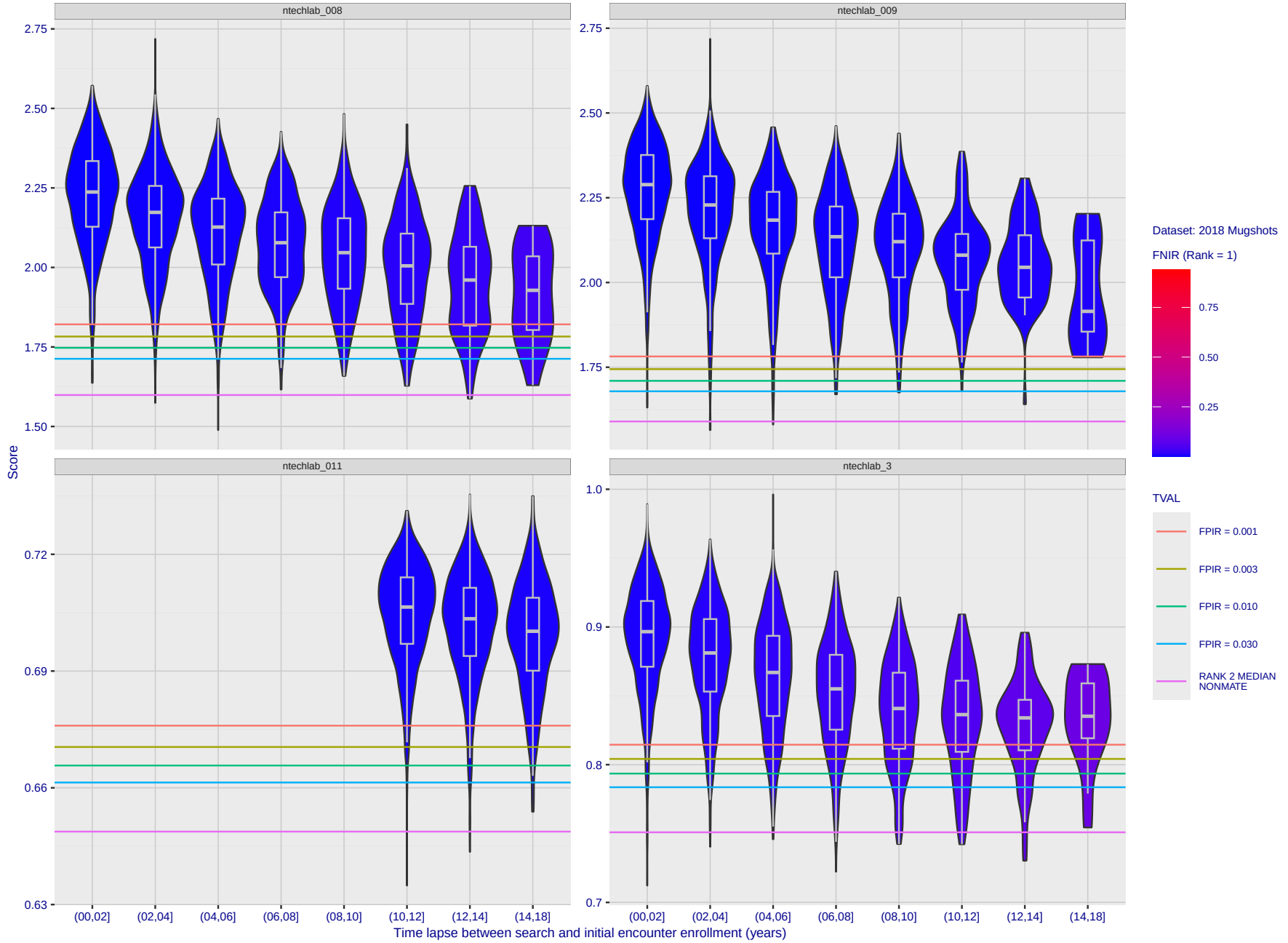


*Figure 407:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32
FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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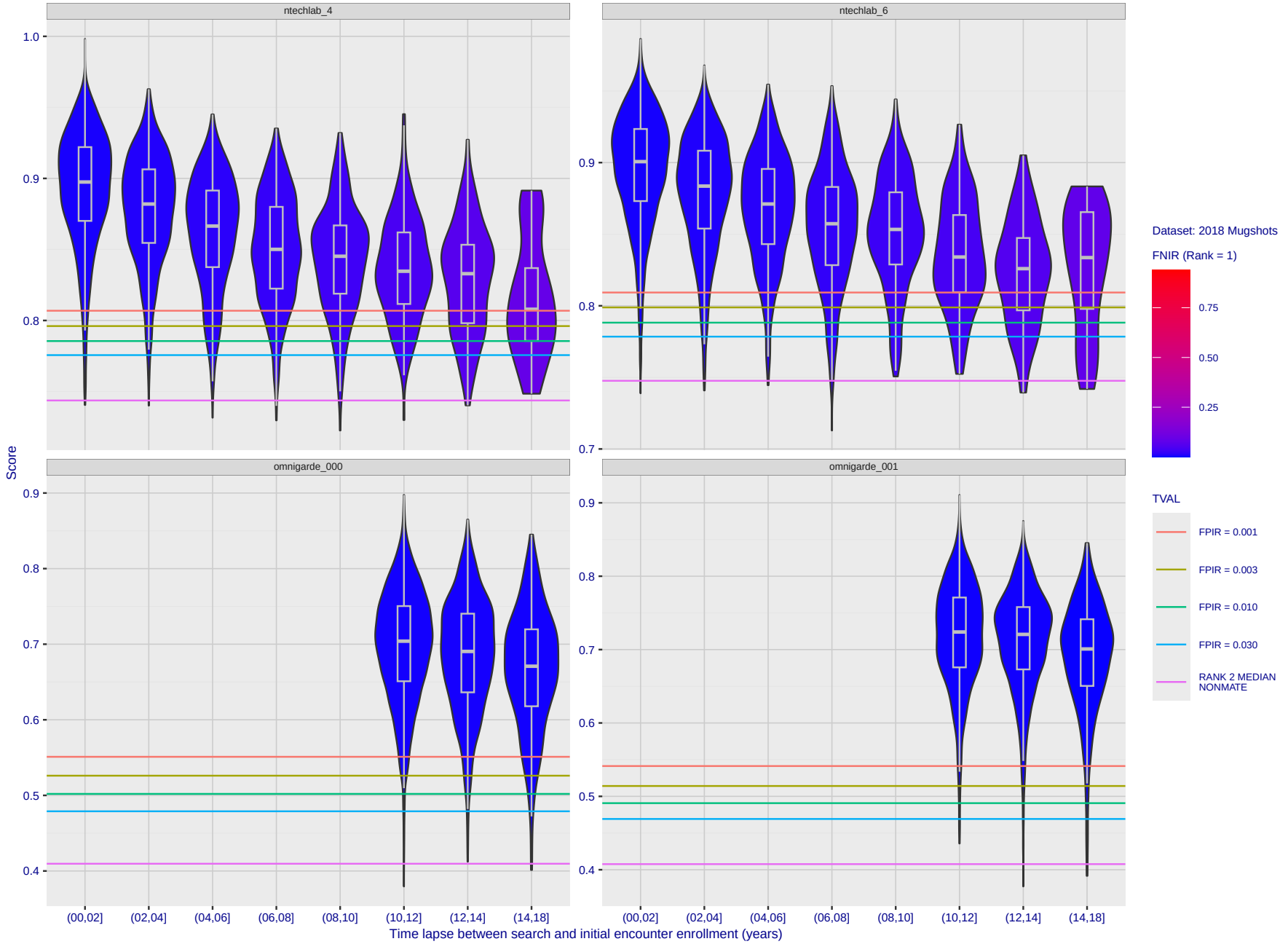


*Figure 408:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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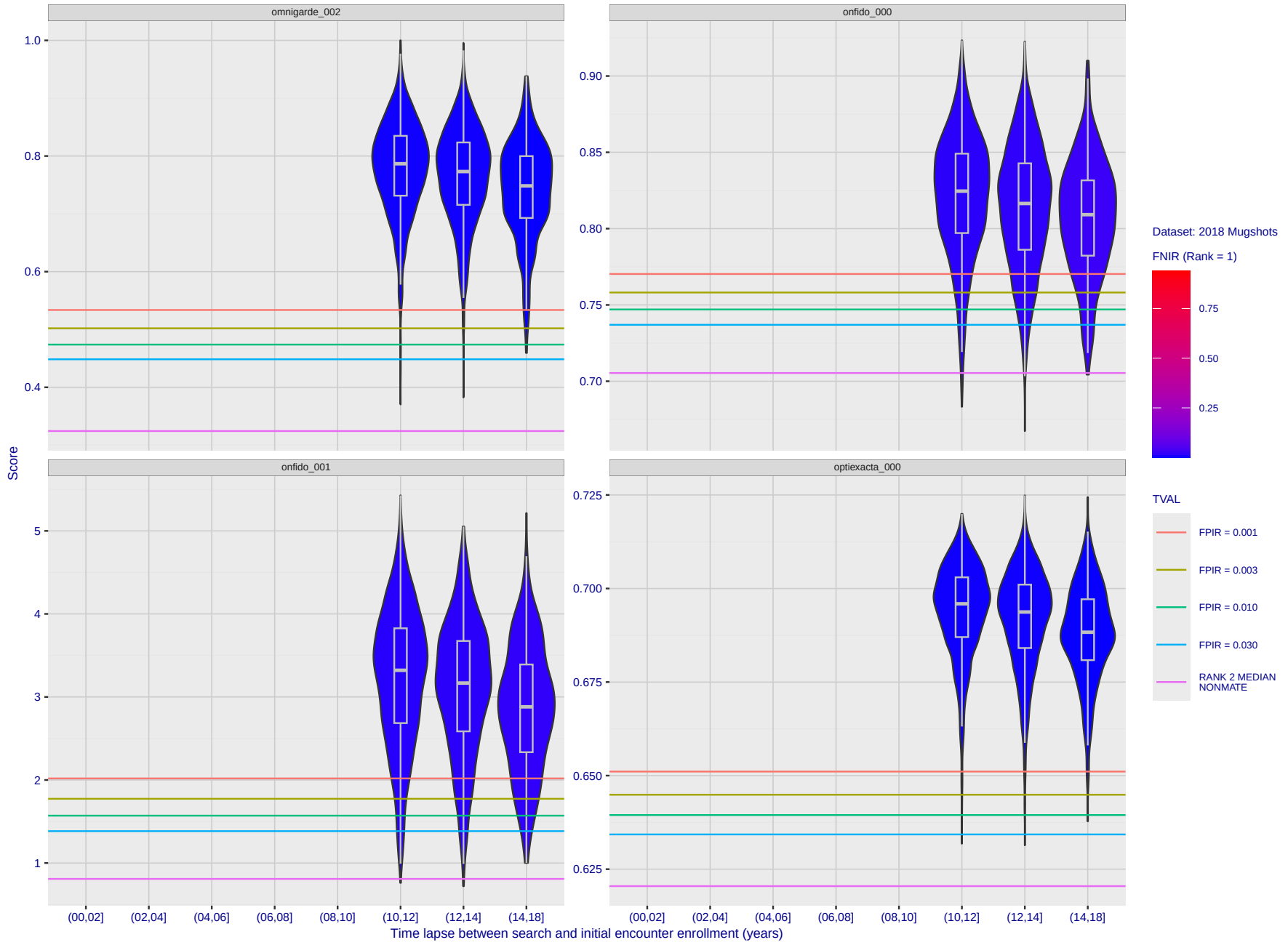


*Figure 409:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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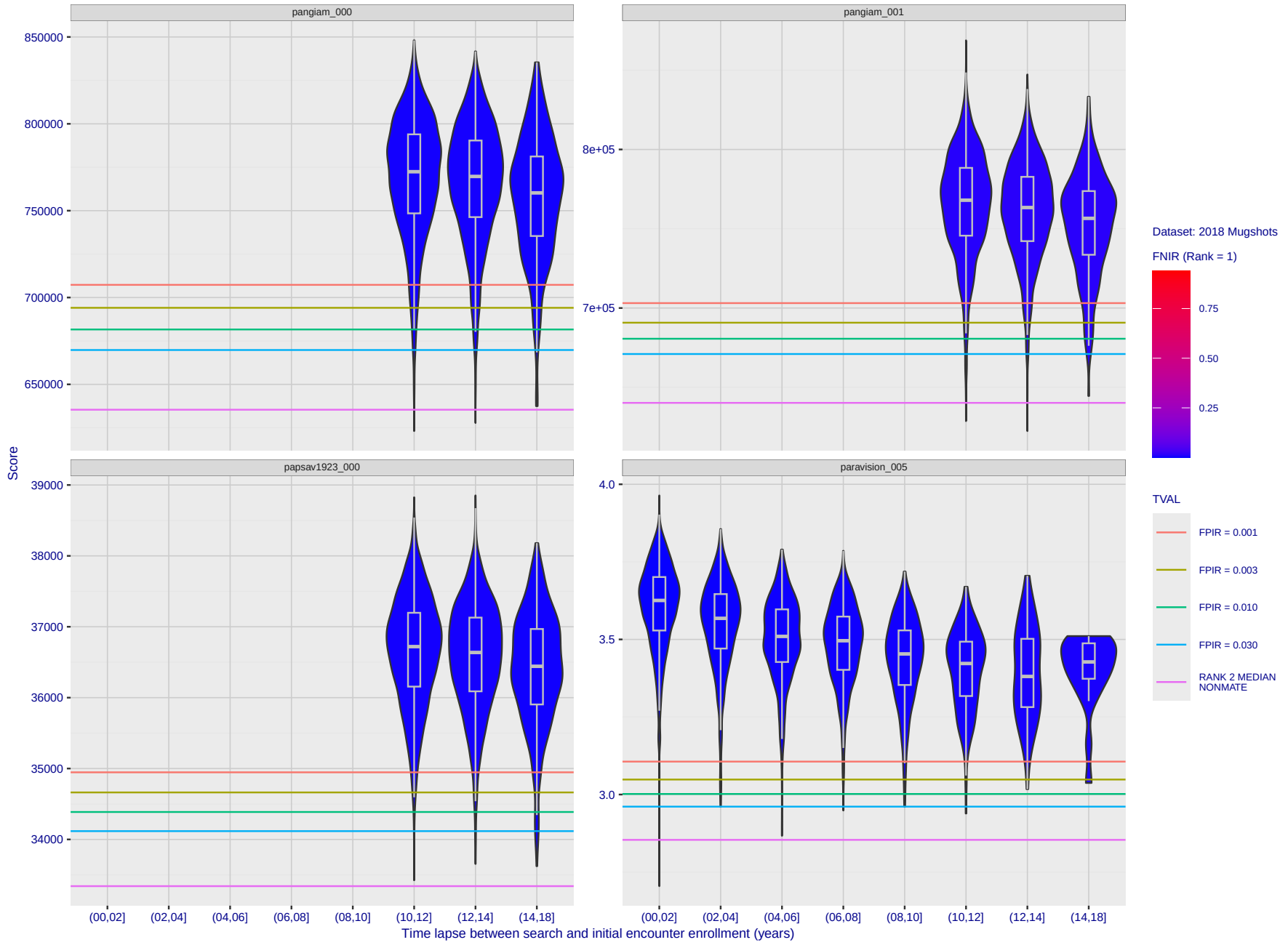


*Figure 410:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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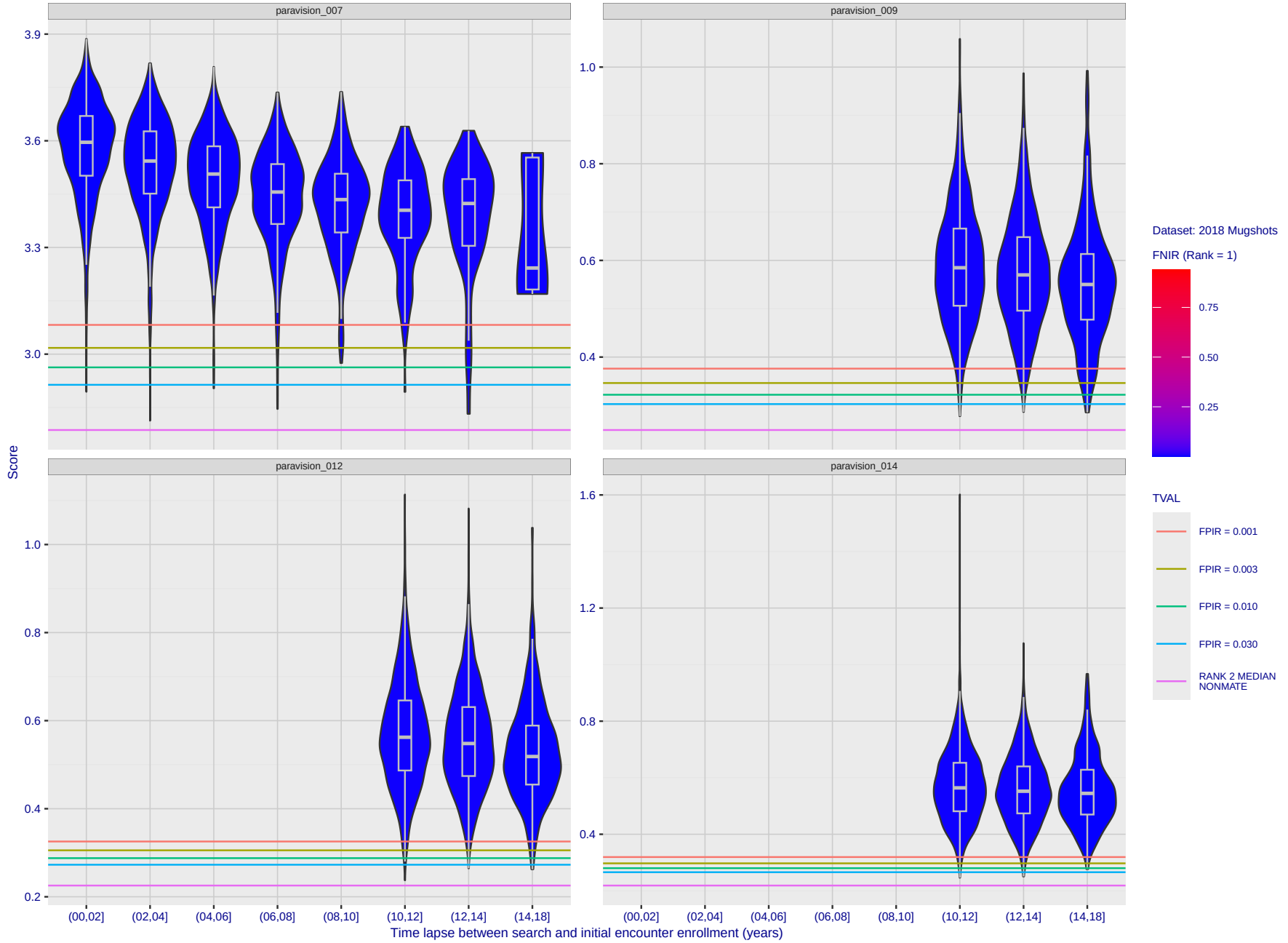


*Figure 411:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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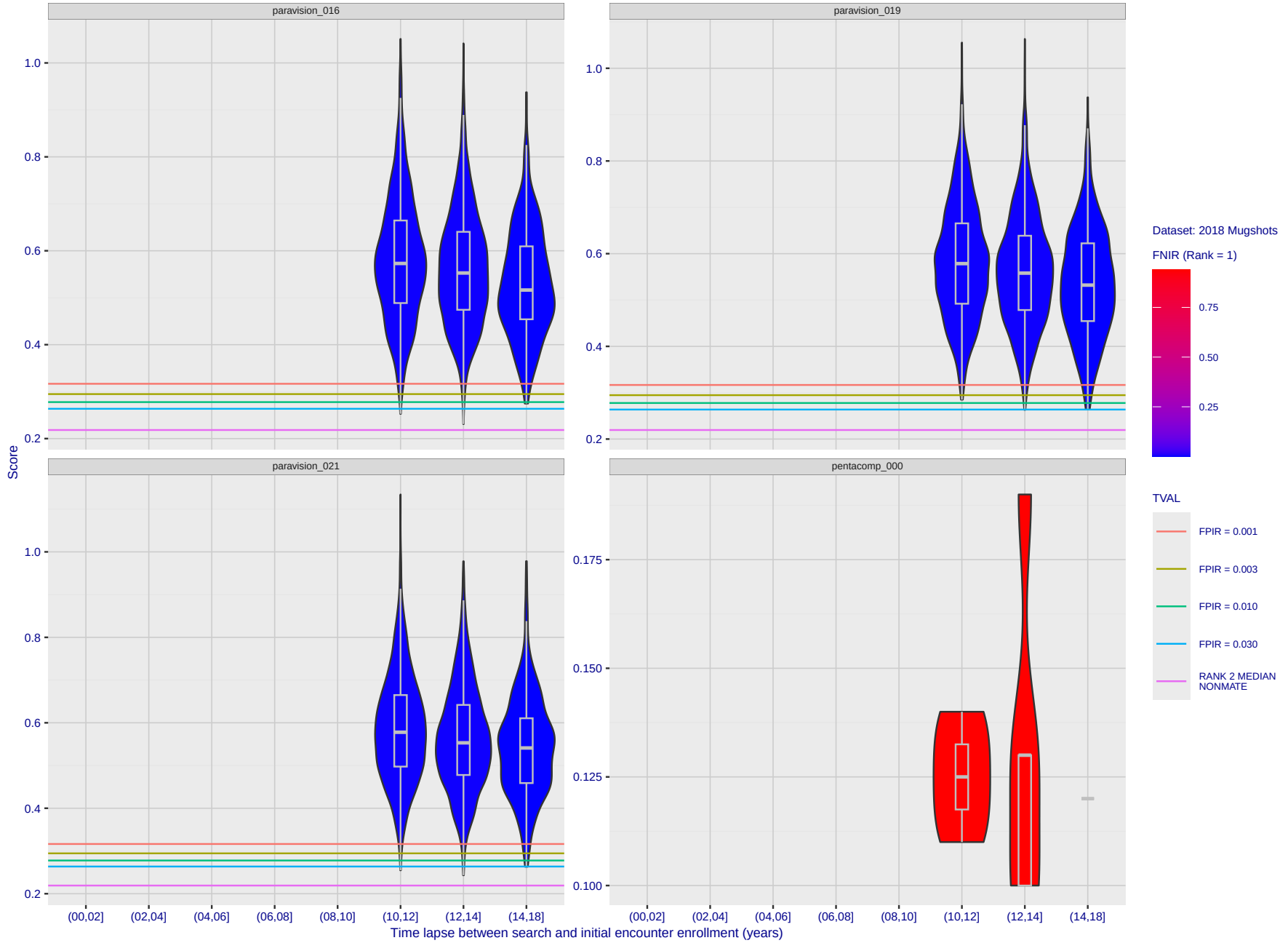


*Figure 412:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
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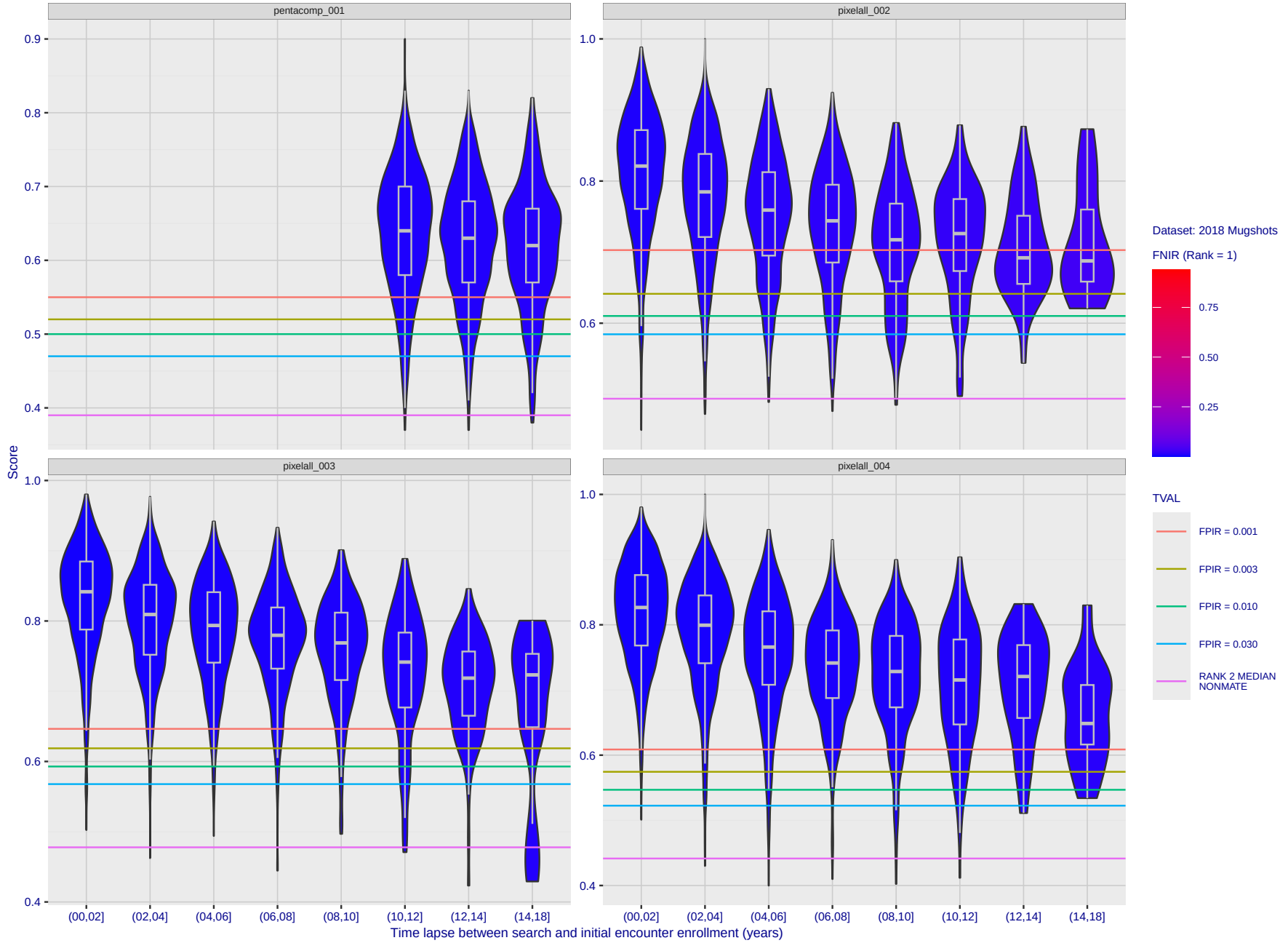


*Figure 413:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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R = Num. candidates examined
T = Threshold
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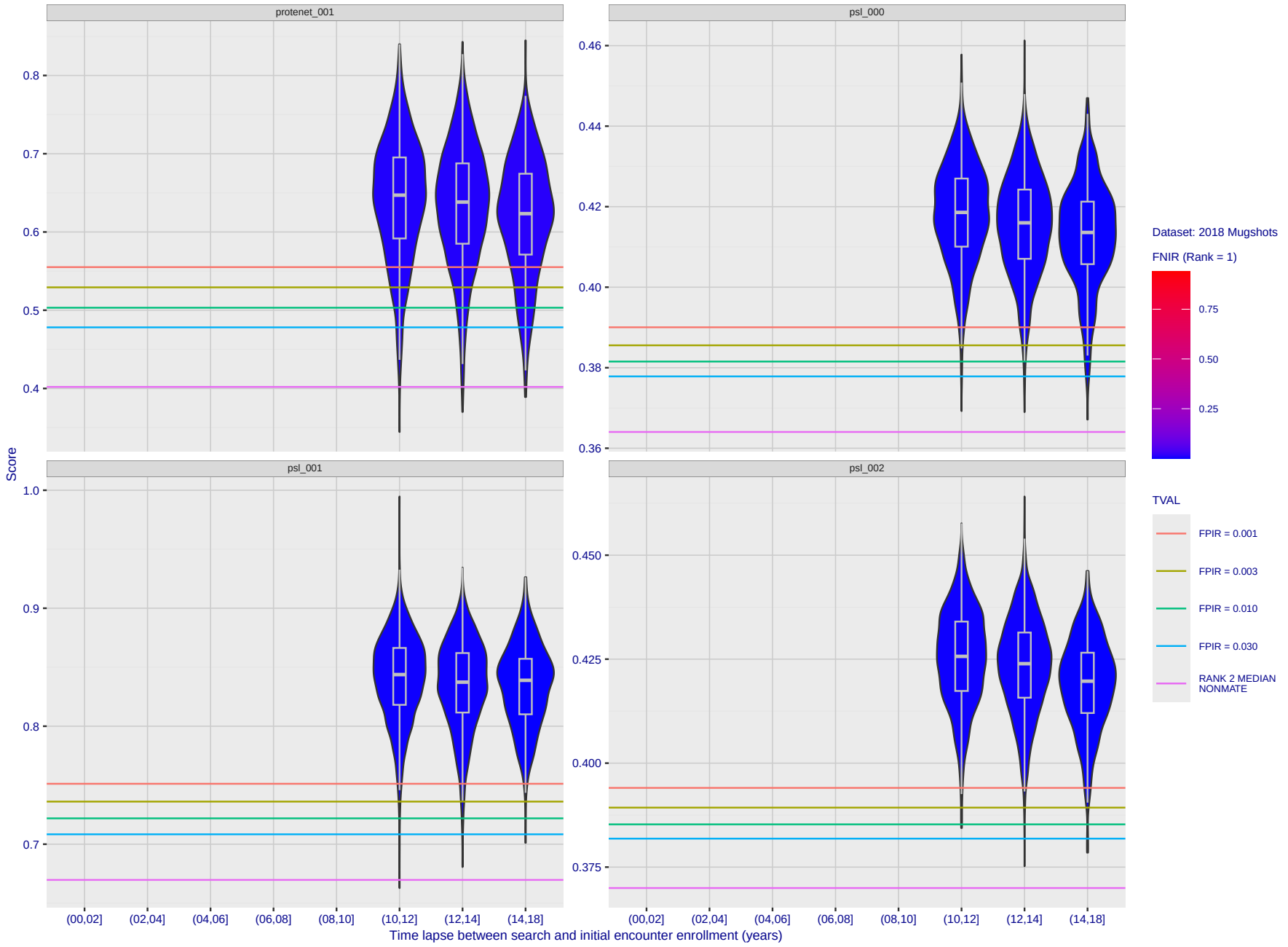


*Figure 414:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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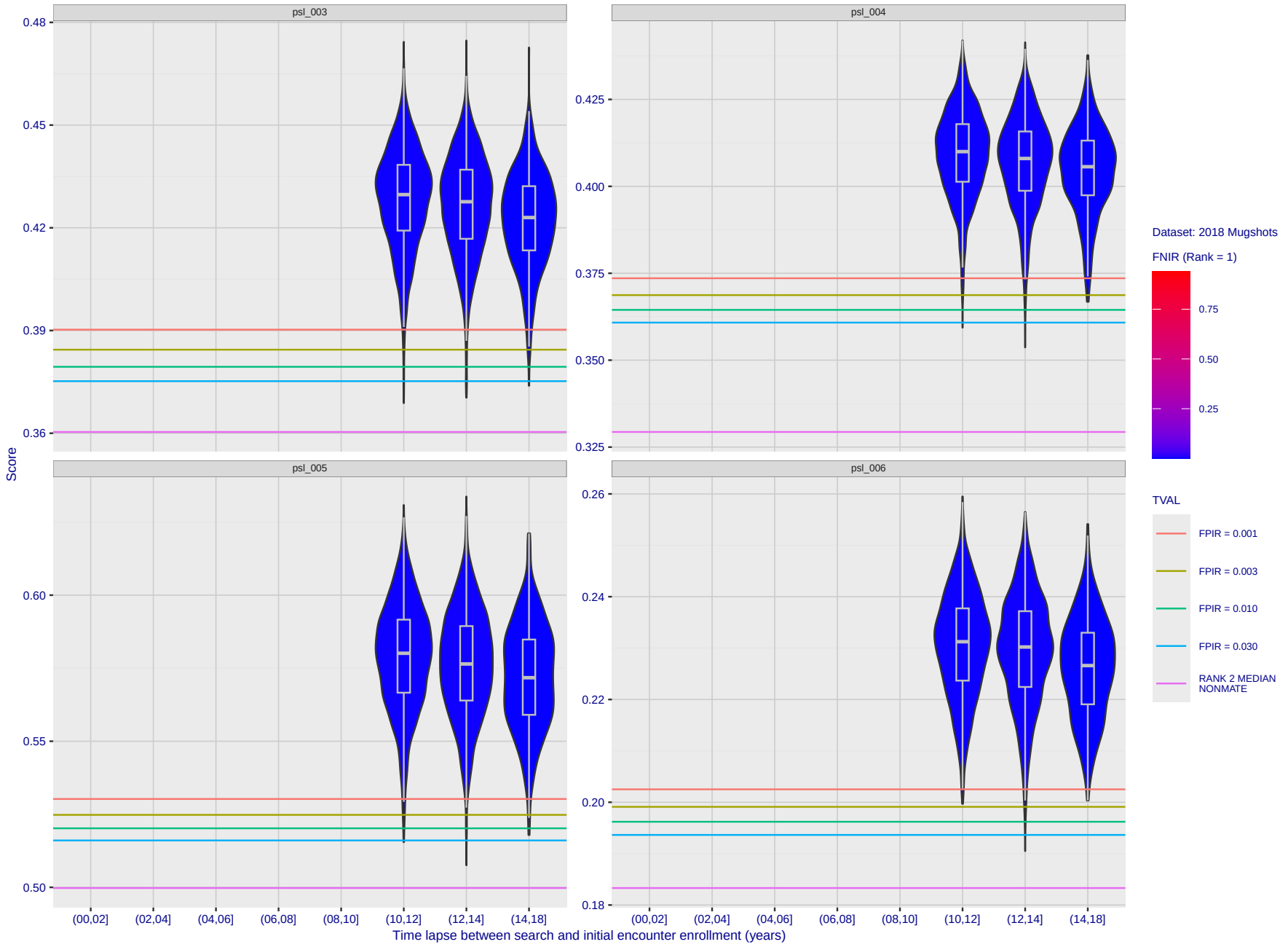


*Figure 415:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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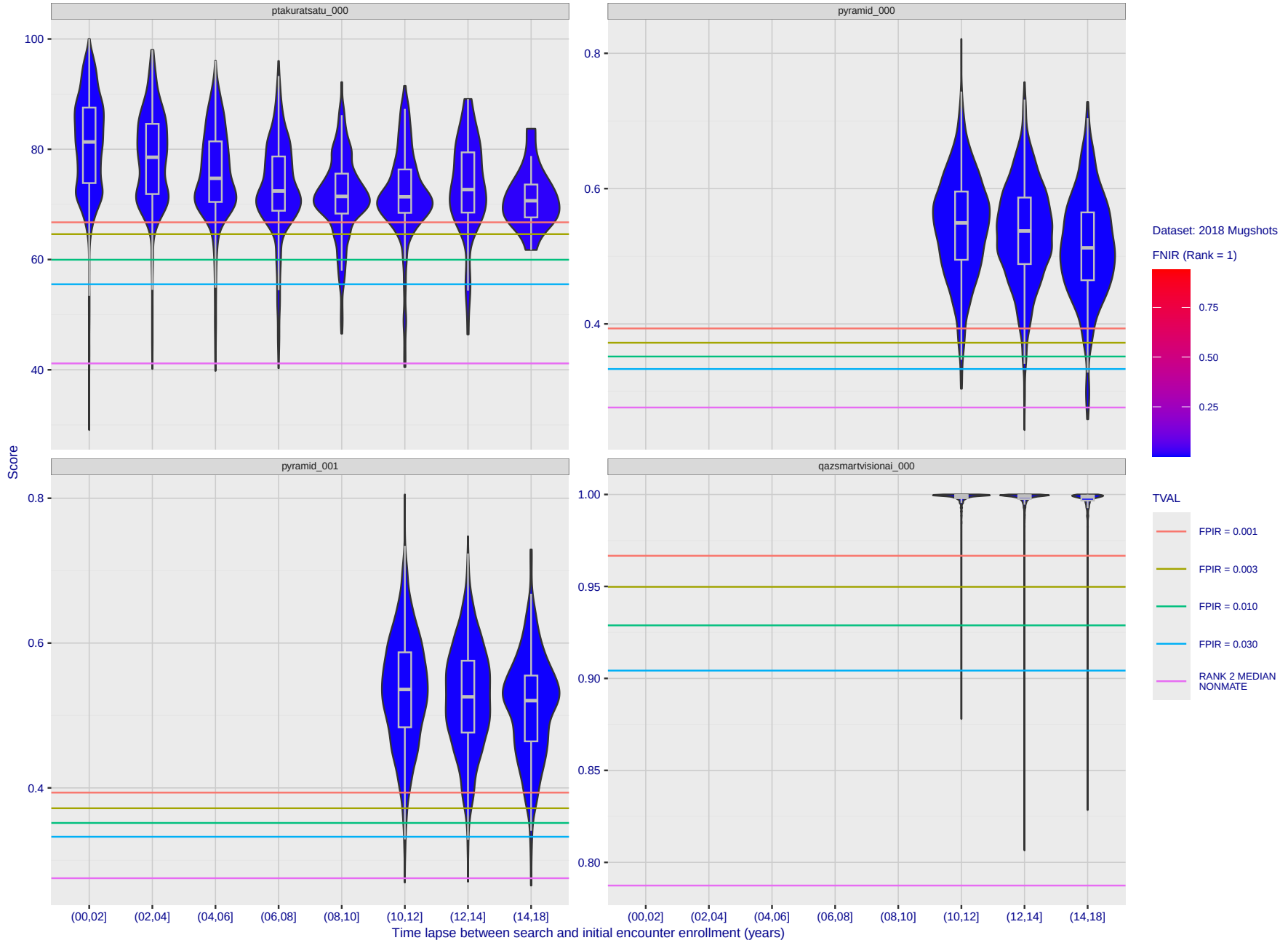


*Figure 416:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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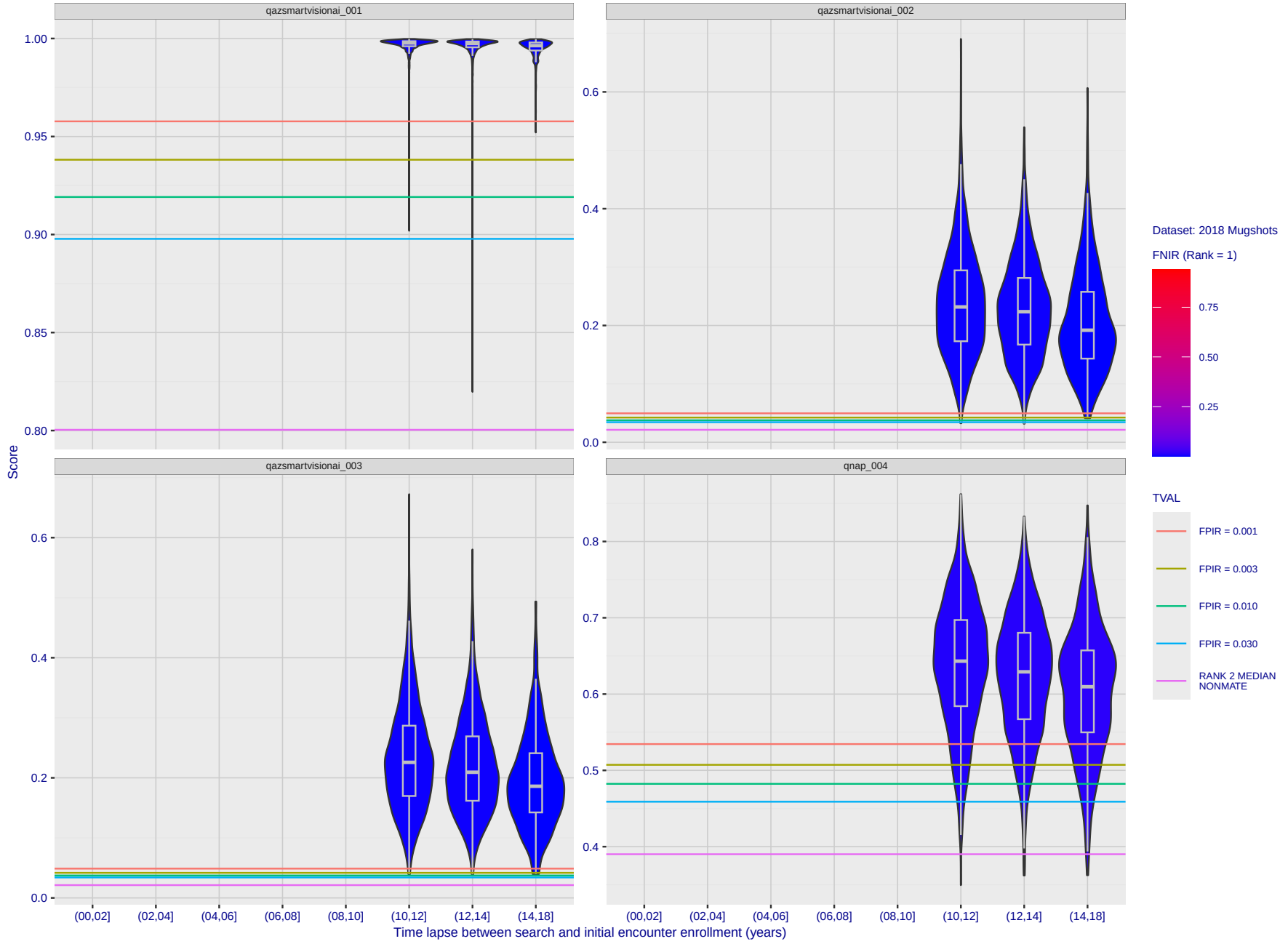


*Figure 417:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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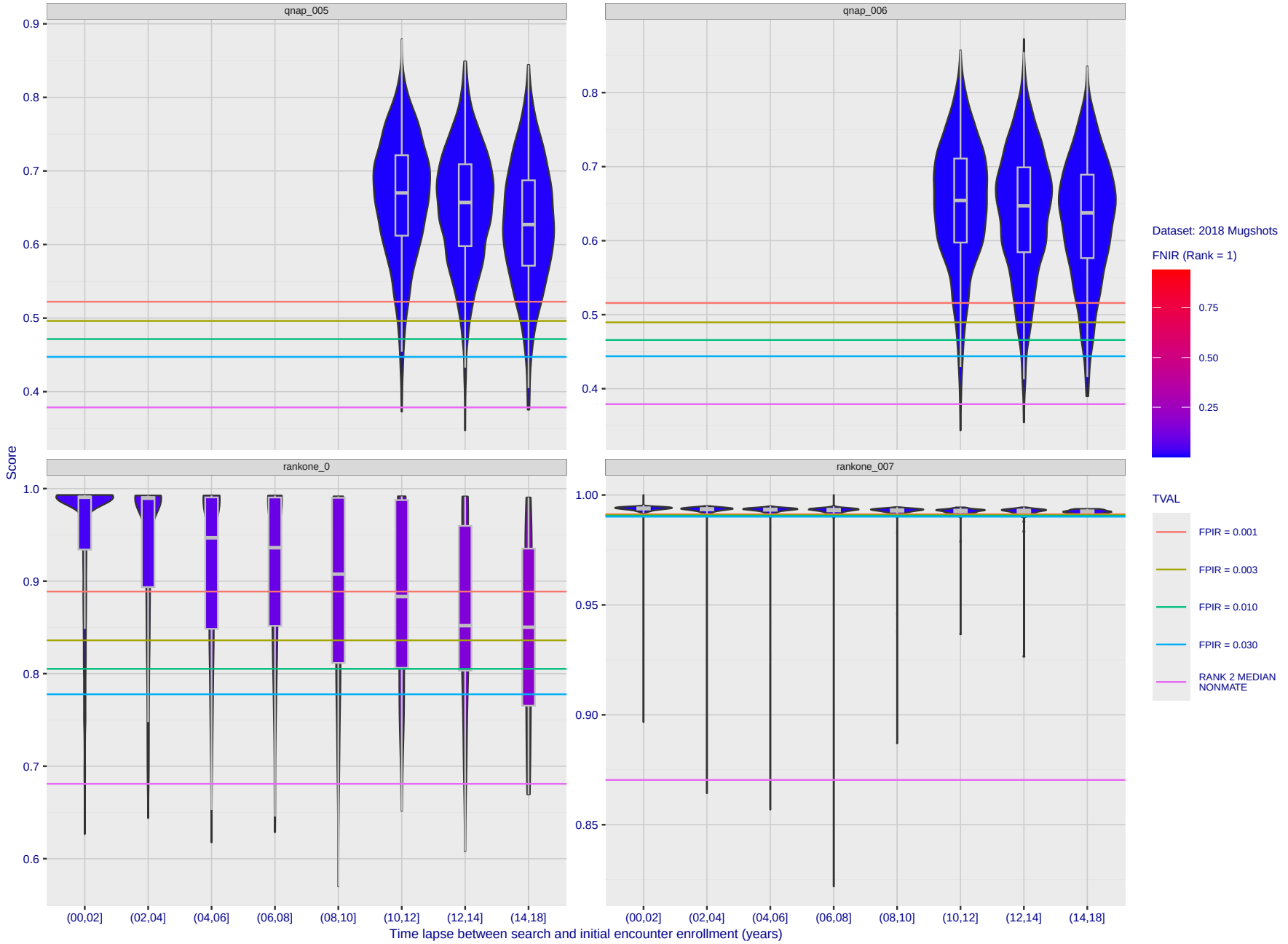


*Figure 418:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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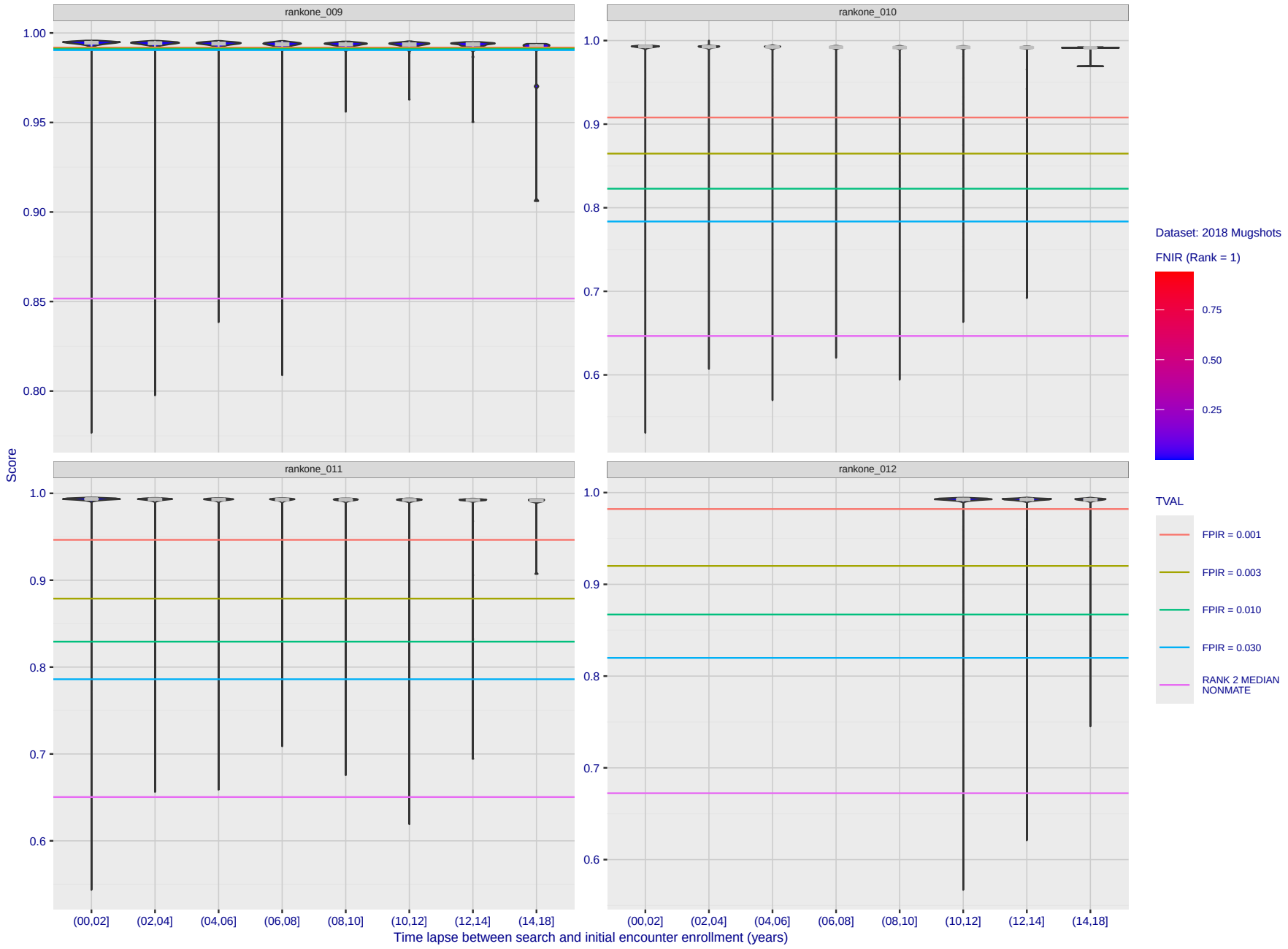


*Figure 419:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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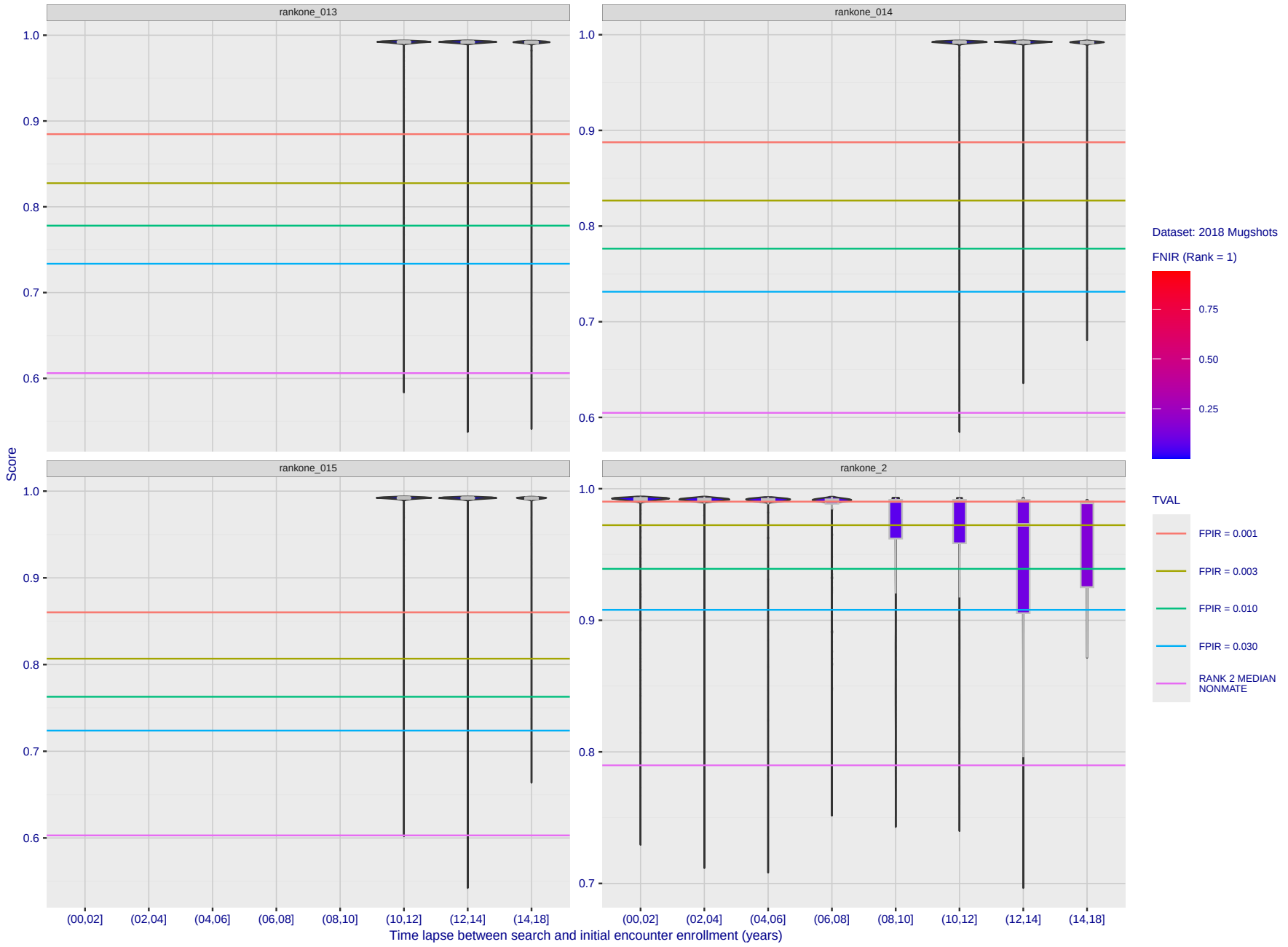


*Figure 420:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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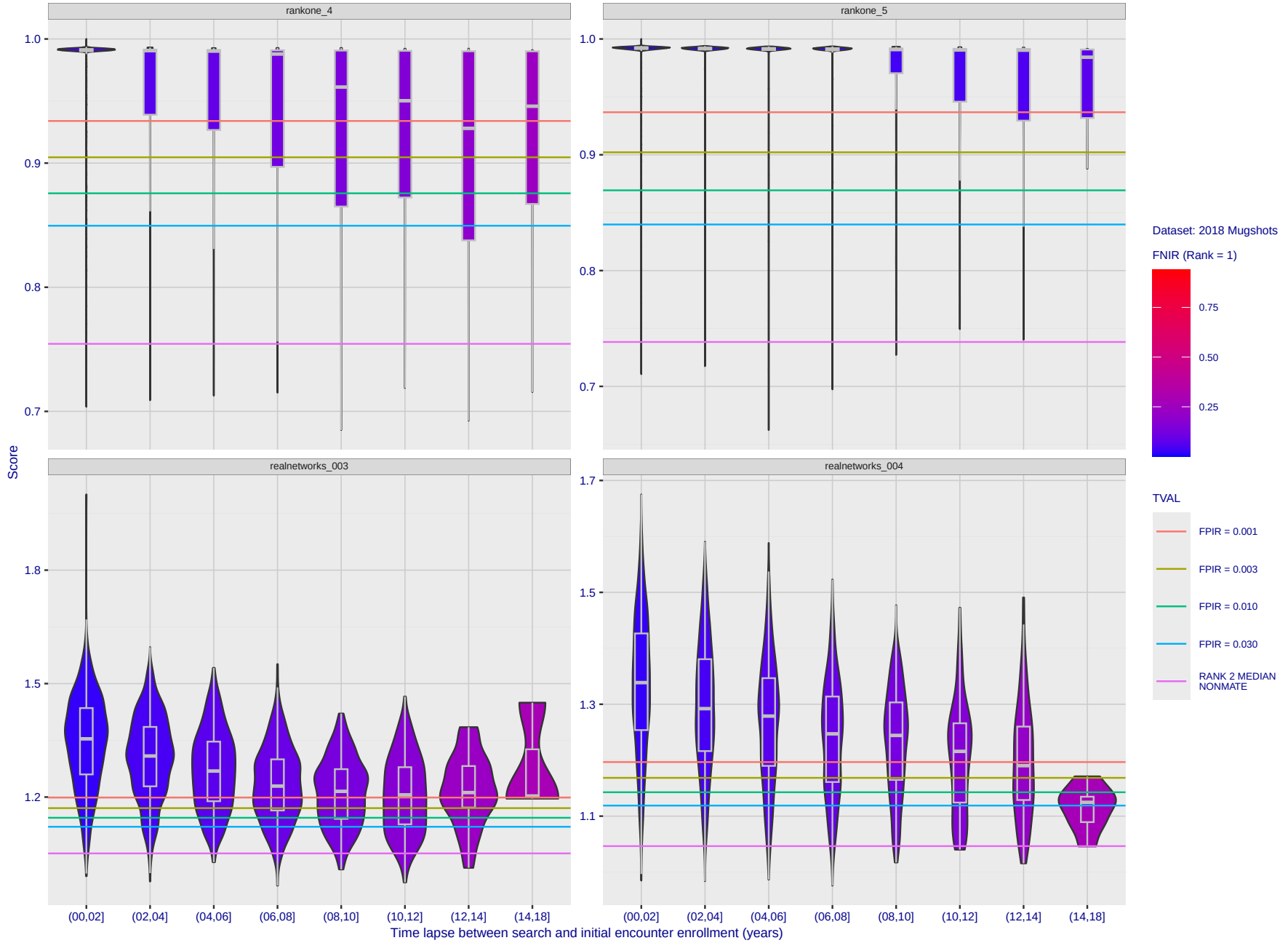


*Figure 421:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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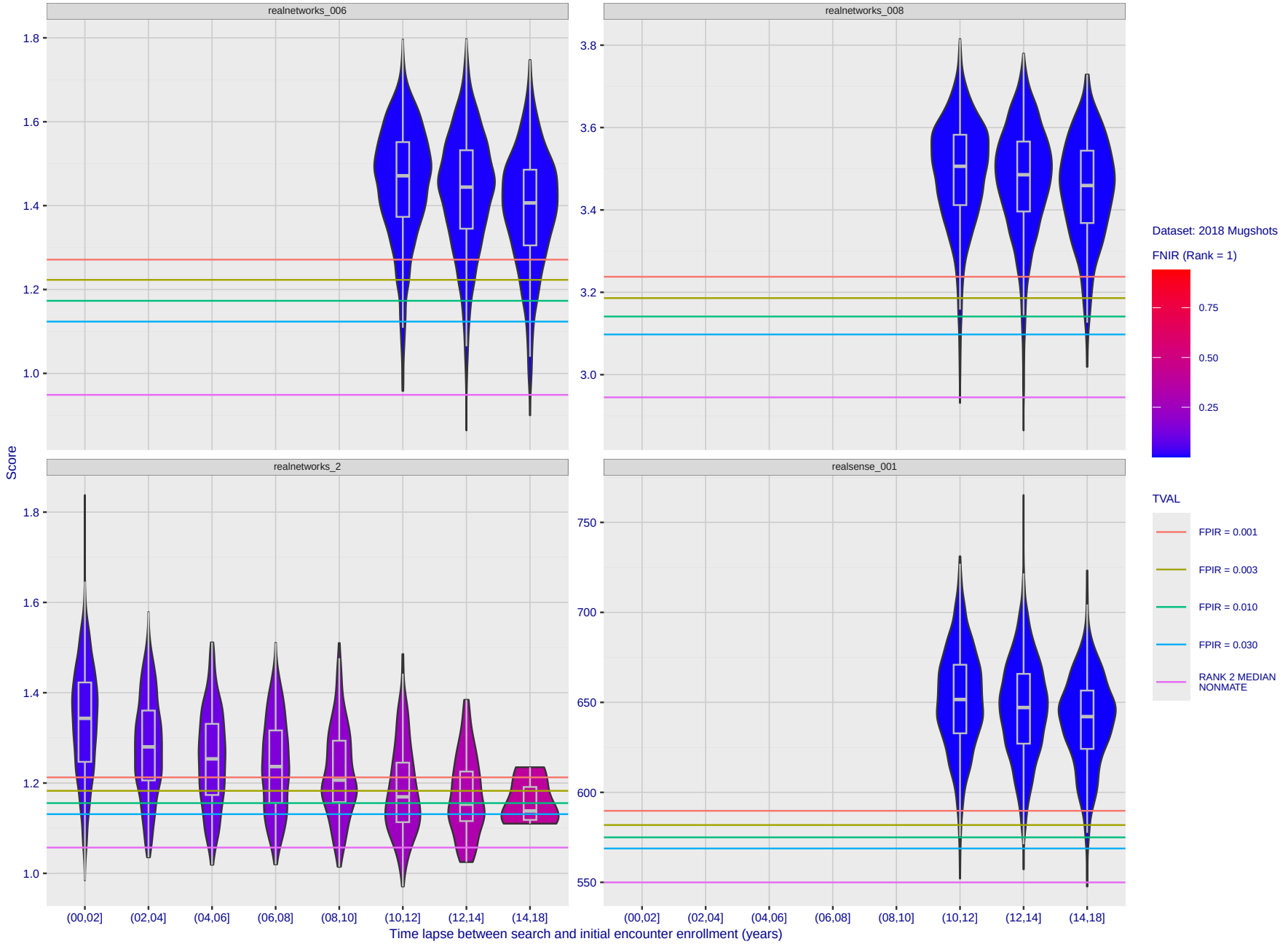


*Figure 422:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
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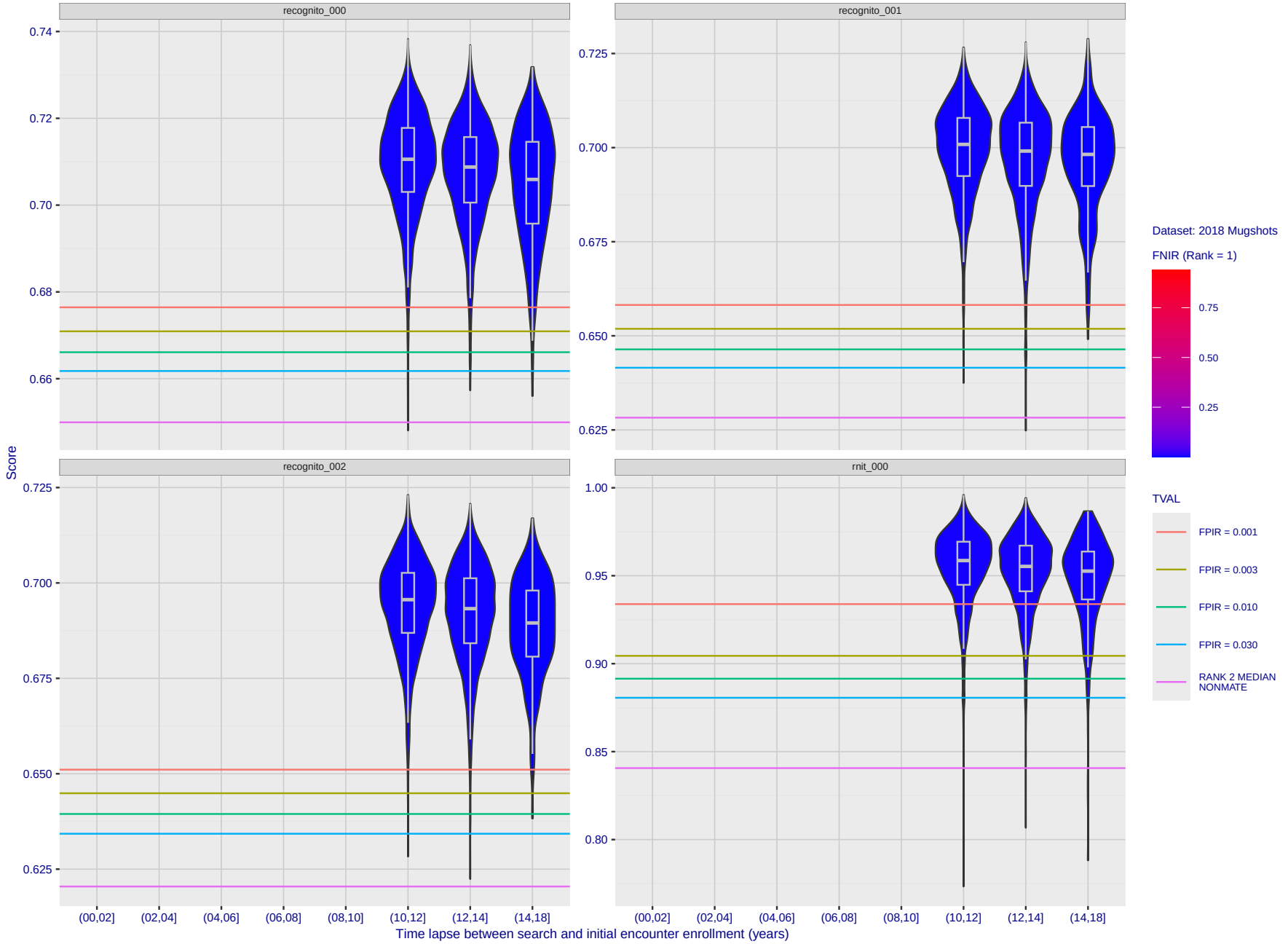


*Figure 423:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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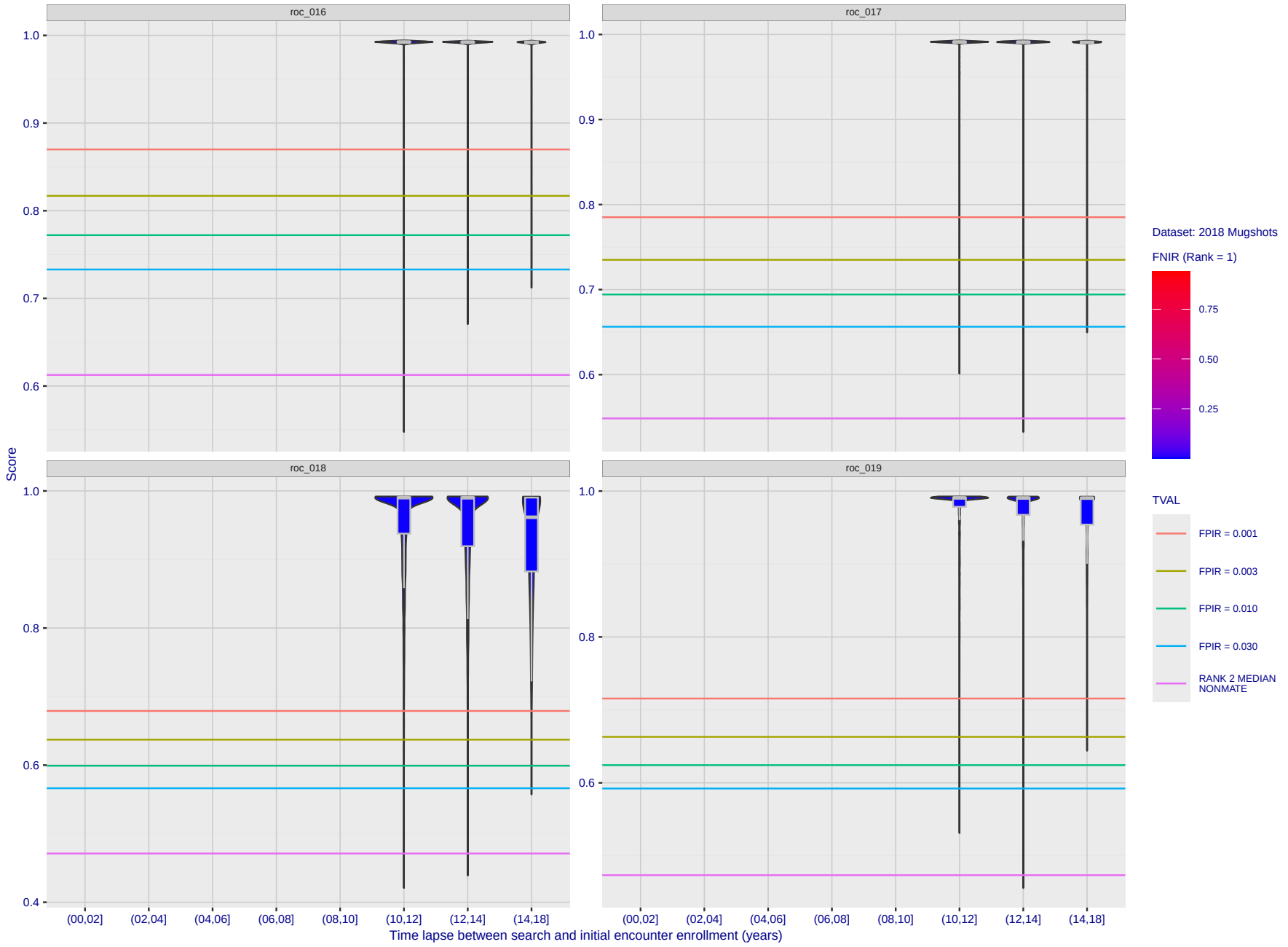


*Figure 424:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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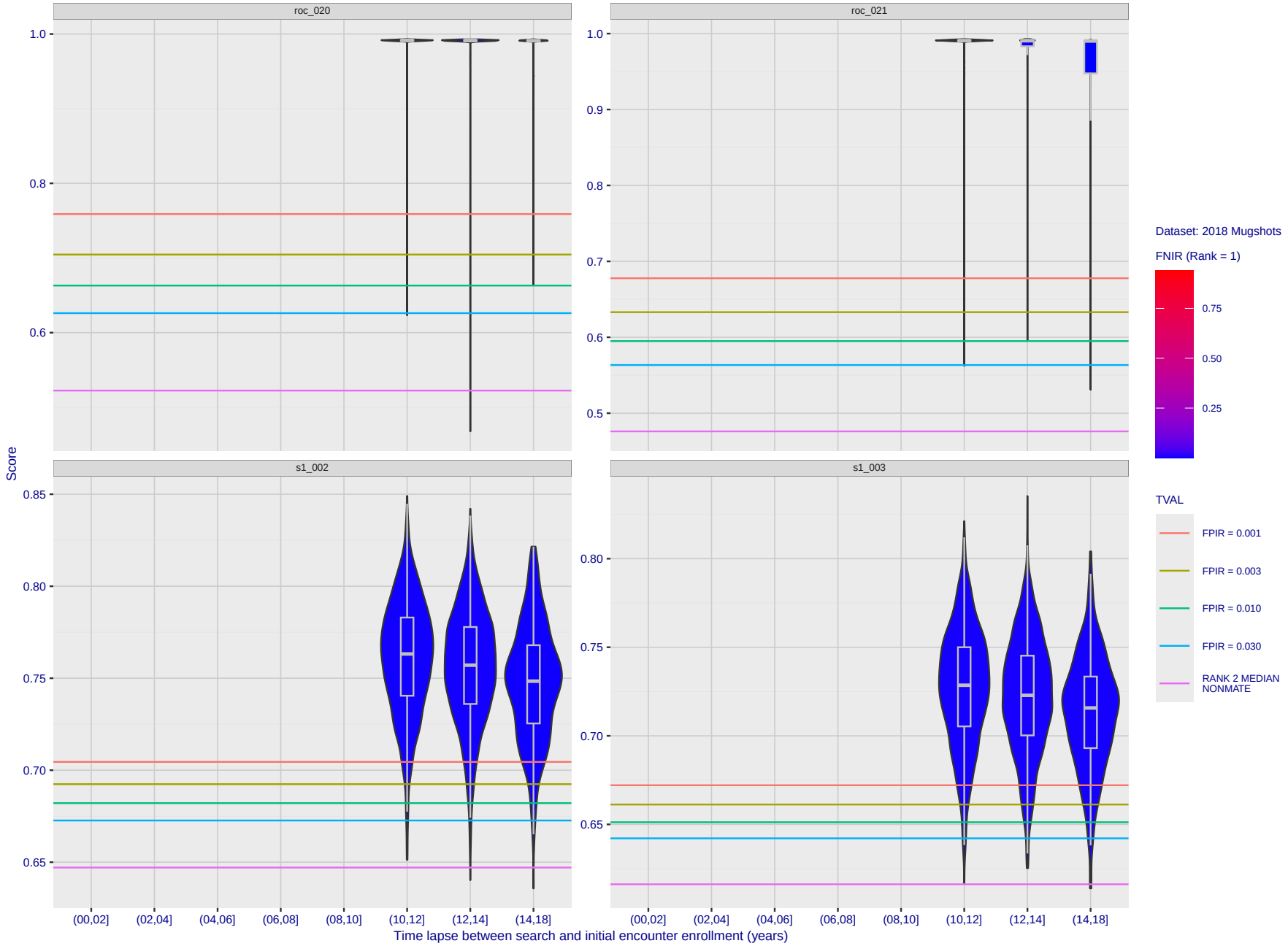


*Figure 425:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
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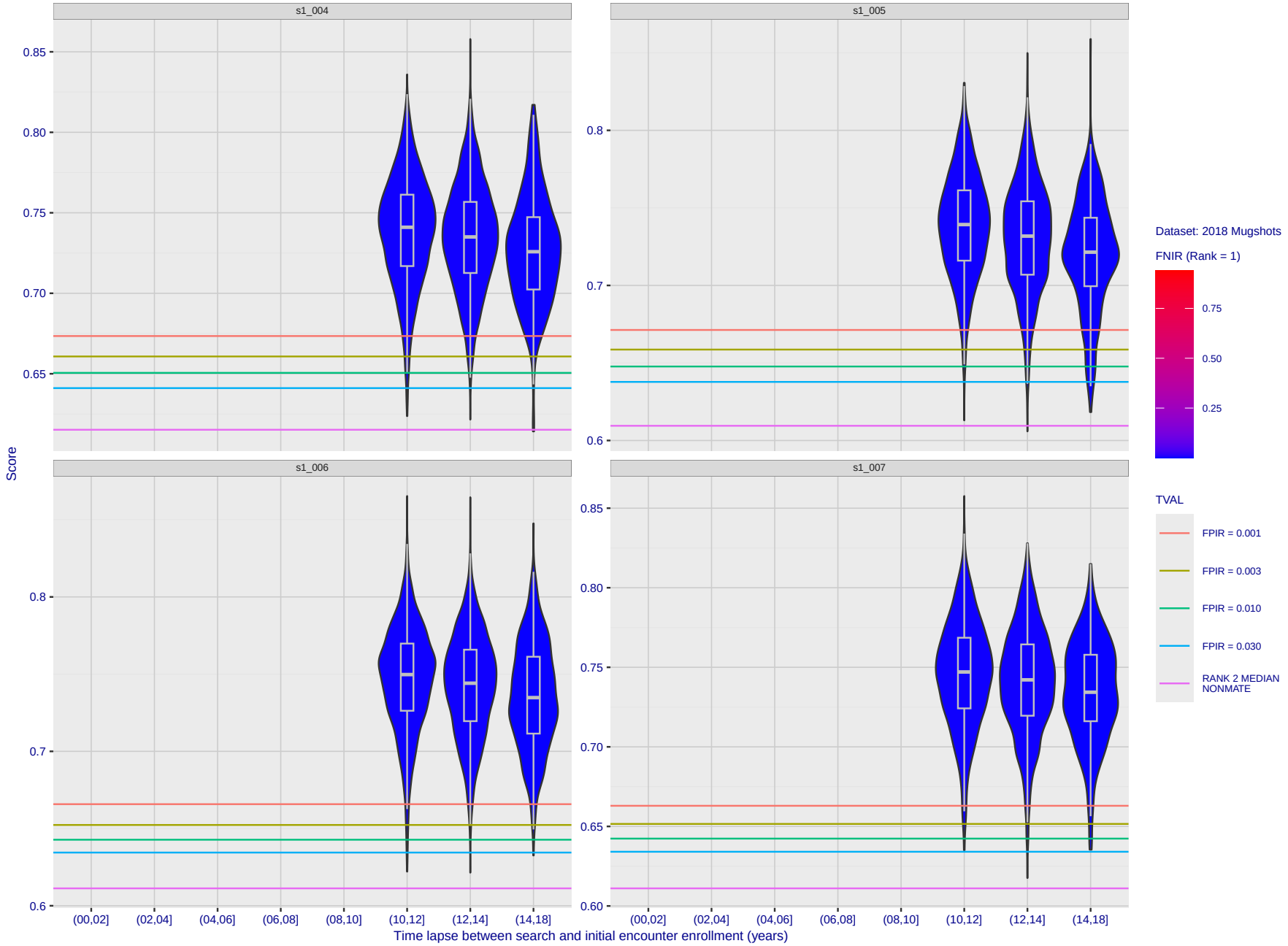


*Figure 426:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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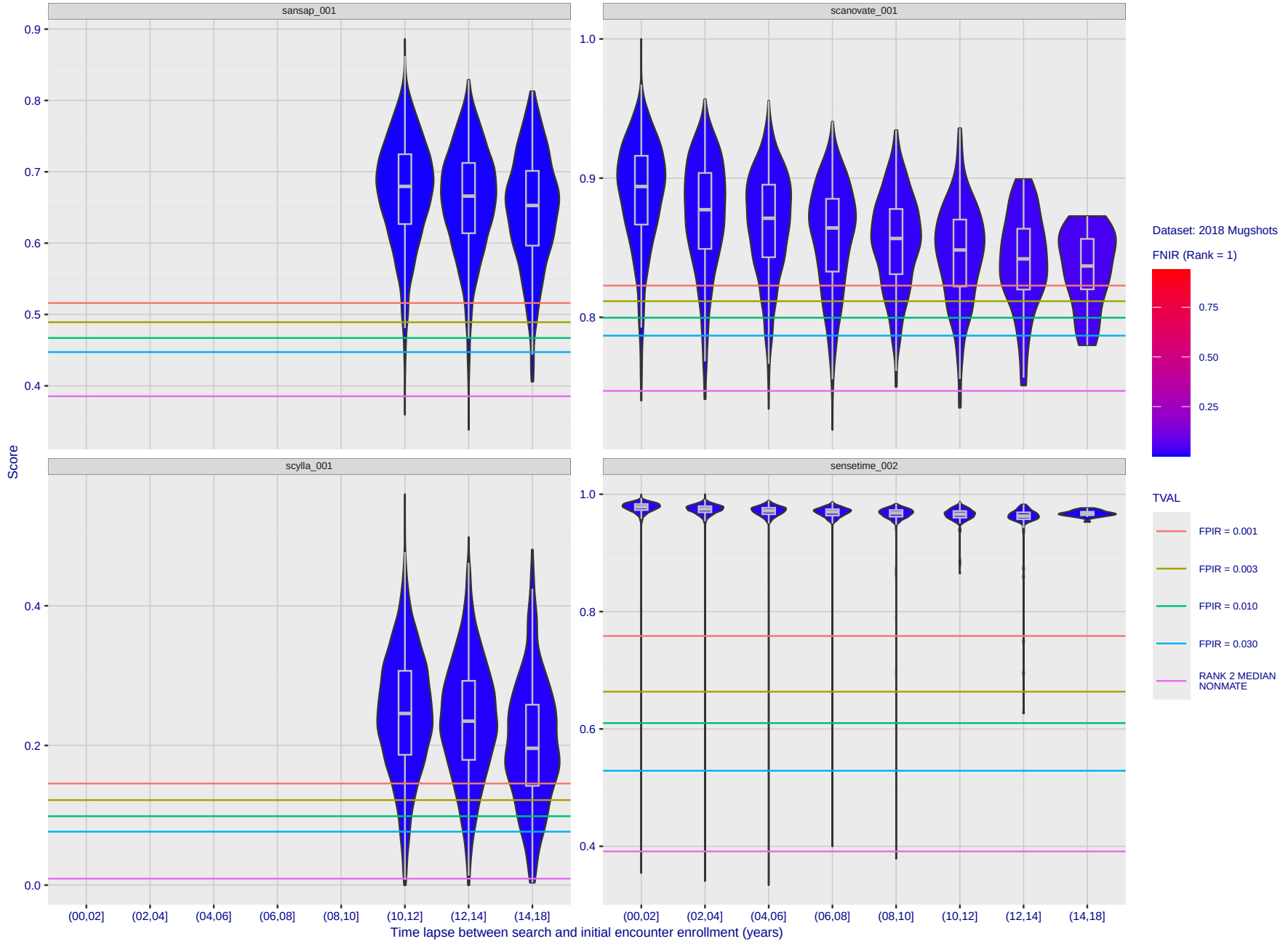


*Figure 427:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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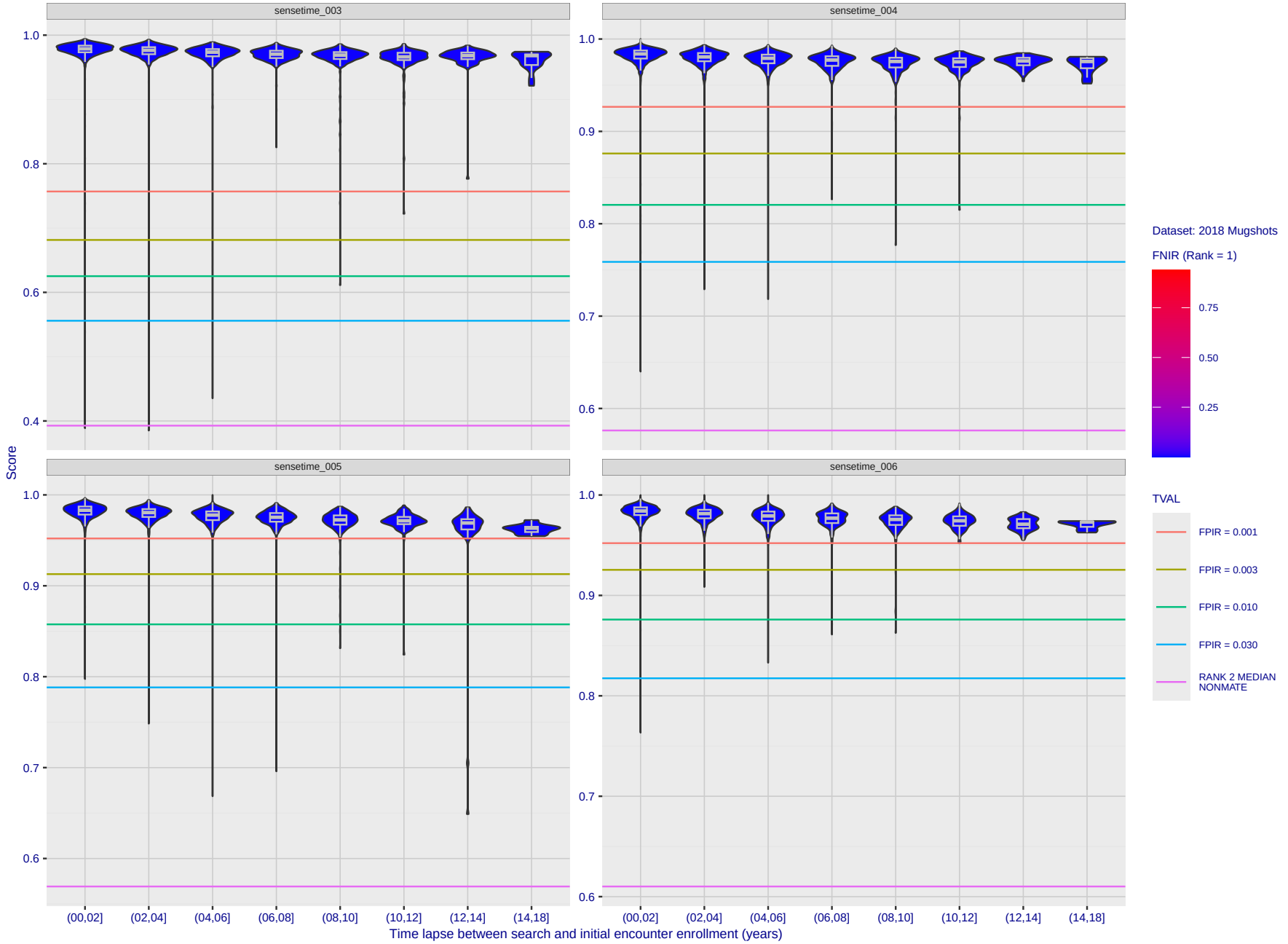


*Figure 428:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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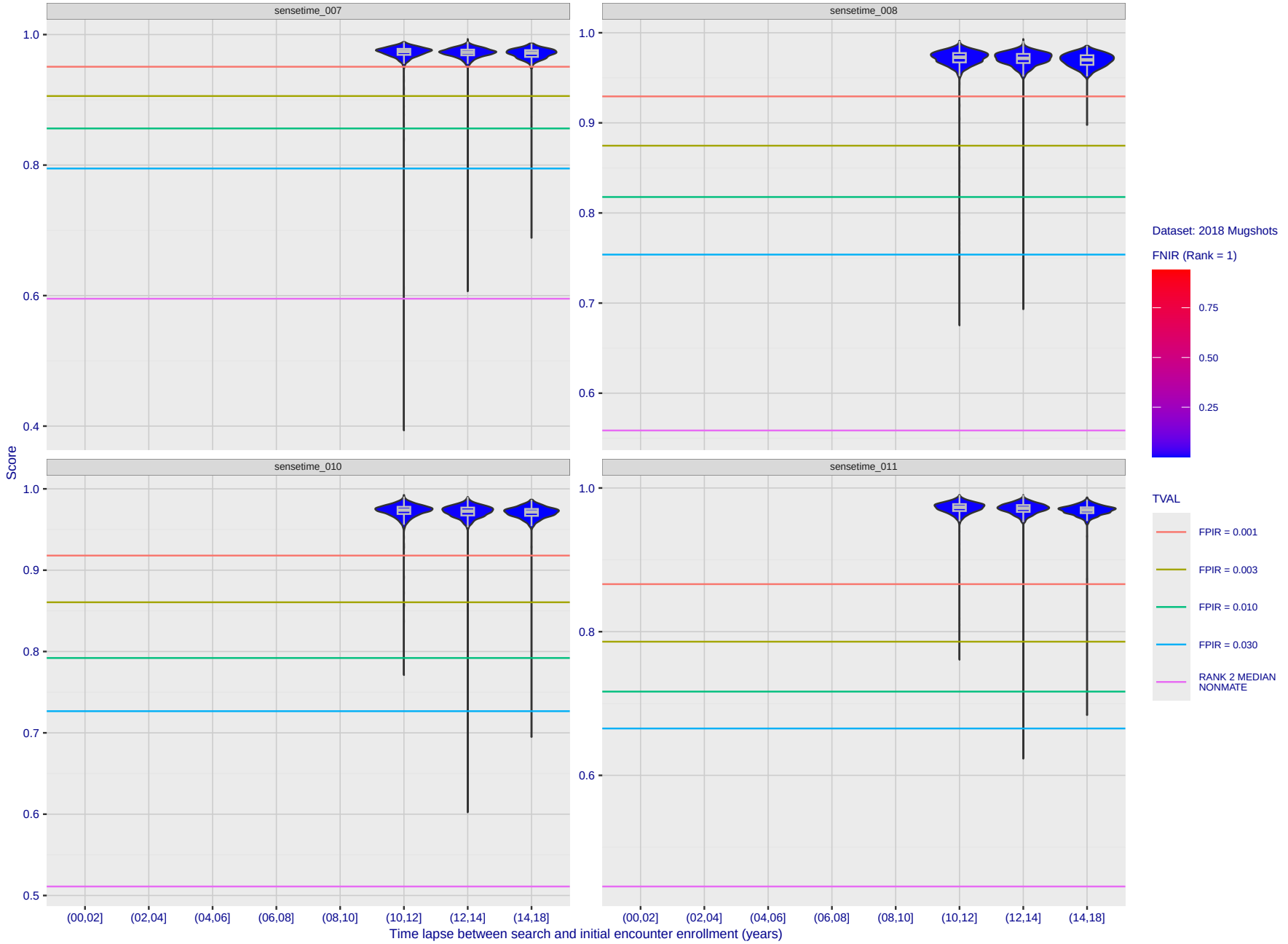


*Figure 429:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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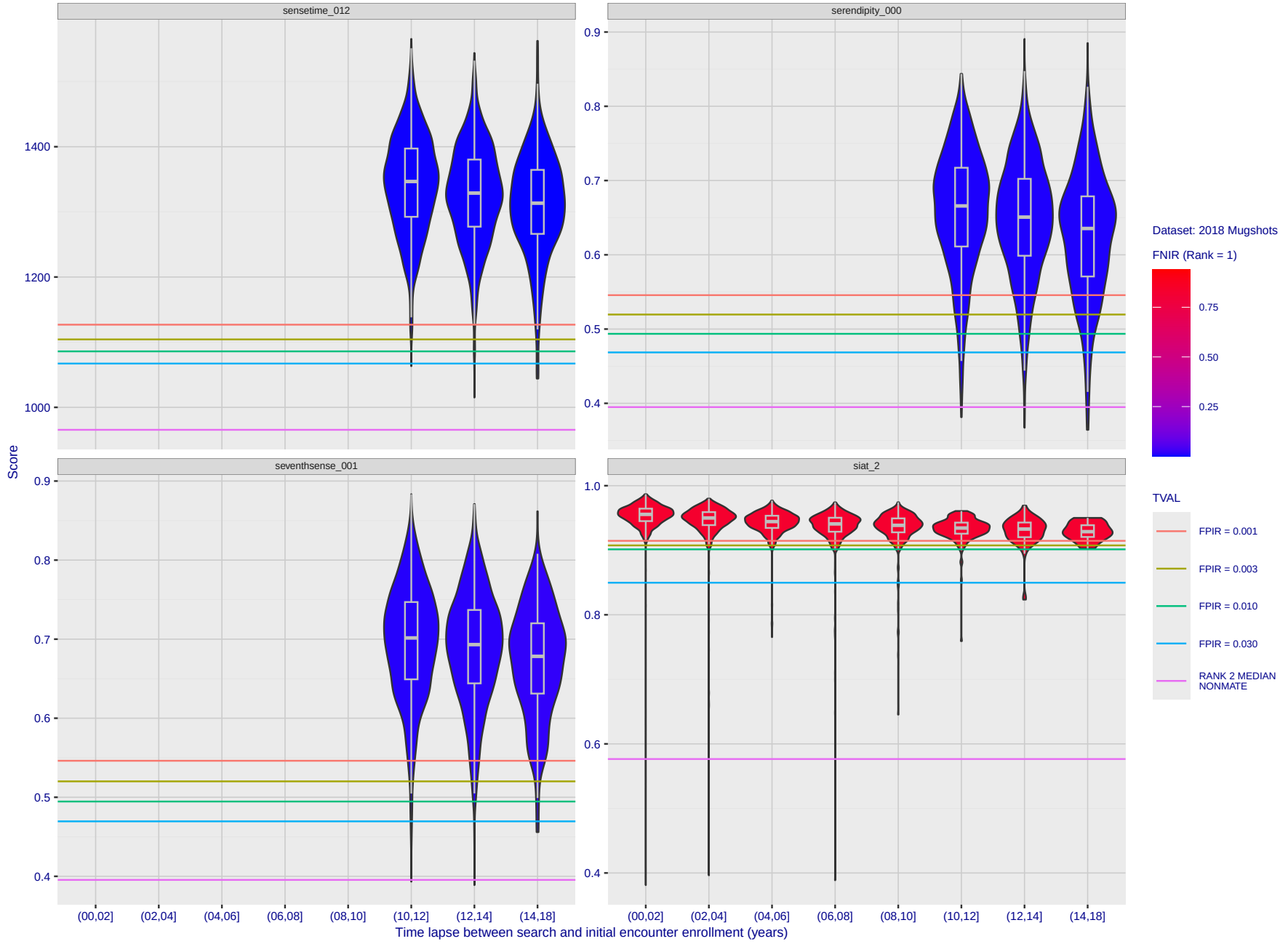


*Figure 430:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



2026/09/04 10:23:32
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T > 0 → Identification

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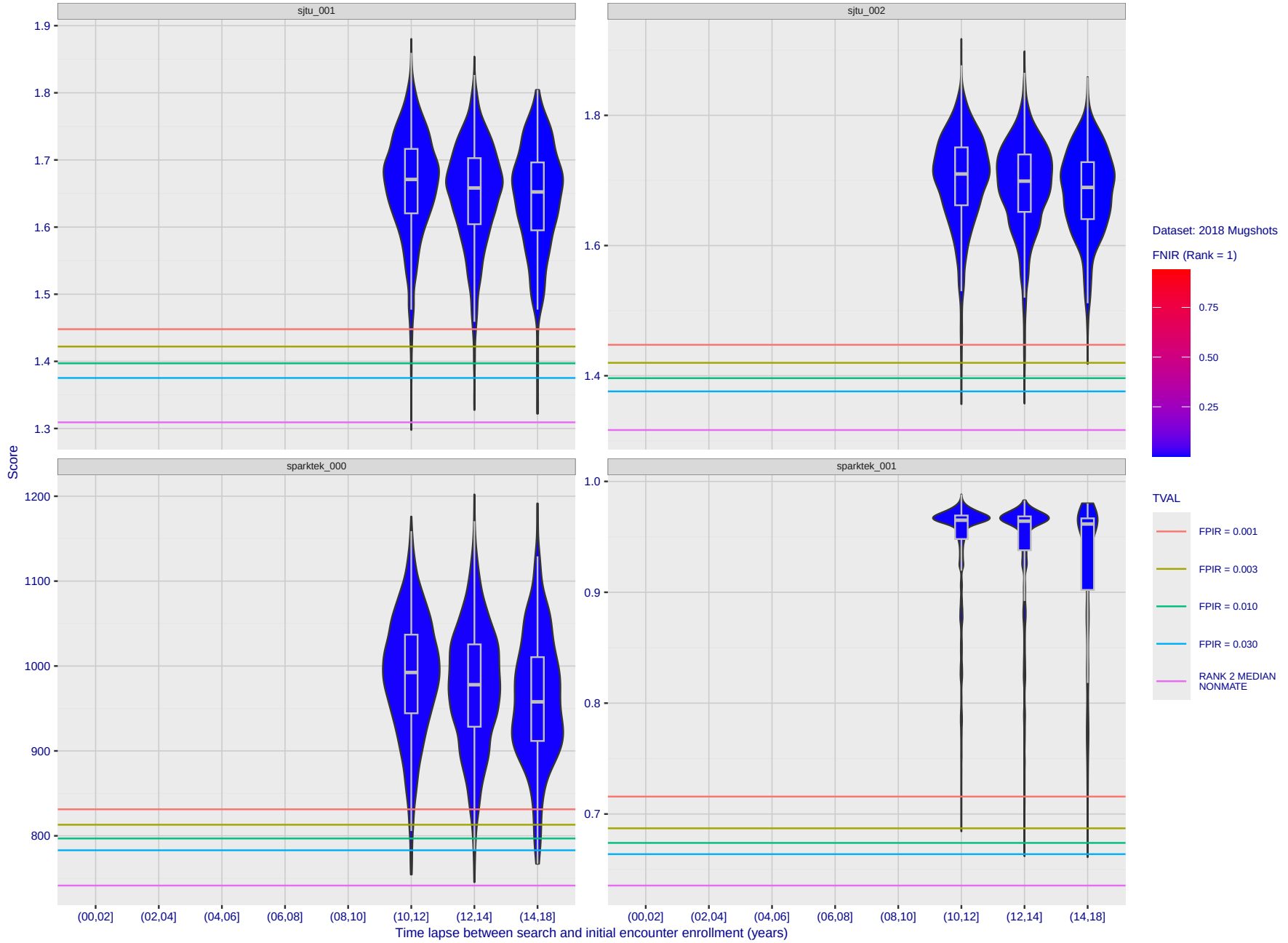


*Figure 431:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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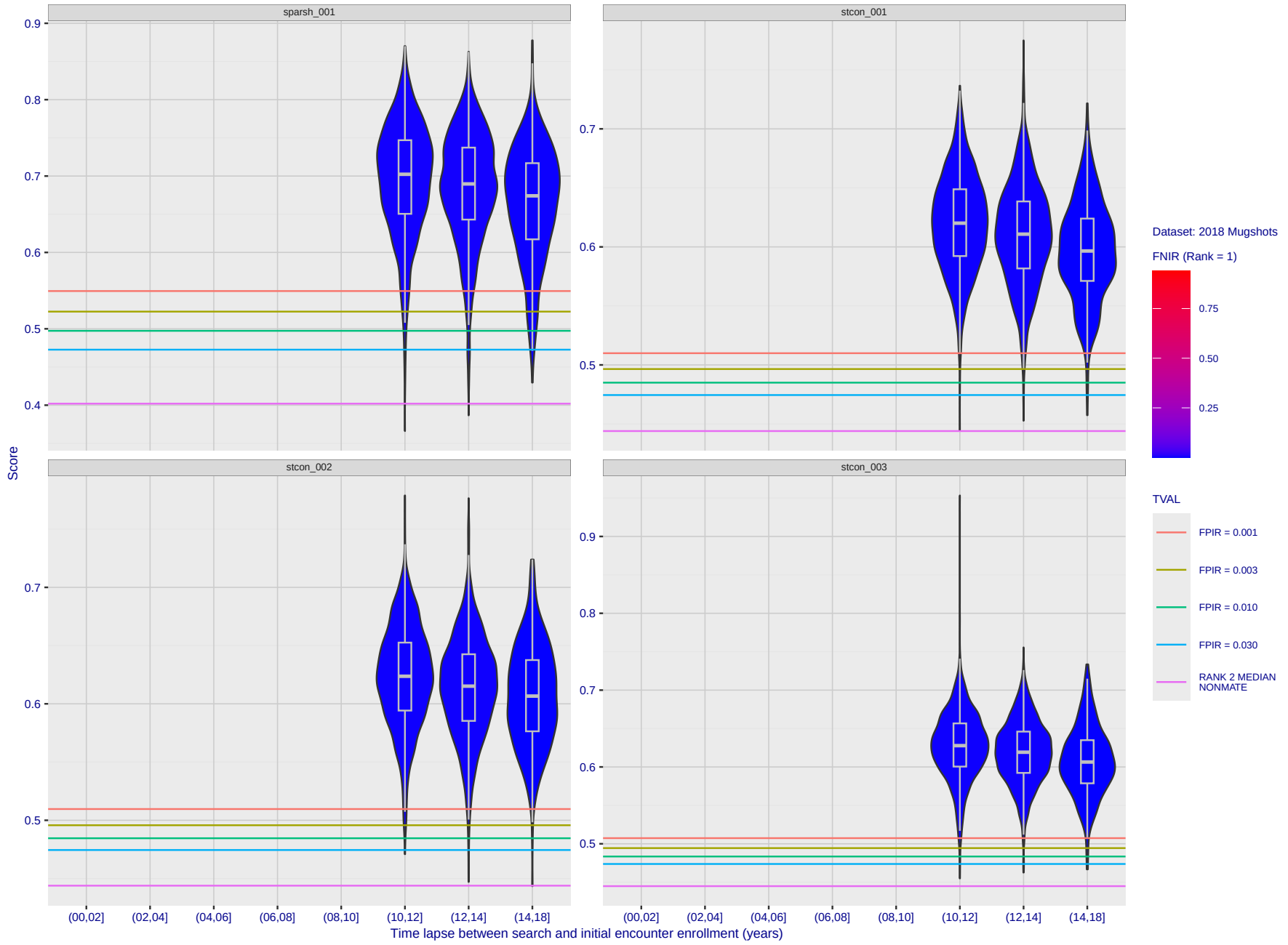


*Figure 432:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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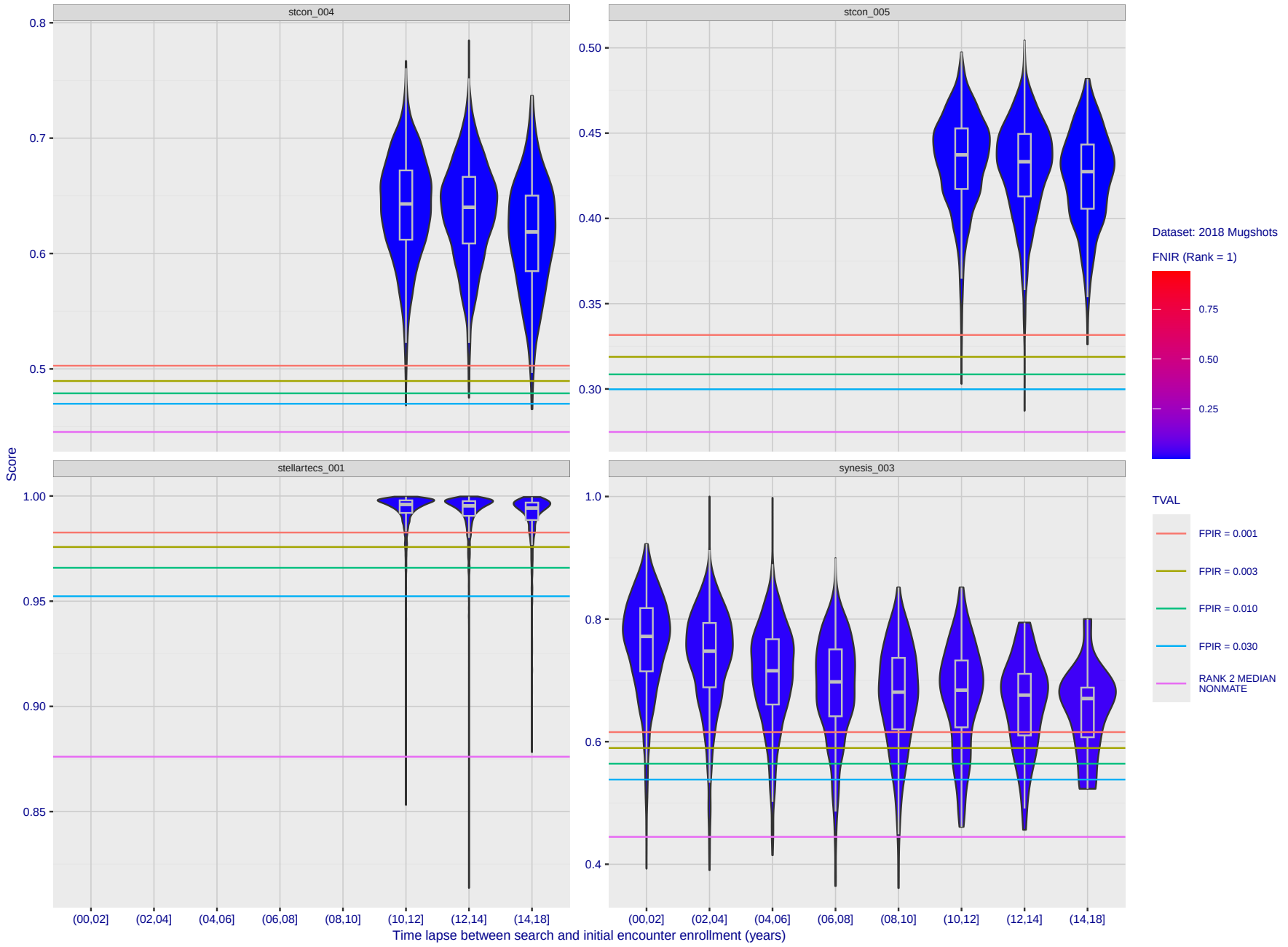


*Figure 433:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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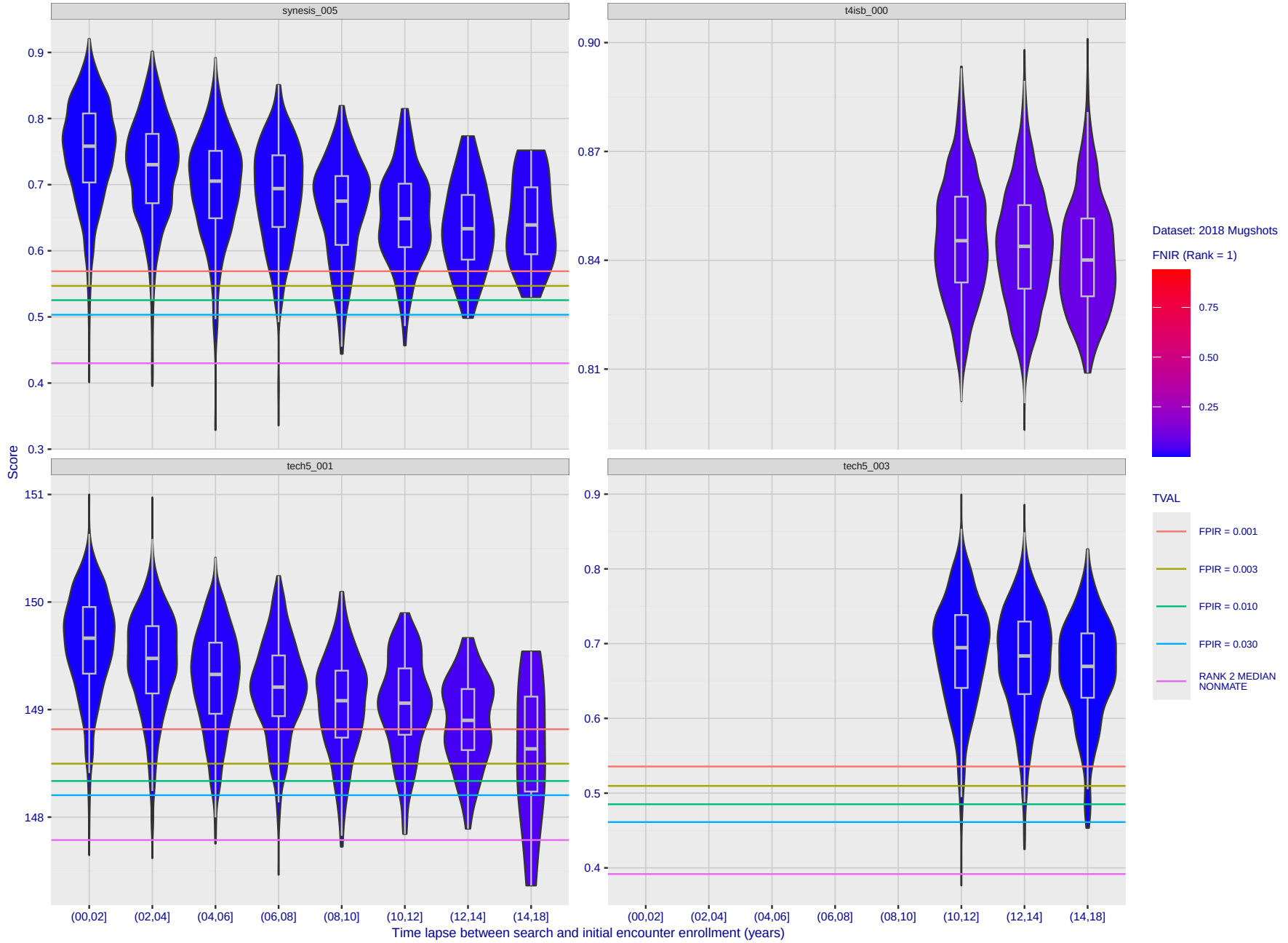


*Figure 434:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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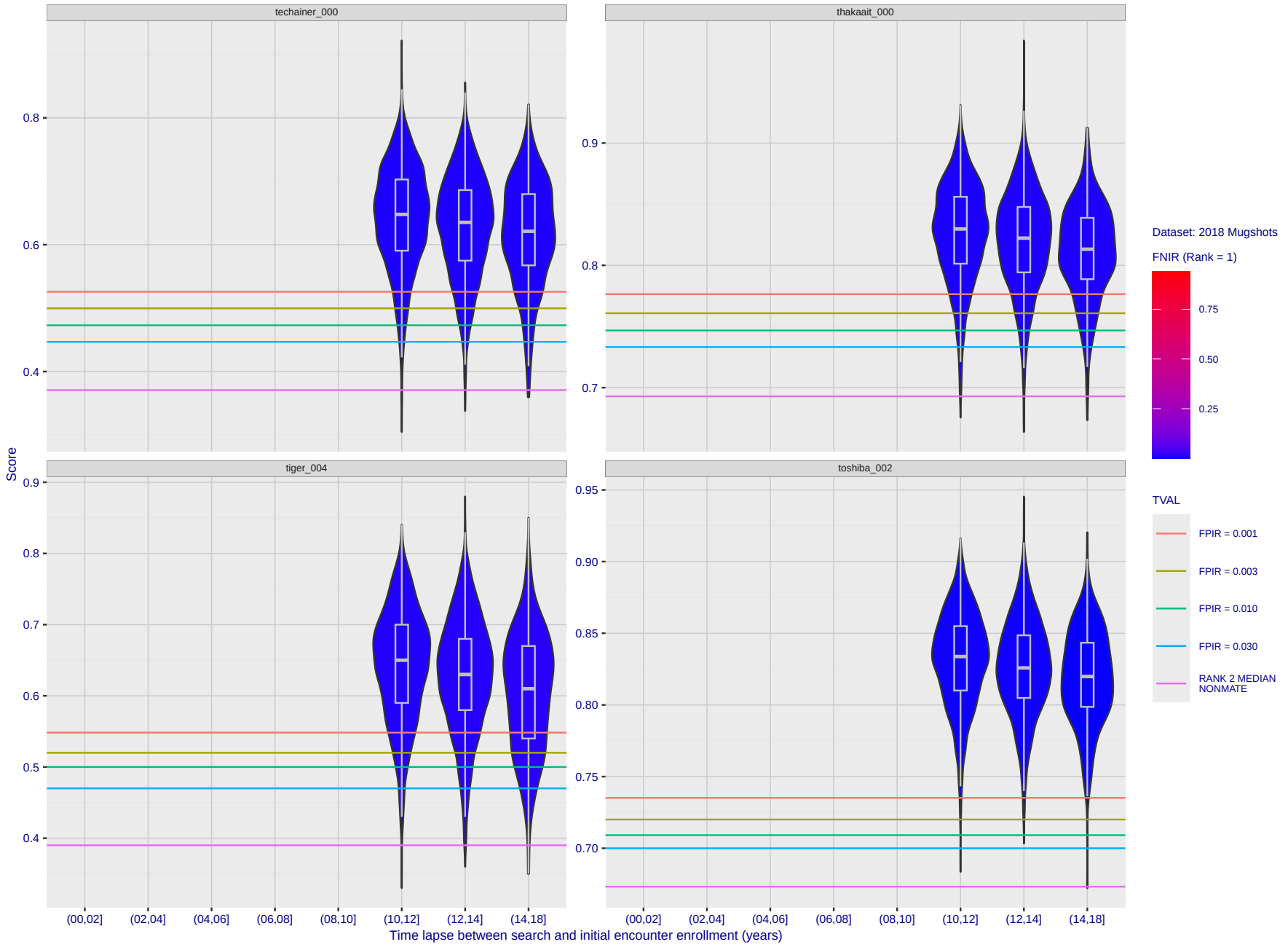


*Figure 435:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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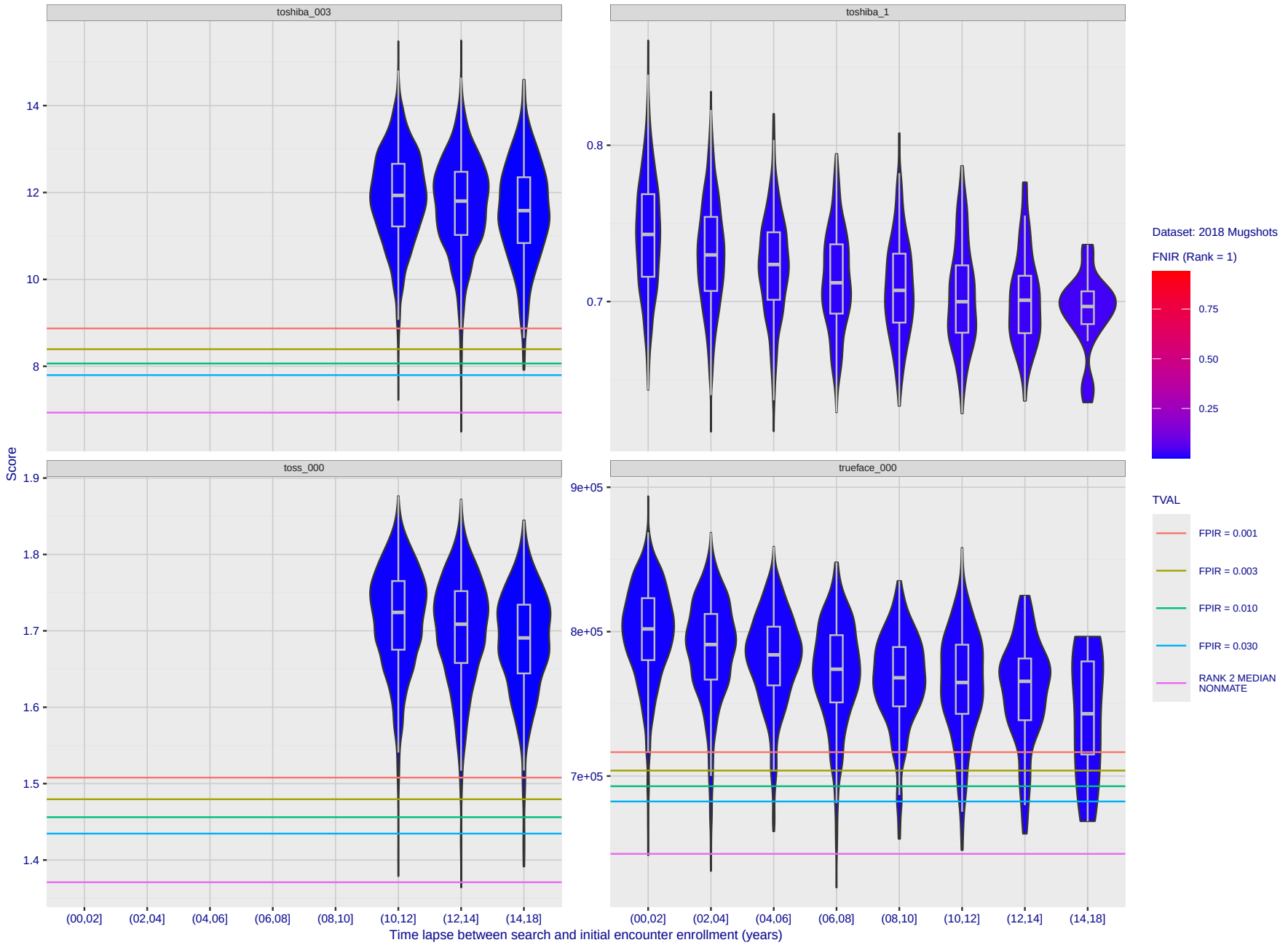


*Figure 436:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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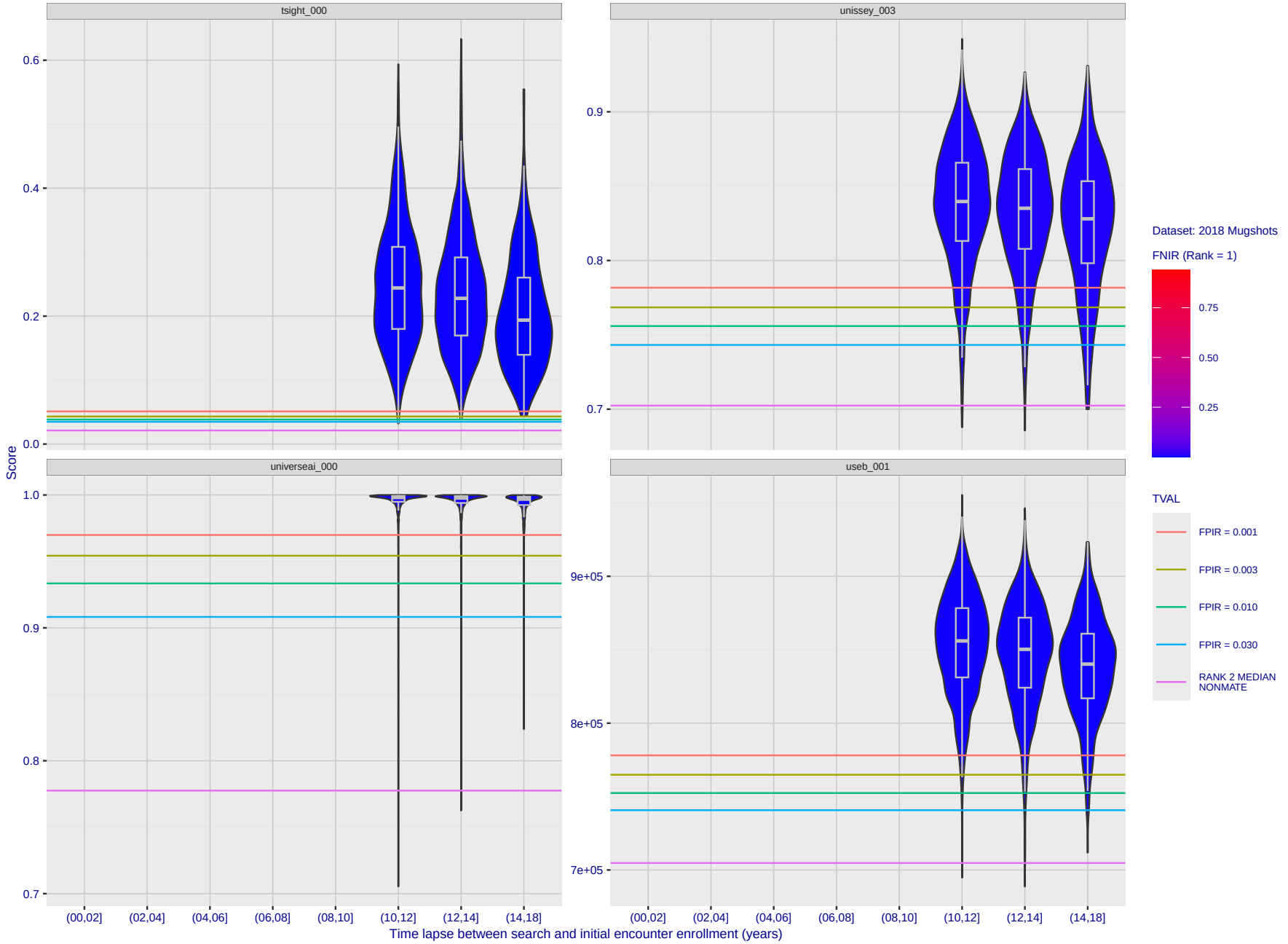


*Figure 437:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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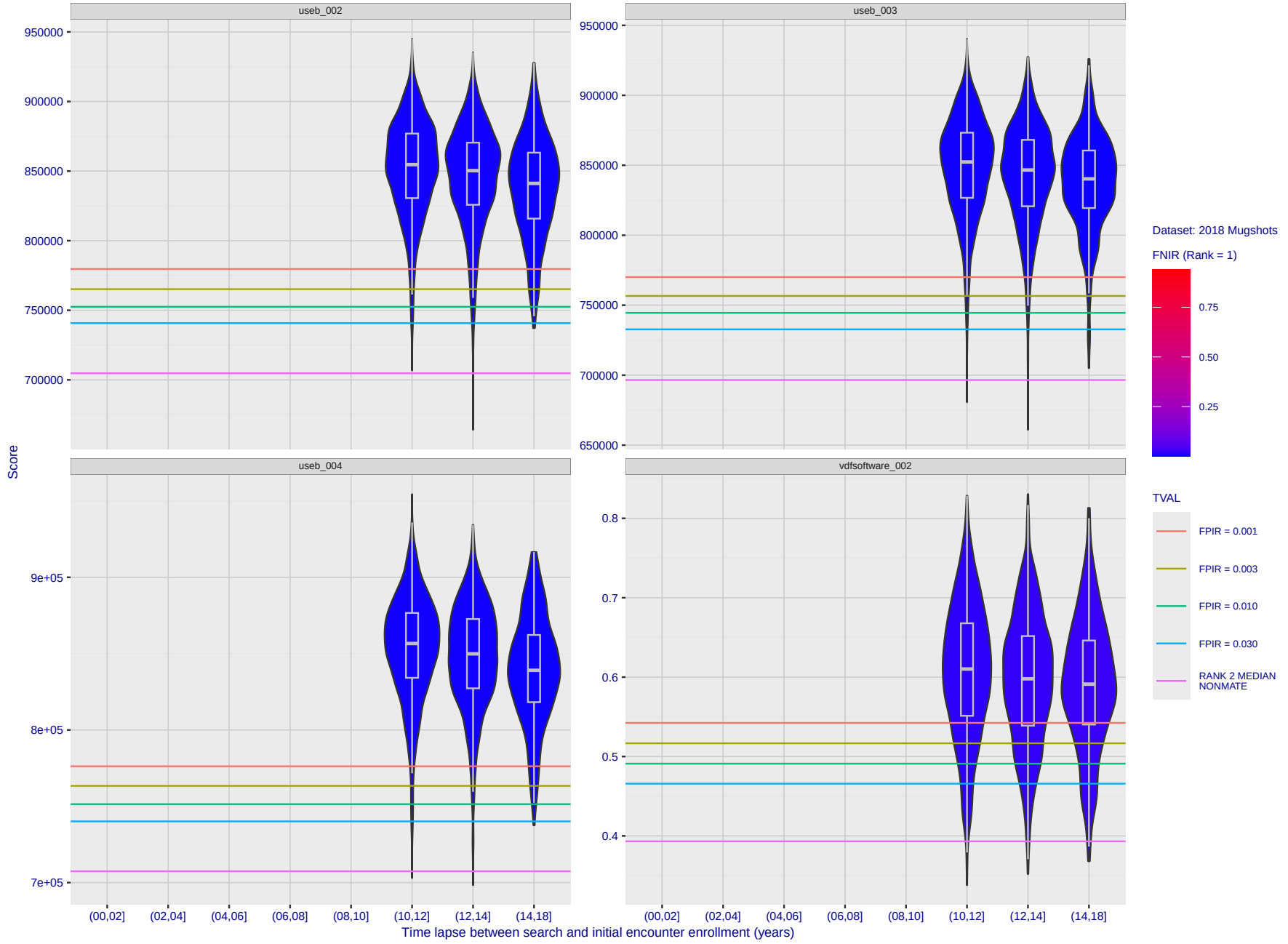


*Figure 438:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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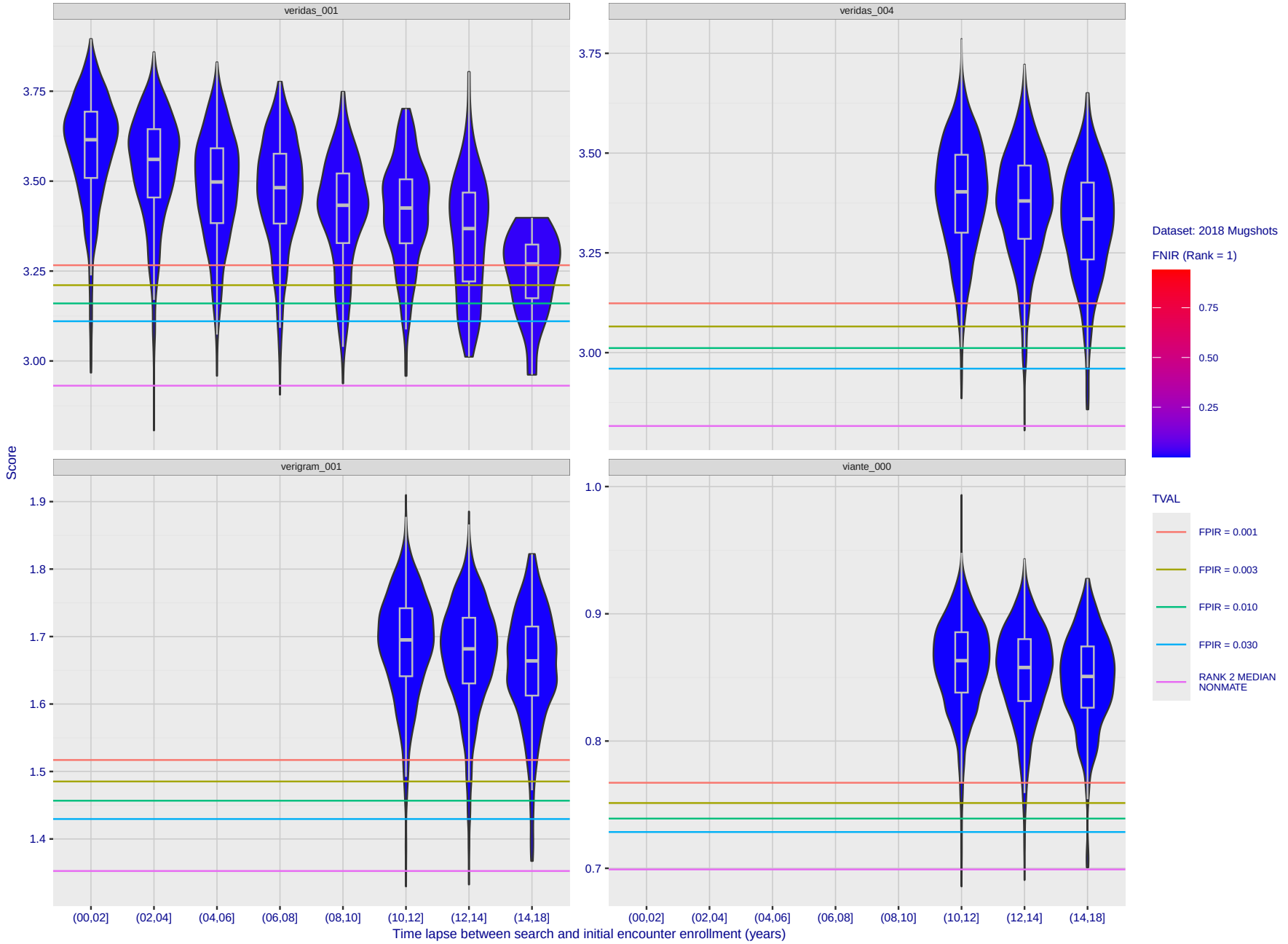


*Figure 439:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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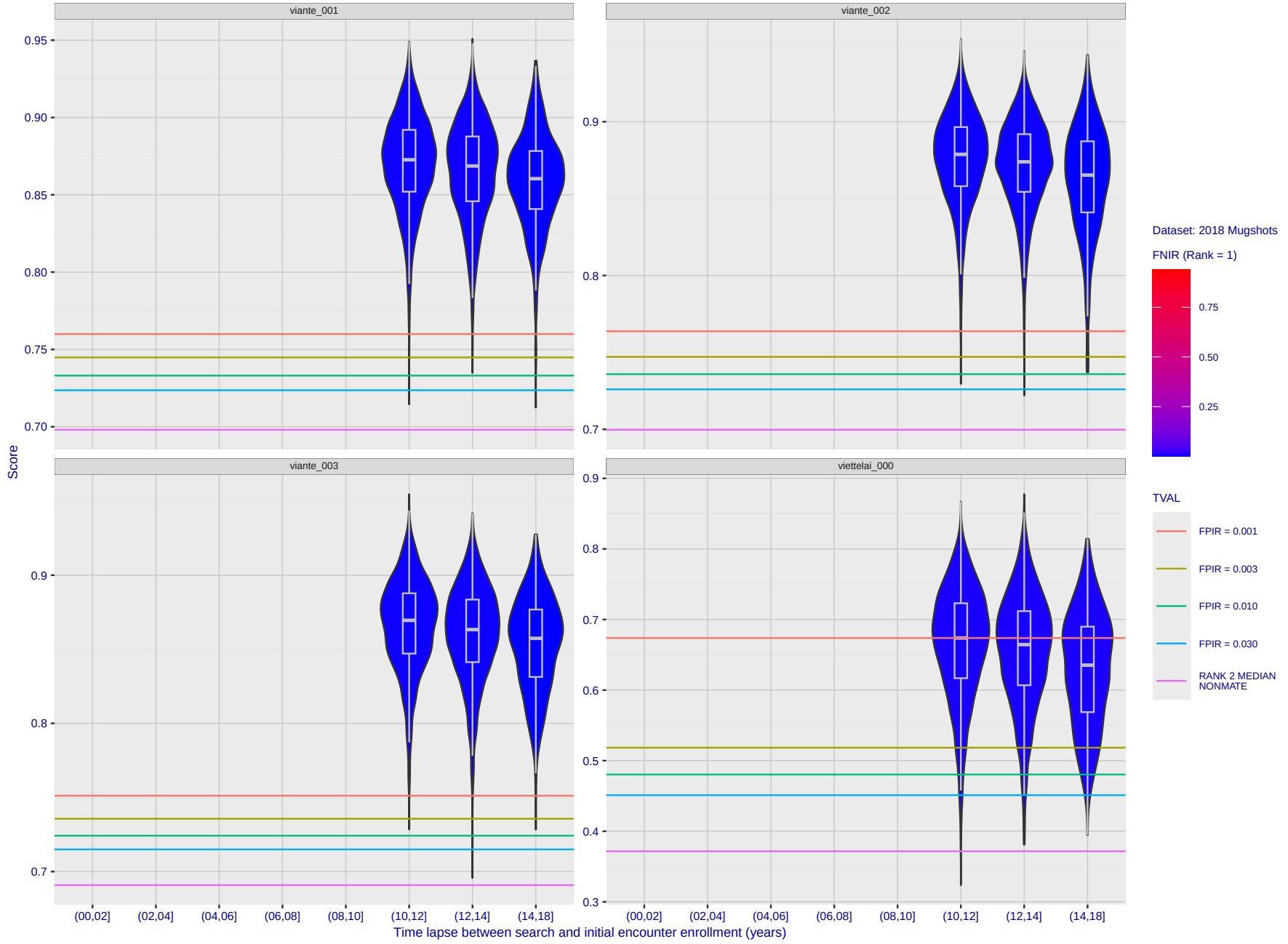


*Figure 440:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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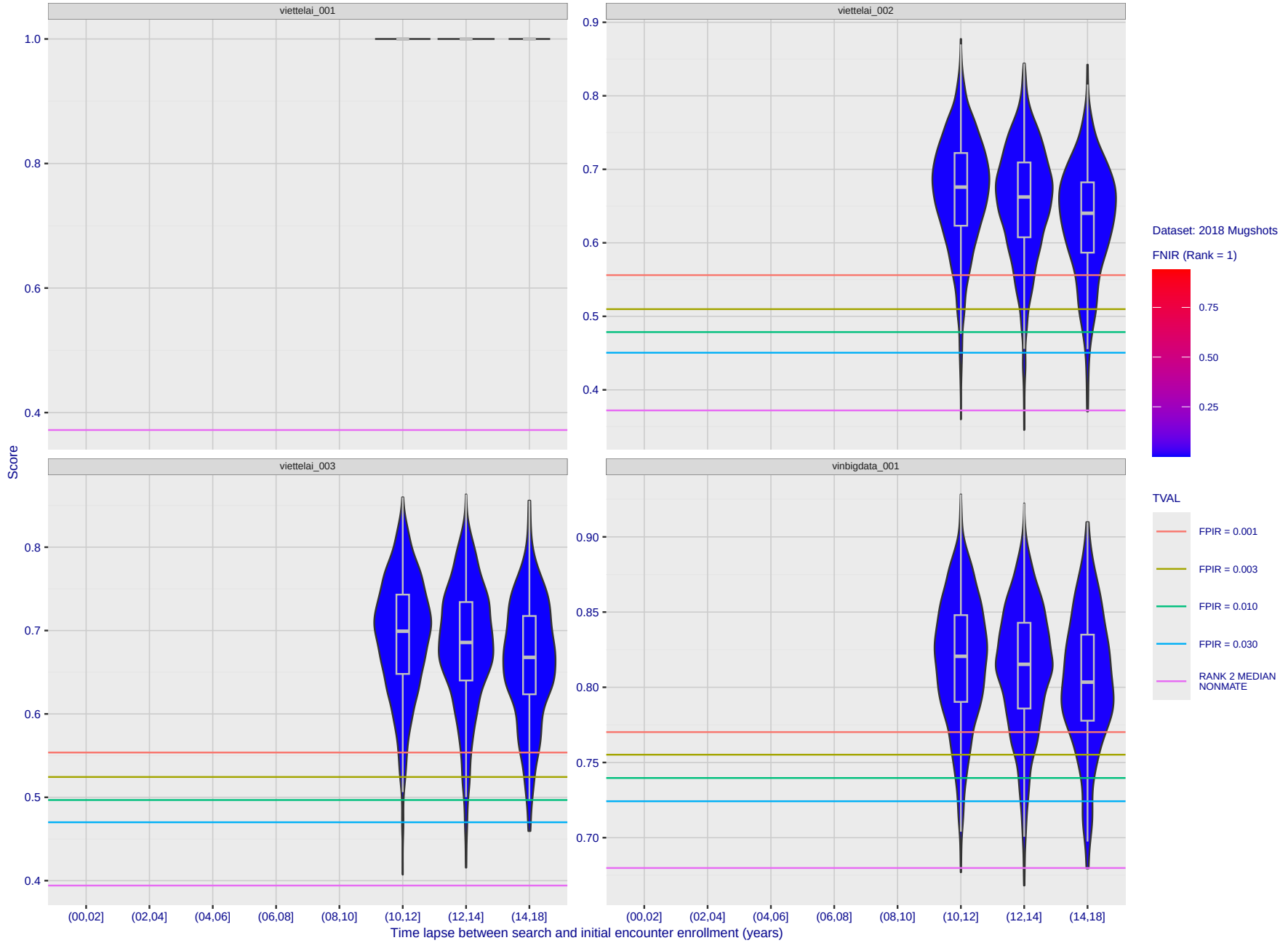


*Figure 441:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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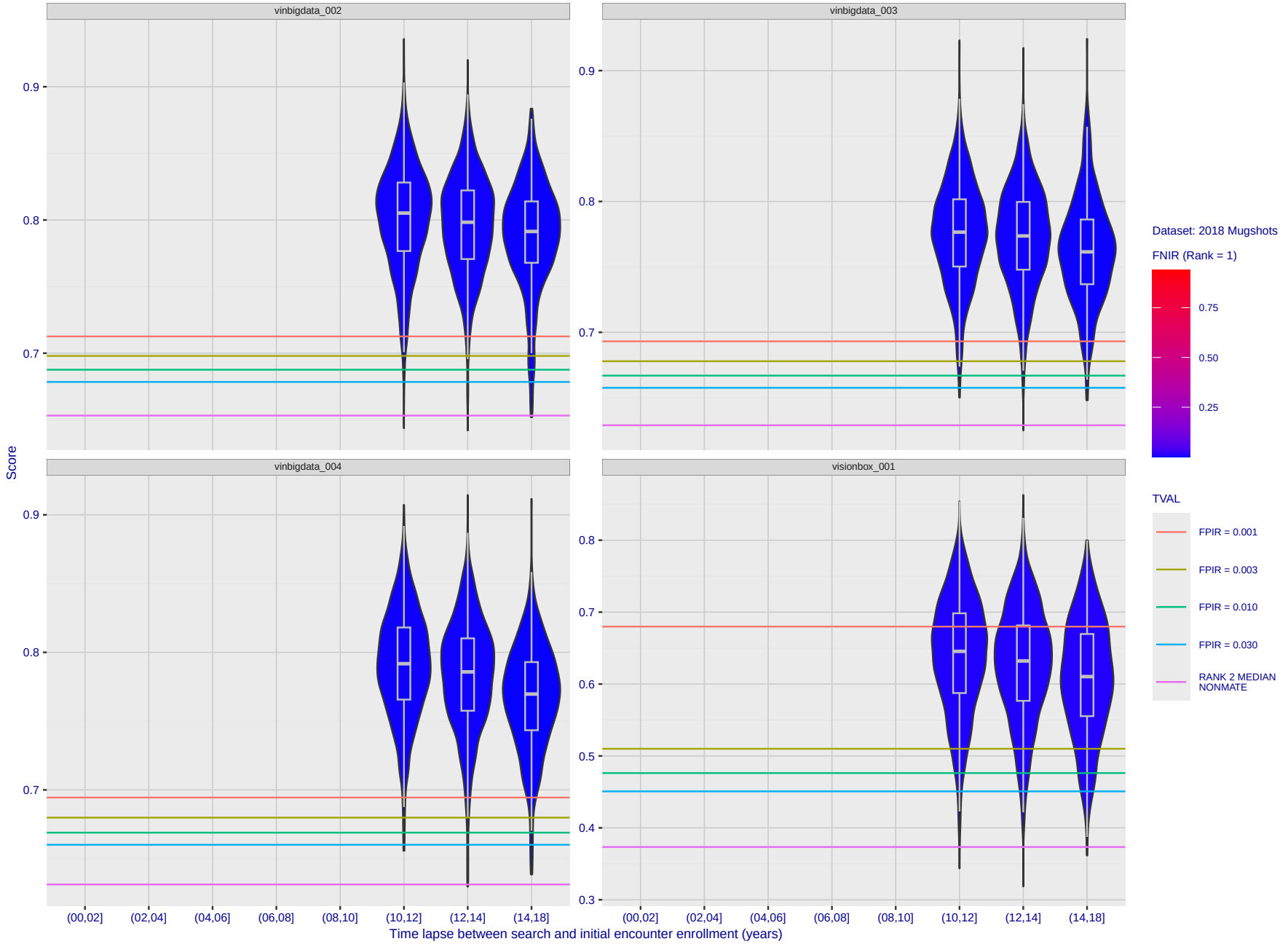


*Figure 442:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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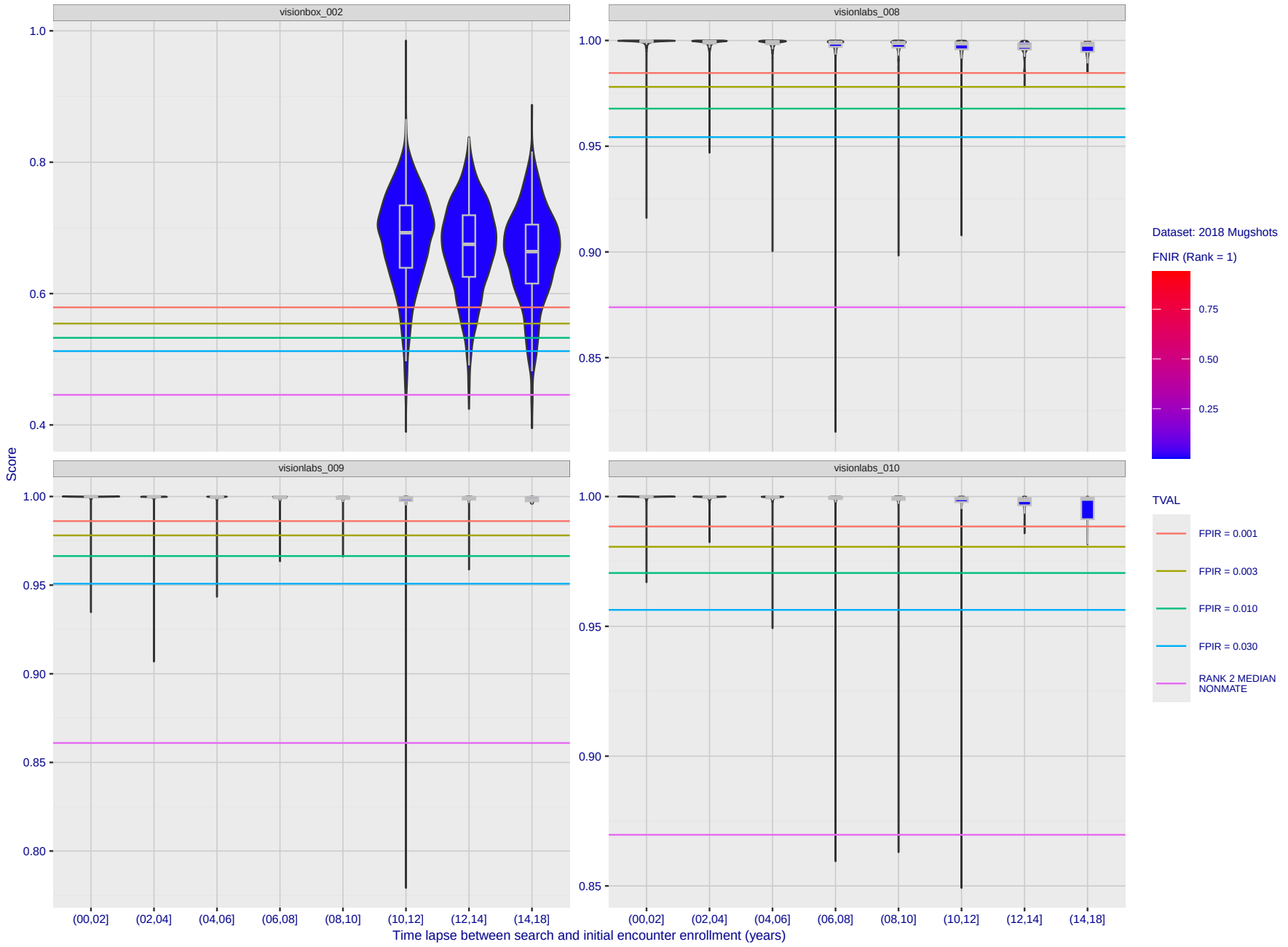


*Figure 443:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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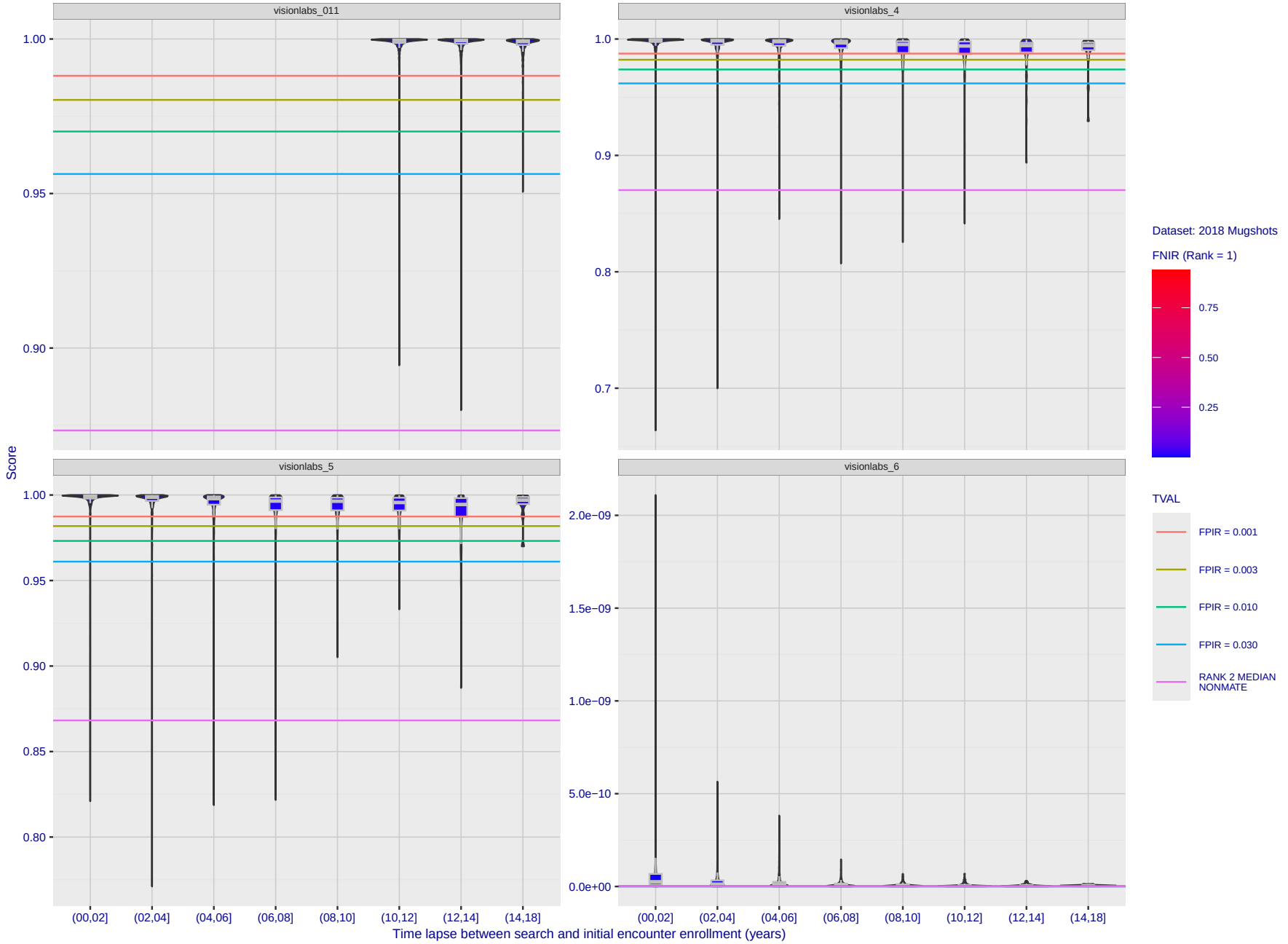


*Figure 444:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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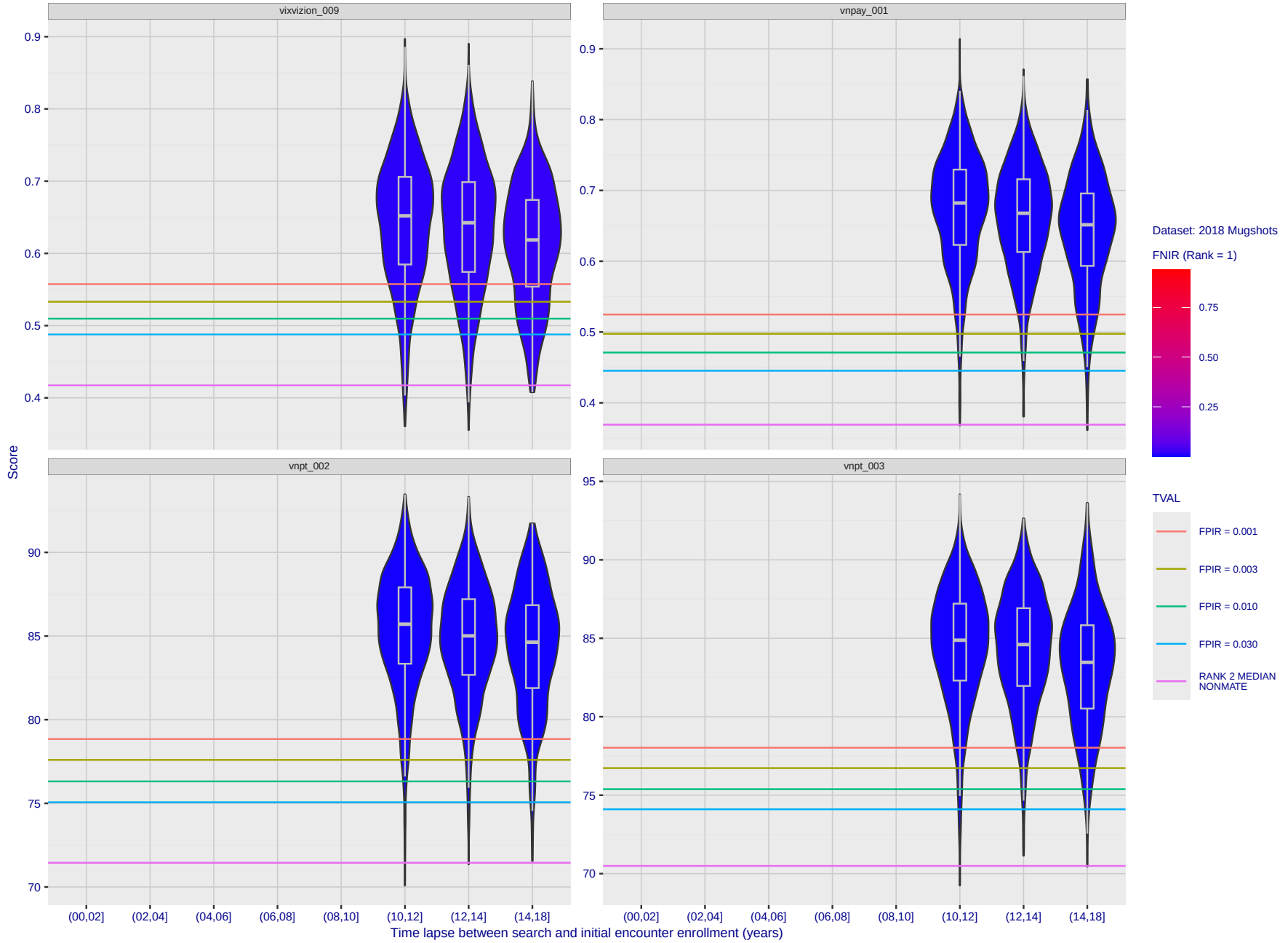


*Figure 445:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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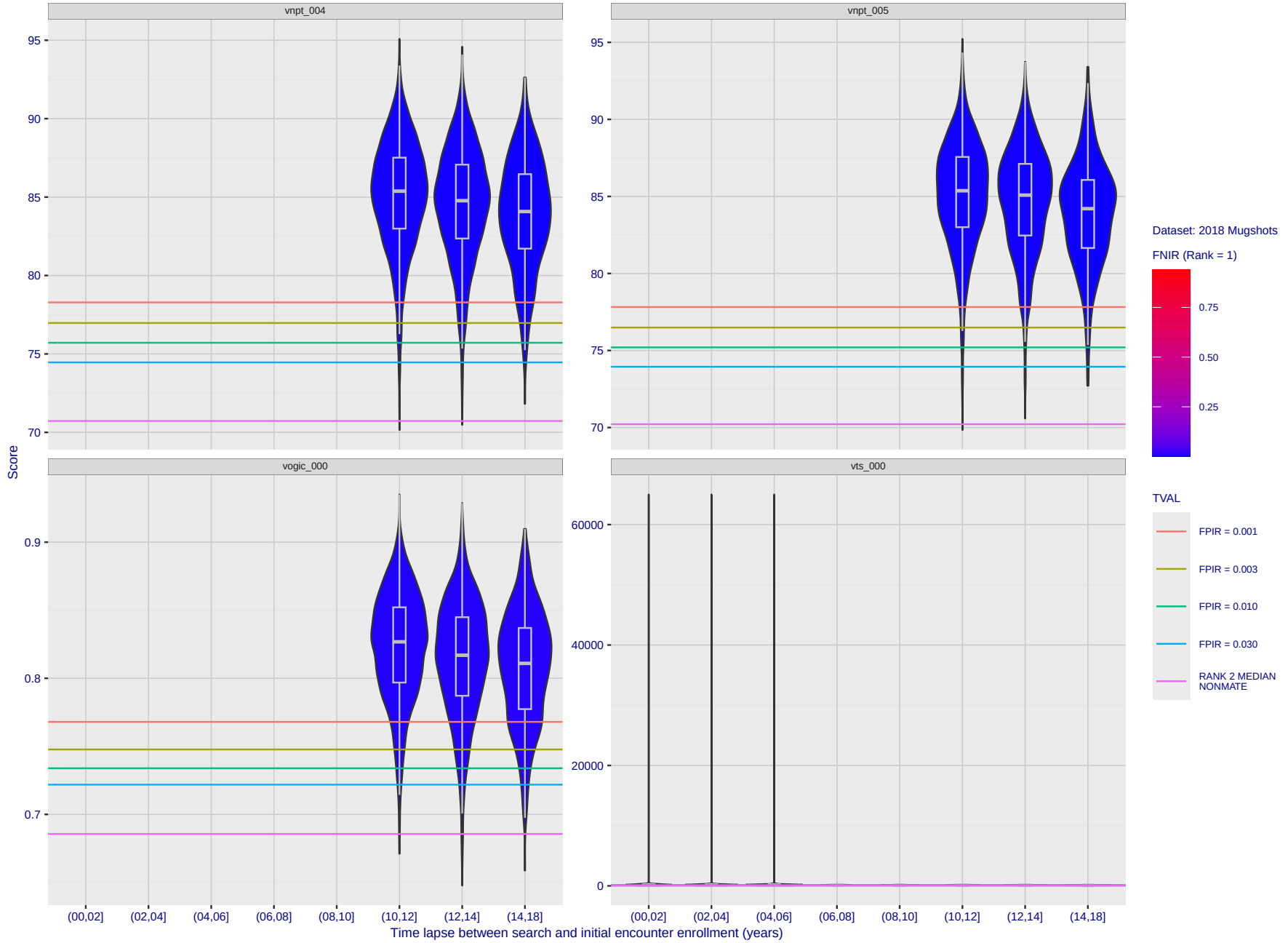


*Figure 446:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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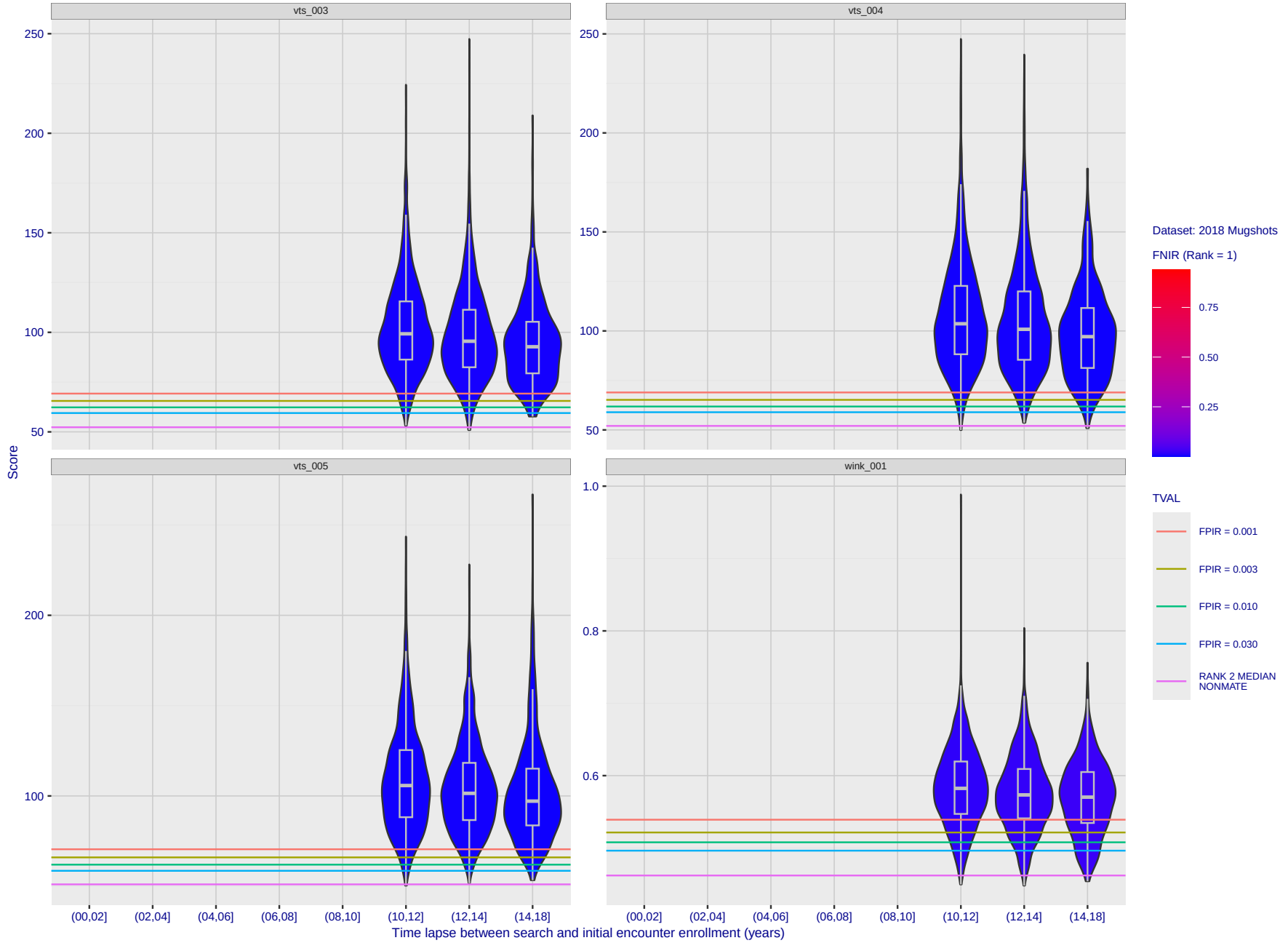


*Figure 447:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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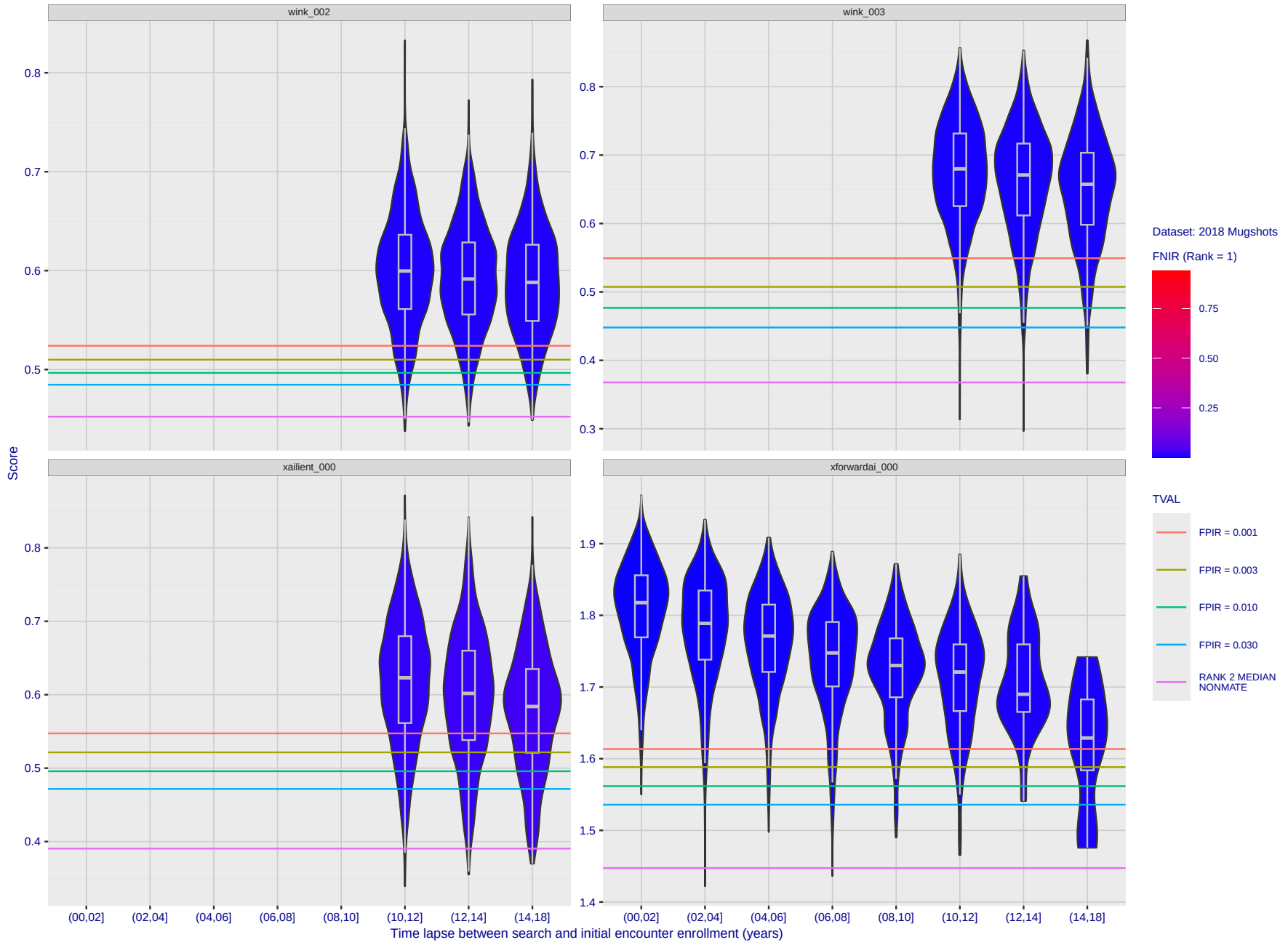


*Figure 448:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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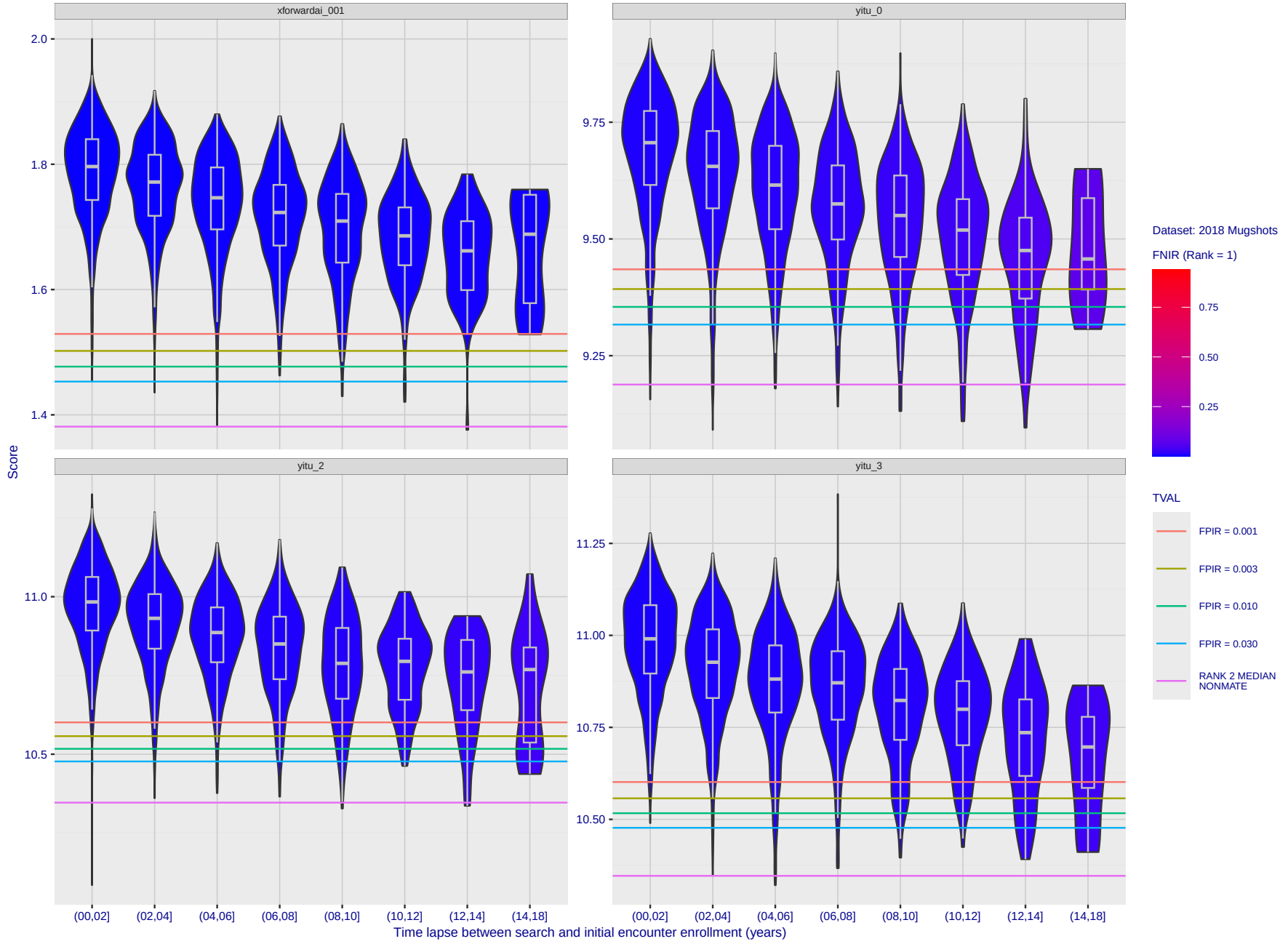


*Figure 449:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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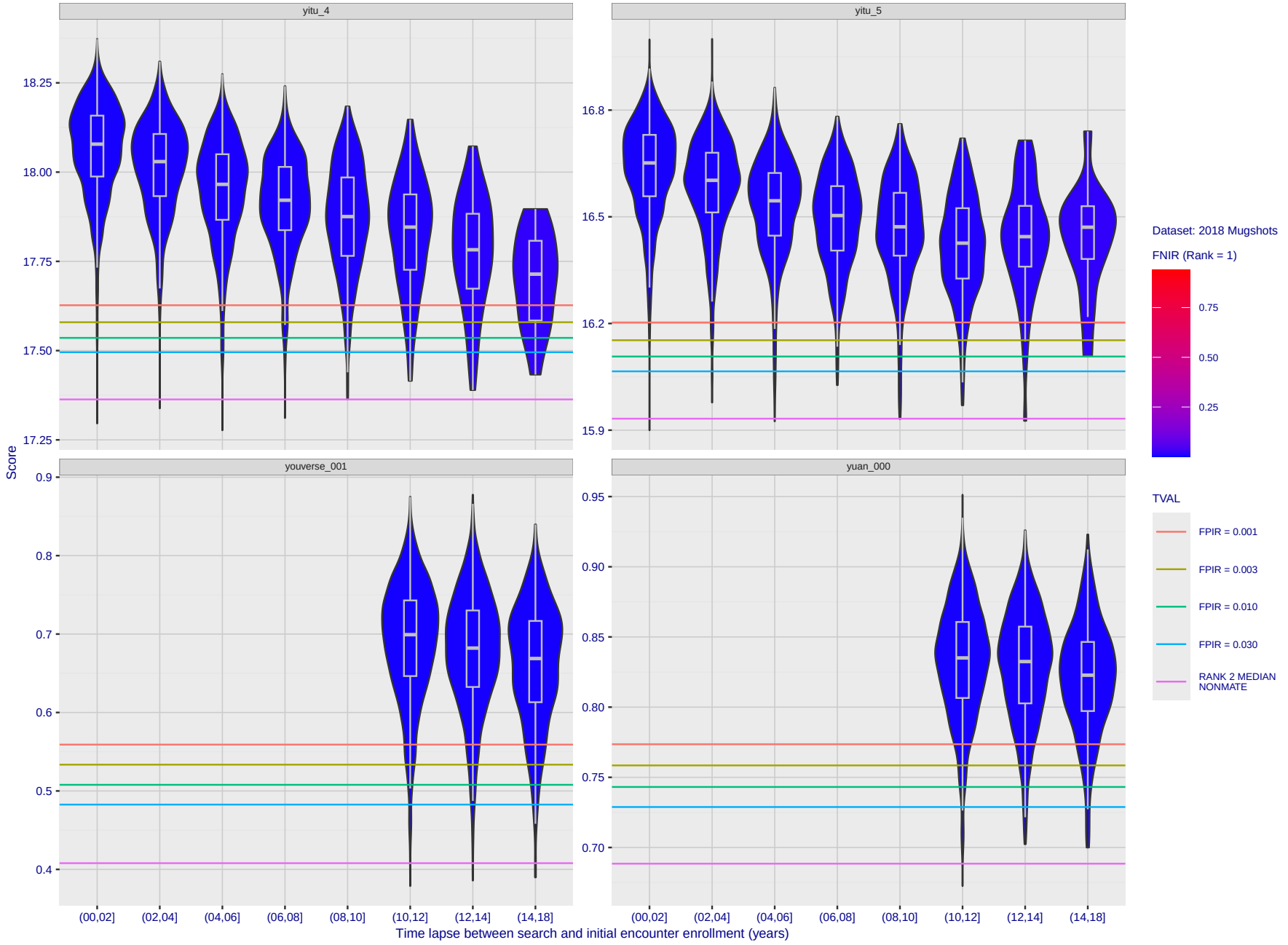


*Figure 450:* **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*



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# Appendix C Effect of enrolling multiple images

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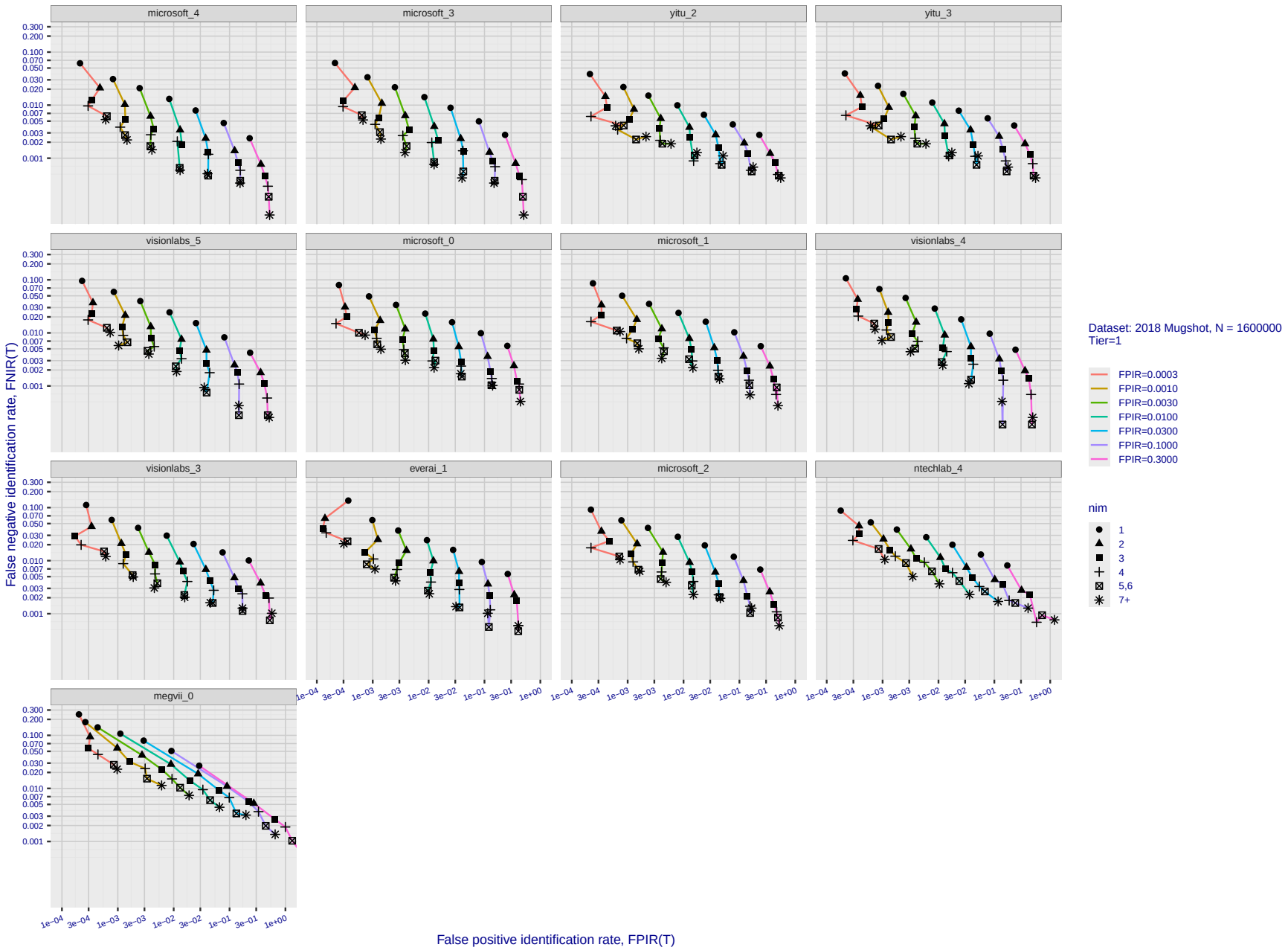


*Figure 451:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

2026/09/04 10:23:32

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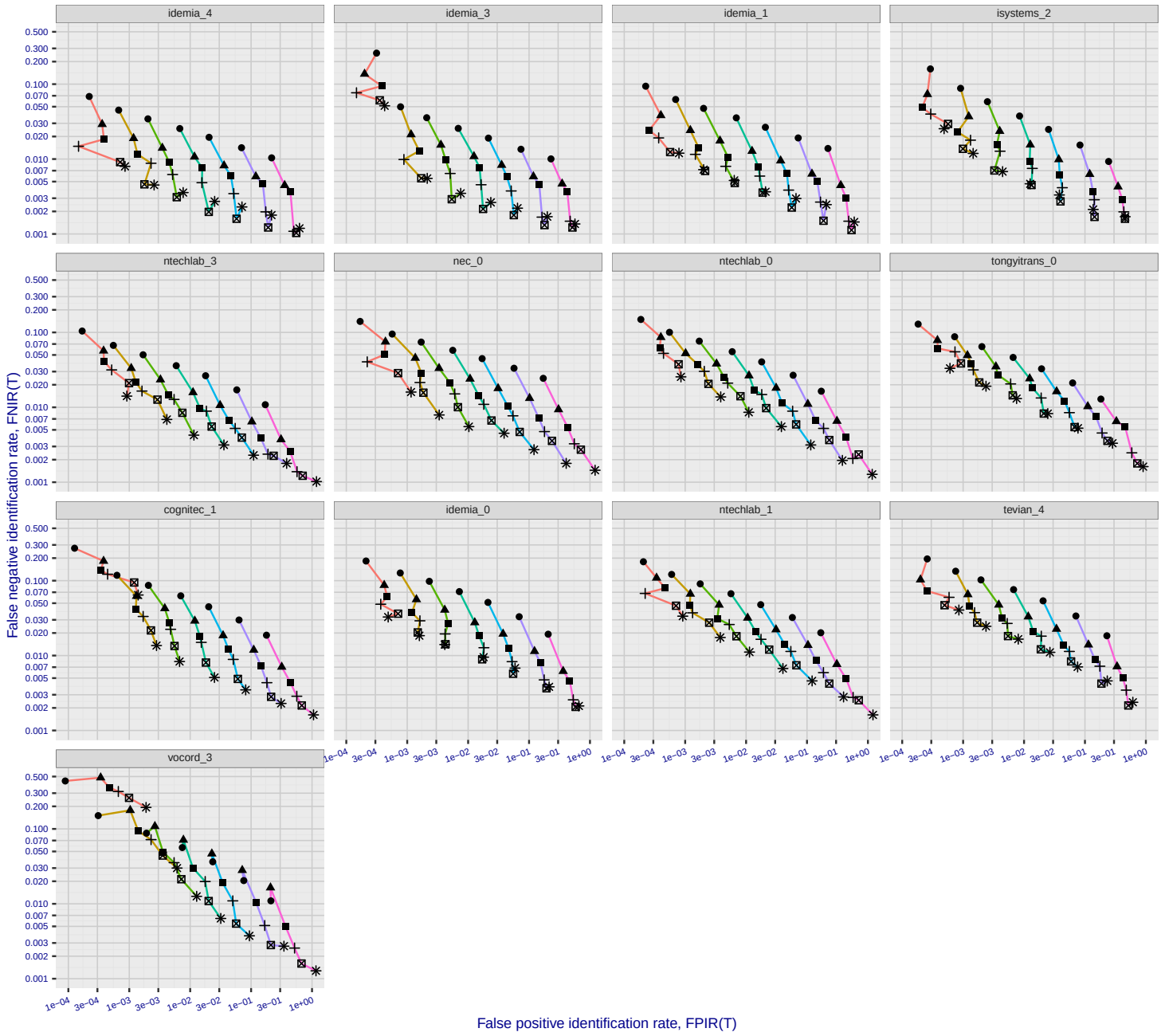


*Figure 452:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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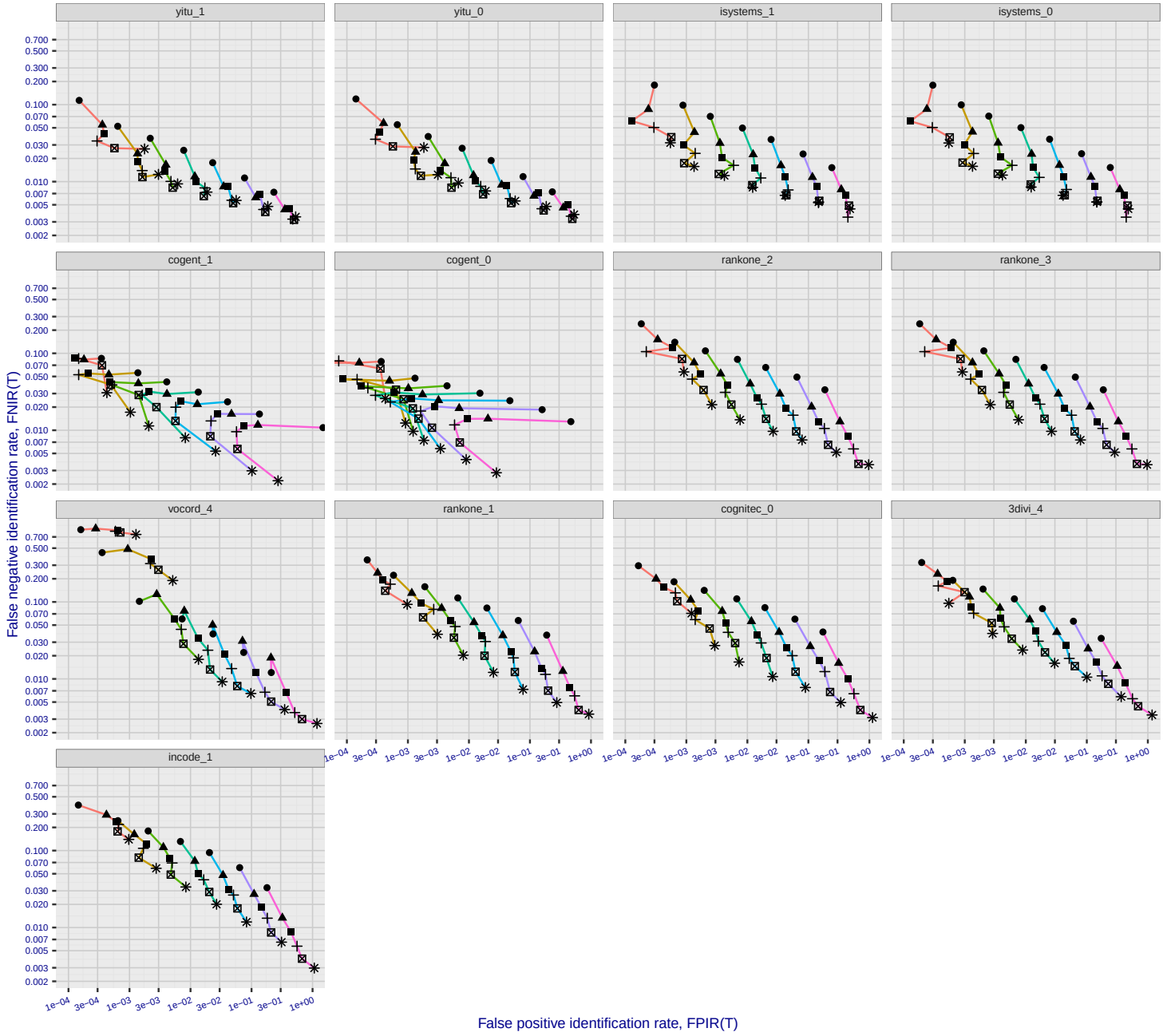


*Figure 453:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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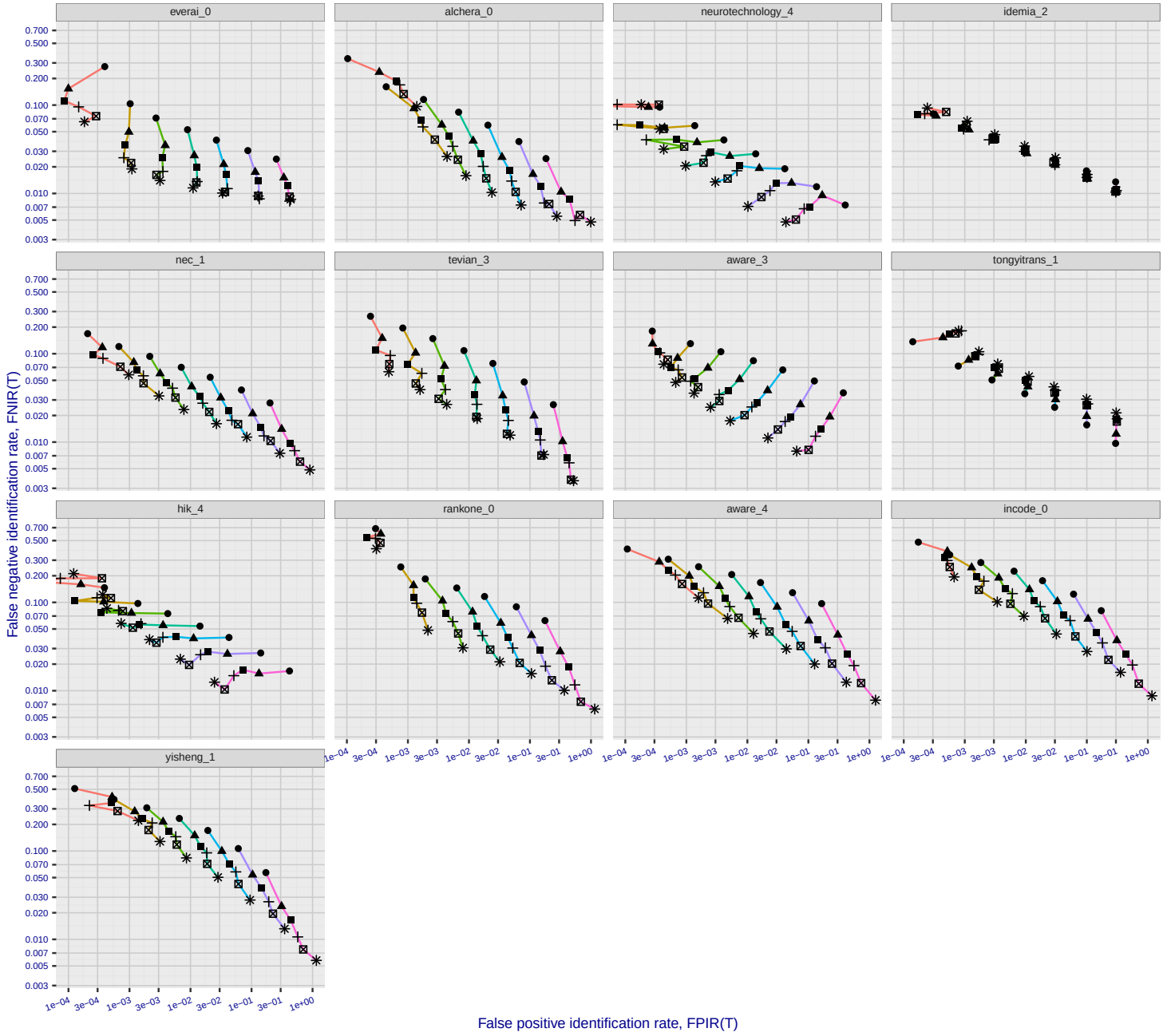


*Figure 454:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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T = 0 → Investigation
T > 0 → Identification

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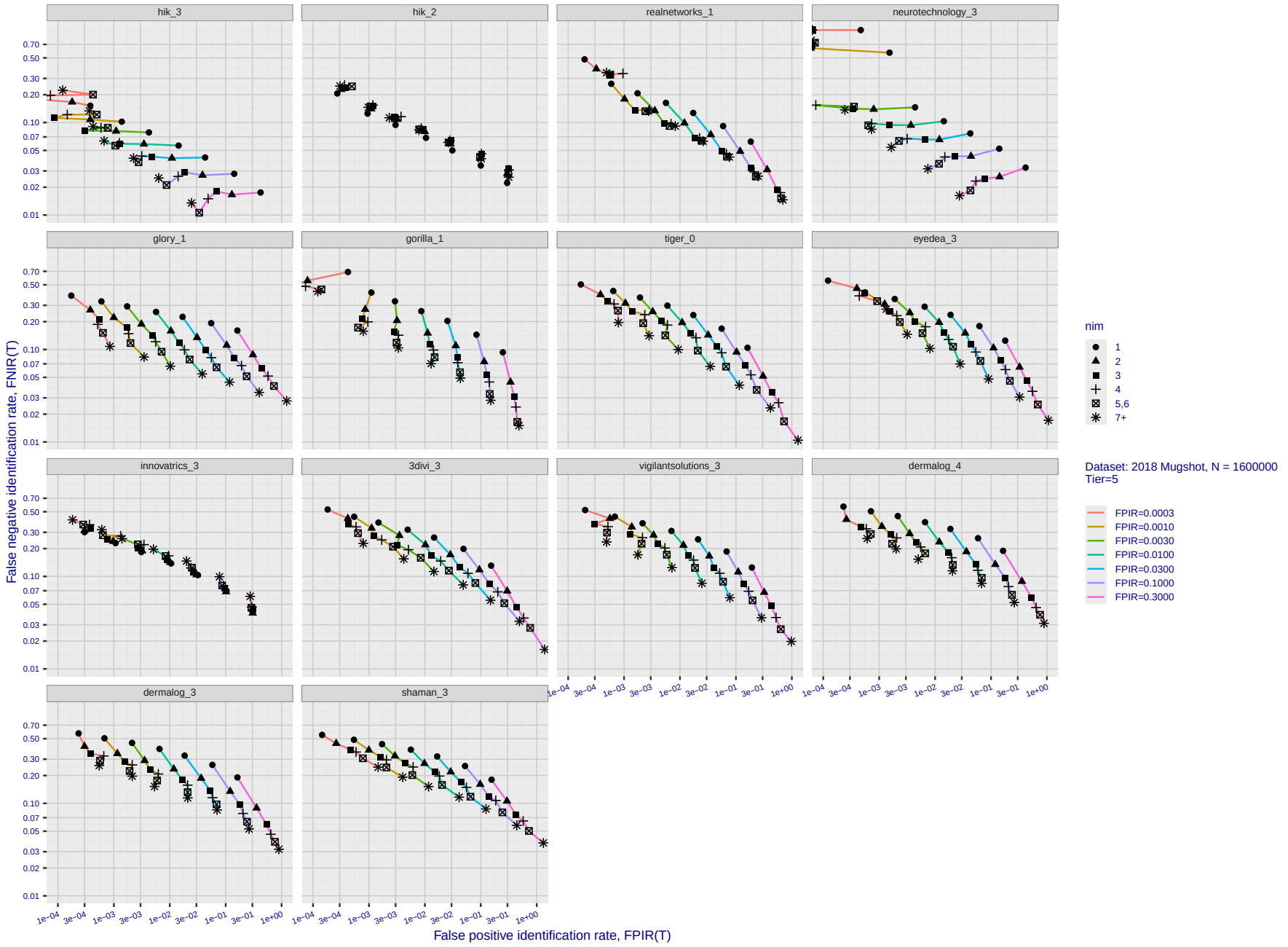


*Figure 455:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*



2026/09/04 10:23:32
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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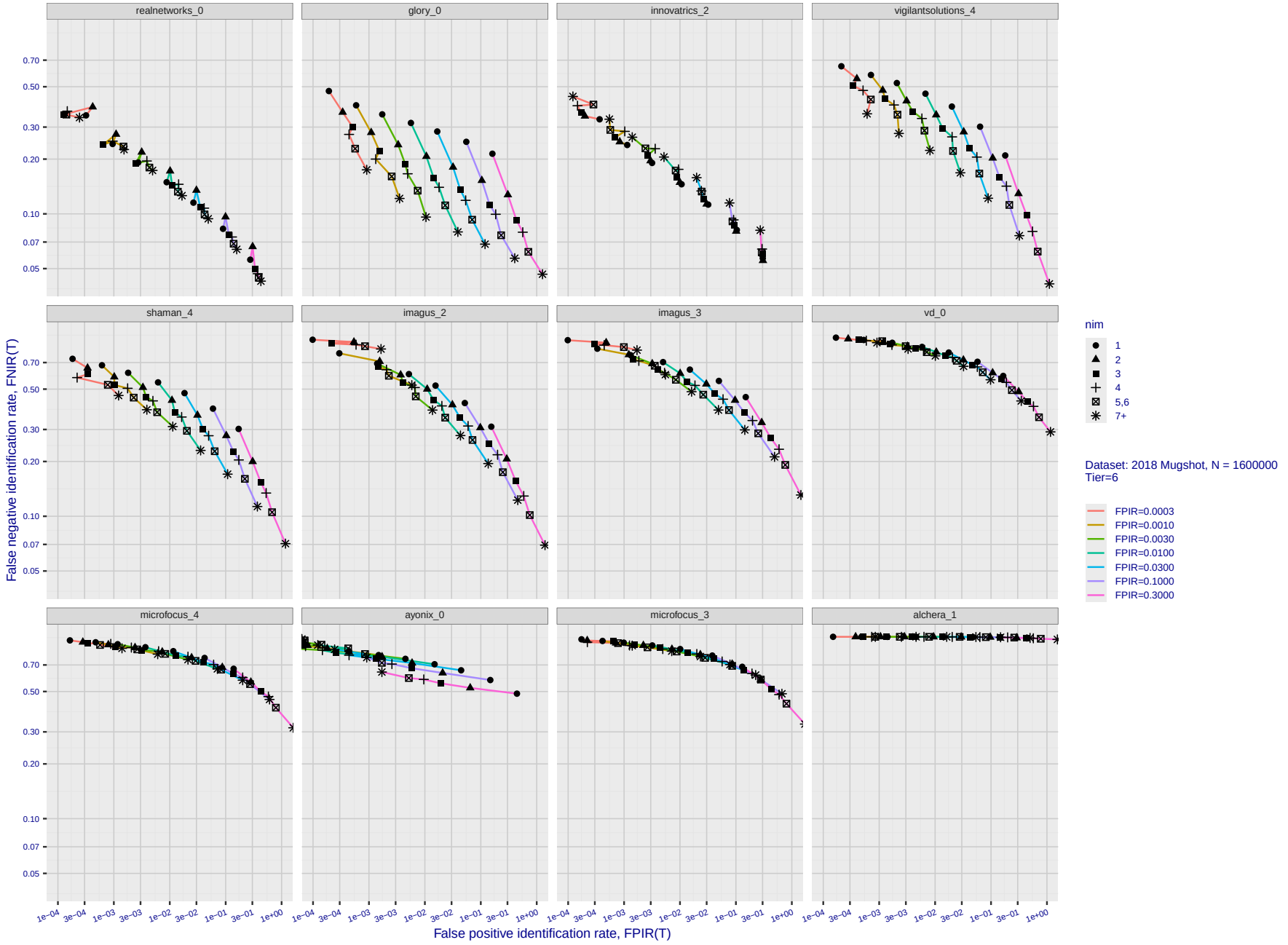


*Figure 456:* **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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# Appendix D Accuracy with poor quality webcam images

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|---|---|---|---|---|---|
| 2026/09/04 | FNIR(N, R, T) = | False neg. identification rate | N = Num. enrolled subjects | T = Threshold | $T = 0 \rightarrow$ Investigation |
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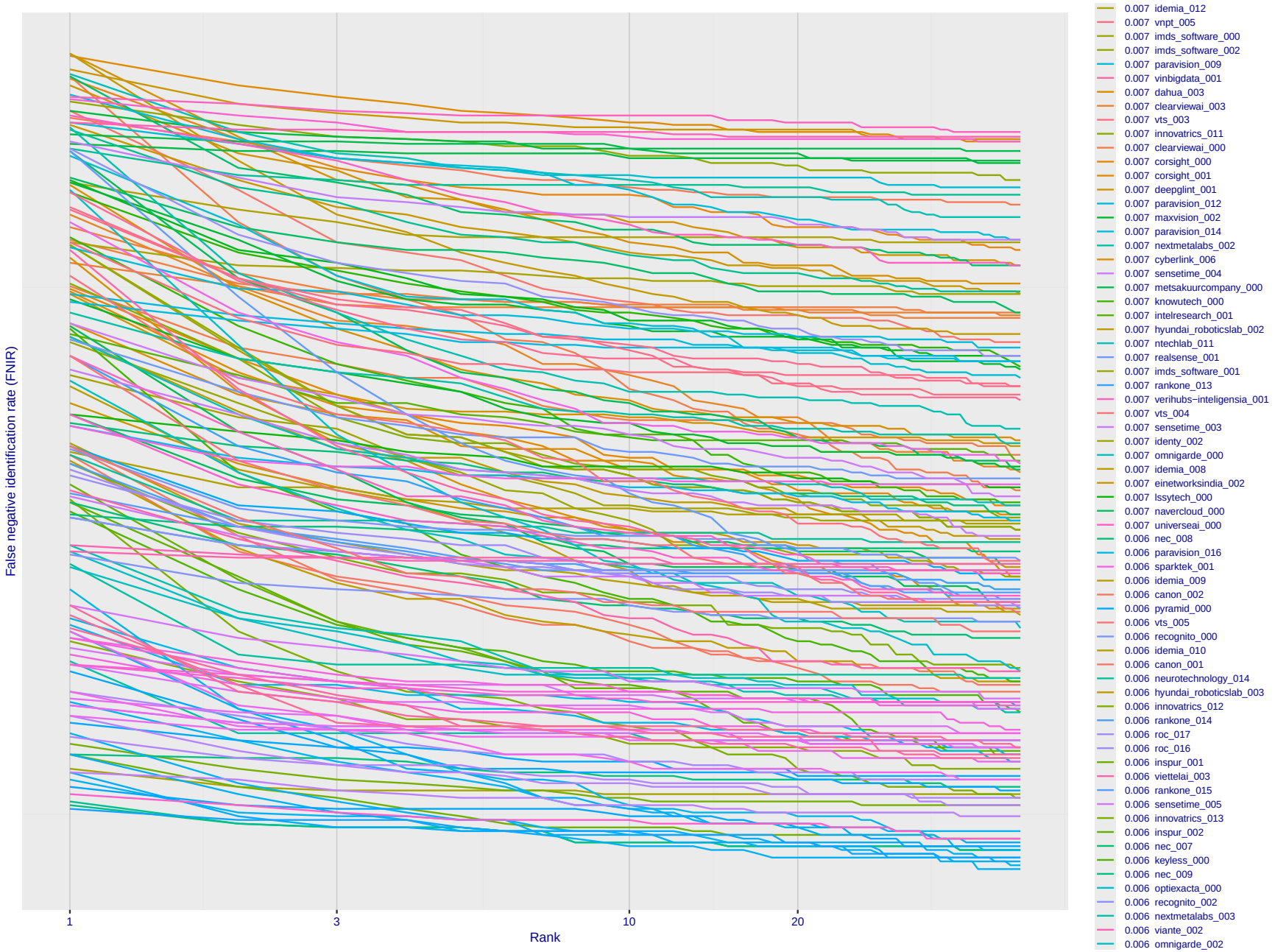


*Figure 457:* **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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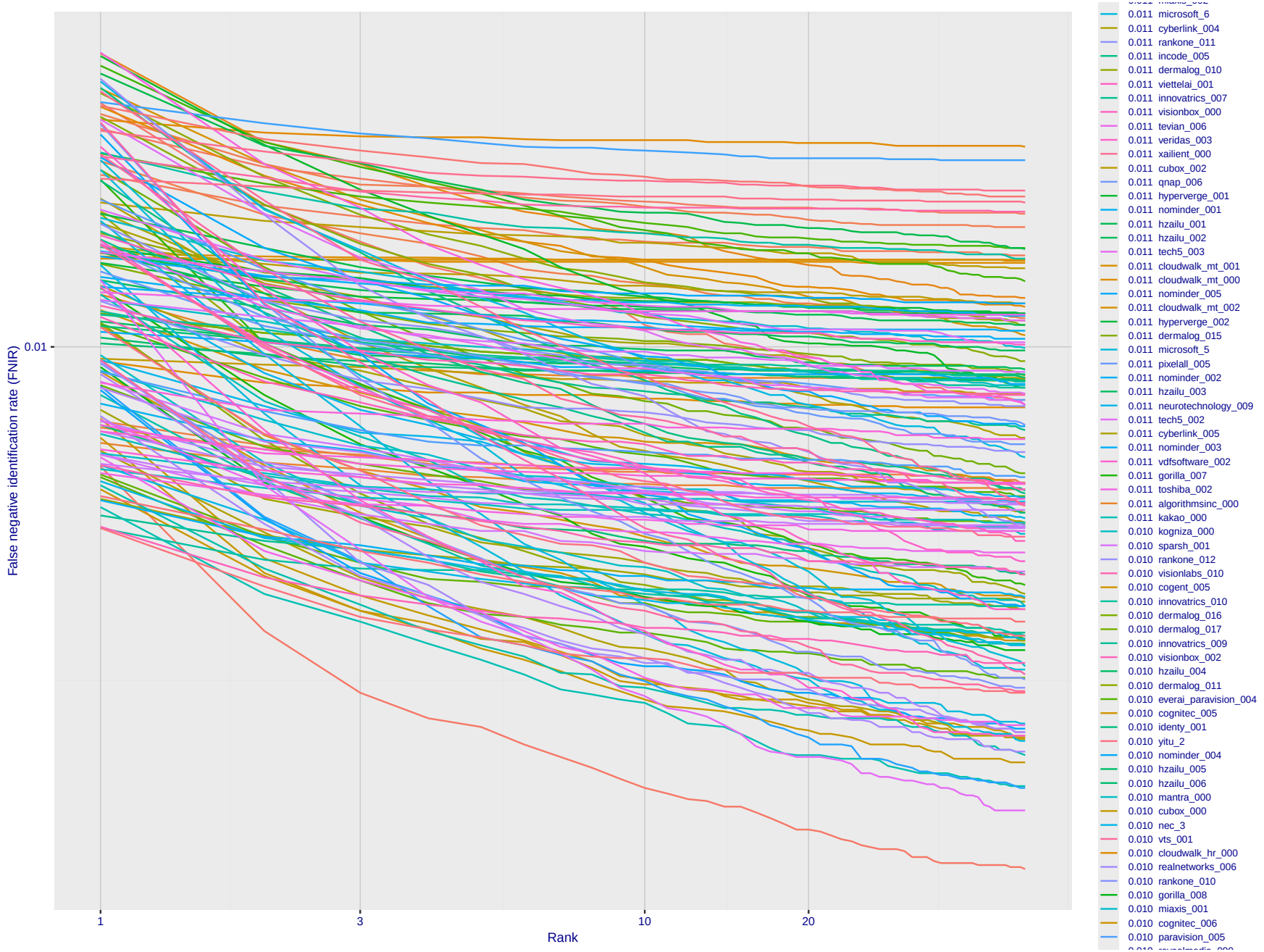


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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
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*Figure 458:* **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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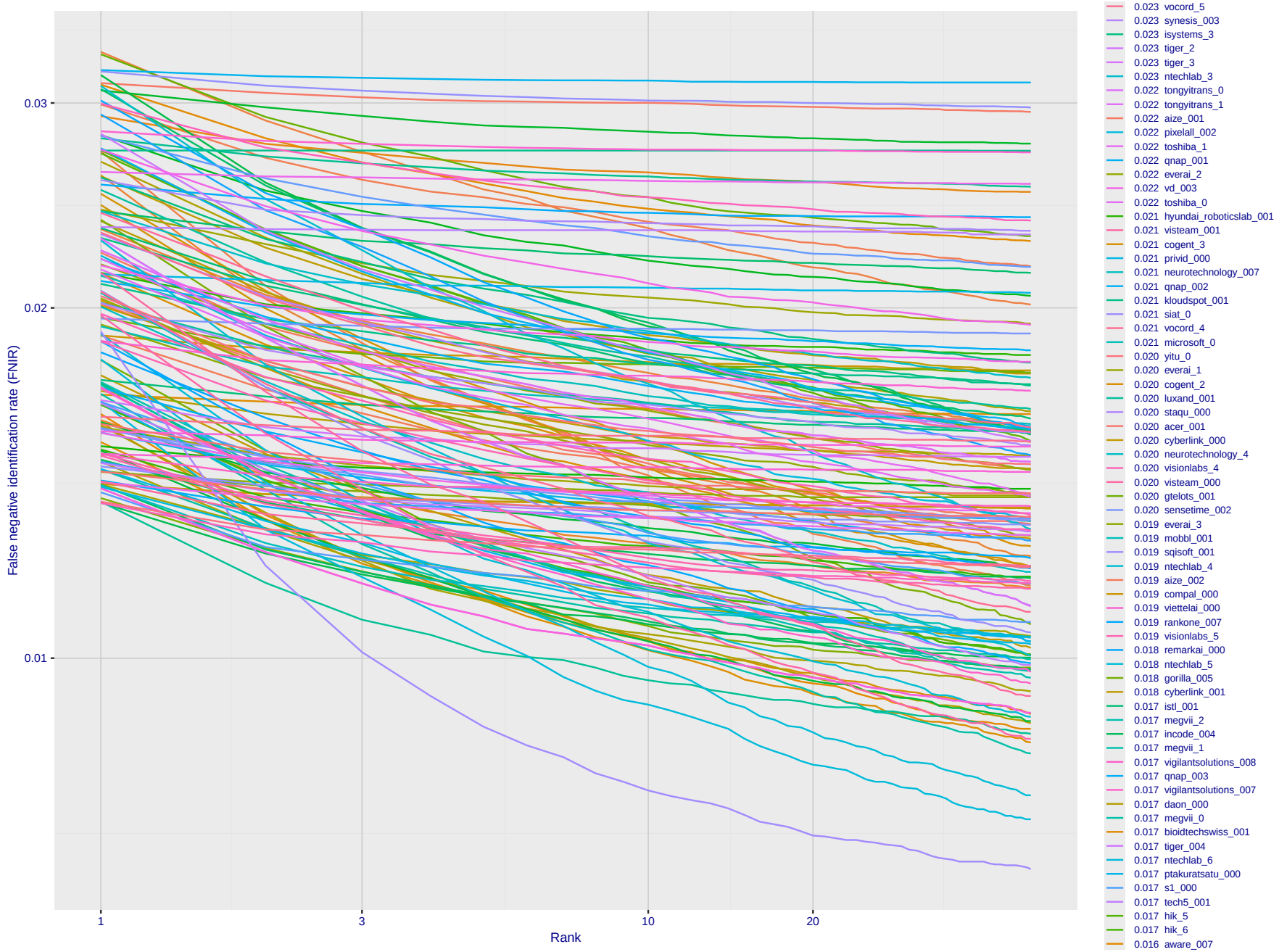


*Figure 459:* **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*



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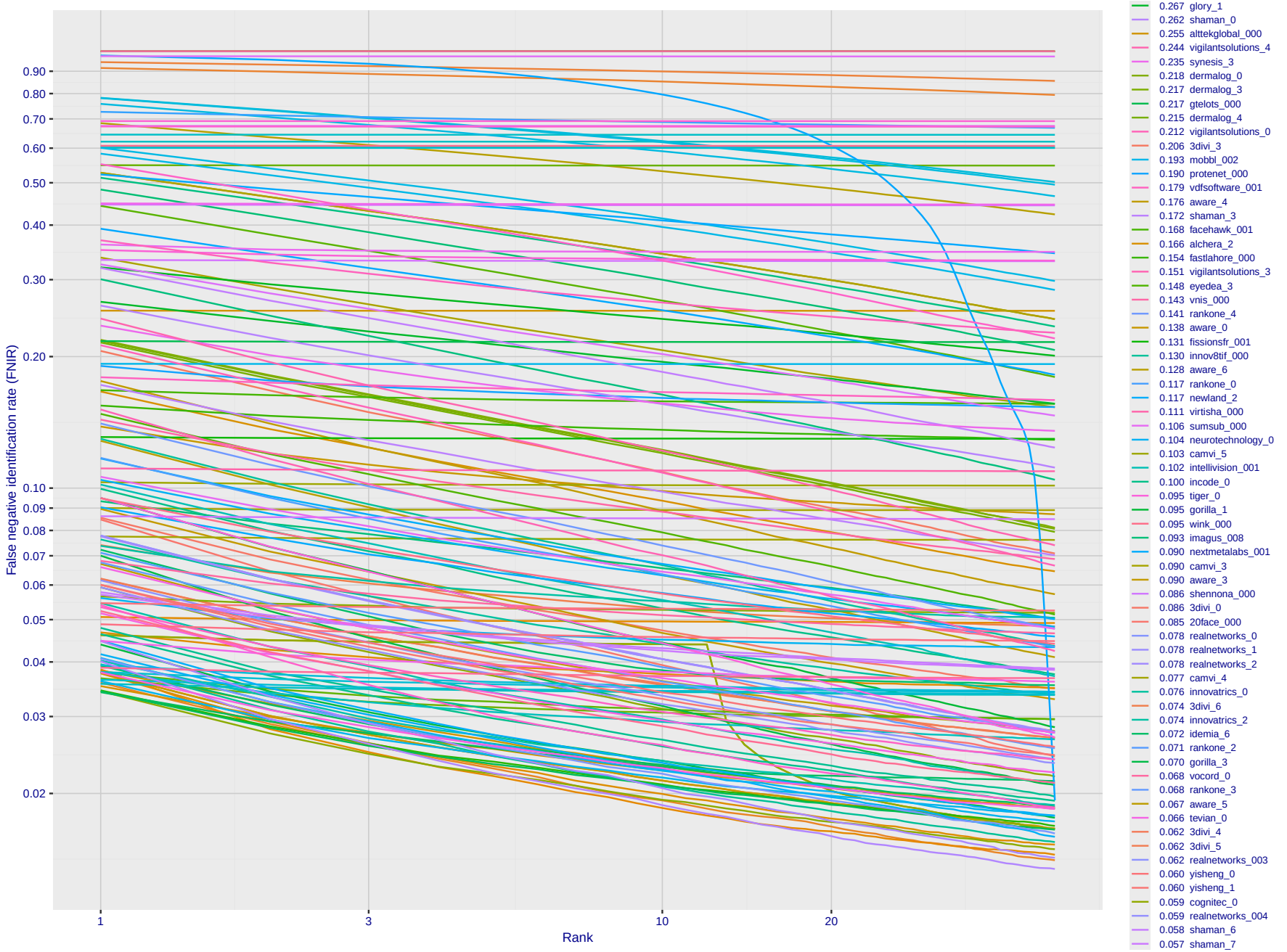


*Figure 460:* **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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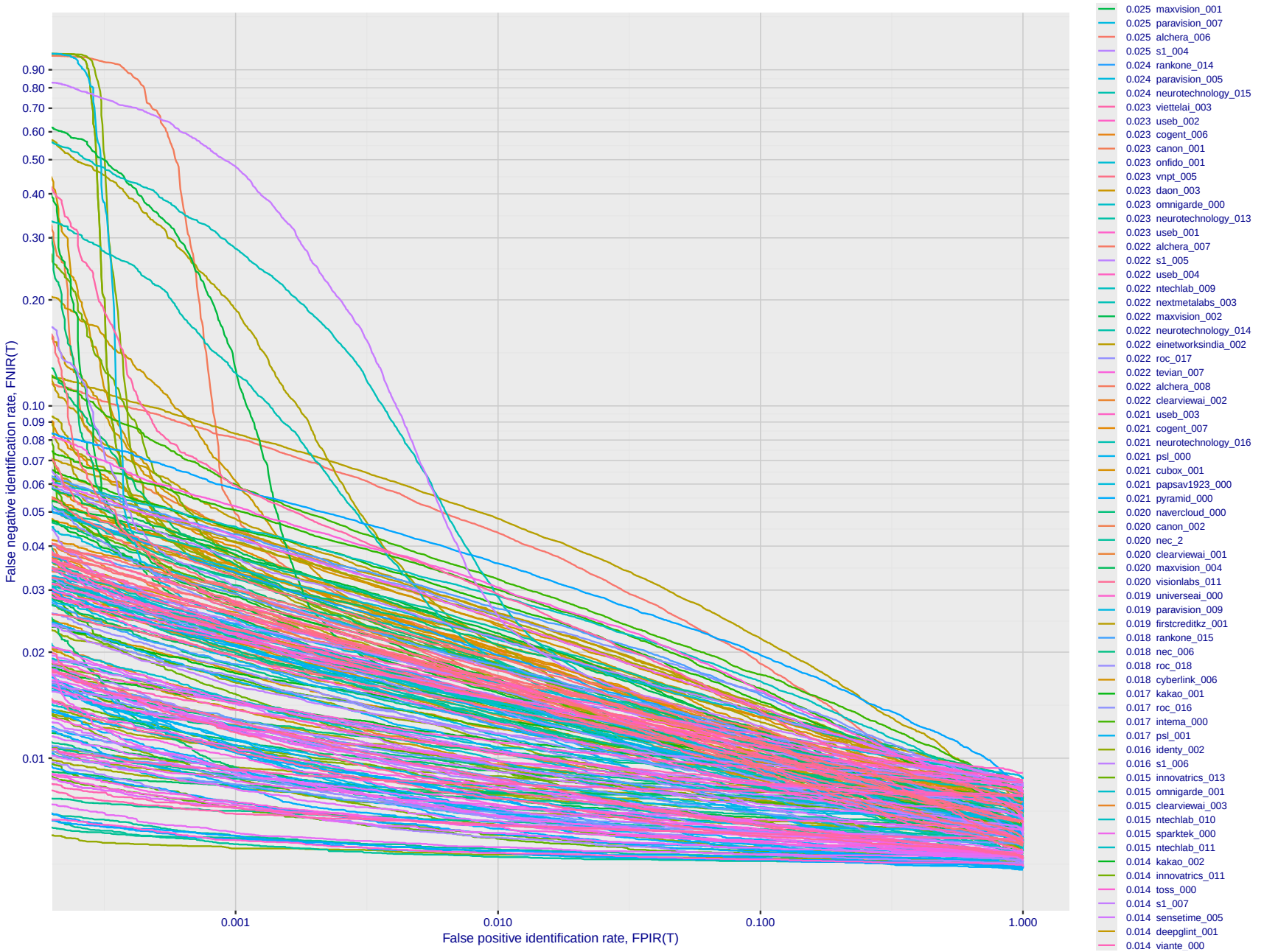


*Figure 461:* **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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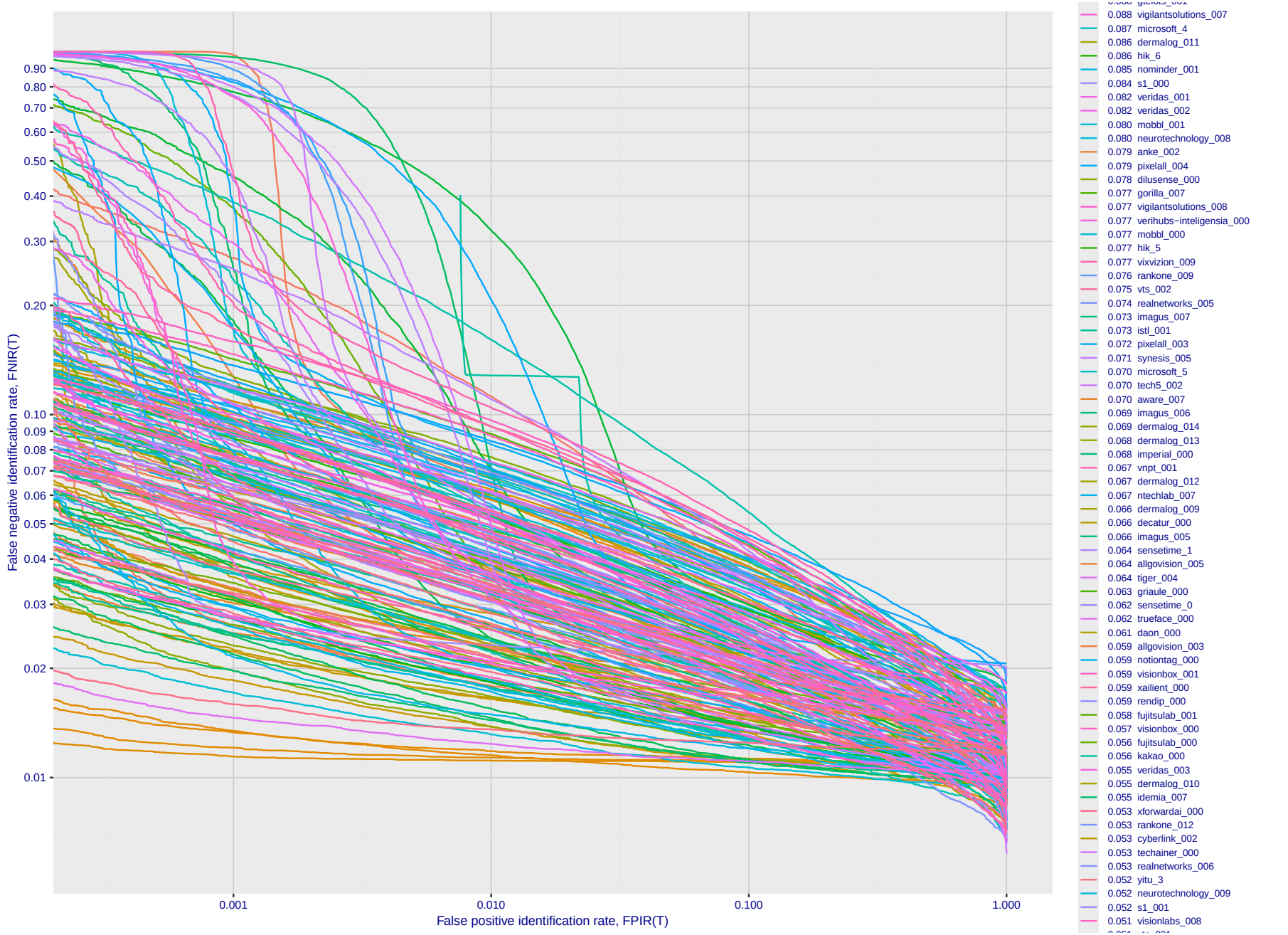


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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
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T = Threshold

T = 0 → Investigation
T > 0 → Identification

*Figure 462:* **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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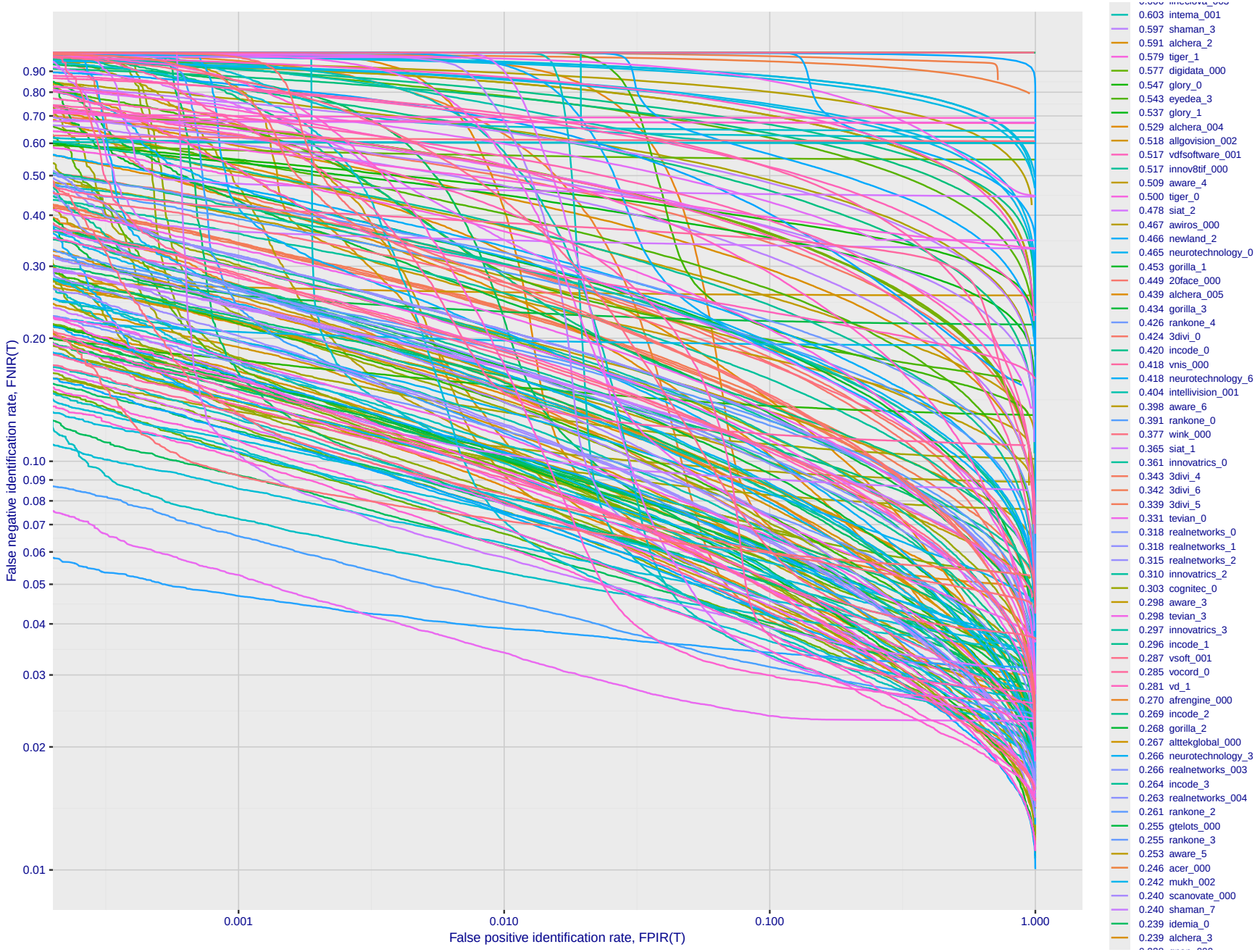


*Figure 463:* **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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# Appendix E Accuracy for profile-view to frontal recognition

Figures 464 - 466 gives accuracy results for searching 100 000 mated and 100 000 non-mated profile-view images against the same FRVT 2018 frontal enrollment dataset, N = 1 600 000, used in the main mugshot trials. This experiment corresponds to row-13 of Table 1. An example of profile-view image is given in Figure 7.

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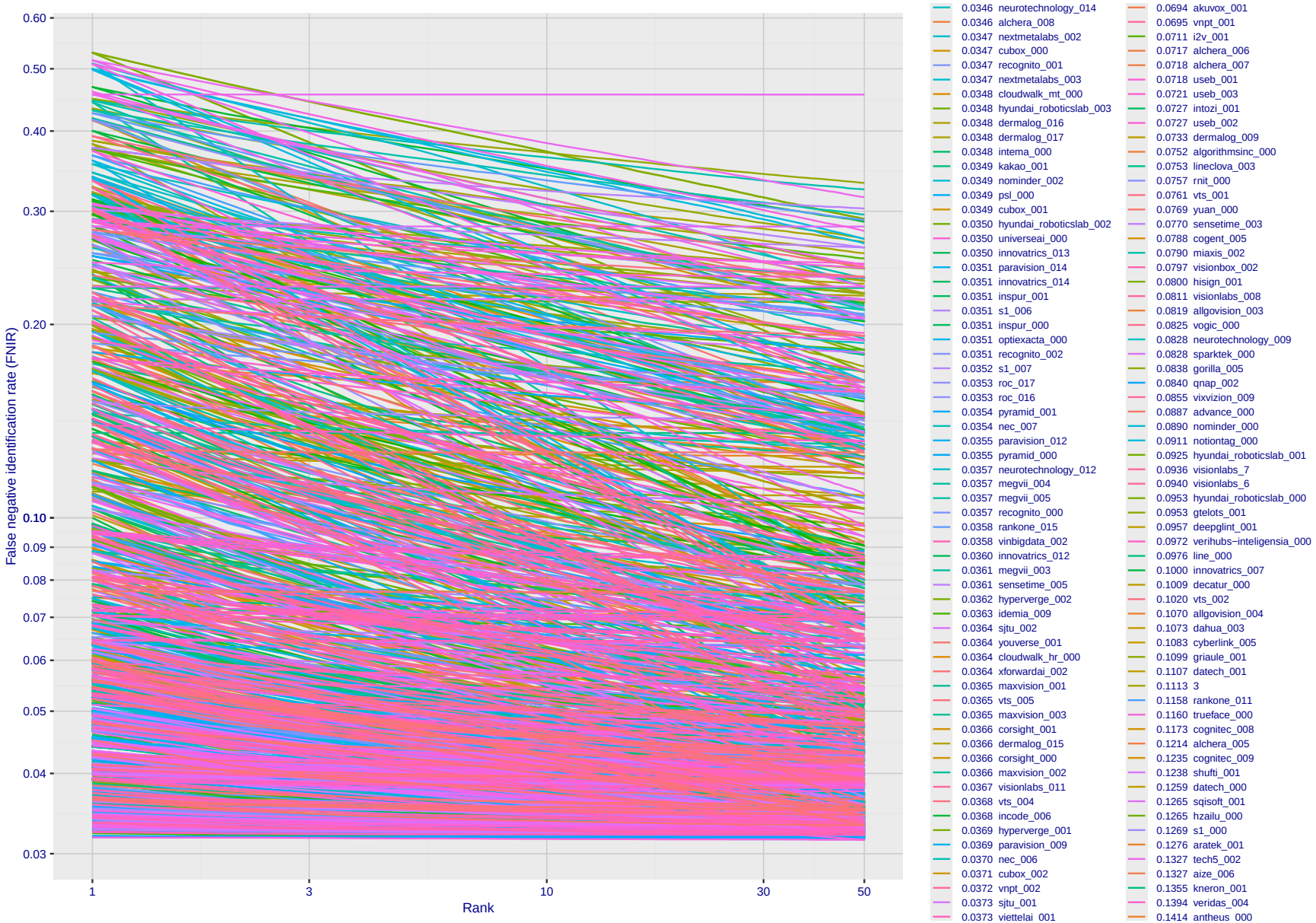


*Figure 464:* **[Mugshot and profile-view dataset] Rank-based accuracy**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Note that some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms. The best result is that 91% of profile-view searches yield the correct mate at rank 1, and better than 94% in the top-50 candidates.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

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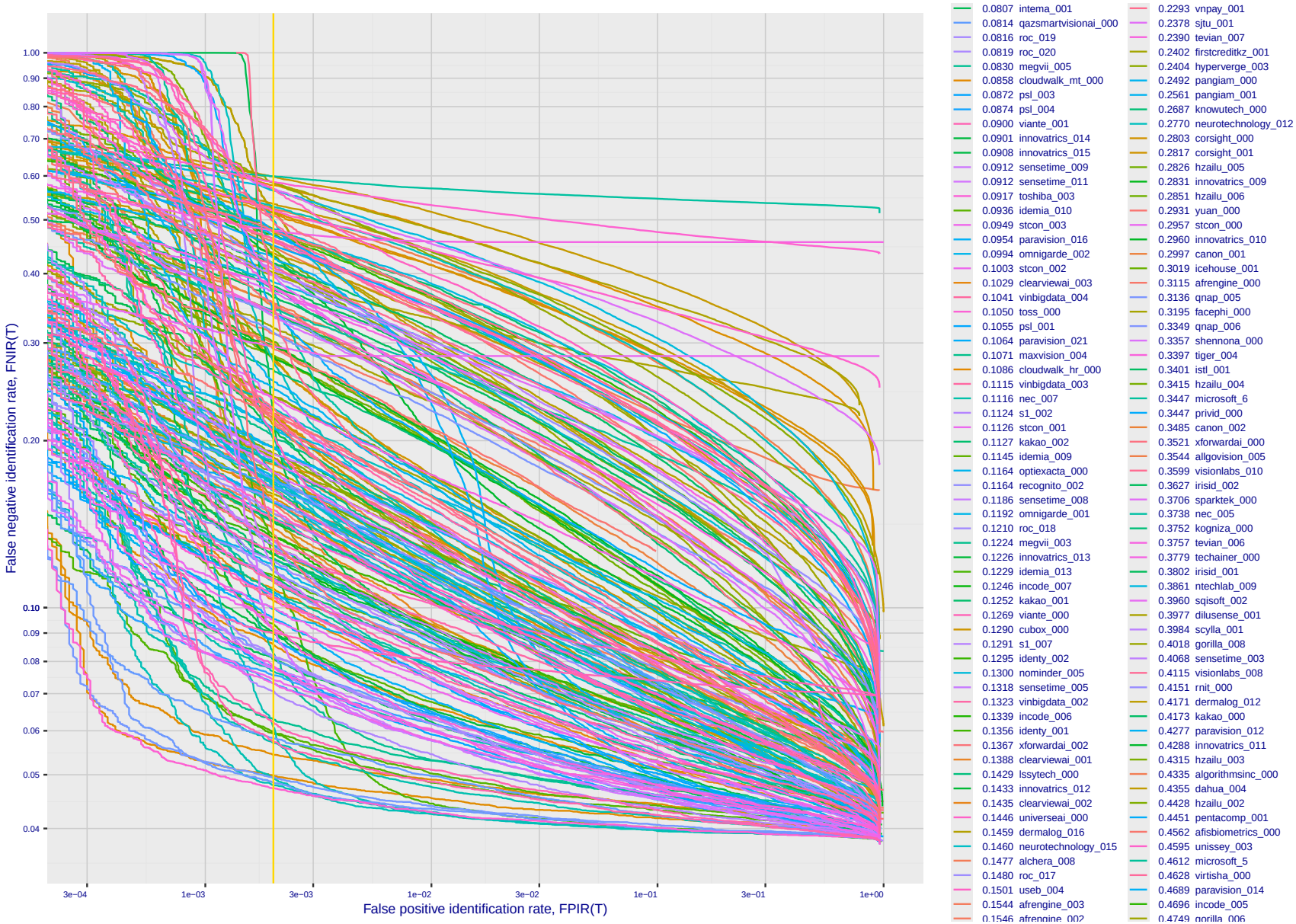


*Figure 465:* **[Mugshot and profile-view dataset] Threshold-based accuracy**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Note that some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms.*



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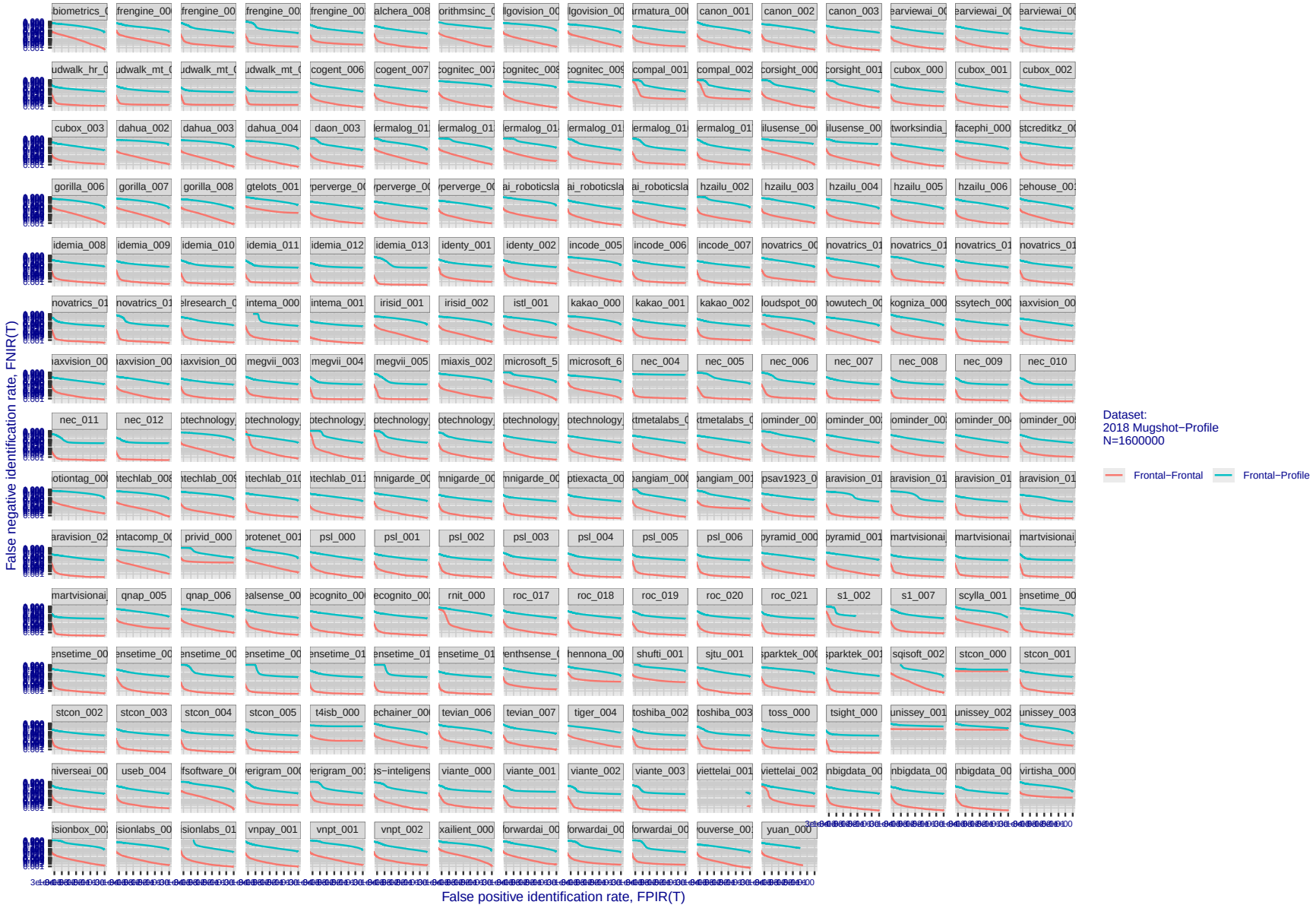


*Figure 466:* **[Mugshot and profile-view dataset] Speed-accuracy tradeoff**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms. Blue lines connect points of equal threshold from which it is evident that some algorithms would give markedly higher false positive outcomes if profile-view images were searched in a system configured for frontal searches. This would be a vulnerability in an access control system.*



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10:23:32

FNIR(N, R, T) = False neg. identification rate
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T > 0 → Identification

# Appendix F Search duration

As in and prior tests, this section documents search speeds spanning three orders of magnitude. In applications where search volumes are high enough, this will have implications for hardware requirements especially for large N or when search duration is appreciably larger than the time it takes to prepare a template from the search image(s). Further, given very large (and growing) operational databases, the scalability of algorithms is important. It has been reported previously [8] that search duration can scale sublinearly with enrolled population size N. Further there has been considerable recent research on indexing, exact [13] and approximate nearest neighbor search [1,13] and fast-search [14,16].

Figure 467 charts the search duration measurements presented earlier in Tables 2 - 4.

- Most algorithms scale linearly. For those in that category, there is a wide range in speed with search durations ranging from 82 milliseconds for a 12 million gallery (for NEC-3) to more than 40 seconds (for Yitu-3, Toshiba-2) and even higher for less accurate algorithms.
- Some developers (Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs) provide algorithms whose template search durations grow approimately logarithmically i.e. $T(N)\tilde{a}\log N$ with the constant $a$ varying between implementations. In the figure this model is fit using the point $T(1) = 0$, and $T(640\,000)$. This very sublinear behaviour affords extremely fast search times in very large galleries. One caveat for the sublinear algorithms is that their fast-search data structures can require considerable computation time - on the order of hours - for N in the millions, and this scales mildy super-linearly, i.e. $O(N^b), b > 1$. There are exceptions: the Camvi algorithms take minutes; and Innvovatrics' scale sublinearly.

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| | | | | | |
|---|---|---|---|---|---|
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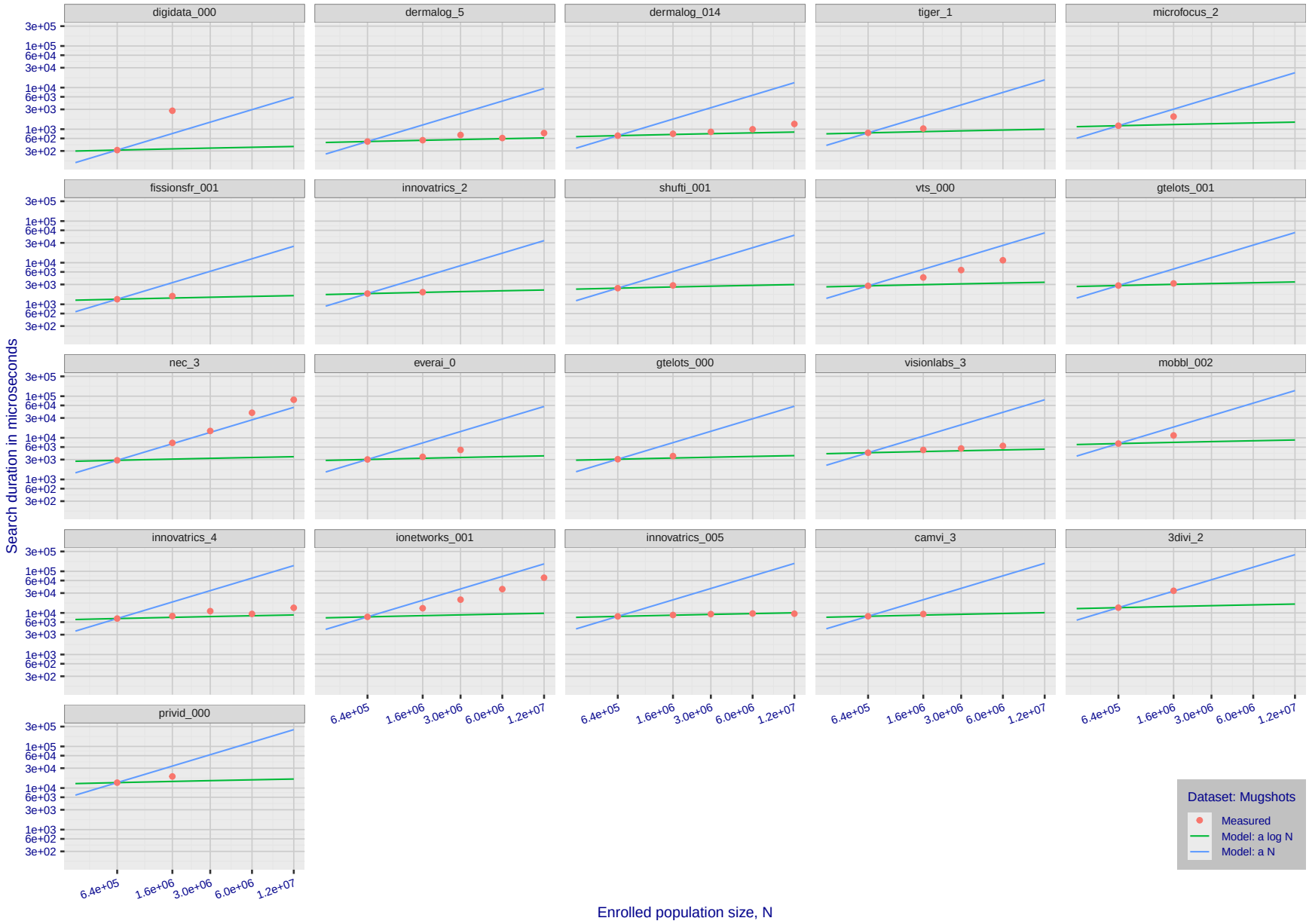


*Figure 467:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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FNIR(N, R, T) = False neg. identification rate
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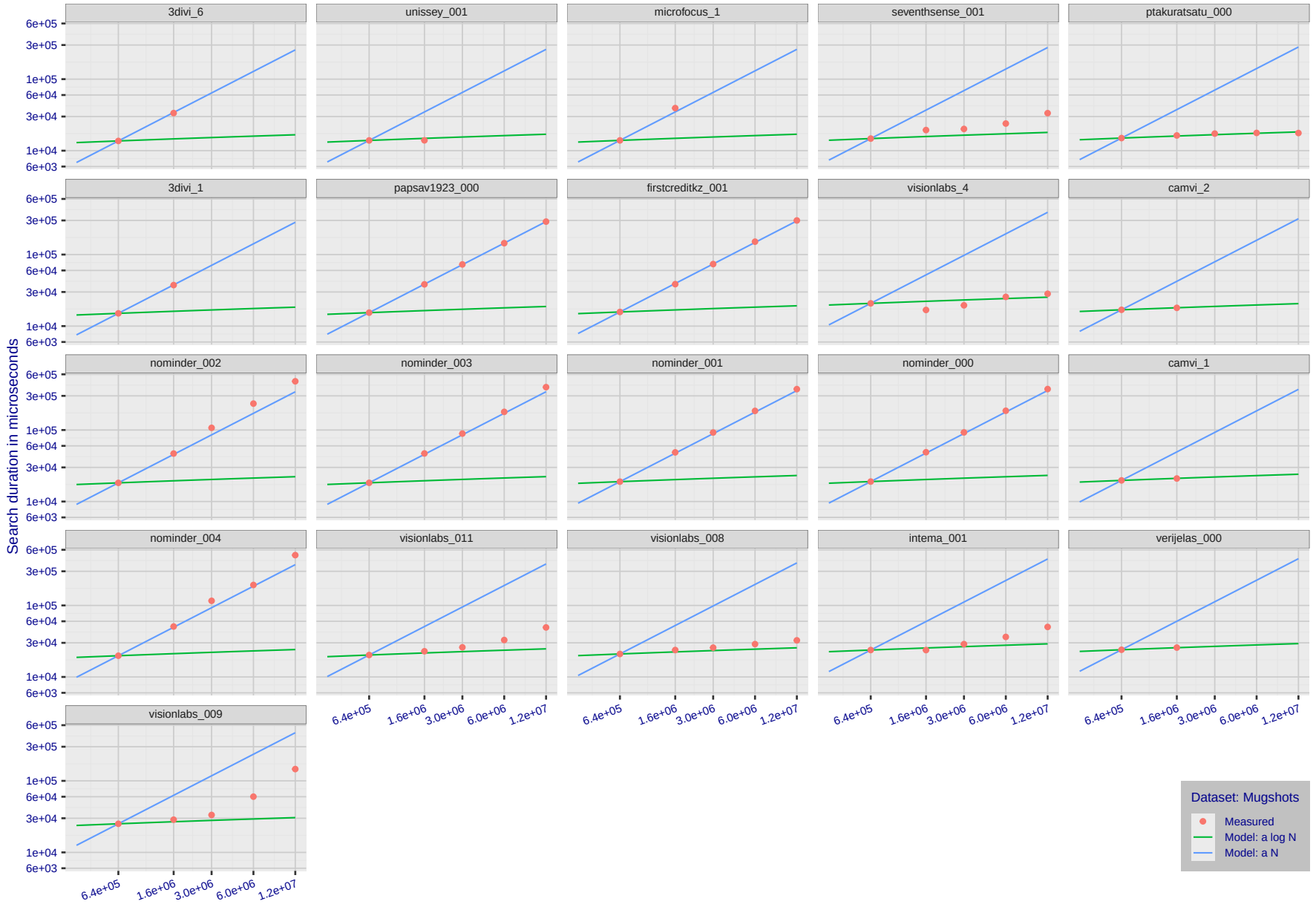


*Figure 468:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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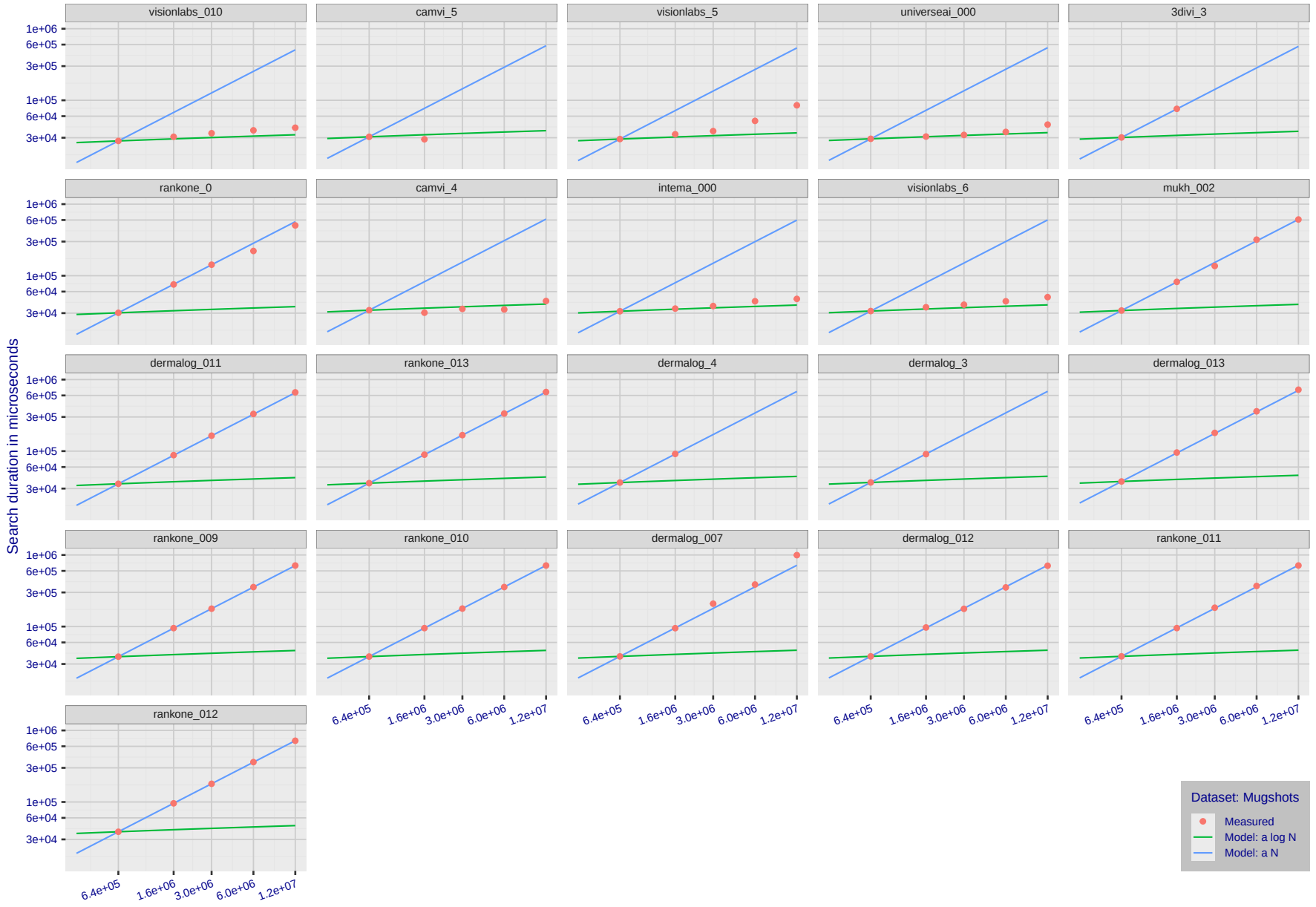


*Figure 469:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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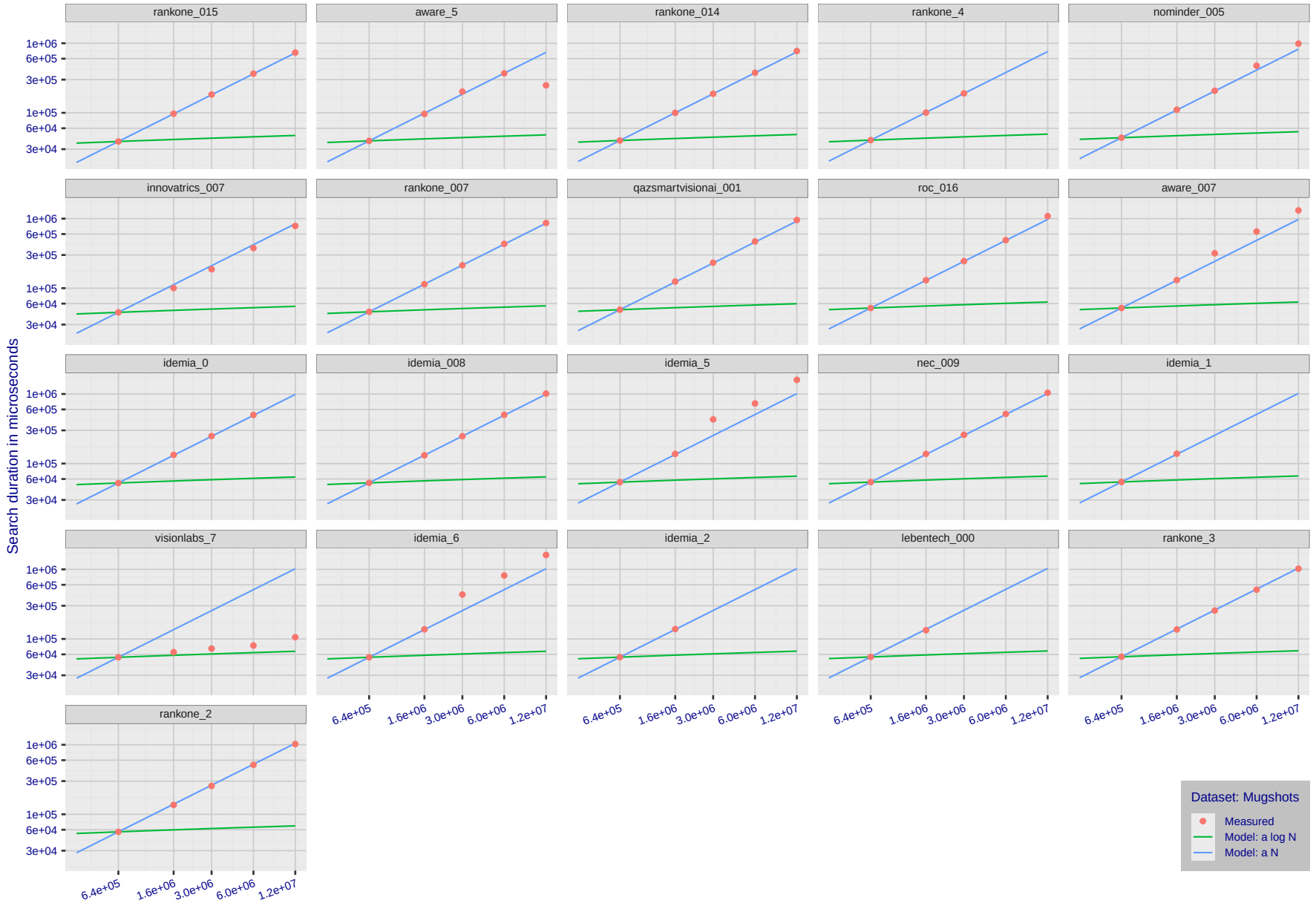


*Figure 470:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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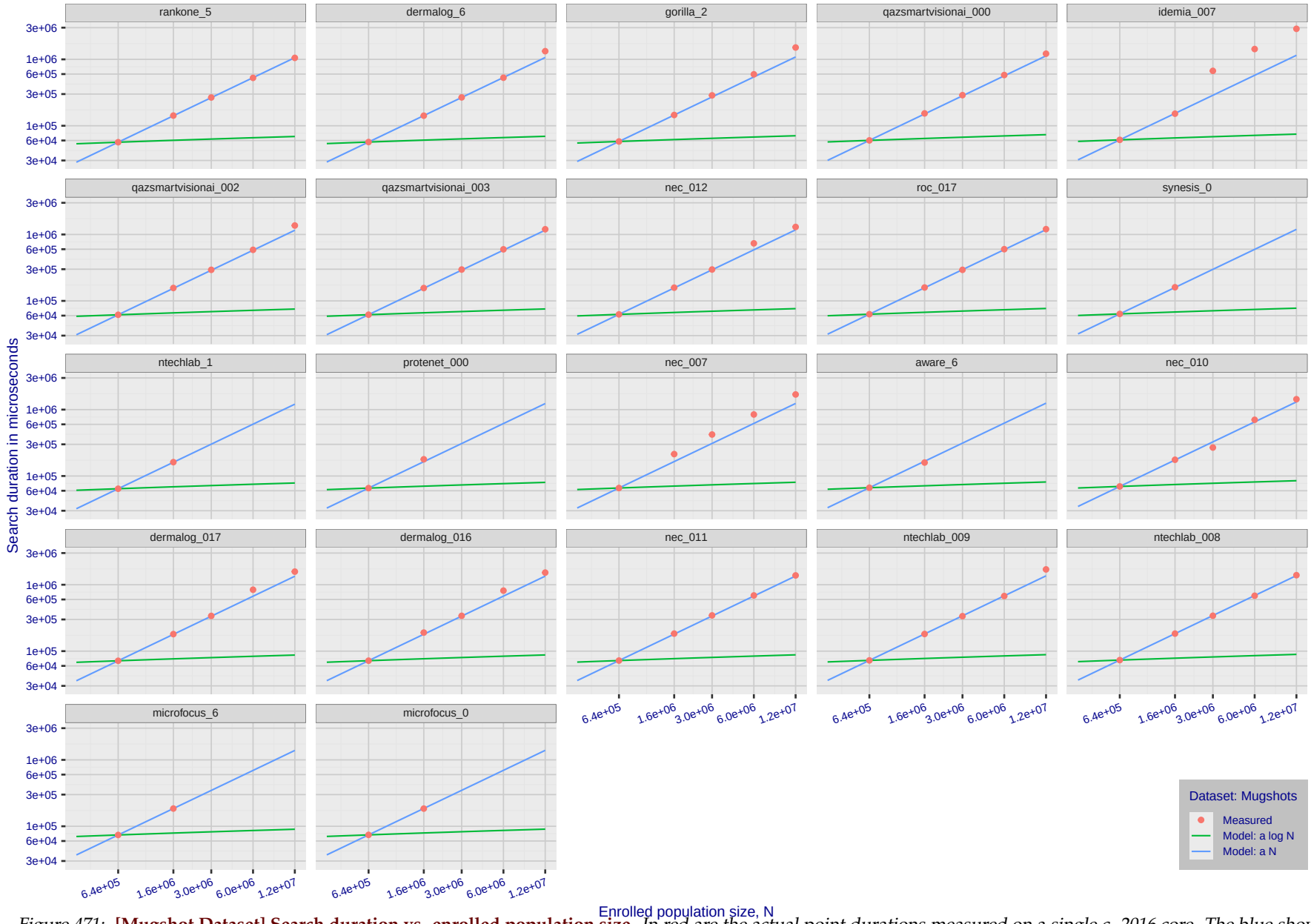


*Figure 471:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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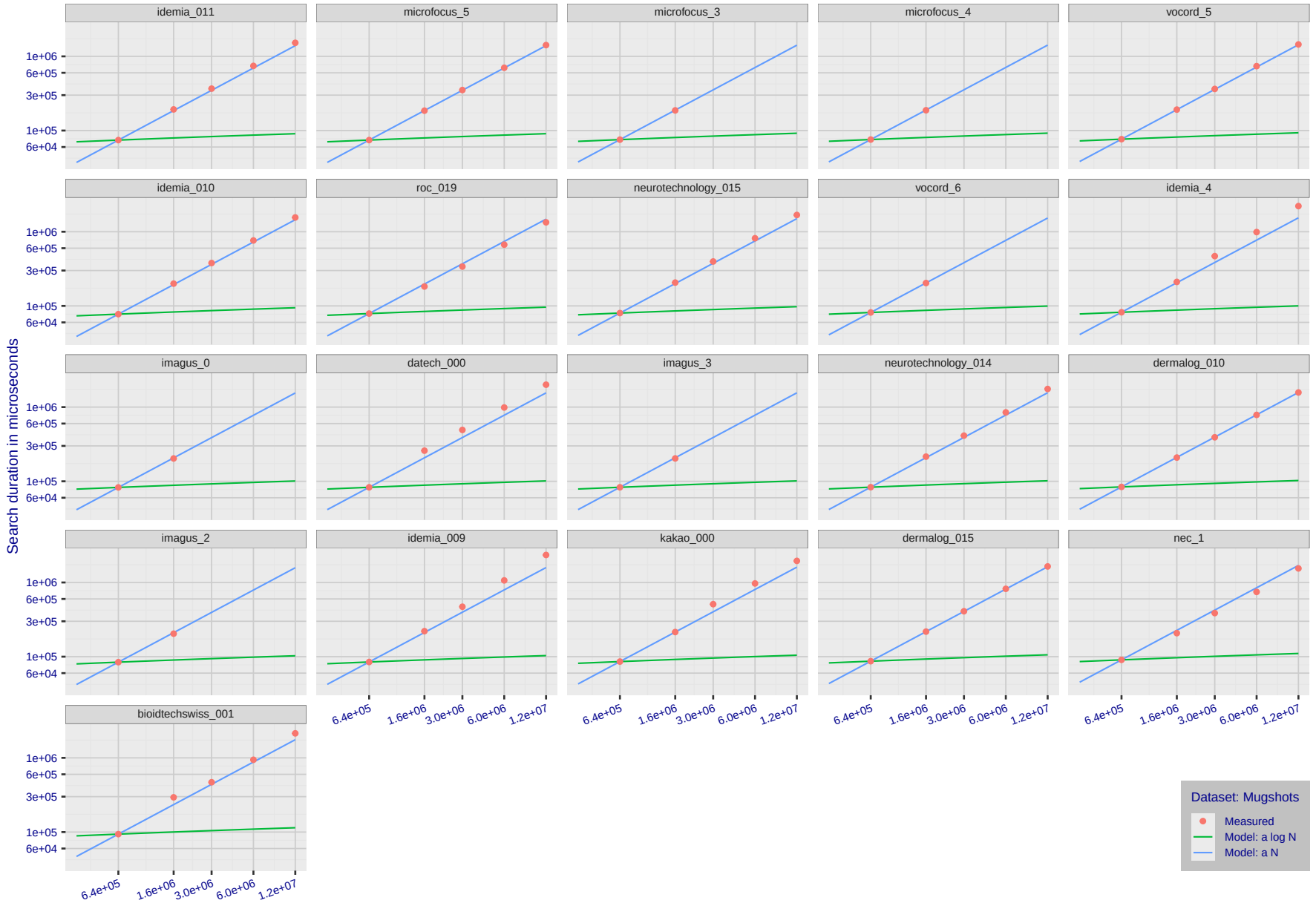


*Figure 472:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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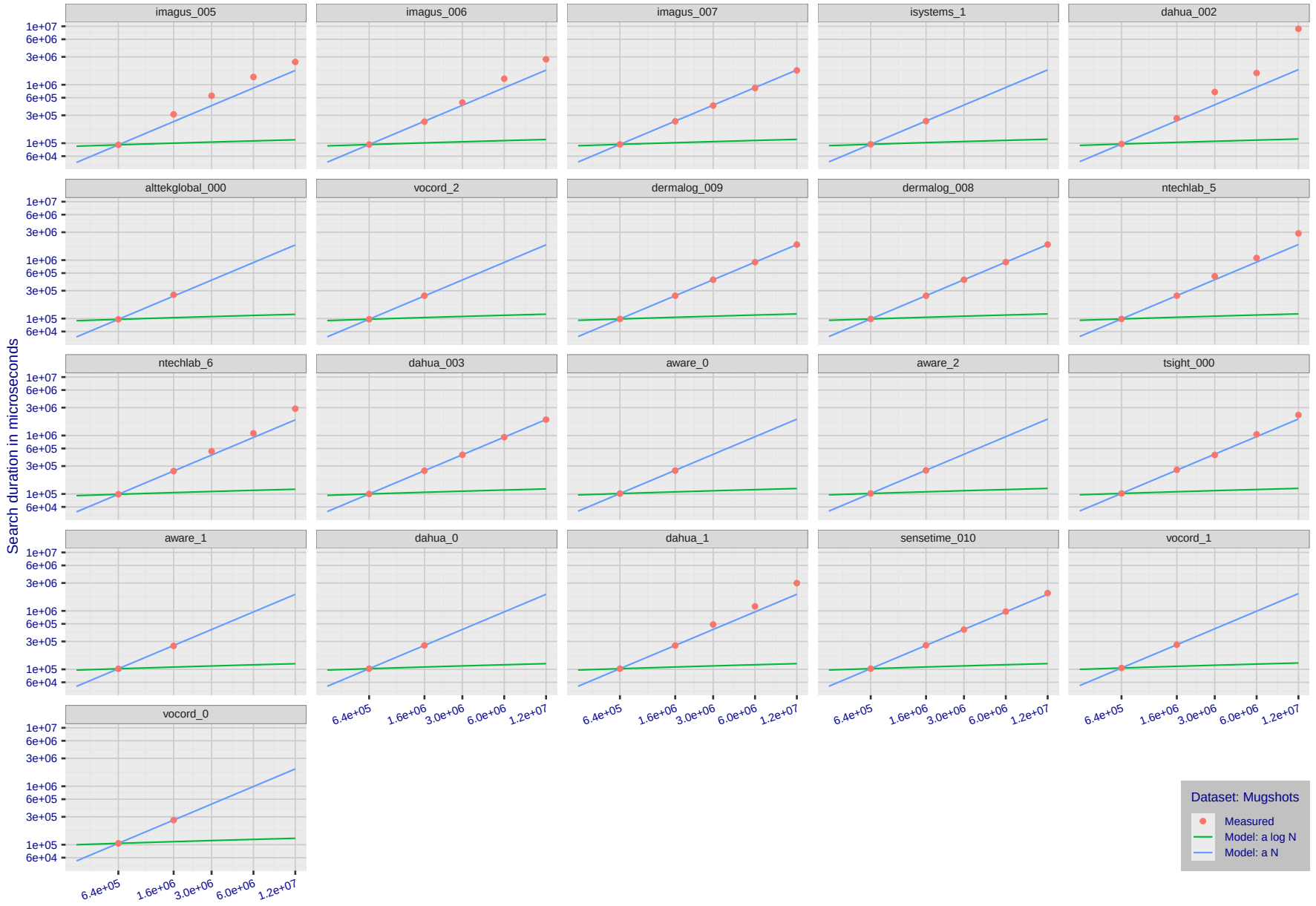


*Figure 473:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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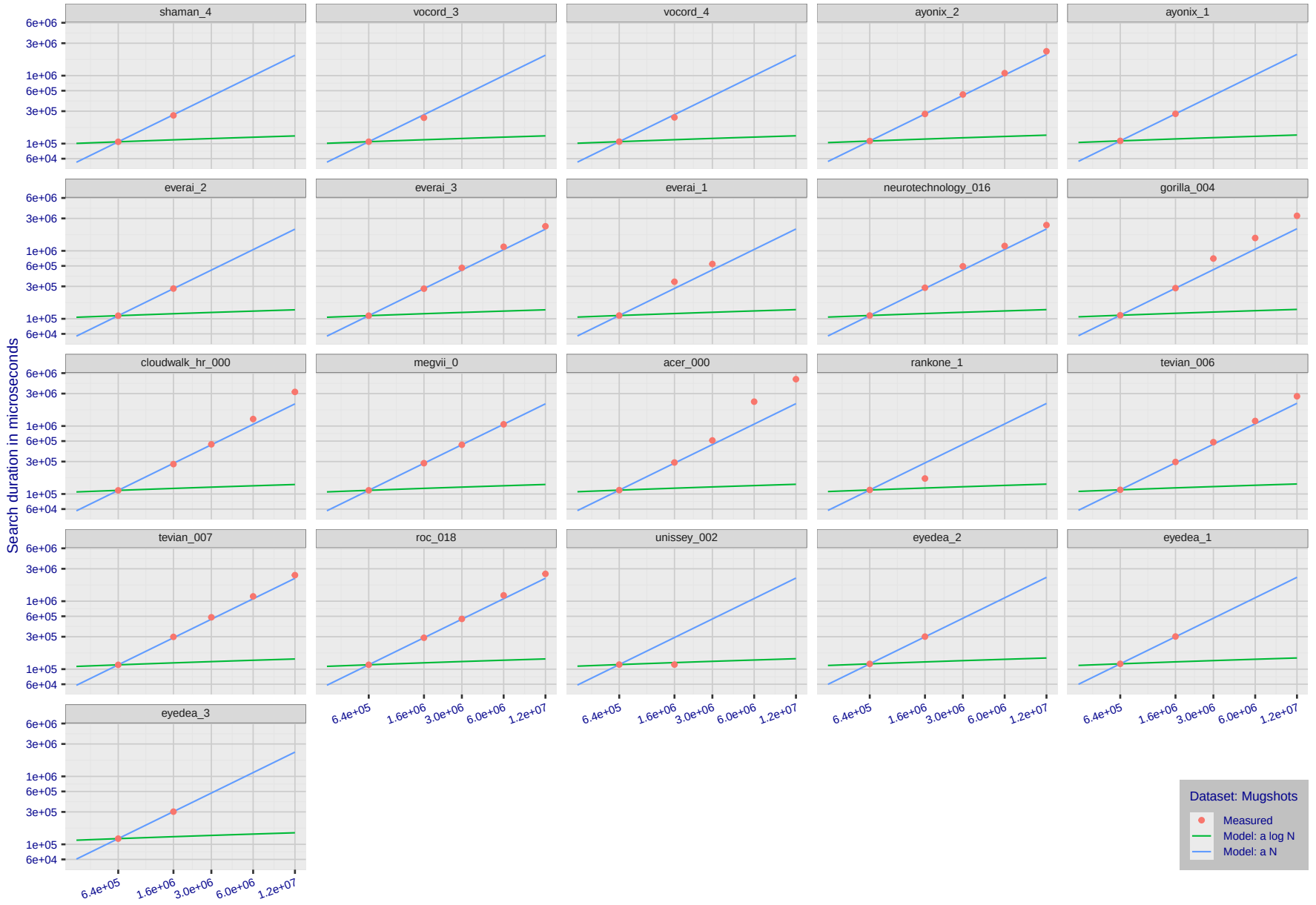


*Figure 474:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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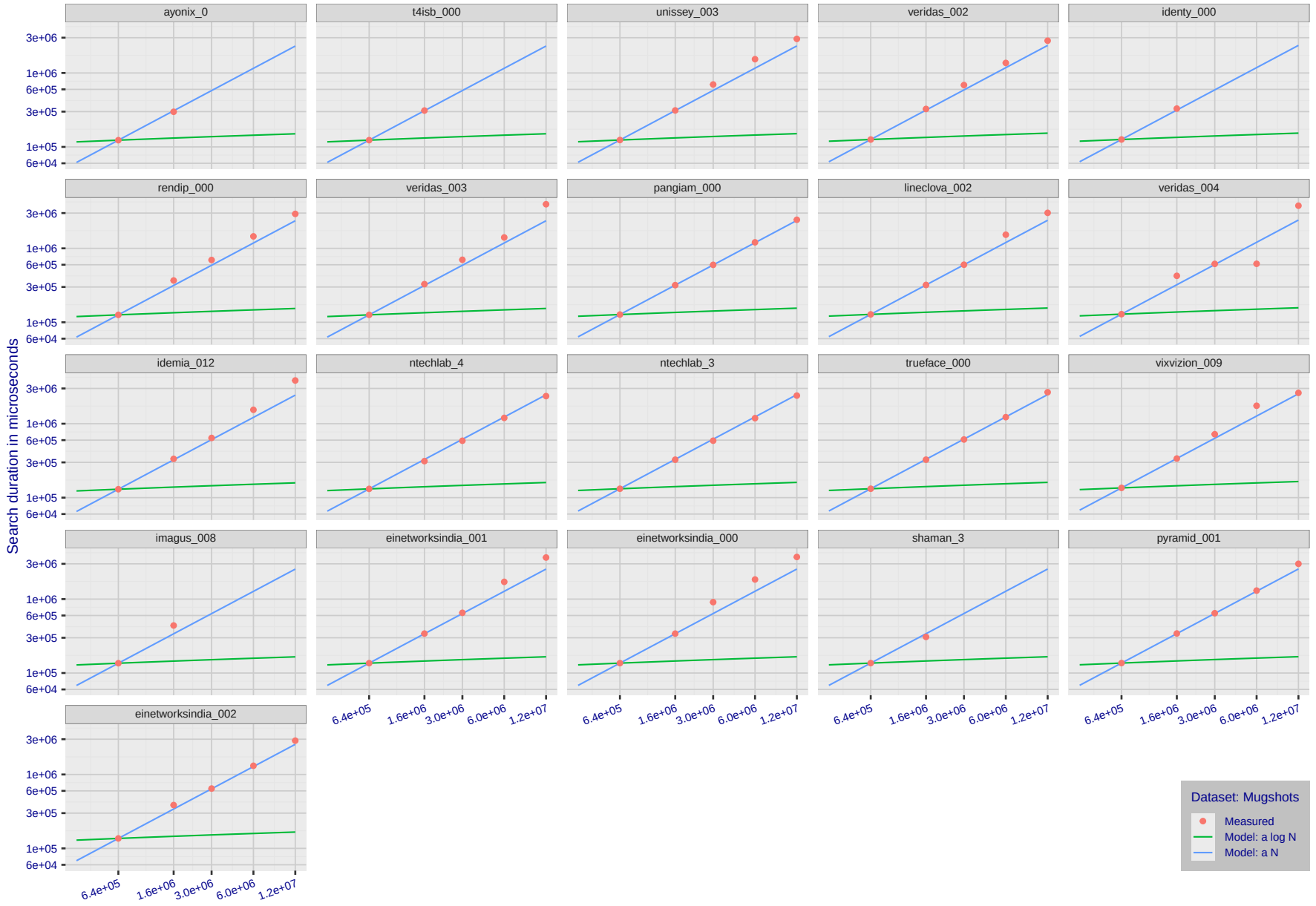


*Figure 475:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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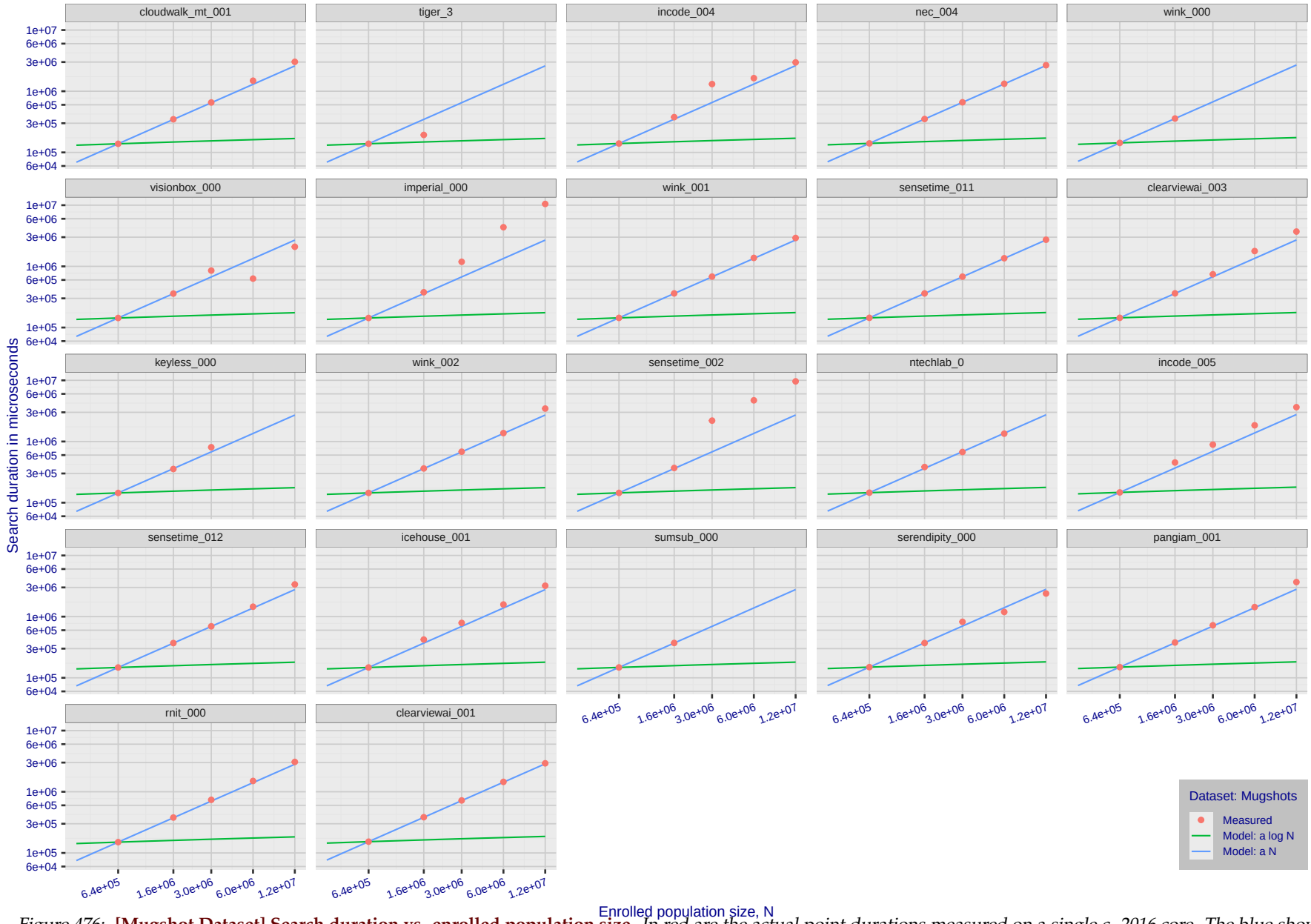


*Figure 476:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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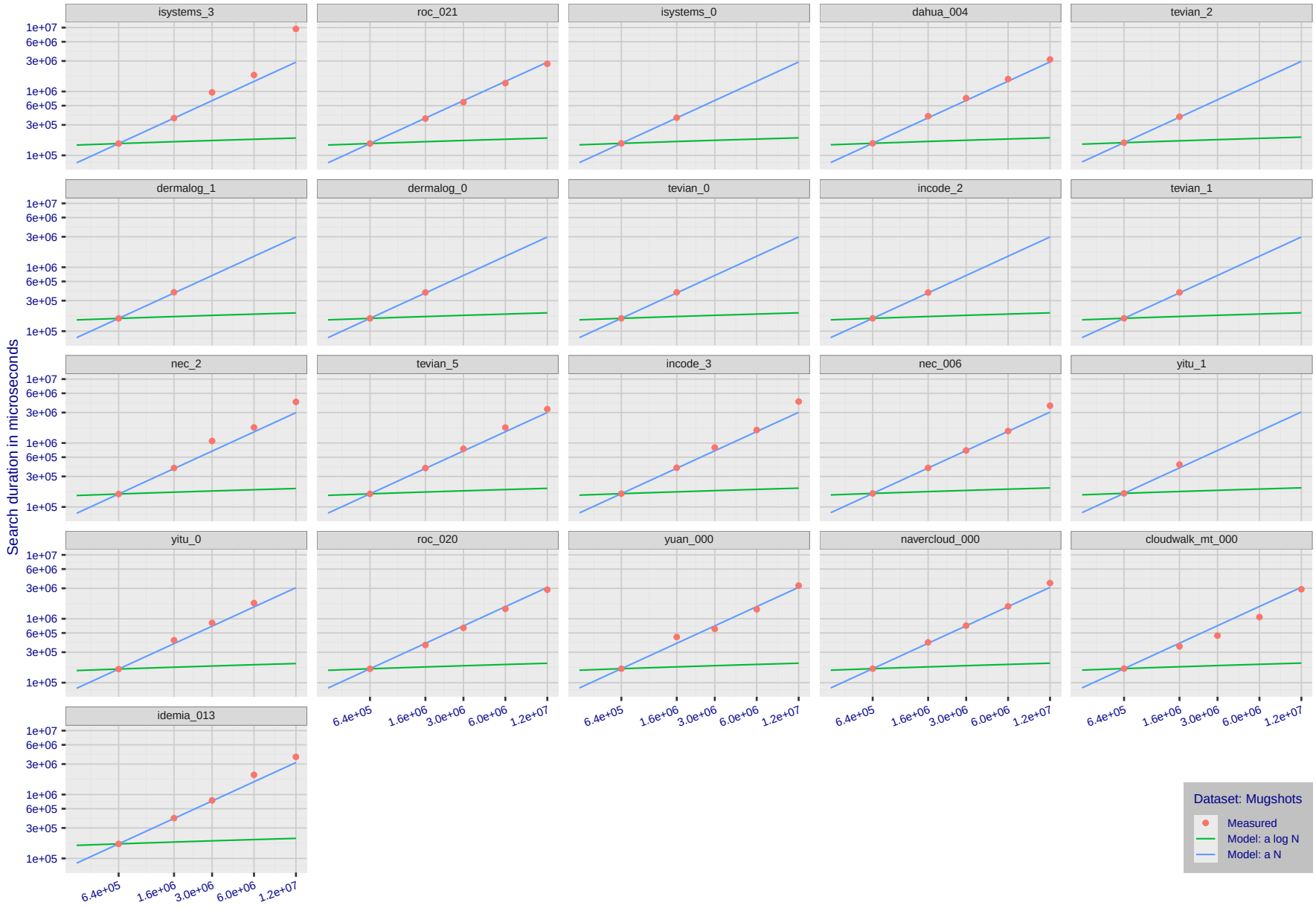


*Figure 477:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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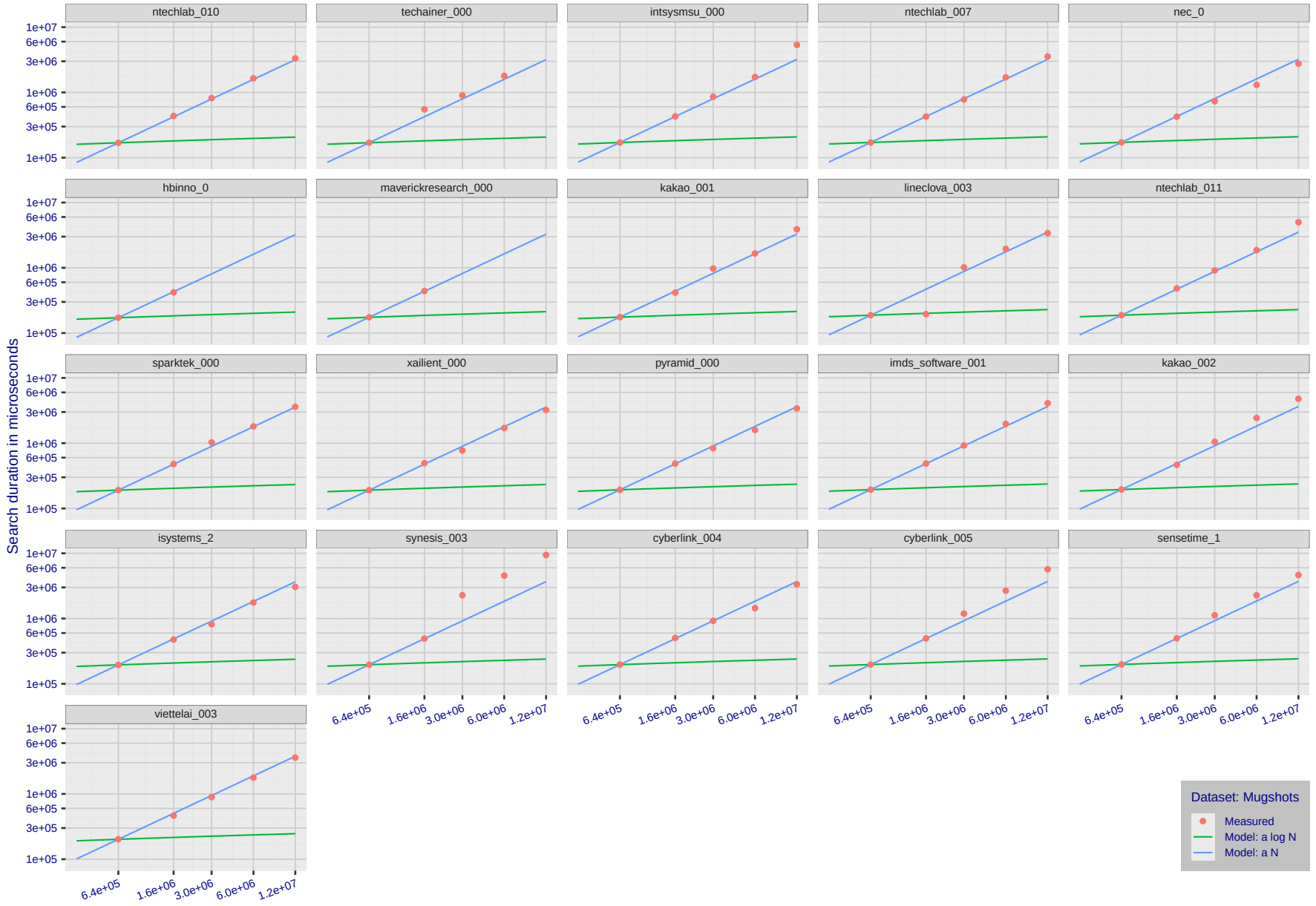


*Figure 478:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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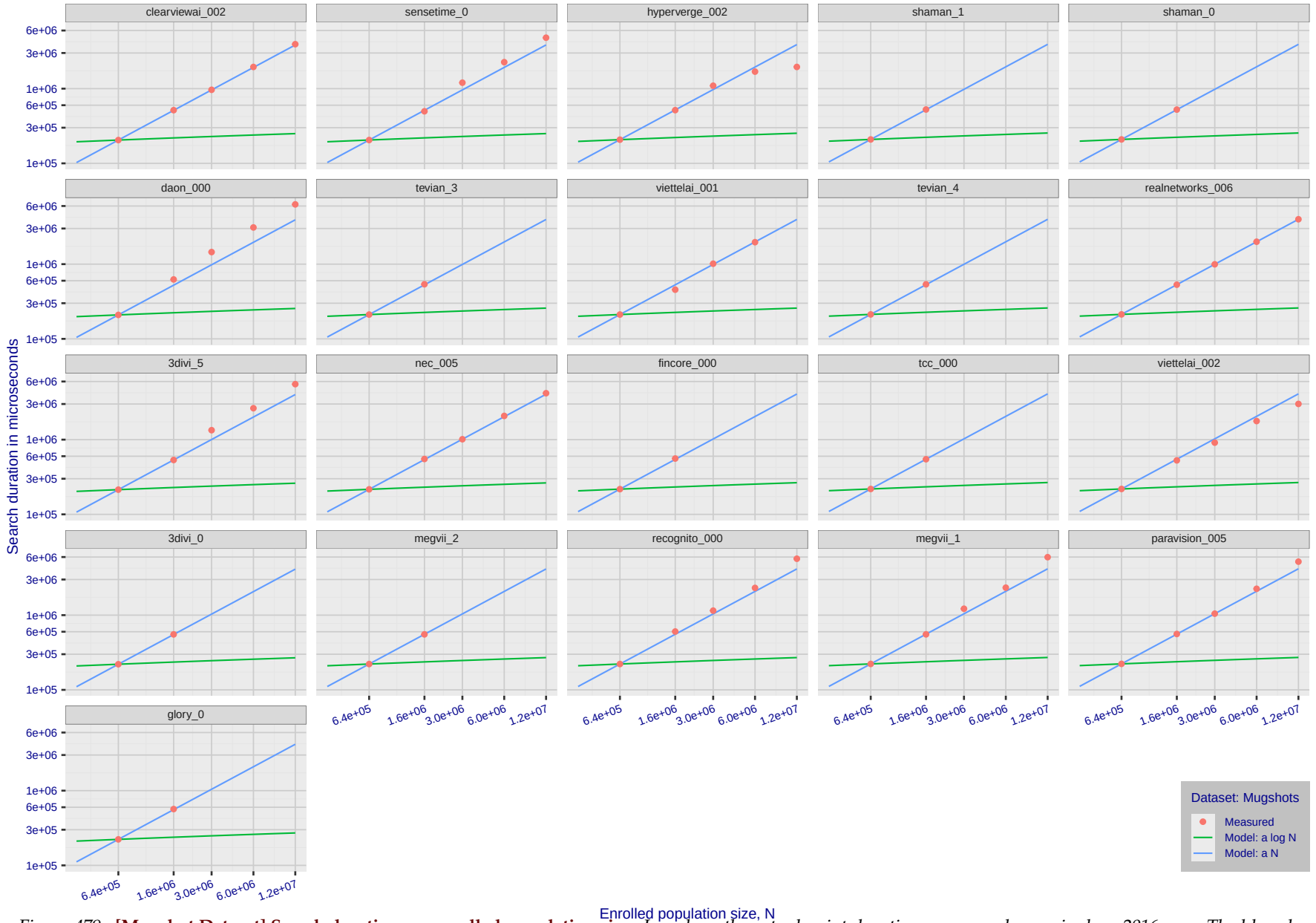


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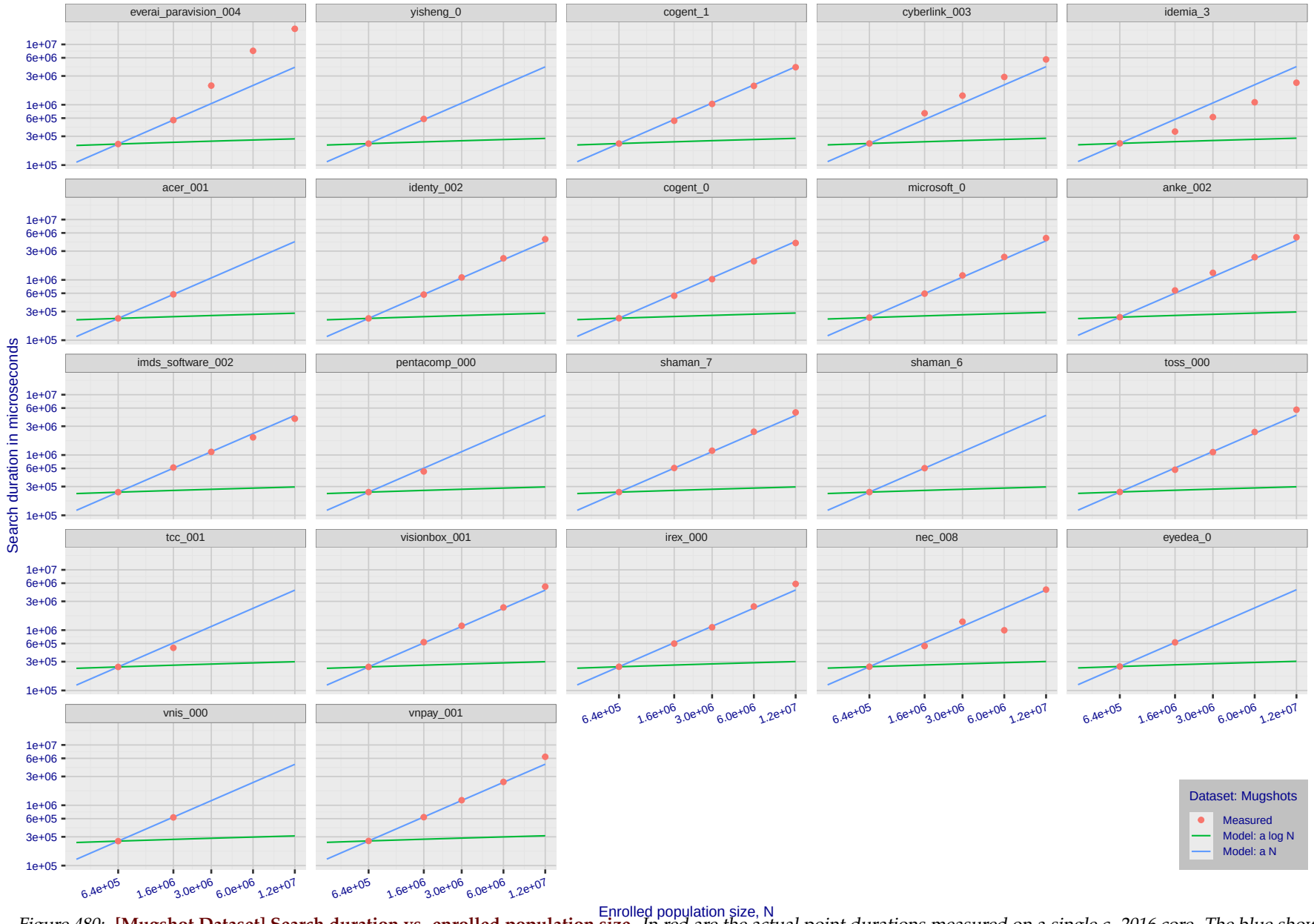


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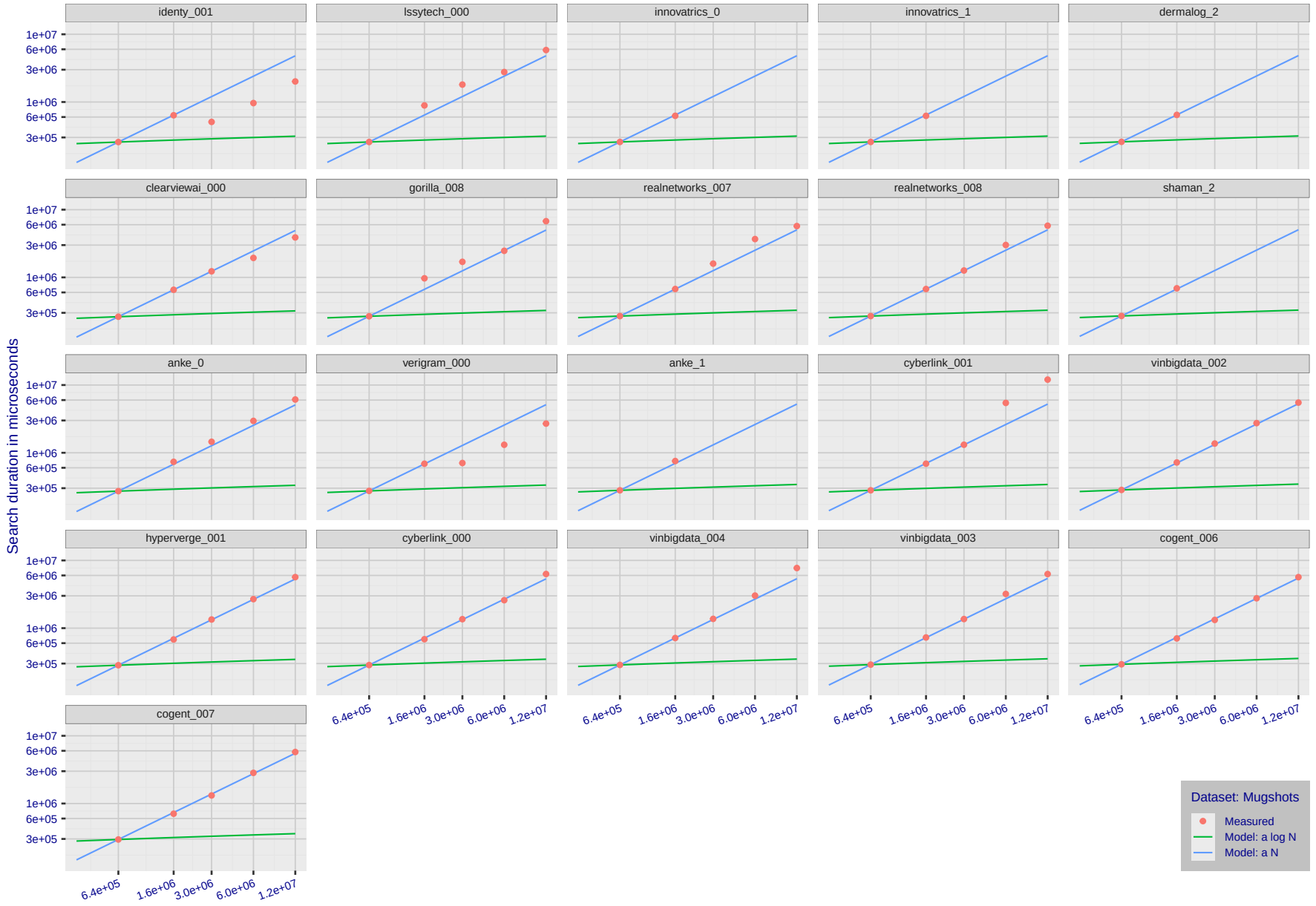


*Figure 481:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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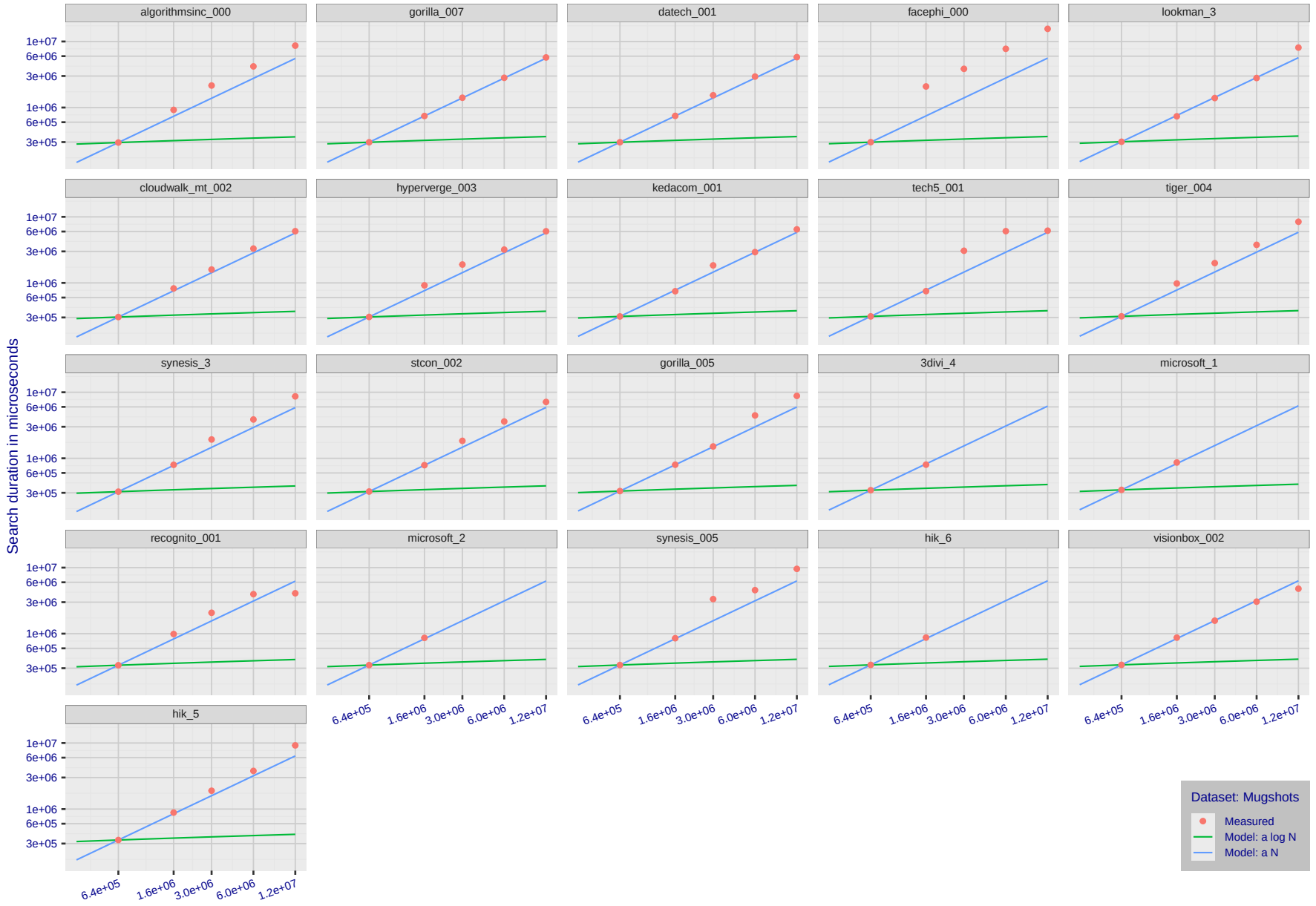


*Figure 482:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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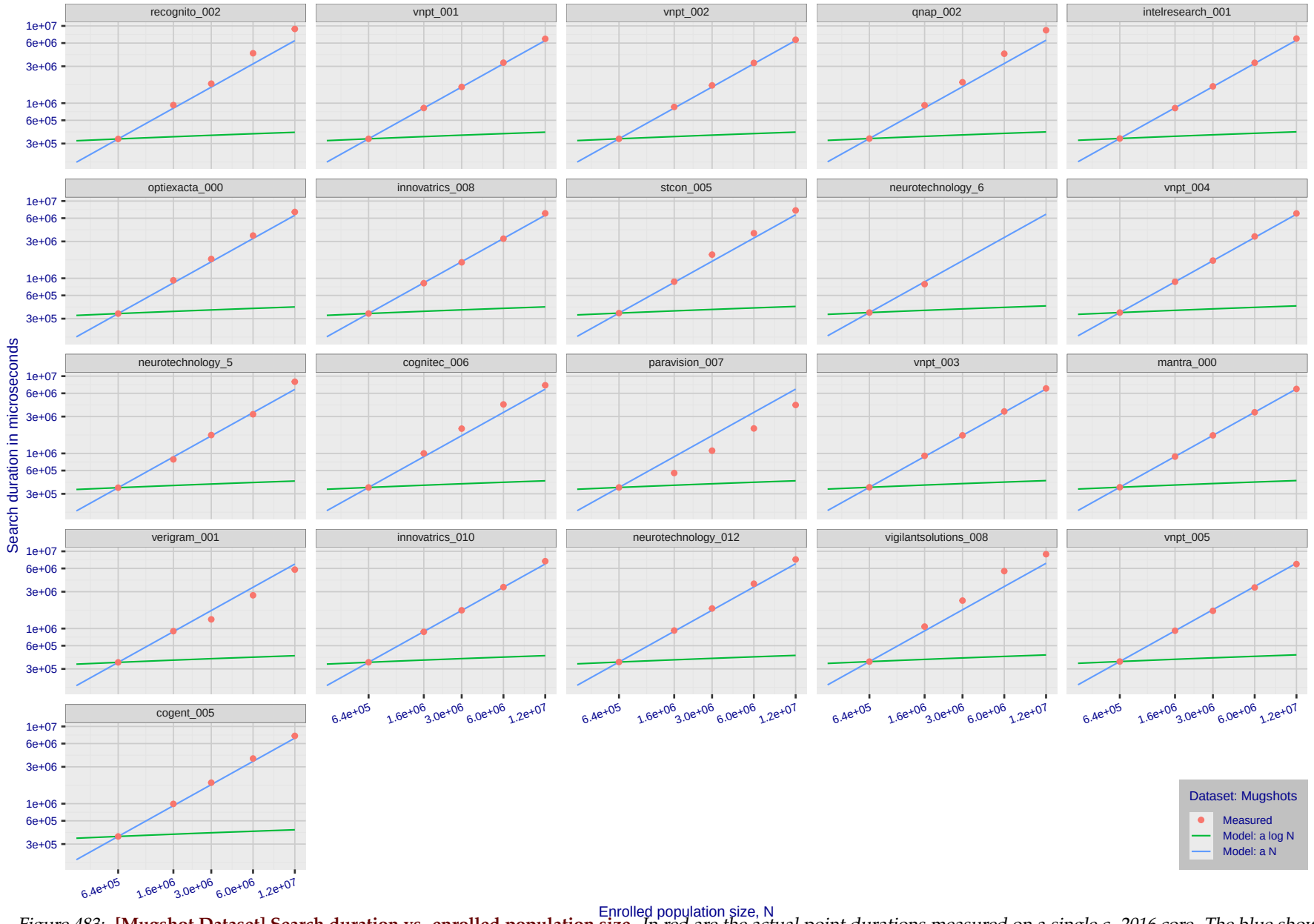


*Figure 483:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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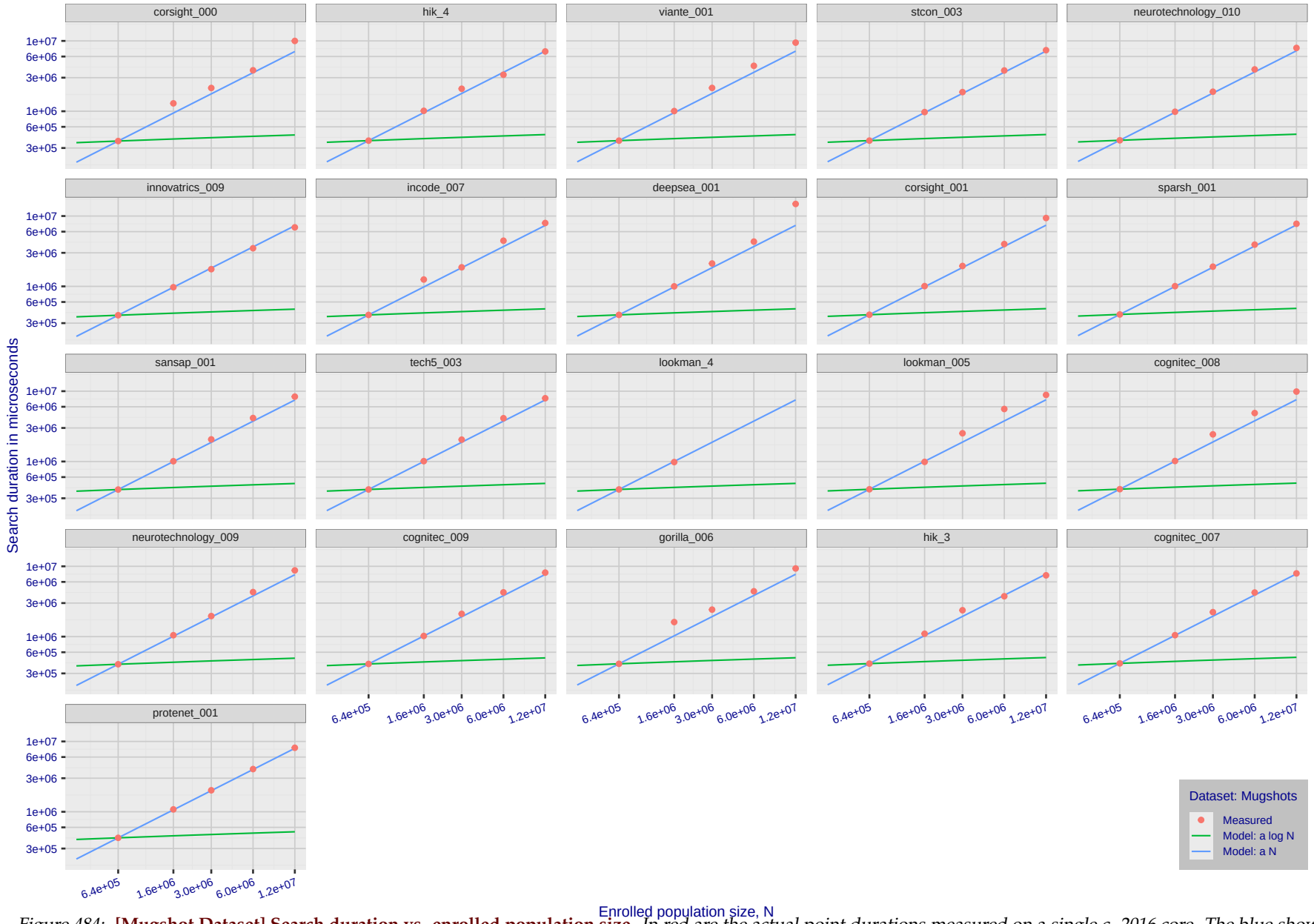


*Figure 484:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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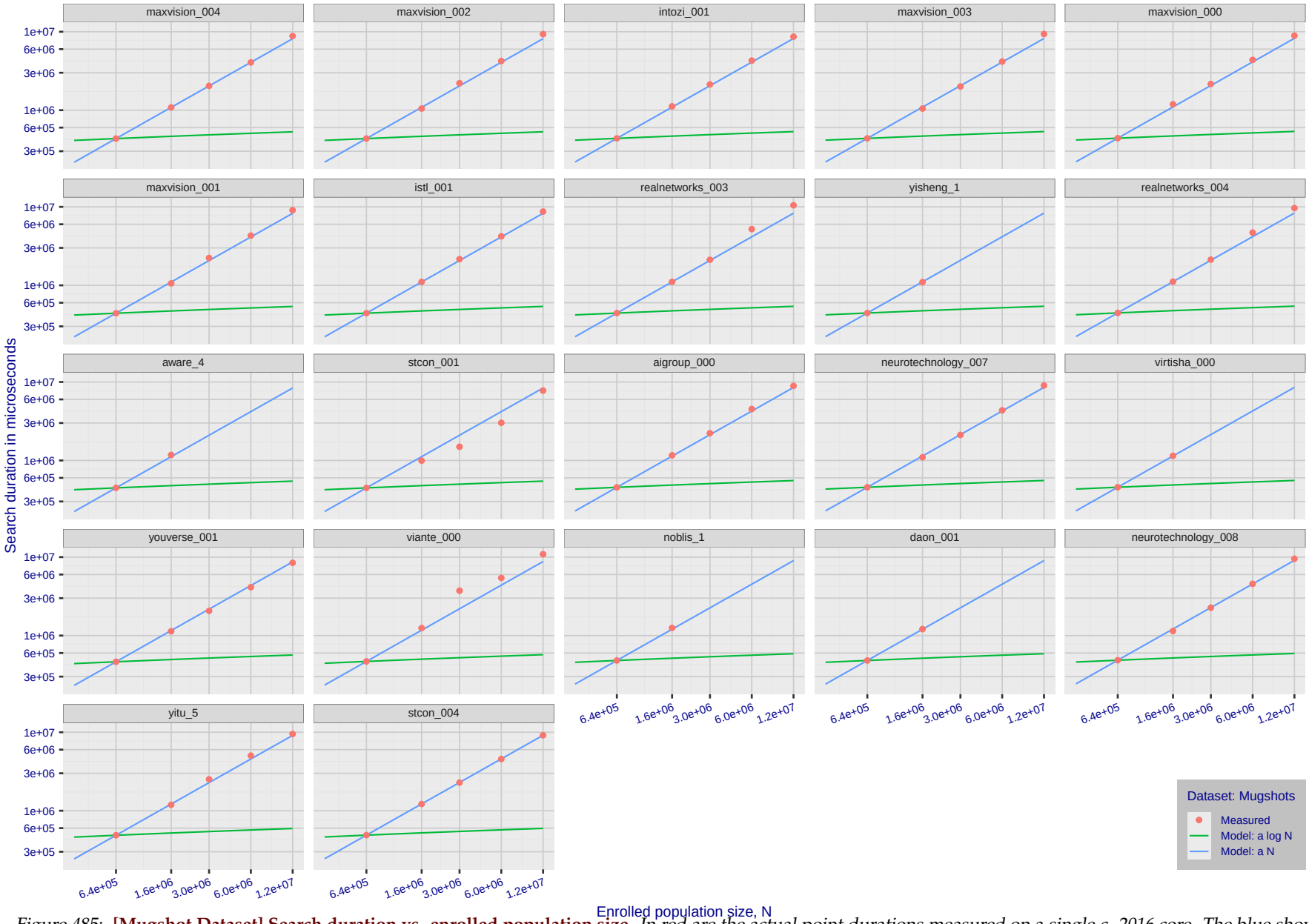


*Figure 485:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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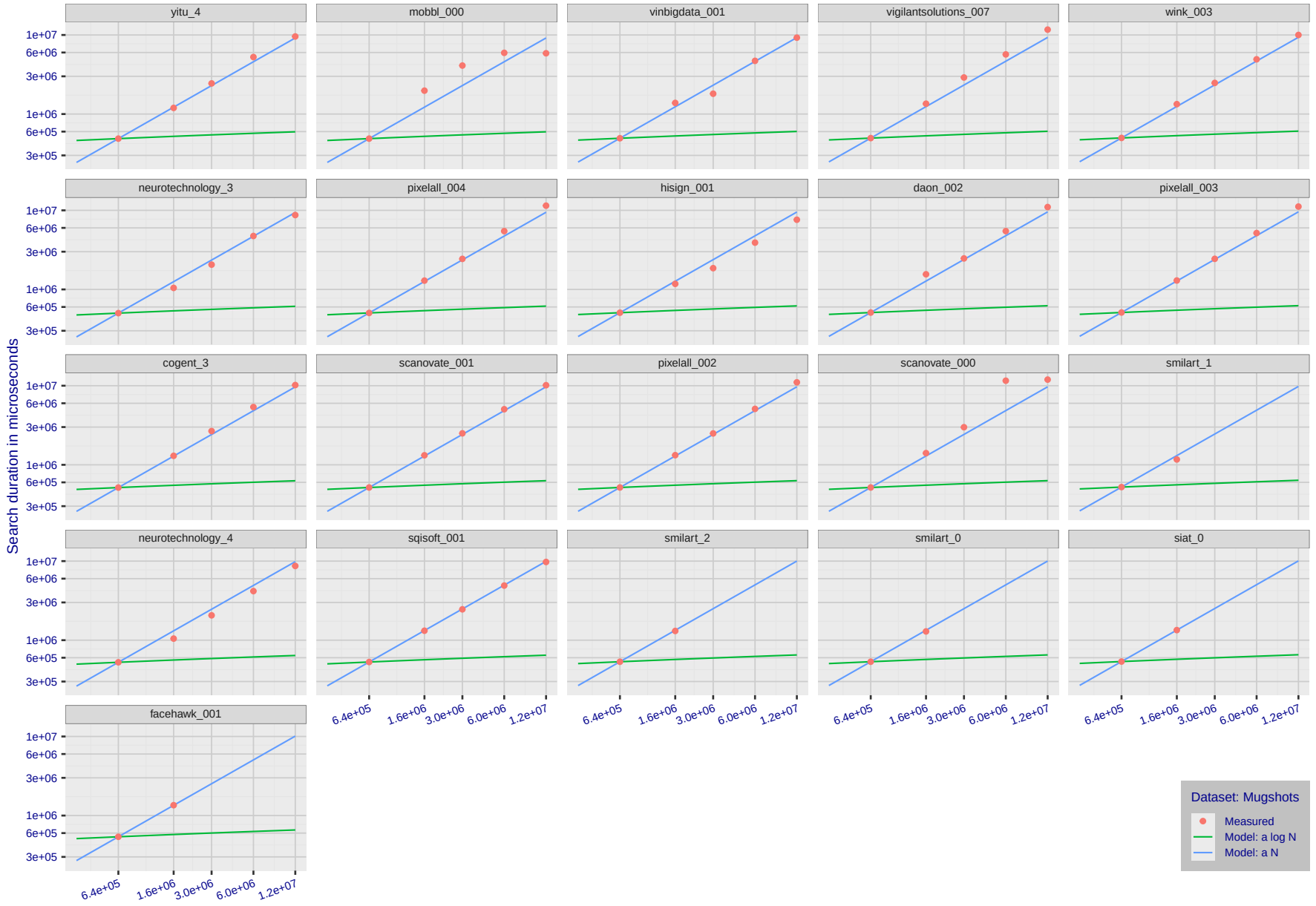


*Figure 486:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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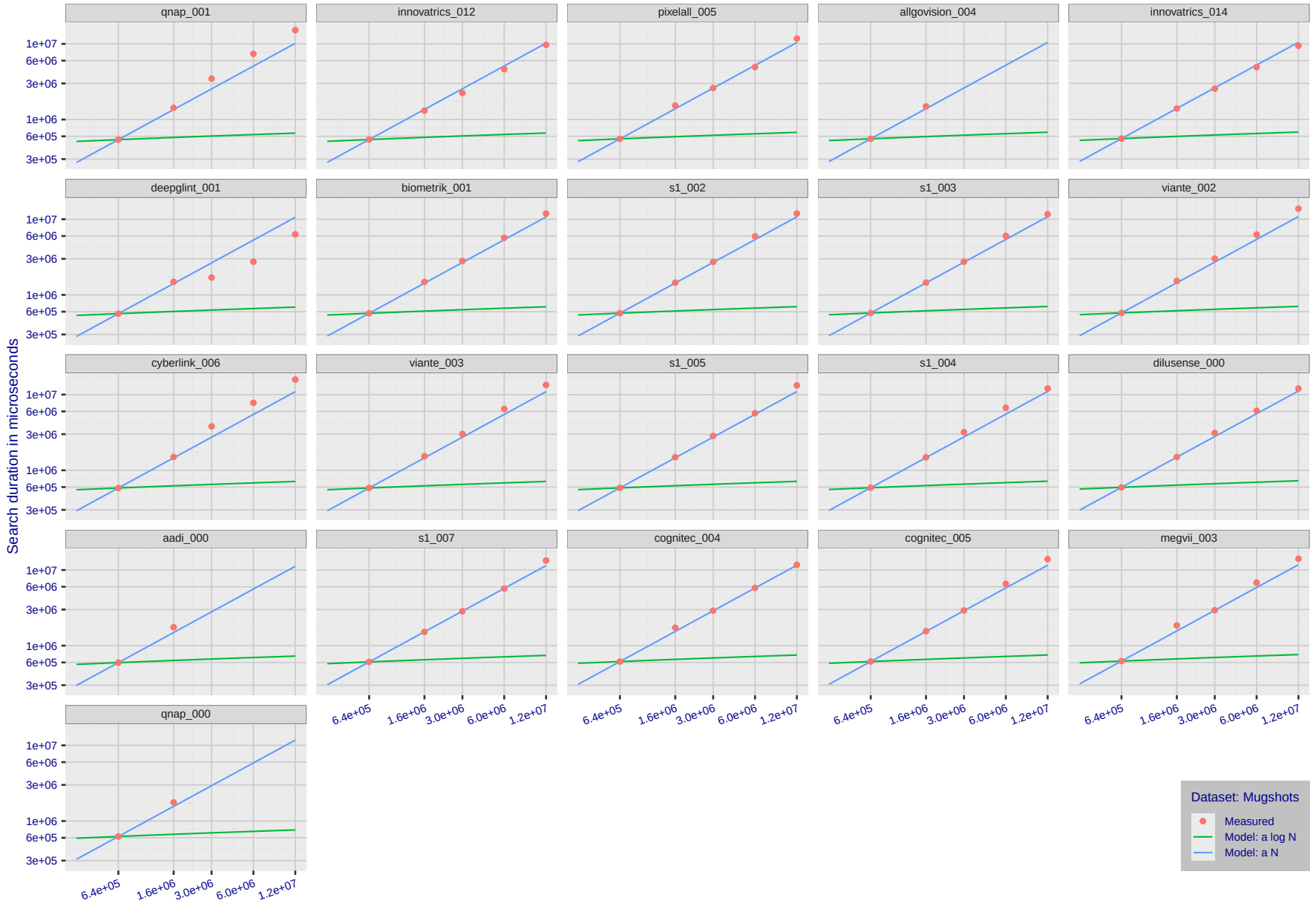


*Figure 487:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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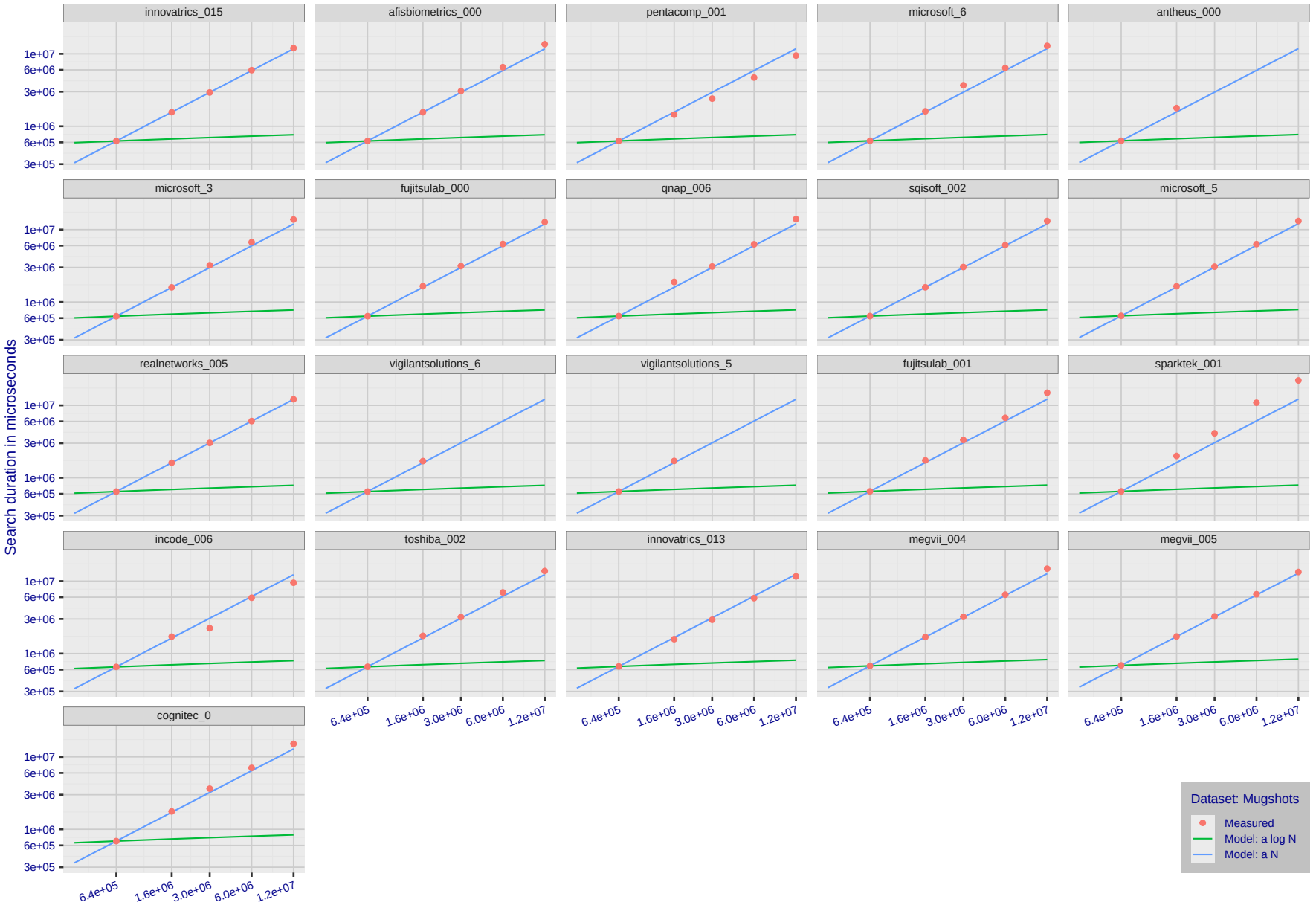


*Figure 488:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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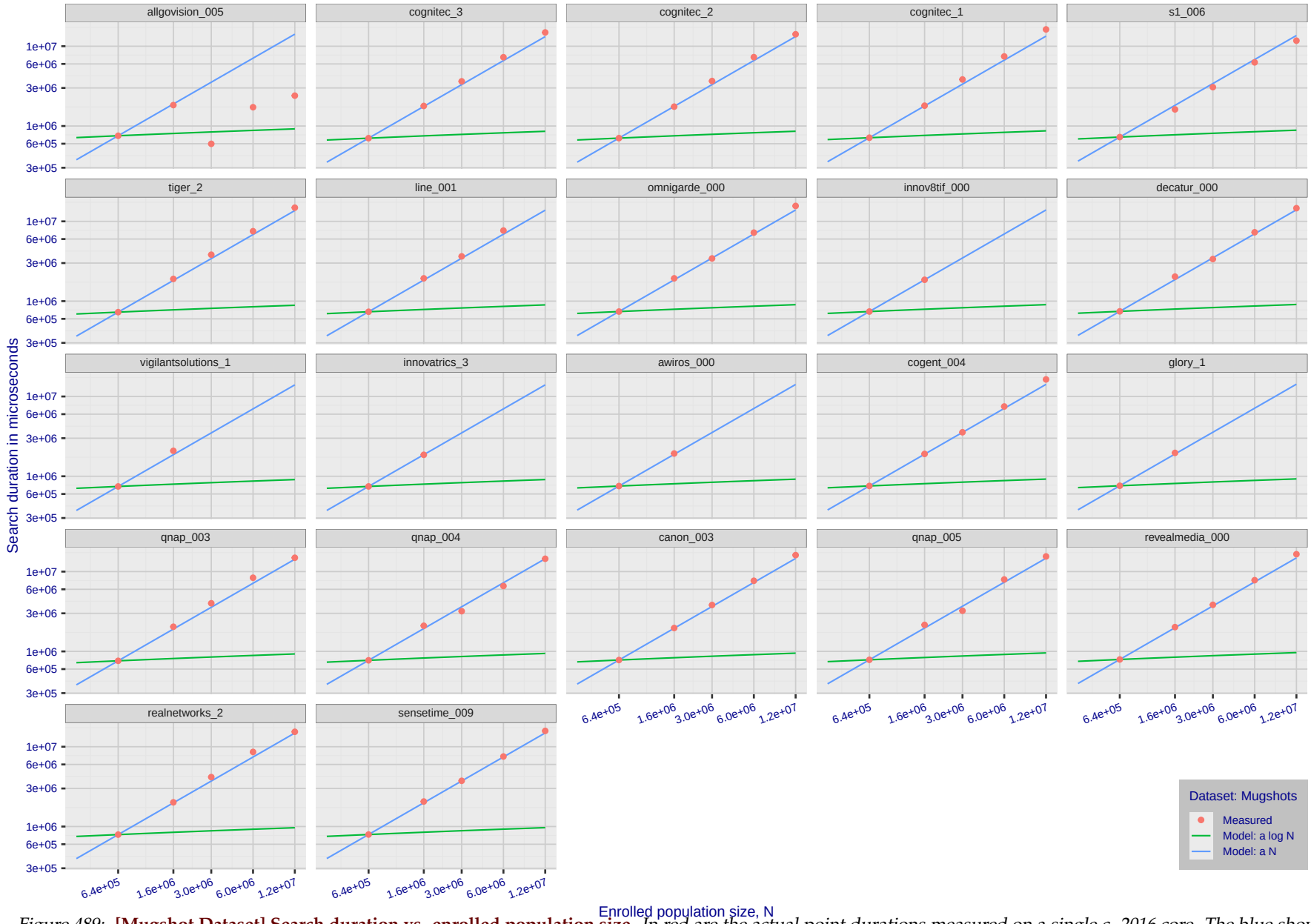


*Figure 489:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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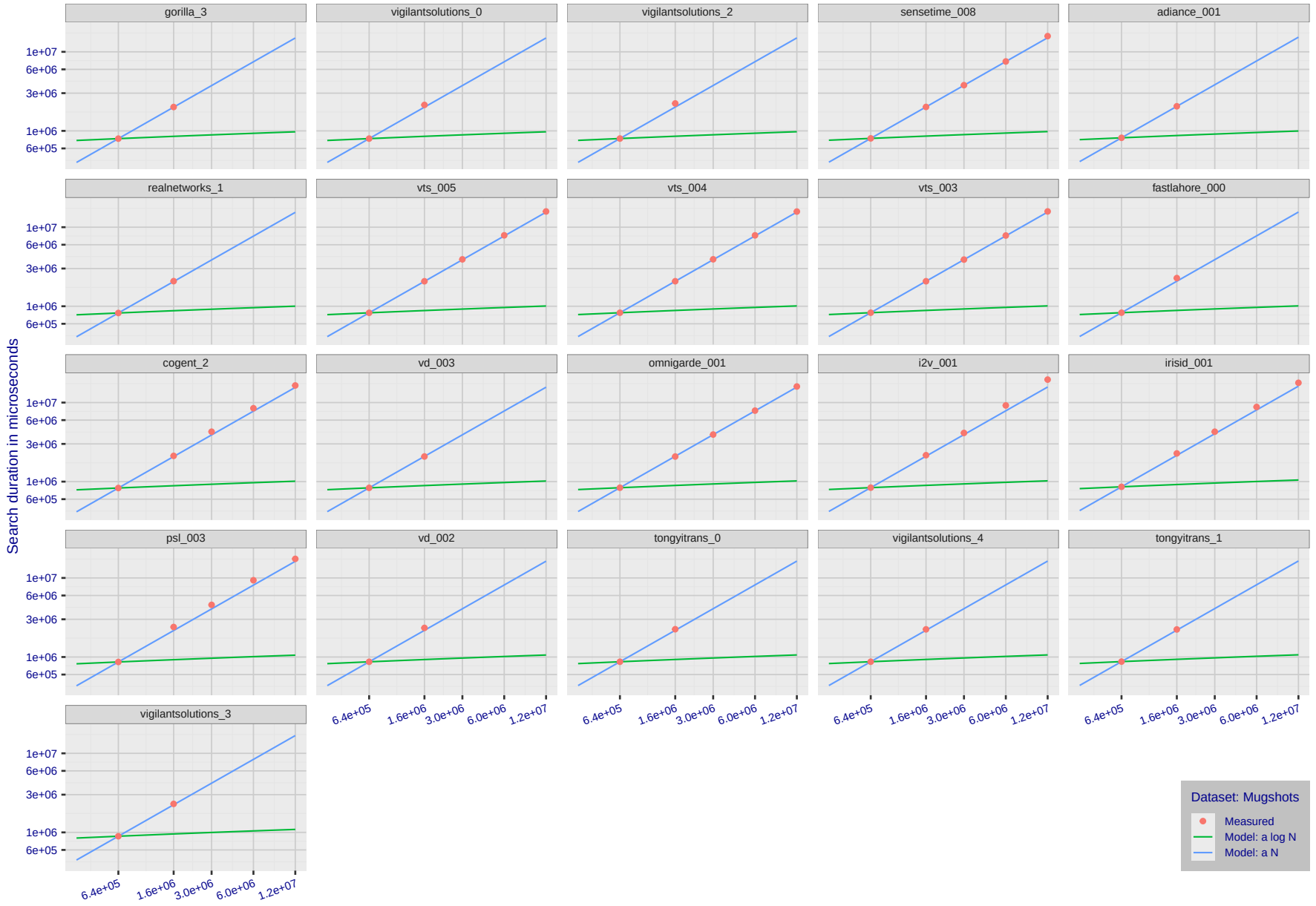


*Figure 490:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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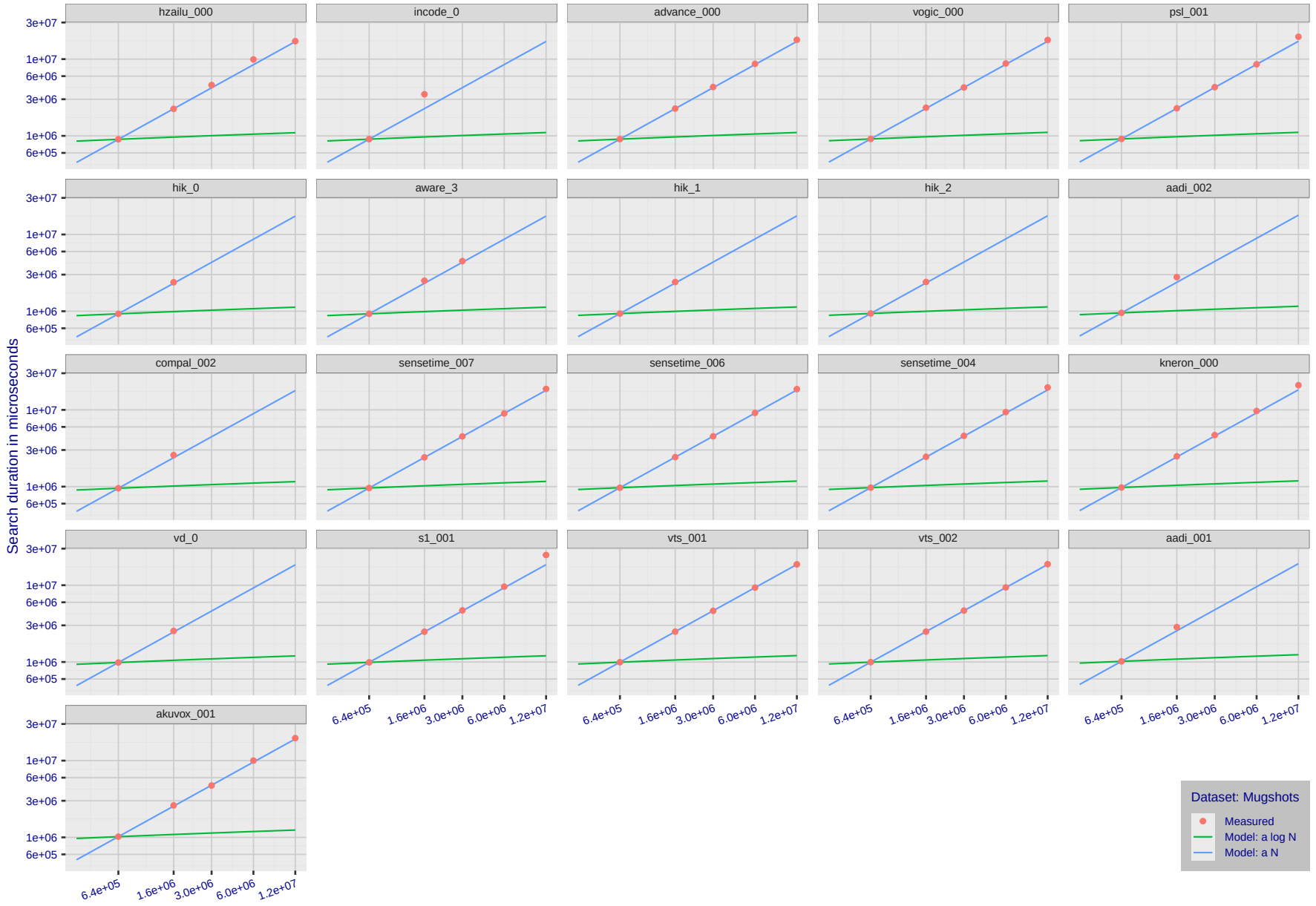


*Figure 491:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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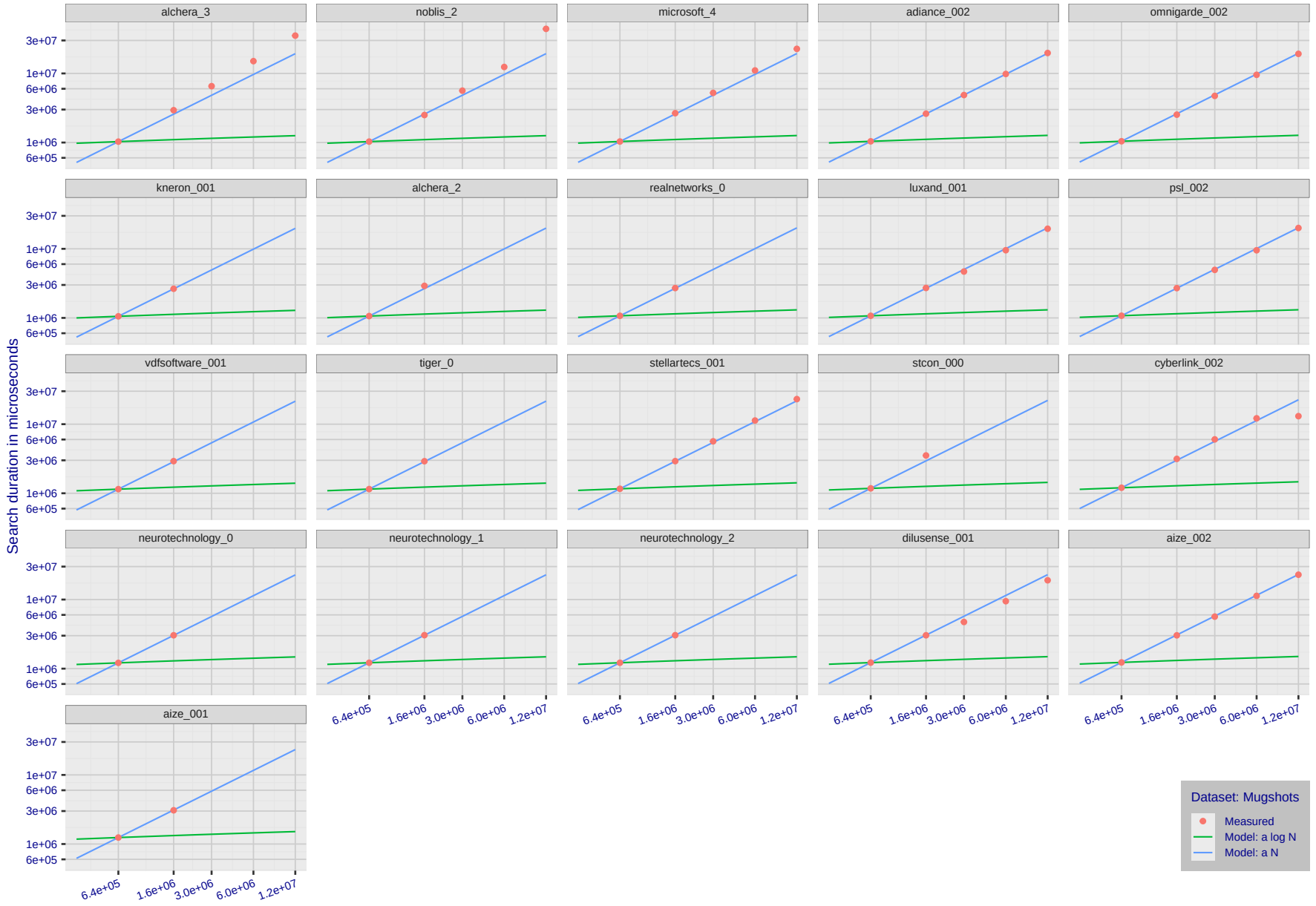


*Figure 492:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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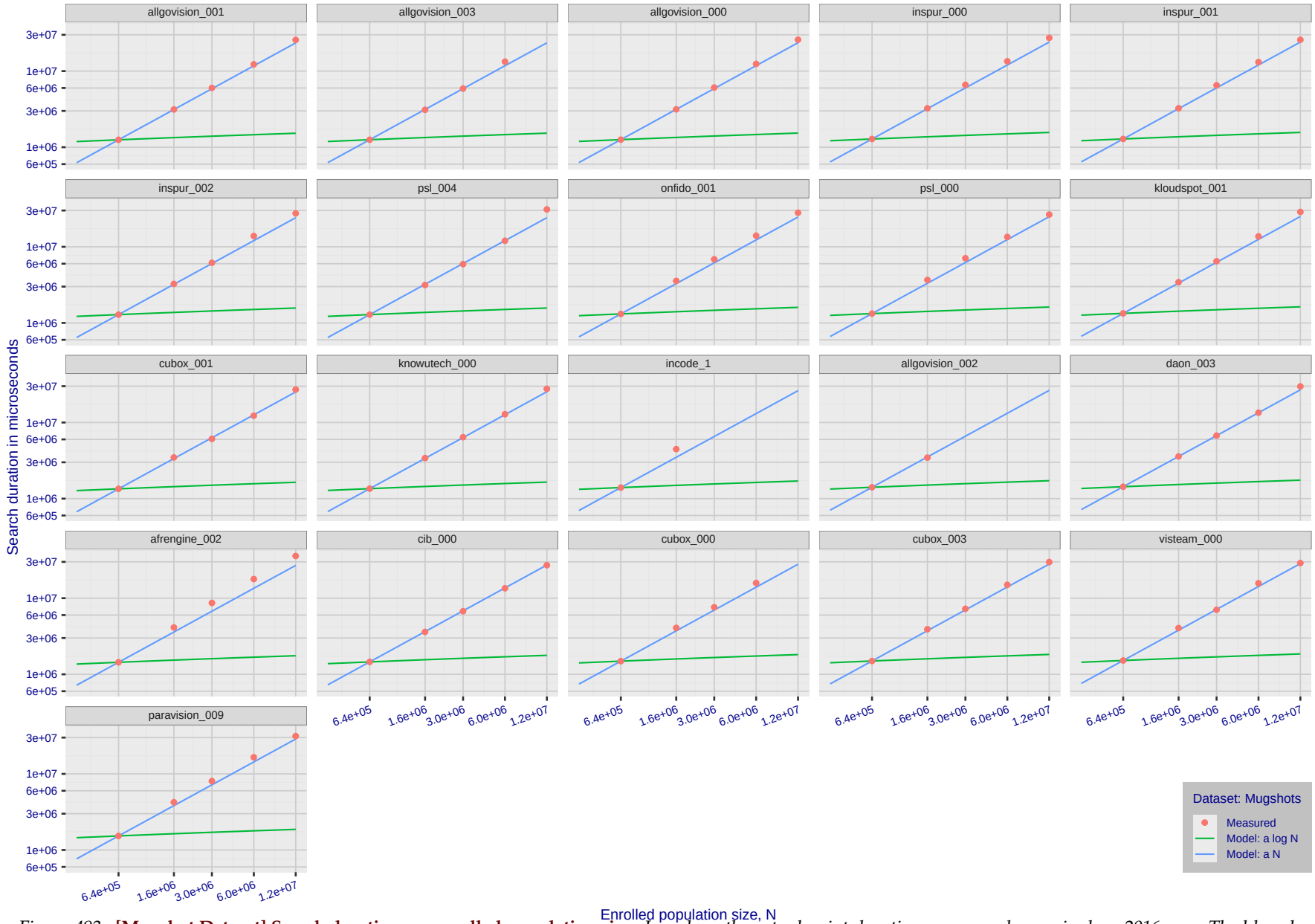


*Figure 493:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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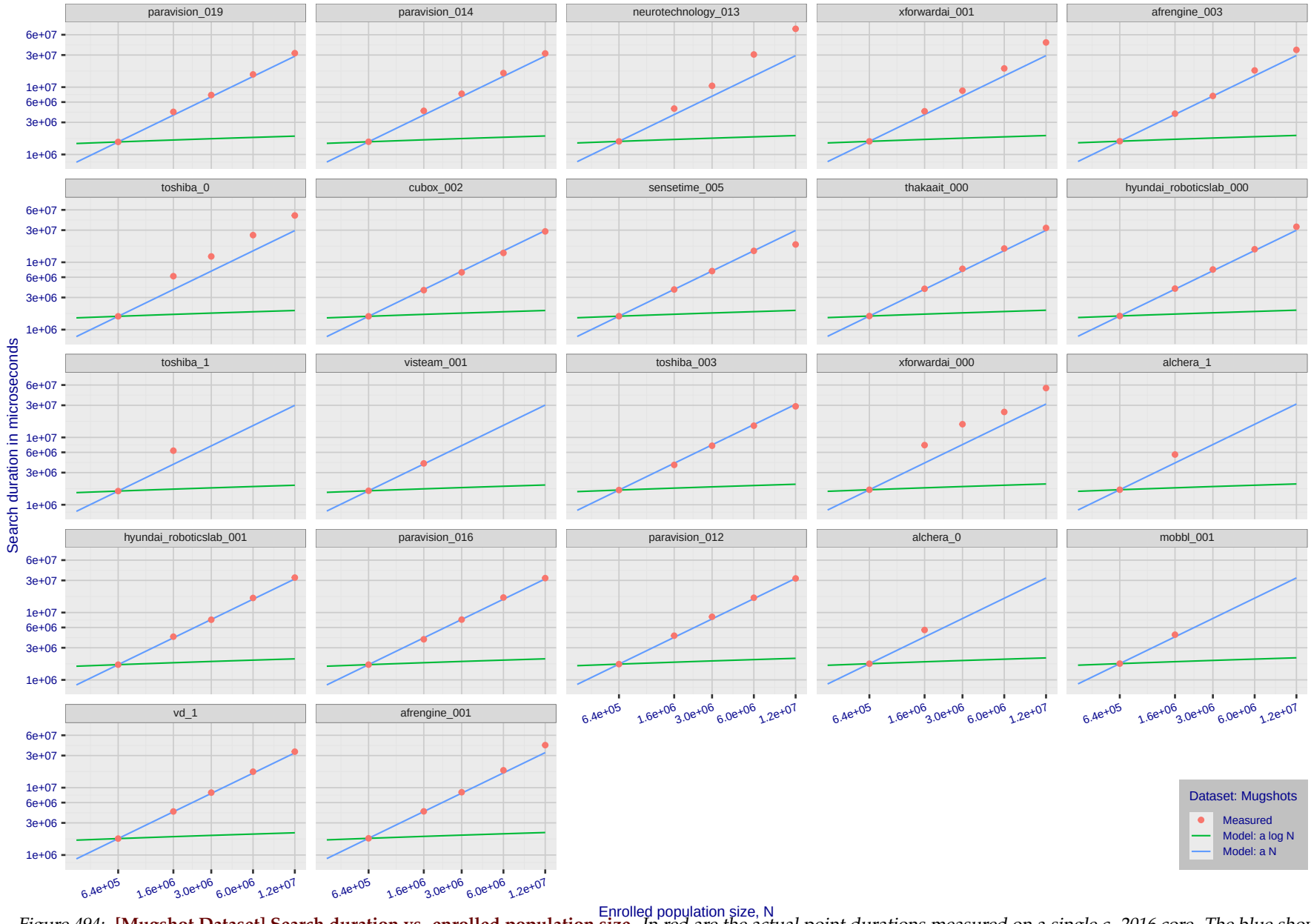


*Figure 494:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*

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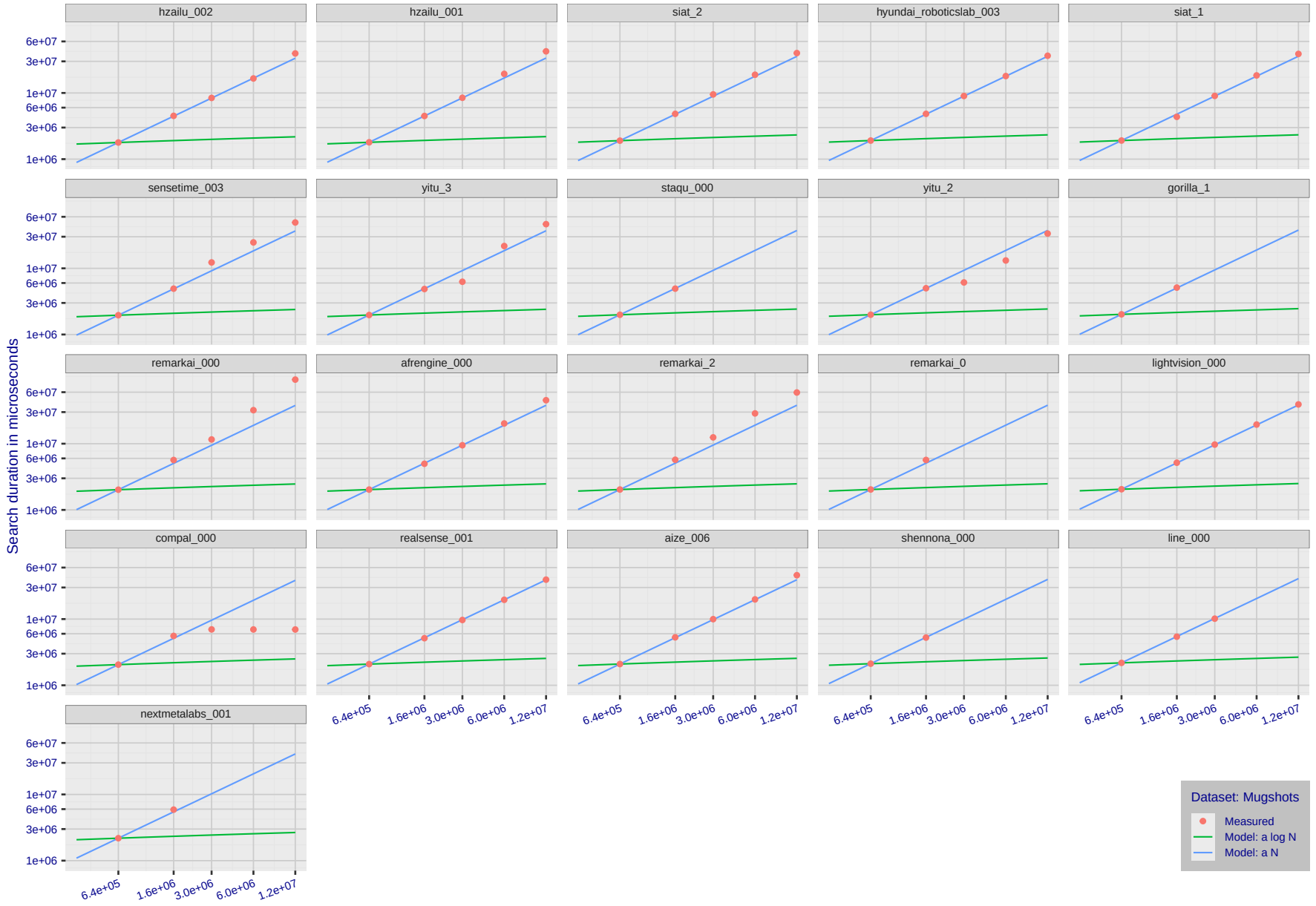


*Figure 495:* **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 51.*



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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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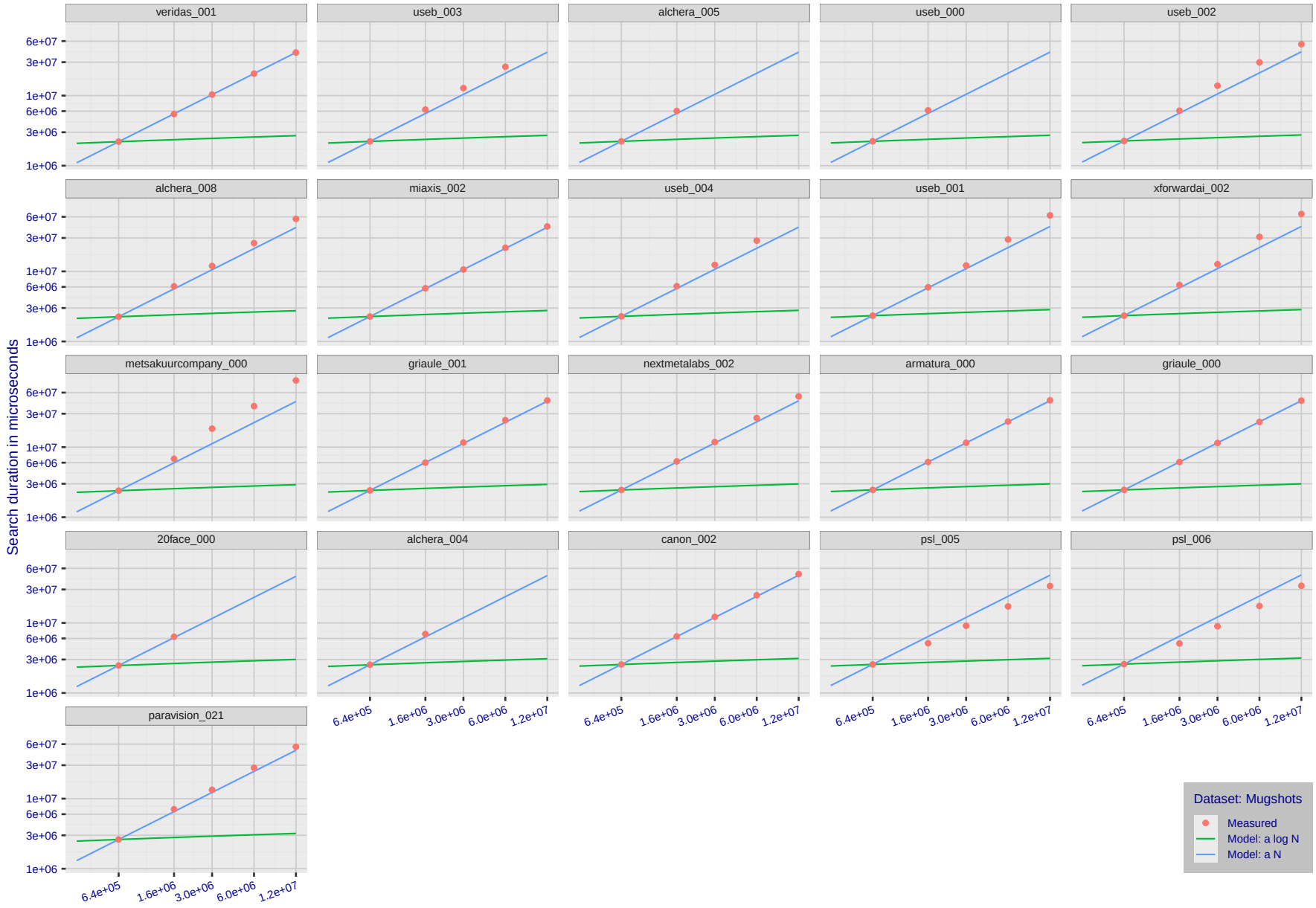


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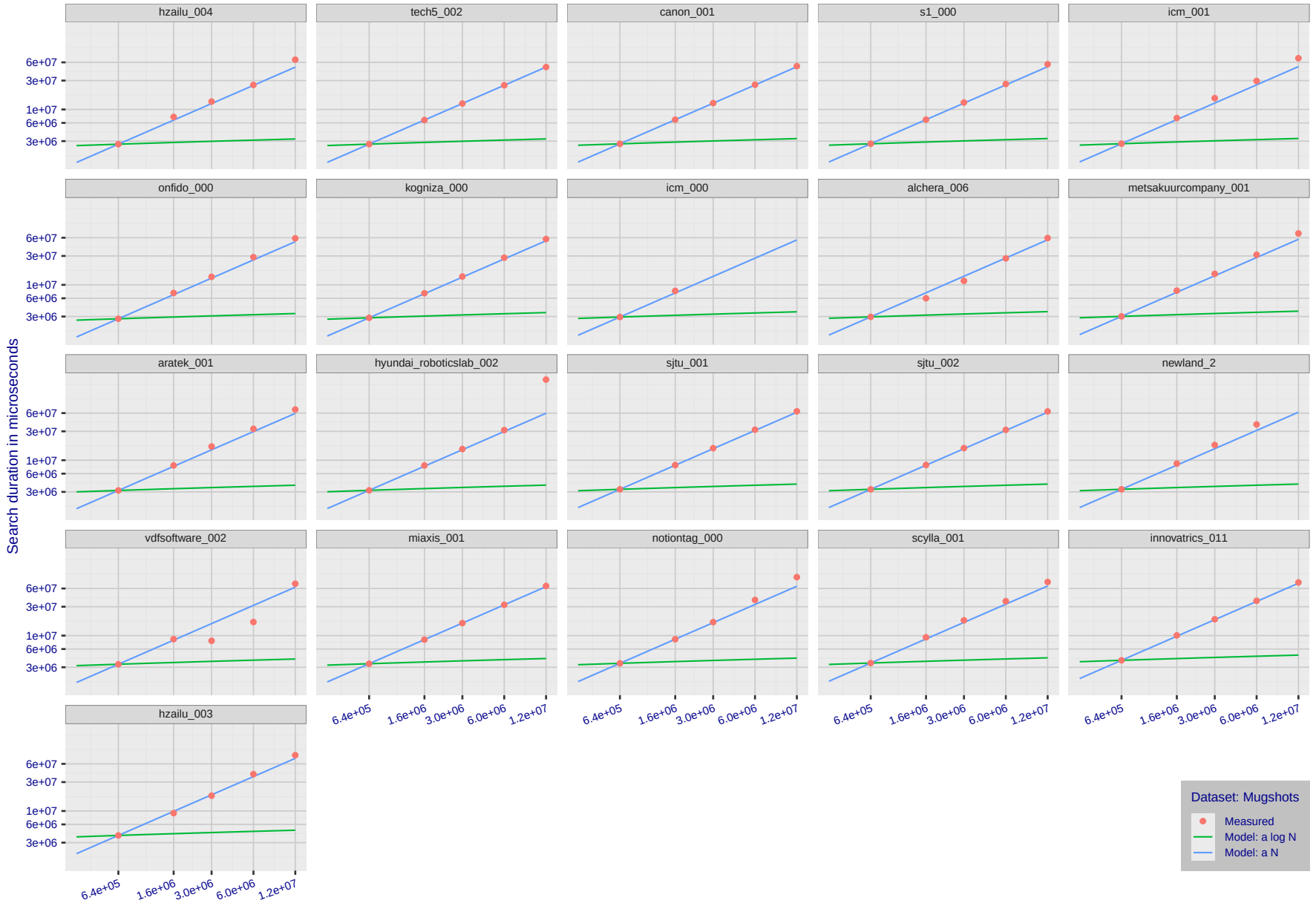


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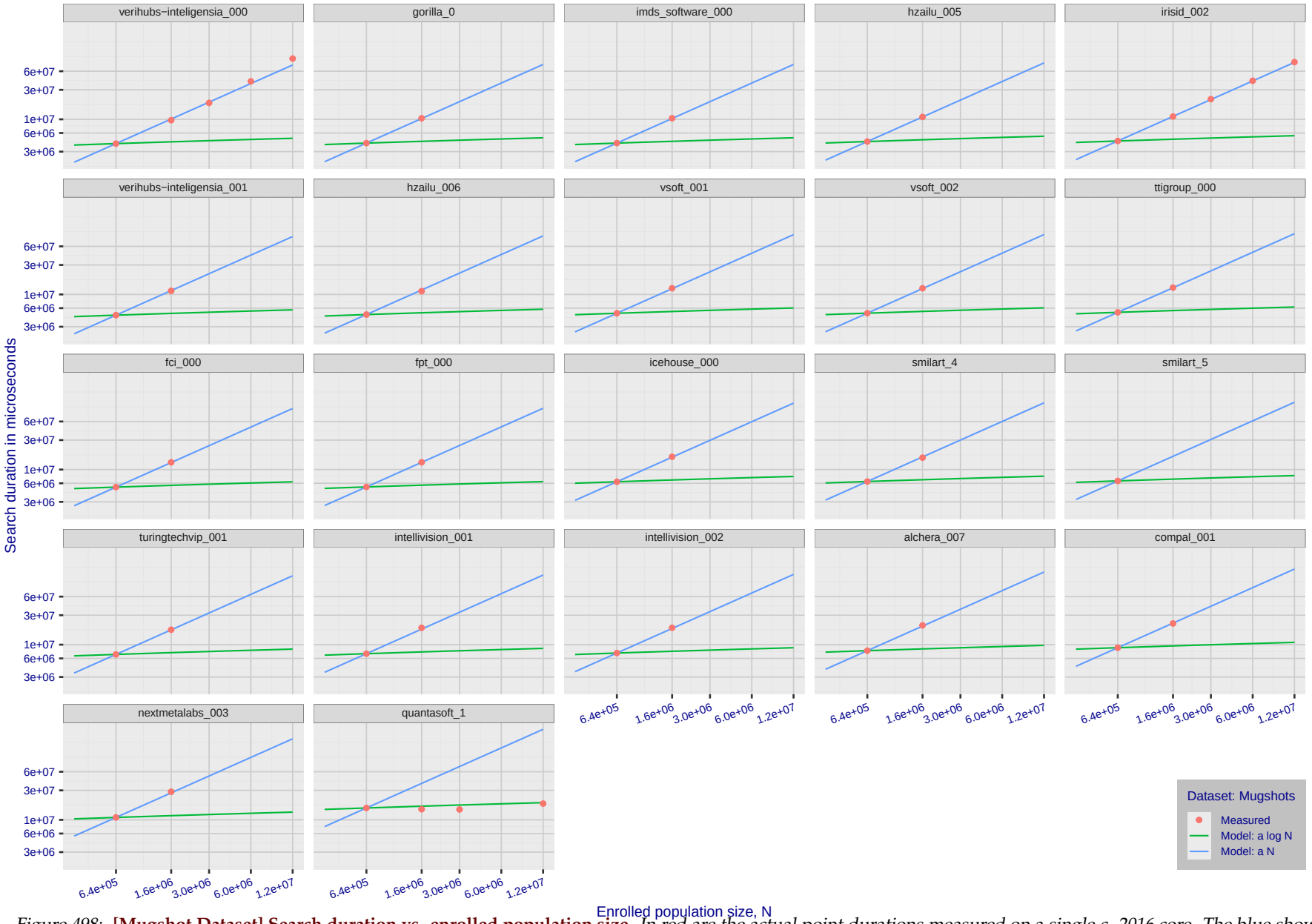


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# Appendix G Gallery Insertion Timing

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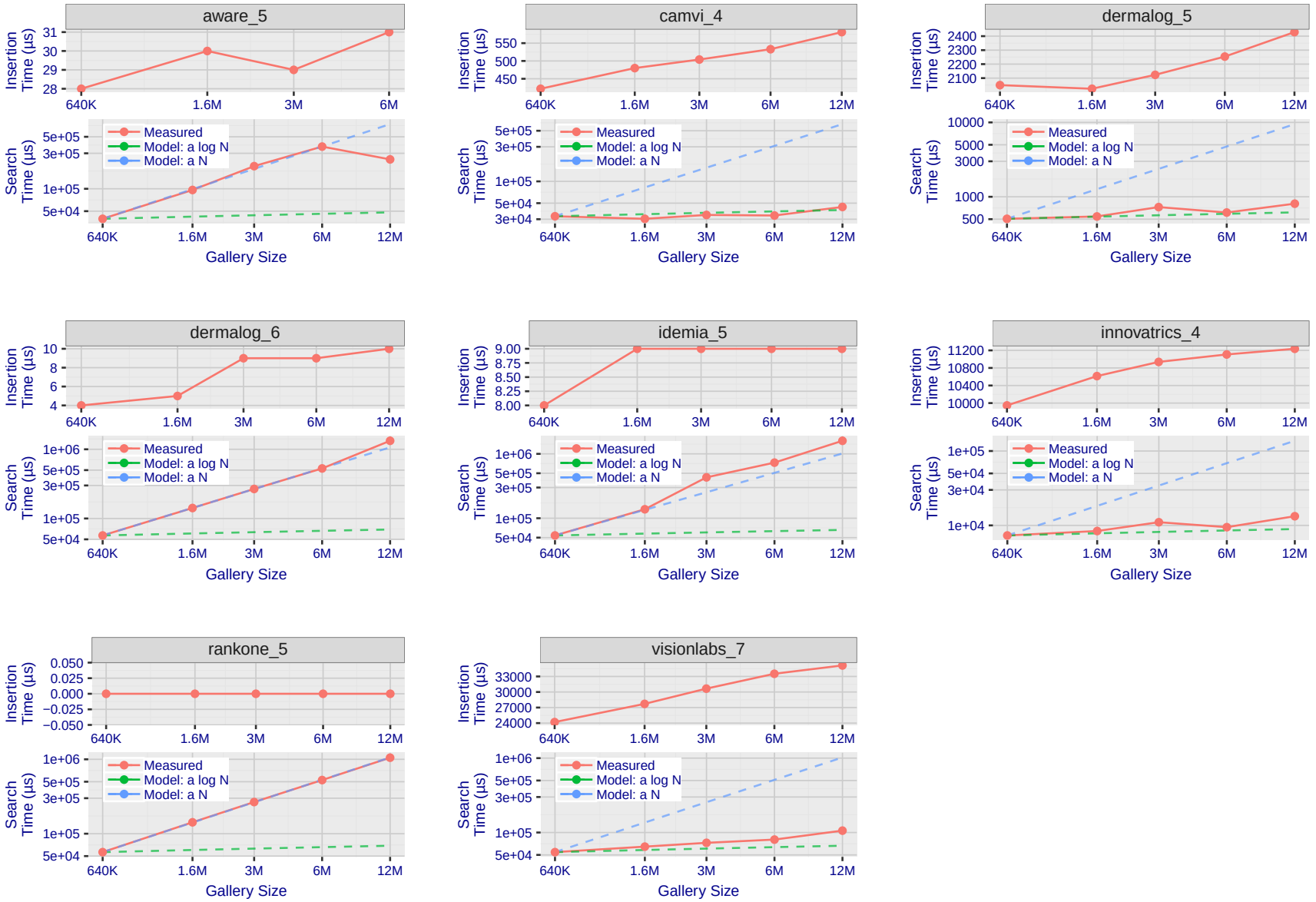


*Figure 500:* **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with $N$ up to* 12 000 000. *Generally, only the more accurate algorithms were run on galleries with $N$ up to* 12 000 000.



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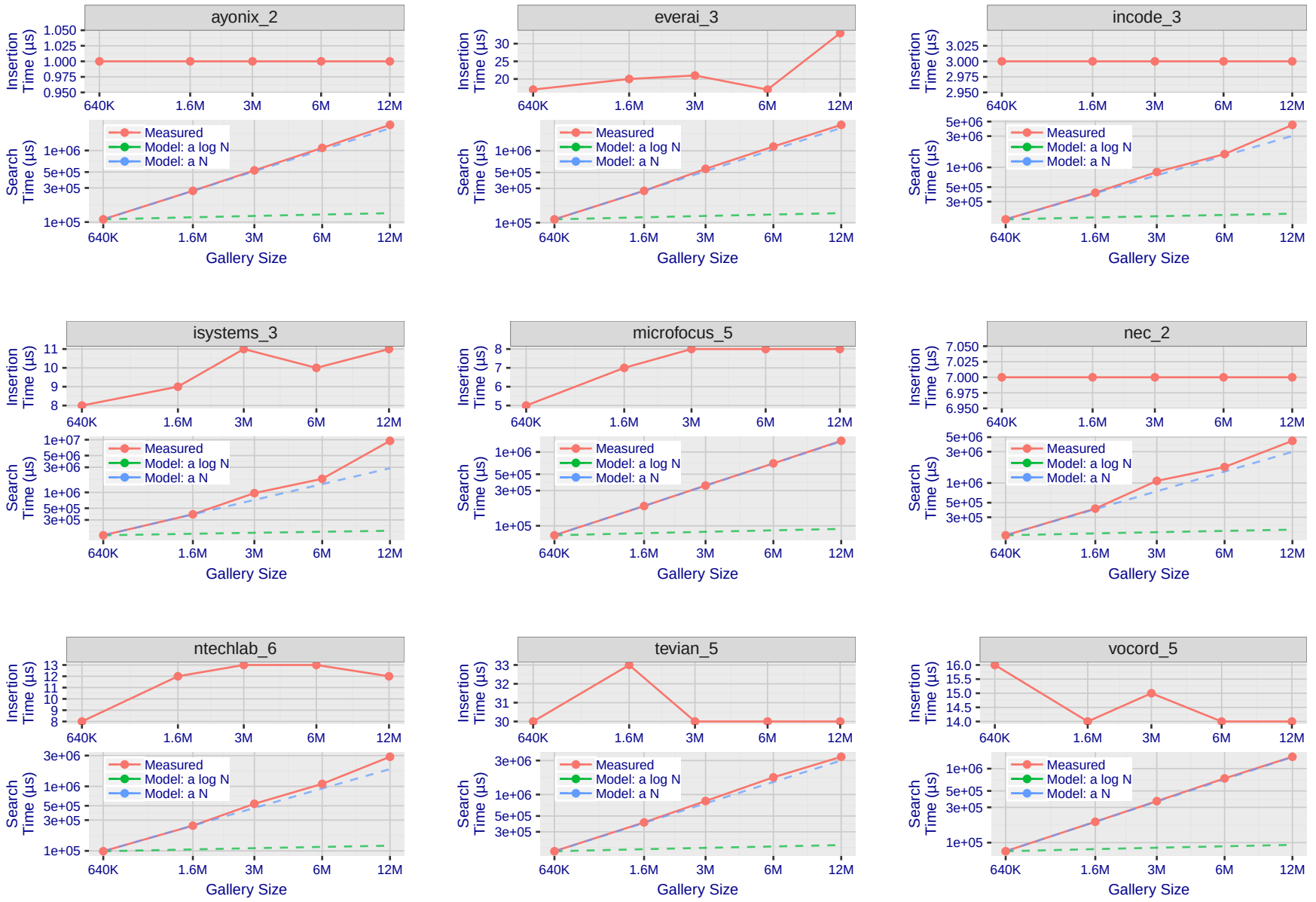


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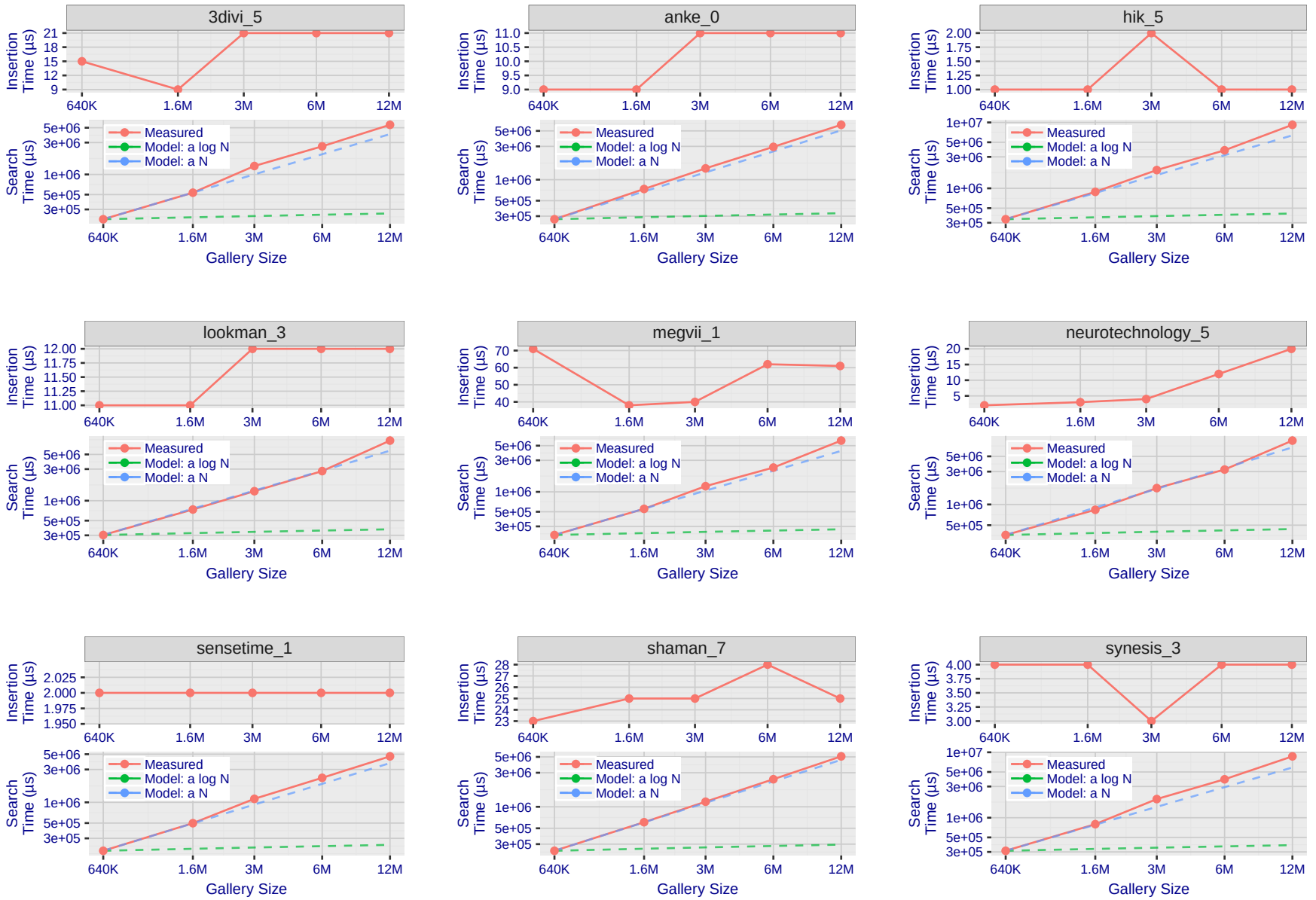


*Figure 502:* **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with $N$ up to* 12 000 000. *Generally, only the more accurate algorithms were run on galleries with $N$ up to* 12 000 000.

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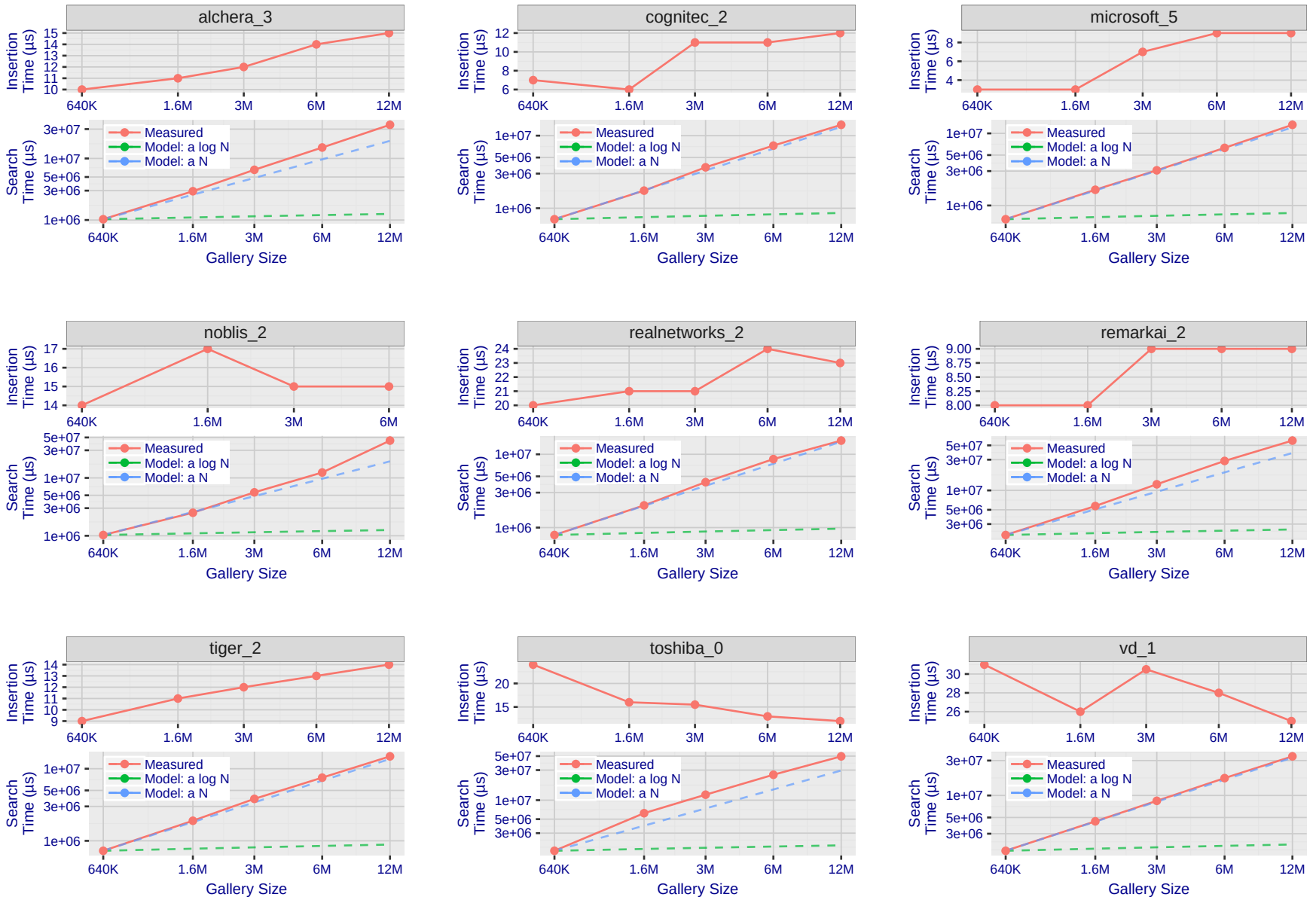


*Figure 503:* **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with $N$ up to* 12 000 000. *Generally, only the more accurate algorithms were run on galleries with $N$ up to* 12 000 000.



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